

Intelligent Security API

Developer Guide

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1 Reading Guide

Chapter	Description
Overview	Includes the ISAPI overview, applicable products, terms and definitions, abbreviations, and update history.
ISAPI Framework	Read the chapter to take a quick look at the ISAPI framework and basic functions.
Quick Start Guide	Read the chapter to quickly understand the programming process of basic functions such as authentication, message parsing, real-time live view, playback, and event uploading.
API Reference	Start programming according to API definitions.
How-To Video Guidance	How-to videos demonstrate detailed steps of different integration tasks.

2 Overview

2.1 Introduction

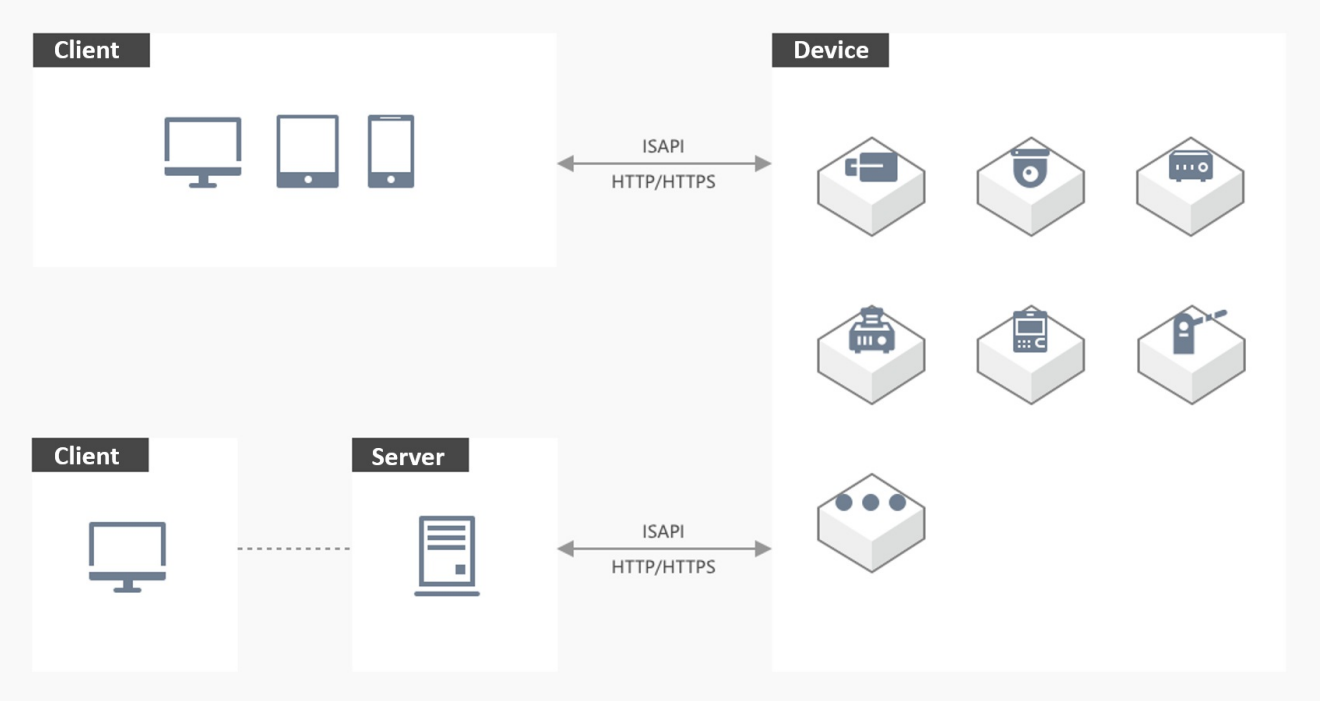
Intelligent Security API (hereinafter referred to as ISAPI) is an application layer protocol based on HTTP (Hypertext Transfer Protocol) and adopts the REST (Representational State Transfer) architecture for communication between security devices (cameras, DVRs, NVRs, etc.) and the platform or client software.

Since established in 2013, ISAPI has included more than 11,000 APIs for different functions, including device management, vehicle recognition, parking lot management, intelligent facial application, access control management, interrogation management, and recording management. It is applicable to industries such as traffic, fire protection, education, and security inspection.

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2.1.1 Application Scenario

When you integrate devices via ISAPI, the device acts as the server to listen on the fixed port and the user's application acts as the client to actively log in to the device for communication. To achieve the above goals, the device should be configured with a fixed IP address and the requests from the client can reach the server.



2.1.2 Layers in the Network Model

ISAPI is an application layer protocol based on HTTP, thereby it inherits all specifications and properties from HTTP.

Protocols frequently used along with ISAPI include SADP (Search Active Device Protocol) based on multicast for discovering and activating devices, RTSP (Real-Time Streaming Protocol) based on TCP/UDP for live view and video playback of devices, etc.

Application Layer	Device Discovery SADP	Signaling Interaction ISAPI (HTTP)	Media Stream RTSP/RTP
Transport Layer	MCAST	TCP	TCP/UDP
Network Layer	IP		
Network Interface	Hardware Drive Interface		

2.2 Product Scope

- Face Recognition Terminals
  - Value Series

    DS-K1A330, DS-K1A330W, DS-K1A340FWX, DS-K1A340FX, DS-K1A340WX, DS-K1A340X, DS-K1T320EFWX, DS-K1T320EFWX-B, DS-K1T320EFX, DS-K1T320EWX, DS-K1T320EX, DS-K1T320MFWX, DS-K1T320MFWX-B, DS-K1T320MFX, DS-K1T320MWX, DS-K1T320MX, DS-K1T321EFWX, DS-K1T321EFWX-B, DS-K1T321EFX, DS-K1T321EWX, DS-K1T321EX, DS-K1T321MFWX, DS-K1T321MFWX-B, DS-K1T321MFX, DS-K1T321MWX, DS-K1T321MX, DS-K1T323EBFWX-E1, DS-K1T323EBWX-E1, DS-K1T323EBWX-QRE1, DS-K1T323MBFWX-E1, DS-K1T323MBWX-E1, DS-K1T323MBWX-QRE1, DS-K1T331, DS-K1T331W, DS-K1T341A, DS-K1T341AM, DS-K1T341AMF, DS-K1T341B, DS-K1T341BMI-T, DS-K1T341BMW, DS-K1T341BMWI-T, DS-K1T341CM, DS-K1T341CMF, DS-K1T341CMFW, DS-K1T341CMW, DS-K1T341M, DS-K1T342DWX, DS-K1T342DWX-E1, DS-K1T342DX, DS-K1T342DX-E1, DS-K1T342EFWX, DS-K1T342EFWX-E1, DS-K1T342EWX, DS-K1T342EWX-E1, DS-K1T342EX, DS-K1T342EX-E1, DS-K1T342MFWX, DS-K1T342MFWX-E1, DS-K1T342MFX, DS-K1T342MFX-E1, DS-K1T342MWX, DS-K1T342MWX-E1, DS-K1T342MX, DS-K1T342MX-E1, DS-K1T343EFWX, DS-K1T343EFWX-B, DS-K1T343EFX, DS-K1T343EWX, DS-K1T343EX, DS-K1T343MFWX, DS-K1T343MFWX-B, DS-K1T343MFX, DS-K1T343MWX, DS-K1T343MX, DS-K1T344EBFWX-E1, DS-K1T344EBWX-E1, DS-K1T344EBWX-QRE1, DS-K1T344EX-E1, DS-K1T344MBFWX-E1, DS-K1T344MBWX-E1, DS-K1T344MBWX-QRE1, DS-K1T344MFX-E1, DS-K1T344MX-E1, DS-K1T604, DS-K1T605, DS-K1T606, DS-K1T607, DS-K1T607E, DS-K1T607EF, DS-K1T607M, DS-K1T607MF, DS-K1T607MFW, DS-K1T607MW, DS-K1T607P, DS-K1T607PE, DS-K1T607PEF, DS-K1T607PMFW, DS-K1T607PMW, DS-K1T607T, DS-K1T607TE, DS-K1T607TEF, DS-K1T607TM, DS-K1T607TMF, DS-K1T607TMFW, DS-K1T640, DS-K1T642, DS-K1T642E, DS-K1T642EF, DS-K1T642EFW, DS-K1T642EW, DS-K1T642M, DS-K1T642MF, DS-K1T642MFW, DS-K1T642MW, DS-K1T643MWX-T, DS-K1T643MX-T, DS-K1T670MFWX, DS-K1T670MWX-QR, DS-K1T670MWX-WBQR, DS-K1T670MWX-WEQR, DS-K1T670MX, DS-K1T670MX-QR, DS-K1T670MX-WB, DS-K1T670MX-WE, DS-K1T671, DS-K1T671AM, DS-K1T671AMW, DS-K1T671BM, DS-K1T671BMF, DS-K1T671BMFW, DS-K1T671BMW, DS-K1T671BTM, DS-K1T671BTMF, DS-K1T671BTMFW, DS-K1T671BTMW, DS-K1T671M, DS-K1T671MF, DS-K1T671T, DS-K1T671TM, DS-K1T671TM-3XF, DS-K1T671TMF, DS-K1T671TMFW, DS-K1T671TMW, DS-K1T672, DS-K1T672DWX-T, DS-K1T672DX-T, DS-K1T672E, DS-K1T672M, DS-K1T672MW, DS-K1T673DBWFX, DS-K1T673DBWFX-QR, DS-K1T673DG1X-E1, DS-K1T673DGX, DS-K1T673DGX-E1, DS-K1T673DWX, DS-K1T673DWX-E1, DS-K1T673DWX-PRO, DS-K1T673DWX-PROE1, DS-K1T673DX, DS-K1T673DX-E1, DS-K1T673TDGX, DS-K1T673TDWX, DS-K1T673TDWX-E1, DS-K1T673TDWX-PRO, DS-K1T673TDWX-PROE1, DS-K1T673TDX, DS-K1T673TDX-E1, DS-K1T673TDX-T, DS-K1T674MWX, DS-K1T680D, DS-K1T680D-E1, DS-K1T680DF, DS-K1T680DF-E1, DS-K1T680DFG, DS-K1T680DFW, DS-K1T680DG, DS-K1T680DW, DS-

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2.3 Terms And Definitions

2.3.3 Event

It refers to the information uploaded by the device. The event is uploaded by the device in real time for the immediate response from the client or the platform. If the device is offline, the event will be stored in the cache first and then it will be uploaded again when the connection is restored.

2.3.2 Arming

Arming means that the client establishes connection with the device so that events can be uploaded to the client via the connection. The client can subscribe to some event types, and the device will upload the specified events only, otherwise the device will upload all types of events to the client.

2.3.4 Listen

After the platform starts listening service, when an event occurs, the event information will be sent to the listening port of the platform based on the IP address and port No., then the connection will be closed.

2.3.5 Listening Host

The listening host service can be enabled to receive the event information from devices.

2.3.16 Person-Based Access Control

The person ID is the unique identifier for person management, and the card, fingerprint, face picture are the person's credentials which are linked via the person ID. Permissions are also linked via the person ID.

2.3.17 Access Permission

Users can set the access permission that which person can open which doors at what time.

2.3.10 Credential Type

Credentials are the data which are used for recognizing specific persons. Cards, fingerprints, and face pictures can be the credentials and linked to the specified persons. When the device detects the credential, it can recognize the person whom the credential links to via the comparison algorithm.

2.3.11 Person Type

Persons can be divided into normal persons, visitors, and blocklist persons according to different areas. In the access control system, normal persons have permanent permissions to access the specified areas, visitors have temporary permissions to access the specified areas, and blocklist persons do not have the permissions to access the specified areas.

2.3.12 Group

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A group of persons which is used in multi-factor authentication. The same person can belong to different groups at the same time. Up to 4 groups can be added at the same time.

2.3.13 Multi-Factor Authentication

Persons in the group can only open the door by the configured authentication rule, such as swiping card, authenticated by fingerprint, face picture, or iris.

2.3.18 Device Number

Generally refers to the private SIP number of devices in the building industry. It is an 11-digit number composed of the phase number, building number, unit number, floor number, serial number, and community No. It is a unique ID for devices during call intercom via the private SIP protocol.

2.3.19 New Device Number

For further expansions and differences in device number rules across various industries (such as health care and prisons), the new device number is a 16-digit number composed of four factors: industry, device type, version, and original device number. The format is: industry # device type # version # device number.

2.3.20 Alarm Receiving Center

The ARC receives alarm information from devices and provides alarm services.

2.3.21 Event Code

An event code is used to identify a specific event and corresponds to a CID code.

2.4 Symbols And Acronyms

ISAPI: Intelligent Security API.

ISUP: Intelligent Security Uplink Protocol.

HTTP: Hypertext Transfer Protocol

FDLib: Face Picture Library

FDID: Face Picture Library ID

PID: Face Picture ID in the Face Picture Library

ACS: Access Control System. The access control system controls the entrance and exit channels. The system consists of card readers, access controllers, electric locks, exit buttons, cards, application software, etc.

ARC: Alarm Receiving Center

CID: Event Code (Contac ID)

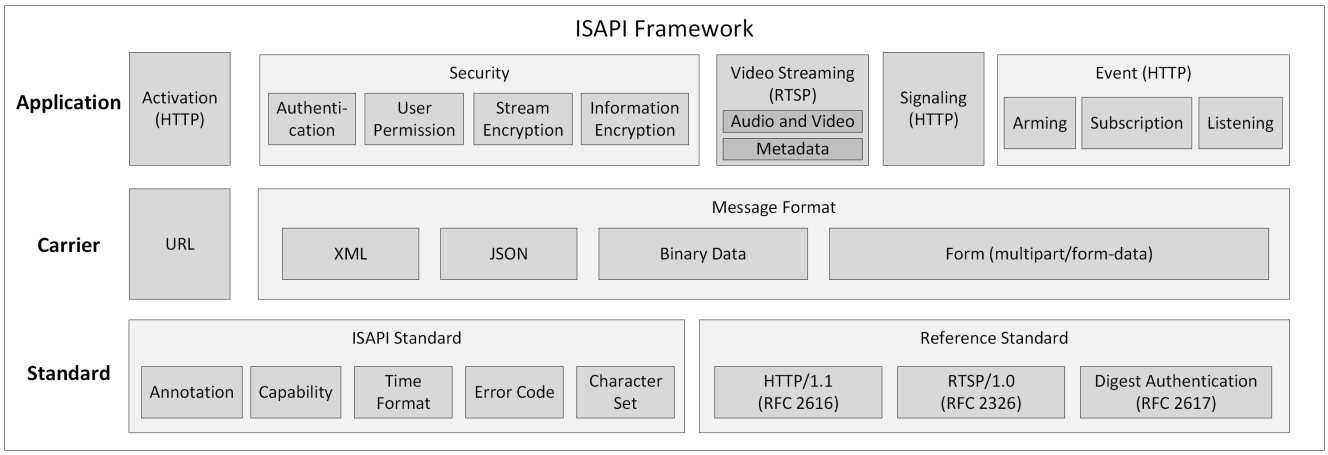
2.5 Update History

No update record

3 ISAPI Framework

3.1 Overview

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Notes:

In general, ISAPI refers to the communication protocol based on the HTTP standard. As ISAPI is usually used along with RTSP (Real-Time Streaming Protocol), the RTSP standard is brought into the ISAPI system.

The metadata scheme for transmitting additional information of the stream is extended based on the RTSP standard to transmit the video stream and the structured intelligent information of the stream simultaneously. It is compatible with the RTSP standard.

3.2 Activation

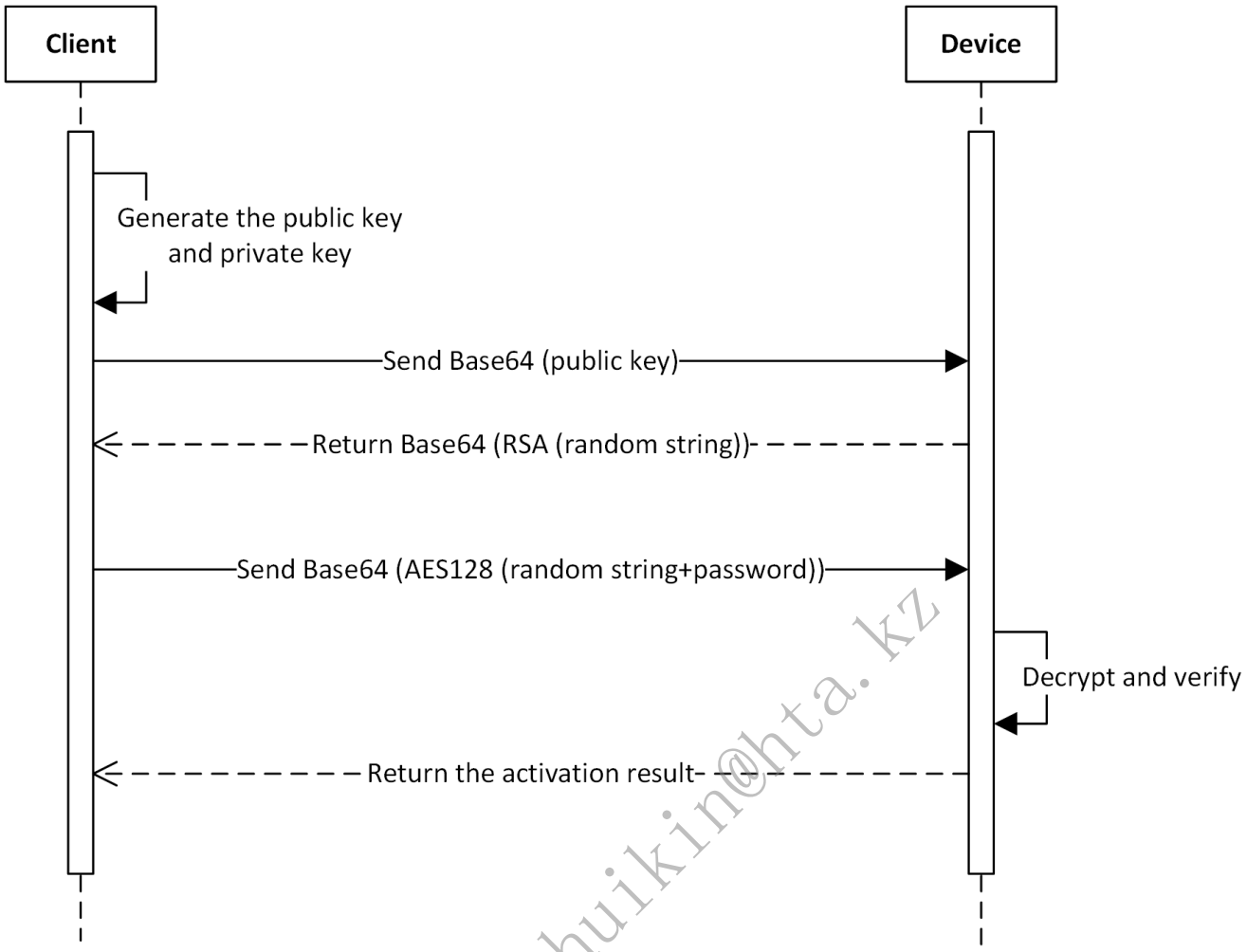
The purpose of activation is to ensure that the user can set the password for the device and the password meets the security requirement. After the device is activated, you can use the related functions.

ISAPI is a communication protocol running on the application layer. When activating the device via ISAPI, you should know the device's IP address and make sure that the device is connected to the client.

The web application built in the device supports activating the device via ISAPI. When you enter the device's IP address in the address bar of the web browser on the PC, you can activate the device according to the activation guide.

If you want to activate the device on your own application, you need to integrate the activation function via ISAPI. The API calling flow and related APIs are shown below.

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Firstly, two operations are defined:

- bytesToHexstring: it is used to convert a byte array (the length is N) to a hexadecimal string (the length is 2N). For example, `127,10,23` -> `7f0a17`
- hexStringToBytes: it is used to convert a hexadecimal string (the length is 2N) to a byte array (the length is N). For example, `7f0a17` -> `127,10,23`

1. The client generates a public and private key pair (1024 bits), and gets the 128-byte modulus in the public key (hereinafter referred to as public key modulus). If the length is longer than 128, the leading 0 needs to be removed.
2. The client converts the public key modulus to a 256-byte public key string via bytesToHexstring and sends the public key string to the device in XML message (related URI: `POST /ISAPI/Security/challenge`) after being encoded by Base64.
3. The device parses the request to obtain a 256-byte public key string decoded by Base64 and converts it to a 128-byte public key modulus via hexStringToBytes. The complete public key is the combination of obtained public key modulus and public exponent (the default value is `'010001'`).
4. The device generates a 32-byte hexadecimal random string, calls the RSA API to encrypt the random string with the private key, converts the encrypted data to a string via bytesToHexstring, encodes the string by Base64, and then sends it to the client.
5. The client decodes the string from the device by Base64, converts it via hexStringToBytes to get the encrypted data, decrypts the encrypted data with the private key via RSA to obtain a 32-byte hexadecimal random string, converts the obtained string via hexStringToBytes to get a 16-byte AES key. Then the client uses the AES key to encrypt the `"string consisting of the first 16 characters of the random string and the real password"` by AES128 ECB mode (with zero-padding method) to get a ciphertext, and converts the ciphertext via bytesToHexstring, encodes it by Base64, and sends it to the device in XML message (related URI: `PUT /ISAPI/System/activate`). Note: If the first 16 characters of the random string are `aaaabbbbccccdddd` and the real password is `Abc12345`, the data before encryption is `aaaabbbbccccddddAbc12345`. This can ensure that the client uses the random string as the key for encryption.
6. The device decodes the string by Base64, converts it via hexStringToBytes to get the ciphertext, uses the AES key to decrypt the ciphertext by AES128 ECB mode, and gets the real password via removing the first 16 characters.

7. The device verifies the password and returns the activation result.

Notes:

- You can get the device's activation status by calling the URI `GET /SDK/activateStatus` which requires no authentication.
- Devices also support to be activated via SADP (Search Active Device Protocol) which is based on the communication protocol of the data link layer. With SADP, you do not have to know the IP address of the device but need to ensure that the device and the application running SADP are connected to the same router. SADP also supports discovering devices in the LAN, changing the password of the devices, and so on. The HCSadpSDK is provided for SADP integration, including the developer guide, plug-in, and sample demo which can be used as a simple SADP tool.

3.3 Security Mechanism

3.3.1 Authentication

When the client applications send requests to devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Currently, all mainstream request class libraries of HTTP have encapsulated digest authentication. See details in Authentication of Quick Start Guide.

3.3.2 User Permission

There are three kinds of users with different permissions for access control and management.

Administrator: Has the permission to access all supported resources and should keep activated all the time. It is also known as "admin".

Operator: Has the permission to access general resources and a part of advanced resources.

Normal User: Only has the permission to access general resources.

3.3.3 Information Encryption

During ISAPI integration, the HTTPS service of devices is enabled by default. When the client applications communicate with devices via HTTPS, the information can be transmitted securely.

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3.4 Video Streaming

3.4.1 Audio and Video Stream

ISAPI supports getting and setting stream media parameters of the device, such as video resolution, encoding format, and stream.

Cameras support standard RTSP (Real-Time Streaming Protocol, see details in RFC 7826). Client applications can get the stream from devices via RTSP.

For details about real-time streaming and video playback, refer to **Real-Time Live View** and **Playback** in Quick Start Guide.

3.4.2 Metadata

The metadata is the structured intelligent information generated by intelligent devices. When the client applications get the audio and/or video stream from devices via RTSP, the metadata will be returned by the device at the same time. For example, to display the face target frame, face information, vehicle target frame, license plate number, vehicle information, and other information on the video stream, the client applications can overlay the above information on the video image.

Before using the metadata, you need to enable the metadata function of the device and then get the stream from the device via RTSP. Some devices support subscribing to the metadata by type. For details about the process of integrating the metadata function, refer to **Metadata Management**.

4 Quick Start Guide

4.1 Authentication

When the client applications send requests to the devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Client applications only need to call APIs of the class library to implement the digest authentication. The sample code is shown below.

4.1.1 C/C++ (libcurl)

```
// #include <curl/curl.h>
// Callback Function
static size_t OnWriteData(void* buffer, size_t size, size_t nmemb, void* lpVoid)
{
    std::string* str = dynamic_cast<std::string*>((std::string *)lpVoid);
    if( NULL == str || NULL == buffer )
    {
        return -1;
    }

    char* pData = (char*)buffer;
    str->append(pData, size * nmemb);
    return nmemb;
}

std::string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
std::string strResponseData;
CURL *pCurlHandle = curl_easy_init();
curl_easy_setopt(pCurlHandle, CURLOPT_CUSTOMREQUEST, "GET");
curl_easy_setopt(pCurlHandle, CURLOPT_URL, strUrl.c_str());
// Set the user name and password
curl_easy_setopt(pCurlHandle, CURLOPT_USERPWD, "admin:admin12345");
// Set the authentication method to the digest authentication
curl_easy_setopt(pCurlHandle, CURLOPT_HTTPAUTH, CURLAUTH_DIGEST);
// Set the callback function
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEFUNCTION, OnWriteData);
// Set the parameters of the callback function to get the returned information
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEDATA, &strResponseData);
// Timeout settings for receiving the data. If receiving data is not completed within 5 seconds, the application will exit directly
curl_easy_setopt(pCurlHandle, CURLOPT_TIMEOUT, 5);
// Set the redirection times to avoid too many redirections
curl_easy_setopt(pCurlHandle, CURLOPT_MAXREDIRS, 1);
// Connection timeout duration. If the duration is too short, the client application will be disconnected before the data request sent by the application
reaches the device
curl_easy_setopt(pCurlHandle, CURLOPT_CONNECTTIMEOUT, 5);
CURLcode nRet = curl_easy_perform(pCurlHandle);
if (0 == nRet)
{
    // Output the received message
    std::cout << strResponseData << std::endl;
}
curl_easy_cleanup(pCurlHandle);
```

4.1.2 C# (WebClient)

```
// using System.Net;
// using System.Net.Security;
try
{
    string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
    WebClient client = new WebClient();
    // Set the user name and password
    client.Credentials = new NetworkCredential("admin", "admin12345");
    byte[] responseData = client.DownloadData(strUrl);
    string strResponseData = Encoding.UTF8.GetString(responseData);
    // Output received information
    Console.WriteLine(strResponseData);
}
catch (Exception ex)
{
    Console.WriteLine(ex.Message);
}
```

4.1.3 Java (HttpClient)

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```java
// import org.apache.commons.httpclient.HttpClient;
String url = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
HttpClient client = new HttpClient();
// Set the user name and password
UsernamePasswordCredentials creds = new UsernamePasswordCredentials("admin", "admin12345");
client.getState().setCredentials(AuthScope.ANY, creds);
GetMethod method = new GetMethod(url);
method.setDoAuthentication(true);
int statusCode = client.executeMethod(method);
byte[] responseData = method.getResponseBodyAsString().getBytes(method.getResponseCharSet());
String strResponseData = new String(responseData, "utf-8");
method.releaseConnection();
// Output received information
System.out.println(strResponseData);
```

4.1.4 Python (requests)

```python
# - *- coding: utf-8 -*-

import requests
request_url = 'http://192.168.18.84:80/ISAPI/System/deviceInfo'
# Set the authentication information
auth = requests.auth.HTTPDigestAuth('admin', 'admin12345')
# Send the request and receive response
response = requests.get(request_url, auth=auth)
# Output response content
print(response.text)
```

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4.2 Message Parsing

4.2.1 Message Format

During the process of communication and interaction via ISAPI, the request and response messages are often text data in XML or JSON format. Besides that, the data of firmware packages and configuration files is in binary format. A request can also be in form format with multiple formats of data (multipart/form-data).

4.2.1.1 XML

Generally, the `Content-Type` in the headers of the HTTP request is `application/xml; charset="UTF-8"`.

Request and response messages in XML format are all encoded with UTF-8 standards in ISAPI.

The namespace `http://www.isapi.org/ver20/XMLSchema` and ISAPI version number `2.0` of XML messages are configured by default, see the example below.

```xml
<?xml version="1.0" encoding="UTF-8"?>
<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <Node>
    <id>1</id>
    <enabled>true</enabled>
    <nodeName>nodeName</nodeName>
    <level>level1</level>
  </Node>
</NodeList>
```

4.2.1.2 JSON

The `Content-Type` in the headers of the HTTP request is often `application/json`.

To distinguish between APIs with XML messages and those with JSON messages, ISAPI adds the query parameter format=json to all request URLs with JSON messages, e.g., `http://192.168.1.1:80/ISAPI/System/Sensor/thermometrySensor?format=json`. Messages of request URLs without the query parameter format=json are usually in XML format. However, there may be some exceptions, and the message format is subject to the API definition.

Request and response messages in JSON format are all encoded by UTF-8 in ISAPI.

4.2.1.3 Binary Data

For the firmware and configuration files, the `Content-Type` in the header of an HTTP request is often `application/octet-stream`.

4.2.1.4 Form (multipart/form-data)

When multiple pieces of data are submitted at the same time in an ISAPI request (e.g., the person information and face picture need to be submitted at the same time when a face record is added to the face picture library), the `Content-Type` in the header of the corresponding HTTP request is usually `multipart/form-data, boundary=AaB03x`, where the boundary is a variable used to separate the entire HTTP body into multiple units and each unit is a piece of data with its own headers and body. In `Content-Disposition` of form unit headers, the `name` property refers to the form unit name, which is required for all form units; the `filename` property refers to the file name of form unit body, which is required only when the form unit body is a file. In headers of form units, `Content-Length` refers to the body length, which starts after CRLF(`\r\n`) and ends before two hyphens (`--`) of next form. There should be a CRLF used as the delimiter of two form units before two hyphens (`--`), and the `Content-Length` of previous form unit does not include the CRLF length. For the detailed format description, refer to RFC 1867 (Form-Based File Upload in HTML). Pay attention to two hyphens (`--`) before and after the boundary.

Notes

- In RFC specifications, it is strongly recommended to contain the field `Content-Length` in the entity header, and there is no requirement that the field `Content-Length` should be contained in the header of each form element. The absence of field `Content-Length` in the header should be considered when the client and device programs parse the form data.
- To avoid the conflict between message content and boundary value, it is recommended to use a longer and more complex string as the boundary value.

The example of ISAPI form data submitted by a client to a device is as follows.

```
POST /ISAPI/Intelligent/FDLib/pictureUpload
Content-Type: multipart/form-data; boundary=e5c2f8c5461142aea117791dade6414d
Content-Length: 56789

--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="PictureUploadData";
Content-Type: application/xml
Content-Length: 1234

<PictureUploadData/>
--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="face_picture"; filename="face_picture.jpg";
Content-Type: image/jpeg
Content-Length: 34567

Picture Data
--e5c2f8c5461142aea117791dade6414d--
```

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The example of ISAPI form data responded from a device to a client is as follows.

In ISAPI messages, when there are multiple form units, three nodes (`pid`, `contentid`, and `filename`) are used for linking form units. The corresponding relations are as follows:

Node Name	Form Field	Description
pid	name	pid in XML/JSON messages corresponds to the name property of Content-Disposition in form headers.
contentid	Content-ID	contentid in XML/JSON messages corresponds to Content-ID in form headers.
filename	filename	filename in XML/JSON messages corresponds to filename property of Content-Disposition in form headers.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=136a73438ecc4618834b999409d05bb9
Content-Length: 56789

--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="mixedTargetDetection";
Content-Type: application/json
Content-Length: 811

{
    "ipAddress": "172.6.64.7",
    "macAddress": "01:17:24:45:D9:F4",
    "channelID": 1,
    "dateTime": "2009-11-14T15:27+08:00",
    "eventType": "mixedTargetDetection",
    "eventDescription": "Mixed target detection",
    "deviceID": "123456789",
    "CaptureResult": [{
        "targetID": 1,
        "Human": {
            "Rect": {
                "height": 1.0,
                "width": 1.0,
                "x": 0.0,
                "y": 0.0
            },
            "contentID1": "humanImage", /*human body thumbnail*/
            "contentID2": "humanBackgroundImage", /*human body background picture*/
            "pId1": "9d48a26f7b8b4f2390c16808f93f3534", /*human body thumbnail ID */
            "pId2": "5EE7078E07BB47CF860DE8E4E9A85F28" /*ID of human body background picture*/
        }
    }]
}
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="9d48a26f7b8b4f2390c16808f93f3534"; filename="humanImage.jpg";
Content-Type: image/jpeg
Content-Length: 34567
Content-ID: humanImage

Picture Data
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="5EE7078E07BB47CF860DE8E4E9A85F28"; filename="humanBackgroundImage.jpg";
Content-Type: image/jpeg
Content-Length: 345678
Content-ID: humanBackgroundImage

Picture Data
--136a73438ecc4618834b999409d05bb9--
```

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4.2.2 Annotation

The field descriptions of ISAPI request and response messages are marked as annotations in the example messages as shown below.

```
<?xml version="1.0" encoding="UTF-8"?>

<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, node list, attr:version{req, string, version No., range:[,]}-->
  <Node>
    <!--ro, opt, object, node information-->
    <id>
      <!--ro, req, int, node No., range:[,], step:, unit:, unitType:-->1
    </id>
    <enabled>
      <!--ro, opt, bool, whether to enable-->true
    </enabled>
    <nodeName>
      <!--ro, req, string, node name, range:[1,32]-->test
    </nodeName>
    <level>
      <!--ro, opt, enum, level, subType:string,
      [level1#Level 1,level2#Level 2,level3#Level 3]-->level1
    </level>
  </Node>
</NodeList>
```

```
{
    "name":  "test",
    /*ro, req, string, name, range:[1,32]*/
    "type":  "type1",
    /*ro, req, enum, type, subType:string, [type1#type 1,type2#type 2]*/
    "enabled":  true,
    /*ro, opt, bool, enable or not, desc:xxxxxxx*/
    "NodeList": {
    /*opt, object, node list, dep:and,{$.enabled,eq,true}*/
        "scene":  1,
        /*req, enum, scene, subType:int, [1#scene 1; 2#scene 2; 3#scene 3]*/
        "ID":  1
        /*req, int, No., range:[1,8], step:, unit:, unitType:*/
    }
}
```

Key annotations are shown in the table below.

Annotation	Description	Remark
ro	Attribute: Read-Only	This field can only be obtained and cannot be edited.
wo	Attribute: Write-Only	This field can only be edited and cannot be obtained.
req	Attribute: Required	This field is required for request messages sent to the device and response messages returned from the device.
opt	Attribute: Optional	This field is optional for request messages sent to the device and response messages returned from the device.
dep	Attribute: Dependent	This field is valid and required when specific conditions are satisfied.
object	Field Type: Object	The field of type object contains multiple sub-fields.
list	Field Type: List	The subType following it refers to the data type of sub-items in the list.
subType	Field Type: String	The range following it refers to the maximum and the minimum string size of the field.
int	Field Type: Int	The range following it refers to the maximum and the minimum value of the field.
float	Field Type: Float	The range following it refers to the maximum and the minimum value of the field.
bool	Field Type: Boolean	The value can be true or false.
enum	Field Type: Enumeration	The subType following it indicates that the enumerators are of type string or int. The [] following the subType contains all enumerators.
subType	Sub-Type of Field	When the type of field is list or enum, the value of subType is the data type of each sub-object.
desc	Field Description	The detailed description of the field.

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4.2.3 Capability Set

ISAPI has designed capability sets for almost all functions, APIs, and fields. URLs for getting the capability set end with `/capabilities`. Some URLs may contain query parameters in the format: `/capabilities?format=json&type=xxx`.

There are two types of fields in the capability message of ISAPI: whether the device supports a function and the value range of a field in an API.

Whether the device supports a function: it is often in the format `isSupportXxxxxxxx`, which indicates that whether the device supports a function and a set of APIs for implementing this function.

The capability message example in JSON format is shown below:

```
{
    "isSupportMap":  true,
    /*ro, opt, bool, whether it supports the e-map function, desc:/ISAPI/SDT/Management/map/capabilities?format=json*/
    "isSupportAlgTrainResourceInfo":  true,
    /*ro, opt, bool, whether it supports only getting the resource information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/ResourceInfo?format=json*/
    "isSupportAlgTrainAuthInfo":  true,
    /*ro, opt, bool, whether it supports ony getting the authorization information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/SoftLock/AuthInfo?format=json*/
    "isSupportAlgTrainNodeList":  true,
    /*ro, opt, bool, whether it supports only getting the node information of the algorithm training platform, desc:/ISAPI/SDT/algorithmTraining/NodeList?
format=json*/
    "isSupportNAS":  true
    /*ro, opt, bool, whether it supports mounting and unmounting NAS, desc:/ISAPI/SDT/Management/NAS/capabilities?format=json*/
}
```

The capability message example in XML format is shown below:

```
<isSupportNetworkStatus>
    <!--ro, opt, bool, whether it supports searching the network status, desc: related API (/ISAPI/System/Network/status?format=json)-->true
</isSupportNetworkStatus>
```

The value range of the field: the maximum value, minimum value, the maximum size, the minimum size, options, and so on of each field of the API.

The example of JSON format is shown below:

```
{
    "boolType": {
        /*req, object, example of the capability of type bool*/
        "@opt": [true, false]
        /*req, array, options, subType: bool*/
    },
    "integerType": {
        /*req, object, example of the capability of type integer*/
        "@min": 0,
        /*ro, req, int, the minimum value*/
        "@max": 100
        /*ro, req, int, the maximum value*/
    },
    "stringType": {
        /*req, object, example of the capability of type string*/
        "@min": 0,
        /*ro, req, int, the minimum string size*/
        "@max": 32
        /*ro, req, int, the maximum string size*/
    },
    "enumType": {
        /*req, object, capability example of type enum*/
        "@opt": ["enum1", "enum2", "enum3"]
        /*req, array, options, subType: string*/
    }
}
```

The example of XML format is shown below:

```
<boolType opt="true,false" def="true">
    <!--ro, opt, bool, example of the capability of type bool-->true
</boolType>
<integerType min="0" max="100">
    <!--ro, opt, int, example of the capability of type int-->50
</integerType>
<stringType min="0" max="64">
    <!--ro, opt, string, example of the capability of type string-->test
</stringType>
<enumType opt="red,white,black" def="red">
    <!--ro, opt, string, example of the capability of type enum-->white
</enumType>
```

Note: For the same capability set, devices of different models and versions may return different results. The values shown in this document are only examples for reference. The capability set actually returned by the device takes precedence.

4.2.4 Time Format

ISAPI adopts ISO 8601 Standard Time Format, which is the same as W3C Standard Date and Time Formats.

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Format: `YYYY-MM-DDThh:mm:ss.sTZD`

```
YYYY = the year consisting of four decimal digits
MM = the month consisting of two decimal digits (01-January, 02-February, and so forth)
DD = the day consisting of two decimal digits (01 to 31)
hh = the hour consisting of two decimal digits (00 to 23, a.m. and p.m. are not allowed)
mm = the minute consisting of two decimal digits (00 to 59)
ss = the second consisting of two decimal digits (00 to 59)
s = one or more digits representing the fractional part of a second
TZD = time zone identifier (Z or +hh:mm or -hh:mm)
```

Example: 2017-08-16T20:17:06.123+08:00 refers to 20:17:06.123 on August 16, 2017 (local time which is 8 hours ahead of UTC). The plus sign (+) indicates that the local time is ahead of UTC, and the minus sign (-) means that the local time is behind UTC.

After the DST is enabled, the local time and time difference will change compared with UTC, and the values of related fields also need to be changed. Disabling the DST will bring into the opposite effect.

Example: In 1986, the DST was in effect from May 4 at 2:00 a.m. (GMT+8). During the DST period, the clocks were moved one hour ahead, which means that there was one less hour on that day. When the DST ends at 2:00 a.m. on September 14, 1986, the clocks were moved one hour back and there was an extra hour on that day. The changes of the time are as follows:

- DST Starts: 1986-05-04T02:00:00+08:00 --> 1986-05-04T03:00:00+09:00
- DST Ends: 1986-09-14T02:00:00+09:00 --> 1986-09-14T01:00:00+08:00

Notes:

- The time difference cannot be simply used to determine the time zone. Because when the DST starts, the time difference will change and it cannot represent the actual time zone.
- Both TZ (UTC time, e.g., 1986-05-03T18:00:00Z) and TD (local time and time difference, e.g., 1986-05-04T02:00:00+08:00) meet the time format standards of ISO 8601. In ISAPI, the TD format is recommended to be used in messages sent from the user applications and the devices.
- A few old-version devices will return the time in TZ format. For representing the time difference information and forward compatibility, an extra field `timeDiff` is added as shown in the example below. User applications need to support both TD format and TZ format when parsing the time in the messages returned by devices.

```
{
    "dateTime": "1986-05-03T18:00:00Z", /*device time. The value in TZ format is the UTC time and the value in TD format is the time difference between the device's local time and UTC*/
    "timeDiff": "+08:00" /*optional, time difference between the local time and UTC time. If this field does not exist, the user application will convert the dateTime into the local time for use*/
}
```

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4.2.5 Character Set

To prevent characters not commonly used from resulting in exceptions in device programs and user applications, ISAPI limits the valid field values of type string to a specific range of characters. Character sets allowed to be used in the fields of type string in ISAPI are listed below.

- Single-byte character set: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and special characters (see details in the table below).
- Multi-byte character set: language characters based on Unicode and encoded by UTF-8 (UTF-8 encoding is a transformation format of Unicode character set. For details, refer to RFC 2044).

No.	Name	Special Character	No.	Name	Special Character
1	Open Parenthesis	(	18	Dollar Sign	$
2	Close Parenthesis	)	19	Percent Sign	%
3	Plus Sign	+	20	Ampersand	&
4	Comma	,	21	Close Single Quotation Mark	'
5	Minus Sign	-	22	Asterisk	*
6	Period	.	23	Slash	/
7	Semicolon	;	24	Smaller Than	<
8	Equal Sign	=	25	Greater Than	>
9	At Sign	@	26	Question Mark	?
10	Open Square Bracket	[	27	Caret	^
11	Close Square Bracket	]	28	Open Single Quotation Mark	`
12	Underscore	_	29	Vertical Bar	|
13	Open Brace	{	30	Tilde	~
14	Close Brace	}	31	Double Quotation Marks	"
15	Space		32	Colon	:
16	Exclamation Mark	!	33	Backslash	|
17	Octothorpe	#			

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The valid characters that can be used in some special fields are listed below.

- User name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 30 in the special character table.
- Password: User Name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 33 in the special character table.
- Names displayed on the UI (device name, person name, face picture library name, etc.): lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters.
- Normal fields of type string support lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters by default.

4.2.6 Error Processing

When requesting via ISAPI failed (the HTTP status code is not 200), the device will return the HTTP status code and ISAPI error code. For HTTP status codes, refer to 10 Status Code Definitions in RFC 2616. For ISAPI error codes, refer to Error Code Dictionary.

Message Example:

```
HTTP/1.1 403 Forbidden
Content-Type: application/json; charset="UTF-8"
Date: Thu, 15 Jul 2021 20:43:30 GMT
Content-Length: 229
Connection: Keep-Alive

{
    "requestURL": "/ISAPI/Event/triggers/notifications/channels/whiteLightAlarm",
    "statusCode": 4,
    "statusString": "Invalid Operation",
    "subStatusCode": "notSupport",
    "errorCode": 1073741825,
    "errorMsg": "notSupport"
}
```

4.3 Event Uploading

When the rules configured on the device are triggered, the device will generate event messages (e.g., motion detection, etc.) and actively upload them to the client. ISAPI supports three methods to receive event messages uploaded by the device, that is, in arming mode, in listening mode, and via subscription.

4.3.1 Arming

The client establishes a HTTP persistent connection with the device to receive event messages from the device.

There are two methods (arming with subscription and arming without subscription) to receive events from the device. The arming without subscription is to get all event messages from the device via HTTP GET method, while the arming with subscription is to get messages of subscribed events via HTTP POST method.

Notes

- ISAPI arming (with or without subscription) uses the HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send event messages continuously, while it's not supported for clients to send any message to the device via the connection.
- When the heartbeat timed out and no message is received from the device, you should terminate the arming connection and try establishing a new one.

4.3.1.1 Arming without Subscription

1. Establish the connection of arming without subscription: `GET /ISAPI/Event/notification/alertStream` and keep the connection alive via configuring `Connection: keep-alive` in HTTP headers on the client.
2. Receive events sent by the device. The event message will be separated and parsed by boundary. For parsing details, see **Event Message Parsing** below.
3. Terminate the arming connection when no event message needs to be received.

Event Message Parsing:

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```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

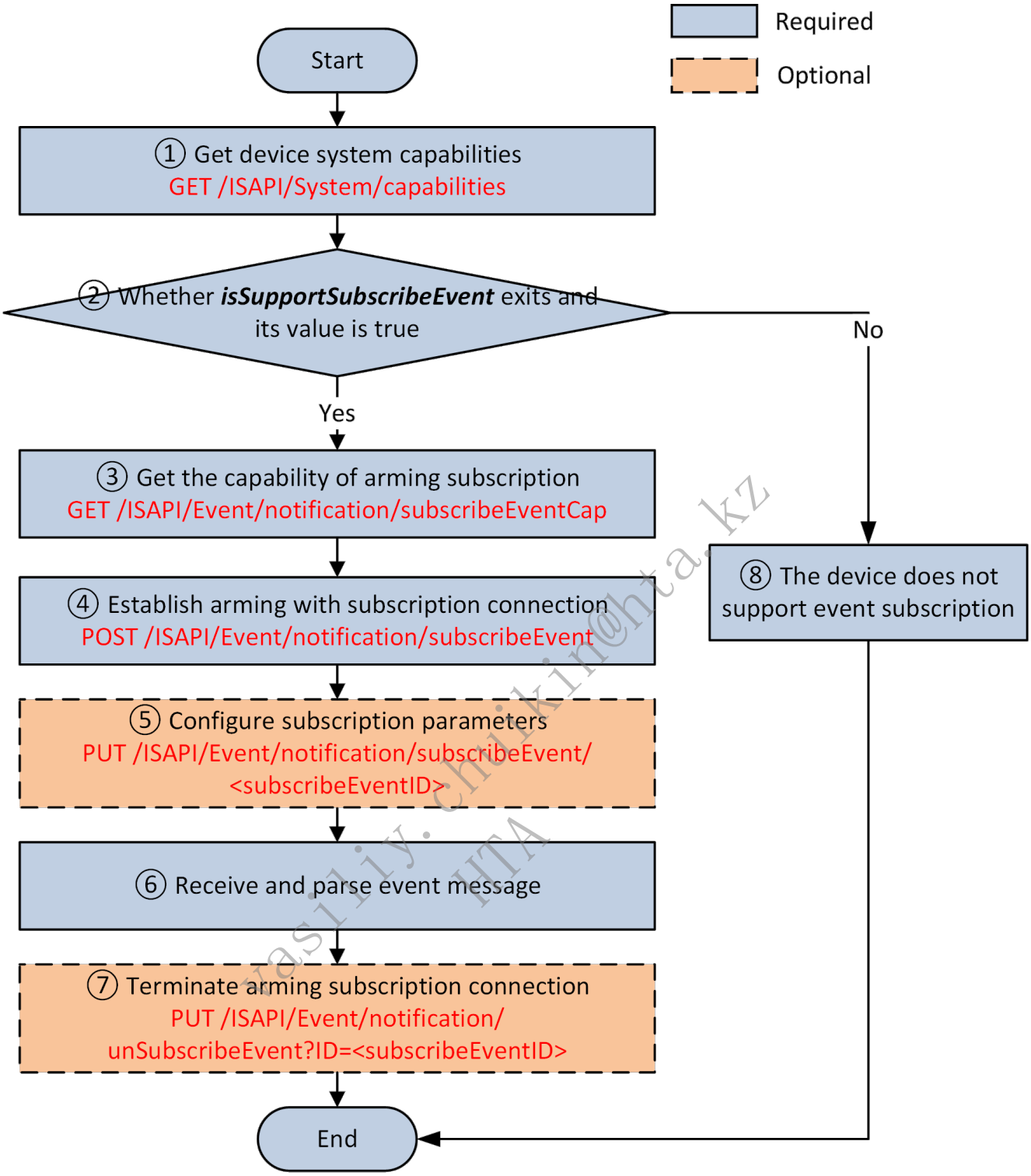
<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

Note: `<data_gateway_ip>` and `<frontier>` are variables, `[Picture Data]` indicates the raw data of a picture.

4.3.1.2 Arming with Subscription

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by the boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note：

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

Event Messages Parsing:

Client Creates Arming with Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

Server Responds to Request and Push Event Message

When the client sends a request to the device for establishing an arming connection, the device sends data in HTTP form format (multipart/form-data). In HTTP request headers, the `Content-Type` is `multipart/form-data, boundary=AaB03x`, of which the boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Please note two hyphens `--` before and after boundary. In normal conditions, devices will not close the arming link actively, and the end mark `--AaB03x--` will not be sent on the arming link.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Picture Data
--AaB03x--
```

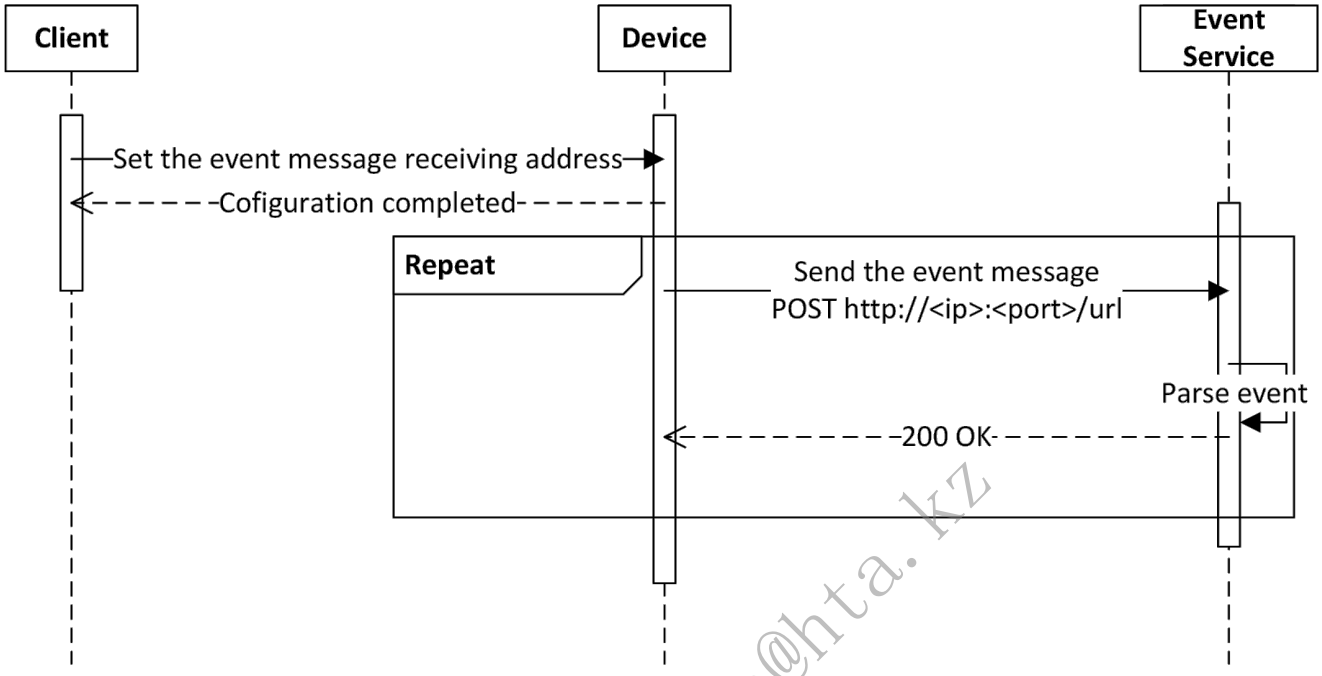
The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. A form message which starts with `--boundary` and ends with `--boundary--`.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description. `form-data` is a piece of form data.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next `\r\n` to the next `--boundary`.

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4.3.2 Listening

After a client enables the listening service, when an event occurs, the device will send the event information actively to the configured event receiving address. The event receiving address should be valid and configured on the device.

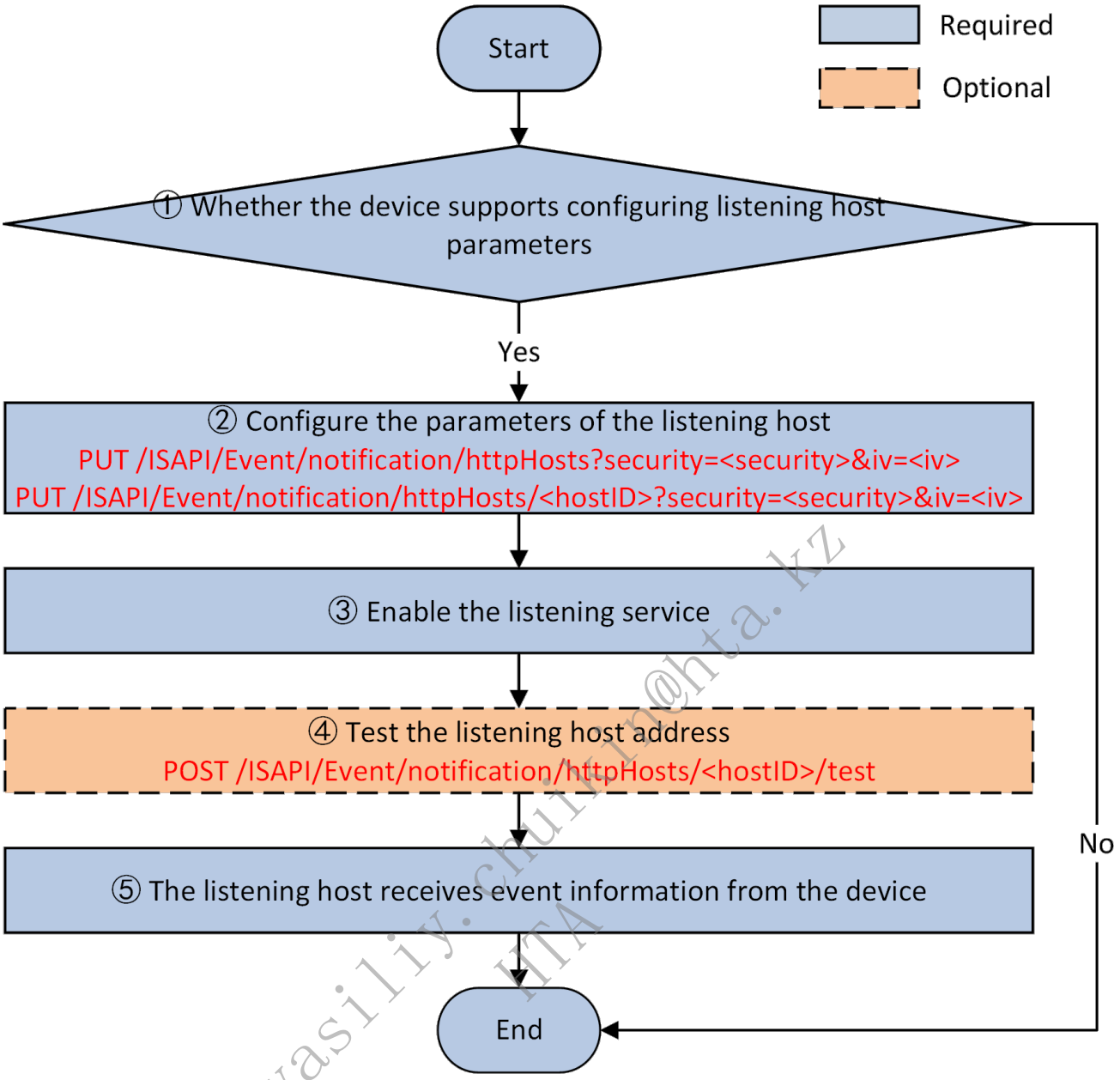


Notes:

- The client and event service can be the same program.
- In listening mode, no heartbeat information is generated on devices.

4.3.2.1 API Calling Flow

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1. Check whether the device supports configuring listening host parameters.

   Get the configuration capability of the listening host: `GET /ISAPI/Event/notification/httpHosts/capabilities`. If the node `<HttpHostNotificationCap>` is returned and its value is true, it indicates that the device supports configuring listening host parameters.

2. Configure parameters of the listening host.

   Configure parameters of all listening hosts: `PUT /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

   Get parameters of all listening hosts: `GET /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

   Configure parameters of a listening host: `PUT /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`;

   Get parameters of a listening host: `GET /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`.

3. Enable the listening service.

   You need to enable the listening service of the listening host.

4. (Optional) Test the listening service.

   The platform applies the command to the device to test whether the listening host is available for the device: `POST /ISAPI/Event/notification/httpHosts/<hostID>/test`.

5. The listening host receives event information from the device.

   When an event occurs, the device creates an connection with the client and uploads alarm information actively. Meanwhile, the listening host receives data from the device. See details in **Event Messages**.

Note: You can also configure the listening parameters such as the time out via URL `/ISAPI/Event/notification/httpHosts/<hostID>/uploadCtrl`.

4.3.2.2 Event Messages

When an event occurs or an alarm is triggered in listening mode, the event/alarm information can be uploaded with binary data (such as pictures) and without binary data.

1. **Without Binary Data:**

The `Content-Type` in headers of the HTTP request sent by the device is usually `application/xml` or `application/json` as follows:

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--Request_URI, related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: data_gateway_ip:port <!--HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: en-us
Date: YourDate
Content-Type: application/xml; <!--Content Type, which is used for the upper Layer to distinguish different formats when parsing the message-->
Content-Length: text_length
Connection: keep-alive  <!--maintain the connection between the device and the server for better transmission performance-->

<EventNotificationAlert/>
```

Response by the Listening Host

```
HTTP/1.1 200 OK
Date: YourDate
Connection: close
```

2. **With Binary Data:**

The format of the data sent by the device is HTTP form (multipart/form-data). The `Content-Type` in headers of the HTTP request is usually `multipart/form-data, boundary=<frontier>`, of which boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. See details in RFC 1867 (Form-based File Upload in HTML). An example is shown below. Please note two hyphens `--` before and after the boundary.

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--Request_URI, , related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: device_ip:port <!--HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: en-us
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive <!--maintain the connection between the device and the server for better transmission performance-->

--<frontier>
Content-Disposition: form-data; name="Event_Type"
Content-Type: text/xml <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message format accroding to
Content-Type when parsing event messages-->

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[Picture Data]
--<frontier>--
```

Response by the Listening Host

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```
HTTP/1.1 200 OK
Date: YourDate
Connection: close
```

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=frontier	Content type, multipart/form-data refers to data in form format.
boundary	frontier	Delimiter of the form message. A form message which starts with `--boundary` and ends with `--boundary--`.
Content-Disposition	form-data; name="Picture_Name";	Content description. `form-data` is a piece of form data.
filename	"Picture_Name"	File name. The file refers to the form message.
Content-Length	10	Content length, starting from the next `\r\n` to the next `--boundary`.

4.3.2.3 Exception Handling

Error Codes

statusCode	statusString	subStatusCode	errorCode	Description
6	Invalid Content	eventNotSupport	0x60001024	

5 Device (General)

5.1 Arming and Subscription

5.1.1 Introduction to the Function

With arming and subscription, the client can establish HTTP persistent connection with the device, and continuously receive the event messages from the device.

For ISAPI event arming, the client can receive all types of events by GET method, or receive the subscribed events only by POST method.

5.1.2 API Calling Flow

5.1.2.1 Without Subscription

1. Establish a connection for arming: `GET /ISAPI/Event/notification/alertStream`. You need to set `Connection: keep-alive` in HTTP Headers.
2. When receiving events sent by the device, the event messages can be separated and parsed by boundary. See "Parsing Event Messages" below for details.
3. Disable the arming connection when you do not need to receive event messages.

5.1.2.1.2 Syntax

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```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/form-data; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!-- Some alarms are in JSON format, so the upper layer should parse based on the Content-Type field -->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/jpeg
Content-Length: image_length

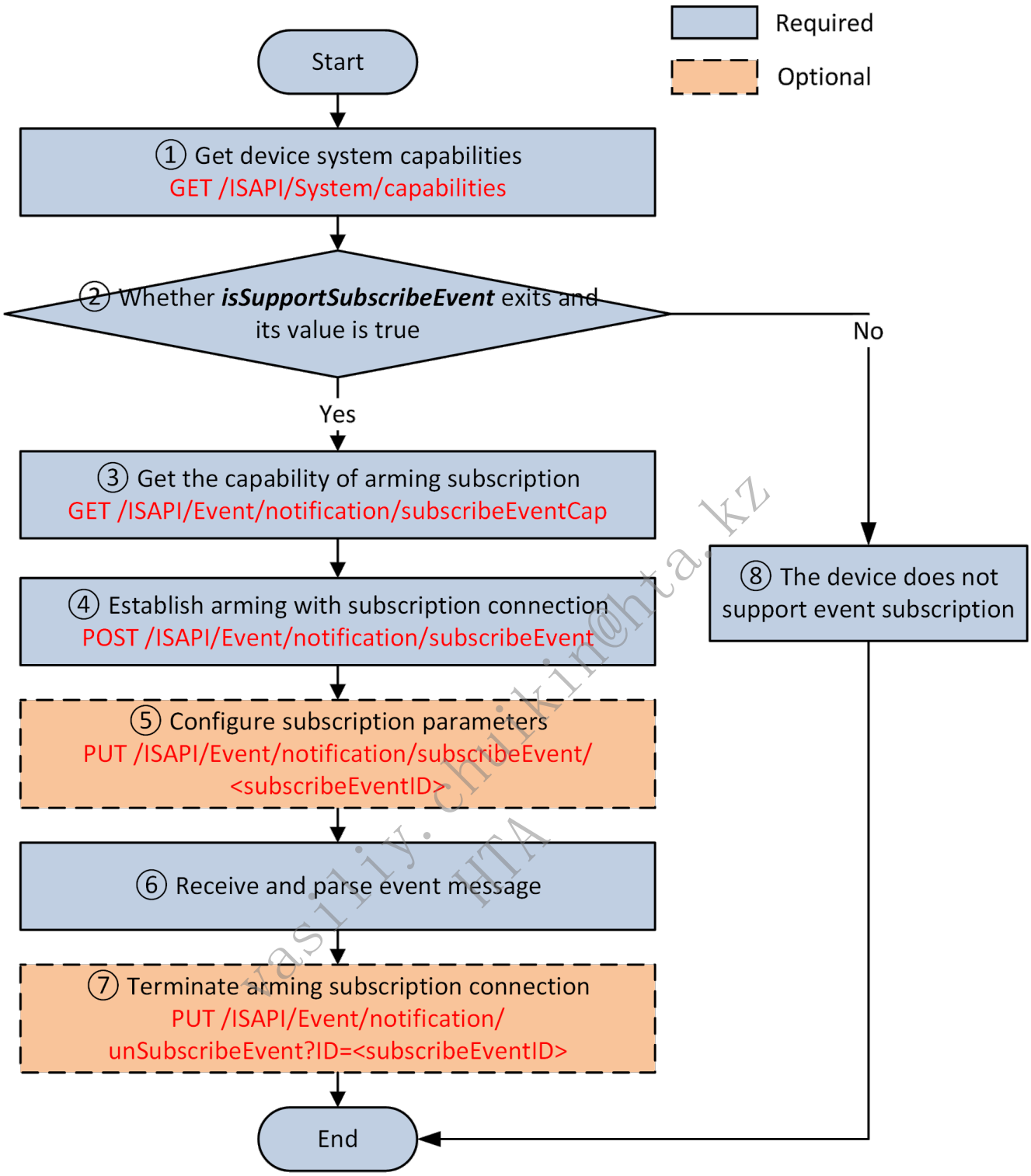
[Image Data]
--<frontier>
```

Note: `<data_gateway_ip>` and `<frontier>` are variables, and `[Image Data]` is an abbreviated representation indicating the raw data of an image at this location.

5.1.2.2 Subscription

5.1.2.2.1 API Calling Flow

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP Headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note:

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

5.1.2.2.2 Example

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Host: device_ip
Accept-Language: zh-cn
Date: YourDate
Content-Type: application/xml;
Content-Length: text_length
Connection: Keep-Alive

<SubscribeEvent/>
HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

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5.1.2.3 Event Messages Parsing

After the arming connection with the device is established, the data sent by the device is in HTTP form format (multipart/form-data). In an HTTP request, `Content-Type` in Headers is usually `multipart/form-data, boundary=AaB03x`, and boundary is a variable used to divide HTTP Body into multiple units, each being a set of data and has its own Headers and Body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Note the dash `--` before and after boundary. Under normal circumstances, the device will not actively close the arming connection, so the device will not send the form format end symbol `--AaB03x--` on the arming connection.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Image Data
--AaB03x--
```

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. --boundary is the start of a form. --boundary-- is the end of the whole HTTP form message.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next \r\n to the next --boundary.

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5.1.3 Restriction Description

Note that ISAPI arming (with or without subscription) uses HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send out event messages continuously, while you cannot send any message to the device via the connection.

After the heartbeat time, if you do not receive any message from the device, you should disable the arming connection and try establishing a new one.

5.1.4 Sample Messages

5.1.4.1 Establish Arming Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

5.1.4.2 The Device Responses and Uploads an Event Message

```
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <ipAddress>10.17.133.46</ipAddress>
    <portNo>80</portNo>
    <protocol>HTTP</protocol>
    <macAddress>44:19:b6:6d:24:85</macAddress>
    <channelID>1</channelID>
    <dateTime>2017-05-04T11:20:02+08:00</dateTime>
    <activePostCount>0</activePostCount>
    <eventType>heartBeat</eventType>
    <eventState>active</eventState>
    <eventDescription>heartBeat</eventDescription>
</EventNotificationAlert>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length
Content-ID: image_ID

[Picture Data]
--<frontier>
```

5.2 Calling Flow of Device Packet Capture

5.2.1 Function Introduction

When problems arise after a device is deployed on site, interaction message between the device and the external network is necessary to help developers for troubleshooting. Packet capture can be stored on the local device, and packet capture files can be exported after capture is complete. Also, packet capture files can be uploaded to cloud storage, and packet capture data can be obtained in real-time even if the device does not have the storage space.

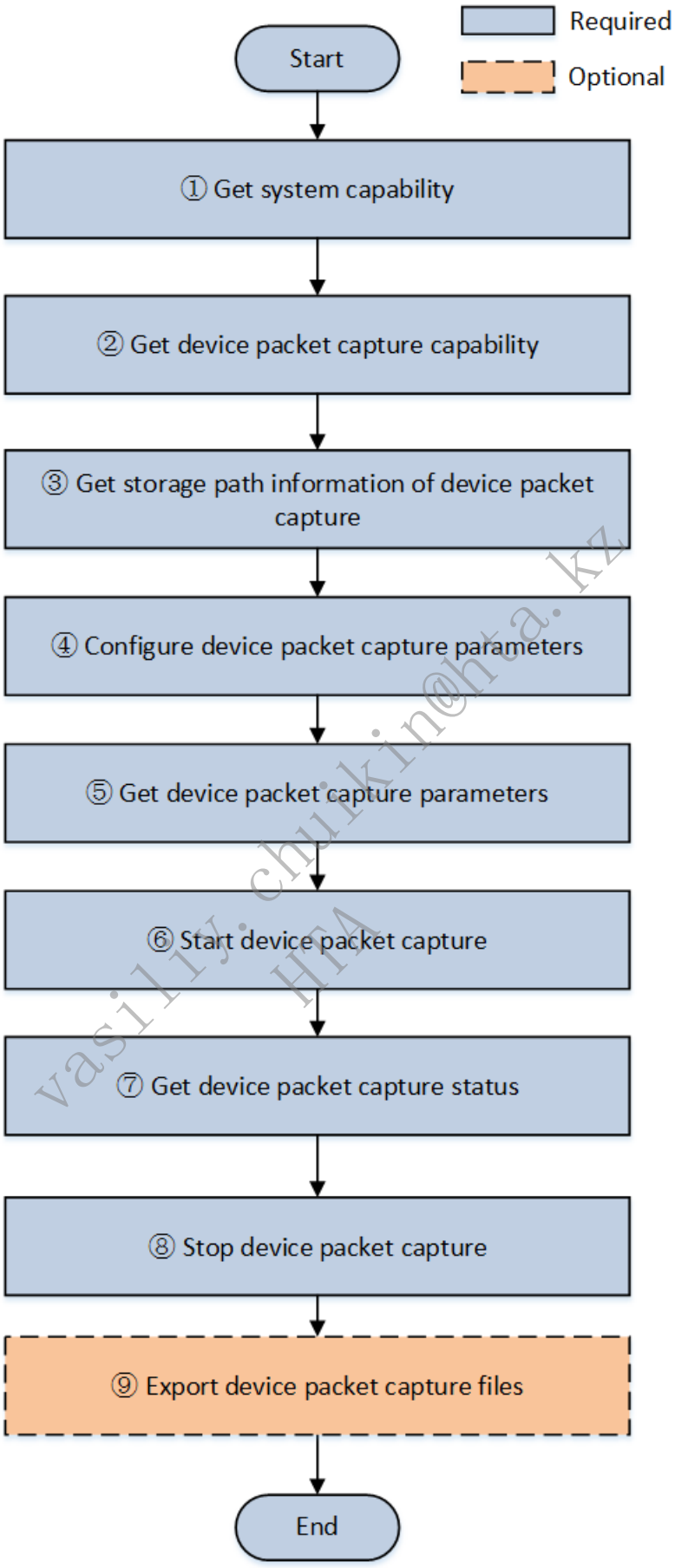
1. Device packet capture: save packet capture files on the local device, and export the files after capture is complete. Also, uploading packet capture files to cloud storage after capture is complete is supported. Then the client can obtain the storage URL and download packet capture files from the cloud storage.

2. Device real-time packet capture: after it is enabled, the device returns an URI for downloading packet capture data. The client can submit this URI to the browser to download the packet data. The device transmits packet data via HTTP Chunked, and users can store the packet data through the browser.

5.2.2 API Calling Flow

5.2.2.1 Device Packet Capture

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`. Get to know if the device supports packet capture by the field: `<isSupportNetworkCapture>true</isSupportNetworkCapture>`.

2. Check if the device supports packet captures: `GET /ISAPI/System/networkCapture/capabilities?format=json`. If `isSupportManualControl` is true, the device supports packet capture. If `isSupportManualControlAsyn` is true, the device supports asynchronous packet capture.

3. Get storage path information of device packet capture: `GET /ISAPI/System/networkCapture/StoragePathInfo?format=json`.

4. Configure device packet capture parameters such as capture duration, storage path, port, and address: `PUT /ISAPI/System/networkCapture/captureParams?format=json`.

5. Get device packet capture parameters such as capture duration, storage path, port, and address: `GET /ISAPI/System/networkCapture/captureParams?format=json`.

6. Start device packet capture: depending on the parameters, packet capture files can be saved on the local device for export after capture is complete, or packet capture files can be uploaded to cloud storage and downloaded via the storage URL, or the packet capture data can be returned in real-time.

Start device packet capture: `PUT /ISAPI/System/networkCapture/manualStart?format=json&asyn=<asyn>&realTime=<realTime>`.

Note:

- If the API does not contain URL parameters, the packet capture file is saved on the local device.
- If the API contains `asyn=true`, the packet capture file is automatically uploaded to cloud storage after capture is complete.

7. After starting capture, you can repeatedly get capture status, including whether the capture is ongoing, the size of the packet capture data, and the progress and storage URL for uploading the data to cloud storage.

Get status of device packet capture: `GET /ISAPI/System/networkCapture/manualStatus?format=json`.

8. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

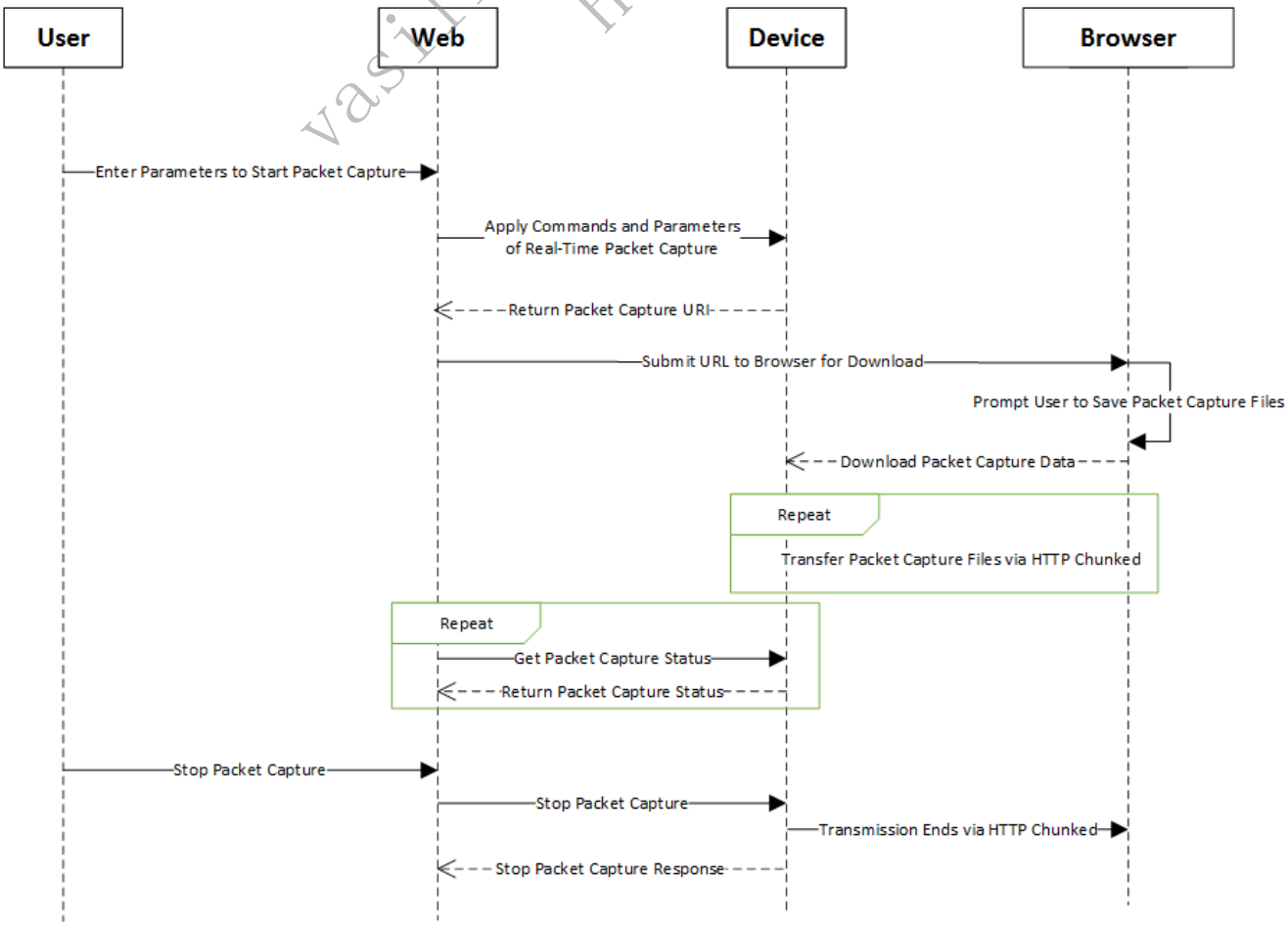
Stop device packet capture: `PUT /ISAPI/System/networkCapture/manualStop?format=json`.

9. (Optional) If packet capture data is stored on the local device, packet capture files need to be exported. If packet capture files are saved to cloud storage or captured in real-time, there is no need to export packet capture files.

Export device packet capture files: `GET /ISAPI/System/networkCapture/exportFile?format=json`.

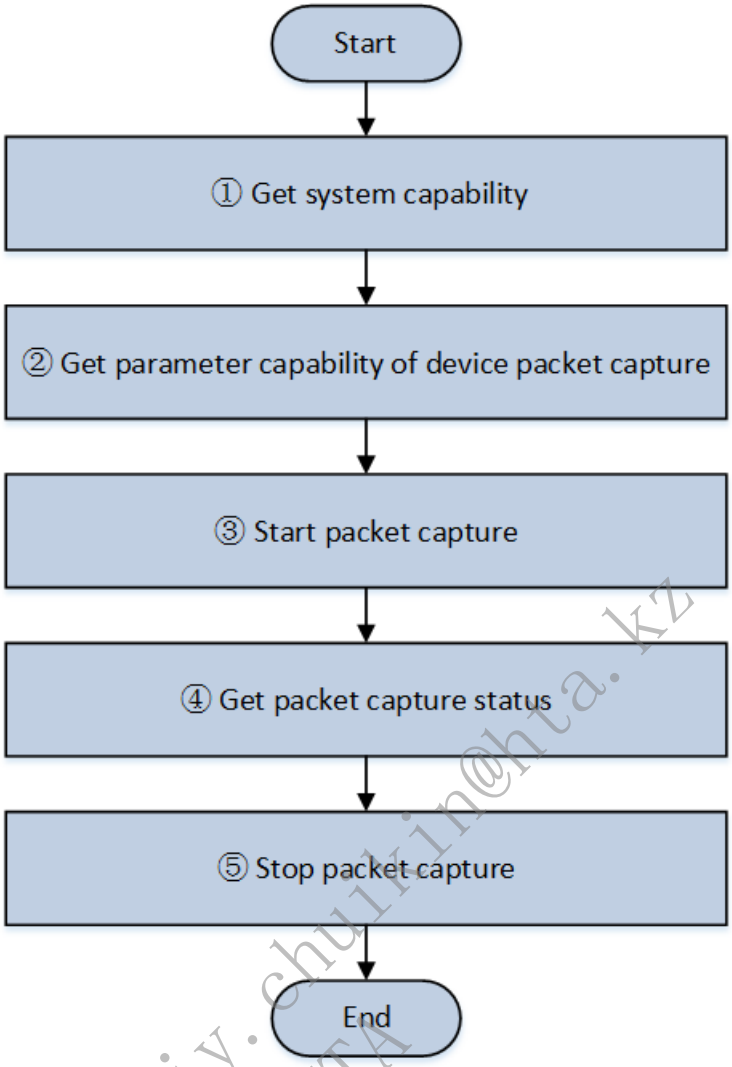
5.2.2.2 Device Real-Time Packet Capture

The process of real-time packet capture is shown as the following:



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See the following figure for the calling flow:



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1. Get device system capabilities: `GET /ISAPI/System/capabilities`. `<isSupportStartNetworkCapture>true</isSupportStartNetworkCapture>` indicates the device supports starting packet capture. `<isSupportStopNetworkCapture>true</isSupportStopNetworkCapture>` indicates the device supports stopping packet capture. `<isSupportGetNetworkCaptureStatus>true</isSupportGetNetworkCaptureStatus>` indicates the device supports getting packet capture status.

2. Get capabilities of packet capture parameters: `GET /ISAPI/System/NetworkCaptureParams/capabilities?format=json`. The `realTimeEnabled` field indicates whether the device supports real-time packet capture.

3. Set the field realTimeEnabled as true in the parameters applied to the device to start device packet capture. The device returns an URI, and the client can download the real-time packet capture data from the device through a browser.

Start packet capture: `POST /ISAPI/System/StartNetworkCapture?format=json&security=<security>&iv=<iv>`.

Note:

- If `realTimeEnabled=true` is contained when starting device packet capture, it indicates packet capture data is uploaded in real-time by HTTP Chunked. The URL for downloading the returned packet capture data is valid for 30 seconds by default. If the download is attempted after this time, the device should return an HTTP 404 status code.

4. After starting packet capture, you can repeatedly get packet capture status, including whether packet capture is ongoing and the size of the packet capture data.

Get packet capture status: `GET /ISAPI/System/GetNetworkCaptureStatus?format=json`.

5. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

Stop device packet capture: `POST /ISAPI/System/StopNetworkCapture?format=json`.

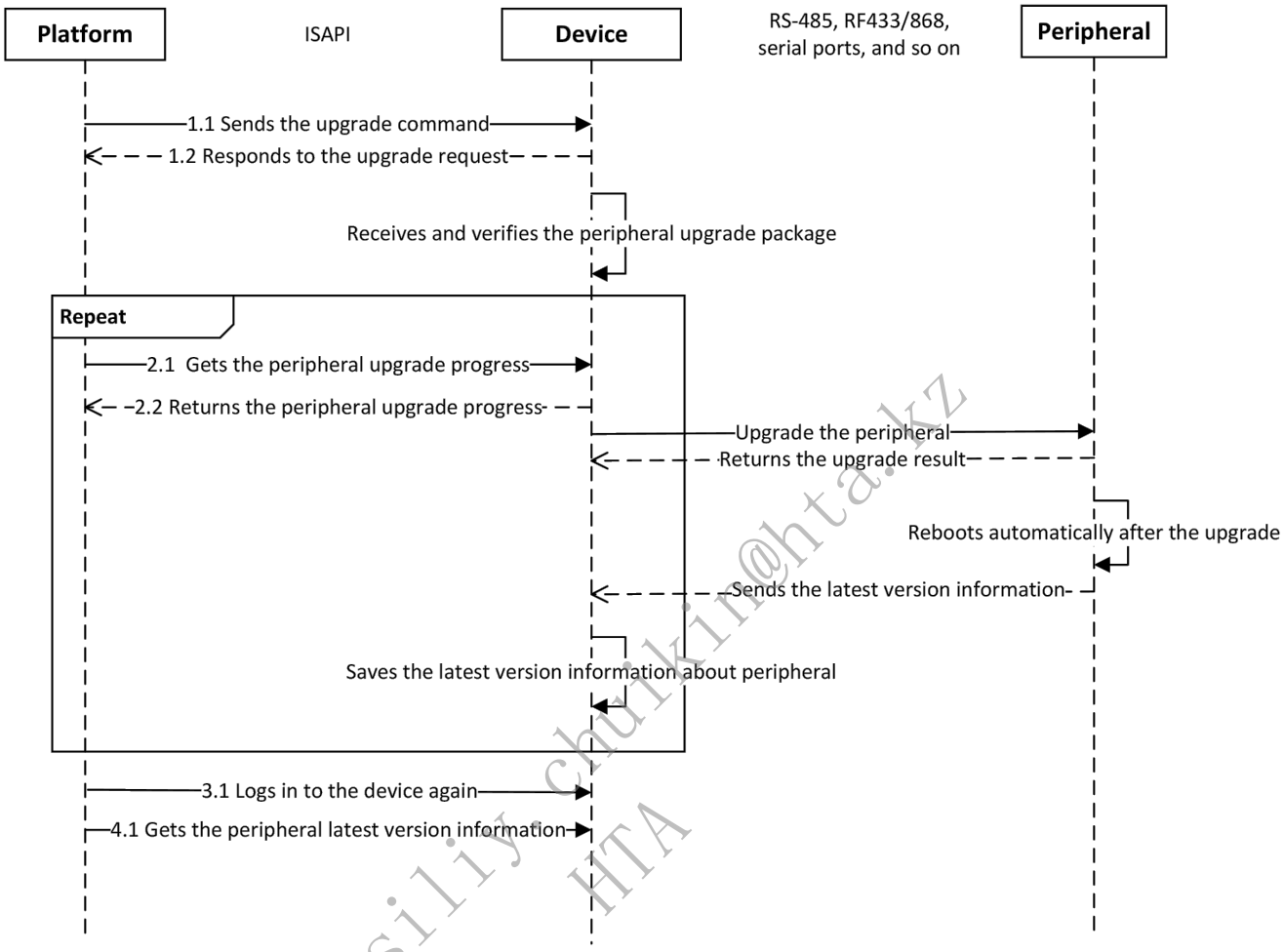
5.3 Device Peripherals Upgrade

5.3.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades device peripherals via ISAPI.

5.3.2 API Calling Flow

The sequence diagram of upgrading device peripherals by the platform is shown below.



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1. Get the device system capability `GET /ISAPI/System/capabilities` and check whether the device supports upgrading peripherals. If the field `isSupportAcsUpdate` is returned and its value is true, it indicates that the device supports this function, otherwise, the device does not support this function.
2. Get the capability of upgrading the peripherals module `GET /ISAPI/System/AcsUpdate/capabilities`, and get the types and IDs of peripherals that support upgrading.
3. The platform sends the upgrade command `POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>`. In the URL `type` refers to the peripheral type, `moduleAddress` refers to the peripheral module address, and `indexID` refers to the ID of peripheral to be upgraded. The platform will apply the upgrade peripheral package to the device.
4. Get the peripheral upgrade progress `GET /ISAPI/System/upgradeStatus?type=<Type>`.
5. Log in to the device again.
6. Get the peripheral latest version information.

5.4 Device Time Sync

5.4.1 Introduction to the Function

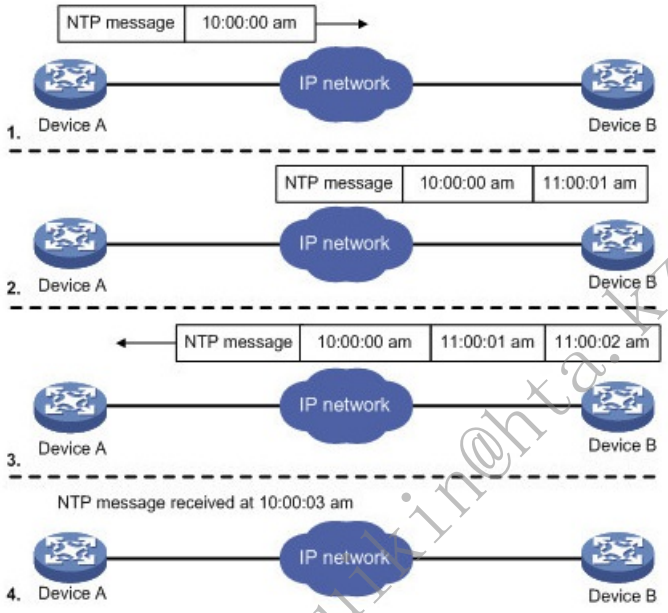
Time sync is a method to synchronize the time of all devices connecting to the NTP server, so that all devices can share the same clock time for providing related functions based on time. Supported time sync types: NTP time sync, manual sync, satellite time sync, platform time synchronization, etc. The following describes the method of NTP time sync.

5.4.1.1 NTP Time Sync

The local system of running NTP can receive sync from other clock sources (self as client), other clocks can sync from the local system (self as server), and sync with other devices.

The basic working principle of NTP is shown in the picture. Device A and Device B are connected via the network, and their systems follow their own independent system time. To auto sync their time, you can set device time auto sync via NTP. For example:

Before time sync between Device A and Device B, the time of Device A is 10:00:00 am, and that of Device B is 11:00:00 am. Device B is set as the server of NTP server, so that the time of Device A should be synchronized with that of Device B. The time of NTP message transmitted between Device A and Device B is 1 second.



The working process of system clock synchronization is as follows:

Device A sends an NTP message to Device B with a timestamp of 10:00:00 am (T1) that is when it leaves Device A.

When the NTP message reaches Device B. Device B will add its own timestamp, which is 11:00:01 am (T2).

Then the NTP message leaves Device B with Device B's timestamp, which is 11:00:02 am (T3).

Device A receives the response message, and the local time of Device A is 10:00:03 am (T4).

Above all, Device A can calculate two important parameters:

Round-trip delay of NTP message: Delay = (T4-T1) - (T3-T2) = 2 seconds.

Time difference between Device A and Device B: offset = ((T2-T1)+(T3-T4))/2=1 h.

Device A can sync its own time with that of Device B according to calculation results.

5.4.2 API Calling Flow

5.4.2.1 Time Sync Configuration

1. Get the Capability of Device Time synchronization Management

You can call this API to get the time sync types currently supported by the device, such as NTP time sync, manual time sync, satellite time sync, EZ platform time sync.

Get the capability: `GET /ISAPI/System/time/capabilities`.

2. Set device time synchronization management parameters

You can configure the time synchronization mode as follows：

Get device time synchronization management parameters: `GET /ISAPI/System/time`;

Set device time synchronization management parameters: `PUT /ISAPI/System/time`;

NTP time synchronization: See 4.2.2 NTP Time Sync (Client).

Manual time synchronization: Set the value of `timeMode` to `manual`, and set the device local time in nodes `localTime`、`timeZone`.

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Satellite time synchronization: Set the value of `timeMode` to `satellite`, and set the device local time in nodes `satelliteInterval`.

Platform time synchronization: Set the value of `timeMode` to `platform`.

Note： For manual time synchronization (time offset including time zone offset): Set manual time synchronization: localTime refers to the local time on device (time offset excluded, in format like 2019-02-28T10:50:44); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00); Get manual time synchronization: localTime refers to the local time on device (time offset included, in format like 2019-02-28T10:50:44+8:30); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00).

5.4.2.2 Time Zone Configuration

1. Get device time zone configuration capability

Call `GET /ISAPI/System/capabilities` to get the system capability. When `isSupportTimeZone` is returned, the time zone configuration is supported by the device.

2. Configure time zone parameters

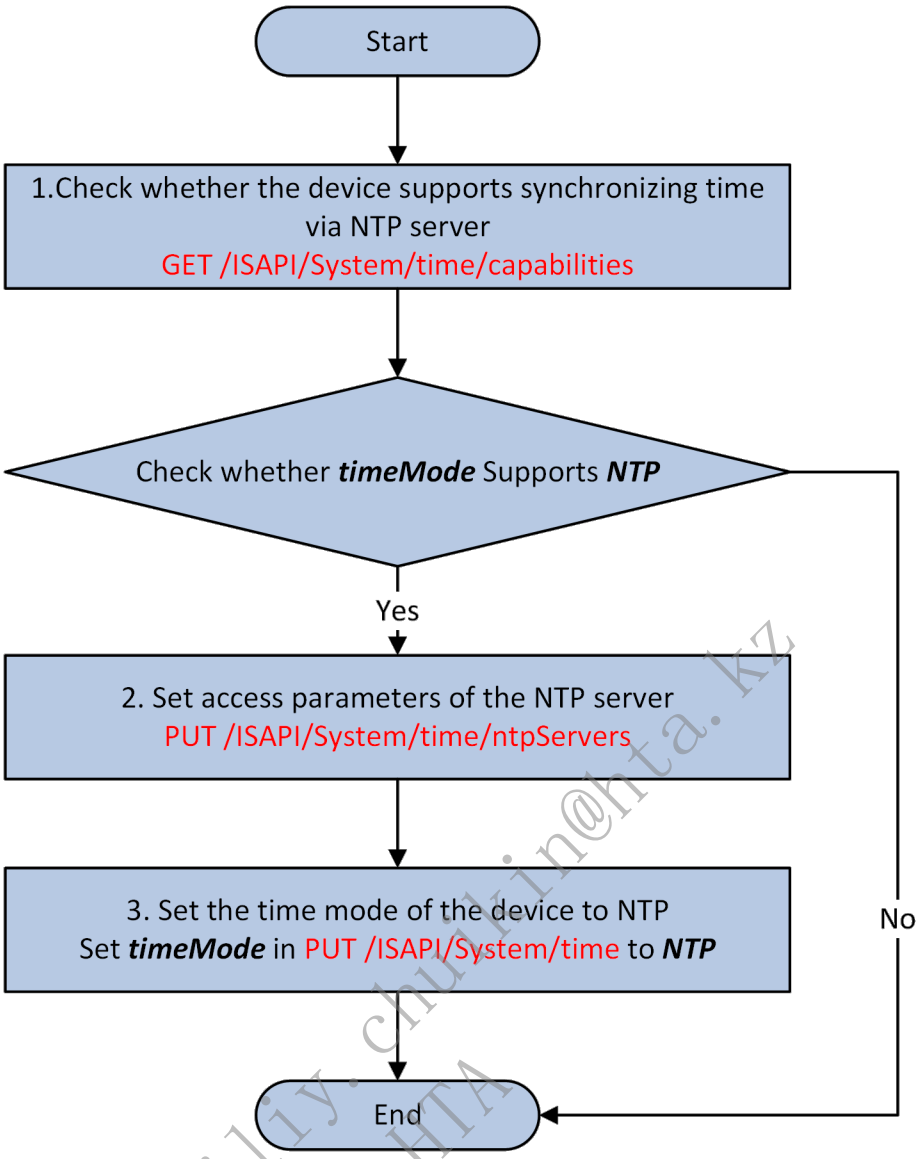
Get the device's time zone parameters: `GET /ISAPI/System/time/timeZone`. Set the device's time zone parameters: `PUT /ISAPI/System/time/timeZone`.

If DST (Daylight Saving Time) is disabled, the example of returned time zone parameters is: CST-8:00:00. It refers to UTC+8, and -8:00:00 is the UTC local time. If DST (Daylight Saving Time) is enabled, the example of returned time zone parameters is: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.2/02:00:00. It refers to UTC+8, the DST time is 30 minutes ahead of local time, the DST starts at 02:00:00 on the first Sunday of April and ends at 02:00:00 on the fifth Tuesday of October. MX.Y.Z: X is the month, Y is the week number in the month, Z is the day of a week (0-Sunday, 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday, 6-Saturday).

5.4.2.3 NTP Time Sync (Client)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as client):

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1. Check whether the device supports synchronizing time via NTP server Get the capability of the device: `GET /ISAPI/System/time/capabilities`, and check whether `timeMode` supports `NTP`.

2. Set access parameters of the NTP server

Supports accessing the NTP server by IP address to synchronize the device time.

Get the access parameter capability of the NTP server: `GET /ISAPI/System/time/ntpServers/capabilities`

Set access parameters of the NTP server: `PUT /ISAPI/System/time/ntpServers`

Get access parameters of the NTP server: `GET /ISAPI/System/time/ntpServers`

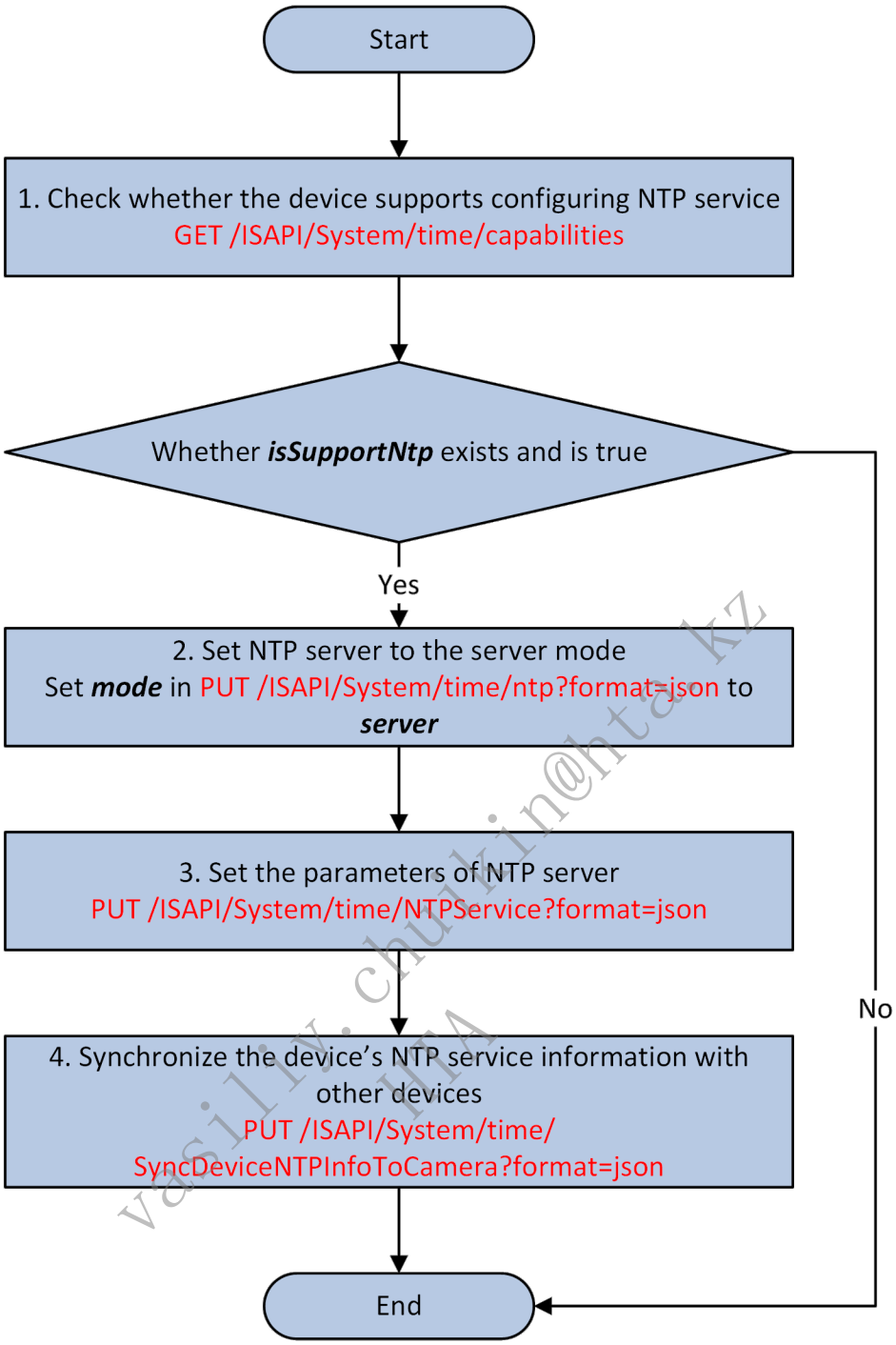
3. Set the time mode of the device to NTP

Supports setting the value of `timeMode` to `NTP`.

Get device time synchronization management parameters: `GET /ISAPI/System/time` Set device time synchronization management parameters: `PUT /ISAPI/System/time`

5.4.2.4 NTP Time Sync (Server Mode)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as server):



1. Check whether the device supports configuring NTP service Get the capability of device time synchronization management: `GET /ISAPI/System/time/capabilities`; If `isSupportNtp` is returned, it indicates that the device supports time synchronization management.

2. Set NTP server to the server mode

Supports setting the value of `mode` to `server`.

Get the capability of server mode: `GET /ISAPI/System/time/ntp/capabilities?format=json`

Set NTP to server mode: `PUT /ISAPI/System/time/ntp?format=json`

Get parameters of NTP server mode: `GET /ISAPI/System/time/ntp?format=json`

3. Set the parameters of NTP server

Supports setting the IP address of the NTP server.

Get the capability of NTP server: `GET /ISAPI/System/time/NTPService/capabilitis?format=json`

Set the NTP server parameters: `PUT /ISAPI/System/time/NTPService?format=json`

Get the parameters of the NTP server: `GET /ISAPI/System/time/NTPService?format=json`

4. Synchronize the device's NTP service information with other devices

Supports synchronizing the time information to the camera.

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Get the capability set of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json`

Synchronize device's NTP service information with the camera: `PUT /ISAPI/System/time/SyncDeviceNTPInfoToCamera?format=json`

Get the progress of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/Progress?format=json`

Search for the results of synchronizing device's NTP service information with the camera: `POST /ISAPI/System/time/SyncDeviceNTPInfoToCamera/SearchResult?format=json`

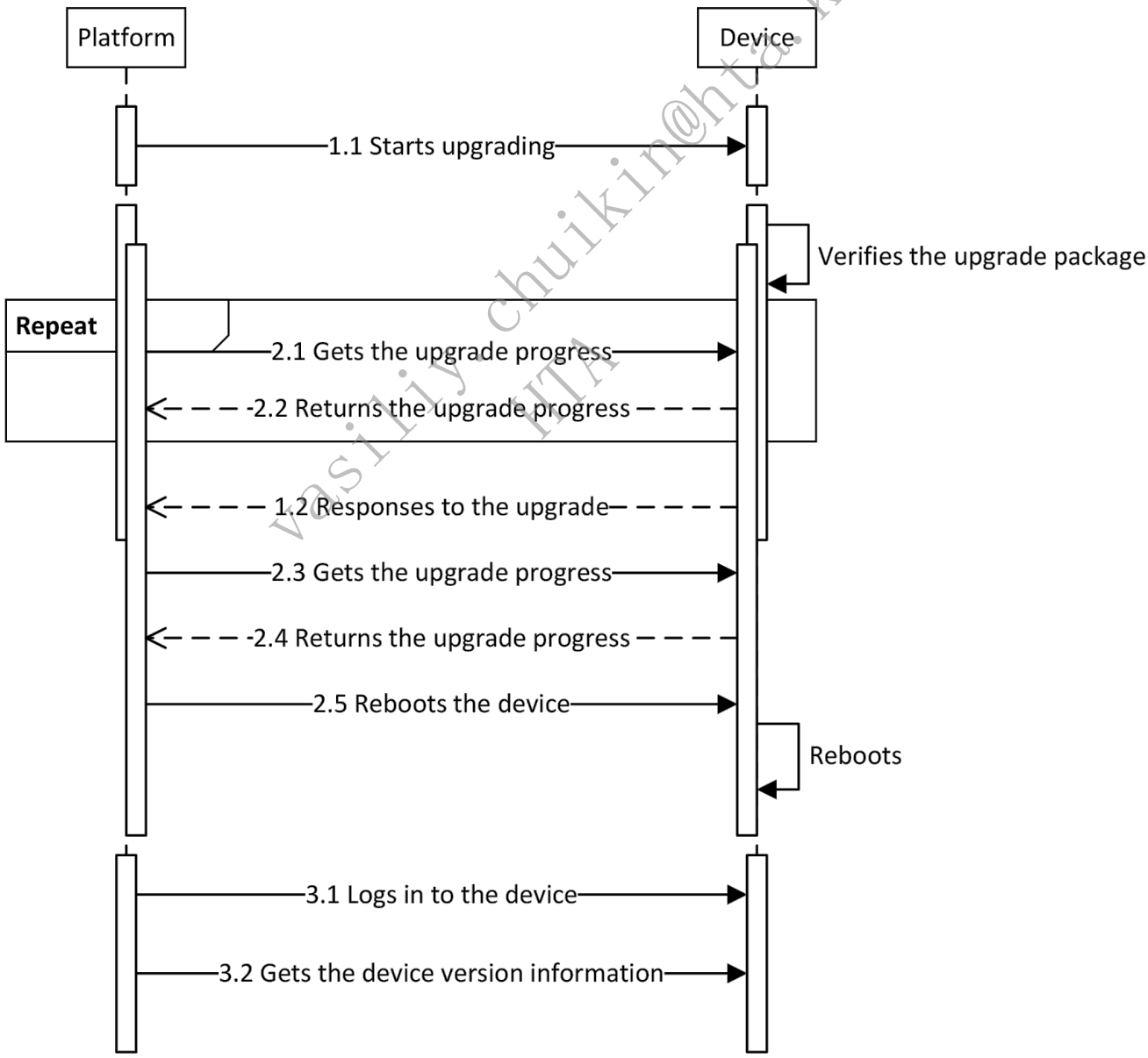
5.5 Device Upgrade

5.5.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades devices via ISAPI.

5.5.2 API Calling Flow

The sequence diagram of upgrading devices by the platform is shown below.



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1. **Upgrade devices.**

   Upgrade the device firmware: `POST /ISAPI/System/updateFirmware`.

2. **Get the device upgrade progress.**

Get the device upgrade progress: `GET /ISAPI/System/upgradeStatus`.

3. **Reboot devices.**

   Reboot devices: `PUT /ISAPI/System/reboot`.

5.6 Mutually Exclusive Functions

5.6.1 Introduction to the Function

Some functions are mutually exclusive due to the device performance (for example, function A and function B cannot run at the same time, i.e, only one of them is allowed at one time).

5.6.2 API Calling Flow

The following three APIs are available for the integration of mutually exclusive functions:

1. Get the information of mutually exclusive functions: `GET /ISAPI/System/mutexFunction/capabilities?format=json`. Call this URL to get the list of existing mutually exclusive functions supported by the device. Note: NVR devices only support setting exlusive function "perimeter" (perimeter protection), and do not support "linedetection" (line crossing detection), "fielddetection" (intrusion detection), "regionEntrance" (region entrance), or "regionExiting" (region exiting).
2. Search for the functions that are mutually exclusive with a specified function: `POST /ISAPI/System/mutexFunction?format=json`. Based on the list of mutually exclusive functions returned by `GET /ISAPI/System/mutexFunction/capabilities?format=json`, you can search for the mutual exclusion status of a specified function and see whether to change the settings and disbale the mutually exclusive function.
3. Get the mutual exclusion information when device function exception occurs: `GET /ISAPI/System/mutexFunctionErrorMsg`. After getting the error code, you can call this API to get the current mutually exclusive functions.

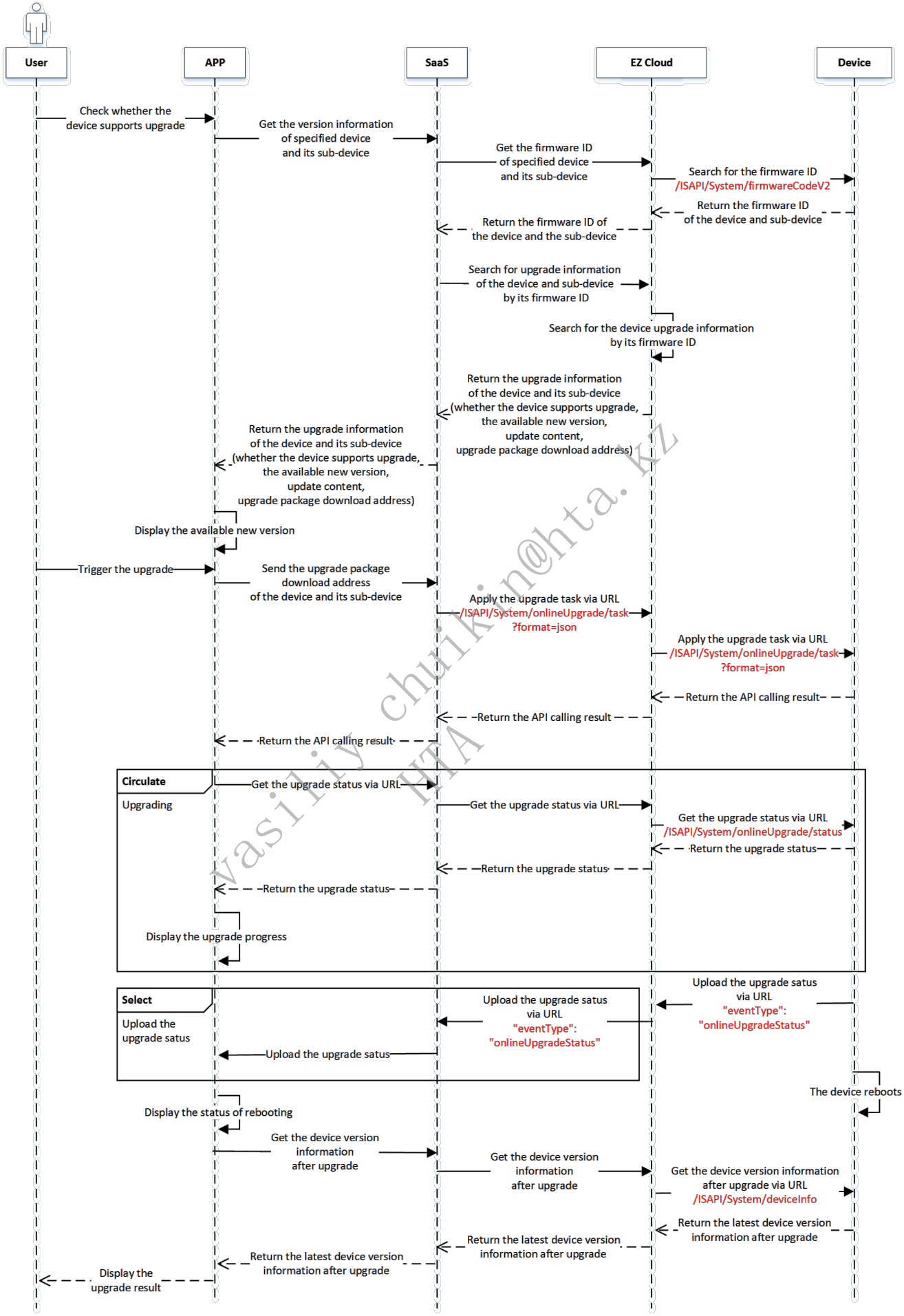
5.7 Remote Online Upgrade Based on The EZ2.0

5.7.1 Introduction to the Function

Considering that the device is connected to EZ via EZ 2.0 Protocol, EZ OTA upgrade solution is not supported. This solution applies to specified version upgrade.

5.7.2 API Calling Flow

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1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by its firmware ID. If there is an available new version, apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade devices (e.g., NVR) or sub-devices (e.g., IPC) via the node

device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.

3. During the device upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` and reboot automatically.
5. After both upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

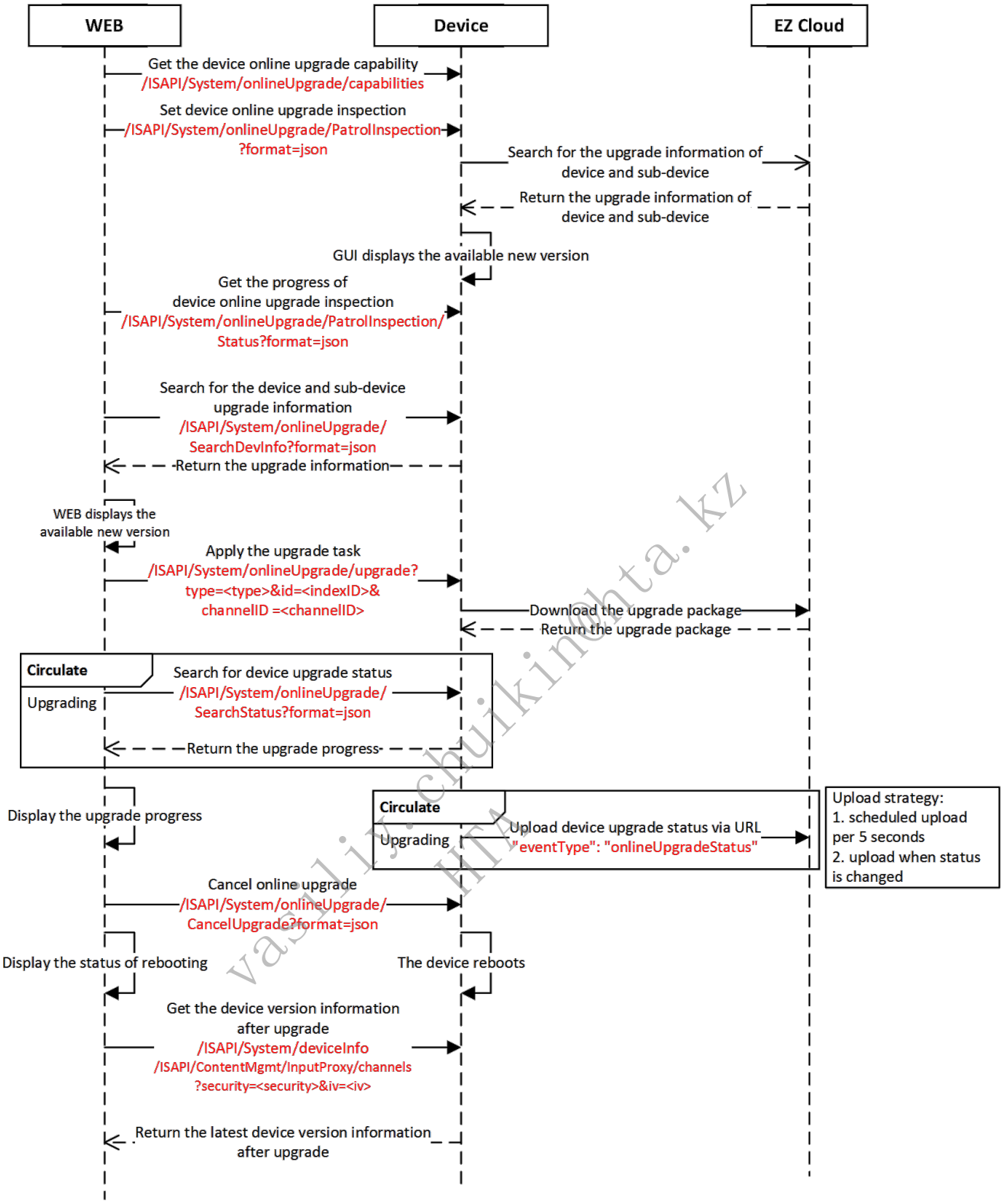
5.8 Remote Online Upgrade Based on the WEB Client

5.8.1 Introduction to the Function

This solution applies to remote online upgrade of the devices and sub-devices connected to WEB client.

5.8.2 API Calling Flow

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1. Get device online upgrade capability: `GET /ISAPI/System/onlineUpgrade/capabilities`.
2. Manually trigger the device online upgrade inspection: `POST /ISAPI/System/onlineUpgrade/PatrolInspection?format=json`. After it is enabled, the device will search for the available new version.
3. During the inspection, the WEB client can get the real-time inspection progress: `GET /ISAPI/System/onlineUpgrade/PatrolInspection/Status?format=json`.
4. After the device inspection is finished, search for the device and sub-device upgrade information: `GET /ISAPI/System/onlineUpgrade/SearchDevInfo?format=json`.
5. Apply the upgrade task: `PUT /ISAPI/System/onlineUpgrade/upgrade?type=<type>&id=<indexID>&channelID =<channelID>`. You can upgrade a specified device via the node `<channelID>` in the URL. Specify sub-device upgrade by configuring `<type>`.

6. You can search for online upgrade status during the upgrade: `GET /ISAPI/System/onlineUpgrade/SearchStatus?format=json`.
7. You can cancel the upgrade during the upgrade: `PUT /ISAPI/System/onlineUpgrade/CancelUpgrade?format=json`.
8. After the upgrade and reboot are finished, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

5.9 Serial Port Accessed External Device Management

5.9.1 Introduction to the Function

Information management of device accessed the serial are as follows: 1. Configure manufacturer, type, and model information of the specific serial port access device. 2. Search for the device type or model supported by the specific serial port.

5.9.2 API Calling Flow

1. Check whether the device supports information management of serial port. Get the capability of the device serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportDeviceInfo>` is returned, the device supports information configuration of devices access the serial port.

2. Set the information of serial port:

- Get the capability of device information parameters of a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Get device information parameters access single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Set device information parameter of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.

3. Check whether the device supports linking information of devices access the serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportSearchDeviceInfoRelations>` is returned, it indicates that the device supports searching for linked information od devices access the serial port.

4. Search for linked information of devices access the serial port.

- Get the capability of searching for linked parameters of information of devices access a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations/capabilities?format=json`.
- Search for linked parameters of information of devices access a single serial port:`POST /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations?format=json`.

5.10 Serial Port Data Transparent Transmission

5.10.1 Introduction to the Function

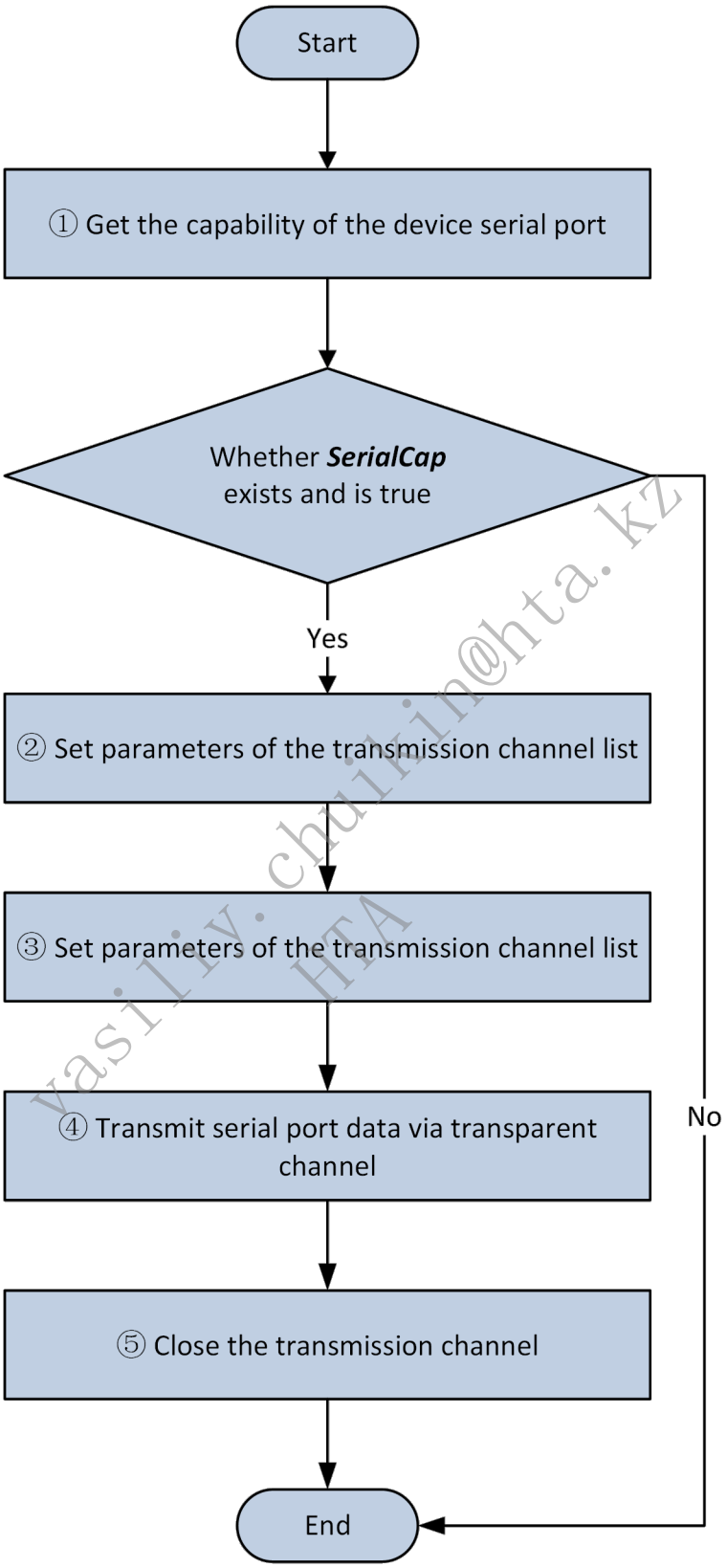
RS485, RS422 and RS232 serial ports external to the device are used as transparent channels to transmit serial port data. Supports the client sending serial data to the device, which then forwards the data to the serial port; conversely, when the serial port sends data to the device, which transparently transmits the data to the client. Note that the device's transparent transmission of serial data is half-duplex communication, meaning that bidirectional communication can be implemented between the device and the client, but the communication data is not necessarily in a question-and-answer format, and the client must match the request and response relationship on its own.

Since most standard HTTP request libraries do not support the persistent connection for receiving and sending data in real time, it is recommended that the client uses two TCP clients to implement the sending and receiving of

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transparently transmitted serial data.

5.10.2 API Calling Flow



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1. Check whether the device supports serial port data transmission.

Get the capability of the device serial port: `GET /ISAPI/System/capabilities`. If `SerialCap` is returned and the value is true, it indicates that the device supports the functions of the serial port.

2. Set parameters of the transmission channel list.

- Get parameters of the specific transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.
- Configure parameters of the specific transmission channel: `GET`

`/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.

3. Open the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/open`;

4. Transmit serial port data via transparent channel.

- Receive data uploaded by device serial port through transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.
- Send data to device serial port through transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.

5. Close the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/close`.

5.11 Serial Port Parameter Configuration

5.11.1 Introduction to the Function

Serial port parameter configuration.

5.11.2 API Calling Flow

1. Check whether the device supports configuring serial port parameters.

Get the capability of device serial port: `GET /ISAPI/System/capabilities. If is returned and its value is true, the device supports functions of serial port.

3. Get parameters of all serial ports.

- Get the capability of all serial ports: `GET /ISAPI/System/Serial/capabilities`.
- Get control parameters of all serial ports: `GET /ISAPI/System/Serial/ports?permissionController=<indexID>`.

4. Set control parameters of a single serial port.

- Get control parameters of single serial port: `GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.
- Configure control parameters of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.

5. Get the status of single serial port:`GET /ISAPI/System/Serial/ports/<portID>/status`.

5.12 Sub-device Batch Upgrade

5.12.1 Introduction to the Function

Sub-device Batch Upgrade (Single Task): applicable to situations of upgrading multiple sub-devices using one upgrade package. Application scenarios include: when the UWB positioning anchor connects to the web, upgrading multiple tags (sub-devices) through the positioning engine (gateway), with the same upgrade package.

Management of Sub-device Batch Upgrade Tasks: There are many types of sub-devices, such as LoRa nodes, which have a slow upgrade process and different models of LoRa nodes in the field also require different upgrade packages. Therefore, it is necessary to support the creation of multiple upgrade tasks at once, entrusting the upgrade to the devices, to enhance the practicality of batch upgrading sub-devices.

5.12.2 API Calling Flow

Sub-device Batch Upgrade (Single Task)

Get the device capability: `/ISAPI/System/capabilities`.

If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that the device support the function.

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If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that the device support getting the progress of batch upgraading sub-devices.

Sub-device batch upgrade: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.

Batch sub-device upgrade progress search: `POST /ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json`.

Sub-device Batch Upgrade Task Management

Get device system capabilities `/ISAPI/System/capabilities`, If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that batch upgrade of sub devices is supported,

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that progress of batch upgrading sub devices can be searched,

If the node `isSupportSearchBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading devices can be searched,

If the node `isSupportModifyBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading sub devices can be edited,

If the node `isSupportDeleteBulkUpgradeChildDeviceTask` exists and is true, it indicates that task of batch upgrading sub devices can be deleted.

Get the capability of batch upgrading sub-device: `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`.

Batch upgrade sub devices: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.****

Search for tasks of batch sub-device upgrade: `POST /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`.

Edit tasks of batch sub-device upgrade: **`PUT /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`**.****

Delete tasks of batch sub-device upgrade: `PUT /ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json`.

Note: 1. The two methods for batch upgrading child devices are the same API: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`. If the `BulkUpgradeChildDeviceListCap.taskID` is returned in `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`, it indicates that the task method is supported in the sub device upgrade. 2. To determine which sub devices can be upgraded, you can use the API `POST /ISAPI/IoTGateway/Childmanage/SearchChild?format=json`. Sub devices with the `tag` `isUpgradable` and `value` `true` indicate that the sub devices support upgrades.

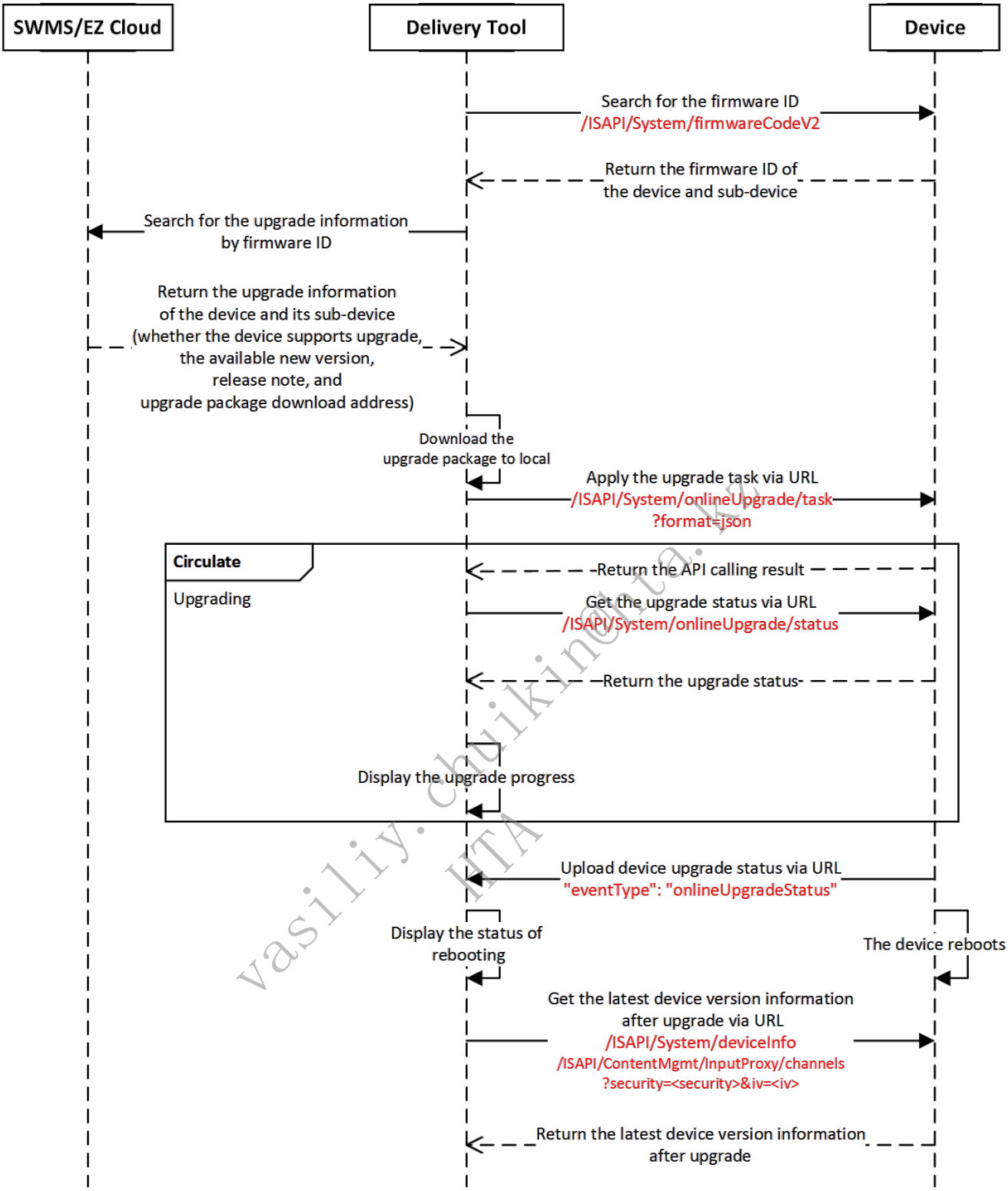
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5.13 Upgrade Based on Interactive Tools Local Deployment

5.13.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI. If the device is connected via ISAPI protocols, it is required to set up HTTP(s) service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK, you can implement the device upgrade via transmitting ISAPI.

5.13.2 API Calling Flow



1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by firmware ID on the SWMS/EZ Cloud, download upgrade package to local storage, and apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., camera) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress: `GET /ISAPI/System/onlineUpgrade/status`.
4. After upgrade is completed, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and reboot automatically.
5. After the device rebooting, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=`

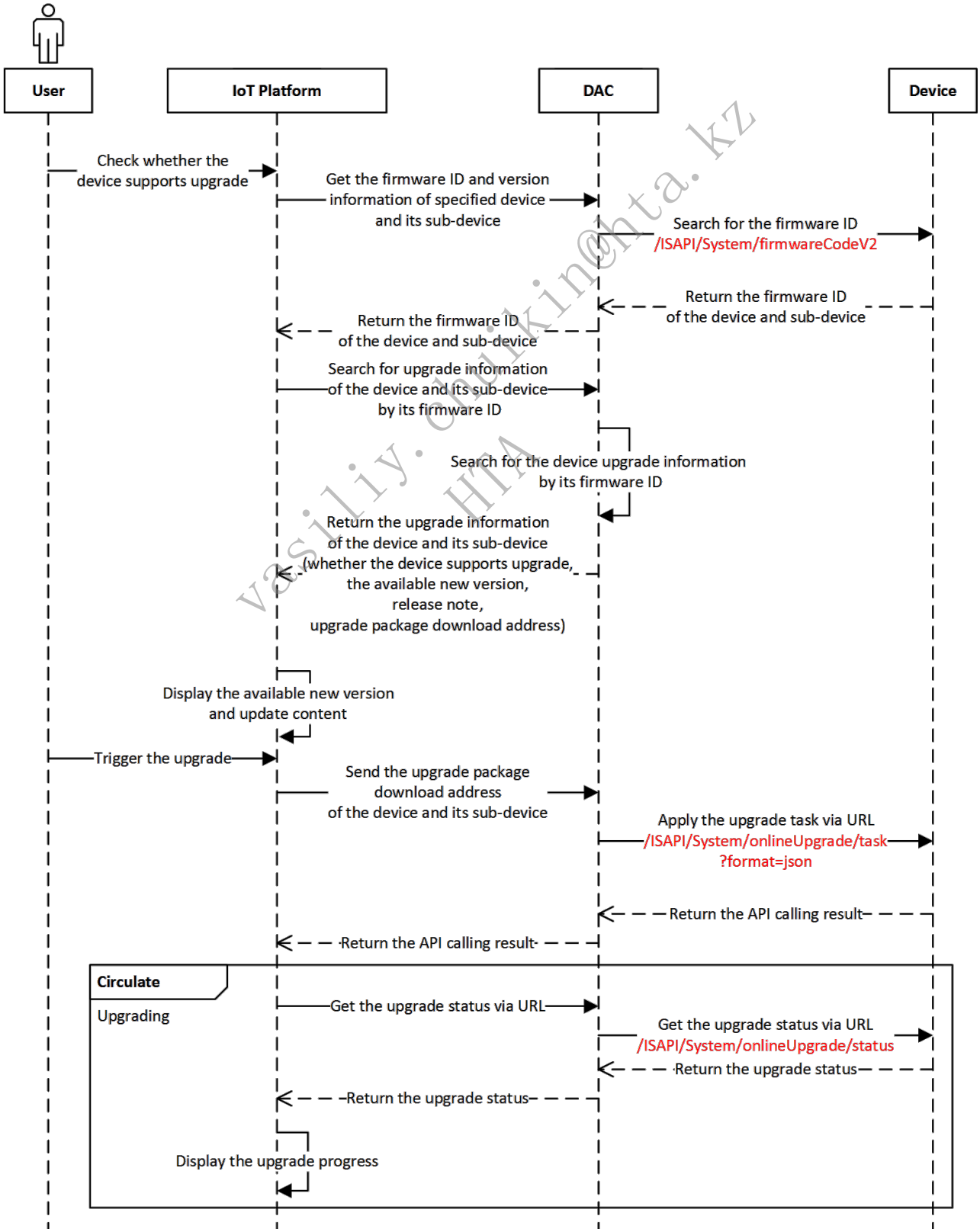
`<security>&iv=<iv>`.

5.14 Upgrade Based on Platform Local Deployment

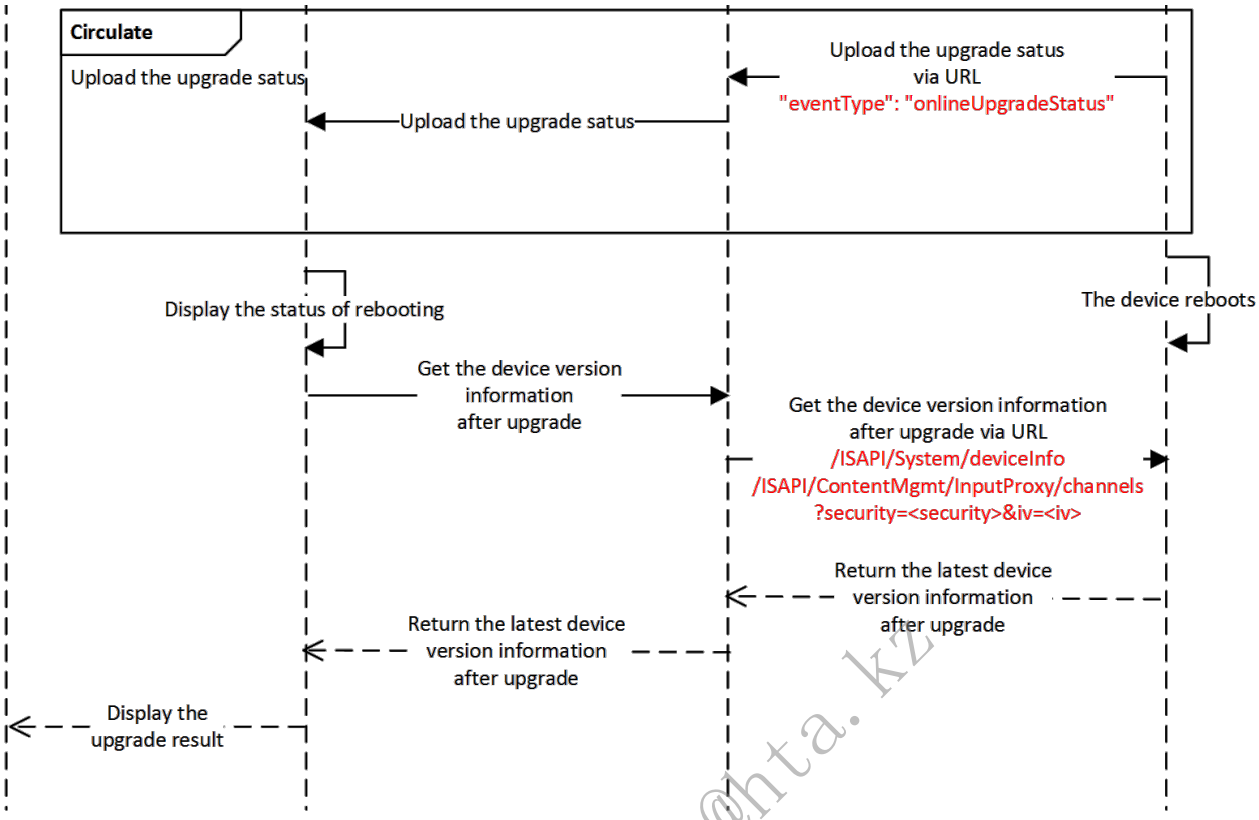
5.14.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI and ISUP. If via ISAPI, it is required to set up HTTP(s) service or transfer binary upgrade package files so as to store device upgrade package and generate URL for downloading upgrade package. If via ISUP, it is required to set up FTP service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK protocols, you can realize device upgrade via transmitting ISAPI.

5.14.2 API Calling Flow



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1. Search for device firmware ID on the platform: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information in DAC by device firmware ID on the platform. If there is an available new version, DAC will trigger an upgrade task and apply it to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and will reboot automatically.
5. After the upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

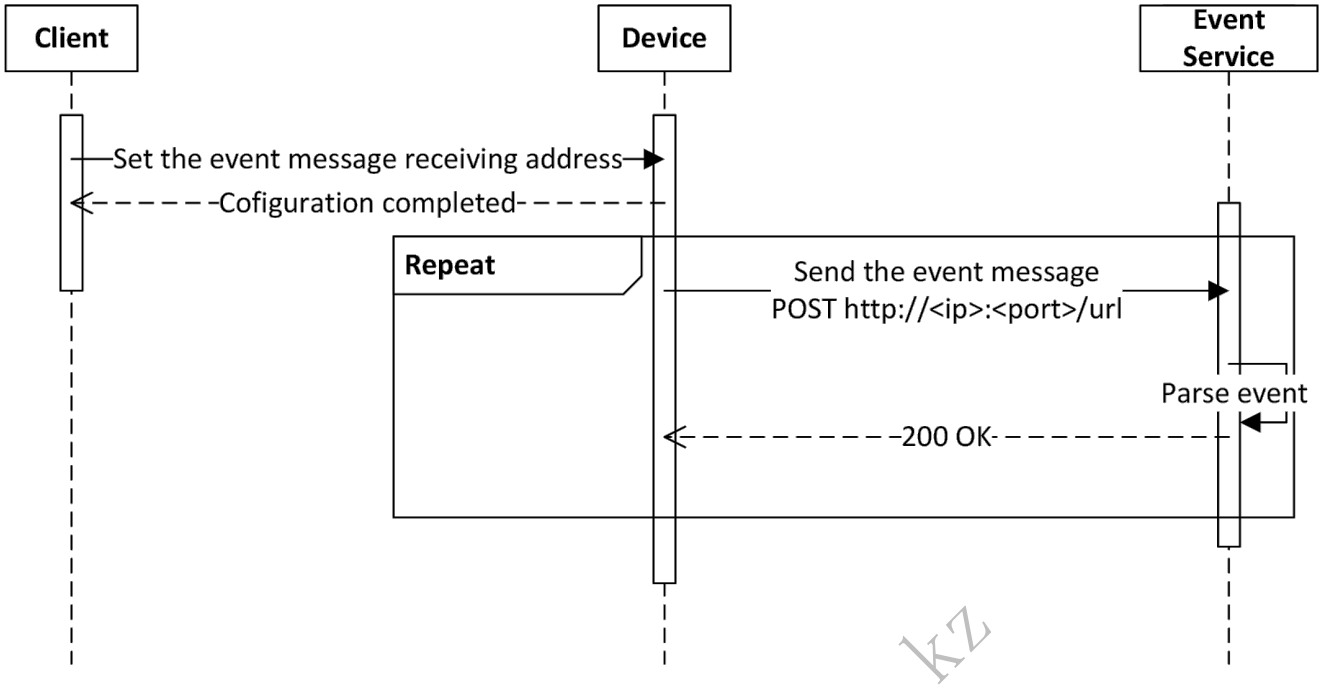
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6 Network Configuration

6.1 Listening Service

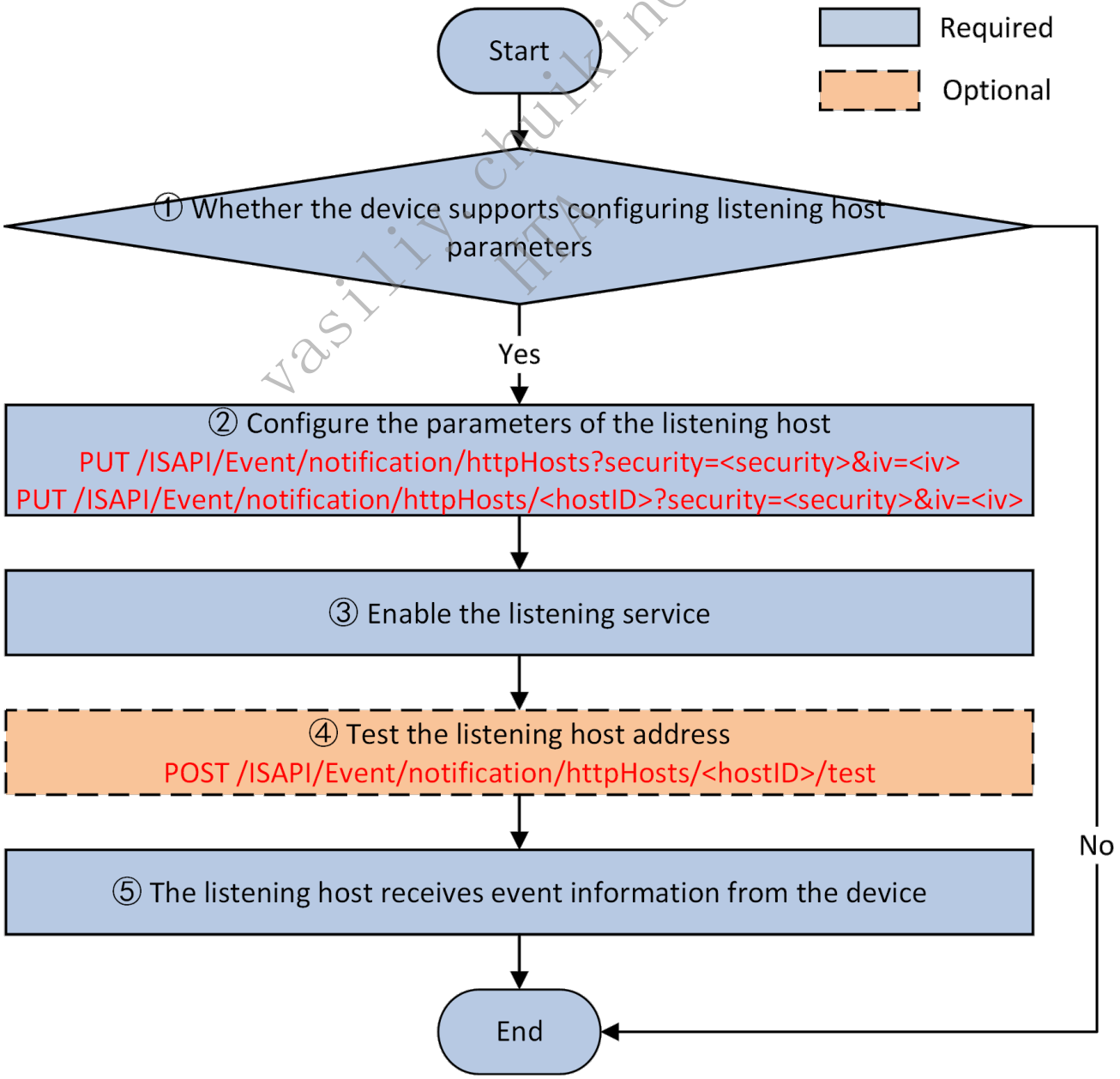
6.1.1 Introduction to the Function

When an event occurs, the device creates connection with the client and uploads alarm information. Meanwhile, the listening host receives data from the device. The IP address and the port No. of the listening host should be configured for the device. The HTTP listening service supports subscribing to specific events when adding or editing the listening host. Only the specified events will be uploaded by the device.(currently not available for the device)



6.1.2 API Calling Flow

6.1.2.1 Listening Service



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1. **Check whether the device supports configuring listening host parameters:**

   Get the configuration capability of the listening host: `GET /ISAPI/Event/notification/httpHosts/capabilities`. If the node `<HttpHostNotificationCap>` exists in the returned message and its value is true, it indicates that the device supports configuring listening host parameters.

2. **Configure the parameters of the listening host:**

   Configure the parameters of all listening hosts: `PUT /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

   Get the parameters of all listening hosts: `GET /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

   Configure the parameters of a listening host: `PUT /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`;

   Get the parameters of a listening host: `GET /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`;

3. **Enable the listening service:**

   The user needs to enable the listening service of the listening host.

4. **(Optional) Test the listening service:**

   The platform applies the command to the device to test whether the listening host is available for the device: `POST /ISAPI/Event/notification/httpHosts/<hostID>/test`.

5. **The listening host receives event information from the device:**

   When an event occurs, the device creates connection with the client and uploads alarm information. Meanwhile, the listening host receives data from the device. See details in **Event Message Grammar**.

Remark: You can also configure the listening parameters such as the timeout.

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6.1.2.2 Event Message Grammar

When an event occurs or an alarm is triggered, the event/alarm information can be with binary data (such as pictures) and without binary data.

1. **Without Binary Data:**

The `Content-Type` in the Headers of the HTTP request sent by the device is usually `application/xml` or `application/json` as follows:

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--/Request_URI, related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: data_gateway_ip:port <!--Host: HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: zh-cn
Date: YourDate
Content-Type: application/xml; <!--content type, which is used for the upper layer to distinguish different formats when parsing the message-->
Content-Length: text_length
Connection: keep-alive  <!--maintain the connection between the device and the server for better transmission performance-->

<EventNotificationAlert/>
```

Response by the Listening Host Notes:

1. `Content-Length` is required and its value is 0.
2. `Connection` is recommended to use keep-alive to reduce the port usage.

```
HTTP/1.1 200 OK
Content-Length: 0
Connection: keep-alive
```

2. **With Binary Data**

The format of the data sent by the device is HTTP form (multipart/form-data). The `Content-Type` in the Headers of the HTTP request sent by the device is usually `multipart/form-data, boundary=<frontier>`: boundary is a variable, which is used to divide the HTTP Body into multiple units and each unit has its Headers and Body. See details in RFC 1867 (Form-based File Upload in HTML). Please note the `--` before and after the boundary.

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--Request_URI, related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: device_ip:port <!--Host: HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: zh-cn
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive <!--maintain the connection between the device and the server for better transmission performance-->

--<frontier>
Content-Disposition: form-data; name="Event_Type"
Content-Type: text/xml <!--Some alarms use JSON format, so the upper-layer application can distinguish based on the Content-Type during parsing-->

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[picture data]
--<frontier>--
```

Response by the Listening Host

Notes:

1. `Content-Length` is required and its value is 0.
2. `Connection` is recommended to use keep-alive to reduce the port usage.

```
HTTP/1.1 200 OK
Content-Length: 0
Connection: keep-alive
```

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Here are the descriptions of the main keywords.

keyword	example	description
Content-Type	multipart/form-data; boundary=frontier	Content type, multipart/form-data refers to data in form format.
boundary	frontier	Separator of the form message. A form message starts with --boundary and ends with --boundary--.
Content-Disposition	form-data; name="Picture_Name";	Content description. form-data refers to data in the form format.
filename	"Picture_Name"	File name. The file refers to the form message.
Content-Length	10	Content length. The length of the content which starts from \r\n to the next --boundary.

6.1.3 Exception Handling

6.1.3.1 Error Codes

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Remarks
6	Invalid Content	eventNotSupport	0x60001024		Event subscription is not supported.	

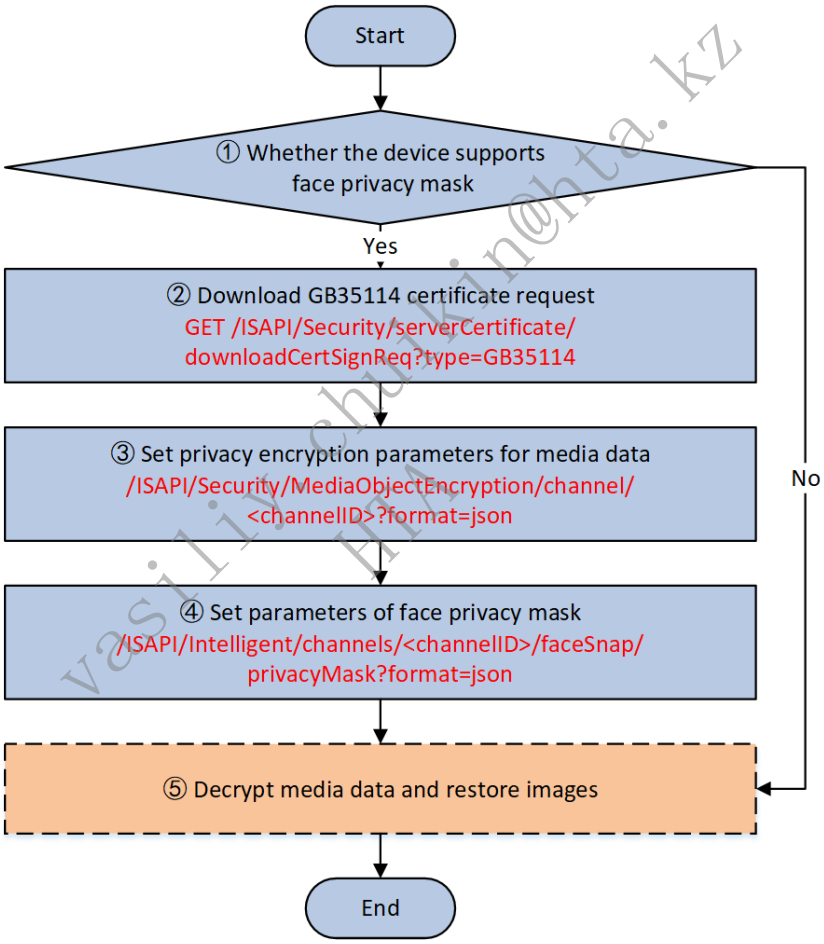
7 Face Capture

7.1 Face Capture

7.1.1 Introduction to the Function

Face capture is to detect whether face pictures show up in images. If set conditions are met and there are faces showing up, the face pictures will be captured and uploaded as alarms.

7.1.2 API Calling Flow



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1. **Whether the device supports face capture:**

   Get the system capabilities: `GET /ISAPI/System/capabilities`. If the node (its parent node is) is returned as true, it indicates that the device supports face capture.

2. **Whether the device supports multi-scene face capture:**

   Get the intelligent capabilities of device: `GET /ISAPI/Intelligent/channels/<channelID>/capabilities`. If the node is returned and its value is true, it indicates that the device supports face capture.

3. **Set face picture capturing rules of multiple scenes:**

   If multi-scene capture of a channel is required, you can set the face picture capturing rules for multiple scenes.

Set the face capture rules of a specified scene: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/<SID>`;

Get the face capture rules of a specified scene: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/<SID>`;

Set the face picture capturing rules of multiple scenes for a specified channel: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene`;

Get the face picture capturing rules of multiple scenes for a specified channel: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene`.

4. **Set face picture capturing rules of a specific scene.**

   If specific-scene capture of a channel is required, you can set the face picture capturing rules.

   Get the capabilities of captured face picture rules `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/capabilities`;

   Set the face capture rule parameters: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule`;

   Get the face capture rule parameters: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule`;

   Get the default parameters of face capture rule: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule?defualt=<defualt>`.

5. **Set linkage mode of face picture capture.**

   Set the linkage parameters of face capture: `PUT /ISAPI/Event/triggers/faceSnap-<channelID>`;

   Get the linkage parameters of face capture: `GET /ISAPI/Event/triggers/faceSnap-<channelID>`;

   Delete the linkage of face capture: `DELETE /ISAPI/Event/triggers/faceSnap-<channelID>`.

6. **Configure the arming schedule of face capture.**

   Set the arming schedule of face capture for a specified channel: `PUT /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>`;

   Get the arming schedule of face capture for a specified channel: `GET /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>`;

   Set the arming schedule of face capture: `PUT /ISAPI/Event/schedules/faceSnap`;

   Get the arming schedule of face capture: `GET /ISAPI/Event/schedules/faceSnap`.

7. **Receive face capture and recognition events.**

   If the wanted face shows up in the set area, face recognition event will be reported. The event information includes face pictures and attribute information, etc. `"eventType": "faceCapture"`.

7.2 Privacy Mask on Faces

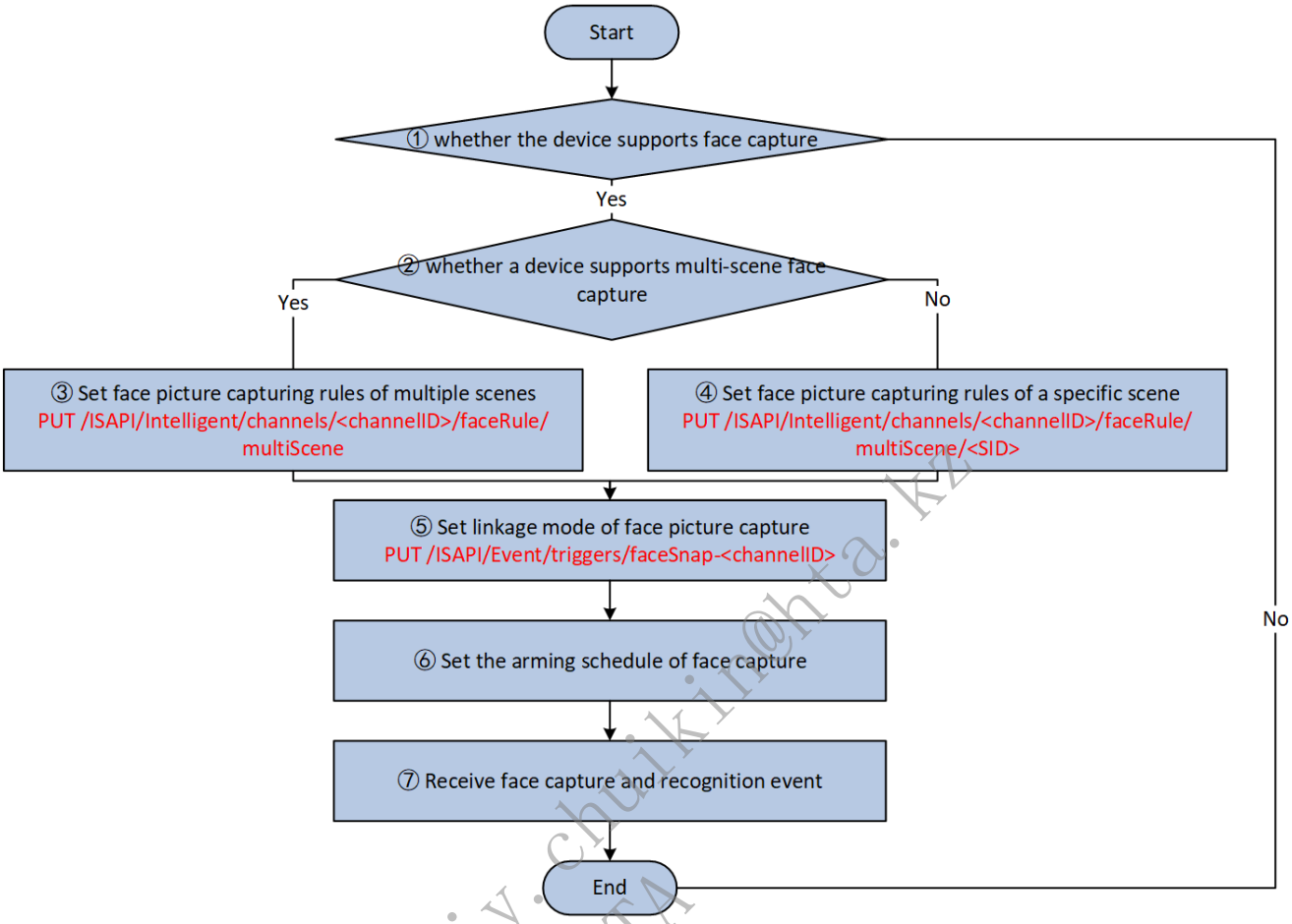
7.2.1 Introduction to the Function

Privacy mask on faces is to encrypt privacy contents for face capture and media data (see "Video Privacy Mask" API calling flow); it is the process of privacy masking for all faces of cameras and the encrypted images can be restored. First of all, all faces need to be identified, the original media data of faces in the images and masking block data will be replaced, the original media data of faces will be added to the private data of the live view stream and encrypted, then reported to the user synchronously; and the user can decrypt the private data to get the original media data and restore the face image. **Note: The current image restoration only supports private streams, which needs to be restored**

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by integrating the Playback Library. The following figure shows the effect of face privacy mask by mosaic processing.

7.2.2 API Calling Flow



1. **Whether the video channel supports face privacy mask.** Get the channel intelligence capabilities: `GET /ISAPI/Intelligent/channels/<channelID>/capabilities`, if the node `<isSupportFacePrivacyMask>` is returned as true, it indicates that the device supports face privacy masking.

2. **Set privacy encryption parameters in media data.** Users apply the encryption parameters including secondary key `secretKey` and the update time of the primary key `secretKeyUpdateFrequency` to devices, etc. Get the encryption capabilities of media data privacy: `GET /ISAPI/Security/MediaObjectEncryption/channel/<channelID>/capabilities?format=json`. Set the encryption parameters of media data privacy `PUT /ISAPI/Security/MediaObjectEncryption/channel/<channelID>?format=json`.Get the encryption parameters of media data privacy: `GET /ISAPI/Security/MediaObjectEncryption/channel/<channelID>?format=json`.

3. **Set parameters of face privacy mask.** Users can enable face privacy mask for a channel `enabled`. Get the capability of face privacy masking: `GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json`;
   Set the parameters of face privacy mask: `PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json`; Get the parameters of face privacy mask: `GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json`;

4. **(Optional) Decrypt media data and restore images**. The device generates the primary key and updates the primary key periodically according to the primary key update time in the media data privacy encryption parameters, then replaces the original media data with the masking block in the specified sensitive area in the live view screen, after which the images in the sensitive area are encrypted by the primary key and added to the private data of the live view stream, and at the same time the primary key is encrypted by the secondary key and added to the private data of the preview stream. The key is also encrypted by the secondary key and added to the private data of the live view stream. The user can parse the private data in the live view stream and decrypt it with the

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secondary key to get the primary key, then decrypt the private data with the primary key to get the original media data and restore the image. **Note**: At present, image restoration is only supported by private streams. You need to get the private data in the live view stream by integrating Playback Library, and get the primary key by decrypting the secondary key, then decrypt the private data by the primary key to get the original media data; and finally call the Playback Library API to restore the image. You can be refer to Playback library interface integration manual for playback library related APIs.

8 VCA

8.1 Face Picture Library Management

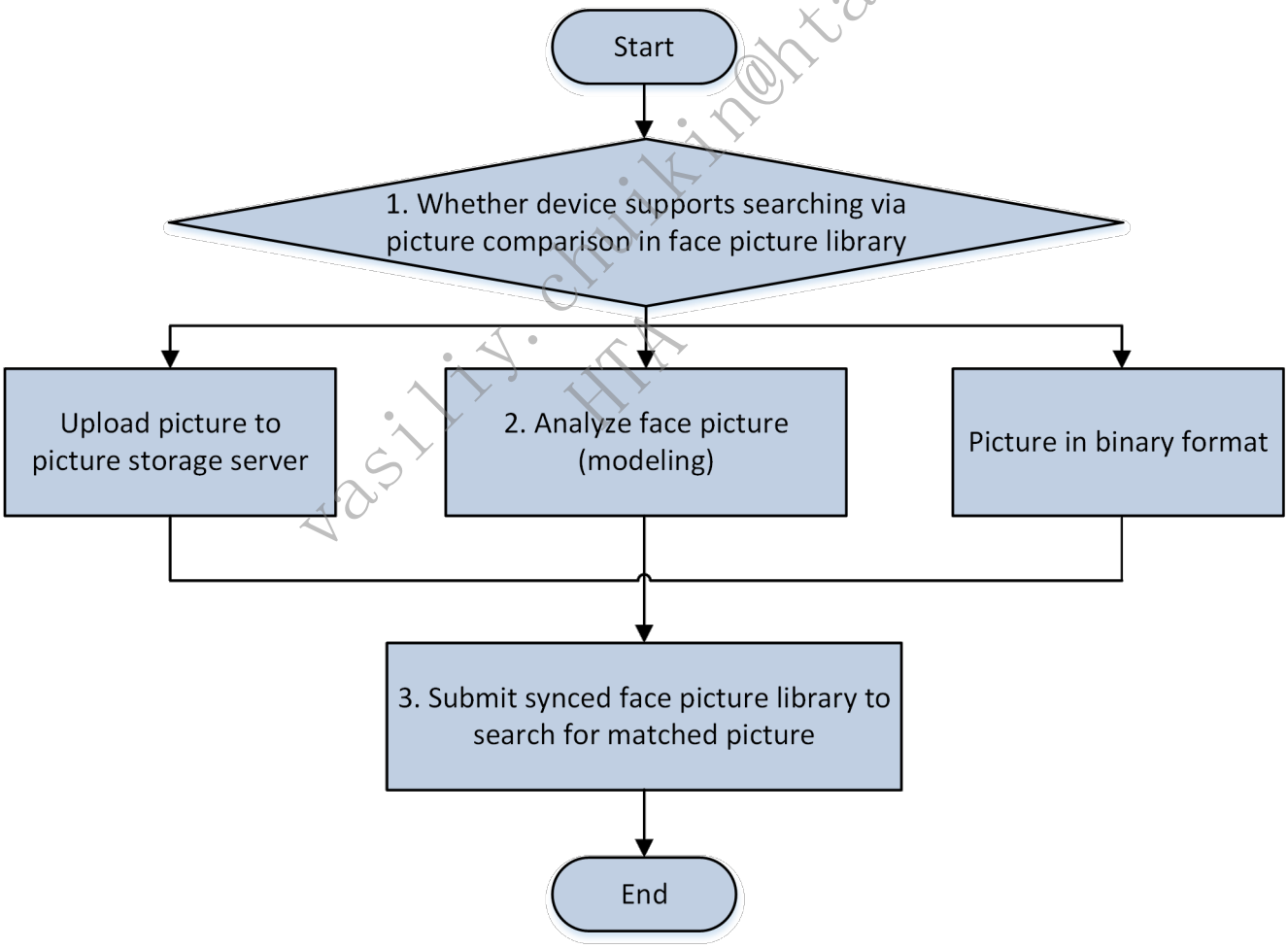
8.1.1. Search via Picture Comparison in Face Capture Library

8.1.1.1 Introduction to the Function

Picture comparison with captured picture library

Scene application: Search for similar person in face picture library, which can be used for identity confirmation.

8.1.1.2 API Calling Flow



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1. **Check whether the device supports searching by picture synchronously.**

   Get the capability of searching via picture comparison: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`. If the node `isSuportFSSearchByPic` is returned and its value is `true`, it indicates that searching via picture comparison synchronously is supported.

   Note: For some devices supporting searching via picture comparison synchronously, the API for getting the capability is invalid, you can call `POST /ISAPI/Intelligent/FDLib/searchByPic?format=json` directly.

2. **Three methods are supported when searching picture.**

By URL:

Based on the picture storage server, picture can be uploaded and you can get the picture URL to download the picture file via the URL.

By model data:

Get face picture capability: `GET /ISAPI/SDT/Face/pictureAnalysis/capabilities`;

Analyze (model) face pictures: `POSTT /ISAPI/SDT/Face/pictureAnalysis`; Return picture modeling data: `targetModelData`.

By binary picture:

Read binary picture files.

3. **Submit synced face picture library to use picture to search picture**

   Search via picture comparison in face capture library: `POST /ISAPI/Intelligent/FDLib/searchByPic?format=json`;

   Note: Input parameter node `dataType` include three modes. Picture URL corresponds to `faceURL`, model data corresponds to `targetModelData`, binary data mode is in form of form, and its JSON data is followed by binary picture data.

   The request message example of binary picture data is as follows:

```
POST /ISAPI/Intelligent/FDLib/searchByPic?format=json
Host: device_ip:port
Accept-Language: zh-cn
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive

--<frontier>
Content-Disposition: form-data; name=""
Content-Type: application/json

[JSON message]
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[Picture data]
--<frontier>--
```

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9 Two-Way Audio

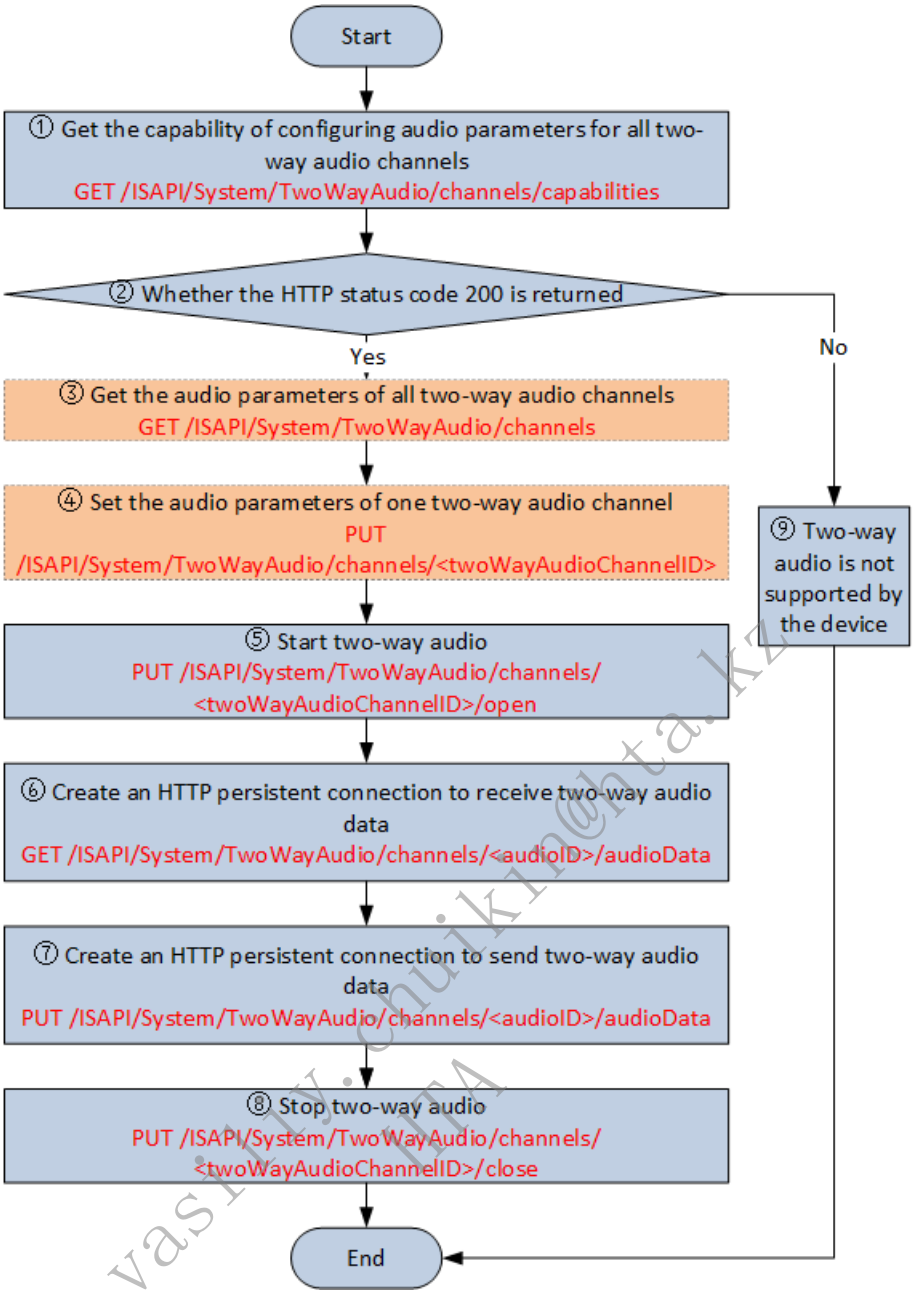
9.1 Two-Way Audio

9.1.1 Introduction to the Function

This function is for the two-way audio between the client and device. When you call the API to implement the two-way audio, it requires the client to collect and encode the local audios and to decode the device audio data.

Note: Before starting two-way audio, check whether the way of collecting device audio data, audio encoding format, volume, etc., is correct or not. The audio encoding format of the client should be consistent with the device audio encoding format.

9.1.2 API Calling Flow



1. Get the capability of configuring audio parameters for all two-way audio channels: `GET /ISAPI/System/TwoWayAudio/channels/capabilities`. Check whether the device supports two-way audio. If the HTTP status code `200 OK` is returned, then the device supports two-way audio.
2. Get the audio parameters of all two-way audio channels: `GET /ISAPI/System/TwoWayAudio/channels`. Get the No. of two-way audio channel for response and parsing `<id>`, audio type `<audioCompressionType>`, frame rate `<audioBitRate>`, sampling rate `<audioSamplingRate>` and so on, to ensure the consistent of audio format during the two-way audio process.
3. Set the audio parameters of one two-way audio channel: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>`. This step is optional, and the `<audioID>` is the channel No. `<id>` responded and parsed in step 2.
4. Start two-way audio: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/open`.
5. Create an HTTP persistent connection to receive the two-way audio data: `GET /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData`.
6. Create an HTTP persistent connection to send the two-way audio data: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData`.
7. When the two-way audio ends, the client disables the persistent connections in steps 6 and 7 and sends the message of stopping the two-way audio: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/close`.

Note:

- Two-way audio over ISAPI supports digest authentication.

- It supports the two-way audio mode or sending only mode (the platform only call `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData` to send the audio data to devices).
- The `<audioID>` in step 3, 4, 5, 6, and 7 is the two-way audio channel No. which starts from 1. It can be parsed from the `<id>` in the device response message of the step 2.
- The audio type `<audioCompressionType>`, frame rate `<audioBitRate>`, and sampling rate `<audioSamplingRate>` can be parsed from the device response message of the step 2 for audio playing and collection in the step 5 and 6.
- The step 6 and 7 use persistent connections. In the HTTP request headers, you don't need to set `Content-Legth`, but need to set the `Connection: keep-alive` and `Content-Type: application/octet-stream`.
- When the encoding format of audio data in the step 5 and 6 is AAC/MP2L2/MP3, the audio data length is variable, so the frame header should be lengthened by 4 bytes. When the encoding format is G.722.1/G.711alaw/G.711ulaw/G.726/G.729/G.729a/G.729b/PCM/ADPCM/Opus, the audio data length is fixed, there is no need to lengthen the frame header. For example, if the content length of a frame of MP3 audio data is 576 bytes and the header length is 4 bytes, the big end data is 0x00000240, the total length is 580 bytes.
- The damaged audio file in the AC3 format is usually incompatible when using. Currently, the audio algorithm library does not support this format.

Statistics of audio types with fixed lengths:

Audio Type	Fixed Length	Frame Interval
G.722.1	80	40ms
G.711alaw	160/320	20ms/40ms
G.711ulaw	160/320	20ms/40ms
G.726	80	40ms
G.729	10	10ms
G.729a	10	10ms
G.729b	10	10ms
PCM	1920	None
ADPCM	80	20ms
Opus	32/64	20ms/40ms

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9.1.3 Exception Handling

9.1.3.1 Error Codes

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Re
4	Invalid Operation	twoWayAudioInProgressPleaseWait	0x40002068	Two-way audio in progress...Please wait.	Two-way audio in progress...Please wait.	Th op is all Tw au in pr Pl ag af tw au st

9.1.4 Message Format and Example

9.1.4.1 Receive Two-Way Audio Data

```
GET /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
Host: 10.17.115.128
Connection: keep-alive
Content-Type: application/octet-stream

HTTP/1.1 200 OK
Connection: keep-alive
Content-Type: application/octet-stream

//The following are the two-way audio data sent by the device.
```

9.1.4.2 Send Two-Way Audio Data

```
PUT /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
Host: 10.17.115.128
Connection: keep-alive
Content-Type: application/octet-stream

//The following are the two-way audio data sent by the client.
```

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10 Access Control (General)

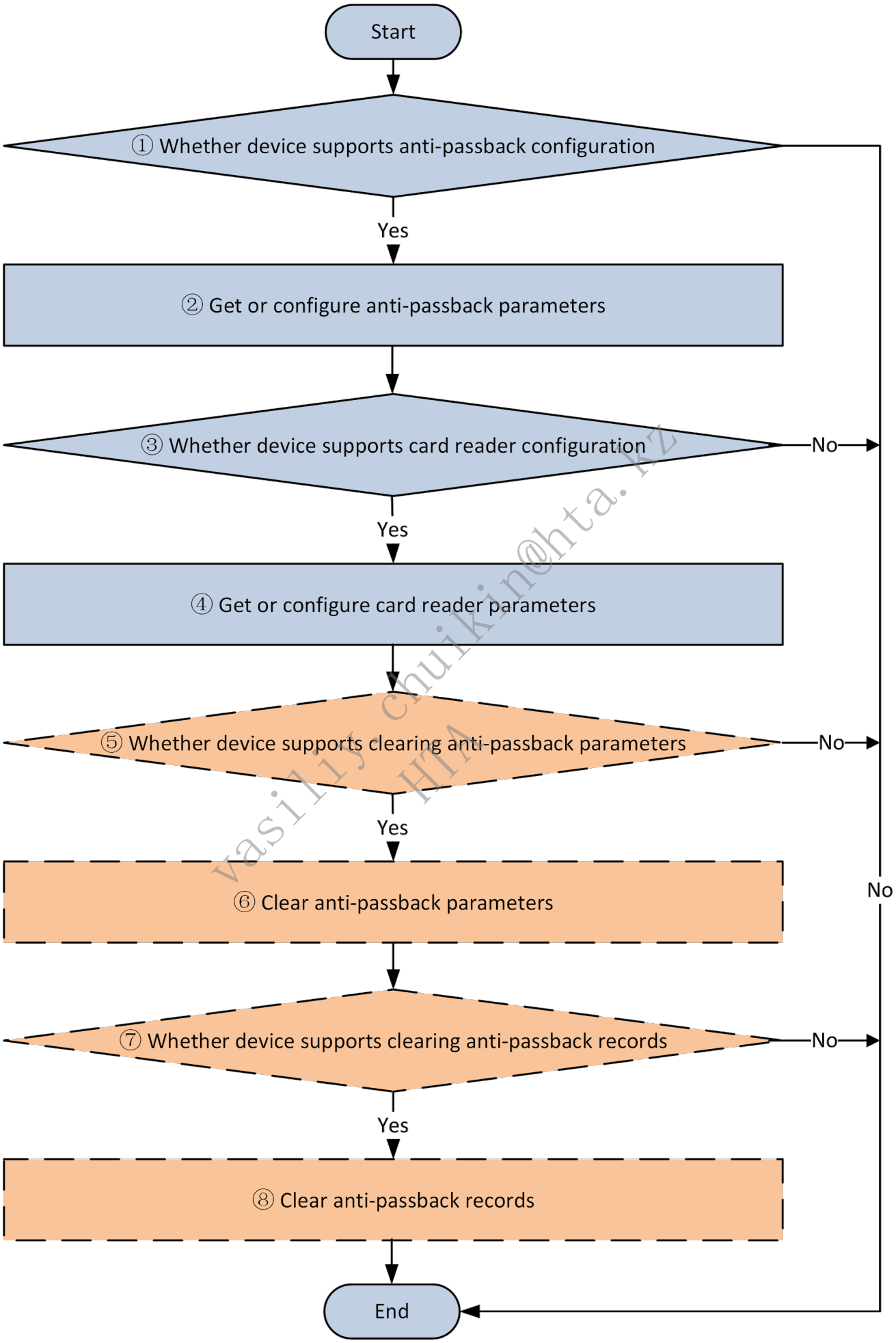
10.1 . Configure Parameters of Stand-Alone Anti-Passback

10.1.1 Introduction to the Function

Anti-passback function allows person to pass entrance and exit by specific route. For areas with multiple entrance/exit, person need to authenticate and pass specific doors. Only one time of authentication and passing is allowed for each door, that is the person need to follow the specific order to pass the doors. Stand-alone anti-passback is designed to minimize the misuse or fraudulent use of access credentials such as passing back card to an unauthorized person, or tailed access. It is applicable to exhibitions, scenic spots, or metro entrances where one card one person is required.

10.1.2 API Calling Flow

Calling Flow:



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ISAPI Protocol Calling Flow:

1. 1. Get the capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportAntiSneakCfg is returned and its value is "true", it indicates that the device supports configuring

parameters of stand-alone anti-passback.

2. Get the parameters of anti-passback configuration: `GET /ISAPI/AccessControl/AntiSneakCfg?format=json`; Set the anti-passing back parameters: `PUT /ISAPI/AccessControl/AntiSneakCfg?format=json`; enable the anti-passback function and the first card reader (first entrance), see details in `GET /ISAPI/AccessControl/AntiSneakCfg/capabilities?format=json`.
3. Get the capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCardReaderAntiSneakCfg is returned and its value is "true", it indicates that the device supports configuring parameters of card readers.
4. Get the anti-passing back configuration parameters of a specified card reader: `GET /ISAPI/AccessControl/CardReaderAntiSneakCfg/<cardReaderID>?format=json`; Set anti-passing back parameters of a card reader: `PUT /ISAPI/AccessControl/CardReaderAntiSneakCfg/<cardReaderID>?format=json`; the node cardReaderID refers to the card reader No. The person's passing route will follow the card reader No. in the API. Note: the anti-passback route should be closed-loop. Improper configuration will affect normal door opening. For example: card reader 1 -> card reader 2 -> card reader 3. The card reader 1 should be set after card reader 3, or authentication in card reader 1 after one loop will fail. Get the configuration capability of anti-passing back parameters of card readers: `GET /ISAPI/AccessControl/CardReaderAntiSneakCfg/capabilities?format=json`.
5. Get the capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportClearAntiSneakCfg is returned and its value is "true", it indicates that the device supports clearing parameters of anti-passback.
6. Clear anti-passing back parameters: `PUT /ISAPI/AccessControl/ClearAntiSneakCfg?format=json`; Get the capability of clearing anti-passback parameters: `GET /ISAPI/AccessControl/ClearAntiSneakCfg/capabilities?format=json`, set the value of antiSneak to false to disable the anti-passback function.
7. Get the capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportClearAntiSneak is returned and its value is "true", it indicates that the device supports clearing records of anti-passback.
8. Clear anti-passback records in the device: `PUT /ISAPI/AccessControl/ClearAntiSneak?format=json`; It supports clearing anti-passback records by person ID. Get the capability of clearing anti-passback records: `GET /ISAPI/AccessControl/ClearAntiSneak/capabilities?format=json`.

Note: clear the historic anti-passback parameters before configuring new parameters.

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10.2 Access Point Unit Management

10.2.1 Introduction to the Function

Configurations related with access point unit, including door parameters configuration, secure door control unit configuration, and configuration of access devices linked to the access point.

10.2.2 API Calling Flow

10.2.2 Door Parameter Configuration

1. Get access control capability: `/ISAPI/AccessControl/capabilities`. If the node `isSupportDoorCfg` exists and is true, it means the device supports door parameters configuration. If the node `isSupportSearchDoorParams` exists and is true, it means the device supports searching for door parameters.

2. Get the parameters of the specified door: `GET /ISAPI/AccessControl/Door/param/<doorID>?security=<security>&iv=<iv>`; configure the parameters of the specified door, such as the type of magnetic contact, exit button type, and door opening duration: `PUT /ISAPI/AccessControl/Door/param/<doorID>?security=<security>&iv=<iv>`. Batch configure door parameters: `PUT /ISAPI/AccessControl/Door/Param?format=json&security=<security>&iv=<iv>`. Supports configuring all doors' parameters, and configuring door parameters by door No. list or by area. Get door parameter capability: `GET /ISAPI/AccessControl/Door/param/<doorID>/capabilities`.

3. Search door parameters: `POST /ISAPI/AccessControl/Door/SearchParams?format=json`. Get the capability of searching door parameters: `GET /ISAPI/AccessControl/Door/SearchParams/capabilities?format=json`.

10.2.2.2 Pairing Parameters of Secure Door Control Unit

Secure door control unit is connected to access control devices via RS485 to access locks, preventing illegal persons from damaging the access control devices and replacing them with other access control devices, which may cause owner to lose access control permissions. The pairing mechanism ensures that the secure door control unit can only be controlled and used by the unique-paired access control device. 2. Exit button can unpair the access control device and secure door control unit.

1. Get access control capability: `/ISAPI/AccessControl/capabilities`. If the node `isSupportDoorSecurityModulePairParams` exists and is true, it indicates that the device supports the function.

2. Get the pairing parameters of secure door control unit: `GET /ISAPI/AccessControl/Door/DoorSecurityModulePairParams?format=json`. Set the parameters to control the enabling status of the pairing function: `PUT /ISAPI/AccessControl/Door/DoorSecurityModulePairParams?format=json`. Get the capability of configuring the pairing parameters of secure door control unit: `GET /ISAPI/AccessControl/Door/DoorSecurityModulePairParams/capabilities?format=json`.

10.2.2.3 Switch Parameters of Secure Door Control Unit

This function is for connecting multiple secure door control units to an access control device, so as to configure the doors of secure door control units.

1. Get access control capability: `/ISAPI/AccessControl/capabilities`. If the node `isSupportDoorSecurityModuleSwitchParams` exists and is true, it indicates that the device supports the function.

2. Get the switch parameters of secure door control unit: `GET /ISAPI/AccessControl/Door/DoorSecurityModuleSwitchParams?format=json`; configure parameters to switch the doors accessed by secure door control unit: `PUT /ISAPI/AccessControl/Door/DoorSecurityModuleSwitchParams?format=json`. Get the switch parameter capability of secure door control unit: `GET /ISAPI/AccessControl/Door/DoorSecurityModuleSwitchParams/capabilities?format=json`.

10.2.2.4 Get Secure Door Control Unit Status

1. Get access control capability: `/ISAPI/AccessControl/capabilities`. If the node `isSupportModuleStatus` exists and is true, it indicates that the device supports the function.

2. Get secure door control unit status: `GET /ISAPI/AccessControl/DoorSecurityModule/moduleStatus`. Get parameter capability of secure door control unit status: `GET /ISAPI/AccessControl/DoorSecurityModule/moduleStatus/capabilities`.

10.2.2.5 Parameters of Door Lock Status When Powered Off

1. Get access control capability: `/ISAPI/AccessControl/capabilities`. If the node `isSupportLockTypeCfg` exists and is true, it indicates that the device supports the function.

2. Get the parameters of the door lock status when the device is powered off: `GET /ISAPI/AccessControl/Configuration/lockType?format=json&doorID=<doorID>`; Configure the lock status when the device is powered off: `PUT /ISAPI/AccessControl/Configuration/lockType?format=json&doorID=<doorID>`. Get the capability of configuring the door lock status when the device is powered off: `GET /ISAPI/AccessControl/Configuration/lockType/capabilities?format=json`.

10.2.2.6 Definition Rule of Magnetic Contact Status

1. Get access control capability: `/ISAPI/AccessControl/capabilities`. If the node `isSupportDoorMagneticDefiniteRule` exists and is true, it indicates that the device supports the function.

2. Get the definition rule of magnetic contact status: `GET /ISAPI/AccessControl/doorMagneticDefiniteRule?format=json`; set the definition rule of magnetic contact status: `PUT /ISAPI/AccessControl/doorMagneticDefiniteRule?format=json`. Get the capability of the definition rule of magnetic contact status: `GET /ISAPI/AccessControl/doorMagneticDefiniteRule/capabilities?format=json`.

10.2.2.7 Parameters Configuration of Unlocking Door

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The decoder's web client supports unlocking the door station, including lock 1, lock 2, and lock 1 and lock 2. Application scenario: When the door station initiates a call to the analog handset, after the analog handset dials in, the door station can start two-way audio communication with the 2-wire analog handset. After the analog handset dials in and sends signaling to the decoder, the decoder unlocks the door station by sending the lock ID.

1. Get access control capability: `/ISAPI/AccessControl/capabilities`. If the node `isSupportOpenDoorParams` exists and is true, it indicates that the device supports the function.

2. Get door-open parameters: `GET /ISAPI/AccessControl/Door/OpenDoorParams?format=json`; set door-open parameters: `PUT /ISAPI/AccessControl/Door/OpenDoorParams?format=json` 3. Get the capability of configuring the parameters of unlocking doors: `GET /ISAPI/AccessControl/Door/OpenDoorParams/capabilities?format=json`.

10.2.2.8 Get Lock Parameters

1. Get access control capability: `/ISAPI/AccessControl/capabilities`. If `isSupportGetSmartLockParam` and `isSupportWorkStatus` exist and the value is true, it indicates the support for getting smart lock information list and lock status.

2. Get the list of smart lock information: `GET /ISAPI/VideoIntercom/SmartLock/lockParam`. Get the capability of smart lock information list: `GET /ISAPI/VideoIntercom/SmartLock/lockParam/capabilities`.

3. Get lock status: `GET /ISAPI/VideoIntercom/WorkStatus`.

10.2.2.9 Access Devices Linked to Access Point

Configure access devices linked to access point, including but not limited to card readers, doorbells, door locks, exit buttons, and relays. Supports binding and unbinding access devices.

1. Get access control capability: `/ISAPI/AccessControl/capabilities`. If `isSupportDoorLinkageDeviceParams`、`isSupportGetDoorLinkageDeviceNum`、`isSupportSearchAvailableLinkageDoor`、`isSupportDoorLinkageDevice`、`isSupportOneKeyUnbindDoorLinkageDevice`、`isSupportSearchDoorLinkageDeviceList` is true, it indicates that the device supports getting the parameters and number of access devices linked to access points, searching for the information of doors which can be linked to access device, configuring and quickly unkinking access devices, and searching for the list of access devices linked to access points.

2. Get the parameters of access device linked to access point: `GET /ISAPI/AccessControl/Door/DoorLinkageDeviceParams?format=json`. It can be used to determine the supported number of access devices which can be linked to single access point.

3. Get the number of access devices linked to access points: `POST /ISAPI/AccessControl/Door/GetDoorLinkageDeviceNum?format=json`. It can be used to search for the number of access devices (abnormal ones included) linked to the door. Get parameter capability of the number of access devices linked to the access point: `GET /ISAPI/AccessControl/Door/GetDoorLinkageDeviceNum/capabilities?format=json`.

4. Search for the list of access devices linked to the access point: `GET /ISAPI/AccessControl/Door/door/<doorID>/SearchDoorLinkageDeviceList?format=json`.

5. Configure access devices linked to the access point: `PUT /ISAPI/AccessControl/Door/door/<doorID>/DoorLinkageDevice?format=json`. It can be used to link or unlink access devices. Get the parameter capability of access devices linked to the access point: `GET /ISAPI/AccessControl/Door/door/<doorID>/DoorLinkageDevice/capabilities?format=json`.

6. Supports quickly unlinking the access devices with the access point: `PUT /ISAPI/AccessControl/Door/door/<doorID>/OneKeyUnbindDoorLinkageDevice?format=json`.

7. Search for the information of doors which can be linked to access device: `POST /ISAPI/AccessControl/Door/SearchAvailableLinkageDoor?format=json`. Get parameter capability of searching for the information of doors which can be linked to access device: **** `GET /ISAPI/AccessControl/Door/SearchAvailableLinkageDoor/capabilities?format=json`.

10.3 Arming Information

10.3.1 Introduction to the Function

The device supports getting the arming information, such as the armed device IP and port, arming type, and protocol

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type.

10.3.2 API Calling Flow

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportDeployInfo is returned and its value is "true", it indicates that the device supports getting arming information.
2. Get arming information capability: `GET /ISAPI/AccessControl/DeployInfo/capabilities`.
3. Get arming information: `GET /ISAPI/AccessControl/DeployInfo`.

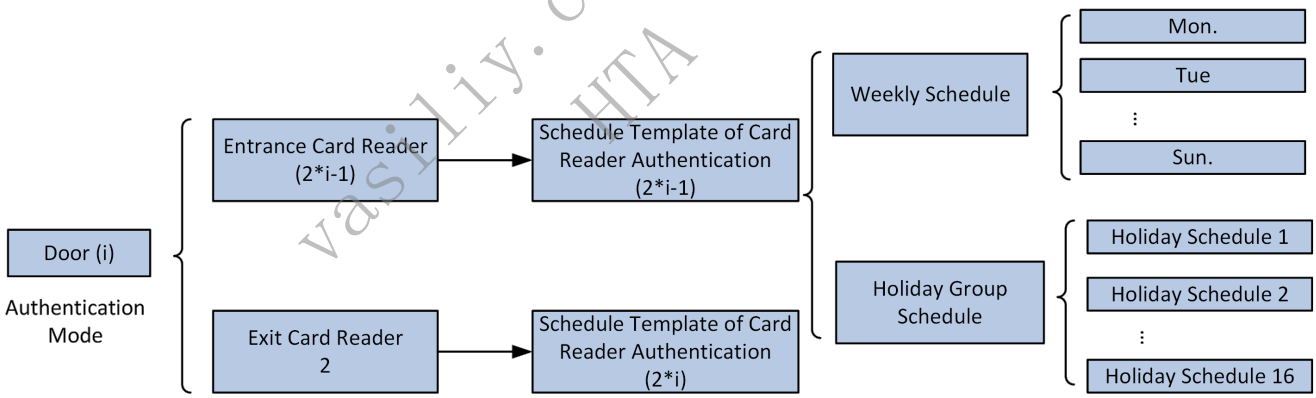
10.4 Authentication Schedule Management

10.4.1 Introduction to the Function

By calling the following APIs, the entrance&exit time schedule and authentication mode can be applied to the card reader. An access control terminal is controlled by two card readers. For example: access control terminal 1 is controlled by entrance card reader 1 and exit card reader 2; access control terminal 2 is controlled by entrance card reader 3 and exit card reader 4, etc. Card reader n is corresponding to schedule template n.

1 weekly schedule and 4 holiday groups can be added in each schedule template. The priority of holiday schedule is higher than that of weekly schedule. A weekly schedule can be configured by date of a week and 8 different time periods of a day. 16 holiday schedules can be added to a holiday group schedule. Each holiday schedule has its start and end date, and the time period is same in the range (8 time periods can be added). The access control can follow the schedule template to manage the time of person's permissions.

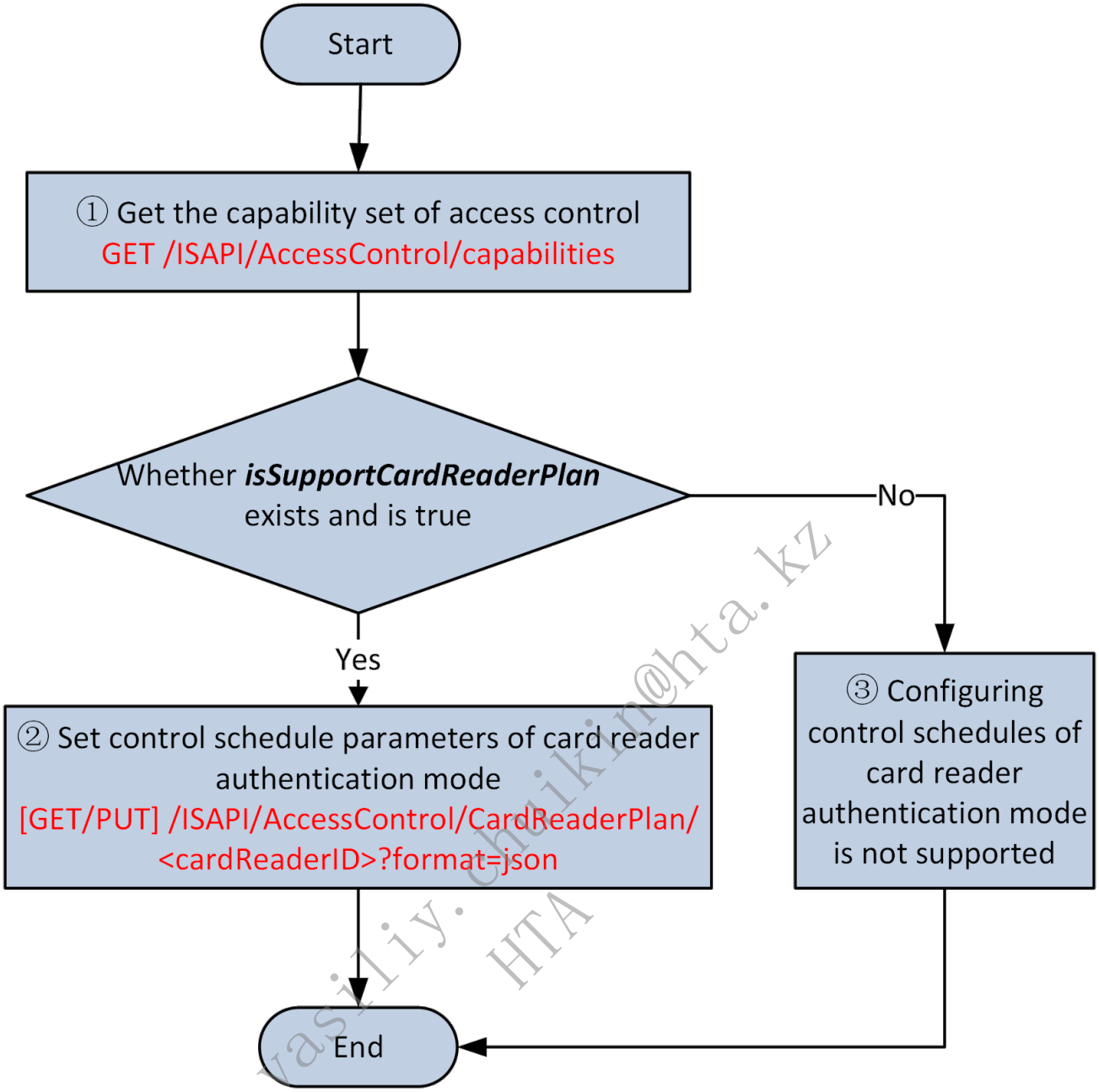
For Person Management of Person and Credential Management, the priority of the authentication method for person is higher than that of the authentication schedule. If the authentication method (the node is userVerifyMode) is applied to a person, the person can access according to the authentication method for person.



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10.4.2 API Calling Flow

10.4.2.1 Card Reader's Authentication Schedule Configuration

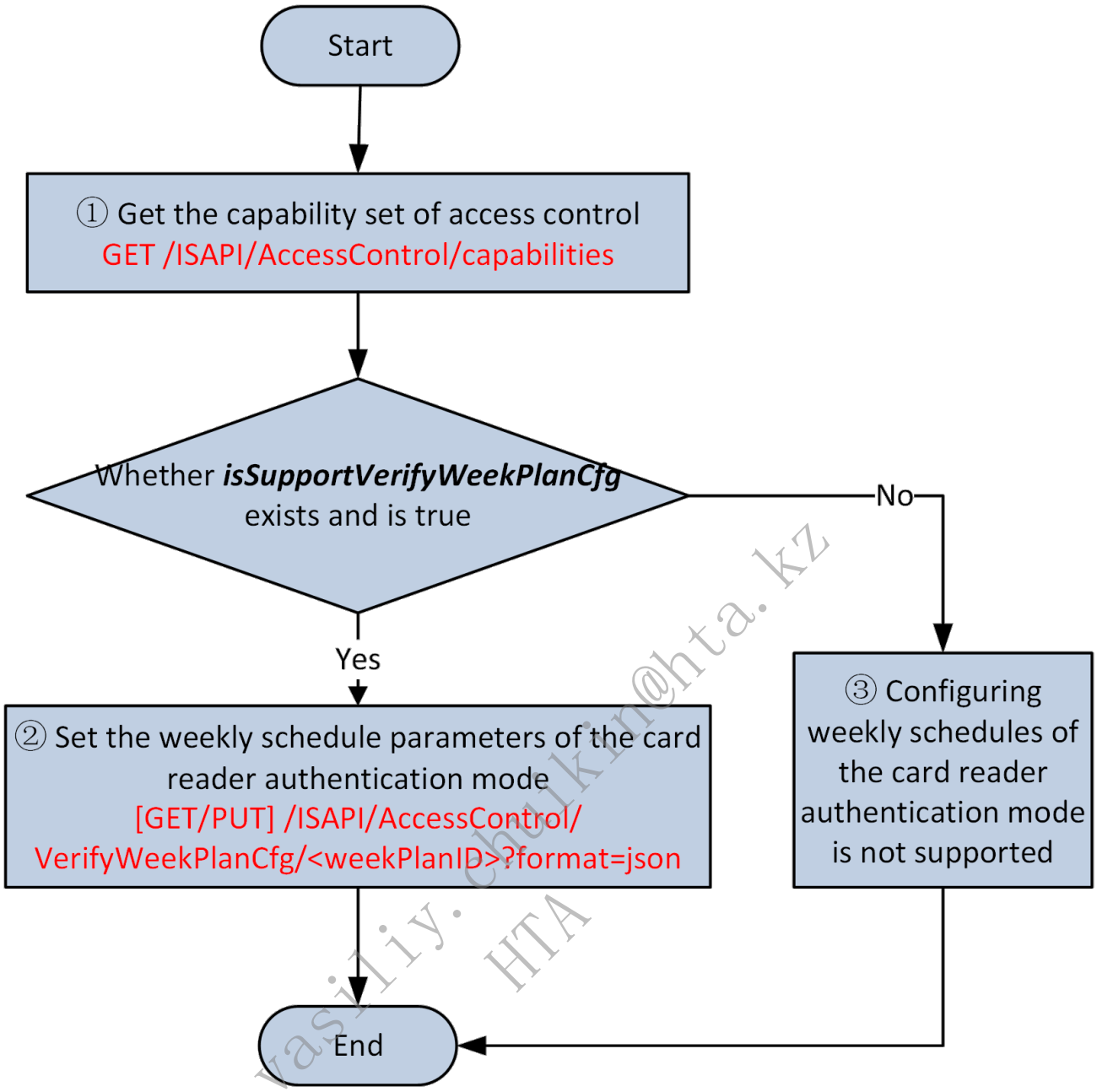


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The API calling flow is as follows:

1. Check whether the device supports configuring control schedules of card reader authentication mode: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCardReaderPlan is returned and its value is "true", it indicates that the device supports configuring acontrol schedules of card reader authentication mode (and the device also supports configuring schedule templates of the card reader authentication mode).
2. Set control schedule parameters of card reader authentication mode: `[GET/PUT] /ISAPI/AccessControl/CardReaderPlan/<cardReaderID>?format=json`; the value of cardReaderID should be the same as the value of templateNo.
3. If the node isSupportCardReaderPlan is returned and its value is "false", it indicates that the device does not support configuring control schedules of card reader authentication mode.

10.4.2.2 Authentication Schedule Template Configuration

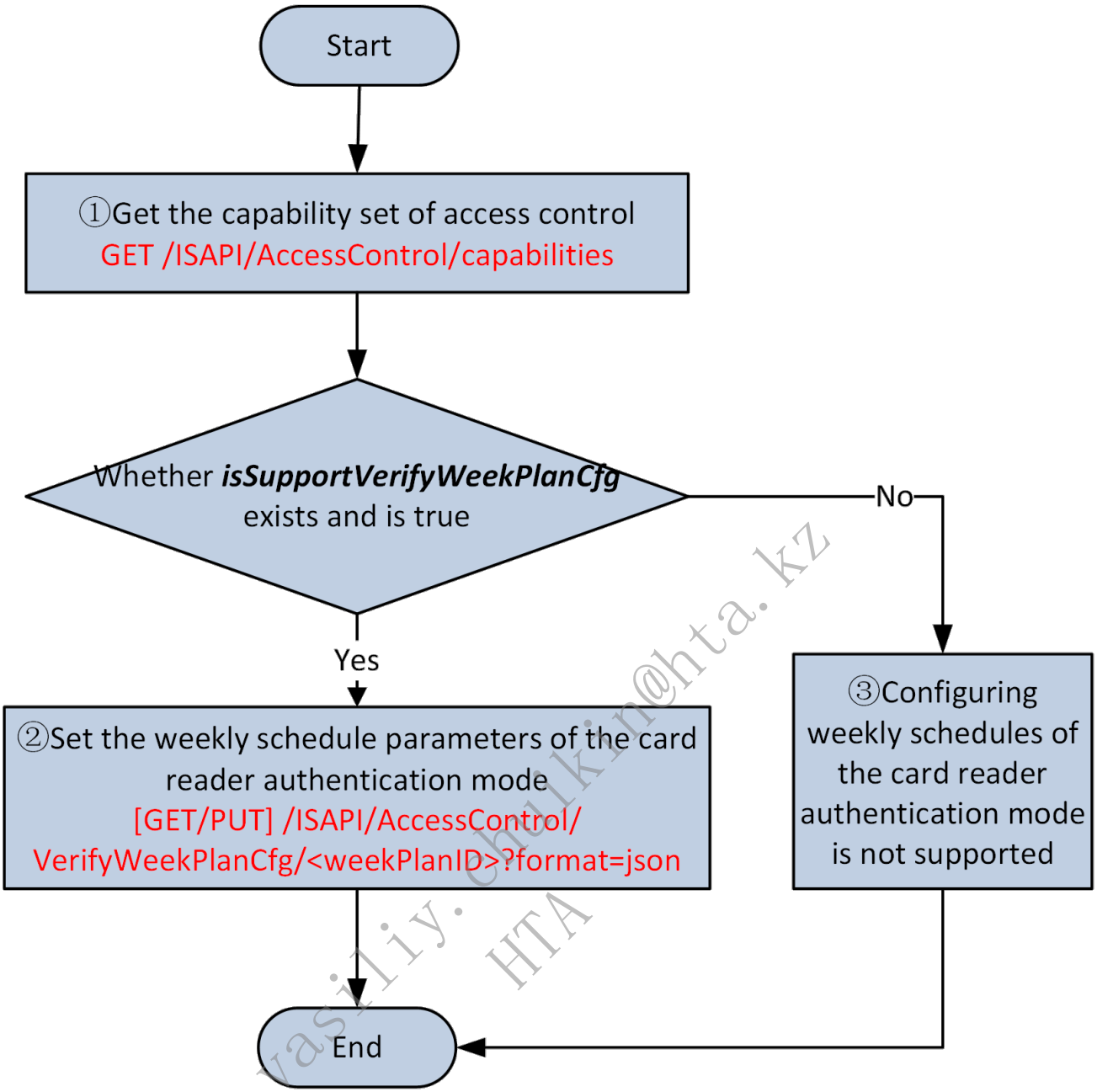


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The API calling flow is as follows:

1. Check whether the device supports configuring schedule templates of the card reader authentication mode: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportVerifyPlanTemplate is returned and its value is "true", it indicates that the device supports configuring schedule templates of the card reader authentication mode (and the device also supports configuring weekly schedules of the card reader authentication mode).
2. Set the schedule template parameters of the card reader authentication mode: `[GET/PUT] /ISAPI/AccessControl/VerifyPlanTemplate/<planTemplateID>?format=json`.
3. If the node isSupportVerifyPlanTemplate is returned and its value is "false", it indicates that the device does not support configuring schedule templates of the card reader authentication mode.

10.4.2.3 Weekly Authentication Schedule Template Configuration

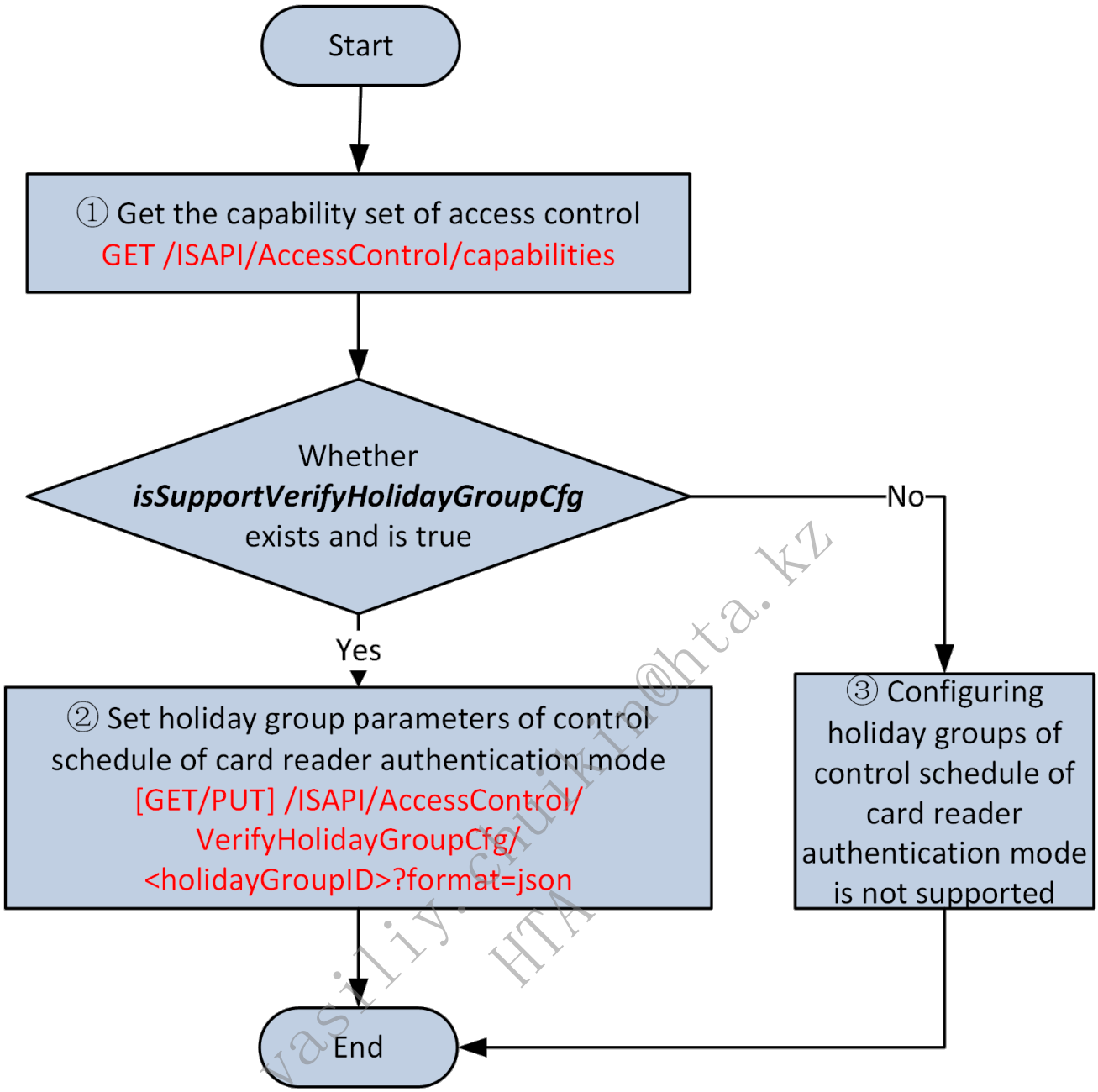


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The API calling flow is as follows:

1. Check whether the device supports configuring weekly schedules of the card reader authentication mode: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCardRightWeekPlanCfg is returned and its value is "true", it indicates that the device supports configuring weekly schedules of the card reader authentication mode.
2. Set the weekly schedule parameters of the card reader authentication mode: `[GET/PUT] /ISAPI/AccessControl/VerifyWeekPlanCfg/<weekPlanID>?format=json`.
3. If the node isSupportVerifyWeekPlanCfg is returned and its value is "false", it indicates that the device does not support configuring weekly schedules of the card reader authentication mode.

10.4.2.4 Holiday Authentication Group Configuration

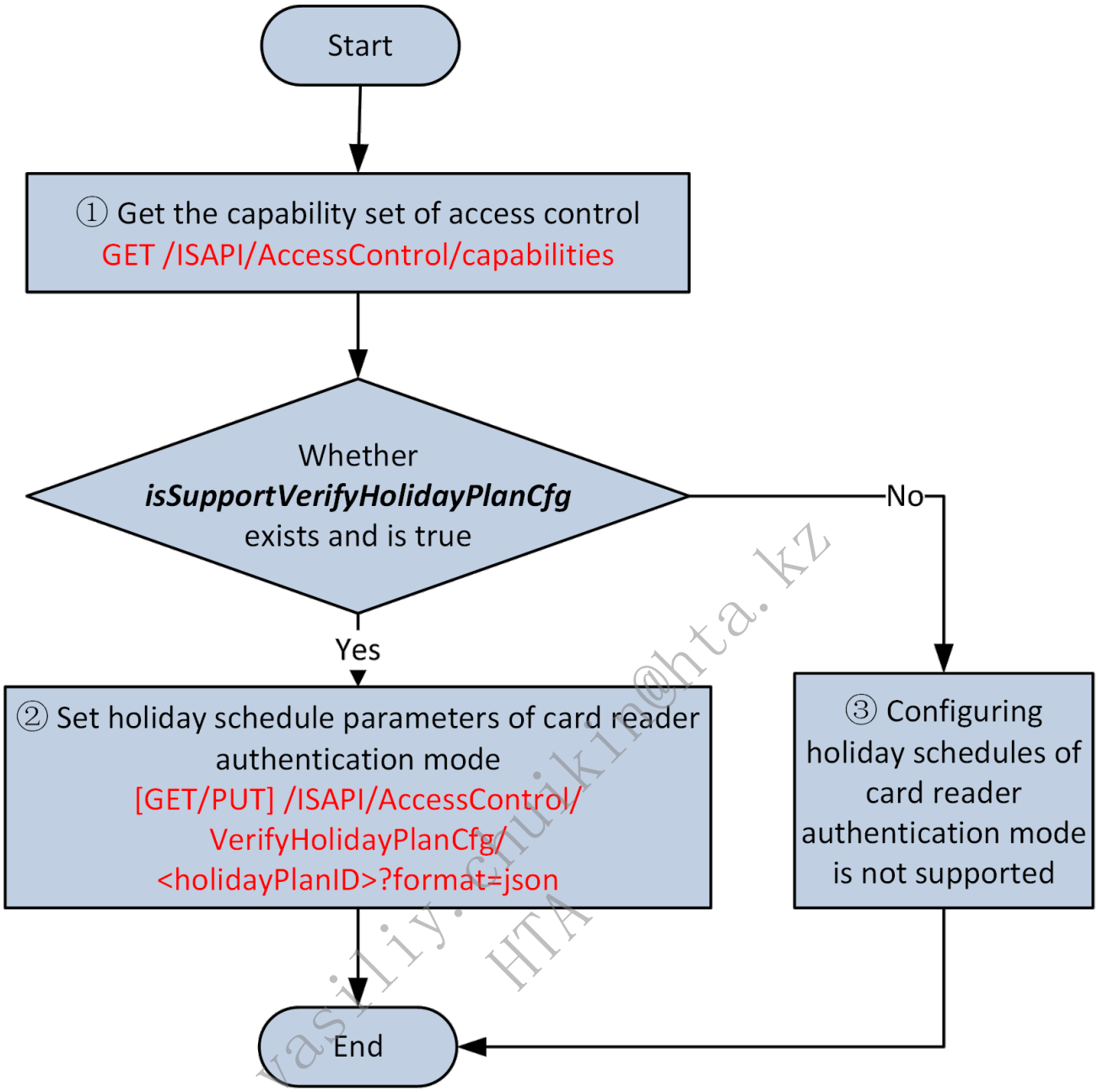


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The API calling flow is as follows:

1. Check whether the device supports configuring holiday groups of control schedule of card reader authentication mode: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportVerifyHolidayGroupCfg is returned and its value is "true", it indicates that the device supports configuring holiday groups of control schedule of card reader authentication mode (and the device also supports configuring holiday schedules of card reader authentication mode).
2. Set holiday group parameters of control schedule of card reader authentication mode: `[GET/PUT] /ISAPI/AccessControl/VerifyHolidayGroupCfg/<holidayGroupID>?format=json`.
3. If the node isSupportVerifyHolidayGroupCfg is returned and its value is "false", it indicates that the device does not support configuring holiday groups of control schedule of card reader authentication mode.

10.4.2.5 Holiday Authentication Schedule Configuration



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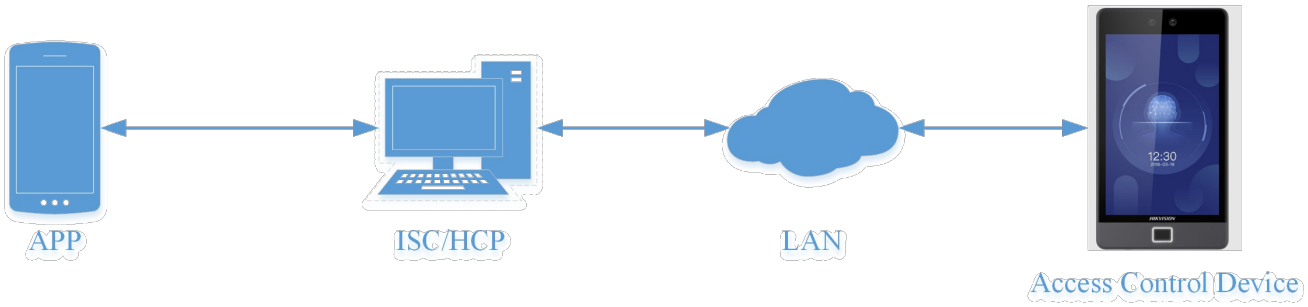
The API calling flow is as follows:

1. Check whether the device supports configuring holiday schedules of card reader authentication mode: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportVerifyHolidayPlanCfg is returned and its value is "true", it indicates that the device supports configuring holiday schedules of card reader authentication mode.
2. Set holiday schedule parameters of card reader authentication mode: `[GET/PUT] /ISAPI/AccessControl/VerifyHolidayPlanCfg/<holidayPlanID>?format=json`.
3. If the node isSupportVerifyHolidayPlanCfg is returned and its value is "false", it indicates that the device does not support configuring holiday schedules of card reader authentication mode.

10.5 Authentication via QR Code

10.5.1 Generate QR Code by Platform

10.5.1.1 Introduction to the Function



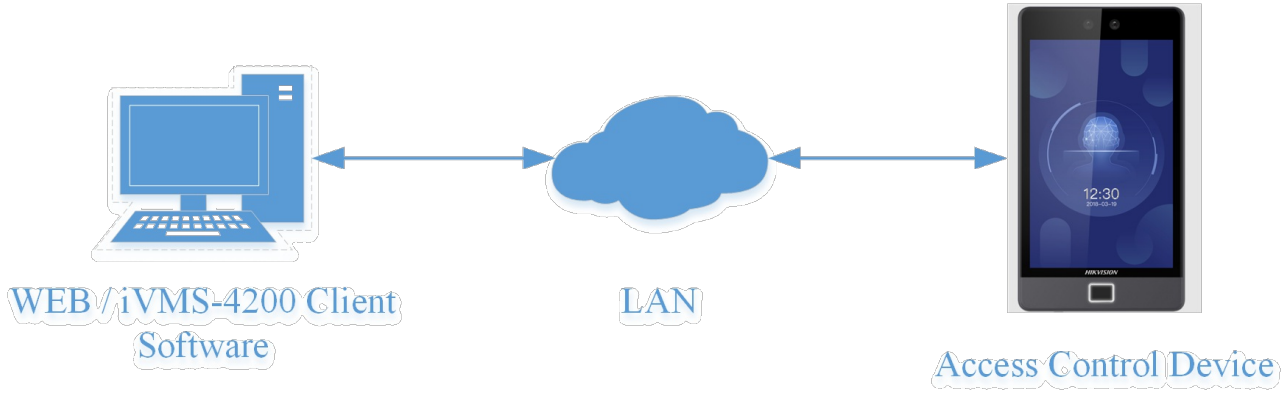
1. After an access control device is deployed, you should add it to the platform and set the QR code key which will be saved to the database of the platform. **Notes: For security, the QR code key should be encrypted before been saved to the platform and the device; if you need to open multiple doors by scanning the QR codes, the device key of the multiple doors should be the same.**
2. The platform applies persons' permissions to the device via person-based function. The QR code is linked with employee No., so the permission of opening a door is decided by whether a person's employee No. is authenticated.
3. Before entering the controlled areas, the visitors need to register on the platform. The platform will generate QR code strings according to the registered information, QR code key, and QR code protocol, for converting QR code strings to QR code pictures, and sending to visitors' phones.
4. When the visitors enter the controlled areas, they can scan the QR codes via the devices. Then, the device can get the encrypted data, decrypt the data by QR code key and protocol, and check whether the door can be opened. If yes, the door will open; if no, the door will not open and the device will prompt the message "No permission".

10.5.1.2 API Calling Flow

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportQRCodeEncryption is returned and its value is "true", it indicates that the device supports QR code encryption.
2. Get the capability of QR code encryption: `GET /ISAPI/AccessControl/QRCodeEncryption/capabilities?format=json`.
3. Set the parameters of QR code encryption: `PUT /ISAPI/AccessControl/QRCodeEncryption?format=json&security=<security>&iv=<iv>`, the parameters include encryption type, encryption key, and initialization vector.
4. Get the related APIs of Person and Credential Management for applying certain persons' permissions to open the doors. So the permission of opening a door is decided by whether a person's employee No. is authenticated.
5. After generating QR code strings by Base64 and generating QR code pictures based on QR code strings, the platform for sending the pictures to visitors' phones. The visitors can access the area after scanning the QR code pictures.

10.5.2 Generate QR Code by Device

10.5.2.1 Introduction to the Function

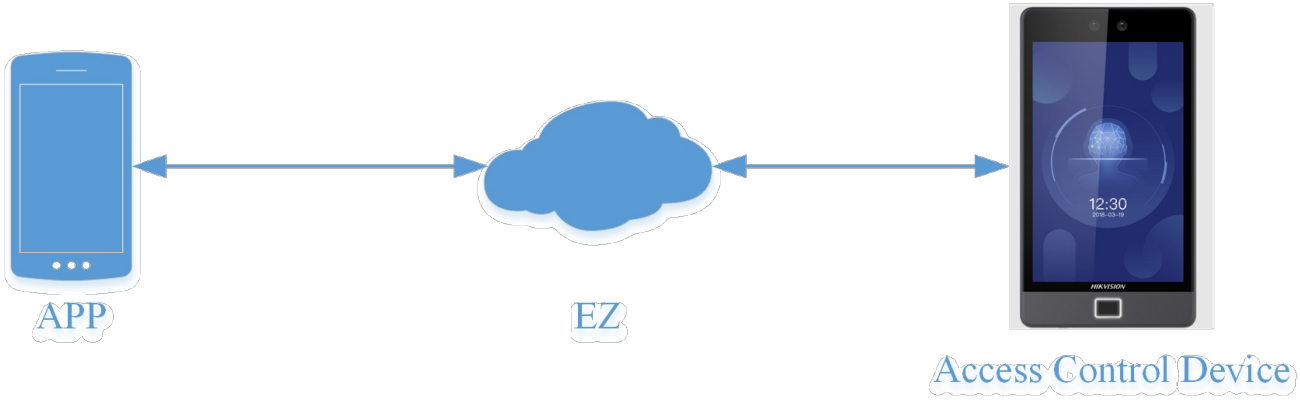


1. After an access control device is deployed, you can log in to the device via the Web / iVMS-4200 client software,

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and set a QR code key for the device (saving the QR code key is not necessary). **Notes: to ensure security, the QR code key should be encrypted before saving to the device; if multiple doors need to be opened via the QR code, the devices of the multiple doors should be configured a same key.**

2. By calling the APIs of person-based functions, the Web / iVMS-4200 client software applies persons' permission to the device. So the permission of opening a door is decided by whether a person's employee No. is authenticated.



3. Before visitors enter the controlled areas, the inviters need to upload the visitor information to EZ and apply the information to the device. The device will generate QR code strings according to the visitor information, QR code key, and QR code protocol, and send back to the inviters. The inviters' phones will convert QR code strings to QR code pictures, and forward the pictures to visitors' phones.
4. When the visitors enter the controlled areas, they can scan the QR codes via the devices. Then, the device can get the encrypted data, decrypt the data by QR code key and protocol, and check whether the door can be opened. If yes, the door will open; if no, the door will not open and prompt the message "No permission". **Note: for usability and security, APIs of generating QR codes by the platform are not accessible, only APIs of generating QR code by device are provided.**

10.5.2.2 API Calling Flow

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the nodes isSupportQRCodeEncryption and isSupportQRCodeInfo are returned and the value is "true", it indicates that the device supports QR code encryption and generating QR code.
2. Get the capability of QR code encryption: `GET /ISAPI/AccessControl/QRCodeEncryption/capabilities?format=json`. Get the capability of generating QR code: `GET /ISAPI/AccessControl/QRCodeInfo/capabilities?format=json`.
3. Set the parameters of QR code encryption: `PUT /ISAPI/AccessControl/QRCodeEncryption?format=json&security=<security>&iv=<iv>`, the parameters include encryption type, encryption key, and initialization vector.
4. Get the related APIs of Person and Credential Management for applying certain persons' permissions to open the doors. The permission of opening a door will be decided by whether a person's employee No. is authenticated.
5. Generate QR code strings encoded via base64: `POST /ISAPI/AccessControl/QRCodeInfo?format=json&security=<security>&iv=<iv>`; then the device will send the QR code strings to the phones of the inviters.
6. Inviters' phones convert the QR code strings to QR code pictures, and send to visitors' phones.
7. The visitors can access the area by scanning the QR code pictures in their phones.

10.5.3 Remarks

The device will send AccessControllerEvent (contains QRCodeInfo) for QR code authentication. If the authentication succeed, the returned event type will be 0x9c, if the authentication failed, the returned event type will be 0x9d.

10.6 Card Management

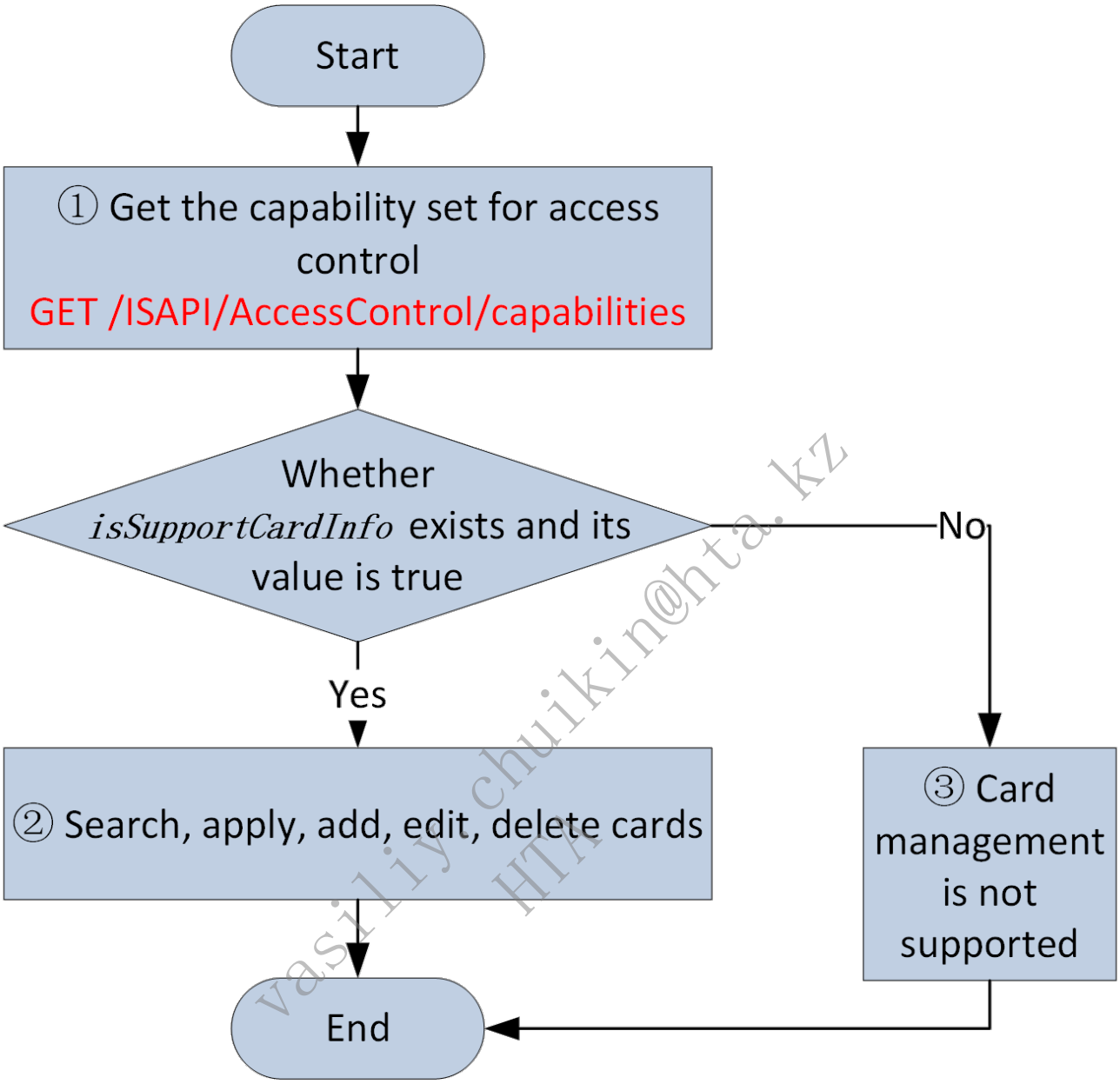
10.6.1 Introduction to the Function

Card management includes searching, applying, adding, editing, deleting, and collecting cards.

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10.6.2 API Calling Flow

10.6.2.1 Check Whether the Device Supports Card Management



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Before calling the API for card management, make sure that the device supports card management.

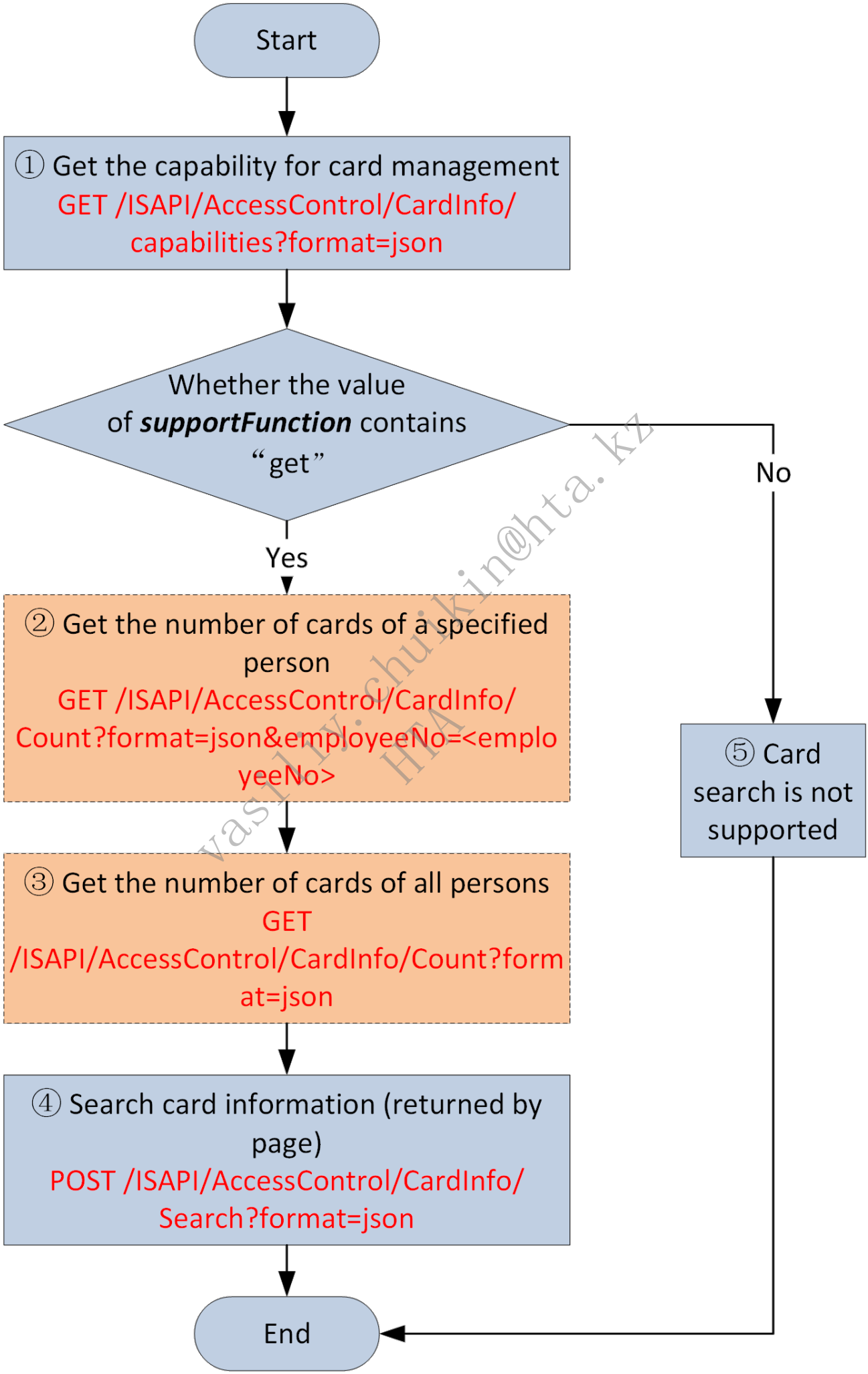
1. Check whether the device supports card management: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCardInfo is returned and its value is "true", it indicates that the device supports card management.
2. Search, apply, add, edit, and delete cards.
3. If the node isSupportCardInfo is returned and its value is "false", it indicates that the device does not support card management.

Note:

- Before applying, adding, or editing cards on the device, make sure that the related person information linked to the person ID has been applied to the device.
- The value of the node numberPerPerson returned by calling `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json` is the maximum number of cards supported per person. If the value returned is 255, it indicates that the number of cards per person is unlimited. If the node is not returned, it indicates that the maximum number of cards can be applied is 5.
- Manage cards of different card number lengths by calling `[GET/PUT] /ISAPI/AccessControl/CardVerificationRule?format=json`.

10.6.2.2 Card Search

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The card search function is for searching the number of cards and card information applied to the device.

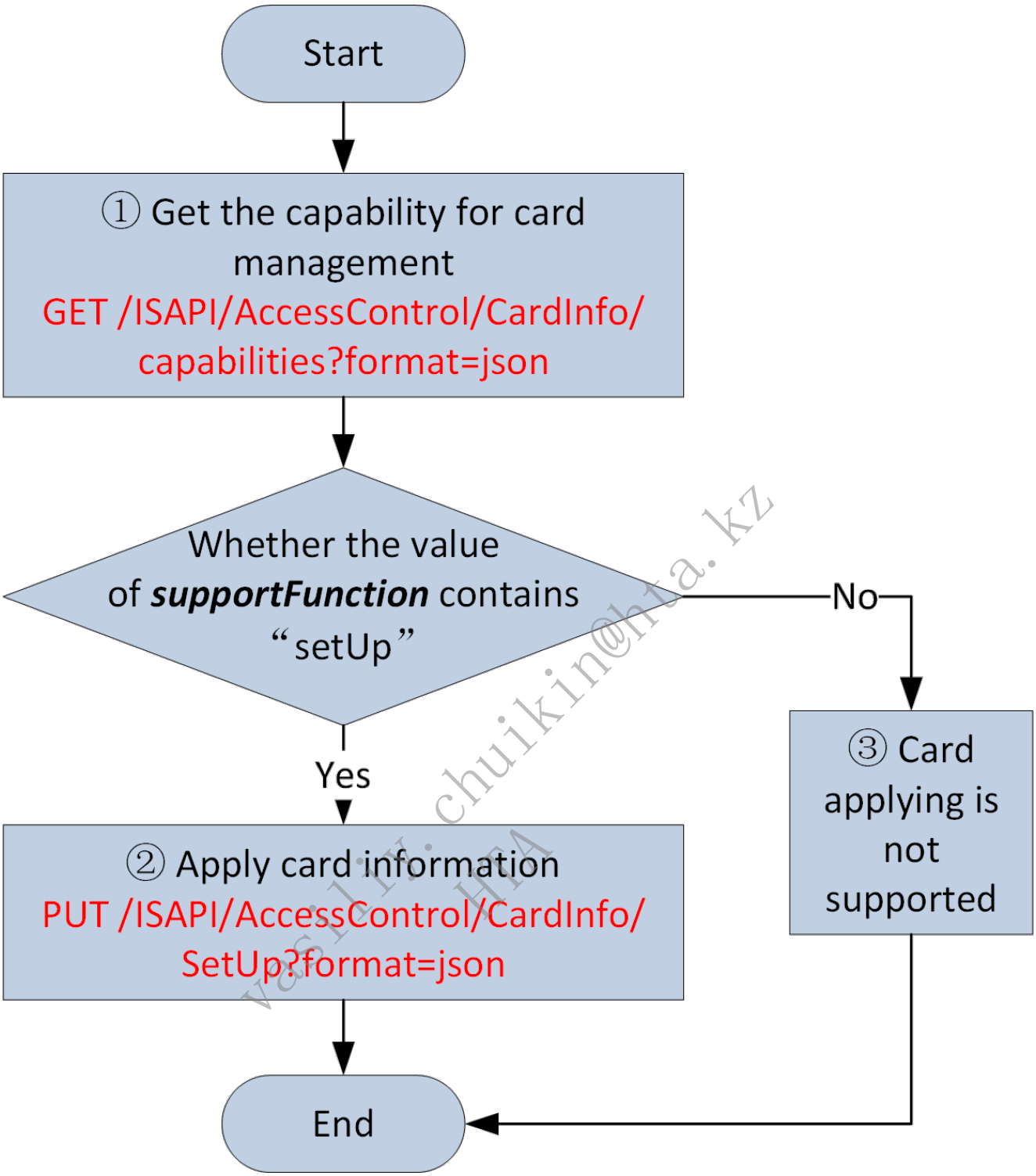
1. Check whether the device supports card search: `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json`; if the value of the node supportFunction contains "get", it indicates that the device supports card search.
2. Search the number of specified persons' cards: `GET /ISAPI/AccessControl/CardInfo/Count?format=json&employeeNo=<employeeNo>`; the returned value of the node cardNumber is the number of the cards added to the specified persons.
3. Search the number of all persons' cards: `GET /ISAPI/AccessControl/CardInfo/Count?format=json`; the returned value of the node cardNumber is the number of the cards added to all persons.
4. Search card information: `POST /ISAPI/AccessControl/CardInfo/Search?format=json`; the card information is returned by page.
5. If the value of the node supportFunction does not contain "get", it indicates that the device does not support card search.

Note:

The value of the node maxRecordNum returned by calling `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json` is the maximum number of cards supported by the device.

10.6.2.3 Card Applying

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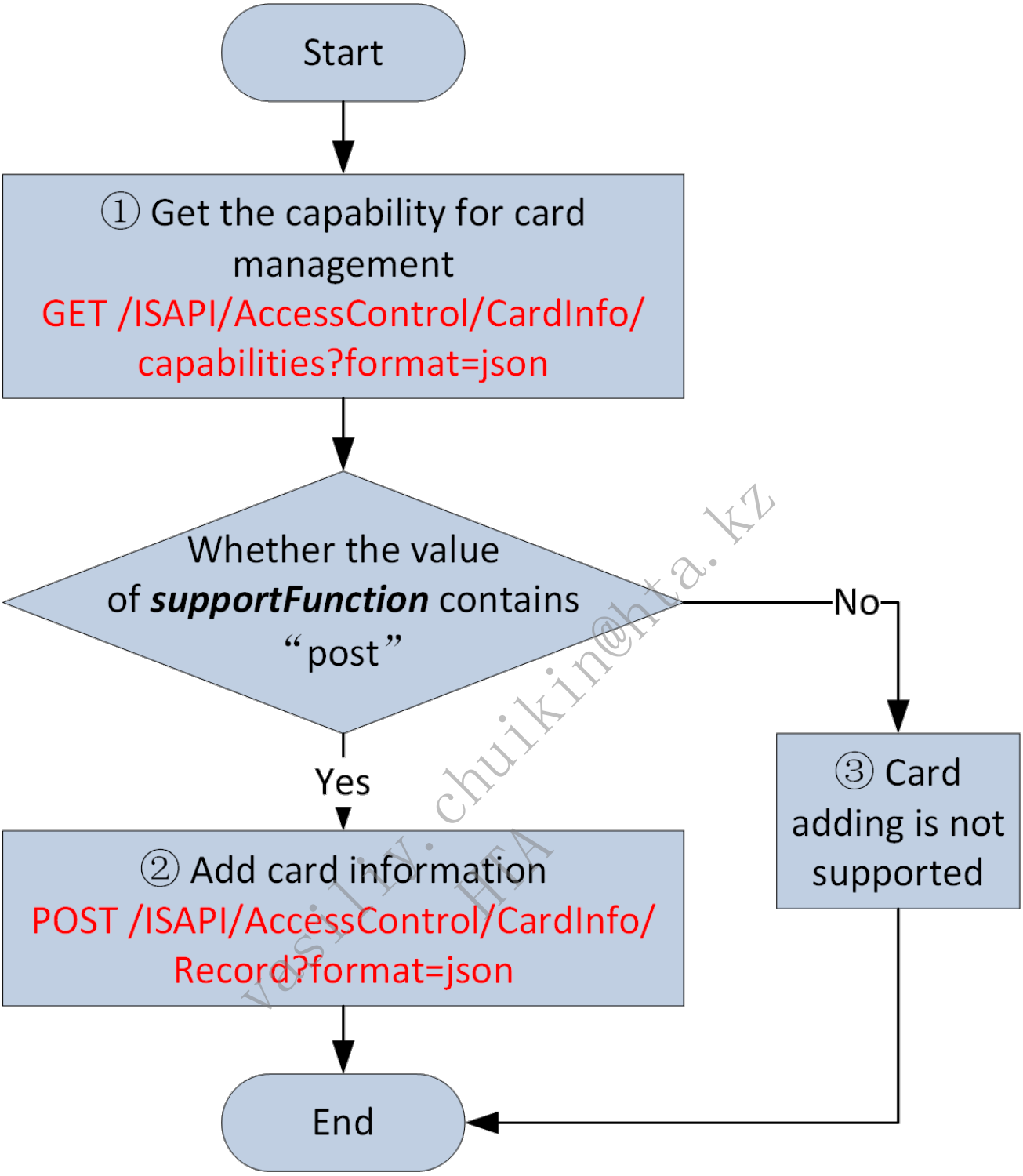
Card information can be applied to the device via the card applying function. If the card has been added to the device, the card information will be edited; if the card has not been added to the device, the card information will be added to the device.

1. Check whether the device supports card applying: `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json`; if the value of the node supportFunction contains "setUp", it indicates that the device supports card applying.
2. Apply card information: `PUT /ISAPI/AccessControl/CardInfo/SetUp?format=json`.
3. If the value of the node supportFunction does not contain "setUp", it indicates that the device does not support card applying.

Note:

Check whether the card has been added to the device via the node cardNo returned after calling the API for card applying.

10.6.2.4 Card Adding



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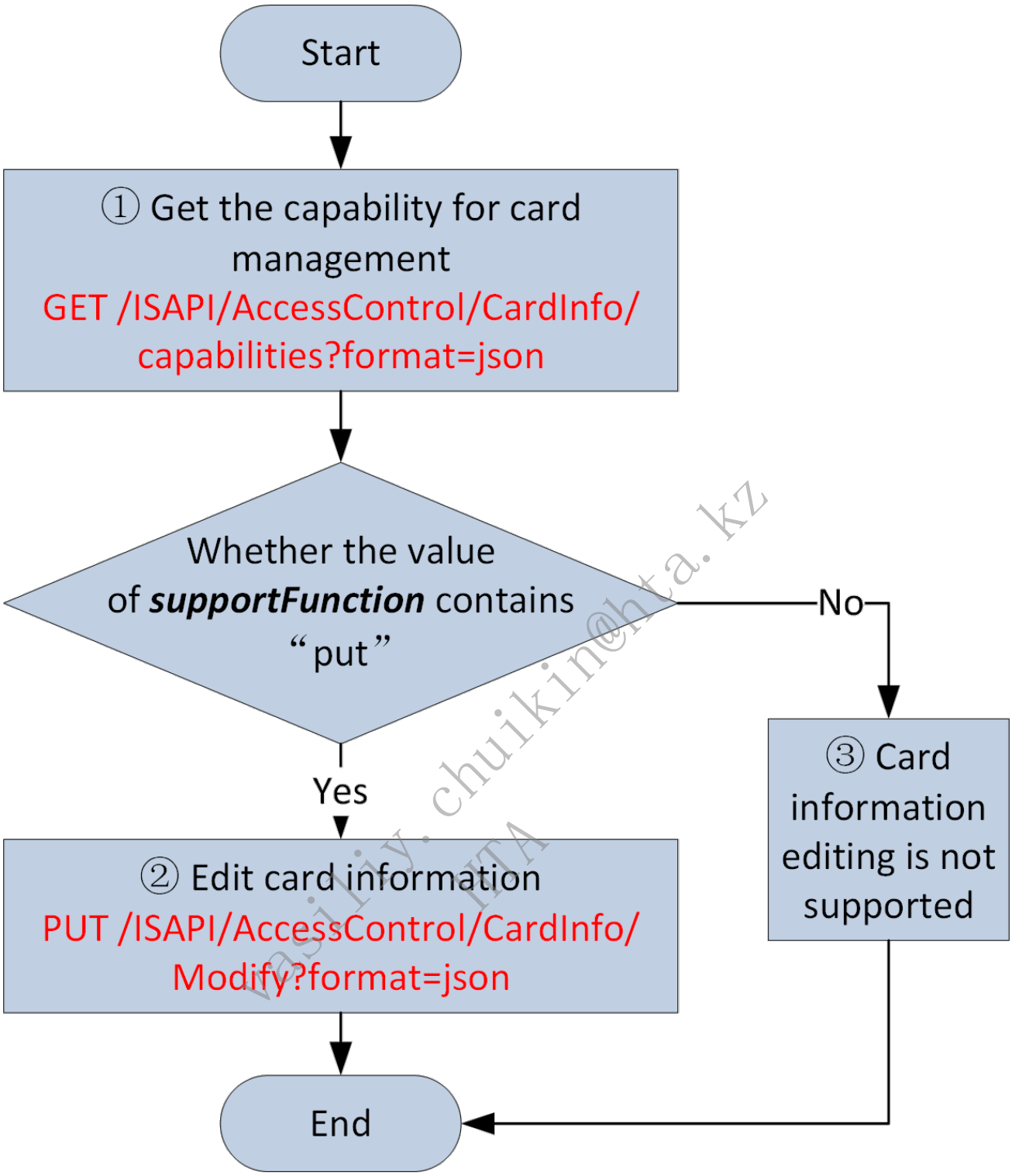
Card information can be added to the device via the card adding function. If the card has been added to the device, the device will report an error; if the card has not been added to the device, the card information will be added to the device.

1. Check whether the device supports card adding: `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json`; if the value of the node supportFunction contains "post", it indicates that the device supports card adding.
2. Add card information: `POST /ISAPI/AccessControl/CardInfo/Record?format=json`.
3. If the value of the node supportFunction does not contain "post", it indicates that the device does not support card adding.

Note:

Check whether the card has been added to the device via the node cardNo returned after calling the API for card adding.

10.6.2.5 Card Information Editing



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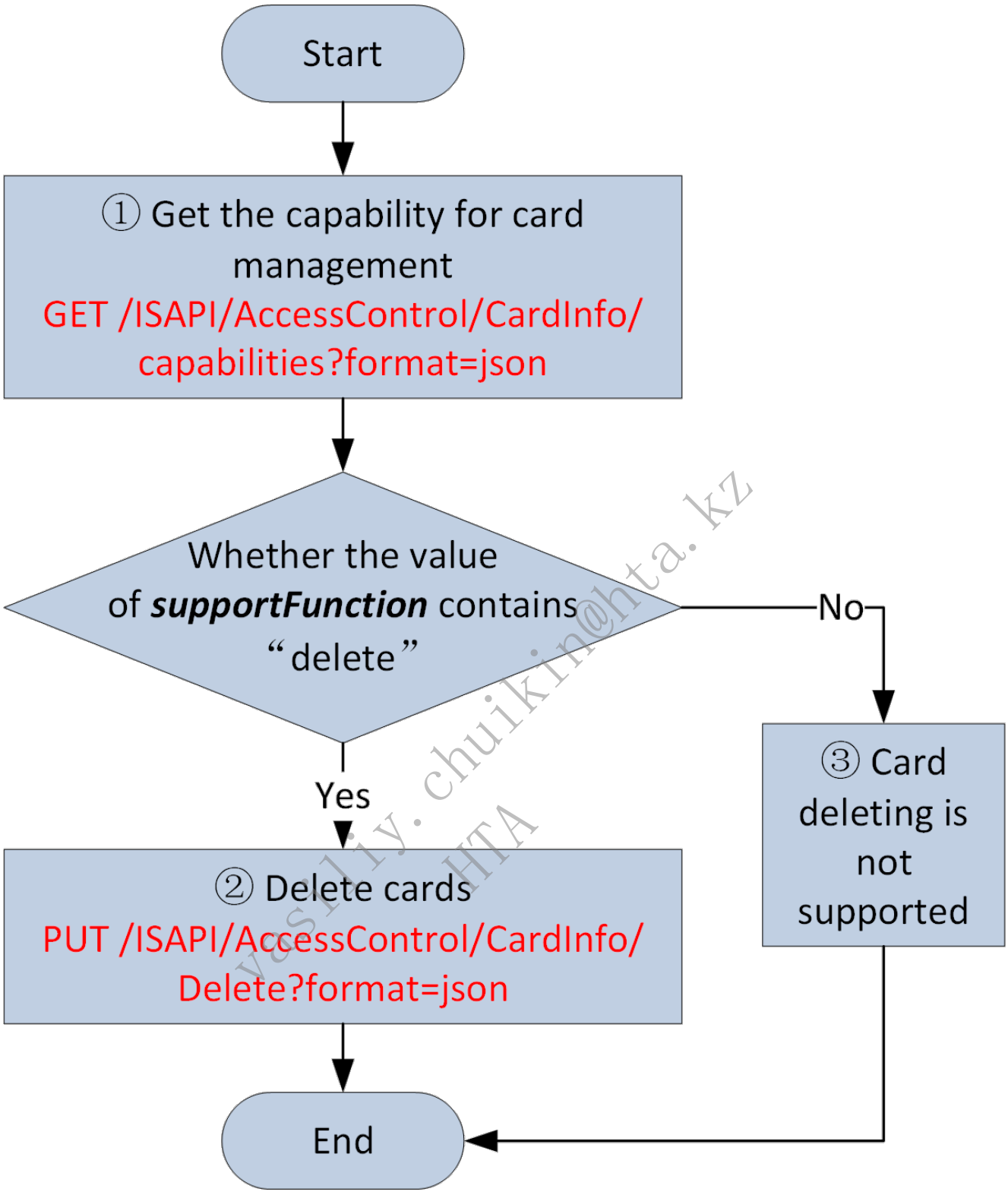
Card information on the device can be edited via the card information editing function. If the card has been added to the device, the card information will be edited; if the card has not been added to the device, the device will report an error.

1. Check whether the device supports card information editing: `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json`; if the value of the node supportFunction contains "put", it indicates that the device supports card information editing.
2. Edit card information: `PUT /ISAPI/AccessControl/CardInfo/Modify?format=json`.
3. If the value of the node supportFunction does not contain "put", it indicates that the device does not support card information editing.

Note:

Check whether the card has been added to the device via the node cardNo returned after calling the API for card information editing.

10.6.2.6 Card Deleting

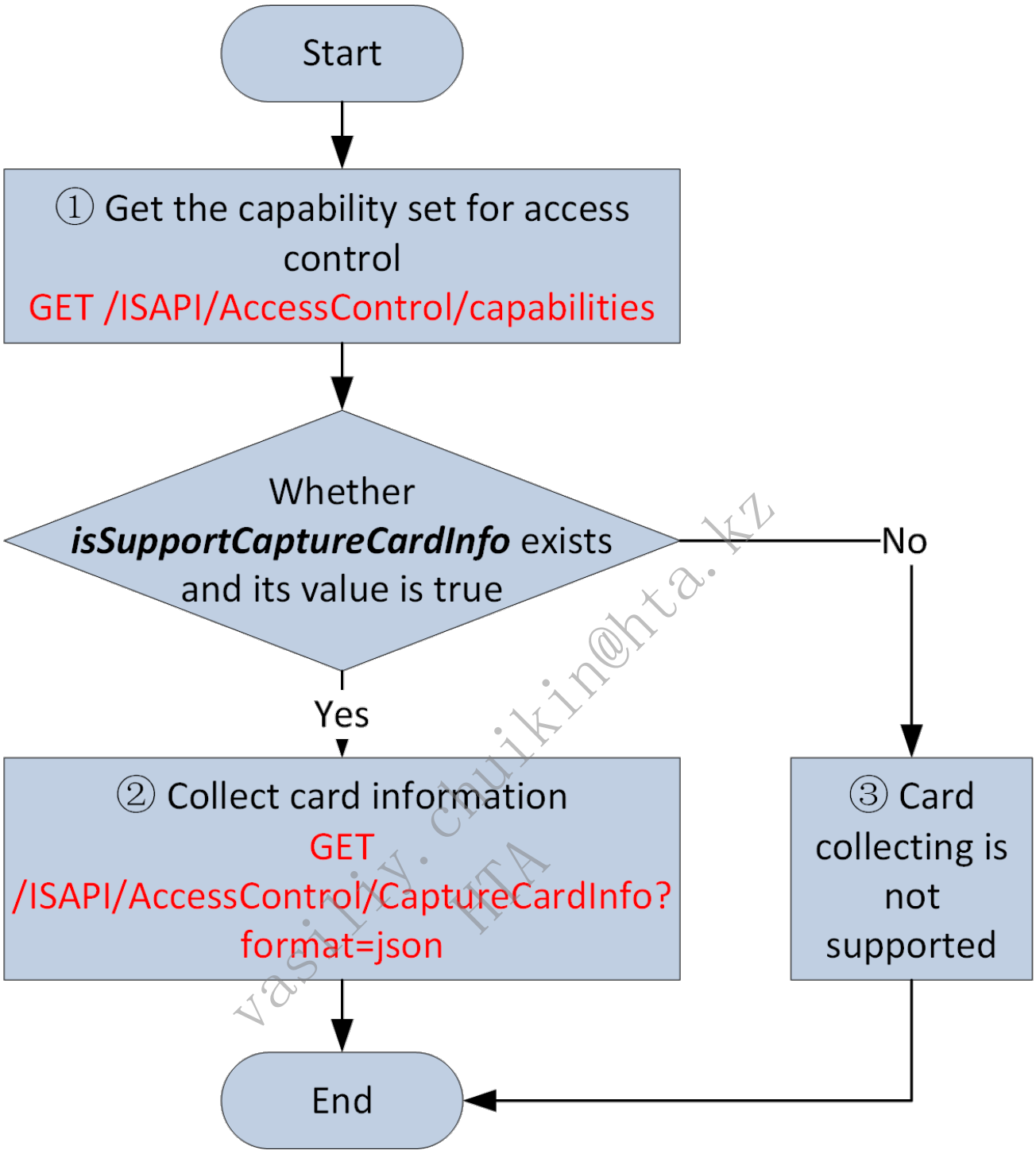


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The card information on the device can be deleted via the card deleting function. The device will not report an error if the card information to be deleted is not added to the device.

1. Check whether the device supports card deleting: `GET /ISAPI/AccessControl/CardInfo/capabilities?format=json`; if the value of the node supportFunction contains "delete", it indicates that the device supports card deleting.
2. Delete cards: `PUT /ISAPI/AccessControl/CardInfo/Delete?format=json`; if calling succeeded, it indicates that the device has deleted the cards.
3. If the value of the node supportFunction does not contain "delete", it indicates that the device does not support card deleting.

10.6.2.7 Card Collecting



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The card collecting function is for collecting the card No., card type, etc.

1. Check whether the device supports card collecting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCaptureCardInfo is returned and its value is "true", it indicates that the device supports card collecting.
2. Collect card information: `GET /ISAPI/AccessControl/CaptureCardInfo?format=json`.
3. If the node isSupportCaptureCardInfo is returned and its value is "false", it indicates that the device does not support card collecting.

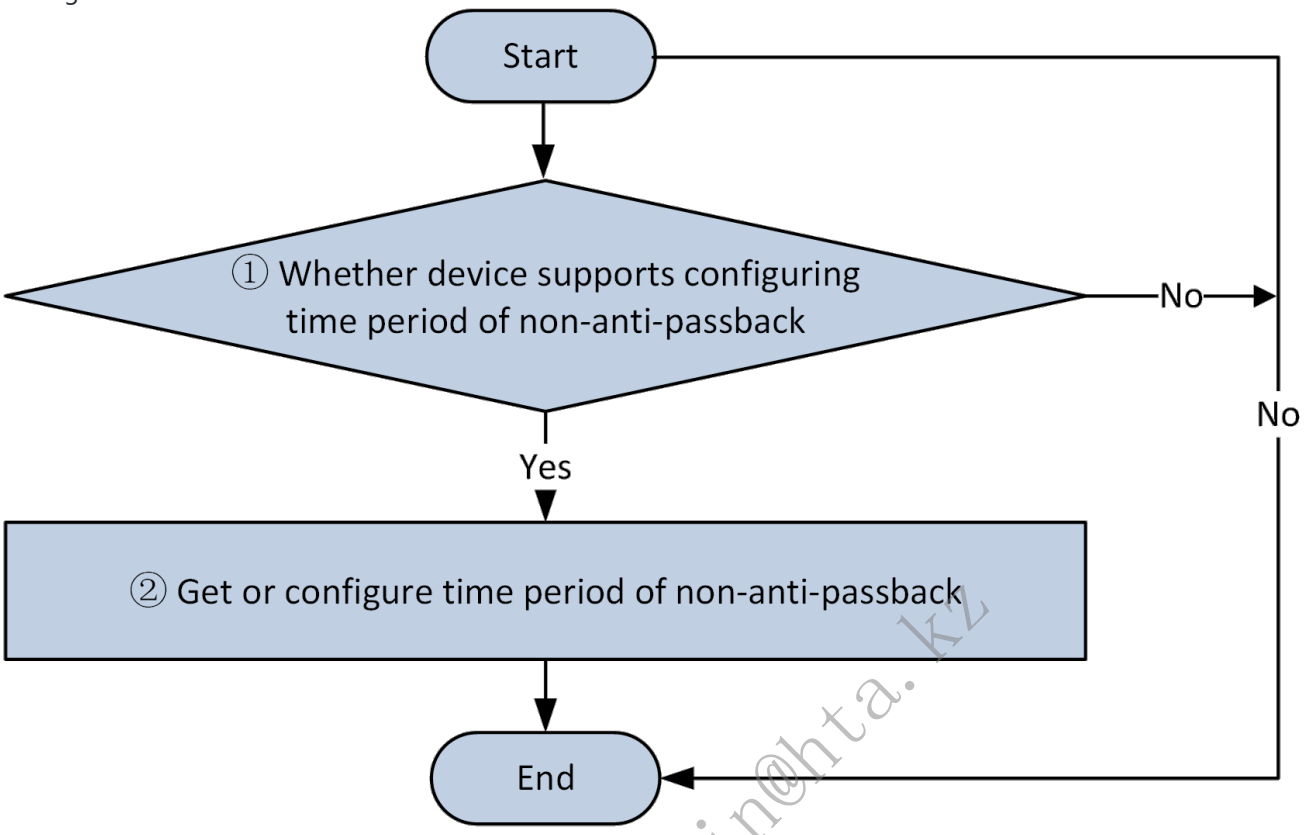
10.7 Configure Non-anti-passback Time Period

Non-anti-passback time period: anti-passback is not triggered in the set time period. Application Scenarios: In rush hour, anti-passback can always happens since the person might follow others in the people flow. Non-anti-passback can help normal entry&exit in rush hour.

10.7.1 Introduction to the Function

10.7.2 API Calling Flow

Calling Flow:



ISAPI Protocol Calling Flow:

1. Get the capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportAntiPassbackTimeRange is returned and its value is "true", it indicates that the device supports configuring time period of anti-passback.
2. Get time period parameters of non anti-passback: `GET /ISAPI/AccessControl/AntiPassback/timeRange?format=json`; configure time period parameters of non anti-passback: `PUT /ISAPI/AccessControl/AntiPassback/timeRange?format=json`; it can specify the time period of anti-passback; Get the capacity of configuring time period of anti-passback: `GET /ISAPI/AccessControl/AntiPassback/timeRange/capabilities?format=json`.

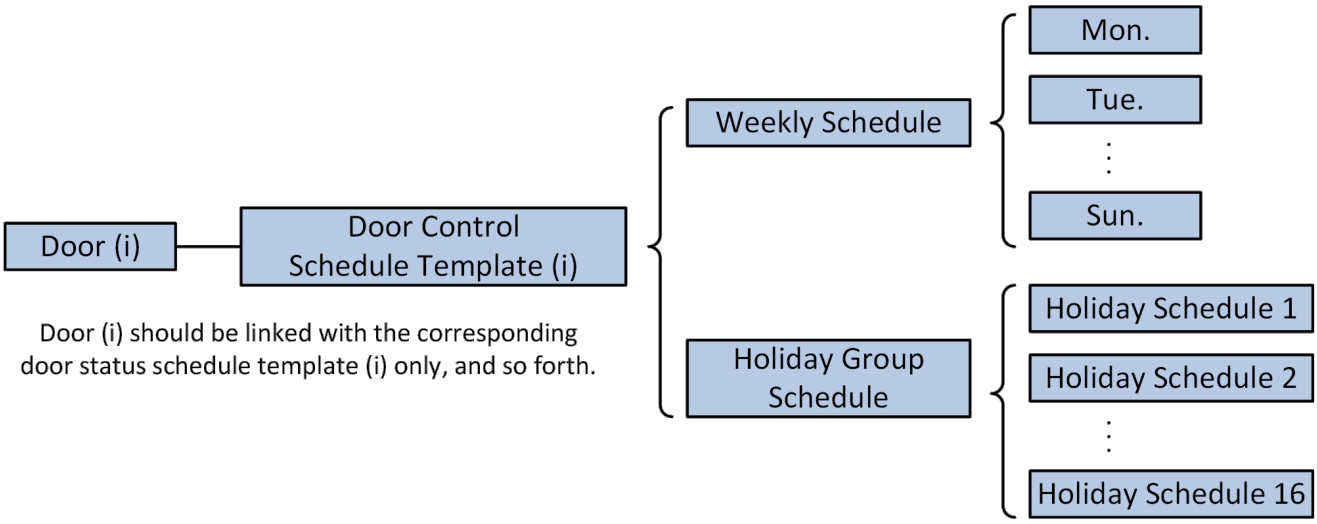
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10.8 Door Control Schedule Management

10.8.1 Introduction to the Function

For doors of access control or floors of elevator control, you can set schedule templates. The schedule template information include time period and status (remain open, remain closed, sleep, and normal). Configuring door control schedule is not required. If it is not configured, No device configuration has permission for all the doors by default. The priority of remote door control is higher than that of door control schedule. The operation of remote door control can take effect when the door is in the status of remain open/closed, sleep, and normal.

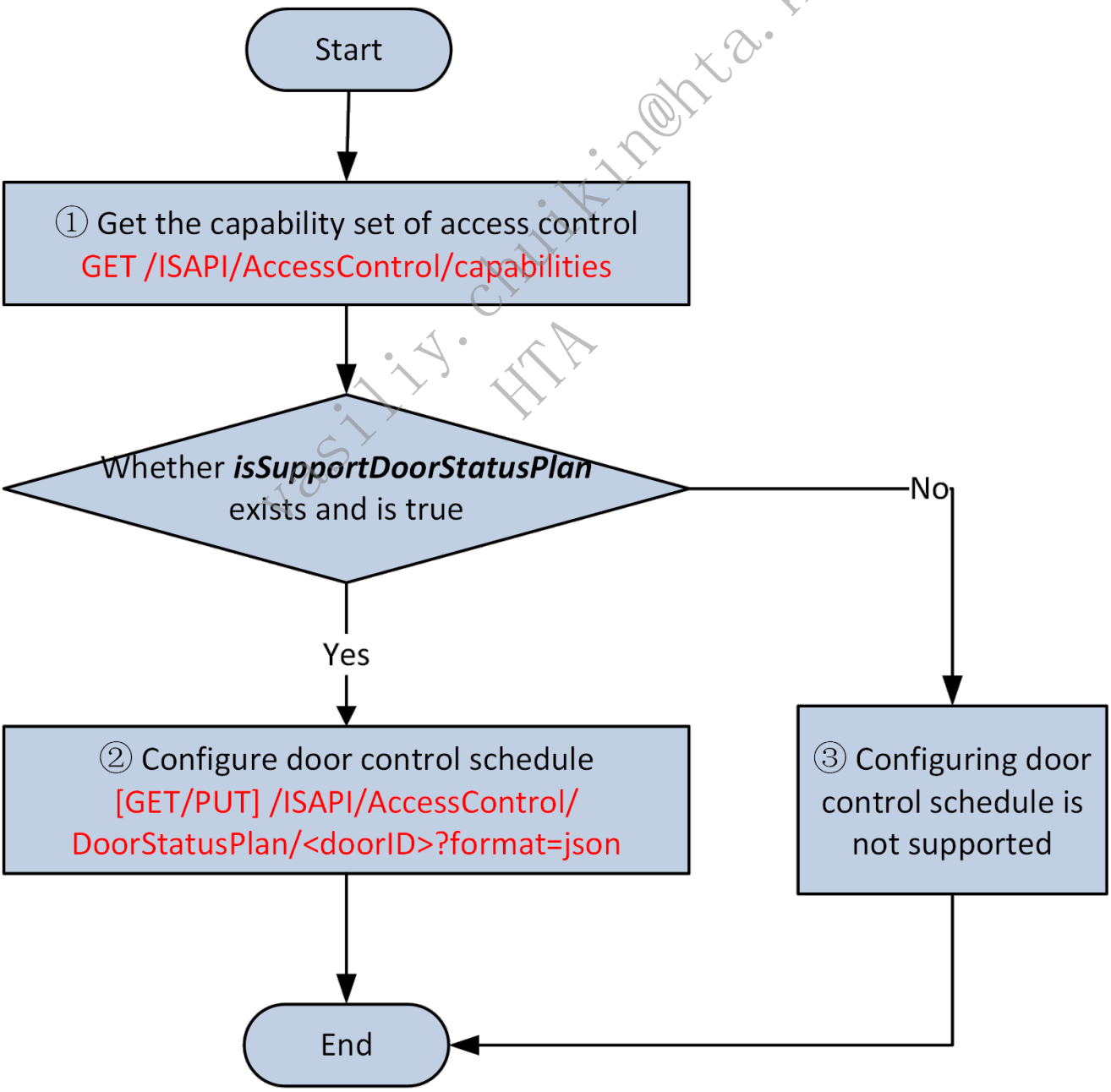
Each schedule template can be linked to one week schedule and four holiday group schedules. Holiday schedule priority is higher than that of weekly schedule. Weekly schedule can be configured with time periods from Monday to Sunday, and 8 different time periods are supported each day. Holiday group schedule can be linked to 16 different holiday schedules. Each holiday schedule can be configured with one start and end date of the holiday, and the access time period is the same for each day (up to 8 different time periods can be configured). This schedule template is configured to manage access control permission.



Door (i) should be linked with the corresponding door status schedule template (i) only, and so forth.

10.8.2 API Calling Flow

10.8.2.1 Configure Door Control Schedule



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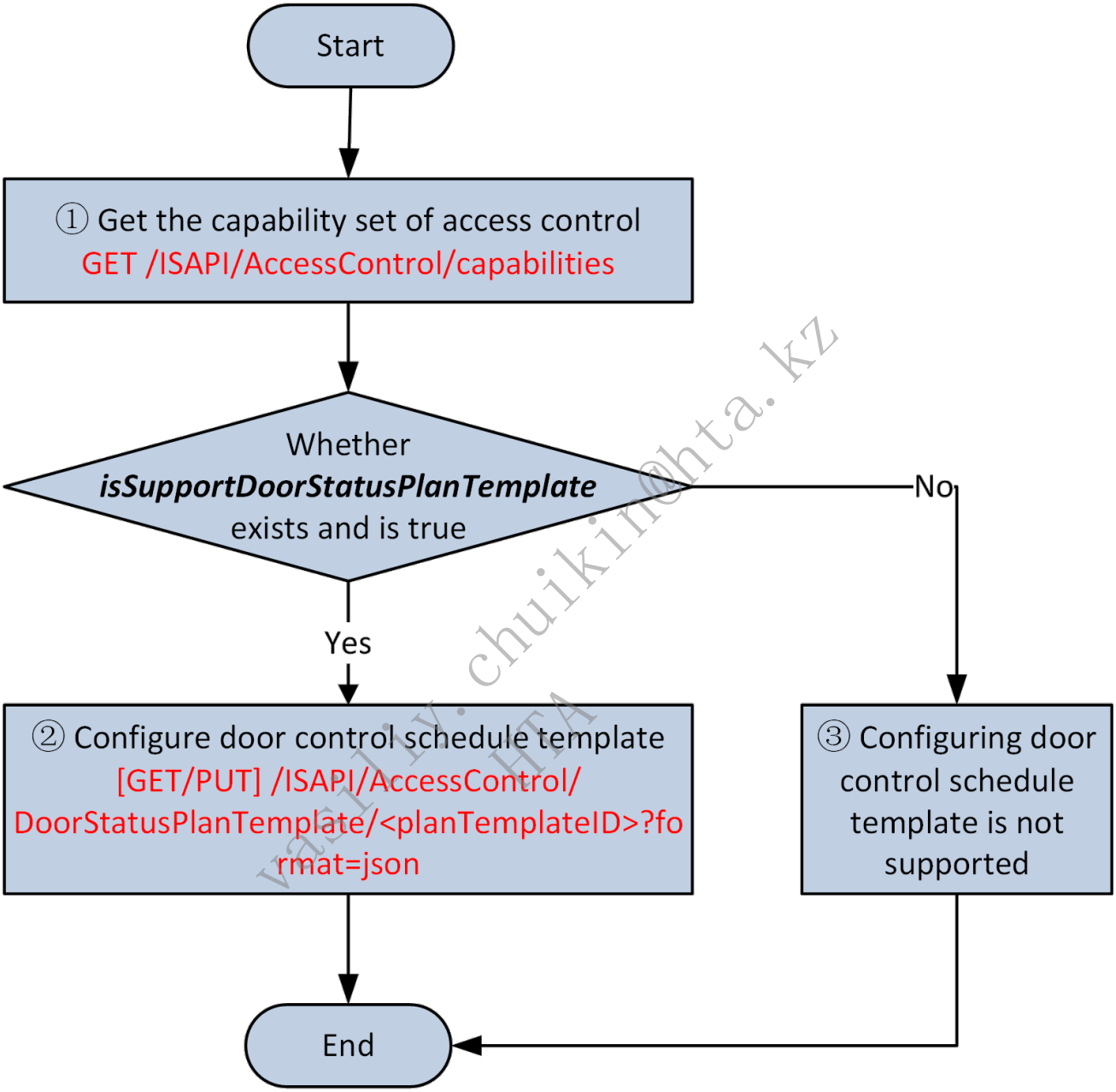
The API calling flow is as follow:

1. Check whether the device supports configuring door control schedule: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportDoorStatusPlan is returned and its value is "true", it indicates that the device supports

configuring door control schedule (and the device also supports configuring schedule template of door control).

2. Set door control schedule: `[GET/PUT] /ISAPI/AccessControl/DoorStatusPlan/<doorID>?format=json`.
3. If the node isSupportDoorStatusPlan is returned and its value is "false", it indicates that the device does not support configuring door control schedule.

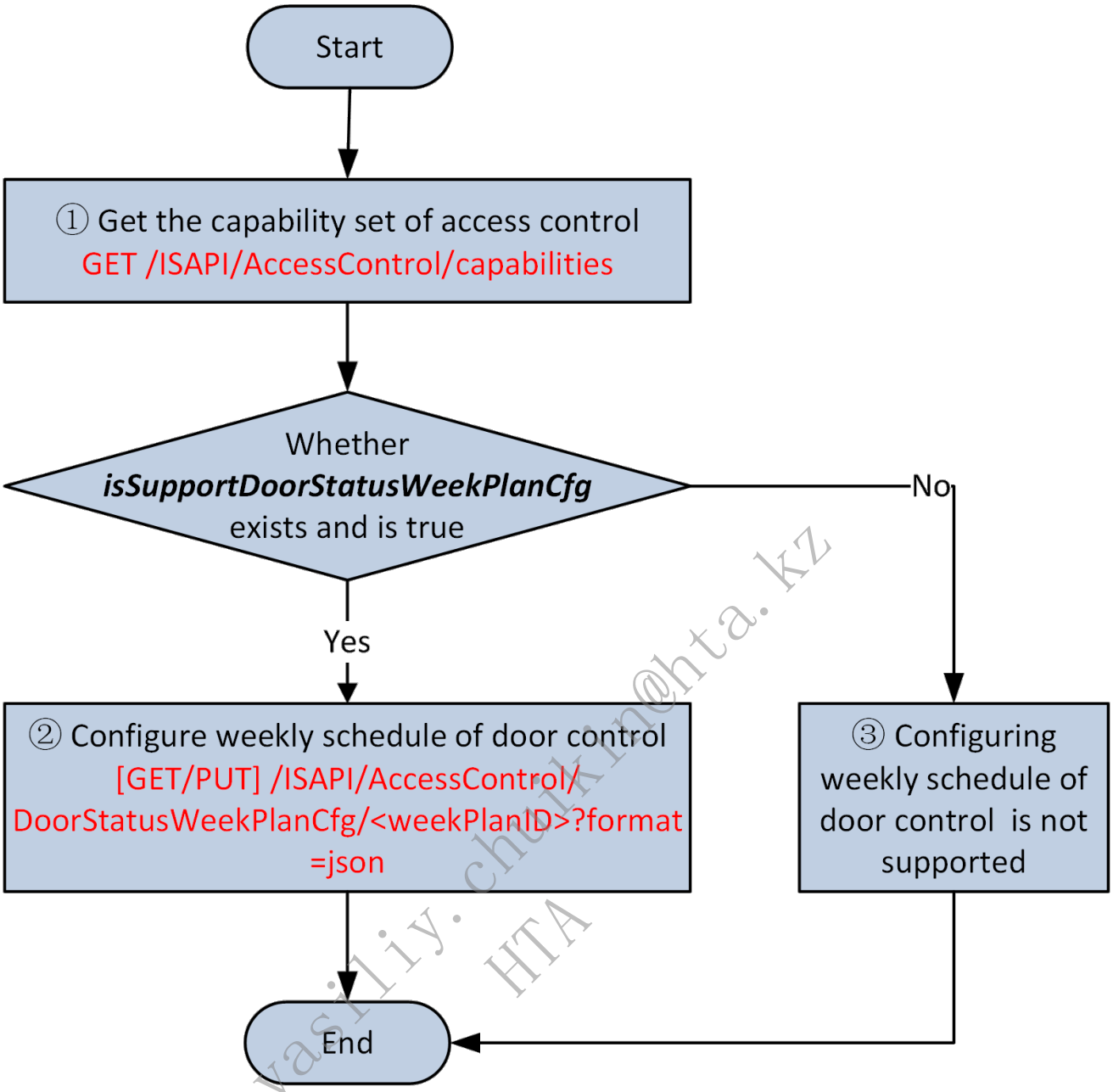
10.8.2.2 Configure Door Control Schedule Template



The API calling flow is as follow:

1. Check whether the device supports configuring door control schedule template:`GET /ISAPI/AccessControl/capabilities`; if the node isSupportDoorStatusPlanTemplate is returned and its value is "true", it indicates that the device supports configuring door control schedule template (and the device should also supports configuring door status schedule template).
2. Get and set parameters of door control schedule template: `[GET/PUT] /ISAPI/AccessControl/DoorStatusPlanTemplate/<planTemplateID>?format=json`.
3. If the node isSupportDoorStatusPlanTemplate is returned and its value is "false", it indicates that the device does not support configuring door control schedule template.

10.8.2.3 Configure Door Control Weekly Schedule

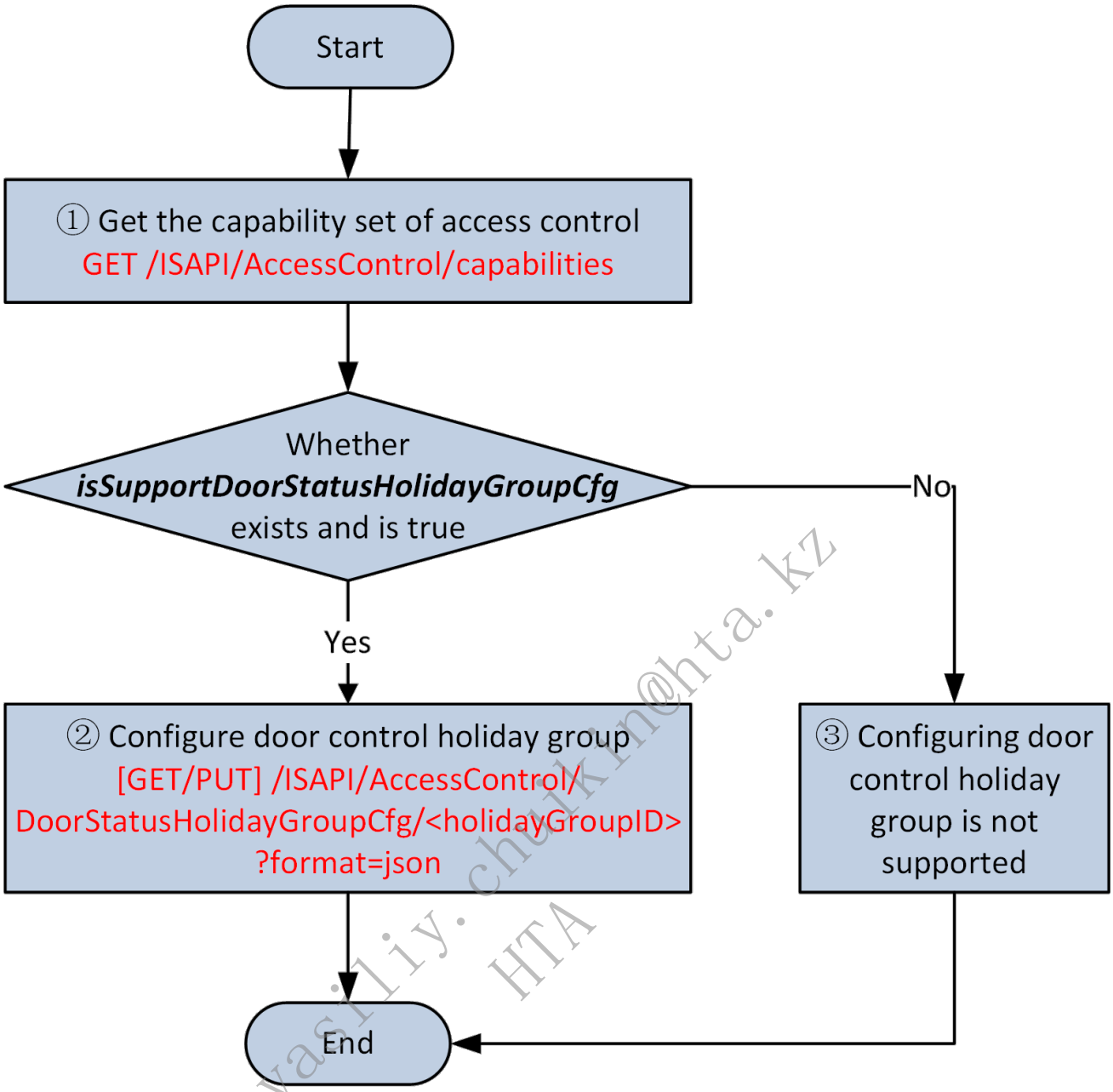


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The API calling flow is as follow:

1. Check whether the device supports configuring door control weekly schedule: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportDoorStatusWeekPlanCfg is returned and its value is "true", it indicates that the device supports this function.
2. Set parameters of door control weekly schedule: `[GET/PUT] /ISAPI/AccessControl/DoorStatusWeekPlanCfg/<weekPlanID>?format=json`.
3. If the node isSupportDoorStatusWeekPlanCfg is returned and its value is "false", it indicates that the device does not support configuring this function.

10.8.2.4 Configure Door Control Holiday Group

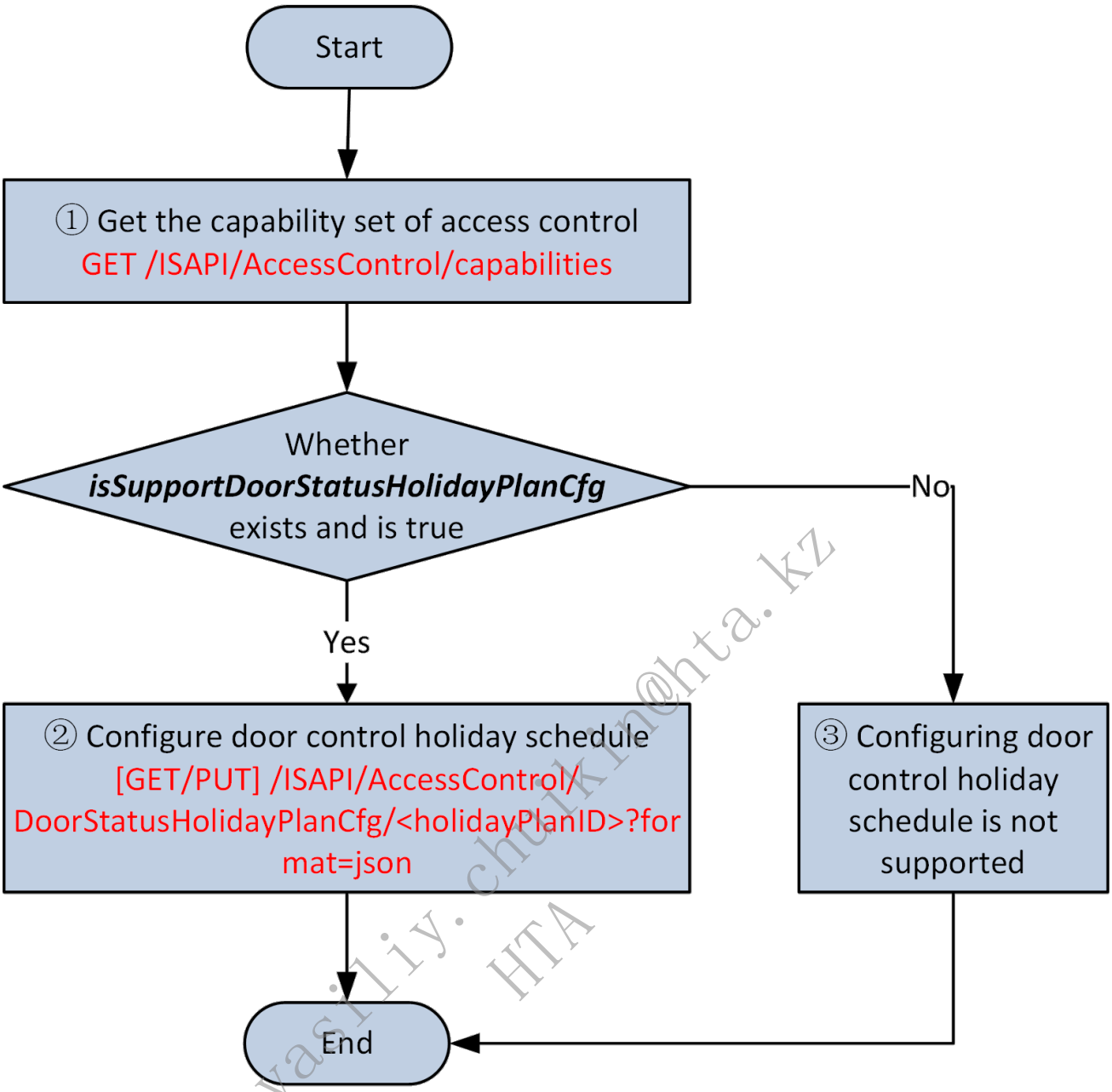


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The API calling flow is as follow:

1. Check whether the device supports configuring door control holiday group: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportDoorStatusHolidayGroupCfg is returned and its value is "true", it indicates that the device supports configuring door control holiday group (and the device also supports configuring door control holiday schedule).
2. Set the holiday group configuration parameters of the door control schedule: `[GET/PUT] /ISAPI/AccessControl/DoorStatusHolidayGroupCfg/<holidayGroupID>?format=json`.
3. If the node isSupportDoorStatusHolidayGroupCfg is returned and its value is "false", it indicates that the device does not support this function.

10.8.2.5 Configure Door Control Holiday Schedule



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The API calling flow is as follow:

1. Check whether the device supports configuring door control holiday schedule: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportDoorStatusHolidayPlanCfg is returned and its value is "true", it indicates that the device supports this function.
2. Set parameters of door control holiday schedule: `[GET/PUT] /ISAPI/AccessControl/DoorStatusHolidayPlanCfg/<holidayPlanID>?format=json`.
3. If the node isSupportDoorStatusHolidayPlanCfg is returned and its value is "false", it indicates that the device does not support this function.

10.9 Event and Card Linkage Parameters

10.9.1 Introduction to the Function

The device supports linking specific actions when access control events triggered. The linkage actions include event, card No., MAC address, and employee No. Take event for example, if a person authenticated by the device, the door will open for the person.

10.9.2 API Calling Flow

10.9.2.1 Configure Parameters of Event and Card Linkage

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportEventCardLinkageCfg is returned and its value is "true", it indicates that the device supports configuring the parameters of event and card linkage.
2. Get the configuration capability of the event and card linkage: `GET /ISAPI/AccessControl/EventCardLinkageCfg/capabilities?format=json`.
3. Get and set the parameters of event and card linkage: `GET|PUT /ISAPI/AccessControl/EventCardLinkageCfg/<ACEID>?format=json`.

10.9.2.2 Search for Parameters of Event and Card Linkage

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportEventCardLinkageCfgSearch is returned and its value is "true", it indicates that the device supports searching for the parameters of event and card linkage.
2. Get the capability of searching for parameters of event and card linkage: `GET /ISAPI/AccessControl/EventCardLinkageCfg/search/capabilities?format=json`.
3. Search for parameters of event and card linkage: `POST /ISAPI/AccessControl/EventCardLinkageCfg/search?format=json`.

10.9.2.3 Delete Parameters of Specific Event and Card Linkage

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportEventCardLinkageCfgDelete is returned and its value is "true", it indicates that the device supports deleting parameters of event and card linkage.
2. Delete parameters of specific event and card linkage: `PUT /ISAPI/AccessControl/EventCardLinkageCfgDelete?format=json`.

10.9.2.4 Get List of Event and Card Linkage ID

1. Call the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportEventCardNoList is returned and its value is true, it indicates that the device supports getting events and the card linkage ID list.
2. Get the capability of the list of event and card linkage ID: `GET /ISAPI/AccessControl/EventCardNoList/capabilities?format=json`.
3. Get the list of event and card linkage ID: `GET /ISAPI/AccessControl/EventCardNoList?format=json`.

10.9.2.5 Optimize Event

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node iisSupportEventOptimizationCfg is returned and its value is "true", it indicates that the device supports event optimization.
2. Get the configuration capability of event optimization: `GET /ISAPI/AccessControl/EventOptimizationCfg/capabilities?format=json`.
3. Get the event optimization configuration parameters: `GET|PUT /ISAPI/AccessControl/EventOptimizationCfg?format=json`.

10.10 Face Picture Management

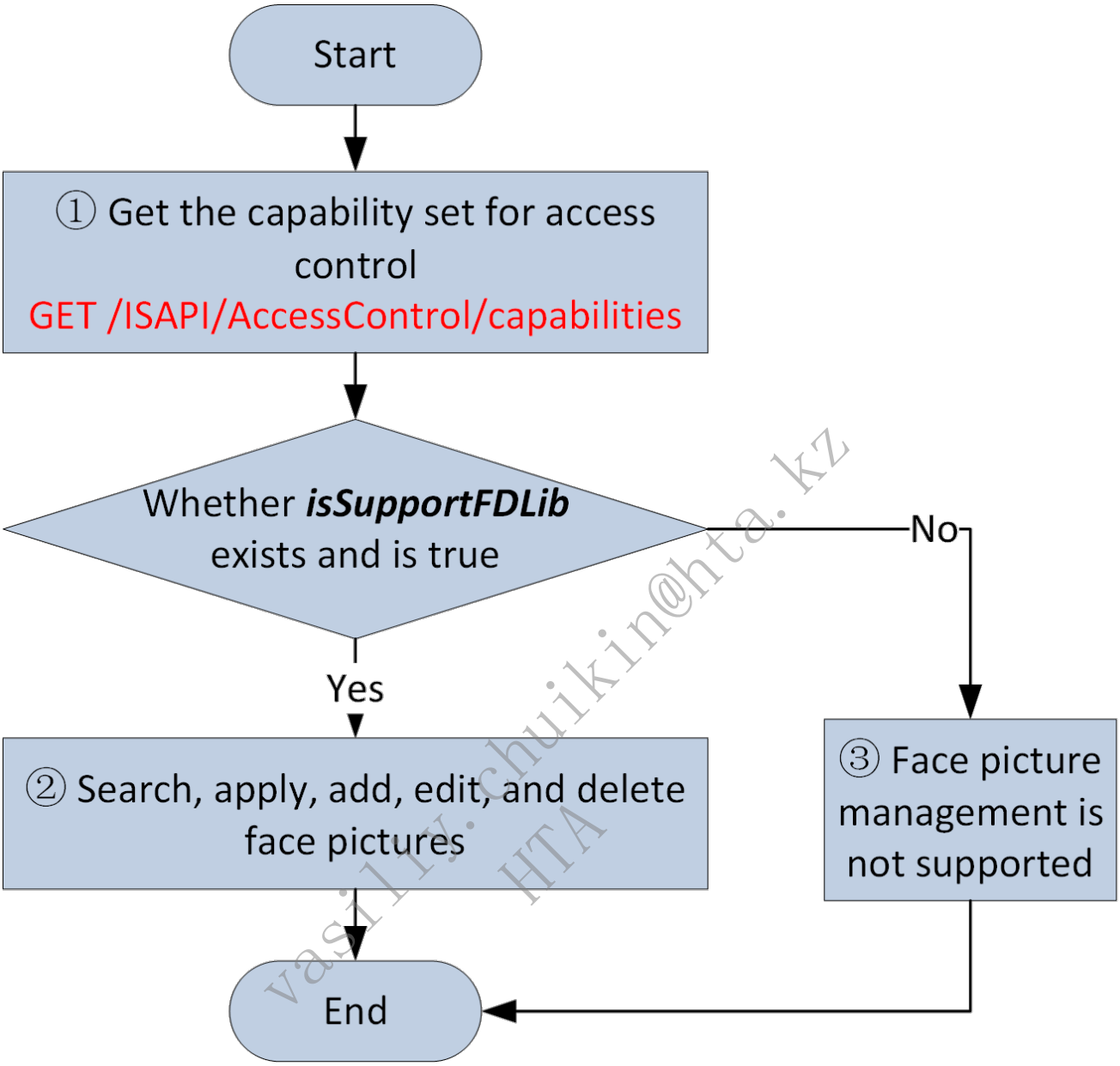
10.10.1 Introduction to the Function

Face picture management includes searching, applying, adding, editing, deleting, and collecting face pictures.

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10.10.2 API Calling Flow

10.10.2.1 Check Whether the Device Supports Face Picture Management



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Before calling the API for face picture management, make sure that the device supports face picture management.

1. Check whether the device supports face picture management: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportFDLib is returned and its value is "true", it indicates that the device supports face picture management.
2. Search, apply, add, edit, and delete face pictures.
3. If the node isSupportFDLib is returned and its value is "false", it indicates that the device does not support face picture management.

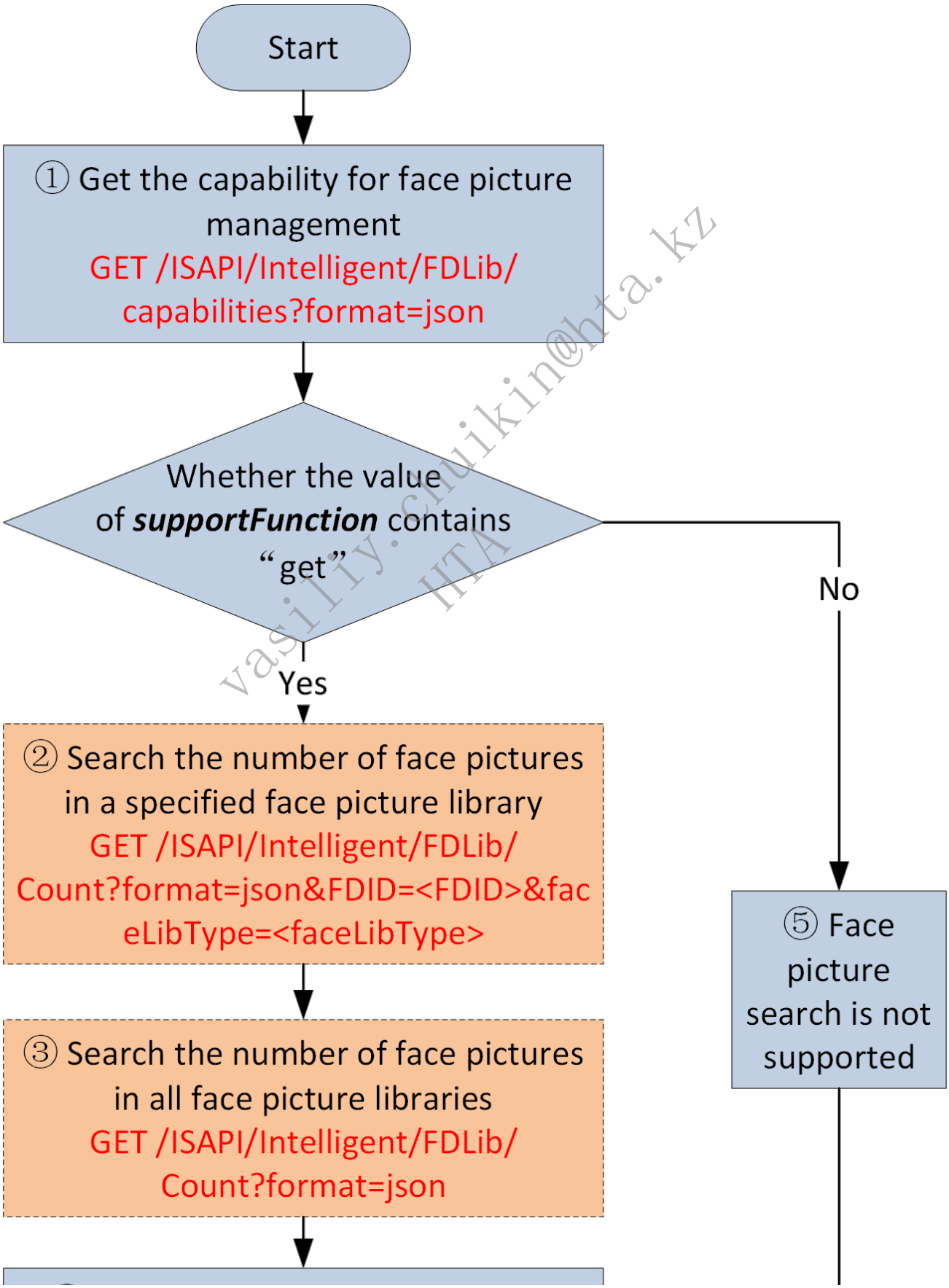
Note:

- Before applying, adding, or editing face picture information on the device, make sure that the related person information linked to the person ID has been applied to the device, and make sure that the device has its face picture library by calling `GET /ISAPI/Intelligent/FDLib?format=json` (if the device has no face picture library, then create the face picture library by calling `POST /ISAPI/Intelligent/FDLib?format=json`), and the ID of the library of face pictures captured in visible light (FDID) is 1.
- If the value of the node mode returned by calling `GET /ISAPI/AccessControl/FaceRecognizeMode/capabilities?format=json` contains "deepMode", it indicates that the device supports the deep mode, which compares face

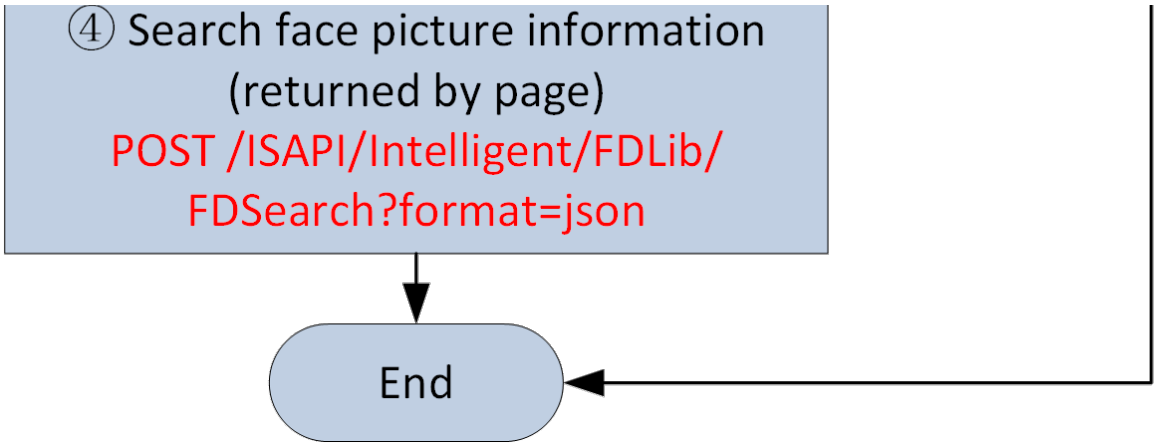
pictures captured in infrared light. For devices which support the deep mode, if the face picture library ID (FDID) is 2, face pictures captured in infrared light will be applied to the face picture library and be used for face picture comparison; if the face picture library ID (FDID) is 1, face pictures captured in visible light will be applied to the face picture library and be displayed on the device.

- Switch between the deep mode and normal mode: `[GET/PUT] /ISAPI/AccessControl/FaceRecognizeMode?format=json`; the modes can be switched via the node mode.

10.10.2.2 Face Picture Search



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The face picture search function is for searching the number of face pictures and face picture information added to the device.

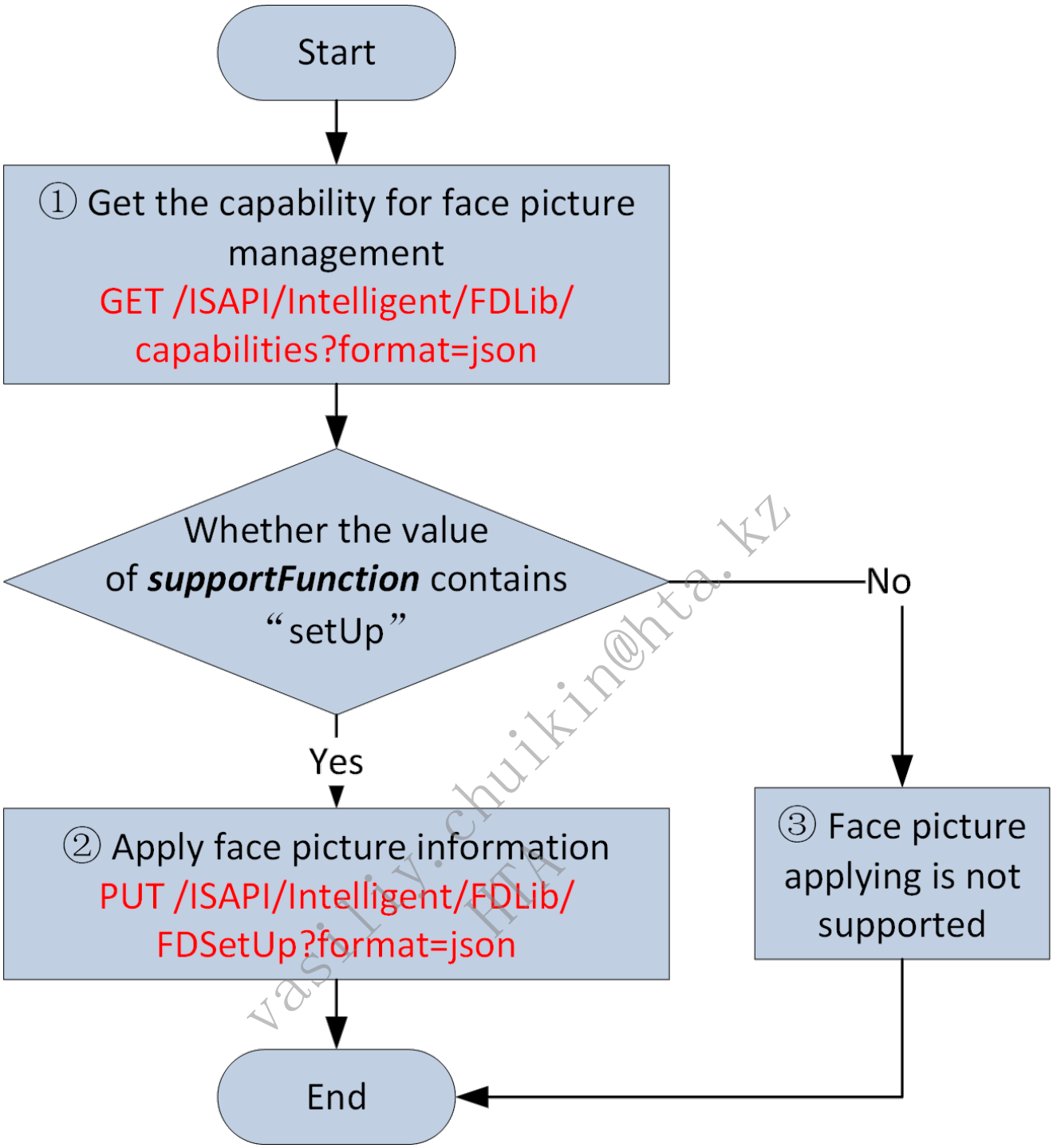
1. Check whether the device supports face picture search: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`; if the value of the node supportFunction contains "get", it indicates that the device supports face picture search.
2. Search the number of face pictures in the specified face picture libraries: `GET /ISAPI/Intelligent/FDLib/Count?format=json&FDID=<FDID>&faceLibType=<faceLibType>`; the returned value of the node recordDataNumber is the number of the added face pictures of the specified face picture libraries.
3. Search the number of face pictures in all face picture libraries: `GET /ISAPI/Intelligent/FDLib/Count?format=json`; the returned value of the node recordDataNumber is the number of face pictures in all face picture libraries.
4. Search face picture information: `POST /ISAPI/Intelligent/FDLib/FDSearch?format=json`; the face picture information is returned by page.
5. If the value of the node supportFunction does not contain "get", it indicates that the device does not support face picture search.

Note:

The value of the node FDRecordDataMaxNum returned by calling `GET /ISAPI/Intelligent/FDLib/capabilities?format=json` is the maximum number of face pictures supported by the device.

10.10.2.3 Face Picture Applying

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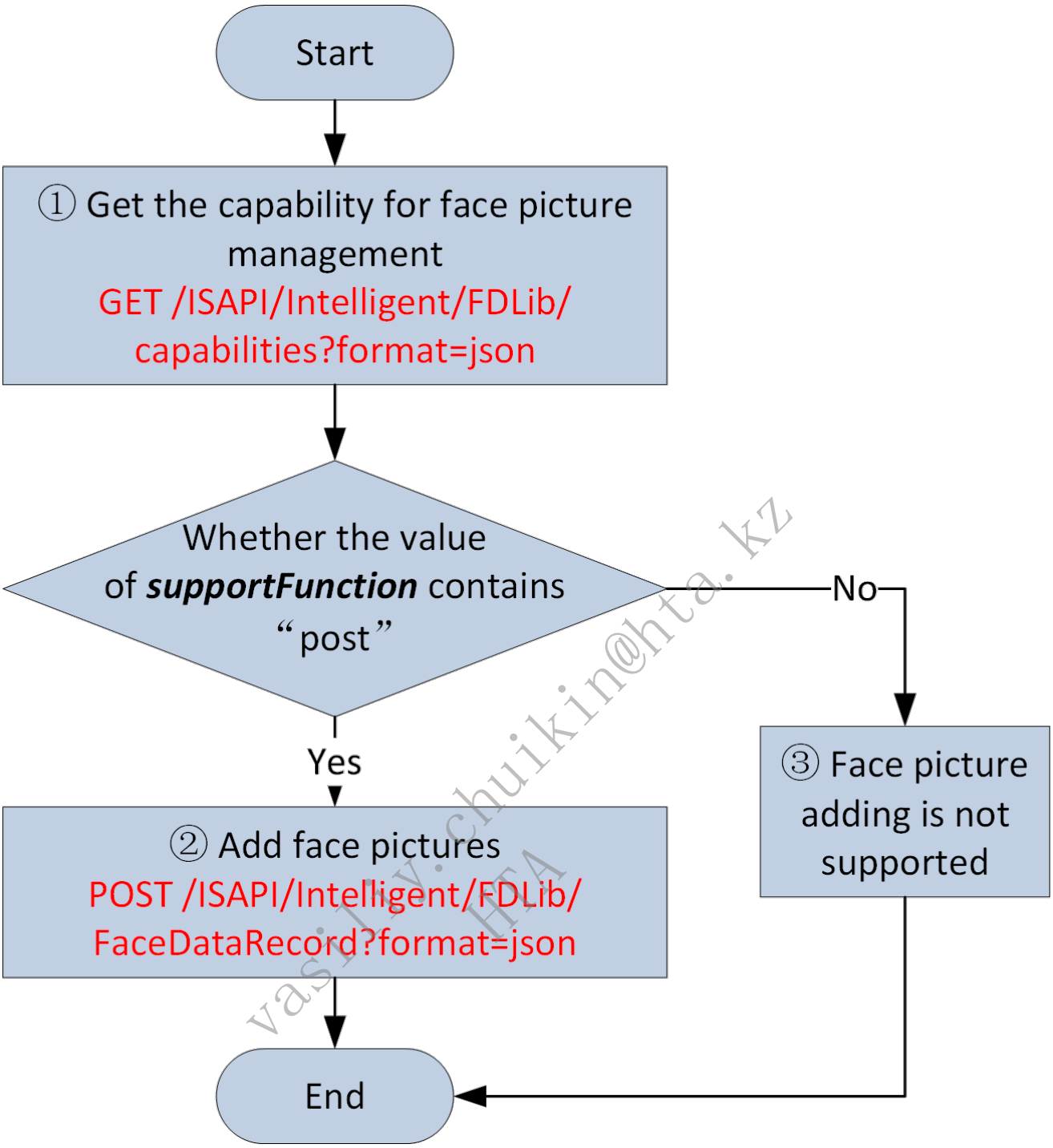
Face picture information can be applied to the device via the face picture applying function. If the face picture has been added to the device, the face picture information will be edited; if the face picture has not been added to the device, the face picture information will be added to the device.

1. Check whether the device supports face picture applying: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`; if the value of the node supportFunction contains "setUp", it indicates that the device supports face picture applying.
2. Apply face picture information: `PUT /ISAPI/Intelligent/FDLib/FDSetUp?format=json`.
3. If the value of the node supportFunction does not contain "setUp", it indicates that the device does not support face picture applying.

Note:

Check whether the face picture has been added to the device via the node FPID returned by calling the API for face picture applying, and link the face picture to the person information via the node FPID in face picture management and the node employeeNo in person management.

10.10.2.4 Face Picture Adding



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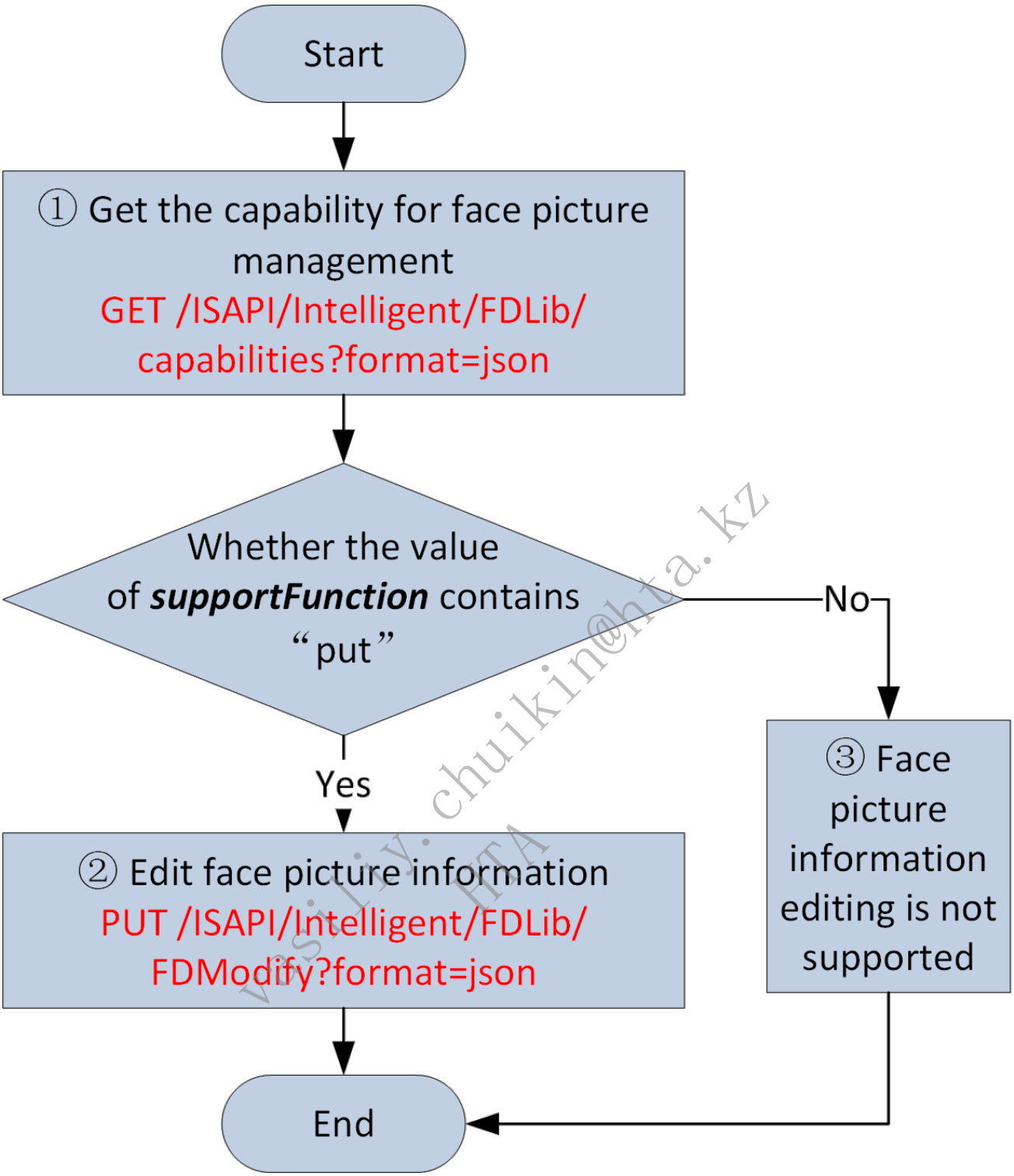
Face picture information can be added to the device via the face picture adding function. If the face picture has been added to the device, the device will report an error; if the face picture has not been added to the device, the face picture information will be added to the device.

1. Check whether the device supports face picture adding: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`; if the value of the node supportFunction contains "post", it indicates that the device supports face picture adding.
2. Add face picture information: `POST /ISAPI/Intelligent/FDLib/FaceDataRecord?format=json`.
3. If the value of the node supportFunction does not contain "post", it indicates that the device does not support face picture adding.

Note:

Check whether the face picture has been added to the device via the node FPID returned by calling the API for face picture adding, and link the face picture to the person information via the node FPID in face picture management and the node employeeNo in person management.

10.10.2.5 Face Picture Information Editing



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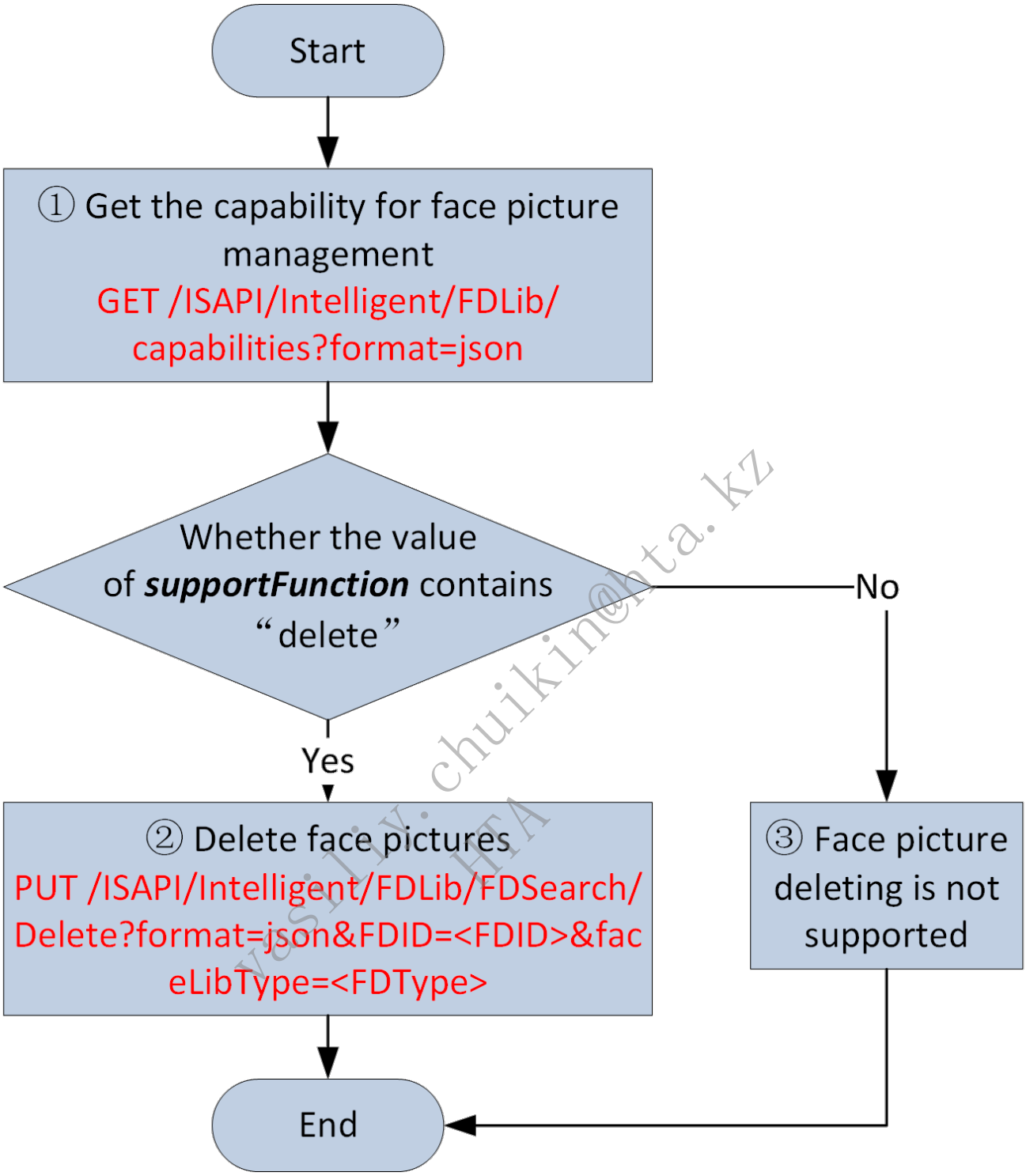
Face picture information on the device can be edited via the face picture information editing function. If the face picture has been added to the device, the face picture information will be edited; if the face picture has not been added to the device, the device will report an error.

1. Check whether the device supports face picture information editing: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`; if the value of the node supportFunction contains "put", it indicates that the device supports face picture information editing.
2. Edit face picture information: `PUT /ISAPI/Intelligent/FDLib/FDModify?format=json`.
3. If the value of the node supportFunction does not contain "put", it indicates that the device does not support face picture information editing.

Note:

Check whether the face picture has been added to the device via the node FPID returned by calling the API for face picture information editing, and link the face picture to the person information via the node FPID in face picture management and the node employeeNo in person management.

10.10.2.6 Face Picture Deleting



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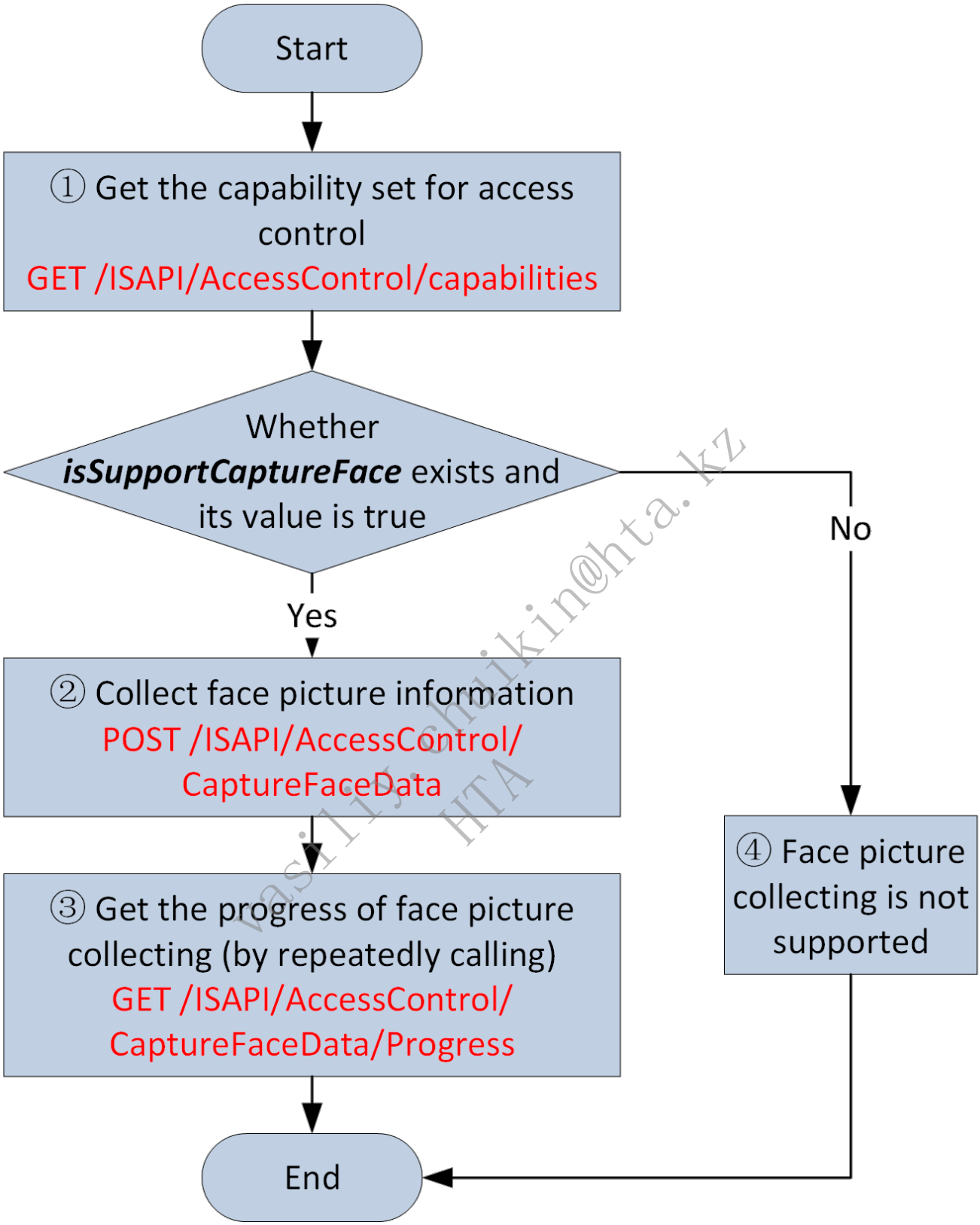
The face picture information on the device can be deleted via the face picture deleting function. The device will not report an error if the face picture to be deleted is not added to the device.

1. Check whether the device supports face picture deleting: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`; if the value of the node supportFunction contains "delete", it indicates that the device supports face picture deleting.
2. Delete face pictures: `PUT /ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&FDID=<FDID>&faceLibType=<FDType>`; if calling succeeded, it indicates that the device has deleted the face pictures.
3. If the value of the node supportFunction does not contain "delete", it indicates that the device does not support face picture deleting.

Note:

All the face picture libraries and the face picture information in the libraries on the device can be deleted by calling `DELETE /ISAPI/Intelligent/FDLib?format=json`.

10.10.2.7 Face Picture Collecting



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Face picture data, face picture quality grades, etc., can be collected via the face picture collecting function.

1. Check whether the device supports face picture collecting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCaptureFace is returned and its value is "true", it indicates that the device supports face picture (captured in visible light) collecting. If the node isSupportCaptureInfraredFace is returned and its value is "true", it indicates that the device supports face picture (captured in infrared light) collecting.
2. Collect face picture information: `POST /ISAPI/AccessControl/CaptureFaceData`.

- If the node captureProgress is returned, and the value is 100, it indicates that the face picture has been collected, and the binary data or URL of the collected face picture will be parsed.
- If the node captureProgress is returned and the value is 0, it indicates that the face picture has not been collected,

and you need to get the progress of face picture collecting.

3. Get the progress of face picture collecting: `GET /ISAPI/AccessControl/CaptureFaceData/Progress`; repeatedly call this API to get the progress of face picture collecting.

- Repeatedly call this API until the node captureProgress is returned and its value is 100, which indicates that the face picture has been collected and the binary data and URL of the face picture will be parsed.
- If the value of the node captureProgress is 0 and the value of the isCurRequestOver is true, which indicates that the face picture collecting failed, stop calling the API.

4. If the node isSupportCaptureFace is returned and its value is "false", it indicates that the device does not support face picture collecting.

10.11 Fingerprint Management

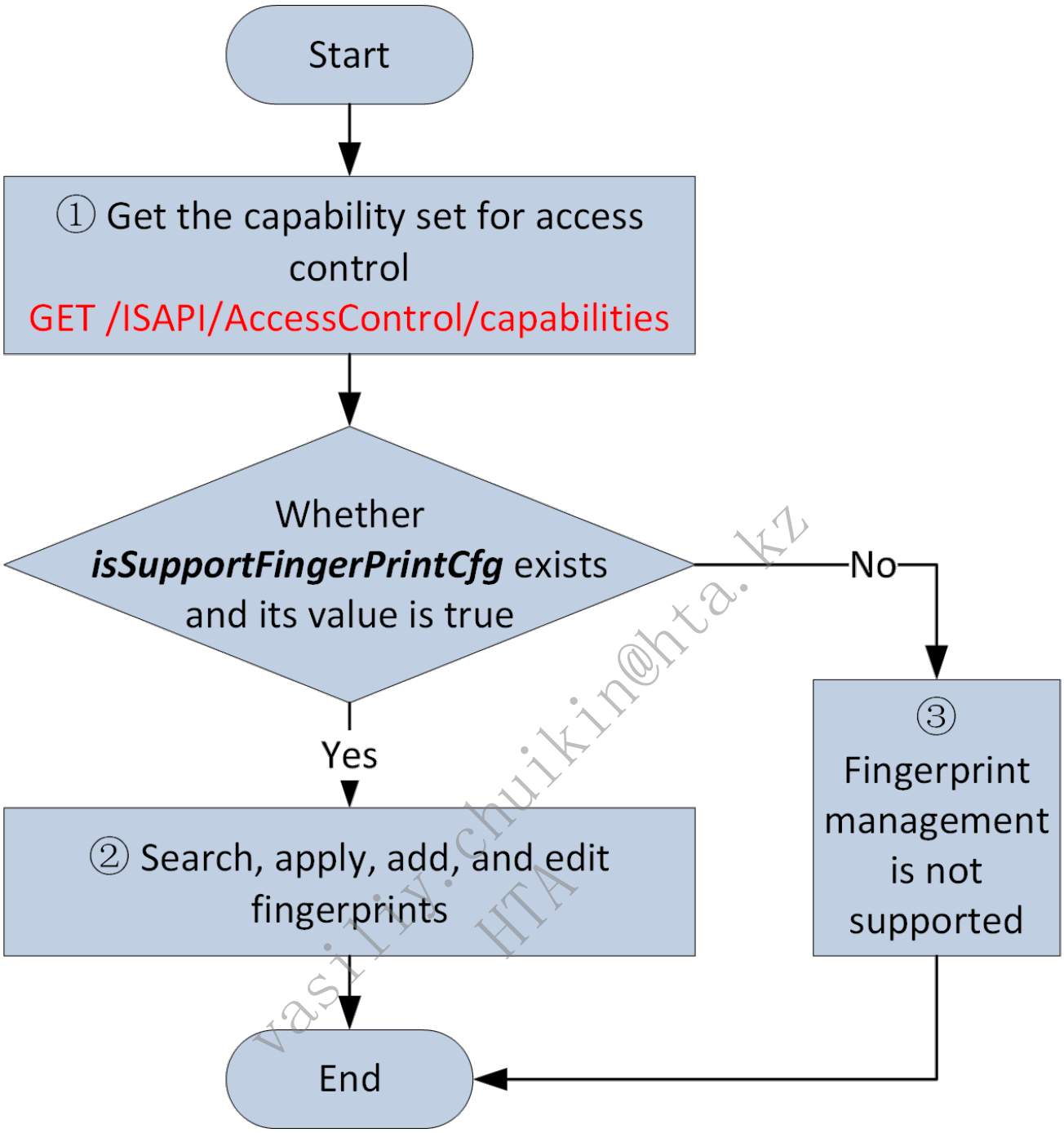
10.11.1 Introduction to the Function

Fingerprint management includes searching, applying, deleting, and collecting fingerprints.

10.11.2 API Calling Flow

10.11.2.1 Check Whether the Device Supports Fingerprint Management

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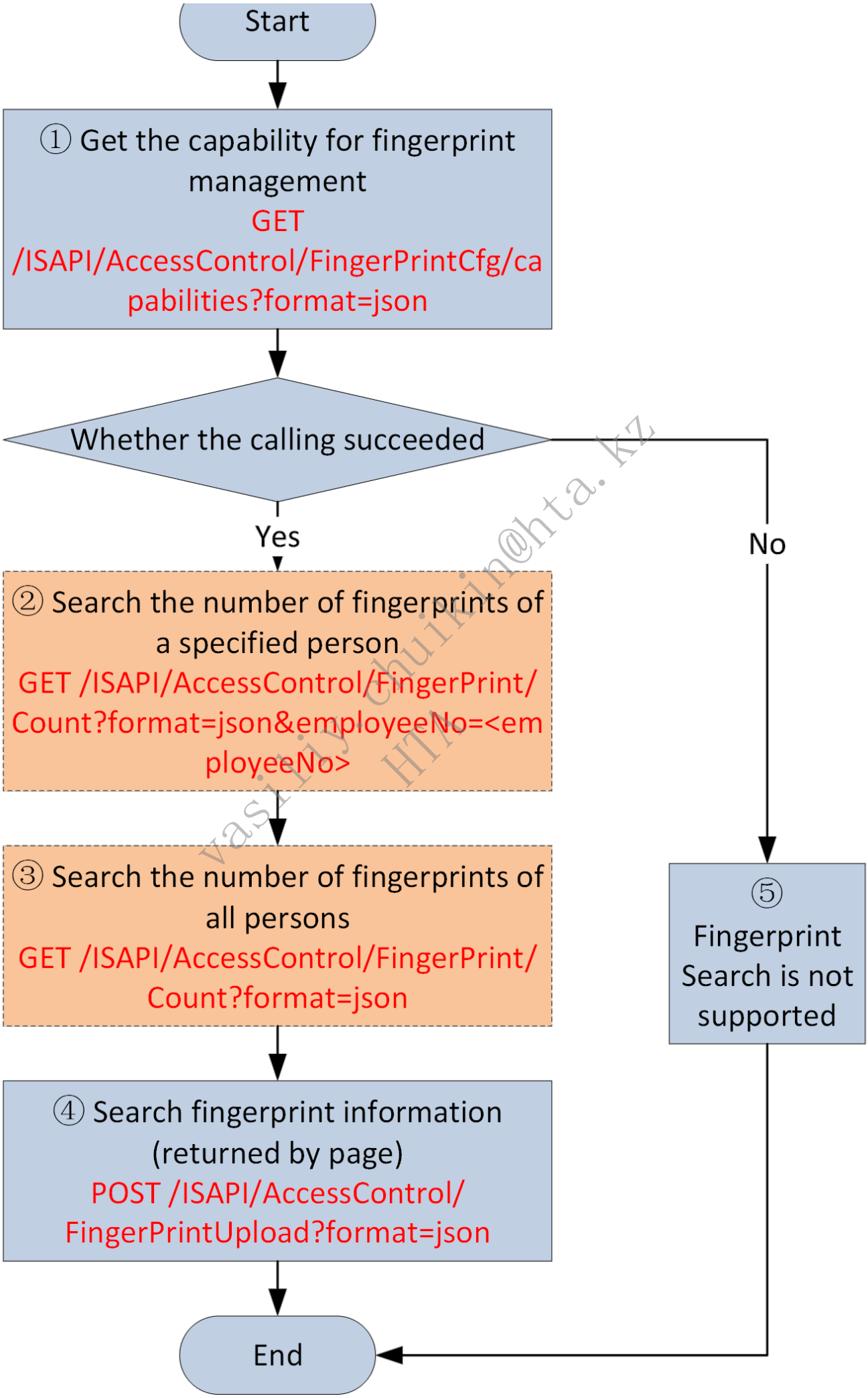
Before calling the API for fingerprint management, make sure that the device supports fingerprint management.

1. Check whether the device supports fingerprint management: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportFingerPrintCfg is returned and its value is "true", it indicates that the device supports fingerprint management.
2. Search, apply, add, and edit fingerprints.
3. If the node isSupportFingerPrintCfg is returned and its value is "false", it indicates that the device does not support fingerprint management.

Note:

- Before applying the fingerprint information to the device, make sure that the related person information linked to the person ID has been applied to the device.
- The maximum number of fingerprints that can be applied to the device per person is 10 (the 10 fingerprints of a person).

10.11.2.2 Fingerprint Search


Start
① Get the capability for fingerprint management
GET /ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json
Whether the calling succeeded
Yes
No
② Search the number of fingerprints of a specified person
GET /ISAPI/AccessControl/FingerPrint/Count?format=json&employeeNo=<employeeNo>
③ Search the number of fingerprints of all persons
GET /ISAPI/AccessControl/FingerPrint/Count?format=json
④ Search fingerprint information (returned by page)
POST /ISAPI/AccessControl/FingerPrintUpload?format=json
⑤ Fingerprint Search is not supported
End

The fingerprint search function is for searching the number of fingerprints and fingerprint information added to the device.

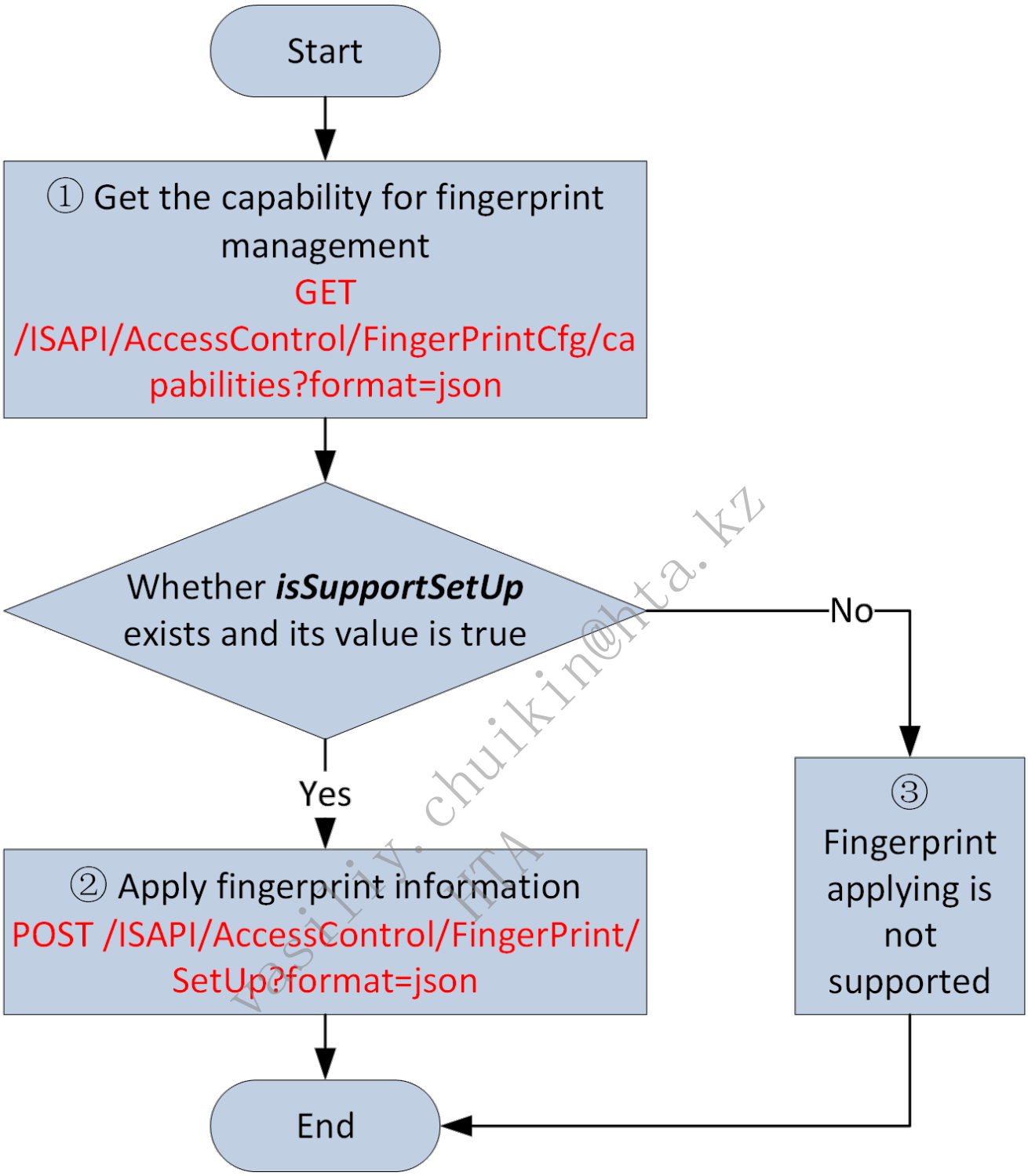
1. Check whether the device supports fingerprint search: `GET /ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json`; if calling succeeded, it indicates that the device supports fingerprint search.
2. Search the number of the specified persons' fingerprints: `GET /ISAPI/AccessControl/FingerPrint/Count?format=json&employeeNo=<employeeNo>`; the returned value of the node numberOfFP is the number of the added fingerprints of the specified persons.
3. Search the number of fingerprints of all persons: `GET /ISAPI/AccessControl/FingerPrint/Count?format=json`; the returned value of the node numberOfFP is the number of the added fingerprints of all persons.
4. Search fingerprint information: `POST /ISAPI/AccessControl/FingerPrintUpload?format=json`; the fingerprint information is returned by page. If the value of the child node status of the node FingerPrintInfo is "NoFP", it indicates that all fingerprint information are returned.
5. If calling failed, it indicates that the device does not support fingerprint search.

Note:

- The value of the node fingerPrintCapacity returned by calling `GET /ISAPI/AccessControl/CardReaderCfg/<cardReaderID>?format=json` is the maximum number of fingerprints supported by the card reader.
- The value of the node fingerPrintNum returned by calling `GET /ISAPI/AccessControl/CardReaderCfg/<cardReaderID>?format=json` is the number of fingerprints added to the card reader.

10.11.2.3 Fingerprint Applying

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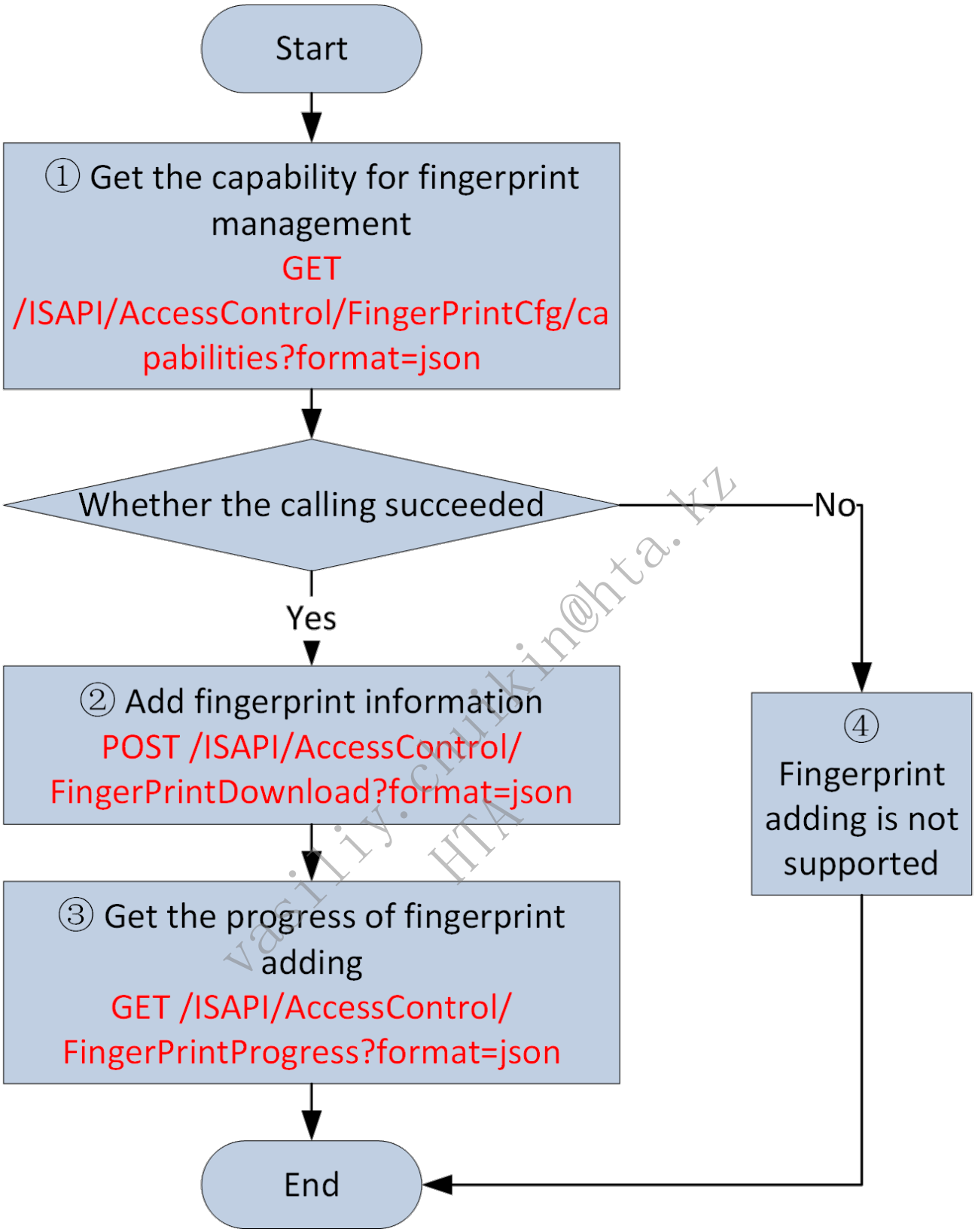
Fingerprint information can be applied to the device via the fingerprint applying function. If the fingerprint has been added to the device, the fingerprint information will be edited; if the fingerprint has not been added to the device, the fingerprint will be added to the device.

1. Check whether the device supports fingerprint applying: `GET /ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json`; if the node isSupportSetUp is returned and its value is "true", it indicates that the device supports fingerprint applying.
2. Apply fingerprint information: `POST /ISAPI/AccessControl/FingerPrint/SetUp?format=json`.
3. If the node isSupportSetUp is returned and its value is false, it indicates that the device does not support fingerprint applying.

Note:

Check whether the fingerprint has been added to the device via the nodes employeeNo and fingerPrintID returned after calling the API for fingerprint applying.

10.11.2.4 Fingerprint Adding



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Fingerprint information can be added to the device via the fingerprint adding function. If the fingerprint has been added to the device, the device will report an error; if the fingerprint has not been added to the device, the fingerprint will be added to the device.

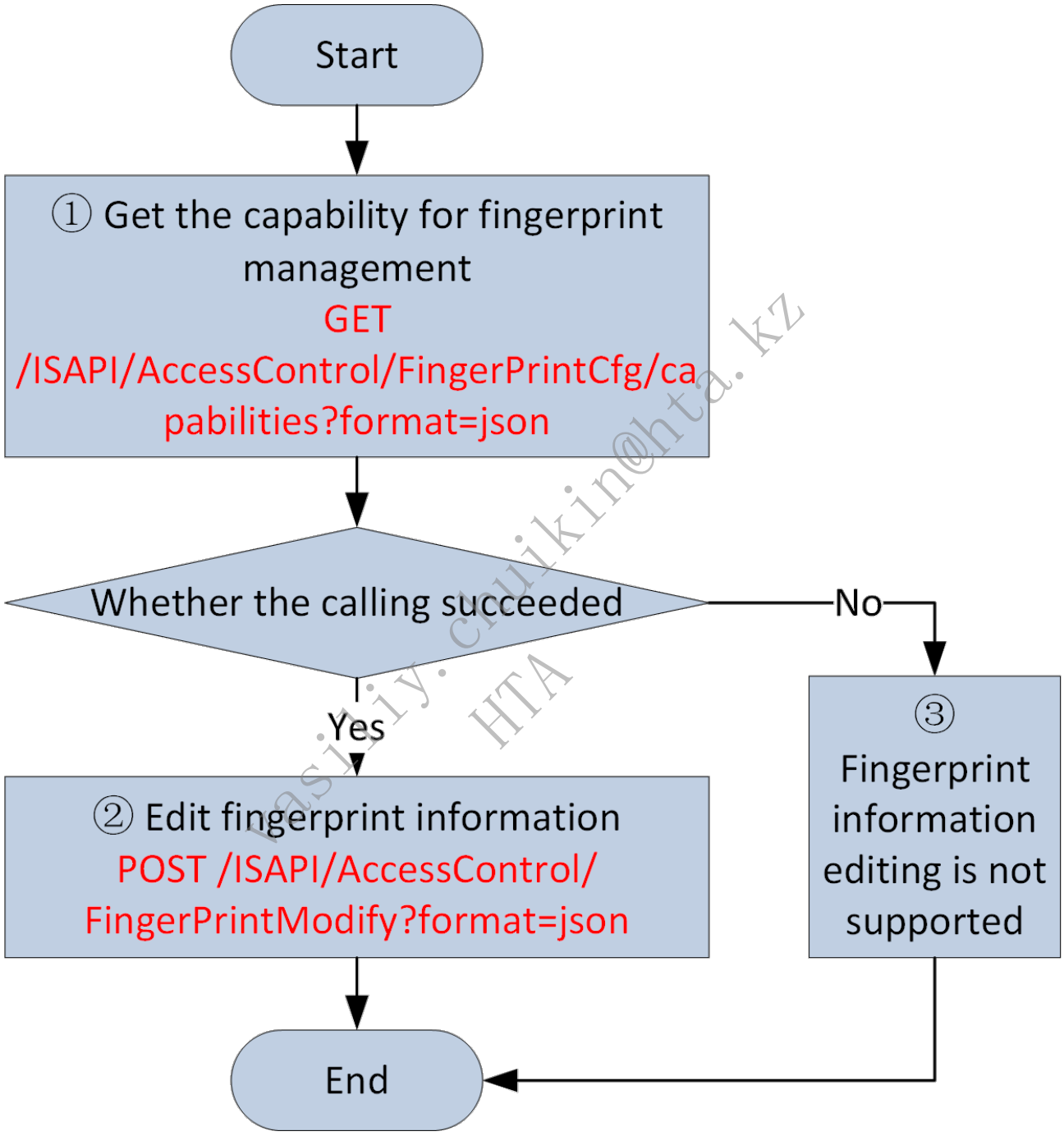
1. Check whether the device supports fingerprint adding: `GET /ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json`; if calling succeeded, it indicates that the device supports fingerprint adding.
2. Add fingerprint information: `POST /ISAPI/AccessControl/FingerPrintDownload?format=json`; if calling succeeded, it indicates that the device has started to execute fingerprint adding, but it does not indicate that the device has added the fingerprint.
3. Get the progress of fingerprint adding: `GET /ISAPI/AccessControl/FingerPrintProgress?format=json`; repeatedly call this API to get the progress of fingerprint adding.

4. If calling failed, it indicates that the device does not support fingerprint adding.

Note:

Check whether the fingerprint has been added to the device via the nodes employeeNo and fingerPrintID returned after calling the API for fingerprint adding.

10.11.2.5 Fingerprint Information Editing



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The fingerprint information on the device can be edited via the fingerprint information editing function. If the fingerprint has been added to the device, the fingerprint information will be edited; if the fingerprint has not been added to the device, the device will report an error.

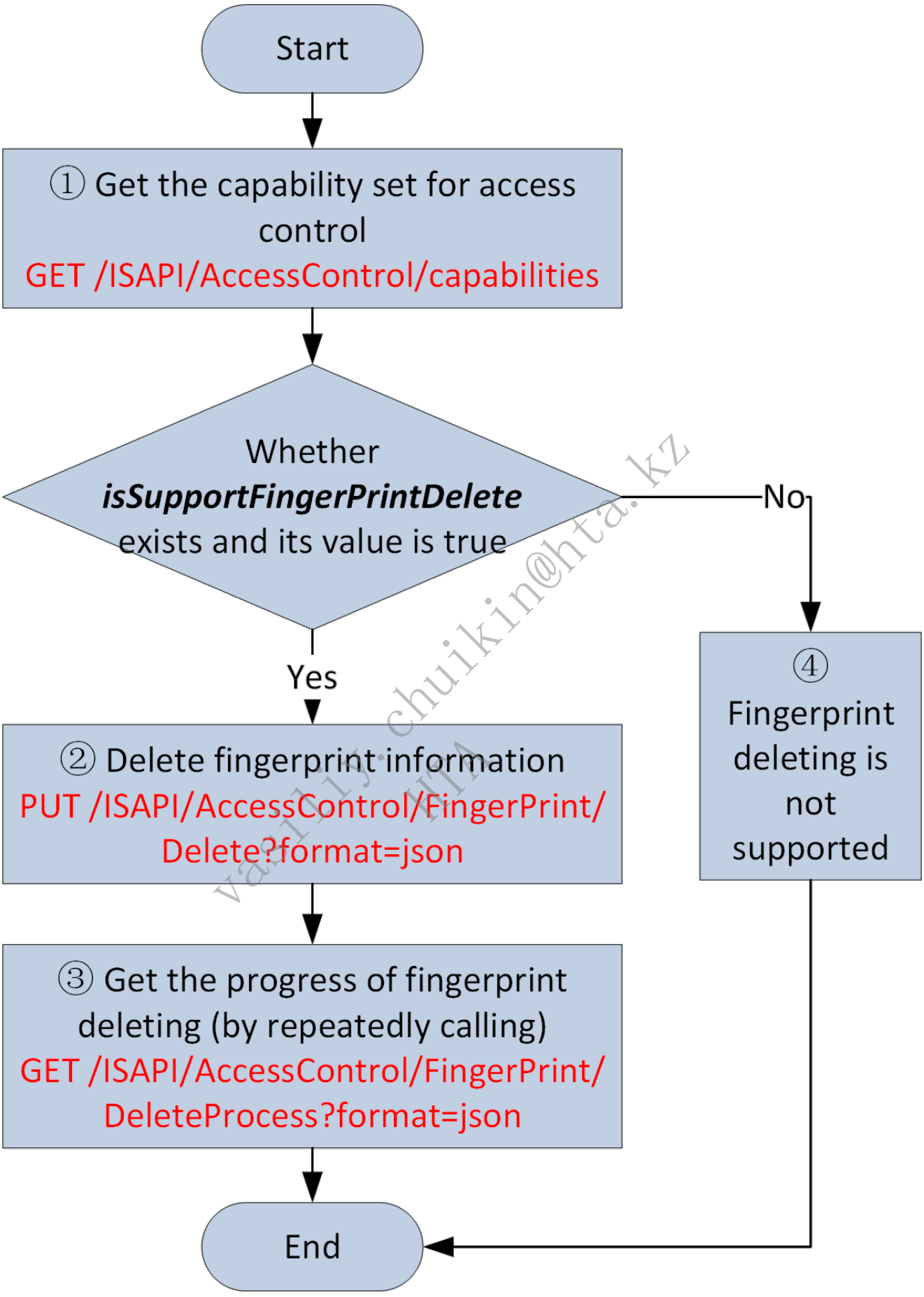
1. Check whether the device supports fingerprint information editing: `GET /ISAPI/AccessControl/FingerPrintCfg/capabilities?format=json`; if calling succeeded, it indicates that the device supports fingerprint information editing.
2. Edit fingerprint information: `POST /ISAPI/AccessControl/FingerPrintModify?format=json`.
3. If calling failed, it indicates that the device does not support fingerprint information editing.

Note:

- Check whether the fingerprint has been added to the device via the nodes employeeNo and fingerPrintID returned after calling the API for fingerprint information editing.
- When the fingerprint information is edited, only the fingerprint parameters will be edited; the fingerprint data will not be edited.

10.11.2.6 Fingerprint Deleting

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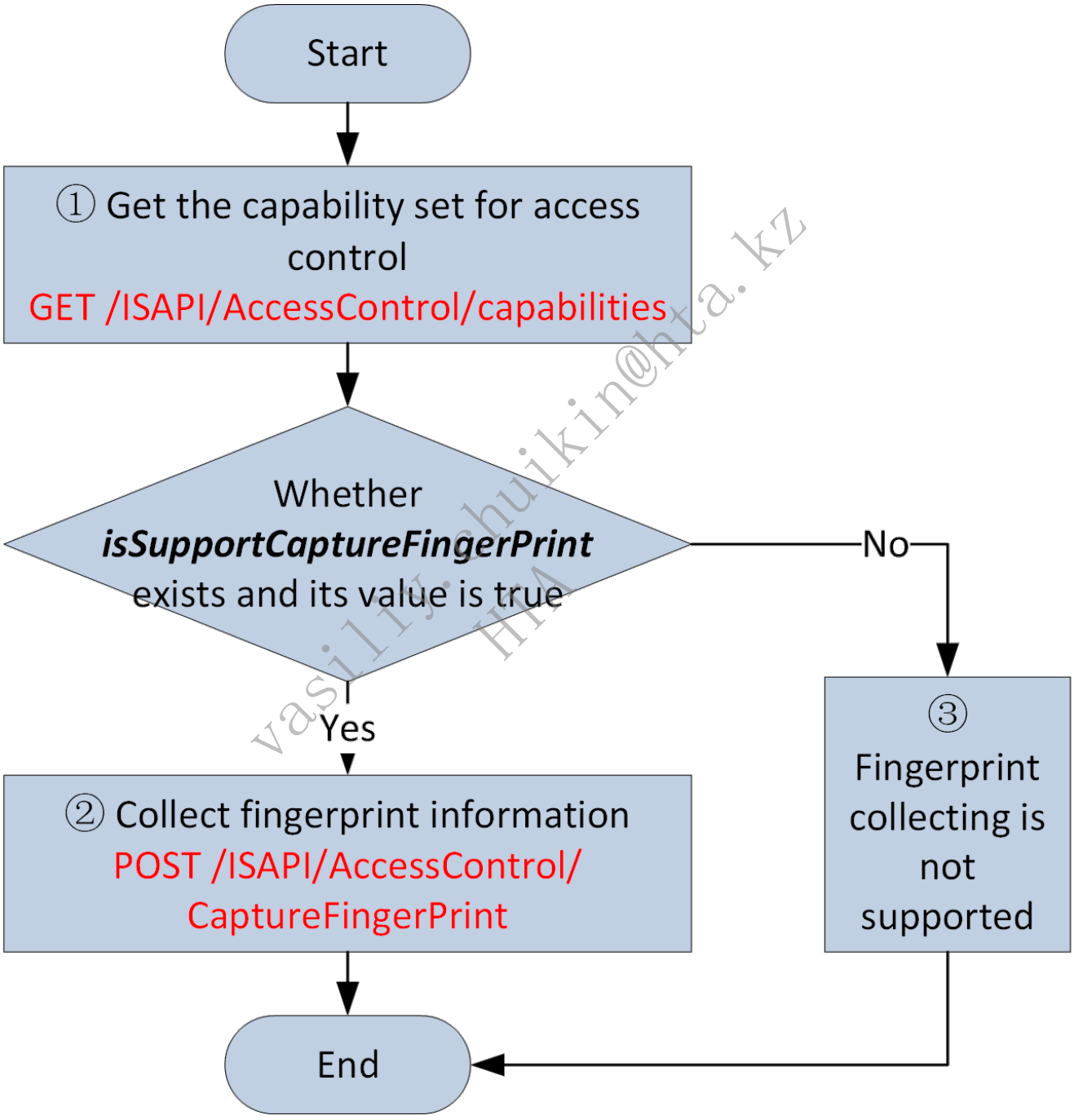
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The fingerprint information on the device can be deleted via the fingerprint deleting function. The device will not report an error if the fingerprint information to be deleted is not added to the device.

1. Check whether the device supports fingerprint deleting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportFingerPrintDelete is returned and its value is "true", it indicates that the device supports fingerprint deleting.

2. Delete fingerprint information: `PUT /ISAPI/AccessControl/FingerPrint/Delete?format=json`; if calling succeeded, it indicates that the device has started to execute fingerprint deleting, but it does not indicate that the device has deleted the fingerprints.
3. Get the progress of fingerprint deleting: `GET /ISAPI/AccessControl/FingerPrint/DeleteProcess?format=json`; repeatedly call this API to get the progress of fingerprint deleting.
4. If the node isSupportFingerPrintDelete is returned and its value is "false", it indicates that the device does not support fingerprint deleting.

10.11.2.7 Fingerprint Collecting



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The fingerprint collecting function is for collecting the fingerprint data, fingerprint quality, etc.

1. Check whether the device supports fingerprint collecting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCaptureFingerPrint is returned and its value is "true", it indicates that the device supports fingerprint collecting.
2. Collect fingerprint information: `POST /ISAPI/AccessControl/CaptureFingerPrint`.
3. If the node isSupportCaptureFingerPrint is returned and its value is "false", it indicates that the device does not support fingerprint collecting.

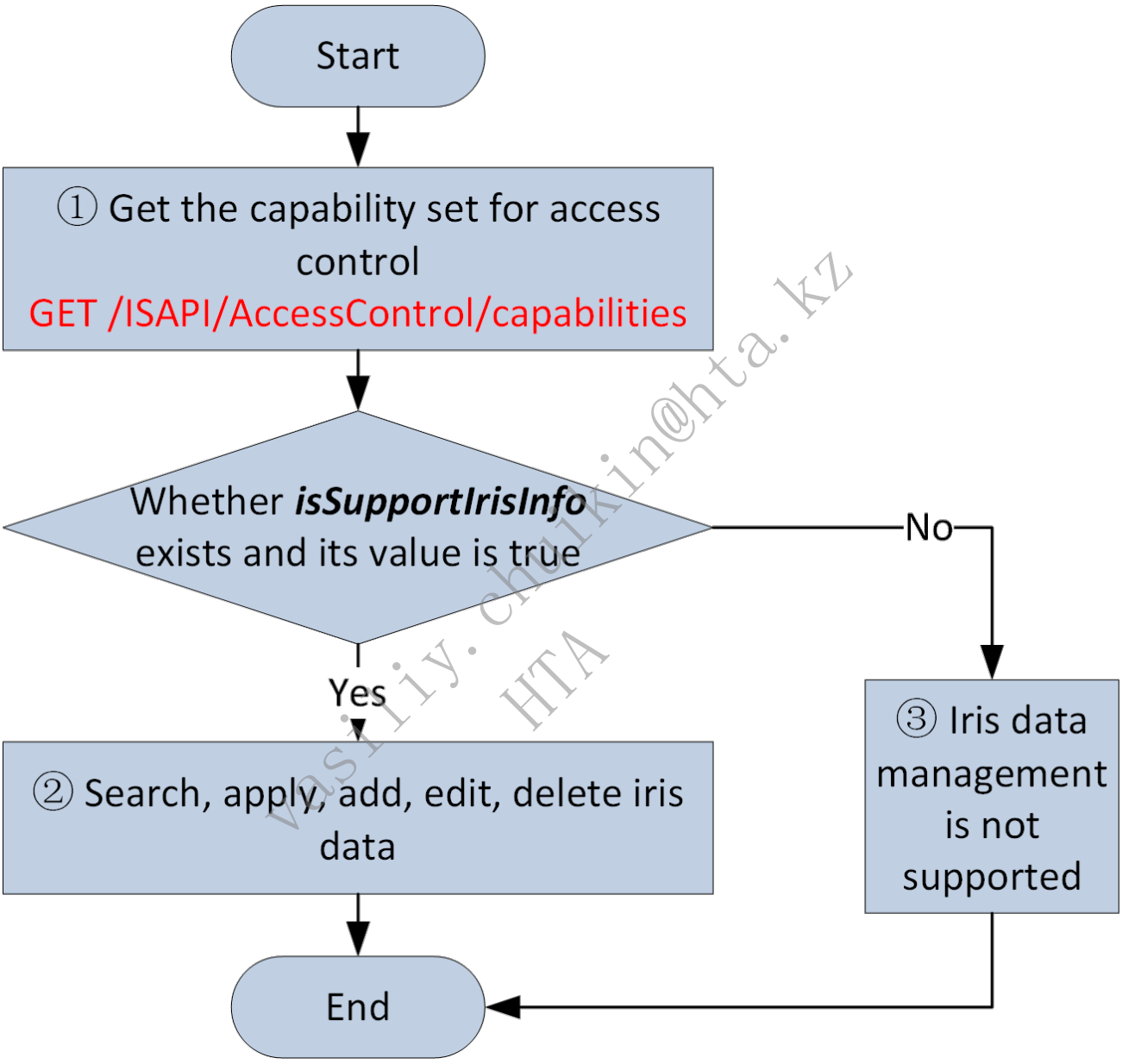
10.12 Iris Data Management

10.12.1 Introduction to the Function

Iris data management includes searching, applying, adding, editing, deleting, and collecting iris data.

10.12.2 API Calling Flow

10.12.2.1 Check Whether the Device Supports Iris Data Management



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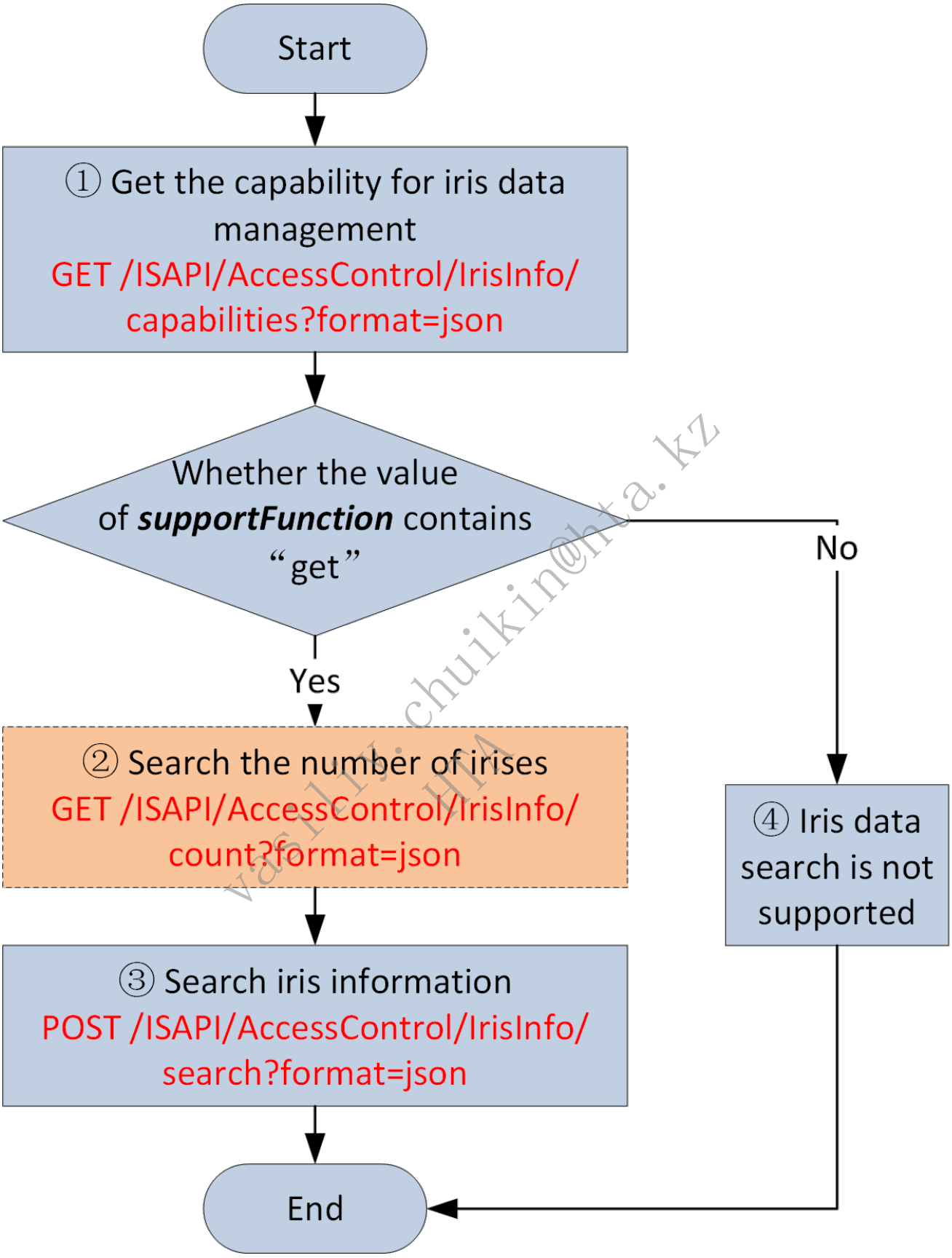
Before calling the API for iris data management, make sure that the device supports iris data management:

1. Check whether the device supports iris data management: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportIrisInfo is returned and its value is true, it indicates that the device supports iris data management.
2. Search, apply, add, edit, and delete iris data.
3. If the node isSupportIrisInfo is returned and its value is false, it indicates that the device does not support iris data management.

Note:

- Before applying, adding, and editing the iris data on the device, make sure that the related person information linked to the person ID has been applied to the device.
- Data for up to two irises of each person (two eyes of each person) can be applied to the device.

10.12.2.2 Iris Data Search



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The iris data search function is for searching the number of irises and iris information added to the device.

1. Check whether the device supports iris data search: `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json`; if the node supportFunction is returned, and the value contains "get", it indicates that the device supports iris data search.
2. Search the number of irises: `GET /ISAPI/AccessControl/IrisInfo/count?format=json`; the value of the node IrisNumber is the number of added irises on the device.
3. Search iris information: `POST /ISAPI/AccessControl/IrisInfo/search?format=json`; the searched iris

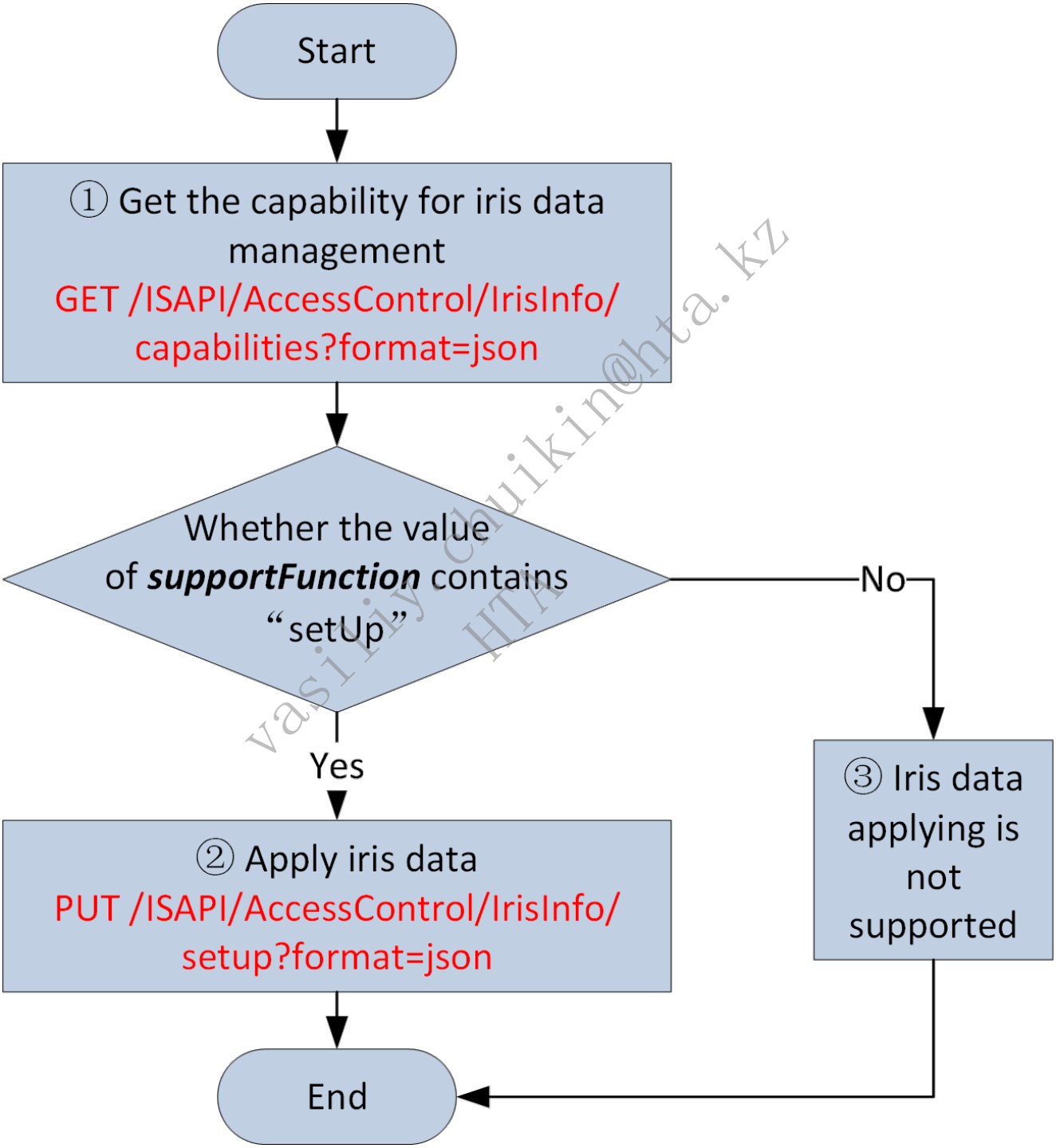
information will be returned by page.

4. If the value of the node supportFunction does not contain "get", it indicates that the device does not support iris data search.

Note:

The value of the node maxRecordNum returned by calling `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json` is the maximum number of irises supported by the device.

10.12.2.3 Iris Data Applying



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The iris data applying function is for applying iris information to the device. If the iris data has already been applied to the device, the information about the iris will be edited; if the iris data has not been applied to the device, the iris information will be added to the device.

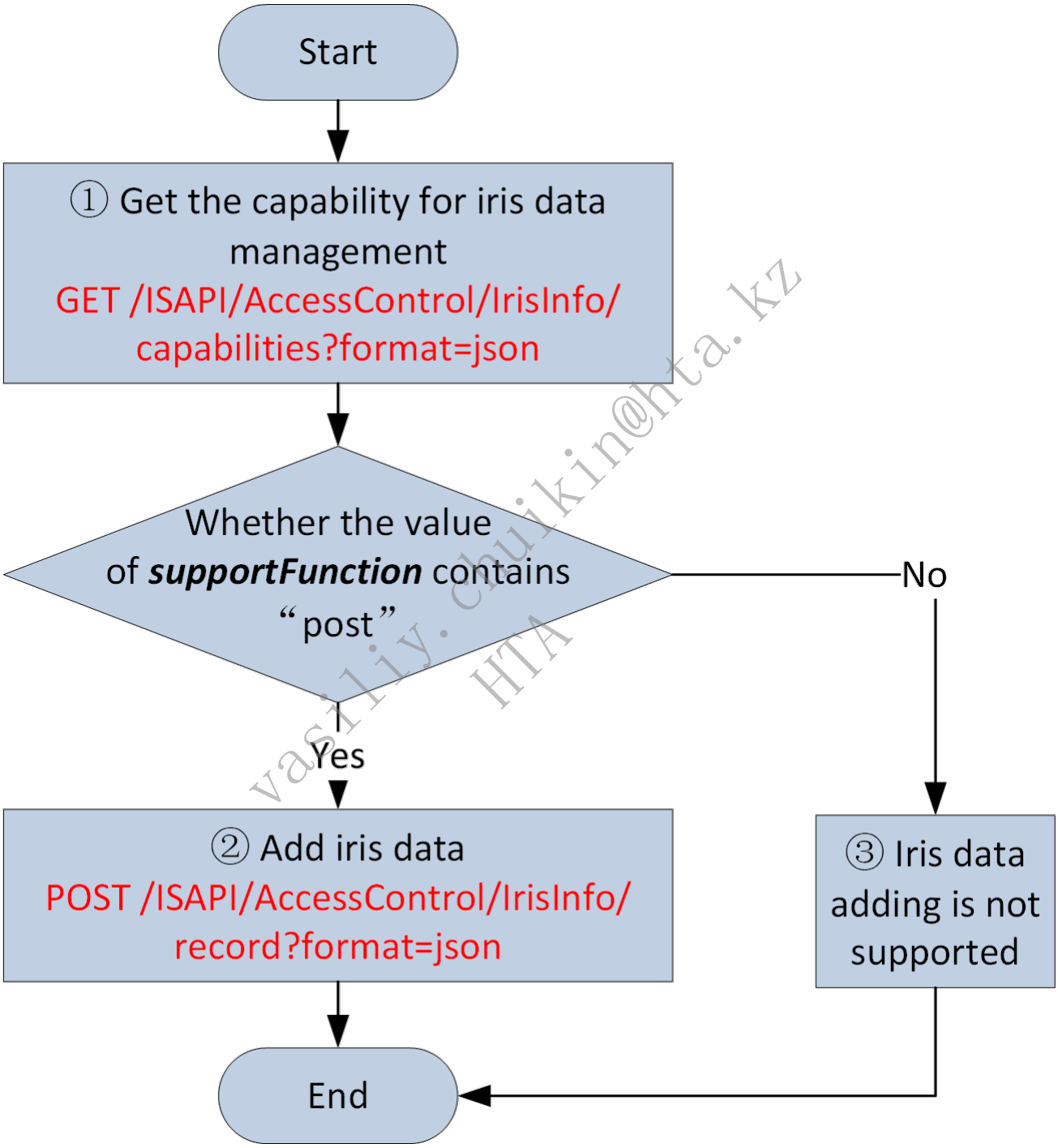
1. Check whether the device supports iris data applying: `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json`; if the node supportFunction is returned and the value contains "setUp", it indicates that the device supports iris data applying.
2. Iris Data Applying: `PUT /ISAPI/AccessControl/IrisInfo/setup?format=json`.

3. If the value of the node supportFunction does not contain “setUp”, it indicates that the device does not support iris data applying.

Note:

Check whether the iris data have been applied to the device via the nodes employeeNo and id returned by calling the API for iris data applying.

10.12.2.4 Iris Data Adding



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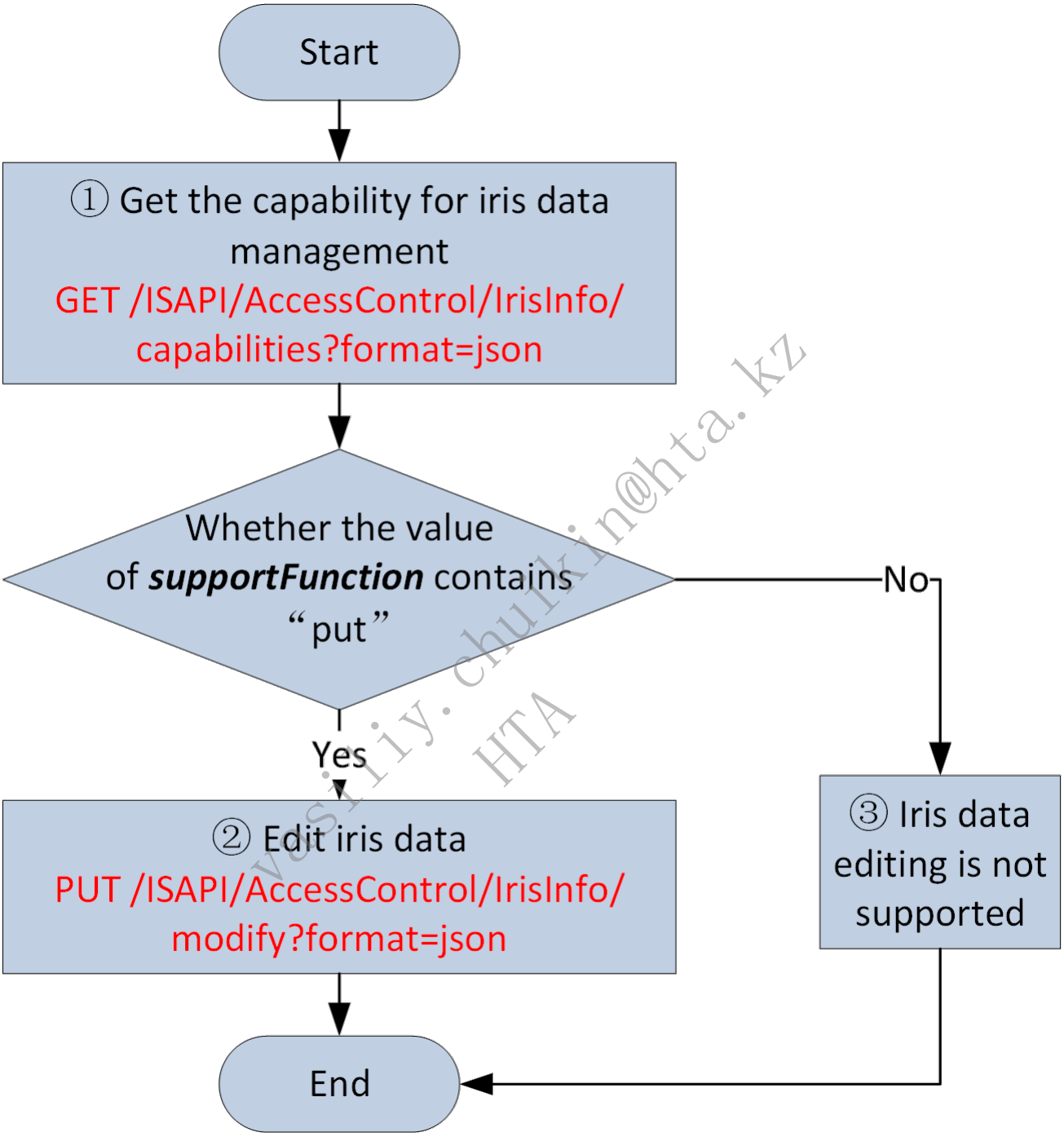
The iris data adding function is for adding the iris data to the device. If the iris data have already been added to the device, the device will report an error; if the iris data have not been added to the device, the iris data will be added to the device.

1. Check whether the device supports iris data adding: `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json`; if the value of the node supportFunction contains “post”, it indicates that the device supports iris data applying.
2. Add iris data: `POST /ISAPI/AccessControl/IrisInfo/record?format=json`.
3. If the value of the node supportFunction does not contain “post”, it indicates that the device does not support iris data applying.

Note:

Check whether the iris data have been applied to the device via the nodes employeeNo and id returned by calling the API for iris data adding.

10.12.2.5 Iris Data Editing



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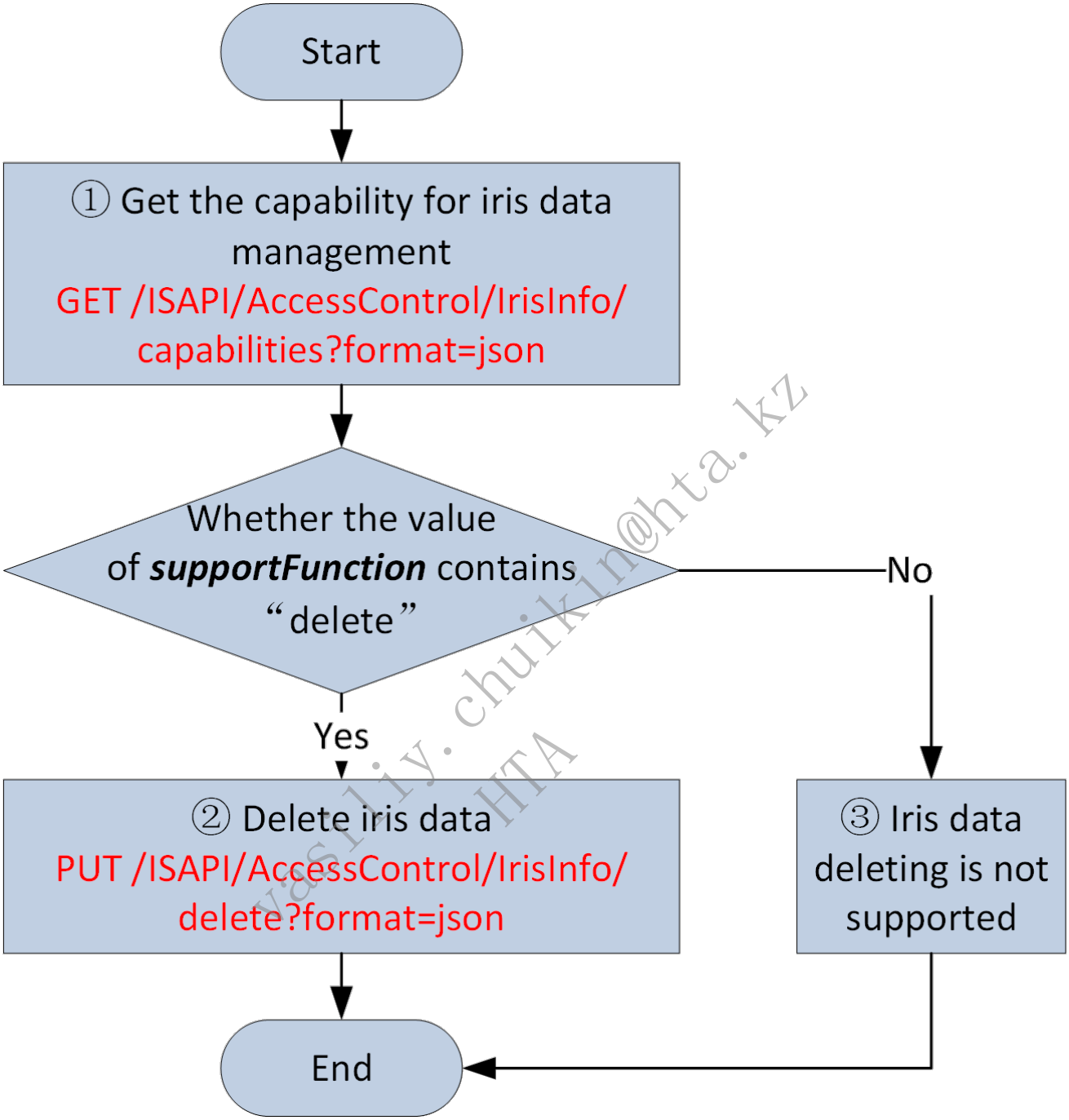
The iris data editing function is for editing the applied iris information on the device. If the iris data have already been added to the device, the iris information will be edited; if the iris data have not been added to the device, the device will report an error.

1. Check whether the device supports iris data editing: `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json`; if the value of the node supportFunction contains "put", it indicates that the device supports iris data editing.
2. Edit iris information: `PUT /ISAPI/AccessControl/IrisInfo/modify?format=json`.
3. If the value of the node supportFunction does not contain "put", it indicates that the device does not support iris data editing.

Note:

Check whether the iris data have been applied to the device via the nodes employeeNo and id returned by calling the API for iris data editing.

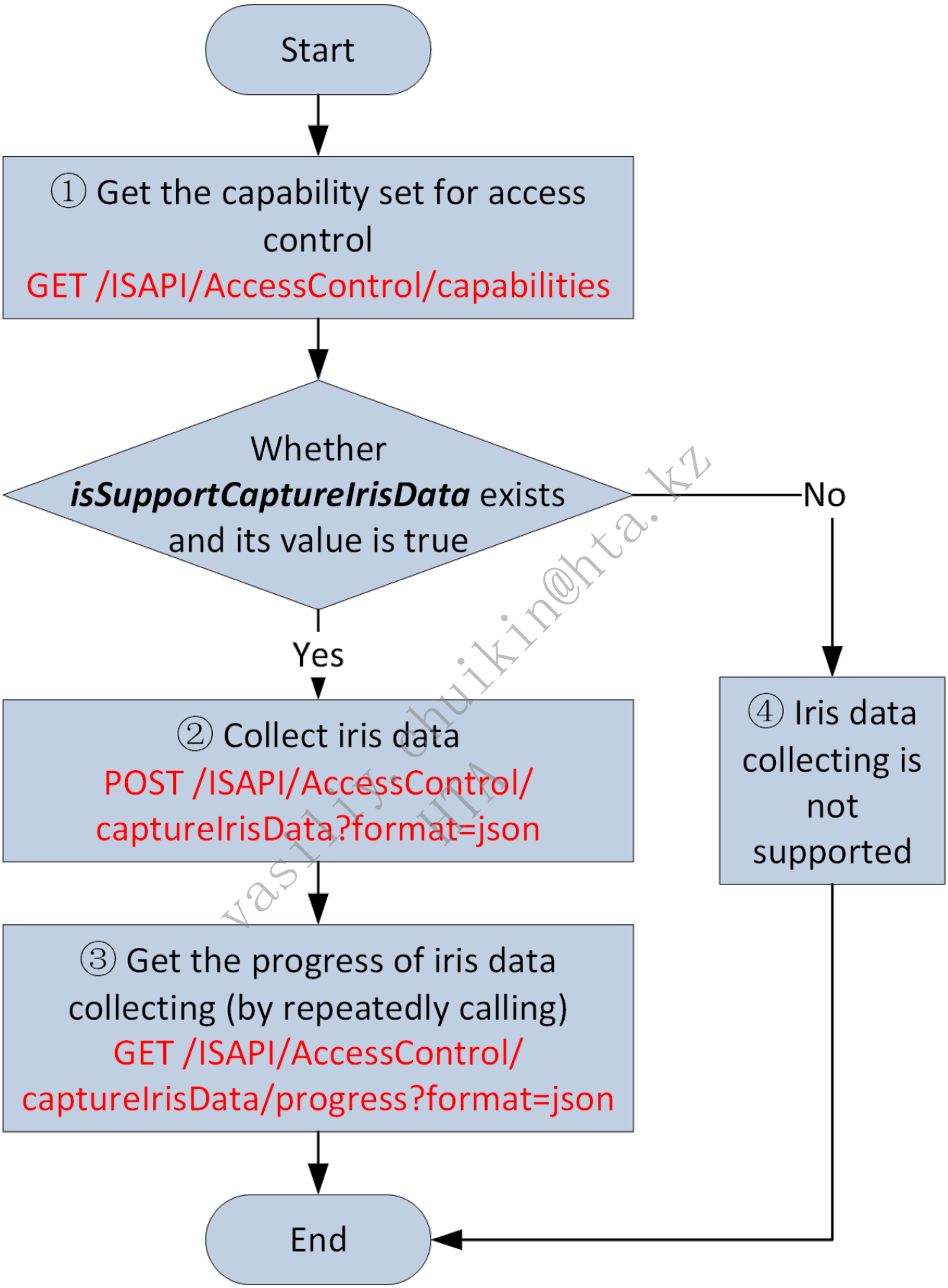
10.12.2.6 Iris Data Deleting



The iris data deleting function is for deleting the applied iris information on the device. If the iris data to be deleted have not been applied to the device, the device will not report an error.

1. Check whether the device supports iris data deleting: `GET /ISAPI/AccessControl/IrisInfo/capabilities?format=json`; if the value of the node supportFunction contains "delete", it indicates that the device supports iris data deleting.
2. Delete iris information: `PUT /ISAPI/AccessControl/IrisInfo/delete?format=json`; if calling succeeded, it indicates that the iris information has been deleted.
3. If the value of the node supportFunction does not contain "delete", it indicates that the device does not support iris data deleting.

10.12.2.7 Iris Data Collecting



The iris data collecting function is for collecting iris data and information.

1. Check whether the device supports iris data collecting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCaptureIrisData is returned and its value is "true", it indicates that the device supports iris data collecting.
2. Collect iris information: `POST /ISAPI/AccessControl/captureIrisData?format=json`; if calling succeeded, it indicates that the device has started to execute the collection.
3. Get the progress of iris data collecting: `GET /ISAPI/AccessControl/captureIrisData/progress?format=json`; repeatedly call this API until the value of captureProgress is returned and is 100, which indicates that the collecting

completed.

4. If the node isSupportCaptureIrisData is returned and its value is "false", it indicates that the device does not support iris data collecting.

10.13 Multi-Factor Authentication

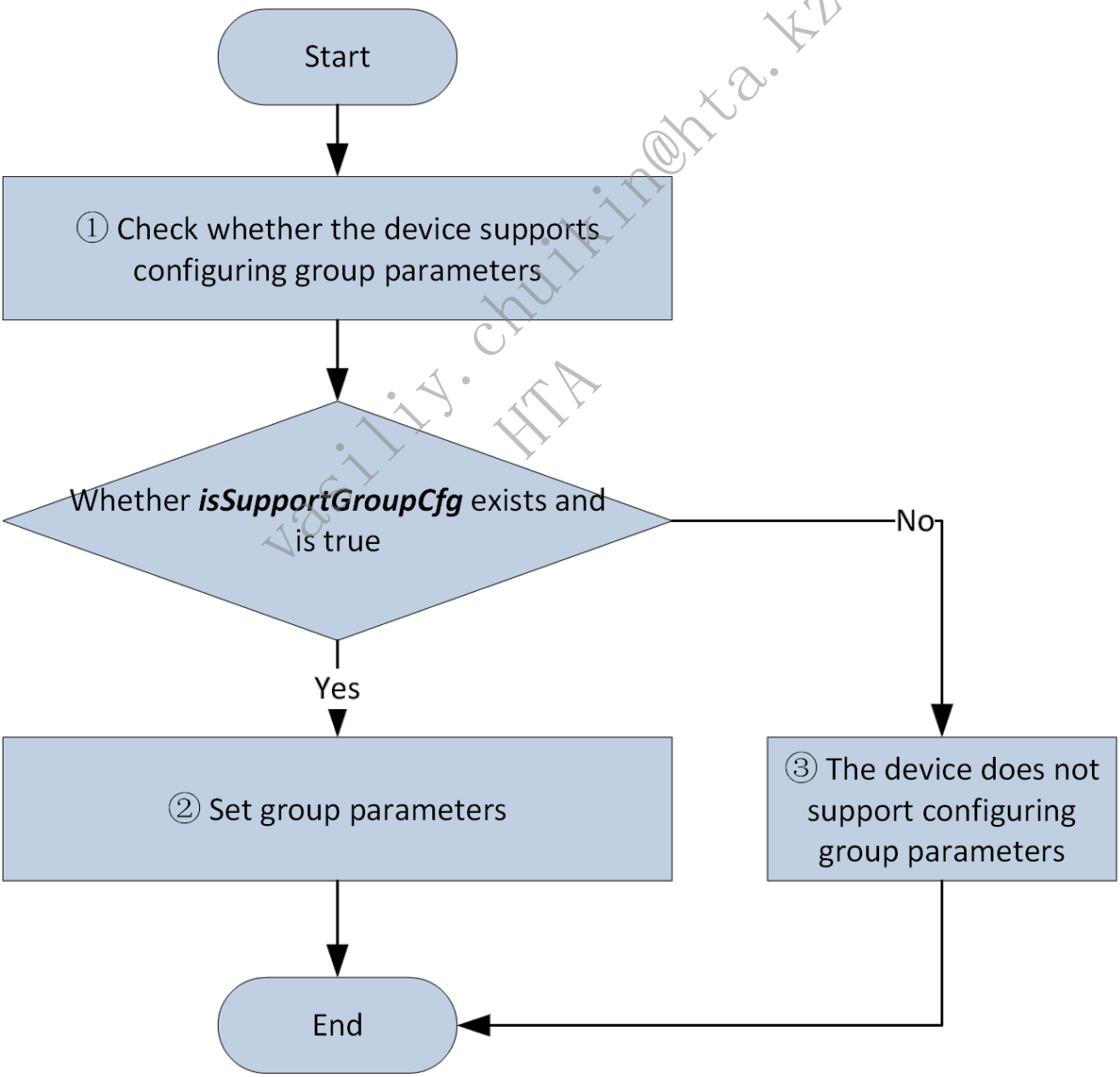
10.13.1 Introduction to the Function

In some scenarios with higher security levels, you can set rules that the door will only open when different persons authenticate in the access control point during fixed time.

For example, in a bank, a door will only open after two or more persons are authenticated (such as swiping card, authenticated by fingerprint, face picture, iris, etc.). If a door is configured multi-factor authentication, the number authentication persons, authentication order, and authentication methods should follow the rules.

10.13.2 API Calling Flow

10.13.2.1 Group Parameter Configuration



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The API calling flow is as follow:

1. Check whether the device supports configuring group parameters: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportGroupCfg is returned and its value is "true", it indicates that the device supports configuring group parameters.

2. Set group parameters: `[GET/PUT] /ISAPI/AccessControl/GroupCfg/<groupID>?format=json`.
3. If the node isSupportGroupCfg is returned and its value is "false", it indicates that the device does not support configuring group parameters.

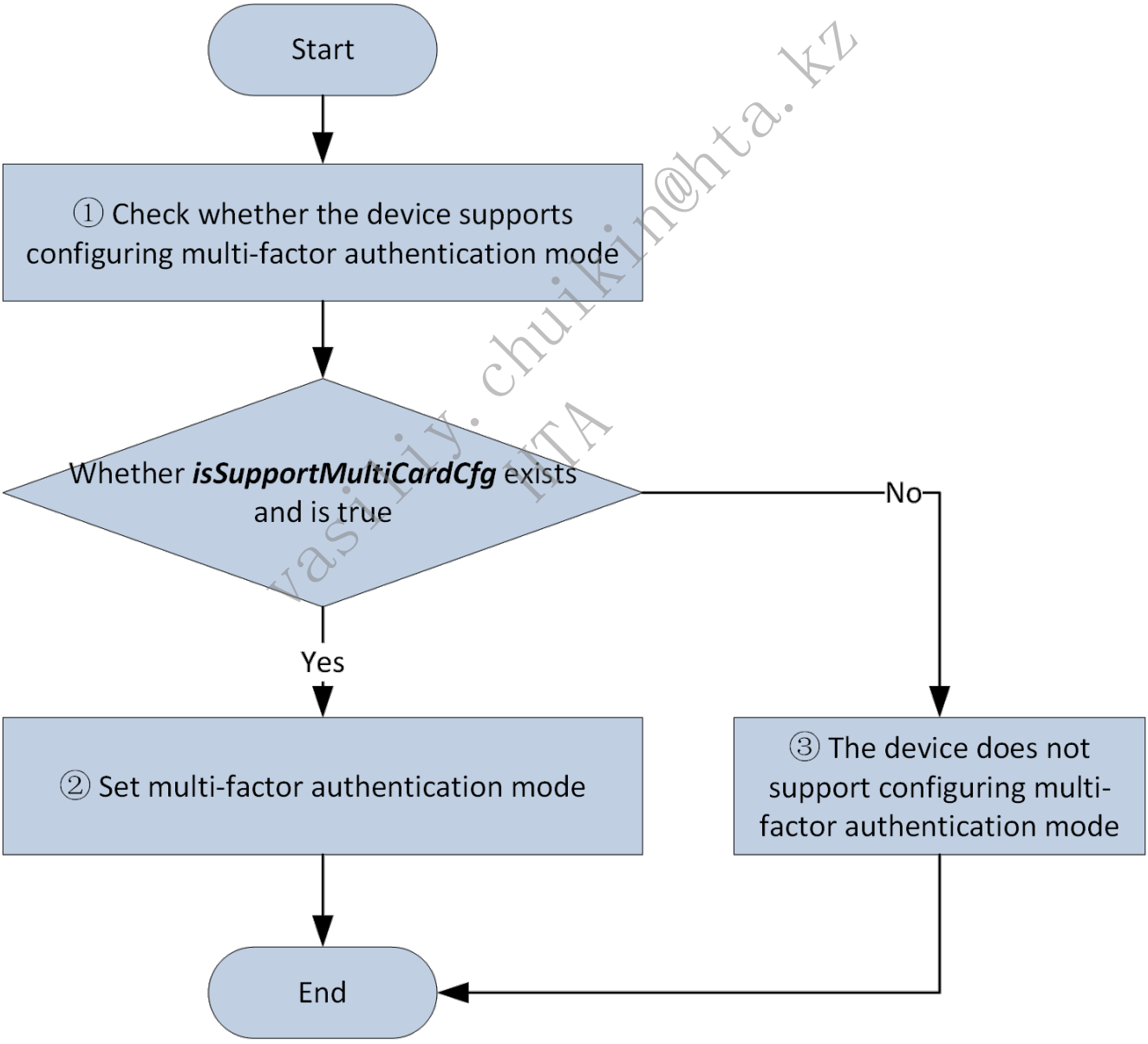
10.13.2.2 Add Person to Group

After configuring the group parameters, the person and group will be linked through ****person management in the process of person and credential management integration (through the belongGroup field). The corresponding interface is as follows:

1. Set person information: `PUT /ISAPI/AccessControl/UserInfo/SetUp?format=json`
2. Add person information: `POST /ISAPI/AccessControl/UserInfo/Record?format=json`
3. Edit person information: `PUT /ISAPI/AccessControl/UserInfo/Modify?format=json`

No more than 4 groups can be linked to one person.

10.13.2.3 Multi-Factor Authentication Mode Configuration



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The API calling flow is as follow:

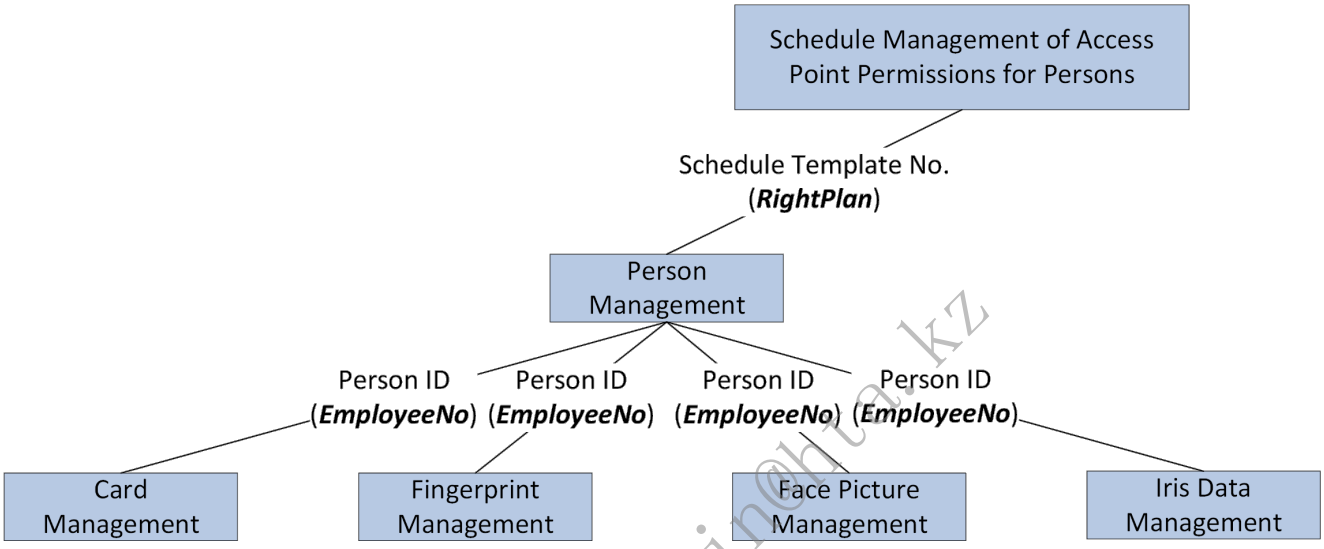
1. Check whether the device supports configuring multi-factor authentication mode: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportMultiCardCfg is returned and its value is "true", it indicates that the device supports configuring multi-factor authentication mode (and the device also supports configuring group parameters).
2. Set multi-factor authentication mode: `[GET/PUT] /ISAPI/AccessControl/MultiCardCfg/<doorID>?format=json`
3. If the node isSupportVerifyWeekPlanCfg is returned and its value is "false", it indicates that the device does not

support configuring multi-factor authentication mode.

10.14 Person and Credential Management

10.14.1 Introduction to the Function

The person and credential management function is person-based, and is for managing persons, credentials (cards, fingerprints, face pictures, and iris data), and permission schedules which control the permissions for persons to enter and exit the controlled areas. Its architecture is shown below.



This document mainly introduces the calling flows for person management and credential management (card, fingerprint, face picture, iris data management). For details about the calling flow for permission schedule management, refer to the "Management of Permission Schedules for Persons and Access Points".

10.15 Person Management

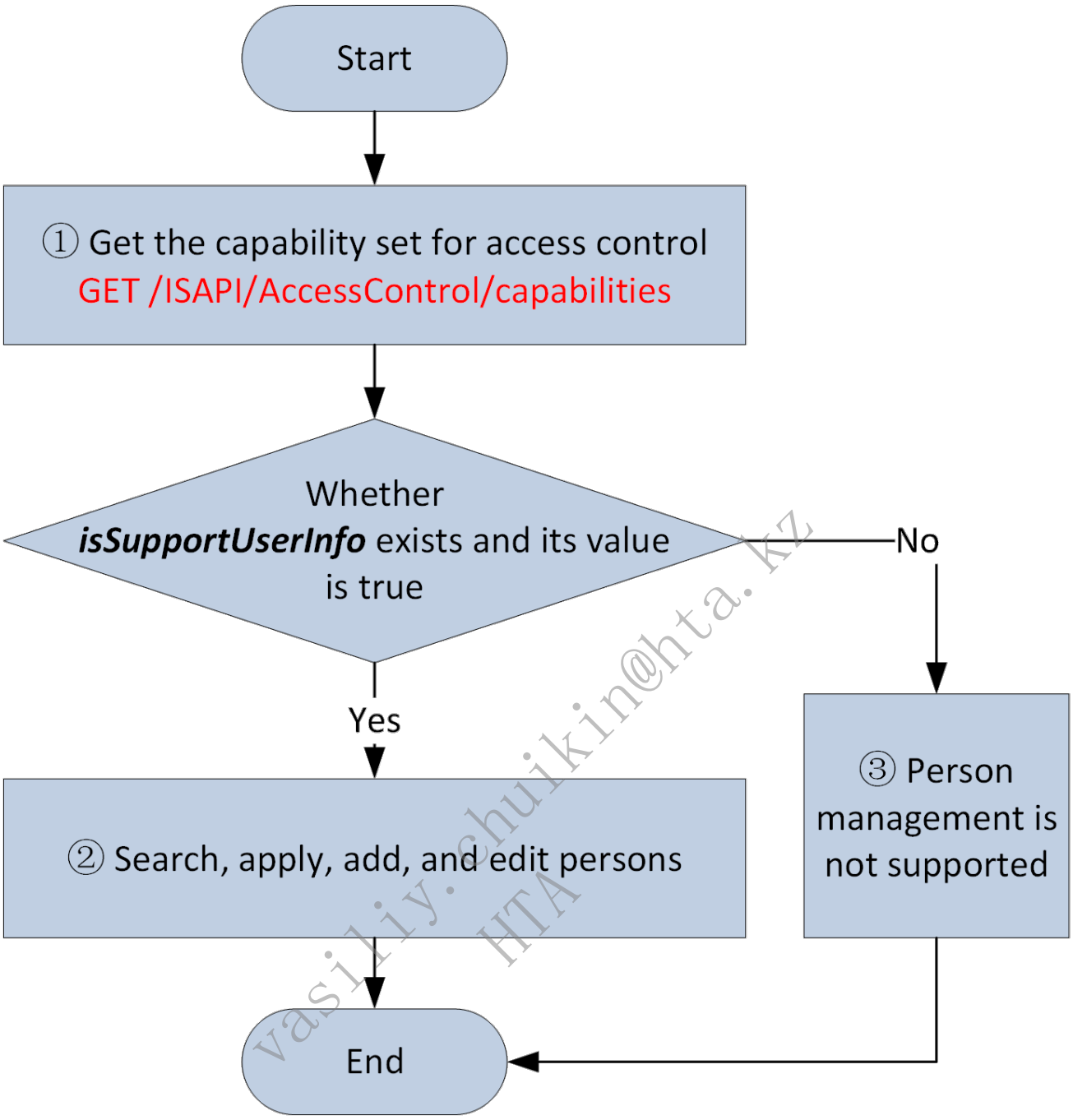
10.15.1 Introduction to the Function

Person management includes searching, applying, adding, editing, and deleting persons.

10.15.2 API Calling Flow

10.15.2.1 Check Whether the Device Supports Person Management

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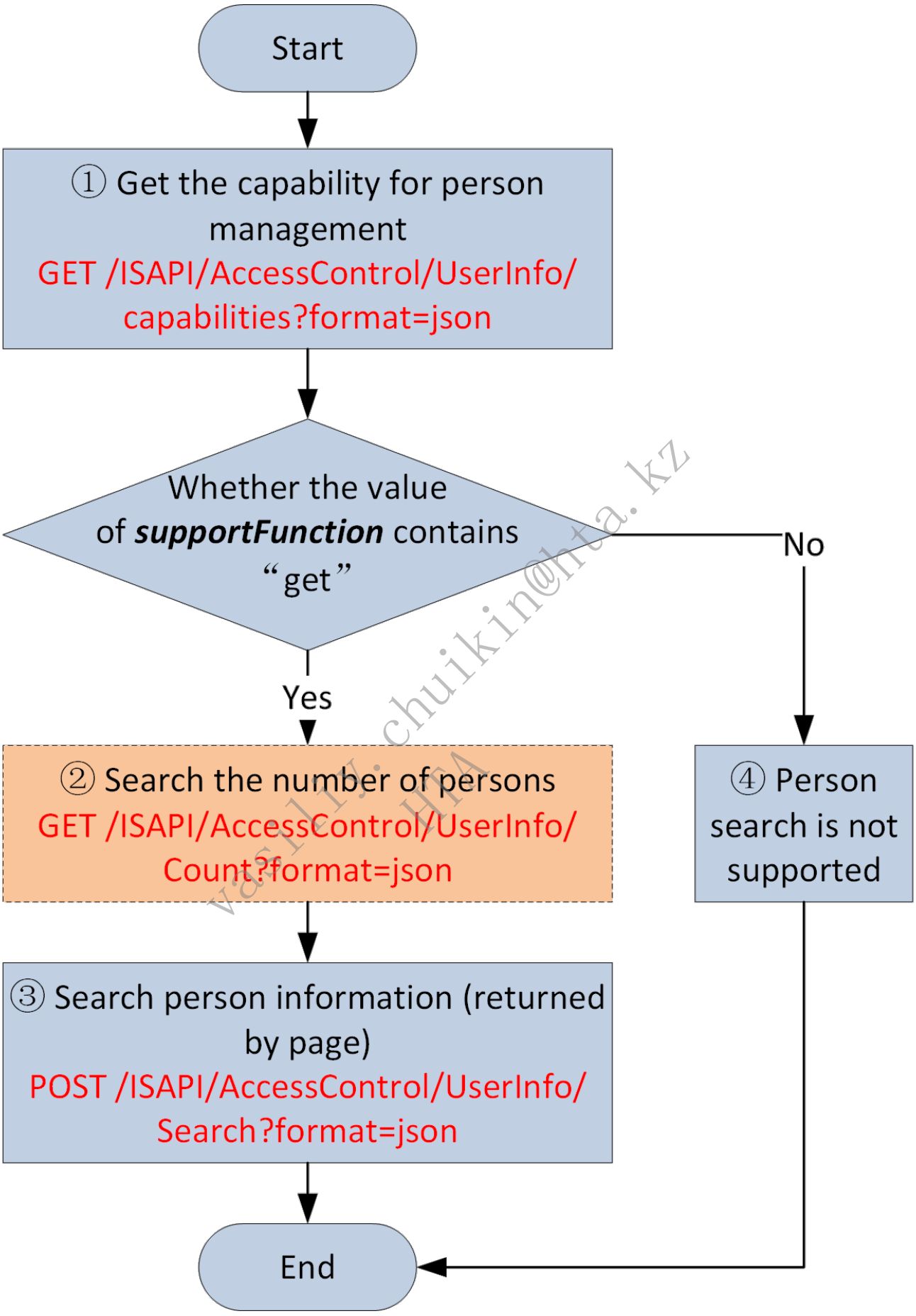
Before calling the API for person management, make sure that the device supports person management.

1. Check whether the device supports person management: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportUserInfo is returned and its value is true, it indicates that the device supports person management.
2. Search, apply, add, and edit persons.
3. If the node isSupportUserInfo is returned and its value is false, it indicates that the device does not support person management.

Note:

The person ID (EmployeeNo) is the unique identifier for person and credential management. After calling `GET /ISAPI/AccessControl/capabilities`, through the child nodes of EmployeeNoInfo which are employeeNo, characterType, and isSupportCompress, the maximum string length and character types of the person ID supported by the device can be checked. Generally, devices support up to 32 bytes and any type of characters. But for access controllers and distribution-type access control devices, check through the child nodes mentioned above.

10.15.2.2 Person Search



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The person search function is for searching the number of persons and person information added to the device.

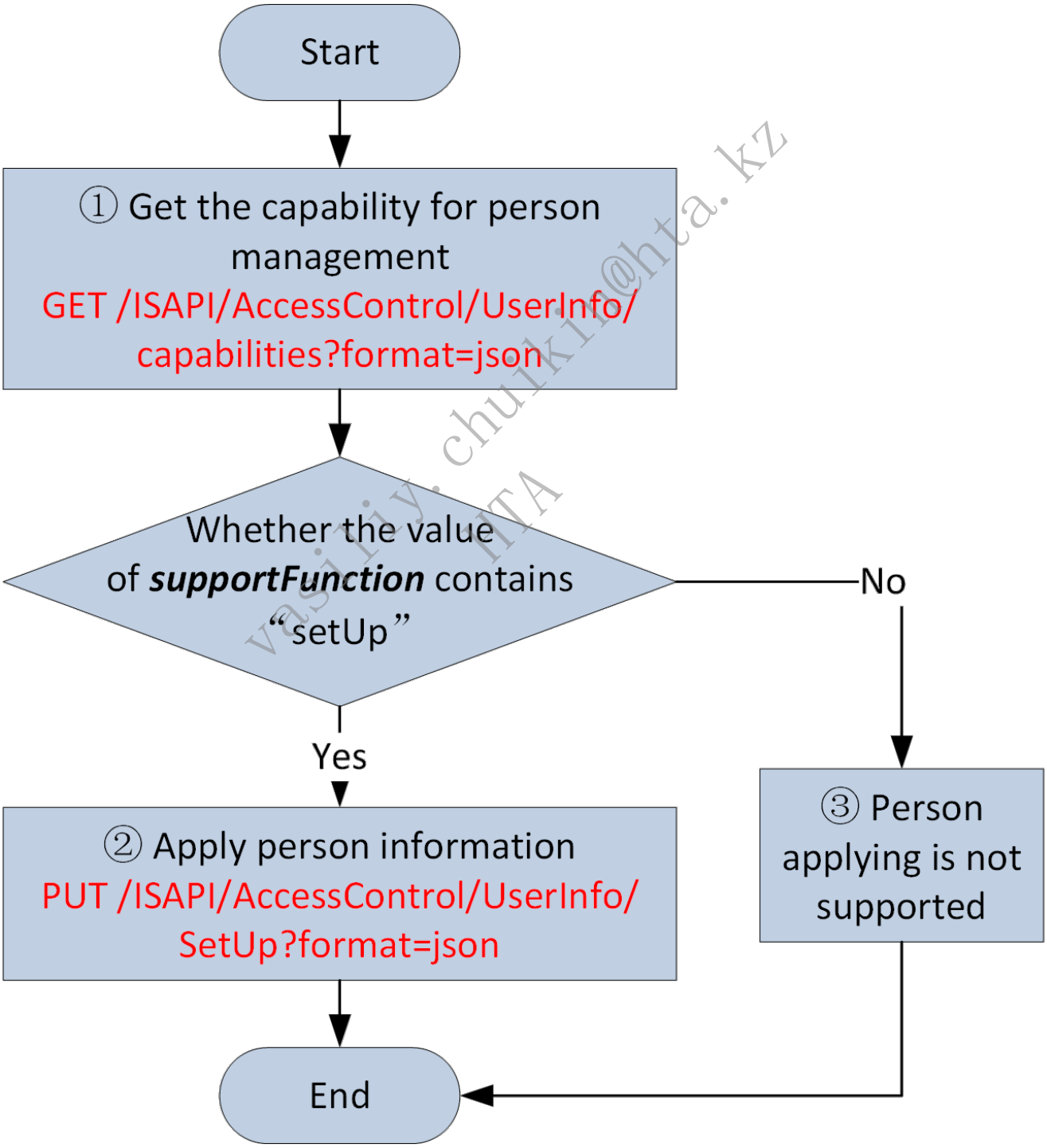
1. Check whether the device supports person search: `GET /ISAPI/AccessControl/UserInfo/capabilities?format=json`; if the value of the node supportFunction contains "get", it indicates that the device supports person search.

2. Search the number of persons: `GET /ISAPI/AccessControl/UserInfo/Count?format=json`; the returned value of the node userNumber is the number of the persons added to the device.
3. Search person information: `POST /ISAPI/AccessControl/UserInfo/Search?format=json`; the person information is returned by page.
4. If the value of the node supportFunction does not contain "get", it indicates that the device does not support person search.

Note:

The value of the node maxRecordNum returned by calling `GET /ISAPI/AccessControl/UserInfo/capabilities?format=json` is the maximum number of persons supported by the device.

10.15.2.3 Person Applying



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Person information can be applied to the device via the person applying function. If the person has been added to the device, the person information will be edited; if the person has not been added to the device, the person information will be applied to the device.

1. Check whether the device supports person applying: `GET /ISAPI/AccessControl/UserInfo/capabilities?`

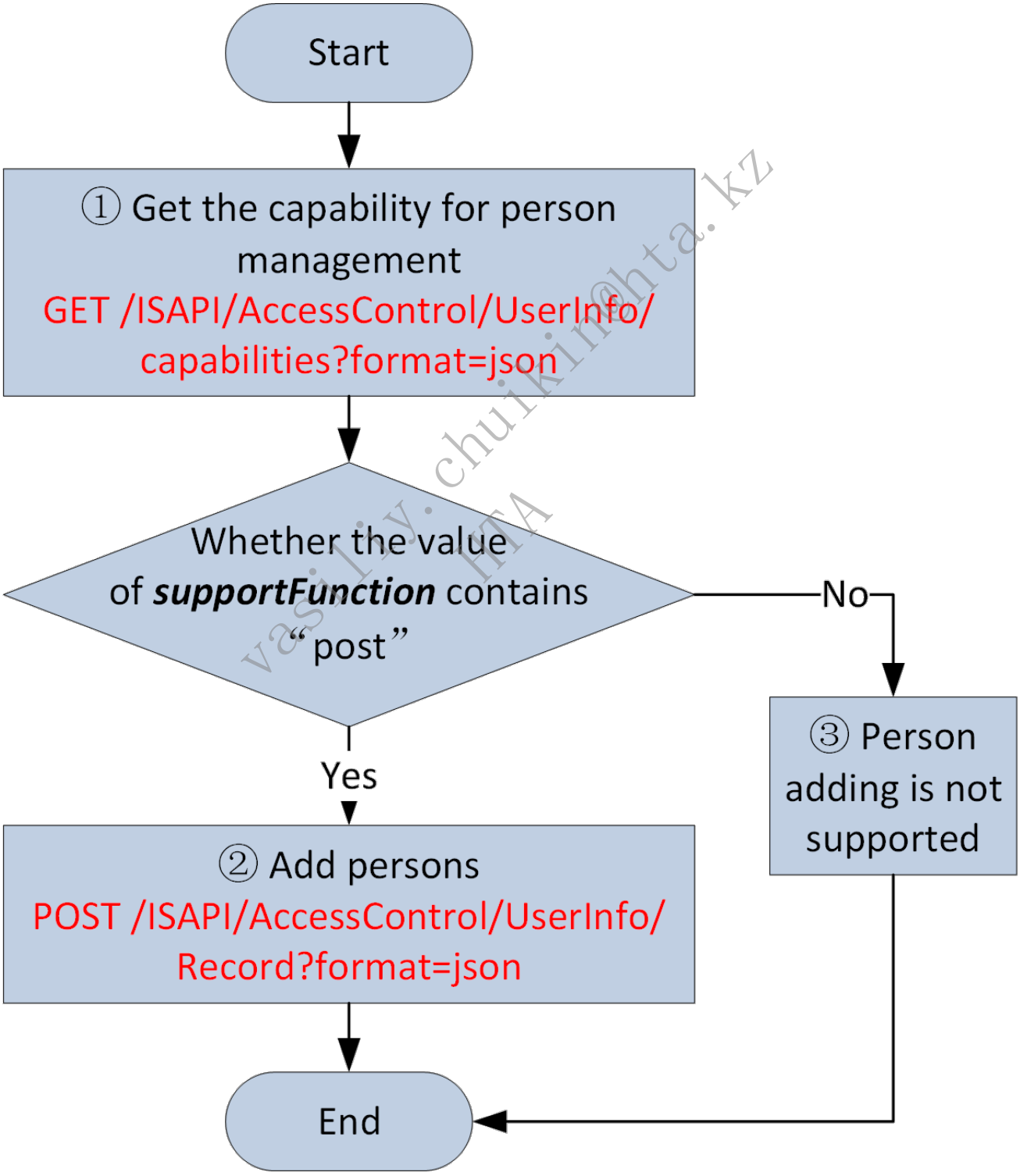
`format=json`; if the value of the node supportFunction contains "setUp", it indicates that the device supports person applying.

2. Apply person information: `PUT /ISAPI/AccessControl/UserInfo/SetUp?format=json`.
3. If the value of the node supportFunction does not contain setUp, it indicates that the device does not support person applying.

Note:

Check whether the person has been added to the device via the node employeeNo returned after calling the API for person applying.

10.15.2.4 Person Adding



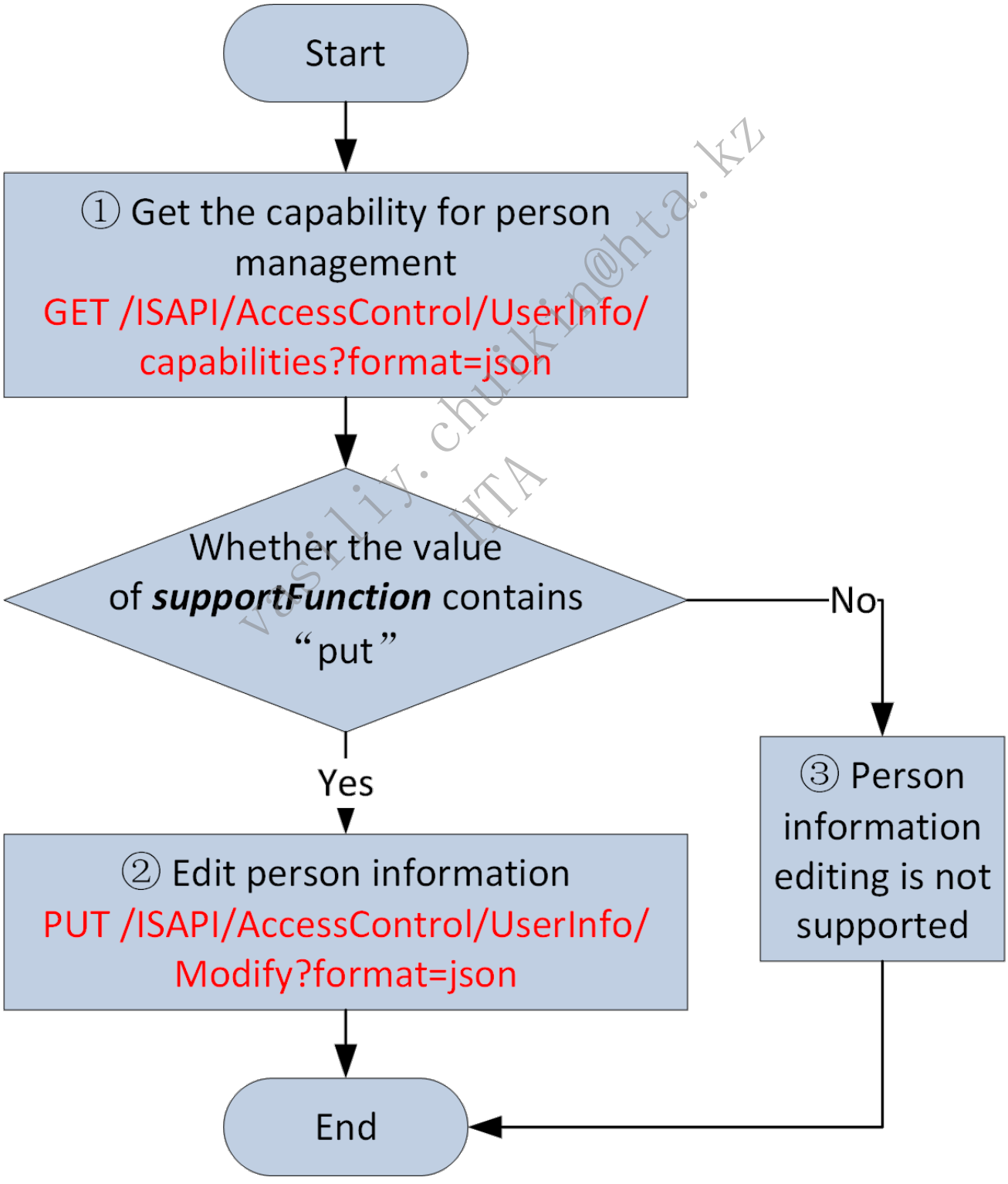
Person can be added to the device via the person adding function. If the person has been added to the device, the device will report an error; if the person has not been added to the device, the person will be added to the device.

1. Check whether the device supports person adding: `GET /ISAPI/AccessControl/UserInfo/capabilities?format=json`; if the value of the node supportFunction contains "post", it indicates that the device supports person adding.
2. Add persons: `POST /ISAPI/AccessControl/UserInfo/Record?format=json`.
3. If the value of the node supportFunction does not contain "post", it indicates that the device does not support person adding.

Note:

Check whether the person has been added to the device via the node employeeNo returned after calling the API for person adding.

10.15.2.5 Person Information Editing



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Person information added to the device can be edited via the person information editing function. If the person has been added to the device, the person information will be edited; if the person has not been added to the device, the device will report an error.

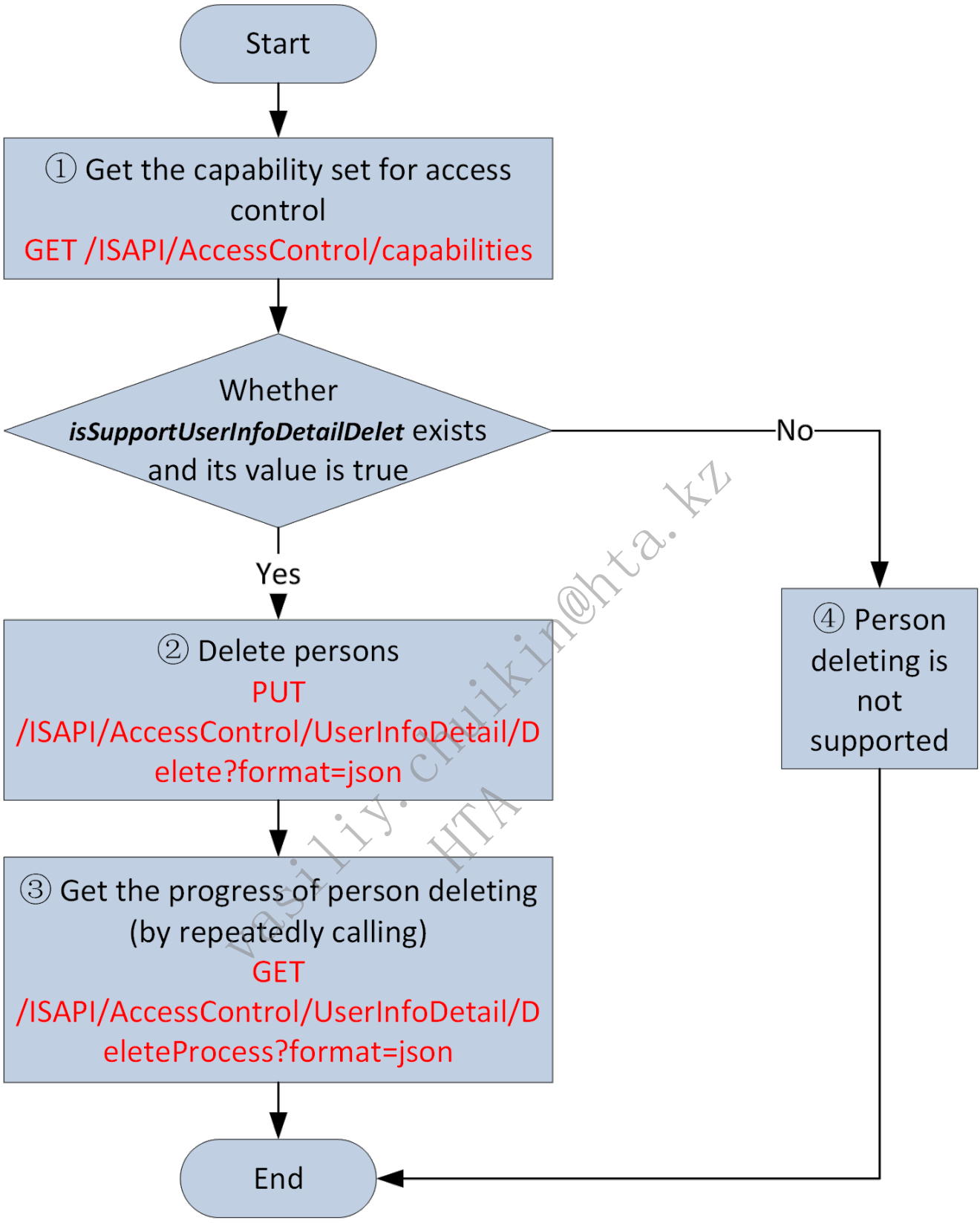
1. Check whether the device supports person information editing: `GET /ISAPI/AccessControl/UserInfo/capabilities?format=json`; if the value of the node supportFunction contains "put", it indicates that the device supports person information editing.
2. Edit Person Information: `PUT /ISAPI/AccessControl/UserInfo/Modify?format=json`.
3. If the value of the node supportFunction does not contain "put", it indicates that the device does not support person information editing.

Note:

Check whether the person has been added to the device via the node employeeNo returned after calling the API for person information editing.

10.15.2.6 Person Deleting

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The person added to the device can be deleted via the person deleting function. The device will not report an error if the person to be deleted is not added to the device.

1. Check whether the device supports person deleting: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportUserInfoDetailDelete is returned and its value is "true", it indicates that the device supports person deleting.
2. Delete persons: `PUT /ISAPI/AccessControl/UserInfoDetail/Delete?format=json`; if calling succeeded, it indicates that the device has started to execute person deleting, but it does not indicate that the device has deleted the person.
3. Get the progress of deleting person information: `GET /ISAPI/AccessControl/UserInfoDetail/DeleteProcess`; repeatedly call this API to get the progress of person deleting.
4. If the node isSupportUserInfoDetailDelete is returned and its value is "false", it indicates that the device does not

support person deleting.

Note:

When the person is deleted, the information on the credentials (the card, fingerprint, face picture, and iris data) linked via the person ID will also be deleted.

10.16 Remote Verification

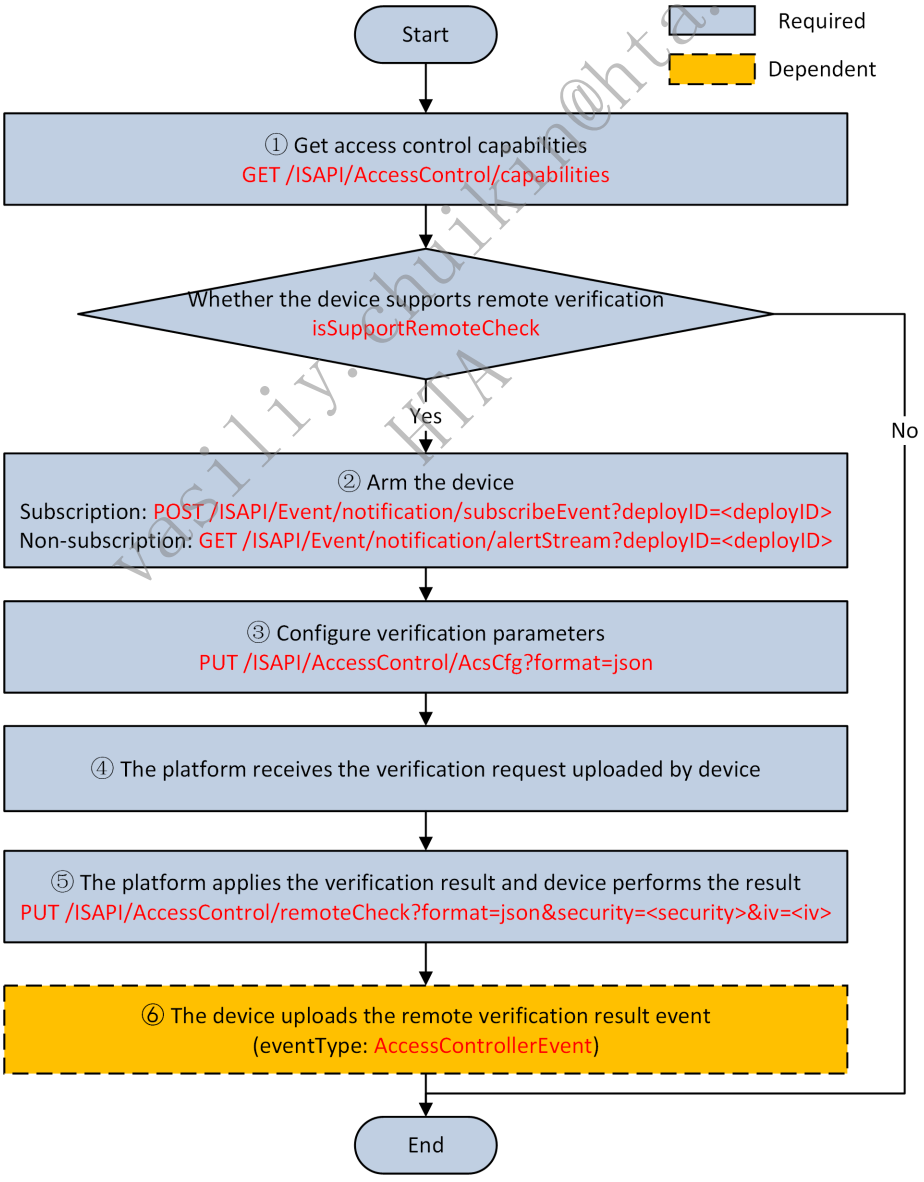
10.16.1 Function Introduction

In high-security scenarios, local device authentication alone is inadequate. Devices need to upload authentication information to the platform, which then determines whether to unlock the door.

For example, in epidemic prevention scenarios, after a device collects person authentication and temperature data, it uploads the information to the platform as an event. The platform then checks the person's recent travel history. Only after confirming no issues does it send an approval result to unlock the door.

10.16.2 Remote Verification in Arming Method

10.16.2.1 API Calling Flow



1. Get access control capabilities to check if the remote verification is supported: `GET /ISAPI/AccessControl/capabilities`. Related field: `isSupportRemoteCheck`. If this field returns `true`, the device supports remote verification.

2. Arm the device.

- Subscription mode: `POST /ISAPI/Event/notification/subscribeEvent?deployID=<deployID>`.
- Non-subscription mode: `GET /ISAPI/Event/notification/alertStream?deployID=<deployID>`.

Note: For detailed arming process, refer to the **Event Arming Management**. The key difference is that non-subscription mode uses `HTTP GET` to receive all event messages, while subscription mode uses `HTTP POST` to receive only subscribed events. Both HTTP connections are established by the platform.

3. Configure remote verification parameters: `PUT /ISAPI/AccessControl/AcsCfg?format=json`.
4. The platform receives verification requests from the device. When the device uploads an event (`AccessControllerEvent`, `IDCardInfoEvent`, `QRCodeEvent`, or `FaceTemperatureMeasurementEvent`) with the field **`remoteCheck`** set to `true`, it indicates remote verification is required.
5. The platform sends remote verification results and the device controls door based on the results: `PUT /ISAPI/AccessControl/remoteCheck?format=json&security=<security>&iv=<iv>`.
6. (Dependent) If the device initially uploaded an access control event (`AccessControllerEvent`), it can upload a verification event after remote verification is complete. This event must be of type `AccessControllerEvent` and include the `remoteCheckResult` field. **Note:** Uploading verification results is not supported by all devices. Generally, only devices with firmware from 2024 or later support this feature.

10.16.2.2 Examples

10.16.2.2.1 Configure Remote Verification Parameters

Recommended API calling sequence:

1. Call `GET /ISAPI/AccessControl/capabilities` to check if `isSupportAcsCfg` is `true`; if true, verification parameter configuration is supported.
2. Call `GET /ISAPI/AccessControl/AcsCfg/capabilities?format=json` to get supported parameters and ranges.
3. Call `GET /ISAPI/AccessControl/AcsCfg?format=json` to get current configurations.
4. Call `PUT /ISAPI/AccessControl/AcsCfg?format=json` to edit parameters.

```
//Verification parameters in arming method

{
    "AcsCfg":{
        "remoteCheckDoorEnabled": true,//The main switch for remote verification, set to true
        "checkChannelType": "ISAPI",//Verification channel type
        "needDeviceCheck": true,//Whether the remote verification requires device authentication; true by default
        "remoteCheckTimeout": 5,//Remote verification timeout (seconds); if no result is received within this time, the verification fails.
        "remoteCheckVerifyMode": 7,//Credential type that require remote verification
        "offlineDevCheckOpenDoorEnabled": false,//Whether to allow door opening based on local device authentication when offline; false by default
        "remoteCheckUserTypeList": ["normal", "visitor", "unregistered"]//List of person types requiring remote verification; if absent or empty, all person
types require remote verification.
    }
}
```

Note: How to understand and set the `remoteCheckVerifyMode` field for credential type requiring remote verification?

`remoteCheckVerifyMode` descriptions:

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Value	Description
1	Face and ID card comparison
2	Face recognition
3	Face and ID card comparison or face recognition
4	Card swiping
5	Face and ID card comparison or card swiping
6	Face recognition or card swipe
7	Face and ID card comparison, card swipe, or face recognition
8	QR code
9	Face and ID card comparison or QR code
10	Face recognition or QR code
11	Face and ID card comparison, face recognition, or QR code
12	Card swipe or QR code
13	Face and ID card comparison, card swipe, or QR code
14	Face recognition, card swipe, or QR code
15	Face and ID card comparison, card swipe, face recognition, or QR code

Remarks:

1. Face and ID card comparison: Refers to face recognition + ID card swiping by default.
2. When remote verification requires credentials local authentication (i.e., `needDeviceCheck` is `true`), if `remoteCheckVerifyMode` is set to **2#Face recognition**, it means face recognition triggers remote verification after device local authentication; other methods only require local authentication.
3. When remote verification requires credentials local authentication (i.e., `needDeviceCheck` is `false`), if `remoteCheckVerifyMode` is set to **2#Face recognition**, it means face recognition directly triggers remote verification without device local authentication; other methods do not trigger remote verification and depend on the device local authentication.

10.16.2.2.2 Send Verification Results

Recommended API calling sequence:

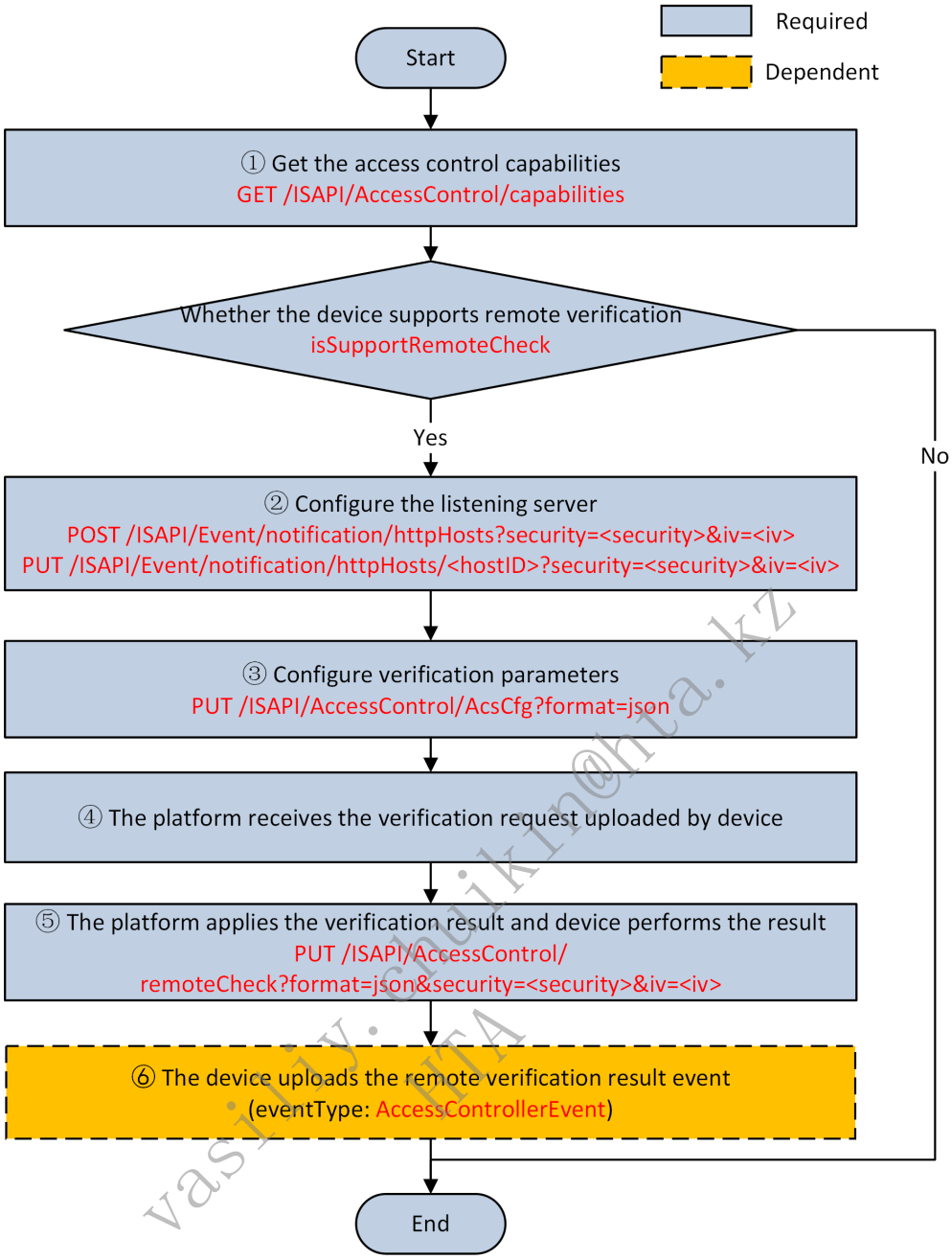
1. Call `GET /ISAPI/AccessControl/capabilities` to check if `isSupportRemoteCheck` is `true`; if true, sending remote verification results is supported.
2. Call `GET /ISAPI/AccessControl/remoteCheck/capabilities?format=json` to get supported parameters and ranges.
3. Call `PUT /ISAPI/AccessControl/remoteCheck?format=json&security=<security>&iv=<iv>` to send verification results.

```
//Example of sending verification results
{
    "RemoteCheck":{
        "serialNo":  1,//Event serial number, matching the device-uploaded event serial number
        "checkResult":  "success",//Verification succeeded
    }
}
```

10.16.3 Remote Verification in Asynchronous Listening Method

10.16.3.1 API Calling Flow

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1. Get access control capabilities to check if the remote verification is supported: `GET /ISAPI/AccessControl/capabilities`. Related field: `isSupportRemoteCheck`. If this field returns `true`, the device supports remote verification.

2. Configure listening server parameters.
   - `POST /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`
   - `PUT /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`.

   Note:
   - For detailed listening server configuration, refer to the API calling flow in the **HTTP Listening Service** domain.
   - By default, access control devices support only one HTTP(S) listening server. In listening mode, the platform opens a listening service on its own IP and port. The device then establishes a connection to send messages, this connection can be closed after transmission or kept open persistently.

3. Configure remote verification parameters: `PUT /ISAPI/AccessControl/AcsCfg?format=json`.

4. The platform receives remote verification requests.

The device sends requests via `HTTP(S) POST`. Each request contains one of the following event types, with `remoteCheck` field set to `true`.

- Access control event (`AccessControllerEvent`)
- ID card event (`IDCardInfoEvent`)
- QR code event (`QRCodeEvent`)
- Face temperature measurement event (`FaceTemperatureMeasurementEvent`)

5. The platform sends remote verification results and the device controls door based on the results: `PUT /ISAPI/AccessControl/remoteCheck?format=json&security=<security>&iv=<iv>`.

6. (Dependent) If the device initially uploaded an access control event (`AccessControllerEvent`), it can upload a verification event after remote verification is complete. This event must be of type `AccessControllerEvent` and include the `remoteCheckResult` field. **Note:** Uploading verification results is not supported by all devices. Generally, only devices with firmware from 2024 or later support this feature.

10.16.3.2 Examples

10.16.3.2.1 Add a Listening Server

```
POST /ISAPI/Event/notification/httpHosts?security=1&iv=cd128e54036c0038a083992406de5320

<?xml version="1.0" encoding="UTF-8"?>
<HttpHostNotification version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <id>1</id>
    <url>/ISAPI/AddHttpHost/Test</url>
    <protocolType>HTTP</protocolType>
    <parameterFormatType>XML</parameterFormatType>
    <addressingFormatType>ipaddress</addressingFormatType>
    <ipAddress>10.8.98.62</ipAddress>
    <portNo>7333</portNo>
    <httpAuthenticationMethod>none</httpAuthenticationMethod>
</HttpHostNotification>
```

Explanation

- `url`: URL configured by the platform and used for device to upload remote verification requests.
- `ipAddress`: Platform (server) IP.
- `portNo`: Platform (server) listening port.
- `httpAuthenticationMethod`: HTTP(S) communication authentication type (e.g., `none` for no authentication). MD5 digest authentication is recommended for higher security.

10.16.3.2.2 Configure Remote Verification Parameters

Recommended API calling sequence:

1. Call `GET /ISAPI/AccessControl/capabilities` to check if `isSupportAcsCfg` is `true`; if true, verification parameter configuration is supported.
2. Call `GET /ISAPI/AccessControl/AcsCfg/capabilities?format=json` to get supported parameters and ranges.
3. Call `GET /ISAPI/AccessControl/AcsCfg?format=json` to get current configurations.
4. Call `PUT /ISAPI/AccessControl/AcsCfg?format=json` to edit parameters.

```
//Verification parameters in ISAPI asynchronous listening method

{
    "AcsCfg":{
        "remoteCheckDoorEnabled": true,//The main switch for remote verification, set to true
        "checkChannelType": "ISAPIListen",//Verification channel type
        "needDeviceCheck": true,//Whether the remote verification requires credentials local authentication; true by default
        "remoteCheckTimeout": 5,//Remote verification timeout (seconds); if no result is received within this time, the verification fails.
        "remoteCheckVerifyMode": 7,//Credential type that require remote verification
        "offlineDevCheckOpenDoorEnabled": false,//Whether to allow door opening based on local device authentication when offline; false by default
        "remoteCheckUserTypeList": ["normal", "visitor", "unregistered"],//List of person types requiring remote verification; if absent or empty, all
person types require remote verification.
        "remoteCheckWithISAPIListen": "async",//Result return mode under ISAPI listening mode: "async" (asynchronous)
    }
}
```

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10.16.3.2.3 Receive Verification Requests from Device

Example (Access Control Event)

```
POST /ISAPI/AddHttpHost/Test HTTP/1.1
Accept: text/html, application/xhtml+xml, */*
Accept-Language: zh-cn
Content-Type: multipart/form-data;boundary=boundary //Note: Form data format
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)
Accept-Encoding: gzip, deflate
Host: 10.21.84.13:7333
Content-Length: 92406
Connection: Keep-Alive
Cache-Control: no-cache

--boundary
Content-Disposition: form-data; name="AccessControllerEvent"
Content-Type: application/json; charset="UTF-8"
Content-Length: 202

{
    "ipAddress": "10.6.123.199",
    "portNo": 80,
    "channelID": 6,
    "dateTime": "2022-11-14T13:41:56+08:00",
    "activePostCount": 1,
    "eventType": "AccessControllerEvent",
    "eventState": "active",
    "eventDescription": "Access Controller Event",
    "AccessControllerEvent": {
        "deviceName": "Access Controller",
        "majorEventType": 5,
        "subEventType": 75,
        "time": "2022-11-14T13:41:56+08:00",
        "name":  "test123",
        "employeeNoString":  "test",
        "serialNo": 1866,
        "currentEvent": true,
        "remoteCheck":  true,
        "frontSerialNo": 1865,
        "picturesNumber":  1
    }
}
--boundary
Content-Disposition: form-data; name="Picture";filename="Picture.jpg";
Content-Type: image/jpeg
Content-Length: 92204

[Binary image data]
--boundary--
HTTP/1.1 200 OK
Connection:close
```

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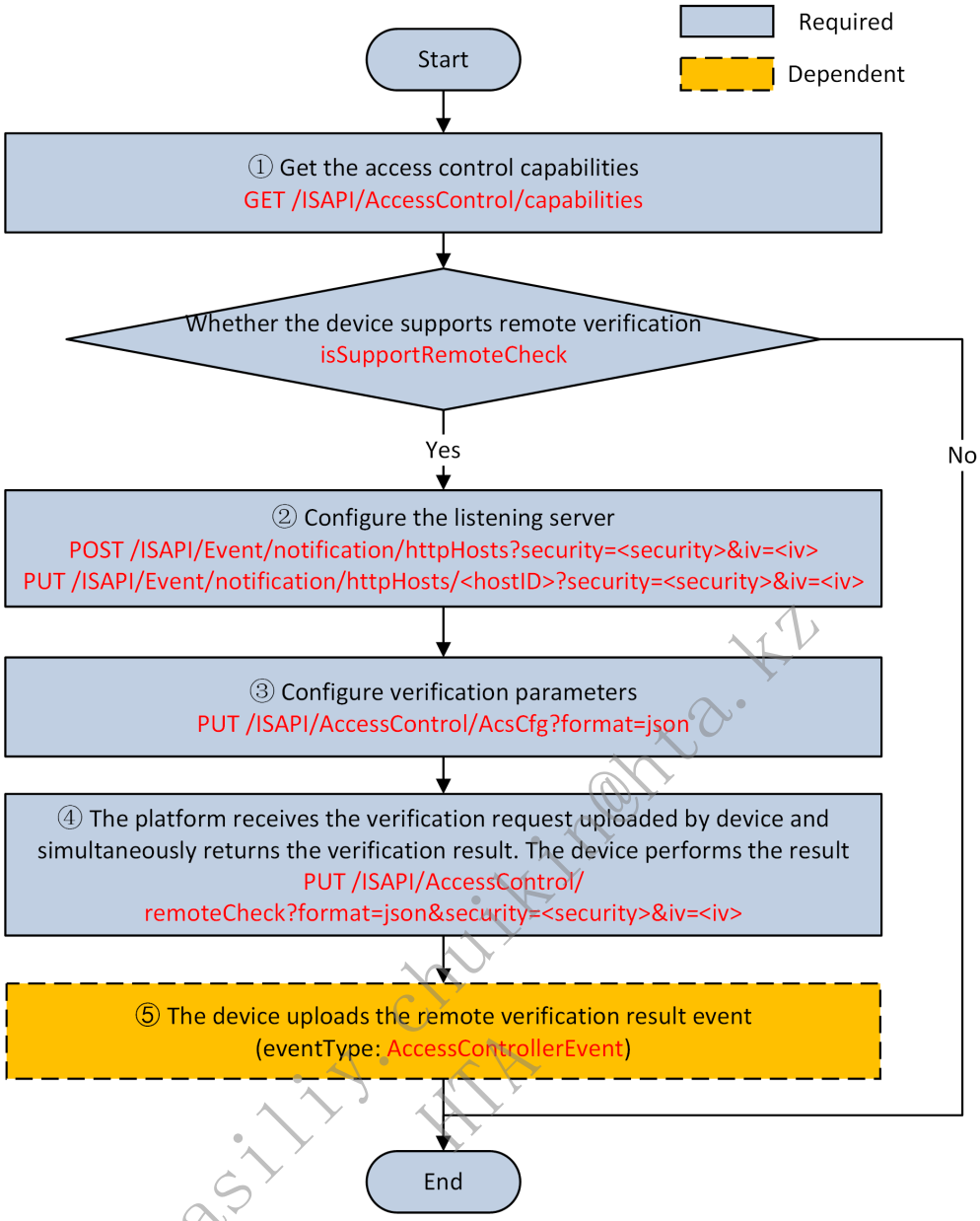
Note: For field "remoteCheck", true indicates remote verification is required.

10.16.3.2.4 Send Verification Results

Refer to the example of Remote Verification in Arming Method.

10.16.4 Remote Verification in Synchronous Listening Method

10.16.4.1 API Calling Flow



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1. Get access control capabilities to check if the remote verification is supported: `GET /ISAPI/AccessControl/capabilities`. Related field: `isSupportRemoteCheck`. If this field returns `true`, the device supports remote verification.

2. Configure listening server parameters.
   - `POST /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`
   - `PUT /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`.

   Note:
   - For detailed listening server configuration, refer to the API calling flow in the **HTTP Listening Service** domain.
   - By default, access control devices support only one HTTP(S) listening server. In listening mode, the platform opens a listening service on its own IP and port. The device then establishes a connection to send messages, this connection can be closed after transmission or kept open persistently.

3. Configure remote verification parameters: `PUT /ISAPI/AccessControl/AcsCfg?format=json`.

4. The platform receives remote verification requests and sends back results synchronously.

   a. The device sends requests via **`HTTP(S) POST`**. Each request contains one of the following event types, with `remoteCheck` field set to `true`.
   - Access control event (`AccessControllerEvent`)

- ID card event (`IDCardInfoEvent`)
- QR code event (`QRCodeEvent`)
- Face temperature measurement event (`FaceTemperatureMeasurementEvent`)

b. The platform synchronously sends results in **`HTTP(S) Response Body`**. That is, the actual return data from platform to device is the request message of URL `PUT /ISAPI/AccessControl/remoteCheck?format=json`.

5. The platform sends remote verification results and the device controls door based on the results: `PUT /ISAPI/AccessControl/remoteCheck?format=json&security=<security>&iv=<iv>`.

6. (Dependent) If the device initially uploaded an access control event (`AccessControllerEvent`), it can upload a verification event after remote verification is complete. This event must be of type `AccessControllerEvent` and include the `remoteCheckResult` field. The platform should return **`200 OK`** in **HTTP(S) Headers** upon receiving the result event. **Note:** Uploading verification results is not supported by all devices. Generally, only devices with firmware from 2024 or later support this feature.

10.16.4.2 Examples

10.16.4.2.1 Add a Listening Server

Refer to examples in Remote Verification in Asynchronous Listening Method.

10.16.4.2.2 Configure Remote Verification Parameters

Recommended API calling sequence:

1. Call `GET /ISAPI/AccessControl/capabilities` to check if `isSupportAcsCfg` is `true`; if true, verification parameter configuration is supported.
2. Call `GET /ISAPI/AccessControl/AcsCfg/capabilities?format=json` to get supported parameters and ranges.
3. Call `GET /ISAPI/AccessControl/AcsCfg?format=json` to get current configurations.
4. Call `PUT /ISAPI/AccessControl/AcsCfg?format=json` to edit parameters.

```
//Verification parameters in ISAPI synchronous listening method

{
    "AcsCfg":{
        "remoteCheckDoorEnabled": true,//The main switch for remote verification, set to true
        "checkChannelType": "ISAPIListen",//Verification channel type, set to ISAPIListen for ISAPI listen channel
        "needDeviceCheck": true,//Whether the remote verification requires credentials local authentication; true by default
        "remoteCheckTimeout": 5,//Remote verification timeout (seconds); if no result is received within this time, the verification fails.
        "remoteCheckVerifyMode": 7,//Credential type that require remote verification
        "offlineDevCheckOpenDoorEnabled": false,//Whether to allow door opening based on local device authentication when offline; false by default
        "remoteCheckUserTypeList": ["normal", "visitor", "unregistered"],//List of person types requiring remote verification; if absent or empty, all
person types require remote verification.
        "remoteCheckWithISAPIListen": "sync",//Result return mode under ISAPI listening mode: "sync" (synchronous)
    }
}
```

10.16.4.2.3 Receive Verification Requests and Return Results Synchronously

Example (Access Control Event):

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```
POST /ISAPI/AddHttpHost/Test HTTP/1.1
Accept: text/html, application/xhtml+xml, */*
Accept-Language: zh-cn
Content-Type: multipart/form-data;boundary=boundary //Note: Form data format
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)
Accept-Encoding: gzip, deflate
Host: 10.21.84.13:7333
Content-Length: 92406
Connection: Keep-Alive
Cache-Control: no-cache

--boundary
Content-Disposition: form-data; name="AccessControllerEvent"
Content-Type: application/json; charset="UTF-8"
Content-Length: 202

{
    "ipAddress": "10.6.123.199",
    "portNo": 80,
    "channelID": 6,
    "dateTime": "2022-11-14T13:41:56+08:00",
    "activePostCount": 1,
    "eventType": "AccessControllerEvent",
    "eventState": "active",
    "eventDescription": "Access Controller Event",
    "AccessControllerEvent": {
        "deviceName": "Access Controller",
        "majorEventType": 5,
        "subEventType": 75,
        "time": "2022-11-14T13:41:56+08:00",
        "name":  "test123",
        "employeeNoString":  "test",
        "serialNo": 1866,
        "currentEvent": true,
        "remoteCheck":  true,
        "frontSerialNo": 1865,
        "picturesNumber":  1
    }
}
--boundary
Content-Disposition: form-data; name="Picture";filename="Picture.jpg";
Content-Type: image/jpeg
Content-Length: 92204

[Binary image data]
--boundary--
HTTP/1.1 200 OK
Date: Thu, 14 Nov 2022 13:42:40 GMT
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
X-XSS-Protection: 1; mode=block
Content-Length: 81
Connection: close
Pragma: no-cache
Cache-Control: no-cache
Content-Type: application/json

{
    "RemoteCheck":{
        "serialNo":  1,
        "checkResult":  "success"
    }
}
```

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Note: For field "remoteCheck", true indicates remote verification is required.

10.16.5 Remote Verification by Platform Only

Device does not perform device local authentication and only central verification is required. This is mainly for scenarios where person information is managed on the platform rather than on devices. Key differences from general remote verification:

1. When configuring the verification parameters, ensure that the `needDeviceCheck` field in `PUT /ISAPI/AccessControl/AcsCfg?format=json` is set to `false`.
2. When applying verification results, include the field `userInfo` in the input message of `PUT /ISAPI/AccessControl/remoteCheck?format=json&security=<security>&iv=<iv>` to display the person information on the device's local UI.

Example of sending verification results:

```
{
    "RemoteCheck":{
        "serialNo":  1,//Event serial number, matching the device-uploaded event serial number
        "checkResult":  "success",//Verification succeeded
        "userInfo": { //Person information
        "name":  "zhangsan", //Person name
        "employeeNo":  "123456", //Employee ID
        "QRCodeInfo":  "test45678", //QR code information
                "filePathType":  "URL", //Face image file type: URL
                "filePath":  "test123" //Face image file URL path
        }
    }
}
```

10.16.6 Group Ticket Remote Verification

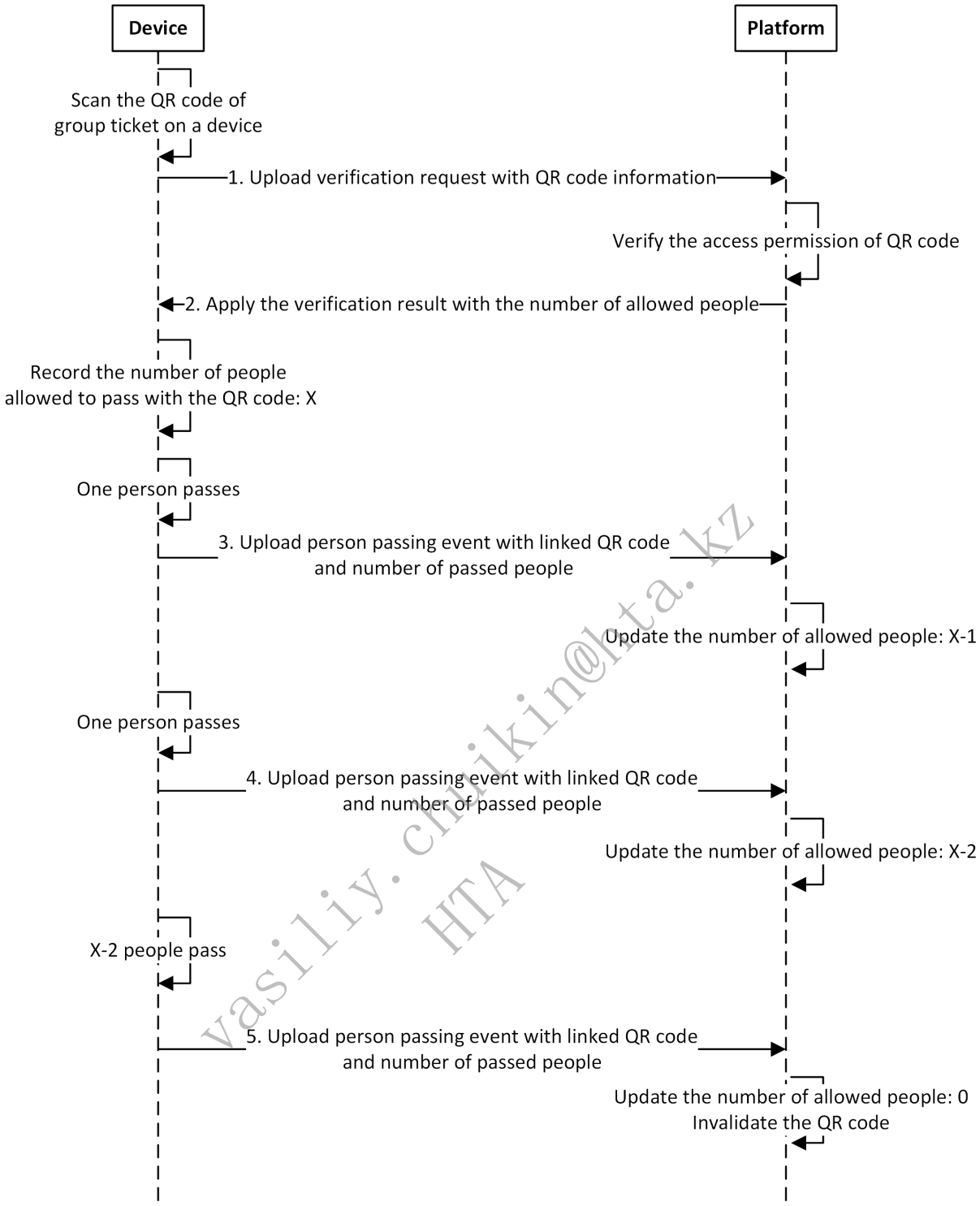
10.16.6.1 Function Introduction

In tourist attractions, access control gates verify visitor tickets purchased through a platform. Both individual and group tickets (allowing multiple entries per ticket) are supported. Credential types include QR code, card swiping, or face and ID card comparison. The platform can configure which credential type requires remote verification. After successful remote verification, the gate's local interface may display the permitted and actual passage counts. The platform then synchronizes ticket status in real-time based on person passing events.

10.16.6.2 API Calling Flow

For group tickets, the remote verification process between the platform and the gate is as follows:

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1. Upload a remote verification request.

   Access control event: `AccessControllerEvent`. When `remoteCheck` field is set to `true`, it indicates that the platform remote verification is required.

2. The platform applies remote verification results.

- URL: `PUT /ISAPI/AccessControl/remoteCheck?format=json&security=<security>&iv=<iv>`.
  - Field `passableUserNum`: Specifies the number of allowable passages for the group ticket.
  - Field `QRCodeInfo`: Associate the group ticket information for later uploaded passing events.

3. Upload person passing event.
   - After passage, the gate uploads an access control event: `AccessControllerEvent`. (major type: `0x5`, minor type: `0xda`).
   - The event includes information:

- Associated QR code: `QRCodeInfo` field.
- Allowable passages of associated QR code: `passableUserNum` field.
- Passed passages of associated QR code: `passedUserNum` field.

Note: The platform can configure whether the gate allows multiple verifications simultaneously by calling `PUT /ISAPI/AccessControl/ChannelControllerCfg?personnelChannelID=<personnelChannelID>`. Set `memoryModeEnabled` to `true` to enable simultaneous verifications without waiting for previous ones to complete.

10.16.7 Text QR Code Remote Verification

10.16.7.1 Function Introduction

Text QR code remote verification enable users to scan a text-based QR code on a device. The device uploads the QR code information to the platform for identity verification, and then controls door opening upon successful verification. For example, in a courthouse, lawyers can scan lawyer codes for access.

10.16.7.2 API Calling Flow

Prerequisites

- Set the working mode (access control mode / permission free mode) via `PUT /ISAPI/AccessControl/IdentityTerminal` (field `workMode`).
- Enable or disable the text QR code by either:
  - Call `PUT /ISAPI/AccessControl/AcsCfg?format=json` and set `QRCodeEnabled` to `true`.
  - Call `PUT /ISAPI/AccessControl/IdentityTerminal` and set `twoDimensionCode` to `true`.
- Enable temperature measurement for devices that will upload the verification request only after temperature measurement.

  Call `PUT /ISAPI/AccessControl/AcsCfg?format=json` and set `thermalEnabled` to `true`.

10.16.7.2.1 When Text QR Code is Enabled

For access control devices supporting working mode switching, verification varies according to current working mode.

- Access control mode: Users must first scan the QR code and then complete credential authentication before the device uploads a verification request as an access control event (`AccessControllerEvent`).

  Thus, under access control mode, for text QR code remote verification, the remote verification credential type should not be set to QR code only, that is, the `remoteCheckVerifyMode` field in `PUT /ISAPI/AccessControl/AcsCfg?format=json` should not be set to `8#QR code`.
- Permission free mode: After users scan the QR code, the device immediately uploads a verification request as a QR code event (`QRCodeEvent`).

10.16.7.2.2 When Text QR Code is Disabled

- For card number QR codes or visitor QR codes (QR code 1.0, 2.0, 3.0), which the device can parse directly, after scanning, the device parses the QR code data and immediately uploads a verification request as an access control event (`AccessControllerEvent`).
- For text QR codes that the device cannot parse, after scanning, the device immediately uploads a verification request as a QR code event (`QRCodeEvent`).

10.16.7.2.3 Notes

Devices like turnstiles that do not support enabling/disabling text QR code will default to supporting QR code scanning if a QR code reader is connected.

10.17 Reset Anti-Passback Rule (Additional Function)

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10.17.1 Introduction to the Function

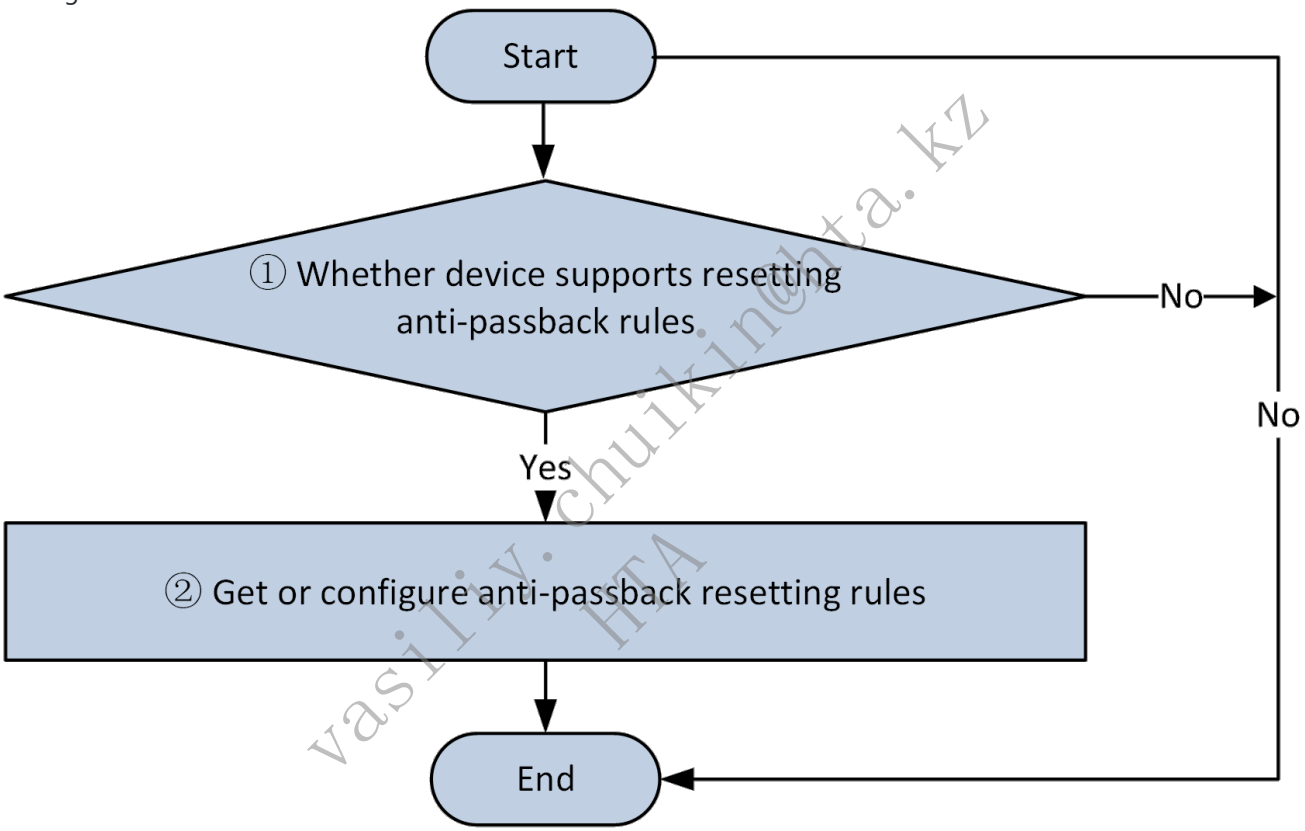
Anti-passback rules which can be reset:

1. Reset by authentication interval. This function will take effect in specific time period after the anti-passbak is triggered. If the user trigger the function by swiping a card by route, the anti-passback flag will be reset in certain time.
2. Reset by time. The anti-passback flag will be reset automatically in certain time.
3. Invalid mode. The resetting rule is disabled.

Application scenarios: The anti-passback function will be help to reduce the cost of manual monitoring. Anti-passback by time period and by time cannot set at the same time.

10.17.2 API Calling Flow

Calling Flow:



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ISAPI Protocol Calling Flow:

1. Get the capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportAntiPassbackResetRules is returned and its value is "true", it indicates that the device supports resetting rules of anti-passback.
2. Get resetting rules of anti-passback: `GET /ISAPI/AccessControl/AntiPassback/resetRules?format=json`; configure resetting rules of anti-passback: `PUT /ISAPI/AccessControl/AntiPassback/resetRules?format=json`.

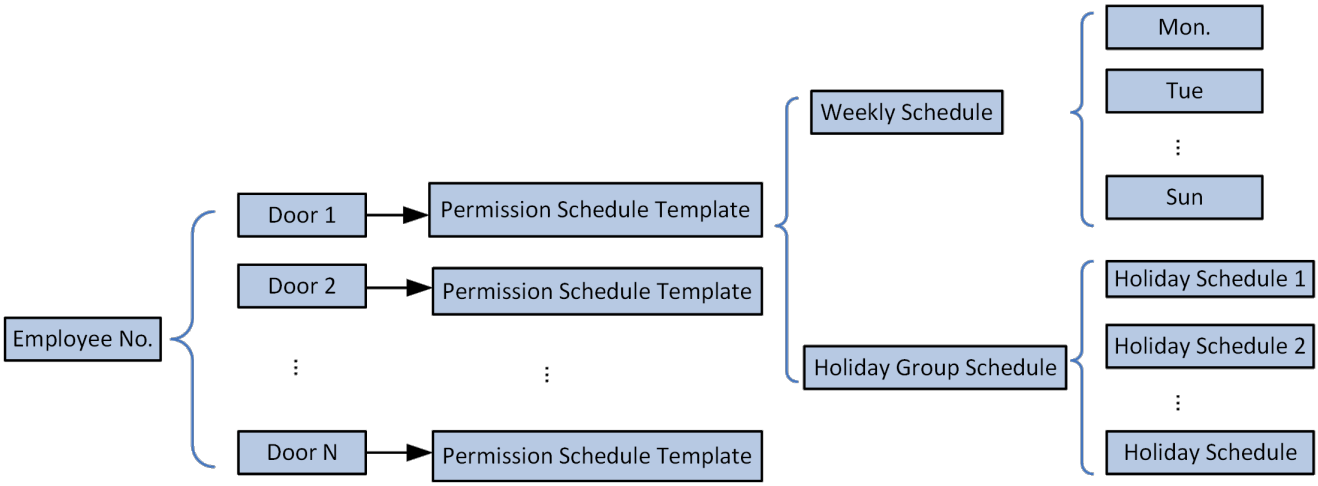
10.18 Schedules Management of Persons' Access Permission

10.18.1 Introduction to the Function

It is required to connect to door permission and schedule template of access permission related to each door before applying permissions to persons. For applying permissions to persons, see calling flow of *Person Management* of *Person and Credential Management*. Configuring schedules of persons' access permission is required, or the related persons cannot access.

1 weekly schedule and 4 holiday groups can be added in each schedule template. The priority of holiday schedule is higher than that of weekly schedule. Weekly schedule can be configured by day of a week and 8 different time period of a day. 16 holiday schedules can be added to a holiday group schedule. Each holiday schedule has its start and end day, and the time period is same in the holiday range (8 time periods can be added). The access control can follow the

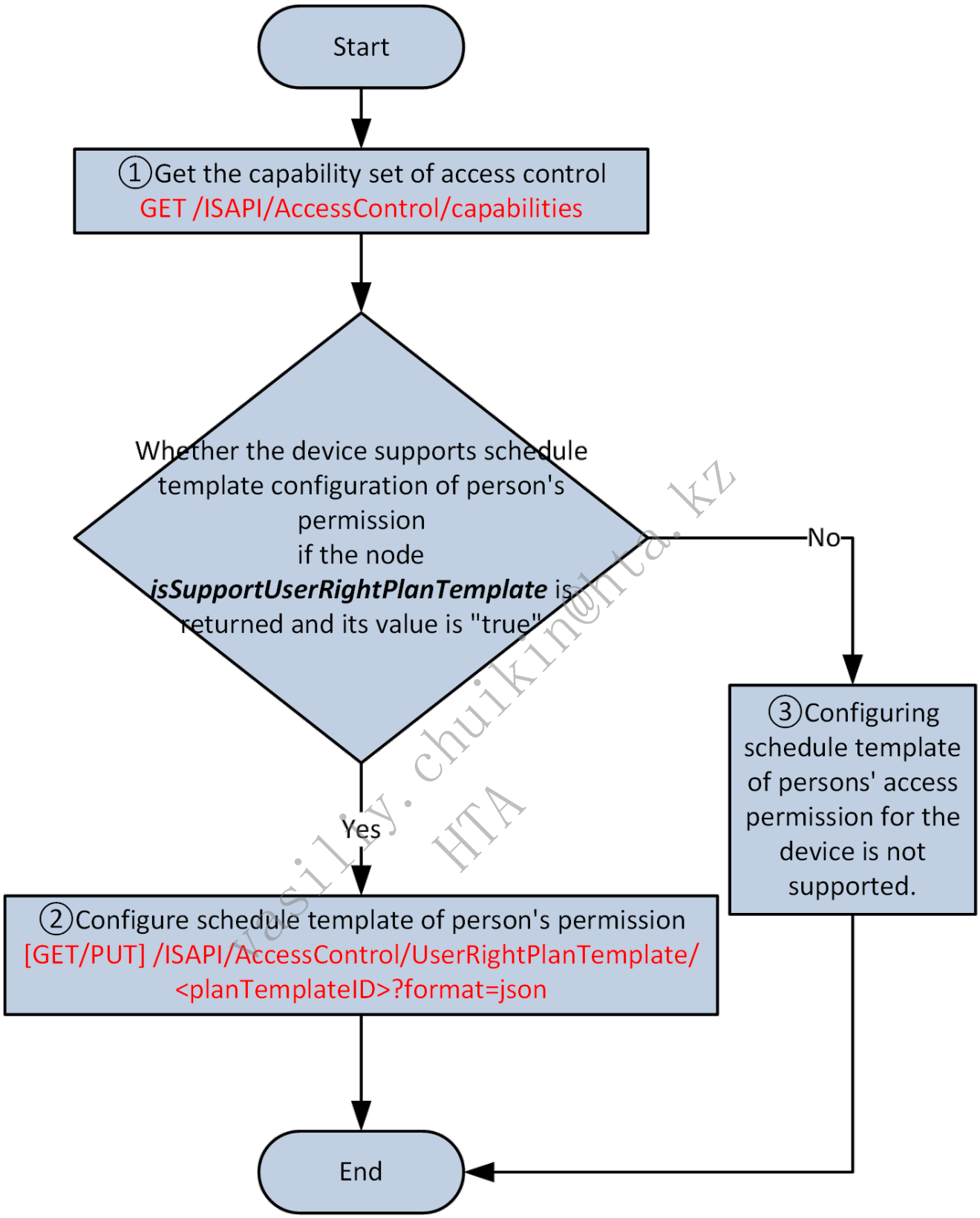
schedule template to manage person's permissions by time.



10.18.2 API Calling Flow

10.18.2.1 Schedule Template of Persons' Access Permission

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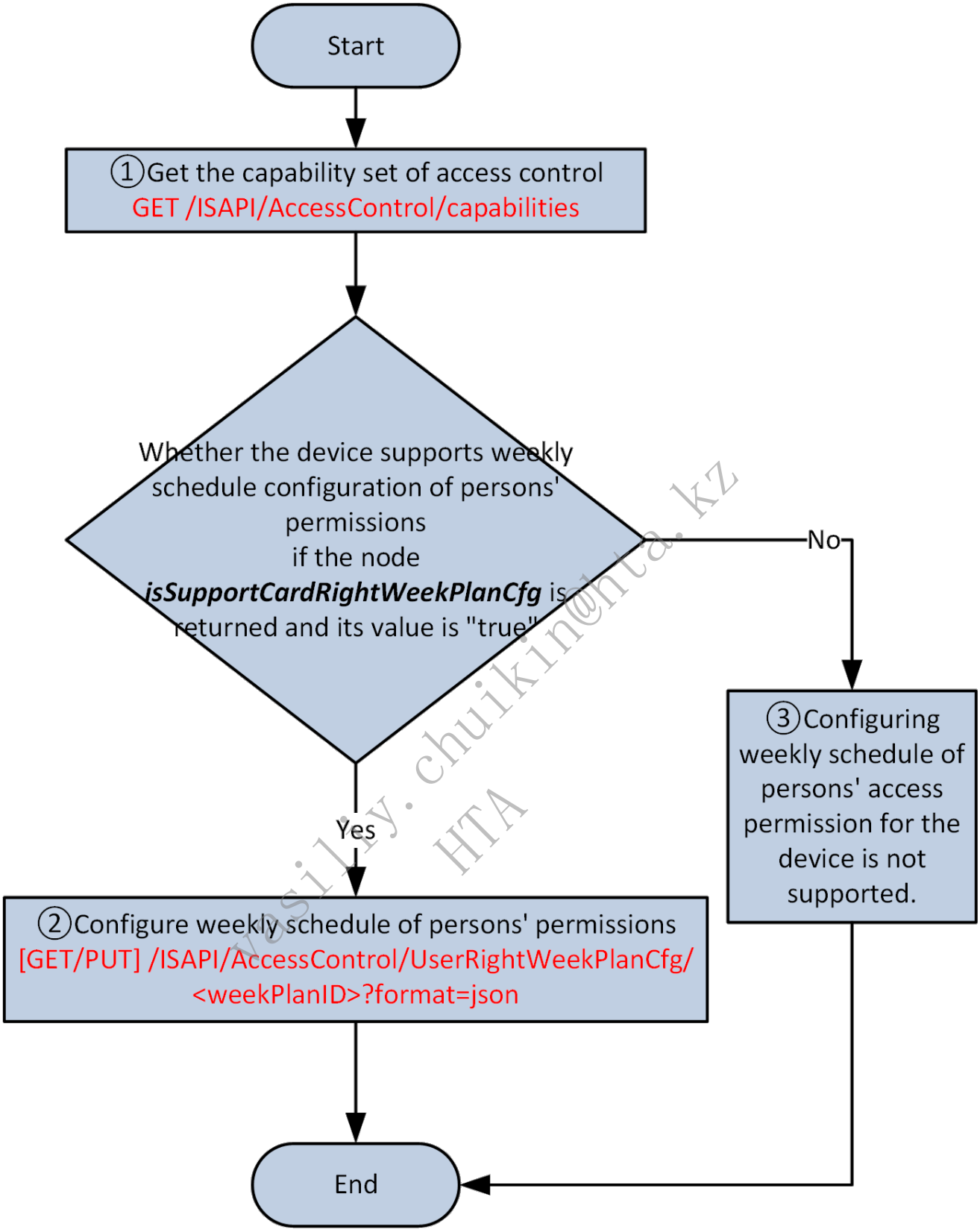


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Calling Flow:

1. Check whether the device supports schedule template configuration of person's permission: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportUserRightPlanTemplate is returned and its value is "true", it indicates that the device supports schedule template configuration of person's permission (if it supports, it also supports weekly schedule configuration of persons' permission).
2. Schedule template configuration of persons' permission: `[GET/PUT] /ISAPI/AccessControl/UserRightPlanTemplate/<planTemplateID>?format=json`.
3. Configuring schedule template of persons' access permission control for the device is not supported.

10.18.2.2 Weekly Schedule of Persons' Access Permission

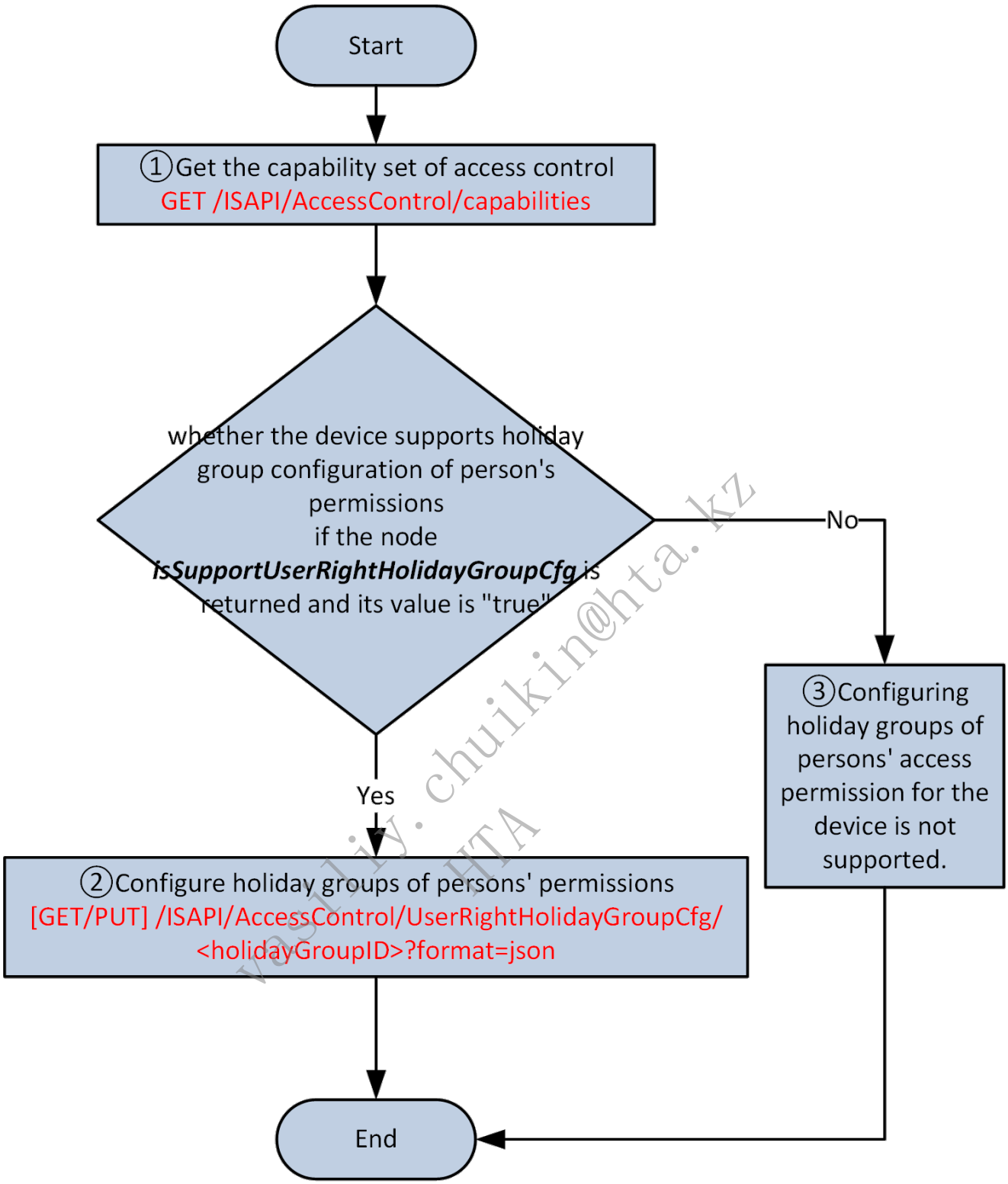


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Calling Flow:

1. Check whether the device supports weekly schedule configuration of persons' permissions: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCardRightWeekPlanCfg is returned and its value is "true", it indicates that the device supports weekly schedule configuration of person's permissions.
2. Weekly schedule configuration of persons' permissions:`[GET/PUT] /ISAPI/AccessControl/UserRightWeekPlanCfg/<weekPlanID>?format=json`.
3. Configuring weekly schedule of persons' access permission control for the device is not supported.

10.18.2.3 Holiday Groups of Persons' Access Permission

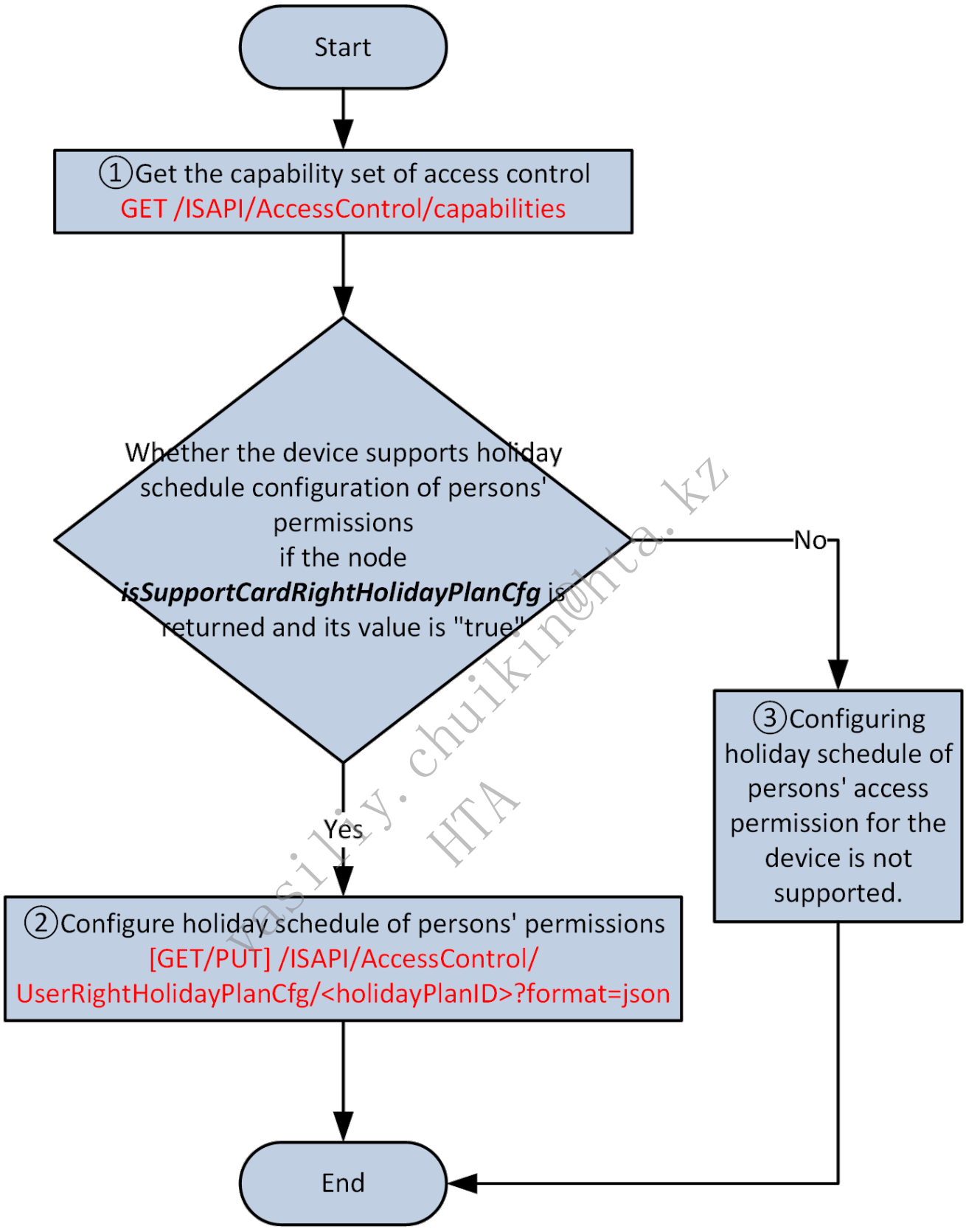


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Calling Flow:

1. Check whether the device supports holiday group configuration of person's permissions: `GET /ISAPI/AccessControl/capabilities`; if the nodeisSupportUserRightHolidayGroupCfg is returned and its value is "true", it indicates that the device supports holiday group configuration of person's permission (if it supports, it also supports holiday schedule configuration of persons' permissions).
2. Holiday group configuration of persons' permissions:`[GET/PUT] /ISAPI/AccessControl/UserRightHolidayGroupCfg/<holidayGroupID>?format=json`.
3. Configuring holiday groups of persons' access permission control for the device is not supported.

10.18.2.4 Holiday Schedule of Persons' Access Permission



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Calling Flow:

1. Check whether the device supports holiday schedule configuration of persons' permissions: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportCardRightHolidayPlanCfg is returned and its value is "true", it indicates that the device supports holiday schedule configuration of person's permissions.
2. Holiday schedule configuration of persons' permissions: `[GET/PUT] /ISAPI/AccessControl/UserRightHolidayPlanCfg/<holidayPlanID>?format=json`.
3. Configuring holiday schedule of persons' access permission control for the device is not supported.

10.19 Search for QR Code Scanning Event

1. Get the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportQRCodeEvent is returned and its value is "true", it indicates that the device supports searching for QR code scanning events.
2. Get the capability of searching for QR code scanning events: `GET /ISAPI/AccessControl/QRCodeEvent/capabilities?format=json`.
3. Search for QR code scanning events: `POST /ISAPI/AccessControl/QRCodeEvent?format=json`. **Notes: the device displays QR code rather than authentication information.**

10.20 Store and Search for Access Control Event

10.20.1 Introduction to the Function

The device supports configuring storage parameters of access control event, searching for access control events, and searching for the amount of access control events. There are three storage modes: deleting old events periodically, deleting old events by specified time and overwriting.

10.20.2 API Calling Flow

10.20.2.1 Configure Storage Parameters of Access Control Events

1. Call the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportEventStorageCfg is returned and its value is "true", it indicates that the device supports configuring the storage parameters of access control events.
2. Get the configuration capability of storing access control events: `GET /ISAPI/AccessControl/AcsEvent/StorageCfg/capabilities?format=json`.
3. Get and set storage parameters of access control events: `GET|PUT /ISAPI/AccessControl/AcsEvent/StorageCfg?format=json`.

10.20.2.2 Search for Access Control Events

1. Call the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportAcsEvent is returned and its value is "true", it indicates that the device supports searching for access control events.
2. Get the capability of searching for access control events: `GET /ISAPI/AccessControl/AcsEvent/capabilities?format=json`.
3. Search for access control events: `POST /ISAPI/AccessControl/AcsEvent?format=json`.

10.20.2.3 Get Total Number of Access Control Events

1. Call the functional capability of access control: `GET /ISAPI/AccessControl/capabilities`; if the node isSupportAcsEventTotalNum is returned and its value is "true", it indicates that the device supports getting the total number of access control events.
2. Get the capability of getting total number of access control events by specific conditions: `GET /ISAPI/AccessControl/AcsEventTotalNum/capabilities?format=json`.
3. Get the total number of access control events by specific conditions: `POST /ISAPI/AccessControl/AcsEventTotalNum?format=json`.

11 Video Intercom (General)

11.1 Call Interaction

11.1.1 Function Introduction

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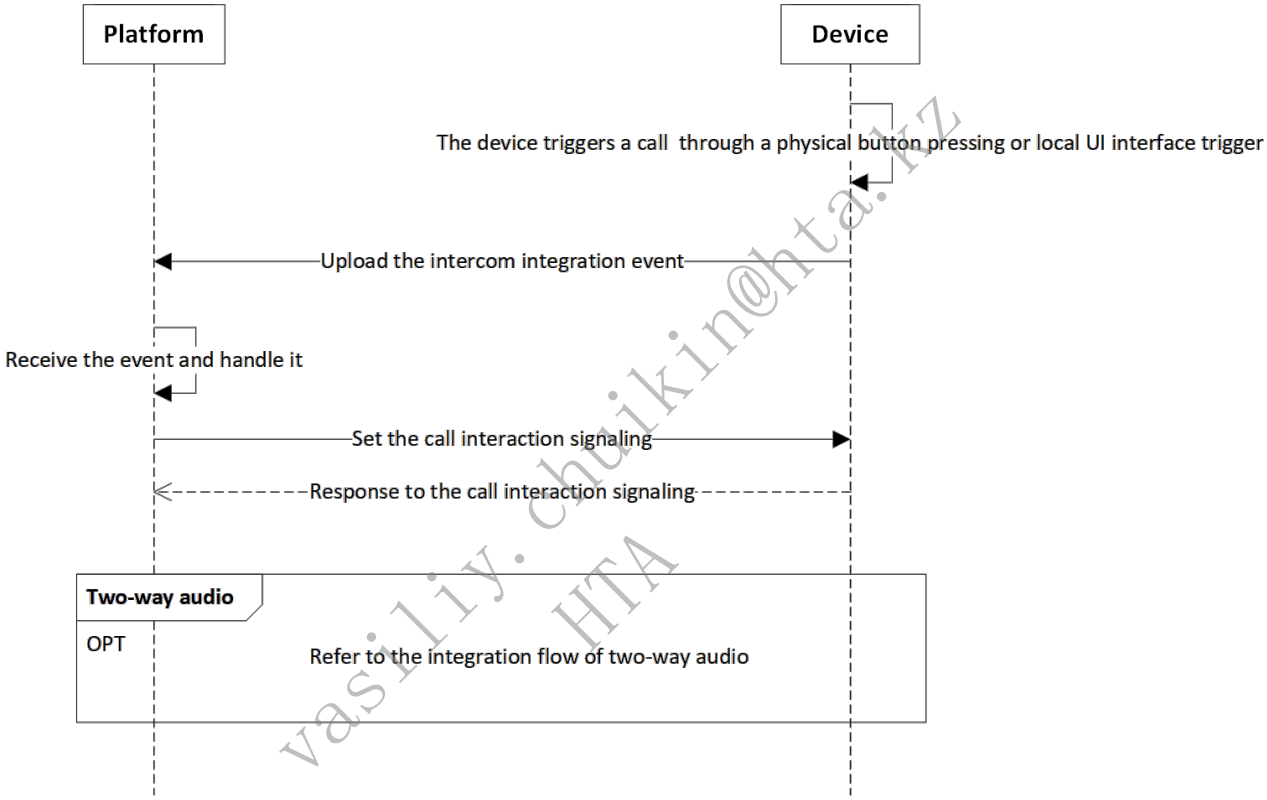
The device can initiate a call request to the platform. The platform responds with a call interaction signaling (such as answer, reject, etc.). Upon sending an answer signal, the platform creates a two-way audio intercom link, enabling the voice data transmission between both parties. (Refer to the two-way audio integration flow.)

Notes:

1. Call interaction signaling can also be used for two-way audio between devices. The two-way audio between devices is implemented based on a private SIP protocol.
2. For two-way audio between devices, if a main station that uses the new device number rules is involved as an intercom party, it is necessary to pre-configure the peer device number parsing rules on the main station to ensure the compatibility with both old and new device numbers.

11.1.2 Integration Flow

11.1.2.1 Sequence Diagram



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11.1.2.2 Integration Steps

1. The device triggers a call, which is triggered by a physical button pressing or local UI interface.
2. The device uploads an intercom interaction event `(eventType: voiceTalkEvent)`.

   Example message for device calling the platform:

```
{
    "ipAddress": "172.6.64.7",
    "ipv6Address": "1080:0:0:0:8:800:200C:417A",
    "portNo": 80,
    "protocol": "HTTP",
    "macAddress": "01:17:24:45:D9:F4",
    "channelID": 1,
    "dateTime": "2004-05-03T17:30:08+08:00",
    "activePostCount": 1,
    "eventType": "voiceTalkEvent",
    "eventState": "active",
    "eventDescription": "Voice Talk Interactive Event",
    "deviceID": "test0123",
    "VoiceTalkEvent":{
        "deviceName": "Door Station",
        "deviceId": "10000000101",
        "cmdType": "request",
        "src":{
            "periodNumber": 1,
            "buildingNumber": 1,
            "unitNumber": 1,
            "floorNumber": 1,
            "roomNumber": 1,
            "devIndex": 1,
            "unitType": "outdoor"
        },
        "target":{
            "unitType": "center",
            "callType": "voice"
        },
        "serialNo": 2,
        "currentEvent": true,
        "frontSerialNo": 1,
        "pictureURL": "test",
        "picturesNumber": 1
    }
}
```

Note: You can get the device's capability to check whether the intercom interaction event (`eventType: voiceTalkEvent`) is supported.

Request:

```
GET /ISAPI/System/capabilities
```

Response:

```
<isSupportVoiceTalkEvent>true</isSupportVoiceTalkEvent>
```

When the device's `isSupportVoiceTalkEvent` field is returned with `true` value, it indicates that the device supports the intercom interaction event.

3. The platform sets the call signaling.

   Request:

```
PUT /ISAPI/VideoIntercom/callSignal?format=json

// Example: Answer the current call
{
  "CallSignal": {
    "cmdType": "answer"
  }
}

// Example: Reject the current call
{
  "CallSignal": {
    "cmdType": "reject"
  }
}
```

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Note: Before setting the call signaling, the platform can first call the API `GET /ISAPI/VideoIntercom/callSignal/capabilities?format=json` to obtain the detailed parameters supported by the device for the call interaction signaling.

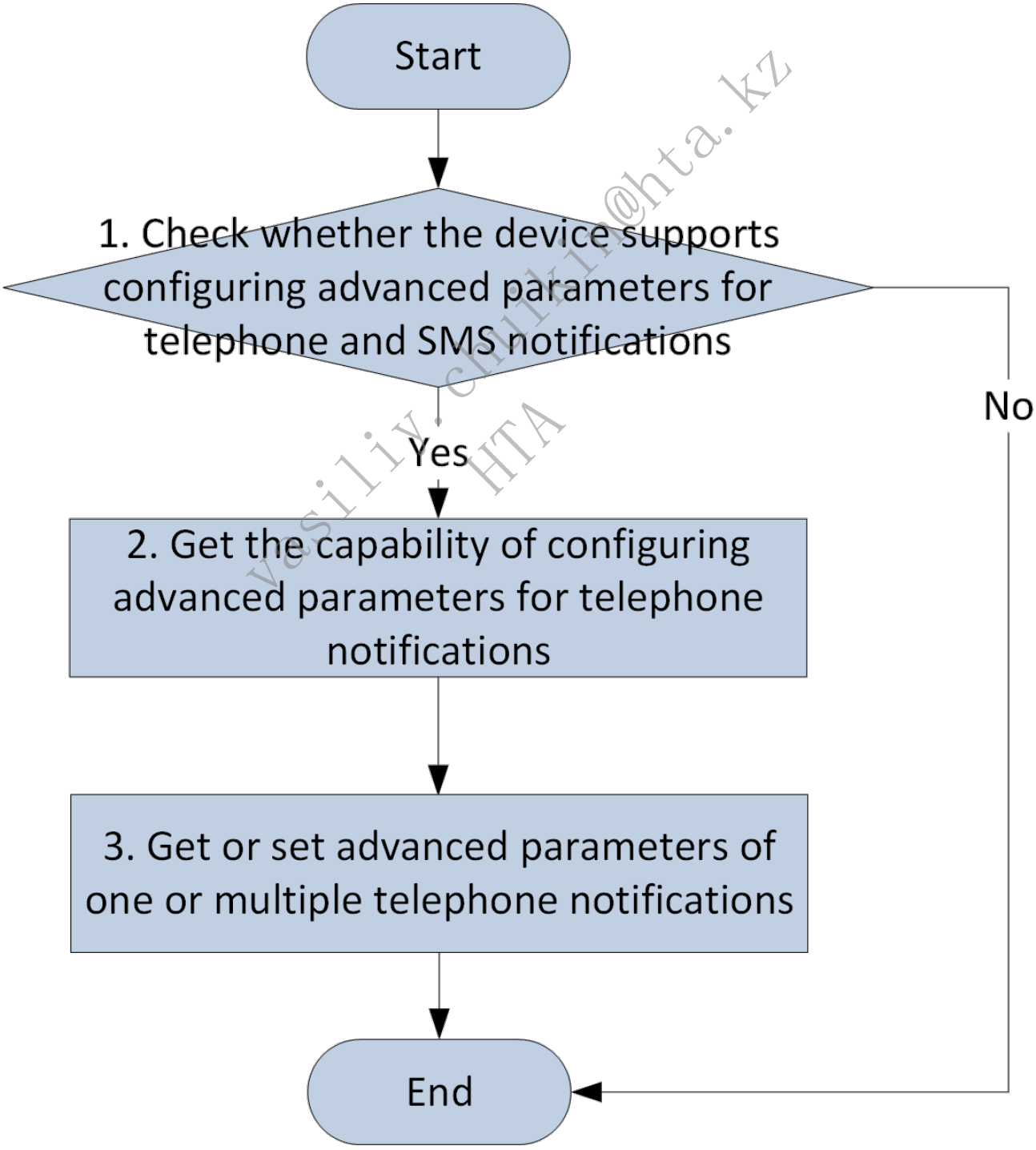
4. Perform the two-way audio between the platform and the device, refer to the API calling flow of two-way audio.

12 Security Control Device (General)

12.1 Advanced Configuration for Telephone Notification

You can configure advanced parameters for telephone notifications, including effective time period, arming / disarming / alarm clearing permission for each telephone number, and event/alarm type (emergency alarm, medical alarm, and gas alarm).

API Calling Flow:



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API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports configuring advanced parameters for telephone and SMS notifications: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node isSptPhoneAnvanced is true, the device supports this function.

2. Get the capability of configuring advanced parameters for telephone notifications: `GET /ISAPI/SecurityCP/Configuration/messageSendPhoneAnvanced/capabilities?format=json`.

3. Get advanced parameters of all telephone notifications: `GET /ISAPI/SecurityCP/Configuration/messageSendPhoneAnvanced?format=json`; set advanced parameters of a single telephone notification: `PUT /ISAPI/SecurityCP/Configuration/messageSendPhoneAnvanced/<phoneID>?format=json`. The phoneID in the URL is the telephone number ID. Up to 8 telephone numbers are supported for a wireless security control panel.

12.2 ARC Management

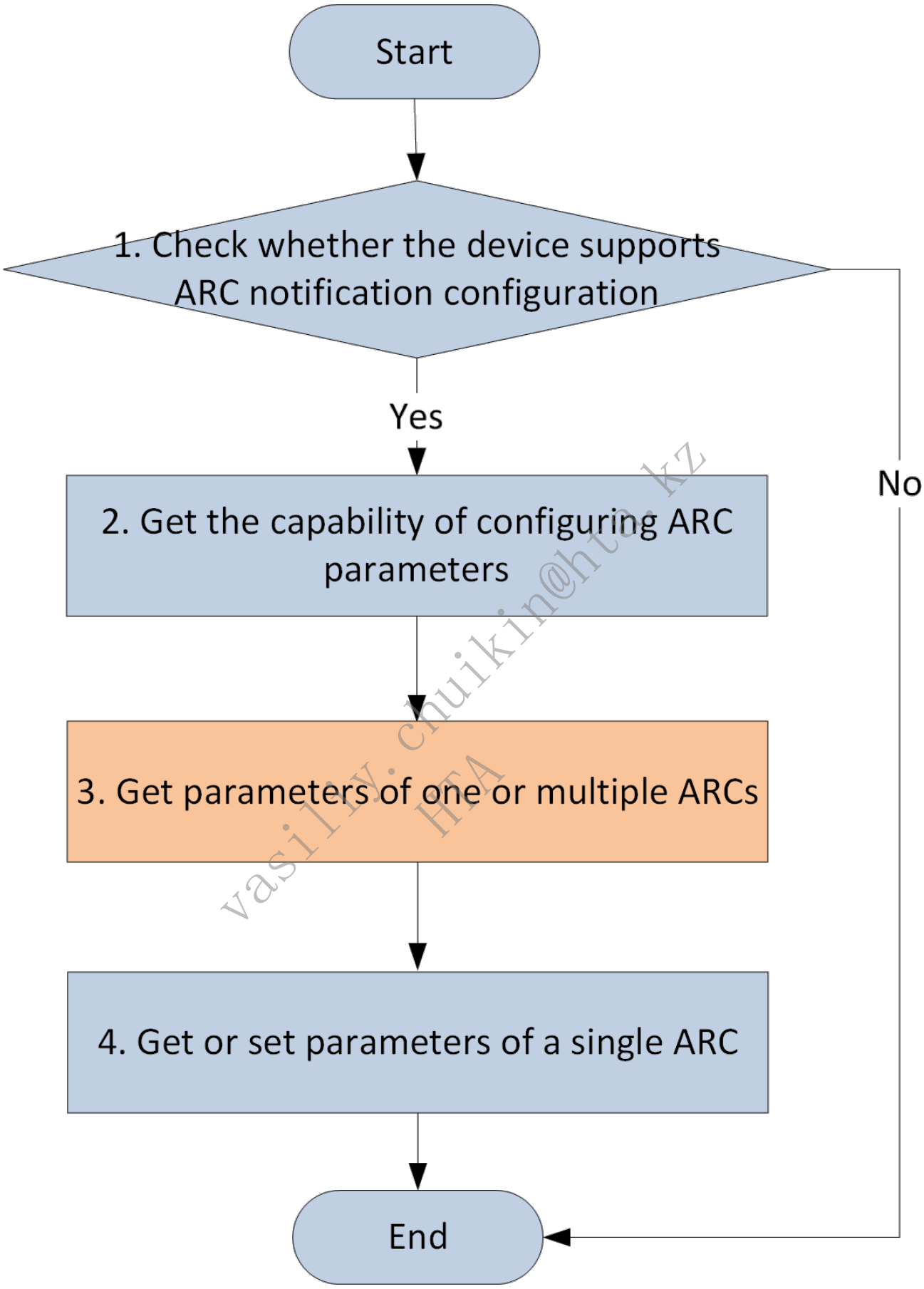
12.2.1 Introduction to the Function

If you have enabled the ARC function and configured events to be received, when the events occur, the corresponding alarm messages will be pushed to the ARC platform. Currently up to 4 ARCs are supported. By default, No. 1 and 3 are the main ARCs and No. 2 and 4 are the spare ARCs. Only when the main ARCs fail to push alarm messages, will the spare ARCs take the charge.

12.2.2 ARC Parameter Configuration

API Calling Flow:

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HTA



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API Calling Steps: 1. Get the configuration capability of the security control panel to check whether the device supports ARC notification configuration: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node isSptARC is true, the device supports this function.

2. Get the capability of configuring ARC parameters: `GET /ISAPI/SecurityCP/Configuration/ARC/capabilities?format=json`.

3. Get parameters of one or multiple ARCs: `GET /ISAPI/SecurityCP/Configuration/ARC?format=json&security=<security>&iv=<iv>`. The userName and password nodes should be encrypted.

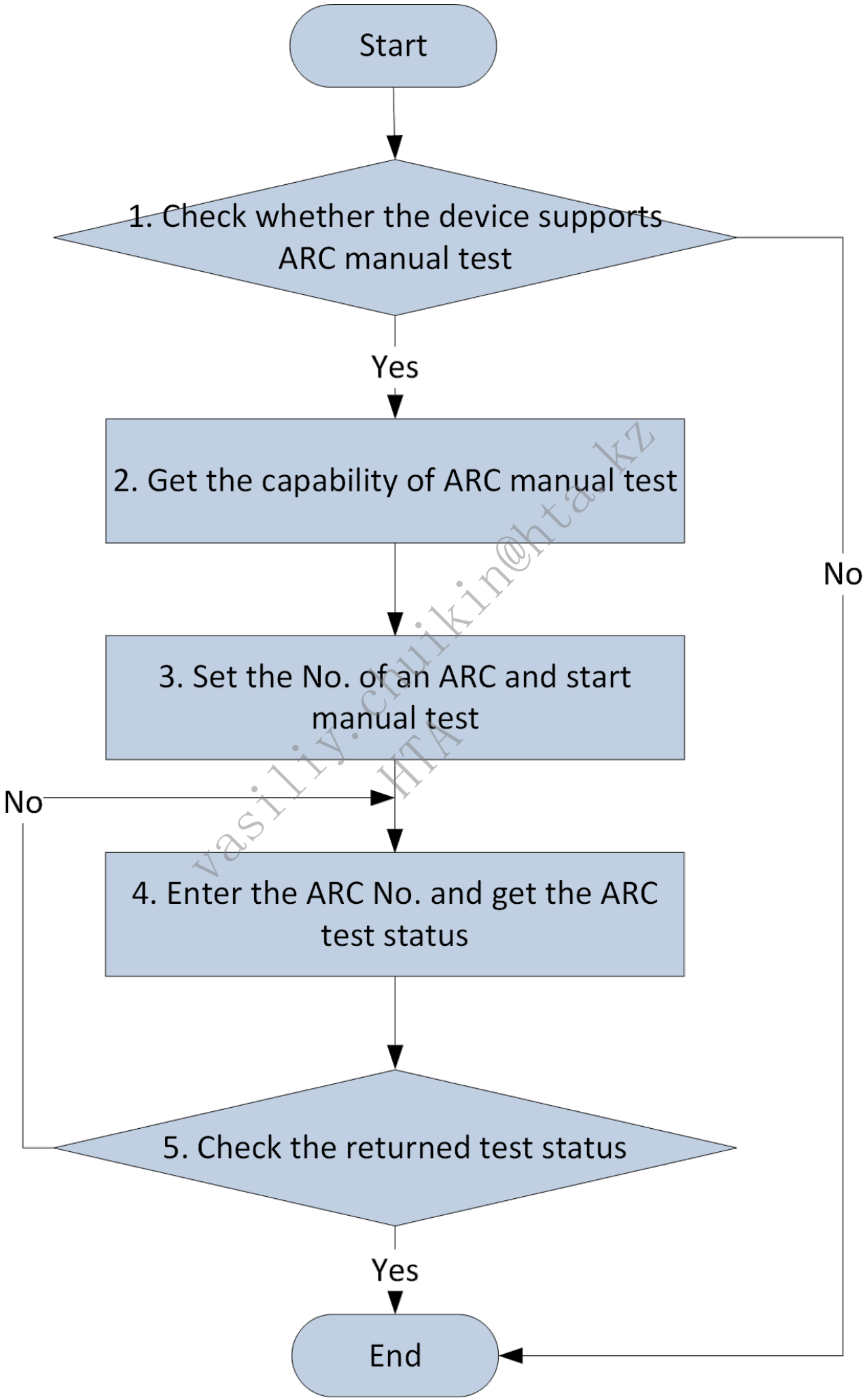
4. Get or set parameters of a single ARC: `PUT /ISAPI/SecurityCP/Configuration/ARC/<indexID>?format=json&security=<security>&iv=<iv>`. The nodes userName and password should be encrypted. The indexID in the URL is the ARC No.

12.2.3 Manual Test of ARC

After configuring ARC parameters, you can check whether the parameter values are valid by testing the link.

API Calling Flow:

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HTA



vasiliy.chuikin@hta.kz
HTA

API Calling Steps: 1. Get the configuration capability of the security control panel to check whether the device supports manual test of ARC: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node isSptARC is true, the device supports this function.

2. Get the capability of manual test of ARC: `GET /ISAPI/SecurityCP/Configuration/ARC/manualTest/capabilities?format=json`.

3. Set the No. of an ARC and start manual test: `PUT /ISAPI/SecurityCP/Configuration/ARC/manualTest?format=json`.

4. Enter the ARC No. and get the ARC test status: `POST /ISAPI/SecurityCP/Configuration/ARC/manualTest/status?format=json`.

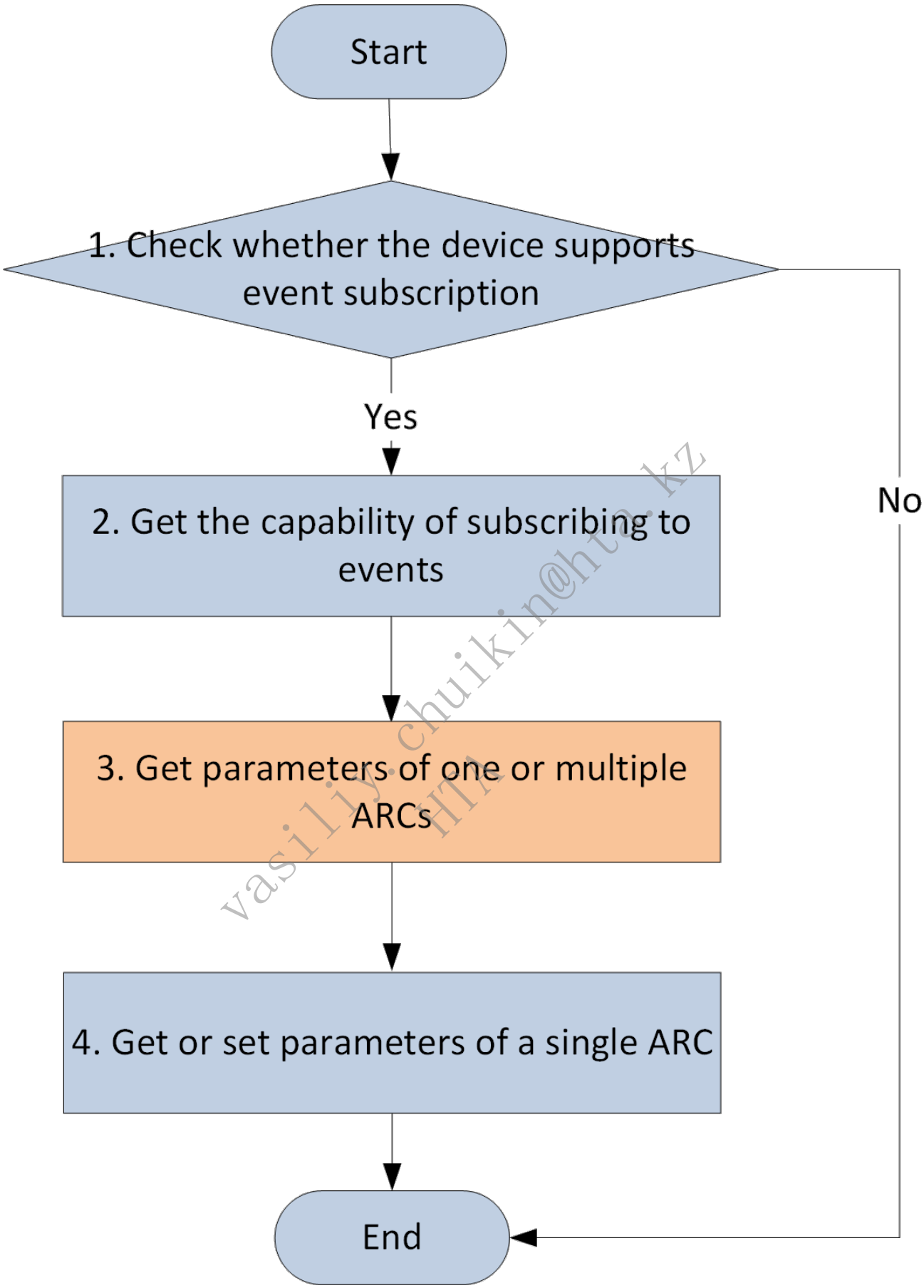
5. Check the returned test status: `success` for success and `failed` for failure. If `processing` is returned within the configured timeout threshold, you need to call the API in the previous step again.

12.2.4 Event Subscription

Event/alarm types include tampering event, life security event, system status event, operation event, emergency alarm, medical alarm, and gas alarm. When the alarms are triggered or the events occur, the corresponding alarm/event details will be uploaded by the device.

API Calling Flow:

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API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports event subscription: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node isSptARC is true, the device supports this function.

2. Get the capability of subscribing to events: `GET /ISAPI/SecurityCP/Configuration/messageSendARC/capabilities?format=json`.

3. Get parameters of one or multiple ARCs: `GET /ISAPI/SecurityCP/Configuration/ARC?format=json&security=<security>&iv=<iv>`.

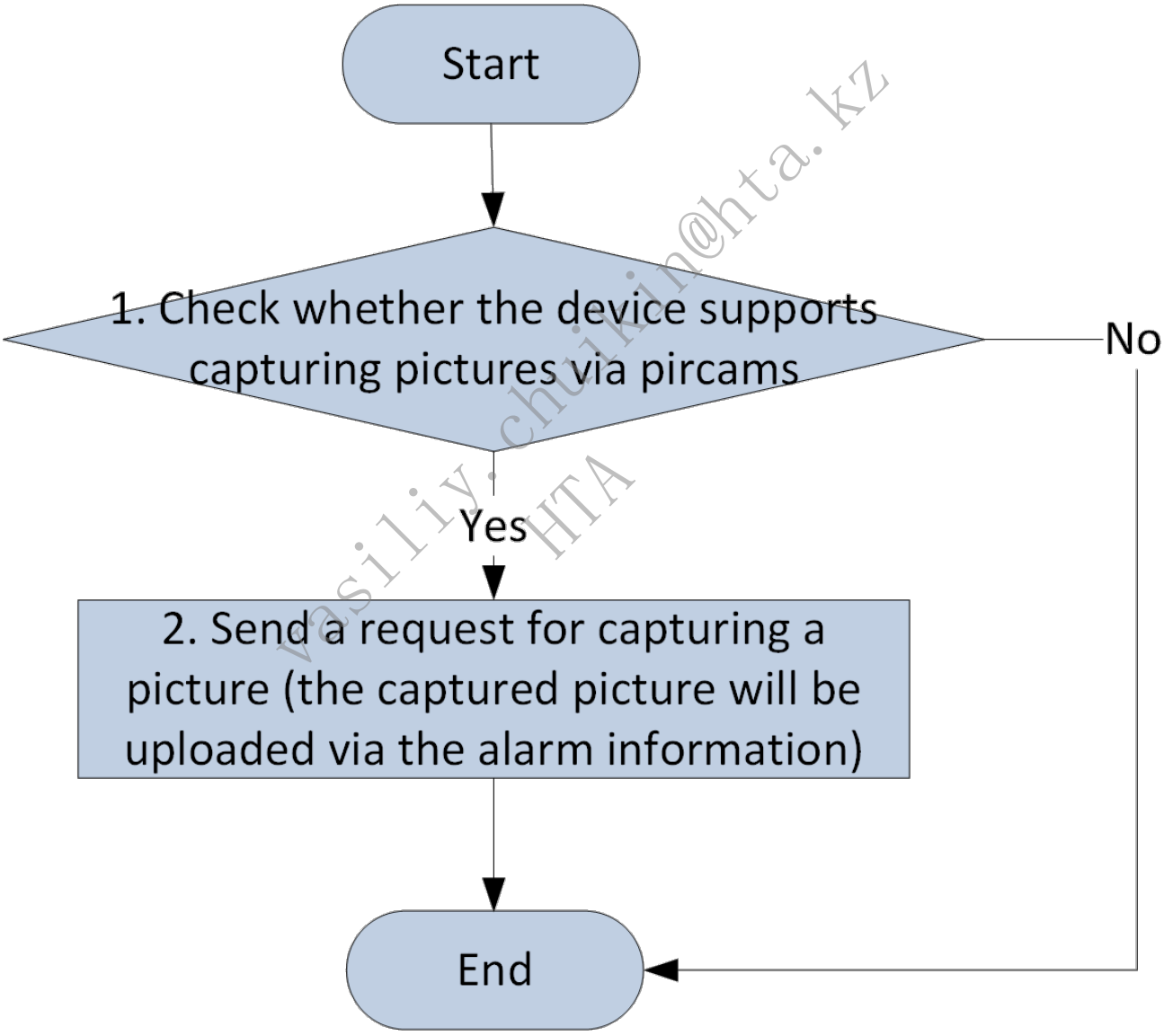
4. Get parameters of a single ARC: `GET /ISAPI/SecurityCP/Configuration/messageSendARC?id=<indexID>&format=json`; set parameters of a single ARC: `PUT /ISAPI/SecurityCP/Configuration/messageSendARC?id=<indexID>&format=json`. The indexID in the URL is the ARC No.

12.3 Capture Pictures of Specific Zones

There are two ways to get the picture captured by pircams: 1) send a request for capturing a picture and get the picture URL in the uploaded alarm (CID event) information; 2) send a request for capturing a picture and call another API to get the picture URL.

12.3.1 Upload Pictures by Pircams

After the pircam captures a picture, the picture will be uploaded via the alarm (CID event) information.



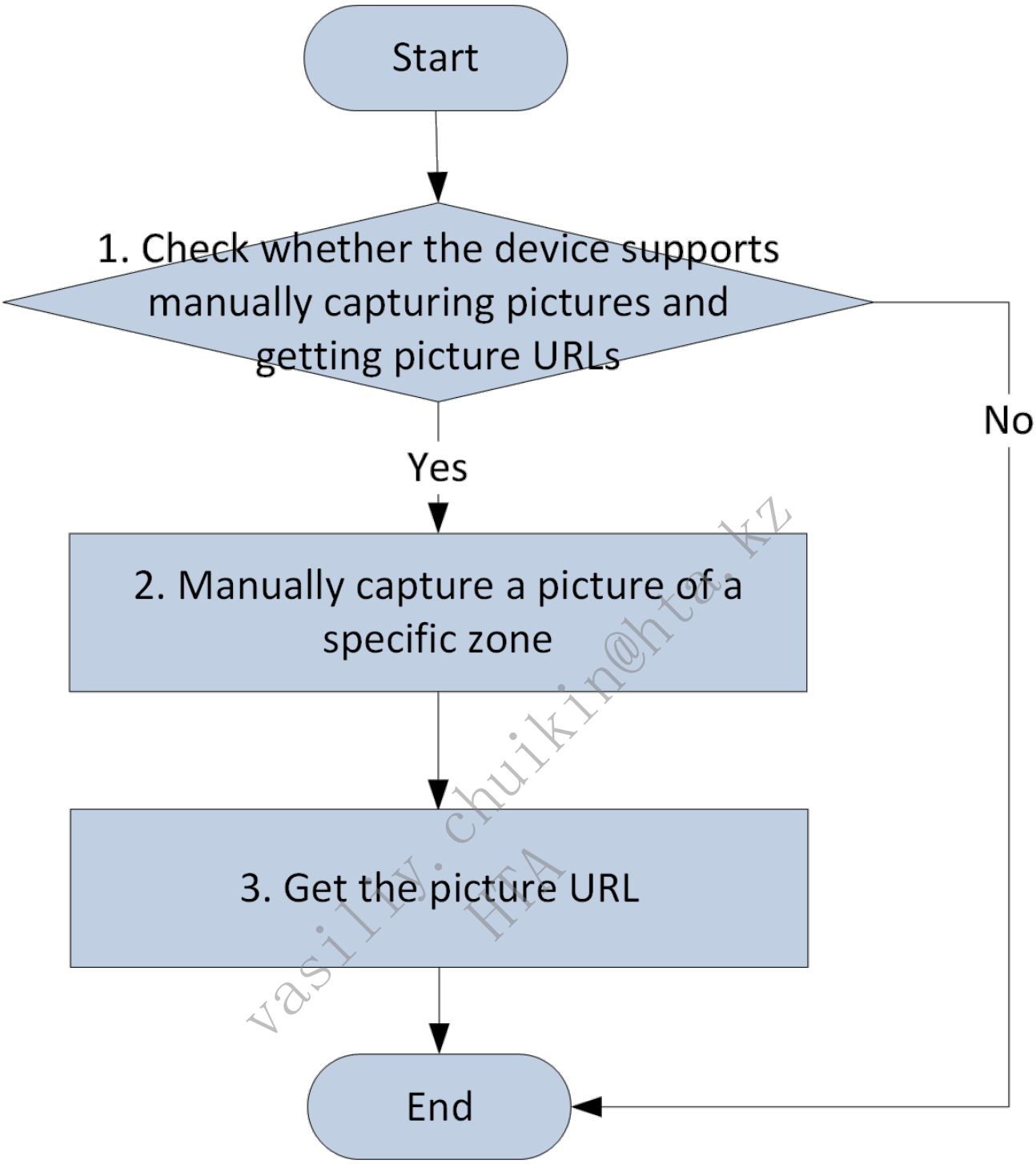
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API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports capturing pictures via pircams: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node isSptPircamCapture is true, the device supports this function.

2. Send a request for capturing a picture (the captured picture will be uploaded via the alarm information): `GET /ISAPI/Streaming/channels/<zoneID>/picture/devicePush`. The zoneID in the URL refers to zone No.

12.3.2 Get Captured Pictures in Asynchronous Mode



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API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports manually capturing pictures and getting picture URLs: `GET /ISAPI/SecurityCP/capabilities?format=json`. If the returned values of the node isSptManualControlCapture and isSptGetPictureByURL is true, the device supports the functions above.

2. Manually capture a picture of a specific zone: `PUT /ISAPI/SecurityCP/Zone/<zoneID>/Capture/ManualControlCapture?format=json`. The zoneID in the URL refers to zone No.

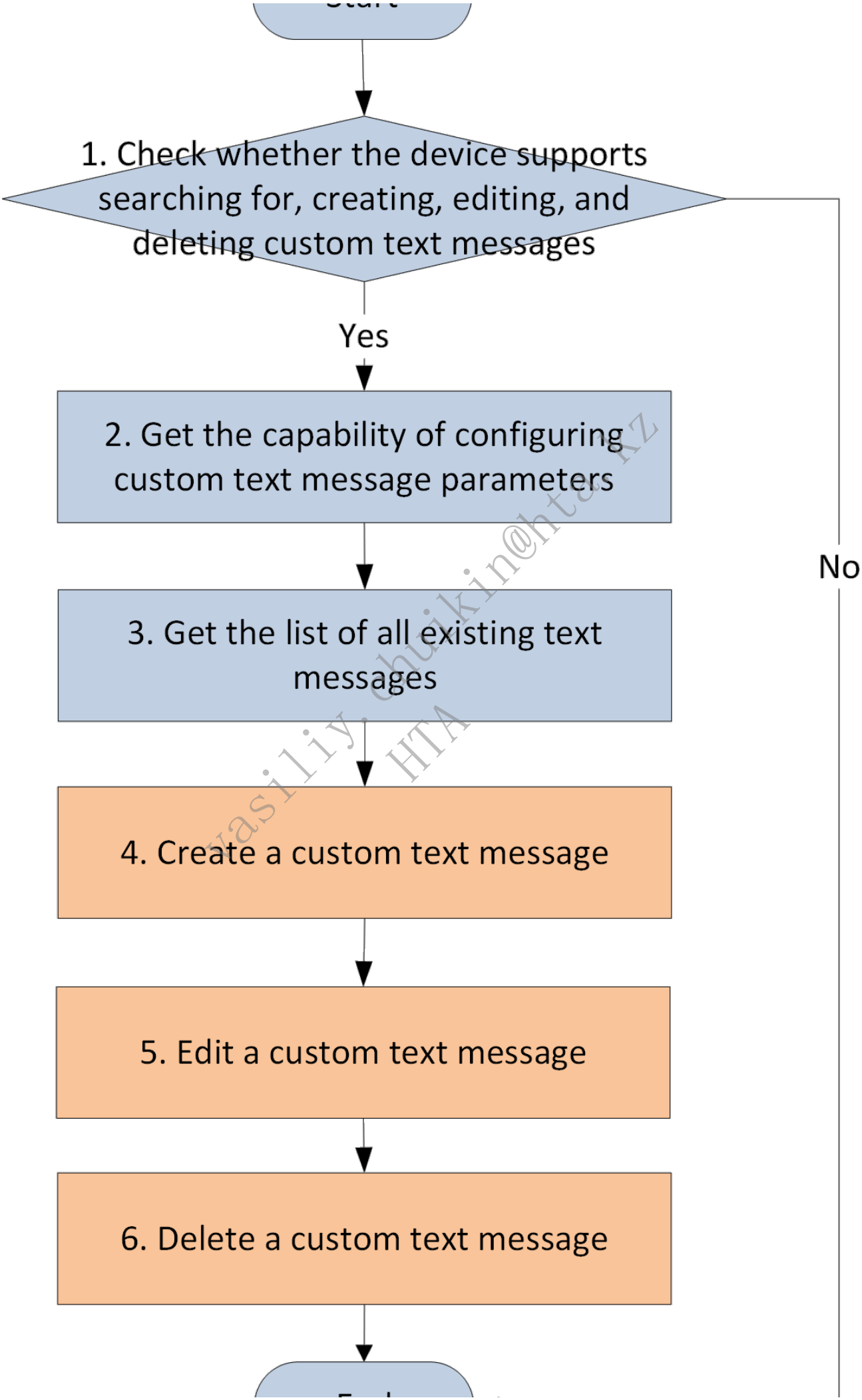
3. Get the picture URL: `GET /ISAPI/SecurityCP/Zone/<zoneID>/Capture/GetPictureByURL?format=json`.

12.4 Custom Text Message Management

Telephone Notifications support both custom audios and custom message texts.

API Calling Flow:





Vasiliy.chuikin@hta.kz
HTA



API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports searching for, creating, editing, and deleting custom text messages: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned values of isSupportSearchCustomAudioList, isSupportAddCustomMessage, isSupportModifyCustomMessage, and isSupportDeleteCustomMessage are all true, the device supports all the functions above.

2. Get the capability of configuring custom text message parameters: `GET /ISAPI/SecurityCP/customMessage/capabilities?format=json`.

3. Get the list of all existing text messages: `GET /ISAPI/SecurityCP/customMessage/searchCustomMessageList?format=json&security=<security>&iv=<iv>`. The node customMessageContent should be encrypted.

4. (Optional) Create a custom text message: `POST /ISAPI/SecurityCP/customMessage/addCustomMessage?format=json&security=<security>&iv=<iv>`.

5. (Optional) Edit a custom text message: `PUT /ISAPI/SecurityCP/customMessage/modifyCustomMessage?format=json&security=<security>&iv=<iv>`.

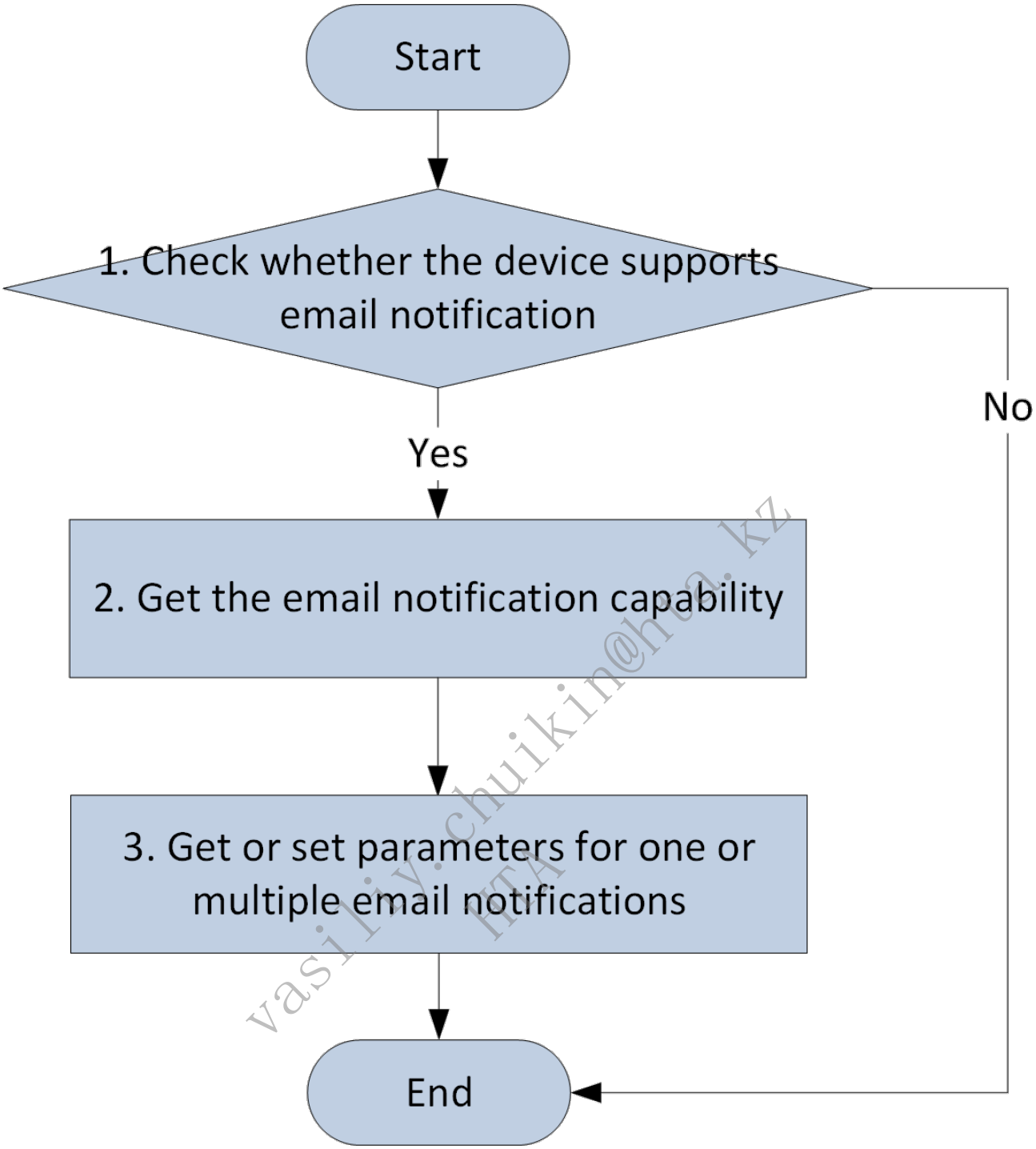
6. (Optional) Delete a custom text message: `POST /ISAPI/SecurityCP/customMessage/deleteCustomMessage?format=json`.

12.5 Email Notification Management

If the device supports email notification, you can enable the event notification and configure related email notification parameters.

API Calling Flow:

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vasiliy.chuikin@nca.kz
НТА

API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports email notification: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node isSptMail is true, the device supports this function.

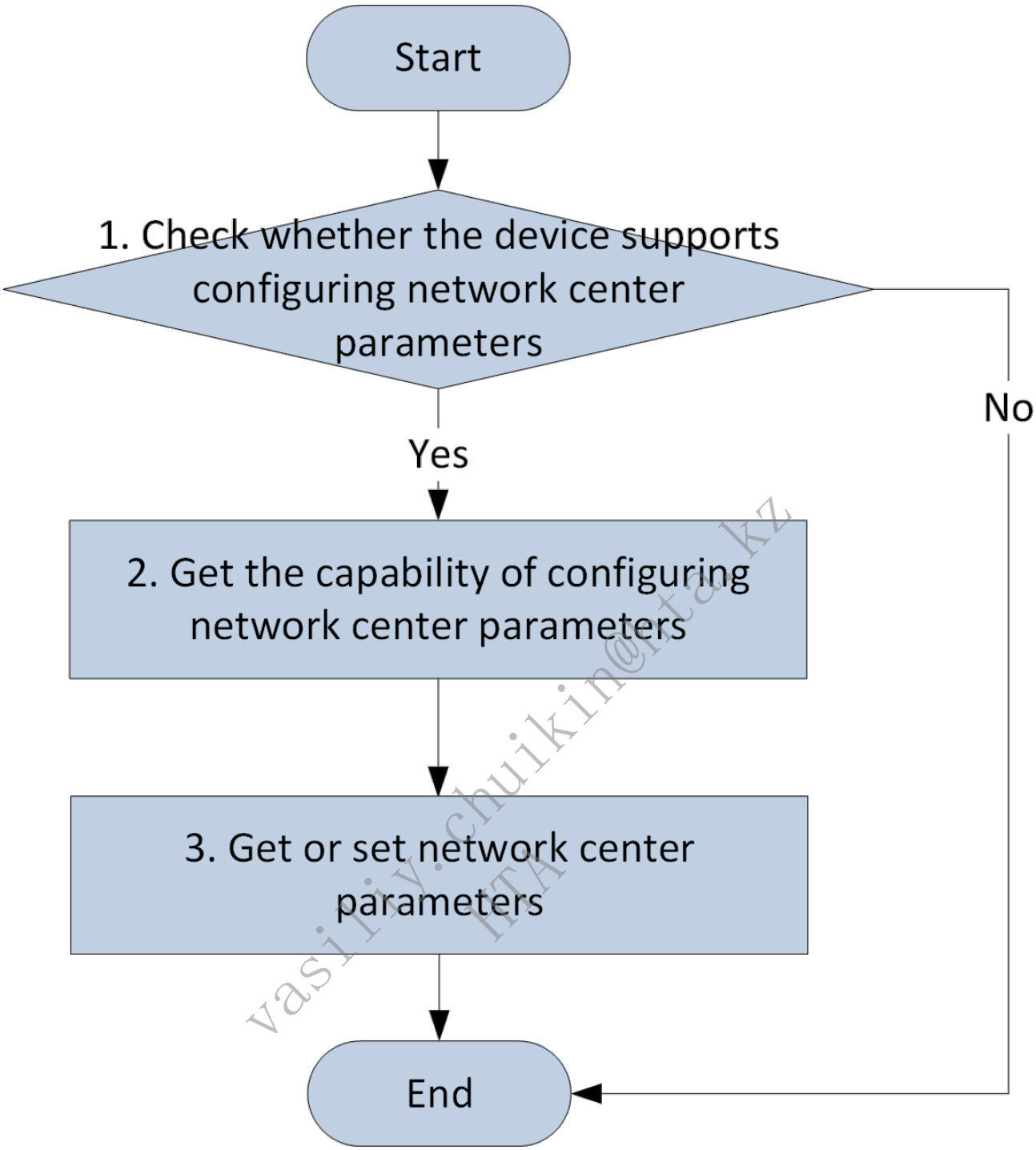
2. Get the email notification capability: `GET /ISAPI/SecurityCP/Configuration/messageSendMail/capabilities?format=json`.

3. Get parameters for multiple email notifications: `GET /ISAPI/SecurityCP/Configuration/messageSendMail?format=json`; set parameters for a single email notification: `PUT /ISAPI/SecurityCP/Configuration/messageSendMail/<mailID>?format=json`. The mailID in the URL is the email ID.

12.6 Network Center Parameter Configuration

For each NIC of the device, you can configure parameters of one or multiple network centers (one NIC can correspond to multiple network centers). Private protocol, NAL2300 protocol, and ISUP are supported currently.

API Calling Flow:



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API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports configuring network center parameters: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node isSptNetCfg is true, the device supports this function.

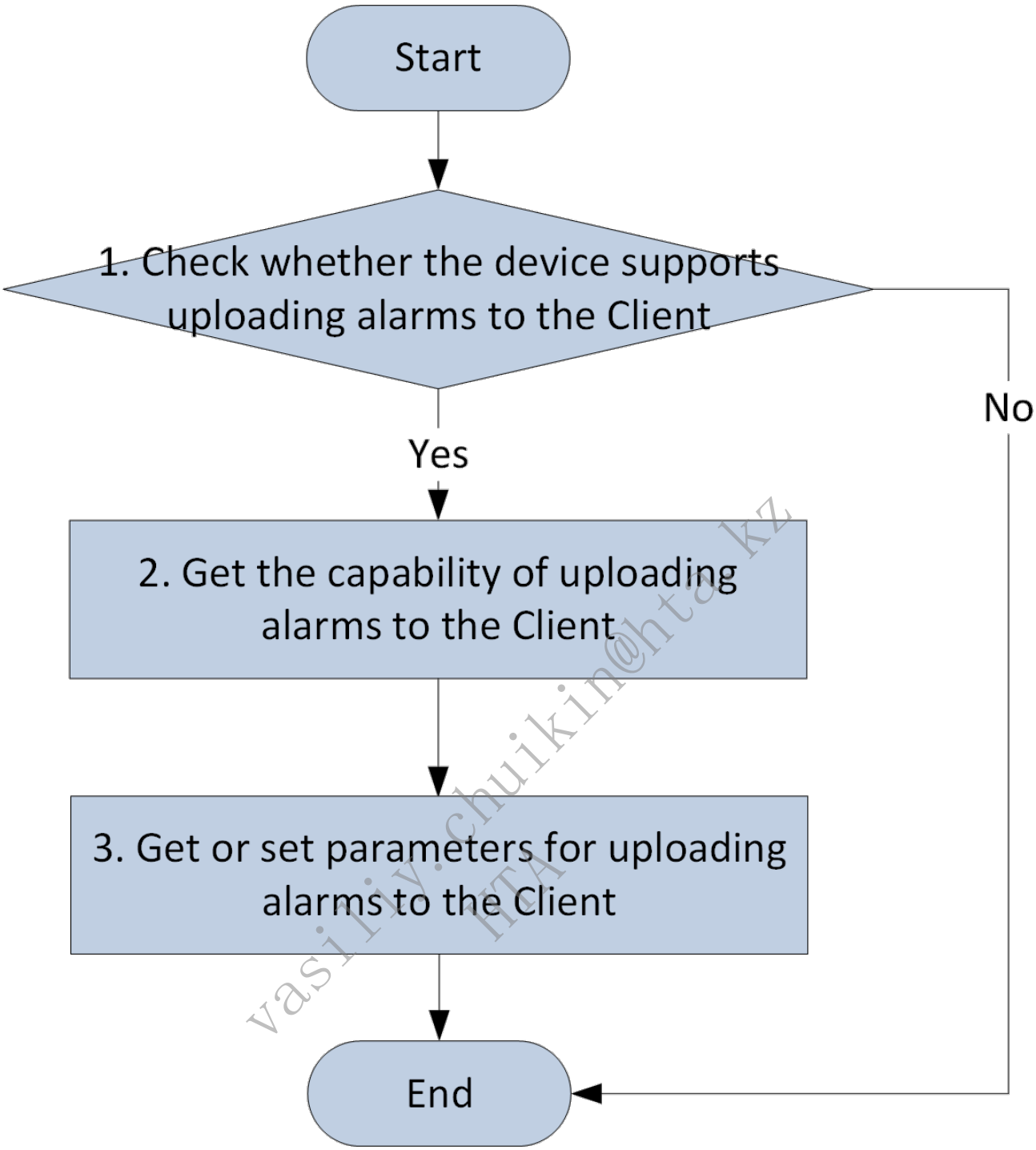
2. Get the capability of configuring network center parameters: `GET /ISAPI/SecurityCP/NetCfg/capabilities?format=json`.

3. Get network center parameters: `GET /ISAPI/SecurityCP/NetCfg/<interfaceID>?format=json`; configure network center parameters: `PUT /ISAPI/SecurityCP/NetCfg/<interfaceID>?format=json`. The interfaceID in the URL refers to the NIC No.: 1 for the main NIC and 2 for the extended NIC.

12.7 Upload Alarms to Client

Events/alarms that can be uploaded to the Client (such as iVMS-4200) include tampering event, life security event, system status event, operation event, emergency alarm, medical alarm, and gas alarm.

API Calling Flow:



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API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports uploading alarms to the Client: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node isSptDirect is true, the device supports this function.

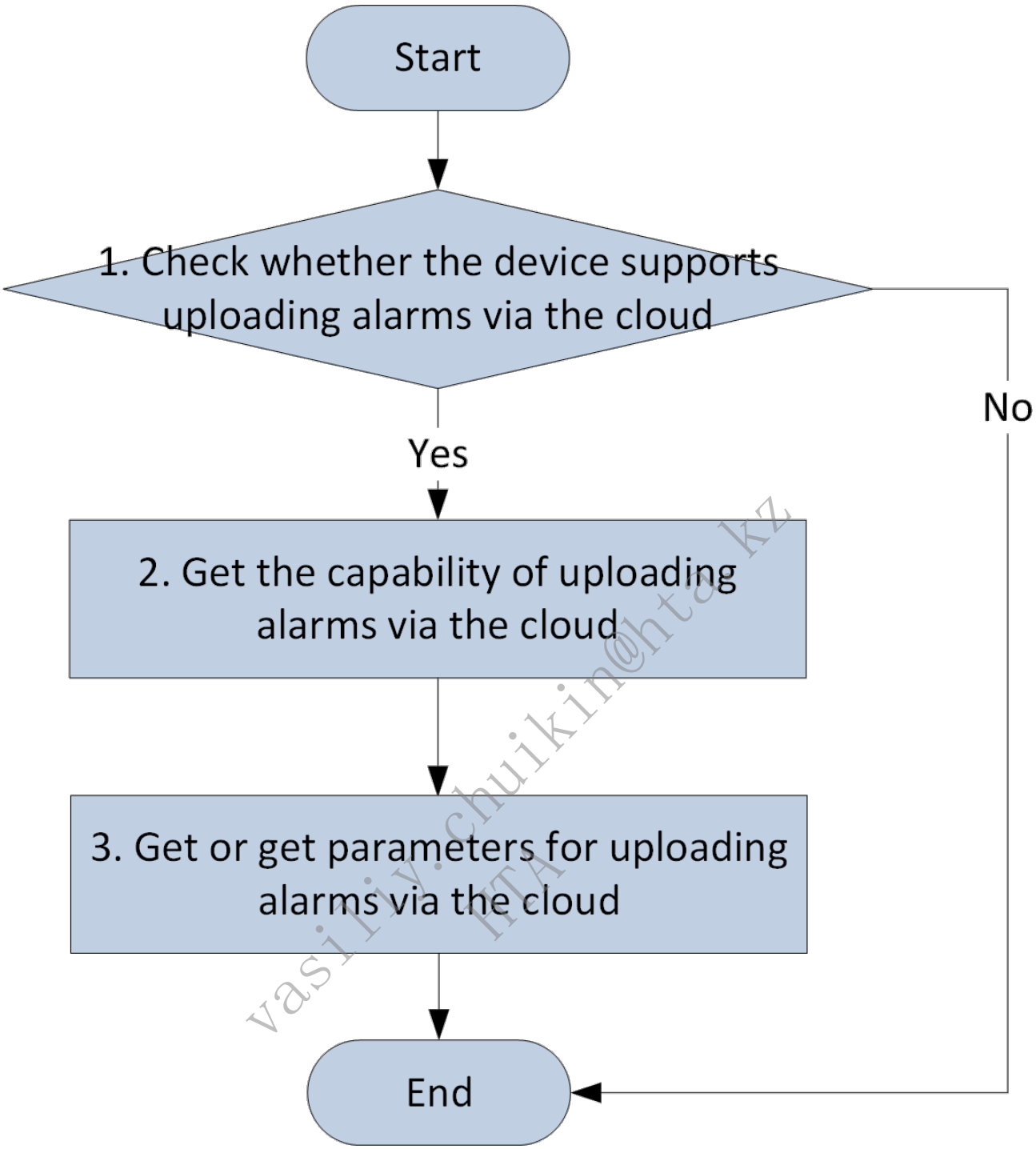
2. Get the capability of uploading alarms to the Client: `GET /ISAPI/SecurityCP/Configuration/messageSendDirect/capabilities?format=json`.

3. Get parameters for uploading alarms to the Client: `GET /ISAPI/SecurityCP/Configuration/messageSendDirect?format=json`; set parameters: `PUT /ISAPI/SecurityCP/Configuration/messageSendDirect?format=json`.

12.8 Upload Alarms via Cloud

Events/alarms that can be uploaded to apps via the cloud include tampering event, life security event, system status event, operation event, emergency alarm, medical alarm, and gas alarm.

API Calling Flow:



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API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports uploading alarms via the cloud: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node isSptCloud is true, the device supports this function.

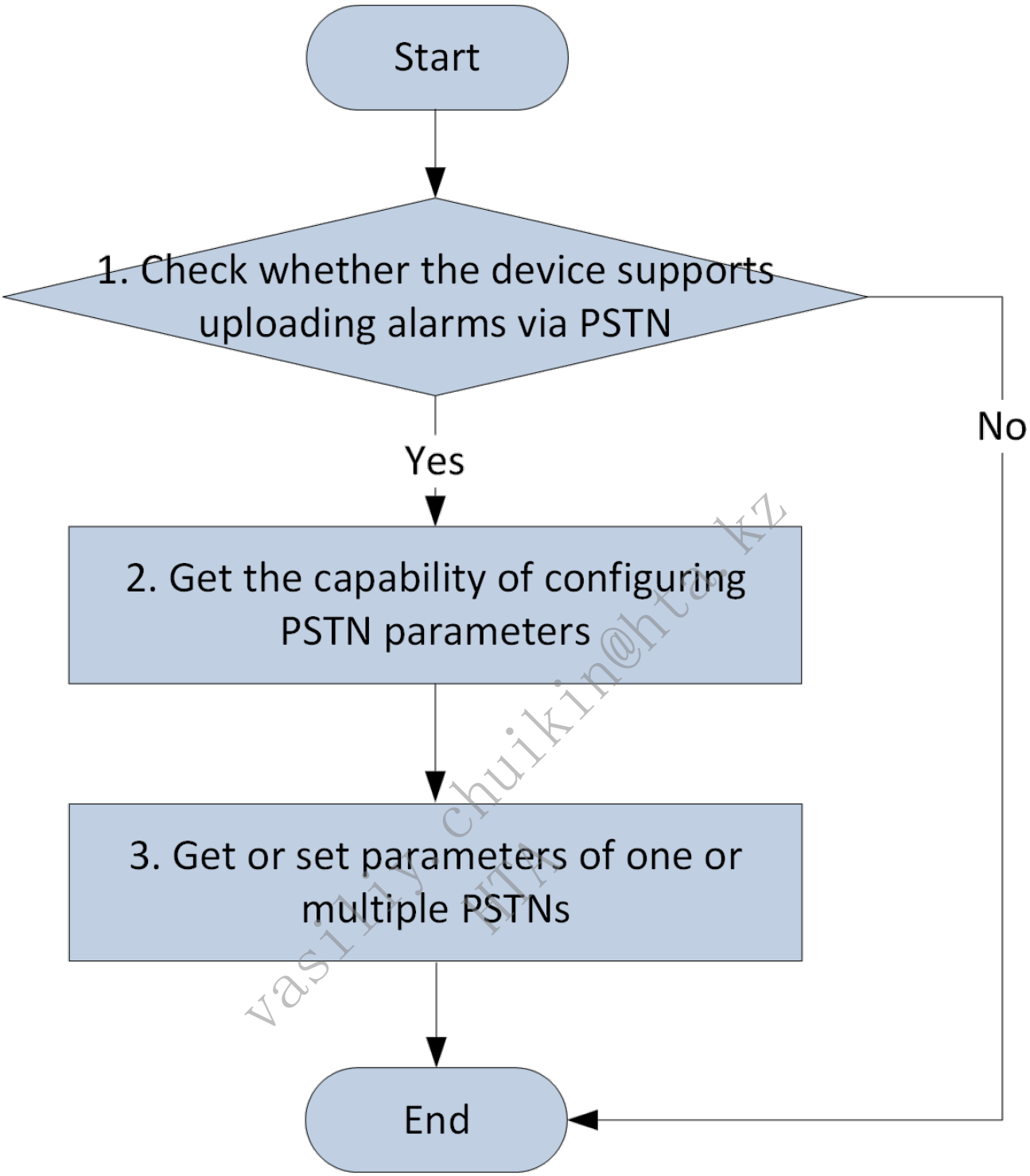
2. Get the capability of uploading alarms via the cloud: `GET /ISAPI/SecurityCP/Configuration/messageSendCloud/capabilities?format=json`.

3. Get parameters for uploading alarms via the cloud: `GET /ISAPI/SecurityCP/Configuration/messageSendCloud?format=json`; set parameters: `PUT /ISAPI/SecurityCP/Configuration/messageSendCloud?format=json`.

12.9 Upload Alarms via PSTN

If the device supports Public Switched Telephone Network (PSTN), the alarms can be uploaded by dial-up and the dial-up target can be an ARC or a person.

API Calling Flow:



vasiliy chuikin@hte.kz

API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports uploading alarms via PSTN: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node isSptPSTNCfg is true, the device supports this function.

2. Get the capability of configuring PSTN parameters: `GET /ISAPI/SecurityCP/Configuration/PSTNCfg/capabilities?format=json`.

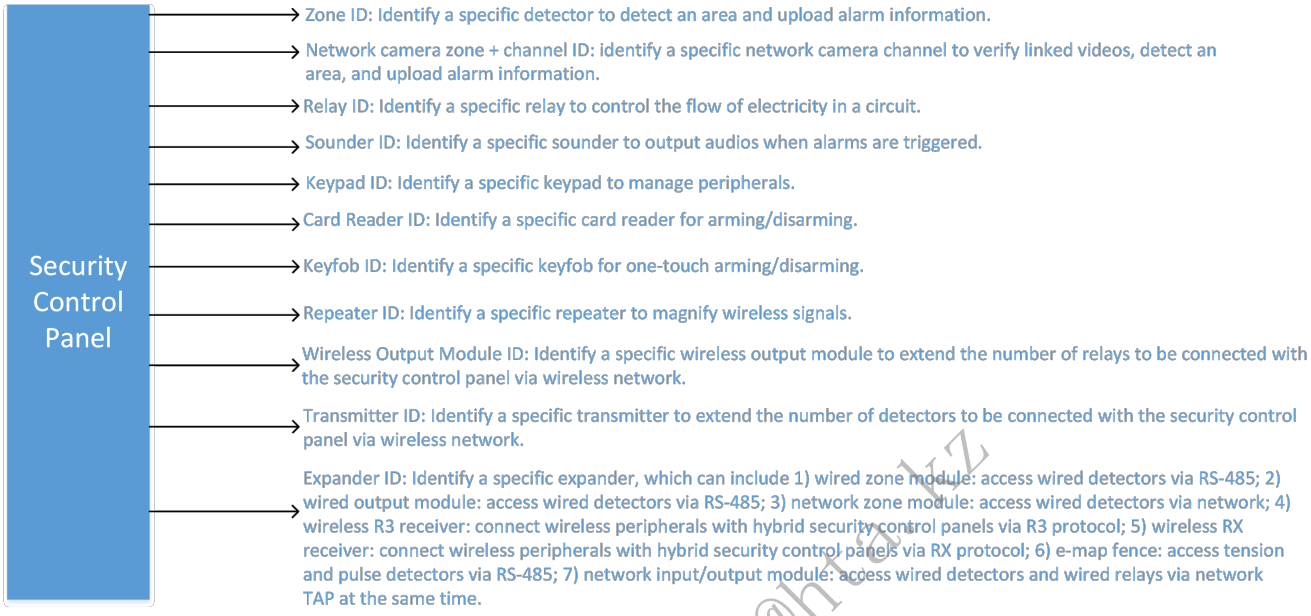
3. Get parameters of all PSTNs: `GET /ISAPI/SecurityCP/Configuration/PSTNCfg?format=json`; set parameters of a single PSTN: `PUT /ISAPI/SecurityCP/Configuration/PSTNCfg/<indexID>?format=json`. Available parameters include whether to enable PSTN link, telephone of the target, communication protocol (CID), and transmission method (0#DTMF 5/S, 1#DTMF 10/S). The indexID in the URL is the PSTN No.

13 Zone Alarm

13.1 Peripheral Management

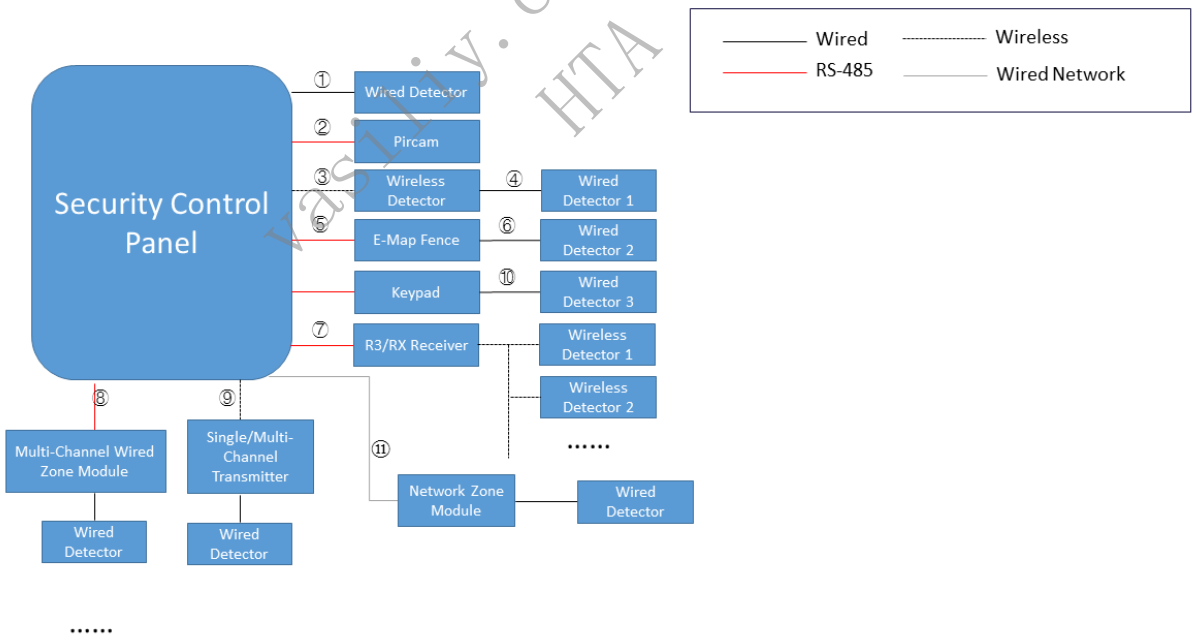
13.1.1 Peripheral No.

The peripheral No. is used to specify a certain peripheral (such as detector, network camera, relay, and sounder) when you perform operations including configuring parameters and upgrading devices. For details about rules, see the picture below:



13.2 Topology of Detectors and Peripherals

13.2.1 Topology of Detectors



The hybrid security control panel can access detectors in the following ways:

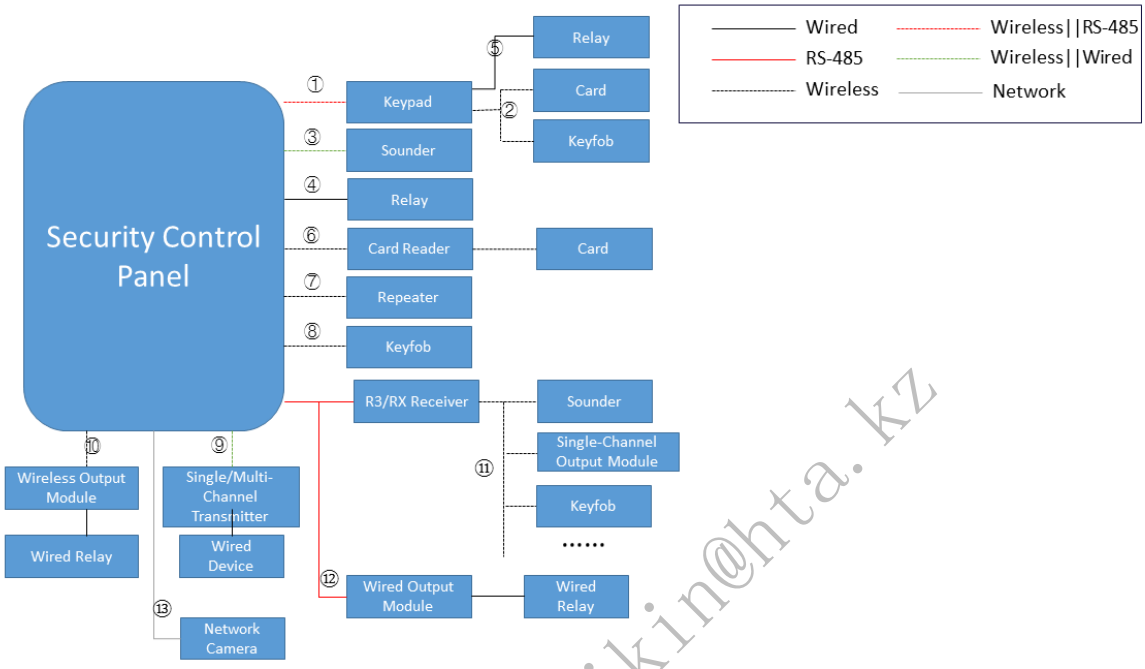
1 (access wired detectors via onboard zones), 2 (access pircams via the RS-485 bus), 5 (access 1 to 2 detectors by connecting the RS-485 bus with e-map fences), 6 (access 1 detector by enabling the function of extending zones for e-map fences), 7 (access wireless detectors by connecting the RS-485 bus with R3/RX receivers), 8 (access wired detectors via multi-channel wired zone modules), 10 (access wired detectors via keypads), 11 (access wired detectors via network zone modules)

The wireless security control panel can access detectors in the following ways:

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1 (access wired detectors via onboard transmitters), 3 (access wireless detectors via internal wireless receivers), 4 (access 1 to 2 wired detectors by enabling the function of extending zones for some magnetic contact detectors), 9 (access wired detectors by connecting wireless receivers with single-channel / multi-channel transmitters)

13.2.2 Topology of Peripherals



The hybrid security control panel can access peripherals in the following ways:

1 (access keypads via the RS-485 bus), 2 (access wireless cards and keyfobs via keypads), 3 (access wired sounders via onboard sounder modules), 4 (access wired relays via onboard relay modules), 5 (access wired relays via keypads), 11 (access wireless sounders, single-channel output modules (wireless relays), and keyfobs via the receivers connected to the RS-485 bus), 12 (access wired relays via the wired output module connected to the RS-485 bus), 13 (access network cameras by login via the network)

The wireless security control panel can access peripherals in the following ways:

1 (access keypas via wireless network), 3 (access wireless sounders via internal wireless receivers), 4 (access wired relays via onboard transmitters), 6 (access card readers via wireless network), 7 (access repeaters via wireless network), 8 (access keyfobs via wireless network), 9 (access wired relays by connecting wireless receivers with single-channel / multi-channel transmitters), 10 (access wired relays by connecting wireless receivers with wireless output modules), 13 (access network cameras by login via the network)

14 API Reference

14.1 Device (General)

14.1.1 Event Subscription

14.1.1.1 Establish arming connection

Request URL

GET /ISAPI/Event/notification/alertStream?deployID=<deployID>

Query Parameter

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Parameter Name	Parameter Type	Description
deployID	string	0 (client-side arming), 1 (real-time arming). Defaults to the client-side arming when delpoyID does not exist.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, event massage, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPV4 address of the device that triggers an alarm-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPV6 address of the device that triggers an alarm-->test
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm, range:[0,65535]-->0
  </portNo>
  <protocol>
    <!--ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address, range:[0,17]-->00-16-EA-AE-3C-40
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers an alarm, range:[0,65535]-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm triggering time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, the uploading times of the same alarm, range:[0,65535]-->0
  </activePostCount>
  <eventType>
    <!--ro, req, string, heartbeat type-->test
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:"active" (triggered), "inactive" (not triggered, the heartbeat data)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, range:[0,128]-->test
  </eventDescription>
</EventNotificationAlert>
```

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14.1.1.2 Set alarm/event subscription parameters

Request URL

PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all"(upload all alarms/events), "list"(upload specified alarm/event)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, "all"-upload all alarms/events,"list"-upload specified alarm/event, subType:string, desc:"all"(upload all alarms/events), "list"(upload
specified alarm/event)-->list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
```

```
      <!--opt, object, uploading mode of specified alarm/event-->
      <type>
        <!--req, enum, alarm/event types, subType:string, desc:see details in event types: "ADAS"(advanced driving assistance system), "ADASAlarm"(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent", "CDsStatus", "DBD"(driving behavior detection) "GPSUpload", "HFPD"(frequently appeared person), "IO"(I/O Alarm), "IOTD", "LES", "LFPD"(low frequency person detection), "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection), "listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison"-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, operation sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, event sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4..."-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--opt, enum, event sub-type, subType:string, desc:"success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--opt, array, subType:object-->
        <childDevID>
          <!--opt, string, range:[1,128]-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--opt, string-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--opt, string-->1,2,3,4
      </audioChannels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4...", desc:event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4..."-->1,2,3,4
  </channels>
  <pictureURLType>
```

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```
    <!--opt, enum, alarm picture format, subType:string, desc:"binary"(binary), "localURL"(device local URL), "cloudStorageURL"(cloud storage URL)-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:5 minute (default). If GUID is not reconnected in the internal, a new GUID will be generated since the device starts a new arm period-->300
    </interval>
    <StatusSub>
      <!--opt, object, status description-->
      <all>
        <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--opt, bool, status of subscribed channels, desc:reporting is not required if all is true-->true
      </channel>
      <hd>
        <!--opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <level>
    <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--opt, array, subType:object-->
    <childDevID>
      <!--opt, string, range:[1,128]-->test
    </childDevID>
  </childDevIDList>
  <radarChannels>
    <!--opt, string-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object-->
    <defaultLevel>
      <!--ro, req, enum, subType:int-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, subType:object-->
      <event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, string, range:[0,64]-->test
        </eventType>
        <level>
          <!--ro, req, enum, subType:int-->1
        </level>
        <channels>
          <!--ro, opt, string, range:[0,256]-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
```

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```
  </qosLevel>
  <audioChannels>
    <!--opt, string-->1,2,3,4
  </audioChannels>
</SubscribeEvent>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

14.1.1.3 Get alarm/event subscription parameters

Request URL

GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), the (subscription ID) will be returned in message SubscribeEventResponse. When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), and if subscribeEventID is applied, it can be used to replace the in SubscribeEventResponse.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--ro, opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--ro, opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel
list)-->list
  </channelMode>
  <eventMode>
    <!--ro, req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)--
>list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--ro, opt, object, subscription of a specified alarm/event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:alarm/event types, which are obtained from the capability, refer to Alarm/Event Types
(eventType)-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
```

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```
      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--ro, opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--ro, opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--ro, opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--ro, opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--ro, opt, enum, event sub-type, subType:string, desc:"success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--ro, opt, array, subType:object-->
        <childDevID>
          <!--ro, opt, string, range:[1,128]-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--ro, opt, string-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--ro, opt, string-->1,2,3,4
      </audioChannels>
      <childDeviceFormat>
        <!--ro, opt, bool-->true
      </childDeviceFormat>
    </Event>
  </EventList>
  <channels>
    <!--ro, opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is all, this node should not be applied-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--ro, opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "localURL" (device local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, the <pictureURLType> of the <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--ro, opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--ro, opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--ro, opt, object, status description-->
      <all>
        <!--ro, opt, bool, whether to subscribe to all channels, HDDs, and capability set changes-->false
      </all>
      <channel>
        <!--ro, opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
      </channel>
      <hd>
        <!--ro, opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
      </hd>
      <capability>
        <!--ro, opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the value of <all> is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will
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```
be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm--
>true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no
related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--ro, opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--ro, opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--ro, opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <level>
    <!--ro, opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--ro, opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--ro, opt, array, subType:object-->
    <childDevID>
      <!--ro, opt, string, range:[1,128]-->test
    </childDevID>
  </childDevIDList>
  <radarChannels>
    <!--ro, opt, string-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object-->
    <defaultLevel>
      <!--ro, req, enum, subType:int-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, subType:object-->
      <event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, string, range:[0,64]-->test
        </eventType>
        <level>
          <!--ro, req, enum, subType:int-->1
        </level>
        <channels>
          <!--ro, opt, string, range:[0,256]-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
  <audioChannels>
    <!--ro, opt, string-->1,2,3,4
  </audioChannels>
  <retransmissionID>
    <!--ro, opt, string, range:[1,64]-->123456
  </retransmissionID>
  <retransmissionTimeoutDuration>
    <!--ro, opt, int, range:[1,168], unit:h-->1
  </retransmissionTimeoutDuration>
  <SubscribeISAPIMessage>
    <!--ro, opt, object-->
    <EventTypeList>
      <!--ro, req, array, subType:object-->
      <eventType>
        <!--ro, req, enum, subType:string-->AIOP_Video
      </eventType>
      <uploadPicEnabled>
        <!--ro, opt, bool-->false
      </uploadPicEnabled>
    </EventTypeList>
    <EventList>
      <!--ro, req, array, subType:object, range:[0,200]-->
      <Event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, enum, subType:string-->AIOP_Video
        </eventType>
        <uploadPicEnabled>
          <!--ro, opt, bool-->false
        </uploadPicEnabled>
      </Event>
    </EventList>
  </SubscribeISAPIMessage>
```

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```
</SubscribeEvent>
```

14.1.1.4 Subscribe to events/alarms in arming mode

Request URL

POST /ISAPI/Event/notification/subscribeEvent?deployID=<deployID>

Query Parameter

Parameter Name	Parameter Type	Description
deployID	enum	0 (device local arming), 1 (real-time arming by platform). If the deployID is absent, it defaults to device local arming.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel
list)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)--
>list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--opt, object, subscription of a specified alarm/event-->
      <type>
        <!--req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
"faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "CertificateCaptureEvent"(person ID
capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event),
"TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status),
"certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison), "regionTargetNumberCounting" (regional target statistics)--
>mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, minor operation type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, minor event type, desc:this node is required when type is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
```

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```
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel check-in.
3. When this node is not applied, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
        <childDevID>
          <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--opt, string, arm by radar channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </audioChannels>
      <childDeviceFormat>
        <!--opt, bool, upload by sub device, desc:upload by sub device-->true
      </childDeviceFormat>
      <unitChannels>
        <!--opt, string, arm by optical fiber channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </unitChannels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is all, this node should not be applied-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "localURL" (device local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--opt, object, status information-->
      <all>
        <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--opt, bool, status of subscribed channels, desc:it is not required if the value of <all> is true-->true
      </channel>
      <hd>
        <!--opt, bool, subscribe to HDD status, desc:it is not required if the value of <all> is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:it is not required if the value of <all> is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
```

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```
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no
related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <subscribeEventID>
    <!--opt, string, custom subscription ID. The platform should guarantee the uniqueness of ID when applying it. The automatic network replenishment of
events will be enabled by default after the node is applied, range:[1,128], desc:when an event/alarm is subscribed (related URL: POST
/ISAPI/Event/notification/subscribeEvent), if this node is applied, it can be used to replace <id> (subscription ID) returned in message
SubscribeEventResponse-->test
  </subscribeEventID>
  <level>
    <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
    <childDevID>
      <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
    </childDevID>
  </childDevIDList>
  <SubscribeISAPIMessage>
    <!--opt, object-->
    <EventTypeList>
      <!--opt, array, event list, subType:object, desc:event list-->
      <eventType>
        <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
      </eventType>
      <uploadPicEnabled>
        <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
      </uploadPicEnabled>
    </EventTypeList>
    <EventList>
      <!--req, array, event list, subType:object, range:[0,200]-->
      <Event>
        <!--opt, object, a single event-->
        <eventType>
          <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
        </eventType>
        <uploadPicEnabled>
          <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
        </uploadPicEnabled>
      </Event>
    </EventList>
  </SubscribeISAPIMessage>
  <radarChannels>
    <!--opt, string, batch arming by radar channel, desc:if this field is applied, radarChannels in Event will not take effect. To arm all radar channels,
this field should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object, subscription events qos level, desc:subscription events qos level-->
    <defaultLevel>
      <!--ro, req, enum, default qos level, subType:int, desc:default qos level-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, custom event qos level list, subType:object-->
      <event>
        <!--ro, opt, object, custom event qos level-->
        <eventType>
          <!--ro, req, string, event type, range:[0,64], desc:its values refer to event types of different domains-->test
        </eventType>
        <level>
          <!--ro, req, enum, qos level of this event type, subType:int, desc:qos level of this event type-->1
        </level>
        <channels>
          <!--ro, opt, string, video resource channel No., range:[0,256], desc:no such node means it applies to all channels; otherwise it applies to the
channels listed in the array, and channels not listed use default qos level-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
  <audioChannels>
    <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
  </audioChannels>
  <unitChannels>
    <!--opt, string, batch arming by optical fiber channel, desc:if this field is applied, unitChannels in Event will not take effect. To arm all fiber
channels, this node should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
  </unitChannels>
  <retransmissionID>
```

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```
    <!--opt, string, ANR arming ID, range:[1,64], desc:UUID (an unique ID generated by the client). If this field is applied during establishing the event arming/subscription connection, the ANR of subscribed event will be enabled by default; when the network reconnected, the platform can apply the same UUID to tell the device to re-upload saved event information during network disconnection period-->123456
  </retransmissionID>
  <retransmissionTimeoutDuration>
    <!--opt, int, data transmission timeout for ANR, range:[1,168], unit:h, desc:data transmission timeout for ANR-->1
  </retransmissionTimeoutDuration>
  <subscribeWithoutPic>
    <!--opt, bool-->false
  </subscribeWithoutPic>
</SubscribeEvent>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEventResponse xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, subscription result, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, subscription ID, which is returned by the device. The device should guarantee the uniqueness of ID, range:[1,64]-->1
  </id>
  <FailedEventList>
    <!--ro, opt, array, list of events, of which subscription failed, subType:object, desc:when subscription of one or multiple events failed, this node should be returned. The client software can check if all events are subscribed to by checking whether the node <FailedEventList> is returned-->
    <Event>
      <!--ro, opt, object, event, of which the subscription failed. The event/alarm type name can be obtained via the capability set-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:event type-->ADAS
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL).-->binary
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel No., desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <subStatusCode>
        <!--ro, req, enum, error code, subType:string, desc:refer to the error code dictionary-->activeNumMax
      </subStatusCode>
      <minorType>
        <!--ro, opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel check-in.
3. When this node is not returned, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
    </Event>
  </FailedEventList>
</SubscribeEventResponse>
```

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14.1.1.5 Get the alarm/event subscription capability

Request URL

GET /ISAPI/Event/notification/subscribeEventCap

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SubscribeEventCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat min="1" max="180">
    <!--ro, opt, int, heartbeat interval time, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->1
  </heartbeat>
  <channelMode opt="all,list">
    <!--ro, opt, enum, channel subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all channels), "list" (subscribe to
channels according to channel list)-->list
  </channelMode>
  <eventMode opt="all,list,allAndChild">
    <!--ro, req, enum, event subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all alarms/events), "list" (subscribe to
specified alarms/events)-->list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--ro, opt, object, subscription of a specified alarm/event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
"faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent",
"IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison
event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event),
"TransactionRecordEvent", "HealthInfoSyncQuery" (health information search event), "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"
(searching the consumption status), "certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison),
"regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
      </type>
      <minorAlarm opt="0x400,0x401,0x402,0x403">
        <!--ro, opt, string, minor alarm type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"--
>0x400,0x401
      </minorAlarm>
      <minorException opt="0x400,0x401,0x402,0x403">
        <!--ro, opt, string, minor exception type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401
      </minorException>
      <minorOperation opt="0x400,0x401,0x402,0x403">
        <!--ro, opt, string, minor operation type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401
      </minorOperation>
      <minorEvent opt="0x01,0x02,0x03,0x04">
        <!--ro, opt, string, minor event type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"--
>0x400,0x401
      </minorEvent>
      <pictureURLType opt="binary,localURL,cloudStorageURL,multipart" def="cloudStorageURL">
        <!--ro, opt, enum, alarm picture format, subType:string, attr:opt{req, string},def{req, string}, desc:"binary" (binary), "localURL" (device local
URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
    </Event>
  </EventList>
  <isSupportModifySubscribeEvent>
    <!--ro, opt, bool, whether the device supports the management of arming subscription, desc:related API: /ISAPI/Event/notification/subscribeEvent/<ID>--
>true
  </isSupportModifySubscribeEvent>
</SubscribeEventCap>
```

14.1.1.6 Cancel subscribing alarm/event

Request URL

PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	Subscription event ID, which should match the id field in the SubscribeEventResponse of the response message from the POST /ISAPI/Event/notification/subscribeEvent?deployID=.This field is optional. When the protocol is /ISAPI/Event/notification/unSubscribeEvent, it indicates that all subscription links are to be unsubscribed. When the protocol is /ISAPI/Event/notification/unSubscribeEvent?ID=XXX, it indicates that a specific subscription link is to be unsubscribed.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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14.1.1.7 Event subscription heartbeat

EventType:heartBeat

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, alarm message, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, transmission communication protocol type, range:[0,64], desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, string, event type-->heartBeat
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->heartBeat
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
</EventNotificationAlert>
```

14.1.2 Online Upgrading Management

14.1.2.1 Get the device online upgrade capability

Request URL

GET /ISAPI/System/onlineUpgrade/capabilities?childDevID=<childDevID>

Query Parameter

Parameter Name	Parameter Type	Description
childDevID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, online upgrade, attr:version{req, string, protocolVersion}-->
  <firmwareNum max="10">
    <!--ro, req, int, number of supported online upgrade packages, attr:max{req, int}-->1
  </firmwareNum>
  <firmwareCode max="256">
    <!--ro, req, int, maximum length of firmware ID, attr:max{req, int}-->1
  </firmwareCode>
  <firmwareVersion max="64">
    <!--ro, req, int, maximum length of version, attr:max{req, int}-->1
  </firmwareVersion>
  <firmwareCodeNumOnce max="10">
    <!--ro, req, int, maximum number of obtained firmware IDs, attr:max{req, int}-->1
  </firmwareCodeNumOnce>
  <upgradePercent min="0" max="100">
```

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```
    <!--ro, req, int, upgrade progress capability in percentage, attr:min{req, int},max{req, int}-->1
  </upgradePercent>
  <Version>
    <!--ro, opt, object, upgrade package version information-->
    <newVersion max="64">
      <!--ro, req, string, version information of the new upgrade package, attr:max{req, int}-->test
    </newVersion>
    <changeLog max="64">
      <!--ro, req, string, log of the new upgrade package, attr:max{req, int}-->test
    </changeLog>
  </Version>
  <rebootAfterUpgrade>
    <!--ro, opt, enum, upgrade options after device reboot, subType:string, desc:"auto" (auto upgrade), "manual" (manual upgrade)-->auto
  </rebootAfterUpgrade>
  <DeviceParameter>
    <!--ro, opt, object, online upgrade parameters-->
    <isSupportAutoDownloadPackage>
      <!--ro, opt, bool, whether it supports automatic download of upgrade package-->true
    </isSupportAutoDownloadPackage>
    <notSupportAutoUpgrade>
      <!--ro, opt, bool, whether it no longer supports automatic download of upgrade package and automatic upgrade, desc:corresponding URL: PUT
/ISAPI/System/onlineUpgrade/upgrade-->true
    </notSupportAutoUpgrade>
    <isSupportTimingUpgrade>
      <!--ro, opt, bool, whether it supports scheduled upgrade-->true
    </isSupportTimingUpgrade>
  </DeviceParameter>
  <ManualDownloadPackage>
    <!--ro, opt, object, manual download of upgrade package-->
    <supportOperation opt="start,cancel,pause,resume">
      <!--ro, opt, enum, supported operations, subType:string, attr:opt{req, string}, desc:"start", "cancel", "pause", "resume"-->start
    </supportOperation>
  </ManualDownloadPackage>
  <isSupportIgnoreCurrentVersion>
    <!--ro, opt, bool, whether it supports ignoring the current version-->true
  </isSupportIgnoreCurrentVersion>
  <UpgradePackageTask>
    <!--ro, opt, object, whether the device supports applying upgrade tasks, desc:if this node is returned, it indicates that the following URLs are also
supported: /ISAPI/System/onlineUpgrade/task?format=json, /ISAPI/System/onlineUpgrade/status/<ID>-->
    <isSupportUpgradeModules>
      <!--ro, opt, bool, whether the device supports upgrading sub modules-->true
    </isSupportUpgradeModules>
    <isSupportUpgradeChannels>
      <!--ro, opt, bool, whether the device supports upgrading sub channels-->true
    </isSupportUpgradeChannels>
    <isSupportDownloadStrategy>
      <!--ro, opt, bool, whether the device supports strategy parameters of downloading upgrade package, desc:corresponding filed is "downloadStrategy" of
URL (/ISAPI/System/onlineUpgrade/task?format=json)-->true
    </isSupportDownloadStrategy>
    <isSupportDelayedUpgradeByDateTime>
      <!--ro, opt, bool, whether the device supports delaying the upgrade by time, desc:corresponding filed is "delayedUpgradeDateTime" of URL
(/ISAPI/System/onlineUpgrade/task?format=json)-->true
    </isSupportDelayedUpgradeByDateTime>
    <upgradePackageAddressType opt="URL, ftpIPAddress, ftpDomainAddress">
      <!--ro, opt, string, attr:opt{req, string}-->URL
    </upgradePackageAddressType>
  </UpgradePackageTask>
  <isSupportPatrolInspection>
    <!--ro, opt, bool, whether the device supports patrol inspection-->true
  </isSupportPatrolInspection>
  <isSupportBatchUpgrade>
    <!--ro, opt, bool, whether the device supports batch online upgrades-->true
  </isSupportBatchUpgrade>
  <isSupportCancelUpgrade>
    <!--ro, opt, bool, whether the device supports canceling online upgrade-->true
  </isSupportCancelUpgrade>
  <isSupportPackagePause>
    <!--ro, opt, bool, whether the device supports pausing upgrade package download, desc:/ISAPI/System/onlineUpgrade/PackagePause?format=json-->true
  </isSupportPackagePause>
  <isSupportPackageResume>
    <!--ro, opt, bool, whether the device supports resuming upgrade package download, desc:/ISAPI/System/onlineUpgrade/PackageResume?format=json-->true
  </isSupportPackageResume>
  <isSupportCancelUpgradeByTaskID>
    <!--ro, opt, bool, whether the device supports canceling online upgrade by task ID, desc:corresponding to the field assignByTaskID of operationType in
URL PUT  /ISAPI/System/onlineUpgrade/CancelUpgrade?format=json-->true
  </isSupportCancelUpgradeByTaskID>
</OnlineUpgradeCap>
```

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14.1.2.2 Get the online upgrade package downloading progress

Request URL

GET /ISAPI/System/onlineUpgrade/downloadPackage/status?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DownloadPackageStatus": {
    /*ro, req, object, upgrade package downloading status*/
        "status":  "notDownload",
        /*ro, req, enum, download status, subType:string, desc:"notDownload", "downloading", "downloadFailed", "pause", "finish", "incorrectPackage"
(incorrect upgrade package format), "hdOperationFailed" (HD operation failed)*/
        "total":  0.0,
        /*ro, opt, float, upgrade package total size, desc:unit: MB, corrects to one decimal place*/
        "remain":  0.0,
        /*ro, opt, float, remaining space, desc:unit: MB, corrects to one decimal place*/
        "speed":  0.0,
        /*ro, opt, float, download speed, desc:unit: MB, corrects to one decimal place*/
        "remainTime":  0,
        /*ro, opt, int, estimated remaining time, unit:s*/
        "progress":  0
        /*ro, req, int, progress, range:[0,100]*/
    }
}
```

14.1.2.3 Start to download upgrade package to device

Request URL

PUT /ISAPI/System/onlineUpgrade/downloadPackage?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.1.2.4 Get the online upgrade progress of device

Request URL

GET /ISAPI/System/onlineUpgrade/status

Query Parameter

Parameter Name	Parameter Type	Description
childDevID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, online upgrade status, attr:version{req, string, protocolVersion}-->
  <status>
    <!--ro, req, enum, online upgrade status, subType:string, desc:"notUpgrade" (not upgraded), "upgrading", "successful" (upgraded), "LanguageMismatch"
(language mismatch), "writeFlashError" (failed to write to flash), "packageTypeMismatch" (upgrade package type mismatch), "packageVersionMismatch" (upgrade
version mismatch), "netUnreachable" (network disconnected), "unknownError" (unknown error), "engineVersionMismatch" (mismatch of upgrade package engine
version)-->notUpgrade
  </status>
  <percent>
    <!--ro, req, int, current upgrade progress in percentage, range:[0,100]-->1
  </percent>
  <onlineUpdateDevInfoList>
    <!--ro, opt, object, online upgrade device information, desc:containing the online upgrade status of sub devices, sub modules, or sub channels-->
    <onlineUpdateDevInfo>
      <!--ro, opt, object, online upgrade device information-->
      <moduleType>
        <!--ro, req, enum, module type, subType:string, desc:"keypad", "detector", "cardReader" (card reader), "siren" (sounder), "relay", "transmitter"
(transmitter), "remoteCtrl" (remote control),"repeater" (repeater), "SSD", "extensionModule" (extension module)-->keypad
      </moduleType>
      <moduleID>
        <!--ro, req, int, module ID, range:[1,128], step:1-->1
      </moduleID>
      <model>
        <!--ro, opt, string, model, range:[1,32]-->test
      </model>
      <status>
        <!--ro, opt, enum, upgrade status, subType:string, desc:"allowance" (upgradeable, the upgrade command is not triggered), "ready" (waiting for
upgrade, the upgrade command has been triggered), "upgriding" (upgrading), "success" (upgraded), "failed" (upgrade failed)-->allowance
      </status>
      <percent>
        <!--ro, opt, int, current upgrade percentage, range:[0,100]-->1
      </percent>
      <updateTime>
        <!--ro, opt, datetime, upgrade completion time-->2004-05-03T17:30:08+08:00
      </updateTime>
    </onlineUpdateDevInfo>
  </onlineUpdateDevInfoList>
</OnlineUpgradeStatus>
```

14.1.2.5 Get all of the online upgrade task progress

Request URL

GET /ISAPI/System/onlineUpgrade/status/task?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "onlineUpgradeStatusList": [
    /*ro, req, array, online upgrade status list, subType:object*/
        {
            "status":  "notUpgrade",
            /*ro, req, enum, upgrade status, subType:string, desc:“notUpgrade”, “upgrading”, “successful” (upgrading succeeded), “languageMismatch”,
“writeFlashError”, “packageTypeMismatch”, “packageVersionMismatch”, “netUnreachable” (network disconnected), “unknownError”, “waitManualTrigger”*/
            "percent":  0,
            /*ro, req, int, upgrade progress percentage, range:[0,100]*/
            "taskID":  "test",
            /*ro, opt, string, task ID, range:[1,64], desc:generated when creating upgrade tasks, related URI: /ISAPI/System/onlineUpgrade/task?
format=json*/
            "moudleType":  "stationHostAndScreen",
            /*ro, opt, enum, component type, subType:string, desc:"stationHostAndScreen" (dock station upgrade package, including host and screen),
"stationHost" (dock station host), "spellingControllerMainAndSubBoard" (main board and sub board of splicing controller), "spellingControllerMainBoard"
(main board of splicing controller), "spellingControllerSubBoard" (sub board of splicing controller), "bodyCamera" (body camera)*/
            "packageName":  "ZCS-H1_target_V1.0.1_build201123.zip",
            /*ro, opt, string, upgrade package name, range:[1,128]*/
            "taskTime":  "1970-01-01T00:00:00+08:00",
            /*ro, opt, datetime, applying upgrade task time*/
            "taskParam": {
            /*ro, opt, object, task parameters, desc:corresponding URL: POST /ISAPI/System/onlineUpgrade/task?format=json*/
                "URL":  "test",
                /*ro, req, string, upgrade package address, range:[1,1024]*/
                "delayedUpgrade":  true,
                /*ro, opt, bool, whether to delay the upgrade, desc:if this node is not applied, it indicates that the upgrade will be started immediately;
if this node is applied and the value is false, it indicates that the upgrade will be delayed*/
                "delayedUpgradeDateTime":  "1970-01-01T00:00:00+08:00"
                /*ro, opt, datetime, time of starting the delayed upgrade, dep:or,{$.PackageDescription.delayedUpgrade,eq,false}*/
            }
        }
    ]
}
```

14.1.2.6 Apply the task of online upgrade

Request URL

POST /ISAPI/System/onlineUpgrade/task?format=json

Query Parameter

None

Request Message

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```
{
    "PackageDescription": {
    /*req, object, description of the upgrade package*/
        "name":  "DSJ_N3_GML_DEBUG_STD_1.2.0_190524.zip",
        /*opt, string, upgrade package name, range:[1,256]*/
        "URL":  "test",
        /*req, string, URL of the upgrade package, range:[1,1024], desc:the task ID is contained in the URL. The uploading and downloading of the upgrade package is based on the object storage protocol (version 2.0). If this node is null, the FTP address will take effect. If both the URL and FTP address are configured, the URL has priority over FTP address*/
        "MD5":  "test",
        /*opt, string, range:[0,256]*/
        "moduleType":  "cardReader",
        /*opt, enum, module type, subType:string, desc:“cardReader” (card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm programma of card reader), "dockStationHost" (dock station host), "dockStationScreen" (dock station screen), "stationHostAndScreen" (dock station host and screen)*/
        "moduleID":  1,
        /*opt, int, module ID*/
        "spellingControllerSubBoardList": [
        /*opt, array, splicing controller sub board list, subType:object, range:[1,1024], dep:or, {$.PackageDescription.moduleType,eq,spellingControllerMainAndSubBoard},{$.PackageDescription.moduleType,eq,spellingControllerSubBoard}, desc:it is valid when the value of the node moduleType is "spellingControllerMainAndSubBoard" or "spellingControllerSubBoard". If it is not configured, it indicates that all the sub boards will be updated*/
            {
                "subBoardID":  1,
                /*opt, int, sub board ID, range:[1,1024]*/
                "ipV4Address":  "test"
                /*ro, opt, string, IPv4, range:[1,16], desc:when an exception occurs on the main board, the sub board can be updated through the IP address. The default sub board interface will be used*/
            }
        ],
        "channel":  1,
        /*opt, int, channel No., desc:this node is used for upgrading the IP channel of the NVR*/
        "childDevID":  "3434645611",
        /*opt, string, range:[9,32]*/
        "customTaskID":  "test",
        /*opt, string, user-defined task ID, range:[1,128], desc:if this node exists, it will be returned in the result*/
        "delayedUpgrade":  true,
        /*opt, bool, whether to delay upgrading the package, desc:if this node is not applied, it indicates that the package will be upgraded immediately; if this node is applied and the value is false, it indicates that the upgrade will be delayed. User can decide the time to upgrade the package of the dock station or body camera if the upgrade is delayed*/
        "delayedUpgradeDateTime":  "1970-01-01T00:00:00+08:00",
        /*opt, datetime, time of starting the delayed upgrade, dep:or,{$.PackageDescription.delayedUpgrade,eq,false}, desc:the related capability is "isSupportDelayedUpgradeByDateTime" in GET  /ISAPI/System/onlineUpgrade/capabilities*/
        "ftpAddress":  "test",
        /*opt, string, FTP address of the upgrade package, desc:this node will take effect when URL is null and the format is: ftp://[username]:[password]@[IP address]:[port No.]*/
        "downloadStrategy": {
        /*opt, object, device's upgrade package downloading strategy parameters*/
            "sequelEnabled":  true,
            /*opt, bool, whether to enable ANR (Automatic Network Replenishment), desc:if enabled, when downloading is interrupted, the device will try downloading the package again; if disabled, the default strategy of the device will be adopted*/
            "sequelTimeout":  3000,
            /*opt, int, timeout of a single ANR attempt, step:1, unit:ms, desc:when a single ANR attempt exceeds the timeout period but the download does not continue, the ANR attempt will be recognized as failed*/
            "maxRetryTimes":  5
            /*opt, int, ANR maximum attempt times, desc:when the device ANR exceeds the maximum attempt times but the download does not continue, the download and the upgrade will be recognized as failed*/
        },
        "upgradePackageURLType":  "full"
        /*opt, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1-succeeded, other value-error occurred. This field is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"OK" (succeeded), other value (error occurred). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded), other value (error occurred). This field is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error message, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "test"
    /*ro, opt, string, ID of the upgrade task, desc:it can be used to get the status and progress of the task*/
}
```

14.1.3 Port Settings

14.1.3.1 Get the serial port capability of the device

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Request URL

GET /ISAPI/System/Serial/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SerialCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, range of RS-485 serial port numbers supported by the device, attr:version{req, string, protocolVersion}-->
  <rs485PortNums>
    <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->0
  </rs485PortNums>
  <supportRS232Config>
    <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
  </supportRS232Config>
  <rs422PortNums>
    <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->0
  </rs422PortNums>
  <rs232PortNums>
    <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->0
  </rs232PortNums>
  <rs485WorkMode opt="Led, CaptureTrigger,transparent">
    <!--ro, opt, string, range of RS-485 serial port No. supported by the device, attr:opt{req, string}-->test
  </rs485WorkMode>
  <rs232SerialNumber opt="1,2,3">
    <!--ro, opt, int, range of RS-232 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs232SerialNumber>
  <rs485SerialNumber opt="4,5,6">
    <!--ro, opt, int, range of RS-485 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs485SerialNumber>
  <isSupportAuthenticationService>
    <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
  </isSupportAuthenticationService>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports configuring the information about the serial port device-->true
  </isSupportDeviceInfo>
  <isSupportSearchDeviceInfoRelations>
    <!--ro, opt, bool, whether the device supports searching the linkage information of the serial port device-->true
  </isSupportSearchDeviceInfoRelations>
</SerialCap>
```

14.1.3.2 Get the capability of a specific serial port

Request URL

GET /ISAPI/System/Serial/ports/<portID>/capabilities?permissionController=<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port No., attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID-->0
  </id>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <encryptKey min="0" max="10">
    <!--ro, opt, string, key, attr:min{req, int},max{req, int}, desc:key-->test
  </encryptKey>
  <serialNumber opt="1,2,3,4,5">
    <!--ro, opt, int, serial No., attr:opt{req, string}-->0
```

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```
    </serialNumber>
    <serialPortType opt="RS485,RS422,RS232">
      <!--ro, opt, string, serial port type, attr:opt{req, string}-->RS485
    </serialPortType>
    <serialAddress min="0" max="10">
      <!--ro, opt, int, serial port address, attr:min{req, int},max{req, int}, desc:serial port address-->1
    </serialAddress>
    <duplexMode opt="half,full">
      <!--ro, opt, string, duplex mode of the serial port, attr:opt{req, string}-->half
    </duplexMode>
    <direction opt="monodirectional,bdirectional">
      <!--ro, opt, string, attr:opt{req, string}-->monodirectional
    </direction>
    <baudRate opt="1200,2400,4800,9600,19200,38400,57600,115200">
      <!--ro, opt, int, attr:opt{req, string}-->1200
    </baudRate>
    <dataBits opt="6,7,8">
      <!--ro, opt, int, attr:opt{req, string}-->6
    </dataBits>
    <parityType opt="none,even,odd,mark,space">
      <!--ro, opt, string, attr:opt{req, string}-->none
    </parityType>
    <stopBits opt="1,1.5,2">
      <!--ro, opt, string, stop bit, attr:opt{req, string}-->1
    </stopBits>
    <workMode opt="console,transparent,audiomixe,screenCtrl,ptzCtrl,keyboard,matrix,audioMixers">
      <!--ro, opt, string, working mode, attr:opt{req, string}-->console
    </workMode>
    <flowCtrl opt="none,software,hardware">
      <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
    </flowCtrl>
    <deviceName min="0" max="32">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </deviceName>
    <deviceProtocol min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </deviceProtocol>
    <mode opt="readerMode,clientMode,externMode,stairsControl,accessControlHost,disabled,custom,cardReceiver,QRCodeReader">
      <!--ro, opt, string, working mode, attr:opt{req, string}-->readerMode
    </mode>
    <modeChangeReboot>
      <!--ro, opt, bool-->true
    </modeChangeReboot>
    <outputDataType opt="cardNo,employeeNo,auto">
      <!--ro, opt, string, output data type, attr:opt{req, string}, desc:data type output from the door station to the elevator controller: floorNumber (floor
No.,default),cardNo (card No.)-->cardNo
    </outputDataType>
    <copyToAll opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </copyToAll>
    <isVariable opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </isVariable>
    <PTZInfo>
      <!--ro, opt, object-->
      <bindChannelNo min="0" max="10">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->0
      </bindChannelNo>
      <PTZProtocolList>
        <!--ro, req, object-->
        <PTZProtocol>
          <!--ro, opt, object-->
          <id>
            <!--ro, req, int, protocol index-->0
          </id>
          <protoDesc>
            <!--ro, req, string, protocol description-->test
          </protoDesc>
        </PTZProtocol>
      </PTZProtocolList>
    </PTZInfo>
    <ScreenCtrlInfo>
      <!--ro, opt, object, ScreenCtrlInfo, desc:this field is valid when the value of workMode is "screenCtrl"-->
      <ScreenCtrlProtoList>
        <!--ro, req, object, ScreenCtrlProtoList-->
        <ScreenCtrlProto>
          <!--ro, opt, object, ScreenCtrlProto-->
          <id>
            <!--ro, req, int, protocol index-->0
          </id>
          <protoDesc>
            <!--ro, req, string, protocol description-->test
          </protoDesc>
        </ScreenCtrlProto>
      </ScreenCtrlProtoList>
    </ScreenCtrlInfo>
    <MatrixCtrlInfo>
      <!--ro, opt, object, MatrixCtrlInfo-->
      <MatrixCtrlProtoList>
        <!--ro, req, object, MatrixCtrlProtoList-->
        <MatrixCtrlProto>
          <!--ro, opt, object, MatrixCtrlProto-->
          <id>
```

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          <!--ro, req, int, protocol index-->0
        </id>
        <protoDesc>
          <!--ro, req, string, protocol description-->test
        </protoDesc>
      </MatrixCtrlProto>
    </MatrixCtrlProtoList>
  </MatrixCtrlInfo>
  <KeyBoardCtrlInfo>
    <!--ro, opt, object, KeyBoardCtrlInfo-->
    <KeyBoardCtrlProtoList>
      <!--ro, req, object, KeyBoardCtrlProtoList-->
      <KeyBoardCtrlProto>
        <!--ro, opt, object, KeyBoardCtrlProto-->
        <id>
          <!--ro, req, int, protocol index-->0
        </id>
        <protoDesc>
          <!--ro, req, string, protocol description-->test
        </protoDesc>
      </KeyBoardCtrlProto>
    </KeyBoardCtrlProtoList>
  </KeyBoardCtrlInfo>
  <ControlAddress>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <address min="0" max="255">
      <!--ro, req, int, IP address, attr:min{req, int},max{req, int}-->0
    </address>
  </ControlAddress>
  <rs485WorkMode
opt="Led,sensor,CaptureTrigger,transparent,GPS,ModbusRTU,supplementLight,audibleVisualAlarm,columnSpeaker,commModeUsing433MHz,signalLampControl">
    <!--ro, opt, string, working mode of RS-485 serial port, dep:and,{$.SerialPortList.SerialPort.serialPortType,eq,RS485}, attr:opt{req, string},
desc:working mode of RS-485 serial port,which is used for LED display or triggering transmission of captured pictures. This node is valid only when
<serialPortType> is "RS485"-->sensor
  </rs485WorkMode>
  <SensorCtrlList>
    <!--ro, opt, object-->
    <Sensor>
      <!--ro, opt, object-->
      <id min="1" max="3">
        <!--ro, req, int, index, attr:min{req, int},max{req, int}-->1
      </id>
      <name min="1" max="32">
        <!--ro, req, string, attr:min{req, int},max{req, int}-->visibilitySensor
      </name>
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable, attr:opt{req, string}, desc:whether to enable the serial port: "true,false"-->false
      </enabled>
    </Sensor>
  </SensorCtrlList>
  <stairsControl>
    <!--ro, opt, object, elevator control parameters-->
    <outputDataType opt="floorNumber,cardNo">
      <!--ro, opt, enum, data type output from the door station to the elevator controller, subType:string, attr:opt{req, string}, desc:floorNumber (floor
No.,default),cardNo (card No.)-->floorNumber
    </outputDataType>
  </stairsControl>
  <comMode opt="active,passive">
    <!--ro, opt, enum, communication mode, subType:string, attr:opt{req, string}, desc:communication mode-->active
  </comMode>
  <ModbusRTU>
    <!--ro, opt, object-->
    <ModbusRTUBaudRate opt="1200,2400,4800,9600,19200" def="9600">
      <!--ro, opt, int, attr:opt{req, string},def{req, string}-->9600
    </ModbusRTUBaudRate>
    <ModbusRTUDataBits opt="5,6,7,8" def="8">
      <!--ro, opt, int, attr:opt{req, string},def{req, string}-->6
    </ModbusRTUDataBits>
    <ModbusRTUParityType opt="none,even,odd" def="none">
      <!--ro, opt, string, attr:opt{req, string},def{req, string}-->none
    </ModbusRTUParityType>
    <ModbusRTUStopBits opt="1,2" def="1">
      <!--ro, opt, string, stop bit, attr:opt{req, string},def{req, string}-->1
    </ModbusRTUStopBits>
    <ModbusRTUFlowCtrl opt="none,software,hardware">
      <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
    </ModbusRTUFlowCtrl>
    <decoderAddress min="0" max="255" def="0">
      <!--ro, opt, int, decoder IP address, attr:min{req, int},max{req, int},def{req, int}-->1
    </decoderAddress>
    <online opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </online>
  </ModbusRTU>
  <accessName min="1" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </accessName>
  <serialPortUse opt="acquAndCtrl,swipe,strongElectric">
    <!--ro, opt, string, attr:opt{req, string}-->acquisition
  </serialPortUse>
  <acquisitionProtocol opt="modbus,custom">
```

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    <acquisitionProtocol opt="Modbus,Custom">
      <!--ro, opt, string, dep:and,{$.SerialPort.serialPortUse,eq,acquisition}, attr:opt{req, string}-->modbus
    </acquisitionProtocol>
    <serialProtocol opt="FECbus,GB26875">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->GB26875
    </serialProtocol>
    <SupplementLightList size="5">
      <!--ro, opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}, attr:size{req, int}-->
      <SupplementLight>
        <!--ro, opt, object-->
        <SupplementLightID min="1" max="5">
          <!--ro, req, int, range:[1,5], attr:min{req, int},max{req, int}-->1
        </SupplementLightID>
      </SupplementLight>
    </SupplementLightList>
    <permissionController opt="0,1">
      <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
    </permissionController>
    <deviceModel min="0" max="32">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </deviceModel>
    <deviceSoftwareVersion min="0" max="32">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </deviceSoftwareVersion>
    <audibleVisualAlarmType opt="CIRCLE,AFE80,AFE82,XY">
      <!--ro, opt, enum, CIRCLE; AFE80; AFE82; XY, subType:string, attr:opt{req, string}, desc:CIRCLE; AFE80; AFE82; XY-->CIRCLE
    </audibleVisualAlarmType>
    <cardReaderType opt="K1108AM">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </cardReaderType>
    <strongElectricType opt="19M04-RO/T,19M08-RO/T">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </strongElectricType>
    <signalLampControl>
      <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,signalLampControl}-->
      <protocolType opt="GB920,Hisense,LESER,JERE">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->GB920
      </protocolType>
    </signalLampControl>
  </SerialPort>
```

14.1.3.3 Set the parameters of a specific serial port supported by the device

Request URL

PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	--
childDevID	string	--
deviceIndex	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, serial port ID-->0
  </id>
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
  <encryptKey>
    <!--opt, string-->test
  </encryptKey>
  <serialNumber>
    <!--opt, int, serial No.-->0
  </serialNumber>
  <serialPortType>
    <!--opt, enum, serial port type: "RS485","RS422","RS232", subType:string, desc:serial port type: "RS485","RS422","RS232"-->RS485
  </serialPortType>
  <serialAddress>
    <!--opt, int-->1
  </serialAddress>
  <duplexMode>
    <!--opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"-->half
```

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  </duplexMode>
  <direction>
    <!--opt, enum, "monodirectional,bdirectional", subType:string, desc:"monodirectional, bdirectional"-->monodirectional
  </direction>
  <baudRate>
    <!--opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--opt, int-->6
  </dataBits>
  <parityType>
    <!--opt, enum, parity type, subType:string, desc:"none, even, odd, mark, space"-->none
  </parityType>
  <stopBits>
    <!--opt, string, stop bit: "1,1.5,2"-->1
  </stopBits>
  <workMode>
    <!--opt, enum, work mode, subType:string, desc:working mode: "console","transparent","audiomixer","stairsControl"-elevator control,"cardReader"-card
reader,"disabled","custom". This node is required only when <serialPortType> is set to "RS232"-->console
  </workMode>
  <flowCtrl>
    <!--opt, enum, "none,software,hardware", subType:string, desc:"none, software, hardware"-->none
  </flowCtrl>
  <rs485WorkMode>
    <!--opt, enum, working mode of RS-485 serial port, which is used for LED display or triggering transmission of captured pictures: "Led, CaptureTrigger".
This node is valid only when <serialPortType> is "RS485", subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}, desc:working mode of RS-485 serial
port,which is used for LED display or triggering transmission of captured pictures: "Led,CaptureTrigger". This node is valid only when <serialPortType> is
"RS485"-->sensor
  </rs485WorkMode>
  <copyToAll>
    <!--opt, bool-->true
  </copyToAll>
  <deviceName>
    <!--ro, opt, string-->test
  </deviceName>
  <deviceProtocol>
    <!--ro, opt, int-->1
  </deviceProtocol>
  <mode>
    <!--opt, enum, work mode, subType:string, desc:deq,working mode: "readerMode,clientMode,externMode,accessControlHost,disabled",this node is valid only
when <serialPortType> is "RS485"-->readerMode
  </mode>
  <outputDataType>
    <!--opt, enum, output data type, subType:string, dep:and,{$.SerialPort.mode,eq,accessControlHost}, desc:"cardNo,employeeNo", this node is valid when
<mode>is "accessControlHost"-->cardNo
  </outputDataType>
  <isVariable>
    <!--opt, bool-->true
  </isVariable>
  <PTZInfo>
    <!--opt, object, PTZ information-->
    <bindChannelNo>
      <!--req, int-->0
    </bindChannelNo>
    <PTZProtocolList>
      <!--req, object-->
      <PTZProtocol>
        <!--opt, object-->
        <id>
          <!--req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--req, string-->test
        </protoDesc>
      </PTZProtocol>
    </PTZProtocolList>
  </PTZInfo>
  <ScreenCtrlInfo>
    <!--opt, object-->
    <ScreenCtrlProtoList>
      <!--req, object-->
      <ScreenCtrlProto>
        <!--opt, object-->
        <id>
          <!--req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--req, string-->test
        </protoDesc>
      </ScreenCtrlProto>
    </ScreenCtrlProtoList>
  </ScreenCtrlInfo>
  <MatrixCtrlInfo>
    <!--opt, object-->
    <MatrixCtrlProtoList>
      <!--req, object-->
      <MatrixCtrlProto>
        <!--opt, object-->
        <id>
          <!--req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--req, string-->test
```

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        </protoDesc>
      </MatrixCtrlProto>
    </MatrixCtrlProtoList>
  </MatrixCtrlInfo>
  <KeyBoardCtrlInfo>
    <!--opt, object-->
    <KeyBoardCtrlProtoList>
      <!--req, object-->
      <KeyBoardCtrlProto>
        <!--opt, object-->
        <id>
          <!--req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--req, string-->test
        </protoDesc>
      </KeyBoardCtrlProto>
    </KeyBoardCtrlProtoList>
  </KeyBoardCtrlInfo>
  <ControlAddress>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <address>
      <!--req, int, address, range:[0,255]-->1
    </address>
  </ControlAddress>
  <SensorCtrlList>
    <!--opt, object-->
    <Sensor>
      <!--opt, object-->
      <id>
        <!--req, int, ID, range:[1,3]-->1
      </id>
      <name>
        <!--req, string, sensor name, range:[1,32]-->visibilitySensor
      </name>
      <enabled>
        <!--req, bool, whether to enable the function, desc:whether to enable the serial port: "true,false"-->false
      </enabled>
    </Sensor>
  </SensorCtrlList>
  <stairsControl>
    <!--opt, object, elevator control parameters, this node is valid when the value of the node <mode> is stairsControl, dep:and,
{$.SerialPort.mode,eq,stairsControl}-->
    <outputDataType>
      <!--opt, enum, output data type, subType:string, desc:"cardNo,employeeNo", this node is valid when <mode>is "accessControlHost"-->floorNumber
    </outputDataType>
  </stairsControl>
  <comMode>
    <!--opt, enum, subType:string-->active
  </comMode>
  <ModbusRTU>
    <!--opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
    <ModbusRTUBaudRate>
      <!--opt, enum, subType:string-->9600
    </ModbusRTUBaudRate>
    <ModbusRTUDataBits>
      <!--opt, enum, subType:int-->6
    </ModbusRTUDataBits>
    <ModbusRTUParityType>
      <!--opt, enum, subType:string-->none
    </ModbusRTUParityType>
    <ModbusRTUStopBits>
      <!--opt, enum, subType:string-->1
    </ModbusRTUStopBits>
    <ModbusRTUFlowCtrl>
      <!--opt, enum, subType:string-->none
    </ModbusRTUFlowCtrl>
    <decoderAddress>
      <!--opt, int, range:[0,255]-->0
    </decoderAddress>
  </ModbusRTU>
  <accessName>
    <!--opt, string, range:[1,32]-->test
  </accessName>
  <serialPortUse>
    <!--opt, enum, subType:string-->acquisition
  </serialPortUse>
  <acquisitionProtocol>
    <!--opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,acquisition}-->modbus
  </acquisitionProtocol>
  <serialProtocol>
    <!--opt, enum, subType:string-->GB26875
  </serialProtocol>
  <SupplementLightList>
    <!--opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}-->
    <SupplementLight>
      <!--opt, object-->
      <SupplementLightID>
        <!--req, int, range:[1,5]-->1
      </SupplementLightID>
    </SupplementLight>
```

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```
  </SupplementLightList>
  <cardReaderType>
    <!--opt, enum, subType:string-->K1108AM
  </cardReaderType>
</SerialPort>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error reason description in detail, desc:error reason description in detail-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.3.4 Get the parameters of a specific port supported by the device

Request URL

GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	0 (main permission controller), 1 (sub permission controller) When this parameter does not exist, it indicates getting the serial port information of the main permission controller (i.e., the device serial port information).
childDevID	string	--
deviceIndex	string	For C60S devices connected to multiple LED sending cards, the URL needs to include deviceIndex= to distinguish which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, serial port ID-->0
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable the serial port-->true
  </enabled>
  <encryptKey>
    <!--ro, opt, string-->test
  </encryptKey>
  <serialNumber>
    <!--ro, opt, int, serial No.-->0
  </serialNumber>
  <serialPortType>
    <!--ro, opt, enum, serial port type, subType:string, desc:"RS485", "RS422", "RS232”-->RS485
  </serialPortType>
  <serialAddress>
```

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```
    <!--ro, opt, int-->1
  </serialAddress>
  <duplexMode>
    <!--ro, opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"-->half
  </duplexMode>
  <direction>
    <!--ro, opt, enum, "monodirectional,bdirectional", subType:string, desc:"monodirectional", "bdirectional"-->monodirectional
  </direction>
  <baudRate>
    <!--ro, opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--ro, opt, int-->6
  </dataBits>
  <parityType>
    <!--ro, opt, enum, parity type, subType:string, desc:parity type: "none,even,odd,mark,space"-->none
  </parityType>
  <stopBits>
    <!--ro, opt, enum, stop bit, subType:string, desc:stop bit-->1
  </stopBits>
  <workMode>
    <!--ro, opt, enum, working mode, subType:string, desc:"console", "transparent", "audiomixer", "screenCtrl", "ptzCtrl", "keyboard", "matrix",
"audioMixers"-->console
  </workMode>
  <flowCtrl>
    <!--ro, opt, enum, "none,software,hardware", subType:string, desc:"none", "software", "hardware"-->none
  </flowCtrl>
  <rs485WorkMode>
    <!--ro, opt, enum, working mode of RS-485 serial port, subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}, desc:working mode of RS-485 serial
port,which is used for LED display or triggering transmission of captured pictures: "Led,CaptureTrigger". This node is valid only when <serialPortType> is
"RS485"-->sensor
  </rs485WorkMode>
  <copyToAll>
    <!--ro, opt, bool-->true
  </copyToAll>
  <deviceName>
    <!--ro, opt, string-->test
  </deviceName>
  <deviceProtocol>
    <!--ro, opt, int-->1
  </deviceProtocol>
  <mode>
    <!--ro, opt, enum, working mode, subType:string, desc:deq,working mode: "readerMode,clientMode,externMode,accessControlHost,disabled",this node is valid
only when <serialPortType> is "RS485"-->readerMode
  </mode>
  <outputDataType>
    <!--ro, opt, enum, output data type, subType:string, dep:and,{$.SerialPort.mode,eq,accessControlHost}, desc:output data type: "cardNo,employeeNo",this
node is valid when <mode>is "accessControlHost"-->cardNo
  </outputDataType>
  <isVariable>
    <!--ro, opt, bool-->true
  </isVariable>
  <PTZInfo>
    <!--ro, opt, object, PTZ information-->
    <bindChannelNo>
      <!--ro, req, int-->0
    </bindChannelNo>
    <PTZProtocolList>
      <!--ro, req, object-->
      <PTZProtocol>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--ro, req, string-->test
        </protoDesc>
      </PTZProtocol>
    </PTZProtocolList>
  </PTZInfo>
  <ScreenCtrlInfo>
    <!--ro, opt, object-->
    <ScreenCtrlProtoList>
      <!--ro, req, object-->
      <ScreenCtrlProto>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--ro, req, string-->test
        </protoDesc>
      </ScreenCtrlProto>
    </ScreenCtrlProtoList>
  </ScreenCtrlInfo>
  <MatrixCtrlInfo>
    <!--ro, opt, object-->
    <MatrixCtrlProtoList>
      <!--ro, req, object-->
      <MatrixCtrlProto>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, serial port ID-->0
```

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```
        </id>
        <protoDesc>
          <!--ro, req, string-->test
        </protoDesc>
      </MatrixCtrlProto>
    </MatrixCtrlProtoList>
  </MatrixCtrlInfo>
  <KeyBoardCtrlInfo>
    <!--ro, opt, object-->
    <KeyBoardCtrlProtoList>
      <!--ro, req, object-->
      <KeyBoardCtrlProto>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, serial port ID-->0
        </id>
        <protoDesc>
          <!--ro, req, string-->test
        </protoDesc>
      </KeyBoardCtrlProto>
    </KeyBoardCtrlProtoList>
  </KeyBoardCtrlInfo>
  <ControlAddress>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable the serial port-->true
    </enabled>
    <address>
      <!--ro, req, int, address, range:[0,255]-->0
    </address>
  </ControlAddress>
  <SensorCtrlList>
    <!--ro, opt, object-->
    <Sensor>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int, serial port ID, range:[1,10]-->1
      </id>
      <name>
        <!--ro, req, enum, sensor name, subType:string, desc:sensor name-->visibilitySensor
      </name>
      <enabled>
        <!--ro, req, bool, whether to enable the function, desc:whether to enable the serial port: "true,false"-->false
      </enabled>
      <senorOnlineStatus>
        <!--ro, opt, enum, subType:string-->online
      </senorOnlineStatus>
    </Sensor>
  </SensorCtrlList>
  <stairsControl>
    <!--ro, opt, object, elevator control parameters, dep:and,{$.SerialPort.mode,eq,stairsControl}-->
    <outputDataType>
      <!--ro, opt, enum, output data type, subType:string, desc:output data type: "cardNo,employeeNo",this node is valid when <mode>is "accessControlHost"--
>floorNumber
    </outputDataType>
  </stairsControl>
  <comMode>
    <!--ro, opt, enum, subType:string-->active
  </comMode>
  <ModbusRTU>
    <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
    <ModbusRTUBaudRate>
      <!--ro, opt, enum, subType:string-->9600
    </ModbusRTUBaudRate>
    <ModbusRTUDataBits>
      <!--ro, opt, enum, subType:int-->6
    </ModbusRTUDataBits>
    <ModbusRTUParityType>
      <!--ro, opt, enum, subType:string-->none
    </ModbusRTUParityType>
    <ModbusRTUStopBits>
      <!--ro, opt, enum, subType:string-->1
    </ModbusRTUStopBits>
    <ModbusRTUFlowCtrl>
      <!--ro, opt, enum, subType:string-->none
    </ModbusRTUFlowCtrl>
    <decoderAddress>
      <!--ro, opt, int, range:[0,255]-->0
    </decoderAddress>
    <online>
      <!--ro, opt, bool-->true
    </online>
  </ModbusRTU>
  <accessName>
    <!--ro, opt, string, range:[1,32]-->test
  </accessName>
  <serialPortUse>
    <!--ro, opt, enum, subType:string-->acquAndCtrl
  </serialPortUse>
  <serialProtocol>
    <!--ro, opt, enum, subType:string-->GB26875
  </serialProtocol>
  <SupplementLightList>
    <!--ro, opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}-->
```

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```
    <SupplementLight>
      <!--ro, opt, object-->
      <SupplementLightID>
        <!--ro, req, int, range:[1,5]-->1
      </SupplementLightID>
    </SupplementLight>
  </SupplementLightList>
  <cardReaderType>
    <!--ro, opt, enum, subType:string-->K1108AM
  </cardReaderType>
  <strongElectricType>
    <!--ro, opt, enum, subType:string-->19M04-RO/T
  </strongElectricType>
  <signalLampControl>
    <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,signalLampControl}-->
    <protocolType>
      <!--ro, opt, enum, subType:string-->GB920
    </protocolType>
  </signalLampControl>
  <passingMode>
    <!--ro, opt, enum, subType:string, dep:or,{$.SerialPort.mode,eq,fingerprintReader}-->entrance
  </passingMode>
  <modbusProtocol>
    <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
    <checkBitOrder>
      <!--ro, opt, enum, subType:string-->bigEndian
    </checkBitOrder>
    <deviceMode>
      <!--ro, opt, enum, subType:string-->subordinate
    </deviceMode>
  </modbusProtocol>
  <serialPortName>
    <!--ro, opt, string, range:[1,32]-->test
  </serialPortName>
  <accessProtocolType>
    <!--ro, opt, enum, subType:string-->PELCO
  </accessProtocolType>
  <weightSenseInfo>
    <!--ro, opt, object-->
    <countTimes>
      <!--ro, opt, enum, subType:float-->0.01
    </countTimes>
    <countUnit>
      <!--ro, opt, enum, subType:string-->kg
    </countUnit>
  </weightSenseInfo>
  <PDUCtrlType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PDUCtrl}-->DS-65A2T0008
  </PDUCtrlType>
  <paperLessOfficeType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,paperless}-->DS-65PL0015-T
  </paperLessOfficeType>
  <videoWallControllerlType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,spellingCtrl}-->matrixProtocol
  </videoWallControllerlType>
  <audioCtrlType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,audioCtrl}-->DS-QAPDM16081C
  </audioCtrlType>
  <interactiveTabletType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,interactiveTablet}-->interactiveTabletProtocol
  </interactiveTabletType>
  <humanSensorType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,humanSensor}-->HM-TD1017-2/QW-HS121
  </humanSensorType>
  <transparentOverWebsocket>
    <!--ro, opt, bool-->true
  </transparentOverWebsocket>
  <PTZCtrlType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PTZCtrl}-->VISCA
  </PTZCtrlType>
  <airConditionerCtrlType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,airConditionerCtrl}-->GL
  </airConditionerCtrlType>
  <acquisitionLinkageType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,acquisitionLinkage}-->general
  </acquisitionLinkageType>
  <externalInterfaceBoard>
    <!--ro, opt, object, dep:or,{$.SerialPort.mode,eq,externalInterfaceBoard}-->
    <serialPortList>
      <!--ro, opt, array, subType:object-->
      <serialPort>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->0
        </id>
        <serialPortType>
          <!--ro, opt, enum, subType:string-->RS485
        </serialPortType>
        <RS485serialProtocol>
          <!--ro, opt, enum, subType:string-->private485
        </RS485serialProtocol>
        <private485Mode>
          <!--ro, opt, enum, subType:string-->cardReceiver
```

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```
        </private485Mode>
        <RS232Mode>
          <!--ro, opt, enum, subType:string-->readerMode
        </RS232Mode>
      </serialPort>
    </serialPortList>
  </externalInterfaceBoard>
</SerialPort>
```

14.1.3.5 Get the list of serial ports supported by the device

Request URL

GET /ISAPI/System/Serial/ports?permissionController=<indexID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	0 (main permission controller), 1 (sub permission controller) If this parameter does not exist, it indicates getting the serial port information of main permission controller, that is, the serial port information of the device.
deviceIndex	string	For C60S device connecting multiple LED sending cards, the URL needs to include the parameter deviceIndex= to specify which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPortList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <SerialPort>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->0
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <encryptKey>
      <!--ro, opt, string-->test
    </encryptKey>
    <serialNumber>
      <!--ro, opt, int-->0
    </serialNumber>
    <serialPortType>
      <!--ro, opt, enum, subType:string-->RS485
    </serialPortType>
    <serialAddress>
      <!--ro, opt, int-->1
    </serialAddress>
    <duplexMode>
      <!--ro, opt, enum, subType:string-->half
    </duplexMode>
    <direction>
      <!--ro, opt, enum, subType:string-->monodirectional
    </direction>
    <baudRate>
      <!--ro, opt, enum, subType:int-->2400
    </baudRate>
    <dataBits>
      <!--ro, opt, int-->6
    </dataBits>
    <parityType>
      <!--ro, opt, enum, subType:string-->none
    </parityType>
    <stopBits>
      <!--ro, opt, enum, subType:string-->1
    </stopBits>
    <workMode>
      <!--ro, opt, enum, subType:string-->console
    </workMode>
    <flowCtrl>
      <!--ro, opt, enum, subType:string-->none
    </flowCtrl>
```

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```
<rs485WorkMode>
  <!--ro, opt, enum, subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}-->sensor
</rs485WorkMode>
<copyToAll>
  <!--ro, opt, bool-->true
</copyToAll>
<deviceName>
  <!--ro, opt, string-->test
</deviceName>
<deviceProtocol>
  <!--ro, opt, int-->1
</deviceProtocol>
<mode>
  <!--ro, opt, enum, subType:string-->readerMode
</mode>
<outputDataType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.mode,eq,accessControlHost}-->cardNo
</outputDataType>
<isVariable>
  <!--ro, opt, bool-->true
</isVariable>
<PTZInfo>
  <!--ro, opt, object-->
  <bindChannelNo>
    <!--ro, req, int-->0
  </bindChannelNo>
  <PTZProtocolList>
    <!--ro, req, object-->
    <PTZProtocol>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->0
      </id>
      <protoDesc>
        <!--ro, req, string-->test
      </protoDesc>
    </PTZProtocol>
  </PTZProtocolList>
</PTZInfo>
<ScreenCtrlInfo>
  <!--ro, opt, object-->
  <ScreenCtrlProtoList>
    <!--ro, req, object-->
    <ScreenCtrlProto>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->0
      </id>
      <protoDesc>
        <!--ro, req, string-->test
      </protoDesc>
    </ScreenCtrlProto>
  </ScreenCtrlProtoList>
</ScreenCtrlInfo>
<MatrixCtrlInfo>
  <!--ro, opt, object-->
  <MatrixCtrlProtoList>
    <!--ro, req, object-->
    <MatrixCtrlProto>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->0
      </id>
      <protoDesc>
        <!--ro, req, string-->test
      </protoDesc>
    </MatrixCtrlProto>
  </MatrixCtrlProtoList>
</MatrixCtrlInfo>
<KeyBoardCtrlInfo>
  <!--ro, opt, object-->
  <KeyBoardCtrlProtoList>
    <!--ro, req, object-->
    <KeyBoardCtrlProto>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->0
      </id>
      <protoDesc>
        <!--ro, req, string-->test
      </protoDesc>
    </KeyBoardCtrlProto>
  </KeyBoardCtrlProtoList>
</KeyBoardCtrlInfo>
<ControlAddress>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <address>
    <!--ro, req, int, range:[0,255]-->1
  </address>
</ControlAddress>
```

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```
<SensorCtrlList>
  <!--ro, opt, object-->
  <Sensor>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int, range:[1,10]-->1
    </id>
    <name>
      <!--ro, req, enum, subType:string-->visibilitySensor
    </name>
    <enabled>
      <!--ro, req, bool-->false
    </enabled>
    <senorOnlineStatus>
      <!--ro, opt, enum, subType:string-->online
    </senorOnlineStatus>
  </Sensor>
</SensorCtrlList>
<stairsControl>
  <!--ro, opt, object, dep:and,{$.SerialPort.mode,eq,stairsControl}-->
  <outputDataType>
    <!--ro, opt, enum, subType:string-->floorNumber
  </outputDataType>
</stairsControl>
<comMode>
  <!--ro, opt, enum, subType:string-->active
</comMode>
<ModbusRTU>
  <!--ro, opt, object, dep:and,{$.SerialPortList.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
  <ModbusRTUBaudRate>
    <!--ro, opt, enum, subType:string-->9600
  </ModbusRTUBaudRate>
  <ModbusRTUDataBits>
    <!--ro, opt, enum, subType:int-->6
  </ModbusRTUDataBits>
  <ModbusRTUParityType>
    <!--ro, opt, enum, subType:string-->none
  </ModbusRTUParityType>
  <ModbusRTUStopBits>
    <!--ro, opt, enum, subType:string-->1
  </ModbusRTUStopBits>
  <ModbusRTUFlowCtrl>
    <!--ro, opt, enum, subType:string-->none
  </ModbusRTUFlowCtrl>
  <decoderAddress>
    <!--ro, opt, int, range:[0,255]-->0
  </decoderAddress>
  <online>
    <!--ro, opt, bool-->true
  </online>
</ModbusRTU>
<accessName>
  <!--ro, opt, string, range:[1,32]-->test
</accessName>
<serialPortUse>
  <!--ro, opt, enum, subType:string-->acquAndCtrl
</serialPortUse>
<serialProtocol>
  <!--ro, opt, enum, subType:string-->GB26875
</serialProtocol>
<accessProtocolType>
  <!--ro, opt, enum, subType:string-->PELCO
</accessProtocolType>
<SupplementLightList>
  <!--ro, opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}-->
  <SupplementLight>
    <!--ro, opt, object-->
    <SupplementLightID>
      <!--ro, req, int, range:[1,5]-->1
    </SupplementLightID>
  </SupplementLight>
</SupplementLightList>
<cardReaderType>
  <!--ro, opt, enum, subType:string-->K1108AM
</cardReaderType>
<strongElectricType>
  <!--ro, opt, enum, subType:string-->19M04-RO/T
</strongElectricType>
<passingMode>
  <!--ro, opt, enum, subType:string, dep:or,{$.SerialPortList.SerialPort.mode,eq,fingerprintReader}-->entrance
</passingMode>
<modbusProtocol>
  <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
  <checkBitOrder>
    <!--ro, opt, enum, subType:string-->bigEndian
  </checkBitOrder>
  <deviceMode>
    <!--ro, opt, enum, subType:string-->subordinate
  </deviceMode>
</modbusProtocol>
<serialPortName>
  <!--ro, opt, string, range:[1,32]-->test
</serialPortName>
<PDUCtrlType>
```

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```
      <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PDUCtrl}-->DS-65A2T0008
    </PDUCtrlType>
    <paperLessOfficeType>
      <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,paperless}-->DS-65PL0015-T
    </paperLessOfficeType>
    <videoWallControllerlType>
      <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,spellingCtrl}-->matrixProtocol
    </videoWallControllerlType>
    <audioCtrlType>
      <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,audioCtrl}-->DS-QAPDM16081C
    </audioCtrlType>
    <interactiveTabletType>
      <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,interactiveTablet}-->interactiveTabletProtocol
    </interactiveTabletType>
    <humanSensorType>
      <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,humanSensor}-->HM-TD1017-2/QW-HS121
    </humanSensorType>
    <transparentOverWebsocket>
      <!--ro, opt, bool-->true
    </transparentOverWebsocket>
    <PTZCtrlType>
      <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PTZCtrl}-->VISCA
    </PTZCtrlType>
    <airConditionerCtrlType>
      <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,airConditionerCtrl}-->GL
    </airConditionerCtrlType>
    <acquisitionLinkageType>
      <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,acquisitionLinkage}-->general
    </acquisitionLinkageType>
    <externalInterfaceBoard>
      <!--ro, opt, object, dep:or,{$.SerialPort.mode,eq,externalInterfaceBoard}-->
      <serialPortList>
        <!--ro, opt, array, subType:object-->
        <serialPort>
          <!--ro, opt, object-->
          <id>
            <!--ro, req, int-->0
          </id>
          <serialPortType>
            <!--ro, opt, enum, subType:string-->RS485
          </serialPortType>
          <RS485serialProtocol>
            <!--ro, opt, enum, subType:string-->private485
          </RS485serialProtocol>
          <private485Mode>
            <!--ro, opt, enum, subType:string-->cardReceiver
          </private485Mode>
          <RS232Mode>
            <!--ro, opt, enum, subType:string-->readerMode
          </RS232Mode>
        </serialPort>
      </serialPortList>
    </externalInterfaceBoard>
  </SerialPort>
</SerialPortList>
```

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14.1.4 System Maintenance

14.1.4.1 Get the storage capability of the device

Request URL

GET /ISAPI/ContentMgmt/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<RacmCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device storage capability, attr:version{req, string, protocolVersion}-->
  <isSupportZeroChan>
    <!--ro, opt, bool, whether the device supports channel-zero, desc:related URI: /ISAPI/ContentMgmt/ZeroVideo/channels-->true
  </isSupportZeroChan>
  <inputProxyNums>
    <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->0
  </inputProxyNums>
  <eSATANums>
    <!--ro, opt, int, supported number of eSATA interfaces-->0
  </eSATANums>
  <miniSASNums>
    <!--ro, opt, int, supported number of miniSAS interfaces-->0
```

```
    <!--ro, opt, int, supported number of multiSAS interfaces-->0
  </miniSASNums>
  <nasNums>
    <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->0
  </nasNums>
  <ipSanNums>
    <!--ro, opt, int, supported number of IPSANs-->0
  </ipSanNums>
  <isSupportRaid>
    <!--ro, opt, bool, whether the device supports RAID-->true
  </isSupportRaid>
  <isSupportExtHdCfg>
    <!--ro, opt, bool, whether the device supports advanced HDD management-->true
  </isSupportExtHdCfg>
  <isSupportTransCode>
    <!--ro, opt, bool, whether the device supports live view transcoding-->true
  </isSupportTransCode>
  <isSupportIpcImport>
    <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/ipcConfig-->true
  </isSupportIpcImport>
  <NasMountType>
    <!--ro, opt, bool, NAS type-->true
    <isNFSSupportAuthentication>
      <!--ro, opt, bool, whether the device supports NFS authentication-->true
    </isNFSSupportAuthentication>
    <isCIFSSupportAuthentication>
      <!--ro, opt, bool, whether the device supports CIFS authentication-->true
    </isCIFSSupportAuthentication>
  </NasMountType>
  <isSupportIpcStreamType>
    <!--ro, opt, bool, whether the device supports the stream type of the network camera-->true
  </isSupportIpcStreamType>
  <isSupportIOInputProxy>
    <!--ro, opt, bool, whether the device supports dynamic input channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/inputs-->true
  </isSupportIOInputProxy>
  <isSupportIOOutputProxy>
    <!--ro, opt, bool, whether the device supports dynamic output channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/outputs-->true
  </isSupportIOOutputProxy>
  <isSupportPTZRs485Proxy>
    <!--ro, opt, bool, whether the device supports dynamic PTZ RS-485 channels, desc:related URI: /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities-->true
  </isSupportPTZRs485Proxy>
  <isSupportSrcIDSearch>
    <!--ro, opt, bool, whether the device supports stream ID search-->true
  </isSupportSrcIDSearch>
  <isSupportReversePlayback>
    <!--ro, opt, bool, whether the device supports reverse playback-->true
  </isSupportReversePlayback>
  <isSupportSMARTTest>
    <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
  </isSupportSMARTTest>
  <isSupportDownloadByTime>
    <!--ro, opt, bool, whether the device supports downloading by time, desc:related URI: /ISAPI/ContentMgmt/download/capabilities-->true
  </isSupportDownloadByTime>
  <pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
personQueueCounting,personQueueTime,violentMotion,leavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,r
oadblock,
abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminite,OPTEX,securityControlPanel,ge
tUp,toiletTarry,
playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePh
one,fatigueDriving,
notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringW
arning,
PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideS
hop,stallOccupyingRoad,     illegalHeap,illegalParkingofNon-
MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel,
ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,temperatureIntervalMeasurement,AIOP,tail,abnormalFace,normalFace,multiFace,callFace,sunglasses
Face,humanEnter,operateOverTime,stickUp,installScanner,objectPlacedOrTaken,fakeCard,securityChecking,pothole,crack,roadMaint,imageCapture,methaneConcentrati
onException,methaneLightIntensityException,bannerDetection,variableDataEvent,shipChannelAbnormal,radarVideoDetection,smokeAndFireDetection,smokeDetection,se
nsorAlarm,alarmResult,AIOP_Video,AIOP_Polling_Snap,swipePatrol,BatteryCarInfoEvent,indoorPasswayBlock,detectorTemperatureAlarm,fireNoRegulation,detectorMoti
onAlarm,detectorShelterAlarm,detectorRunningStatus,faceCapture,UVPhotonDetection,DevPowerStatusReport,HFPD,audioAnalysis,IndoorPasswayRecovery,BatteryCarRec
overy,DetectorTemperatureRecovery,RadarFallenObjectsOnTracks,VegetationSpectrumDetection,personDensityDetection,ChannelClearEvent,channelOccupy,ChannelPassi
ngEvent,FallenObjectsOnTracksEvent,audioexception,CarWindow,shipsDetection,InsulatorAbnormalAlarm,PackingSpaceRecognition,smokeAlarm,SmokeAlarmRecovery,fire
SmartFireDetect,FirePointAlarmRecovery,offDuty,backDuty,sleepDuty,batteryCarDisobey,SmokingDetectionRecoverEvent,noFireExtinguisherRecog,FireExtinguisherRec
overyEvent,TelephoningBehaviorAlarmEvent,TelephoningBehaviorRecoveryEvent,GasTankMonitoringEvent,TemperatureMeasurementPreviewRecovery,TMPA,TMA,TemperatureM
easurementPreviewAlarmRecovery,TemperatureMeasurementAlarmRecovery,DetectorShelterAlarmRecovery,TemperatureUp,TemperatureDown,TemperatureEvent,TemperatureUp
Recover,TemperatureDownRecover,TemperatureEventRecover,ElectricFenceAlarmEvent,cityManagement,flowVelocity,volumeFlowRate,rainfall,partialDischarge,gasLeaka
ge,manualSnapshot,AlarmTriggeredByPanicButton,detectorSmokeAlarm,smokeAndFireSample,synthesis,FireStatusSwitch,FlameCombustionCounting,linkedTracking,smartS
hipDetection,targetModeling,SafetyWorkwearDetectionEvent,SafetyWorkwearRecoveryEvent,DeskMatOffset">
    <!--ro, opt, enum, picture search type, subType:string, attr:opt{req, string}, desc:"AllEvent", "CMR", "MOTION", "ALARM", "EDR", "ALARMANDMOTION",
"pir:PIR", "wlsensor",  "callhelp", "facedetection", "FieldDetection", "scenechangedetection", "LineDetection", "regionEntrance", "regionExiting",
"loitering", "group", "rapidMove", "parking", "unattendedBaggage", "attendedBaggage", "vehicleDetection", "manual", "manualSnapShot",  "playSnapShot",
"safetyHelmet", "personQueueCounting", "personQueueTime", "violentMotion", "leavePosition", "advReachHeight", "peopleNumChange", "mixedTargetDetection",
"illegalParking", "pedestrian", "construction", "roadblock", "evidence", "crosslane", "vehicleexist", "lanechange", "wrongdirection", "congestion",
"turnround", "parallelParking", "abandonedObject", "trafficAccident", "fogDetection", "smokeDetection", "personDensity", "failDown", "accessController",
"videoInterCom", "GJD", "Luminite", "OPTEX", "securityControlPanel", "getUp", "toiletTarry", "playCellphone", "vehicleMonitor", "running", "retention",
"spacingChange", "yardTarry", "keyPersonGetUp", "sitQuietly", "standup", "plateRecognition", "smoking", "onThePhone", "fatigueDriving",
"notLookingStraightAhead", "LeaveThePost", "seatbelt", "sunglasses", "identityAbnormal", "ForwardCollisionWarning", "LaneDepartureWarning",
"BlindSpotDetection", "HeadwayMonitoringWarning", "PedestrianCollisionWarning", "SuddenSpeedUp", "SuddenSpeedDown", "SuddenLeftTurn", "SuddenRightTurn",
"Turnover", "NotYieldToPedestrian;allPic", "ATMPanel", "ATMSurround", "ATMFace", "ATMSafetyCabin", "smokeDetectAlarm", "temperatureIntervalMeasurement",
"AIOP", "tail", "abnormalFace", "normalFace;callFace", "sunglassesFace", "humanEnter", "operateOverTime", "stickup", "installScanner",
```

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```
"objectPlacedOrTaken", "fakeCard", "securityChecking", "pothole", "crack", "roadMaint", "imageCapture", "ReID", "thermalVehicleDetection",
"methaneConcentrationException", "methaneLightIntensityException", "bannerDetection", "waterQualityDetection", "waterLevelDetection", "parallelParking",
"methaneConcentrationException", "variableDataEvent", "VOCsGasDetection", "shipChannelAbnormal", "radarVideoDetection", "AIDfogDetection:AID",
"smokeAndFireDetection", "smokeDetection", "sensorAlarm", "GasTankDetection", "alarmResult", "AIOP_Video", "AIOP_Polling_Snap", "swipePatrol",
"BatteryCarInfoEvent", "indoorPasswayBlock", "detectorTemperatureAlarm", "fireNoRegulation", "detectorMotionAlarm", "detectorShelterAlarm",
"detectorRunningStatus", "faceCapture", "UVPhotonDetection", "DevPowerStatusReport", 'HFPD"-->AllEvent
  </pictureSearchType>
  <recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofN
on-MotorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
,temperatureIntervalMeasurement,AIOP,securityChecking,scheduledRecordUpload,variableDataEvent,smokeAndFireDetection,smokeDetection,AIOP_Video,AIOP_Polling_S
nap,BatteryCarInfoEvent,indoorPasswayBlock,detectorTemperatureAlarm,fireNoRegulation,detectorMotionAlarm,detectorShelterAlarm,detectorRunningStatus,faceCapt
ure,UVPhotonDetection,DevPowerStatusReport,ChannelClearEvent,channelOccupy,ChannelPassingEvent,personQueueCounting,personQueueTime,leavePosition,localButton
Record,callRecord,partialDischarge,gasLeakage,smokeAndFireSample,passback,linkedTracking,CallRecordAnswered,CallRecordNotAnswered,lensMotorLocking">
    <!--ro, opt, enum, video search type, subType:string, attr:opt{req, string}, desc:"AllEvent" (all events), "CMR" (capture by schedule), "MOTION" (motion
detection), "ALARM", "EDR", "ALARMANDMOTION", "pir", "wlsensor", "callhelp", "facedetection", "FieldDetection", "scenechangedetection", "LineDetection",
"regionEntrance", "regionExiting", "loitering", "group", "rapidMove", "parking", "unattendedBaggage", "attendedBaggage", "vehicleDetection", "manual",
"manualSnapShot", "playSnapShot", "safetyHelmet",  "personDensityDetection", "smokeDetection", "ATMPanel", "ATMSurround", "ATMFace", "ATMSafetyCabin",
"temperatureIntervalMeasurement", "AIOP", "securityChecking", "scheduledRecordUpload", "thermalVehicleDetection", "waterQualityDetection",
"waterLevelDetection", "methaneConcentrationException", "variableDataEvent"-->AllEvent
  </recordSearchType>
  <isSupportActivateIpc>
    <!--ro, opt, bool, whether the device supports activating the network camera-->true
  </isSupportActivateIpc>
  <isSupportCheckIpcSecurity>
    <!--ro, opt, object, whether the device supports IPC security verification-->
    <isSupportCheckPassword>
      <!--ro, opt, bool, whether the device supports password detection-->true
    </isSupportCheckPassword>
  </isSupportCheckIpcSecurity>
  <isSupportMainAndSubRecord>
    <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
  </isSupportMainAndSubRecord>
  <isSupportSyncIPCPassword>
    <!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true
  </isSupportSyncIPCPassword>
  <isSupportTransferIPC>
    <!--ro, opt, bool, whether the device supports transparent transmission of the network camera-->true
  </isSupportTransferIPC>
  <isSupportSmartSearch>
    <!--ro, opt, bool, whether the device supports video dual-VCA, desc:related URI: /ISAPI/ContentMgmt/SmartSearch/capabilities-->true
  </isSupportSmartSearch>
  <isSupportPOS>
    <!--ro, opt, bool, whether the device supports POS-->true
  </isSupportPOS>
  <isSupportLinkNum>
    <!--ro, opt, bool, whether the device supports getting the number of links, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/linkNum-->true
  </isSupportLinkNum>
  <isSupportPassBackBasicCfg>
    <!--ro, opt, bool, whether the device supports configuring basic parameters of the CVR copy-back function, desc:related URI:
/ISAPI/ContentMgmt/record/passback/basicCfg/capabilities-->true
  </isSupportPassBackBasicCfg>
  <PassBackTaskConfig>
    <!--ro, opt, object, capability of managing tasks of the CVR copy-back function-->
    <isSupportAddTask>
      <!--ro, opt, bool, whether the device supports adding a task, desc:related URI: /ISAPI/ContentMgmt/record/passback/addTask/capabilities-->true
    </isSupportAddTask>
    <isSupportSearchTask>
      <!--ro, opt, bool, whether the device supports searching tasks, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchTask/capabilities-->true
    </isSupportSearchTask>
    <isSupportControlTask>
      <!--ro, opt, bool, whether the device supports controlling the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/controlTask/capabilities--
>true
    </isSupportControlTask>
    <isSupportDeleteTask>
      <!--ro, opt, bool, whether the device supports deleting the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/deleteTask/capabilities-->true
    </isSupportDeleteTask>
  </PassBackTaskConfig>
  <PassBackPlanConfig>
    <!--ro, opt, object, capability of configuring the CVR copy-back schedule-->
    <isSupportAddPlan>
      <!--ro, opt, bool, whether the device supports adding a schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/addPlan/capabilities-->true
    </isSupportAddPlan>
    <isSupportSearchPlan>
      <!--ro, opt, bool, whether the device supports searching schedules, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchPlan/capabilities--
>true
    </isSupportSearchPlan>
    <isSupportDeletePlan>
      <!--ro, opt, bool, whether the device supports deleting the schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/deletePlan/capabilities--
>true
    </isSupportDeletePlan>
  </PassBackPlanConfig>
  <IRAIDCap>
    <!--ro, opt, object, network array capability-->
    <isSupportIRAIDRapidConfiguration>
      <!--ro, opt, bool, whether the device supports quick configuration of iRAID, desc:related URI:
/ISAPI/ContentMgmt/Storage/IRAID/RapidConfiguration/capabilities-->true
    </isSupportIRAIDRapidConfiguration>
  </IRAIDCap>
```


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  <isSupportStorageExtraInfo>
    <!--ro, opt, bool, whether the device supports configuring additional storage information, desc:related URI:
/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities-->true
  </isSupportStorageExtraInfo>
  <isSupportRecordStatus>
    <!--ro, opt, bool, whether the device supports getting the recording status, desc:related URI: /ISAPI/ContentMgmt/channels/<ID>/record/capabilities--
>true
  </isSupportRecordStatus>
  <isSupportRacmChannelsCap opt="true,false">
    <!--ro, opt, bool, whether the device supports the Rcam capability by channel, attr:opt{opt, string}, desc:related URI:
/ISAPI/ContentMgmt/channels/<ID>/capabilities-->true
  </isSupportRacmChannelsCap>
  <LockCap>
    <!--ro, opt, object, capability of locking and unlocking video, desc:/ISAPI/ContentMgmt/record/control/locks/capabilities-->
    <LockByName>
      <!--ro, opt, object, lock/unlock video by file name-->
      <TrackId>
        <!--ro, req, string, channel stream ID, desc:channel No. * 100 + stream type (-1: main stream, -2: sub-stream)-->101
      </TrackId>
      <recordName min="1" max="100">
        <!--ro, req, string, video file name, range:[1,100], attr:min{req, int},max{req, int}-->test
      </recordName>
      <command opt="lock,unlock">
        <!--ro, req, enum, operation command, subType:string, attr:opt{req, string}, desc:"Lock", "unlock"-->lock
      </command>
      <duration min="1" max="10">
        <!--ro, opt, int, lock time, unit:s, dep:and,{$.LockByName.command,eq,lock}, attr:min{req, int},max{req, int}, desc:0: lock permanently-->1
      </duration>
      <recorderCode min="1" max="100">
        <!--ro, opt, string, video recorder No., attr:min{req, int},max{req, int}-->test
      </recorderCode>
      <policeCode min="1" max="100">
        <!--ro, opt, string, police No., attr:min{req, int},max{req, int}-->test
      </policeCode>
    </LockByName>
    <LockByTime>
      <!--ro, opt, object, lock/unlock video by time-->
      <TrackId>
        <!--ro, req, string, channel stream ID, desc:channel No. * 100 + stream type (-1: main stream, -2: sub-stream)-->101
      </TrackId>
      <recordType opt="ALL,MANUAL,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,COMMAND,SMART">
        <!--ro, req, enum, recording type, subType:string, attr:opt{req, string}, desc:"ALL" (all videos), "MANUAL" (manual recording), "CMR" (capture by
schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture alarm pictures), "EDR"(capture alarm pictures or motion detection pictures),
"ALARMANDMOTION" (capture alarm pictures and motion detection pictures), "Command" (command-based recording), "SMART" (smart recording)-->ALL
      </recordType>
      <startDateTime>
        <!--ro, req, datetime, video segment start time-->1970-01-01T00:00:00+08:00
      </startDateTime>
      <endDateTime>
        <!--ro, req, datetime, video segment end time-->1970-01-01T00:00:00+08:00
      </endDateTime>
      <command opt="lock,unlock">
        <!--ro, req, enum, operation command, subType:string, attr:opt{req, string}, desc:“lock”, “unlock”-->lock
      </command>
      <duration min="1" max="10">
        <!--ro, opt, int, lock time, unit:s, dep:and,{$.LockByName.command,eq,lock}, attr:min{req, int},max{req, int}, desc:0: lock permanently-->1
      </duration>
    </LockByTime>
  </LockCap>
  <isSupportForamtAll>
    <!--ro, opt, bool, whether the device supports formatting all HDDs, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/format-->true
  </isSupportForamtAll>
  <isSupportExtendCabinetCfg>
    <!--ro, opt, bool, whether it supports enclosure configuration, desc:related URI: /ISAPI/ContentMgmt/Storage/ExtendCabinet/capabilities-->true
  </isSupportExtendCabinetCfg>
  <diskGroupNums>
    <!--ro, opt, int, supported number of disk groups, desc:related URI: /ISAPI/ContentMgmt/Storage/diskGroup-->0
  </diskGroupNums>
  <ExternalDevice>
    <!--ro, opt, object, external device storage-->
    <USB>
      <!--ro, opt, object, USB information-->
      <debugLogOutput>
        <!--ro, opt, bool, whether to enable the debugging log, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
      </debugLogOutput>
      <isSupportCapturePackage>
        <!--ro, opt, bool, whether the device supports configuring capture and filter function of a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/capturePackage/capabilities?format=json-->true
      </isSupportCapturePackage>
      <isSupportStreamStorage>
        <!--ro, opt, bool, whether the device supports saving stream to a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/streamStorage/capabilities?format=json-->true
      </isSupportStreamStorage>
    </USB>
  </ExternalDevice>
  <customProtocolNums>
    <!--ro, opt, int, number of custom protocols for accessing network cameras, desc:related URI: /ISAPI/ContentMgmt/InputProxy/customProtocols-->0
  </customProtocolNums>
  <isSupportIPCTiming>
    <!--ro, opt, bool, whether the device supports synchronizing the time of the network camera, desc:relate to enableTiming on URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>-->true
  </isSupportIPCTiming>
  <isSupportLogConfig>
    <!--ro, opt, bool, whether the device supports log configuration, desc:related URI: /ISAPI/ContentMgmt/LogConfig/capabilities-->true
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  </isSupportLogConfig>
  <SecurityLog>
    <!--ro, opt, object, security log, desc:security log-->
    <isSupportSecurityLog>
      <!--ro, opt, bool, whether the device supports the security log-->true
    </isSupportSecurityLog>
    <isSupportLogServer>
      <!--ro, opt, bool, whether it supports security log server configuration-->true
    </isSupportLogServer>
    <isSupportLogServerTest>
      <!--ro, opt, bool, whether it supports security log server test-->true
    </isSupportLogServerTest>
    <SecurityLogTypeList>
      <!--ro, req, array, list of security log types, subType:object-->
      <SecurityLogType>
        <!--ro, opt, object, list of log types-->
        <primaryType>
          <!--ro, req, enum, main log type, subType:string, desc:"Event", "Operation", "Other" (other types), "All" (all types)-->Event
        </primaryType>
        <secondaryType opt="test">
          <!--ro, req, enum, sub type, subType:string, attr:opt{req, string}, desc:see other types on the appendix. "all" (all types)-->all
        </secondaryType>
      </SecurityLogType>
    </SecurityLogTypeList>
    <logMinItemNum>
      <!--ro, opt, int, range:[1,2147483646]-->100
    </logMinItemNum>
  </SecurityLog>
  <isSupportGetBackupDevice>
    <!--ro, opt, bool, whether the device supports getting the backup device, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/device?format=json-->true
  </isSupportGetBackupDevice>
  <isSupportAutoBackup>
    <!--ro, opt, bool, whether the device supports automatic backup, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/auto/capabilities?format=json--
>true
  </isSupportAutoBackup>
  <certificateValidationEnabled>
    <!--ro, opt, bool, whether to enable the certification validation, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <certificateValidationEnabled> node supports-->true
  </certificateValidationEnabled>
  <defaultAdminPortEnabled>
    <!--ro, opt, bool, whether to enable the default communication port, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <defaultAdminPortEnabled> node supports-->true
  </defaultAdminPortEnabled>
  <iSptInputProxyChanCap>
    <!--ro, opt, bool, whether the device supports getting the capability set of the digital channels, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/capabilities-->true
  </iSptInputProxyChanCap>
  <isSupportLogDataPackage>
    <!--ro, opt, bool, whether the device supports exporting the log files, desc:related URI: /ISAPI/ContentMgmt/LogSearch/dataPackage-->true
  </isSupportLogDataPackage>
  <logSearchTimeSpanNums>
    <!--ro, opt, int, number of time periods for log search, desc:related URI: /ISAPI/ContentMgmt/logSearch; corresponding node: <timeSpanList> in
<CMSearchDescription>-->0
  </logSearchTimeSpanNums>
  <CloudStorageServerCap>
    <!--ro, opt, object, capability of the cloud storage service-->
    <isSupportCloudStorageParameter>
      <!--ro, opt, bool, whether the device supports configuring the cloud storage parameters, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/capabilities?format=json-->true
    </isSupportCloudStorageParameter>
    <isSupportCloudStoragePool>
      <!--ro, opt, bool, whether the device supports configuring the cloud storage pool, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/pool/capabilities?format=json-->true
    </isSupportCloudStoragePool>
    <supportChannelsNum>
      <!--ro, opt, int, number of channels supported by the cloud storage-->0
    </supportChannelsNum>
    <protocolType opt="videoStreamingOS">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->videoStreamingOS
    </protocolType>
  </CloudStorageServerCap>
  <isSupportWebPrivatePlaybackByUTC>
    <!--ro, opt, bool, whether it supports playback (via private protocol) according to UTC-->true
  </isSupportWebPrivatePlaybackByUTC>
  <isSupportFindCommonFileByUTC>
    <!--ro, opt, bool, whether it supports searching for ordinary files (private protocols) by UTC time-->true
  </isSupportFindCommonFileByUTC>
  <isSupportFindEventFileByUTC>
    <!--ro, opt, bool, whether it supports searching for event related videos (via private protocol) according to UTC-->true
  </isSupportFindEventFileByUTC>
  <isSupportSmartSearchRecordByUTC>
    <!--ro, opt, bool, whether it supports VCA search by UTC time-->true
  </isSupportSmartSearchRecordByUTC>
  <isSupportMRDSearchByTimeZone>
    <!--ro, opt, bool, whether it supports searching for files (private protocols) by monthly calendar by UTC time-->true
  </isSupportMRDSearchByTimeZone>
  <isSupportSearchRecordLabelByUTC>
    <!--ro, opt, bool, whether it supports searching for video tags by UTC time-->true
  </isSupportSearchRecordLabelByUTC>
  <isSupportSearchPictureByUTC>
    <!--ro, opt, bool, whether it supports searching for pictures by UTC time-->true
  </isSupportSearchPictureByUTC>
  <isSupportSmartSearchPictureByUTC>
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    <!--ro, opt, bool, whether it supports smart pictures search by UTC time-->true
  </isSupportSmartSearchPictureByUTC>
  <isSupportGetFigureByUTC>
    <!--ro, opt, bool-->true
  </isSupportGetFigureByUTC>
  <isSupportFindLogByUTC>
    <!--ro, opt, bool, whether it supports searching log by UTC time-->true
  </isSupportFindLogByUTC>
  <isSupportUploadRecordByUTC>
    <!--ro, opt, bool, whether it supports uploading files to cloud storage by UTC time-->true
  </isSupportUploadRecordByUTC>
  <isSupportPlaybackByUTC>
    <!--ro, opt, bool, whether it supports playback and locating by UTC time-->true
  </isSupportPlaybackByUTC>
  <isSupportFaceDetecttionAlarmByTimeZone>
    <!--ro, opt, bool, whether it supports reporting face detection by UTC time, desc:the time should be before isSupportCountingSearchByUTC and after
isSupportPlaybackByUTC-->true
  </isSupportFaceDetecttionAlarmByTimeZone>
  <isSupportSourceModify>
    <!--ro, opt, bool, whether it supports configuring network camera parameters on the LAN, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/sourceModify/capabilities-->true
  </isSupportSourceModify>
  <isSupportCountingSearchByUTC>
    <!--ro, opt, bool, whether it supports searching people counting results by UTC time-->true
  </isSupportCountingSearchByUTC>
  <isSupportCluster>
    <!--ro, opt, bool, whether it supports cluster function, desc:related URI: /ISAPI/ContentMgmt/Cluster-->true
  </isSupportCluster>
  <supportAIDTFSType opt="illegalParking,wrongdirection,crosslane,vehicleexist,lanechange,turnround,evidence">
    <!--ro, opt, enum, supported AIDTFS types, subType:string, attr:opt{req, string}, desc:"illegalParking", "wrongdirection" (wrong-way driving),
"crosslane"(driving on the lane line), "vehicleexist" (vehicle on non-motor vehicle lane), "Lanechange" (illegal lane change detection), "turnround"
(illegal U-turning detection), "evidence" (face picture collection)-->illegalParking
  </supportAIDTFSType>
  <isSupportMisinfoStateSearch>
    <!--ro, opt, bool, whether it supports searching by false alarm status-->true
  </isSupportMisinfoStateSearch>
  <isSupportOneKeyOperationStorageData>
    <!--ro, opt, bool, whether it supports one-touch configuration for storage data, desc:related URI: /ISAPI/ContentMgmt/Storage/oneKey/capabilities?
format=json-->true
  </isSupportOneKeyOperationStorageData>
  <eventRecordSearch>
    <!--ro, opt, object, whether the device supports searching videos by event, desc:related URI: /ISAPI/ContentMgmt/eventRecordSearch?format=json-->
    <eventType
opt="all,batch,alarmResult,faceSnap,vehicleMatchResult,hawkResult,VMD,alarmIn,inquestAlarm,behavior,posAlarm,accessControllerEvent,IOTD,trialAlarm,rapidMove
,defocus,scenechangedetection,PIR,fireSmartFireDetect,shipsDetection,TMA,TDA,framesPeopleCounting,personQueueTime,peopleDetection,safetyHelmetDetection,mixe
dTargetDetection,TMPA,smokeDetection,smokeAndFireDetection,abnormalReboot,audioexception,ANPR,objectsThrownDetection,audioAnalysis,HFPD,shelteralarm,cityMan
agement,hideStart,hideStop,sleepOnduty,PerimeterTargetDetEvent">
      <!--ro, opt, enum, event type, subType:string, attr:opt{req, string}, desc:“vehicleMatchResult” (vehicle comparison), “hawkResult” (video analysis
task), “faceContrast” (face picture comparison), “faceSnap” (face capture)-->alarmResult
    </eventType>
    <recordChannelListSize>
      <!--ro, opt, int, maximum length of the value of recordChannelList (recoding channel list)-->16
    </recordChannelListSize>
    <triggerChannelsSize>
      <!--ro, opt, int, maximum length of the value of triggerChannels (list of video channel numbers which trigger recording)-->16
    </triggerChannelsSize>
    <alarmResultAuxEventType opt="all,alarmResultFailure,alarmResultSuccess,whiteList">
      <!--ro, opt, enum, sub-type of face picture comparison alarm, subType:string, dep:and,{$.RacmCap.eventRecordSearch.eventType,eq,alarmResult},
attr:opt{req, string}, desc:"alarmResultFailure" (face picture comparison failed), "alarmResultSuccess" (face picture comparison succeeded), "whiteList"
(allowList alarm, i.e., strangers alarm)-->alarmResultFailure
    </alarmResultAuxEventType>
    <inquestParam>
      <!--ro, opt, object, interrogation information, dep:and,{$.EventSearchDescription.eventType,eq,inquestAlarm}-->
      <inquestEventType opt="any,inquestStart,inquestStop,inquestTag,inquestSegment">
        <!--ro, opt, enum, interrogation information, subType:string, attr:opt{req, string}, desc:“any”, “inquestStart” (interrogation start), ‘inquestStop”
(interrogation stop), “inquestTag” (marked information), “inquestSegment” (interrogation segment status)-->any
      </inquestEventType>
      <roomIndex min="1" max="65535">
        <!--ro, opt, int, interrogation room No., attr:min{req, int},max{req, int}, desc:it starts from 1-->1
      </roomIndex>
    </inquestParam>
    <behavior>
      <!--ro, opt, object, abnormal event detection, dep:and,{$.EventSearchDescription.eventType,eq,behavior}-->
      <behaviorEventType
opt="any,linedetection,regionEntrance,regionExiting,fielddetection,loitering,parking,violentMotion,group,unattendedBaggage,attendedBaggage,audioAbnormal,lea
vePosition,peopleNumChange,retention,failDown,running,spacingChange,playCellphone,objectPlacedOrTaken,tailing,advReachHeight">
        <!--ro, req, enum, abnormal event detection event type, subType:string, attr:opt{req, string}, desc:"any", "Linedetection" (line crossing
detection), "regionEntrance", "regionExiting", "fielddetection" (intrusion detection), "Loitering" (loitering detection), "parking" (parking detection),
"violentMotion", "group" (people gathering), "unattendedBaggage", "attendedBaggage" (object removal), "audioAbnormal" (sudden change of sound intensity
detection)-->any
      </behaviorEventType>
      <falseAlarmType opt="all,falseAlarmFilterate" def="all">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->all
      </falseAlarmType>
    </behavior>
    <posAlarm>
      <!--ro, opt, object, pos event, dep:and,{$.EventSearchDescription.eventType,eq,posAlarm}, desc:it is recorded in the device, not in the internet-->
      <caseSensitive opt="true,false">
        <!--ro, req, bool, whether it is case-sensitive, attr:opt{req, string}-->false
      </caseSensitive>
      <combinateMode opt="and,or">
        <!--ro, req, enum, keyword combination, subType:string, attr:opt{req, string}, desc:"and", "or"-->and
      </combinateMode>
      <keyword size="3">
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        <!--ro, opt, string, keyword, attr:size{req, int}-->test
      </keyword>
      <keywordLen min="0" max="16">
        <!--ro, opt, string, keyword length range, range:[0,16], attr:min{req, int},max{req, int}-->test
      </keywordLen>
    </posAlarm>
    <trialParam>
      <!--ro, opt, object, trial event, dep:and,{$.EventSearchDescription.eventType,eq,trialAlarm}-->
      <trialEventType opt="any,trialStart,trialStop,trialTag,trialSegment,trialJudge">
        <!--ro, opt, enum, trial information, subType:string, attr:opt{req, string}, desc:"any", "trialStart" (trial start), "trialStop" (trial stop),
"trialTag" (marked information), "trialSegment" (trial segment status), "trialJudge" (trial case information)-->any
      </trialEventType>
      <caseNo min="0" max="56">
        <!--ro, req, string, case No., range:[0,56], attr:min{req, int},max{req, int}-->test
      </caseNo>
      <caseName>
        <!--ro, opt, string, case name, range:[0,100]-->test
      </caseName>
      <litigant1 min="0" max="32">
        <!--ro, opt, string, litigant 1, range:[0,32], attr:min{req, int},max{req, int}-->test
      </litigant1>
      <litigant2 min="0" max="32">
        <!--ro, opt, string, litigant 2, range:[0,32], attr:min{req, int},max{req, int}-->test
      </litigant2>
      <chiefJudge min="0" max="32">
        <!--ro, opt, string, judge, range:[0,32], attr:min{req, int},max{req, int}-->test
      </chiefJudge>
      <caseType opt="0,1,2">
        <!--ro, opt, enum, case type, subType:int, attr:opt{req, string}, desc:0 (all), 1 (criminal case), 2 (civil case)-->0
      </caseType>
    </trialParam>
    <accessControllerEvent>
      <!--ro, opt, object, access control event, dep:and,{$.EventSearchDescription.eventType,eq,accessControllerEvent}-->
      <majorEventType>
        <!--ro, opt, int, alarm main type, desc:see details in macro definition-->0
      </majorEventType>
      <subEventType>
        <!--ro, opt, int, alarm sub type, desc:see details in macro definition-->0
      </subEventType>
      <cardNo min="0" max="32">
        <!--ro, opt, string, card No., range:[0,32], attr:min{req, int},max{req, int}-->test
      </cardNo>
      <name min="0" max="32">
        <!--ro, opt, string, name, range:[0,32], attr:min{req, int},max{req, int}-->test
      </name>
      <MACAddr min="0" max="12">
        <!--ro, opt, string, MAC address, range:[0,12], attr:min{req, int},max{req, int}-->test
      </MACAddr>
      <inductiveEventType opt="authenticated,authenticationFailed">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->authenticated
      </inductiveEventType>
    </accessControllerEvent>
    <IOTD>
      <!--ro, opt, object, upload events related with IoT devices, dep:and,{$.EventSearchDescription.eventType,eq,IOTD}-->
      <DeviceEventTypeList>
        <!--ro, req, array, sub event type list, subType:object-->
        <DeviceEventType>
          <!--ro, opt, object, sub event type-->
          <subEventType>
            <!--ro, opt, string, sub event type, desc:see details in the node IoTEventType of Data Dictionary-->test
          </subEventType>
        </DeviceEventType>
      </DeviceEventTypeList>
      <IoTchannelID size="256">
        <!--ro, opt, string, channel No., attr:size{req, int}, desc:search all channels if the node does not exists-->test
      </IoTchannelID>
      <caseSensitive opt="true,false">
        <!--ro, req, bool, whether it is case-sensitive, attr:opt{req, string}-->false
      </caseSensitive>
      <combinateMode opt="and,or">
        <!--ro, req, enum, keyword combination, subType:string, attr:opt{req, string}, desc:"and", "or"-->and
      </combinateMode>
      <keyword size="3">
        <!--ro, opt, string, keyword, attr:size{req, int}-->test
      </keyword>
      <zonNo min="0" max="32">
        <!--ro, opt, int, zone No., attr:min{req, int},max{req, int}, desc:it is valid only when deviceType==2&&subEventType=="zone"-->1
      </zonNo>
    </IOTD>
    <AudioAnalysis>
      <!--ro, opt, object, audio analysis event, dep:and,{$.EventSearchDescription.eventType,eq,audioAnalysis}-->
      <targetType
opt="all,robbery,help,remittance,comment,fraud,cardNo,confirm,certificateID,certificateNumber,passWord,breakevenInterest,invest,noRisk,riskLow,riskSmall,tra
nsfer,transferMoney,sureEarnNoLose,onlyEarnNoLose,financing,noLose,borrowMoney,hightReturn,highInterestRates,highYield">
        <!--ro, req, enum, audio analysis type, subType:string, attr:opt{req, string}, desc:"all", "robbery", "help", "remittance", "comment", "fraud",
"cardNo" (card No.), "confirm", "certificateID", "certificateNumber", "passWord", "breakevenInterest" (break-even), "invest", "noRisk", "riskLow" (low
risk), "riskSmall" (small risk), "transfer", "transferMoney", "sureEarnNoLose" (earn without losing), "onlyEarnNoLose" (earn without losing), "financing",
"noLose" (not lose), "borrowMoney", "hightReturn", "highInterestRates" (high interest rate), "highYield"-->all
      </targetType>
    </AudioAnalysis>
    <objectsThrownDetection>
      <!--ro, opt, object, objects thrown from heights detection, dep:and,{$.EventSearchDescription.eventType,eq,objectsThrownDetection}-->
      <alarmLevel opt="1,2,3">
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        <!--ro, req, enum, alarm level, subType:string, attr:opt{req, string}, desc:"1" (low), "2" (medium), "3" (high). It is categorized into three levels
according to the target track and the actual event credibility. The higher the level, the greater the credibility of the alarm being triggered by an actual
object thrown from building event. This node should be configured in the rule to filter alarm event uploading.-->1
      </alarmLevel>
      <floorID min="-100" max="100">
        <!--ro, opt, int, floor range, attr:min{req, int},max{req, int}-->1
      </floorID>
    </objectsThrownDetection>
    <HFPDSearch>
      <!--ro, opt, object, dep:and,{$.EventSearchDescription.eventType,eq,HFPD}-->
      <conditionType opt="greaterThan,equal,lessThan,notEqual,greaterThanOrEqual,lessThanOrEqual">
        <!--ro, req, enum, search operation type, subType:string, attr:opt{req, string}, desc:"greaterThan", "equal", "lessThan", "notEqual",
"greaterThanOrEqual" (greater than or equal), "lessThanOrEqual" (less than or equal)-->greaterThan
      </conditionType>
      <occurrences min="0" max="100" def="1">
        <!--ro, opt, int, Frequency Range, attr:min{req, int},max{req, int},def{req, int}, desc:number of appearances for the target-->1
      </occurrences>
    </HFPDSearch>
    <cityManagement>
      <!--ro, opt, object, City Management, dep:and,{$.RacmCap.eventRecordSearch.eventType,eq,cityManagement}-->
      <subEventType opt="unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad">
        <!--ro, opt, string, sub event type, attr:opt{req, string}-->unregisteredStreetVendor
      </subEventType>
    </cityManagement>
    <secondEventType
opt="alarmResult,faceSnap,vehicleMatchResult,hawkResult,VMD,alarmIn,inquestAlarm,behavior,posAlarm,accessControllerEvent,IOTD,trialAlarm,rapidMove,defocus,s
cenechangedetection,PIR,fireSmartFireDetect,shipsDetection,TMA,TDA,framesPeopleCounting,personQueueTime,peopleDetection,safetyHelmetDetection,mixedTargetDet
ection,TMPA,smokeDetection,smokeAndFireDetection,abnormalReboot,audioexception,ANPR,objectsThrownDetection,audioAnalysis,HFPD,shelteralarm,FPEvent,fireDetec
tion,MuckTruckTargetAnalysisEvent,headRiseRate,vacantSeatRate,studentBehavior">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->alarmResult
    </secondEventType>
    <secondVerifyAlarmEnabled>
      <!--ro, opt, bool-->false
    </secondVerifyAlarmEnabled>
    <secondVerifyAlarmType opt="all,succ,fail,analyzed">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->all
    </secondVerifyAlarmType>
    <ANPRSearch>
      <!--ro, opt, object-->
      <licensePlate min="1" max="32">
        <!--ro, opt, string, license plate number, attr:min{req, int},max{req, int}-->A283KY77
      </licensePlate>
      <province opt="0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34">
        <!--ro, opt, string, N/A, attr:opt{req, string}-->0
      </province>
      <country opt="0,1,2,3,4,5,6,7,8,9,10,11,12,169,172,277,255">
        <!--ro, opt, int, country/region, attr:opt{req, string}-->0
      </country>
    </ANPRSearch>
    <PerimeterTargetDetEvent>
      <!--ro, opt, object-->
      <analysisStatusList opt="analyzed,unanalyzed,analysisException">
        <!--ro, opt, string, analysis status, attr:opt{req, string}-->test
      </analysisStatusList>
      <taskID min="1" max="64">
        <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
      </taskID>
    </PerimeterTargetDetEvent>
    <alarmResult>
      <!--ro, opt, object, capability of face picture comparison search condition-->
      <FDID min="1" max="64">
        <!--ro, req, string, dep:and,{$.RacmCap.eventRecordSearch.eventType,eq,alarmResult}, attr:min{req, int},max{req, int}-->test
      </FDID>
    </alarmResult>
    <streamList size="1024">
      <!--ro, opt, string, attr:size{req, int}-->test
      <streamID min="32" max="32">
        <!--ro, req, string, range:[32,32], attr:min{req, int},max{req, int}-->test
      </streamID>
    </streamList>
  </eventRecordSearch>
  <isSupportManualRecord>
    <!--ro, opt, bool, whether the device supports manual recording, desc:related URI: /ISAPI/ContentMgmt/record/control/manual/capabilities?format=json--
>true
  </isSupportManualRecord>
  <isSupportRemark>
    <!--ro, opt, bool, whether the device supports file remarks-->true
  </isSupportRemark>
  <FileUpload>
    <!--ro, opt, object, automatically upload file-->
    <enabled>
      <!--ro, req, bool, whether the device supports uploading the file-->true
    </enabled>
  </FileUpload>
  <PoliceInfoUpload>
    <!--ro, opt, object, automatically upload police officer information-->
    <enabled>
      <!--ro, req, bool, whether the device supports uploading the police officer information-->true
    </enabled>
  </PoliceInfoUpload>
  <PoliceInfo>
    <!--ro, opt, object, police officer information-->
    <policeID>
      <!--ro, opt, bool, whether the device supports police officer ID-->true
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      <!--ro, opt, bool, whether the device supports police officer ID-->true
    </policeID>
    <policeCode>
      <!--ro, opt, bool, whether the device supports police officer number-->true
    </policeCode>
    <policeName>
      <!--ro, opt, bool, whether the device supports police officer name-->true
    </policeName>
    <password>
      <!--ro, opt, bool, whether the device supports police officer password-->true
    </password>
    <deviceID>
      <!--ro, opt, bool, whether the device supports device ID-->true
    </deviceID>
    <isSupportPoliceFacePicture>
      <!--ro, opt, bool, whether the device supports managing the face picture of police officer,
desc:related URL:
/ISAPI/ContentMgmt/police/<policeID>?format=json,
/ISAPI/ContentMgmt/police?format=json,
/ISAPI/ContentMgmt/police/deleteFacePicture?format=json-->true
    </isSupportPoliceFacePicture>
  </PoliceInfo>
  <isSupportAssignChannelID>
    <!--ro, opt, bool, whether the device supports specifying the access channel No. when adding the network camera, desc:related to channelID on URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>-->true
  </isSupportAssignChannelID>
  <isSupportAssignStreamID>
    <!--ro, opt, bool, whether the device supports specifying stream ID of the channel when adding the network camera, desc:related to streamID on URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>-->true
  </isSupportAssignStreamID>
  <PictureSearchSubTypeList>
    <!--ro, opt, object, list of picture searching sub type-->
    <PictureSearchSubType>
      <!--ro, opt, object, the sub type of picture search-->
      <mainType>
        <!--ro, req, enum, main event type of the searched picture, subType:string, desc:"vehicleDetection", "HVTVehicleDetection" (mixed-traffic
detection)-->vehicleDetection
      </mainType>
      <subType opt="all,motorVehicle,nonMotorVehicle,pedestrian">
        <!--ro, req, enum, sub type of each main event type, subType:string, attr:opt{req, string}, desc:"all", "motorVehicle", "nonMotorVehicle",
"pedestrian"-->all
      </subType>
    </PictureSearchSubType>
  </PictureSearchSubTypeList>
  <isSupportPlaybackStreamIDByUTC>
    <!--ro, opt, bool, whether it supports playback by stream ID according to UTC time-->true
  </isSupportPlaybackStreamIDByUTC>
  <isSupportTimeLockStreamIDByUTC>
    <!--ro, opt, bool, whether it supports locking and unlocking the video segment by steam ID according to UTC time-->true
  </isSupportTimeLockStreamIDByUTC>
  <isSupportTimeSearch>
    <!--ro, opt, bool, whether it supports searching for start and end time of recording by channel, desc:/ISAPI/ContentMgmt/time/search?format=json-->true
  </isSupportTimeSearch>
  <behaviorEventPicSearch opt="allBehaviorEvent,running,group,violentMotion,failDown,playCellphone,peopleNumChange,leavePosition,retention,sleepOnduty">
    <!--ro, opt, string, abnormal event detection events supported by picture search, attr:opt{req, string}, desc:"allBehaviorEvent" represents all events
which are listed in the node "behaviorEventPicSearch"-->test
  </behaviorEventPicSearch>
  <perimeterEventPicSearch opt="allPerimeterEvent,linedetection,fielddetection,regionEntrance,regionExiting,loitering">
    <!--ro, opt, string, perimeter protection events supported by picture search, attr:opt{req, string}, desc:"allPerimeterEvent" represents all events
which are listed in the node "perimeterEventPicSearch"-->test
  </perimeterEventPicSearch>
  <isSupportSourceStatus>
    <!--ro, opt, bool, whether it supports getting the status of access source-->true
  </isSupportSourceStatus>
  <isSupportSSDSMARTTest>
    <!--ro, opt, bool, whether the device supports S.M.A.R.T. detection on SSD, desc:related URI: /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start-->true
  </isSupportSSDSMARTTest>
  <MobileStorage>
    <!--ro, opt, object, storage capability of the mobile HDD-->
    <isSupportGetFileTime>
      <!--ro, opt, bool, whether it supports getting the file time of the mobile HDD, desc:it supports mobile HDD currently.
/ISAPI/ContentMgmt/mobileStorage/fileTime/capabilities?format=json-->true
    </isSupportGetFileTime>
  </MobileStorage>
  <isSupportGetChannelList>
    <!--ro, opt, bool, whether it supports getting channels of the HDD, desc:it supports HDD currently.
/ISAPI/ContentMgmt/Storage/hdd/channelList/capabilities?format=json-->true
  </isSupportGetChannelList>
  <isSupportCascade>
    <!--ro, opt, bool, whether it supports configuring parameters for NVR cascading topology-->true
  </isSupportCascade>
  <isSupportFileManagement>
    <!--ro, opt, bool, whether it supports configuring file management, desc:related URI: /ISAPI/ContentMgmt/fileManagement/capabilities?format=json-->true
  </isSupportFileManagement>
  <isSupportPictureInfo>
    <!--ro, opt, bool, whether it supports picture search on the SD card of the capture camera-->true
  </isSupportPictureInfo>
  <isSupportChanStatusPage>
    <!--ro, opt, string, whether it supports searching the digital channel by page, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/statusPage/capabilities-->test
  </isSupportChanStatusPage>
  <isSupportRecordSearchByTargetType>
    <!--ro, opt, bool, whether it supports searching for event recordings by target types (pedestrians and vehicles)-->true
  </isSupportRecordSearchByTargetType>
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  <storIpcStartNo>
    <!--ro, opt, int, the first IP channel No. of the recording, desc:it is supported by Hikmed Imaging Technology-->0
  </storIpcStartNo>
  <maxRecordIpcNum>
    <!--ro, opt, int, number of the IP channels which support recording, desc:it is supported by Hikmed Imaging Technology-->0
  </maxRecordIpcNum>
  <maxConferenceIpcNum>
    <!--ro, opt, int, number of the meeting IP channels, desc:it is supported by Hikmed Imaging Technology-->0
  </maxConferenceIpcNum>
  <decIpcStartNo>
    <!--ro, opt, int, start No. of screen mirroring IP channels, desc:it is supported by Hikmed Imaging Technology-->0
  </decIpcStartNo>
  <maxDecodeIpcNum>
    <!--ro, opt, int, number of screen mirroring IP channels, desc:it is supported by Hikmed Imaging Technology-->0
  </maxDecodeIpcNum>
  <pipIpcStartNo>
    <!--ro, opt, int, start No. of the integrated IP channels, desc:it is supported by Hikmed Imaging Technology-->0
  </pipIpcStartNo>
  <maxPipIpcNum>
    <!--ro, opt, int, number of the integrated IP channels, desc:it is supported by Hikmed Imaging Technology-->0
  </maxPipIpcNum>
  <isSupportLogSearch>
    <!--ro, opt, bool, whether it supports log search-->true
  </isSupportLogSearch>
  <isSupportRecordSearch>
    <!--ro, opt, bool, whether it supports video search-->true
  </isSupportRecordSearch>
  <isSupportRecordDownload>
    <!--ro, opt, bool, whether it supports video downloading-->true
  </isSupportRecordDownload>
  <isSupportPnp opt="true,false">
    <!--ro, opt, bool, whether it supports configuring plug-and-play for the network camera, attr:opt{req, string}-->true
  </isSupportPnp>
  <esataUseage>
    <!--ro, opt, enum, storage strategy, subType:string, desc:"backup", "record"-->backup
  </esataUseage>
  <subType opt="all,leave,on,sleep">
    <!--ro, opt, enum, absence sub type, subType:string, attr:opt{req, string}, desc:"all", "leave" (absence), "on" (on duty), "sleep" (sleeping on duty)--
>all
  </subType>
  <isSupportManualBackup>
    <!--ro, opt, bool, whether it supports manual backup, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/manual/capabilities?format=json-->true
  </isSupportManualBackup>
  <isSupportLinkToIPCWeb>
    <!--ro, opt, bool, whether it supports going to the network camera web page (the digital channel management page), desc:N/A-->true
  </isSupportLinkToIPCWeb>
  <isSupportIPCConfigCSV>
    <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/ipcConfigCSV?format=json-->true
  </isSupportIPCConfigCSV>
  <isSupportExperimentExam>
    <!--ro, opt, bool, whether it supports the related function of the experiment exam, desc:related URI: /ISAPI/ContentMgmt/experimentExam/capabilities?
format=json-->true
  </isSupportExperimentExam>
  <isSupportPOSRecordSearch>
    <!--ro, opt, bool, whether it supports POS video research, desc:related URI: /ISAPI/ContentMgmt/POSRecord/search/capabilities-->true
  </isSupportPOSRecordSearch>
  <isSupportSpare>
    <!--ro, opt, bool, whether the device supports hot spare configuration, desc:related capability set: /ISAPI/ContentMgmt/spare/capabilities?format=json--
>true
  </isSupportSpare>
  <isSupportPTEventTableTemplate>
    <!--ro, opt, bool-->true
  </isSupportPTEventTableTemplate>
  <isSupportPTEventTableFile>
    <!--ro, opt, bool-->true
  </isSupportPTEventTableFile>
  <isSupportTracksDates>
    <!--ro, opt, bool, whether it supports configuring recording schedule by calendar, desc:related URI:
/ISAPI/ContentMgmt/record/tracks/dates/capabilities?format=json-->true
  </isSupportTracksDates>
  <isSupportRecordAudio>
    <!--ro, opt, bool, whether the device supports audio recording, desc:only the audio will be recorded and the video will not be recorded. Related URI:
/ISAPI/ContentMgmt/recordAudio/<audioID>/capabilities?format=json-->true
  </isSupportRecordAudio>
  <isSupportSearchAudio>
    <!--ro, opt, bool, whether the device supports searching the audio, desc:related URI: /ISAPI/ContentMgmt/recordAudio/search/capabilities?format=json--
>true
  </isSupportSearchAudio>
  <isSupportAudioDailyDistribution>
    <!--ro, opt, bool, whether it supports searching for audio file information by monthly calendar, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json-->true
  </isSupportAudioDailyDistribution>
  <isSupportDownloadAudio>
    <!--ro, opt, bool, whether it supports downloading audio files by file name, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/fileNameDownload/capabilities?format=json-->true
  </isSupportDownloadAudio>
  <isSupportDeleteAudio>
    <!--ro, opt, bool, whether the device supports deleting the recorded audio file, desc:Related URI: /ISAPI/ContentMgmt/recordAudio/deleteFile?
format=json-->true
  </isSupportDeleteAudio>
  <isSupportEventRecordCfg>
    <!--ro, opt, bool, whether it supports configuring parameters of event recordings, desc:related URI: /ISAPI/ContentMgmt/eventRecordCfg?format=json--
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>true
  </isSupportEventRecordCfg>
  <isSupportSearchRemoteAlbum>
    <!--ro, opt, bool, whether the device supports searching the remote album information, desc:related URI: /ISAPI/ContentMgmt/searchRemoteAlbum?
format=json-->true
  </isSupportSearchRemoteAlbum>
  <isNotSupportTracks>
    <!--ro, opt, bool, whether it does not support recording schedule, desc:true will be returned if it is not supported-->true
  </isNotSupportTracks>
  <isSupportTFCardConfig>
    <!--ro, opt, bool, whether the device supports function switch of the extended TF card: /ISAPI/ContentMgmt/TFCardConfig/capabilities?format=json-->true
  </isSupportTFCardConfig>
  <isSupportDockStationFileStorageDuration>
    <!--ro, opt, bool, whether the device supports configuring the retention period of files, desc:related URI:
/ISAPI/ContentMgmt/dockStationFile/DockStationFileStorageDuration?format=json-->true
  </isSupportDockStationFileStorageDuration>
  <isSupportDockStationFileAcquisitionMode>
    <!--ro, opt, bool, whether the device supports configuring the file collection mode, desc:related URI:
/ISAPI/ContentMgmt/dockStationFile/DockStationFileAcquisitionMode?format=json-->true
  </isSupportDockStationFileAcquisitionMode>
  <isSupportSSDAbnormalCfg>
    <!--ro, opt, bool, whether the device supports SSD exception detection parameters, desc:/ISAPI/ContentMgmt/Storage/ssd/SSDAbnormalCfg/capabilities?
format=json-->true
  </isSupportSSDAbnormalCfg>
  <isSupportAudioQuota>
    <!--ro, opt, bool, whether the device supports audio recording quota, desc:GET/PUT /ISAPI/ContentMgmt/channels/<audioID>/AudioQuota?format=json-->true
  </isSupportAudioQuota>
  <targetRegionEventPicSearch opt="linedetection,fielddetection,regionEntrance,regionExiting,VMD">
    <!--ro, opt, string, event type supported to return target region in the picture when searching for pictures, attr:opt{req, string}-->test
  </targetRegionEventPicSearch>
  <isSupportAudioChanCap>
    <!--ro, opt, bool, whether the device supports managing audio channel resources, desc:related URL: /ISAPI/ContentMgmt/audioChannel/capabilities?
format=json-->true
  </isSupportAudioChanCap>
  <isSupportImportEnforcementWorkFile>
    <!--ro, opt, bool, whether the device supports importing the law enforcement working file, desc:POST /ISAPI/ContentMgmt/ImportEnforcementWorkFile?
format=json-->true
  </isSupportImportEnforcementWorkFile>
  <isSupportEnforcementProjectInfo>
    <!--ro, opt, bool, whether the device supports getting the project information of the body camera for law enforcement, desc:GET/PUT
/ISAPI/ContentMgmt/EnforcementProjectInfo?format=json-->true
  </isSupportEnforcementProjectInfo>
  <isSupportTriggerVoiceMonitoringHistoryEventRetrans>
    <!--ro, opt, bool, whether the device supports reloading audio files of the history event of triggering human voice monitoring, desc:POST
/ISAPI/ContentMgmt/channels/<audioID>/TriggerVoiceMonitoringHistoryEventRetrans?format=json-->true
  </isSupportTriggerVoiceMonitoringHistoryEventRetrans>
  <isSupportDockStationAudioFileToText>
    <!--ro, opt, bool, whether the device supports converting dock station's audio to text, desc:POST /ISAPI/ContentMgmt/DockStationAudioFileToText?
format=json-->true
  </isSupportDockStationAudioFileToText>
  <isSupportDockStationAudioFileAnalysisKeyWord>
    <!--ro, opt, bool, whether the device supports keyword analysis of dock station's audio file, desc:GET/PUT
/ISAPI/ContentMgmt/DockStationAudioFileAnalysisKeyWord?format=json-->true
  </isSupportDockStationAudioFileAnalysisKeyWord>
  <isSupportStartingUpRecordParams>
    <!--ro, opt, bool-->true
  </isSupportStartingUpRecordParams>
  <isSupportSendLawEnforcementFileUploadInfo>
    <!--ro, opt, bool-->true
  </isSupportSendLawEnforcementFileUploadInfo>
  <isSupportQuotaParams>
    <!--ro, opt, bool-->true
  </isSupportQuotaParams>
  <isSupportSearchAudioRecordPlan>
    <!--ro, opt, bool-->true
  </isSupportSearchAudioRecordPlan>
  <isSupportAddQuotaParams>
    <!--ro, opt, bool-->true
  </isSupportAddQuotaParams>
  <isSupportModifyQuotaParams>
    <!--ro, opt, bool-->true
  </isSupportModifyQuotaParams>
  <isSupportGetQuotaParams>
    <!--ro, opt, bool-->true
  </isSupportGetQuotaParams>
  <isSupportDeleteQuotaParams>
    <!--ro, opt, bool-->true
  </isSupportDeleteQuotaParams>
  <isSupportAutoRecordSynthesisParams>
    <!--ro, opt, bool-->true
  </isSupportAutoRecordSynthesisParams>
  <isSupportGetSSDSystemConfigRecoverProgress>
    <!--ro, opt, bool-->true
  </isSupportGetSSDSystemConfigRecoverProgress>
  <isSupportSSDSystemConfigRecover>
    <!--ro, opt, bool-->true
  </isSupportSSDSystemConfigRecover>
  <isSupportSSDSystemConfigState>
    <!--ro, opt, bool-->true
  </isSupportSSDSystemConfigState>
  <isSupportPicSearchByCoordinate>
    <!--ro, opt, bool, whether the device supports AcuSearch configurations, desc:GET or PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json-->true
  </isSupportPicSearchByCoordinate>
  <AcuSearchCap>
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    <!--ro, opt, object, AcuSearch capability-->
    <targetType opt="face,human,vehicle">
      <!--ro, opt, enum, AcuSearch supported target type, subType:string, attr:opt{req, string}, desc:"face", "human" (human body), "vehicle"-->face
    </targetType>
    <dataType opt="url,binary,modelData">
      <!--ro, opt, enum, supported picture transmission format, subType:string, attr:opt{req, string}, desc:"URL", (picture URL), "modelData" (model data),
"binary" (picture binary form)-->url
    </dataType>
    <maxPicSize>
      <!--ro, opt, int, maximum size of image can be searched, range:[0,5], desc:unit: MB-->1
    </maxPicSize>
    <maxPicWidthSize>
      <!--ro, opt, int, maximum width of image can be searched, range:[0,4096]-->4096
    </maxPicWidthSize>
    <maxPicHeightSize>
      <!--ro, opt, int, maximum height of image can be searched, range:[0,4096]-->4096
    </maxPicHeightSize>
  </AcuSearchCap>
  <isSupportRecordAsyncParams>
    <!--ro, opt, bool-->true
  </isSupportRecordAsyncParams>
  <isSupportRedirectStorageUrl>
    <!--ro, opt, bool-->true
  </isSupportRedirectStorageUrl>
  <isSupportUploadFIileToCloud>
    <!--ro, opt, bool-->true
  </isSupportUploadFIileToCloud>
  <isSupportAddRecordAgeParams>
    <!--ro, opt, bool-->true
  </isSupportAddRecordAgeParams>
  <isSupportDeleteRecordAgeParams>
    <!--ro, opt, bool-->true
  </isSupportDeleteRecordAgeParams>
  <isSupportSearchRecordAgeParams>
    <!--ro, opt, bool-->true
  </isSupportSearchRecordAgeParams>
  <isSupportModifyRecordAgeParams>
    <!--ro, opt, bool-->true
  </isSupportModifyRecordAgeParams>
  <isSupportEstimateAgeRecordCycle>
    <!--ro, opt, bool-->true
  </isSupportEstimateAgeRecordCycle>
  <isSupportSendDiskPowerOnCtrl>
    <!--ro, opt, bool-->true
  </isSupportSendDiskPowerOnCtrl>
  <isSupportSearchAgeCondChannel>
    <!--ro, opt, bool-->true
  </isSupportSearchAgeCondChannel>
  <isSupportDiskMaterielInfoDetect>
    <!--ro, opt, bool-->true
  </isSupportDiskMaterielInfoDetect>
  <isNotSupportHDDBadSectorsTestStart>
    <!--ro, opt, bool-->false
  </isNotSupportHDDBadSectorsTestStart>
  <isSupportSearchAcuCollectPics>
    <!--ro, opt, bool-->true
  </isSupportSearchAcuCollectPics>
  <isSupportModifyAcuCollectPics>
    <!--ro, opt, bool-->true
  </isSupportModifyAcuCollectPics>
  <isSupportSearchRAIDStoragePerformance>
    <!--ro, opt, bool-->true
  </isSupportSearchRAIDStoragePerformance>
  <isSupportSearchDiskConfigurationSpecificationList>
    <!--ro, opt, bool-->true
  </isSupportSearchDiskConfigurationSpecificationList>
  <isSupportOneKeyRebuildArray>
    <!--ro, opt, bool-->true
  </isSupportOneKeyRebuildArray>
  <isSupportSearchRecordUploadTaskProgress>
    <!--ro, opt, bool-->true
  </isSupportSearchRecordUploadTaskProgress>
  <isSupportSendRecordUploadTask>
    <!--ro, opt, bool-->true
  </isSupportSendRecordUploadTask>
  <isSupportRecordStorageCalculate>
    <!--ro, opt, bool-->true
  </isSupportRecordStorageCalculate>
  <isSupportStartStorageEnvDetection>
    <!--ro, opt, bool-->true
  </isSupportStartStorageEnvDetection>
  <isSupportSearchStorageEnvDetectionProcess>
    <!--ro, opt, bool-->true
  </isSupportSearchStorageEnvDetectionProcess>
  <isSupportSearchAddRaidNeedDiskNum>
    <!--ro, opt, bool-->true
  </isSupportSearchAddRaidNeedDiskNum>
  <isSupportInitQuotaParams>
    <!--ro, opt, bool-->true
  </isSupportInitQuotaParams>
  <isSupportAvailableStoragePathSpaceParams>
    <!--ro, opt, bool-->true
  </isSupportAvailableStoragePathSpaceParams>
```

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```
<isSupportInitAvailableStoragePathSpace>
  <!--ro, opt, bool-->true
</isSupportInitAvailableStoragePathSpace>
<isSupportDelAvailableStoragePathSpace>
  <!--ro, opt, bool-->true
</isSupportDelAvailableStoragePathSpace>
<isSupportAddAvailableStoragePathSpace>
  <!--ro, opt, bool-->true
</isSupportAddAvailableStoragePathSpace>
<isSupportSearchAvailableStoragePathSpace>
  <!--ro, opt, bool-->true
</isSupportSearchAvailableStoragePathSpace>
<isSupportRecordLabelAsyncParams>
  <!--ro, opt, bool-->true
</isSupportRecordLabelAsyncParams>
<isSupportGetAllAvailableStoragePathSpace>
  <!--ro, opt, bool-->true
</isSupportGetAllAvailableStoragePathSpace>
<isSupportFormatArray>
  <!--ro, opt, bool-->true
</isSupportFormatArray>
<isSupportEvaluateModQuotaPerformance>
  <!--ro, opt, bool-->true
</isSupportEvaluateModQuotaPerformance>
<isSupportSearchPathSpaceInitStatus>
  <!--ro, opt, bool-->true
</isSupportSearchPathSpaceInitStatus>
<isSupportFastStorageIndexStoragePathParams>
  <!--ro, opt, bool-->true
</isSupportFastStorageIndexStoragePathParams>
<isSupportUploadFIilesToCloud>
  <!--ro, opt, bool-->true
</isSupportUploadFIilesToCloud>
<isSupportSearchUploadFIilesProgress>
  <!--ro, opt, bool-->true
</isSupportSearchUploadFIilesProgress>
<isSupportSendAcuManualModel>
  <!--ro, opt, bool-->true
</isSupportSendAcuManualModel>
<isSupportSearchAcuManualModel>
  <!--ro, opt, bool-->true
</isSupportSearchAcuManualModel>
<isSupportSyncPicSearchByText>
  <!--ro, opt, bool-->true
</isSupportSyncPicSearchByText>
<isSupportPicSearchByText>
  <!--ro, opt, bool-->true
</isSupportPicSearchByText>
<isSupportKeyChannelCfg>
  <!--ro, opt, bool-->true
</isSupportKeyChannelCfg>
<isSupportAsynSearchByTextTask>
  <!--ro, opt, bool-->true
</isSupportAsynSearchByTextTask>
<isSupportSearchTextTaskStatus>
  <!--ro, opt, bool-->true
</isSupportSearchTextTaskStatus>
<isSupportSearchRecordPlanParams>
  <!--ro, opt, bool-->true
</isSupportSearchRecordPlanParams>
<isSupportStorageDirectoryCfg>
  <!--ro, opt, bool-->true
</isSupportStorageDirectoryCfg>
<isSupportDeleteAsynSearchByTextTask>
  <!--ro, opt, bool-->true
</isSupportDeleteAsynSearchByTextTask>
<isSupportAvailableStoragePathSpaceRangeParams>
  <!--ro, opt, bool-->true
</isSupportAvailableStoragePathSpaceRangeParams>
<isSupportRecordIntegrityTest>
  <!--ro, opt, bool-->true
</isSupportRecordIntegrityTest>
<isSupportRecoverbBackupsInfo>
  <!--ro, opt, bool-->true
</isSupportRecoverbBackupsInfo>
<isSupportDeleteBackupsInfo>
  <!--ro, opt, bool-->true
</isSupportDeleteBackupsInfo>
<isSupportSearchBackupsInfo>
  <!--ro, opt, bool-->true
</isSupportSearchBackupsInfo>
<isSupportCreateCluster>
  <!--ro, opt, bool-->true
</isSupportCreateCluster>
<isSupportModifyClusterInfo>
  <!--ro, opt, bool-->true
</isSupportModifyClusterInfo>
<isSupportSearchClusterInfo>
  <!--ro, opt, bool-->true
</isSupportSearchClusterInfo>
<isSupportNPlusZeroParams>
  <!--ro, opt, bool-->true
</isSupportNPlusZeroParams>
<isSupportSearchClusteProgress>
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<isSupportSearchClusterProgress>
<!--ro, opt, bool-->true
</isSupportSearchClusterProgress>
<isSupportSSDTemperatureAbnormalTrigger>
<!--ro, opt, bool-->true
</isSupportSSDTemperatureAbnormalTrigger>
<isSupportSSDLifeTimeAbnormalTrigger>
<!--ro, opt, bool-->true
</isSupportSSDLifeTimeAbnormalTrigger>
<isSupportSearchcRedundantResources>
<!--ro, opt, bool-->true
</isSupportSearchcRedundantResources>
<isSupportAddStoragePoolList>
<!--ro, opt, bool-->true
</isSupportAddStoragePoolList>
<isSupportResetPassword>
<!--ro, opt, bool-->true
</isSupportResetPassword>
<isSupportSearchChannelList>
<!--ro, opt, bool-->true
</isSupportSearchChannelList>
<isSupportRecordSynthesisPlan>
<!--ro, opt, bool-->true
</isSupportRecordSynthesisPlan>
<isSupportInitQuotaDefaultParam>
<!--ro, opt, bool-->true
</isSupportInitQuotaDefaultParam>
<isSupportStorageAutoPreallocation>
<!--ro, opt, bool-->true
</isSupportStorageAutoPreallocation>
<isSupportSearchAcuSeekChannel>
<!--ro, opt, bool-->true
</isSupportSearchAcuSeekChannel>
<isSupportModifyAcuSeekChannel>
<!--ro, opt, bool-->true
</isSupportModifyAcuSeekChannel>
<isSupportTestPicSearchByCoordinateChannel>
<!--ro, opt, bool-->true
</isSupportTestPicSearchByCoordinateChannel>
<isSupportGetAcuSeekChannelStatus>
<!--ro, opt, bool-->true
</isSupportGetAcuSeekChannelStatus>
<isSupportStopRedirectStorageUrl>
<!--ro, opt, bool-->true
</isSupportStopRedirectStorageUrl>
<isSupportArrayExpansionEstimation>
<!--ro, opt, bool-->true
</isSupportArrayExpansionEstimation>
<isSupportArrayExpansion>
<!--ro, opt, bool-->true
</isSupportArrayExpansion>
<isSupportSearchPassbackStatistics>
<!--ro, opt, bool-->true
</isSupportSearchPassbackStatistics>
<isSupportRAIDOnekeyTemplate>
<!--ro, opt, bool-->true
</isSupportRAIDOnekeyTemplate>
<isSupportPassbackTaskOverview>
<!--ro, opt, bool-->true
</isSupportPassbackTaskOverview>
<isSupportRAIDOnekeyTemplateEstimation>
<!--ro, opt, bool-->true
</isSupportRAIDOnekeyTemplateEstimation>
<isSupportArchiveStorageAlarm>
<!--ro, opt, bool-->true
</isSupportArchiveStorageAlarm>
<isSupportArchiveRecordPlan>
<!--ro, opt, bool-->true
</isSupportArchiveRecordPlan>
<isSupportSearchArchiveRecordPlan>
<!--ro, opt, bool-->true
</isSupportSearchArchiveRecordPlan>
<isSupportArchiveExecutionPlan>
<!--ro, opt, bool-->true
</isSupportArchiveExecutionPlan>
<isSupportSearchArchiveExecutionPlan>
<!--ro, opt, bool-->true
</isSupportSearchArchiveExecutionPlan>
<isSupportClusterDurationParams>
<!--ro, opt, bool-->true
</isSupportClusterDurationParams>
<isSupportSearchArchiveRecordStatus>
<!--ro, opt, bool-->true
</isSupportSearchArchiveRecordStatus>
<isSupportAddArchiveRecord>
<!--ro, opt, bool-->true
</isSupportAddArchiveRecord>
<isSupportDeleteArchiveRecord>
<!--ro, opt, bool-->true
</isSupportDeleteArchiveRecord>
<isSupportGetArchiveRecordInfo>
<!--ro, opt, bool-->true
</isSupportGetArchiveRecordInfo>
<isSupportArchiveRecordOverview>

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```
    <!--ro, opt, bool-->true
  </isSupportArchiveRecordOverview>
  <isSupportGetAddClusterDevicesProcess>
    <!--ro, opt, bool-->true
  </isSupportGetAddClusterDevicesProcess>
  <isSupportDeleteStorageDirectory>
    <!--ro, opt, bool-->true
  </isSupportDeleteStorageDirectory>
  <isSupportFormatSSDFileSystem>
    <!--ro, opt, bool-->true
  </isSupportFormatSSDFileSystem>
  <isSupportGetSSDFileSystemFormatStatus>
    <!--ro, opt, bool-->true
  </isSupportGetSSDFileSystemFormatStatus>
  <isSupportCreateCloudUploadAsyncTask>
    <!--ro, opt, bool-->true
  </isSupportCreateCloudUploadAsyncTask>
  <isSupportDeleteCloudUploadAsyncTask>
    <!--ro, opt, bool-->true
  </isSupportDeleteCloudUploadAsyncTask>
  <isSupportGetCloudUploadAsyncTaskResult>
    <!--ro, opt, bool-->true
  </isSupportGetCloudUploadAsyncTaskResult>
  <isSupportSearchDeviceFolderInfo>
    <!--ro, opt, bool-->true
  </isSupportSearchDeviceFolderInfo>
</RacmCap>
```

14.1.4.2 Get the system security capability

Request URL

GET /ISAPI/Security/capabilities?username=<userName>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
userName	string	user name
deviceIndex	string	For centralized splicing controllers connecting to LED sending cards, the URL needs to include the parameter deviceIndex= to specify LED sending card The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SecurityCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, system security capability, attr:version{req, string, protocolVersion}-->
  <supportUserNums>
    <!--ro, opt, int, supported maximum number of users-->0
  </supportUserNums>
  <userBondIpNums>
    <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->0
  </userBondIpNums>
  <userBondMacNums>
    <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->0
  </userBondMacNums>
  <isSupCertificate>
    <!--ro, opt, bool, whether the device supports authentication-->true
  </isSupCertificate>
  <issupIllegalLoginLock>
    <!--ro, opt, bool, whether the device supports locking login-->true
  </issupIllegalLoginLock>
  <isSupportOnlineUser>
    <!--ro, opt, bool, whether the device supports the online user configuration-->true
  </isSupportOnlineUser>
  <isSupportAnonymous>
    <!--ro, opt, bool, whether the device supports anonymous login-->true
  </isSupportAnonymous>
  <isSupportStreamEncryption>
    <!--ro, opt, bool, whether the device supports stream encryption-->true
  </isSupportStreamEncryption>
  <securityVersion opt="1,2,3,4">
    <!--ro, opt, string, encryption capability set, attr:opt{req, string}, desc:the encryption capability of each version consists of two parts: encryption
algorithm and the range of encrypted nodes currently 1 refers to AES128 encryption and 2 refers to AES256 encryption, the range of encrypted nodes is
```

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```
described in each protocol-->1
  </securityVersion>
  <keyIterateNum>
    <!--ro, opt, int, secret key iteration times, dep:or,{$.SecurityCap.securityVersion,eq,1},{$.SecurityCap.securityVersion,eq,2}, desc:this node depends
on the node securityVersion, the range is between 100 and 1000-->100
  </keyIterateNum>
  <isSupportUserCheck>
    <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, dep:or,
{$.SecurityCap.securityVersion,eq,0},{$.SecurityCap.securityVersion,eq,1}, desc:it is an added capability, which indicates that whether supporting the login
password verification for editing/adding/deleting user parameters, this node depends on the node securityVersion, which means that it is only valid for the
versions that support encrypting the sensitive information-->true
  </isSupportUserCheck>
  <isSupportGUIDFileDataExport>
    <!--ro, opt, bool, whether the device supports exporting the device's GUID file, desc:related URI: /ISAPI/Security/GUIDFileData-->true
  </isSupportGUIDFileDataExport>
  <isSupportSecurityQuestionConfig>
    <!--ro, opt, bool, whether the device supports answering security questions, desc:related URI: /ISAPI/Security/questionConfiguration-->true
  </isSupportSecurityQuestionConfig>
  <supportSecurityNode opt="wirelessServer,wirelessDial">
    <!--ro, opt, string, additional node for supporting sensitive information encryption, attr:opt{req, string},
desc:in the earlier version of the solution for encrypting the sensitive information via ISAPI (i.e., the solution for transmitting plaintext via ISAPI),
some URIs containing sensitive information are left out now these URIs need to be encrypted for transmission, this capability node is added to be compatible
with old devices _x000D_
          if wirelessServer in this node is returned, it indicates that the following sensitive information should be encrypted _x000D_
          /ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities, /ISAPI/System/Network/interfaces/<ID>/wirelessServer, if wirelessDial in this
node is returned, it indicates that the following sensitive information should be encrypted _x000D_
          /ISAPI/System/Network/WirelessDial/Interfaces,_x000D_
    /ISAPI/System/Network/WirelessDial/Interfaces/<ID>-->wirelessServer
  </supportSecurityNode>
  <isSupportGetOnlineUserListSC>
    <!--ro, opt, bool, whether the device supports searching the online user list, desc:related URI: /ISAPI/Security/onlineUser-->true
  </isSupportGetOnlineUserListSC>
  <SecurityLimits>
    <!--ro, opt, object, capability of configuring security limit parameters-->
    <LoginPasswordLenLimit min="1" max="16">
      <!--ro, opt, string, length limit of the user's login password, attr:min{req, int},max{req, int}-->1
    </LoginPasswordLenLimit>
    <SecurityAnswerLenLimit min="1" max="128">
      <!--ro, opt, string, length limit of the security question's answer, attr:min{req, int},max{req, int}-->1
    </SecurityAnswerLenLimit>
  </SecurityLimits>
  <RSAKeyLength opt="512,1024,2048" def="2048">
    <!--ro, opt, enum, HTTPS certificate length, subType:int, attr:opt{req, string},def{req, string}, desc:512, 1024, 2048-->2048
  </RSAKeyLength>
  <isSupportONVIFUserManagement>
    <!--ro, opt, bool, whether the device supports user management of Open Network Video Interface Protocol-->true
  </isSupportONVIFUserManagement>
  <WebCertificateCap>
    <!--ro, opt, object, HTTP authentication capability, desc:if this node is not returned, it indicates that device supports basic and digest
authentication-->
    <CertificateType opt="basic,digest,digest/basic">
      <!--ro, req, enum, certificate type, subType:string, attr:opt{req, string}, desc:“basic” (basic authentication), “digest” (digest authentication),
“digest/basic” (digest/basic authentication)-->basic
    </CertificateType>
    <SecurityAlgorithm>
      <!--ro, opt, object, security algorithm, dep:or,{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest},
{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
      <algorithmType opt="MD5,SHA256,MD5/SHA256">
        <!--ro, opt, enum, algorithm type, subType:string, attr:opt{req, string}, desc:“MD5”, “SHA256”, “MD5/SHA256”-->MD5
      </algorithmType>
    </SecurityAlgorithm>
  </WebCertificateCap>
  <isSupportConfigFileImport>
    <!--ro, opt, bool, whether the device supports importing the configuration file, desc:true (support), this node is not returned (not support)-->true
  </isSupportConfigFileImport>
  <isSupportConfigFileExport>
    <!--ro, opt, bool, whether the device supports exporting the configuration file, desc:true (support), this node is not returned (not support)-->true
  </isSupportConfigFileExport>
  <cfgFileSecretKeyLenLimit min="0" max="16">
    <!--ro, opt, string, length limit of the configuration file's verification key, attr:min{req, int},max{req, int}-->1
  </cfgFileSecretKeyLenLimit>
  <supportIPCActivatePassword>
    <!--ro, opt, bool, whether the device supports configuring the password for activating the network camera-->true
  </supportIPCActivatePassword>
  <isIrreversible>
    <!--ro, opt, bool, whether the device supports irreversible password storage, desc:If this function is not supported, the plaintext password of the user
information will be stored in the device; otherwise, the password will be hashed for storage in the device.-->true
  </isIrreversible>
  <salt>
    <!--ro, opt, string, the specific salt used by the user to log in-->test
  </salt>
  <isSupportOnvifInfo>
    <!--ro, opt, bool, whether the device supports getting the Open Network Video Interface Protocol, desc:true (support), this node is not returned (not
support)-->true
  </isSupportOnvifInfo>
  <isSupportPictureURlCertificate opt="true,false">
    <!--ro, opt, bool, whether the device supports configuring parameters for picture URL authentication, attr:opt{req, string}, desc:if this node is not
returned, it indicates the digest authentication for picture URL by default-->true
  </isSupportPictureURlCertificate>
  <isSupportGB35114Certificate>
    <!--ro, opt, bool, whether supports GB35114 protocol and CA/SIP authentication certificate-->true
  </isSupportGB35114Certificate>
  <isSupportAuthenticationMode opt="uKey,private">
    <!--ro, opt, enum, verification mode, subType:string, attr:opt{req, string}, desc:“uKey”, “private”-->private
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```
  </isSupportAuthenticationMode>
  <isSupportUserWithIssuerDN>
    <!--ro, opt, bool-->true
  </isSupportUserWithIssuerDN>
  <isSupportUnloggedUserPermissionConfig opt="true,false">
    <!--ro, opt, bool, whether the device supports configuring permission for users that have not logged in, attr:opt{req, string}-->true
  </isSupportUnloggedUserPermissionConfig>
  <isSupportUserNamePasswordCheckUpgrade>
    <!--ro, opt, bool, whether the device supports upgrading the verification of the user name and password, desc:if this node is not returned, it indicates
that device does not support this function-->true
  </isSupportUserNamePasswordCheckUpgrade>
  <isSupportDeviceCertificatesManagement>
    <!--ro, opt, bool, whether the device supports device certificate management, desc:if this node is not returned, it indicates that device does not
support this function-->true
  </isSupportDeviceCertificatesManagement>
  <isSupportSIPCertificatesManagement>
    <!--ro, opt, bool, whether the device supports SIP certificate management, desc:this node is returned if the device supports this function and not
returned if device does not support this function, related URI: /ISAPI/Security/SIPCertificate, additional node: <SIPCertificatesManagementCap>-->true
  </isSupportSIPCertificatesManagement>
  <isSupportDeviceSelfSignCertExport>
    <!--ro, opt, bool, whether the device supports exporting self-signed certificate-->true
  </isSupportDeviceSelfSignCertExport>
  <isSupportSecurityEmail>
    <!--ro, opt, bool, whether the device supports configuring the security email, desc:if this node is not returned, it indicates that device does not
support this function-->true
  </isSupportSecurityEmail>
  <isSupportRTSPCertificate opt="true,false">
    <!--ro, opt, bool, whether the device supports RTSP certificate management, attr:opt{req, string}-->true
  </isSupportRTSPCertificate>
  <maxIllegalLoginTimes min="3" max="20" def="5">
    <!--ro, opt, int, maximum failed illegal login attempts, range:[3,20], attr:min{req, int},max{req, int},def{req, int}, desc:this node is valid when the
value of the node issupportIllegalLoginLock is true and will be returned when the maximum failed attempts of illegally logging in to the device reach the
limit-->3
  </maxIllegalLoginTimes>
  <SecurityAdvanced>
    <!--ro, opt, object, security advanced parameters-->
    <securityEnhanced>
      <!--ro, opt, bool, whether the device supports security reinforcement-->true
    </securityEnhanced>
    <noOperationEnabled>
      <!--ro, req, bool, whether to enable the control timeout when there is no operation-->true
    </noOperationEnabled>
    <noOperationTime min="1" max="60" def="15">
      <!--ro, req, int, control timeout when there is no operation, range:[1,60], unit:min, attr:min{req, int},max{req, int},def{req, int}-->15
    </noOperationTime>
    <isSupportDigestStatus>
      <!--ro, opt, bool, digest status-->true
    </isSupportDigestStatus>
    <passwordLimitLevel opt="middling,high">
      <!--ro, opt, enum, password strength limit level, subType:string, attr:opt{req, string},
desc:the configured password is required by the following limit _x000D_
“middling” (middle level), “high” (high level), middle level password contains 8 to 16 characters, including at least 2 types of the following characters:
digits, lower case letters, upper case letters, and special characters; high level password contains 8 to 16 characters, including at least 3 types of the
following characters: digits, lower case letters, upper case letters, and special characters-->middling
    </passwordLimitLevel>
  </SecurityAdvanced>
  <LoginLinkNum>
    <!--ro, opt, object, number of controls logging in at the same time-->
    <maxLinkNum min="1" max="128" def="50">
      <!--ro, req, int, maximum number of controls logging in at the same time, range:[1,128], attr:min{req, int},max{req, int},def{req, int}-->50
    </maxLinkNum>
  </LoginLinkNum>
  <isSupportCCClientCertificate>
    <!--ro, opt, bool, whether the device supports the CC (Client Certificate) function-->true
  </isSupportCCClientCertificate>
  <passwordValidity min="0" max="365">
    <!--ro, opt, string, range of the password validity period, range:[0,365], attr:min{req, int},max{req, int}, desc:if this node is returned, it indicates
that the device supports configuring the password validity period, and only the administrator can edit the configuration-->1
  </passwordValidity>
  <isSupportResetChannelPassword>
    <!--ro, opt, bool, whether the device supports resetting the password of the channel-->true
  </isSupportResetChannelPassword>
  <DoubleVerificationCap>
    <!--ro, opt, object, secondary authentication capability-->
    <isSupportUsersConfig>
      <!--ro, opt, bool, whether the device supports configuring users for secondary authentication-->true
    </isSupportUsersConfig>
    <isSupportUsersPermissionConfig>
      <!--ro, opt, bool, whether the device supports configuring user permission for secondary authentication-->true
    </isSupportUsersPermissionConfig>
    <isSupportFactoryReset>
      <!--ro, opt, bool, whether the device supports restoring secondary authentication to defaults-->true
    </isSupportFactoryReset>
  </DoubleVerificationCap>
  <isSupportUserSearch>
    <!--ro, opt, bool, whether the device supports searching users-->true
  </isSupportUserSearch>
  <keypadPassword min="1" max="10">
    <!--ro, opt, int, keypad password length, attr:min{req, int},max{req, int}, desc:When the password length of different user types is different, this
node only indicates the keypad password length of the administrator; otherwise, the keypad password length of all user types is the same and defined by the
value of this node-->1
  </keypadPassword>
  <installerKeypadPassword min="1" max="10">
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    <!--ro, opt, int, keypad password length of the installer, attr:min{req, int},max{req, int}-->1
  </installerKeypadPassword>
  <operatorKeypadPassword min="1" max="10">
    <!--ro, opt, int, keypad password length of the operator, attr:min{req, int},max{req, int}-->1
  </operatorKeypadPassword>
  <userOperateType opt="1,2,3">
    <!--ro, opt, enum, user type, subType:int, attr:opt{req, string}, desc:1 (network user), 2 (keypad user), 3 (network user and keypad user)-->1
  </userOperateType>
  <isSptUserEnabled>
    <!--ro, opt, bool, whether the device supports configuring parameters for enabling the user-->true
  </isSptUserEnabled>
  <isSptAdminCap>
    <!--ro, opt, bool, whether the device supports getting the admin permission capability-->true
  </isSptAdminCap>
  <isSupportBackLogCfg>
    <!--ro, opt, bool, whether the device supports configuring basic parameters of the background log-->true
  </isSupportBackLogCfg>
  <isSupportBackLogExport>
    <!--ro, opt, bool, whether the device supports exporting the background log-->true
  </isSupportBackLogExport>
  <isSupportModifyInnerPassword opt="true,false">
    <!--ro, opt, bool, whether the device supports editing its built-in passwords,, attr:opt{req, string}, desc:such as the database password, system
account password-->true
  </isSupportModifyInnerPassword>
  <isSupportEncryptCertificate>
    <!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
  </isSupportEncryptCertificate>
  <maxIllegalLoginLockTime min="1" max="120" def="30.00">
    <!--ro, opt, int, lock time range, range:[1,120], unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
  </maxIllegalLoginLockTime>
  <cloudSensitiveInfoEncryptType>
    <!--ro, opt, string, encryption type of the cloud sensitive information, desc:the current device returns the value "2.1", which indicates that the
method of generating the key for encrypting the sensitive information is SHA256 (calc_sha256 (user, salt, SHA256 (user+salt2+SHA256 (user+salt+password))) +
AaBbCcDd1234!@#$)-->2.1
  </cloudSensitiveInfoEncryptType>
  <isSupportCertificateCustomID>
    <!--ro, opt, bool, whether the device supports using the user's custom ID to configure the certificate-->true
  </isSupportCertificateCustomID>
  <isSupportResetLANPassword>
    <!--ro, opt, bool, whether the device supports resetting the device's password in the LAN, desc:related URI:
/ISAPI/Security/resetPassword/certification?format=json-->true
  </isSupportResetLANPassword>
  <isSupportPicHttpsCertificatesManagement>
    <!--ro, opt, bool, whether the device supports managing the HTTPS certificate for pictures, desc:true (support), this node is not returned (not
support)-->true
  </isSupportPicHttpsCertificatesManagement>
  <isSupportAKSKImport>
    <!--ro, opt, bool, whether the device supports importing AKSK, desc:true (support), this node is not returned (not support)-->true
  </isSupportAKSKImport>
  <isSupportAKSKExport>
    <!--ro, opt, bool, whether the device supports exporting AKSK, desc:true (support), this node is not returned (not support)-->true
  </isSupportAKSKExport>
  <isSupportAKSKReset>
    <!--ro, opt, bool, whether the device supports resetting AKSK, desc:true (support), this node is not returned (not support)-->true
  </isSupportAKSKReset>
  <ExportCertificateFile>
    <!--ro, opt, object, whether the device supports exporting the certificate, desc:related URI: POST
/ISAPI/Security/serverCertificate/exportCertificateFile-->
    <type opt="GB35114,GB35114Service">
      <!--ro, req, enum, types of certificates that can be exported, subType:string, attr:opt{req, string}, desc:“GB35114” (support exporting GB35114
certificate for the client), “GB35114Service” (support exporting GB35114 certificate for the server)-->GB35114
    </type>
  </ExportCertificateFile>
  <isSupportGB351114FDWSFCertificate>
    <!--ro, opt, bool, whether the device supports FDWSF certificate management, desc:true (support), this node is not returned (not support), related URIs:
/ISAPI/Security/GB351114/FDWSFCertificate/capabilities?format=json, POST /ISAPI/Security/GB351114/FDWSFCertificate?format=json, GET
/ISAPI/Security/GB351114/FDWSFCertificate?format=json, and DELETE /ISAPI/Security/GB351114/FDWSFCertificate?format=json-->true
  </isSupportGB351114FDWSFCertificate>
  <isSupportMediaObjectEncryption>
    <!--ro, opt, bool, whether supports setting encryption parameters of the private target in media data for the specified video channel, desc:if this
function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI:
/ISAPI/Security/MediaObjectEncryption/capabilities?format=json-->true
  </isSupportMediaObjectEncryption>
  <isSupportDisclaimer>
    <!--ro, opt, bool, whether support configuring the product disclaimer, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/disclaimer/capabilities?format=json-->true
  </isSupportDisclaimer>
  <isSupportSoftwareLicense>
    <!--ro, opt, bool, whether support configuring the product open source statement, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/softwareLicense?format=json-->true
  </isSupportSoftwareLicense>
  <isSupportInstallerCap>
    <!--ro, opt, bool, whether the device supports getting the installer permission, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/UserPermission/installer/capabilities?format=json-->true
  </isSupportInstallerCap>
  <isSupportResetPasswordFile>
    <!--ro, opt, bool, whether the device supports exporting and importing the secret key file of device in the LAN, desc:Related URI:
/ISAPI/Security/resetPassword/keyFile/capabilities?format=json-->true
  </isSupportResetPasswordFile>
  <isNotSupportLogSeverCertificate>
    <!--ro, opt, bool, whether the device support importing log server certificate, desc:related URI: /ISAPI/Security/serverCertificate/certificate-->true
  </isNotSupportLogSeverCertificate>
  <isSupportLocalLoginCfg>
    <!--ro, opt, bool, whether it supports the capability of device local login parameters, desc:GET/PUT /ISAPI/Security/LocalLoginCfg?format=json-->true
```

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```
    </isSupportLocalLoginCfg>
    <isSupportGetProtocolFirewallRulesFileImportStatus>
      <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallRulesFileImportStatus?format=json-->true
    </isSupportGetProtocolFirewallRulesFileImportStatus>
    <isSupportImportProtocolFirewallRulesFile>
      <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/ImportProtocolFirewallRulesFile?format=json-->true
    </isSupportImportProtocolFirewallRulesFile>
    <isSupportGetProtocolFirewallEngineUpgradeStatus>
      <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallEngineUpgradeStatus?format=json-->true
    </isSupportGetProtocolFirewallEngineUpgradeStatus>
    <isSupportUpgradeProtocolFirewallEngine>
      <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/UpgradeProtocolFirewallEngine?format=json-->true
    </isSupportUpgradeProtocolFirewallEngine>
    <isSupportProtocolFirewallVersionInfo>
      <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/ProtocolFirewallVersionInfo?format=json-->true
    </isSupportProtocolFirewallVersionInfo>
    <isSupportProtocolFirewallParams>
      <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/Security/ProtocolFirewallParams?format=json-->true
    </isSupportProtocolFirewallParams>
    <isSupportResetPasswordBySecurityPhoneNo>
      <!--ro, opt, bool, whether it supports resetting device password with phone number verification,
desc:1. attendance check devices support applying a preset phone number during device activation. The user can reset the device password with the preset
phone number, which is similar with password reset via email. The phone number can be changed on the website or local UI of the device _x000D_
2. if the node is not returned or the returned value is false, it indicates "no"-->true
    </isSupportResetPasswordBySecurityPhoneNo>
    <isSupportGetIntrusionPreventionEngineUpgradeStatus>
      <!--ro, opt, bool, whether the device supports getting the upgrade status of intrusion prevention engine library, desc:GET
/ISAPI/Security/GetIntrusionPreventionEngineUpgradeStatus?format=json-->true
    </isSupportGetIntrusionPreventionEngineUpgradeStatus>
    <isSupportIntrusionPreventionEngineVersionInfo>
      <!--ro, opt, bool, whether the device supports version information of intrusion prevention engine, desc:GET
/ISAPI/Security/IntrusionPreventionEngineVersionInfo?format=json-->true
    </isSupportIntrusionPreventionEngineVersionInfo>
    <isSupportIntrusionPreventionParams>
      <!--ro, opt, bool, whether the device supports intrusion prevention parameters, desc:GET/PUT /ISAPI/Security/IntrusionPreventionParams?format=json--
>true
    </isSupportIntrusionPreventionParams>
    <isSupportUpgradeIntrusionPreventionEngine>
      <!--ro, opt, bool, whether the device supports upgrading intrusion prevention engine library, desc:PUT /ISAPI/Security/UpgradeIntrusionPreventionEngine?
format=json-->true
    </isSupportUpgradeIntrusionPreventionEngine>
    <notSupportImportCertificateType opt="CA,CCClentCertificate,GB35114,GB35114Service">
      <!--ro, opt, string, file type that cannot be imported, attr:opt{req, string}, desc:if this node does not exist, all supported certificate types can be
imported-->test
    </notSupportImportCertificateType>
    <isSupportPasswordProtection>
      <!--ro, opt, bool, whether it supports configuring SDK protocol authentication mode parameters, desc:PUT/GET
/ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
    </isSupportPasswordProtection>
    <isSupportGMT0028SecretStandardParam>
      <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/Security/GMT0028SecretStandard/GMT0028SecretStandardParam?format=json-->true
    </isSupportGMT0028SecretStandardParam>
    <isSupportCipherCardRunningPolicy>
      <!--ro, opt, bool-->true
    </isSupportCipherCardRunningPolicy>
    <isSupportGetSecurityScanStatus>
      <!--ro, opt, bool-->true
    </isSupportGetSecurityScanStatus>
    <isSupportStartSecurityScan>
      <!--ro, opt, bool-->true
    </isSupportStartSecurityScan>
    <isSupportStopSecurityScan>
      <!--ro, opt, bool-->true
    </isSupportStopSecurityScan>
    <isSupportModifySecurityItems>
      <!--ro, opt, bool-->true
    </isSupportModifySecurityItems>
  </SecurityCap>
```

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14.1.4.3 Get device system capability

Request URL

GET /ISAPI/System/capabilities?type=<all>

Query Parameter

Parameter Name	Parameter Type	Description
all	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device system capability, attr:version{opt, string, protocolVersion}-->
  <isSupportPreview>
    <!--ro, opt, bool, whether the device supports live view-->true
  </isSupportPreview>
  <isSupportSystemMaintain>
    <!--ro, opt, bool, whether the device supports system maintenance, desc:for traffic devices, this node is required and must be true-->true
  </isSupportSystemMaintain>
  <isSupportReboot>
    <!--ro, opt, bool, whether the device supports rebooting, desc:for traffic devices, this node is required and must be true.-->true
  </isSupportReboot>
  <isSupportFactoryReset>
    <!--ro, opt, bool, whether the device supports restoring to default settings, desc:for traffic devices, this node is required and must be true-->true
  </isSupportFactoryReset>
  <isSupportConfigurationData>
    <!--ro, opt, bool, whether the device supports importing and exporting the configuration files safely, desc:for traffic devices, this node is required
and must be true-->true
  </isSupportConfigurationData>
  <isSupportUpdatefirmware>
    <!--ro, opt, bool, whether the device supports upgrading, desc:for traffic devices, this node is required and must be true-->true
  </isSupportUpdatefirmware>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports getting the device information, desc:for traffic devices, this node is required and must be true-->true
  </isSupportDeviceInfo>
  <isSupportTime>
    <!--ro, opt, bool, whether the device supports time configuration, desc:for traffic device, this node is required and must be true. For other devices,
the isSupportTimeCap is used to configure time-->true
  </isSupportTime>
  <SysCap>
    <!--ro, opt, object, system capability-->
    <isSupportDst>
      <!--ro, opt, bool, whether the device supports DST (Daylight Saving Time)-->true
    </isSupportDst>
    <NetworkCap>
      <!--ro, opt, object, network capability, desc:related URI: /ISAPI/System/Network/capabilities-->
      <isSupportWireless>
        <!--ro, req, bool, whether the device supports wireless network-->true
      </isSupportWireless>
      <isSupportWired>
        <!--ro, opt, bool, whether it supports wired network-->true
      </isSupportWired>
      <isSupportWAN>
        <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
      </isSupportWAN>
      <isSupportPPPoE>
        <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
      </isSupportPPPoE>
      <isSupportBond>
        <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
      </isSupportBond>
      <isSupport802_1x>
        <!--ro, req, bool, whether the device supports 802.1x protocol-->true
      </isSupport802_1x>
      <isSupportNtp>
        <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
      </isSupportNtp>
      <isSupportFtp>
        <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
      </isSupportFtp>
      <isSupportUpnp>
        <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play ) protocol-->true
      </isSupportUpnp>
      <isSupportPNP>
        <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
      </isSupportPNP>
      <isSupportDdns>
        <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
      </isSupportDdns>
      <isSupportHttps>
        <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
      </isSupportHttps>
      <isSupport28181>
        <!--ro, opt, bool, N/A-->true
      </isSupport28181>
      <SnmpCap>
        <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
        <isSupport>
          <!--ro, req, bool, whether the device supports SNMP-->true
        </isSupport>
      </SnmpCap>
      <isSupportExtNetCfg>
        <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
      </isSupportExtNetCfg>
      <isSupportIPFilter>
        <!--ro, opt, bool, whether the device supports IP filtering-->true
      </isSupportIPFilter>
      <isSupportSSH>
        <!--ro, opt, bool, whether the device supports SSH-->true
      </isSupportSSH>
      <isSupportNetPreviewStrategy>
        <!--ro, opt, bool, whether the device supports network preview strategy-->true
      </isSupportNetPreviewStrategy>
```

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```
<isSupportEZVIZ>
  <!--ro, opt, bool, whether the device supports EZ protocol-->true
</isSupportEZVIZ>
<isSupportEhome>
  <!--ro, opt, bool, whether the device supports ISUP server configuration-->true
</isSupportEhome>
<isSupportWirelessDial>
  <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
</isSupportWirelessDial>
<isSupportWirelessServer>
  <!--ro, opt, bool, whether the device supports wireless server-->true
</isSupportWirelessServer>
<isSupportWPS>
  <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
</isSupportWPS>
<isWirelessMutexWithWirelessServer>
  <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
</isWirelessMutexWithWirelessServer>
<isSupportMACFilter>
  <!--ro, opt, bool, whether the device supports physical address filtering-->true
</isSupportMACFilter>
<GB28181Cap>
  <!--ro, opt, object, N/A-->
  <isSupportGB28181Service>
    <!--ro, opt, bool, N/A-->true
  </isSupportGB28181Service>
</GB28181Cap>
<WPS>
  <!--ro, opt, object, Wi-Fi Protected Setup-->
  <NetworkInterfaceList size="2">
    <!--ro, opt, array, list of network nodes, subType:object, attr:size{opt, string}-->
    <NetworkInterface>
      <!--ro, opt, object, network node-->
      <id>
        <!--ro, req, string, NIC index-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable or not-->true
      </enabled>
      <isSupportAutoConnect>
        <!--ro, opt, bool, whether the device supports automatic connection-->true
      </isSupportAutoConnect>
      <isSupportDevicePinCode>
        <!--ro, opt, bool, whether the device supports PIN code-->true
      </isSupportDevicePinCode>
      <isSupportDevicePinCodeUpdate>
        <!--ro, opt, bool, whether the device supports updating the PIN code-->true
      </isSupportDevicePinCodeUpdate>
      <ApPinCode>
        <!--ro, opt, object, PIN code of the access point-->
        <ssid min="1" max="1">
          <!--ro, opt, string, SSID (Service Set Identifier), attr:min{opt, string},max{opt, string}-->test
        </ssid>
        <pinCode min="1" max="1">
          <!--ro, opt, string, PIN code, attr:min{opt, string},max{opt, string}-->test
        </pinCode>
      </ApPinCode>
    </NetworkInterface>
  </NetworkInterfaceList>
</WPS>
<isSupportRFIDData>
  <!--ro, opt, bool, whether the device supports RFID collection configuration-->true
</isSupportRFIDData>
<isSupportwifiProbeSSID>
  <!--ro, opt, bool, whether the device supports configuration of Wi-Fi signal detection-->true
</isSupportwifiProbeSSID>
<isSupportWifiProbe>
  <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
</isSupportWifiProbe>
<verificationCode min="1" max="1">
  <!--ro, opt, string, verification code, attr:min{opt, string},max{opt, string}-->test
</verificationCode>
<WPSCap>
  <!--ro, opt, object, capability of configuring Wi-Fi Protected Setup-->
  <isSupport>
    <!--ro, req, bool, whether the device supports Wi-Fi Protected Setup-->true
  </isSupport>
  <isSupportAutoConnect>
    <!--ro, req, bool, whether the device supports automatic connection-->true
  </isSupportAutoConnect>
</WPSCap>
<NetWorkMode>
  <!--ro, opt, object, network mode configuration-->
  <workMode>
    <!--ro, opt, string, network mode-->true
  </workMode>
</NetWorkMode>
<VerificationCodeModification>
  <!--ro, opt, object, device verification code configuration-->
  <verificationCodeType opt="normal,empty">
    <!--ro, opt, string, verification code type, attr:opt{opt, string}-->empty
  </verificationCodeType>
  <isSupportDeclarationURL>
```

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```
        <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
      </isSupportDeclarationURL>
      <isSupportPrivacyPolicyURL>
        <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
      </isSupportPrivacyPolicyURL>
      <verificationCodeModify opt="true,false">
        <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{opt, string}-->true
      </verificationCodeModify>
      <Hyperlinks>
        <!--ro, opt, object, hyperlink configuration-->
        <declarationURL>
          <!--ro, opt, string, hyperlink to the Terms of Service-->test
        </declarationURL>
        <privacyPolicyURL>
          <!--ro, opt, string, hyperlink to the Privacy Policy-->test
        </privacyPolicyURL>
      </Hyperlinks>
      <isSupportVerificationCodeCheck>
        <!--ro, opt, bool, whether the device supports verifying the verification code-->true
      </isSupportVerificationCodeCheck>
      <isSupportOldVerificationCode>
        <!--ro, opt, bool, whether the device supports the EZ password configuration of old version (the password of old version consists of six
characters)-->true
      </isSupportOldVerificationCode>
      <maintenanceVerificationCodeType opt="normal,empty">
        <!--ro, opt, string, maintenance verification code type, attr:opt{opt, string}, desc:used to identify the verificationCode type in
ISAPI/System/Network/EZVIZ?security=<security>&iv=<iv>&moduleType=<moduleType>, valid only when the moduleType in the URL is maintenanceModule-->empty
      </maintenanceVerificationCodeType>
      <maintenanceVerificationCodeModify opt="true,false">
        <!--ro, opt, bool, whether the device maintenance verification code has been modified, attr:opt{opt, string}, desc:used to identify the
verificationCode type in ISAPI/System/Network/EZVIZ?security=<security>&iv=<iv>&moduleType=<moduleType>, valid only when the moduleType in the URL is
maintenanceModule-->true
      </maintenanceVerificationCodeModify>
    </VerificationCodeModification>
    <isSupportIntegrate>
      <!--ro, opt, bool, whether the device supports access protocol configuration-->true
    </isSupportIntegrate>
    <isSupportPlatformAccess>
      <!--ro, opt, bool, whether the device supports platform access, desc:GET /ISAPI/System/Network/platformAccessFilter/capabilities (platform access IP
address filtering configuration (Ehome and GB28181 allowlist configuration))-->true
    </isSupportPlatformAccess>
    <isSupportIntelligentBoost>
      <!--ro, opt, bool, whether the device supports intelligent acceleration-->true
    </isSupportIntelligentBoost>
    <isSupportResourceStatistics>
      <!--ro, opt, bool, whether the device supports getting the network resource information-->true
    </isSupportResourceStatistics>
    <isSupportBandwidthLimit>
      <!--ro, opt, bool, whether the device supports bandwidth limit-->true
    </isSupportBandwidthLimit>
    <isSupportPOEPortsDisableAdaptiveServer>
      <!--ro, opt, bool, whether the device supports the POE protocol-->true
    </isSupportPOEPortsDisableAdaptiveServer>
    <isSupportPOEConfiguration>
      <!--ro, opt, bool, whether the device supports configuring the POE NIC parameters-->true
    </isSupportPOEConfiguration>
    <isSupportGetLinkSocketIP>
      <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
    </isSupportGetLinkSocketIP>
    <isSupportWebSocket>
      <!--ro, opt, bool, whether the device supports WebSocket-->true
    </isSupportWebSocket>
    <isSupportWebSocketS>
      <!--ro, opt, bool, whether the device supports WebSocketS-->true
    </isSupportWebSocketS>
    <Adaption>
      <!--ro, opt, object, network self-adaption configuration-->
      <streamType opt="0,1,2,3,4,5,7,8,9,10">
        <!--ro, opt, string, stream type that supports network self-adaption during live view, attr:opt{opt, string}, desc:0 (main stream), 1 (sub-
stream), 2 (third stream), 3 (fourth stream/virtual stream), 4 (fifth stream), 5 (sixth stream), 7 (seventh stream), 8 (eighth stream), 9 (ninth stream), 10
(tenth stream), and so forth. The index value here is the same as that in the live view structure NET_DVR_PREVIEWINFO of Device Network SDK. Note that 6 is
an invalid value, and from the seventh stream (index value: 7), the index value is the same as that for live view via RTSP-->test
      </streamType>
      <isSupportPlayback>
        <!--ro, opt, bool, whether the device supports playback-->true
      </isSupportPlayback>
    </Adaption>
    <isSupportVideoImgDB>
      <!--ro, opt, bool, whether the device supports image and video library configuration-->true
    </isSupportVideoImgDB>
    <isSupportVideoImgLibService>
      <!--ro, opt, bool, whether the device supports image and video library service-->true
    </isSupportVideoImgLibService>
    <isSupportWirelessWifi>
      <!--ro, opt, bool, whether the device supports Wi-Fi status-->true
    </isSupportWirelessWifi>
    <isSupportPrivatePlatom>
      <!--ro, opt, bool, whether the device supports accessing third party platform-->true
    </isSupportPrivatePlatom>
    <isSupportGantryETCServer>
      <!--ro, opt, bool-->true
    </isSupportGantryETCServer>
    <isSupportEventDataOverWebSocket opt="true,false">
      <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{opt, string}-->true
```

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```
        <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, desc:opt(opt, string)-->true
      </isSupportEventDataOverWebSocket>
      <isSupportDynamicHostName>
        <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) configuration-->true
      </isSupportDynamicHostName>
      <isSupportTCPAcceleration>
        <!--ro, opt, bool, whether the device supports TCP acceleration-->true
      </isSupportTCPAcceleration>
      <isSupportFlowControl>
        <!--ro, opt, bool, whether the device supports traffic control-->true
      </isSupportFlowControl>
      <isSupportEZVIZTiming>
        <!--ro, opt, bool, whether the device supports synchronizing time with EZ platform, desc:related URI: enabledTiming in the message of
/ISAPI/System/Network/EZVIZ-->true
      </isSupportEZVIZTiming>
      <isSupportTrafficMonitor>
        <!--ro, opt, bool, whether the device supports traffic monitoring, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json-->true
      </isSupportTrafficMonitor>
      <isSupportDMSAuthInfo>
        <!--ro, opt, bool, whether the device supports DMS authentication information management-->true
      </isSupportDMSAuthInfo>
      <isSupportWiredandWirelessTrafficMonitor>
        <!--ro, opt, bool, whether the device supports traffic monitoring (wireless NIC and wired NIC), desc:related URI:
/ISAPI/System/Network/Interfaces/<ID>/trafficMonitor?format=json-->true
      </isSupportWiredandWirelessTrafficMonitor>
      <isSupport4G>
        <!--ro, opt, bool, whether the device supports 4G module-->true
      </isSupport4G>
      <isSupport4GConfig>
        <!--ro, opt, bool, whether the device supports 4G configuration-->true
      </isSupport4GConfig>
      <isSupportGRIDServer>
        <!--ro, opt, bool, N/A-->true
      </isSupportGRIDServer>
      <isSupportGRIDInfo>
        <!--ro, opt, bool, N/A-->true
      </isSupportGRIDInfo>
      <isSupportSipTransmit>
        <!--ro, opt, bool, whether the device supports configuring SIP transmission parameters-->true
      </isSupportSipTransmit>
      <isSupportSip>
        <!--ro, opt, bool, whether the device supports private SIP (VOIP)-->true
      </isSupportSip>
      <isSupportWifiCascade>
        <!--ro, opt, bool, whether the device supports enabling automatic Wi-Fi cascading, desc:related URI: /ISAPI/System/Network/wifiCascade?format=json--
>true
      </isSupportWifiCascade>
      <isSupportRelativeInfo>
        <!--ro, opt, bool, whether the device supports getting the information about the connected devices, desc:related URI:
/ISAPI/System/Network/wifiCascade/relativeInfo?format=json-->true
      </isSupportRelativeInfo>
      <isSupportEmailEncrypt>
        <!--ro, opt, bool, whether the device supports email encryption configuration-->true
      </isSupportEmailEncrypt>
      <isSupportPortFilter>
        <!--ro, opt, bool, whether the device supports port filtering-->true
      </isSupportPortFilter>
      <isSupportPingDeny>
        <!--ro, opt, bool, whether the device supports configuring parameters to prevent the device from being tested using ping command-->true
      </isSupportPingDeny>
      <isSupportDOSDeny>
        <!--ro, opt, bool, whether the device supports configuring parameters to prevent the DoS attack-->true
      </isSupportDOSDeny>
      <isSupportNetworkStatus>
        <!--ro, opt, bool, whether the device supports searching for the network status, desc:related URI: /ISAPI/System/Network/status?format=json-->true
      </isSupportNetworkStatus>
      <isSupportEZVIZQRCode>
        <!--ro, opt, bool, whether the device supports getting the EZ QR code, desc:related URI: /ISAPI/System/Network/EZVIZ/QRCode?format=json-->true
      </isSupportEZVIZQRCode>
      <isSupportEZVIZUnbind>
        <!--ro, opt, bool, whether the device supports unbinding from EZ, desc:/ISAPI/System/Network/EZVIZ/unbind?format=json-->true
      </isSupportEZVIZUnbind>
      <isSupportPingTest>
        <!--ro, opt, bool, whether the device supports ping test, desc:Related URI: /ISAPI/System/Network/pingtest/capabilities-->true
      </isSupportPingTest>
      <isSupportMOTCloudPlateform>
        <!--ro, opt, bool, whether the device supports configuring parameters for accessing the cloud platform service, desc:related URI:
/ISAPI/System/Network/Traffic/MOTCloudPlateform/capabilities?format=json-->true
      </isSupportMOTCloudPlateform>
      <isSupportSpectrumScan>
        <!--ro, opt, bool, whether the device supports frequency spectrum scanning, desc:related URI:
/ISAPI/System/Network/interface/<netCardID>/spectrumScan/capabilities?format=json-->true
      </isSupportSpectrumScan>
      <isSupportFRP>
        <!--ro, opt, bool, whether the device supports FRP (Fast Reverse Proxy), desc:related URI: /ISAPI/Syetem/Network/FRP?format=json-->true
      </isSupportFRP>
      <isSupport5GModule>
        <!--ro, opt, bool, whether the device supports 5G module configuration, desc:related URI: /ISAPI/System/Network/5GModule/capabilities?format=json--
>true
      </isSupport5GModule>
      <isSupportWirelessDialRoute>
        <!--ro, opt, bool, whether the device supports configuring parameters of route forwarding rules of wireless dial-up, desc:related URI:
/ISAPI/System/Network/WirelessDia/route/capabilities?format=json-->true
```

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```
        </isSupportWirelessDialRoute>
        <isSupportNPlusOne>
          <!--ro, opt, bool, whether the device supports N+1 protocol-->true
        </isSupportNPlusOne>
        <supportEnvironmentalStandards opt="HJ212">
          <!--ro, opt, string, environmental monitoring standard, attr:opt{req, string}, desc:N/A-->true
        </supportEnvironmentalStandards>
        <isSupportIntranetSegment>
          <!--ro, opt, bool, whether the device supports LAN segment configuration, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI:
/ISAPI/System/Network/intranetSegment/capabilities?format=json-->true
        </isSupportIntranetSegment>
        <isSupportDHCPServer>
          <!--ro, opt, bool, whether the device supports DHCP service configuration, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI: /ISAPI/System/Network/DHCPServer/capabilities?
format=json-->true
        </isSupportDHCPServer>
        <isSupportClusterIntercom>
          <!--ro, opt, bool, whether the 7200 intercom protocol is transmitted, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is true or not be returned.-->true
        </isSupportClusterIntercom>
        <isSupportHydrologicMonitoring>
          <!--ro, opt, bool, whether the device supports hydrologic monitoring data communication, desc:if this function is supported, this node will be
returned and its value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI:
/ISAPI/System/Network/HydrologicMonitoring/capabilities?format=json-->true
        </isSupportHydrologicMonitoring>
        <isWirelessDialMutexWithWlanAndWifi>
          <!--ro, opt, bool, whether the wireless dial-up is mutually exclusive with the WLAN hotspot and Wi-Fi, desc:before a function is enabled, another
one must be disabled first-->true
        </isWirelessDialMutexWithWlanAndWifi>
        <isSupportSpeechTranscription>
          <!--ro, opt, bool, whether the device supports accessing the speech-to-text transcription platform, desc:if this function is supported, this node
will be returned and its value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI:
/ISAPI/System/Network/speechTranscriptionPlatform/capabilities?format=json-->true
        </isSupportSpeechTranscription>
        <isSupportAsynPingTest>
          <!--ro, opt, bool, whether the device supports asynchronous network detection, desc:network detection means testing the network stability and packet
loss rate of the network using the ping command for knowing about the network condition. Related URI: /ISAPI/System/Network/asynPingTest/capabilities?
format=json-->true
        </isSupportAsynPingTest>
        <isSupportBasebandCapture>
          <!--ro, opt, bool, whether the device supports capturing packets of the baseband, desc:you can capture packets transmitted between the 4G module and
the base station using the ping command. Related URI: /ISAPI/System/Network/cellularNetwork/basebandCapture/capabilities?format=json-->true
        </isSupportBasebandCapture>
        <isSupportCellularNetworkSpeedTest>
          <!--ro, opt, bool, whether the device supports testing the connection speed between the device and the cellular network, desc:you can test the
connection speed of the network where the current device is. Related URI: /ISAPI/System/Network/cellularNetwork/speedTest/capabilities?format=json-->true
        </isSupportCellularNetworkSpeedTest>
        <isSupportCellularNetworkCellInfo>
          <!--ro, opt, bool, whether the device supports getting a cell's information of the cellular network, desc:you can get the information of the
cellular network where the device is. Related URI: /ISAPI/System/Network/cellularNetwork/cellInfo?format=json&ecurity=<security>&iv=<iv>-->true
        </isSupportCellularNetworkCellInfo>
        <isSupportMultiBandFrequencyScanning>
          <!--ro, opt, bool, whether the device supports scanning all frequency bands of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/multiBandFrequencyScanning/capabilities?format=json-->true
        </isSupportMultiBandFrequencyScanning>
        <isSupportCellularNetworkLockBand>
          <!--ro, opt, bool, whether the device accessing the locked frequency band of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/lockBand/capabilities?format=json-->true
        </isSupportCellularNetworkLockBand>
        <isSupportRadioFrequencyFluctuation>
          <!--ro, opt, bool, whether the device supports configuring RF parameters of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/radioFrequencyFluctuation/capabilities?format=json-->true
        </isSupportRadioFrequencyFluctuation>
        <isSupportBasebandBlock>
          <!--ro, opt, bool, whether the device supports the unauthorized list of the base station, desc:related URI:
/ISAPI/System/Network/cellularNetwork/basebandBlock/capabilities?format=json-->true
        </isSupportBasebandBlock>
        <isSupportRFIDTagCheckTest>
          <!--ro, opt, bool, whether the device supports the RFID tag disk storage performance test, desc:related URI: GET
/ISAPI/System/Network/RFIDData/GetTagCheckTestInfo?format=json-->true
        </isSupportRFIDTagCheckTest>
        <isSupportQoSDSCPtoLoaclPriorityMapping>
          <!--ro, opt, bool, whether the device supports DSCP and local mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/DSCPtoLoaclPriorityMapping/capabilities?format=json-->true
        </isSupportQoSDSCPtoLoaclPriorityMapping>
        <isSupportQoS802.1pto802.1pPriorityMapping>
          <!--ro, opt, bool, whether the device supports 802.1p mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/802.1pto802.1pPriorityMapping/capabilities?format=json-->true
        </isSupportQoS802.1pto802.1pPriorityMapping>
        <isSupportLinkTestStart>
          <!--ro, opt, bool, whether the network link test is enabled, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Start?format=json-->true
        </isSupportLinkTestStart>
        <isSupportLinkTestResult>
          <!--ro, opt, bool, whether the device supports getting the result of network link test, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Start?format=json-->true
        </isSupportLinkTestResult>
        <isSupportNetworkModuleUpgrade>
          <!--ro, opt, bool, whether it supports upgrading network module, desc:including importing upgrade package
(/ISAPI/System/Network/WirelessDial/ImportUpgradePackage?format=json), getting importing progress
(/ISAPI/System/Network/WirelessDial/GetUpgradePackageImportStatus?format=json), starting upgrade
(/ISAPI/System/Network/WirelessDial/StartUpgradeNetworkModule?format=json)-->true
        </isSupportNetworkModuleUpgrade>
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<isSupportNetworkModuleUpgrade>
<isSupportNetworkModuleVersionInfo>
<!--ro, opt, bool, whether it supports getting the version information of network module, desc:GET
/ISAPI/System/Network/WirelessDial/NetworkModuleVersionInfo?format=json-->true
</isSupportNetworkModuleVersionInfo>
<isSupportGreenShieldPlatform>
<!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/GreenShieldPlatform/GreenShieldPlatform?format=json-->true
</isSupportGreenShieldPlatform>
<isSupportFusionUniversalApplicationPlatCfg>
<!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/FusionUniversalApplicationPlatform/FusionUniversalApplicationPlatCfg?format=json-->true
</isSupportFusionUniversalApplicationPlatCfg>
<isSupportModbusAccess>
<!--ro, opt, bool, whether the device supports Modbus accessing parameters, desc:GET/PUT /ISAPI/System/Network/ModbusAccess?format=json-->true
</isSupportModbusAccess>
<isSupportModbusTestSubordinateStatus>
<!--ro, opt, bool, whether it supports Modbus testing specified client status, desc:POST /ISAPI/System/Network/ModbusTestSubordinateStatus?
format=json-->true
</isSupportModbusTestSubordinateStatus>
<isSupportGetModbusAndRegisterMap>
<!--ro, opt, bool, whether it supports getting the mapping relationship between Modbus uploaded content and register quantity, desc:GET
/ISAPI/System/Network/GetModbusAndRegisterMap?format=json-->true
</isSupportGetModbusAndRegisterMap>
<isSupportTSOverWebSocket>
<!--ro, opt, bool, whether the device supports transparent transmission via the WebSocket serial port, desc:(TS:TransportSerial)-->true
</isSupportTSOverWebSocket>
<isSupportRequestOutlookAuth>
<!--ro, opt, bool, whether the device supports applying authentication request of Outlook email, desc:POST
/ISAPI/System/Network/Email/RequestOutLookAuth?format=json-->true
</isSupportRequestOutlookAuth>
<isSupportOutlookEmailList>
<!--ro, opt, bool, whether the device supports Outlook email information, desc:GET or PUT /ISAPI/System/Network/Email/OutlookEmailList?format=json--
>true
</isSupportOutlookEmailList>
<isSupportDSTPServerCfg>
<!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET or PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
format=json-->true
</isSupportDSTPServerCfg>
<isSupportDSTPProtocolsCap>
<!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
</isSupportDSTPProtocolsCap>
<isSupportSearchOutlookAuthResult>
<!--ro, opt, bool-->true
</isSupportSearchOutlookAuthResult>
<RTDROverWebSocket>
<!--ro, opt, object-->
<isSupportGasSpectrumRealTimeData>
<!--ro, opt, bool-->true
</isSupportGasSpectrumRealTimeData>
<isSupportDangerousGasSensorRealTimeData>
<!--ro, opt, bool-->true
</isSupportDangerousGasSensorRealTimeData>
<isSupportComposeWindConditionRealTimeData>
<!--ro, opt, bool-->true
</isSupportComposeWindConditionRealTimeData>
<isSupportMeteorologySensorRealTimeData>
<!--ro, opt, bool-->true
</isSupportMeteorologySensorRealTimeData>
<isSupportSPLData>
<!--ro, opt, bool-->true
</isSupportSPLData>
</RTDROverWebSocket>
<isSupportServerCfg>
<!--ro, opt, bool-->true
</isSupportServerCfg>
<isSupportDelimiterCfg>
<!--ro, opt, bool-->true
</isSupportDelimiterCfg>
<isSupportStartCellNetSelfDiagnosis>
<!--ro, opt, bool-->true
</isSupportStartCellNetSelfDiagnosis>
<isSupportCellNetSelfDiagnosisStatus>
<!--ro, opt, bool-->true
</isSupportCellNetSelfDiagnosisStatus>
<isSupportVoiceTalkOverWebSocket>
<!--ro, opt, bool-->true
</isSupportVoiceTalkOverWebSocket>
<isSupportPlaybackOverWebSocket>
<!--ro, opt, bool-->true
</isSupportPlaybackOverWebSocket>
<WebSocketPlaybackCap>
<!--ro, opt, object-->
<cmdType opt="speed,pause,resume,single,stop,seek,startExtracting,stopExtracting,transcoding">
<!--ro, req, string, attr:opt{opt, string}-->speed
</cmdType>
<isSupportPlaybackSeekOverStart>
<!--ro, opt, bool-->false
</isSupportPlaybackSeekOverStart>
</WebSocketPlaybackCap>
<isSupportHTTPAndWSPortMultiplex>
<!--ro, opt, bool, dep:and,{$.DeviceCap.SysCap.NetworkCap.isSupportWebSocket,eq,true}-->true
</isSupportHTTPAndWSPortMultiplex>
<isSupportHTTPSAndWSSPortMultiplex>
<!--ro, opt, bool, dep:and,{$.DeviceCap.SysCap.NetworkCap.isSupportWebSocketS,eq,true}-->true
</isSupportHTTPSAndWSSPortMultiplex>
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      <isSupportReverseOverWebSocket>
        <!--ro, opt, bool-->true
      </isSupportReverseOverWebSocket>
    </NetworkCap>
    <IOCap>
      <!--ro, opt, object, I/O capability, desc:related URI: /ISAPI/System/IO/capabilities-->
      <IOInputPortNums>
        <!--ro, opt, int, number of local input ports of the device-->true
      </IOInputPortNums>
      <IOOutputPortNums>
        <!--ro, opt, int, number of local output ports of the device-->true
      </IOOutputPortNums>
      <isSupportStrobeLamp>
        <!--ro, opt, bool, whether the device supports configuring the output of the gate's light, desc:related URI:
/ISAPI/System/IO/outputs/strobeLampConf-->true
      </isSupportStrobeLamp>
      <SoftIOInputPortNums>
        <!--ro, opt, int, number of soft input ports of the device, desc:this node depends on whether the URI /ISAPI/System/IO/softInputs/capabilities?
format=json is supported-->true
      </SoftIOInputPortNums>
      <isSupportIOOutputAdvanceParameter>
        <!--ro, opt, bool, whether the device supports configuring advanced parameters of the output port, desc:this node determines whether the URI
/ISAPI/System/IO/outputs/advanceParameter/capabilities?format=json is supported-->true
      </isSupportIOOutputAdvanceParameter>
      <isSupportSetAllIOOutput>
        <!--ro, opt, bool, whether the device supports configuring all output ports, desc:this node determines whether the PUT method of the URI
/ISAPI/System/IO/outputs is supported-->true
      </isSupportSetAllIOOutput>
      <isSupportCombinationAlarm>
        <!--ro, opt, bool, whether the device supports the combined alarm, desc:CombinationAlarmCap in /ISAPI/System/IO/inputs/capabilities-->true
      </isSupportCombinationAlarm>
      <enabledIOOutputPortNums>
        <!--ro, opt, int, number of enabled alarm outputs-->0
      </enabledIOOutputPortNums>
      <isSupportAlarmKeyParam>
        <!--ro, opt, bool, whether supports configuring the physical keys of one-push alarm, desc:related URI:
/ISAPI/System/IO/inputs/<IOInputID>/AlarmKeyParam/capabilities?format=json-->true
      </isSupportAlarmKeyParam>
      <isSupportWebDisplayAlarmStatus>
        <!--ro, opt, bool, whether the web supports displaying the alarm output status, desc:used for web interface display, true indicates that the web
supports displaying the alarm output status; otherwise, it does not support-->true
      </isSupportWebDisplayAlarmStatus>
    </IOCap>
    <SerialCap>
      <!--ro, opt, object, range of RS-485 serial port numbers supported by the device-->
      <rs485PortNums>
        <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->true
      </rs485PortNums>
      <supportRS232Config>
        <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
      </supportRS232Config>
      <rs422PortNums>
        <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->true
      </rs422PortNums>
      <rs232PortNums>
        <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->true
      </rs232PortNums>
      <rs485WorkMode opt="Led,CaptureTrigger">
        <!--ro, opt, string, RS-485 working modes supported by the device, attr:opt{opt, string}, desc:Led (external LED screen, which is used for LED
display), CaptureTrigger (external device, which is used to trigger the transmission of the captured data)-->test
      </rs485WorkMode>
      <rs232SerialNumber opt="1,2,3">
        <!--ro, opt, string, range of RS-232 serial port numbers supported by the device, attr:opt{opt, string}-->test
      </rs232SerialNumber>
      <rs485SerialNumber opt="4,5,6">
        <!--ro, opt, string, range of RS-485 serial port numbers supported by the device, attr:opt{opt, string}-->test
      </rs485SerialNumber>
      <isSupportAuthenticationService>
        <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
      </isSupportAuthenticationService>
      <isSupportDeviceInfo>
        <!--ro, opt, bool, whether the device supports configuring the information about the serial port device-->true
      </isSupportDeviceInfo>
      <isSupportSearchDeviceInfoRelations>
        <!--ro, opt, bool, whether the device supports searching the linkage information of the serial port device-->true
      </isSupportSearchDeviceInfoRelations>
    </SerialCap>
    <VideoCap>
      <!--ro, opt, object, video encoding capability, desc:related URI: /ISAPI/System/Video/capabilities-->
      <videoInputPortNums>
        <!--ro, opt, int, number of video input ports-->true
      </videoInputPortNums>
      <videoOutputPortNums>
        <!--ro, opt, int, number of video output ports-->true
      </videoOutputPortNums>
      <menuNums>
        <!--ro, opt, int, number of local menus that can be displayed-->true
      </menuNums>
      <externalChannelNum>
        <!--ro, opt, bool, number of extended analog channels-->true
      </externalChannelNum>
      <isSupportHeatmap>
        <!--ro, opt, bool, whether the device supports heat map-->true
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        </isSupportHeatmap>
        <isSupportCounting>
          <!--ro, opt, bool, whether the device supports people counting-->true
        </isSupportCounting>
        <countingType>
          <!--ro, opt, string, statistics type, desc:human (people counting); object (counting)-->test
        </countingType>
        <isSupportPicture>
          <!--ro, opt, bool, whether the device supports OSD picture overlay-->true
        </isSupportPicture>
        <isSupportCapturePicOverlays>
          <!--ro, opt, bool, whether the device supports text overlay on a single picture-->true
        </isSupportCapturePicOverlays>
        <isSupportMergePicOverlays>
          <!--ro, opt, bool, whether the device supports text overlay on composite picture-->true
        </isSupportMergePicOverlays>
        <isSupportPreviewSwitch>
          <!--ro, opt, bool, whether the device supports switching live view-->true
        </isSupportPreviewSwitch>
        <isSupportMultiChannelCounting>
          <!--ro, opt, bool, whether the device supports multichannel statistics-->true
        </isSupportMultiChannelCounting>
        <isSupportCountingCollection>
          <!--ro, opt, bool, whether the device supports people counting ANR-->true
        </isSupportCountingCollection>
        <isSupportHeatmapCollection>
          <!--ro, opt, bool, whether the device supports heat map data ANR-->true
        </isSupportHeatmapCollection>
        <channelFlexible opt="name,online,linknum">
          <!--ro, opt, string, this node is located after the node <isSupportHeatmapCollection>., attr:opt{opt, string}-->true
        </channelFlexible>
        <isSupportOutputsResource>
          <!--ro, opt, bool-->true
        </isSupportOutputsResource>
        <isSupportMixedChannel>
          <!--ro, opt, bool, this node is located after the node <channelFlexible>.-->true
        </isSupportMixedChannel>
        <isSupportMixedChannelStatus>
          <!--ro, opt, bool, this node is located after the node <isSupportMixedChannel>.-->true
        </isSupportMixedChannelStatus>
        <isSupportRecodStatus>
          <!--ro, opt, bool, whether the device supports searching for the recording status-->true
        </isSupportRecodStatus>
        <isSupportPrivacyMask>
          <!--ro, opt, bool, whether the device supports privacy mask-->true
        </isSupportPrivacyMask>
        <isSupportBinocularPreviewSwitch>
          <!--ro, opt, bool, whether the device supports switching live view of the dual-lens camera-->true
        </isSupportBinocularPreviewSwitch>
        <isSupportCalibCheck>
          <!--ro, opt, bool, whether the device supports calibration verification-->true
        </isSupportCalibCheck>
        <isSupportPIP>
          <!--ro, opt, bool, whether the device supports PIP-->true
        </isSupportPIP>
        <isSupportFocusVideoMode>
          <!--ro, opt, bool, whether the device supports the focus mode of the video-->true
        </isSupportFocusVideoMode>
        <isSupportExternalChannel>
          <!--ro, opt, bool, whether the device supports extended analog channels-->true
        </isSupportExternalChannel>
        <isSupportCompressedTranscode>
          <!--ro, opt, bool, whether the device supports configuring the encoding compression of transcoding stream, desc:related URL:
/ISAPI/Streaming/channels/<channelID>/TranscodeCfg/capabilities?format=json-->true
        </isSupportCompressedTranscode>
        <isSupportCompressedStream>
          <!--ro, opt, bool, whether the device supports transcoding compressed stream, desc:it is valid when isSupportCompressedTranscode is true. The device
supports getting real-time stream and real-time compressed stream. The streaming channel stream type can be obtained via id when the serviceType is
transcode (URL: /ISAPI/ContentMgmt/StreamingProxy/channels)-->true
        </isSupportCompressedStream>
        <isSupportLocalStreamCfg>
          <!--ro, opt, bool, whether the device supports configuring the streaming parameters of local device, desc:related URL:
/ISAPI/System/Video/inputs/channels/<channelID>/LocalStreamCfg/capabilities?format=json-->true
        </isSupportLocalStreamCfg>
        <isSupportMenuStatus>
          <!--ro, opt, bool, whether it supports getting the menu status, desc:corresponding URI: /ISAPI/System/Video/Menu/<indexID>/status?format=json, the
range of indexID comes from VideoCap.menuNums-->true
        </isSupportMenuStatus>
        <isSupportVideoOutputMode>
          <!--ro, opt, bool, whether the device supports video output pattern parameters-->true
        </isSupportVideoOutputMode>
        <pictureChannelList>
          <!--ro, opt, array, subType:object-->
          <channelID>
            <!--ro, opt, int, channel No.-->0
          </channelID>
        </pictureChannelList>
        <transcodeStreamType opt="previewTranscode,recordTranscode">
          <!--ro, opt, string, attr:opt{req, string}-->true
        </transcodeStreamType>
        <isSupportImportPanoramaIOverlayPicture>
          <!--ro, opt, bool, POST /ISAPI/System/Video/channels/<channelID>/ImportPanoramaIOverlayPicture?format=json, desc:POST
/ISAPI/System/Video/channels/<channelID>/ImportPanoramaIOverlayPicture?format=json-->true
        </isSupportImportPanoramaIOverlayPicture>
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      <isSupportPanoramaIOverlayPictureParams>
        <!--ro, opt, bool, GET or PUT /ISAPI/System/Video/channels/<channelID>/PanoramaIOverlayPictureParams?format=json, desc:GET or PUT
/ISAPI/System/Video/channels/<channelID>/PanoramaIOverlayPictureParams?format=json-->true
      </isSupportPanoramaIOverlayPictureParams>
    </VideoCap>
    <AudioCap>
      <!--ro, opt, object, audio encoding capability, desc:related URL: /ISAPI/System/Audio/capabilities-->
      <audioInputNums>
        <!--ro, req, int, number of audio inputs-->1
      </audioInputNums>
      <audioOutputNums>
        <!--ro, req, int, number of audio outputs-->1
      </audioOutputNums>
      <mixAudioInSet>
        <!--ro, opt, bool, N/A-->true
      </mixAudioInSet>
      <mixAudioOutSet>
        <!--ro, opt, bool, N/A-->true
      </mixAudioOutSet>
      <isSupportAudioMixing>
        <!--ro, opt, bool, whether the device supports audio mixing-->true
      </isSupportAudioMixing>
      <isSupportRemoveFrequencyPoint>
        <!--ro, opt, bool, whether the device supports clearing the howling frequency point-->true
      </isSupportRemoveFrequencyPoint>
      <isSupportAudioInConfig>
        <!--ro, opt, bool, whether the device supports configuring parameters of all audio inputs, desc:related URI:
/ISAPI/System/Audio/AudioIn/capabilities-->true
      </isSupportAudioInConfig>
      <muteDuringPanTilt opt="MicIn">
        <!--ro, opt, string, whether the device supports muting during motion, attr:opt{opt, string}, desc:muteDuringPanTilt in
/ISAPI/System/TwoWayAudio/channels/ID-->true
      </muteDuringPanTilt>
      <isSupportAudioOutConfig>
        <!--ro, opt, bool, whether the device supports configuring parameters of all audio outputs, desc:related URI:
/ISAPI/System/Audio/AudioOut/capabilities?format=json-->true
      </isSupportAudioOutConfig>
      <isSupportSAIP>
        <!--ro, opt, bool, whether the device supports connecting to the independent audio track channel-->true
      </isSupportSAIP>
      <SAIPChannelsNum>
        <!--ro, opt, int, number of devices that supports connecting to the independent audio track channel, dep:and,{$.AudioCap.isSupportSAIP,eq,true}-->1
      </SAIPChannelsNum>
      <isSupportDataStorage>
        <!--ro, opt, bool, whether the device supports uploading the audio file, desc:related URL: /ISAPI/System/Audio/dataStorage?format=json-->true
      </isSupportDataStorage>
      <isSupportDataRelease>
        <!--ro, opt, bool, whether the device supports releasing the audio file, desc:related URL: /ISAPI/System/Audio/dataRelease?format=json-->true
      </isSupportDataRelease>
      <isSupportAudioPlan>
        <!--ro, opt, bool, whether the device supports the audio scheme plan, desc:related URL: /ISAPI/System/Audio/plan?format=json-->true
      </isSupportAudioPlan>
      <isSupportAudioTask>
        <!--ro, opt, bool, whether the device supports the audio scheme task, desc:related URL: /ISAPI/System/Audio/plan/<PlanID>/task?format=json-->true
      </isSupportAudioTask>
      <isSupportaudioCollection>
        <!--ro, opt, bool, whether the device supports collecting audio files, desc:related URL: /ISAPI/System/Audio/audioCollection/capabilities?
format=json; maintenance function, it is used for collecting audio to analyze audio questions-->true
      </isSupportaudioCollection>
      <isSupportWirelessPickUp>
        <!--ro, opt, bool, whether the device supports connecting wireless pickup, desc:related URL: /ISAPI/System/Audio/wirelessPickUp/capabilities?
format=json-->true
      </isSupportWirelessPickUp>
      <audioOutputBroadcastNums opt="1,2,3,4">
        <!--ro, opt, int, number of audio broadcast outputs, attr:opt{opt, string}, desc:at present, this node is only supported by broadcast audio devices
when opt="1,2,3,4" is returned and , the output depends on the actual node; when not returned, the output starts from 1 by default according to business
type, the audio devices classify the output ports into broadcast output port and audio output port considering the start No. of every output starts from 1,
it should return the specific number of supported broadcast outputs if this node is returned, it indicates that the device supports the audio output volume
configuration; otherwise, the device does not support. Corresponding URI: /ISAPI/System/Audio/AudioOut/broadcast/capabilities?format=json-->0
      </audioOutputBroadcastNums>
      <isSupportAudioOutMixing>
        <!--ro, opt, bool, whether the device supports outputting audio mix, desc:related URL: /ISAPI/System/Audio/AudioOut/mixing/capabilities?format=json-
->true
      </isSupportAudioOutMixing>
      <isSupportRTSPAudio>
        <!--ro, opt, bool, whether the device supports listening to the audio transmitted via RTSP in real time, desc:RTSP URI:
/ISAPI/System/Audio/AudioIn/channels/<audioID>-->true
      </isSupportRTSPAudio>
      <isSupportRTSPAudioPlaybackByFile>
        <!--ro, opt, bool, whether the device supports playing back audio transmitted via RTSP according to files, desc:the RTSP streaming URL is
playbackURI returned by /ISAPI/ContentMgmt/recordAudio/search?format=json-->true
      </isSupportRTSPAudioPlaybackByFile>
      <isSupportDeviceAudioCapture>
        <!--ro, opt, bool, whether the device supports audio collection, desc:GET/PUT /ISAPI/System/Audio/DeviceAudioCapture?format=json-->true
      </isSupportDeviceAudioCapture>
    </AudioCap>
    <isSupportHolidy>
      <!--ro, opt, bool, this node is located after the node <AudioCap>-->true
    </isSupportHolidy>
    <RebootConfigurationCap>
      <!--ro, opt, object, capability of configuring parameters for rebooting the device, desc:related URI: /ISAPI/System/reboot/capabilities. This node is
located before the node <isSupportSubscribeEvent> and after the node <isSupportHolidy>-->
      <Genetec>
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        <!--ro, opt, bool, whether to enable the Genetec protocol-->true
      </Genetec>
      <ONVIF>
        <!--ro, opt, bool, whether to enable the ONVIF protocol-->true
      </ONVIF>
      <RTSP>
        <!--ro, opt, bool, whether to enable the RTSP protocol-->true
      </RTSP>
      <HTTP>
        <!--ro, opt, bool, whether to enable the HTTP protocol-->true
      </HTTP>
      <SADP>
        <!--ro, opt, object, SADP (Search Active Devices Protocol, including the link layer and multicast discovery)-->
        <ISDiscoveryMode>
          <!--ro, opt, bool, mode of the SADP multicast discovery protocol-->true
        </ISDiscoveryMode>
        <PcapMode>
          <!--ro, opt, bool, mode of the SADP link layer discovery protocol-->true
        </PcapMode>
      </SADP>
      <IPCAddStatus>
        <!--ro, opt, bool, IP address status-->true
      </IPCAddStatus>
    </RebootConfigurationCap>
    <isSupportExternalDevice>
      <!--ro, opt, bool, whether the device supports connecting to peripherals: true (support), desc:related URI: /ISAPI/System/externalDevice/capabilities-
->true
    </isSupportExternalDevice>
    <isSupportSoundCfg>
      <!--ro, opt, bool, whether the device supports the sound configuration, desc:related URI: /ISAPI/System/SoundCfg/capabilities-->true
    </isSupportSoundCfg>
    <isSupportSubscribeEvent>
      <!--ro, opt, bool, whether the device supports subscribing to events, desc:related URI: /ISAPI/Event/notification/subscribeEventCap-->true
    </isSupportSubscribeEvent>
    <isSupportSubscribeIOTInfo>
      <!--ro, opt, bool, whether the device supports subscribing to the IoT information-->true
    </isSupportSubscribeIOTInfo>
    <isSupportDiagnosedData>
      <!--ro, opt, bool, whether the device supports exporting the device diagnosis data, desc:true (support), this node is not returned (not support).
Related URI: /ISAPI/System/diagnosedData-->true
    </isSupportDiagnosedData>
    <isSupportChangedUpload>
      <!--ro, opt, bool, whether the device supports uploading changes-->true
    </isSupportChangedUpload>
    <isSupportGettingWorkingStatus>
      <!--ro, opt, bool, whether the device supports getting the device status-->true
    </isSupportGettingWorkingStatus>
    <isSupportGettingChannelInfoByCondition>
      <!--ro, opt, bool, whether the device supports getting the channel information by condition-->true
    </isSupportGettingChannelInfoByCondition>
    <isSupportDiagnosedDataParameter>
      <!--ro, opt, bool, whether the device supports enabling or disabling exporting the diagnosis information-->true
    </isSupportDiagnosedDataParameter>
    <isSupportSimpleDevStatus>
      <!--ro, opt, bool, whether the device supports getting the device's simple status, desc:related URI: /ISAPI/System/workingstatus?format=json-->true
    </isSupportSimpleDevStatus>
    <isSupportFlexible>
      <!--ro, opt, bool, whether the device supports the flexible combination configuration. Related URI: /ISAPI/System/Video/inputs/channels?format=json--
>true
    </isSupportFlexible>
    <isSupportPTZChannels>
      <!--ro, opt, bool, whether the device supports returning the PTZ channel (it is distinguished from the video channel)-->true
    </isSupportPTZChannels>
    <isSupportMetadata>
      <!--ro, opt, bool, whether the device supports metadata, desc:related URI: /ISAPI/Streaming/channels/<ID>/metadata/capabilities-->true
    </isSupportMetadata>
    <isSupportTimeCap>
      <!--ro, opt, bool, whether the device supports the time configuration, desc:related URI: /ISAPI/System/time/capabilities-->true
    </isSupportTimeCap>
    <isSupportThermalStreamData>
      <!--ro, opt, bool, whether the device supports uploading thermal imaging stream data in real time-->true
    </isSupportThermalStreamData>
    <isSupportShutdown>
      <!--ro, opt, bool, whether the device supports the shutdown configuration, desc:related URI: /ISAPI/System/shutdown?format=json-->true
    </isSupportShutdown>
    <isSupportConsumptionMode>
      <!--ro, opt, bool, whether the device supports switching the consumption mode, desc:related URI: /ISAPI/System/consumptionMode/capabilities?
format=json-->true
    </isSupportConsumptionMode>
    <isSupportManualPowerConsumption>
      <!--ro, opt, bool, whether the device supports controlling the device consumption manually-->true
    </isSupportManualPowerConsumption>
    <isSupportPostUpdateFirmware>
      <!--ro, opt, bool, whether the device supports upgrading firmware by POST method-->true
    </isSupportPostUpdateFirmware>
    <isSupportPostConfigData>
      <!--ro, opt, bool, whether the device supports importing and exporting the configuration file by POST method-->true
    </isSupportPostConfigData>
    <isSupportUserLock>
      <!--ro, opt, bool, whether the device supports locking the user, desc:related URI: /ISAPI/System/userLock/config/capabilities?format=json-->true
    </isSupportUserLock>
    <isSupportModuleLock>
      <!--ro, opt, bool, whether the device supports locking the module, desc:related URI: /ISAPI/System/moduleLock-->true
    </isSupportModuleLock>
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    <supportSmartOverlapChannles opt="1">
      <!--ro, opt, bool, whether the device supports configuring the smart event stream, attr:opt{opt, string}, desc:if this function is supported, this node and the corresponding channel ID will be returned; otherwise, this node will not be returned-->true
    </supportSmartOverlapChannles>
    <isSupportNetworkCapture>
      <!--ro, opt, bool, whether the device supports capturing network packets, desc:this node is used to determine whether the device supports capturing communication packets transmitted between the device and the external. Related URI: /ISAPI/System/networkCapture/capabilities?format=json-->true
    </isSupportNetworkCapture>
    <isSupportOperationRecord>
      <!--ro, opt, bool, whether the device supports the device operation records, desc:the charging voltage and monthly flow can be recorded in the device in real-time low-consumption mode and full-consumption mode. Related URI: /ISAPI/System/operationRecord/capabilities?format=json-->true
    </isSupportOperationRecord>
    <isSupportSearchModuleLock>
      <!--ro, opt, bool, whether the device support search module locking parameters, desc:/ISAPI/System/moduleLock/searchconfig?format=json-->true
    </isSupportSearchModuleLock>
    <devSupportUpdateVersion>
      <!--ro, opt, string, supported version of local device upgrade, desc:the device returns the upgrade version to guide tool users (e.g. service technician) to select an upgrade package for local upgrade-->1
    </devSupportUpdateVersion>
    <notSupportCatalogUpgrade>
      <!--ro, opt, bool-->true
    </notSupportCatalogUpgrade>
  </SysCap>
  <voicetalkNums>
    <!--ro, opt, int, number of two-way audio channels-->2
  </voicetalkNums>
  <isSupportSnapshot>
    <!--ro, opt, bool, whether the device supports capturing pictures-->true
  </isSupportSnapshot>
  <SecurityCap>
    <!--ro, opt, object, encryption capability set-->
    <supportUserNums>
      <!--ro, opt, int, supported maximum number of users-->1
    </supportUserNums>
    <userBondIpNums>
      <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->1
    </userBondIpNums>
    <userBondMacNums>
      <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->1
    </userBondMacNums>
    <isSupCertificate>
      <!--ro, opt, bool, whether the device supports authentication-->true
    </isSupCertificate>
    <issupIllegalLoginLock>
      <!--ro, opt, bool, whether the device supports locking login-->true
    </issupIllegalLoginLock>
    <isSupportOnlineUser>
      <!--ro, opt, bool, whether the device supports the online user configuration-->true
    </isSupportOnlineUser>
    <isSupportAnonymous>
      <!--ro, opt, bool, whether the device supports anonymous login-->true
    </isSupportAnonymous>
    <isSupportStreamEncryption>
      <!--ro, opt, bool, whether the device supports stream encryption-->true
    </isSupportStreamEncryption>
    <securityVersion opt="1,2,3,4,7">
      <!--ro, opt, string, encryption capability, attr:opt{opt, string}, desc:the encryption capability of each version consists of two parts: encryption algorithm and the range of encrypted nodes. Currently 1 refers to AES128 encryption and 2 refers to AES256 encryption. The range of encrypted nodes is described in each protocol-->test
    </securityVersion>
    <keyIterateNum>
      <!--ro, opt, int, iteration times, desc:the value is usually between 100 and 1000-->100
    </keyIterateNum>
    <isSupportUserCheck>
      <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, desc:this node depends on the node <securityVersion>, which means that it is only valid for the versions that support encrypting the sensitive information-->true
    </isSupportUserCheck>
    <isSupportGUIDFileDataExport>
      <!--ro, opt, bool, whether the device supports exporting the device's GUID file-->true
    </isSupportGUIDFileDataExport>
    <isSupportSecurityQuestionConfig>
      <!--ro, opt, bool, whether the device supports answering security questions-->true
    </isSupportSecurityQuestionConfig>
    <supportSecurityNode opt="wirelessServer,wirelessDial">
      <!--ro, opt, string, additional node for supporting sensitive information encryption, attr:opt{opt, string}, desc:in the earlier version of the solution for encrypting the sensitive information via ISAPI (i.e., the solution for transmitting plaintext via ISAPI), some URIs containing sensitive information are left out. Now these URIs need to be encrypted for transmission. This capability node is added to be compatible with old devices-->test
    </supportSecurityNode>
    <isSupportGetOnlineUserListSC>
      <!--ro, opt, bool, whether the device supports searching the online user list-->true
    </isSupportGetOnlineUserListSC>
    <SecurityLimits>
      <!--ro, opt, object, capability of configuring security limit parameters-->
      <LoginPasswordLenLimit min="1" max="16">
        <!--ro, opt, string, length limit of the user's login password, attr:min{opt, string},max{opt, string}-->test
      </LoginPasswordLenLimit>
      <SecurityAnswerLenLimit min="1" max="128">
        <!--ro, opt, string, length limit of the security question's answer, attr:min{opt, string},max{opt, string}-->test
      </SecurityAnswerLenLimit>
    </SecurityLimits>
    <RSAKeyLength opt="512,1024,2048,3072" def="3072">
      <!--ro, opt, enum, HTTPS certificate length, subType:string, attr:opt{opt, string},def{opt, string}, desc:512, 1024, 2048-->3072
    </RSAKeyLength>
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    <isSupportONVIFUserManagement>
      <!--ro, opt, bool, whether the device supports ONVIF user management-->true
    </isSupportONVIFUserManagement>
    <WebCertificateCap>
      <!--ro, opt, object, HTTP authentication capability-->
      <CertificateType opt="basic,digest,digest/basic">
        <!--ro, req, string, certificate type: basic (authentication), attr:opt{opt, string}-->test
      </CertificateType>
      <SecurityAlgorithm>
        <!--ro, opt, object, encryption algorithm, dep:or,{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,degist},
{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
        <algorithmType opt="MD5,SHA256,MD5/SHA256">
          <!--ro, opt, string, algorithm type, attr:opt{req, string}-->test
        </algorithmType>
      </SecurityAlgorithm>
    </WebCertificateCap>
    <isSupportConfigFileImport>
      <!--ro, opt, bool, whether the device supports importing the configuration file-->true
    </isSupportConfigFileImport>
    <isSupportConfigFileExport>
      <!--ro, opt, bool, whether the device supports exporting the configuration file-->true
    </isSupportConfigFileExport>
    <cfgFileSecretKeyLenLimit min="0" max="16">
      <!--ro, opt, string, length limit of the configuration file's verification key, attr:min{opt, string},max{opt, string}-->0
    </cfgFileSecretKeyLenLimit>
    <supportIPCActivatePassword>
      <!--ro, opt, bool, whether the device supports configuring the password for activating the network camera-->true
    </supportIPCActivatePassword>
    <isIrreversible>
      <!--ro, opt, bool, whether the device supports irreversible password storage, desc:if this function is not supported, the plaintext password of the
user information will be stored in the device; otherwise, the password will be hashed for storage in the device-->true
    </isIrreversible>
    <salt>
      <!--ro, opt, string, the specific salt used by the user to log in-->test
    </salt>
    <isSupportOnvifInfo>
      <!--ro, opt, bool, whether the device supports getting the ONVIF protocol information-->true
    </isSupportOnvifInfo>
    <isSupportUserNamePasswordCheckUpgrade>
      <!--ro, opt, bool, whether the device supports upgrading the verification of the user name and password-->true
    </isSupportUserNamePasswordCheckUpgrade>
    <isSupportDeviceCertificatesManagement>
      <!--ro, opt, bool, whether the device supports certificate management-->true
    </isSupportDeviceCertificatesManagement>
    <isSupportSIPCertificatesManagement>
      <!--ro, opt, bool, whether the device supports SIP certificate management-->true
    </isSupportSIPCertificatesManagement>
    <isSupportDeviceSelfSignCertExport>
      <!--ro, opt, bool, whether the device supports exporting the self-signed certificate-->true
    </isSupportDeviceSelfSignCertExport>
    <isSupportSecurityEmail>
      <!--ro, opt, bool, whether the device supports configuring the security email-->true
    </isSupportSecurityEmail>
    <isSupportRTSPCertificate opt="true,false">
      <!--ro, opt, bool, whether the device supports RTSP authentication, attr:opt{opt, string}-->true
    </isSupportRTSPCertificate>
    <isSupportPictureURlCertificate opt="true,false">
      <!--ro, opt, bool, whether the device supports configuring parameters for picture URL authentication, attr:opt{opt, string}-->true
    </isSupportPictureURlCertificate>
    <isSupportGB35114Certificate>
      <!--ro, opt, bool, N/A-->true
    </isSupportGB35114Certificate>
    <isSupportAuthenticationMode opt="uKey,private">
      <!--ro, opt, enum, verification mode, subType:string, attr:opt{req, string}, desc:"uKey", "private"-->private
    </isSupportAuthenticationMode>
    <isSupportUnloggedUserPermissionConfig opt="true,false">
      <!--ro, opt, bool, whether the device supports configuring permission for users that have not logged in, attr:opt{opt, string}-->true
    </isSupportUnloggedUserPermissionConfig>
    <maxIllegalLoginTimes min="3" max="20" def="5">
      <!--ro, opt, string, maximum failed illegal login attempts, attr:min{opt, string},max{opt, string},def{opt, string}, desc:this node is valid when the
value of the node <issupportIllegalLoginLock> is true and will be returned when the maximum failed attempts of illegally logging in to the device reach the
limit-->1
    </maxIllegalLoginTimes>
    <SecurityAdvanced>
      <!--ro, opt, object, advanced security configuration-->
      <securityEnhanced>
        <!--ro, opt, bool, whether the device supports security reinforcement-->true
      </securityEnhanced>
      <noOperationEnabled>
        <!--ro, req, bool, whether to enable the control timeout when there is no operation-->true
      </noOperationEnabled>
      <noOperationTime min="1" max="60" def="15">
        <!--ro, req, int, control timeout when there is no operation, unit:分钟, attr:min{opt, string},max{opt, string},def{opt, string}-->1
      </noOperationTime>
      <isSupportDigestStatus>
        <!--ro, opt, bool, digest status-->true
      </isSupportDigestStatus>
    </SecurityAdvanced>
    <LoginLinkNum>
      <!--ro, opt, object, number of users logging in at the same time-->
      <maxLinkNum min="1" max="128" def="50">
        <!--ro, req, int, number of controls logging in at the same time, attr:min{opt, string},max{opt, string},def{opt, string}-->64
      </maxLinkNum>
    </LoginLinkNum>
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    <isSupportCCClientCertificate>
      <!--ro, opt, bool, whether the device supports the CC (Client Certificate) function-->true
    </isSupportCCClientCertificate>
    <passwordValidity min="0" max="365">
      <!--ro, opt, string, range of the password validity period, attr:min{opt, string},max{opt, string}, desc:unit: day. If this node is returned, it
indicates that the device supports configuring the password validity period, and only the administrator can edit the configuration-->1
    </passwordValidity>
    <isSupportResetChannelPassword>
      <!--ro, opt, bool, whether the device supports resetting the password of the channel-->true
    </isSupportResetChannelPassword>
    <DoubleVerificationCap>
      <!--ro, opt, object, secondary authentication capability-->
      <isSupportUsersConfig>
        <!--ro, opt, bool, whether the device supports configuring users for secondary authentication-->true
      </isSupportUsersConfig>
      <isSupportUsersPermissionConfig>
        <!--ro, opt, bool, whether the device supports configuring user permission for secondary authentication-->true
      </isSupportUsersPermissionConfig>
      <isSupportFactoryReset>
        <!--ro, opt, bool, whether the device supports restoring secondary authentication to defaults-->true
      </isSupportFactoryReset>
      <factoryResetMode opt="full,basic,other">
        <!--ro, opt, string, attr:opt{req, string}-->full
      </factoryResetMode>
      <isSupportQuestionConfiguration>
        <!--ro, opt, bool, whether it supports configuring a single security question for secondary authentication, desc:PUT
/ISAPI/Security/questionConfiguration/<indexID>-->true
      </isSupportQuestionConfiguration>
    </DoubleVerificationCap>
    <isSupportUserSearch>
      <!--ro, opt, bool, whether the device supports searching users-->true
    </isSupportUserSearch>
    <keypadPassword min="1" max="10">
      <!--ro, opt, int, keypad password length, attr:min{req, int},max{req, int}, desc:when the password length of different user types is different, this
node only indicates the keypad password length of the administrator; otherwise, the keypad password length of all user types is the same and defined by the
value of this node-->1
    </keypadPassword>
    <installerKeypadPassword min="1" max="10">
      <!--ro, opt, int, keypad password length of the installer, attr:min{req, int},max{req, int}-->1
    </installerKeypadPassword>
    <operatorKeypadPassword min="1" max="10">
      <!--ro, opt, int, keypad password length of the operator, attr:min{req, int},max{req, int}-->1
    </operatorKeypadPassword>
    <userOperateType opt="1,2,3">
      <!--ro, opt, enum, user type: 1 (network user), subType:int, attr:opt{req, string}, desc:1 (network user), 2 (keypad user), 3 (network user and keypad
user)-->1
    </userOperateType>
    <isSptUserEnabled>
      <!--ro, opt, bool, whether the device supports configuring parameters for enabling the user-->true
    </isSptUserEnabled>
    <isSptAdminCap>
      <!--ro, opt, bool, whether the device supports getting the admin permission-->true
    </isSptAdminCap>
    <isSupportBackLogCfg>
      <!--ro, opt, bool, whether the device supports configuring basic parameters of the background log-->true
    </isSupportBackLogCfg>
    <isSupportBackLogExport>
      <!--ro, opt, bool, whether the device supports exporting the background log-->true
    </isSupportBackLogExport>
    <isSupportModifyInnerPassword opt="true,false">
      <!--ro, opt, bool, whether the device supports editing the built-in password, attr:opt{req, string}, desc:the device's built-in passwords include the
database password, system account password and so on-->true
    </isSupportModifyInnerPassword>
    <isSupportEncryptCertificate>
      <!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
    </isSupportEncryptCertificate>
    <maxIllegalLoginLockTime min="1" max="120" def="30">
      <!--ro, opt, int, lock time range, range:[1,120], unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
    </maxIllegalLoginLockTime>
    <cloudSensitiveInfoEncryptType>
      <!--ro, opt, string, encryption type of the cloud sensitive information, desc:the current device returns the value "2.1", which means that the method
of generating the key for encrypting the sensitive information is SHA256(calc_sha256(user, salt, SHA256(user+salt2+SHA256(user+salt+password)))+
AaBbCcDd1234!@#$).-->2.1
    </cloudSensitiveInfoEncryptType>
    <isSupportCertificateCustomID>
      <!--ro, opt, bool, whether the device supports using the user's custom ID to configure the certificate-->true
    </isSupportCertificateCustomID>
    <isSupportResetLANPassword>
      <!--ro, opt, bool, whether the device supports resetting the device's password in the LAN, desc:related URI:
/ISAPI/Security/resetPassword/certification?format=json-->true
    </isSupportResetLANPassword>
    <isSupportPicHttpsCertificatesManagement>
      <!--ro, opt, bool, whether the device supports managing the HTTPS certificate for pictures, desc:if this function is supported, this node will be
returned and its value is true; otherwise, this node will not be returned-->true
    </isSupportPicHttpsCertificatesManagement>
    <isSupportAKSKImport>
      <!--ro, opt, bool, whether the device supports importing AKSK, desc:if this function is supported, this node will be returned and its value is true;
otherwise, this node will not be returned-->true
    </isSupportAKSKImport>
    <isSupportAKSKExport>
      <!--ro, opt, bool, whether the device supports exporting AKSK, desc:If this function is supported, this node will be returned and its value is true;
otherwise, this node will not be returned.-->true
    </isSupportAKSKExport>
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    <isSupportAKSKReset>
      <!--ro, opt, bool, whether the device supports resetting AKSK, desc:If this function is supported, this node will be returned and its value is true;
otherwise, this node will not be returned.-->true
    </isSupportAKSKReset>
    <ExportCertificateFile>
      <!--ro, opt, object, whether the device supports exporting the certificate, desc:related URI: POST
/ISAPI/Security/serverCertificate/exportCertificateFile-->
    </ExportCertificateFile>
    <isSupportGB351114FDWSFCertificate>
      <!--ro, opt, bool, whether the device supports FDWSF certificate management, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned. Related URIs: /ISAPI/Security/GB351114/FDWSFCertificate/capabilities?format=json, POST
/ISAPI/Security/GB351114/FDWSFCertificate?format=json, GET /ISAPI/Security/GB351114/FDWSFCertificate?format=json, and DELETE
/ISAPI/Security/GB351114/FDWSFCertificate?format=json.-->true
    </isSupportGB351114FDWSFCertificate>
    <isSupportMediaObjectEncryption>
      <!--ro, opt, bool, whether the device supports setting encryption parameters of the private target in media data for the specified video channel,
desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI:
/ISAPI/Security/MediaObjectEncryption/capabilities?format=json-->true
    </isSupportMediaObjectEncryption>
    <isSupportDisclaimer>
      <!--ro, opt, bool, whether support configuring the product disclaimer, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/disclaimer/capabilities?format=json-->true
    </isSupportDisclaimer>
    <isSupportSoftwareLicense>
      <!--ro, opt, bool, whether support configuring the product open source statement, desc:if this function is supported, this node will be returned and
its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/softwareLicense?format=json-->true
    </isSupportSoftwareLicense>
    <isSupportInstallerCap>
      <!--ro, opt, bool, whether the device supports getting the installer permission, desc:if this function is supported, this node will be returned and
its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/UserPermission/installer/capabilities?format=json-->true
    </isSupportInstallerCap>
    <isSupportResetPasswordFile>
      <!--ro, opt, bool, whether the device supports exporting and importing the secret key file of device in the LAN, desc:Related URI:
/ISAPI/Security/resetPassword/keyFile/capabilities?format=json-->true
    </isSupportResetPasswordFile>
    <isNotSupportLogSeverCertificate>
      <!--ro, opt, bool, whether the device support importing log server certificate, desc:related URI: /ISAPI/Security/serverCertificate/certificate-->true
    </isNotSupportLogSeverCertificate>
    <isSupportLocalLoginCfg>
      <!--ro, opt, bool, whether it supports the capability of device local login parameters, desc:GET/PUT /ISAPI/Security/LocalLoginCfg?format=json-->true
    </isSupportLocalLoginCfg>
    <isSupportGetProtocolFirewallRulesFileImportStatus>
      <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallRulesFileImportStatus?format=json-->true
    </isSupportGetProtocolFirewallRulesFileImportStatus>
    <isSupportImportProtocolFirewallRulesFile>
      <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/ImportProtocolFirewallRulesFile?format=json-->true
    </isSupportImportProtocolFirewallRulesFile>
    <isSupportGetProtocolFirewallEngineUpgradeStatus>
      <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallEngineUpgradeStatus?format=json-->true
    </isSupportGetProtocolFirewallEngineUpgradeStatus>
    <isSupportUpgradeProtocolFirewallEngine>
      <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/UpgradeProtocolFirewallEngine?format=json-->true
    </isSupportUpgradeProtocolFirewallEngine>
    <isSupportProtocolFirewallVersionInfo>
      <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/ProtocolFirewallVersionInfo?format=json-->true
    </isSupportProtocolFirewallVersionInfo>
    <isSupportProtocolFirewallParams>
      <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/Security/ProtocolFirewallParams?format=json-->true
    </isSupportProtocolFirewallParams>
    <isSupportResetPasswordBySecurityPhoneNo>
      <!--ro, opt, bool, whether it supports resetting device password with phone number verification,
desc:1. attendance check devices support applying a preset phone number during device activation. The user can reset the device password with the preset
phone number, which is similar with password reset via email. The phone number can be changed on the website or local UI of the device
2. if the node is not returned or the returned value is false, it indicates "no"-->true
    </isSupportResetPasswordBySecurityPhoneNo>
    <isSupportGetIntrusionPreventionEngineUpgradeStatus>
      <!--ro, opt, bool, whether the device supports getting the upgrade status of intrusion prevention engine library, desc:GET
/ISAPI/Security/GetIntrusionPreventionEngineUpgradeStatus?format=json-->true
    </isSupportGetIntrusionPreventionEngineUpgradeStatus>
    <isSupportIntrusionPreventionEngineVersionInfo>
      <!--ro, opt, bool, whether the device supports version information of intrusion prevention engine, desc:GET
/ISAPI/Security/IntrusionPreventionEngineVersionInfo?format=json-->true
    </isSupportIntrusionPreventionEngineVersionInfo>
    <isSupportIntrusionPreventionParams>
      <!--ro, opt, bool, whether the device supports intrusion prevention parameters, desc:GET/PUT /ISAPI/Security/IntrusionPreventionParams?format=json--
>true
    </isSupportIntrusionPreventionParams>
    <isSupportUpgradeIntrusionPreventionEngine>
      <!--ro, opt, bool, whether the device supports upgrading intrusion prevention engine library, desc:PUT
/ISAPI/Security/UpgradeIntrusionPreventionEngine?format=json-->true
    </isSupportUpgradeIntrusionPreventionEngine>
    <notSupportImportCertificateType opt="CA,CCClentCertificate,GB35114,GB35114Service,https">
      <!--ro, opt, string, file type that cannot be imported, attr:opt{req, string}, desc:if this node does not exist, all supported certificate types can
be imported-->test
    </notSupportImportCertificateType>
    <isSupportPasswordProtection>
      <!--ro, opt, bool, whether it supports parameter configuration of SDK protocol authentication mode, desc:PUT/GET
/ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
    </isSupportPasswordProtection>
  </SecurityCap>
  <EventCap>
    <!--ro, opt, object, event capability-->
    <isSupportHDFull>
      <!--ro, opt, bool, whether the device supports HDD full alarm-->true
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    <!--ro, opt, bool, whether the device supports HDD full alarm-->true
  </isSupportHDFull>
  <isSupportHDError>
    <!--ro, opt, bool, whether the device supports HDD error alarm-->true
  </isSupportHDError>
  <isSupportHDBadBlock>
    <!--ro, opt, bool-->true
  </isSupportHDBadBlock>
  <isSupportNicBroken>
    <!--ro, opt, bool, whether the device supports the network disconnected event-->true
  </isSupportNicBroken>
  <isSupportIpConflict>
    <!--ro, opt, bool, whether the device supports the IP address conflict event-->true
  </isSupportIpConflict>
  <isSupportIllAccess>
    <!--ro, opt, bool, whether the device supports the illegal login event-->true
  </isSupportIllAccess>
  <isSupportViException>
    <!--ro, opt, bool, whether the device supports video input exception (only for analog channels)-->true
  </isSupportViException>
  <isSupportViMismatch>
    <!--ro, opt, bool, whether the device supports the event of video standard mismatch-->true
  </isSupportViMismatch>
  <isSupportRecordException>
    <!--ro, opt, bool, whether the device supports the recording exception event-->true
  </isSupportRecordException>
  <isSupportRaidException>
    <!--ro, opt, bool, whether the device supports the array exception event-->true
  </isSupportRaidException>
  <isSupportViResMismatch>
    <!--ro, opt, bool, whether the device supports the event of video resolution mismatch-->true
  </isSupportViResMismatch>
  <isSupportPOCException>
    <!--ro, opt, bool, whether the device supports the event of powering over coaxial exception-->true
  </isSupportPOCException>
  <isSupportSmartDetection>
    <!--ro, opt, bool, whether the device supports the Smart event-->true
  </isSupportSmartDetection>
  <isSupportSpareException>
    <!--ro, opt, bool, whether the device supports hot spare exception-->true
  </isSupportSpareException>
  <isSupportPoePowerException>
    <!--ro, opt, bool, whether the device supports the POE exception event-->true
  </isSupportPoePowerException>
  <isSupportTriggerFocus>
    <!--ro, opt, bool, whether the device supports defocus exception-->true
  </isSupportTriggerFocus>
  <isSupportMotionDetection>
    <!--ro, opt, bool, whether the device supports motion detection-->true
  </isSupportMotionDetection>
  <isSupportVideoLoss>
    <!--ro, opt, bool, whether the device supports video loss events-->true
  </isSupportVideoLoss>
  <isSupportTamperDetection>
    <!--ro, opt, bool, whether the device supports tampering detection-->true
  </isSupportTamperDetection>
  <isSupportFireDetection>
    <!--ro, opt, bool, whether the device supports fire detection events-->true
  </isSupportFireDetection>
  <isSupportRegionEntrance>
    <!--ro, opt, bool, whether the device supports region entrance events-->true
  </isSupportRegionEntrance>
  <isSupportRegionExiting>
    <!--ro, opt, bool, whether the device supports region exiting events-->true
  </isSupportRegionExiting>
  <isSupportLoitering>
    <!--ro, opt, bool, whether the device supports loitering events-->true
  </isSupportLoitering>
  <isSupportGroup>
    <!--ro, opt, bool, whether the device supports people gathering events-->true
  </isSupportGroup>
  <isSupportRapidMove>
    <!--ro, opt, bool, whether the device supports fast moving events-->true
  </isSupportRapidMove>
  <isSupportParking>
    <!--ro, opt, bool, whether the device supports parking detection events-->true
  </isSupportParking>
  <isSupportUnattendedBaggage>
    <!--ro, opt, bool, whether the device supports unattended baggage events-->true
  </isSupportUnattendedBaggage>
  <isSupportAttendedBaggage>
    <!--ro, opt, bool, whether the device supports object removal events-->true
  </isSupportAttendedBaggage>
  <isSupportHumanAttribute>
    <!--ro, opt, bool, whether the device supports human body features-->true
  </isSupportHumanAttribute>
  <isSupportHumanRecognition>
    <!--ro, opt, bool, whether the device supports target capture events-->true
  </isSupportHumanRecognition>
  <isSupportFaceContrast>
    <!--ro, opt, bool, whether the device supports face picture comparison events-->true
  </isSupportFaceContrast>
  <isSupportFaceLib>
    <!--ro, opt, bool, Front-end devices support linkage by face picture library., desc:related URI: /ISAPI/Event/schedules/faceLib/<ID>/<FDID>-->true
```

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    </isSupportFaceLib>
    <isSupportBlackListFaceContrast>
      <!--ro, opt, bool, whether the device supports events of blocklist face picture comparison-->true
    </isSupportBlackListFaceContrast>
    <isSupportWhiteListFaceContrast>
      <!--ro, opt, bool, whether the device supports events of allowlist face picture comparison-->true
    </isSupportWhiteListFaceContrast>
    <isSupportillegalParking>
      <!--ro, opt, bool, whether the device supports illegal parking detection-->true
    </isSupportillegalParking>
    <isSupportPedestrian>
      <!--ro, opt, bool, whether the device supports pedestrian detection-->true
    </isSupportPedestrian>
    <isSupportTrafficAccident>
      <!--ro, opt, bool, whether the device supports traffic incident detection-->true
    </isSupportTrafficAccident>
    <isSupportConstruction>
      <!--ro, opt, bool, whether the device supports construction detection-->true
    </isSupportConstruction>
    <isSupportRoadBlock>
      <!--ro, opt, bool, whether the device supports roadblock detection-->true
    </isSupportRoadBlock>
    <isSupportAbandonedObject>
      <!--ro, opt, bool, whether the device supports fallen object detection-->true
    </isSupportAbandonedObject>
    <isSupportParallelParking>
      <!--ro, opt, bool, whether the device supports parallel parking detection-->true
    </isSupportParallelParking>
    <isSupportParkingState>
      <!--ro, opt, bool, whether the device supports parking space status detection-->true
    </isSupportParkingState>
    <isSupportTrafficJam>
      <!--ro, opt, bool, whether the device supports traffic jam detection (old version)-->true
    </isSupportTrafficJam>
    <isSupportCongestion>
      <!--ro, opt, bool, whether the device supports traffic jam detection-->true
    </isSupportCongestion>
    <isSupportStudentsStoodUp>
      <!--ro, opt, bool, whether the device supports the student standing up detection event-->true
    </isSupportStudentsStoodUp>
    <isSupportFramesPeopleCounting>
      <!--ro, opt, bool, whether the device supports the regional people counting event-->true
    </isSupportFramesPeopleCounting>
    <isSupportFaceSnap>
      <!--ro, opt, bool, whether the device supports face capture events-->true
    </isSupportFaceSnap>
    <isSupportPersonDensityDetection>
      <!--ro, opt, bool, whether the device supports people density detection-->true
    </isSupportPersonDensityDetection>
    <HFPD>
      <!--ro, opt, bool, whether the device supports frequently appeared person detection-->true
    </HFPD>
    <isSupportPersonQueueDetection>
      <!--ro, opt, bool, whether the device supports queue detection events-->true
    </isSupportPersonQueueDetection>
    <isSupportVehicleStatistics>
      <!--ro, opt, bool, whether the device supports vehicle statistics events-->true
    </isSupportVehicleStatistics>
    <isSupportFaceThermometry>
      <!--ro, opt, bool, whether the device supports face temperature screening-->true
    </isSupportFaceThermometry>
    <isSupportSafetyHelmetDetection>
      <!--ro, opt, bool, whether the device supports hard hat detection-->true
    </isSupportSafetyHelmetDetection>
    <isSupportPowerSupplyCabinet>
      <!--ro, opt, bool, whether the device supports cabinet powering detection-->true
    </isSupportPowerSupplyCabinet>
    <isSupportSensorDetection>
      <!--ro, opt, bool, whether the device supports sensor detection-->true
    </isSupportSensorDetection>
    <isSupportSignAbnormal>
      <!--ro, opt, bool, whether the device supports the physical sign exception detection-->true
    </isSupportSignAbnormal>
    <isSupportTeacherBehaviorDetect>
      <!--ro, opt, bool, whether the device supports teacher's behavior detection-->true
    </isSupportTeacherBehaviorDetect>
    <isSupportCityManagement>
      <!--ro, opt, bool, whether the device supports the city management function-->true
    </isSupportCityManagement>
    <isSupportMixedTargetDetection>
      <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
    </isSupportMixedTargetDetection>
    <isSupportFaceSnapModeling>
      <!--ro, opt, bool, whether the device supports face modeling events-->true
    </isSupportFaceSnapModeling>
    <isSupportIntersectionAnalysis>
      <!--ro, opt, bool, whether the device supports the intersection analysis-->true
    </isSupportIntersectionAnalysis>
    <isSupportTriggerCapCheck>
      <!--ro, opt, bool, whether the device supports verifying the capability of triggering linkage actions, desc:this node indicates that the capability
message returned by calling the URI GET /ISAPI/Event/triggersCap has been verified, and the event linkage actions can be displayed exactly according to the
capability-->true
    </isSupportTriggerCapCheck>
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<isSupportLuma>
  <!--ro, opt, bool, VQD brightness exception-->true
</isSupportLuma>
<isSupportChroma>
  <!--ro, opt, bool, VQD image color cast-->true
</isSupportChroma>
<isSupportSnow>
  <!--ro, opt, bool, VQD snow noise-->true
</isSupportSnow>
<isSupportStreak>
  <!--ro, opt, bool, VQD stripe noise-->true
</isSupportStreak>
<isSupportFreeze>
  <!--ro, opt, bool, VQD image freeze-->true
</isSupportFreeze>
<isSupportSigLose>
  <!--ro, opt, bool, VQD signal loss-->true
</isSupportSigLose>
<isSupportClarity>
  <!--ro, opt, bool, VQD resolution exception-->true
</isSupportClarity>
<isSupportJitter>
  <!--ro, opt, bool, VQD image flutter-->true
</isSupportJitter>
<isSupportBlock>
  <!--ro, opt, bool, VQD video tampering-->true
</isSupportBlock>
<isSupportFlowers>
  <!--ro, opt, bool, VQD blurred image alarm-->true
</isSupportFlowers>
<isSupportNoise>
  <!--ro, opt, bool, VQD blurred screen-->true
</isSupportNoise>
<isSupportGhost>
  <!--ro, opt, bool, VQD image noise-->true
</isSupportGhost>
<isSupportPurple>
  <!--ro, opt, bool, VQD purple fringing-->true
</isSupportPurple>
<isSupportICR>
  <!--ro, opt, bool, VQD ICP-->true
</isSupportICR>
<isSupportProtectiveFilm>
  <!--ro, opt, bool, VQD protective film unremoved-->true
</isSupportProtectiveFilm>
<isSupportVoltageInstable>
  <!--ro, opt, bool, whether the device supports the voltage instable exception detection-->true
</isSupportVoltageInstable>
<isSupportCertificateRevocation>
  <!--ro, opt, bool, whether the device supports the certificate expiry alarm-->true
</isSupportCertificateRevocation>
<isSupportSysStorFull>
  <!--ro, opt, bool, whether the device supports the storage full alarm of the cluster system-->true
</isSupportSysStorFull>
<isSupportVersionAbnormal>
  <!--ro, opt, bool, whether the device supports cluster version exception-->true
</isSupportVersionAbnormal>
<isSupportNodeOffline>
  <!--ro, opt, bool, whether the device supports device offline detection-->true
</isSupportNodeOffline>
<isSupportRecordCycleAbnormal>
  <!--ro, opt, bool, whether the device supports the event of insufficient recording period-->true
</isSupportRecordCycleAbnormal>
<isSupportIpcTransferAbnormal>
  <!--ro, opt, bool, whether the device supports the event of network camera migration failure-->true
</isSupportIpcTransferAbnormal>
<isSupportDrivingBehaviorDetect>
  <!--ro, opt, bool, whether the device supports driving behavior detection-->true
</isSupportDrivingBehaviorDetect>
<isSupportWasteGasDetection>
  <!--ro, opt, bool, whether the device supports waste gas monitoring-->true
</isSupportWasteGasDetection>
<isSupportGreyScaleAlarm>
  <!--ro, opt, bool, whether the device supports the gray scale alarm-->true
</isSupportGreyScaleAlarm>
<isSupportPTLocking>
  <!--ro, opt, bool, whether the device supports the PTZ locking exception-->true
</isSupportPTLocking>
<isSupportNoMaskDetection>
  <!--ro, opt, bool, whether the device supports not wearing mask detection-->true
</isSupportNoMaskDetection>
<isSupportTMPA>
  <!--ro, opt, bool, whether the device supports the warning linkage of face temperature screening-->true
</isSupportTMPA>
<RuleScheduleCap>
  <!--ro, opt, object, capability of configuring the arming schedule by rule-->
  <isSupportCityManagement>
    <!--ro, opt, bool, whether the device supports configuring the arming schedule by rule for the city management event-->true
  </isSupportCityManagement>
</RuleScheduleCap>
<RegionScheduleCap>
  <!--ro, opt, object, capability of configuring the arming schedule by area-->
  <isSupportPersonDensityDetection>
    <!--ro, opt, bool, whether the device supports configuring the arming schedule by area for the people density detection-->true
```

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        </isSupportPersonDensityDetection>
        <isSupportPersonQueueDetection>
          <!--ro, opt, bool, whether the device supports configuring the arming schedule by area for the queue detection (area popularity analytics)-->true
        </isSupportPersonQueueDetection>
      </RegionScheduleCap>
      <isSupportThermalCalibrationFileException>
        <!--ro, opt, bool, whether the device supports the temperature calibration file exception-->true
      </isSupportThermalCalibrationFileException>
      <isSupportTemperatureIntervalMeasurement>
        <!--ro, opt, bool, whether the device supports the temperature range detection-->true
      </isSupportTemperatureIntervalMeasurement>
      <isSupportThermalVehicleDetection>
        <!--ro, opt, bool, whether the device supports thermal imaging vehicle detection-->true
      </isSupportThermalVehicleDetection>
      <isSupportWaterQualityDetection>
        <!--ro, opt, bool, whether the device supports water quality detection-->true
      </isSupportWaterQualityDetection>
      <isSupportWaterLevelDetection>
        <!--ro, opt, bool, whether the device supports water level detection-->true
      </isSupportWaterLevelDetection>
      <isSupportDeviceTemperatureAbnormal>
        <!--ro, opt, bool, whether the device supports temperature exception detection-->true
      </isSupportDeviceTemperatureAbnormal>
      <isSupportObjectAccumulationDetection>
        <!--ro, opt, bool, whether the device supports object accumulation detection-->true
      </isSupportObjectAccumulationDetection>
      <isSupportAppAbnormal>
        <!--ro, opt, bool, whether the device supports the App exception alarm or event-->true
      </isSupportAppAbnormal>
      <isSupportPTEventCfg>
        <!--ro, opt, bool, whether the device supports transparent transmission event configuration event, desc:/ISAPI/Event/PTEventCfg/capabilities?
format=json-->true
      </isSupportPTEventCfg>
      <isSupportLensTemperatureException>
        <!--ro, opt, bool, whether the device supports uploading lens abnormal temperature event-->true
      </isSupportLensTemperatureException>
      <isSupportPTZFXYFinish>
        <!--ro, opt, bool, the event of railway robot and PTZ going to the preset, desc:PTZFXYFinish-->true
      </isSupportPTZFXYFinish>
      <isSupportNormalTemperatureAlarm>
        <!--ro, opt, bool, whether the device supports normal temperature alarm-->true
      </isSupportNormalTemperatureAlarm>
      <isSupportLowTemperatureAlarm>
        <!--ro, opt, bool, whether the device supports low temperature alarm-->true
      </isSupportLowTemperatureAlarm>
      <isSupportLowCoolerLifeEvent>
        <!--ro, opt, bool, whether it supports short life exception event of refrigerating machine-->true
      </isSupportLowCoolerLifeEvent>
      <isSupportObstacleAvoidanceAlarm>
        <!--ro, opt, bool, whether it supports obstacle avoidance alarm-->true
      </isSupportObstacleAvoidanceAlarm>
      <isSupportPTZRotationAbnormal>
        <!--ro, opt, bool, whether it supports PTZ rotation exception event-->true
      </isSupportPTZRotationAbnormal>
      <isSupportBlackListVehContrast>
        <!--ro, opt, bool, whether the vehicle blocklist event is supported-->true
      </isSupportBlackListVehContrast>
      <isSupportWhiteListVehContrast>
        <!--ro, opt, bool, whether the vehicle allowlist event is supported-->true
      </isSupportWhiteListVehContrast>
    </EventCap>
    <ITCCap>
      <!--ro, opt, object, related node and URIs: the node <isSupportVehicleDetection> in /ISAPI/ITC/capability and /ISAPI/ITC/capabilities-->
      <isSupportITC>
        <!--ro, opt, bool, whether the device supports ITC-->true
      </isSupportITC>
      <isSupportIllegalDictionary>
        <!--ro, opt, bool, whether the device supports the violation dictionary-->true
      </isSupportIllegalDictionary>
      <isSupportVehicleDetection>
        <!--ro, opt, bool, whether the device supports vehicle detection-->true
      </isSupportVehicleDetection>
      <isSupportHVTVehicleDetection>
        <!--ro, opt, bool, whether the device supports mixed-traffic vehicle detection-->true
      </isSupportHVTVehicleDetection>
      <isSupportlicencePlateAuditData>
        <!--ro, opt, bool, whether the device supports importing or exporting the license plate blocklist and allowlist data-->true
      </isSupportlicencePlateAuditData>
      <isSupportSearchLPListAudit>
        <!--ro, opt, bool, whether the device supports searching for the license plate blocklist and allowlist-->true
      </isSupportSearchLPListAudit>
      <isSupportVehicleDetectionManualTrigger opt="false,true">
        <!--ro, opt, bool, whether the device supports manually triggering vehicle detection, attr:opt{opt, string}-->true
      </isSupportVehicleDetectionManualTrigger>
      <isSupportSyncSignalOutput>
        <!--ro, opt, bool, whether the device supports configuring output parameters-->true
      </isSupportSyncSignalOutput>
    </ITCCap>
    <ImageCap>
      <!--ro, opt, object, capability of configuring image parameters-->
      <isSupportRegionalExposure>
        <!--ro, opt, bool, whether the device supports regional exposure-->true
      </isSupportRegionalExposure>
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    <isSupportRegionalFocus>
      <!--ro, opt, bool, whether the device supports regional focus-->true
    </isSupportRegionalFocus>
  </ImageCap>
  <RacmCap>
    <!--ro, opt, object, UI before picture search-->
    <isSupportZeroChan>
      <!--ro, opt, bool, whether the device supports channel-zero, desc:related URI: /ISAPI/ContentMgmt/ZeroVideo/channels-->true
    </isSupportZeroChan>
    <inputProxyNums>
      <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->1
    </inputProxyNums>
    <eSATANums>
      <!--ro, opt, int, supported number of eSATA interfaces-->1
    </eSATANums>
    <miniSASNums>
      <!--ro, opt, int, supported number of miniSAS interfaces-->1
    </miniSASNums>
    <nasNums>
      <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->1
    </nasNums>
    <ipSanNums>
      <!--ro, opt, int, supported number of IPSANs-->1
    </ipSanNums>
    <isSupportRaid>
      <!--ro, opt, bool, whether the device supports RAID-->true
    </isSupportRaid>
    <isSupportExtHdCfg>
      <!--ro, opt, bool, whether the device supports advanced HDD management-->true
    </isSupportExtHdCfg>
    <isSupportTransCode>
      <!--ro, opt, bool, whether the device supports live view transcoding-->true
    </isSupportTransCode>
    <isSupportIpcImport>
      <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/ipcConfig-->true
    </isSupportIpcImport>
    <NasMountType>
      <!--ro, opt, object, NAS mounting type-->
      <isNFSSupportAuthentication>
        <!--ro, opt, bool, whether the device supports CIFS authentication-->true
      </isNFSSupportAuthentication>
      <isCIFSSupportAuthentication>
        <!--ro, opt, bool, whether the device supports CIFS authentication-->true
      </isCIFSSupportAuthentication>
    </NasMountType>
    <isSupportIpcStreamType>
      <!--ro, opt, bool, whether the device supports the stream type of the network camera-->true
    </isSupportIpcStreamType>
    <isSupportIOInputProxy>
      <!--ro, opt, bool, whether the device supports dynamic input channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/inputs-->true
    </isSupportIOInputProxy>
    <isSupportIOOutputProxy>
      <!--ro, opt, bool, whether the device supports dynamic output channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/outputs-->true
    </isSupportIOOutputProxy>
    <isSupportPTZRs485Proxy>
      <!--ro, opt, bool, whether the device supports dynamic PTZ RS-485 channels, desc:related URI: /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities-->true
    </isSupportPTZRs485Proxy>
    <isSupportSrcIDSearch>
      <!--ro, opt, bool, whether the device supports stream ID search-->true
    </isSupportSrcIDSearch>
    <isSupportReversePlayback>
      <!--ro, opt, bool, whether the device supports reverse playback-->true
    </isSupportReversePlayback>
    <isSupportSMARTTest>
      <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
    </isSupportSMARTTest>
    <isSupportDownloadByTime>
      <!--ro, opt, bool, whether the device supports downloading by time, desc:related URI: /ISAPI/ContentMgmt/download/capabilities-->true
    </isSupportDownloadByTime>
    <pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
personQueueCounting,personQueueTime,violentMotion,leavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,r
oadblock,
abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminite,OPTEX,securityControlPanel,ge
tUp,toiletTarry,
playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePh
one,fatigueDriving,
notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringW
arning,
PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideS
hop,stallOccupyingRoad,     illegalHeap,illegalParkingofNon-
MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel,
ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,alarmResult">
      <!--ro, opt, string, picture search type, attr:opt{opt, string}-->test
    </pictureSearchType>
    <recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofN
on-MotorVehicle
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on-motorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
">
      <!--ro, opt, string, video search type, attr:opt{opt, string}, desc:"llEvent" (all events), "CMR" (capture by schedule), "MOTION" (capture motion
detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures
and motion detection pictures), "Command", "pir" (pir alarm), "wlsensor" (wireless alarm), "callhelp" (panic alarm), "facedetection" (face detection),
"FieldDetection" (intrusion detection), "scenechangedetection" (scene change detection), "LineDetection" (line crossing detection)-->test
    </recordSearchType>
    <isSupportActivateIpc>
      <!--ro, opt, bool, whether the device supports activating the network camera-->true
    </isSupportActivateIpc>
    <isSupportCheckIpcSecurity>
      <!--ro, opt, object, device password detection-->
      <isSupportCheckPassword>
        <!--ro, opt, bool, whether the device supports password detection-->true
      </isSupportCheckPassword>
    </isSupportCheckIpcSecurity>
    <isSupportMainAndSubRecord>
      <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
    </isSupportMainAndSubRecord>
    <isSupportSyncIPCPassword>
      <!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true
    </isSupportSyncIPCPassword>
    <isSupportTransferIPC>
      <!--ro, opt, bool, whether the device supports transparent transmission of the network camera-->true
    </isSupportTransferIPC>
    <isSupportSmartSearch>
      <!--ro, opt, bool, whether the device supports video dual-VCA, desc:related URI: /ISAPI/ContentMgmt/SmartSearch/capabilities-->true
    </isSupportSmartSearch>
    <isSupportPOS>
      <!--ro, opt, bool, whether the device supports POS-->true
    </isSupportPOS>
    <isSupportLinkNum>
      <!--ro, opt, bool, whether the device supports getting the number of links, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/linkNum--
>true
    </isSupportLinkNum>
    <isSupportPassBackBasicCfg>
      <!--ro, opt, bool, whether the device supports configuring basic parameters of the CVR copy-back function, desc:related URI:
/ISAPI/ContentMgmt/record/passback/basicCfg/capabilities-->true
    </isSupportPassBackBasicCfg>
    <PassBackTaskConfig>
      <!--ro, opt, object, capability of managing tasks of the CVR copy-back function-->
      <isSupportAddTask>
        <!--ro, opt, bool, whether the device supports adding a task, desc:related URI: /ISAPI/ContentMgmt/record/passback/addTask/capabilities-->true
      </isSupportAddTask>
      <isSupportSearchTask>
        <!--ro, opt, bool, whether the device supports searching tasks, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchTask/capabilities-->true
      </isSupportSearchTask>
      <isSupportControlTask>
        <!--ro, opt, bool, whether the device supports controlling the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/controlTask/capabilities--
>true
      </isSupportControlTask>
      <isSupportDeleteTask>
        <!--ro, opt, bool, whether the device supports deleting the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/deleteTask/capabilities--
>true
      </isSupportDeleteTask>
    </PassBackTaskConfig>
    <PassBackPlanConfig>
      <!--ro, opt, object, capability of configuring the CVR copy-back schedule-->
      <isSupportAddPlan>
        <!--ro, opt, bool, whether the device supports adding a schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/addPlan/capabilities-->true
      </isSupportAddPlan>
      <isSupportSearchPlan>
        <!--ro, opt, bool, whether the device supports searching schedules, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchPlan/capabilities--
>true
      </isSupportSearchPlan>
      <isSupportDeletePlan>
        <!--ro, opt, bool, whether the device supports deleting the schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/deletePlan/capabilities--
>true
      </isSupportDeletePlan>
    </PassBackPlanConfig>
    <IRAIDCap>
      <!--ro, opt, object, network array capability-->
      <isSupportIRAIDRapidConfiguration>
        <!--ro, opt, bool, whether the device supports the rapid IRAID configuration, desc:related URI:
/ISAPI/ContentMgmt/Storage/IRAID/RapidConfiguration/capabilities-->true
      </isSupportIRAIDRapidConfiguration>
    </IRAIDCap>
    <isSupportStorageExtraInfo>
      <!--ro, opt, bool, whether the device supports configuring additional storage information, desc:related URI:
/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities-->true
    </isSupportStorageExtraInfo>
    <isSupportRecordStatus>
      <!--ro, opt, bool, whether the device supports getting the recording status, desc:related URI: /ISAPI/ContentMgmt/channels/<ID>/record/capabilities--
>true
    </isSupportRecordStatus>
    <isSupportRacmChannelsCap opt="true,false">
      <!--ro, opt, bool, whether the device supports the Rcam capability by channel, attr:opt{opt, string}, desc:related URI:
/ISAPI/ContentMgmt/channels/<ID>/capabilities-->true
    </isSupportRacmChannelsCap>
    <LockCap>
      <!--ro, opt, object, video locking and unlocking, desc:related URI: /ISAPI/ContentMgmt/record/control/locks/capabilities-->
    </LockCap>
    <isSupportForamtAll>
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      <!--ro, opt, bool, whether the device supports formatting all HDDs, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/format-->true
    </isSupportForamtAll>
    <isSupportExtendCabinetCfg>
      <!--ro, opt, bool, whether the device supports configuring parameters of the extended cabinet. Related URI:
/ISAPI/ContentMgmt/Storage/ExtendCabinet/capabilities-->true
    </isSupportExtendCabinetCfg>
    <diskGroupNums>
      <!--ro, opt, int, supported number of disk groups, desc:related URI: /ISAPI/ContentMgmt/Storage/diskGroup-->1
    </diskGroupNums>
    <ExternalDevice>
      <!--ro, opt, object, external device storage-->
      <USB>
        <!--ro, opt, object, USB information-->
        <debugLogOutput>
          <!--ro, opt, bool, whether to enable the debugging log, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
        </debugLogOutput>
        <isSupportCapturePackage>
          <!--ro, opt, bool, whether the device supports configuring capture and filter function of a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/capturePackage/capabilities?format=json-->true
        </isSupportCapturePackage>
        <isSupportStreamStorage>
          <!--ro, opt, bool, whether the device supports saving stream to a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/streamStorage/capabilities?format=json-->true
        </isSupportStreamStorage>
      </USB>
    </ExternalDevice>
    <customProtocolNums>
      <!--ro, opt, int, number of custom protocols connected via IPC, desc:related URI: /ISAPI/ContentMgmt/InputProxy/customProtocols, supported number of
custom protocols-->1
    </customProtocolNums>
    <isSupportIPCTiming>
      <!--ro, opt, bool, whether the device supports synchronizing the time of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <enableTiming> node supports-->true
    </isSupportIPCTiming>
    <isSupportLogConfig>
      <!--ro, opt, bool, whether the device supports log configuration, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
    </isSupportLogConfig>
    <SecurityLog>
      <!--ro, opt, object, device's capability of security log-->
      <isSupportSecurityLog>
        <!--ro, opt, bool, whether the device supports the security log-->true
      </isSupportSecurityLog>
      <isSupportLogServer>
        <!--ro, opt, bool, whether it supports security log server configuration-->true
      </isSupportLogServer>
      <isSupportLogServerTest>
        <!--ro, opt, bool, whether it supports security log server test-->true
      </isSupportLogServerTest>
      <SecurityLogTypeList>
        <!--ro, req, array, list of security log types, subType:object-->
        <SecurityLogType>
          <!--ro, opt, object, security log type-->
          <primaryType>
            <!--ro, req, string, main log type, desc:"Event", "Operation", "Other", "All"-->test
          </primaryType>
          <secondaryType opt="all">
            <!--ro, req, string, minor log type, attr:opt{opt, string}, desc:for other types, refer to the appendix-->test
          </secondaryType>
        </SecurityLogType>
      </SecurityLogTypeList>
      <isSupportLogServerList>
        <!--ro, opt, bool, whether the device supports batch configuring security log server-->true
      </isSupportLogServerList>
      <logMinItemNum>
        <!--ro, opt, int, range:[1,2147483646]-->100
      </logMinItemNum>
    </SecurityLog>
    <isSupportGetBackupDevice>
      <!--ro, opt, bool, whether the device supports getting the backup device, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/device?format=json--
>true
    </isSupportGetBackupDevice>
    <isSupportAutoBackup>
      <!--ro, opt, bool, whether the device supports automatic backup, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/auto/capabilities?format=json--
>true
    </isSupportAutoBackup>
    <certificateValidationEnabled>
      <!--ro, opt, bool, whether to enable the certification validation, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <certificateValidationEnabled> node supports-->true
    </certificateValidationEnabled>
    <defaultAdminPortEnabled>
      <!--ro, opt, bool, whether to enable the default communication port, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <defaultAdminPortEnabled> node supports-->true
    </defaultAdminPortEnabled>
    <isSupportLogDataPackage>
      <!--ro, opt, bool, whether the device supports exporting the log files, desc:related URI: /ISAPI/ContentMgmt/LogSearch/dataPackage-->true
    </isSupportLogDataPackage>
    <logSearchTimeSpanNums>
      <!--ro, opt, int, number of time periods for log search, desc:related URI: /ISAPI/ContentMgmt/logSearch; corresponding node: timeSpanList in
CMSearchDescription-->1
    </logSearchTimeSpanNums>
    <CloudStorageServerCap>
      <!--ro, opt, object, capability of configuring the cloud storage server-->
      <isSupportCloudStorageParameter>
        <!--ro, opt, bool, whether the device supports configuring the cloud storage parameters, desc:related URI:
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        <!--ro, opt, bool, whether the device supports configuring the cloud storage parameters, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/capabilities?format=json-->true
      </isSupportCloudStorageParameter>
      <isSupportCloudStoragePool>
        <!--ro, opt, bool, whether the device supports configuring the cloud storage pool, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/pool/capabilities?format=json-->true
      </isSupportCloudStoragePool>
      <supportChannelsNum>
        <!--ro, opt, int, number of channels supported by the cloud storage, desc:if this node is not returned, all channels of the device support; if this
node is returned, several channels of the device supports cloud storage (at present, it will not return the specific number of the channels supporting cloud
storage)-->1
      </supportChannelsNum>
    </CloudStorageServerCap>
    <isSupportWebPrivatePlaybackByUTC>
      <!--ro, opt, bool, whether it supports playback (via private protocol) by UTC-->true
    </isSupportWebPrivatePlaybackByUTC>
    <isSupportFindCommonFileByUTC>
      <!--ro, opt, bool, whether it supports searching for ordinary files (private protocols) by UTC time-->true
    </isSupportFindCommonFileByUTC>
    <isSupportFindEventFileByUTC>
      <!--ro, opt, bool, whether it supports searching for event files (private protocols) by UTC time-->true
    </isSupportFindEventFileByUTC>
    <isSupportSmartSearchRecordByUTC>
      <!--ro, opt, bool, whether it supports smart video search by UTC time-->true
    </isSupportSmartSearchRecordByUTC>
    <isSupportMRDSearchByTimeZone>
      <!--ro, opt, bool, whether it supports searching for files (private protocols) by monthly calendar by UTC time-->true
    </isSupportMRDSearchByTimeZone>
    <isSupportSearchRecordLabelByUTC>
      <!--ro, opt, bool, whether it supports searching for video tags by UTC time-->true
    </isSupportSearchRecordLabelByUTC>
    <isSupportSearchPictureByUTC>
      <!--ro, opt, bool, whether it supports searching for pictures by UTC time-->true
    </isSupportSearchPictureByUTC>
    <isSupportSmartSearchPictureByUTC>
      <!--ro, opt, bool, whether it supports smart pictures search by UTC time-->true
    </isSupportSmartSearchPictureByUTC>
    <isSupportFindLogByUTC>
      <!--ro, opt, bool, whether it supports searching log by UTC time-->true
    </isSupportFindLogByUTC>
    <isSupportUploadRecordByUTC>
      <!--ro, opt, bool, whether it supports uploading files to cloud storage by UTC time-->true
    </isSupportUploadRecordByUTC>
    <isSupportPlaybackByUTC>
      <!--ro, opt, bool, whether it supports playback and locating by UTC time-->true
    </isSupportPlaybackByUTC>
    <isSupportFaceDetecttionAlarmByTimeZone>
      <!--ro, opt, bool, no file is found-->true
    </isSupportFaceDetecttionAlarmByTimeZone>
    <isSupportSourceModify>
      <!--ro, opt, bool, whether it supports configuring the parameters of the LAN IPC, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/sourceModify/capabilities-->true
    </isSupportSourceModify>
    <isSupportCountingSearchByUTC>
      <!--ro, opt, bool, whether it supports searching people counting results by UTC time-->true
    </isSupportCountingSearchByUTC>
    <isSupportCluster>
      <!--ro, opt, bool, whether the device supports the cluster function, desc:related URI: /ISAPI/ContentMgmt/Cluster-->true
    </isSupportCluster>
    <supportAIDTFSType opt="illegalParking,wrongdirection,crosslane,vehicleexist,lanechange,turnround,evidence">
      <!--ro, opt, string, the event picture search type supported by AID and TFS, attr:opt{opt, string}-->test
    </supportAIDTFSType>
    <isSupportMisinfoStateSearch>
      <!--ro, opt, bool, whether it supports searching by false alarm status-->true
    </isSupportMisinfoStateSearch>
    <isSupportOneKeyOperationStorageData>
      <!--ro, opt, bool, whether it supports one-touch configuration for storage data, desc:related URI: /ISAPI/ContentMgmt/Storage/oneKey/capabilities?
format=json-->true
    </isSupportOneKeyOperationStorageData>
    <eventRecordSearch>
      <!--ro, opt, object, whether the device supports searching for videos by event, desc:related URI: /ISAPI/ContentMgmt/eventRecordSearch?format=json-->
      <eventType opt="faceContrast,faceSnap,vehicleMatchResult,hawkResult">
        <!--ro, opt, string, event type, attr:opt{opt, string}-->test
      </eventType>
      <auxEventType opt="faceContrastFailure,faceContrastSuccess">
        <!--ro, opt, string, search sub type of the event, attr:opt{opt, string}-->test
      </auxEventType>
    </eventRecordSearch>
    <isSupportManualRecord>
      <!--ro, opt, bool, whether the device supports manual recording, desc:related URI: /ISAPI/ContentMgmt/record/control/manual/capabilities?format=json--
>true
    </isSupportManualRecord>
    <isSupportRemark>
      <!--ro, opt, bool, whether it supports file remarks-->true
    </isSupportRemark>
    <FileUpload>
      <!--ro, opt, object, automatically upload file-->
      <enabled>
        <!--ro, req, bool, whether it supports uploading the file-->true
      </enabled>
    </FileUpload>
    <PoliceInfoUpload>
      <!--ro, opt, object, automatically upload police officer information-->
      <enabled>
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        <!--ro, req, bool, whether the device supports uploading the police officer information-->true
      </enabled>
    </PoliceInfoUpload>
    <PoliceInfo>
      <!--ro, opt, object, police officer information-->
      <policeID>
        <!--ro, opt, bool, whether the device supports police officer ID-->true
      </policeID>
      <policeCode>
        <!--ro, opt, bool, whether the device supports police officer number-->true
      </policeCode>
      <policeName>
        <!--ro, opt, bool, whether the device supports police officer name-->true
      </policeName>
      <password>
        <!--ro, opt, bool, whether the device supports police officer password-->true
      </password>
      <deviceID>
        <!--ro, opt, bool, whether the device supports device ID-->true
      </deviceID>
    </PoliceInfo>
    <isSupportAssignChannelID>
      <!--ro, opt, bool, whether the device supports specifying the access channel No. when adding the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <defaultAdminPortEnabled> node supports-->true
    </isSupportAssignChannelID>
    <isSupportAssignStreamID>
      <!--ro, opt, bool, whether it supports specifying stream ID of the channel when adding the network camera in channel mode, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <defaultAdminPortEnabled> node supports-->true
    </isSupportAssignStreamID>
    <PictureSearchSubTypeList>
      <!--ro, opt, array, list of picture search sub type, subType:object-->
      <PictureSearchSubType>
        <!--ro, opt, object, the sub type of picture search-->
        <mainType>
          <!--ro, req, string, main event type of the searched picture-->test
        </mainType>
        <subType opt="all,motorVehicle,nonMotorVehicle,pedestrian">
          <!--ro, req, string, sub type of each main event type, attr:opt{opt, string}-->test
        </subType>
      </PictureSearchSubType>
    </PictureSearchSubTypeList>
    <isSupportPlaybackStreamIDByUTC>
      <!--ro, opt, bool, whether it supports playback by stream ID according to UTC time-->true
    </isSupportPlaybackStreamIDByUTC>
    <isSupportTimeLockStreamIDByUTC>
      <!--ro, opt, bool, whether it supports locking and unlocking the video segment by steam ID according to UTC time-->true
    </isSupportTimeLockStreamIDByUTC>
    <isSupportTimeSearch>
      <!--ro, opt, bool, whether it supports searching for recording start and end time by channel, desc:related URI: /ISAPI/ContentMgmt/time/search?
format=json-->true
    </isSupportTimeSearch>
    <behaviorEventPicSearch opt="allBehaviorEvent,running,group,violentMotion,failDown,playCellphone,peopleNumChange,leavePosition,retention,sleepOnduty">
      <!--ro, opt, string, abnormal event detection supported by picture search, attr:opt{opt, string}-->test
    </behaviorEventPicSearch>
    <perimeterEventPicSearch opt="allPerimeterEvent,linedetection,fielddetection,regionEntrance,regionExiting,loitering">
      <!--ro, opt, string, perimeter protection events supported by picture search, attr:opt{opt, string}-->test
    </perimeterEventPicSearch>
    <isSupportSourceStatus>
      <!--ro, opt, bool, whether it supports getting the status of access source-->true
    </isSupportSourceStatus>
    <isSupportSSDSMARTTest>
      <!--ro, opt, bool, whether the device supports S.M.A.R.T. detection on SSD, desc:related URI: /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start--
>true
    </isSupportSSDSMARTTest>
    <MobileStorage>
      <!--ro, opt, object, storage capability of the mobile HDD-->
      <isSupportGetFileTime>
        <!--ro, opt, bool, whether it supports getting the file time of the mobile HDD, desc:it supports mobile HDD currently. Related URI:
/ISAPI/ContentMgmt/mobileStorage/fileTime/capabilities?format=json-->true
      </isSupportGetFileTime>
    </MobileStorage>
    <isSupportGetChannelList>
      <!--ro, opt, bool, whether it supports getting channels of the HDD, desc:it supports mobile HDD currently. Related URI:
/ISAPI/ContentMgmt/Storage/hdd/channelList/capabilities?format=json-->true
    </isSupportGetChannelList>
    <isSupportCascade>
      <!--ro, opt, bool, whether it supports configuring parameters for NVR cascading topology-->true
    </isSupportCascade>
    <isSupportPictureInfo>
      <!--ro, opt, bool, whether it supports picture search on the SD card of the capture camera-->true
    </isSupportPictureInfo>
    <isSupportCloudStorageModel>
      <!--ro, opt, bool, whether the device supports configuring the cloud storage mode,
desc:/ISAPI/ContentMgmt/Storage/CloudStorageModel/<channelID>/capabilities?format=json-->true
    </isSupportCloudStorageModel>
    <isSupportCloudStorageModelList>
      <!--ro, opt, bool, whether the device supports configuring cloud storage modes in a batch,
desc:/ISAPI/ContentMgmt/Storage/CloudStorageModel/<channelID>?format=json-->true
    </isSupportCloudStorageModelList>
    <isSupportRecordAudio>
      <!--ro, opt, bool, whether the device supports recording the audio, desc:only the audio will be recorded and the video will not be recorded. Related
URI: /ISAPI/ContentMgmt/recordAudio/<audioID>/capabilities?format=json-->true
    </isSupportRecordAudio>
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    <isSupportSearchAudio>
      <!--ro, opt, bool, whether the device supports searching for the audio, desc:related URI: /ISAPI/ContentMgmt/recordAudio/search/capabilities?
format=json-->true
    </isSupportSearchAudio>
    <isSupportAudioDailyDistribution>
      <!--ro, opt, bool, whether it supports searching for audio file information by monthly calendar, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json-->true
    </isSupportAudioDailyDistribution>
    <isSupportDownloadAudio>
      <!--ro, opt, bool, whether the device supports downloading the audio by file name, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/fileNameDownload/capabilities?format=json-->true
    </isSupportDownloadAudio>
    <isSupportDeleteAudio>
      <!--ro, opt, bool, whether the device supports deleting the recorded audio file, desc:related URI: /ISAPI/ContentMgmt/recordAudio/deleteFile?
format=json-->true
    </isSupportDeleteAudio>
    <isSupportSearchRemoteAlbum>
      <!--ro, opt, bool, whether the device supports remotely searching for album information, desc:related URI: /ISAPI/ContentMgmt/searchRemoteAlbum?
format=json-->true
    </isSupportSearchRemoteAlbum>
    <isSupportStreamParam>
      <!--ro, opt, bool, whether the device supports configuring the access stream parameters of all digital channels, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/streamParam/capabilities?format=json-->true
    </isSupportStreamParam>
    <isSupportFusionChan>
      <!--ro, opt, bool, whether the device supports fusion channel, desc:if this node does not exist, it indicates that the device does not support the
function; interaction IP channel: it is a concept from the education sharing system; the education sharing system will allocate IP channels to images of the
connected sub classrooms or meeting rooms. fusion channel: a channel generated when adding the NVR locally or remotely; it supports live view, playback,
storage, and recording schedule/linkage configuration; the channel will take effect by enabling the fusion channel-->true
    </isSupportFusionChan>
    <isSupportRecordMultipleDownload>
      <!--ro, opt, bool, whether the device supports downloading videos in multiples, desc:if this node does not exist, it indicates that the device does
not support the function by default-->true
    </isSupportRecordMultipleDownload>
  </RacmCap>
  <isSupportMI3704>
    <!--ro, opt, bool-->true
  </isSupportMI3704>
  <isSupportLogServer>
    <!--ro, opt, bool, whether it supports security log server configuration-->true
  </isSupportLogServer>
  <isSupportLogServerTest>
    <!--ro, opt, bool, whether it supports security log server test-->true
  </isSupportLogServerTest>
  <PTZCtrlCap>
    <!--ro, opt, object, device's PTZ capability-->
    <isSupportPatrols>
      <!--ro, opt, bool, whether the device supports patrol-->true
    </isSupportPatrols>
    <isSupportCombinedPath>
      <!--ro, opt, bool, whether the device supports the PTZ combined path-->true
    </isSupportCombinedPath>
    <isSupportRailwayRobot>
      <!--ro, opt, bool, whether the device supports railway robot, desc:/ISAPI/PTZCtrl/channels/<channelID>/RailwayRobot/capabilities?format=json-->true
    </isSupportRailwayRobot>
  </PTZCtrlCap>
  <SmartCap>
    <!--ro, opt, object, compact and smart capability, desc:/ISAPI/Smart/capabilities-->
    <isSupportROI>
      <!--ro, opt, bool, whether the device supports ROI encoding configuration-->true
    </isSupportROI>
    <isSupportFaceDetect>
      <!--ro, opt, bool, whether the device supports face detection-->true
    </isSupportFaceDetect>
    <isSupportIntelliTrace>
      <!--ro, opt, bool, whether the device supports smart tracking-->true
    </isSupportIntelliTrace>
    <isSupportFieldDetection>
      <!--ro, opt, bool, whether the device supports intrusion detection-->true
    </isSupportFieldDetection>
    <isSupportDefocusDetection>
      <!--ro, opt, bool, whether the device supports defocus detection-->true
    </isSupportDefocusDetection>
    <isSupportAudioDetection>
      <!--ro, opt, bool, whether the device supports audio detection-->true
    </isSupportAudioDetection>
    <isSupportSceneChangeDetection>
      <!--ro, opt, bool, whether the device supports scene change detection-->true
    </isSupportSceneChangeDetection>
    <isSupportLineDetection>
      <!--ro, opt, bool, whether the device supports line crossing detection-->true
    </isSupportLineDetection>
    <isSupportRegionEntrance>
      <!--ro, opt, bool, whether the device supports region entrance detection-->true
    </isSupportRegionEntrance>
    <isSupportRegionExiting>
      <!--ro, opt, bool, whether the device supports region exiting detection-->true
    </isSupportRegionExiting>
    <isSupportLoitering>
      <!--ro, opt, bool, whether the device supports loitering detection-->true
    </isSupportLoitering>
    <isSupportGroup>
      <!--ro, opt, bool, whether the device supports people gathering detection-->true
    </isSupportGroup>
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      <isSupportRapidMove>
        <!--ro, opt, bool, whether the device supports fast moving detection-->true
      </isSupportRapidMove>
      <isSupportParking>
        <!--ro, opt, bool, whether the device supports parking detection-->true
      </isSupportParking>
      <isSupportUnattendedBaggage>
        <!--ro, opt, bool, whether the device supports unattended baggage detection-->true
      </isSupportUnattendedBaggage>
      <isSupportAttendedBaggage>
        <!--ro, opt, bool, whether the device supports object removal detection-->true
      </isSupportAttendedBaggage>
      <isSupportPeopleDetection>
        <!--ro, opt, bool, whether the device supports human body detection-->true
      </isSupportPeopleDetection>
      <isSupportSmartCalibration>
        <!--ro, opt, bool, whether the device supports calibration filter size-->true
      </isSupportSmartCalibration>
      <isSupportStorageDetection>
        <!--ro, opt, bool, whether the device supports health monitoring storage-->true
      </isSupportStorageDetection>
      <isSupportShipsDetection>
        <!--ro, opt, bool, whether the device supports ship detection-->true
      </isSupportShipsDetection>
      <isSupportShield>
        <!--ro, opt, bool, whether the device supports configuring the compact and smart shielded regions-->true
      </isSupportShield>
      <isSupportAlgVersion>
        <!--ro, opt, bool, whether the device supports algorithm version-->true
      </isSupportAlgVersion>
      <isSupportVideoOverlap>
        <!--ro, opt, bool, whether the device supports configuring information overlay for video stream-->true
      </isSupportVideoOverlap>
      <isSupportParkingState>
        <!--ro, opt, bool, whether the device supports parking status-->true
      </isSupportParkingState>
      <isSupportChannelResource>
        <!--ro, opt, bool, whether the device supports channel resources-->true
      </isSupportChannelResource>
      <isSupportAnalysisUnitSwitch opt="true,false">
        <!--ro, opt, bool, whether the device supports switching analysis unit, attr:opt{opt, string}-->true
      </isSupportAnalysisUnitSwitch>
      <isSupportHFPD>
        <!--ro, opt, bool, whether the device supports frequently appeared person detection, desc:related URI: /ISAPI/SDT/HFPD?format=json-->true
      </isSupportHFPD>
      <isSupportImageROI>
        <!--ro, opt, bool, whether the device supports smartJpeg (image ROI)-->true
      </isSupportImageROI>
      <isSupportLFPD>
        <!--ro, opt, bool, whether the device supports rarely appeared person detection, desc:related URI: /ISAPI/SDT/LFPD?format=json-->true
      </isSupportLFPD>
      <isSupportSmartOverlapParams>
        <!--ro, opt, bool, whether it supports smart overlay parameters, desc:related URI: /ISAPI/Smart/channels/<channelID>/SmartOverlapParams?format=json--
>true
      </isSupportSmartOverlapParams>
    </SmartCap>
    <isSupportEhome>
      <!--ro, opt, bool, whether the device supports ISUP functions-->true
    </isSupportEhome>
    <isSupportStreamingEncrypt>
      <!--ro, opt, bool, whether the device supports stream encryption-->true
    </isSupportStreamingEncrypt>
    <TestCap>
      <!--ro, opt, object, test capability-->
      <isSupportFTPTest>
        <!--ro, opt, bool, whether it supports FTP test-->true
      </isSupportFTPTest>
      <isSupportPingTest>
        <!--ro, opt, bool, whether it supports PING test-->true
      </isSupportPingTest>
      <isSupportNTPTest>
        <!--ro, opt, bool, whether it supports NTP test-->true
      </isSupportNTPTest>
      <isSupportNASTest>
        <!--ro, opt, bool, whether it supports NAS test-->true
      </isSupportNASTest>
      <isSupportEmailTest>
        <!--ro, opt, bool, whether the device supports email test-->true
      </isSupportEmailTest>
    </TestCap>
    <ImageMiscCap>
      <!--ro, opt, object, DPC capability-->
      <isSupportDPC>
        <!--ro, opt, bool, whether the device supports DPC-->true
      </isSupportDPC>
    </ImageMiscCap>
    <PanoramaCap>
      <!--ro, opt, object, capability of panoramic channels, desc:the object cannot be extended. You can extend it in /ISAPI/Panorama/capabilities-->
      <isSupportGeneratePanorama>
        <!--ro, opt, bool, whether the device supports generating panoramic images-->true
      </isSupportGeneratePanorama>
      <isSupportPanoramaPosition3D>
        <!--ro, opt, bool, whether the device supports panoramic 3D position-->true
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    </isSupportPanoramaPosition3D>
    <isSupportPreset>
      <!--ro, opt, bool, whether the device supports configuring panoramic preset-->true
    </isSupportPreset>
    <isSupportGetPanoramicSplicingStatus>
      <!--ro, opt, bool-->true
    </isSupportGetPanoramicSplicingStatus>
    <isSupportStopPanoramicSplicing>
      <!--ro, opt, bool-->true
    </isSupportStopPanoramicSplicing>
    <isSupportSplicingChannelMotionControl>
      <!--ro, opt, bool-->true
    </isSupportSplicingChannelMotionControl>
    <isSupportGetPanoramaSplicingPictures>
      <!--ro, opt, bool-->true
    </isSupportGetPanoramaSplicingPictures>
    <isSupportImportPanoramaSplicingParamFile>
      <!--ro, opt, bool-->true
    </isSupportImportPanoramaSplicingParamFile>
    <isSupportExportPanoramaSplicingParamFile>
      <!--ro, opt, bool-->true
    </isSupportExportPanoramaSplicingParamFile>
    <isSupportImportPanoramicSplicingPicture>
      <!--ro, opt, bool-->true
    </isSupportImportPanoramicSplicingPicture>
    <isSupportExportPanoramicSplicingPicture>
      <!--ro, opt, bool-->true
    </isSupportExportPanoramicSplicingPicture>
  </PanoramaCap>
  <ThermalCap>
    <!--ro, opt, object, thermal capability, desc:related URI: /ISAPI/Thermal/capabilities-->
    <isSupportFireDetection>
      <!--ro, opt, bool, whether the device supports fire detection, desc:related URI: /ISAPI/Thermal/channels/<ID>/fireDetection/capabilities-->true
    </isSupportFireDetection>
    <isFireFocusZoomSupport>
      <!--ro, opt, bool, whether the device supports visible light lens zooming, desc:related URI: /ISAPI/Thermal/channels/<ID>/fireFocusZoom-->true
    </isFireFocusZoomSupport>
    <isSupportThermometry>
      <!--ro, opt, bool, whether the device supports thermography, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities-->true
    </isSupportThermometry>
    <isSupportRealtimeThermometry>
      <!--ro, opt, bool, whether the device supports uploading real-time temperature measurement data-->true
    </isSupportRealtimeThermometry>
    <isSupportThermIntell>
      <!--ro, opt, bool, whether the device supports configuring smart thermal mutual exclusion, desc:related URI:
/ISAPI/Thermal/channels/<ID>/thermIntell/capabilities-->true
    </isSupportThermIntell>
    <isSupportNVR>
      <!--ro, opt, bool, it is used for client to check-->true
    </isSupportNVR>
    <isSupportThermometrySchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of temperature measurement, desc:related URI: /ISAPI/Event/schedules/thermometry--
>true
    </isSupportThermometrySchedule>
    <isSupportTemperatureSchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of temperature difference, desc:related URI: /ISAPI/Event/schedules/temperature--
>true
    </isSupportTemperatureSchedule>
    <isSupportFireDetectionSchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of fire detection, desc:related URI: /ISAPI/Event/schedules/fireDetections-->true
    </isSupportFireDetectionSchedule>
    <isSupportPower>
      <!--ro, opt, bool, whether the device supports power on/off capability, desc:related URI: /ISAPI/Thermal/Power/capabilities-->true
    </isSupportPower>
    <isSupportRealtimeTempHumi>
      <!--ro, opt, bool, whether the device supports real-time detection of temperature and humidity, desc:related URI:
/ISAPI/Thermal/TempHumi/capabilities-->true
    </isSupportRealtimeTempHumi>
    <ManualThermCap>
      <!--ro, opt, object, the capability of manual temperature measurement-->
      <manualThermRuleNum>
        <!--ro, opt, int, if this node is not returned, it indicates manual temperature measurement is not supported, desc:related URI:
/ISAPI/Thermal/manualThermometry/capabilities-->true
      </manualThermRuleNum>
    </ManualThermCap>
    <isSupportManualThermBasic>
      <!--ro, opt, bool, whether the device supports basic configuration of manual temperature measurement, desc:related URI:
/ISAPI/Thermal/manualThermBasic/capabilities-->true
    </isSupportManualThermBasic>
    <isSupportFireShieldMask>
      <!--ro, opt, bool, whether the device supports configuring the parameters of fire shielded regions, desc:related URI:
/ISAPI/Thermal/channels/<ID>/fireShieldMask/capabilities-->true
    </isSupportFireShieldMask>
    <isSupportSmokeShieldMask>
      <!--ro, opt, bool, whether the device supports configuring the parameters of smoke shielded regions, desc:related URI:
/ISAPI/Thermal/channels/<ID>/smokeShieldMask/capabilities-->true
    </isSupportSmokeShieldMask>
    <isSupportThermometryMode>
      <!--ro, opt, bool, whether the device supports configuring temperature measurement mode, desc:related URI:
/ISAPI/Thermal/channels/<ID>/thermometryMode/capabilities-->true
    </isSupportThermometryMode>
    <isSupportThermalPip>
      <!--ro, opt, bool, whether the device supports the PIP configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/pip/capabilities-->true
    </isSupportThermalPip>
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    <isSupportThermalIntelRuleDisplay>
      <!--ro, opt, bool, whether the device supports the PIP configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/IntelRuleDisplay/capabilities-->true
    </isSupportThermalIntelRuleDisplay>
    <AlgVersionInfo>
      <!--ro, opt, object, whether the device supports getting the version information of thermal algorithms library, desc:related URI: /ISAPI/Thermal/AlgVersion-->
      <thermometryAlgName min="1" max="128">
        <!--ro, opt, string, version information of temperature measurement algorithms library, attr:min{opt, string},max{opt, string}-->test
      </thermometryAlgName>
      <shipsAlgName min="1" max="128">
        <!--ro, opt, string, version name of ship detection algorithms library, attr:min{opt, string},max{opt, string}-->test
      </shipsAlgName>
      <fireAlgName min="1" max="128">
        <!--ro, opt, string, algorithm library version name of fire detection, attr:min{opt, int},max{opt, int}-->test
      </fireAlgName>
      <smokeAlgName min="1" max="128">
        <!--ro, opt, string, algorithm library version name of smoke detection, attr:min{opt, int},max{opt, int}-->test
      </smokeAlgName>
      <wasteGasAlgName min="1" max="128">
        <!--ro, opt, string, algorithm library version name of waste gas detection, attr:min{opt, string},max{opt, string}-->test
      </wasteGasAlgName>
      <greyScaleAlarmAlgName min="1" max="128">
        <!--ro, opt, string, algorithm library version name of greyscale alarm, attr:min{opt, string},max{opt, string}-->test
      </greyScaleAlarmAlgName>
    </AlgVersionInfo>
    <isSupportFaceThermometry>
      <!--ro, opt, bool, whether supports temperature screening configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/faceThermometry/capabilities-->true
    </isSupportFaceThermometry>
    <isSupportThermalBlackBody>
      <!--ro, opt, bool, whether the device supports configuring black body parameters of thermal imaging, desc:related URI: /ISAPI/Thermal/channels/<ID>/blackBody/capabilities-->true
    </isSupportThermalBlackBody>
    <isSupportThermalStreamParam>
      <!--ro, opt, bool, whether the device supports configuring stream parameters of thermal imaging, desc:related URI: /ISAPI/Thermal/channels/<ID>/streamParam/capabilities-->true
    </isSupportThermalStreamParam>
    <isSupportBodyTemperatureCompensation>
      <!--ro, opt, bool, whether the device supports temperature compensation configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation/capabilities-->true
    </isSupportBodyTemperatureCompensation>
    <isSupportClickToThermometry>
      <!--ro, opt, bool, whether the device supports click-to-thermometry, desc:related URI: ISAPI/Thermal/channels/<ID>/clickToThermometry/initialization-->true
    </isSupportClickToThermometry>
    <isSupportThermometryHistorySearch>
      <!--ro, opt, bool, whether the device supports searching for historical temperature data, desc:related URI: /ISAPI/Thermal/channels/<ID>/historyTemperature/capabilities-->true
    </isSupportThermometryHistorySearch>
    <isSupportThermometryShieldMask>
      <!--ro, opt, bool, whether the device supports configuring shield parameters of thermography region, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometryShieldMask/capabilities-->true
    </isSupportThermometryShieldMask>
    <isSupportTemperatureCorrection>
      <!--ro, opt, bool, whether device supports temperature measurement correction, desc:related URI: /ISAPI/Thermal/channels/<ID>/CorrectionParam/capabilities-->true
    </isSupportTemperatureCorrection>
    <isSupportBurningPrevention>
      <!--ro, opt, bool, whether device supports burning prevention, desc:related URI: /ISAPI/Thermal/channels/<ID>/burningPrevention/capabilities-->true
    </isSupportBurningPrevention>
    <isSupportTemperatureCollection>
      <!--ro, opt, bool, whether device supports temperature ANR, desc:related URI: ISAPI/Thermal/temperature/collection/capabilities?format=json-->true
    </isSupportTemperatureCollection>
    <isSupportJpegPicWithAppendData>
      <!--ro, opt, bool, whether the device supports capturing pictures, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/jpegPicWithAppendData?format=json-->true
    </isSupportJpegPicWithAppendData>
    <isSupportRealTimethermometryForHTTP>
      <!--ro, opt, bool, whether device supports real-time temperature measurement, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/realTimethermometry/rules?format=json-->true
    </isSupportRealTimethermometryForHTTP>
    <isSupportthermometryOffLineCapture>
      <!--ro, opt, bool, whether the device supports capturing pictures offline, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/OffLineCapture?format=json-->true
    </isSupportthermometryOffLineCapture>
    <isSupportShipsDetectionWithScene>
      <!--ro, opt, bool, whether the device supports ship detection by scene, desc:related URI: /ISAPI/Thermal/channels/<ID>/shipsDetection/capabilities?format=json-->true
    </isSupportShipsDetectionWithScene>
    <isSupportWasteGasDetection>
      <!--ro, opt, bool, whether the device supports waste gas monitoring-->true
    </isSupportWasteGasDetection>
    <isSupportGreyScaleAlarm>
      <!--ro, opt, bool, whether the device supports the gray scale alarm-->true
    </isSupportGreyScaleAlarm>
    <isSupportFaceSnapThermometry>
      <!--ro, opt, bool, whether the device supports temperature screening when capturing face pictures-->true
    </isSupportFaceSnapThermometry>
    <isSupportThermalTemperatureCorrect>
      <!--ro, opt, bool, whether device supports temperature calibration, desc:related URI: /ISAPI/Thermal/channels/<ID>/temperatureCorrect?format=json-->true
    </isSupportThermalTemperatureCorrect>
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    <isSupportThermalVehicleDetection>
      <!--ro, opt, bool, whether the device supports thermal imaging vehicle detection-->true
    </isSupportThermalVehicleDetection>
    <isSupportSensorParam>
      <!--ro, opt, bool, whether the device supports configuring the sensor temperature measurement parameters, desc:related URI:
/ISAPI/Thermal/channels/<channelID>/thermometry/sensorParam/capabilities?format=json-->true
    </isSupportSensorParam>
    <isSupportTemperatureIntervalMeasurement>
      <!--ro, opt, bool, whether the device supports temperature range detection-->true
    </isSupportTemperatureIntervalMeasurement>
    <isSupportIndustryThermometryTemperatureRise>
      <!--ro, opt, bool, whether the device supports setting temperature rise compensation in thermal industrial temperature measurement, desc:related URI:
/ISAPI/Thermal/channels/<ID>/industryThermometry/temperatureRise?format=json-->true
    </isSupportIndustryThermometryTemperatureRise>
    <isSupportThermalDataLinkInfo>
      <!--ro, opt, bool, whether the device supports getting the temperature data link information, desc:related URI: /ISAPI/Thermal/thermalDataLinkInfo?
format=json; /ISAPI/Thermal/thermalDataLinkStatus?format=json-->true
    </isSupportThermalDataLinkInfo>
    <isSupportTemperatureCompensationCalib>
      <!--ro, opt, bool, whether the device supports temperature compensation, desc:related URI:
/ISAPI/Thermal/channels/<channelID>/temperatureCompensationCalibration?format=json-->true
    </isSupportTemperatureCompensationCalib>
    <isSupportWebDisplayPixelToPixelParam>
      <!--ro, opt, bool, whether the web needs to display full-screen temperature parameters, desc:related URI:
/ISAPI/Thermal/channels/<channelID>/thermometry/pixelToPixelParam/capabilities-->true
    </isSupportWebDisplayPixelToPixelParam>
    <isSupportThermalInspection>
      <!--ro, opt, bool, whether the device supports thermal patrol, desc:related URI: /ISAPI/Thermal/inspection/capabilities?format=json-->true
    </isSupportThermalInspection>
    <isSupportBlackBodyTemperature>
      <!--ro, opt, bool, whether the device supports searching for black body center point temperature, desc:related URI:
/ISAPI/Thermal/channels/<channelID>/blackBodyTemperature?format=json-->true
    </isSupportBlackBodyTemperature>
    <isSupportExportThermalCalibrationFile>
      <!--ro, opt, bool, whether the device supports exporting the temperature calibration file, desc:POST /ISAPI/Thermal/ExportThermalCalibrationFile?
format=json-->true
    </isSupportExportThermalCalibrationFile>
    <isSupportSearchSmokeAndFireDetection>
      <!--ro, opt, bool, whether the device supports searching for history records of smoke and fire detection, desc:POST
/ISAPI/Thermal/fireDetection/SearchSmokeAndFireDetection?format=json-->true
    </isSupportSearchSmokeAndFireDetection>
    <isSupportColumnOven>
      <!--ro, opt, bool, whether the device supports column oven parameters, desc:GET/PUT /ISAPI/OPCA/GC/ColumnOven?format=json-->true
    </isSupportColumnOven>
    <isSupportImportThermalCalibrationFile>
      <!--ro, opt, bool, whether the device supports importing the temperature calibration file, desc:POST /ISAPI/Thermal/ImportThermalCalibrationFile?
format=json-->true
    </isSupportImportThermalCalibrationFile>
    <isSupportThermalCalibrationFileCreatParam>
      <!--ro, opt, bool, whether the device supports the parameters of generating temperature calibration file, desc:GET
/ISAPI/Thermal/ThermalCalibrationFileCreatParam?format=json-->true
    </isSupportThermalCalibrationFileCreatParam>
    <isSupportNormalRuleParams>
      <!--ro, opt, bool, whether the device supports the parameters of smoke and fire detection normal rules, desc:GET/PUT
/ISAPI/Thermal/fireDetection/channels/<channelID>/NormalRuleParams?format=json-->true
    </isSupportNormalRuleParams>
    <isSupportEnvironmentTemperatureMonitorScheduleParams>
      <!--ro, opt, bool, whether the device supports the parameters of environment temperature monitor arming schedule, desc:GET/PUT
/ISAPI/Thermal/EnvironmentTemperatureMonitorScheduleParams?format=json-->true
    </isSupportEnvironmentTemperatureMonitorScheduleParams>
    <isSupportEnvironmentTemperatureMonitorParam>
      <!--ro, opt, bool, whether the device supports the parameters of environment temperature monitoring, desc:GET/PUT
/ISAPI/Thermal/EnvironmentTemperatureMonitorParam?format=json-->true
    </isSupportEnvironmentTemperatureMonitorParam>
    <isSupportEnvironmentTemperatureMonitorTriggerParams>
      <!--ro, opt, bool, whether the device supports the parameters of environment temperature monitoring linkage action, desc:GET/PUT
/ISAPI/Thermal/EnvironmentTemperatureMonitorTriggerParams?format=json-->true
    </isSupportEnvironmentTemperatureMonitorTriggerParams>
    <isSupportSearchAsynThermalInspectionTaskTestPoint>
      <!--ro, opt, bool, whether the device supports searching for information of specified asynchronous thermal inspection point, desc:POST
/ISAPI/Thermal/SearchAsynThermalInspectionTaskTestPoint?format=json-->true
    </isSupportSearchAsynThermalInspectionTaskTestPoint>
    <isSupportSearchAsynThermalInspectionTask>
      <!--ro, opt, bool, whether the device supports searching for specified asynchronous thermal inspection task, desc:POST
/ISAPI/Thermal/SearchAsynThermalInspectionTask?format=json-->true
    </isSupportSearchAsynThermalInspectionTask>
    <isSupportAddAsynThermalInspectionTaskTestPoint>
      <!--ro, opt, bool, whether the device supports adding information of specified asynchronous thermal inspection point, desc:POST
/ISAPI/Thermal/AddAsynThermalInspectionTaskTestPoint?format=json-->true
    </isSupportAddAsynThermalInspectionTaskTestPoint>
    <isSupportAddAsynThermalInspectionTask>
      <!--ro, opt, bool, whether the device supports adding asynchronous thermal inspection task, desc:POST /ISAPI/Thermal/AddAsynThermalInspectionTask?
format=json-->true
    </isSupportAddAsynThermalInspectionTask>
    <isSupportImportAsynThermalInspectionTaskBaseMap>
      <!--ro, opt, bool, whether the device supports importing asynchronous thermal inspection task base map, desc:POST
/ISAPI/Thermal/ImportAsynThermalInspectionTaskBaseMap?format=json-->true
    </isSupportImportAsynThermalInspectionTaskBaseMap>
    <isSupportShutterFreeze>
      <!--ro, opt, bool, whether the device supports freezing the shutter, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ShutterFreeze?format=json-->true
    </isSupportShutterFreeze>
  </ThermalCap>
  <WLAlarmCap>
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    <!--ro, opt, object, wireless alarm capability-->
    <isSupportTeleControl>
      <!--ro, opt, bool, whether the device supports remote control-->true
    </isSupportTeleControl>
    <isSupportPIR>
      <!--ro, opt, bool, whether the device supports PIR configuration-->true
    </isSupportPIR>
    <isSupportWLSensors>
      <!--ro, opt, bool, N/A-->true
    </isSupportWLSensors>
    <isSupportCallHelp>
      <!--ro, opt, bool, whether the device supports calling for help-->true
    </isSupportCallHelp>
    <WLSensorsNum>
      <!--ro, opt, int, N/A-->1
    </WLSensorsNum>
  </WLAlarmCap>
  <isSupportGIS>
    <!--ro, opt, bool, whether the device supports GIS, desc:related URI: /ISAPI/GIS/channels-->true
  </isSupportGIS>
  <isSupportCompass>
    <!--ro, opt, bool, whether the device supports compass configuration, desc:related URI: /ISAPI/Compass/channels/<ID>/capabilities-->true
  </isSupportCompass>
  <isSupportRoadInfoOverlays>
    <!--ro, opt, bool, whether the device supports overlaying the lane information-->true
  </isSupportRoadInfoOverlays>
  <isSupportFaceCaptureStatistics>
    <!--ro, opt, bool, whether the device supports face capture statistics, desc:related URI: /ISAPI/Intelligent/channels/<ID>/faceCaptureStatistics/search-
->true
  </isSupportFaceCaptureStatistics>
  <isSupportElectronicsEnlarge>
    <!--ro, opt, bool, whether the device supports electronics enlarge-->true
  </isSupportElectronicsEnlarge>
  <isSupportRemoveStorage>
    <!--ro, opt, bool, whether the device supports remote storage-->true
  </isSupportRemoveStorage>
  <isSupportCloud>
    <!--ro, opt, bool, whether the device supports cloud storage-->true
  </isSupportCloud>
  <isSupportRecordHost>
    <!--ro, opt, bool, whether the device supports configuring the education sharing server, desc:related URI: /ISAPI/RecordHost/capabilities-->true
  </isSupportRecordHost>
  <isSupportMetalDetection>
    <!--ro, opt, bool-->true
  </isSupportMetalDetection>
  <isSupportViewshedOverlay>
    <!--ro, opt, bool, whether the device supports overlaying viewshed camera target, desc:related URI:
/ISAPI/System/Video/inputs/channels/<ID>/viewshedOverlay-->true
  </isSupportViewshedOverlay>
  <isSupportGPSLabelTracking>
    <!--ro, opt, bool, whether the device supports GPS tag tracking, desc:related URI:/ISAPI/GIS/channels/<ID>/GPSLabelTracking?format=json-->true
  </isSupportGPSLabelTracking>
  <isSupportEagleEye>
    <!--ro, opt, bool, whether the device supports the PanoVu series camera-->true
  </isSupportEagleEye>
  <isSupportPanorama>
    <!--ro, opt, bool, whether the device supports panoramic mode-->true
  </isSupportPanorama>
  <isSupportFirmwareVersionInfo>
    <!--ro, opt, bool, whether the device supports displaying the firmware version information-->true
  </isSupportFirmwareVersionInfo>
  <isSupportExternalWirelessServer>
    <!--ro, opt, bool, whether the device supports external wireless servers: true (support)-->true
  </isSupportExternalWirelessServer>
  <isSupportSetupCalibration>
    <!--ro, opt, bool, whether the device supports configuring calibration-->true
  </isSupportSetupCalibration>
  <isSupportGetmutexFuncErrMsg>
    <!--ro, opt, bool, whether the device supports the mutual exclusion information-->true
  </isSupportGetmutexFuncErrMsg>
  <isSupportTokenAuthenticate>
    <!--ro, opt, bool, whether the device supports token authentication-->true
  </isSupportTokenAuthenticate>
  <isSupportStreamDualVCA>
    <!--ro, opt, bool, whether the device supports overlaying intelligent private data on the stream, desc:related URI: in the
/ISAPI/Intelligent/channels/<ID>/intelliResource; corresponding node: <isOverlapIntelli>-->true
  </isSupportStreamDualVCA>
  <isSupportLaserSpotManual>
    <!--ro, opt, bool, whether the device supports configuring laser light spot-->true
  </isSupportLaserSpotManual>
  <isSupportRTMP>
    <!--ro, opt, bool, whether the device supports RTMP configuration, desc:related URI: /ISAPI/Streaming/channels/<ID>/RTMPCfg/capabilities-->true
  </isSupportRTMP>
  <isSupportTraffic>
    <!--ro, opt, bool, traffic service capability-->true
  </isSupportTraffic>
  <isSupportLaserSpotAdjustment>
    <!--ro, opt, bool, adjust the size of laser spot, desc:related URI: /ISAPI/Image/channels/<ID>/laserLight/laserSpotAdjustment/capabilities-->true
  </isSupportLaserSpotAdjustment>
  <VideoIntercomCap>
    <!--ro, opt, object, video intercom capability, desc:related URI: /ISAPI/VideoIntercom/capabilities-->
    <isSupportSceneManage>
      <!--ro, opt, bool, scene management (the configuration port needs to be switched according to various scenes)-->true
    </isSupportSceneManage>
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<isSupportDeviceId>
  <!--ro, opt, bool, whether the device supports device ID-->true
</isSupportDeviceId>
<isSupportOperationTime>
  <!--ro, opt, bool, whether the device supports the operation time-->true
</isSupportOperationTime>
<isSupportCallerInfo>
  <!--ro, opt, bool, whether the device supports the information management of the calling persons-->true
</isSupportCallerInfo>
<isSupportCallSignal>
  <!--ro, opt, bool, whether the device supports the call signaling interaction, desc:related URI: /ISAPI/VideoIntercom/callSignal?format=json-->true
</isSupportCallSignal>
<isSupportCallStatus>
  <!--ro, opt, bool, whether the device supports getting call status, desc:related URI: /ISAPI/VideoIntercom/callStatus?format=json-->true
</isSupportCallStatus>
<isSupportCallElevator>
  <!--ro, opt, bool, whether the device supports calling for elevator-->true
</isSupportCallElevator>
<isSupportRelatedDeviceAdress>
  <!--ro, opt, bool, whether the device supports linking device address-->true
</isSupportRelatedDeviceAdress>
<isSupportCardSectorCheck>
  <!--ro, opt, bool, sector encryption verification-->true
</isSupportCardSectorCheck>
<isSupportRemoteOpenDoor>
  <!--ro, opt, bool, whether it supports remote unlocking-->true
</isSupportRemoteOpenDoor>
<isSupportKeyCfg>
  <!--ro, opt, bool, configuration for pressing button to call for resident-->true
</isSupportKeyCfg>
<isSupportWorkModeCfg>
  <!--ro, opt, bool, whether the device supports configuring the working mode, desc:related URI: URL/ISAPI/VideoIntercom/workMode/capabilities-->true
</isSupportWorkModeCfg>
<isSupportSystemSwitchCfg>
  <!--ro, opt, bool, configuring the switch of video intercom system-->true
</isSupportSystemSwitchCfg>
<isSupportPrivilegePasswordStatus>
  <!--ro, opt, bool, getting the status of the device's permission password-->true
</isSupportPrivilegePasswordStatus>
<isSupportSmartHomeRoomsCfg>
  <!--ro, opt, bool, whether the device supports configuring room information-->true
</isSupportSmartHomeRoomsCfg>
<isSupportSmartHomeDevices>
  <!--ro, opt, bool, whether the device supports intelligent devices management-->true
</isSupportSmartHomeDevices>
<isSupportScenesCfg>
  <!--ro, opt, bool, whether the device supports scene configuration (adding, deleting, editing and searching for)-->true
</isSupportScenesCfg>
<isSupportNameModify>
  <!--ro, opt, bool, whether the device supports editing the zone name-->true
</isSupportNameModify>
<isSupportElevatorControlCfg>
  <!--ro, opt, bool, whether the device supports elevator control-->true
</isSupportElevatorControlCfg>
<isSupportPrivilegePassword>
  <!--ro, opt, bool, whether the device supports configuring passwords for different scenes-->true
</isSupportPrivilegePassword>
<isSupportSubModules>
  <!--ro, opt, bool, whether the device supports sub-module management-->true
</isSupportSubModules>
<isSupportIssueCards>
  <!--ro, opt, bool, whether the device supports configuring the card issuing function of the master card-->true
</isSupportIssueCards>
<isSupportSendCardCfg>
  <!--ro, opt, bool, whether the door station supports configuring card issuing function-->true
</isSupportSendCardCfg>
<isSupportRingBackTone>
  <!--ro, opt, bool, whether the device supports importing the ring back tone-->true
</isSupportRingBackTone>
<isSupportCallPriority>
  <!--ro, opt, bool, whether the device supports calling permission-->true
</isSupportCallPriority>
<isSupportPhoneCfg>
  <!--ro, opt, bool, whether the device supports configuring resident phones-->true
</isSupportPhoneCfg>
<isSupportWatchPoint>
  <!--ro, opt, bool, whether the device supports watch point, desc:/ISAPI/VideoIntercom/WatchPoint-->true
</isSupportWatchPoint>
<isSupportVoiceConfiguration>
  <!--ro, opt, bool, whether the device supports the configuration of enabling the voice recognition and prompt-->true
</isSupportVoiceConfiguration>
<isSupportSubModulesCfg>
  <!--ro, opt, bool, whether the device supports configuring editing the sub module parameters-->true
</isSupportSubModulesCfg>
<isSupportDisplayRegion>
  <!--ro, opt, bool, whether the device supports configuring the area to be displayed-->true
</isSupportDisplayRegion>
<isSupportUpdateData>
  <!--ro, opt, bool, whether the device supports importing upgrade files-->true
</isSupportUpdateData>
<isSupportUpdateStrategy>
  <!--ro, opt, bool, whether the device supports upgrade configuration-->true
</isSupportUpdateStrategy>
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    <isSupportMetaInfo>
      <!--ro, opt, bool, whether the device supports getting the information of upgrade files-->true
    </isSupportMetaInfo>
    <isSupportStreamAbility>
      <!--ro, opt, bool, whether the device supports IPC management-->true
    </isSupportStreamAbility>
    <platform opt="R0,R1,S29,N10,N8,N4">
      <!--ro, opt, enum, platform type, subType:string, attr:opt{req, string}, desc:"R0", "R1"; it will be returned if the device supports-->R0
    </platform>
    <isSupportSecurityMode>
      <!--ro, opt, bool, whether the device supports switching the version of security protocols-->true
    </isSupportSecurityMode>
    <isSupportDeviceCommunication>
      <!--ro, opt, bool, whether the device supports configuring the parameters of device group-net communication-->true
    </isSupportDeviceCommunication>
    <isSupportAutoSync>
      <!--ro, opt, bool, whether the device supports automatically synchronous parameters configuration-->true
    </isSupportAutoSync>
    <isSupportDifferentKeystrokes>
      <!--ro, opt, bool, whether the device supports distinguishing long press and short press-->true
    </isSupportDifferentKeystrokes>
    <isSupportKeyCfgWeekPlan>
      <!--ro, opt, bool, the week schedule configuration for pressing to call-->true
    </isSupportKeyCfgWeekPlan>
    <isSupportKeyCfgHolidayPlan>
      <!--ro, opt, bool, the holiday schedule configuration for pressing to call-->true
    </isSupportKeyCfgHolidayPlan>
    <isSupportKeyCfgHolidayGroup>
      <!--ro, opt, bool, the holiday group configuration for pressing to call-->true
    </isSupportKeyCfgHolidayGroup>
    <isSupportKeyCfgPlanTemplate>
      <!--ro, opt, bool, the schedule template configuration for pressing to call-->true
    </isSupportKeyCfgPlanTemplate>
    <isSupportCallCfg>
      <!--ro, opt, bool, device call configuration-->true
    </isSupportCallCfg>
    <isSupportAccessParams>
      <!--ro, opt, bool, the application access parameters-->true
    </isSupportAccessParams>
    <isSupportTransmitRequestEventConfirm>
      <!--ro, opt, bool, response to the transparent transmission request-->true
    </isSupportTransmitRequestEventConfirm>
    <isSupportIncomingCallLinkage>
      <!--ro, opt, bool, whether the device supports configuring incoming calls linkage-->true
    </isSupportIncomingCallLinkage>
    <isSupportAdbControl>
      <!--ro, opt, bool, whether the device supports remotely enabling and disabling ADB-->true
    </isSupportAdbControl>
    <isSupportProtocolFileManage>
      <!--ro, opt, bool, whether the device supports the management of the protocol notice files, desc:/ISAPI/VideoIntercom/protocolFileManage/capabilities?
format=json-->true
    </isSupportProtocolFileManage>
    <isSupportStreamSwitch>
      <!--ro, opt, bool, whether the device supports stream switch, desc:/ISAPI/VideoIntercom/streamSwitch/capabilities?format=json-->true
    </isSupportStreamSwitch>
    <isSupportKeywordAudioRecognition>
      <!--ro, opt, bool, whether the device supports keyword voice recognition, desc:/ISAPI/VideoIntercom/keywordAudioRecognition/capabilities?format=json--
>true
    </isSupportKeywordAudioRecognition>
    <isSupportOneWaySpeaking>
      <!--ro, opt, bool, whether the device supports speaking loud, desc:true (support), this node is not returned (not support)-->true
    </isSupportOneWaySpeaking>
    <isSupportPositionInfo>
      <!--ro, opt, bool, whether the device supports configuring the device location information, desc:/ISAPI/VideoIntercom/positionInfo?format=json-->true
    </isSupportPositionInfo>
    <isSupportElevatorControlDisplay>
      <!--ro, opt, bool, whether the device supports configuring displaying elevator control, desc:/ISAPI/VideoIntercom/elevatorControlDisplay?format=json--
>true
    </isSupportElevatorControlDisplay>
    <isSupportRingManage>
      <!--ro, opt, bool, whether the device supports ring tone management, desc:related URI: /ISAPI/VideoIntercom/ringTone?format=json;
/ISAPI/VideoIntercom/ringTone/<ringID>?format=json-->true
    </isSupportRingManage>
    <isSupportPasswordAuthentication>
      <!--ro, opt, bool, whether the device supports verifying the video intercom password, desc:related URI:
/ISAPI/VideoIntercom/passwordAuthentication/capabilitie-->true
    </isSupportPasswordAuthentication>
    <isSupportAlarmControlByPhone>
      <!--ro, opt, bool, whether the device supports phone arming control, desc:related URI: /ISAPI/SecurityCP/AlarmControlByPhone/capabilities-->true
    </isSupportAlarmControlByPhone>
    <isSupportZoneCfgByScene>
      <!--ro, opt, bool, whether the device supports configuring the zones parameter by scene, desc:related URI:
/ISAPI/SecurityCP/BasicParam/sceneZoneCfg/capabilities-->true
    </isSupportZoneCfgByScene>
    <isSupportAppKeyConfiguration>
      <!--ro, opt, bool, whether the device supports configuring APPKey, desc:related URI: /ISAPI/VideoIntercom/AppKeyConfiguration/capabilities-->true
    </isSupportAppKeyConfiguration>
    <isSupportIntercomProtocolChange>
      <!--ro, opt, bool, whether the device supports configuring the switch of device VoIP protocols, desc:related URI:
/ISAPI/VideoIntercom/IntercomProtocolType/capabilities-->true
    </isSupportIntercomProtocolChange>
    <isSupportGetAnnouncementMessage>
      <!--ro, opt, bool, whether the device supports getting announcement message, desc:related URI: /ISAPI/VideoIntercom/AnnouncementMessage/capabilities--
>true
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    </isSupportGetAnnouncementMessage>
    <isSupportGetDeployInfo>
      <!--ro, opt, bool, whether the device supports getting arming information, desc:related URI: /ISAPI/AccessControl/DeployInfo/capabilities-->true
    </isSupportGetDeployInfo>
    <detectorType opt="panicButton,magneticContact,smokeDetector,activeInfraredDetector,passiveInfraredDetector,combustibleGasDetector,doorbell">
      <!--ro, opt, string, detector types supported by the device, attr:opt{req, string}, desc:"panicButton" (panic button), "magneticContact" (magnetic
contact), "smokeDetector" (smoke detector), "activeInfraredDetector" (active infrared detecor), "passiveInfraredDetector" (passice infrared detector),
"combustibleGasDetector" (combustible gas detector)-->test
    </detectorType>
    <isSupportBroadcastRealtime>
      <!--ro, opt, bool, whether the device supports real-time audio broadcast-->true
    </isSupportBroadcastRealtime>
    <isSupportOrganizationNodes>
      <!--ro, opt, bool, whether the device supports organization architecture management-->true
    </isSupportOrganizationNodes>
    <isSupportAudioPlayStatus>
      <!--ro, opt, bool, whether the device supports uploading the audio playing status-->true
    </isSupportAudioPlayStatus>
    <isSupportDeviceIdAnalysisRuleCfg>
      <!--ro, opt, bool, whether the device supports configuring the analysis rule of the device No. (related URI:
/ISAPI/VideoIntercom/AnalysisRuleCfg/capabilities?format=json)-->true
    </isSupportDeviceIdAnalysisRuleCfg>
    <isSupportOperationTimeByChannel>
      <!--ro, opt, bool, whether the device supports configuring the operation time by channel-->true
    </isSupportOperationTimeByChannel>
    <isSupportCallAuthorization>
      <!--ro, opt, bool, whether the device supports configuring permission for incoming calls, desc:related URI:
/ISAPI/VideoIntercom/CallAuthorization/capbilities?format=json-->true
    </isSupportCallAuthorization>
    <isSupportRecordControl>
      <!--ro, opt, bool, whether the device supports audio/video control, desc:related URI: /ISAPI/VideoIntercom/recordControl/capabilities?format=json--
>true
    </isSupportRecordControl>
    <isSupportTrusteeship>
      <!--ro, opt, bool, whether the device supports configuring trusteeship, desc:related URI: /ISAPI/VideoIntercom/trusteeship/capabilities?format=json--
>true
    </isSupportTrusteeship>
    <isSupportAnalogIndoorStation>
      <!--ro, opt, bool, whether the device supports the management of the analog indoor station, desc:related URI:
/ISAPI/VideoIntercom/analogIndoorStation/capabilities?format=json-->true
    </isSupportAnalogIndoorStation>
    <isSupportImportPartitions>
      <!--ro, opt, bool, whether the device supports importing the terminal list, desc:(partition list for the paging microphone devices distributed by the
platform) (Related URI: /ISAPI/VideoIntercom/ImportTerminals/capabilities?format=json)-->true
    </isSupportImportPartitions>
    <isSupportBroadcastPlan>
      <!--ro, opt, bool, whether the device supports full application of schedule broadcast, desc:related URI:
ISAPI/VideoIntercom/broadcast/plan/capabilities?format=json-->true
    </isSupportBroadcastPlan>
    <isSupportBroadcastPlanSearch>
      <!--ro, opt, bool, whether the device supports searching for full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/planSearch?
format=json-->true
    </isSupportBroadcastPlanSearch>
    <isSupportBroadcastPlanPause>
      <!--ro, opt, bool, whether the device supports pausing full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/pause?
format=json-->true
    </isSupportBroadcastPlanPause>
    <isSupportBroadcastPlanResume>
      <!--ro, opt, bool, whether the device supports restoring full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/resume?
format=json-->true
    </isSupportBroadcastPlanResume>
    <isSupportBroadcastPlanTest>
      <!--ro, opt, bool, whether the device supports the test run of full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/test?
format=json-->true
    </isSupportBroadcastPlanTest>
    <isSupportBroadcastPlanEx>
      <!--ro, opt, bool, whether the device supports incremental application of schedule broadcast, desc:related URI:
/ISAPI/VideoIntercom/broadcast/planEx/capabilities?format=json-->true
    </isSupportBroadcastPlanEx>
    <isSupportBroadcastPlanExSearch>
      <!--ro, opt, bool, whether the device supports searching for incremental schedule broadcast, desc:related URI:
/ISAPI/VideoIntercom/broadcast/planExSearch?format=json-->true
    </isSupportBroadcastPlanExSearch>
    <isSupportBroadcastPlanModify>
      <!--ro, opt, bool, whether the device supports editing a single schedule broadcast, desc:it is valid only during incremental application-->true
    </isSupportBroadcastPlanModify>
    <isSupportBroadcastPlanDelete>
      <!--ro, opt, bool, whether the device supports deleting a single schedule broadcast, desc:it is valid only during incremental application-->true
    </isSupportBroadcastPlanDelete>
    <isSupportAudioInPlayCfg>
      <!--ro, opt, bool, whether the device supports the configuration of playing the audio input sources of the broadcast, desc:URI for getting the
capability: /ISAPI/VideoIntercom/broadcast/AudioInPlayCfg/capabilities?format=json-->true
    </isSupportAudioInPlayCfg>
    <isSupportSearchWardBedInfo>
      <!--ro, opt, bool, whether the device supports searching for ward bed information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/searchWardBedInfo/capabilities?format=json. It is used for the main station to search for whether the bed connected to the sub machine
in the ward is already occupied-->true
    </isSupportSearchWardBedInfo>
    <isSupportPatient>
      <!--ro, opt, bool, whether the device supports applying patient information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/medical/patient/capabilities?format=json-->true
    </isSupportPatient>
    <isSupportIntroduction>
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      <!--ro, opt, bool, whether the device supports applying the medical introduction, desc:URI for getting the capability:
/ISAPI/VideoIntercom/medical/introduction/capabilities?format=json-->true
    </isSupportIntroduction>
    <isSupportConstantVoltage>
      <!--ro, opt, bool, whether the device supports constant voltage broadcast, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/ConstantVoltage/capabilities?format=json-->true
    </isSupportConstantVoltage>
    <isSupportRecordMode>
      <!--ro, opt, bool, whether it supports configuring recording mode, desc:URI for getting the capability: /ISAPI/VideoIntercom/RecordMode/capabilities?
format=json-->true
    </isSupportRecordMode>
    <isSupportCallForwarding>
      <!--ro, opt, bool, whether the device supports active call transfer, desc:URI for getting the capability:
/ISAPI/VideoIntercom/callForwarding/capabilities?format=json-->true
    </isSupportCallForwarding>
    <isSupportMediaLibrary>
      <!--ro, opt, bool, whether the device supports applying media library information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/MediaLibrary/capabilities?format=json-->true
    </isSupportMediaLibrary>
    <isSupportBroadcastControl>
      <!--ro, opt, bool, whether the device supports configuring broadcast control parameters, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/control/capabilities?format=json-->true
    </isSupportBroadcastControl>
    <isSupportGetImportPartitions>
      <!--ro, opt, bool, whether the device supports getting partition list, desc:POST /ISAPI/VideoIntercom/broadcast/GetImportPartitions?format=json-->true
    </isSupportGetImportPartitions>
    <isSupportSearchMediaLibrary>
      <!--ro, opt, bool, whether the device supports searching media library, desc:POST /ISAPI/VideoIntercom/broadcast/SearchMediaLibrary?format=json-->true
    </isSupportSearchMediaLibrary>
    <isSupportImportPartitionsBatch>
      <!--ro, opt, bool, whether the device supports batch importing of partitions, desc:PUT /ISAPI/VideoIntercom/broadcast/ImportPartitionsBatch?
format=json-->true
    </isSupportImportPartitionsBatch>
    <isSupportMediaLibraryBatch>
      <!--ro, opt, bool, whether the device supports batch applying of media library information, desc:PUT /ISAPI/VideoIntercom/broadcast/MediaLibraryBatch?
format=json-->true
    </isSupportMediaLibraryBatch>
    <isSupportPlanScheme>
      <!--ro, opt, bool, whether the device supports broadcast schedules, desc:GET /ISAPI/VideoIntercom/broadcast/PlanScheme/capabilities?format=json-->true
    </isSupportPlanScheme>
    <isSupportRealTimeInsertion>
      <!--ro, opt, bool, whether the device supports real-time cut-in, desc:POST /ISAPI/VideoIntercom/broadcast/RealTimeInsertion?format=json-->true
    </isSupportRealTimeInsertion>
  </VideoIntercomCap>
  <SecurityCPCapabilities>
    <!--ro, opt, object, [TBC], desc:related URI: /ISAPI/SecurityCP/capabilities?format=json-->true
  </SecurityCPCapabilities>
  <isSupportSafetyCabin>
    <!--ro, opt, bool, safety cabinet controller, desc:related URI: /ISAPI/SafetyCabin/capabilities-->true
  </isSupportSafetyCabin>
  <isSupportAcsUpdate>
    <!--ro, opt, bool, whether the device supports peripheral module upgrade (true, support), desc:related URI: /ISAPI/System/AcsUpdate/capabilities-->true
  </isSupportAcsUpdate>
  <isSupportSecurityCP>
    <!--ro, opt, bool, N/A-->true
  </isSupportSecurityCP>
  <isSupportPEA>
    <!--ro, opt, bool, the host capability of one-push alarm-->true
  </isSupportPEA>
  <isSupportCurrentLock>
    <!--ro, opt, bool, whether the device supports the configuration of locking current-->true
  </isSupportCurrentLock>
  <isSupportGuardAgainstTheft>
    <!--ro, opt, bool, whether the device supports configuring device guard against theft, desc:related URI: /ISAPI/System/guardAgainstTheft/capabilities--
>true
  </isSupportGuardAgainstTheft>
  <isSupportPicInfoOverlap>
    <!--ro, opt, bool, whether the device supports overlaying pictures and texts-->true
  </isSupportPicInfoOverlap>
  <isSupportPlay>
    <!--ro, opt, bool, shielded live view-->true
  </isSupportPlay>
  <isSupportPlayback>
    <!--ro, opt, bool, shielded playback-->true
  </isSupportPlayback>
  <UHFRFIDReader>
    <!--ro, opt, object, capability supported by the RFID reader-->
    <isSupportBasicInformation>
      <!--ro, opt, bool, whether the device supports the basic parameters of the reader-->true
    </isSupportBasicInformation>
    <isSupportHardDiskStorageTest>
      <!--ro, opt, bool, whether the device supports disk storage performance test of the reader-->true
    </isSupportHardDiskStorageTest>
    <isSupportCarReaderConfig>
      <!--ro, opt, bool, whether the device supports configuring the parameters of the electronic license plate reader, desc:related URI:
/ISAPI/Traffic/UHFRFIDReader/carReaderConfig/capabilities?format=json-->true
    </isSupportCarReaderConfig>
  </UHFRFIDReader>
  <isSupportMasterSlaveConfig>
    <!--ro, opt, bool, whether the device supports configuring main and sub device, desc:related URI: /ISAPI/System/masterSlaveConfig?format=json-->true
  </isSupportMasterSlaveConfig>
  <isSupportIntelligentStructureAnalysis>
    <!--ro, opt, bool, whether the device supports intelligent structuralization analysis, desc:related URI:
/ISAPI/Intelligent/structureAnalysis/capabilities-->true
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/ISAPI/Intelligent/structureAnalysis/capabilities-->true
  </isSupportIntelligentStructureAnalysis>
  <isSupportIntelligentAnalysisEngines>
    <!--ro, opt, bool, whether the device supports configuring the intelligent engine, desc:related URI: /ISAPI/Intelligent/analysisEngines/capabilities--
>true
  </isSupportIntelligentAnalysisEngines>
  <PreviewDisplayNum>
    <!--ro, opt, int, number of the browser live view window-->1
  </PreviewDisplayNum>
  <isSupportBoard opt="true,false">
    <!--ro, opt, bool, whether the device supports related sub-board protocols, attr:opt{opt, string}-->true
  </isSupportBoard>
  <ResourceSwitch>
    <!--ro, opt, object, configuration of resource conversion-->
    <workMode opt="4KPreview,educationRecord">
      <!--ro, req, string, the working mode: education record mode, attr:opt{opt, string}-->test
    </workMode>
  </ResourceSwitch>
  <isSupportCustomStream>
    <!--ro, opt, bool, whether the device supports custom stream, desc:related URI: /ISAPI/Streaming/channels/<ID>/customStream/capabilities-->true
  </isSupportCustomStream>
  <isSupportTriggerCapCheck>
    <!--ro, opt, bool, it indicates that the capability of the URL (/ISAPI/Event/triggersCap) is already verified-->true
  </isSupportTriggerCapCheck>
  <isSupportChannelEventCap>
    <!--ro, opt, bool, whether the device supports getting the event capability by channel, desc:related URI: /ISAPI/Event/channels/<ID>/capabilities-->true
  </isSupportChannelEventCap>
  <SensorCap>
    <!--ro, opt, object, capability of configuring sensor parameters, desc:related URI: /ISAPI/System/Sensor/capabilities-->
    <SensorNums>
      <!--ro, opt, int, the maximum number of sensors that can be added-->1
    </SensorNums>
    <isSupportSensorPortConfiguration>
      <!--ro, opt, bool, supports configuring the detection parameters-->true
    </isSupportSensorPortConfiguration>
    <isSupportSensorPortControl>
      <!--ro, opt, bool, supports detection control-->true
    </isSupportSensorPortControl>
    <isSupportAcquireSensorStatus>
      <!--ro, opt, bool, supports getting the real-time status of the sensor-->true
    </isSupportAcquireSensorStatus>
    <isSupportSearchSensorData>
      <!--ro, opt, bool, supports searching the sensor data-->true
    </isSupportSearchSensorData>
  </SensorCap>
  <isSupportEncryption>
    <!--ro, opt, bool, stream encryption capability-->true
  </isSupportEncryption>
  <isSupportTrial>
    <!--ro, opt, bool, whether the device supports configuring inquest device-->true
  </isSupportTrial>
  <isSupportSignInstrument>
    <!--ro, opt, bool, the capability of physical sign monitor-->true
  </isSupportSignInstrument>
  <isSupportLocate>
    <!--ro, opt, bool, whether the device supports location configuration-->true
  </isSupportLocate>
  <isSupportVibration>
    <!--ro, opt, bool, whether the device supports configuring vibration parameters-->true
  </isSupportVibration>
  <isSupportMixedTargetDetection>
    <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
  </isSupportMixedTargetDetection>
  <SDCardCtrlCap>
    <!--ro, opt, object, the remote control capability of the memory card-->
    <isSupportIOCtrl>
      <!--ro, opt, bool, whether the device supports the capability of control and command upgrade-->true
    </isSupportIOCtrl>
    <isSupportReadCardStatus>
      <!--ro, opt, bool, whether the device supports the capability of card recognition operation-->true
    </isSupportReadCardStatus>
    <isSupportStartUpdate>
      <!--ro, opt, bool, whether the device supports the capability of upgrade start control-->true
    </isSupportStartUpdate>
    <isSupportEndUpdate>
      <!--ro, opt, bool, whether the device supports the capability of upgrade end control-->true
    </isSupportEndUpdate>
    <isSupportPowerCtrl>
      <!--ro, opt, bool, whether the device supports the capability of power on and off control-->true
    </isSupportPowerCtrl>
  </SDCardCtrlCap>
  <isSupportMutiChannelCalibration>
    <!--ro, opt, bool, whether the device supports multi-channel calibration-->true
  </isSupportMutiChannelCalibration>
  <isSupportElevatorFloorCalibration>
    <!--ro, opt, bool, whether the device supports configuring floor display-->true
  </isSupportElevatorFloorCalibration>
  <isSupportPictureCaptureComparision>
    <!--ro, opt, bool, whether the device supports 1V1 comparison of the pictures in the library and from camera-->true
  </isSupportPictureCaptureComparision>
  <isSupportGPSCalibratation>
    <!--ro, opt, bool, whether the device supports the GPS calibration function-->true
  </isSupportGPSCalibratation>
  <isSupportCityManagement>
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    <!--ro, opt, bool, whether the device supports city management detection event-->true
  </isSupportCityManagement>
  <isSupportBVCorrect>
    <!--ro, opt, bool, whether the device supports configuring the parameters of camera correction-->true
  </isSupportBVCorrect>
  <isSupportSteamingCodingModeSwitch>
    <!--ro, opt, bool, encoding output mode (true-support), desc:related URI: /ISAPI/System/streamingCoding/mode/capabilities?format=json-->true
  </isSupportSteamingCodingModeSwitch>
  <supportImageChannel opt="1,2,3,4,5,6">
    <!--ro, opt, string, whether the device supports configuring the image channel, attr:opt{req, string},
desc:the configuration of actual image channel supported by the device (URI for picture parameters: /ISAPI/Image/channels/<channelID>), if this node is not
returned, the image channel is the same with the encoding channel (/ISAPI/Streaming/channels) by default
    this node is mainly for the situation when the image channel is inconsistent with the actual stream channel. For example, some device, which altogether
has four sensors and they are all spliced into one encoding channel for output. Every sensor supports configuring image parameters; by default, the image
channel 1 is only valid for the images of sensor1-->test
  </supportImageChannel>
  <isSupportPanoramaCalibration>
    <!--ro, opt, bool, whether the device supports configuring the calibration of the panoramic camera-->true
  </isSupportPanoramaCalibration>
  <isSupportPanoramaMode>
    <!--ro, opt, bool, whether the device supports the mode switch of the panoramic camera-->true
  </isSupportPanoramaMode>
  <supportSnapshotChannel opt="1">
    <!--ro, opt, string, the capability supported by the capture schedule channel, attr:opt{opt, string}-->test
  </supportSnapshotChannel>
  <isSupportPanoramaCorrections>
    <!--ro, opt, bool, whether the device supports configuring the correction algorithm parameters-->true
  </isSupportPanoramaCorrections>
  <isSupportActiveMulticast>
    <!--ro, opt, bool, whether the device supports multicast configuration (/ISAPI/Streaming/channels/<ID>/capabilities)-->true
  </isSupportActiveMulticast>
  <isSupportChannelEventListCap>
    <!--ro, opt, bool, it indicates that whether the device supports getting the event capability of all channels of the device-->true
  </isSupportChannelEventListCap>
  <VCAResourceChannelsCap>
    <!--ro, opt, array, it indicates that the device supports the independent switch of intelligent resources by channel, subType:object-->
    <ChannelsList>
      <!--ro, opt, object, channel list-->
      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
    </ChannelsList>
  </VCAResourceChannelsCap>
  <isSupportTimerVCAResource>
    <!--ro, opt, bool, whether the device supports scheduled switch of intelligent resources-->true
  </isSupportTimerVCAResource>
  <MixedTargetDetection>
    <!--ro, opt, object, the capability of recognizing specified target among mixed targets supported by the device-->
    <isSupportFaceRecognition>
      <!--ro, opt, bool, whether the device supports facial recognition-->true
    </isSupportFaceRecognition>
    <isSupportHumanRecognition>
      <!--ro, opt, bool, whether the device supports human body recognition-->true
    </isSupportHumanRecognition>
    <isSupportVehicleRecognition>
      <!--ro, opt, bool, whether the device supports vehicle recognition-->true
    </isSupportVehicleRecognition>
  </MixedTargetDetection>
  <isSupportDiscoveryMode>
    <!--ro, opt, bool, whether the device supports discovery mode, desc:related URI: /ISAPI/System/discoveryMode/capabilities-->true
  </isSupportDiscoveryMode>
  <isSupportDumpData>
    <!--ro, opt, bool, whether the device supports exporting the Dump file-->true
  </isSupportDumpData>
  <isSupportCameraAccessConfiguration>
    <!--ro, opt, bool, whether the device supports the camera access configuration, desc:related URI: /ISAPI/System/cameraAccess/configuration/capabilities?
format=json-->true
  </isSupportCameraAccessConfiguration>
  <isSupportIntelligentSearch>
    <!--ro, opt, bool, whether the device supports intelligent search, desc:related URI: /ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json--
>true
  </isSupportIntelligentSearch>
  <isSupportAutoMaintenance>
    <!--ro, opt, bool, whether the device supports automatic maintenance, desc:related URI: /ISAPI/System/autoMaintenance/capabilities?format=json-->true
  </isSupportAutoMaintenance>
  <isSupportRtspOverHTTPS>
    <!--ro, opt, bool, supports tspOverHTTPS-->true
  </isSupportRtspOverHTTPS>
  <isSupportIntersectionAnalysis>
    <!--ro, opt, bool, whether the device supports the intersection analysis, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/intersectionAnalysis/capabilities?format=json-->true
  </isSupportIntersectionAnalysis>
  <isSupportAudioInteract>
    <!--ro, opt, bool, whether the device supports audio interaction, desc:related URI: /ISAPI/Streaming/interact/capabilities?format=json-->true
  </isSupportAudioInteract>
  <isSupportChannelFullEventListCap>
    <!--ro, opt, bool, it indicates that whether the device supports getting the event capability of all channels of the device-->true
  </isSupportChannelFullEventListCap>
  <isSupportAUXInfoCap>
    <!--ro, opt, bool, it indicates that whether the device supports getting the capability of getting the attributes of device's all channels-->true
  </isSupportAUXInfoCap>
  <supportPanoramaChannel opt="2">
    <!--ro, opt, string, the panoramic channels supported by the device, attr:opt{opt, string}-->test
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  </supportPanoramaChannel>
  <regionClip>
    <!--ro, opt, object, the channels and streams that supports area cropping-->
    <channels opt="101">
      <!--ro, opt, string, the channels and streams that supports area cropping, attr:opt{opt, string}, desc:101: channel No.*100+stream type-->test
    </channels>
  </regionClip>
  <AIDEventSupport
opt="abandonedObject,pedestrian,congestion,roadBlock,construction,trafficAccident,fogDetection,wrongDirection,illegalParking,SSharpDriving,lowSpeed,dragRacing,obstacle,vehNoYieldPedest,illegalMannedVeh,illegalMannedNonMotorVeh,umbrellaTentInstall,nonMotorVehOnVehLane,wearingNoHelmet,pedestRedLightRunning,pedestOnNonMotorVehLane,pedestOnVehLane,conflagration,smoke,VehicleSpeedDrop,lowVisibility,animalsOccupy,targetSpeedDetection,emergencyLane,gasser,changeLaneContinuously,RegionalVehicleStatistics">
    <!--ro, opt, string, supported AID event, attr:opt{opt, string}, desc:supported AID event-->test
  </AIDEventSupport>
  <TFSEventSupport opt="illegalParking
,wrongDirection,crossLane,laneChange,vehicleExist,turnRound,parallelParking,notKeepDistance,notSlowZebraCrossing,overtakeRightSide,lowSpeed,dragRacing,changeLaneContinuously,SSharpDriving,largeVehicleOccupyLine,jamCrossLine,obstacle,blackSmokeVehicle,turnRightStop,occupyDedicatedLane,notDriveInDedicatedLane,TFS_nonZipperMerge,overSpeed,emergencyLane,gasser">
    <!--ro, opt, string, supported TFS, attr:opt{opt, string}-->test
  </TFSEventSupport>
  <isVehicleStatisticsSupport>
    <!--ro, opt, bool, whether the device supports configuring the parameters for traffic data statistics-->true
  </isVehicleStatisticsSupport>
  <isSupportElevatorFloorName>
    <!--ro, opt, bool, whether the device supports configuring floor custom name-->true
  </isSupportElevatorFloorName>
  <isSupportVehicleFaceRecognition>
    <!--ro, opt, bool, whether the device supports checkpoint vehicle recognition, desc:background: the traffic devices support "facial recognition" (driver), however, the corresponding node is not added to the device to notify the integration of this capability-->true
  </isSupportVehicleFaceRecognition>
  <VQDEventSupport opt="luma,chroma,snow,streak,freeze,sigLose,clarity,jitter,block,flowers,noise,ghost,purple,ICR,protectiveFilm,blur,contrast">
    <!--ro, opt, string, VQD event list, attr:opt{opt, string}, desc:"Luma" (brightness exception); "Chroma" (image color cast); "Snow" (snow noise); "Streak" (strip noise); "Freeze" (frozen screen); "SigLose" (signal loss); "Clarity" (resolution exception); "JItter" (image flutter); "Block" (video tampering); "Flowers" (blurred screen); "Noise" (image noise); "Ghost" (ghost image); "Purple" (purple fringing); "ICR" (ICP); "protectiveFilm" (protective film unremoved)-->test
  </VQDEventSupport>
  <IOTCap>
    <!--ro, opt, object, supports searching for the device status list by page, desc:not returned (not support, get the device status list via GET)-->
    <supportChannelNum opt="5000">
      <!--ro, opt, int, the number of supported IOT device channels, attr:opt{opt, string}-->1
    </supportChannelNum>
    <startChannelNo>
      <!--ro, opt, int, the channel No. of start IOT device-->1
    </startChannelNo>
    <isSupportlinkageChannelsSearch>
      <!--ro, opt, bool, true (support), desc:related URI: (POST /ISAPI/System/IOT/linkageChannels?format=json)-->true
    </isSupportlinkageChannelsSearch>
    <isSupportIOTChannelCap>
      <!--ro, opt, bool, whether the device supports the channel configuration capability, desc:it returns true for support, related URI: (GET /ISAPI/System/IOT/channels/<IoTChannelID>/capabilities?format=json)-->true
    </isSupportIOTChannelCap>
    <satusSearchTaskCap>
      <!--ro, opt, object, task search capability-->
      <maxResults>
        <!--ro, opt, int, the maximum number of returned results-->1
      </maxResults>
      <filterType opt="all,online,offline,updateFailed,notPartitioned">
        <!--ro, opt, enum, filtering type, subType:string, attr:opt{req, string}, desc:all, online, offline, updateFailed (upgrading failed)-->all
      </filterType>
      <deviceStatus opt="calling,noCall,medicationChangePrompt,alarming,offline,free,callWaiting">
        <!--ro, opt, string, the value range for the device status search condition, range:[0,32], attr:opt{req, string}, desc:"calling" (calling), "noCall" (non-calling status), "medicationChangePrompt", "alarming", "offline", "free"; if multiple conditions are specified, their relationship is "and" by default-->calling
      </deviceStatus>
      <logicRelationship opt="and,or">
        <!--ro, opt, string, the value range for the logic relationship of device status search condition, range:[0,16], attr:opt{req, string}, desc:and- the relationship between them is "and" (by default), or- the relationship between them is "or"; it is valid when deviceStatus exists-->and
      </logicRelationship>
      <alarmType opt="SOS,toilet">
        <!--ro, opt, string, value range of alarm type, range:[0,16], attr:opt{req, string}, desc:SOS-SOS call, toilet-toilet alarm. It is valid when the deviceStatus contains Alarming-->test
      </alarmType>
      <callWaitingType opt="normal,cisit">
        <!--ro, opt, string, alarm type, attr:opt{req, string}, desc:normal-normal call waiting, visit-visit call waiting. It is valid when the deviceStatus contains callWaiting-->test
      </callWaitingType>
    </satusSearchTaskCap>
    <channelConfigList>
      <!--ro, opt, array, capability list returned according to unitType, subType:object-->
      <channelConfig>
        <!--ro, opt, object, device type capability-->
        <unitType>
          <!--ro, opt, enum, unit type, subType:string, desc:"indoor" (indoor station), "villa" (villa door station), "confirm" (doorphone), "outdoor" (door station), "fence" (outer door station), "doorbell" (doorbell), "manage" (main station), "acs" (access control devices), "wardStation", "bedheadExtension", "bedsideExtension", "terminal", "netAudio" (network sound box), "analogIndoor" (analog indoor station), "decoder" (decoder), "networkFireAlarmMatrix", "amplifier" (network amplifier), "pagingMicrophone" (SIP phone)-->outdoor
        </unitType>
        <floorNumber min="-10" max="999">
          <!--ro, opt, int, floor No., attr:min{req, int},max{req, int}-->0
        </floorNumber>
        <roomNumber min="0" max="999">
          <!--ro, opt, int, room No., attr:min{req, int},max{req, int}-->0
        </roomNumber>
        <isSupportSlaveDevice>
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          <!--ro, opt, bool, whether the device supports sub-devices-->true
        </isSupportSlaveDevice>
        <isSupportZeroFloorNumber>
          <!--ro, opt, bool, whether the zero floor is supported, desc:it is used together with node floorNumber-->true
        </isSupportZeroFloorNumber>
        <periodNumber min="1" max="999">
          <!--ro, opt, int, community number, attr:min{req, int},max{req, int}-->0
        </periodNumber>
        <buildingNumber min="1" max="999">
          <!--ro, opt, int, building number, attr:min{req, int},max{req, int}-->0
        </buildingNumber>
        <unitNumber min="0" max="999">
          <!--ro, opt, int, unit number, attr:min{req, int},max{req, int}-->0
        </unitNumber>
        <deviceIndex min="1" max="32">
          <!--ro, opt, string, device No. length range, attr:min{req, int},max{req, int}-->test
        </deviceIndex>
        <industryType opt="builidings,prison,medicalTreatment,broadcasting,general">
          <!--ro, opt, string, industry (scene) type, range:[0,32], attr:opt{req, string}, desc:"builidings" (building), "prison", "medicalTreatment"
(medical), "broadcasting", "general"-->test
        </industryType>
        <alarmEnabled opt="true,false">
          <!--ro, opt, bool, whether to enable the arming function of IoT devices, attr:opt{req, string}, desc:this field is used to check if arming is
supported when adding devices. (Related field: alarmEnabled; related URI: /ISAPI/System/IOT/channels?format=json)-->true
        </alarmEnabled>
        <priority min="1" max="15">
          <!--ro, opt, int, calling priority, attr:min{req, int},max{req, int}-->0
        </priority>
        <transportType opt="unicast,multicast">
          <!--ro, opt, string, broadcasting method, attr:opt{req, string}, desc:unicast, multicast-->test
        </transportType>
        <audioVolume min="0" max="15">
          <!--ro, opt, int, volume, attr:min{req, int},max{req, int}-->0
        </audioVolume>
        <callForwardingEnabled>
          <!--ro, opt, bool, whether to enable call transfer, desc:true (enable), false (disable). The default value is false-->true
        </callForwardingEnabled>
        <callForwardingType opt="unconditional,busy,noReply,noReachable,calling">
          <!--ro, opt, string, call transfer type, range:[0,32], attr:opt{req, string}, desc:"unconditional" (call transfer under any condition), "busy"
(call transfer when busy), "noReply" (call transfer when there is no reply), "noReachable" (call transfer when it is unreachable), "calling" (call transfer
when calling)-->test
        </callForwardingType>
        <isSupportCallTerminalNumbe>
          <!--ro, opt, bool, whether the device supports configuring the target master station No. for call forwarding-->true
        </isSupportCallTerminalNumbe>
        <customCommunicationName min="0" max="128">
          <!--ro, opt, string, custom contact name, range:[0,128], attr:min{req, int},max{req, int}, desc:when this node is configured, the custom contact
name will be displayed during device calling prior to the terminal number-->test
        </customCommunicationName>
        <isSupportMoreFeatures>
          <!--ro, opt, bool, whether the device supports more functions, desc:it is not supported by default if this node is not returned-->true
        </isSupportMoreFeatures>
        <isNotSupportRegPassword>
          <!--ro, opt, bool, whether the password registration configuration is not supported, desc:it is supported if this node is not returned or false is
returned-->true
        </isNotSupportRegPassword>
        <mappingRoomNumber min="0" max="5">
          <!--ro, opt, int, mapping room No., attr:min{req, int},max{req, int}, desc:the mapping room No. is unique and can be called without exposing the
actual room No., for privacy requirement for overseas users-->0
        </mappingRoomNumber>
        <deviceID min="0" max="64">
          <!--ro, opt, string, device ID, attr:min{req, int},max{req, int}, desc:supported for ISUP protocol access-->test
        </deviceID>
        <ISUPkey min="1" max="32">
          <!--ro, opt, string, ISUP Key, attr:min{req, int},max{req, int}, desc:supported for ISUP protocol access-->test
        </ISUPkey>
        <dialUpAddress min="1" max="1024">
          <!--ro, opt, string, dial address, attr:min{req, int},max{req, int}, desc:this node is required when <unitType> is "analogIndoor"-->test
        </dialUpAddress>
      </channelConfig>
    </channelConfigList>
    <isSupportConfigTemplate>
      <!--ro, opt, bool, whether the device supports exporting the device list template, desc:related URL: GET /ISAPI/System/IOT/channelConfig/template?
format=json-->true
    </isSupportConfigTemplate>
    <isSupportConfigReport>
      <!--ro, opt, bool, whether the device supports getting the importing details, desc:related URL: GET /ISAPI/System/IOT/channelConfig/report?
format=json-->true
    </isSupportConfigReport>
    <unitType
opt="indoor,villa,confirm,outdoor,fence,doorbell,manage,acs,interactive,wardStation,bedheadExtension,bedsideExtension,analogIndoor,decoder,networkFireAlarmM
atrix,amplifier,pagingMicrophone,PDA,netSwitch,entrywayStation,alarmManage,alarmTerminal">
      <!--ro, opt, string, device type, range:[0,32], attr:opt{req, string}, desc:terminal, netAudio (network sound box)-->test
    </unitType>
    <serialNumber min="1" max="32">
      <!--ro, opt, string, the value range of device serial No. length, range:[0,32], attr:min{req, int},max{req, int}-->test
    </serialNumber>
    <isSupportAsyncStatus>
      <!--ro, opt, bool, whether the device supports asynchronously getting the status of non-video channels, desc:URI for device update: PUT
/ISAPI/System/IOT/channels/updateStatusTask?format=json; URI for getting the update progress: GET /ISAPI/System/IOT/channels/updateStatusProgress?
format=json-->true
    </isSupportAsyncStatus>
    <isSupportAddIOTChannelList>
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      <!--ro, opt, bool, whether the device supports batch adding channels of IoT devices, desc:URI for batch adding channels of IoT devices: POST
/ISAPI/System/IOT/channels/AddIOTChannelList?format=json-->true
    </isSupportAddIOTChannelList>
    <isSupportSearchMoreIOTSource>
      <!--ro, opt, bool, whether the device supports searching for the IoT devices which can be added (by multiple conditions), desc:related URI: POST
/ISAPI/System/IOT/channels/AddIOTChannelList?format=json-->true
    </isSupportSearchMoreIOTSource>
    <channelConfigReportMode>
      <!--ro, opt, enum, mode of importing the list of IoT devices, subType:string, desc:different devices (main station and door station) process the
imported device list in a different way, which should be returned to the users for prompt. URL for importing the device list:
/ISAPI/System/IOT/channelConfig?format=json; cover; add-->cover
    </channelConfigReportMode>
    <isSupportEncryptImportIOTChannelFile>
      <!--ro, opt, bool, whether it supports encrypting the imported non-video channel files, desc:corresponding URI: PUT /ISAPI/System/IOT/channelConfig?
format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned-->true
    </isSupportEncryptImportIOTChannelFile>
    <isSupportEncryptExportIOTChannelFile>
      <!--ro, opt, bool, whether it supports encrypting exported non-video channel files, desc:corresponding URI: GET /ISAPI/System/IOT/channelConfig?
format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned-->true
    </isSupportEncryptExportIOTChannelFile>
    <isSupportExportAbnormalReport>
      <!--ro, opt, bool, whether it supports exporting error reports for non-video channel devices, desc:corresponding URI: GET
/ISAPI/System/IOT/ChannelConfig/ExportAbnormalReport?format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned--
>true
    </isSupportExportAbnormalReport>
    <isSupportSyncDeviceInfoTask>
      <!--ro, opt, bool-->true
    </isSupportSyncDeviceInfoTask>
  </IOTCap>
  <isSupportLCDScreen>
    <!--ro, opt, bool, whether the device supports LCD display screen-->true
  </isSupportLCDScreen>
  <isSupportBluetooth>
    <!--ro, opt, bool, whether the device supports bluetooth, desc:related URI: /ISAPI/System/Bluetooth/capabilities-->true
  </isSupportBluetooth>
  <isSupportNetScreen>
    <!--ro, opt, bool, whether the device supports network screen (Nova screen), desc:related URI: /ISAPI/System/netScreens/capabilities?format=json-->true
  </isSupportNetScreen>
  <isSupportLocation>
    <!--ro, opt, bool, whether the device supports getting the device location information, desc:related URI: /ISAPI/System/location?format=json-->true
  </isSupportLocation>
  <isSupportDebugInfoExport>
    <!--ro, opt, bool, whether the device supports exporting the debugging information, desc:related URI: /ISAPI/System/debugInfoExport/capabilities?
format=json-->true
  </isSupportDebugInfoExport>
  <isSupportIoTGateway>
    <!--ro, opt, bool, whether the device supports IoT gateway functions, desc:related URI: /ISAPI/System/IoTGateway/capabilities?format=json-->true
  </isSupportIoTGateway>
  <isSupportImageCap>
    <!--ro, opt, bool, whether the device supports configuring image parameters-->true
  </isSupportImageCap>
  <isSupportLogToUSB opt="true,false">
    <!--ro, opt, bool, whether the device supports exporting logs to USB, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/exportLogToUSB/capabilities?format=json-->true
  </isSupportLogToUSB>
  <isSupportWiresharkToUSB opt="true,false">
    <!--ro, opt, bool, whether the device supports exporting packet capture information to USB, attr:opt{opt, string}, desc:related URI:
ISAPI/System/wiresharkToUSB/capabilities?format=json-->true
  </isSupportWiresharkToUSB>
  <CDBurningStreamEncryptionChannels opt="1,2">
    <!--ro, opt, string, return the capability of device recording stream encryption channel, attr:opt{opt, string}, desc:related URI:
/ISAPI/Streaming/channels/<ID>/capabilities-->true
  </CDBurningStreamEncryptionChannels>
  <isSupportOCR>
    <!--ro, opt, bool, whether the device supports OCR detection, desc:related URI: /ISAPI/Intelligent/channels/<ID>/OCR/capabilities?format=json-->true
  </isSupportOCR>
  <isSupportInstallationAngleCalibration>
    <!--ro, opt, bool, whether the device supports the capability of installation angle calibration, desc:related URI:
/ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities-->true
  </isSupportInstallationAngleCalibration>
  <isSupportZeroBiasCalibration>
    <!--ro, opt, bool, whether the device supports the zero angle calibration capability node, desc:related URI:
/ISAPI/System/zeroBiasCalibration/channels/<ID>?format=json-->true
  </isSupportZeroBiasCalibration>
  <isSupportCalibrationFile>
    <!--ro, opt, bool, whether the device supports importing the calibration file, desc:related URI: /ISAPI/System/configurationData-->true
  </isSupportCalibrationFile>
  <isSupportDisplayTrajectory>
    <!--ro, opt, bool, whether the device supports target track overlay (playing library)-->true
  </isSupportDisplayTrajectory>
  <maximumSuperPositionTime opt="5,10,20,30">
    <!--ro, opt, string, the maximum overlay time, attr:opt{opt, string}, desc:unit: second; it is only returned when the target track overlay is supported-
->true
  </maximumSuperPositionTime>
  <isSupportUnitConfig>
    <!--ro, opt, bool, whether the device supports the unit configuration, desc:related URI: /ISAPI/System/unitConfig/capabilities?format=json-->true
  </isSupportUnitConfig>
  <isSupportShipsDetectionWithScene>
    <!--ro, opt, bool, whether the device supports ship detection by scene-->true
  </isSupportShipsDetectionWithScene>
  <isSupportPictureServer>
    <!--ro, opt, bool, whether the device supports configuring the picture storage server, desc:related URI: /ISAPI/System/PictureServer/capabilities?
format=json-->true
  </isSupportPictureServer>
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<isSupportEmmcInfo>
<!--ro, opt, bool, the capability of supporting the EMMC configuration, desc:related URI: /ISAPI/System/emmcInfo/capabilities?format=json-->true
</isSupportEmmcInfo>
<isSupportVehicleIllegalType opt="1,2...29">
<!--ro, opt, string, the vehicle violation types supported by the device, attr:opt{opt, string}, desc:the content value is the same as the index value,
refer to NET_ITS_PLATE_RESULT.NET_DVR_VEHICLE_INFO.byIllegalType the capture camera is connected to the new platform, and the device supports "traffic
violation alarm"; however, the corresponding node is not added to the device to notify the integration of this capability; that is, the traffic violation
alarm capability lacks details. If later the platform needs to develop capability with details, the current device will be incompatible with the capability;
and detailed capability is hidden in the configuration of all trigger modes, which will influence overall access-->test
</isSupportVehicleIllegalType>
<isSupportRFIDChannels>
<!--ro, opt, bool, whether the device supports getting the RFID channel, desc:related URI: /ISAPI/System/RFIDChannels?format=json-->true
</isSupportRFIDChannels>
<isSupportRfidCollection>
<!--ro, opt, bool, whether the device supports the event of uploading RFID tag information, desc:related URI:
/ISAPI/System/Network/RFIDData/<channelID>/capabilities, eventType: rfidCollection-->true
</isSupportRfidCollection>
<DevInfoSearch opt="tiltAngle">
<!--ro, opt, string, whether the device supports searching for the device information, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/workingstatus/devInfoSearch?format=json-->test
</DevInfoSearch>
<isSupportAIOpenPlatform>
<!--ro, opt, bool, whether the device supports AI open platform device capability, desc:related URI: /ISAPI/Intelligent/AIOpenPlatform/capabilities?
format=json-->true
</isSupportAIOpenPlatform>
<isSupportPictureDownloadError>
<!--ro, opt, bool, whether the device supports uploading the exception that downloading the picture failed, desc:eventType: pictureDownloadError-->true
</isSupportPictureDownloadError>
<isSupportStudentBehavior>
<!--ro, opt, bool, whether the device supports uploading the data of students in-class behavior, desc:eventType: studentBehavior-->true
</isSupportStudentBehavior>
<guideEventSupport opt="linkageCapture">
<!--ro, opt, string, the list of web quick configuration guide, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/guideConfig/<guideEvent>/capabilities?format=json-->test
</guideEventSupport>
<supportRemoteCtrl opt="up,down,left,right,enter,menu,num,power,esc,edit,F1,.prev,rec,play,stop,notSupport">
<!--ro, opt, string, whether the device supports remote control, attr:opt{opt, string}, desc:related URI: /ISAPI/System/remoteCtrl/XXX-->test
</supportRemoteCtrl>
<isSupportSSDFileSystemUpgrade>
<!--ro, opt, bool, whether the device supports updating the SSD file system, desc:related URI: /ISAPI/System/SSDFileSystem/upgrade?format=json-->true
</isSupportSSDFileSystemUpgrade>
<isSupportSSDFileSystemFormat>
<!--ro, opt, bool, whether the device supports formatting SSD, desc:related URI: /ISAPI/System/SSDFileSystem/format?format=json-->true
</isSupportSSDFileSystemFormat>
<FaceCounting>
<!--ro, opt, object, whether the device supports the customer traffic (face)-->
<ChannelsList>
<!--ro, opt, array, channel list, subType:object-->
<channelsID>
<!--ro, req, int, No. of channels supported by the device-->1
</channelsID>
</ChannelsList>
</FaceCounting>
<isSupportCluster>
<!--ro, opt, bool, related URI:, desc:/ISAPI/ContentMgmt/Cluster/capabilities-->true
</isSupportCluster>
<DeviceConfigurationSupport opt="time,network,accessProtocol,recordSchedule,channel,customProtocol">
<!--ro, opt, string, content supported by device parameters configuration, attr:opt{opt, string}, desc:related URI: /ISAPI/System/deviceConfiguration--
>test
</DeviceConfigurationSupport>
<behaviorEventSupport opt="getUp,toiletTarry,playCellphone,advReachHeight">
<!--ro, opt, string, abnormal event detection types, attr:opt{opt, string}-->test
</behaviorEventSupport>
<isSupportBVCorrectPassthrough>
<!--ro, opt, bool, whether the device supports transparent transmission of camera correction data, desc:related URI:
/ISAPI/System/BVCorrect/Passthrough/channels/<ID>/capabilities?format=json-->true
</isSupportBVCorrectPassthrough>
<isSupportChangedCallStatus>
<!--ro, opt, bool-->true
</isSupportChangedCallStatus>
<VehiclePositionControl>
<!--ro, opt, object, whether the device supports vehicle position arming-->
<license min="0" max="16">
<!--ro, req, string, license plate number, attr:min{opt, string},max{opt, string}-->test
</license>
<intervalTime min="0" max="65535" def="1">
<!--ro, req, int, uploading interval, unit:s, attr:min{opt, string},max{opt, string},def{opt, string}-->1
</intervalTime>
</VehiclePositionControl>
<supportGISEvent opt="AID,TPS,ANPR,mixedTargetDetection">
<!--ro, opt, string, the capability of supporting getting the GIS information, attr:opt{opt, string}-->test
</supportGISEvent>
<isSupportGPSPTZCalibratation>
<!--ro, opt, bool, whether the device supports GPS PTZ calibration, desc:related URI: /ISAPI/System/GPSPTZCalibratation/channels/<ID>/points?
format=json-->true
</isSupportGPSPTZCalibratation>
<isSupportRadar>
<!--ro, opt, bool, whether the device supports radar functions, desc:related URI: /ISAPI/Radar/capabilities?format=json-->true
</isSupportRadar>
<isSupportISUPHttpPassthrough>
<!--ro, opt, bool, whether the device supports HTTP transparent transmission via ISUP V5.0-->true
</isSupportISUPHttpPassthrough>
<isSupportMaterialListQuery>
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    <!--ro, opt, bool, whether the device supports uploading the material list, desc:event: materialListQuery-->true
  </isSupportMaterialListQuery>
  <isSupportDebugLogAsync>
    <!--ro, opt, bool, whether the device supports exporting the debugging log asynchronously-->true
  </isSupportDebugLogAsync>
  <isSupportAlgorithmsInfo>
    <!--ro, opt, bool, whether the device supports getting the algorithm version of the device-->true
  </isSupportAlgorithmsInfo>
  <isSupportSecurityChecking>
    <!--ro, opt, bool, whether the device supports configuring security checking, desc:true (support); otherwise, not support-->true
  </isSupportSecurityChecking>
  <isSupportSecurityCheckException>
    <!--ro, opt, bool, whether the device supports the security checking exception event, desc:eventType: securityCheckException-->true
  </isSupportSecurityCheckException>
  <isSupportPictureManage>
    <!--ro, opt, bool, whether the device supports picture management function, desc:related URI: /ISAPI/System/isSupportPictureManage/capabilities?
format=json-->true
  </isSupportPictureManage>
  <isSupportAutoSwitch>
    <!--ro, opt, bool, whether the device supports automatically powering on and powering off, desc:related URI: /ISAPI/System/autoSwitch/capabilities?
format=json-->true
  </isSupportAutoSwitch>
  <SHMCap>
    <!--ro, opt, object, SHM alarm-->
    <isSupportHighHDTemperature>
      <!--ro, opt, bool, whether the device supports HDD high temperature detection, desc:eventType: highHDTemperature-->true
    </isSupportHighHDTemperature>
    <isSupportLowHDTemperature>
      <!--ro, opt, bool, whether the device supports HDD low temperature detection, desc:eventType: lowHDTemperature-->true
    </isSupportLowHDTemperature>
    <isSupportHDImpact>
      <!--ro, opt, bool, whether the device supports HDD impact detection, desc:eventType: hdImpact-->true
    </isSupportHDImpact>
    <isSupportHDBadBlock>
      <!--ro, opt, bool, whether the device supports HDD bad sector detection, desc:eventType: hdBadBlock-->true
    </isSupportHDBadBlock>
    <isSupportSevereHDFailure>
      <!--ro, opt, bool, whether the device supports HDD major fault detection, desc:eventType: severeHDFailure-->true
    </isSupportSevereHDFailure>
  </SHMCap>
  <isSupportLinkStatusEvent>
    <!--ro, opt, bool, whether the device supports uploading the information about the network connection status of the device, desc:eventType: linkEvent--
>true
  </isSupportLinkStatusEvent>
  <isSupportVideoDealStatus>
    <!--ro, opt, bool, whether the device supports uploading the information about the video data processing status, desc:eventType: videoDealStatus-->true
  </isSupportVideoDealStatus>
  <isSupportAudioDealStatus>
    <!--ro, opt, bool, whether the device supports uploading the information about the audio data processing status, desc:eventType: audioDealStatus-->true
  </isSupportAudioDealStatus>
  <isSupportNetPackLossStatus>
    <!--ro, opt, bool, whether the device supports uploading the information about the network lost package status, desc:eventType: netPacketLoss-->true
  </isSupportNetPackLossStatus>
  <isSupportVideoConferenceStatus>
    <!--ro, opt, bool, whether the device supports uploading the information about the video conference status, desc:eventType: videoConference-->true
  </isSupportVideoConferenceStatus>
  <isSupportDataPrealarm>
    <!--ro, opt, bool, whether the device supports network traffic pre-alarm event, desc:eventType: dataPrealarm-->true
  </isSupportDataPrealarm>
  <isSupportClearCache>
    <!--ro, opt, bool, whether the device supports clearing the cache, desc:related URI: /ISAPI/System/clearCache?format=json-->true
  </isSupportClearCache>
  <isSupportIntelligentMode>
    <!--ro, opt, bool, whether the device supports switching the intelligent mode, desc:related URI: /ISAPI/System/IntelligentMode?format=json-->true
  </isSupportIntelligentMode>
  <isSupportFaceTemperatureMeasurementEvent>
    <!--ro, opt, bool, whether the device supports the skin-surface temperature measurement event, desc:eventType: FaceTemperatureMeasurementEvent-->true
  </isSupportFaceTemperatureMeasurementEvent>
  <isSupportQRCodeEvent>
    <!--ro, opt, bool, whether the device supports the QR code event, desc:eventType: QRCodeEvent-->true
  </isSupportQRCodeEvent>
  <isSupportVisitorEvent>
    <!--ro, opt, bool, whether the device supports the visitor event, desc:eventType: VisitorEvent-->true
  </isSupportVisitorEvent>
  <isSupportEmployeeAuthEvent>
    <!--ro, opt, bool, whether the device supports the employee authorization event, desc:eventType: EmployeeAuthEvent-->true
  </isSupportEmployeeAuthEvent>
  <isSupportVisitorInfoSearchEvent>
    <!--ro, opt, bool, whether the device supports the visitor information search event, desc:eventType: VisitorInfoSearchEvent-->true
  </isSupportVisitorInfoSearchEvent>
  <isSupportConsumptionEvent>
    <!--ro, opt, bool, whether the device supports the transaction record event, desc:eventType: ConsumptionEvent-->true
  </isSupportConsumptionEvent>
  <isSupportTransactionRecordEvent>
    <!--ro, opt, bool, whether the device supports the transaction record event, desc:eventType: TransactionRecordEvent-->true
  </isSupportTransactionRecordEvent>
  <isSupportFaceContrastResultDisplay>
    <!--ro, opt, bool, whether the device supports displaying the face picture comparison result, desc:related URI:
/ISAPI/Intelligent/faceContrast/resultDisplay/capabilities?format=json-->true
  </isSupportFaceContrastResultDisplay>
  <isSupportPatchUpgrade>
    <!--ro, opt, bool, whether it supports incremental upgrade, desc:related URI: /ISAPI/Custom/OpenPlatform/patchUpgrade/capabilities?format=json-->true
  </isSupportPatchUpgrade>
  <isSupportSnapshotAsync>
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<isSupportSnapshotAsync>
<!--ro, opt, bool, whether the device supports capturing pictures asynchronously (the picture URL will be returned), desc:related URI: /ISAPI/Streaming/channels/<ID>/picture/async?format=json-->true
</isSupportSnapshotAsync>
<isSupportLogExportByUSB>
<!--ro, opt, bool, exporting logs to PC from USB flash, desc:related URI: /ISAPI/System/exportUSBLog?format=json-->true
</isSupportLogExportByUSB>
<isSupportShellCmd>
<!--ro, opt, bool, whether the device supports shell command control, desc:related URI: /ISAPI/System/shellCmd?format=json-->true
</isSupportShellCmd>
<faceMonitor opt="dailyFaceMonitor,dailyFaceTracking,VCAFaceMonitor,VCAFaceTracking,manualFaceMonitor,advanceConfiguration,">
<!--ro, opt, bool, whether the device supports face arming, attr:opt{opt, string}, desc:related URI: /ISAPI/Intelligent/channels/<ID>/dailyFaceMonitor?format=json-->true
</faceMonitor>
<isSupportRelatedVCAResource>
<!--ro, opt, bool, whether the device supports the system-level intelligent resources configuration protocol, desc:related URI: /ISAPI/System/RelatedVCAResource/capabilities?format=json; this protocol conficts with thew previous intelligent resources switch protocol-->true
</isSupportRelatedVCAResource>
<isSupportVehicleMonitor>
<!--ro, opt, bool, whether the device supports intelligent arming and tracking of vehicles, desc:related URI: /ISAPI/Traffic/channels/<ID>/vehicleMonitor/capabilities?format=json-->true
</isSupportVehicleMonitor>
<isSupportManualVehicleMonitor>
<!--ro, opt, bool, whether the device supports manual arming and tracking of vehicles, desc:related URI: /ISAPI/Traffic/channels/<ID>/vehicleMonitor/manual/capabilitis?format=json-->true
</isSupportManualVehicleMonitor>
<isSupportDailyVehicleMonitor>
<!--ro, opt, bool, whether the device supports daily arming and tracking of vehicles, desc:related URI: /ISAPI/Traffic/channels/<ID>/vehicleDetect/capabilities-->true
</isSupportDailyVehicleMonitor>
<isSupportInputProxyDeviceInfo>
<!--ro, opt, bool, whether the device supports getting the information of the devices added to the proxy device channel, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities-->true
</isSupportInputProxyDeviceInfo>
<isSupportAccessControlCap>
<!--ro, opt, bool, access control capability, desc:related URI: /ISAPI/AccessControl/capabilities-->true
</isSupportAccessControlCap>
<isSupportDebugLog>
<!--ro, opt, bool, whether the device supports exporting the debugging log, desc:related URI: /ISAPI/System/debugLog?format=json-->true
</isSupportDebugLog>
<isSupportFaceAppear>
<!--ro, opt, bool, whether the device supports face appearing detection event, desc:eventType: faceAppear-->true
</isSupportFaceAppear>
<isSupportFaceDisappear>
<!--ro, opt, bool, whether the device supports face disappearing detection event, desc:eventType: faceDisappear-->true
</isSupportFaceDisappear>
<isSupportTeacherBehavior>
<!--ro, opt, bool, whether the device supports the abnormal event detection of teachers, desc:eventType: teacherBehavior-->true
</isSupportTeacherBehavior>
<isSupportHeadRiseRate>
<!--ro, opt, bool, whether the device supports face appearing detection event, desc:eventType: headRiseRate-->true
</isSupportHeadRiseRate>
<isSupportIPCUpgrade>
<!--ro, opt, bool, whether the platform supports upgrading network cameras via NVR and other devices, desc:related URI: /ISAPI/System/IPCUpgrade/<ID>?format=json-->true
</isSupportIPCUpgrade>
<isSupportFaceLibRebuildEvent>
<!--ro, opt, bool, whether the device supports uploading the modeling rebuilding of the face picture library event, desc:eventType: faceLibRebuild-->true
</isSupportFaceLibRebuildEvent>
<isSupportCalibrationStream>
<!--ro, opt, bool, supported capability of calibration stream, desc:related URI: /ISAPI/Streaming/calibStream/capabilities?format=json-->true
</isSupportCalibrationStream>
<isSupportChannelOccupy>
<!--ro, opt, bool, whether the device supports outdoor fire engine access occupied by vehicles, desc:eventType: channelOccupy-->true
</isSupportChannelOccupy>
<isSupportOffDuty>
<!--ro, opt, bool, whether to support fire control room person absence detection, desc:eventType: offDuty-->true
</isSupportOffDuty>
<isSupportNoCertificate>
<!--ro, opt, bool, whether to support fire control room person without certificate detection, desc:eventType: noCertificate-->true
</isSupportNoCertificate>
<isSupportSmokeAlarm>
<!--ro, opt, bool, whether the device supports smoke alarm, desc:eventType: smokeAlarm-->true
</isSupportSmokeAlarm>
<isSupportBatteryCarDisobey>
<!--ro, opt, bool, whether the device supports illegal parking detection of electric moped, desc:eventType: batteryCarDisobey-->true
</isSupportBatteryCarDisobey>
<isSupportNoFireExtinguisherRecog>
<!--ro, opt, bool, whether the device supports fire extinguisher loss detection, desc:eventType: noFireExtinguisherRecog-->true
</isSupportNoFireExtinguisherRecog>
<isSupportIndoorPasswayBlock>
<!--ro, opt, bool, whether the device supports indoor access blocking detection, desc:eventType: indoorPasswayBlock-->true
</isSupportIndoorPasswayBlock>
<isSupportFireSmartFireDetect>
<!--ro, opt, bool, whether the device supports fire detection, desc:eventType: fireSmartFireDetect-->true
</isSupportFireSmartFireDetect>
<isSupportDetectorRunningStatus>
<!--ro, opt, bool, whether the device supports getting the detector running status, desc:eventType: detectorRunningStatus-->true
</isSupportDetectorRunningStatus>
<isSupportDetectorOperationStatus>
<!--ro, opt, bool, whether the device supports getting the detector operation status, desc:eventType: detectorOperationStatus-->true
</isSupportDetectorOperationStatus>
<isSupportDetectorTemperatureAlarm opt="highTemperature,riseTemperature,flame">

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    <!--ro, opt, bool, whether the device supports the detector temperature alarm, attr:opt{req, string}-->true
  </isSupportDetectorTemperatureAlarm>
  <isSupportDetectorShelterAlarm>
    <!--ro, opt, bool, whether the device supports detector sheltering alarm, desc:eventType: detectorShelterAlarm-->true
  </isSupportDetectorShelterAlarm>
  <isSupportDetectorMotionAlarm>
    <!--ro, opt, bool, whether the device supports the detector moving alarm, desc:eventType: detectorMotionAlarm-->true
  </isSupportDetectorMotionAlarm>
  <isSupportDetectorTamperAlarm>
    <!--ro, opt, bool, whether the device supports detector tampering alarm, desc:eventType: detectorTamperAlarm-->true
  </isSupportDetectorTamperAlarm>
  <isSupportDetectorEmergencyAlarm>
    <!--ro, opt, bool, whether the device supports the detector emergency alarm, desc:eventType: detectorEmergencyAlarm-->true
  </isSupportDetectorEmergencyAlarm>
  <isSupportDetectorSmokeAlarm>
    <!--ro, opt, bool, whether the device supports the detector smoke alarm, desc:eventType: detectorSmokeAlarm-->true
  </isSupportDetectorSmokeAlarm>
  <isSupportDetectorCombustibleGasAlarm>
    <!--ro, opt, bool, whether the device supports the detector combustible gas alarm, desc:eventType: detectorCombustibleGasAlarm-->true
  </isSupportDetectorCombustibleGasAlarm>
  <isSupportFireControlData>
    <!--ro, opt, bool, whether the device supports uploading real-time fire control data, desc:eventType:FireControlData-->true
  </isSupportFireControlData>
  <isSupportFireNoRegulation>
    <!--ro, opt, bool, whether to support people leaving event uploading, desc:eventType: fireNoRegulation-->true
  </isSupportFireNoRegulation>
  <isSupportSmokeFireRecognize>
    <!--ro, opt, bool, whether the device supports smoke and fire recognition event uploading, desc:eventType: smokeFireRecognize-->true
  </isSupportSmokeFireRecognize>
  <isSupportClientProxyWEB>
    <!--ro, opt, bool, whether to support WEB page jumping, desc:whether to support WEB page jumping-->true
  </isSupportClientProxyWEB>
  <WEBLocation>
    <!--ro, opt, string, WEB page placement location, range:[1,32], desc:local-local server, remote-remote server; (if this node is not returned, the WEB
page will be placed in the local server by default)-->local
  </WEBLocation>
  <isSupportWEBLocalPlug>
    <!--ro, opt, bool, whether to support local download of web plug-in, desc:true-yes (by default), false-no; when this node is not returned, it is
supported by default-->true
  </isSupportWEBLocalPlug>
  <deviceId>
    <!--ro, opt, string, device ID, range:[1,32], desc:if WEBLocation is "remote", the unique device ID should be returned to identify the unique
corresponding web package. This device ID is managed by the Web configuration package. For a specified device of different versions, the same deviceID will
be returned-->R0101webs_4.0_isapi
  </deviceId>
  <isSupportTimeZone>
    <!--ro, opt, bool, whether the device supports configuring the DST (Daylight Saving Time), desc:related URI: /ISAPI/System/time/timeZone-->true
  </isSupportTimeZone>
  <isSptHealthDetection>
    <!--ro, opt, bool, whether the device supports health monitoring, desc:related URI: /ISAPI/System/healthDetection-->true
  </isSptHealthDetection>
  <isSptDiagnosis>
    <!--ro, opt, bool, whether the device supports diagnosis, desc:related URI: /ISAPI/System/diagnosis-->true
  </isSptDiagnosis>
  <isSptSerialLogCfg>
    <!--ro, opt, bool, whether the device supports configuring parameters of serial port log redirection, desc:related URI: /ISAPI/System/serialLogCfg?
format=json-->true
  </isSptSerialLogCfg>
  <isSptFileExport>
    <!--ro, opt, bool, whether the device supports exporting the device file, desc:related URI: /ISAPI/System/fileExport?format=json-->true
  </isSptFileExport>
  <isSptCertificationStandard>
    <!--ro, opt, bool, whether the device supports configuring host verification standard, desc:related URI: /ISAPI/SecurityCP/certificationStandard-->true
  </isSptCertificationStandard>
  <isSptKeypadLock>
    <!--ro, opt, bool, whether the device supports locking the keypad, desc:related URI: /ISAPI/System/keypadLock/config-->true
  </isSptKeypadLock>
  <isSupportVibrationDetection>
    <!--ro, opt, bool, whether the device supports the vibration detection, desc:related URI: /ISAPI/System/channels/<ID>/vibrationDetection?format=json--
>true
  </isSupportVibrationDetection>
  <isSupportSmokingDetectAlarm>
    <!--ro, opt, bool, whether the device supports the smoking alarm, desc:eventType: smokeDetectAlarm-->true
  </isSupportSmokingDetectAlarm>
  <isSupportRadarChannels>
    <!--ro, opt, bool, whether the device supports getting the radar channel, desc:related URI: /ISAPI/System/RadarChannels?format=json-->true
  </isSupportRadarChannels>
  <radarIPDForm>
    <!--ro, opt, enum, radar-camera status, subType:string, desc:"single" (single radar); "double_diagonal" (double radars arranged diagonally);
"double_vertical" (double radars arranged vertically)-->single
  </radarIPDForm>
  <isSupportRadarFieldDetection>
    <!--ro, opt, bool, whether the device supports radar's area intrusion detection, desc:related URI:
/ISAPI/Smart/RadarChannels/<radarChannelID>/radarFieldDetection/capabilities?format=json-->true
  </isSupportRadarFieldDetection>
  <isSupportRadarLineDetection>
    <!--ro, opt, bool, whether the device supports radar line crossing detection, desc:related URI:
/ISAPI/Smart/RadarChannels/<radarChannelID>/radarLineDetection/capabilities?format=json-->true
  </isSupportRadarLineDetection>
  <mixedTargetDetectionWebNoDisplay>
    <!--ro, opt, bool, multiple targets recognition is not displayed-->true
  </mixedTargetDetectionWebNoDisplay>
  <isSupportRemovePlaybackRecord>
    <!--ro, opt, bool, whether to support removing playback (the device does not support playback), desc:if this function is supported, this node will be
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    <!--ro, opt, bool, whether to support removing playback (the device does not support playback), desc:if this function is supported, this node will be
returned and its value is true; otherwise, this node will not be returned. This node will be returned when the value of isSupportRemoveStorage is true;
otherwise, this node will not be returned-->true
  </isSupportRemovePlaybackRecord>
  <isSupportAudioUpload>
    <!--ro, opt, bool, whether the device supports uploading the audio file, desc:related URI: /ISAPI/System/Audio/audioUpload?format=json-->true
  </isSupportAudioUpload>
  <VCAResourceList>
    <!--ro, opt, array, supported smart resources list, subType:object-->
    <VCAResource>
      <!--ro, opt, object, smart resources-->
      <VCAResourceType>
        <!--ro, opt, string, smart resources type-->test
      </VCAResourceType>
      <ChannleVCAResourceList>
        <!--ro, opt, array, supported smart resources list, subType:object-->
        <ChannleVCAResource>
          <!--ro, opt, object, smart channel resources-->
          <channel>
            <!--ro, opt, int, channel No.-->1
          </channel>
          <channleVCAResourceType>
            <!--ro, opt, string, smart channel resources type-->test
          </channleVCAResourceType>
        </ChannleVCAResource>
      </ChannleVCAResourceList>
    </VCAResource>
  </VCAResourceList>
  <isSupportSlotInsertion>
    <!--ro, opt, bool, whether to support slot plugging in and out, desc:eventType: slotInsertion-->true
  </isSupportSlotInsertion>
  <isSupportNetworkLoop>
    <!--ro, opt, bool, whether to support network loop alarm, desc:eventType: networkLoop-->true
  </isSupportNetworkLoop>
  <isSupportApOnOffLine>
    <!--ro, opt, bool, whether to support AP online and offline alarm event uploading, desc:eventType: apOnOffLine-->true
  </isSupportApOnOffLine>
  <isSupportAbnormalWANPort>
    <!--ro, opt, bool, whether to support WAN abnormal alarm event uploading, desc:eventType: abnormalWANPort-->true
  </isSupportAbnormalWANPort>
  <isSupportAPException>
    <!--ro, opt, bool, whether to support AP exception alarm information uploading, desc:eventType: APException-->true
  </isSupportAPException>
  <isSupportACException>
    <!--ro, opt, bool, whether to support AC exception alarm information uploading, desc:eventType: ACException-->true
  </isSupportACException>
  <isSupportCartridgeCapacity>
    <!--ro, opt, bool, whether the device supports the configuration of notifying cartridge capacity, desc:related URL:
/ISAPI/System/printer/cartridgeCapacity/capabilities?format=json-->true
  </isSupportCartridgeCapacity>
  <isSupportCDCakeBoxCapacity>
    <!--ro, opt, bool, whether the device supports notifying CD capacity, desc:related URL: /ISAPI/System/printer/CDCakeBoxCapacity/capabilities?
format=json-->true
  </isSupportCDCakeBoxCapacity>
  <isSupportSleep>
    <!--ro, opt, bool, whether the device supports configuring sleep, desc:related URL: /ISAPI/System/sleep/capabilities?format=json-->true
  </isSupportSleep>
  <isSupportContainerDetection>
    <!--ro, opt, bool, whether the device supports container detection, desc:/ISAPI/Intelligent/containerDetection/capabilities?format=json (for devices
that does not return this node, you can check according to the node returned by the following protocol:
/ISAPI/Traffic/ContentMgmt/InputProxy/channels/<ID>/ocrScene/capabilities)-->true
  </isSupportContainerDetection>
  <isSupportRadarVideoDetection>
    <!--ro, opt, bool, related URI: /ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json-->true
  </isSupportRadarVideoDetection>
  <isSupportRadarVideoDetectionRealtime>
    <!--ro, opt, bool, whether the device supports radar video detection, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json-->true
  </isSupportRadarVideoDetectionRealtime>
  <isSupportBatteryStatusUpload>
    <!--ro, opt, bool, whether the device supports uploading the battery status, desc:eventType: batteryStatus-->true
  </isSupportBatteryStatusUpload>
  <isSupportPowerConsumptionStatusUpload>
    <!--ro, opt, bool, whether the device supports uploading device consumption status-->true
  </isSupportPowerConsumptionStatusUpload>
  <isSupportOnlineUpgradeTask>
    <!--ro, opt, bool, whether device online upgrading is supported-->true
  </isSupportOnlineUpgradeTask>
  <isSupportPersonArmingTrack>
    <!--ro, opt, bool, whether the device supports person arming configuration, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/personArmingTrack/capabilities?format=json-->true
  </isSupportPersonArmingTrack>
  <isSupportManualPersonArmingTrack>
    <!--ro, opt, bool, whether the device supports manual person arming, desc:related URI: /ISAPI/Intelligent/channels/<ID>/manualPersonArmingTrack?
format=json-->true
  </isSupportManualPersonArmingTrack>
  <isSupportGPSVerification>
    <!--ro, opt, bool, whether the device supports GPS verification points, desc:related URI: /ISAPI/System/GPSVerification/channels/<ID>/points?
format=json-->true
  </isSupportGPSVerification>
  <isSupportHBDLib>
    <!--ro, opt, bool, whether the device supports the human body library, desc:related URI: /ISAPI/Intelligent/HBDLib/capabilities?format=json-->true
  </isSupportHBDLib>
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  <isSupportDisinfectionInfo>
    <!--ro, opt, bool, whether the device supports uploading disinfection information, desc:eventType: disinfectionInfo-->true
  </isSupportDisinfectionInfo>
  <isSupportTempQueryDelete>
    <!--ro, opt, bool, whether the device supports deleting temperature query, desc:eventType: tempQueryDelete-->true
  </isSupportTempQueryDelete>
  <isSupportMobilePayment>
    <!--ro, opt, bool, whether the device supports mobile payment, desc:/ISAPI/System/mobilePayment?format=json-->true
  </isSupportMobilePayment>
  <isSupportFileSearch>
    <!--ro, opt, bool, whether the device supports searching files, desc:related URI: /ISAPI/System/fileSearch?format=json-->true
  </isSupportFileSearch>
  <isSupportFileExportAsync>
    <!--ro, opt, bool, whether the device supports exporting files asynchronously, desc:related URI: /ISAPI/System/fileExport/async?format=json-->true
  </isSupportFileExportAsync>
  <isSupportSSDFileSystemCapacity>
    <!--ro, opt, bool, whether the device supports getting the capacity distribution of the SSD system, desc:related URI:
/ISAPI/System/SSDFileSystem/capacity?format=json-->true
  </isSupportSSDFileSystemCapacity>
  <isSupport4GSafetyHelmetInfo>
    <!--ro, opt, bool, whether the device supports the hard hat event, desc:eventType: 4GSafetyHelmetInfo-->true
  </isSupport4GSafetyHelmetInfo>
  <isSupportICCID>
    <!--ro, opt, bool, whether the device supports getting the ICCID information, desc:related URI: /ISAPI/System/deviceInfo/ICCID?format=json-->true
  </isSupportICCID>
  <isSupportFireEscapeDetection>
    <!--ro, opt, bool, whether the device supports fire escape detection, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/fireEscapeDetection/capabilities?format=json-->true
  </isSupportFireEscapeDetection>
  <isSupportTakingElevatorDetection>
    <!--ro, opt, bool, whether the device supports taking elevator detection, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/takingElevatorDetection/capabilities?format=json-->true
  </isSupportTakingElevatorDetection>
  <isSupportVehicleMatchResult>
    <!--ro, opt, bool, whether the device supports license plate comparison, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/licensePlateContrast/capabilities?format=json-->true
  </isSupportVehicleMatchResult>
  <isSupportDataAware>
    <!--ro, opt, bool, the data aware capability, desc:related URI: /ISAPI/System/dataAware/capabilities?format=json-->true
  </isSupportDataAware>
  <isSupportAbnormalReboot>
    <!--ro, opt, bool, whether the device supports abnormal reboot event, desc:eventType: abnormalReboot-->true
  </isSupportAbnormalReboot>
  <isSupportVehicleThermometry>
    <!--ro, opt, bool, whether the device supports vehicle temperature measurement detection, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleDetect/thermometry/capabilities?format=json-->true
  </isSupportVehicleThermometry>
  <AssociationScenesList>
    <!--ro, opt, array, the list of scene adaptive linkage, subType:object, desc:it indicates the device supports the scene adaptive linkage between
channels. The current device has two channels: main channel (box camera) and sub channel (radar camera), however, the device only opens a main channel to
users for scene configuration; the scene configuration of the sub channel is not open for the users; the scene of the sub channel is automatically connected
to the main channel scene; currently, the front-end returns this capability-->
    <AssociationScenes>
      <!--ro, req, object, the channel list of scene adaptive linkage between channels-->
      <channelID>
        <!--ro, req, int, supported scene channel ID, desc:supported scene channel ID-->1
      </channelID>
      <isSupportScene>
        <!--ro, opt, bool, whether the device supports the scene configuration, desc:whether the current channel is the main channel and supports scene
configuration , true (main channel and support), this node is not returned (sub channel and not support)-->true
      </isSupportScene>
    </AssociationScenes>
  </AssociationScenesList>
  <supportRelationVideoEvent opt="PIR,cityManagement,roadMaint">
    <!--ro, opt, string, supports uploading the video of linked triggering event, attr:opt{req, string}, desc:"PIR" (PIR event), "cityManagement" (city
management event), "roadMaint" (road maintenance event)-->test
  </supportRelationVideoEvent>
  <isSupportVCAResourceList>
    <!--ro, opt, bool, whether the device supports configuring intelligent resources in a batch, desc:related URI:
/ISAPI/System/Video/inputs/channels/VCAResourceList?format=json-->true
  </isSupportVCAResourceList>
  <isSupportConvergenceCloud>
    <!--ro, opt, bool, whether the device supports the URI for EZ interconnection channels, desc:URI for capability of all channels:
/ISAPI/Event/convergenceCloud/channels/capabilities?format=json; URI for capability of specified channel:
/ISAPI/Event/convergenceCloud/channels/<ID>/capabilities?format=json-->true
  </isSupportConvergenceCloud>
  <isSupportInterconnection>
    <!--ro, opt, bool, whether the device supports the URI for EZ interconnection devices, desc:related URI: /ISAPI/Event/convergenceCloud/capabilities?
format=json-->true
  </isSupportInterconnection>
  <isSupportHPVehicleStatus>
    <!--ro, opt, bool, whether the device supports the platform vehicle status event, desc:eventType: HPVehicleStatus-->true
  </isSupportHPVehicleStatus>
  <isSupportHPWorkingStatus>
    <!--ro, opt, bool, whether the device supports the scheduled platform working status event, desc:eventType: HPWorkingStatus-->true
  </isSupportHPWorkingStatus>
  <isSupportHPPeopleDetection>
    <!--ro, opt, bool, whether the device supports the platform people detection event, desc:eventType: HPPeopleDetection-->true
  </isSupportHPPeopleDetection>
  <isSupportHPVehicleDoorDetection>
    <!--ro, opt, bool, whether the device supports the platform door detection event, desc:eventType: HPVehicleDoorDetection-->true
  </isSupportHPVehicleDoorDetection>
  <isSupportRoadMaint>
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    <!--ro, opt, bool, whether the device supports road maintenance, desc:related URI: /ISAPI/Intelligent/channels/<ID>/roadMaint?format=json-->true
  </isSupportRoadMaint>
  <isSupportDockStationInfo>
    <!--ro, opt, bool, whether the device supports the event of uploading the dock station information to the media, desc:eventType: dockStationInfo; this
node is not returned if not supported-->true
  </isSupportDockStationInfo>
  <isSupportBodyCameraManualForensics>
    <!--ro, opt, bool, whether the device supports the event of uploading the manual evidence information of the body camera, desc:eventType:
bodyCameraManualForensics; this node is not returned if not supported-->true
  </isSupportBodyCameraManualForensics>
  <isSupportBodyCameraLogin>
    <!--ro, opt, bool, whether the device supports the event of police officer logging in to the body camera, desc:eventType: bodyCameraLogin; this node is
not returned if not supported-->true
  </isSupportBodyCameraLogin>
  <isSupportBodyCameraRunStatus>
    <!--ro, opt, bool, whether the device supports the status of body camera, desc:eventType: bodyCameraRunStatus; this node is not returned if not
supported-->true
  </isSupportBodyCameraRunStatus>
  <isSupportBodyCameraSignIn>
    <!--ro, opt, bool, whether the device supports the event of police officer signing in the body camera, desc:eventType: bodyCameraSignIn; this node is
not returned if not supported-->true
  </isSupportBodyCameraSignIn>
  <isSupportAlertDistribution>
    <!--ro, opt, bool, whether the device supports the alert distribution, desc:eventType: alertDistribution; this node is not returned if not supported--
>true
  </isSupportAlertDistribution>
  <isSupportImageCapture>
    <!--ro, opt, bool, whether the device supports the event of scheduled capture, desc:related URI: /ISAPI/System/Video/inputs/channels/<ID>/imageCapture?
format=json-->true
  </isSupportImageCapture>
  <isSupportScheduledRecordUpload>
    <!--ro, opt, bool, whether the device supports the event of channel scheduled recording, desc:related URI:
/ISAPI/System/Video/inputs/channels/<ID>/scheduledRecordUpload?format=json-->true
  </isSupportScheduledRecordUpload>
  <isSupportTargetPatternCollect>
    <!--ro, opt, bool, whether the device supports the target track collection event, desc:eventType: targetPatternCollect-->true
  </isSupportTargetPatternCollect>
  <isSupportAudioAnalysis>
    <!--ro, opt, bool, whether the device supports the audio analysis event, desc:eventType: audioAnalysis; this node is not returned if not supported--
>true
  </isSupportAudioAnalysis>
  <isSupportDebuggingDataExport>
    <!--ro, opt, bool, whether the device supports importing the parameters (in-out parameters, rule configuration parameters), desc:related URI:
/ISAPI/System/debuggingData/parameters?format=json; this node is not returned if not supported-->true
  </isSupportDebuggingDataExport>
  <isSupportInternalParaAdjust>
    <!--ro, opt, bool, whether the device supports trinocular device, whether the device supports editing in-parameters, desc:related URI:
/ISAPI/System/internalParameters/adjustment?format=json; this node is not returned if not supported-->true
  </isSupportInternalParaAdjust>
  <isSupportManualImageCapture>
    <!--ro, opt, bool, whether the device supports manual capture of the video channel, desc:related URI:
/ISAPI/System/Video/inputs/channels/<ID>/manualImageCapture?format=json; this node is not returned if not supported-->true
  </isSupportManualImageCapture>
  <characteristicCode min="1" max="128">
    <!--ro, opt, string, device attribute code, range:[1,128], attr:min{opt, string},max{opt, string}, desc:related URI:
/ISAPI/System/deviceInfo/characteristicCode?format=json-->test
  </characteristicCode>
  <isSupportSIMCardStatus>
    <!--ro, opt, bool, whether the device supports getting the information of SIM card status, desc:related URI: /ISAPI/System/SIMCardStatus?format=json--
>true
  </isSupportSIMCardStatus>
  <OpenPlatformCap>
    <!--ro, opt, object, detailed HEOP capability supported by the device, desc:related URI: /ISAPI/Custom/OpenPlatform/capabilities-->
    <isSupportOpenPlatform>
      <!--ro, req, bool, whether the device supports HEOP (Embedded Open Platform)-->true
    </isSupportOpenPlatform>
    <maxAPPNum>
      <!--ro, opt, int, the maximum number of third-party application packages allowed to be installed-->0
    </maxAPPNum>
  </OpenPlatformCap>
  <isSupportLogException>
    <!--ro, opt, bool, whether the device supports the log exception alarm, desc:eventType: logException-->true
  </isSupportLogException>
  <isSupportMEF>
    <!--ro, opt, bool, whether the device supports Mobile electronic fence - mobile electronic fence data detection, desc:related URI:
/ISAPI/System/Network/MEF/<ID>?format=json-->true
  </isSupportMEF>
  <manualEvent
opt="fielddetection,linedetection,ANPR,faceCapture,mixedTargetDetection,studentsStoodUp,headRiseRate,teacherBehavior,ElevatorRunningDetectionEvent,offDuty,b
atteryCarDisobey,fireExtinguisherRecog,indoorPasswayBlock,fireSmartFireDetect,smokeAlarm,gasTankMonitoring,smokeDetectAlarm,TelephoningBehaviorAlarmEvent,sl
eepDuty,playCellphone,audioAnalysis,TemperatureEvent,TemperatureEventUp,TemperatureEventDown,TMA">
    <!--ro, opt, string, whether the device supports URI for mannual triggering event, attr:opt{req, string}, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/manualEvent/<eventType>?format=json-->test
  </manualEvent>
  <isSupportAppCfg>
    <!--ro, opt, bool, whether the device supports application management, desc:related URI: /ISAPI/System/app/capabilities?format=json-->true
  </isSupportAppCfg>
  <isSupportCleanUp>
    <!--ro, opt, bool, whether the device supports one-click clearing, desc:related URI: /ISAPI/System/cleanup?format=json-->true
  </isSupportCleanUp>
  <isSupportEndUserCheckMsg opt="sleep,reboot,factoryReset,cleanup,parentPickUpQuery,schoolAttendanceQuery,screenShare,screenMirror">
    <!--ro, opt, bool, whether the device supports uploading the confirmation information of end users, attr:opt{req, string}, desc:eventType:
endUserCheckMsg-->true
  </isSupportEndUserCheckMsg>
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  <isSupportScreenConfig>
    <!--ro, opt, bool, whether the device supports configuring screen parameters, desc:related URI: /ISAPI/System/screenConfig/capabilities?format=json-->true
  </isSupportScreenConfig>
  <isSupportTemperatureMonitor>
    <!--ro, opt, bool, whether the device supports temperature monitoring configuration, desc:related URI: /ISAPI/System/temperatureMonitor/capabilities?format=json-->true
  </isSupportTemperatureMonitor>
  <isSupportObjectServer>
    <!--ro, opt, bool, whether the device supports the object storage service, desc:related URI: /ISAPI/System/objectServer/capabilities?format=json-->true
  </isSupportObjectServer>
  <isSupportCDsStatus>
    <!--ro, opt, bool, whether the device supports notifying disk status, desc:related URI for enabling the function: the bEnableCDsStatus node of /ISAPI/Trial/roompara/capabilities; corresponding event: CDsStatus-->true
  </isSupportCDsStatus>
  <isSupportCoalGangueDetection>
    <!--ro, opt, bool, whether the device supports coal gangue detection, desc:related URI: /ISAPI/Intelligent/channels/<channelID>/coalGangueDetection; this node is not returned if not supported-->true
  </isSupportCoalGangueDetection>
  <isSupportVacantSeatRate>
    <!--ro, opt, bool, whether the device supports seat vacancy detection, desc:related URI: /ISAPI/Intelligent/channels/<channelID>/vacantSeatRate?format=json; corresponding event: VacantSeatRate; this node is not returned if not supported-->true
  </isSupportVacantSeatRate>
  <isSupportObjectsThrownDetection>
    <!--ro, opt, bool, whether the device supports objects thrown detection, desc:related URI: /ISAPI/Intelligent/channels/<channelID>/objectsThrownDetection?format=json; corresponding event: objectsThrownDetection; this node is not returned if not supported-->true
  </isSupportObjectsThrownDetection>
  <isSupportBVCorrectDataSearch>
    <!--ro, opt, bool, whether the device supports searching for the parameters of camera correction, desc:related URI: /ISAPI/System/BVCorrectData/search?format=json; this node is not returned if not supported-->true
  </isSupportBVCorrectDataSearch>
  <isSupportSlagTruckWashingDetection>
    <!--ro, opt, bool, whether the device supports dump truck cleaning detection event, desc:eventType: slagTruckWashingDetection-->true
  </isSupportSlagTruckWashingDetection>
  <isSupportWOL>
    <!--ro, opt, bool, whether the device supports configuring the LAN wakeup parameters, desc:it can be realized via the WOL (Wake On-Lan) function. Related URL: /ISAPI/System/WOL/capabilities?format=json-->true
  </isSupportWOL>
  <isSupportPersonQueueTimingStatistics>
    <!--ro, opt, bool, whether the device supports counting the number of persons in each queue, desc:eventType: isSupportPersonQueueTimingStatistics-->true
  </isSupportPersonQueueTimingStatistics>
  <isSupportLensParamFile>
    <!--ro, opt, bool, whether the device supports importing and exporting camera parameters files-->true
  </isSupportLensParamFile>
  <isSupportVehiclePassingInParkingLot>
    <!--ro, opt, bool, whether the device supports the detection of vehicle passing in parking lot entrance and exit, desc:eventType: vehiclePassingInParkingLot-->true
  </isSupportVehiclePassingInParkingLot>
  <isSupportPowerOutput>
    <!--ro, opt, bool, whether power supply output configuration is supported, desc:related URL: /ISAPI/System/powerOutput/capabilities?format=json-->true
  </isSupportPowerOutput>
  <isSupportIPCUpgradeByFTP>
    <!--ro, opt, bool, it indicates that the device supports upgrading the proxy device channel, such as IPC upgrade via NVR, desc:related URI: /ISAPI/System/IPCUpgrade/<channelID>/FTP?format=json&security=<security>&iv=<iv>-->true
  </isSupportIPCUpgradeByFTP>
  <isSupportMethaneConcentrationException>
    <!--ro, opt, bool, whether the device supports methane concentration exception detection, desc:eventType: methaneConcentrationException-->true
  </isSupportMethaneConcentrationException>
  <isSupportMethaneLightIntensityException>
    <!--ro, opt, bool, whether the device supports methane light intensity exception detection, desc:eventType: methaneLightIntensityException-->true
  </isSupportMethaneLightIntensityException>
  <isSupportLocalButtonPermission>
    <!--ro, opt, bool, whether the device supports controlling local button permission, desc:related URI: /ISAPI/System/localButtonPermission/capabilities?format=json-->true
  </isSupportLocalButtonPermission>
  <isSupportLocalButtonRecord>
    <!--ro, opt, bool, whether the device supports button triggering record, desc:eventType: localButtonRecord-->true
  </isSupportLocalButtonRecord>
  <isSupportLocalButtonSnapshot>
    <!--ro, opt, bool, whether the device supports button triggering snapshot, desc:eventType: localButtonSnapshot-->true
  </isSupportLocalButtonSnapshot>
  <isSupportVoiceTalkEvent>
    <!--ro, opt, bool, whether the device supports voice talk, desc:eventType: voiceTalkEvent-->true
  </isSupportVoiceTalkEvent>
  <isSupportPowerSupply>
    <!--ro, opt, bool, whether the device supports external power supply, desc:related URI: /ISAPI/System/powerSupply/capabilities?format=json-->true
  </isSupportPowerSupply>
  <isSupportbarrierGateStatus>
    <!--ro, opt, bool, whether the device supports barrier gate status, desc:eventType: barrierGateStatus-->true
  </isSupportbarrierGateStatus>
  <DeployCap>
    <!--ro, opt, object, the capability of arming non-subscription events-->
    <deployID opt="1,2">
      <!--ro, opt, enum, arming type, subType:int, attr:opt{req, string}, desc:0 (client arming); 1 (real-time arming)-->0
    </deployID>
  </DeployCap>
  <isSupportSignalControllerStatus>
    <!--ro, opt, bool, whether the device supports uploading the signal controller status, desc:eventType: signalControllerStatus-->true
  </isSupportSignalControllerStatus>
  <isSupportSignalControllerFault>
    <!--ro, opt, bool, whether the device supports uploading the signal controller fault, desc:eventType: signalControllerFault-->true
  </isSupportSignalControllerFault>
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  <isSupportNTPConfig>
    <!--ro, opt, bool, whether the device supports configuring NTP parameters, desc:/ISAPI/System/time/ntp/capabilities?format=json-->true
  </isSupportNTPConfig>
  <isSupportConsumptionQuery>
    <!--ro, opt, bool, whether the device supports uploading the request for searching for consumption, desc:eventType: ConsumptionQuery-->true
  </isSupportConsumptionQuery>
  <StreamingCodingRegionPositionCap>
    <!--ro, opt, object, whether the device supports getting all stream splice area positions information capability,
desc:/ISAPI/System/streamingCoding/channels/<channelID>/regionPosition-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
    </ChannelsList>
  </StreamingCodingRegionPositionCap>
  <isSupportLowPowerPlan>
    <!--ro, opt, bool, whether the device supports configuring low consumption mode schedule, desc:related URI:
/ISAPI/System/consumptionMode/lowPowerPlan/capablities?format=json-->true
  </isSupportLowPowerPlan>
  <isSupportPlateQuaAlarm>
    <!--ro, opt, bool, whether the device supports plate quality detection, desc:eventType: plateQuaAlarm-->true
  </isSupportPlateQuaAlarm>
  <isSupportWiegand>
    <!--ro, opt, bool, whether the device supports the Wiegand protocol, desc:related URI: /ISAPI/System/Wiegand/<wiegandID>/capabilities?format=json-->true
  </isSupportWiegand>
  <zoomFocusWebDisplay
opt="ROI,roadTrafficDetection,SMD,mixedTargetDetection,faceCapture,personCounting,multidimensionalPeopleCounting,AIOP,heelPDC,packingSpaceRecognition,tunnel
Detection,cityManagement">
    <!--ro, opt, string, whether the web supports displaying the zoomfocus page, attr:opt{req, string},
desc:ROI page path: Configuration—Vedio/Audio—ROI
roadTrafficDetection:Configuration—Road Traffic—Detection Configuration; SMD; mixedTargetDetection; faceCapture-->ROI
  </zoomFocusWebDisplay>
  <isSupportInternalParamsAdjustment>
    <!--ro, opt, bool, whether the device supports in-parameters correction by region, desc:true (support), false or this node is not returned (not
support). Related URI: /ISAPI/System/internalParameters/regionAdjustment/capabilities?format=json-->true
  </isSupportInternalParamsAdjustment>
  <isSupportDiagnosticReport>
    <!--ro, opt, bool, whether the device supports diagnostic report function, desc:true (support), false or this node is not returned (not support).
Related URI: /ISAPI/System/diagnosticReport/capabilities?format=json-->true
  </isSupportDiagnosticReport>
  <isSupportWaterQualitySearch>
    <!--ro, opt, bool, whether the device supports the function of searching for water quality, desc:true (support), false or this node is not returned (not
support). Related URI: /ISAPI/Event/waterQualityDetection/search?format=json-->true
  </isSupportWaterQualitySearch>
  <isSupportExhaustDetection>
    <!--ro, opt, bool, whether the device supports configuring gas detection parameters, desc:true (support), false or this node is not returned (not
support). Related URI: /ISAPI/Event/exhaustDetection/channels/<channelID>/capabilities?format=json-->true
  </isSupportExhaustDetection>
  <isSupportTPMSecretKey>
    <!--ro, opt, bool, whether the device supports enabling stream encryption for all TPM keys lists, desc:true (support), false or this node is not
returned (not support). Related URI: /ISAPI/Streaming/encryption/TPMSecretKeyList?format=json&security=<security>&iv=<iv>-->true
  </isSupportTPMSecretKey>
  <isSupportWeatherStation>
    <!--ro, opt, bool, whether the device supports uploading weather station information, desc:eventType: weatherStation; this node is not returned if not
supported-->true
  </isSupportWeatherStation>
  <isSupportRoadVisibilityAnalysis>
    <!--ro, opt, bool, whether the device supports road visibility detection analysis, desc:eventType: weatherStation; this node is not returned if not
supported-->true
  </isSupportRoadVisibilityAnalysis>
  <isSupportRoadWeatherAnalysis>
    <!--ro, opt, bool, whether the device supports road weather detection analysis, desc:eventType: roadWeatherAnalysis; this node is not returned if not
supported-->true
  </isSupportRoadWeatherAnalysis>
  <isSupportRoadVisibilityAbnormal>
    <!--ro, opt, bool, whether the device supports uploading road visibility detection exception, desc:eventType: roadVisibilityAbnormal; this node is not
returned if not supported-->true
  </isSupportRoadVisibilityAbnormal>
  <isSupportRoadWeatherAbnormal>
    <!--ro, opt, bool, whether the device supports uploading road weather detection exception, desc:eventType: roadWeatherAbnormal; this node is not
returned if not supported-->true
  </isSupportRoadWeatherAbnormal>
  <isSupportChannelsTimeSync>
    <!--ro, opt, bool, whether the device supports batch device time synchronization of the IPC/IOT channel, desc:related URL:
/ISAPI/System/time/channelsTimeSync/capabilities?format=json-->true
  </isSupportChannelsTimeSync>
  <isSupport433Sensor>
    <!--ro, opt, bool, whether the device supports configuring 433 sensor transmission,
desc:related URLs:   /ISAPI/System/Sensor/433sensor/433wirelessTransfer/capabilities?format=json
    /ISAPI/System/Sensor/433sensor/capabilities?format=json
    /ISAPI/System/Sensor/433sensor/sensorDetection/capabilities?format=json
    /ISAPI/System/Sensor/433sensor/sensorPicOverlay/capabilities?format=json
    /ISAPI/System/Sensor/433sensor/rfParamTemplate/capabilities?format=json-->true
  </isSupport433Sensor>
  <isSupportSignalSource>
    <!--ro, opt, bool, whether the device supports configuring signal source parameters, desc:related URL: /ISAPI/System/signalSource/capabilities?
format=json-->true
  </isSupportSignalSource>
  <isSupportUSBConfig>
    <!--ro, opt, bool, whether the device supports configuring USB parameters, desc:related URL: /ISAPI/System/USBConfig/capabilities?format=json-->true
  </isSupportUSBConfig>
  <isSupportStandbyStrategy>
    <!--ro, opt, bool, whether the device supports configuring standby strategy parameters, desc:related URL: /ISAPI/System/standbyStrategy/capabilities?
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<!--ro, opt, bool, whether the device supports configuring standby strategy parameters, desc:related URL: /ISAPI/System/standbyStrategy/capabilities?format=json-->true
</isSupportStandbyStrategy>
<isSupportWakeupStrategy>
<!--ro, opt, bool, whether the device supports configuring wakeup strategy parameters, desc:related URL: /ISAPI/System/wakeupStrategy/capabilities?format=json-->true
</isSupportWakeupStrategy>
<isSupportSeachMutexFunction>
<!--ro, opt, bool, whether the device supports searching for disabling the mutual exclusion function, desc:related URL: GET /ISAPI/System/mutexFunction/capabilities?format=json; function description: whether the device supports searching for mutual exclusion function-->true
</isSupportSeachMutexFunction>
<isSupportMultiDeviceID>
<!--ro, opt, bool, whether the device supports multiple long numbers configuration, desc:related URL: /ISAPI/VideoIntercom/deviceId/batchCfg/capabilities?format=json-->true
</isSupportMultiDeviceID>
<isNotSupportMultiDeviceIDCustomName>
<!--ro, opt, bool, whether the device supports configuring custom name for multiple long numbers, desc:if this function is not supported, the user cannot configure custom name for the device via the long number configuration API; true (not support), false or this node is not returned (support)-->true
</isNotSupportMultiDeviceIDCustomName>
<isSupportMixedSecurityChecking>
<!--ro, opt, bool, whether the device supports uploading the event of mixed security checking, desc:eventType: mixedSecurityChecking-->true
</isSupportMixedSecurityChecking>
<isSupportVisiblePackageEvent>
<!--ro, opt, bool, whether the device supports visible package event, desc:eventType: visiblePackageEvent-->true
</isSupportVisiblePackageEvent>
<isSupportBarCodeEvent>
<!--ro, opt, bool, whether the device supports the barcode event-->true
</isSupportBarCodeEvent>
<isSupportGaugeReading>
<!--ro, opt, bool, whether the device supports the gauge reading function, desc:related URL: GET /ISAPI/Event/channels/<channelID>/gaugeReading/capabilities?format=json; this node is not returned if not supported-->true
</isSupportGaugeReading>
<isSupportUncertificateCompareEvent>
<!--ro, opt, bool, whether the device supports the UncertificateCompareEvent event, desc:eventType: UncertificateCompareEvent; this node is not returned if not supported-->true
</isSupportUncertificateCompareEvent>
<isSupportAudioDiacritical>
<!--ro, opt, bool, whether the device supports audio voice changing function, desc:related URL: /ISAPI/System/Video/inputs/channels/<channelID>/audioDiacritical/capabilities?format=json-->true
</isSupportAudioDiacritical>
<isSupportMEFReportSearch>
<!--ro, opt, bool, whether the device supports searching for electronic fence data statistics report, desc:URL for getting the capability: /ISAPI/System/Network/MEF/reportSearch/capabilities?format=json; this node is not returned if not supported-->true
</isSupportMEFReportSearch>
<isSupportBroadcastTriggerEvent>
<!--ro, opt, bool, whether the device supports broadcast triggering event, desc:after the event is uploaded to the platform via the paging microphone, the platform will broadcast and check according to the uploaded terminal No. list; eventType: broadcastTriggerEvent. URL for checking: /ISAPI/VideoIntercom/broadcastTriggerConfirm?format=json-->true
</isSupportBroadcastTriggerEvent>
<isSupportFishingShipDetection>
<!--ro, opt, bool, whether the device supports fishing ship detection event, desc:eventType: fishingShipDetection; related URL: /ISAPI/Thermal/channels/<channelID>/fishingShipDetection/<SID>?format=json-->true
</isSupportFishingShipDetection>
<isSupportHPVehicleLoadingRate>
<!--ro, opt, bool, whether the device supports the platform vehicle loading rate event, desc:eventType: vehicleLoadingRate, this node is not returned if not supported. Related URL: /ISAPI/Intelligent/channels/<channelID>/HPDetection/capabilities?format=json-->true
</isSupportHPVehicleLoadingRate>
<isSupportVideoBarrierGateDetection>
<!--ro, opt, bool, whether the device supports port and gate video detection, desc:eventType: videoBarrierGateDetection, this node is not returned if not supported. Related URL: /ISAPI/Traffic/ContentMgmt/videoBarrierGate/capabilities?format=json-->true
</isSupportVideoBarrierGateDetection>
<isSupportLogicalResource>
<!--ro, opt, bool, whether the device supports camera organization and management, desc:related URL: /ISAPI/System/LogicalResource/capabilities?format=json, this node is not returned if not supported-->true
</isSupportLogicalResource>
<isSupportSlagTruckCoverPlate>
<!--ro, opt, bool, whether the device supports detecting whether back cover the slag truck is covered, desc:eventType: slagTruckCoverPlateDetection, this node is not returned if not supported. Related URL: /ISAPI/Event/channels/<channelID>/slagTruckCoverPlate/capabilities?format=json-->true
</isSupportSlagTruckCoverPlate>
<isSupportTrainDetection>
<!--ro, opt, bool, whether the device supports train detection, desc:If train detection is not supported, this node will not be returned. Corresponding eventType: trainDetection-->true
</isSupportTrainDetection>
<isNotSupportVideoOverlays>
<!--ro, opt, bool, whether the device does not support configuring OSD overlay on the device web configuration page, desc:this node is only valid for checking whether the OSD overlay configuration of the image menu on the person and vehicle verification station is supported or not (this configuration is supported by default in the protocol and it does not have node control capability). If true is returned, it indicates that the OSD overlay is not supported; if false or no node is returned, it indicates the OSD overlay function is supported-->true
</isNotSupportVideoOverlays>
<isSupportScheduledCaptureTask>
<!--ro, opt, bool, whether the device supports scheduled capturing tasks, desc:related URI: /ISAPI/Sytem/Video/scheduledCapture/capabilities-->true
</isSupportScheduledCaptureTask>
<isSupportGuideCfg>
<!--ro, opt, bool, whether the device supports the wizard configuration, desc:this node is only valid for the indoor station to check whether to enable wizard configuration, which is provided for the users to perform basic function configuration after activating and logging in to the device (including Internet parameters configuration, dialog configuration and batch adding analog indoor stations). If true is returned, it indicates wizard configuration is supported; if false or no node is returned, it indicates the function is not supported-->true
</isSupportGuideCfg>
<isSupportCameraGroupPeopleCounting>
<!--ro, opt, bool, whether the device supports passenger flow statistics of multiple cameras, desc:/ISAPI/Intelligent/counting/cameraGroup/capabilities?format=json-->true
</isSupportCameraGroupPeopleCounting>
<isSupportAngleDeviationDetection>
<!--ro, opt, bool, whether the device supports tilting detection, desc:related URL:

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/ISAPI/Event/channels/<channelID>/angleDeviationDetection/capabilities?format=json; this node is not returned if not supported-->true
  </isSupportAngleDeviationDetection>
  <isSupportCityCode>
    <!--ro, opt, bool, whether the device supports getting city code, desc:related URI: /ISAPI/System/cityCode?format=json-->true
  </isSupportCityCode>
  <isSupportAlcoholDetectionResult>
    <!--ro, opt, bool, whether the device supports uploading alcohol test, desc:eventType: alcoholDetectionResult-->true
  </isSupportAlcoholDetectionResult>
  <isSupportAlcoholDetectionRodBind>
    <!--ro, opt, bool, whether the device supports the request of binding alcohol test stick, desc:eventType: alcoholDetectionRodBind-->true
  </isSupportAlcoholDetectionRodBind>
  <isSupportAlcoholDetectionRodBindQuery>
    <!--ro, opt, bool, whether the device supports the query for the request of binding alcohol test stick, desc:eventType: alcoholDetectionRodBindQuery--
>true
  </isSupportAlcoholDetectionRodBindQuery>
  <isSupportAlcoholDetectionRodUnbind>
    <!--ro, opt, bool, whether the device supports the event of unbinding alcohol test stick, desc:eventType: alcoholDetectionRodUnbind-->true
  </isSupportAlcoholDetectionRodUnbind>
  <isSupportAbnormalDriving>
    <!--ro, opt, bool, whether the device supports abnormal driving behaviors detection, desc:eventType: abnormalDriving-->true
  </isSupportAbnormalDriving>
  <isSupportEmanationsEvent>
    <!--ro, opt, bool, whether the device supports the event of emanation test, desc:eventType: emanationsEvent; the security inspection system is equipped
with an emanation tester, and when the emanation exceeds the threshold, an alarm event will be uploaded. This event is only triggered by the emanation
tester, and its configuration is not required-->true
  </isSupportEmanationsEvent>
  <isSupportClientUpdatePackage>
    <!--ro, opt, bool, whether the device supports the client upgrade package management, desc:related URL: /ISAPI/System/clientUpdatePackage/capabilities?
format=json-->true
  </isSupportClientUpdatePackage>
  <isSupportDeviceStatusUploadStrategy>
    <!--ro, opt, bool, whether the device supports configuring the parameters of device status uploading strategy, desc:related URL:
/ISAPI/System/status/uploadStrategy/capabilities?format=json-->true
  </isSupportDeviceStatusUploadStrategy>
  <isSupportDialIndicatorReading>
    <!--ro, opt, bool, whether the devices supports dial indicator reading, desc:related URL:
/ISAPI/Intelligent/channels/<channelID>/dialIndicatorReading/capabilities?format=json; the function refers to reading single pointer dial and uploading the
result (currently, it can only recognize dials that do not accumulate the readings such as thermometer and barometer)-->true
  </isSupportDialIndicatorReading>
  <isNotSupportAudioVideoCompress>
    <!--ro, opt, bool, whether the device does not support audio/video encoding, desc:this node is only valid for the person and vehicle verification
station; if true is returned, the video/audio page will not be displayed on the device web page. If false or no node is returned, the video/audio page will
be displayed on the device web page by default-->true
  </isNotSupportAudioVideoCompress>
  <isSupportUPLCExceptionEvent>
    <!--ro, opt, bool, whether the device supports the event of liquid chromatograph system exception, desc:eventType: UPLCException-->true
  </isSupportUPLCExceptionEvent>
  <isSupportUrgentTurn>
    <!--ro, opt, bool, whether the device supports the sharp turn event, desc:eventType: urgentTurn-->true
  </isSupportUrgentTurn>
  <isSupportUrgentBrake>
    <!--ro, opt, bool, whether the device supports the sudden brake event, desc:eventType: urgentBrake-->true
  </isSupportUrgentBrake>
  <isSupportUrgentAcceleration>
    <!--ro, opt, bool, whether the device supports the rapid acceleration event, desc:eventType: urgentAcceleration-->true
  </isSupportUrgentAcceleration>
  <isSupportVehiclePanicButton>
    <!--ro, opt, bool, whether the device supports the alarm/event of mobile emergency button, desc:eventType: vehiclePanicButton-->true
  </isSupportVehiclePanicButton>
  <isSupportOverSpeed>
    <!--ro, opt, bool, whether the device supports the vehicle speeding alarm/event, desc:eventType: overSpeed-->true
  </isSupportOverSpeed>
  <isSupportAbnormalAcceleration>
    <!--ro, opt, bool, whether the device supports the event of abnormal vehicle custom acceleration alarm, desc:eventType: abnormalAcceleration-->true
  </isSupportAbnormalAcceleration>
  <isSupportRollover>
    <!--ro, opt, bool, whether the device supports the vehicle turnover alarm/event, desc:eventType: rollover-->true
  </isSupportRollover>
  <isSupportCollision>
    <!--ro, opt, bool, whether the device supports the vehicle collision alarm/event, desc:eventType: collision-->true
  </isSupportCollision>
  <isSupportVehicleOBUInfo>
    <!--ro, opt, bool, whether the device supports uploading the vehicle OBU information, desc:eventType: vehicleOBUInfo-->true
  </isSupportVehicleOBUInfo>
  <isSupportPowerUpPicture>
    <!--ro, opt, bool, whether the device supports importing and exporting pictures displayed during startup,
desc:/ISAPI/System/powerUpPicture/capabilities?format=json-->true
  </isSupportPowerUpPicture>
  <isSupportPowerUpPictureCfg>
    <!--ro, opt, bool, whether the device supports configuring parameters of pictures displayed during startup,
desc:/ISAPI/System/powerUpPictureCfg/capabilities?format=json-->true
  </isSupportPowerUpPictureCfg>
  <isSupportAlgoPackageScheduling>
    <!--ro, opt, bool, whether the device supports algorithm package scheduling, desc:Related URIs:  /ISAPI/System/AlgoPackageScheduling/capabilities?
format=json and /ISAPI/System/AlgoPackageScheduling/task/capabilities?format=json-->true
  </isSupportAlgoPackageScheduling>
  <isSupportVoiceTalkRequestingAlarm>
    <!--ro, opt, bool, whether the device supports the two-way audio request alarm, desc:eventType: voiceTalkRequestingAlarm-->true
  </isSupportVoiceTalkRequestingAlarm>
  <isSupportSetupStructure>
    <!--ro, opt, bool, whether the device supports configuring construction structure parameters, desc:related URI: /ISAPI/System/setupStructure?
format=json-->true
  </isSupportSetupStructure>
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  <isSupportPassNumInfoEvent>
    <!--ro, opt, bool, whether the device supports uploading the number of people passed, desc:eventType: passNumInfoEvent-->true
  </isSupportPassNumInfoEvent>
  <isShowClientRemote>
    <!--ro, opt, bool, whether the device supports displaying remote configuration on the client,
desc:1. If this node is not returned or it is returned and its value is true, it indicates that the device supports this function.
2. If this node is returned and its value is false, it indicates that the device does not support this function.
iVMS-4200 client software supports remote configuration by default. However, there are still some devices that do not support remote configuration via iVMS-4200 and they can only be configured remotely via the web browser. The capability determined by this node is used to control whether to display the remote configuration on iVMS-4200 client software-->true
  </isShowClientRemote>
  <isSupportUserManualQRCode>
    <!--ro, opt, bool, whether the device supports getting the QR code of the user manual, desc:related URI: /ISAPI/System/userManualQRCode/capabilities?format=json-->true
  </isSupportUserManualQRCode>
  <isSupportUnclosedFrontPanel>
    <!--ro, opt, bool, whether the device supports uploading the event of the front panel not closed, desc:related URI for getting the capability: /ISAPI/Trial/roompara/capabilities; corresponding node: < unclosedFrontPanelEnabled>. Related event: unclosedFrontPanel-->true
  </isSupportUnclosedFrontPanel>
  <isSupportVehicleWeight>
    <!--ro, opt, bool, whether the device supports configuring parameters for vehicle weight and axle detection, desc:/ISAPI/Traffic/channels/<channelID>/vehicleWeight?format=json-->true
  </isSupportVehicleWeight>
  <isSupportWeightInterface>
    <!--ro, opt, bool, whether it supports integrated overload control parameters, desc:/ISAPI/Traffic/ContentMgmt/WeightInterface?format=json-->true
  </isSupportWeightInterface>
  <isSupportSecurityPanicEvent>
    <!--ro, opt, bool, whether the device supports emergency events of security inspection, desc:eventType: securityPanicEvent-->true
  </isSupportSecurityPanicEvent>
  <isSupportTrafficLightMalfunction>
    <!--ro, opt, bool, whether the device supports the event of detecting faults of traffic lights, desc:eventType: trafficLightMalfunction-->true
  </isSupportTrafficLightMalfunction>
  <isSupportTOFShelterAlarm>
    <!--ro, opt, bool, whether the device supports TOF tampering alarm, desc:/ISAPI/Event/channels/<channelID>/TOFShelterAlarm/capabilities?format=json-->true
  </isSupportTOFShelterAlarm>
  <isSupportSetMealQuery>
    <!--ro, opt, bool, whether the device supports uploading the event of searching set meals, desc:eventType: SetMealQuery-->true
  </isSupportSetMealQuery>
  <isSupportMultiScreenControlAlarm>
    <!--ro, opt, bool, whether the device supports uploading the multi-screen controller alarm, desc:eventType: multiScreenControlAlarm-->true
  </isSupportMultiScreenControlAlarm>
  <isSupportKeyFrameLoss>
    <!--ro, opt, bool, whether the device supports uploading the event of key frame loss, desc:related URI: /ISAPI/Event/keyFrameLoss/capabilities?format=json; eventType: keyFrameLoss.-->true
  </isSupportKeyFrameLoss>
  <isSupportWorkStatusCfg>
    <!--ro, opt, bool, whether the device supports working status configuration, desc:/ISAPI/System/WorkStatusCfg?format=json-->true
  </isSupportWorkStatusCfg>
  <isSupportHealthInfoSyncQuery>
    <!--ro, opt, bool, whether the device supports the event of searching for health information, desc:eventType: HealthInfoSyncQuery-->true
  </isSupportHealthInfoSyncQuery>
  <isSupportVisitorPictureSearchEvent>
    <!--ro, opt, bool, whether the device supports the event of searching for visitor profiles, desc:eventType: VisitorPictureSearchEvent-->true
  </isSupportVisitorPictureSearchEvent>
  <isSupportEZVIZParameterTest>
    <!--ro, opt, bool, whether the device supports testing EZ parameters, desc:related URI: /ISAPI/System/Network/EZVIZ/parameterTest?format=json-->true
  </isSupportEZVIZParameterTest>
  <isSupportUnattendedPackageEvent>
    <!--ro, opt, bool, whether the device supports the unattended package event, desc:eventType: unattendedPackageEvent-->true
  </isSupportUnattendedPackageEvent>
  <isSupportPackageFragmentEvent>
    <!--ro, opt, bool, whether the device supports the event of uploading package fragment data, desc:eventType: packageFragmentEvent-->true
  </isSupportPackageFragmentEvent>
  <isSupportPackageBarcodeMismatchEvent>
    <!--ro, opt, bool, whether the device supports the event of package and barcode mismatching, desc:eventType: packageBarcodeMismatchEvent-->true
  </isSupportPackageBarcodeMismatchEvent>
  <isSupportCheckingModeSwitchEvent>
    <!--ro, opt, bool, whether the device supports the event of switching the image checking mode, desc:eventType: checkingModeSwitchEvent-->true
  </isSupportCheckingModeSwitchEvent>
  <FaceSnapPrivacyMask>
    <!--ro, opt, object, whether the device supports privacy mask on faces, desc:based on face picture capture, all faces will be masked to protect the privacy. Related URI: /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
    </ChannelsList>
  </FaceSnapPrivacyMask>
  <FaceContrastPrivacyMask>
    <!--ro, opt, object, whether the device supports privacy mask on faces of key persons, desc:based on face picture comparison, the faces of key persons will be masked to protect the privacy. Related URI: /ISAPI/Intelligent/channels/<channelID>/faceContrast/privacyMask/capabilities?format=json-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
    </ChannelsList>
  </FaceContrastPrivacyMask>
  <isSupportHumanBodyComparison>
    <!--ro, opt, bool, whether the device supports human body comparison events, desc:related URI: /ISAPI/Event/channels/<channelID>/humanBodyComparison/capabilities?format=json; eventType: humanBodyComparison-->true
  </isSupportHumanBodyComparison>
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  <isSupportMedicalInfoSearchEvent>
    <!--ro, opt, bool, whether the device supports the event of searching for medical information, desc:medicalInfoSearchEvent-->true
  </isSupportMedicalInfoSearchEvent>
  <isSupportConsumptionStatusQuery>
    <!--ro, opt, bool, whether the device supports uploading the request for searching for consumption status, desc:eventType: ConsumptionStatusQuery-->true
  </isSupportConsumptionStatusQuery>
  <isSupportShipChannelAbnormal>
    <!--ro, opt, bool, whether the device supports uploading ship channel exception detection event, desc:eventType: shipChannelAbnormal-->true
  </isSupportShipChannelAbnormal>
  <isSupportDepositRetrieveEvent>
    <!--ro, opt, bool, whether the device supports the event of object access in the intelligent cabinet-->true
  </isSupportDepositRetrieveEvent>
  <isSupportDepositTimeoutEvent>
    <!--ro, opt, bool, whether the device supports the event of overtime object deposit in the intelligent cabinet-->true
  </isSupportDepositTimeoutEvent>
  <isSupportCabinetStatusEvent>
    <!--ro, opt, bool, whether the device supports the event of intelligent cabinet's changed status-->true
  </isSupportCabinetStatusEvent>
  <isSupportCabinetAuthorityExpired>
    <!--ro, opt, bool, whether the device supports the event of intelligent cabinet's permission expiration-->true
  </isSupportCabinetAuthorityExpired>
  <isSupportT1Test>
    <!--ro, opt, bool, whether the device supports the configuration of controlling T1 tool-->true
  </isSupportT1Test>
  <isSupportDrivingDirectionAtIntersection>
    <!--ro, opt, bool, whether the device supports intersection driving direction detection, desc:eventType: drivingDirectionAtIntersection; this node is
not returned if not supported-->true
  </isSupportDrivingDirectionAtIntersection>
  <isSupportPlaceSceneExecuteStatus>
    <!--ro, opt, bool, whether the device supports the event of scene execution status, desc:eventType: placeSceneExecuteStatus; this node is not returned
if not supported-->true
  </isSupportPlaceSceneExecuteStatus>
  <isSupportSceneRecordingCtrl>
    <!--ro, opt, bool, whether the device supports the event of controlling scene recording, desc:eventType: sceneRecordingCtrl; this node is not returned
if not supported-->true
  </isSupportSceneRecordingCtrl>
  <isSupportStreamRecordStatus>
    <!--ro, opt, bool, whether the device supports getting the stream ID record status, desc:/ISAPI/System/Video/inputs/streams/recordStatus-->true
  </isSupportStreamRecordStatus>
  <isSupportManualSnapPicture>
    <!--ro, opt, bool, whether the device supports manual picture capturing, desc:related URI:
/ISAPI/Streaming/channels/<trackStreamID>/picture/capabilities?format=json-->true
  </isSupportManualSnapPicture>
  <isSupportVerificationResultEvent>
    <!--ro, opt, bool, whether the device supports the event of human and vehicle verification result, desc:eventType: verificationResultEvent-->true
  </isSupportVerificationResultEvent>
  <OTAP>
    <!--ro, opt, object, the capability of Open Things Access Protocol, desc:OTAP capability-->
    <isSupportOTAPModel>
      <!--ro, opt, bool, whether the device supports getting the virtual No. and device No. of the OTAP, desc:GET /ISAPI/System/Network/OTAP/Model?
format=json-->true
    </isSupportOTAPModel>
  </OTAP>
  <isSupportForeignObjectInRoadDetection>
    <!--ro, opt, bool, whether the device supports the detection of foreign object in road, desc:eventType: foreignObjectInRoadDetection-->true
  </isSupportForeignObjectInRoadDetection>
  <isSupportUPLCMethodOperatingStatus>
    <!--ro, opt, bool, whether the device supports the event of UPLC method operating status, desc:eventType: UPLCMethodOperatingStatus-->true
  </isSupportUPLCMethodOperatingStatus>
  <isSupportMaterialScanTaskStatus>
    <!--ro, opt, bool, whether the device supports material scanning task status, desc:eventType: materialScanTaskStatus-->true
  </isSupportMaterialScanTaskStatus>
  <isSupportSearchInputProxy>
    <!--ro, opt, bool, whether the device supports searching for accessible IPC of device in the LAN, desc:related URI: GET
/ISAPI/ContentMgmt/InputProxy/search?security=<security>&iv=<iv>; true (support), this node is not returned (call the specific API to confirm whether it
supports or not)-->true
  </isSupportSearchInputProxy>
  <isSupportPostureDetection>
    <!--ro, opt, bool, whether the device supports people posture detection, desc:eventType: MeetingLock-->true
  </isSupportPostureDetection>
  <isSupportFans>
    <!--ro, opt, bool, whether the device supports  configuring fan, desc:/ISAPI/System/Fans?format=json-->true
  </isSupportFans>
  <pickUpCalibration>
    <!--ro, opt, object, whether the device supports wireless pickup calibration, desc:related URI:
/ISAPI/System/PickUpCalibration/channels/<channelID>/capabilities?format=json-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
    </ChannelsList>
  </pickUpCalibration>
  <isSupportSerialLogUpload>
    <!--ro, opt, bool, whether the device supports uploading serial port log, desc:true (support), false (not support)-->true
  </isSupportSerialLogUpload>
  <isSupportLiquidDetectionEvent>
    <!--ro, opt, bool, whether the device supports liquid detection, desc:eventType: liquidDetectionEvent-->true
  </isSupportLiquidDetectionEvent>
  <isSupportETCRSU>
    <!--ro, opt, bool, whether the device supports  ETCRSU configuration, desc:related URI: /ISAPI/Traffic/ETCRSU level protocol-->true
  </isSupportETCRSU>
  <isSupportVideoCodeConvertTool>
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    <!--ro, opt, bool, whether the device supports getting the video encoding conversion tool, desc:related URI: /ISAPI/System/videoCodeConvertTool?
format=json protocol-->true
  </isSupportVideoCodeConvertTool>
  <supportQuickConfigType opt="icr,dayNightGate,syncSignalOutput">
    <!--ro, opt, string, the list of supported functions of the quick configuration pop-up on the web, attr:opt{opt, string},
desc:the web quick configuration does not distinguish the configuration order, and it displays all supported function lists on the pop-up on the tab page
    icr refers to ICR switch configuration (related URL: /ISAPI/Image/channels/<channelID>/); icr
   dayNightGate refers to day and night mode switching (related URL: /ISAPI/Image/channels/<channelID>/dayNightGate);
    IOOutputs refers to synchronous signal output (related URL: /ISAPI/System/syncSignalOutput)-->syncSignalOutput
  </supportQuickConfigType>
  <isSupportVisualLocation>
    <!--ro, opt, bool, whether the device supports configuring visual location service, desc:related URI:
/ISAPI/System/LocationMgr/visualLocation/capabilities?format=json-->true
  </isSupportVisualLocation>
  <isSupportVideoAccessAbnormal>
    <!--ro, opt, bool, whether the device supports abnormal video access, desc:eventType: VideoAccessAbnormal; this node is not returned if not supported--
>true
  </isSupportVideoAccessAbnormal>
  <isSupportSubSystemAbnormal>
    <!--ro, opt, bool, whether the device supports abnormal subsystem, desc:eventType: SubSystemAbnormal; this node is not returned if not supported-->true
  </isSupportSubSystemAbnormal>
  <isSupportSearchFileList>
    <!--ro, opt, bool, whether the device supports searching for list of device files, desc:related URI: /ISAPI/System/fileExport/SearchFileList?
format=json-->true
  </isSupportSearchFileList>
  <isSupportDistanceDetectionEvent>
    <!--ro, opt, bool, whether the device supports distance detection, desc:eventType: distanceDetectionEvent; currently it is only used for devices test
tools-->true
  </isSupportDistanceDetectionEvent>
  <isSupportMeetingQueryEvent>
    <!--ro, opt, bool, whether the device supports searching for meetings, desc:eventType: MeetingElementCancel-->true
  </isSupportMeetingQueryEvent>
  <isSupportMeetingLockEvent>
    <!--ro, opt, bool, whether the device supports locking meetings, desc:eventType: MeetingLock-->true
  </isSupportMeetingLockEvent>
  <isSupportMeetingAddEvent>
    <!--ro, opt, bool, whether the device supports reserving meetings, desc:eventType: MeetingAdd-->true
  </isSupportMeetingAddEvent>
  <isSupportMeetingModifyEvent>
    <!--ro, opt, bool, whether the device supports modifying meetings, desc:eventType: MeetingModify-->true
  </isSupportMeetingModifyEvent>
  <isSupportMeetingCancelEvent>
    <!--ro, opt, bool, whether the device supports canceling meetings, desc:eventType: ScreenBindIPC-->true
  </isSupportMeetingCancelEvent>
  <isSupportScreenBindIPCEvent>
    <!--ro, opt, bool, whether the device supports the screen binding IPC, desc:eventType: ScreenBindIPC-->true
  </isSupportScreenBindIPCEvent>
  <isSupportSensorDetection>
    <!--ro, opt, bool, whether the device supports uploading sensor data, desc:eventType: sensorDetection-->true
  </isSupportSensorDetection>
  <isSupportClearLogCache>
    <!--ro, opt, bool, whether the device supports clearing the log cache, desc:related URI: /ISAPI/ContentMgmt/ClearLogCache?format=json-->true
  </isSupportClearLogCache>
  <isSupportRoadOccupyDetection>
    <!--ro, opt, bool, whether the device supports the entrance and exit block detection, desc:related URI:
/ISAPI/ITC/Entrance/channels/<channelID>/roadOccupyDetection/capabilities?format=json-->true
  </isSupportRoadOccupyDetection>
  <isSupportMeetingRoomSignInEvent>
    <!--ro, opt, bool, whether the device supports the MeetingRoomSignIn event, desc:eventType: MeetingRoomSignIn-->true
  </isSupportMeetingRoomSignInEvent>
  <isSupportMeetingRoomSignOutEvent>
    <!--ro, opt, bool, whether the device supports the MeetingRoomSignOut event, desc:eventType: MeetingRoomSignOut-->true
  </isSupportMeetingRoomSignOutEvent>
  <isSupportTimeTypeSTD>
    <!--ro, opt, bool, whether the device supports the time type of ISO8601, desc:if "true" is returned, it indicates that all the devices containing
ISO8601 parameters support adding the timeType=STD, such as /ISAPI/ContentMgmt/search?timeType=STD; the time will be implemented according to the standard--
>true
  </isSupportTimeTypeSTD>
  <isSupportScanner>
    <!--ro, opt, bool, whether the device supports scanning events and configuration, desc:if true is returned, it indicates the device supports the
/ISAPI/System/scanner/capabilities?format=json and scannerInfo event-->true
  </isSupportScanner>
  <isSupportRegionTargetNumberCounting>
    <!--ro, opt, bool, whether the device supports the region target counting, desc:related URI:
/ISAPI/Event/channels/<channelID>/RegionTargetNumberCounting/Capabilities?format=json-->true
  </isSupportRegionTargetNumberCounting>
  <depthMapMaintainMode>
    <!--ro, opt, object, the maintenance mode of depth image, desc:it will influence the smart functions when using depth images for tests, such as vertical
customer traffic, object accumulation detection and so on enabling the maintenance mode will change the audio and video parameters of the channels related
URI: /ISAPI/Intelligent/channels/<channelID>/DepthMapMaintainModeCfg/capabilities?format=json-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
      <channelID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelID>
    </ChannelsList>
  </depthMapMaintainMode>
  <isSupportExportPipCalibrationFile>
    <!--ro, opt, bool, whether it supports exporting visual and thermal image fusion calibration file, desc:related URI:
/ISAPI/System/ExportPipCalibrationFile?format=json-->true
  </isSupportExportPipCalibrationFile>
  <isSupportCourseUploadInfo>
    <!--ro, opt, bool, whether it supports notifying the course uploading completion, desc:eventType: courseUploadInfo-->true
  </isSupportCourseUploadInfo>
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<isSupportPresetSwitchingInformation>
<!--ro, opt, bool, whether the device supports uploading the preset switching information, desc:eventType: presetSwitchingInformation-->true
</isSupportPresetSwitchingInformation>
<isSupportCabinetChargingQuery>
<!--ro, opt, bool, whether it supports the request of searching for the charging of intelligent cabinet, desc:eventType: cabinetChargingQuery-->true
</isSupportCabinetChargingQuery>
<isSupportSearchVoltageAndPowerConsumptionReport>
<!--ro, opt, bool, whether the device supports searching for voltage and consumption report, desc:related URI:
/ISAPI/System/SearchVoltageAndPowerConsumptionReport/capabilities?format=json-->true
</isSupportSearchVoltageAndPowerConsumptionReport>
<isSupportDeviceDrop>
<!--ro, opt, bool, whether the device supports the deviceDrop event, desc:corresponding event: deviceDrop-->true
</isSupportDeviceDrop>
<isSupportDeviceNotMoved>
<!--ro, opt, bool, whether the device supports the deviceNotMoved event, desc:corresponding event: deviceNotMoved-->true
</isSupportDeviceNotMoved>
<isSupportWalkWrongWayOnEscalator>
<!--ro, opt, bool, whether the device supports the walkWrongWayOnEscalator event, desc:corresponding event: walkWrongWayOnEscalator-->true
</isSupportWalkWrongWayOnEscalator>
<isSupportFallOnEscalator>
<!--ro, opt, bool, whether the device supports the fallOnEscalator event, desc:corresponding event: fallOnEscalator-->true
</isSupportFallOnEscalator>
<isSupportCarryLargeLuggageOnEscalator>
<!--ro, opt, bool, whether the device supports the carryLargeLuggageOnEscalator event, desc:corresponding event: carryLargeLuggageOnEscalator-->true
</isSupportCarryLargeLuggageOnEscalator>
<isSupportPushStrollerOnEscalator>
<!--ro, opt, bool, whether the device supports the pushStrollerOnEscalator event, desc:corresponding event: pushStrollerOnEscalator-->true
</isSupportPushStrollerOnEscalator>
<isSupportTeachingSpeechDetection>
<!--ro, opt, bool, whether it supports teachingSpeechDetection event, desc:corresponding event: teachingSpeechDetection-->true
</isSupportTeachingSpeechDetection>
<updatefirmwareType opt="warningScreenFirmware,warningScreenFont,warningScreenVoice,ledReceiverCard">
<!--ro, opt, string, RS-485 working modes supported by the device, attr:opt{opt, string}, desc:corresponding to <type> in /ISAPI/System/updateFirmware?
type=<type>; "warningScreenFirmware", "warningScreenFont", "warningScreenVoice"; note that <isSupportUpdatefirmware> is supported when this node is
supported-->warningScreenFirmware
</updatefirmwareType>
<isSupportPaymentQRCodeQuery>
<!--ro, opt, bool, whether the device supports the paymentQRCodeQuery event, desc:corresponding event: paymentQRCodeQuery-->true
</isSupportPaymentQRCodeQuery>
<isSupportTopUpEvent>
<!--ro, opt, bool, whether the device supports the topUpEvent event, desc:corresponding event: topUpEvent-->true
</isSupportTopUpEvent>
<isSupportTopUpRequest>
<!--ro, opt, bool, whether the device supports the topUpRequest event, desc:corresponding event: topUpRequest-->true
</isSupportTopUpRequest>
<isSupportTopUpStatusQuery>
<!--ro, opt, bool, whether the device supports the topUpStatusQuery event, desc:corresponding event: topUpStatusQuery-->true
</isSupportTopUpStatusQuery>
<isSupportLogOffRefundRequest>
<!--ro, opt, bool, whether the device supports the logOffRefundRequest event, desc:corresponding event: logOffRefundRequest-->true
</isSupportLogOffRefundRequest>
<isSupportLogOffRefundStatusQuery>
<!--ro, opt, bool, whether the device supports searching for the refund status of logged-off persons, desc:corresponding event: logOffRefundStatusQuery-
->true
</isSupportLogOffRefundStatusQuery>
<isSupportAuthenticationTimeoutEvent>
<!--ro, opt, bool, whether the device supports the logOffRefundStatusQuery event, desc:corresponding event: logOffRefundStatusQuery-->true
</isSupportAuthenticationTimeoutEvent>
<isSupportCertificateCapture>
<!--ro, opt, bool, whether the device supports the certificateCapture events, desc:corresponding event: certificateCapture-->true
</isSupportCertificateCapture>
<isSupportDeviceBatchActivation>
<!--ro, opt, bool, whether the device supports batch activation, desc:POST /ISAPI/System/activate/DeviceBatchActivation?format=json-->true
</isSupportDeviceBatchActivation>
<isSupportPlaceEnvironmentMonitoringRule>
<!--ro, opt, bool, whether the device supports place environment monitoring rule parameters,
desc:/ISAPI/Event/channels/<channelID>/PlaceEnvironmentMonitoringRule?format=json-->true
</isSupportPlaceEnvironmentMonitoringRule>
<isSupportEngineResource>
<!--ro, opt, bool, whether the device supports engine resources, desc:/ISAPI/System/EngineResource?format=json-->true
</isSupportEngineResource>
<isSupportPTModulePowerCfg>
<!--ro, opt, bool, whether the device supports modifying the power supply parameters of PTZ base, desc:PUT /ISAPI/System/ModifyPTModulePowerCfg?
format=json POST /ISAPI/System/GetPTModulePowerCfg?format=json-->true
</isSupportPTModulePowerCfg>
<isSupportGetBoxDoorDetectionCfg>
<!--ro, opt, bool, whether the device supports getting chassis door status detection configuration, desc:POST /ISAPI/System/GetBoxDoorDetectionCfg?
format=json-->true
</isSupportGetBoxDoorDetectionCfg>
<isSupportModifyBoxDoorDetectionCfg>
<!--ro, opt, bool, whether the device supports modifying chassis door status configuration, desc:PUT /ISAPI/System/ModifyBoxDoorDetectionCfg?
format=json-->true
</isSupportModifyBoxDoorDetectionCfg>
<isSupportSmartBoxStatus>
<!--ro, opt, bool, whether the device supports intelligent chassis status, desc:/ISAPI/System/SmartBoxStatus?format=json-->true
</isSupportSmartBoxStatus>
<isSupportBatteryStatusCtrl>
<!--ro, opt, bool, whether the device supports battery status event upload control parameters, desc:/ISAPI/System/BatteryStatusCtrl?format=json-->true
</isSupportBatteryStatusCtrl>
<isSupportCooler>
<!--ro, opt, bool, whether the device supports cooling management, desc:/ISAPI/System/Cooler/-->true
</isSupportCooler>
<isSupportBoxDoorDetectionCfg>
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<!--ro, opt, bool, whether the device supports configuring chassis door status detection, desc:/ISAPI/System/BoxDoorDetectionCfg?format=json-->true
</isSupportBoxDoorDetectionCfg>
<isSupportBoxDoorStatus>
<!--ro, opt, bool, whether the device supports chassis door status, desc:/ISAPI/System/BoxDoorStatus?format=json-->true
</isSupportBoxDoorStatus>
<isSupportLightningProtectionStatus>
<!--ro, opt, bool, whether the device supports lightning protection status, desc:/ISAPI/System/LightningProtectionStatus?format=json-->true
</isSupportLightningProtectionStatus>
<isSupportDeviceVisualLocalizationDetectionEvent>
<!--ro, opt, bool, whether the device supports the DeviceVisualLocalizationDetectionEvent event, desc:corresponding event:
DeviceVisualLocalizationDetectionEvent-->true
</isSupportDeviceVisualLocalizationDetectionEvent>
<isSupportDeviceVisualLocalizationAnalysisEvent>
<!--ro, opt, bool, whether the device supports the DeviceVisualLocalizationAnalysisEvent event, desc:corresponding event:
DeviceVisualLocalizationAnalysisEvent-->true
</isSupportDeviceVisualLocalizationAnalysisEvent>
<isSupportGetPowerPortWorkParamList>
<!--ro, opt, bool, whether the device supports getting the working parameter list of power supply interface, desc:POST
/ISAPI/System/GetPowerPortWorkParamList?format=json-->true
</isSupportGetPowerPortWorkParamList>
<isSupportUsersLoginNotification>
<!--ro, opt, bool, whether the device supports parameters of notifications on user login, desc:/ISAPI/System/UsersLoginNotification?format=json-->true
</isSupportUsersLoginNotification>
<isSupportLiquidLevelDetectorInfo>
<!--ro, opt, bool, whether the device supports uploading information of liquid level detector, desc:eventType: LiquidLevelDetectorInfo-->true
</isSupportLiquidLevelDetectorInfo>
<isSupportWaterPressureDetectorInfo>
<!--ro, opt, bool, whether the device supports uploading information of water pressure detector, desc:eventType: WaterPressureDetectorInfo-->true
</isSupportWaterPressureDetectorInfo>
<isSupportEncryptionFileUpgradeMgr>
<!--ro, opt, bool, whether the device supports advanced management of encrypted files, desc:GET /ISAPI/System/EncryptionFileUpgradeMgr/capabilities?
format=json-->true
</isSupportEncryptionFileUpgradeMgr>
<isSupportInformedConsentTextParam>
<!--ro, opt, bool, whether the device supports text parameters of consent confirmation document,
desc:/ISAPI/System/InformedConsentMgr/InformedConsentTextParam?format=json-->true
</isSupportInformedConsentTextParam>
<isSupportSendConsolePassword>
<!--ro, opt, bool, whether the device supports applying control panel password, desc:POST /ISAPI/System/SendConsolePassword?format=json-->true
</isSupportSendConsolePassword>
<isSupportAutoLaserRanging>
<!--ro, opt, bool, whether to support auto laser ranging parameter configuration, desc:/ISAPI/System/AutoLaserRanging?format=json-->true
</isSupportAutoLaserRanging>
<isSupportDelayedReboot>
<!--ro, opt, bool, whether the device supports delayed reboot after power disconnection, desc:POST /ISAPI/System/DelayedReboot?format=json-->true
</isSupportDelayedReboot>
<isSupportFalseAlarmMaterialLibMgr>
<!--ro, opt, bool, whether the device supports false alarm library management, desc:GET /ISAPI/Intelligent/falseAlarmMaterialLibMgr/capabilities?
format=json-->true
</isSupportFalseAlarmMaterialLibMgr>
<isSupportGetServerRandom>
<!--ro, opt, bool, whether the device supports getting random number of Chinese cryptographic server, desc:POST
/ISAPI/Security/ProtocolCertificate/GetServerRandom?format=json-->true
</isSupportGetServerRandom>
<isSupportCustomPreviewSplitCfg>
<!--ro, opt, bool, whether the device supports custom parameters of divided window in live view, desc:GET/PUT /ISAPI/System/Video/CustomPreviewSplitCfg?
format=json-->true
</isSupportCustomPreviewSplitCfg>
<isSupportSendTelnetTestCmd>
<!--ro, opt, bool, whether the device supports applying telnet testing command, desc:POST /ISAPI/System/Network/NetworkAdvancedCfg/SendTelnetTestCmd?
format=json-->true
</isSupportSendTelnetTestCmd>
<isSupportTelnetTestCmdStatus>
<!--ro, opt, bool, whether the device supports the execution status of telnet testing command, desc:GET
/ISAPI/System/Network/NetworkAdvancedCfg/TelnetTestCmdStatus?format=json-->true
</isSupportTelnetTestCmdStatus>
<isSupportImportUSBKeyAuthInfo>
<!--ro, opt, bool, whether the device supports Chinese cryptographic USBKey authentication information, desc:POST
/ISAPI/Security/ProtocolCertificate/ImportUSBKeyAuthInfo?format=json-->true
</isSupportImportUSBKeyAuthInfo>
<isSupportSendSupervisionRecordCollection>
<!--ro, opt, bool, whether the device supports sending completing status of supervision recording replenishment, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SendSupervisionRecordCollection?format=json-->true
</isSupportSendSupervisionRecordCollection>
<isSupportSearchPersonnelOperationSupervisionTask>
<!--ro, opt, bool, whether it supports searching for operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SearchPersonnelOperationSupervisionTask?format=json-->true
</isSupportSearchPersonnelOperationSupervisionTask>
<isSupportDeletePersonnelOperationSupervisionTask>
<!--ro, opt, bool, whether it supports deleting operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/DeletePersonnelOperationSupervisionTask?format=json-->true
</isSupportDeletePersonnelOperationSupervisionTask>
<isSupportAddPersonnelOperationSupervisionTask>
<!--ro, opt, bool, whether it supports adding operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/AddPersonnelOperationSupervisionTask?format=json-->true
</isSupportAddPersonnelOperationSupervisionTask>
<isSupportUserLoginStatusChangeEvent>
<!--ro, opt, bool, whether the device supports the UserLoginStatusChangeEvent event, desc:eventType: UserLoginStatusChangeEvent-->true
</isSupportUserLoginStatusChangeEvent>
<isSupportObstacleAvoidance>
<!--ro, opt, bool, whether it supports obstacle avoidance management, desc:GET/PUT /ISAPI/System/ObstacleAvoidance/-->true
</isSupportObstacleAvoidance>
<isSupportManualSnapFishEyeWallPanoramaView>
<!--ro, opt, bool, whether the device supports manually capturing expanded panorama images by wall-mounted fisheye, desc:GET

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    <!--ro, opt, bool, whether the device supports manually capturing expanded panorama images by wall-mounted fisheye, desc:GET
/ISAPI/Streaming/channels/<channelID>/ManualSnapFishEyeWallPanoramaView?format=json-->true
  </isSupportManualSnapFishEyeWallPanoramaView>
  <isSupportPreviewSplitSwitchCfg>
    <!--ro, opt, bool, whether the device supports custom parameters of divided window switch, desc:GET/PUT /ISAPI/System/Video/PreviewSplitSwitchCfg?
format=json-->true
  </isSupportPreviewSplitSwitchCfg>
  <isSupportCountingTargetParam>
    <!--ro, opt, bool, whether the device supports target parameters of people counting, desc:GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/counting/CountingTargetParam?format=json-->true
  </isSupportCountingTargetParam>
  <isSupportDebugLogModuleType opt="playService,communicationService,attendanceService,faceService,LCD">
    <!--ro, opt, bool, whether the device supports exporting debug log by module, attr:opt{req, string}-->true
  </isSupportDebugLogModuleType>
  <isSupportLoginPasswordCfg>
    <!--ro, opt, bool, whether the device supports login password parameters, desc:GET/PUT /ISAPI/Security/users/LoginPasswordCfg?format=json-->true
  </isSupportLoginPasswordCfg>
  <isSupportUserDataFile>
    <!--ro, opt, bool, whether the device supports importing and exporting the user data file, desc:POST /ISAPI/Security/users/UserDataFile/capabilities?
format=json-->true
  </isSupportUserDataFile>
  <isSupportNtripAccessCfg>
    <!--ro, opt, bool, whether the device supports NTRIP protocol access parameters, desc:GET/PUT /ISAPI/System/Network/Ntrip/NtripAccessCfg?format=json--
>true
  </isSupportNtripAccessCfg>
  <isSupportNetworkWatchdogParams>
    <!--ro, opt, bool, whether the device supports parameters of watchdog for network performance, desc:GET/PUT /ISAPI/System/NetworkWatchdogParams?
format=json-->true
  </isSupportNetworkWatchdogParams>
  <isSupportAudioRecordPlan>
    <!--ro, opt, bool, whether it supports recording schedule, desc:GET/PUT /ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan?format=json-->true
  </isSupportAudioRecordPlan>
  <isSupportStartVoiceMonitor>
    <!--ro, opt, bool, whether it supports enabling audio listening, desc:POST
/ISAPI/System/TwoWayAudio/channels/channels/<twoWayAudioChannelID>/StartVoiceMonitor?format=json-->true
  </isSupportStartVoiceMonitor>
  <isSupportStopVoiceMonitor>
    <!--ro, opt, bool, whether it disabling audio listening-->true
  </isSupportStopVoiceMonitor>
  <voiceMonitorChannelCap>
    <!--ro, opt, object-->
    <audioIDList>
      <!--ro, opt, array, camera list, subType:object-->
      <audioID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </audioID>
    </audioIDList>
  </voiceMonitorChannelCap>
  <isSupportWebPTZCfg>
    <!--ro, opt, bool, whether it supports displaying the web page for PTZ configuration, desc:this node is to determine whether to display the web page for
PTZ configuration and is irrelevant to whether PTZ is supported; default process logic: when this node does not exist, whether to display the web page for
PTZ configuration is subject to the device type-->true
  </isSupportWebPTZCfg>
  <isSupportGaugeReadingEvent>
    <!--ro, opt, bool, whether the device supports the GaugeReadingEvent event, desc:eventType: GaugeReadingEvent-->true
  </isSupportGaugeReadingEvent>
  <isSupportVNCServerCfg>
    <!--ro, opt, bool, whether it supports VNC service configuration, desc:GET/PUT /ISAPI/System/VNCServerCfg?format=json-->true
  </isSupportVNCServerCfg>
  <isSupportResources>
    <!--ro, opt, bool, whether it supports resources, desc:GET /ISAPI/System/DynamicResource/Resources?format=json, Resources-->true
  </isSupportResources>
  <isSupportDeviceGroupMgr>
    <!--ro, opt, bool, whether it supports device application resources group management, desc:GET /ISAPI/System/DeviceGroupMgr/capabilities?format=json--
>true
  </isSupportDeviceGroupMgr>
  <isSupportAsyncResetSubDevicePassword>
    <!--ro, opt, bool, whether the device supports resetting the password of sub-device asynchronously,
desc:/ISAPI/Security/resetPassword/AsyncResetSubDevicePassword?format=json-->true
  </isSupportAsyncResetSubDevicePassword>
  <isSupportAsyncResetSubDevicePasswordStatusList>
    <!--ro, opt, bool, whether the device supports the status list of asynchronously resetting sub-devices' passwords,
desc:/ISAPI/Security/resetPassword/AsyncResetSubDevicePasswordStatusList?format=json-->true
  </isSupportAsyncResetSubDevicePasswordStatusList>
  <isSupportEDRAccessList>
    <!--ro, opt, bool, whether the device supports accessing address parameters of EDR service, desc:GET/PUT /ISAPI/System/Network/EDRAccessList?
format=json-->true
  </isSupportEDRAccessList>
  <isSupportBatteryCarInfoEvent>
    <!--ro, opt, bool, whether to support the BatteryCarInfoEvent event, desc:eventType: BatteryCarInfoEvent-->true
  </isSupportBatteryCarInfoEvent>
  <isSupportChildmanage>
    <!--ro, opt, bool, whether the device supports gateway sub-device topology management, desc:GET /ISAPI/IoTGateway/Childmanage/capabilities?format=json--
>true
  </isSupportChildmanage>
  <isSupportSearchResources>
    <!--ro, opt, bool, whether the device supports searching the resource list, desc:POST /ISAPI/System/DynamicResource/SearchResources?format=json-->true
  </isSupportSearchResources>
  <isSupportResetPasswordPhoneNo>
    <!--ro, opt, bool, whether the device supports the phone No. for resetting password, desc:GET/PUT /ISAPI/Security/resetPassword/ResetPasswordPhoneNo?
format=json-->true
  </isSupportResetPasswordPhoneNo>
  <isSupportGetPhoneNoQRCode>
    <!--ro, opt, bool, whether the device supports getting the QR code to be scanned by phone for resetting password, desc:GET
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/ISAPI/Security/resetPassword/GetPhoneNoQRCode?format=json-->true
  </isSupportGetPhoneNoQRCode>
  <isSupportResetPasswordByPhoneNoSecurityCode>
    <!--ro, opt, bool, whether it supports resetting device password with phone number verification, desc:PUT
/ISAPI/Security/resetPassword/ResetPasswordByPhoneNoSecurityCode?format=json-->true
  </isSupportResetPasswordByPhoneNoSecurityCode>
  <isSupportGaugeReadingOverlapParams>
    <!--ro, opt, bool, whether the device supports water level gauge detection information overlay parameters, desc:GET/PUT
/ISAPI/Event/gaugeReading/channels/<channelID>/GaugeReadingOverlapParams?format=json-->true
  </isSupportGaugeReadingOverlapParams>
  <isSupportGaugeReadingRainfallEvent>
    <!--ro, opt, bool, whether the device supports the GaugeReadingRainfallEvent event, desc:eventType: GaugeReadingRainfallEvent-->true
  </isSupportGaugeReadingRainfallEvent>
  <isSupportSearchTrafficLightMalfunctionResult>
    <!--ro, opt, bool, whether the device supports searching for traffic light fault detection result, desc:POST
/ISAPI/Event/TrafficLightMalfunctionDetection/SearchTrafficLightMalfunctionResult?format=json-->true
  </isSupportSearchTrafficLightMalfunctionResult>
  <isSupportCheckPhoneNoSecurityCode>
    <!--ro, opt, bool, whether the device supports phone security code verification, desc:PUT /ISAPI/Security/resetPassword/CheckPhoneNoSecurityCode?
format=json-->true
  </isSupportCheckPhoneNoSecurityCode>
  <isSupportUserValidExpired>
    <!--ro, opt, bool, whether the device supports the UserValidExpired event, desc:eventType: UserValidExpired-->true
  </isSupportUserValidExpired>
  <isSupportSearchWaterLevelData>
    <!--ro, opt, bool, whether the device supports searching for water level detection data, desc:POST
/ISAPI/Event/waterQualityDetection/SearchWaterLevelData?format=json-->true
  </isSupportSearchWaterLevelData>
  <isSupportSearchRecommendPerspective>
    <!--ro, opt, bool, whether the device supports searching for recommended FoV, desc:POST
/ISAPI/Streaming/channels/channels/<channelID>/SearchRecommendPerspective?format=json-->true
  </isSupportSearchRecommendPerspective>
  <isSupportGetLegendaryPerspectiveInfo>
    <!--ro, opt, bool, whether the device supports getting the information of camera FoV, desc:GET
/ISAPI/Streaming/channels/channels/<channelID>/GetLegendaryPerspectiveInfo?format=json-->true
  </isSupportGetLegendaryPerspectiveInfo>
  <configurationDataType opt="OSD,MODE,IMAGE,IOOUT,OTHERS">
    <!--ro, opt, string, "OSD"-import video and image OSD settings, "MODE"-import application mode, "IMAGE"-import image parameters, "IOONT"-import
supplement Light Parameters, "OTHERS", attr:opt{req, string}-->test
  </configurationDataType>
  <isSupportRestoreVideoAdjustmentParam>
    <!--ro, opt, bool, whether the device supports restoring video image setting parameters, desc:GET/PUT
/ISAPI/System/Video/channels/<channelID>/RestoreVideoAdjustmentParam?format=json-->true
  </isSupportRestoreVideoAdjustmentParam>
  <isSupportWeatherStationInfo>
    <!--ro, opt, bool, whether the device supports weather station information, desc:GET/PUT /ISAPI/Traffic/channels/<channelID>/WeatherStationInfo?
format=json-->true
  </isSupportWeatherStationInfo>
  <isSupportRadarPerimeterAlgCap>
    <!--ro, opt, bool, whether the device supports the capability of radar perimeter algorithm parameters, desc:GET
/ISAPI/Smart/RadarPerimeterAlg/capabilities?format=json-->true
  </isSupportRadarPerimeterAlgCap>
  <isSupportBodyCameraOneKeyAlarmEvent>
    <!--ro, opt, bool, whether the device supports the BodyCameraOneKeyAlarmEvent event, desc:eventType: BodyCameraOneKeyAlarmEvent; this node is not
returned if not supported-->true
  </isSupportBodyCameraOneKeyAlarmEvent>
  <isSupportGetOnlineTrafficLight>
    <!--ro, opt, bool, whether the device supports getting online traffic light information, desc:GET
/ISAPI/Event/TrafficLightMalfunctionDetection/GetOnlineTrafficLight?format=json-->true
  </isSupportGetOnlineTrafficLight>
  <isSupportTrafficLightMalfunctionDefaultParams>
    <!--ro, opt, bool, whether the device supports default parameters of traffic light fault detection rules, desc:GET/PUT
/ISAPI/Event/TrafficLightMalfunctionDetection/TrafficLightMalfunctionDefaultParams?format=json-->true
  </isSupportTrafficLightMalfunctionDefaultParams>
  <isSupportFallenObjectsOnTracksDetection>
    <!--ro, opt, bool, whether the device supports detection of fallen objects on tracks, desc:GET
/ISAPI/Intelligent/channels/<channelID>/FallenObjectsOnTracksDetection/capabilities?format=json-->true
  </isSupportFallenObjectsOnTracksDetection>
  <isSupportMonitorAuthorization>
    <!--ro, opt, bool, whether the device supports device service monitoring authorization information, desc:GET
/ISAPI/Security/DeviceServiceMonitor/MonitorAuthorization?format=json-->true
  </isSupportMonitorAuthorization>
  <isSupportBodyCameraTakeEvent>
    <!--ro, opt, bool, whether the device supports the BodyCameraTakeEvent event, desc:eventType: BodyCameraTakeEvent; this node is not returned if not
supported-->true
  </isSupportBodyCameraTakeEvent>
  <isSupportCollaborationModeCfg>
    <!--ro, opt, bool, whether the device supports configuring multi-lens coordination mode, desc:GET/PUT /ISAPI/Intelligent/CollaborationModeCfg?
format=json-->true
  </isSupportCollaborationModeCfg>
  <isSupportAudioChannelVCAResource>
    <!--ro, opt, bool, whether the device supports switching intelligent resources running on audio channels, desc:GET/PUT
/ISAPI/System/VCAResource/channels/<audioID>/AudioChannelVCAResource?format=json-->true
  </isSupportAudioChannelVCAResource>
  <isSupportGasTankMonitoringEvent>
    <!--ro, opt, bool, whether the device supports the GasTankMonitoringEvent event, desc:eventType: GasTankMonitoringEvent-->true
  </isSupportGasTankMonitoringEvent>
  <isSupportChannelClearEvent>
    <!--ro, opt, bool, whether the device supports the ChannelClearEvent event, desc:eventType: ChannelClearEvent-->true
  </isSupportChannelClearEvent>
  <isSupportBackDuty>
    <!--ro, opt, bool, whether the device supports the backDuty event, desc:eventType: backDuty-->true
  </isSupportBackDuty>
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  <isSupportSleepDuty>
    <!--ro, opt, bool, whether the device supports the sleepDuty event, desc:eventType: sleepDuty-->true
  </isSupportSleepDuty>
  <isSupportSearchChildDevResourcesStatus>
    <!--ro, opt, bool, whether the device supports searching for sub-device resource status, desc:POST
/ISAPI/System/DynamicResource/SearchChildDevResourcesStatus?format=json-->true
  </isSupportSearchChildDevResourcesStatus>
  <isSupportDeleteResources>
    <!--ro, opt, bool, whether the device supports deleting resources, desc:PUT /ISAPI/System/DynamicResource/DeleteResources?format=json-->true
  </isSupportDeleteResources>
  <isSupportMenuFunctionList>
    <!--ro, opt, bool, whether the device supports the menu list, desc:GET /ISAPI/Thermal/MenuFunctionList?format=json-->true
  </isSupportMenuFunctionList>
  <isSupportKeyFunctionList>
    <!--ro, opt, bool, whether the device supports the button list, desc:GET /ISAPI/Thermal/KeyFunctionList?format=json-->true
  </isSupportKeyFunctionList>
  <isSupportKeyValueList>
    <!--ro, opt, bool, whether the device supports button key list, desc:GET /ISAPI/Thermal/KeyValueList?format=json-->true
  </isSupportKeyValueList>
  <isSupportGetPseudoColorBackgroundPic>
    <!--ro, opt, bool, whether the device supports 3-channel palette background picture, desc:GET
/ISAPI/Streaming/channels/<channelID>/GetPseudoColorBackgroundPic?format=json-->true
  </isSupportGetPseudoColorBackgroundPic>
  <isSupportElevatorRunningDetection>
    <!--ro, opt, bool, whether the device supports elevator running status detection, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ElevatorRunningDetection/capabilities?format=json-->true
  </isSupportElevatorRunningDetection>
  <isSupportSearchPersonQueueRealTimeResult>
    <!--ro, opt, bool, whether the device supports searching for history records of person waiting detection event, desc:POST
/ISAPI/Intelligent/personQueueDetection/SearchPersonQueueRealTimeResult?format=json-->true
  </isSupportSearchPersonQueueRealTimeResult>
  <isSupportSearchLeavePositionEventResult>
    <!--ro, opt, bool, whether the device supports searching for history records of absence detection event, desc:POST
/ISAPI/Intelligent/HehaviorAnalysis/SearchLeavePositionEventResult?format=json-->true
  </isSupportSearchLeavePositionEventResult>
  <isSupportImportElectronicBadgeBackgroundImage>
    <!--ro, opt, bool, whether the device supports importing the background picture of nameplate, desc:POST
/ISAPI/Traffic/ImportElectronicBadgeBackgroundImage?format=json-->true
  </isSupportImportElectronicBadgeBackgroundImage>
  <isSupportVoiceMonitoringEvent>
    <!--ro, opt, bool, whether the device supports uploading the VoiceMonitoringEvent event, desc:eventType: VoiceMonitoringEvent-->true
  </isSupportVoiceMonitoringEvent>
  <isSupportPseudoColorBackgroundParam>
    <!--ro, opt, bool, whether the device supports 3-channel palette background parameters, desc:GET/PUT
/ISAPI/Streaming/channels/<channelID>/PseudoColorBackgroundParam?format=json-->true
  </isSupportPseudoColorBackgroundParam>
  <isSupportGotoPerimeterShieldMaskScene>
    <!--ro, opt, bool, whether the device supports jumping to shielded scenes of specified radar perimeter, desc:POST
/ISAPI/Smart/RadarPerimeterAlg/GotoPerimeterShieldMaskScene?format=json-->true
  </isSupportGotoPerimeterShieldMaskScene>
  <isSupportSearchPersonQueueTimingStatistics>
    <!--ro, opt, bool, whether the device supports the SearchPersonQueueTimingStatistics event, desc:POST
/ISAPI/Intelligent/personQueueDetection/SearchPersonQueueTimingStatistics?format=json-->true
  </isSupportSearchPersonQueueTimingStatistics>
  <isSupportLocalUserChange>
    <!--ro, opt, bool, whether the device supports the LocalUserChange event, desc:eventType: LocalUserChange-->true
  </isSupportLocalUserChange>
  <isSupportSendClientTask>
    <!--ro, opt, bool, whether the device supports applying tasks configured on the client, desc:POST /ISAPI/System/RailwayPatrol/SendClientTask?
format=json-->true
  </isSupportSendClientTask>
  <isSupportGetCurrentClientTaskStatus>
    <!--ro, opt, bool, whether the device supports getting the current execution progress of the client task, desc:GET
/ISAPI/System/RailwayPatrol/GetCurrentClientTaskStatus?format=json-->true
  </isSupportGetCurrentClientTaskStatus>
  <isSupportTestEDRServerConnect>
    <!--ro, opt, bool, whether the device supports testing the connection status of the EDR platform, desc:POST /ISAPI/System/Network/TestEDRServerConnect?
format=json-->true
  </isSupportTestEDRServerConnect>
  <isSupportRealTimeStreamInfo>
    <!--ro, opt, bool, whether the device supports getting the real-time stream information, desc:GET /ISAPI/Streaming/RealTimeStreamInfo?format=json-->true
  </isSupportRealTimeStreamInfo>
  <isSupportIrcutFilterStatusChange>
    <!--ro, opt, bool, whether the device supports the IrcutFilterStatusChange event, desc:eventType: IrcutFilterStatusChange-->true
  </isSupportIrcutFilterStatusChange>
  <isSupportSendRoadMaintPictureAnalysisTask>
    <!--ro, opt, bool, whether the device supports applying tasks of road maintenance picture analysis, desc:POST
/ISAPI/Intelligent/SendRoadMaintPictureAnalysisTask?format=json-->true
  </isSupportSendRoadMaintPictureAnalysisTask>
  <isSupportFDRemoveDuplicateParams>
    <!--ro, opt, bool, whether the device supports deduplication parameters of custom face picture library, desc:GET/PUT
/ISAPI/System/Video/inputs/counting/FDRemoveDuplicateParams?format=json-->true
  </isSupportFDRemoveDuplicateParams>
  <isSupportRadarMixedTargetSensitivityParams>
    <!--ro, opt, bool, whether the device supports sensitivity parameters of multi-target-type detection via radar, desc:GET/PUT
/ISAPI/Smart/RadarMixedTargetAlg/RadarMixedTargetSensitivityParams?format=json-->true
  </isSupportRadarMixedTargetSensitivityParams>
  <isSupportMixedTargetRadarSensitivityParams>
    <!--ro, opt, bool, whether the device supports sensitivity parameters of multi-target-type detection via video, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/MixedTargetRadarSensitivityParams?format=json-->true
  </isSupportMixedTargetRadarSensitivityParams>
  <isSupportMixedTargetRadarDebugParams>
    <!--ro, opt, bool, whether the device supports algorithm debugging parameters of multi-target-type detection via video, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/MixedTargetRadarDebugParams?format=json-->true
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  </isSupportMixedTargetRadarDebugParams>
  <isSupportHttpPreview>
    <!--ro, opt, bool, whether the device supports HTTP live view,
desc:related URI: GET /ISAPI/Streaming/channels/<trackStreamID>/httppreview;
there are 2 steps as follows:
1. check if this node is returned by device and if the returned value is true. If yes, the HTTP live view function is supported by device.
2. if this node is not returned by device, you should go through the stream ID in the API (GET /ISAPI/Streaming/channels/<trackStreamID>/httppreview) of all
channels. If notSupport is returned for all channels, HTTP live view is not supported by device; if at least the success code is returned for at least one
channel, HTTP live view is supported by device.
Note: whether the stream ID (trackStreamID) is supported is determined by the response parameters StreamingChannelList, StreamingChannel, and id from GET
/ISAPI/Streaming/channels-->true
  </isSupportHttpPreview>
  <isSupportGetRecordThumbnail>
    <!--ro, opt, bool, whether the device supports getting the video file thumbnail, desc:POST /ISAPI/ContentMgmt/search/GetRecordThumbnail?format=json--
>true
  </isSupportGetRecordThumbnail>
  <isSupportProxyAIOP>
    <!--ro, opt, bool, whether the device supports proxy access of AIOP (AIOP devices connected with network cameras via NVR transparent transmission),
desc:when this node is supported, full AIOP functions (device AIOP capability, model package management, task rule management, detection data, etc.) can be
accessed via proxy transmission of ISAPI sub-device-->true
  </isSupportProxyAIOP>
  <isSupportStaticObjectPositionOffsetComparison>
    <!--ro, opt, bool, whether the device supports position deviation comparison of static objects, desc:GET
/ISAPI/Intelligent/channels/<channelID>/StaticObjectPositionOffsetComparison/capabilities?format=json-->true
  </isSupportStaticObjectPositionOffsetComparison>
  <isSupportClearFDRemoveDuplicateFDLib>
    <!--ro, opt, bool, whether the device supports manually clearing the custom face picture library on device, desc:POST
/ISAPI/System/Video/inputs/counting/ClearFDRemoveDuplicateFDLib?format=json-->true
  </isSupportClearFDRemoveDuplicateFDLib>
  <isSupportDevPowerStatusReportCtrl>
    <!--ro, opt, bool, whether the device supports control parameters of device battery status upload, desc:GET/PUT /ISAPI/System/DevPowerStatusReportCtrl?
format=json-->true
  </isSupportDevPowerStatusReportCtrl>
  <isSupportIndicatorLight>
    <!--ro, opt, bool, whether the device supports indicator light configuration, desc:GET /ISAPI/System/IndicatorLight/capabilities?format=json-->true
  </isSupportIndicatorLight>
  <isSupportEnvironmentTemperatureMonitorEvent>
    <!--ro, opt, bool, whether the device supports the EnvironmentTemperatureMonitorEvent event, desc:eventType: EnvironmentTemperatureMonitorEvent-->true
  </isSupportEnvironmentTemperatureMonitorEvent>
  <isSupportPhysicalTraining>
    <!--ro, opt, bool, whether the device supports physical training, desc:GET /ISAPI/Intelligent/channels/<channelID>/PhysicalTraining/capabilities?
format=json-->true
  </isSupportPhysicalTraining>
  <isSupportPullUp>
    <!--ro, opt, bool, whether the device supports the PullUp event, desc:eventType: PullUp-->true
  </isSupportPullUp>
  <isSupportSitUpWithHandsBehindHead>
    <!--ro, opt, bool, whether the device supports the SitUpWithHandsBehindHead event, desc:eventType: SitUpWithHandsBehindHead-->true
  </isSupportSitUpWithHandsBehindHead>
  <isSupportSitUpWithArmsOnChest>
    <!--ro, opt, bool, whether the device supports the SitUpWithArmsOnChest event, desc:eventType: SitUpWithArmsOnChest-->true
  </isSupportSitUpWithArmsOnChest>
  <isSupportPushUp>
    <!--ro, opt, bool, whether the device supports the PushUp event, desc:eventType: PushUp-->true
  </isSupportPushUp>
  <isSupportPullUpHold>
    <!--ro, opt, bool, whether the device supports the PullUpHold event, desc:eventType: PullUpHold-->true
  </isSupportPullUpHold>
  <isSupportVerificationPlatformParams>
    <!--ro, opt, bool, whether the device supports accessing parameters for the verification and recording platform, desc:GET/PUT
/ISAPI/System/Network/VerificationPlatformAccess/VerificationPlatformParams?format=json-->true
  </isSupportVerificationPlatformParams>
  <isSupportVehicleMaintenanceDetection>
    <!--ro, opt, bool, whether the device supports vehicle maintenance detection, desc:GET /ISAPI/Intelligent/VehicleMaintenanceDetection/capabilities?
format=json; detects the status of vehicles in the maintenance shop and triggers the alarm when the vehicle is lifted (when detecting chassis) or lowered,
or the hood is opened-->true
  </isSupportVehicleMaintenanceDetection>
  <isSupportPhysicalTrainingStatus>
    <!--ro, opt, bool, whether the device supports the PhysicalTrainingStatus event, desc:eventType: PhysicalTrainingStatus-->true
  </isSupportPhysicalTrainingStatus>
  <isSupportDeviceAbnormalDebug>
    <!--ro, opt, bool, whether the device supports device operation exception debugging, desc:GET /ISAPI/System/DeviceAbnormalDebug/capabilities?
format=json; during device operation exception, the device operation debugging data (debugging log, dump file, etc.) will be collected by technical support-
->true
  </isSupportDeviceAbnormalDebug>
  <isSupportSetWaterLevelResetOfWaterTank>
    <!--ro, opt, bool, whether the device supports manually filling up the water tank and uploading the filled tank information for calculating remaining
water via lens wiper usage, desc:PUT /ISAPI/System/Hardware/SetWaterLevelResetOfWaterTank?format=json-->true
  </isSupportSetWaterLevelResetOfWaterTank>
  <isSupportGetRemoteAlbumPicture>
    <!--ro, opt, bool, whether the device supports getting pictures from the remote album, desc:POST /ISAPI/ContentMgmt/GetRemoteAlbumPicture?format=json--
>true
  </isSupportGetRemoteAlbumPicture>
  <isSupportUltravioletImagingFusionParams>
    <!--ro, opt, bool, whether the device supports UV imaging fusion parameters, desc:GET/PUT
/ISAPI/System/Video/channels/<channelID>/UltravioletImagingFusionParams?format=json-->true
  </isSupportUltravioletImagingFusionParams>
  <isSupportAudioAnalysisCfg>
    <!--ro, opt, bool, whether it supports audio analysis parameters, desc:GET/PUT
/ISAPI/Intelligent/SensitiveInfoAnalysis/AudioIn/<AudioInID>/AudioAnalysisCfg?format=json-->true
  </isSupportAudioAnalysisCfg>
  <isSupportRuningLogPlan>
    <!--ro, opt, bool, whether the device supports device operation log schedule, desc:GET/PUT /ISAPI/ContentMgmt/RuningLogPlan?format=json-->true
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    </isSupportRuningLogPlan>
    <isSupportSearchRuningLogData>
      <!--ro, opt, bool, whether the device supports searching for device operation log data, desc:POST /ISAPI/ContentMgmt/SearchRuningLogData?format=json-->true
    </isSupportSearchRuningLogData>
    <isSupportExportRuningLogData>
      <!--ro, opt, bool, whether the device supports exporting device operation log data, desc:POST /ISAPI/ContentMgmt/ExportRuningLogData?format=json-->true
    </isSupportExportRuningLogData>
    <isSupportCertificateCaptureEvent>
      <!--ro, opt, bool, whether the device supports the CertificateCaptureEvent event, desc:corresponding event: CertificateCaptureEvent-->true
    </isSupportCertificateCaptureEvent>
    <isSupportUpgradeKeywordModel>
      <!--ro, opt, bool, whether the device supports upgrading the sensitive information keyword model, desc:PUT /ISAPI/Intelligent/SensitiveInfoAnalysis/UpgradeKeywordModel?format=json-->true
    </isSupportUpgradeKeywordModel>
    <isSupportGetKeywordModelStatus>
      <!--ro, opt, bool, whether the device supports getting algorithm running status of the sensitive information keyword model, desc:GET /ISAPI/Intelligent/SensitiveInfoAnalysis/GetKeywordModelStatus?format=json-->true
    </isSupportGetKeywordModelStatus>
    <isSupportCloseLensPrivacyCover>
      <!--ro, opt, bool, whether the device supports manually closing the lens privacy cap, desc:PUT /ISAPI/System/channels/<channelID>/CloseLensPrivacyCover?format=json-->true
    </isSupportCloseLensPrivacyCover>
    <isSupportOpenLensPrivacyCover>
      <!--ro, opt, bool, whether the device supports manually opening the lens privacy cap, desc:PUT /ISAPI/System/channels/<channelID>/OpenLensPrivacyCover?format=json-->true
    </isSupportOpenLensPrivacyCover>
    <isSupportLensPrivacyCoverModeParams>
      <!--ro, opt, bool, whether the device supports working mode parameters of the lens privacy cap, desc:GET/PUT /ISAPI/System/channels/<channelID>/LensPrivacyCoverModeParams?format=json-->true
    </isSupportLensPrivacyCoverModeParams>
    <isSupportOTAPService>
      <!--ro, opt, bool, whether the device supports OTAP service configuration, desc:/ISAPI/System/Network/OTAPService/capabilities?format=json-->true
    </isSupportOTAPService>
    <isSupportGetPlaybackStreamInfo>
      <!--ro, opt, bool, whether the device supports getting video playback stream information, desc:GET /ISAPI/Streaming/GetPlaybackStreamInfo?format=json-->true
    </isSupportGetPlaybackStreamInfo>
    <isSupportNBIoTPlateformAccessStatus>
      <!--ro, opt, bool, whether the device supports the NB-IoT platform access status, desc:GET /ISAPI/System/Network/NBIoTPlateformAccessStatus?format=json-->true
    </isSupportNBIoTPlateformAccessStatus>
    <isSupportScannedListEvent>
      <!--ro, opt, bool, whether the device supports Bluetooth scanning list event, desc:eventType: ScannedListEvent-->true
    </isSupportScannedListEvent>
    <isSupportDownloadAudioRecordByTime>
      <!--ro, opt, bool, whether the device supports downloading audio files by time, desc:POST /ISAPI/System/RecordAudio/DownloadAudioRecordByTime?format=json-->true
    </isSupportDownloadAudioRecordByTime>
    <isSupportBackUpConfigurationFile>
      <!--ro, opt, bool, whether the device supports local backup of the configuration file, desc:PUT /ISAPI/System/BackUpConfigurationFile?format=json-->true
    </isSupportBackUpConfigurationFile>
    <isSupportRestoreConfigurationFile>
      <!--ro, opt, bool, whether the device supports restoring the device local configuration file, desc:PUT /ISAPI/System/RestoreConfigurationFile?format=json-->true
    </isSupportRestoreConfigurationFile>
    <isSupportGetCfgFileBackUpInfo>
      <!--ro, opt, bool, whether the device supports getting the local configuration file backup information, desc:GET /ISAPI/System/GetCfgFileBackUpInfo?format=json-->true
    </isSupportGetCfgFileBackUpInfo>
    <isSupportSearchVideoCodeMealParams>
      <!--ro, opt, bool, whether the device supports searching for video encoding plan parameters, desc:POST /ISAPI/Streaming/channels/<channelID>/SearchVideoCodeMealParams?format=json-->true
    </isSupportSearchVideoCodeMealParams>
    <isSupportVideoCodeMealParams>
      <!--ro, opt, bool, whether the device supports video encoding plan parameters, desc:GET/PUT /ISAPI/Streaming/channels/<channelID>/VideoCodeMealParams?format=json-->true
    </isSupportVideoCodeMealParams>
    <isSupportFlameCombustionStatusRuleParams>
      <!--ro, opt, bool, whether the device supports flame burning status detection rule parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusRuleParams?format=json-->true
    </isSupportFlameCombustionStatusRuleParams>
    <isSupportFlameCombustionStatusOverlapParams>
      <!--ro, opt, bool, whether the device supports flame burning status detection information overlay parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusOverlapParams?format=json-->true
    </isSupportFlameCombustionStatusOverlapParams>
    <isSupportAcousticLeakDetection>
      <!--ro, opt, bool, whether the device supports acoustic leak detection capability, desc:related URL: /ISAPI/System/AcousticLeakDetection/capabilities?format=json-->true
    </isSupportAcousticLeakDetection>
    <isSupportFlameCombustionStatusSceneList>
      <!--ro, opt, bool, whether the device supports flame burning status detection scene list, desc:GET /ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusSceneList?format=json-->true
    </isSupportFlameCombustionStatusSceneList>
    <isSupportGasLeakageDetection>
      <!--ro, opt, bool, whether the device supports gas leak event, desc:eventType: GasLeakageDetection-->true
    </isSupportGasLeakageDetection>
    <isSupportDischargeDetection>
      <!--ro, opt, bool, whether the device supports discharging event, desc:eventType: DischargeDetection-->true
    </isSupportDischargeDetection>
    <isSupportSearchBaseMapIDList>
      <!--ro, opt, bool, whether the device supports searching for base map ID list, desc:POST /ISAPI/ContentMgmt/SearchBaseMapIDList?format=json-->true
    </isSupportSearchBaseMapIDList>
    <isSupportRestoreDefaultParam>
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    <!--ro, opt, bool, whether the device supports restoring default parameters, desc:PUT /ISAPI/Streaming/RestoreDefaultParam?format=json-->true
  </isSupportRestoreDefaultParam>
  <isSupportSearchEventTargetInfo>
    <!--ro, opt, bool, whether the device supports searching for event detection target information in FoV, desc:GET
/ISAPI/System/Video/inputs/channels/<channelID>/SearchEventTargetInfo?format=json-->true
  </isSupportSearchEventTargetInfo>
  <isSupportElevatorRunningDetectionEvent>
    <!--ro, opt, bool, whether the device supports elevator running event, desc:eventType: ElevatorRunningDetectionEvent-->true
  </isSupportElevatorRunningDetectionEvent>
  <isSupportElevatorOperationStatistics>
    <!--ro, opt, bool, whether the device supports elevator running statistics event, desc:eventType: ElevatorOperationStatistics-->true
  </isSupportElevatorOperationStatistics>
  <isSupportStandingLongJump>
    <!--ro, opt, bool, whether the device supports standing long jump event, desc:eventType: StandingLongJump-->true
  </isSupportStandingLongJump>
  <isSupportJumpRope>
    <!--ro, opt, bool, whether the device supports jumping rope event, desc:eventType: JumpRope-->true
  </isSupportJumpRope>
  <isSupportUserAvailablePwQuery>
    <!--ro, opt, bool, whether the device supports person query through password, desc:eventType:UserAvailablePwQuery-->true
  </isSupportUserAvailablePwQuery>
  <isSupportAddHistoryEventSubscribeCfg>
    <!--ro, opt, bool, whether the device supports adding history event subscription, desc:POST /ISAPI/Event/notification/AddHistoryEventSubscribeCfg?
format=json-->true
  </isSupportAddHistoryEventSubscribeCfg>
  <isSupportHistoryEventSubscribeUploadCtrl>
    <!--ro, opt, bool, whether the device supports history event subscription uploading control, desc:PUT
/ISAPI/Event/notification/HistoryEventSubscribeUploadCtrl?format=json-->true
  </isSupportHistoryEventSubscribeUploadCtrl>
  <isSupportHLSStreamCfg>
    <!--ro, opt, bool, whether the device supports HLS streaming address parameters, desc:GET/PUT /ISAPI/Streaming/channels/<channelID>/HLSStreamCfg?
format=json-->true
  </isSupportHLSStreamCfg>
  <isSupportGetKeywordModelVersionInfo>
    <!--ro, opt, bool, whether the device supports getting sensitive information keyword model version information, desc:GET
/ISAPI/Intelligent/SensitiveInfoAnalysis/GetKeywordModelVersionInfo?format=json-->true
  </isSupportGetKeywordModelVersionInfo>
  <isSupportManualTriggerRemoteAlbumSnapshot>
    <!--ro, opt, bool, whether the device supports manually triggering remote album capture, desc:PUT /ISAPI/ContentMgmt/ManualTriggerRemoteAlbumSnapshot?
format=json-->true
  </isSupportManualTriggerRemoteAlbumSnapshot>
  <isSupportRadarVideoAssistedFocusParams>
    <!--ro, opt, bool, whether the device supports radar-assisted focusing parameters, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusParams?format=json-->true
  </isSupportRadarVideoAssistedFocusParams>
  <isSupportCollectionResultUpload>
    <!--ro, opt, bool, whether the device supports the uploading collected result event, desc:eventType: CollectionResultUpload-->true
  </isSupportCollectionResultUpload>
  <isSupportDeveloperMode>
    <!--ro, opt, bool, whether the device supports developer mode, desc:GET/PUT /ISAPI/System/DeveloperMode?format=json-->true
  </isSupportDeveloperMode>
  <isSupportGetDetailsFunction>
    <!--ro, opt, bool, whether the device supports getting detailed functions supported in T1 test, desc:GET /ISAPI/System/T1Test/DetailsFunction?
format=json-->true
  </isSupportGetDetailsFunction>
  <isSupportSearchDeviceIANAList>
    <!--ro, opt, bool, whether the device supports searching for supported IANA list, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
  </isSupportSearchDeviceIANAList>
  <isSupportModifySpecificROIStreamJointParams>
    <!--ro, opt, bool, whether the device supports setting ROI stream jointing parameters, desc:PUT
/ISAPI/Streaming/channels/<channelID>/ModifySpecificROIStreamJointParams?format=json-->true
  </isSupportModifySpecificROIStreamJointParams>
  <isSupportDeviceTamperParams>
    <!--ro, opt, bool, whether the device supports tampering parameters, desc:GET/PUT /ISAPI/System/DeviceTamperParams?format=json-->true
  </isSupportDeviceTamperParams>
  <isSupportDeviceTamperEvent>
    <!--ro, opt, bool, whether the device supports tampering event, desc:eventType: DeviceTamperEvent-->true
  </isSupportDeviceTamperEvent>
  <isSupportRoadVariableLaneCtrlParams>
    <!--ro, opt, bool, whether the device supports dynamic management parameters of changeable lane, desc:GET/PUT
/ISAPI/Traffic/RoadVariableLaneCtrl/channels/<channelID>/RoadVariableLaneCtrlParams?format=json-->true
  </isSupportRoadVariableLaneCtrlParams>
  <isSupportImportModelFile>
    <!--ro, opt, bool, whether the device supports importing the water quality detection model file, desc:POST
/ISAPI/Event/waterQualityDetection/ImportModelFile?format=json-->true
  </isSupportImportModelFile>
  <isSupportSearchImportStatus>
    <!--ro, opt, bool, whether the device supports getting water quality detection model file importing progress, desc:GET
/ISAPI/Event/waterQualityDetection/SearchImportStatus?format=json-->true
  </isSupportSearchImportStatus>
  <isSupportSafetyWorkwearDetectionOverlapParams>
    <!--ro, opt, bool, whether the device supports parameters of wearing safety work clothes detection information overlay, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionOverlapParams?format=json-->true
  </isSupportSafetyWorkwearDetectionOverlapParams>
  <isSupportSafetyWorkwearDetectionRuleParams>
    <!--ro, opt, bool, whether the device supports detection rule parameters of wearing safety work clothes, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionRuleParams?format=json-->true
  </isSupportSafetyWorkwearDetectionRuleParams>
  <isSupportUpdateFirmwareMode>
    <!--ro, opt, bool, whether the device supports the upgrade mode, desc:GET/PUT /ISAPI/System/UpdateFirmwareMode?format=json-->true
  </isSupportUpdateFirmwareMode>
  <isSupportIndoorPasswayRecovery>
    <!--ro, opt, bool, whether the device supports indoor passway block alarm restoring event, desc:eventType: IndoorPasswayRecovery#-->true
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    </isSupportIndoorPasswayRecovery>
    <isSupportBatteryCarRecovery>
      <!--ro, opt, bool, whether the device supports electric moped parking violation alarm restoring event, desc:eventType: BatteryCarRecovery-->true
    </isSupportBatteryCarRecovery>
    <isSupportDetectorTemperatureRecovery>
      <!--ro, opt, bool, whether the device supports detector temperature alarm restoring event, desc:eventType: DetectorTemperatureRecovery-->true
    </isSupportDetectorTemperatureRecovery>
    <isSupportGetRFIDReaderWorkStatus>
      <!--ro, opt, bool, whether the device supports getting RFID read/write working status, desc:GET /ISAPI/Traffic/UHFRFIDReader/GetRFIDReaderWorkStatus?
format=json-->true
    </isSupportGetRFIDReaderWorkStatus>
    <isSupportRFIDReaderWorkStatusCtrl>
      <!--ro, opt, bool, whether the device supports RFID read/write working status control, desc:PUT /ISAPI/Traffic/UHFRFIDReader/RFIDReaderWorkStatusCtrl?
format=json-->true
    </isSupportRFIDReaderWorkStatusCtrl>
    <isSupportFlameCombustionStatusAlgorithmsVersion>
      <!--ro, opt, bool, whether the device supports getting flame burning status detection algorithm version, desc:GET
/ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusAlgorithmsVersion?format=json-->true
    </isSupportFlameCombustionStatusAlgorithmsVersion>
    <isSupportTestResultUpload>
      <!--ro, opt, bool, whether the device supports test result uploading event, desc:eventType: TestResultUpload-->true
    </isSupportTestResultUpload>
    <isSupportAutoTestParams>
      <!--ro, opt, bool, whether the device supports automation testing parameters, desc:GET/PUT /ISAPI/System/AutoTest/AutoTestParams?format=json-->true
    </isSupportAutoTestParams>
    <isSupportExportATInstructRecord>
      <!--ro, opt, bool, whether the device supports exporting AT command records, desc:POST /ISAPI/System/ExportATInstructRecord?format=json-->true
    </isSupportExportATInstructRecord>
    <isSupportOfflineVideoFileManagement>
      <!--ro, opt, bool, whether the device supports offline video file management, desc:GET /ISAPI/ContentMgmt/OfflineVideoFileManagement/capabilities?
format=json-->true
    </isSupportOfflineVideoFileManagement>
    <isSupportChannelOccupySceneList>
      <!--ro, opt, bool, whether the device supports getting the scene list of fire escape occupation, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupySceneList?format=json-->true
    </isSupportChannelOccupySceneList>
    <isSupportChannelOccupyAlgorithmsVersion>
      <!--ro, opt, bool, whether the device supports getting the algorithm version of fire escape occupation, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyAlgorithmsVersion?format=json-->true
    </isSupportChannelOccupyAlgorithmsVersion>
    <isSupportChannelOccupyOverlapParams>
      <!--ro, opt, bool, whether the device supports overlay parameters of fire escape occupation, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyOverlapParams?format=json-->true
    </isSupportChannelOccupyOverlapParams>
    <isSupportChannelOccupyRuleParams>
      <!--ro, opt, bool, whether the device supports rule parameters of fire escape occupation, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyRuleParams?format=json-->true
    </isSupportChannelOccupyRuleParams>
    <isSupportGotoChannelOccupyScene>
      <!--ro, opt, bool, whether the device supports jumping to the specified fire escape occupation scene, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/GotoChannelOccupyScene?format=json-->true
    </isSupportGotoChannelOccupyScene>
    <isSupportChannelPassingEvent>
      <!--ro, opt, bool, whether the device supports vehicles passing outdoor fire escape, desc:eventType:channelPassingEvent-->true
    </isSupportChannelPassingEvent>
    <isSupportGunsHandoverRuleParams>
      <!--ro, opt, bool, whether the device supports object handover rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/GunsHandoverRuleParams?format=json-->true
    </isSupportGunsHandoverRuleParams>
    <isSupportGunsHandover>
      <!--ro, opt, bool, whether the device supports object handover event, desc:eventType:GunsHandover-->true
    </isSupportGunsHandover>
    <isSupportBulkUpgradeChildDevice>
      <!--ro, opt, bool, whether the device supports batch upgrading sub-devices, desc:POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json-->true
    </isSupportBulkUpgradeChildDevice>
    <isSupportSearchBulkUpgradeChildDeviceProgress>
      <!--ro, opt, bool, whether the device supports checking the progress of batch upgrading sub-devices, desc:PUT
/ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json-->true
    </isSupportSearchBulkUpgradeChildDeviceProgress>
    <isSupportDash>
      <!--ro, opt, bool, whether the device supports the dash event, desc:eventType:Dash-->true
    </isSupportDash>
    <isSupportStopRadarVideoAssistedFocusCalibrating>
      <!--ro, opt, bool, whether the device supports stopping the calibration of radar-assisted focusing, desc:PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StopRadarVideoAssistedFocusCalibrating?format=json-->true
    </isSupportStopRadarVideoAssistedFocusCalibrating>
    <isSupportRadarVideoAssistedFocusCalibrating>
      <!--ro, opt, bool, whether the device supports the calibration of radar-assisted focusing, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusCalibrating?format=json-->true
    </isSupportRadarVideoAssistedFocusCalibrating>
    <isSupportRadarVideoAssistedFocusCalibratingStatus>
      <!--ro, opt, bool, whether the device supports displaying the calibration status of radar-assisted focusing, desc:GET
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusCalibratingStatus?format=json-->true
    </isSupportRadarVideoAssistedFocusCalibratingStatus>
    <isSupportETCMatchResultUpload>
      <!--ro, opt, bool, whether the device supports uploading ETC vehicle comparison results, desc:eventType:ETCMatchResultUpload-->true
    </isSupportETCMatchResultUpload>
    <isSupportPartsLifeRemind>
      <!--ro, opt, bool, whether the device supports prompting the life span of device component, desc:corresponding event: PartsLifeRemind-->true
    </isSupportPartsLifeRemind>
    <isSupportAuthorizeMicAcquisitionParam>
      <!--ro, opt, bool, whether the device supports audio authorization information, desc:GET/PUT
/ISAPI/System/InformedConsentMgr/AuthorizeMicAcquisitionParam?format=json-->true
    </isSupportAuthorizeMicAcquisitionParam>
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  <isSupportTOFDetectEvent>
    <!--ro, opt, bool, whether the device supports TOF detection event, desc:eventType:TOFDetectEvent-->true
  </isSupportTOFDetectEvent>
  <isSupportAudioAnalysisResultOSDOverlay>
    <!--ro, opt, bool, whether the device supports OSD parameters of audio analysis results, desc:GET/PUT
/ISAPI/Intelligent/SensitiveInfoAnalysis/AudioAnalysisResultOSDOverlay?format=json-->true
  </isSupportAudioAnalysisResultOSDOverlay>
  <isSupportChildPasswordStatus>
    <!--ro, opt, bool, whether the device supports getting the activation password status of sub-devices, desc:GET
/ISAPI/System/activate/ChildPasswordStatus?format=json-->true
  </isSupportChildPasswordStatus>
  <isSupportMutexFunctionClose>
    <!--ro, opt, bool, whether the device supports quickly disabling exclusive functions, desc:POST /ISAPI/System/mutexFunctionClose?format=json-->true
  </isSupportMutexFunctionClose>
  <isSupportFirePointAlarmRecovery>
    <!--ro, opt, bool, whether the device supports fire detection alarm restoring event, desc:eventType:FirePointAlarmRecovery-->true
  </isSupportFirePointAlarmRecovery>
  <isSupportSmokeAlarmRecovery>
    <!--ro, opt, bool, whether the device supports smoke alarm restoring event, desc:eventType:FirePointAlarmRecovery-->true
  </isSupportSmokeAlarmRecovery>
  <isSupportTemperatureMeasurementPreviewAlarmRecovery>
    <!--ro, opt, bool, whether the device supports temperature measurement pre-alarm restoring event, desc:eventType:TemperatureMeasurementPreviewRecovery--
>true
  </isSupportTemperatureMeasurementPreviewAlarmRecovery>
  <isSupportTemperatureMeasurementAlarmRecovery>
    <!--ro, opt, bool, whether the device supports temperature measurement alarm restoring event, desc:eventType:TemperatureMeasurementRecovery-->true
  </isSupportTemperatureMeasurementAlarmRecovery>
  <isSupportUVPhotonDetection>
    <!--ro, opt, bool, whether the device supports UV photon detection event, desc:eventType:UVPhotonDetection-->true
  </isSupportUVPhotonDetection>
  <isSupportUVPhotonTimeDetection>
    <!--ro, opt, bool, whether the device supports UV photon scheduled-detection event, desc:eventType:UVPhotonTimeDetection-->true
  </isSupportUVPhotonTimeDetection>
  <isSupportPatientExpensesInfoQuery>
    <!--ro, opt, bool, whether the device supports the patient cost search event, desc:eventType:PatientExpensesInfoQuery-->true
  </isSupportPatientExpensesInfoQuery>
  <isSupportReinforcementCallEvent>
    <!--ro, opt, bool, whether the device supports reinforcement call event, desc:eventType:ReinforcementCallEvent-->true
  </isSupportReinforcementCallEvent>
  <isSupportChargeMode>
    <!--ro, opt, bool, whether the device supports charging mode, desc:GET/PUT /ISAPI/System/ChargeMode?format=json-->true
  </isSupportChargeMode>
  <isSupportCombinationRuleParams>
    <!--ro, opt, bool, whether the device supports combination rule parameters of perimeter detection, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterCombinationDetection/CombinationRuleParams?format=json-->true
  </isSupportCombinationRuleParams>
  <isSupportSwitchHolidayPlanCfg>
    <!--ro, opt, bool, whether the device supports the regular startup/shutdown on holiday, desc:GET/PUT /ISAPI/System/SwitchHolidayPlanCfg?format=json--
>true
  </isSupportSwitchHolidayPlanCfg>
  <isSupportSwitchIrregularHolidayPlanCfg>
    <!--ro, opt, bool, whether the device supports irregular startup/shutdown on holiday, desc:GET/PUT /ISAPI/System/SwitchIrregularHolidayPlanCfg?
format=json-->true
  </isSupportSwitchIrregularHolidayPlanCfg>
  <isSupportSearchPresetConfigInfo>
    <!--ro, opt, bool, whether the device supports searching for the configuration of specified presets, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SearchPresetConfigInfo?format=json-->true
  </isSupportSearchPresetConfigInfo>
  <isSupportSwitchPlanCfg>
    <!--ro, opt, bool, whether the device supports startup/shutdown schedule, desc:GET/PUT /ISAPI/System/SwitchPlanCfg?format=json-->true
  </isSupportSwitchPlanCfg>
  <isSupportVegetationSpectrumDetection>
    <!--ro, opt, bool, whether the device supports vegetation spectrum detection, desc:eventType:VegetationSpectrumDetection-->true
  </isSupportVegetationSpectrumDetection>
  <isSupportCaptureResultAttributeSearch>
    <!--ro, opt, bool, whether the device supports searching for attributes of human body capture, desc:POST
/ISAPI/Intelligent/CaptureResultAttributeSearch?format=json-->true
  </isSupportCaptureResultAttributeSearch>
  <isSupportHealthEducationInfoReadStatus>
    <!--ro, opt, bool, whether the device supports the event of uploading the reading status of health education information,
desc:eventType:HealthEducationInfoReadStatus-->true
  </isSupportHealthEducationInfoReadStatus>
  <isSupportNurseLocationEvent>
    <!--ro, opt, bool, whether the device supports nurse locating event, desc:eventType:NurseLocationEvent-->true
  </isSupportNurseLocationEvent>
  <isSupportAddChildDevResource>
    <!--ro, opt, bool, whether the device supports adding gateway resource information, desc:POST /ISAPI/System/DynamicResource/AddChildDevResource?
format=json-->true
  </isSupportAddChildDevResource>
  <isSupportModifyChildDevResource>
    <!--ro, opt, bool, whether the device supports editing gateway resource information, desc:POST /ISAPI/System/DynamicResource/ModifyChildDevResource?
format=json-->true
  </isSupportModifyChildDevResource>
  <isSupportDiagnosisDataCollectionParams>
    <!--ro, opt, bool, whether the device supports diagnosis data statistics parameters, desc:GET/PUT /ISAPI/System/DiagnosisDataCollectionParams?
format=json-->true
  </isSupportDiagnosisDataCollectionParams>
  <isSupportServicePlatformAccountAuth>
    <!--ro, opt, bool, whether the device supports platform account authentication, desc:GET /ISAPI/Security/ServicePlatformAccountAuth/capabilities?
format=json-->true
  </isSupportServicePlatformAccountAuth>
  <isSupportSyncPatientInfoEvent>
    <!--ro, opt, bool, whether the device supports synchronizing patient information, desc:eventType:SyncPatientInfoEvent-->true
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    </isSupportSyncPatientInfoEvent>
    <isSupportAuthenticationNetworkCfg>
      <!--ro, opt, bool, whether the device supports the authentication parameters of networking method, desc:GET/PUT
/ISAPI/Security/ProtocolCertificate/AuthenticationNetworkCfg?format=json-->true
    </isSupportAuthenticationNetworkCfg>
    <isSupportPersonnelChannelLogParams>
      <!--ro, opt, bool, whether the device supports turnstile log parameters, desc:GET/PUT /ISAPI/ContentMgmt/PersonnelChannelLogParams?format=json-->true
    </isSupportPersonnelChannelLogParams>
    <isSupportElectricSaveModeParams>
      <!--ro, opt, bool, whether the device supports power-saving mode parameters, desc:GET/PUT /ISAPI/System/ElectricSaveModeParams?format=json-->true
    </isSupportElectricSaveModeParams>
    <isSupportChannelOccupyPOSParams>
      <!--ro, opt, bool, whether the device supports overlaying POS information on the outdoor access blocking detection data, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyPOSParams?format=jsonGET or PUT
/ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionRuleParams?format=json-->true
    </isSupportChannelOccupyPOSParams>
    <isSupportChildPowerTimePlanParams>
      <!--ro, opt, bool, whether the device supports power-off schedule parameters of sub-devices, desc:GET/PUT /ISAPI/System/ChildPowerTimePlanParams?
format=json-->true
    </isSupportChildPowerTimePlanParams>
    <isSupportPowerIndicatorLightCtrlParams>
      <!--ro, opt, bool, whether the device supports parameters of device battery indicator control, desc:GET/PUT /ISAPI/System/PowerIndicatorLightCtrlParams?
format=json-->true
    </isSupportPowerIndicatorLightCtrlParams>
    <isSupportRadarVideoMapDetectionParams>
      <!--ro, opt, bool, whether the device supports parameters of radar-assisted map detection, desc:GET
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoMapDetectionParams?format=json-->true
    </isSupportRadarVideoMapDetectionParams>
    <isSupportRadarPOSParams>
      <!--ro, opt, bool, whether the device supports overlaying POS information on the radar detection data, desc:GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/RadarPOSParams?format=json-->true
    </isSupportRadarPOSParams>
    <isSupportVideoIntercomModeEvent>
      <!--ro, opt, bool, whether the device supports the two-way audio mode event, desc:eventType:VideoIntercomModeEvent-->true
    </isSupportVideoIntercomModeEvent>
    <isSupportSoftwareAndHardwareCtrlParams>
      <!--ro, opt, bool, whether the device supports parameters of hardware and software service control, desc:GET/PUT
/ISAPI/System/FunctionSwitchCtrl/SoftwareAndHardwareCtrlParams?format=json-->true
    </isSupportSoftwareAndHardwareCtrlParams>
    <isSupportUserLoginAbnormal>
      <!--ro, opt, bool, whether the device supports user login exception information, desc:GET/PUT /ISAPI/Security/users/UserLoginAbnormal?format=json-->true
    </isSupportUserLoginAbnormal>
    <isSupportIndustryThermometryAdvancedParam>
      <!--ro, opt, bool, whether the device supports the advanced parameters of industrial temperature measurement, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/IndustryThermometryAdvancedParam?format=json-->true
    </isSupportIndustryThermometryAdvancedParam>
    <isSupportGotoVideoRecheckScene>
      <!--ro, opt, bool, whether the device supports jumping to the specified scene of water level recheck detection, desc:POST
/ISAPI/Event/channels/<channelID>/GotoVideoRecheckScene?format=json-->true
    </isSupportGotoVideoRecheckScene>
    <isSupportVideoRecheckRuleParams>
      <!--ro, opt, bool, whether the device supports rule parameters of water level recheck scene, desc:PUT
/ISAPI/Event/channels/<channelID>/VideoRecheckRuleParams?format=json-->true
    </isSupportVideoRecheckRuleParams>
    <isSupportVideoRecheckSceneList>
      <!--ro, opt, bool, whether the device supports getting the list of water level recheck detection scenes, desc:GET
/ISAPI/Event/channels/<channelID>/VideoRecheckSceneList?format=json-->true
    </isSupportVideoRecheckSceneList>
    <isSupportPeopleSedentary>
      <!--ro, opt, bool, whether the device supports sedentary detection, desc:eventType:PeopleSedentary-->true
    </isSupportPeopleSedentary>
    <isSupportPeopleFallingDownDetection>
      <!--ro, opt, bool, whether the device supports person fall detection, desc:eventType: PeopleFallingDownDetection-->true
    </isSupportPeopleFallingDownDetection>
    <isSupportBridgeDeflectionTargetCalibration>
      <!--ro, opt, bool, whether the device supports target calibration of bridge deflection, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetCalibration?format=json-->true
    </isSupportBridgeDeflectionTargetCalibration>
    <isSupportGetBridgeDeflectionTargetCalibrationStatus>
      <!--ro, opt, bool, whether the device supports getting the target calibration status of bridge deflection, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetBridgeDeflectionTargetCalibrationStatus?format=json-->true
    </isSupportGetBridgeDeflectionTargetCalibrationStatus>
    <isSupportBridgeDeflectionMonitoringRuleParams>
      <!--ro, opt, bool, whether the device supports the rule parameters of bridge deflection monitoring, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionMonitoringRuleParams?format=json-->true
    </isSupportBridgeDeflectionMonitoringRuleParams>
    <isSupportHardwareReboot>
      <!--ro, opt, bool, whether the device supports hardware rebooting, desc:PUT /ISAPI/System/HardwareReboot?format=json-->true
    </isSupportHardwareReboot>
    <isSupportSearchSceneCalibrationStatus>
      <!--ro, opt, bool, whether the device supports searching for calibration status of multiple channels by scene, desc:POST
/ISAPI/System/setupCalibration/channels/<channelID>/SearchSceneCalibrationStatus?format=json-->true
    </isSupportSearchSceneCalibrationStatus>
    <isSupportSearchSceneCalibration>
      <!--ro, opt, bool, whether the device supports searching for calibration status of multiple channels by scene, desc:POST
/ISAPI/System/setupCalibration/channels/<channelID>/SearchSceneCalibration?format=json-->true
    </isSupportSearchSceneCalibration>
    <isSupportSoundSourceLocationRuleParams>
      <!--ro, opt, bool, whether the device supports rule parameters of sound source locating, desc:GET/PUT
/ISAPI/System/SoundSourceLocation/AudioIn/<AudioInID>/SoundSourceLocationRuleParams?format=json-->true
    </isSupportSoundSourceLocationRuleParams>
    <isSupportPersonnelChannelLogDebug>
      <!--ro, opt, bool, whether the device supports debugging commands of turnstile logs, desc:GET /ISAPI/ContentMgmt/PersonnelChannelLogDebug/capabilities?
format=json-->true
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format=json-->true
    </isSupportPersonnelChannelLogDebug>
    <isSupportCheckDeviceExistNUCFile>
      <!--ro, opt, bool, whether the device supports checking the locally stored the NUC file, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/CheckDeviceExistNUCFile?format=json-->true
    </isSupportCheckDeviceExistNUCFile>
    <isSupportExportSmartDataData>
      <!--ro, opt, bool, whether the device supports exporting smart data, desc:GET /ISAPI/System/VideoQualityMeasurementCollection/ExportSmartDataData?
format=json-->true
    </isSupportExportSmartDataData>
    <isSupportGetCollectImageCalculateResult>
      <!--ro, opt, bool, whether the device supports getting the calculation result of image collection, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/GetCollectImageCalculateResult?format=json-->true
    </isSupportGetCollectImageCalculateResult>
    <isSupportAIOPAlarmTrans>
      <!--ro, opt, bool, whether the device supports uploading AIOP-related events via ISAPI message transmitted by COMM_VCA_ALARM,
desc:AIOP-related events include the following binary alarm types defined in HCNetSDK:
AIOP-accessed video detection alarm information (binary alarm: COMM_UPLOAD_AIOP_VIDEO, ISAPI alarm: AIOP_Video), AIOP-accessed picture detection alarm
information (binary alarm: COMM_UPLOAD_AIOP_PICTURE, ISAPI alarm: AIOP_Picture), AIOP-accessed capture polling detection alarm information (binary alarm:
COMM_UPLOAD_AIOP_POLLING_SNAP, ISAPI alarm: AIOP_Polling_Snap), AIOP-accessed video polling detection alarm information (binary alarm:
COMM_UPLOAD_AIOP_POLLING_VIDEO, ISAPI alarm: AIOP_Polling_Video)）-->true
    </isSupportAIOPAlarmTrans>
    <isSupportLongDistanceRace>
      <!--ro, opt, bool, whether the device supports long-distance race event, desc:eventType:LongDistanceRace-->true
    </isSupportLongDistanceRace>
    <isSupportIndicatorLightAbnormal>
      <!--ro, opt, bool, whether the device supports abnormal alarm indicator event, desc:eventType:IndicatorLightAbnormal-->true
    </isSupportIndicatorLightAbnormal>
    <isSupportVideoLostPictureStatus>
      <!--ro, opt, bool, whether the device supports the status of custom pictures displayed when video loss occurred, desc:GET
/ISAPI/System/Video/videolosses/VideoLostPictureStatus?format=json-->true
    </isSupportVideoLostPictureStatus>
    <DualVCA>
      <!--ro, opt, object, intelligent information overlay-->
      <isSupportdualVCAList>
        <!--ro, opt, bool, whether it supports setting the parameters for overlaying intelligent information of channels in a batch,
desc:/ISAPI/Streaming/channels/dualVCA?format=json-->true
      </isSupportdualVCAList>
      <enabled opt="true,false" def="true">
        <!--ro, opt, bool, whether to enable the function, attr:opt{req, string},def{req, bool}, desc:"ruleRegion" (overlay only rule area information)-->true
      </enabled>
      <channelID min="1" max="256">
        <!--ro, opt, int, channel No., attr:min{req, int},max{req, int}-->1
      </channelID>
      <overlapRangeType opt="ruleRegion,fullScreen" def="ruleRegion">
        <!--ro, opt, string, type of overlay area, attr:opt{req, string},def{req, string}-->ruleRegion
      </overlapRangeType>
    </DualVCA>
    <isSupportGetScopeCalibrationStatus>
      <!--ro, opt, bool, whether the device supports getting the scope calibration status, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetScopeCalibrationStatus?format=json-->true
    </isSupportGetScopeCalibrationStatus>
    <isSupportSearchShootTableUserGroupParams>
      <!--ro, opt, bool, whether the device supports searching for the parameters of specified zeroing profiles, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SearchShootTableUserGroupParams?format=json-->true
    </isSupportSearchShootTableUserGroupParams>
    <isSupportModifyCurrentShootTableParams>
      <!--ro, opt, bool, whether the device supports editing the information of current valid zeroing profiles, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/ModifyCurrentShootTableParams?format=json-->true
    </isSupportModifyCurrentShootTableParams>
    <isSupportCalibrateScopeTargetPointPosition>
      <!--ro, opt, bool, whether the device supports calibrating the absolute position of the sight cross and drop point, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/CalibrateScopeTargetPointPosition?format=json-->true
    </isSupportCalibrateScopeTargetPointPosition>
    <isSupportGetCurrentShootTableParams>
      <!--ro, opt, bool, whether the device supports getting the information of the current valid zeroing profiles, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetCurrentShootTableParams?format=json-->true
    </isSupportGetCurrentShootTableParams>
    <isSupportVideoGunsightRuleParam>
      <!--ro, opt, bool, whether the device supports rule parameters of video scope, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/VideoGunsightRuleParam?format=json-->true
    </isSupportVideoGunsightRuleParam>
    <isSupportFinishCollectImageData>
      <!--ro, opt, bool, whether it supports completing the collection of image data, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/FinishCollectImageData?format=json-->true
    </isSupportFinishCollectImageData>
    <isSupportStartCollectImageData>
      <!--ro, opt, bool, whether it supports starting collecting image data, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/StartCollectImageData?format=json-->true
    </isSupportStartCollectImageData>
    <isSupportSearchBridgeDeflectionMonitoringEvent>
      <!--ro, opt, bool, whether it supports searching for the event result of bridge deflection detection, desc:POST
/ISAPI/Intelligent/SearchBridgeDeflectionMonitoringEvent?format=json-->true
    </isSupportSearchBridgeDeflectionMonitoringEvent>
    <isSupportGetAllPIPCofigParams>
      <!--ro, opt, bool, whether it supports getting all picture-in-picture configuration parameters, desc:GET /ISAPI/System/Video/GetAllPIPCofigParams?
format=json-->true
    </isSupportGetAllPIPCofigParams>
    <isSupportManualDetectionCarriageClearStatus>
      <!--ro, opt, bool, whether it supports manual detection of the clearing status of the carriage, desc:POST
/ISAPI/System/channels/<channelID>/ManualDetectionCarriageClearStatus?format=json-->true
    </isSupportManualDetectionCarriageClearStatus>
    <isSupportGetCarriageClearCalibrationStatus>
      <!--ro, opt, bool, whether it supports getting the reference calibration carriage clearing status, desc:GET
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/ISAPI/System/channels/<channelID>/GetCarriageClearCalibrationStatus?format=json-->true
  </isSupportGetCarriageClearCalibrationStatus>
  <isSupportModifyCarriageClearCalibration>
    <!--ro, opt, bool, whether it supports setting the reference calibration of carriage clearing, desc:POST
/ISAPI/System/channels/<channelID>/ModifyCarriageClearCalibration?format=json-->true
  </isSupportModifyCarriageClearCalibration>
  <isSupportCarriageClearRuleParams>
    <!--ro, opt, bool, whether it supports setting rule parameters of carriage clearing, desc:GET/PUT
/ISAPI/System/channels/<channelID>/CarriageClearRuleParams?format=json-->true
  </isSupportCarriageClearRuleParams>
  <isSupportRadarFallenObjectsOnTracks>
    <!--ro, opt, bool, whether it supports radar detection of fallen objects on tracks, desc:eventType:RadarFallenObjectsOnTracks-->true
  </isSupportRadarFallenObjectsOnTracks>
  <isSupportTemperatureRiseCalibrationParams>
    <!--ro, opt, bool, whether it supports temperature rise calibration parameters, desc:GET/PUT /ISAPI/Thermal/TemperatureRiseCalibrationParams?
format=json-->true
  </isSupportTemperatureRiseCalibrationParams>
  <isSupportStartTemperatureRiseCalibration>
    <!--ro, opt, bool, whether it supports starting temperature rise calibration, desc:PUT /ISAPI/Thermal/StartTemperatureRiseCalibration?format=json-->true
  </isSupportStartTemperatureRiseCalibration>
  <isSupportGetTemperatureRiseCalibrationStatus>
    <!--ro, opt, bool, whether it supports getting temperature rise calibration status, desc:GET /ISAPI/Thermal/GetTemperatureRiseCalibrationStatus?
format=json-->true
  </isSupportGetTemperatureRiseCalibrationStatus>
  <isSupportExportTemperatureRiseCalibrationFile>
    <!--ro, opt, bool, whether it supports exporting temperature rise calibration file, desc:GET /ISAPI/Thermal/ExportTemperatureRiseCalibrationFile?
format=json-->true
  </isSupportExportTemperatureRiseCalibrationFile>
  <isSupportGetTemperatureRiseCorrectionVersion>
    <!--ro, opt, bool, whether it supports getting temperature rise calibration version information, desc:GET
/ISAPI/Thermal/GetTemperatureRiseCorrectionVersion?format=json-->true
  </isSupportGetTemperatureRiseCorrectionVersion>
  <isSupportSearchImportDeviceDataTaskProgress>
    <!--ro, opt, bool, whether it supports searching for the progress of device data importing task, desc:POST
/ISAPI/System/SearchImportDeviceDataTaskProgress?format=json-->true
  </isSupportSearchImportDeviceDataTaskProgress>
  <isSupportSearchExportDeviceDataTaskProgress>
    <!--ro, opt, bool, whether it supports searching for the progress of device data exporting task, desc:POST
/ISAPI/System/SearchExportDeviceDataTaskProgress?format=json-->true
  </isSupportSearchExportDeviceDataTaskProgress>
  <isSupportAddImportDeviceDataTask>
    <!--ro, opt, bool, whether it supports creating device data importing task, desc:POST /ISAPI/System/AddImportDeviceDataTask?format=json-->true
  </isSupportAddImportDeviceDataTask>
  <isSupportAddExportDeviceDataTask>
    <!--ro, opt, bool, whether it supports creating device data exporting task, desc:POST /ISAPI/System/AddExportDeviceDataTask?format=json-->true
  </isSupportAddExportDeviceDataTask>
  <isSupportSearchRecordStorage>
    <!--ro, opt, bool, whether it supports searching for record storage HDD information, desc:POST /ISAPI/ContentMgmt/search/SearchRecordStorage?
format=json-->true
  </isSupportSearchRecordStorage>
  <isSupportDetectionPathRuleParams>
    <!--ro, opt, bool, whether it supports hazardous gas detection patrol path rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/DetectionPathRuleParams?format=json-->true
  </isSupportDetectionPathRuleParams>
  <isSupportSearchObjectDistanceFocusInfoList>
    <!--ro, opt, bool, whether it supports searching for the list of object distance focus calibration information, desc:POST
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/SearchObjectDistanceFocusInfoList?format=json-->true
  </isSupportSearchObjectDistanceFocusInfoList>
  <isSupportMoveObjectDistanceFocusLocation>
    <!--ro, opt, bool, whether it supports moving to the point where the focus of the object distance is calibrated, desc:PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/MoveObjectDistanceFocusLocation?format=json-->true
  </isSupportMoveObjectDistanceFocusLocation>
  <isSupportAddObjectDistanceFocusInfo>
    <!--ro, opt, bool, whether it supports adding object distance focus calibration information, desc:POST
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/AddObjectDistanceFocusInfo?format=json-->true
  </isSupportAddObjectDistanceFocusInfo>
  <isSupportModifyObjectDistanceFocusInfo>
    <!--ro, opt, bool, whether it supports editing object distance focus calibration information, desc:PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ModifyObjectDistanceFocusInfo?format=json-->true
  </isSupportModifyObjectDistanceFocusInfo>
  <isSupportObjectDistanceCalibrateAutoFocusParams>
    <!--ro, opt, bool, whether it supports auto-focus parameters of object distance calibration, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceCalibrateAutoFocusParams?format=json-->true
  </isSupportObjectDistanceCalibrateAutoFocusParams>
  <isSupportExpresswayNetworkTollAccessParams>
    <!--ro, opt, bool, whether it supports protocol access parameters of network parking fee charging on highway, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceCalibrateAutoFocusParams?format=json-->true
  </isSupportExpresswayNetworkTollAccessParams>
  <isSupportControlCameraDetectAction>
    <!--ro, opt, bool, whether it supports controlling the patrol path of camera, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/ControlCameraDetectAction?format=json-->true
  </isSupportControlCameraDetectAction>
  <isSupportObjectDistanceAccessWay>
    <!--ro, opt, bool, whether it supports the method of getting the object distance, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceAccessWay?format=json-->true
  </isSupportObjectDistanceAccessWay>
  <isSupportCancelTemperatureRiseCalibration>
    <!--ro, opt, bool, whether it supports canceling temperature rise calibration, desc:PUT /ISAPI/Thermal/CancelTemperatureRiseCalibration?format=json--
>true
  </isSupportCancelTemperatureRiseCalibration>
  <isSupportRTCTimeTrigger>
    <!--ro, opt, bool, whether the device supports RTC time exception linkage, desc:GET/PUT /ISAPI/Event/RTCTimeTrigger?format=json-->true
  </isSupportRTCTimeTrigger>
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  <isSupportChargingGunAbnormalEvent>
    <!--ro, opt, bool, whether it supports charging station inserted gun exception event, desc:eventType: ChargingGunAbnormalEvent-->true
  </isSupportChargingGunAbnormalEvent>
  <isSupportVideoChannelEnhance>
    <!--ro, opt, bool, whether it supports enhancement configuration of adding video channel, desc:GET/PUT
/ISAPI/ContentMgmt/InputProxy/VideoChannelEnhance?format=json-->true
  </isSupportVideoChannelEnhance>
  <isSupportDeleteObjectDistanceFocusInfo>
    <!--ro, opt, bool, whether it supports deleting the object distance focus calibration information, desc:POST
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/DeleteObjectDistanceFocusInfo?format=json-->true
  </isSupportDeleteObjectDistanceFocusInfo>
  <isSupportGetStartTemperatureRiseCalibrationParams>
    <!--ro, opt, bool, whether the device supports getting parameter of starting temperature rise calibration, desc:GET
/ISAPI/Thermal/GetStartTemperatureRiseCalibrationParams?format=json-->true
  </isSupportGetStartTemperatureRiseCalibrationParams>
  <isSupportSearchAudioRecordPlan>
    <!--ro, opt, bool, whether the device supports searching all recording schedules, desc:POST /ISAPI/System/RecordAudio/SearchAudioRecordPlan?format=json-
->true
  </isSupportSearchAudioRecordPlan>
  <isSupportCleanerDetectionOverlapParams>
    <!--ro, opt, bool, whether the device supports cleaning liquid detection information overlay parameters, desc:GET/PUT
/ISAPI/System/Hardware/channels/<channelID>/CleanerDetectionOverlapParams?format=json-->true
  </isSupportCleanerDetectionOverlapParams>
  <isSupportWaterLevelVerticalAreaDisplayParams>
    <!--ro, opt, bool, whether it supports the display parameters of water level vertical view monitoring area, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/WaterLevelVerticalAreaDisplayParams?format=json-->true
  </isSupportWaterLevelVerticalAreaDisplayParams>
  <isSupportGetRecordStatusList>
    <!--ro, opt, bool, whether the device supports getting the recording status of all channels, desc:POST /ISAPI/ContentMgmt/search/GetRecordStatusList?
format=json-->true
  </isSupportGetRecordStatusList>
  <isSupportWiperClean>
    <!--ro, opt, bool, whether the device supports wiper cleaning event, desc:eventType: WiperClean-->true
  </isSupportWiperClean>
  <isSupportGetNATStatus>
    <!--ro, opt, bool, whether the device supports getting NAT status, desc:GET /ISAPI/System/Network/NetworkAdvancedCfg/GetNATStatus?format=json-->true
  </isSupportGetNATStatus>
  <isSupportRestoreSensitivityDefaultParams>
    <!--ro, opt, bool, whether the device supports restoring radar perimeter sensitivity parameters to the default, desc:PUT
/ISAPI/Smart/RadarPerimeterAlg/RestoreSensitivityDefaultParams?format=json-->true
  </isSupportRestoreSensitivityDefaultParams>
  <isSupportCarryPassengersPricingStatus>
    <!--ro, opt, bool, whether the device supports displaying the status change of meter, desc:eventType: CarryPassengersPricingStatus-->true
  </isSupportCarryPassengersPricingStatus>
  <isSupportCarryPassengersNoPricing>
    <!--ro, opt, bool, Whether to support event of not using the meter when passengers are on board, desc:eventType: CarryPassengersNoPricing-->true
  </isSupportCarryPassengersNoPricing>
  <isSupportNPlusMultiple>
    <!--ro, opt, bool, whether the device supports N+M mode of host and standby device, desc:whether the device supports setting workMode to N+M. Related
API: PUT /ISAPI/System/Network/NPlusOne/capabilities?format=json-->true
  </isSupportNPlusMultiple>
  <ifPopUpNetWorkCfgSetting>
    <!--ro, opt, bool, whether to pop up network configuration interface, desc:PUT/GET /ISAPI/System/Network/interfaces/<interfaceID>. true (if the default
IP address of the device is 192.168.0.100, the device will pop up the network configuration interface), false or node does not exist (the network
configuration interface will not pop up)-->true
  </ifPopUpNetWorkCfgSetting>
  <AudioOutputChannelCfg>
    <!--ro, opt, object, support linking audio output channel with audio file-->
    <AudioOutputChannelIDList>
      <!--ro, opt, array, audio output channel list, subType:object-->
      <AudioOutputChannelID>
        <!--ro, req, int, audio output channel No.-->1
      </AudioOutputChannelID>
    </AudioOutputChannelIDList>
  </AudioOutputChannelCfg>
  <isSupportImportTemperatureRiseCalibrationFile>
    <!--ro, opt, bool, whether the device supports importing the temperature rise calibration file, desc:PUT
/ISAPI/Thermal/ImportTemperatureRiseCalibrationFile?format=json-->true
  </isSupportImportTemperatureRiseCalibrationFile>
  <isSupportGetAPPQRCodePic>
    <!--ro, opt, bool, whether it supports getting QR code picture of the app accessed by the device, desc:GET /ISAPI/System/GetAPPQRCodePic?format=json--
>true
  </isSupportGetAPPQRCodePic>
  <isSupportPersionQueueRealTimeData>
    <!--ro, opt, bool, whether it supports real-time counting of the number of people in the queuing area, desc:GET
/ISAPI/Intelligent/personQueueDetection/channels/<channelID>/PersionQueueRealTimeData?format=json-->true
  </isSupportPersionQueueRealTimeData>
  <isSupportImageVignettingOptimizeParams>
    <!--ro, opt, bool, whether it supports image dark corner optimization parameters, desc:GET/PUT
/ISAPI/Streaming/channels/<channelID>/ImageVignettingOptimizeParams?format=json-->true
  </isSupportImageVignettingOptimizeParams>
  <isSupportStartMultispectralCalibration>
    <!--ro, opt, bool, whether it supports enabling multispectral calibration, desc:PUT
/ISAPI/System/SpectrumCalibration/channels/<channelID>/StartMultispectralCalibration?format=json-->true
  </isSupportStartMultispectralCalibration>
  <isSupportGetMultispectralCalibrationStatus>
    <!--ro, opt, bool, whether it supports getting the multispectral calibration status, desc:GET
/ISAPI/System/SpectrumCalibration/channels/<channelID>/GetMultispectralCalibrationStatus?format=json-->true
  </isSupportGetMultispectralCalibrationStatus>
  <isSupportStopRadarVideoTargetTracking>
    <!--ro, opt, bool, whether it supports stopping target tracking (radar), desc:PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StopRadarVideoTargetTracking?format=json-->true
  </isSupportStopRadarVideoTargetTracking>
  <isSupportGetRadarVideoTargetTrackingStatus>
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    <!--ro, opt, bool, whether it supports getting the status of target tracking (radar), desc:GET
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/GetRadarVideoTargetTrackingStatus?format=json-->true
  </isSupportGetRadarVideoTargetTrackingStatus>
  <isSupportStartRadarVideoTargetTracking>
    <!--ro, opt, bool, whether it supports starting target tracking (radar), desc:PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StartRadarVideoTargetTracking?format=json-->true
  </isSupportStartRadarVideoTargetTracking>
  <isSupportBatteryChargeCtrl>
    <!--ro, opt, bool, whether it supports battery charging control parameters, desc:GET/PUT /ISAPI/System/BatteryChargeCtrl?format=json-->true
  </isSupportBatteryChargeCtrl>
  <isUpgrageAuxDevice opt="true,false">
    <!--ro, opt, string, whether it supports synchronous upgrade of auxiliary device, attr:opt{opt, string}, desc:it represents support details of
isUpgrageAuxDevice in /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>&childDevID=<devIndex>&isUpgrageAuxDevice=
<isUpgrageAuxDevice>-->true
  </isUpgrageAuxDevice>
  <isSupportClearFile>
    <!--ro, opt, bool, whether it supports clearing files under the specified path of the device, desc:POST
/ISAPI/System/VideoQualityMeasurementCollection/ClearFile?format=json-->true
  </isSupportClearFile>
  <isSupportRestoreDefaultParamByChannel>
    <!--ro, opt, bool, whether it supports restoring default parameters of a single channel, desc:PUT /ISAPI/Streaming/RestoreDefaultParam?
format=json&channelID=<channelID>-->true
  </isSupportRestoreDefaultParamByChannel>
  <isSupportGetMultispectralCalibrationParams>
    <!--ro, opt, bool, whether it supports getting multispectral calibration parameters, desc:GET
/ISAPI/System/SpectrumCalibration/channels/<channelID>/GetMultispectralCalibrationParams?format=json-->true
  </isSupportGetMultispectralCalibrationParams>
  <isSupportGetThermometryCalibrationData>
    <!--ro, opt, bool, whether it supports getting temperature calibration-related data, desc:GET /ISAPI/Thermal/GetThermometryCalibrationData?format=json--
>true
  </isSupportGetThermometryCalibrationData>
  <isSupportImportEncryptedComponentFile>
    <!--ro, opt, bool, whether it supports importing encrypted component files, desc:POST
/ISAPI/System/EncryptionFileUpgradeMgr/ImportEncryptedComponentFile?format=json-->true
  </isSupportImportEncryptedComponentFile>
  <isSupportGetFrameMeanValue>
    <!--ro, opt, bool, whether it supports getting frame average range information, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/GetFrameMeanValue?format=json-->true
  </isSupportGetFrameMeanValue>
  <isSupportGetBaffleImage>
    <!--ro, opt, bool, whether it supports getting image data before and after the image is captured, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/GetBaffleImage?format=json-->true
  </isSupportGetBaffleImage>
  <isSupportThermometryGainMode>
    <!--ro, opt, bool, whether it supports the temperature measurement gain parameters, desc:GET/PUT /ISAPI/Thermal/ThermometryGainMode?format=json-->true
  </isSupportThermometryGainMode>
  <isSupportGetChannelEnhancePreOperationTips>
    <!--ro, opt, bool, whether it supports getting the prompt information before the enhancement configuration of adding the channel, desc:POST
/ISAPI/ContentMgmt/InputProxy/GetChannelEnhancePreOperationTips?format=json-->true
  </isSupportGetChannelEnhancePreOperationTips>
  <isSupportGetDeviceNetworkLinkStatus>
    <!--ro, opt, bool, whether it supports getting the network connection status of the device, desc:GET /ISAPI/System/GetDeviceNetworkLinkStatus?
format=json-->true
  </isSupportGetDeviceNetworkLinkStatus>
  <isSupportManualRefreshOnlineUserList>
    <!--ro, opt, bool, whether the web needs to display a refresh button, desc:used for web interface display, determine whether the device needs to display
the refresh button to refresh the list of online users via this field-->true
  </isSupportManualRefreshOnlineUserList>
  <isSupportVCAAlarmTrans>
    <!--ro, opt, bool, whether it supports uploading events related to abnormal behavior detection through the transmission of ISAPI messages, desc:for all
events related to abnormal behavior detection, refer to HCNetSDK enumeration of VCA_RULE_EVENT_TYPE_EX-->true
  </isSupportVCAAlarmTrans>
  <isSupportHumidityEvent>
    <!--ro, opt, bool, whether it supports humidity alarm events, desc:HumidityEvent-->true
  </isSupportHumidityEvent>
  <isSupportDisplayThermometryFirePrevention>
    <!--ro, opt, bool, whether the web needs to display the temperature measurement function as
temperature measurement & fire prevention,
desc:if this field is true, the web interface display
"Temperature Measurement & Fire Prevention", otherwise, it will be displayed as "Temperature Measurement". If this field is not supported, the web interface
will display "Temperature Measurement"-->true
  </isSupportDisplayThermometryFirePrevention>
  <isSupportPreRemoteCheckEvent>
    <!--ro, opt, bool, whether it supports the collection of pre-verification events, desc:eventType:PreRemoteCheckEvent-->true
  </isSupportPreRemoteCheckEvent>
  <isSupportSearchExportDeviceStatusTaskProgress>
    <!--ro, opt, bool, whether it supports getting the progress of the task to export device information and status, desc:POST
/ISAPI/System/SearchExportDeviceStatusTaskProgress?format=json-->true
  </isSupportSearchExportDeviceStatusTaskProgress>
  <isSupportAddExportDeviceStatusTask>
    <!--ro, opt, bool, whether it supports creating the task to export device information and status, desc:POST /ISAPI/System/AddExportDeviceStatusTask?
format=json-->true
  </isSupportAddExportDeviceStatusTask>
  <isSupportInsulatorThermometryRuleParams>
    <!--ro, opt, bool, whether it supports the insulator temperature measurement rule parameters, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/InsulatorThermometryRuleParams?format=json-->true
  </isSupportInsulatorThermometryRuleParams>
  <isSupportInsulatorCruiseParams>
    <!--ro, opt, bool, whether it supports insulator patrol parameters, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/InsulatorCruiseParams?format=json--
>true
  </isSupportInsulatorCruiseParams>
  <isSupportModifyInsulatorDetectionRegion>
    <!--ro, opt, bool, whether it supports modifying the insulator group detection area, desc:PUT
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/ISAPI/Thermal/channels/<channelID>/ModifyInsulatorDetectionRegion?format=json-->true
  </isSupportModifyInsulatorDetectionRegion>
  <isSupportMoveInsulatorPosition>
    <!--ro, opt, bool, whether it supports movement to the specified insulator string position, desc:POST
/ISAPI/Thermal/channels/<channelID>/MoveInsulatorPosition?format=json-->true
  </isSupportMoveInsulatorPosition>
  <isSupportDeleteInsulatorGroup>
    <!--ro, opt, bool, whether it supports deleting a specified insulator group, desc:POST /ISAPI/Thermal/channels/<channelID>/DeleteInsulatorGroup?
format=json-->true
  </isSupportDeleteInsulatorGroup>
  <isSupportGenerateInsulatorRuleRegion>
    <!--ro, opt, bool, whether it supports generating a specified insulator string rule area, desc:POST
/ISAPI/Thermal/channels/<channelID>/GenerateInsulatorRuleRegion?format=json-->true
  </isSupportGenerateInsulatorRuleRegion>
  <isSupportGetInsulatorAllName>
    <!--ro, opt, bool, whether it supports getting all the names of insulator groups, desc:GET /ISAPI/Thermal/channels/<channelID>/GetInsulatorAllName?
format=json-->true
  </isSupportGetInsulatorAllName>
  <isSupportSearchUploadLog>
    <!--ro, opt, bool, whether it supports searching and uploading log files, desc:PUT /ISAPI/ContentMgmt/SearchUploadLog?format=json-->true
  </isSupportSearchUploadLog>
  <isSupportRainfallTelemeter>
    <!--ro, opt, bool, whether it supports remotely monitoring rainfall, desc:eventType:RainfallTelemeter-->true
  </isSupportRainfallTelemeter>
  <isSupportPosOverlayTest>
    <!--ro, opt, bool, whether it supports the test of enabling POS information overlay configuration, desc:POST
/ISAPI/System/Video/inputs/channels/<channelID>/PosOverlayTest?format=json-->true
  </isSupportPosOverlayTest>
  <isSupportFullScreenPosOverlay>
    <!--ro, opt, bool, whether it supports full-screen POS information overlay parameters, desc:GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/FullScreenPosOverlay?format=json-->true
  </isSupportFullScreenPosOverlay>
  <isSupportFullScreenPosOverlayCap>
    <!--ro, opt, bool, whether to get the dynamic capability of full-screen POS information overlay parameters, desc:GET
/ISAPI/System/Video/inputs/channels/<channelID>/FullScreenPosOverlayCap?format=json-->true
  </isSupportFullScreenPosOverlayCap>
  <isSupportVCAOptimizeDisplay>
    <!--ro, opt, bool, whether it supports the display of the intelligent resource usability page, desc:if the field is true, the user-friendly interface is
displayed; otherwise, the previous one will be displayed; if the field is not supported, it is determined by the device type-->true
  </isSupportVCAOptimizeDisplay>
  <isSupportSearchInsulatorParams>
    <!--ro, opt, bool, whether it supports searching the parameters of a specified insulator group, desc:POST
/ISAPI/Thermal/channels/<channelID>/SearchInsulatorParams?format=json-->true
  </isSupportSearchInsulatorParams>
  <isSupportSetInsulatorParams>
    <!--ro, opt, bool, whether it supports setting insulator group parameters, desc:PUT /ISAPI/Thermal/channels/<channelID>/SetInsulatorParams?format=json--
>true
  </isSupportSetInsulatorParams>
  <isSupportInsulatorAbnormalAlarm>
    <!--ro, opt, bool, whether it supports insulator exception alarm, desc:eventType:InsulatorAbnormalAlarm-->true
  </isSupportInsulatorAbnormalAlarm>
  <isSupportGetFlashLifespanInfo>
    <!--ro, opt, bool, whether it supports getting the lifespan information of the built-in disk storage medium, desc:GET
/ISAPI/System/GetFlashLifespanInfo?format=json-->true
  </isSupportGetFlashLifespanInfo>
  <isSupportDownloadAudioAnalysisFile>
    <!--ro, opt, bool, whether it supports downloading audio analysis files, desc:POST /ISAPI/Intelligent/SensitiveInfoAnalysis/DownloadAudioAnalysisFile?
format=json-->true
  </isSupportDownloadAudioAnalysisFile>
  <isSupportCrowdDispersing>
    <!--ro, opt, bool, whether it supports crowd dispersing, desc:corresponding event: CrowdDispersing-->true
  </isSupportCrowdDispersing>
  <isSupportPeripheralAbnormalEvent>
    <!--ro, opt, bool, whether it supports peripheral fault event, desc:If it is not supported, this node will not be returned. Corresponding eventType:
PeripheralAbnormalEvent-->true
  </isSupportPeripheralAbnormalEvent>
  <isSupportPeripheralAbnormalTrigger>
    <!--ro, opt, bool, whether it supports peripheral fault linkage, desc:GET/PUT /ISAPI/Event/PeripheralAbnormalTrigger?format=json-->true
  </isSupportPeripheralAbnormalTrigger>
  <isSupportBodyCameraAbnormalEvent>
    <!--ro, opt, bool, whether it supports exception events of body cameras, desc:corresponding event: BodyCameraAbnormalEvent-->true
  </isSupportBodyCameraAbnormalEvent>
  <isSupportDeviceWorkModeAsyncStatusEvent>
    <!--ro, opt, bool, whether it supports the event of enforcement device working mode asynchronous status, desc:corresponding event:
DeviceWorkModeAsyncStatusEvent-->true
  </isSupportDeviceWorkModeAsyncStatusEvent>
  <isSupportDeviceUserAsyncStatusEvent>
    <!--ro, opt, bool, whether it supports asynchronous parameter status event for law enforcement device users, desc:corresponding event:
DeviceUserAsyncStatusEvent-->true
  </isSupportDeviceUserAsyncStatusEvent>
  <isSupportVideoCodingAsyncStatusEvent>
    <!--ro, opt, bool, whether it supports video encoding asynchronous parameter status event, desc:corresponding event: VideoCodingAsyncStatusEvent-->true
  </isSupportVideoCodingAsyncStatusEvent>
  <isSupportRecordAsyncStatusEvent>
    <!--ro, opt, bool, whether it supports manual recording asynchronous parameter status event, desc:corresponding event: RecordAsyncStatusEvent-->true
  </isSupportRecordAsyncStatusEvent>
  <isSupportISUPAccessAsyncStatusEvent>
    <!--ro, opt, bool, whether it supports asynchronous parameter status event for ISUP protocol access, desc:corresponding event:
ISUPAccessAsyncStatusEvent-->true
  </isSupportISUPAccessAsyncStatusEvent>
  <isSupportWirelessConfigListAsyncStatusEvent>
    <!--ro, opt, bool, whether it supports asynchronous parameters status event for wireless configuration list, desc:corresponding event:
WirelessConfigListAsyncStatusEvent-->true
  </isSupportWirelessConfigListAsyncStatusEvent>
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  </isSupportWirelessConfigListAsyncStatusEvent>
  <isSupportDeviceLocationAsyncStatusEvent>
    <!--ro, opt, bool, whether it supports asynchronous parameter status event for device location, desc:corresponding event:
DeviceLocationAsyncStatusEvent-->true
  </isSupportDeviceLocationAsyncStatusEvent>
  <isSupportVideoCodingAsyncParams>
    <!--ro, opt, bool, whether it supports video encoding async parameters, desc:GET/PUT /ISAPI/Streaming/VideoCodingAsyncParams?format=json-->true
  </isSupportVideoCodingAsyncParams>
  <isSupportMoveShield>
    <!--ro, opt, bool, whether it supports movement to a specified shielded area, desc:POST /ISAPI/Intelligent/channels/<channelID>/MoveShield?format=json--
>true
  </isSupportMoveShield>
  <isSupportStopAutoGenerateShield>
    <!--ro, opt, bool, whether it supports stopping the automatic recognition of smart shielded area, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/StopAutoGenerateShield?format=json-->true
  </isSupportStopAutoGenerateShield>
  <isSupportStartGenerateParams>
    <!--ro, opt, bool, whether it supports starting generating smart shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/StartGenerateParams?
format=json-->true
  </isSupportStartGenerateParams>
  <isSupportIntelligentShieldAlgorithmsVersion>
    <!--ro, opt, bool, whether it supports getting the version information of the intelligent shielding area recognition algorithm library, desc:GET
/ISAPI/Intelligent/channels/<channelID>/IntelligentShieldAlgorithmsVersion?format=json-->true
  </isSupportIntelligentShieldAlgorithmsVersion>
  <isSupportConvertShield>
    <!--ro, opt, bool, whether it supports converting a specified shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/ConvertShield?format=json-
->true
  </isSupportConvertShield>
  <isSupportDeleteShield>
    <!--ro, opt, bool, whether it supports deleting a specified shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/DeleteShield?format=json--
>true
  </isSupportDeleteShield>
  <isSupportGetGenerateParams>
    <!--ro, opt, bool, whether it supports getting the parameters of generating smart shielded areas, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetGenerateParams?format=json-->true
  </isSupportGetGenerateParams>
  <isSupportManualAddShield>
    <!--ro, opt, bool, whether it supports manually adding shielded areas, desc:PUT /ISAPI/Intelligent/channels/<channelID>/ManualAddShield?format=json--
>true
  </isSupportManualAddShield>
  <isSupportIntelligentShieldParams>
    <!--ro, opt, bool, whether it supports smart shielded area parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/IntelligentShieldParams?
format=json-->true
  </isSupportIntelligentShieldParams>
  <isSupportIntelligentShieldListParams>
    <!--ro, opt, bool, whether it supports the smart shielded area list parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/IntelligentShieldListParams?format=json-->true
  </isSupportIntelligentShieldListParams>
  <isSupportDeicingCfg>
    <!--ro, opt, bool, whether it supports lens heating and de-icing control parameters, desc:GET/PUT /ISAPI/System/Hardware/DeicingCfg?format=json-->true
  </isSupportDeicingCfg>
  <isSupportInspectionOverdueEvent>
    <!--ro, opt, bool, whether it supports annual inspection overdue event for alcohol detection enforcement device, desc:corresponding event:
InspectionOverdueEvent-->true
  </isSupportInspectionOverdueEvent>
  <isSupportChannelsPanoramaCalibration>
    <!--ro, opt, bool, whether it supports batch configuring channel calibration distance, desc:GET/PUT /ISAPI/Panorama/channels/calibration/distance?
format=json-->true
  </isSupportChannelsPanoramaCalibration>
  <isSupportRadarPerimeterShieldOverlapParams>
    <!--ro, opt, bool, whether it supports radar perimeter overlay parameters, desc:GET/PUT
/ISAPI/Smart/RadarPerimeterAlg/Radar/<RadarID>/RadarPerimeterShieldOverlapParams?format=json-->true
  </isSupportRadarPerimeterShieldOverlapParams>
  <isSupportAlarmFilteInterval>
    <!--ro, opt, bool, whether it supports configuring alarm filter time interval, desc:corresponding to the "alarmFilteInterval" node in the POST
/ISAPI/IoTGateway/AlarmRules/SetAlarmRules?format=json-->true
  </isSupportAlarmFilteInterval>
  <isSupportImportLicense>
    <!--ro, opt, bool, whether it supports importing the authorization file, desc:POST /ISAPI/System/SoftwareLicense/ImportLicense?format=json-->true
  </isSupportImportLicense>
  <isSupportShipStatisticsAnalysisParams>
    <!--ro, opt, bool, whether it supports ship traffic statistics parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ShipStatisticsAnalysisParams?format=json-->true
  </isSupportShipStatisticsAnalysisParams>
  <isSupportShipStatisticsEvent>
    <!--ro, opt, bool, whether it supports ship traffic statistics event, desc:corresponding event: ShipStatisticsEvent-->true
  </isSupportShipStatisticsEvent>
  <isSupportPeopleVitalSignsDetection>
    <!--ro, opt, bool, whether it supports vital sign detection event, desc:eventType: PeopleVitalSignsDetection-->true
  </isSupportPeopleVitalSignsDetection>
  <isSupportPeopleVitalSignsStatistic>
    <!--ro, opt, bool, whether it supports vital sign statistics event, desc:eventType: PeopleVitalSignsStatistic-->true
  </isSupportPeopleVitalSignsStatistic>
  <isSupportPeopleBodyMoving>
    <!--ro, opt, bool, whether it supports human body movement detection event, desc:eventType: PeopleBodyMoving-->true
  </isSupportPeopleBodyMoving>
  <isSupportPeopleGetUp>
    <!--ro, opt, bool, whether it supports person get-up detection event, desc:eventType: PeopleGetUp (detected a person who got up from the bed during
posture detection)-->true
  </isSupportPeopleGetUp>
  <isSupportPersonLocationDetection>
    <!--ro, opt, bool, whether it supports person location detection event, desc:eventType: PersonLocationDetection-->true
  </isSupportPersonLocationDetection>
  <isSupportPeopleFallingDownRecovery>
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    <!--ro, opt, bool, whether it supports person fall restoring event, desc:eventType: PeopleFallingDownRecovery-->true
  </isSupportPeopleFallingDownRecovery>
  <isSupportPersonRetentionEvent>
    <!--ro, opt, bool, whether it supports people overstay event, desc:eventType: PersonRetentionEvent-->true
  </isSupportPersonRetentionEvent>
  <isSupportPersonRetentionRecoveryEvent>
    <!--ro, opt, bool, whether it supports people overstay restoring event, desc:eventType: PersonRetentionRecoveryEvent-->true
  </isSupportPersonRetentionRecoveryEvent>
  <isSupportPeopleLongTimeLeaveEvent>
    <!--ro, opt, bool, whether it supports person long-time off-bed event, desc:eventType: PeopleLongTimeLeaveEvent-->true
  </isSupportPeopleLongTimeLeaveEvent>
  <isSupportPeopleLongTimeLeaveRecoveryEvent>
    <!--ro, opt, bool, whether it supports person long-time off-bed restoring event, desc:eventType: PeopleLongTimeLeaveRecoveryEvent-->true
  </isSupportPeopleLongTimeLeaveRecoveryEvent>
  <isSupportPeopleVitalSignsAbnormalEvent>
    <!--ro, opt, bool, whether it supports vital sign exception event, desc:eventType: PeopleVitalSignsAbnormalEvent-->true
  </isSupportPeopleVitalSignsAbnormalEvent>
  <isSupportPeopleVitalSignsRecoveryEvent>
    <!--ro, opt, bool, whether it supports vital sign exception restoring event, desc:eventType: PeopleVitalSignsRecoveryEvent-->true
  </isSupportPeopleVitalSignsRecoveryEvent>
  <isSupportPeopleVitalSignsExportTaskEvent>
    <!--ro, opt, bool, whether it supports vital sign data exporting task result event, desc:eventType: PeopleVitalSignsExportTaskEvent-->true
  </isSupportPeopleVitalSignsExportTaskEvent>
  <isSupportTOFCalibrate>
    <!--ro, opt, bool, whether it supports configuring TOF detection calibration, desc:PUT /ISAPI/System/Video/inputs/tamperDetection/TOFCalibrate?
format=json-->true
  </isSupportTOFCalibrate>
  <isSupportSearchBulkUpgradeChildDeviceTask>
    <!--ro, opt, bool, whether it supports searching for tasks of batch upgrading sub-devices, desc:/ISAPI/System/SearchBulkUpgradeChildDeviceTask?
format=json-->true
  </isSupportSearchBulkUpgradeChildDeviceTask>
  <isSupportModifyBulkUpgradeChildDeviceTask>
    <!--ro, opt, bool, whether it supports modifying the task of batch upgrading sub-devices, desc:/ISAPI/System/ModifyBulkUpgradeChildDeviceTask?
format=json-->true
  </isSupportModifyBulkUpgradeChildDeviceTask>
  <isSupportDeleteBulkUpgradeChildDeviceTask>
    <!--ro, opt, bool, whether it supports deleting tasks of batch upgrading sub-devices, desc:/ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json--
>true
  </isSupportDeleteBulkUpgradeChildDeviceTask>
  <isSupportAddEventTemplate>
    <!--ro, opt, bool, whether it supports adding event message templates, desc:POST /ISAPI/Event/notification/httpHosts/AddEventTemplate?format=json-->true
  </isSupportAddEventTemplate>
  <isSupportTestEventTemplate>
    <!--ro, opt, bool, whether it supports testing event message templates, desc:POST /ISAPI/Event/notification/httpHosts/TestEventTemplate?format=json--
>true
  </isSupportTestEventTemplate>
  <isSupportEventTemplateParams>
    <!--ro, opt, bool, whether it supports event message template parameters, desc:GET /ISAPI/Event/notification/httpHosts/EventTemplateParams?format=json--
>true
  </isSupportEventTemplateParams>
  <isSupportAddCombineEvents>
    <!--ro, opt, bool, whether it supports adding combined event, desc:POST /ISAPI/Event/CombineEvents/AddCombineEvents?format=json-->true
  </isSupportAddCombineEvents>
  <isSupportModifyCombineEventsParams>
    <!--ro, opt, bool, whether it supports modifying combined event parameters, desc:POST /ISAPI/Event/CombineEvents/ModifyCombineEventsParams?format=json--
>true
  </isSupportModifyCombineEventsParams>
  <isSupportSearchCombineEventsParams>
    <!--ro, opt, bool, whether it supports searching for combined event parameters, desc:POST /ISAPI/Event/CombineEvents/SearchCombineEventsParams?
format=json-->true
  </isSupportSearchCombineEventsParams>
  <isSupportDeleteCombineEventsList>
    <!--ro, opt, bool, whether it supports batch deleting combined event, desc:POST /ISAPI/Event/CombineEvents/DeleteCombineEventsList?format=json-->true
  </isSupportDeleteCombineEventsList>
  <isSupportEventRuleAndTriggerParams>
    <!--ro, opt, bool, whether it supports configuring the rule linkage parameters of a specified event, desc:POST
/ISAPI/Event/EventBatchRulesMgr/EventRuleAndTriggerParams?format=json-->true
  </isSupportEventRuleAndTriggerParams>
  <isSupportSearchEventRuleAndTriggerParams>
    <!--ro, opt, bool, whether it supports searching for the rule linkage parameters of a specified event, desc:POST
/ISAPI/Event/EventBatchRulesMgr/SearchEventRuleAndTriggerParams?format=json-->true
  </isSupportSearchEventRuleAndTriggerParams>
  <isSupportCallMessageHandlingEvent>
    <!--ro, opt, bool, whether it supports the call message handling event, desc:eventType: CallMessageHandlingEvent-->true
  </isSupportCallMessageHandlingEvent>
  <isSupportResetBatchChild>
    <!--ro, opt, bool, whether it supports batch restoring sub-devices, desc:PUT /ISAPI/System/ResetBatchChild?format=json-->true
  </isSupportResetBatchChild>
  <isSupportSearchResetBatchChildStatus>
    <!--ro, opt, bool, whether it supports searching for the status of batch restoring sub-devices, desc:POST /ISAPI/System/SearchResetBatchChildStatus?
format=json-->true
  </isSupportSearchResetBatchChildStatus>
  <isSupportSearchRebootBatchChildStatus>
    <!--ro, opt, bool, whether it supports searching for the status of batch rebooting sub-devices, desc:POST /ISAPI/System/SearchRebootBatchChildStatus?
format=json-->true
  </isSupportSearchRebootBatchChildStatus>
  <isSupportRebootBatchChild>
    <!--ro, opt, bool, whether it supports batch restarting sub-devices, desc:PUT /ISAPI/System/RebootBatchChild?format=json-->true
  </isSupportRebootBatchChild>
  <isSupportFlowStatisticsTelemeter>
    <!--ro, opt, bool, whether it supports traffic statistics telemetry event, desc:eventType:FlowStatisticsTelemeter-->true
  </isSupportFlowStatisticsTelemeter>
  <isSupportModifyFlowStatisticsTelemeterUpload>
    <!--ro, opt, bool, whether it supports modifying the parameters of traffic statistics telemetry report strategy, desc:PUT
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    <!--ro, opt, bool, whether it supports modifying the parameters of traffic statistics telemetry report strategy, desc:PUT
/ISAPI/System/Network/NetworkFlowMonitor/ModifyFlowStatisticsTelemeterUpload?format=json-->true
  </isSupportModifyFlowStatisticsTelemeterUpload>
  <isSupportClearFlowStatisticsTelemeter>
    <!--ro, opt, bool, whether it supports clearing the flow statistics telemetry data, desc:PUT
/ISAPI/System/Network/NetworkFlowMonitor/ClearFlowStatisticsTelemeter?format=json-->true
  </isSupportClearFlowStatisticsTelemeter>
  <isSupportFlowStatisticsTelemeterUpload>
    <!--ro, opt, bool, whether it supports the remote telemetry reporting strategy parameters for traffic statistics, desc:GET
/ISAPI/System/Network/NetworkFlowMonitor/FlowStatisticsTelemeterUpload?format=json-->true
  </isSupportFlowStatisticsTelemeterUpload>
  <isSupportScreenGraphicsColorControl>
    <!--ro, opt, bool, whether it supports screen image color control, desc:POST /ISAPI/System/ScreenColorQualityDetection/ScreenGraphicsColorControl?
format=json-->true
  </isSupportScreenGraphicsColorControl>
  <isSupportSearchScreenColorQualityDetectionResult>
    <!--ro, opt, bool, whether it supports searching for the screen image color quality detection results, desc:POST
/ISAPI/System/ScreenColorQualityDetection/SearchScreenColorQualityDetectionResult?format=json-->true
  </isSupportSearchScreenColorQualityDetectionResult>
  <isSupportSearchScreenColorQualityDetectionProcess>
    <!--ro, opt, bool, whether it supports searching for the screen image color quality detection progress, desc:POST
/ISAPI/System/ScreenColorQualityDetection/SearchScreenColorQualityDetectionProcess?format=json-->true
  </isSupportSearchScreenColorQualityDetectionProcess>
  <isSupportScreenColorQualityDetectionParams>
    <!--ro, opt, bool, whether it supports the screen image color quality detection parameters, desc:GET/PUT
/ISAPI/System/ScreenColorQualityDetection/channels/<channelID>/ScreenColorQualityDetectionParams?format=json-->true
  </isSupportScreenColorQualityDetectionParams>
  <isSupportAddInsulatorGroup>
    <!--ro, opt, bool, whether it supports adding insulator list, desc:GET /ISAPI/Thermal/channels/<channelID>/AddInsulatorGroup?format=json-->true
  </isSupportAddInsulatorGroup>
  <isSupportSuperResolutionReboot>
    <!--ro, opt, bool, whether the device supports rebooting super resolution, desc:whether the device supports restart to complete the conversion: true
(yes), false (no). Different devices use different chips. Some devices convert videos/images to ones with super resolution effect when the super resolution
function is enabled, while others require a device restart before the conversion is completed-->true
  </isSupportSuperResolutionReboot>
  <isSupportStartAutoTest>
    <!--ro, opt, bool, whether it supports starting automatic test, desc:POST /ISAPI/System/AutoTest/StartAutoTest?format=json-->true
  </isSupportStartAutoTest>
  <isSupportGetUserManualLink>
    <!--ro, opt, bool, whether it supports getting the device user manual link, desc:GET /ISAPI/System/GetUserManualLink?format=json-->true
  </isSupportGetUserManualLink>
  <isSupportExportSignaturePublicKey>
    <!--ro, opt, bool, whether it supports exporting the signed public key, desc:GET /ISAPI/Security/ProtocolCertificate/ExportSignaturePublicKey?
format=json-->true
  </isSupportExportSignaturePublicKey>
  <isSupportCipherCardRunningStatus>
    <!--ro, opt, bool, whether it supports the password card running status, desc:GET /ISAPI/System/CipherCardRunningStatus?format=json-->true
  </isSupportCipherCardRunningStatus>
  <isSupportBridgeDeflectionMonitoringCapture>
    <!--ro, opt, bool, whether it supports manual triggering of bridge deflection monitoring and capturing, desc:POST
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionMonitoringCapture?format=json-->true
  </isSupportBridgeDeflectionMonitoringCapture>
  <isSupportAdaptiveVideoStreamParams>
    <!--ro, opt, bool, whether it supports adaptive video stream control parameters, desc:GET/PUT /ISAPI/ContentMgmt/InputProxy/AdaptiveVideoStreamParams?
format=json-->true
  </isSupportAdaptiveVideoStreamParams>
  <isSupportSearchAudioDetectionTaskProcess>
    <!--ro, opt, bool, whether it supports searching for the progress of audio detection tasks, desc:POST
/ISAPI/System/DeviceAudioDetection/SearchAudioDetectionTaskProcess?format=json-->true
  </isSupportSearchAudioDetectionTaskProcess>
  <isSupportSendAudioDetectionTask>
    <!--ro, opt, bool, whether it supports applying audio detection tasks, desc:POST /ISAPI/System/DeviceAudioDetection/SendAudioDetectionTask?format=json--
>true
  </isSupportSendAudioDetectionTask>
  <isSupportMedicalImageURLExpiredEvent>
    <!--ro, opt, bool, whether it supports medical image URL expiration event, desc:eventType:MedicalImageURLExpiredEvent-->true
  </isSupportMedicalImageURLExpiredEvent>
  <isSupportBridgeDeflectionMonitoringEvent>
    <!--ro, opt, bool, whether it supports bridge deflection monitoring event, desc:eventType:BridgeDeflectionMonitoringEvent-->true
  </isSupportBridgeDeflectionMonitoringEvent>
  <isSupportBridgeDeflectionTargetLossEvent>
    <!--ro, opt, bool, whether it supports bridge deflection target loss event, desc:eventType:BridgeDeflectionTargetLossEvent-->true
  </isSupportBridgeDeflectionTargetLossEvent>
  <isSupportNobodyDetectionRuleParams>
    <!--ro, opt, bool, whether it supports unmanned detection rule parameters, desc:GET/PUT /ISAPI/Event/channels/<channelID>/NobodyDetectionRuleParams?
format=json-->true
  </isSupportNobodyDetectionRuleParams>
  <isSupportNobodyDetectionRegionShieldParams>
    <!--ro, opt, bool, whether it supports the shielding parameters for unmanned detection areas, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/NobodyDetectionRegionShieldParams?format=json-->true
  </isSupportNobodyDetectionRegionShieldParams>
  <isSupportReduceTemperatureFilterParams>
    <!--ro, opt, bool, whether it supports the cooling filter parameter, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ReduceTemperatureFilterParams?
format=json-->true
  </isSupportReduceTemperatureFilterParams>
  <isSupportGetStreamOverlayLogDebugStatus>
    <!--ro, opt, bool, whether it supports getting the status of overlaying log debugging status on stream, desc:GET
/ISAPI/ContentMgmt/channels/<channelID>/GetStreamOverlayLogDebugStatus?format=json-->true
  </isSupportGetStreamOverlayLogDebugStatus>
  <isSupportStreamOverlayLogDebugParams>
    <!--ro, opt, bool, whether it supports the parameters of overlaying log debugging information on stream, desc:GET/PUT
/ISAPI/ContentMgmt/channels/<channelID>/StreamOverlayLogDebugParams?format=json-->true
  </isSupportStreamOverlayLogDebugParams>
  <isSupportBridgeDeflectionAbnormalRuleParams>
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    <!--ro, opt, bool, whether it supports the bridge deflection exception detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionAbnormalRuleParams?format=json-->true
  </isSupportBridgeDeflectionAbnormalRuleParams>
  <isSupportBridgeDeflectionAbnormalEvent>
    <!--ro, opt, bool, whether it supports bridge deflection exception event, desc:eventType:BridgeDeflectionAbnormalEvent-->true
  </isSupportBridgeDeflectionAbnormalEvent>
  <isSupportRestoreBackup>
    <!--ro, opt, bool, whether it supports data restoring and backup, desc:POST /ISAPI/System/SystemSnapshot/RestoreBackup?format=json-->true
  </isSupportRestoreBackup>
  <isSupportDeleteBackupFile>
    <!--ro, opt, bool, whether it supports deleting backup files, desc:POST /ISAPI/System/SystemSnapshot/DeleteBackupFile?format=json-->true
  </isSupportDeleteBackupFile>
  <isSupportSearchRestoreBackupProgress>
    <!--ro, opt, bool, whether it supports searching for the progress of restoring backup files, desc:POST
/ISAPI/System/SystemSnapshot/SearchRestoreBackupProgress?format=json-->true
  </isSupportSearchRestoreBackupProgress>
  <isSupportGetBackupsFileList>
    <!--ro, opt, bool, whether it supports getting the list of backup files, desc:POST /ISAPI/System/SystemSnapshot/GetBackupsFileList?format=json-->true
  </isSupportGetBackupsFileList>
  <isSupportAutoBackupParams>
    <!--ro, opt, bool, whether it supports automatic backup parameters, desc:GET/PUT /ISAPI/System/SystemSnapshot/AutoBackupParams?format=json-->true
  </isSupportAutoBackupParams>
  <isSupportDeviceSoftwareRunningProgramRestart>
    <!--ro, opt, bool, whether it supports restarting the device software program, desc:POST /ISAPI/System/DeviceSoftwareRunningProgramRestart?format=json--
>true
  </isSupportDeviceSoftwareRunningProgramRestart>
  <isSupportSendScreenColorQualityDetectionRequest>
    <!--ro, opt, bool, whether it supports sending a screen color quality detection request, desc:POST
/ISAPI/System/ScreenColorQualityDetection/SendScreenColorQualityDetectionRequest?format=json-->true
  </isSupportSendScreenColorQualityDetectionRequest>
  <isSupportTelephoningDetectionRuleParams>
    <!--ro, opt, bool, whether it supports phone call detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/telephoningDetect/TelephoningDetectionRuleParams?format=json-->true
  </isSupportTelephoningDetectionRuleParams>
  <isSupportTelephoningBehaviorAlarmEvent>
    <!--ro, opt, bool, whether it supports phone call alarm event, desc:eventType:TelephoningBehaviorAlarmEvent-->true
  </isSupportTelephoningBehaviorAlarmEvent>
  <isSupportTelephoningBehaviorRecoveryEvent>
    <!--ro, opt, bool, whether it supports phone call restoring event, desc:eventType:TelephoningBehaviorRecoveryEvent-->true
  </isSupportTelephoningBehaviorRecoveryEvent>
  <isSupportSleepDutyRecoveryEvent>
    <!--ro, opt, bool, whether the device supports absence or sleeping on duty restoring event, desc:eventType:sleepDutyRecoveryEvent-->true
  </isSupportSleepDutyRecoveryEvent>
  <isSupportGetBridgeDeflectionTargetOffsetResetStatus>
    <!--ro, opt, bool, whether it supports getting the zeroing result of bridge deflection target displacement, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetBridgeDeflectionTargetOffsetResetStatus?format=json-->true
  </isSupportGetBridgeDeflectionTargetOffsetResetStatus>
  <isSupportBridgeDeflectionTargetOffsetReset>
    <!--ro, opt, bool, whether it supports bridge deflection target displacement reset, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetOffsetReset?format=json-->true
  </isSupportBridgeDeflectionTargetOffsetReset>
  <isSupportBridgeDeflectionTargetLossRuleParams>
    <!--ro, opt, bool, whether it supports bridge deflection target loss rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetLossRuleParams?format=json-->true
  </isSupportBridgeDeflectionTargetLossRuleParams>
  <isSupportGetAnalogOutputParams>
    <!--ro, opt, bool, whether it supports getting all analog output parameters, desc:GET /ISAPI/System/AnalogOutput/GetAnalogOutputParams?format=json--
>true
  </isSupportGetAnalogOutputParams>
  <isSupportSystemReset>
    <!--ro, opt, bool, whether the device supports restoring system to default settings, desc:POST /ISAPI/System/SystemReset?format=json-->true
  </isSupportSystemReset>
  <isSupportFakePlateDetection>
    <!--ro, opt, bool, whether it supports fake license plate detection event, desc:corresponding event: FakePlateDetection-->true
  </isSupportFakePlateDetection>
  <isSupportSafetyWorkwearDetectionEvent>
    <!--ro, opt, bool, whether it supports the event of wearing safety work clothes detection, desc:corresponding event: SafetyWorkwearDetectionEvent-->true
  </isSupportSafetyWorkwearDetectionEvent>
  <isSupportVehicleMaintenanceStatusEvent>
    <!--ro, opt, bool, whether it supports vehicle maintenance status events, desc:corresponding event: VehicleMaintenanceStatusEvent-->true
  </isSupportVehicleMaintenanceStatusEvent>
  <isSupportEventRetransmissionInfo>
    <!--ro, opt, bool, whether it supports reloading event information, desc:GET /ISAPI/System/EventRetransmissionInfo?format=json-->true
  </isSupportEventRetransmissionInfo>
  <isSupportGetEnvironmentCaptureStatus>
    <!--ro, opt, bool, whether it supports getting the environmental data collection status, desc:POST
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/GetEnvironmentCaptureStatus?format=json-->true
  </isSupportGetEnvironmentCaptureStatus>
  <isSupportEnvironmentCapture>
    <!--ro, opt, bool, whether it supports environmental data collection, desc:POST
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/EnvironmentCapture?format=json-->true
  </isSupportEnvironmentCapture>
  <isSupportSearchCPUDetailInfo>
    <!--ro, opt, bool, whether it supports searching for detailed CPU performance information, desc:POST /ISAPI/System/SearchCPUDetailInfo?format=json--
>true
  </isSupportSearchCPUDetailInfo>
  <isSupportPanelControlParams>
    <!--ro, opt, bool, whether it supports device hardware panel configuration parameters, desc:GET/PUT /ISAPI/System/PanelControlParams?format=json-->true
  </isSupportPanelControlParams>
  <isSupportSearchGoodsIncAndDecStatistics>
    <!--ro, opt, bool, whether it supports searching for cargo increase/decrease statistics, desc:POST
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/SearchGoodsIncAndDecStatistics?format=json-->true
  </isSupportSearchGoodsIncAndDecStatistics>
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<isSupportCustomAlarmHTTPParams>
<!--ro, opt, bool, whether it supports custom alarm HTTP parameters, desc:GET/PUT /ISAPI/Event/notification/httpHosts/CustomAlarmHTTPParams?format=json-->true
</isSupportCustomAlarmHTTPParams>
<isSupportTestCustomAlarmHTTP>
<!--ro, opt, bool, whether it supports custom alarm HTTP request test, desc:PUT /ISAPI/Event/notification/httpHosts/TestCustomAlarmHTTP?format=json-->true
</isSupportTestCustomAlarmHTTP>
<isSupportCustomAlarmHTTPLinkageParams>
<!--ro, opt, bool, whether it supports customizing the alarm HTTP linkage parameters, desc:GET/PUT /ISAPI/Thermal/CustomAlarmHTTPLinkageParams?format=json-->true
</isSupportCustomAlarmHTTPLinkageParams>
<isSupportAccumulationDetectionPeriodicReport>
<!--ro, opt, bool, whether it supports scheduled reporting of pile detection events, desc:corresponding event: AccumulationDetectionPeriodicReport-->true
</isSupportAccumulationDetectionPeriodicReport>
<isSupportImportDiskCompatibilityCertificate>
<!--ro, opt, bool, whether it supports importing disk compatibility certificate, desc:PUT /ISAPI/Security/serverCertificate/certificate?type=<type>&platformNo=<platformNo>, and type is DiskCompatibility-->true
</isSupportImportDiskCompatibilityCertificate>
<isSupportGetAcumulationDetectionAlgorithmicVersion>
<!--ro, opt, bool, whether it supports getting the version of the stacking detection algorithm, desc:GET /ISAPI/Event/channels/<channelID>/objectAccumulationDetection/GetAcumulationDetectionAlgorithmicVersion?format=json-->true
</isSupportGetAcumulationDetectionAlgorithmicVersion>
<isSupportSnapshotUpgradeDifferenceFile>
<!--ro, opt, bool, whether it supports getting snapshot upgrade difference files, desc:GET /ISAPI/System/SystemSnapshot/SnapshotUpgradeDifferenceFile?format=json-->true
</isSupportSnapshotUpgradeDifferenceFile>
<isSupportGroupRecoveryEvent>
<!--ro, opt, bool, whether it supports people gathering restoring event-->true
</isSupportGroupRecoveryEvent>
<isSupportElectronicEyepieceParams>
<!--ro, opt, bool, whether it supports electron microscope channel parameters, desc:GET/PUT /ISAPI/System/Video/ElectronicEyepieceParams?format=json-->true
</isSupportElectronicEyepieceParams>
<isSupportSmokingDetectionRecoverEvent>
<!--ro, opt, bool, whether it supports smoking detection restoring event, desc:fire alarm upload in progress, eventType: SmokingDetectionRecoverEvent-->true
</isSupportSmokingDetectionRecoverEvent>
<isSupportDeviceRegionParams>
<!--ro, opt, bool, whether it supports device area parameters, desc:GET/PUT /ISAPI/System/DeviceRegionParams?format=json-->true
</isSupportDeviceRegionParams>
<isSupportTemperatureEvent>
<!--ro, opt, bool, whether it supports temperature alarm event, desc:TemperatureEvent-->true
</isSupportTemperatureEvent>
<isSupportTemperatureRecoverEvent>
<!--ro, opt, bool, whether it supports temperature restoring event, desc:TemperatureRecoverEvent-->true
</isSupportTemperatureRecoverEvent>
<isSupportWebSSH>
<!--ro, opt, bool, whether it supports WebSSH, desc:if the node is not returned, it indicates no-->true
</isSupportWebSSH>
<isSupportCreateDebugChannel>
<!--ro, opt, bool, whether it supports creating WebSSH debugging channel, desc:if the node is not returned, it indicates no-->true
</isSupportCreateDebugChannel>
<isSupportExportDebugLog>
<!--ro, opt, bool, whether it supports exporting the debugging log, desc:if the node is not returned, it indicates no-->true
</isSupportExportDebugLog>
<isSupportAcumulationDetectionEffectiveAreaCfgParams>
<!--ro, opt, bool, whether it supports the configuration parameters of pile detection effective area, desc:GET/PUT /ISAPI/Event/channels/<channelID>/objectAccumulationDetection/AcumulationDetectionEffectiveAreaCfgParams?format=json-->true
</isSupportAcumulationDetectionEffectiveAreaCfgParams>
<isSupportGetGoalDryWetBallTempCurve>
<!--ro, opt, bool, whether it supports getting the target dry-wet bulb temperature curve, desc:POST /ISAPI/System/TobaccoBakeMgr/GetGoalDryWetBallTempCurve?format=json-->true
</isSupportGetGoalDryWetBallTempCurve>
<isSupportBakeTobaccoRuleParams>
<!--ro, opt, bool, whether it supports tobacco baking rule parameters, desc:GET/PUT /ISAPI/System/TobaccoBakeMgr/BakeTobaccoRuleParams?format=json-->true
</isSupportBakeTobaccoRuleParams>
<isSupportStartBakeTobacco>
<!--ro, opt, bool, whether it supports starting baking tobacco, desc:POST /ISAPI/System/TobaccoBakeMgr/StartBakeTobacco?format=json-->true
</isSupportStartBakeTobacco>
<isSupportEndBakeTobacco>
<!--ro, opt, bool, whether it supports stopping baking tobacco, desc:POST /ISAPI/System/TobaccoBakeMgr/EndBakeTobacco?format=json-->true
</isSupportEndBakeTobacco>
<isSupportDryWetBallTempEventRuleParams>
<!--ro, opt, bool, whether it supports the dry-wet bulb temperature reporting rule parameters, desc:GET/PUT /ISAPI/System/TobaccoBakeMgr/DryWetBallTempEventRuleParams?format=json-->true
</isSupportDryWetBallTempEventRuleParams>
<isSupportDevAccessServerNetworkStatus>
<!--ro, opt, bool, whether it supports device connection status with multi-access service network, desc:GET /ISAPI/System/DevAccessServerNetworkStatus?format=json-->true
</isSupportDevAccessServerNetworkStatus>
<isSupportMaintenanceServiceAuthorization>
<!--ro, opt, bool, whether it supports device maintenance service authorization information, desc:GET/PUT /ISAPI/System/MaintenanceServiceAuthorization?format=json-->true
</isSupportMaintenanceServiceAuthorization>
<isSupportUploadIndividualCallSignalOperation>
<!--ro, opt, bool, whether it supports uploading single-mode call signaling events, desc:UploadIndividualCallSignalOperation-->true
</isSupportUploadIndividualCallSignalOperation>
<isSupportUploadGroupCallSignalOperation>
<!--ro, opt, bool, whether it supports uploading group call signaling event, desc:UploadGroupCallSignalOperation-->true
</isSupportUploadGroupCallSignalOperation>
<isSupportDrivingBehaviorVoiceReminder>
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    <!--ro, opt, bool, whether it supports configuring intelligent voice reminders for driving behavior, desc:GET/PUT
/ISAPI/Event/DrivingBehaviorVoiceReminder?format=json-->true
  </isSupportDrivingBehaviorVoiceReminder>
  <isSupportImportMultiCameraCalibratationFile>
    <!--ro, opt, bool, whether it supports importing image calibration files of multiple cameras, desc:POST
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/ImportMultiCameraCalibratationFile?format=json-->true
  </isSupportImportMultiCameraCalibratationFile>
  <isSupportDeletePlayerFileList>
    <!--ro, opt, bool, whether it supports batch deleting player files, desc:POST /ISAPI/System/Video/DeletePlayerFileList?format=json-->true
  </isSupportDeletePlayerFileList>
  <isSupportPlayerInfoParams>
    <!--ro, opt, bool, whether it supports player information parameters, desc:GET /ISAPI/System/Video/PlayerInfoParams?format=json-->true
  </isSupportPlayerInfoParams>
  <isSupportBurnPlayerParams>
    <!--ro, opt, bool, whether it supports burning player parameters, desc:GET/PUT /ISAPI/System/Video/BurnPlayerParams?format=json-->true
  </isSupportBurnPlayerParams>
  <isSupportOneKeyVoiceAbnormalDetection>
    <!--ro, opt, bool, whether it supports quick voice exception detection, desc:POST
/ISAPI/System/VoiceAbnormalDetection/channels/<channelID>/OneKeyVoiceAbnormalDetection?format=json-->true
  </isSupportOneKeyVoiceAbnormalDetection>
  <isSupportVoiceAbnormalDetectionRuleParams>
    <!--ro, opt, bool, whether it supports quick voice exception detection parameters, desc:GET/PUT
/ISAPI/System/VoiceAbnormalDetection/channels/<channelID>/VoiceAbnormalDetectionRuleParams?format=json-->true
  </isSupportVoiceAbnormalDetectionRuleParams>
  <isSupportSearchOneKeyVoiceAbnormalDetectionProcess>
    <!--ro, opt, bool, whether it supports searching for the progress of quick voice exception detection, desc:POST
/ISAPI/System/VoiceAbnormalDetection/SearchOneKeyVoiceAbnormalDetectionProcess?format=json-->true
  </isSupportSearchOneKeyVoiceAbnormalDetectionProcess>
  <isSupportSearchOneKeyAudioDetection>
    <!--ro, opt, bool, whether it supports searching for the progress of quick audio detection, desc:POST
/ISAPI/Smart/AudioExceptionDetection/SearchOneKeyAudioDetection?format=json-->true
  </isSupportSearchOneKeyAudioDetection>
  <isSupportOneKeyAudioDetection>
    <!--ro, opt, bool, whether it supports quick audio detection, desc:POST /ISAPI/Smart/AudioExceptionDetection/channels/<channelID>/OneKeyAudioDetection?
format=json-->true
  </isSupportOneKeyAudioDetection>
  <isNotSupportScheduledRecordUpload>
    <!--ro, opt, bool, whether it supports the event of scheduled recording of channel, desc:if this node does not exist or returns false (default value),
it indicates for support; if it returns true, it indicates no-->false
  </isNotSupportScheduledRecordUpload>
  <isSupportDevEnvironmentalCtrlMaterielInfoDetect>
    <!--ro, opt, bool, whether it supports the detection of equipment environmental control material information?, desc:POST
/ISAPI/System/DeviceModuleDiagnosis/DevEnvironmentalCtrlMaterielInfoDetect?format=json-->true
  </isSupportDevEnvironmentalCtrlMaterielInfoDetect>
  <isSupportWaterRescueDetectionRuleParams>
    <!--ro, opt, bool, whether it supports water rescue detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/WaterRescueDetectionRuleParams?format=json-->true
  </isSupportWaterRescueDetectionRuleParams>
  <isSupportWaterRescueDetectionEvent>
    <!--ro, opt, bool, whether it supports water rescue detection event, desc:corresponding event: WaterRescueDetectionEvent-->true
  </isSupportWaterRescueDetectionEvent>
  <isSupportGetActualDryWetBallTempCurve>
    <!--ro, opt, bool, whether it supports getting the actual wet-bulb temperature curve, desc:POST
/ISAPI/System/TobaccoBakeMgr/GetActualDryWetBallTempCurve?format=json-->true
  </isSupportGetActualDryWetBallTempCurve>
  <isSupportPerimeterAlgoTargetFilterParams>
    <!--ro, opt, bool, whether it supports target filtering parameters of perimeter algorithm, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterAlgoMgr/PerimeterAlgoTargetFilterParams?format=json-->true
  </isSupportPerimeterAlgoTargetFilterParams>
  <isSupportSendDeviceMaintenanceStatus>
    <!--ro, opt, bool, whether it supports applying device maintenance status, desc:PUT /ISAPI/System/SendDeviceMaintenanceStatus?format=json-->true
  </isSupportSendDeviceMaintenanceStatus>
  <videoEncodeChannel opt="1,2,3,4,5,6">
    <!--ro, opt, string, channel supporting video parameter display on the web, attr:opt{req, string}, desc:N/A-->1,2
  </videoEncodeChannel>
  <isSupportLocalButtonRecordControl>
    <!--ro, opt, bool, whether it supports controlling local recording button on the device, desc:/ISAPI/System/localButtonRecord?format=json-->true
  </isSupportLocalButtonRecordControl>
  <isSupportLocalButtonSnapshotControl>
    <!--ro, opt, bool, whether it supports controlling local capture button on the device, desc:/ISAPI/System/localButtonSnapshot?format=json-->true
  </isSupportLocalButtonSnapshotControl>
  <isSupportSecurityDisclaimerParams>
    <!--ro, opt, bool, whether it supports the security disclaimer parameters, desc:GET/PUT /ISAPI/System/InformedConsentMgr/SecurityDisclaimerParams?
format=json-->true
  </isSupportSecurityDisclaimerParams>
  <isSupportThermometryGatewayParams>
    <!--ro, opt, bool, whether it supports configuring parameters of the temperature measurement gateway, desc:GET/PUT
/ISAPI/Thermal/ThermometryGatewayParams?format=json-->true
  </isSupportThermometryGatewayParams>
  <isSupportSearchRegionDefaultInfo>
    <!--ro, opt, bool, whether it supports searching for the default parameters for the region, desc:POST /ISAPI/System/SearchRegionDefaultInfo?format=json-
->true
  </isSupportSearchRegionDefaultInfo>
  <isSupportAntiCollisionWarningEvent>
    <!--ro, opt, bool, whether it supports anti-collision pre-alarm event, desc:AntiCollisionWarningEvent-->true
  </isSupportAntiCollisionWarningEvent>
  <isSupportMovementWorkStatus>
    <!--ro, opt, bool, whether it supports gas cloud module working status, desc:GET /ISAPI/System/MovementMgr/MovementWorkStatus?format=json-->true
  </isSupportMovementWorkStatus>
  <isSupportMovementTimingPowerOffParams>
    <!--ro, opt, bool, whether it supports scheduled shutdown parameters of the gas cloud imaging module, desc:GET/PUT
/ISAPI/System/MovementMgr/MovementTimingPowerOffParams?format=json-->true
  </isSupportMovementTimingPowerOffParams>
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  <isSupportFireExtinguisherDetectionRuleParams>
    <!--ro, opt, bool, whether it supports fire extinguisher detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/FireExtinguisherDetectionRuleParams?format=json-->true
  </isSupportFireExtinguisherDetectionRuleParams>
  <isSupportFireExtinguisherRecoveryEvent>
    <!--ro, opt, bool, whether it supports fire extinguisher detection restoring event, desc:eventType:FireExtinguisherRecoveryEvent-->true
  </isSupportFireExtinguisherRecoveryEvent>
  <isSupportGasTankMonitoringRecoveryEvent>
    <!--ro, opt, bool, whether it supports gas tank detection restoring event, desc:eventType:GasTankMonitoringRecoveryEvent-->true
  </isSupportGasTankMonitoringRecoveryEvent>
  <isSupportSearchImportPlayerFileProcess>
    <!--ro, opt, bool, whether it supports searching for the progress of importing player files, desc:POST
/ISAPI/System/Video/SearchImportPlayerFileProcess?format=json-->true
  </isSupportSearchImportPlayerFileProcess>
  <isSupportStatisticalDataFlashStorageParams>
    <!--ro, opt, bool, whether it supports data statistics for Flash storage parameters, desc:GET/PUT
/ISAPI/System/dataAware/StatisticalDataFlashStorageParams?format=json-->true
  </isSupportStatisticalDataFlashStorageParams>
  <eventRecheckCap>
    <!--ro, opt, object, event review capability, desc:low-end cameras depend on powerful NVRs, due to the limitation of computing power for secondary
review of low-end camera alarms to reduce false alarms-->
    <algorithmInfoList>
      <!--ro, opt, array, list of event detection algorithm information, subType:object, range:[1,5]-->
      <algorithmInfoitem>
        <!--ro, req, object, event detection algorithm element-->
        <algorithmType>
          <!--ro, req, enum, event detection algorithm type, subType:string, desc:"VMD", "regionEntrance", "regionExiting", "linedetection" (line crossing),
"fielddetection" (intrusion detection)-->VMD
        </algorithmType>
        <algorithmVersion>
          <!--ro, req, string, event detection algorithm version, desc:N/A-->1B
        </algorithmVersion>
      </algorithmInfoitem>
    </algorithmInfoList>
  </eventRecheckCap>
  <isSupportModifyWindowGroup>
    <!--ro, opt, bool, whether it supports modifying the specified window group of the scope, desc:PUT
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/ModifyWindowGroup?format=json-->true
  </isSupportModifyWindowGroup>
  <isSupportCalibrateWindowGroup>
    <!--ro, opt, bool, whether it supports calibrating specified window group of the scope, desc:PUT
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/CalibrateWindowGroup?format=json-->true
  </isSupportCalibrateWindowGroup>
  <isSupportAllWindowParams>
    <!--ro, opt, bool, whether it supports all scope window parameters, desc:GET /ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/AllWindowParams?
format=json-->true
  </isSupportAllWindowParams>
  <isSupportWindowCalibrationStatus>
    <!--ro, opt, bool, whether it supports getting the scope window calibration status, desc:GET
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/WindowCalibrationStatus?format=json-->true
  </isSupportWindowCalibrationStatus>
  <isSupportOBUModuleInfo>
    <!--ro, opt, bool, whether it supports OBU module information, desc:GET /ISAPI/Traffic/ETC/OBUModuleInfo?format=json-->true
  </isSupportOBUModuleInfo>
  <isSupportExportOBUDetectionLogFile>
    <!--ro, opt, bool, whether it supports exporting the OBU label detection module logs, desc:POST /ISAPI/Traffic/ETC/ExportOBUDetectionLogFile?
format=json-->true
  </isSupportExportOBUDetectionLogFile>
  <isSupportOBUDetectionReboot>
    <!--ro, opt, bool, whether it supports restarting OBU label detection module, desc:PUT /ISAPI/Traffic/ETC/OBUDetectionReboot?format=json-->true
  </isSupportOBUDetectionReboot>
  <isSupportGetOBURebootStatus>
    <!--ro, opt, bool, whether it supports getting the restart status of the OBU module, desc:GET /ISAPI/Traffic/ETC/GetOBURebootStatus?format=json-->true
  </isSupportGetOBURebootStatus>
  <isSupportSearchOBUStatisticsReport>
    <!--ro, opt, bool, whether it supports searching for the statistical data of OBU tag detection, desc:POST /ISAPI/Traffic/ETC/SearchOBUStatisticsReport?
format=json-->true
  </isSupportSearchOBUStatisticsReport>
  <isSupportSearchTargetEnhancementInfo>
    <!--ro, opt, bool, whether it supports searching for target prominence parameters, desc:POST
/ISAPI/System/Video/TargetEnhancement/channels/<channelID>/SearchTargetEnhancementInfo?format=json-->true
  </isSupportSearchTargetEnhancementInfo>
  <isSupportTargetEnhancementParams>
    <!--ro, opt, bool, whether it supports target prominence parameters, desc:GET/PUT
/ISAPI/System/Video/TargetEnhancement/channels/<channelID>/TargetEnhancementParams?format=json-->true
  </isSupportTargetEnhancementParams>
  <isSupportPicSearchByCoordinate>
    <!--ro, opt, bool, whether the device supports configuring AcuSearch parameters, desc:GET/PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json--
>true
  </isSupportPicSearchByCoordinate>
  <AcuSearchCap>
    <!--ro, opt, object, AcuSearch capability-->
    <targetType opt="face,human,vehicle">
      <!--ro, opt, enum, supported target types of AcuSearch, subType:string, attr:opt{req, string}, desc:"face", "human", "vehicle"-->face
    </targetType>
    <dataType opt="url,binary,modelData">
      <!--ro, opt, enum, supported image transmission format types, subType:string, attr:opt{req, string}, desc:"url", "binary", "modelData"-->url
    </dataType>
    <maxPicSize>
      <!--ro, opt, int, maximum size of image can be searched, range:[0,5], desc:unit: MB-->1
    </maxPicSize>
    <maxPicWidthSize>
      <!--ro, opt, int, maximum width of image can be searched, range:[0,4096]-->4096
    </maxPicWidthSize>
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    </maxPicWidthSize>
    <maxPicHeightSize>
      <!--ro, opt, int, maximum height of image can be searched, range:[0,4096]-->4096
    </maxPicHeightSize>
  </AcuSearchCap>
  <isSupportSendDevRemoteGUICtrl>
    <!--ro, opt, bool, whether it supports applying parameter settings of remote GUI control of devices, desc:POST
/ISAPI/System/RemoteCtrlDevGUI/SendDevRemoteGUICtrl?format=json-->true
  </isSupportSendDevRemoteGUICtrl>
  <isSupportStreamTemperatureParams>
    <!--ro, opt, bool, whether the device supports configuring stream temperature parameters of thermal imaging, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/StreamTemperatureParams?format=json-->true
  </isSupportStreamTemperatureParams>
  <moduleTypeSupportCap>
    <!--ro, opt, object, capability supported by device module-->
    <isSupportMaintenanceModule>
      <!--ro, opt, bool, whether it supports maintenance module, desc:the wire products are embedded with the business module (businessModule) and
maintenance module (maintenanceModule), which supports independent access to EZVIZ cloud, etc.; at the same time, they support the cloud platform to send
maintenance instructions to the maintenance module-->true
    </isSupportMaintenanceModule>
  </moduleTypeSupportCap>
  <isSupportSearchEventResources>
    <!--ro, opt, bool, whether it supports searching for event resource information, desc:POST /ISAPI/Event/EventBatchRulesMgr/SearchEventResources?
format=json-->true
  </isSupportSearchEventResources>
  <isSupportRestorePerimeterAlgoDefaultParam>
    <!--ro, opt, bool, whether it supports restoring parameters of perimeter algorithm, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterAlgoMgr/RestorePerimeterAlgoDefaultParam?format=json-->true
  </isSupportRestorePerimeterAlgoDefaultParam>
  <isSupportGetDeviceMaintenanceStatus>
    <!--ro, opt, bool, whether it supports getting device maintenance status, desc:GET /ISAPI/System/GetDeviceMaintenanceStatus?format=json-->true
  </isSupportGetDeviceMaintenanceStatus>
  <isSupportImportConfigurationFileByModel>
    <!--ro, opt, bool, whether it supports importing configuration files by device model, desc:POST /ISAPI/System/ImportConfigurationFileByModel?
format=json-->true
  </isSupportImportConfigurationFileByModel>
  <isSupportRadarVideoDetectionStorageParams>
    <!--ro, opt, bool, whether it supports the parameters of saving radar-assisted target detection result, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/RadarVideoDetectionStorageParams?format=json-->true
  </isSupportRadarVideoDetectionStorageParams>
  <isSupportBodyCameraMediaInfo>
    <!--ro, opt, bool, whether it supports body camera reporting media information, desc:eventType: BodyCameraMediaInfo-->true
  </isSupportBodyCameraMediaInfo>
  <isSupportRadarVideoPOSOverlayParams>
    <!--ro, opt, bool, whether it supports smart radar-assisted POS information overlay parameters, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/RadarVideoPOSOverlayParams?format=json-->true
  </isSupportRadarVideoPOSOverlayParams>
  <isSupportDeleteNetworkCaptureFile>
    <!--ro, opt, bool, whether it supports deleting captured packet files, desc:POST /ISAPI/System/DeleteNetworkCaptureFile?format=json-->true
  </isSupportDeleteNetworkCaptureFile>
  <isSupportRecoverCustomEventTemplate>
    <!--ro, opt, bool, whether it supports restoring a custom event message template, desc:POST
/ISAPI/Event/notification/httpHosts/RecoverCustomEventTemplate?format=json-->true
  </isSupportRecoverCustomEventTemplate>
  <isSupportModCustomEventTemplateParams>
    <!--ro, opt, bool, whether it supports modifying a specified template of custom event message, desc:PUT
/ISAPI/Event/notification/httpHosts/ModCustomEventTemplateParams?format=json-->true
  </isSupportModCustomEventTemplateParams>
  <isSupportSearchAbnormalStreamInfo>
    <!--ro, opt, bool, whether it supports searching for streaming exception information, desc:POST /ISAPI/Streaming/SearchAbnormalStreamInfo?format=json--
>true
  </isSupportSearchAbnormalStreamInfo>
  <isSupportAbnormalStreamEvent>
    <!--ro, opt, bool, whether it supports real-time streaming exception report-->true
  </isSupportAbnormalStreamEvent>
  <isSupportBatchDeleteNetworkCaptureFile>
    <!--ro, opt, bool, whether it supports batch deleting captureed packet files, desc:PUT /ISAPI/System/BatchDeleteNetworkCaptureFile?format=json-->true
  </isSupportBatchDeleteNetworkCaptureFile>
  <isSupportWindowRuleParams>
    <!--ro, opt, bool, whether it supports the scope window rule parameters, desc:GET/PUT
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/WindowRuleParams?format=json-->true
  </isSupportWindowRuleParams>
  <isSupportGetMultiRemoteAlbumPicture>
    <!--ro, opt, bool, whether the device supports getting pictures from the remote album under specified path, desc:POST
/ISAPI/Thermal/GetMultiRemoteAlbumPicture?format=json-->true
  </isSupportGetMultiRemoteAlbumPicture>
  <isSupportGetDeviceTokenInfo>
    <!--ro, opt, bool, whether the device supports getting the device token information, desc:/ISAPI/System/ImportConfigurationFileByURL?format=json
/ISAPI/System/DeviceAuth/GetDeviceTokenInfo?format=json-->true
  </isSupportGetDeviceTokenInfo>
  <isSupportDisplayEnableCloudServicePrompt>
    <!--ro, opt, bool, whether it supports displaying the prompt for enabling cloud services, desc:false by default. When the device returns true, it
indicates that displaying a prompt of enabling cloud services is supported. Scenario example: After your device is connected to the Internet and the cloud
service is enabled, we will provide you with better cloud device management and maintenance upgrade services. Therefore, we will collect relevant device
information such as the device's IP address, device name, and device serial No. If you have any questions, you can contact customer service at any time--
>true
  </isSupportDisplayEnableCloudServicePrompt>
  <isSupportFiredoorDetectionEvent>
    <!--ro, opt, bool, whether it supports fireproof door detection event, desc:eventType:FiredoorDetectionEvent-->true
  </isSupportFiredoorDetectionEvent>
  <isSupportImportConfigurationFileByURL>
    <!--ro, opt, bool, whether it supports importing configuration file via URL, desc:POST /ISAPI/System/ImportConfigurationFileByURL?format=json-->true
  </isSupportImportConfigurationFileByURL>
  <isSupportPlayCellphoneRecoveryEvent>
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    <!--ro, opt, bool, whether it supports playing mobile restoring event, desc:eventType:PlayCellphoneRecoveryEvent-->true
  </isSupportPlayCellphoneRecoveryEvent>
  <isSupportDetectorShelterAlarmRecovery>
    <!--ro, opt, bool, whether it supports detector occlusion detection restoring event, desc:eventType:DetectorShelterAlarmRecovery-->true
  </isSupportDetectorShelterAlarmRecovery>
  <isSupportExportThermalMovementFile>
    <!--ro, opt, bool, whether it supports exporting the device temperature measuring module template, desc:POST /ISAPI/Thermal/ExportThermalMovementFile?
format=json-->true
  </isSupportExportThermalMovementFile>
  <isSupportUPSAbnormalAutoShutDownParams>
    <!--ro, opt, bool, whether it supports parameters of UPS power supply exception and auto power off, desc:GET/PUT
/ISAPI/System/UPSAbnormalAutoShutDownParams?format=json-->true
  </isSupportUPSAbnormalAutoShutDownParams>
  <isSupportThermalLensDriveParams>
    <!--ro, opt, bool, whether it supports configuring thermal lens driver parameters, desc:GET/PUT /ISAPI/Thermal/ThermalLensDriveParams?format=json-->true
  </isSupportThermalLensDriveParams>
  <isSupportBatchDeleteUserInfoCfg>
    <!--ro, opt, bool, whether it supports batch deleting user information configurations, desc:PUT /ISAPI/Security/users/BatchDeleteUserInfoCfg?
format=json-->true
  </isSupportBatchDeleteUserInfoCfg>
  <isSupportAddUserGroup>
    <!--ro, opt, bool, whether it supports adding user groups, desc:POST /ISAPI/System/UserGroupingMgr/AddUserGroup?format=json-->true
  </isSupportAddUserGroup>
  <isSupportModifyUserGroup>
    <!--ro, opt, bool, whether it supports editing user groups, desc:PUT /ISAPI/System/UserGroupingMgr/ModifyUserGroup?format=json-->true
  </isSupportModifyUserGroup>
  <isSupportSearchUserGroup>
    <!--ro, opt, bool, whether it supports searching user groups, desc:POST /ISAPI/System/UserGroupingMgr/SearchUserGroup?format=json-->true
  </isSupportSearchUserGroup>
  <isSupportDeleteUserGroup>
    <!--ro, opt, bool, whether it supports deleting user groups, desc:PUT /ISAPI/System/UserGroupingMgr/DeleteUserGroup?format=json-->true
  </isSupportDeleteUserGroup>
  <isSupportReadUpgradePackageData>
    <!--ro, opt, bool, whether it supports notifying the device to read upgrade package data, desc:PUT /ISAPI/System/SystemUpgrade/ReadUpgradePackageData?
format=json-->true
  </isSupportReadUpgradePackageData>
  <isSupportWebDisplayThermometryDistanceTips>
    <!--ro, opt, bool, whether it supports displaying the prompt of temperature measurement distance on the web page, desc:the temperature measurement
distance not only affects the temperature measurement results, but also affects the accuracy of temperature measurement location-->true
  </isSupportWebDisplayThermometryDistanceTips>
  <isSupportFlashLifespanProtectParams>
    <!--ro, opt, bool, whether it supports flash life protection parameter, desc:GET/PUT /ISAPI/System/FLASHMgr/FlashLifespanProtectParams?format=json--
>true
  </isSupportFlashLifespanProtectParams>
  <isSupportIlluminanceTelemeter>
    <!--ro, opt, bool, whether it supports light intensity telemetry, desc:eventType:IlluminanceTelemeter-->true
  </isSupportIlluminanceTelemeter>
  <isSupportWindDirectionInDegreesTelemeter>
    <!--ro, opt, bool, whether it supports wind direction telemetry, desc:eventType:WindDirectionInDegreesTelemeter-->true
  </isSupportWindDirectionInDegreesTelemeter>
  <isSupportWindSpeedTelemeter>
    <!--ro, opt, bool, whether it supports wind telemetry, desc:eventType:WindSpeedTelemeter-->true
  </isSupportWindSpeedTelemeter>
  <isSupportAirPressureTelemeter>
    <!--ro, opt, bool, whether it supports air pressure telemetry, desc:eventType:AirPressureTelemeter-->true
  </isSupportAirPressureTelemeter>
  <isSupportBatteryPercentageTelemeter>
    <!--ro, opt, bool, whether it supports battery telemetry, desc:eventType:BatteryPercentageTelemeter-->true
  </isSupportBatteryPercentageTelemeter>
  <isSupportTemperaturesTelemeter>
    <!--ro, opt, bool, whether it supports temperature telemetry, desc:eventType:TemperaturesTelemeter-->true
  </isSupportTemperaturesTelemeter>
  <isSupportHumiditysTelemeter>
    <!--ro, opt, bool, whether it supports humidity telemetry, desc:eventType:HumiditysTelemeter-->true
  </isSupportHumiditysTelemeter>
  <isSupportDipAngleTelemeter>
    <!--ro, opt, bool, whether it supports angle telemetry, desc:eventType:DipAngleTelemeter-->true
  </isSupportDipAngleTelemeter>
  <isSupportLocationEvent>
    <!--ro, opt, bool, whether it supports the device location information uploading event, desc:eventType:Location-->true
  </isSupportLocationEvent>
  <isSupportSoilMoistureTelemeter>
    <!--ro, opt, bool, whether it supports soil moisture telemetry, desc:eventType:SoilMoistureTelemeter-->true
  </isSupportSoilMoistureTelemeter>
  <isSupportLitterTelemeter>
    <!--ro, opt, bool, whether it supports litter telemetry, desc:eventType: LitterTelemeter-->true
  </isSupportLitterTelemeter>
  <isSupportSearchRealStreamStatus>
    <!--ro, opt, bool, whether it supports searching for the real-time streaming status, desc:POST /ISAPI/Streaming/SearchRealStreamStatus?format=json--
>true
  </isSupportSearchRealStreamStatus>
  <isSupportSearchStreamBandwidthInfo>
    <!--ro, opt, bool, whether it supports searching for stream bandwidth status, desc:POST /ISAPI/Streaming/SearchStreamBandwidthInfo?format=json-->true
  </isSupportSearchStreamBandwidthInfo>
  <isSupportExportThermometryData>
    <!--ro, opt, bool, whether it supports exporting visible light temperature measurement data, desc:POST
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ExportThermometryData?format=json-->true
  </isSupportExportThermometryData>
  <isSupportModifyClickToThermometry>
    <!--ro, opt, bool, whether it supports configuring parameters of visible light click-to-thermometry rules, desc:PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ModifyClickToThermometry?format=json-->true
  </isSupportModifyClickToThermometry>
  <isSupportExpertThermometryAlarmParams>
    <!--ro, opt, bool, whether it supports alarm parameters of temperature measurement in expert mode, desc:GET/PUT
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    <!--ro, opt, bool, whether it supports alarm parameters of temperature measurement in expert mode, desc:GET/PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ExpertThermometryAlarmParams?format=json-->true
  </isSupportExpertThermometryAlarmParams>
  <isSupportExpertThermometryRuleParams>
    <!--ro, opt, bool, whether it supports the rule parameters of temperature measurement in expert mode, desc:GET/PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ExpertThermometryRuleParams?format=json-->true
  </isSupportExpertThermometryRuleParams>
  <isSupportNormalThermometryRuleParams>
    <!--ro, opt, bool, whether it supports the rule parameters of temperature measurement in normal mode, desc:GET/PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/NormalThermometryRuleParams?format=json-->true
  </isSupportNormalThermometryRuleParams>
  <isSupportModeParams>
    <!--ro, opt, bool, whether it supports mode parameters of visible light radiation temperature measurement, desc:GET/PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/ModeParams?format=json-->true
  </isSupportModeParams>
  <isSupportBasicParams>
    <!--ro, opt, bool, whether it supports basic parameters of visible light radiation temperature measurement, desc:GET/PUT
/ISAPI/Smart/VisibleRadiationThermometry/channels/<channelID>/BasicParams?format=json-->true
  </isSupportBasicParams>
  <isSupportImportPlayerFile>
    <!--ro, opt, bool, whether it supports importing player files, desc:PUT /ISAPI/System/Video/ImportPlayerFile?format=json-->true
  </isSupportImportPlayerFile>
  <isSupportSearchQuickQRCode>
    <!--ro, opt, bool, whether it supports searching for quick configuration codes, desc:POST /ISAPI/System/QuickConfigQRCodeMgr/SearchQuickQRCode?
format=json-->true
  </isSupportSearchQuickQRCode>
  <isSupportQuickQRCodeEncryptionParams>
    <!--ro, opt, bool, whether it supports the encryption parameters of quick configuration codes, desc:PUT
/ISAPI/System/QuickConfigQRCodeMgr/QuickQRCodeEncryptionParams?format=json-->true
  </isSupportQuickQRCodeEncryptionParams>
  <isUnsynchronizedUpgradeKVMIntput>
    <!--ro, opt, bool, whether it supports upgrading KVM input node synchronously, desc:1. For distributed main control, if this field is not returned or
returned as false, the KVM input node added to the main control will be upgraded. 2. If this field is returned as true, the KVM input node will not be
upgraded.-->true
  </isUnsynchronizedUpgradeKVMIntput>
  <isSupportWaterRescueDetectionOverlapParams>
    <!--ro, opt, bool, whether it supports intelligent information overlay parameters of water rescue, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/WaterRescueDetectionOverlapParams?format=json-->true
  </isSupportWaterRescueDetectionOverlapParams>
  <isSupportSearchRecordStatusList>
    <!--ro, opt, bool, whether it supports searching for the channel recording status, desc:POST /ISAPI/ContentMgmt/search/SearchRecordStatusList?
format=json-->true
  </isSupportSearchRecordStatusList>
  <isSupportWiperSelfCheckParams>
    <!--ro, opt, bool, whether it supports the wiper self-checking parameters, desc:GET/PUT
/ISAPI/System/Hardware/channels/<channelID>/WiperSelfCheckParams?format=json-->true
  </isSupportWiperSelfCheckParams>
  <isSupportBlackoutNotification>
    <!--ro, opt, bool, whether it supports power-off notification, desc:PUT /ISAPI/System/channels/<channelID>/BlackoutNotification?format=json-->true
  </isSupportBlackoutNotification>
  <isSupportGotoWaterFlowScene>
    <!--ro, opt, bool, whether it supports jumping to the specified water flow detection scene, desc:POST
/ISAPI/Event/channels/<channelID>/GotoWaterFlowScene?format=json-->true
  </isSupportGotoWaterFlowScene>
  <isSupportHVTVehicleDetectUploadChannel>
    <!--ro, opt, bool, whether it supports configuring the channel of uploading mixed-traffic detection information to the FTP server, desc:GET/PUT
/ISAPI/System/Network/ftp/HVTVehicleDetectUpload?format=json&channelID=<channelID>-->true
  </isSupportHVTVehicleDetectUploadChannel>
  <isSupportWaterFlowAdvancedParams>
    <!--ro, opt, bool, whether it supports advanced parameters of water flow detection, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/WaterFlowAdvancedParams?format=json-->true
  </isSupportWaterFlowAdvancedParams>
  <isSupportWaterFlowSceneList>
    <!--ro, opt, bool, whether it supports water flow detection scene list, desc:GET /ISAPI/Event/channels/<channelID>/WaterFlowSceneList?format=json-->true
  </isSupportWaterFlowSceneList>
  <isSupportWaterFlowSceneRuleParams>
    <!--ro, opt, bool, whether it supports water flow detection scene rule parameters, desc:PUT /ISAPI/Event/channels/<channelID>/WaterFlowSceneRuleParams?
format=json-->true
  </isSupportWaterFlowSceneRuleParams>
  <isSupportRainfallScheduledUpload>
    <!--ro, opt, bool, whether it supports scheduled rainfall information uploading event, desc:RainfallScheduledUpload-->true
  </isSupportRainfallScheduledUpload>
  <isSupportFlowVelocityScheduledUpload>
    <!--ro, opt, bool, whether it supports scheduled water flow speed uploading event, desc:FlowVelocityScheduledUpload-->true
  </isSupportFlowVelocityScheduledUpload>
  <isSupportFlowVelocityAlarm>
    <!--ro, opt, bool, whether it supports water flow speed alarm event, desc:FlowVelocityAlarm-->true
  </isSupportFlowVelocityAlarm>
  <isSupportVolumeFlowRateScheduledUpload>
    <!--ro, opt, bool, whether it supports scheduled water flow rate uploading event, desc:VolumeFlowRateScheduledUpload-->true
  </isSupportVolumeFlowRateScheduledUpload>
  <isSupportVolumeFlowRateAlarm>
    <!--ro, opt, bool, whether it supports water flow rate alarm event, desc:VolumeFlowRateAlarm-->true
  </isSupportVolumeFlowRateAlarm>
  <isSupportSecurityDisclaimerResultEvent>
    <!--ro, opt, bool, whether it supports security disclaimer result uploading, desc:SecurityDisclaimerResultEvent-->true
  </isSupportSecurityDisclaimerResultEvent>
  <isSupportInfusionAlarmEvent>
    <!--ro, opt, bool, whether it supports infusion alarm event, desc:InfusionAlarmEvent-->true
  </isSupportInfusionAlarmEvent>
  <isSupportBodyCameraBatteryTamperEvent>
    <!--ro, opt, bool, whether it supports body camera battery tampering event, desc:eventType:BodyCameraBatteryTamperEvent-->true
  </isSupportBodyCameraBatteryTamperEvent>
  <isSupportShipSafeOperationDetectionEvent>
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    <!--ro, opt, bool, whether it supports ship operation safety event detection, desc:eventType:ShipSafeOperationDetectionEvent-->true
  </isSupportShipSafeOperationDetectionEvent>
  <isSupportTrafficLightStatusDetectionEvent>
    <!--ro, opt, bool, whether it supports traffic light status detection event, desc:eventType:TrafficLightStatusDetectionEvent-->true
  </isSupportTrafficLightStatusDetectionEvent>
  <isSupportSoundPressureDetectionEvent>
    <!--ro, opt, bool, whether it supports triggering overload sound pressure alarm, desc:eventType:SoundPressureDetectionEvent-->true
  </isSupportSoundPressureDetectionEvent>
  <isSupportInvigilatorBehaviorRuleParams>
    <!--ro, opt, bool, whether it supports driving license exam proctoring behavior detection rule parameters, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/InvigilatorBehaviorRuleParams?format=json-->true
  </isSupportInvigilatorBehaviorRuleParams>
  <isSupportInvigilatorBehaviorEvent>
    <!--ro, opt, bool, whether it supports driving license exam proctoring behavior detection event, desc:eventType: InvigilatorBehaviorEvent-->true
  </isSupportInvigilatorBehaviorEvent>
  <isSupportPeopleSleepingDetectionReport>
    <!--ro, opt, bool, whether it supports sleep detection report event, desc:eventType:PeopleSleepingDetectionReport-->true
  </isSupportPeopleSleepingDetectionReport>
  <isSupportPhysicalMatchMgr>
    <!--ro, opt, bool, whether it supports sports competitions management, desc:GET /ISAPI/Intelligent/channels/<channelID>/PhysicalMatchMgr/capabilities?
format=json-->true
  </isSupportPhysicalMatchMgr>
  <isSupportSearchNetworkResources>
    <!--ro, opt, bool, whether it supports searching for network resource information, desc:POST
/ISAPI/System/Network/NetworkAdvancedCfg/SearchNetworkResources?format=json-->true
  </isSupportSearchNetworkResources>
  <isSupportTriggerPeripheralAbnormalDetection>
    <!--ro, opt, bool, whether it supports triggering the device to check component fault, desc:PUT /ISAPI/Event/TriggerPeripheralAbnormalDetection?
format=json-->true
  </isSupportTriggerPeripheralAbnormalDetection>
  <isSupportDeviceCabinetDoorOpenAlarm>
    <!--ro, opt, bool, whether it support device chassis open alarm event, desc:eventType:DeviceCabinetDoorOpenAlarm-->true
  </isSupportDeviceCabinetDoorOpenAlarm>
  <isSupportSurplusPower>
    <!--ro, opt, bool, whether it supports remaining battery information, desc:GET/PUT /ISAPI/System/SurplusPower?format=json-->true
  </isSupportSurplusPower>
  <isSupportPeripheralAbnormalScheduledUpload>
    <!--ro, opt, bool, whether it supports scheduled device component fault uploading event, desc:eventType:PeripheralAbnormalScheduledUpload-->true
  </isSupportPeripheralAbnormalScheduledUpload>
  <isSupportBandwidthLimitWarningParams>
    <!--ro, opt, bool, whether it supports the bandwidth limit alarm parameters, desc:GET/PUT
/ISAPI/System/Network/NetworkFlowMonitor/BandwidthLimitWarningParams?format=json-->true
  </isSupportBandwidthLimitWarningParams>
  <isSupportRoadMaintSceneList>
    <!--ro, opt, bool, whether it supports road maintenance scene list, desc:GET /ISAPI/Intelligent/channels/<channelID>/RoadMaintSceneList?format=json--
>true
  </isSupportRoadMaintSceneList>
  <isSupportSearchIntelligencePerformance>
    <!--ro, opt, bool, whether it supports searching post-intelligence performance parameters, desc:POST
/ISAPI/System/VCAResource/SearchIntelligencePerformance?format=json-->true
  </isSupportSearchIntelligencePerformance>
  <isSupportRainfallRuleParams>
    <!--ro, opt, bool, whether it supports rainfall detection rule parameters, desc:GET/PUT /ISAPI/Event/channels/<channelID>/RainfallRuleParams?
format=json-->true
  </isSupportRainfallRuleParams>
  <isSupportRadarFlowmeterOverlapParams>
    <!--ro, opt, bool, whether it supports radar detection information overlay parameters, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/RadarFlowmeterOverlapParams/?format=json-->true
  </isSupportRadarFlowmeterOverlapParams>
  <isSupportCurrentComStatus>
    <!--ro, opt, bool, whether it supports getting the device communication status, desc:GET /ISAPI/System/CurrentComStatus?format=json-->true
  </isSupportCurrentComStatus>
  <isSupportRoadMaintAlgorithmVersion>
    <!--ro, opt, bool, whether it supports road maintenance algorithm version, desc:GET /ISAPI/Intelligent/channels/<channelID>/RoadMaintAlgorithmVersion?
format=json-->true
  </isSupportRoadMaintAlgorithmVersion>
  <isSupportRoadMaintOverlapParams>
    <!--ro, opt, bool, whether it supports road maintenance overlay parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/RoadMaintOverlapParams?
format=json-->true
  </isSupportRoadMaintOverlapParams>
  <isSupportActionExecuteStatus>
    <!--ro, opt, bool, whether it supports action execution status uploading, desc:eventType: ActionExecuteStatus-->true
  </isSupportActionExecuteStatus>
  <isSupportExportConfigurationFile>
    <!--ro, opt, bool, whether it supports exporting the configuration file, desc:GET /ISAPI/System/ExportConfigurationFile?format=json-->true
  </isSupportExportConfigurationFile>
  <isSupportMultiChannelStreamMergeParams>
    <!--ro, opt, bool, whether it supports the parameters of merging the  encoding of multiple panoramic channels, desc:GET/PUT
/ISAPI/System/Video/inputs/MultiChannelStreamMergeParams?format=json-->true
  </isSupportMultiChannelStreamMergeParams>
  <isSupportAnalysisEngineCfgTest>
    <!--ro, opt, bool, whether it supports engine analysis parameters configuration restarting detection, desc:POST
/ISAPI/System/VCAResource/AnalysisEngineCfgTest?format=json-->true
  </isSupportAnalysisEngineCfgTest>
  <isSupportModifyPlaybackSpeed>
    <!--ro, opt, bool, whether it supports video speed playback of the streaming device, desc:POST
/ISAPI/Streaming/channels/<channelID>/ModifyPlaybackSpeed?format=json-->true
  </isSupportModifyPlaybackSpeed>
  <isSupportSeekToSegsPlayback>
    <!--ro, opt, bool, whether it supports dragging to the time period during streaming video playback, desc:POST
/ISAPI/Streaming/channels/<channelID>/SeekToSegsPlayback?format=json-->true
  </isSupportSeekToSegsPlayback>
  <isSupportIntersectionTrafficFlowStatisticsEvent>
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    <!--ro, opt, bool, whether it supports event for counting motor vehicle traffic at an intersection, desc:eventType:
IntersectionTrafficFlowStatisticsEvent-->true
  </isSupportIntersectionTrafficFlowStatisticsEvent>
  <isSupportLensCoverAbnormalEvent>
    <!--ro, opt, bool, whether it supports lens cover abnormal event, desc:event type: LensCoverAbnormalEvent.-->true
  </isSupportLensCoverAbnormalEvent>
  <isSupportCrackMonitoringEvent>
    <!--ro, opt, bool, whether it supports geological cracks detection event, desc:event type: CrackMonitoringEvent-->true
  </isSupportCrackMonitoringEvent>
  <isSupportExportNetworkCaptureFile>
    <!--ro, opt, bool, whether it supports exporting captured packets, desc:POST /ISAPI/System/ExportNetworkCaptureFile?format=json-->true
  </isSupportExportNetworkCaptureFile>
  <isSupportGetNetworkCaptureStatus>
    <!--ro, opt, bool, whether it supports getting captured packet status, desc:POST /ISAPI/System/GetNetworkCaptureStatus?format=json-->true
  </isSupportGetNetworkCaptureStatus>
  <isSupportStartNetworkCapture>
    <!--ro, opt, bool, whether it supports starting capturing network packets, desc:POST /ISAPI/System/StartNetworkCapture?format=json-->true
  </isSupportStartNetworkCapture>
  <isSupportStopNetworkCapture>
    <!--ro, opt, bool, whether it supports stopping capturing network packets, desc:POST /ISAPI/System/StopNetworkCapture?format=json-->true
  </isSupportStopNetworkCapture>
  <isSupportObjectAccumulationAlgorithmParams>
    <!--ro, opt, bool, whether it supports stacking detection algorithm parameters, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/ObjectAccumulationAlgorithmParams?format=json-->true
  </isSupportObjectAccumulationAlgorithmParams>
  <isSupportGetTobaccoStageStatus>
    <!--ro, opt, bool, whether it supports getting tobacco baking status, desc:POST /ISAPI/System/TobaccoBakeMgr/GetTobaccoStageStatus?format=json-->true
  </isSupportGetTobaccoStageStatus>
  <isSupportMultipleIntelligentParams>
    <!--ro, opt, bool, whether it supports multi-intelligent mode parameters, desc:GET/PUT /ISAPI/System/VCAResource/MultipleIntelligentParams?format=json--
>true
  </isSupportMultipleIntelligentParams>
  <isSupportStartSystemEnvDetection>
    <!--ro, opt, bool, whether it supports starting system environment check, desc:POST /ISAPI/System/StartSystemEnvDetection?format=json-->true
  </isSupportStartSystemEnvDetection>
  <isSupportSearchSystemEnvDetectionProcess>
    <!--ro, opt, bool, whether it supports searching for system environment checking progress, desc:POST /ISAPI/System/SearchSystemEnvDetectionProcess?
format=json-->true
  </isSupportSearchSystemEnvDetectionProcess>
  <isSupportEVACInputParams>
    <!--ro, opt, bool, whether it supports EVAC input parameters, desc:GET/PUT /ISAPI/System/EVACInput/EVACInputParams?format=json-->true
  </isSupportEVACInputParams>
  <isSupportModifyT1TestParams>
    <!--ro, opt, bool, whether it editing T1 test parameters, desc:POST /ISAPI/System/AutoTest/ModifyT1TestParams?format=json-->true
  </isSupportModifyT1TestParams>
  <isSupportDetectionRegionRuleParams>
    <!--ro, opt, bool, whether it supports rule parameters of hazardous gas area scanning detection, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/DetectionRegionRuleParams?format=json-->true
  </isSupportDetectionRegionRuleParams>
  <isSupportFaceApproach>
    <!--ro, opt, bool, whether it supports face approaching event, desc:eventType:FaceApproach-->true
  </isSupportFaceApproach>
  <isSupportHardwareEnergySaving>
    <!--ro, opt, bool, whether it supports device hardware energy-saving parameters, desc:GET/PUT /ISAPI/System/consumptionMode/HardwareEnergySaving?
format=json-->true
  </isSupportHardwareEnergySaving>
  <isSupportDeviceHighTemperature>
    <!--ro, opt, bool, whether it supports device overheat event, desc:eventType:DeviceHighTemperature-->true
  </isSupportDeviceHighTemperature>
  <isSupportExportAudioAndVideoFile>
    <!--ro, opt, bool, whether it supports exporting audio and video files, desc:POST /ISAPI/System/ExportAudioAndVideoFile?format=json-->true
  </isSupportExportAudioAndVideoFile>
  <isSupportGetExportAudioAndVideoFileStatus>
    <!--ro, opt, bool, whether it supports getting the export status of video and audio files, desc:GET /ISAPI/System/GetExportAudioAndVideoFileStatus?
format=json-->true
  </isSupportGetExportAudioAndVideoFileStatus>
  <isSupportStartExportAudioAndVideoFile>
    <!--ro, opt, bool, whether it supports starting exporting audio and video files, desc:PUT /ISAPI/System/StartExportAudioAndVideoFile?format=json-->true
  </isSupportStartExportAudioAndVideoFile>
  <isSupportStopExportAudioAndVideoFile>
    <!--ro, opt, bool, whether it supports stopping exporting audio and video files, desc:PUT /ISAPI/System/StopExportAudioAndVideoFile?format=json-->true
  </isSupportStopExportAudioAndVideoFile>
  <isSupportAsyncExportSearchEventResult>
    <!--ro, opt, bool, whether it supports exporting the results of searching for event-related videos asynchronously, desc:POST
/ISAPI/ContentMgmt/search/AsyncExportSearchEventResult?format=json-->true
  </isSupportAsyncExportSearchEventResult>
  <isRebootAfterUpgradeFailed>
    <!--ro, opt, bool, whether restarting device is required after upgrading failed, desc:if this field does not exist, the device will not be restarted
after upgrading failed-->true
  </isRebootAfterUpgradeFailed>
  <isSupportGetAsyncExportSearchEventResultProgress>
    <!--ro, opt, bool, whether it supports getting the async exporting progress of event videos search result, desc:POST
/ISAPI/ContentMgmt/search/GetAsyncExportSearchEventResultProgress?format=json-->true
  </isSupportGetAsyncExportSearchEventResultProgress>
  <isSupportManualTriggerWipe>
    <!--ro, opt, bool, whether it supports manually starting the wiper, desc:POST /ISAPI/System/Hardware/ManualTriggerWipe?format=json-->true
  </isSupportManualTriggerWipe>
  <isSupportWiperCleanMode>
    <!--ro, opt, bool, whether it supports wiper cleaning mode, desc:GET/PUT /ISAPI/System/Hardware/WiperCleanMode?format=json-->true
  </isSupportWiperCleanMode>
  <isSupportRadarWaterFlowRuleParams>
    <!--ro, opt, bool, whether it supports radar-assisted water flow detection rule parameters, desc:GET/PUT
/ISAPI/Event/Radar/<RadarID>/RadarWaterFlowRuleParams?format=json-->true
  </isSupportRadarWaterFlowRuleParams>
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  <isSupportRadarWaterFlowAdvancedParams>
    <!--ro, opt, bool, whether it supports advanced parameters of radar-assisted water flow detection, desc:GET/PUT
/ISAPI/Event/Radar/<RadarID>/RadarWaterFlowAdvancedParams?format=json-->true
  </isSupportRadarWaterFlowAdvancedParams>
  <isSupportLaserSourceSwitch>
    <!--ro, opt, bool, whether it supports laser light source switching, desc:GET/PUT /ISAPI/System/channels/<channelID>/LaserSourceSwitch?format=json--
>true
  </isSupportLaserSourceSwitch>
  <isSupportLensStainConfigParams>
    <!--ro, opt, bool, whether it supports lens dirt detection configuration parameters, desc:GET/PUT /ISAPI/System/Hardware/LensStainConfigParams?
format=json-->true
  </isSupportLensStainConfigParams>
  <setupParamType opt="PTZLaser">
    <!--ro, opt, object, whether it supports construction parameters of device sub modules, attr:opt{req, string}, desc:whether /ISAPI/System/SetupParam?
type=<type> supports the optional parameter "type"-->
  </setupParamType>
  <isSupportLensStain>
    <!--ro, opt, bool, whether it supports lens stain event, desc:eventType:LensStain-->true
  </isSupportLensStain>
  <isSupportLensStainRecovered>
    <!--ro, opt, bool, whether it supports lens stain restoring event, desc:eventType:LensStainRecovered-->true
  </isSupportLensStainRecovered>
  <isSupportDeviceCircuitFault>
    <!--ro, opt, bool, whether it supports device circuit fault event, desc:eventType:DeviceCircuitFault-->true
  </isSupportDeviceCircuitFault>
  <isSupportDeviceCircuitRestored>
    <!--ro, opt, bool, whether it supports device circuit fault restoring event, desc:eventType:DeviceCircuitRestored-->true
  </isSupportDeviceCircuitRestored>
  <isSupportTurnOffBuiltInBattery>
    <!--ro, opt, bool, whether it supports disabling the built-in battery, desc:PUT /ISAPI/System/TurnOffBuiltInBattery?format=json-->true
  </isSupportTurnOffBuiltInBattery>
  <isSupportElevatorStartMove>
    <!--ro, opt, bool, whether it supports elevator starting running event, desc:eventType:ElevatorStartMove-->true
  </isSupportElevatorStartMove>
  <isSupportElevatorStopMove>
    <!--ro, opt, bool, whether it supports elevator stopping running event, desc:eventType:ElevatorStopMove-->true
  </isSupportElevatorStopMove>
  <isSupportOpticalFiberRasterAlarm>
    <!--ro, opt, bool, whether it supports fiber and raster alarm event, desc:corresponding event: OpticalFiberRasterAlarm-->true
  </isSupportOpticalFiberRasterAlarm>
  <isSupportSensorRealTimeData>
    <!--ro, opt, bool, whether it supports real-time sensor data uploading event, desc:corresponding event: SensorRealTimeData-->true
  </isSupportSensorRealTimeData>
  <isSupportGNSSWorkStatus>
    <!--ro, opt, bool, whether it supports GNSS working status uploading event, desc:corresponding event: GNSSWorkStatus.-->true
  </isSupportGNSSWorkStatus>
  <isSupportDevOnPlatformInfoEvent>
    <!--ro, opt, bool, whether it supports on-platform device information request event, desc:corresponding event: DevOnPlatformInfoEvent-->true
  </isSupportDevOnPlatformInfoEvent>
  <isSupportUpdateScheduleEvent>
    <!--ro, opt, bool, whether it supports request event of updating the schedule, desc:corresponding event: UpdateScheduleEvent-->true
  </isSupportUpdateScheduleEvent>
  <isSupportIntelligentApply>
    <!--ro, opt, bool, whether it support intelligent mode application parameters, desc:GET/PUT /ISAPI/System/VCAResource/IntelligentApply?format=json--
>true
  </isSupportIntelligentApply>
  <isSupportMCUDevSerialWrite>
    <!--ro, opt, bool, whether it supports writing MCU device serial number, desc:POST /ISAPI/System/MCUDevSerialWrite?format=json-->true
  </isSupportMCUDevSerialWrite>
  <isSupportSambaServerParams>
    <!--ro, opt, bool, whether it supports for Samba server parameters, desc:GET/PUT /ISAPI/System/SambaServerManagement/SambaServerParams?format=json--
>true
  </isSupportSambaServerParams>
  <isSupportToolingBoardConfigParams>
    <!--ro, opt, bool, whether it supports tooling plate configuration parameters, desc:PUT /ISAPI/System/AutoTest/ToolingBoardConfigParams?format=json--
>true
  </isSupportToolingBoardConfigParams>
  <isSupportAlumFlocObservationParams>
    <!--ro, opt, bool, whether it supports alum floc detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/AlumFlocObservation/channels/<channelID>/AlumFlocObservationParams?format=json-->true
  </isSupportAlumFlocObservationParams>
  <isDisplayRebootTips>
    <!--ro, opt, bool, whether to prompt users to restart the device when switching non-uniformity correction, desc:For devices with low performance,
switching non-uniformity correction will take effect only after restarting the device. The web should inform users of this. For devices with high
performance, restarting is not required. You can determine whether the web needs to prompt users to restart the device via this field. If this field does
not exist, the web needs to prompt by default.-->true
  </isDisplayRebootTips>
  <isSupportAIOPMPAndResChanged>
    <!--ro, opt, bool, whether the meaning of AIOP model package and resource interface is changed, desc:If the returned value is true, it indicates that
the meaning of the AIOP API has changed, that is, the AIOP API for model package and engine management have replaced that of algorithm package scheduling
and of the algorithm and engine management in algorithm application scheduling. For example,
/ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engine/<engineID>?format=json and /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engines?format=json will
return all the engines for scheduling the algorithm packages and applications, and /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/management?format=json
will return the information of the AIOP model package, algorithm application scheduling, and algorithm package scheduling.-->true
  </isSupportAIOPMPAndResChanged>
  <isSupportRoadVisibilityAnalysisCfg>
    <!--ro, opt, bool, whether it supports road visibility detection analysis parameters, desc:GET/PUT
/ISAPI/Traffic/channels/<channelID>/RoadVisibilityAnalysisCfg?format=json-->true
  </isSupportRoadVisibilityAnalysisCfg>
  <isSupportHotStandbyParams>
    <!--ro, opt, bool, whether it supports hot spare parameters, desc:GET/PUT /ISAPI/System/MainSpareDeviceSettings/HotStandbyParams?format=json-->true
  </isSupportHotStandbyParams>
  <isSupportHotStandbySwitchParams>
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    <!--ro, opt, bool, whether it supports hot spare switching, desc:GET/PUT /ISAPI/System/MainSpareDeviceSettings/HotStandbySwitchParams?format=json-->true
  </isSupportHotStandbySwitchParams>
  <isSupportManualControlHotStandby>
    <!--ro, opt, bool, whether it supports manually taking over hot spare, desc:POST /ISAPI/System/MainSpareDeviceSettings/ManualControlHotStandby?
format=json-->true
  </isSupportManualControlHotStandby>
  <isSupportGetHotStandbyStatus>
    <!--ro, opt, bool, whether it supports getting hot spare status, desc:GET /ISAPI/System/MainSpareDeviceSettings/GetHotStandbyStatus?format=json-->true
  </isSupportGetHotStandbyStatus>
  <isSupportRestoreStreamCollectionInfo>
    <!--ro, opt, bool, whether it supports confirmation for collecting the raw stream data without mosaic mask, desc:GET/PUT
/ISAPI/System/InformedConsentMgr/RestoreStreamCollectionInfo?format=json-->true
  </isSupportRestoreStreamCollectionInfo>
  <isSupportFileNameFormat>
    <!--ro, opt, bool, whether it supports file name format, desc:GET/PUT /ISAPI/System/FileNameFormat?format=json-->true
  </isSupportFileNameFormat>
  <isSupportThermalInspectionParams>
    <!--ro, opt, bool, whether it supports thermal inspection parameters, desc:GET/PUT /ISAPI/Thermal/ThermalInspectionParams?format=json-->true
  </isSupportThermalInspectionParams>
  <isSupportThermalRecordParams>
    <!--ro, opt, bool, whether it supports configuring the recording parameters of specified thermal imaging channel, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/ThermalRecordParams?format=json-->true
  </isSupportThermalRecordParams>
  <isSupportThermalLensParams>
    <!--ro, opt, bool, whether it supports configuring thermal imaging lens parameters, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ThermalLensParams?
format=json-->true
  </isSupportThermalLensParams>
  <isSupportThermalInfoDisplayParam>
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Slovenian, Croatian, Lithuanian, Serbian, BrazilianPortuguese, Indonesian, Ukrainian, EURSpanish, Sovenian, Uzbek, Kazak, Kirghiz, Farsi, Azerbaidzhan,
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  <!--ro, opt, bool-->true
</isSupportPlasmaRFConfig>
<isSupportVacuumProcess>
  <!--ro, opt, bool-->true
</isSupportVacuumProcess>
<isSupportVacuumPumpControl>
  <!--ro, opt, bool-->true
</isSupportVacuumPumpControl>
<isSupportVacuumSwitchControl>
  <!--ro, opt, bool-->true
</isSupportVacuumSwitchControl>
<isSupportLifeData>
  <!--ro, opt, bool-->true
</isSupportLifeData>
<isSupportInstrumentAbnormalInfo>
  <!--ro, opt, bool-->true
</isSupportInstrumentAbnormalInfo>
<isSupportPingResult>
  <!--ro, opt, bool-->true
</isSupportPingResult>
<isSupportGetWhiteLightStatus>
  <!--ro, opt, bool-->true
</isSupportGetWhiteLightStatus>
<isSupportManualCtrlWhiteLight>
  <!--ro, opt, bool-->true
</isSupportManualCtrlWhiteLight>
<isSupportWhiteLightCfg>
  <!--ro, opt, bool-->true
</isSupportWhiteLightCfg>
<isSupportFanScenceParams>
  <!--ro, opt, bool-->true
</isSupportFanScenceParams>
<isSupportManualDefog>
  <!--ro, opt, bool-->true
</isSupportManualDefog>
<isSupportSendLogDownloadReq>
  <!--ro, opt, bool-->true
</isSupportSendLogDownloadReq>
<isSupportGetDiagnosticLogInfo>
  <!--ro, opt, bool-->true
</isSupportGetDiagnosticLogInfo>
<isSupportGetDiagnosticStatus>
  <!--ro, opt, bool-->true
</isSupportGetDiagnosticStatus>
<isSupportSendDiagnosticReq>
  <!--ro, opt, bool-->true
</isSupportSendDiagnosticReq>
<isSupportGetDevDiagnosticInfo>
  <!--ro, opt, bool-->true
</isSupportGetDevDiagnosticInfo>
<isSupportGetFireStatus>
  <!--ro, opt, bool-->true
</isSupportGetFireStatus>
<isSupportScannerInfoRuleParams>
  <!--ro, opt, bool-->true
</isSupportScannerInfoRuleParams>
<isSupportScannerInfoTriggerParams>
  <!--ro, opt, bool-->true
</isSupportScannerInfoTriggerParams>
<isSupportWANIntercomMaintenanceEvent>
  <!--ro, opt, bool-->true
</isSupportWANIntercomMaintenanceEvent>
<isSupportWANIntercomStreamingInfoQuery>
  <!--ro, opt, bool-->true
</isSupportWANIntercomStreamingInfoQuery>
<isSupportGetMixedTargetRuleDiagnosisResult>
  <!--ro, opt, bool-->true
</isSupportGetMixedTargetRuleDiagnosisResult>
<isSupportDeskMatPosOverlay>
  <!--ro, opt, bool-->true
</isSupportDeskMatPosOverlay>
<isSupportAddResetPasswordPhoneNo>
  <!--ro, opt, bool-->true
</isSupportAddResetPasswordPhoneNo>
<isSupportModifyResetPasswordPhoneNo>
  <!--ro, opt, bool-->true
</isSupportModifyResetPasswordPhoneNo>
<isSupportSearchResetPasswordPhoneNo>
  <!--ro, opt, bool-->true
</isSupportSearchResetPasswordPhoneNo>
<isSupportDeleteResetPasswordPhoneNo>
  <!--ro, opt, bool-->true
</isSupportDeleteResetPasswordPhoneNo>
<isSupportMasterSlaveModeParam>
  <!--ro, opt, bool-->true
</isSupportMasterSlaveModeParam>
<isSupportStopWebrtcPreviewPush>
  <!--ro, opt, bool-->true
</isSupportStopWebrtcPreviewPush>
<isSupportSyncMasterParamToSlave>
```

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<isSupportSyncMasterParamToSlave>
<!--ro, opt, bool-->true
</isSupportSyncMasterParamToSlave>
<isSupportBasedFusionImageRuleRegionParam>
<!--ro, opt, bool-->true
</isSupportBasedFusionImageRuleRegionParam>
<isSupportFusionAndOriginalImageParam>
<!--ro, opt, bool-->true
</isSupportFusionAndOriginalImageParam>
<isSupportGetAllSlaveCountingCalibrationInfo>
<!--ro, opt, bool-->true
</isSupportGetAllSlaveCountingCalibrationInfo>
<isSupportSnapshotAsyncTrackID>
<!--ro, opt, bool-->true
</isSupportSnapshotAsyncTrackID>
<isSupportPasswordPeriodPrompt>
<!--ro, opt, bool-->true
</isSupportPasswordPeriodPrompt>
<isSupportGetModelingResult>
<!--ro, opt, bool-->true
</isSupportGetModelingResult>
<isSupportMakeMapModel>
<!--ro, opt, bool-->true
</isSupportMakeMapModel>
<isSupportBallisticParam>
<!--ro, opt, bool-->true
</isSupportBallisticParam>
<isSupportExportGaugeReadingEvent>
<!--ro, opt, bool-->true
</isSupportExportGaugeReadingEvent>
<isSupportSearchPhoneNoQRCode>
<!--ro, opt, bool-->true
</isSupportSearchPhoneNoQRCode>
<isSupportSendPingCommand>
<!--ro, opt, bool-->true
</isSupportSendPingCommand>
<isSupportAbnormalAlarmThreshold>
<!--ro, opt, bool-->true
</isSupportAbnormalAlarmThreshold>
<voickTalkType opt="gateway">
<!--ro, opt, string, attr:opt{req, string}-->gateway
</voickTalkType>
<audioBroadcastMaxNum>
<!--ro, opt, int-->256
</audioBroadcastMaxNum>
<isSupportStartInnerPlayback>
<!--ro, opt, bool-->true
</isSupportStartInnerPlayback>
<isSupportControlInnerPlayback>
<!--ro, opt, bool-->true
</isSupportControlInnerPlayback>
<isSupportSearchFocusCalibrationPoint>
<!--ro, opt, bool-->true
</isSupportSearchFocusCalibrationPoint>
<isSupportDeleteFocusCalibrationPoint>
<!--ro, opt, bool-->true
</isSupportDeleteFocusCalibrationPoint>
<isSupportAddFocusCalibrationPoint>
<!--ro, opt, bool-->true
</isSupportAddFocusCalibrationPoint>
<isSupportFocusCalibrateParams>
<!--ro, opt, bool-->true
</isSupportFocusCalibrateParams>
<isSupportModifyTranscodeAdvancedParams>
<!--ro, opt, bool-->true
</isSupportModifyTranscodeAdvancedParams>
<isSupportGetTranscodeAdvancedParams>
<!--ro, opt, bool-->true
</isSupportGetTranscodeAdvancedParams>
<isSupportRadarPerimeterShieldScenesParams>
<!--ro, opt, bool-->true
</isSupportRadarPerimeterShieldScenesParams>
<isSupportAsyncImportPlayerFile>
<!--ro, opt, bool-->true
</isSupportAsyncImportPlayerFile>
<isSupportGetAsyncImportProcess>
<!--ro, opt, bool-->true
</isSupportGetAsyncImportProcess>
<isSupportRefreshEventRuleAndTriggerParams>
<!--ro, opt, bool-->true
</isSupportRefreshEventRuleAndTriggerParams>
<isSupportGetEventRuleAndTriggerParamsRefreshStatus>
<!--ro, opt, bool-->true
</isSupportGetEventRuleAndTriggerParamsRefreshStatus>
<isSupportWhiteLightStatusChange>
<!--ro, opt, bool-->true
</isSupportWhiteLightStatusChange>
<isSupportCleanConfigurationPartitionFiles>
<!--ro, opt, bool-->true
</isSupportCleanConfigurationPartitionFiles>
<isSupportPingCommandStatus>
<!--ro, opt, bool-->true
</isSupportPingCommandStatus>
<isSupportInstrumentStatus>

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```
    <!--ro, opt, bool-->true
  </isSupportInstrumentStatus>
  <isSupportScannerInfoScheduleParams>
    <!--ro, opt, bool-->true
  </isSupportScannerInfoScheduleParams>
  <isSupportMixedTargetDetectionPOSOverlay>
    <!--ro, opt, bool-->true
  </isSupportMixedTargetDetectionPOSOverlay>
  <isSupportFaceGazeDetectionParams>
    <!--ro, opt, bool-->true
  </isSupportFaceGazeDetectionParams>
  <isSupportFaceGazeCalibrationParams>
    <!--ro, opt, bool-->true
  </isSupportFaceGazeCalibrationParams>
  <isSupportFaceGazePOSOverlay>
    <!--ro, opt, bool-->true
  </isSupportFaceGazePOSOverlay>
  <isSupportFaceGazeAlgorithmVersion>
    <!--ro, opt, bool-->true
  </isSupportFaceGazeAlgorithmVersion>
  <isSupportExportFaceGazeStatisticsData>
    <!--ro, opt, bool-->true
  </isSupportExportFaceGazeStatisticsData>
  <isSupportSearchFaceGazeStatisticsData>
    <!--ro, opt, bool-->true
  </isSupportSearchFaceGazeStatisticsData>
  <isSupportRestoreFaceGazeAlgorithmParams>
    <!--ro, opt, bool-->true
  </isSupportRestoreFaceGazeAlgorithmParams>
  <isSupportFaceGazeDetection>
    <!--ro, opt, bool-->true
  </isSupportFaceGazeDetection>
  <isSupportRawCroppingCfg>
    <!--ro, opt, bool-->true
  </isSupportRawCroppingCfg>
  <isSupportBiometricKeyModeSwitch>
    <!--ro, opt, bool-->true
  </isSupportBiometricKeyModeSwitch>
  <isSupportSelfTestStatus>
    <!--ro, opt, bool-->true
  </isSupportSelfTestStatus>
  <isSupportCreateRecordDiagnosis>
    <!--ro, opt, bool-->true
  </isSupportCreateRecordDiagnosis>
  <isSupportGetRecordDiagnosisResult>
    <!--ro, opt, bool-->true
  </isSupportGetRecordDiagnosisResult>
  <isSupportDeleteRecordDiagnosis>
    <!--ro, opt, bool-->true
  </isSupportDeleteRecordDiagnosis>
  <isSupportImportCfgDataIsEdited>
    <!--ro, opt, bool-->true
  </isSupportImportCfgDataIsEdited>
  <isSupportAddEventRuleAndTriggerAsyn>
    <!--ro, opt, bool-->true
  </isSupportAddEventRuleAndTriggerAsyn>
  <isSupportGetEventRuleAndTriggerAsynStatus>
    <!--ro, opt, bool-->true
  </isSupportGetEventRuleAndTriggerAsynStatus>
  <isSupportDeleteScenesShieldMaskRegion>
    <!--ro, opt, bool-->true
  </isSupportDeleteScenesShieldMaskRegion>
  <isSupportBuzzerParams>
    <!--ro, opt, bool-->true
  </isSupportBuzzerParams>
  <isSupportAlgorithmContainerSubnetCfg>
    <!--ro, opt, bool-->true
  </isSupportAlgorithmContainerSubnetCfg>
  <isSupportFineTuningSpec>
    <!--ro, opt, bool-->true
  </isSupportFineTuningSpec>
  <isSupportAddOrchExportTask>
    <!--ro, opt, bool-->true
  </isSupportAddOrchExportTask>
  <isSupportSearchOrchExportTaskProgress>
    <!--ro, opt, bool-->true
  </isSupportSearchOrchExportTaskProgress>
  <isSupportStopOrchExportTask>
    <!--ro, opt, bool-->true
  </isSupportStopOrchExportTask>
  <isSupportBallisticRuleParams>
    <!--ro, opt, bool-->true
  </isSupportBallisticRuleParams>
  <isSupportBallisticCalibrationStatus>
    <!--ro, opt, bool-->true
  </isSupportBallisticCalibrationStatus>
  <isSupportNetworkConnectionMode>
    <!--ro, opt, bool-->true
  </isSupportNetworkConnectionMode>
  <isSupport64BitTimestamp>
    <!--ro, opt, bool-->true
  </isSupport64BitTimestamp>
  <isSupportImportFineTuningModelResult>
    <!--ro, opt, bool-->true
```

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```
    <!--ro, opt, bool-->true
  </isSupportImportFineTuningModelResult>
  <isSupportImportFineTuningModel>
    <!--ro, opt, bool-->true
  </isSupportImportFineTuningModel>
  <isSupportExportFineTuningModel>
    <!--ro, opt, bool-->true
  </isSupportExportFineTuningModel>
  <isSupportMovementEncryptionParam>
    <!--ro, opt, bool-->true
  </isSupportMovementEncryptionParam>
  <isSupportBackupSpec>
    <!--ro, opt, bool-->true
  </isSupportBackupSpec>
  <isSupportCreateBackupFile>
    <!--ro, opt, bool-->true
  </isSupportCreateBackupFile>
  <isSupportSendBackupFileLock>
    <!--ro, opt, bool-->true
  </isSupportSendBackupFileLock>
  <isSupportLocalContactChange>
    <!--ro, opt, bool-->true
  </isSupportLocalContactChange>
  <isSupportSearchLMMEventLabel>
    <!--ro, opt, bool-->true
  </isSupportSearchLMMEventLabel>
  <isSupportDeviceDescriptionChange>
    <!--ro, opt, bool-->true
  </isSupportDeviceDescriptionChange>
  <isSupportPersonnelFileOperationRecords>
    <!--ro, opt, bool-->true
  </isSupportPersonnelFileOperationRecords>
  <isSupportResourcesCap>
    <!--ro, opt, bool-->true
  </isSupportResourcesCap>
  <isSupportMotionOriginPos>
    <!--ro, opt, bool-->true
  </isSupportMotionOriginPos>
  <isSupportPowerMode>
    <!--ro, opt, bool-->true
  </isSupportPowerMode>
  <isSupportRTMPSchedule>
    <!--ro, opt, bool-->true
  </isSupportRTMPSchedule>
  <isSupportPIPDataCfg>
    <!--ro, opt, bool-->true
  </isSupportPIPDataCfg>
  <isSupportReocrdTranscodeParams>
    <!--ro, opt, bool-->true
  </isSupportReocrdTranscodeParams>
</DeviceCap>
```

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14.1.4.4 Import device configuration file

Request URL

POST /ISAPI/System/configurationData?secretkey=<secretkey>&deviceIndex=<deviceIndex>&mode=<mode>

Query Parameter

Parameter Name	Parameter Type	Description
secretkey	string	The verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.
deviceIndex	string	--
mode	enum	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.1.4.5 Get the languages supported by the device

Request URL

GET /ISAPI/System/DeviceLanguage

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--ro, req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian",
"Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish",
"Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian"-Slovenian,
"Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian", "Uzbek", "Kazak",
"Kirghiz", "Farsi", "Azerbaidzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
</DeviceLanguage>
```

14.1.4.6 Set device language parameters

Request URL

PUT /ISAPI/System/DeviceLanguage

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian", "Bulgarian",
"Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese",
"Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian" (Slovenian), "Croatian",
"Lithuanian", "Serbian", "BrazilianPortuguese" (Brazilian Portuguese), "Indonesian", "Ukrainian", "EURSpanish"-->SimChinese
  </language>
</DeviceLanguage>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.4.7 Get the capability of configuring the device language

Request URL

GET /ISAPI/System/DeviceLanguage/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device language configuration, attr:version{req, string, protocolVersion}-->
  <language
opt="SimChinese,TraChinese,English,Russian,Bulgarian,Hungarian,Greek,German,Italian,Czech,Slovakia,French,Polish,Dutch,Portuguese,Spanish,Romanian,Turkish,J
apanese,Danish,Swedish,Norwegian,Finnish,Korean,Thai,Estonia,Vietnamese,Hebrew,Latvian,Arabic,Sovenian,Croatian,Lithuanian,Serbian,BrazilianPortuguese,Indon
esian,Ukrainian,EURSpanish,Uzbek,Kazak,Kirghiz,Farsi,Azerbaidzhan,Burmese,Mongolian,Anglicism,Estonian">
    <!--ro, req, enum, language, subType:string, attr:opt{req, string}, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese),
"English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish",
"Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic",
"Sovenian"-Slovenian, "Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian",
"Uzbek", "Kazak", "Kirghiz", "Farsi", "Azerbaidzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
  <upgradeFirmWareEnabled>
    <!--ro, opt, bool, whether to enable upgrading the firmware-->true
  </upgradeFirmWareEnabled>
</DeviceLanguage>
```

14.1.4.8 Set restoring devices' default parameters

Request URL

PUT /ISAPI/System/factoryReset?mode=<mode>&childDevID=<devIndex>&loginPassword=<loginPassword>&module=<module>

Query Parameter

Parameter Name	Parameter Type	Description
mode	enum	--
devIndex	string	--
loginPassword	string	--
module	enum	--

Request Message

None

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.4.9 Get the network service capability

Request URL

GET /ISAPI/System/Network/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NetworkCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, get the network service capability, attr:version{opt, string, protocolVersion}-->
  <isSupportWireless>
    <!--ro, req, bool, whether the device supports wireless network-->true
  </isSupportWireless>
  <isSupportWired>
    <!--ro, opt, bool, whether the device supports wired network-->true
  </isSupportWired>
  <isSupportMobileNetwork>
    <!--ro, opt, bool, whether the device supports mobile network-->true
  </isSupportMobileNetwork>
  <isSupportWAN>
    <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
  </isSupportWAN>
  <isSupportPPPoE>
    <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
  </isSupportPPPoE>
  <isSupportBond>
    <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
  </isSupportBond>
  <isSupport802_1x>
    <!--ro, req, bool, whether the device supports 802.1x protocol-->true
  </isSupport802_1x>
  <isSupportNtp>
    <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
  </isSupportNtp>
  <isSupportFtp>
    <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
  </isSupportFtp>
  <isSupportUpnp>
    <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play) protocol-->true
  </isSupportUpnp>
  <isSupportPNP>
    <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
  </isSupportPNP>
  <isSupportDdns>
    <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
  </isSupportDdns>
  <isSupportHttps>
    <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
  </isSupportHttps>
  <isSupport28181>
    <!--ro, opt, bool, N/A-->true
  </isSupport28181>
  <SnmpCap>
    <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
    <isSupport>
```

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```
    <isSupport>
      <!--ro, req, bool, whether the device supports SNMP-->true
    </isSupport>
  </SnmpCap>
  <isSupportExtNetCfg>
    <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
  </isSupportExtNetCfg>
  <isSupportIPFilter>
    <!--ro, opt, bool, whether the device supports IP filtering-->true
  </isSupportIPFilter>
  <isSupportSSH>
    <!--ro, opt, bool, whether the device supports SSH-->true
  </isSupportSSH>
  <isSupportNetPreviewStrategy>
    <!--ro, opt, bool, whether the device supports network live view strategy-->true
  </isSupportNetPreviewStrategy>
  <isSupportEZVIZ>
    <!--ro, opt, bool, whether the device supports EZ protocol-->true
  </isSupportEZVIZ>
  <isSupportEhome>
    <!--ro, opt, bool, whether the device supports EHome (ISUP) protocol-->true
  </isSupportEhome>
  <isSupportWirelessDial>
    <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
  </isSupportWirelessDial>
  <isSupportWirelessServer>
    <!--ro, opt, bool, whether the device supports wireless server-->true
  </isSupportWirelessServer>
  <isSupportWPS>
    <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
  </isSupportWPS>
  <isWirelessMutexWithWirelessServer>
    <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
  </isWirelessMutexWithWirelessServer>
  <isSupportMACFilter>
    <!--ro, opt, bool, whether the device supports physical address filtering-->true
  </isSupportMACFilter>
  <GB28181Cap>
    <!--ro, opt, object, N/A-->
    <isSupportGB28181Service>
      <!--ro, opt, bool, N/A-->true
    </isSupportGB28181Service>
  </GB28181Cap>
  <WPS>
    <!--ro, opt, object, Wi-Fi Protected Setup-->
    <NetworkInterfaceList size="2">
      <!--ro, opt, array, Internet API list, subType:object, attr:size{req, int}-->
      <NetworkInterface>
        <!--ro, opt, object, Internet API-->
        <id>
          <!--ro, req, string, NIC index-->test
        </id>
        <enabled>
          <!--ro, req, bool, whether to enable or not-->true
        </enabled>
        <isSupportAutoConnect>
          <!--ro, opt, bool, whether the device supports automatic connection-->true
        </isSupportAutoConnect>
        <isSupportDevicePinCode>
          <!--ro, opt, bool, whether the device supports PIN code-->true
        </isSupportDevicePinCode>
        <isSupportDevicePinCodeUpdate>
          <!--ro, opt, bool, whether the device supports updating the PIN code-->true
        </isSupportDevicePinCodeUpdate>
        <ApPinCode>
          <!--ro, opt, object, PIN code of the access point-->
          <ssid min="1" max="10">
            <!--ro, opt, string, SSID (Service Set Identifier), attr:min{req, int},max{req, int}-->test
          </ssid>
          <pinCode min="1" max="10">
            <!--ro, opt, string, PIN code, attr:min{req, int},max{req, int}-->test
          </pinCode>
        </ApPinCode>
      </NetworkInterface>
    </NetworkInterfaceList>
  </WPS>
  <isSupportRFIDData>
    <!--ro, opt, bool, whether the device supports RFID collection configuration-->true
  </isSupportRFIDData>
  <isSupportwifiProbeSSID>
    <!--ro, opt, bool, whether the device supports configuration of Wi-Fi signal detection-->true
  </isSupportwifiProbeSSID>
  <isSupportWifiProbe>
    <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
  </isSupportWifiProbe>
  <verificationCode min="1" max="10">
    <!--ro, opt, string, verification code, attr:min{req, int},max{req, int}-->test
  </verificationCode>
  <WPSCap>
    <!--ro, opt, object, Wi-Fi Protected Setup capability-->
    <isSupport>
      <!--ro, req, bool, whether the device supports Wi-Fi Protected Setup-->true
    </isSupport>
    <isSupportAutoConnect>
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      <!--ro, req, bool, whether the device supports automatic connection-->true
    </isSupportAutoConnect>
  </WPSCap>
  <NetWorkMode>
    <!--ro, opt, object, network mode-->
    <workMode>
      <!--ro, opt, enum, network mode, subType:string, desc:"close", "wifi" (wireless access), "wifiAp" (wireless access point)-->close
    </workMode>
  </NetWorkMode>
  <VerificationCodeModification>
    <!--ro, opt, object, verification code editing-->
    <verificationCodeType opt="normal,empty">
      <!--ro, opt, string, verification code type, attr:opt{req, string}-->test
    </verificationCodeType>
    <isSupportDeclarationURL>
      <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
    </isSupportDeclarationURL>
    <isSupportPrivacyPolicyURL>
      <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
    </isSupportPrivacyPolicyURL>
    <verificationCodeModify opt="true,false">
      <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{req, string}-->true
    </verificationCodeModify>
    <Hyperlinks>
      <!--ro, opt, object, hyperlink-->
      <declarationURL>
        <!--ro, opt, string, hyperlink to the Terms of Service-->test
      </declarationURL>
      <privacyPolicyURL>
        <!--ro, opt, string, hyperlink to the Privacy Policy-->test
      </privacyPolicyURL>
    </Hyperlinks>
    <isSupportVerificationCodeCheck>
      <!--ro, opt, bool, whether the device supports verifying the verification code-->true
    </isSupportVerificationCodeCheck>
    <isSupportOldVerificationCode>
      <!--ro, opt, bool, whether the device supports the EZ password configuration of old version (the password of old version consists of six characters)--
>true
    </isSupportOldVerificationCode>
  </VerificationCodeModification>
  <isSupportIntegrate>
    <!--ro, opt, bool, whether the device supports access protocol configuration-->true
  </isSupportIntegrate>
  <isSupportPlatformAccess>
    <!--ro, opt, bool, whether the device supports platform access, desc:GET /ISAPI/System/Network/platformAccessFilter/capabilities (platform access IP
address filtering configuration (Ehome allowlist configuration))-->true
  </isSupportPlatformAccess>
  <isSupportIntelligentBoost>
    <!--ro, opt, bool, whether the device supports intelligent acceleration-->true
  </isSupportIntelligentBoost>
  <WifiRegionConfiguration>
    <!--ro, opt, object, Wi-Fi area configuration, desc:when both are not supported, the <WifiRegionConfiguration> node will not return-->
    <isSupportWifiRegion>
      <!--ro, opt, bool, whether the device supports Wi-Fi area configuration, desc:supported by overseas devices; true (support), this node is not returned
(not support)-->true
    </isSupportWifiRegion>
    <isSupportWifiEnhancement>
      <!--ro, opt, bool, whether the device supports Wi-Fi enhancement, desc:N/A-->true
    </isSupportWifiEnhancement>
  </WifiRegionConfiguration>
  <isSupportResourceStatistics>
    <!--ro, opt, bool, whether the device supports getting the network resource information-->true
  </isSupportResourceStatistics>
  <isSupportBandwidthLimit>
    <!--ro, opt, bool, whether the device supports bandwidth limit-->true
  </isSupportBandwidthLimit>
  <isSupportPOEPortsDisableAdaptiveServer>
    <!--ro, opt, bool, whether the device supports the POE protocol-->true
  </isSupportPOEPortsDisableAdaptiveServer>
  <isSupportPOEConfiguration>
    <!--ro, opt, bool, whether the device supports configuring the POE NIC parameters-->true
  </isSupportPOEConfiguration>
  <isSupportGetLinkSocketIP>
    <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
  </isSupportGetLinkSocketIP>
  <isSupportWebSocket>
    <!--ro, opt, bool, whether the device supports WebSocket-->true
  </isSupportWebSocket>
  <isSupportWebSocketS>
    <!--ro, opt, bool, whether the device supports WebSocketS-->true
  </isSupportWebSocketS>
  <Adaption>
    <!--ro, opt, object, network self-adaption-->
    <streamType opt="0,1,2,3,4,5,7,8,9,10">
      <!--ro, opt, enum, stream type that supports network self-adaption during live view, subType:int, attr:opt{req, string}, desc:stream types that
support network self-adaption during live view: 0 (main stream), 1 (sub-stream), 2 (third stream), 3 (fourth stream / virtual stream), 4 (fifth stream), 5
(sixth stream), 7 (seventh stream), 8 (eighth stream), 9 (ninth stream), 10 (tenth stream), and so on-->0
    </streamType>
    <isSupportPlayback>
      <!--ro, opt, bool, whether the device supports playback-->true
    </isSupportPlayback>
  </Adaption>
  <isSupportVideoImgDB>
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    <!--ro, opt, bool, whether the device supports image and video library configuration-->true
  </isSupportVideoImgDB>
  <isSupportVideoImgLibService>
    <!--ro, opt, bool, whether the device supports image and video library service configuration-->true
  </isSupportVideoImgLibService>
  <isSupportWirelessWifi>
    <!--ro, opt, bool, whether the device supports Wi-Fi configuration-->true
  </isSupportWirelessWifi>
  <isSupportPrivatePlatom>
    <!--ro, opt, bool, whether the device supports third party platform configuration-->true
  </isSupportPrivatePlatom>
  <isSupportEventDataOverWebSocket opt="true,false">
    <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{req, string}-->true
  </isSupportEventDataOverWebSocket>
  <isSupportDynamicHostName>
    <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) configuration-->true
  </isSupportDynamicHostName>
  <isSupportTCPAcceleration>
    <!--ro, opt, bool, whether the device supports TCP acceleration-->true
  </isSupportTCPAcceleration>
  <isSupportFlowControl>
    <!--ro, opt, bool, whether the device supports traffic control-->true
  </isSupportFlowControl>
  <isSupportEZVIZTiming>
    <!--ro, opt, bool, whether the device supports EZ platform synchronization (enabledTiming of message returned by/ISAPI/System/Network/EZVIZ)-->true
  </isSupportEZVIZTiming>
  <isSupportTrafficMonitor>
    <!--ro, opt, bool, whether the device supports traffic monitoring-->true
  </isSupportTrafficMonitor>
  <isSupportPingWatchdog>
    <!--ro, opt, bool, whether the device supports configuring the watchdog parameters-->true
  </isSupportPingWatchdog>
  <isSupportAutoFrequencyHop>
    <!--ro, opt, bool, whether the device supports auto frequency hopping-->true
  </isSupportAutoFrequencyHop>
  <isSupportTDMACfg>
    <!--ro, opt, bool, whether the device supports configuring TDMA-->true
  </isSupportTDMACfg>
  <isSupportDynamicChannel>
    <!--ro, opt, bool, whether the device supports dynamic channel selection-->true
  </isSupportDynamicChannel>
  <isSupportSIMCardSwitch>
    <!--ro, opt, bool, whether the device supports configuring scheduled SIM card switch, desc:related URI:
/ISAPI/System/Network/WirelessDial/SIMCardSwitch?format=json-->true
  </isSupportSIMCardSwitch>
  <isSupportDMSAuthInfo>
    <!--ro, opt, bool, whether the device supports DMS authentication information management-->true
  </isSupportDMSAuthInfo>
  <isSupportWiredandWirelessTrafficMonitor>
    <!--ro, opt, bool, whether the device supports traffic monitoring (wireless NIC and wired NIC)-->true
  </isSupportWiredandWirelessTrafficMonitor>
  <isSupport4G>
    <!--ro, opt, bool, whether the device supports 4G module-->true
  </isSupport4G>
  <isSupport4GConfig>
    <!--ro, opt, bool, whether the device supports 4G configuration-->true
  </isSupport4GConfig>
  <isSupportGRIDServer>
    <!--ro, opt, bool, N/A-->true
  </isSupportGRIDServer>
  <isSupportGRIDInfo>
    <!--ro, opt, bool, N/A-->true
  </isSupportGRIDInfo>
  <isSupportSipTransmit>
    <!--ro, opt, bool, whether the device supports configuring SIP transmission parameters, desc:GET|PUT /ISAPI/System/Network/SIP/transmit?format=json--
>true
  </isSupportSipTransmit>
  <isSupportSip>
    <!--ro, opt, bool, whether the device supports private SIP (VOIP)-->true
  </isSupportSip>
  <isSupportWifiCascade>
    <!--ro, opt, bool, whether the device supports enabling automatic Wi-Fi cascading-->true
  </isSupportWifiCascade>
  <isSupportRelativeInfo>
    <!--ro, opt, bool, whether the device supports getting the information about the connected devices-->true
  </isSupportRelativeInfo>
  <isSupportANR>
    <!--ro, opt, bool, whether the device supports automatic network replenishment, desc:related URI: /ISAPI/System/Network/ANR-->true
  </isSupportANR>
  <isSupportEmailEncrypt>
    <!--ro, opt, bool, whether the device supports email encryption configuration-->true
  </isSupportEmailEncrypt>
  <isSupportPortFilter>
    <!--ro, opt, bool, whether the device supports port filtering-->true
  </isSupportPortFilter>
  <isSupportPingDeny>
    <!--ro, opt, bool, whether the device supports configuring parameters to prevent the device from being tested using ping command-->true
  </isSupportPingDeny>
  <isSupportDOSDeny>
    <!--ro, opt, bool, whether the device supports configuring parameters to prevent the DoS attack-->true
  </isSupportDOSDeny>
  <isSupportNetworkStatus>
    <!--ro, opt, bool, whether the device supports searching for the network status-->true
  </isSupportNetworkStatus>
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  <isSupportEZVIZQRCode>
    <!--ro, opt, bool, whether the device supports getting the EZ QR code, desc:/ISAPI/System/Network/EZVIZ/QRCode?format=json-->true
  </isSupportEZVIZQRCode>
  <isSupportEZVIZUnbind>
    <!--ro, opt, bool, whether the device supports unbinding from EZ, desc:/ISAPI/System/Network/EZVIZ/unbind?format=json-->true
  </isSupportEZVIZUnbind>
  <isSupportPingTest>
    <!--ro, opt, bool, whether the device supports ping test, desc:related URI: /ISAPI/System/Network/pingtest/capabilities-->true
  </isSupportPingTest>
  <isSupportMOTCloudPlateform>
    <!--ro, opt, bool, whether the device supports configuring parameters for accessing the cloud platform service, desc:related URI:
/ISAPI/System/Network/Traffic/MOTCloudPlateform/capabilities?format=json-->true
  </isSupportMOTCloudPlateform>
  <isSupportSpectrumScan>
    <!--ro, opt, bool, whether the device supports frequency spectrum scanning, desc:related URI:
/ISAPI/System/Network/interface/<netCardID>/spectrumScan/capabilities?format=json-->true
  </isSupportSpectrumScan>
  <isSupportFRP>
    <!--ro, opt, bool, whether the device supports FRP (Fast Reverse Proxy), desc:related URI: /ISAPI/Syetem/Network/FRP?format=json-->true
  </isSupportFRP>
  <isSupport5GModule>
    <!--ro, opt, bool, whether the device supports 5G module configuration, desc:related URI: /ISAPI/System/Network/5GModule/capabilities?format=json-->true
  </isSupport5GModule>
  <isSupportWirelessDialRoute>
    <!--ro, opt, bool, whether the device supports configuring parameters of route forwarding rules of wireless dial-up, desc:related URI:
/ISAPI/System/Network/WirelessDia/route/capabilities?format=json-->true
  </isSupportWirelessDialRoute>
  <isSupportNPlusOne>
    <!--ro, opt, bool, whether the device supports N+1 protocol-->true
  </isSupportNPlusOne>
  <supportEnvironmentalStandards opt="HJ212">
    <!--ro, opt, string, environmental monitoring standard, attr:opt{req, string}, desc:N/A-->true
  </supportEnvironmentalStandards>
  <isSupportIntranetSegment>
    <!--ro, opt, bool, whether the device supports LAN segment configuration, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is false or not be returned. Related URI: /ISAPI/System/Network/intranetSegment/capabilities?
format=json-->true
  </isSupportIntranetSegment>
  <isSupportDHCPServer>
    <!--ro, opt, bool, whether the device supports DHCP service configuration, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is false or not be returned. Related URI: /ISAPI/System/Network/DHCPServer/capabilities?
format=json-->true
  </isSupportDHCPServer>
  <isSupportClusterIntercom>
    <!--ro, opt, bool, whether the 7200 intercom protocol is transmitted, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will be returned and its value is false or not be returned. Related URI: /ISAPI/System/Network/clusterIntercom/capabilities?
format=json-->true
  </isSupportClusterIntercom>
  <isSupportHydrologicMonitoring>
    <!--ro, opt, bool, whether the device supports hydrologic monitoring data communication, desc:if this function is supported, this node will be returned
and its value is true; otherwise, this node will be returned and its value is false or not be returned. Related URI:
/ISAPI/System/Network/HydrologicMonitoring/capabilities?format=json-->true
  </isSupportHydrologicMonitoring>
  <isSupportGantryETCServer>
    <!--ro, opt, bool, N/A, desc:N/A-->true
  </isSupportGantryETCServer>
  <RoutePolicy>
    <!--ro, opt, object, policy routing-->
    <isSupportRoutePolicy>
      <!--ro, opt, bool, whether the device supports configuring the policy routing parameters-->true
    </isSupportRoutePolicy>
    <isRebootHint>
      <!--ro, opt, bool, whether to alarm to reboot after configuration-->true
    </isRebootHint>
  </RoutePolicy>
  <isWirelessDialMutexWithWlanAndWifi>
    <!--ro, opt, bool, whether the wireless dial-up is mutually exclusive with the WLAN hotspot and Wi-Fi-->true
  </isWirelessDialMutexWithWlanAndWifi>
  <isSupportDMZCfg>
    <!--ro, opt, bool, whether the device supports configuring DMZ host, desc:related URI: /ISAPI/System/Network/DMZCfg/capabilities?format=json-->true
  </isSupportDMZCfg>
  <isSupportSpeechTranscription>
    <!--ro, opt, bool, whether the device supports accessing the speech-to-text transcription platform, desc:if this function is supported, this node will
be returned and its value is true; otherwise, this node will be returned and its value is false or not be returned. Related URI:
/ISAPI/System/Network/speechTranscriptionPlatform/capabilities?format=json-->true
  </isSupportSpeechTranscription>
  <isSupportAsynPingTest>
    <!--ro, opt, bool, whether the device supports asynchronous network detection, desc:network detection means testing the network stability and packet
loss rate of the network using the ping command for knowing about the network condition. Related URI: /ISAPI/System/Network/asynPingTest/capabilities?
format=json-->true
  </isSupportAsynPingTest>
  <isSupportBasebandCapture>
    <!--ro, opt, bool, whether the device supports capturing packets of the baseband, desc:you can capture packets transmitted between the 4G module and the
base station using the ping command. Related URI: /ISAPI/System/Network/cellularNetwork/basebandCapture/capabilities?format=json-->true
  </isSupportBasebandCapture>
  <isSupportCellularNetworkSpeedTest>
    <!--ro, opt, bool, whether the device supports testing the connection speed between the device and the cellular network, desc:you can test the
connection speed of the network where the current device is. Related URI: /ISAPI/System/Network/cellularNetwork/speedTest/capabilities?format=json-->true
  </isSupportCellularNetworkSpeedTest>
  <isSupportCellularNetworkCellInfo>
    <!--ro, opt, bool, whether the device supports getting a cell's information of the cellular network, desc:you can get the information of the cellular
network where the device is. Related URI: /ISAPI/System/Network/cellularNetwork/cellInfo?format=json&ecurity=<security>&iv=<iv>-->true
  </isSupportCellularNetworkCellInfo>
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  <isSupportMultiBandFrequencyScanning>
    <!--ro, opt, bool, whether the device supports scanning all frequency bands of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/multiBandFrequencyScanning/capabilities?format=json-->true
  </isSupportMultiBandFrequencyScanning>
  <isSupportCellularNetworkLockBand>
    <!--ro, opt, bool, whether the device supports accessing the locked frequency band of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/lockBand/capabilities?format=json-->true
  </isSupportCellularNetworkLockBand>
  <isSupportRadioFrequencyFluctuation>
    <!--ro, opt, bool, whether the device supports configuring RF parameters of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/radioFrequencyFluctuation/capabilities?format=json-->true
  </isSupportRadioFrequencyFluctuation>
  <isSupportBasebandBlock>
    <!--ro, opt, bool, whether the device supports the unauthorized list of the base station, desc:related URI:
/ISAPI/System/Network/cellularNetwork/basebandBlock/capabilities?format=json-->true
  </isSupportBasebandBlock>
  <isSupportWifiQuality>
    <!--ro, opt, bool, whether the device supports getting the device's Wi-Fi quality information, desc:related URI: /ISAPI/System/Network/wifiQuality?
format=json-->true
  </isSupportWifiQuality>
  <isSupportWiredNetworkQuality>
    <!--ro, opt, bool, whether the device supports getting the wired network quality information, desc:related URI:
/ISAPI/System/Network/wiredNetworkQuality?format=json-->true
  </isSupportWiredNetworkQuality>
  <isSupportQoSBehaviorCfg>
    <!--ro, opt, bool, whether the device supports configuring QoS traffic behavior, desc:related URI: /ISAPI/System/Network/qos/BehaviorCfg/capabilities?
format=json-->true
  </isSupportQoSBehaviorCfg>
  <isSupportQoSClassifierCfg>
    <!--ro, opt, bool, whether the device supports configuring QoS traffic types, desc:related URI: /ISAPI/System/Network/qos/ClassifierCfg/capabilities?
format=json-->true
  </isSupportQoSClassifierCfg>
  <isSupportQoSPolicyCfg>
    <!--ro, opt, bool, whether the device supports configuring QoS policy, desc:related URI: /ISAPI/System/Network/qos/PolicyCfg/capabilities?format=json--
>true
  </isSupportQoSPolicyCfg>
  <isSupportQoSPolicyVLANCfg>
    <!--ro, opt, bool, whether the device supports configuring QoS policy VLAN, desc:related URI: /ISAPI/System/Network/qos/PolicyVLANCfg/capabilities?
format=json-->true
  </isSupportQoSPolicyVLANCfg>
  <isSupportQoSPortRateLimiting>
    <!--ro, opt, bool, whether the device supports configuring port speed limit, desc:related URI: /ISAPI/System/Network/qos/portRateLimiting/capabilities?
format=json-->true
  </isSupportQoSPortRateLimiting>
  <isSupportQoSQueueAlgorithmCfg>
    <!--ro, opt, bool, whether the device supports configuring QoS queue algorithm, desc:related URI:
/ISAPI/System/Network/qos/queueAlgorithmCfg/capabilities?format=json-->true
  </isSupportQoSQueueAlgorithmCfg>
  <isSupportQoSPortPriorityCfg>
    <!--ro, opt, bool, whether the device supports QoS port priority configuration, desc:related URI:
/ISAPI/System/Network/qos/portPriorityCfg/capabilities?format=json-->true
  </isSupportQoSPortPriorityCfg>
  <isSupportQoS802.1pPriorityMapping>
    <!--ro, opt, bool, whether the device supports DSCP and 802.1p mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/802.1pPriorityMapping/capabilities?format=json-->true
  </isSupportQoS802.1pPriorityMapping>
  <isSupportQoSDSCPPriorityMapping>
    <!--ro, opt, bool, whether the device supports DSCP mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/DSCPPriorityMapping/capabilities?format=json-->true
  </isSupportQoSDSCPPriorityMapping>
  <isSupportQoSLoaclPriorityMapping>
    <!--ro, opt, bool, whether the device supports 802.1p and local mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/loaclPriorityMapping/capabilities?format=json-->true
  </isSupportQoSLoaclPriorityMapping>
  <isSupportStopPingtest>
    <!--ro, opt, bool, whether to stop PingTest, desc:related URI: /ISAPI/System/Network/StopPingtest?format=json-->true
  </isSupportStopPingtest>
  <isSupportSNMPView>
    <!--ro, opt, bool, whether the device supports SNMP view, desc:related URI: /ISAPI/System/Network/SNMP/ViewCfg/capabilities?format=json-->true
  </isSupportSNMPView>
  <isSupportSNMPUserGroup>
    <!--ro, opt, bool, whether the device supports SNMP user groups, desc:related URI:  /ISAPI/System/Network/SNMP/UserGroup/capabilities?format=json-->true
  </isSupportSNMPUserGroup>
  <isSupportSNMPBasicInfo>
    <!--ro, opt, bool, whether the device supports the basic information of SNMP, desc:related URI: /ISAPI/System/Network/SNMP/BasicInfo/capabilities?
format=json-->true
  </isSupportSNMPBasicInfo>
  <isSupportRFIDTagCheckTest>
    <!--ro, opt, bool, whether the device supports the RFID tag disk storage performance test, desc:related URI:
/ISAPI/System/Network/RFIDData/GetTagCheckTestInfo?format=json-->true
  </isSupportRFIDTagCheckTest>
  <isSupportQoSDSCPtoLoaclPriorityMapping>
    <!--ro, opt, bool, whether the device supports DSCP and local mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/DSCPtoLoaclPriorityMapping/capabilities?format=json-->true
  </isSupportQoSDSCPtoLoaclPriorityMapping>
  <isSupportQoS802.1pto802.1pPriorityMapping>
    <!--ro, opt, bool, whether the device supports 802.1p mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/802.1pto802.1pPriorityMapping/capabilities?format=json-->true
  </isSupportQoS802.1pto802.1pPriorityMapping>
  <isSupportNFCAllowList>
    <!--ro, opt, bool, whether the device supports NFC network allowlist management, desc:related URI: /ISAPI/System/Network/NFC/allowList/capabilities?
format=json-->true
  </isSupportNFCAllowList>
  <isSupportLinkTestStart>
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    <!--ro, opt, bool, whether the device supports enabling network link test, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Start?format=json-->true
  </isSupportLinkTestStart>
  <isSupportLinkTestResult>
    <!--ro, opt, bool, whether the device supports getting the result of network link test, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Result?format=json-->true
  </isSupportLinkTestResult>
  <isSupportGetAsynTestEmailResult>
    <!--ro, opt, bool, whether the device supports getting email testing result asynchronously, desc:POST
/ISAPI/System/Network/Email/GetAsynTestEmailResult?format=json-->true
  </isSupportGetAsynTestEmailResult>
  <isSupportTestAsynEmail>
    <!--ro, opt, bool, whether the device supports asynchronously testing email, desc:POST /ISAPI/System/Network/Email/TestAsynEmail?format=json-->true
  </isSupportTestAsynEmail>
  <isSupportModbusAccess>
    <!--ro, opt, bool, whether the device supports Modbus accessing parameters, desc:GET/PUT /ISAPI/System/Network/ModbusAccess?format=json-->true
  </isSupportModbusAccess>
  <isSupportNetworkModuleUpgrade>
    <!--ro, opt, bool, whether the device supports upgrading network module, desc:including importing upgrade package
(/ISAPI/System/Network/WirelessDial/ImportUpgradePackage?format=json), getting importing progress
(/ISAPI/System/Network/WirelessDial/GetUpgradePackageImportStatus?format=json), starting upgrading
(/ISAPI/System/Network/WirelessDial/StartUpgradeNetworkModule?format=json), get upgrade status
(/ISAPI/System/Network/WirelessDial/GetUpgradeNetworkModuleStatus?format=json)-->true
  </isSupportNetworkModuleUpgrade>
  <isSupportNetworkModuleVersionInfo>
    <!--ro, opt, bool, whether the device supports getting the version information of network module, desc:GET
/ISAPI/System/Network/WirelessDial/NetworkModuleVersionInfo?format=json-->true
  </isSupportNetworkModuleVersionInfo>
  <isSupportGreenShieldPlatform>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/GreenShieldPlatform/GreenShieldPlatform?format=json-->true
  </isSupportGreenShieldPlatform>
  <isSupportFusionUniversalApplicationPlatCfg>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/FusionUniversalApplicationPlatform/FusionUniversalApplicationPlatCfg?format=json-->true
  </isSupportFusionUniversalApplicationPlatCfg>
  <isSupportNPQControlStrategy>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/NPQMgr/NPQControlStrategy?format=json-->true
  </isSupportNPQControlStrategy>
  <isSupportNAT>
    <!--ro, opt, bool, whether NAT is supported, desc:GET/PUT /ISAPI/System/Network/nats-->true
  </isSupportNAT>
  <isSupportGetCurrentlyUsingNetworkCardInfo>
    <!--ro, opt, bool, whether the device supports getting the current NIC information, desc:GET
/ISAPI/System/Network/WirelessDial/GetCurrentlyUsingNetworkCardInfo?format=json-->true
  </isSupportGetCurrentlyUsingNetworkCardInfo>
  <isSupportAutoRecampParams>
    <!--ro, opt, bool, whether the device supports parameters of auto re-camping on a regular basis, desc:GET/PUT
/ISAPI/System/Network/WirelessDial/AutoRecampParams?format=json-->true
  </isSupportAutoRecampParams>
  <isSupportDSTPServerCfg>
    <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET/PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
format=json-->true
  </isSupportDSTPServerCfg>
  <isSupportJTT1008ServerParams>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/JTT1008Access/JTT1008ServerParams?format=json-->true
  </isSupportJTT1008ServerParams>
  <isSupportImportNARIKeyFile>
    <!--ro, opt, bool, N/A, desc:PUT /ISAPI/System/Network/NARIEncryptionKeyMgr/ImportNARIKeyFile?format=json-->true
  </isSupportImportNARIKeyFile>
  <isSupportExportNARIKeyFile>
    <!--ro, opt, bool, N/A, desc:GET /ISAPI/System/Network/NARIEncryptionKeyMgr/ExportNARIKeyFile?format=json-->true
  </isSupportExportNARIKeyFile>
  <isSupportNARIEncryptionParams>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/NARIEncryptionKeyMgr/NARIEncryptionParams?format=json-->true
  </isSupportNARIEncryptionParams>
  <isSupportDSTPProtocolsCap>
    <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
  </isSupportDSTPProtocolsCap>
  <isSupportIPAdaptiveParams>
    <!--ro, opt, bool, whether the device supports self-adaptive IP address parameters, desc:GET/PUT /ISAPI/System/Network/IPAdaptiveParams?format=json--
>true
  </isSupportIPAdaptiveParams>
  <isSupportPowerCenterPlatform>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/PowerCenterAccess/PowerCenterPlatform?format=json-->true
  </isSupportPowerCenterPlatform>
  <isSupportTestPowerCenterPlatformConnect>
    <!--ro, opt, bool, N/A, desc:POST /ISAPI/System/Network/PowerCenterAccess/TestPowerCenterPlatformConnect?format=json-->true
  </isSupportTestPowerCenterPlatformConnect>
  <isSupportModbusTestSubordinateStatus>
    <!--ro, opt, bool, whether the device supports Modbus testing the status of the specified client., desc:POST
/ISAPI/System/Network/ModbusTestSubordinateStatus?format=json-->true
  </isSupportModbusTestSubordinateStatus>
  <isSupportGetModbusAndRegisterMap>
    <!--ro, opt, bool, whether the device supports get Modbus uploaded content and the mapping relation between the uploaded content and register quantity,
desc:GET /ISAPI/System/Network/GetModbusAndRegisterMap?format=json-->true
  </isSupportGetModbusAndRegisterMap>
  <isSupportAddFRPDynamicAllocationMapPort>
    <!--ro, opt, bool, whether the device supports batch adding FRP dynamic allocation mapping ports, desc:POST
/ISAPI/System/Network/NATPenetration/AddFRPDynamicAllocationMapPort?format=json-->true
  </isSupportAddFRPDynamicAllocationMapPort>
  <isSupportGetFRPDynamicAllocationMapPort>
    <!--ro, opt, bool, whether the device supports batch getting FRP dynamic allocation mapping ports, desc:GET
/ISAPI/System/Network/NATPenetration/GetFRPDynamicAllocationMapPort?format=json-->true
  </isSupportGetFRPDynamicAllocationMapPort>
```


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```
  <isSupportUSBNetworkCard>
    <!--ro, opt, bool, whether the device supports USB NIC-->true
  </isSupportUSBNetworkCard>
  <isSupportOptimumOperatorParams>
    <!--ro, opt, bool, whether the device supports optimal carrier parameters, desc:GET/PUT /ISAPI/System/Network/WirelessDial/OptimumOperatorParams?
format=json-->true
  </isSupportOptimumOperatorParams>
  <isSupportERTCJoin>
    <!--ro, opt, bool, whether the device supports ERTC access, desc:PUT ISAPI/System/Network/ERTCAccess/ERTCJoin?format=json-->true
  </isSupportERTCJoin>
  <isSupportERTCQuit>
    <!--ro, opt, bool, whether the device supports ERTC quiting, desc:PUT ISAPI/System/Network/ERTCAccess/ERTCQuit?format=json-->true
  </isSupportERTCQuit>
  <isSupportLoRaCommunicationParams>
    <!--ro, opt, bool, whether the device supports LoRa protocol communication parameters, desc:GET/PUT
/ISAPI/System/Network/LoRaAccess/LoRaCommunicationParams?format=json-->true
  </isSupportLoRaCommunicationParams>
  <isSupportSetSIMCardSwitch>
    <!--ro, opt, bool, whether the device supports SIM NIC switch configuration, desc:PUT /ISAPI/System/Network/WirelessDial/SetSIMCardSwitch?format=json--
>true
  </isSupportSetSIMCardSwitch>
  <isSupportDialSMSAutoSwitchParams>
    <!--ro, opt, bool, whether the device supports configuring the NIC suto-switch parameter configuration, desc:GET/PUT
/ISAPI/System/Network/WirelessDial/DialSMSAutoSwitchParams?format=json, DialSMSAutoSwitchParams-->true
  </isSupportDialSMSAutoSwitchParams>
  <isSupportBatchModifyDeviceIPAddress>
    <!--ro, opt, bool, whether the device supports batch configuring device network addresses, desc:PUT /ISAPI/System/Network/BatchModifyDeviceIPAddress?
format=json-->true
  </isSupportBatchModifyDeviceIPAddress>
  <isSupportWirelessConfigListAsyncParams>
    <!--ro, opt, bool, whether the device supports wireless configuration list async parameters, desc:GET/PUT
/ISAPI/System/Network/WirelessConfigListAsyncParams?format=json-->true
  </isSupportWirelessConfigListAsyncParams>
  <isSupportRestartPaging>
    <!--ro, opt, bool, whether the device supports paging again, desc:PUT /ISAPI/System/Network/LoRaAccess/RestartPaging?format=json-->true
  </isSupportRestartPaging>
  <isSupportGetCurrentWirelessServerMode>
    <!--ro, opt, bool, whether the device supports getting the current network hotspot mode, desc:GET
/ISAPI/System/Network/WLANServer/GetCurrentWirelessServerMode?format=json-->true
  </isSupportGetCurrentWirelessServerMode>
  <isSupportNetworkInterfaceMaterielInfoDetect>
    <!--ro, opt, bool, whether the device supports NIC material information detection, desc:PUT /ISAPI/System/Network/NetworkInterfaceMaterielInfoDetect?
format=json-->true
  </isSupportNetworkInterfaceMaterielInfoDetect>
  <isSupportMQTTAccessCfg>
    <!--ro, opt, bool, whether the device supports MQTT access configuration, desc:GET/PUT /ISAPI/System/Network/MQTTAccess/MQTTAccessCfg?format=json-->true
  </isSupportMQTTAccessCfg>
  <isSupportModifySavedWirelessList>
    <!--ro, opt, bool, whether the device supports editing Wi-Fi list saved to specified network interface, desc:PUT
/ISAPI/System/Network/ModifySavedWirelessList?format=json-->true
  </isSupportModifySavedWirelessList>
  <isSupportGetSavedWirelessList>
    <!--ro, opt, bool, whether the device supports getting Wi-Fi list saved to specified network interface, desc:POST
/ISAPI/System/Network/GetSavedWirelessList?format=json-->true
  </isSupportGetSavedWirelessList>
  <isSupportWirelessCommunication433AccessParams>
    <!--ro, opt, bool, whether the device supports 433 protocol communication parameters, desc:GET/PUT
/ISAPI/System/Network/WirelessCommunication433Access/WirelessCommunication433AccessParams?format=json-->true
  </isSupportWirelessCommunication433AccessParams>
  <isSupportTSOverWebSocket>
    <!--ro, opt, bool, whether the device supports transparent transmission via the WebSocket serial port, desc:(TS:TransportSerial)-->true
  </isSupportTSOverWebSocket>
  <isSupportSouthernPowerGridServer>
    <!--ro, opt, bool, N/A, desc:N/A-->true
  </isSupportSouthernPowerGridServer>
  <isSupportGeologicalHazardMonitoringCfg>
    <!--ro, opt, bool, whether the device supports geological hazard detection service parameters, desc:GET/PUT
/ISAPI/System/Network/GeologicalHazardMonitoring/GeologicalHazardMonitoringCfg?format=json-->true
  </isSupportGeologicalHazardMonitoringCfg>
  <isSupportFixedNetworkInterfaceCfg>
    <!--ro, opt, bool, whether the device supports network interface fixed address allocation parameters, desc:GET/PUT
/ISAPI/System/Network/FixedNetworkInterfaceCfg?format=json-->true
  </isSupportFixedNetworkInterfaceCfg>
  <isSupportSearchWIFIChannelDetectionState>
    <!--ro, opt, bool, whether the device supports searching for Wi-Fi channel detection status, desc:POST
/ISAPI/System/Network/SearchWIFIChannelDetectionState?format=json-->true
  </isSupportSearchWIFIChannelDetectionState>
  <isSupportStartWIFIChannelDetection>
    <!--ro, opt, bool, whether the device supports enabling Wi-Fi channel detection, desc:POST /ISAPI/System/Network/StartWIFIChannelDetection?format=json--
>true
  </isSupportStartWIFIChannelDetection>
  <isSupportStartManualChannelAdaptation>
    <!--ro, opt, bool, whether the device supports enabling manual configuration of channel self-adaption, desc:POST
/ISAPI/System/Network/StartManualChannelAdaptation?format=json-->true
  </isSupportStartManualChannelAdaptation>
  <isSupportSearchManualChannelAdaptationState>
    <!--ro, opt, bool, whether the device supports searching for manual channel configuration self-adaption status, desc:POST
/ISAPI/System/Network/SearchManualChannelAdaptationState?format=json-->true
  </isSupportSearchManualChannelAdaptationState>
  <isSupportLoRaWorkMode>
    <!--ro, opt, bool, whether the device supports LoRa module working mode, desc:GET/PUT /ISAPI/System/Network/LoRaAccess/LoRaWorkMode?format=json-->true
  </isSupportLoRaWorkMode>
  <isSupportPLCParams>
    <!--ro, opt, bool-->true
```

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```
    <!--ro, opt, bool-->true
  </isSupportPLCParams>
  <isSupportWIFIRoamingConfigParams>
    <!--ro, opt, bool-->true
  </isSupportWIFIRoamingConfigParams>
  <isSupportPLCNetworkingStatus>
    <!--ro, opt, bool-->true
  </isSupportPLCNetworkingStatus>
  <isSupportAddPLCDiagnosis>
    <!--ro, opt, bool-->true
  </isSupportAddPLCDiagnosis>
  <isSupportGetPLCDiagnosisProgress>
    <!--ro, opt, bool-->true
  </isSupportGetPLCDiagnosisProgress>
  <isSupportSearchOutlookAuthResult>
    <!--ro, opt, bool-->true
  </isSupportSearchOutlookAuthResult>
  <isSupportRequestOutlookAuth>
    <!--ro, opt, bool, whether the device supports applying authentication request of Outlook email, desc:POST
/ISAPI/System/Network/Email/RequestOutlookAuth?format=json-->true
  </isSupportRequestOutlookAuth>
  <isSupportOutlookEmailList>
    <!--ro, opt, bool, whether the device supports Outlook email information, desc:GET or PUT /ISAPI/System/Network/Email/OutlookEmailList?format=json--
>true
  </isSupportOutlookEmailList>
  <isSupportNARIEncryptionPortParams>
    <!--ro, opt, bool-->true
  </isSupportNARIEncryptionPortParams>
  <isSupportAlphaPlatformParams>
    <!--ro, opt, bool-->true
  </isSupportAlphaPlatformParams>
  <isSupportSouthernUniverseIOTAccessParams>
    <!--ro, opt, bool-->true
  </isSupportSouthernUniverseIOTAccessParams>
  <isSupportModifyGeologicalHazardMonitoringParams>
    <!--ro, opt, bool-->true
  </isSupportModifyGeologicalHazardMonitoringParams>
  <isSupportGeologicalHazardMonitoringListParams>
    <!--ro, opt, bool-->true
  </isSupportGeologicalHazardMonitoringListParams>
  <isSupportServerCfg>
    <!--ro, opt, bool-->true
  </isSupportServerCfg>
  <isSupportDelimiterCfg>
    <!--ro, opt, bool-->true
  </isSupportDelimiterCfg>
  <isSupportNetworkSignalEnhancementParams>
    <!--ro, opt, bool-->true
  </isSupportNetworkSignalEnhancementParams>
  <isSupportISUPServiceParams>
    <!--ro, opt, bool-->true
  </isSupportISUPServiceParams>
  <isSupportWirelessDialSchedule>
    <!--ro, opt, bool-->true
  </isSupportWirelessDialSchedule>
  <isSupportISUPServiceParamsTest>
    <!--ro, opt, bool-->true
  </isSupportISUPServiceParamsTest>
  <isSupportVoiceTalkOverWebSocket>
    <!--ro, opt, bool-->true
  </isSupportVoiceTalkOverWebSocket>
  <isSupportPlaybackOverWebSocket>
    <!--ro, opt, bool-->true
  </isSupportPlaybackOverWebSocket>
  <isSupportALSOverWebSocket>
    <!--ro, opt, bool-->true
  </isSupportALSOverWebSocket>
  <isSupportPUSHCfg>
    <!--ro, opt, bool-->true
  </isSupportPUSHCfg>
  <RTDROverWebSocket>
    <!--ro, opt, object-->
    <isSupportGasSpectrumRealTimeData>
      <!--ro, opt, bool-->true
    </isSupportGasSpectrumRealTimeData>
    <isSupportDangerousGasSensorRealTimeData>
      <!--ro, opt, bool-->true
    </isSupportDangerousGasSensorRealTimeData>
    <isSupportComposeWindConditionRealTimeData>
      <!--ro, opt, bool-->true
    </isSupportComposeWindConditionRealTimeData>
    <isSupportMeteorologySensorRealTimeData>
      <!--ro, opt, bool-->true
    </isSupportMeteorologySensorRealTimeData>
  </RTDROverWebSocket>
  <isSupportWIFIAutoConnectParams>
    <!--ro, opt, bool-->true
  </isSupportWIFIAutoConnectParams>
  <isSupportAccessParams>
    <!--ro, opt, bool-->true
  </isSupportAccessParams>
  <isSupportStartCellNetSelfDiagnosis>
    <!--ro, opt, bool-->true
  </isSupportStartCellNetSelfDiagnosis>
```

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```
    <isSupportCellNetSelfDiagnosisStatus>
      <!--ro, opt, bool-->true
    </isSupportCellNetSelfDiagnosisStatus>
    <isSupportMDNSParams>
      <!--ro, opt, bool-->true
    </isSupportMDNSParams>
    <WebSocketPlaybackCap>
      <!--ro, opt, object-->
      <cmdType opt="speed,pause,resume,single,stop,seek,startExtracting,stopExtracting,transcoding">
        <!--ro, req, string, attr:opt{opt, string}-->speed
      </cmdType>
    </WebSocketPlaybackCap>
    <isSupportDevGetRTCConfigurationOperationResult>
      <!--ro, opt, bool-->true
    </isSupportDevGetRTCConfigurationOperationResult>
    <isSupportAddICECandidate>
      <!--ro, opt, bool-->true
    </isSupportAddICECandidate>
    <isSupportRTCConfiguration>
      <!--ro, opt, bool-->true
    </isSupportRTCConfiguration>
    <isSupportOpenDataChannel>
      <!--ro, opt, bool-->true
    </isSupportOpenDataChannel>
    <isSupportHTTPSecurityHeader>
      <!--ro, opt, bool-->true
    </isSupportHTTPSecurityHeader>
  </NetworkCap>
```

14.1.4.10 Set SSH parameters

Request URL

PUT /ISAPI/System/Network/ssh?readerID=<readerID>&type=<type>&SOCChipID=<SOCChipID>&controllerID=<controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	--
type	enum	--
SOCChipID	string	--
controllerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SSH xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, opt, object, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--wo, req, bool, whether to enable the function-->true
  </enabled>
  <port>
    <!--wo, opt, int-->22
  </port>
</SSH>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, read-only,status description: OK,Device Busy,Device Error,Invalid Operation,Invalid XML Format,Invalid XML Content,Reboot,Additional
Error, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot"
(reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.4.11 Reboot device

Request URL

PUT /ISAPI/System/reboot?childDevID=<devIndex>&module=<module>&loginPassword=<loginPassword>&controllerID=<controllerID>&id=<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
devIndex	string	Reboot a sub-device via specifying sub-device ID. The ID of the sub-device connected to the current device can be searched via /ISAPI/IoTGateway/Childmanage/SearchChild?format=json.
module	enum	The "module" specifies a module to restart, without requiring a device reboot, corresponding to the capability (/ISAPI/System/capabilities/deviceReboot->module). "algorithmProgram (traffic terminal radar-assisted algorithm program, supports independent algorithm program startup), "radar" (restart radar module of the radar-assisted camera).
loginPassword	string	--
controllerID	string	--
indexID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.1.4.12 Get the video analysis task capability

Request URL

GET /ISAPI/System/Video/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <videoInputPortNums>
    <!--ro, opt, int, number of video input ports-->0
  </videoInputPortNums>
  <videoOutputPortNums>
    <!--ro, opt, int, number of video output ports-->0
  </videoOutputPortNums>
  <menuNums>
    <!--ro, opt, int, number of local menus that can be displayed-->0
  </menuNums>
  <externalChannelNum>
    <!--ro, opt, bool, number of extended analog channels-->true
  </externalChannelNum>
  <isSupportHeatmap>
    <!--ro, opt, bool, whether the device supports heat map-->true
  </isSupportHeatmap>
  <isSupportCounting>
    <!--ro, opt, bool, whether the device supports people counting-->true
  </isSupportCounting>
  <countingType>
    <!--ro, opt, enum, statistics type, subType:string, desc:statistics type-->human
  </countingType>
  <isSupportPicture>
    <!--ro, opt, bool, whether the device supports OSD picture overlay, desc:related URI: /ISAPI/System/Video/inputs/channels/<channelID>/image/picture--
>true
  </isSupportPicture>
  <isSupportPreviewSwitch>
    <!--ro, opt, bool, whether the device supports switching live view-->true
  </isSupportPreviewSwitch>
  <isSupportRecodStatus>
    <!--ro, opt, bool, whether the device supports searching for the recording status-->true
  </isSupportRecodStatus>
  <isSupportPrivacyMask>
    <!--ro, opt, bool, whether the device supports privacy mask-->true
  </isSupportPrivacyMask>
  <isSupportBinocularPreviewSwitch>
    <!--ro, opt, bool, whether the device supports switching live view of the dual-lens camera-->true
  </isSupportBinocularPreviewSwitch>
```

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```
    <isSupportCalibCheck>
      <!--ro, opt, bool, whether the device supports calibration verification-->true
    </isSupportCalibCheck>
    <isSupportPIP>
      <!--ro, opt, bool-->true
    </isSupportPIP>
    <isSupportFocusVideoMode>
      <!--ro, opt, bool-->true
    </isSupportFocusVideoMode>
    <isSupportExternalChannel>
      <!--ro, opt, bool, whether the device supports extended analog channels-->true
    </isSupportExternalChannel>
    <isSupportMultiChannelCounting>
      <!--ro, opt, bool-->true
    </isSupportMultiChannelCounting>
    <isSupportCountingCollection>
      <!--ro, opt, bool, whether the device supports people counting ANR-->true
    </isSupportCountingCollection>
    <isSupportHeatmapCollection>
      <!--ro, opt, bool, whether the device supports heat map data ANR-->true
    </isSupportHeatmapCollection>
    <channelFlexible opt="name,enable,online,linknum">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->name
    </channelFlexible>
    <isSupportOutputsResource>
      <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/resource?format=json, desc:related URI: /ISAPI/System/Video/outputs/resource?format=json--
>true
    </isSupportOutputsResource>
    <OSDLanguage opt="GBK,EUC-KR,Hebrew" def="GBK">
      <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->GBK
    </OSDLanguage>
    <isSupportMixedChannel>
      <!--ro, opt, bool-->true
    </isSupportMixedChannel>
    <isSupportMixedChannelStatus>
      <!--ro, opt, bool-->true
    </isSupportMixedChannelStatus>
    <isSupportOutputCourseware>
      <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/courseware/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/courseware/capabilities?format=json-->true
    </isSupportOutputCourseware>
    <isSupportVideoInputMode>
      <!--ro, opt, bool, related URI: /ISAPI/System/Video/inputs/mode/capabilities, desc:related URI: /ISAPI/System/Video/inputs/mode/capabilities-->true
    </isSupportVideoInputMode>
    <isSupportVideoOutputMode>
      <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/mode/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/mode/capabilities?format=json-->true
    </isSupportVideoOutputMode>
    <PreviewMode>
      <!--ro, opt, object-->
      <PIPType>
        <!--ro, opt, object-->
        <mainScreenChannelID>
          <!--ro, opt, int-->0
        </mainScreenChannelID>
        <subScreenChannelID>
          <!--ro, opt, int-->0
        </subScreenChannelID>
      </PIPType>
    </PreviewMode>
    <isSupportViewTag>
      <!--ro, opt, bool-->true
    </isSupportViewTag>
    <isSupportMenuStatus>
      <!--ro, opt, bool-->true
    </isSupportMenuStatus>
  </VideoCap>
```

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14.1.5 System Upgrade

14.1.5.1 Get the capabilities of peripheral module upgrade

Request URL

GET /ISAPI/System/AcsUpdate/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<AcsUpdate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, capabilities of peripheral module upgrade, attr:version{req, string, protocolVersion}-->
  <type
opt="cardReader,FPModule,securityModule,extendModule,channelController,IRModule,lampModule,elevatorController,FPAlgorithmProgram,uboot,keypad,wirelessRecv,w
iredZone,sirenIndoor,sirenOutdoor,sirenAudio,repeater,bluetoothModule,single,wallSwitch,smartPlug,detector,transmitter,remoteCtrl,subModule,dispModule,netRe
ader,lockControlBoard,networkZoneModule,userInterfaceBoard,subPermissionController,electricLock,heatingModule,RS485Module,QRCodeModule,R3WirelessRecv,RXWire
lessRecv,wiredOutput,electricGenie,zigbeeModule,PMRModule,networkInputAndOutputModule,infoModule,keypadAndCardReader,socialSecurityCardModule,localControlle
r">
    <!--ro, opt, enum, upgrade type: upgrade peripheral, subType:string, attr:opt{req, string}, desc:"cardReader" (485 card reader), "FPModule" (fingerprint
module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module),
"LampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm programma of card reader),
"uboot" (uboot upgrade), "keypad" (keypad), "wirelessRecv" (wireless receiver module), "wiredZone" (wired zone module), "sirenIndoor" (indoor sounder),
"sirenOutdoor" (outdoor sounder), "sirenAudio" (sounder two-way audio), "repeater" (repeater), "bluetoothModule" (bluetooth module), "single" (single output
module), "wallSwitch" (wall mounted switch), "smartPlug" (smart plug), "detector" (detector), "transmitter" (transmitter peripheral), "remoteCtrl" (keyfob),
"subModule" (sub module), "dispModule" (display module), "netReader" (network card reader), "faceModule" (face module), "touchScreenModule" (touch screen
module), "temperatureModule" (temperature measurement module), "lockControlBoard" (lock control board), "networkZoneModule" (network zone module),
"userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "electricLock" (electric lock), "heatingModule" (heating
module), "RS485Module" (RS-485 module), "QRCodeModule" (QR code module), "electricGenie" (electric Genie), "zigbeeModule" (zigbee module), "PMRModule" (PMR
module), "networkInputAndOutputModule" (network input and output module via network TAP)-->cardReader
  </type>
  <batchType opt="electricGenie,wiredOutput,networkZoneModule,networkInputAndOutputModule">
    <!--ro, opt, enum, batch upgrade type (peripheral type), subType:string, attr:opt{req, string},
desc:peripheral types which support batch upgrading;
if this node is supported, the following operations are supported: 1. Upgrade devices: /ISAPI/System/updateFirmware?type=<type>; 2. Start upgrading the
specified analysis unit (device of the cluster): /ISAPI/System/upgradeStatus/startUpgrade?format=json; 3. Get the device upgrade progress:
/ISAPI/System/upgradeStatus?format=json-->electricGenie
  </batchType>
  <cardReaderNo min="1" max="10" opt="1,4">
    <!--ro, opt, int, card reader No. range, it is valid when the returned value of type consists cardReader, attr:min{opt, int, step:1},max{opt, int,
step:1},opt{opt, string}-->1
  </cardReaderNo>
  <FPModuleNo min="1" max="10">
    <!--ro, opt, int, fingerprint module No. range, it is valid when the returned value of type consists FPModule, attr:min{req, int},max{req, int}-->1
  </FPModuleNo>
  <securityModuleNo min="1" max="10">
    <!--ro, opt, int, security module No. range, it is valid when the returned value of type consists securityModule, attr:min{req, int},max{req, int}-->1
  </securityModuleNo>
  <extendModuleNo min="1" max="10">
    <!--ro, opt, int, No. range of I/O extended module, it is valid when the returned value of type consists extendModule, attr:min{req, int},max{req, int}-
->1
  </extendModuleNo>
  <channelControllerNo min="1" max="10">
    <!--ro, opt, int, lane controller No. range, it is valid when the returned value of type consists channelController, attr:min{req, int},max{req, int}--
>1
  </channelControllerNo>
  <IRModuleNo min="1" max="10">
    <!--ro, opt, int, infrared module No. range, it is valid when the returned value of type consists IRModule, attr:min{req, int},max{req, int}-->1
  </IRModuleNo>
  <lampModuleNo min="1" max="10">
    <!--ro, opt, int, indicator module No. range, it is valid when the returned value of type consists lampModule, attr:min{req, int},max{req, int}-->1
  </lampModuleNo>
  <elevatorControllerNo min="1" max="10">
    <!--ro, opt, int, sub elevator controller No. range, it is valid when the returned value of type consists elevatorController, attr:min{req,
int},max{req, int}-->1
  </elevatorControllerNo>
  <FPAlgorithmProgramNo min="1" max="10">
    <!--ro, opt, int, No. range of fingerprint algorithm programma of card reader, it is valid when the returned value of type consists FPAlgorithmProgram,
attr:min{req, int},max{req, int}-->1
  </FPAlgorithmProgramNo>
  <faceModuleNo min="1" max="10">
    <!--ro, opt, int, face module No. range, it is valid when the returned value of type consists faceModule, attr:min{req, int},max{req, int}-->1
  </faceModuleNo>
  <touchScreenModuleNo min="1" max="10">
    <!--ro, opt, int, touch screen module No. range, it is valid when the returned value of type consists touchScreenModule, attr:min{req, int},max{req,
int}-->1
  </touchScreenModuleNo>
  <temperatureModuleNo min="1" max="10">
    <!--ro, opt, int, No. range of temperature measurement module, it is valid when the returned value of type consists temperatureModule, attr:min{req,
int},max{req, int}-->1
  </temperatureModuleNo>
  <keypadAddress opt="1,3,5">
    <!--ro, opt, int, keypad address range, it is valid when the returned value of type consists keypad, attr:opt{req, string}-->1
  </keypadAddress>
  <wirelessRecvAddress opt="1,3,5">
    <!--ro, opt, int, No. range of wireless receiver module, it is valid when the returned value of type consists wirelessRecv, attr:opt{req, string}-->1
  </wirelessRecvAddress>
  <wiredZoneAddress opt="1,3,5">
    <!--ro, opt, int, No. range of wired zone module, it is valid when the returned value of type consists wiredZone, attr:opt{req, string}-->1
  </wiredZoneAddress>
  <sirenIndoorNo opt="1,3,5">
    <!--ro, opt, int, indoor sounder No. range, it is valid when the returned value of type consists sirenIndoor, attr:opt{req, string}-->1
  </sirenIndoorNo>
  <sirenOutdoorNo opt="1,3,5">
    <!--ro, opt, int, outdoor sounder No. range, it is valid when the returned value of type consists sirenOutdoor, attr:opt{req, string}-->1
  </sirenOutdoorNo>
  <repeaterNo opt="1,3,5">
    <!--ro, opt, int, repeater No. range, it is valid when the returned value of type consists repeater, attr:opt{req, string}-->1
  </repeaterNo>
  <subModuleNo opt="1,3,5">
    <!--ro, opt, int, sub module No. range, it is valid when the returned value of type consists subModule, attr:opt{req, string}-->1
  </subModuleNo>
  <dispModuleNo opt="1,3,5">
    <!--ro, opt, int, display module No. range, it is valid when the returned value of type consists dispModule, attr:opt{req, string}-->1
```

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```
    </dispModuleNo>
    <single opt="1,3,5">
      <!--ro, opt, int, No. range of single output module, it is valid when the returned value of type consists single, attr:opt{req, string}-->1
    </single>
    <wallSwitch opt="1,3,5">
      <!--ro, opt, int, No. range of wall mounted switch, it is valid when the returned value of type consists wallSwitch, attr:opt{req, string}-->1
    </wallSwitch>
    <smartPlug opt="1,3,5">
      <!--ro, opt, int, smart plug No. range, it is valid when the returned value of type consists smartPlug, attr:opt{req, string}-->1
    </smartPlug>
    <zoneNo opt="1,3,5">
      <!--ro, opt, int, zone  No. range, it is valid when the returned value of type consists detector, attr:opt{req, string}, desc:get the detector via the
zone No.-->1
    </zoneNo>
    <transmitterNo opt="1,3,5">
      <!--ro, opt, int, transmitter No. range, it is valid when the returned value of type consists transmitter, attr:opt{req, string}-->1
    </transmitterNo>
    <remoteCtrlNo opt="1,3,5">
      <!--ro, opt, int, keyfob No. range, it is valid when the returned value of type consists remoteCtrl, attr:opt{req, string}-->1
    </remoteCtrlNo>
    <netReaderNo min="1" max="10">
      <!--ro, opt, int, No. range of network card reader, it is valid when the returned value of type consists netReader, attr:min{req, int},max{req, int}-->1
    </netReaderNo>
    <lockControlBoardLNo opt="1,2,3,4,5,6,7,8">
      <!--ro, opt, int, No. range of the left part of the bracket, it is valid when the value of type returns, attr:opt{req, string}-->1
    </lockControlBoardLNo>
    <lockControlBoardRNo opt="1,2,3,4,5,6,7,8">
      <!--ro, opt, int, No. range of the right part of the bracket, it is valid when the value of type returns, attr:opt{req, string}-->1
    </lockControlBoardRNo>
    <networkZoneModuleNo opt="1,3,5">
      <!--ro, opt, int, No. range of network zone module, it is valid when the returned value of type consists networkZoneModule, attr:opt{req, string}-->1
    </networkZoneModuleNo>
    <userInterfaceBoardNo opt="0,1">
      <!--ro, opt, enum, interface board No. range, it is valid when the returned value of type consists userInterfaceBoard, subType:string, attr:opt{req,
string}, desc:0-main user extended interface board, 1-sub user extended interface board-->0
    </userInterfaceBoardNo>
    <electricLockNo opt="1,3,5">
      <!--ro, opt, int, electric lock No. range, it is valid when the returned value of type consists electricLock, attr:opt{req, string}-->1
    </electricLockNo>
    <heatingModuleNo opt="1,3,5">
      <!--ro, opt, int, heating module No. range, it is valid when the returned value of type consists heatingModule, attr:opt{req, string}, desc:the heating
module starts to work to increase the device temperature when the devices runs under low temperature-->1
    </heatingModuleNo>
    <sirenAudioNo opt="1,3,5">
      <!--ro, opt, int, sounder (two-way audio) No. range, it is valid when the returned value of type consists sirenAudio, attr:opt{req, string}-->1
    </sirenAudioNo>
    <QRCodeModuleNo min="1" max="10">
      <!--ro, opt, int, No. range of QR code module, it is valid when the returned value of type consists QRCodeModule, attr:min{req, int},max{req, int}-->1
    </QRCodeModuleNo>
    <R3WirelessRecvNo opt="1,2,3,4">
      <!--ro, opt, int, No. range of R3 wireless receiver module, it is valid when the returned value of type consists R3WirelessRecv, attr:opt{req, string}--
>1
    </R3WirelessRecvNo>
    <RXWirelessRecvNo opt="1,2,3,4">
      <!--ro, opt, int, No. range of RX wireless receiver module, it is valid when the returned value of type consists RXWirelessRecv, attr:opt{req, string}--
>1
    </RXWirelessRecvNo>
    <wiredOutputNo opt="1,2,3,4">
      <!--ro, opt, int, No. range of wired output module , it is valid when the returned value of type consists wiredOutput, attr:opt{req, string}-->1
    </wiredOutputNo>
    <electricGenieNo opt="1,2,3,4">
      <!--ro, opt, int, No. range of electric Genie, it is valid when the returned value of type or batchType consists electricGenie, attr:opt{req, string}--
>1
    </electricGenieNo>
    <networkInputAndOutputModuleNo opt="1,2,3,4">
      <!--ro, opt, int, No. range of network input and output module, it is valid when the returned value of type consists networkInputAndOutputModule,
attr:opt{req, string}-->1
    </networkInputAndOutputModuleNo>
    <infoModuleNo opt="1,2,3,4">
      <!--ro, opt, int, dep:or,{$.AcsUpdate.type,eq,infoModule}, attr:opt{req, string}-->1
    </infoModuleNo>
    <keypadAndCardReaderModuleNo opt="1,2,3,4">
      <!--ro, opt, int, dep:or,{$.AcsUpdate.type,eq,keypadAndCardReaderModule}, attr:opt{req, string}-->1
    </keypadAndCardReaderModuleNo>
    <threeDimensionalStructuredLightModuleNo opt="1,2,3,4">
      <!--ro, opt, int, dep:or,{$.AcsUpdate.type,eq,threeDimensionalStructuredLightModule}, attr:opt{req, string}-->1
    </threeDimensionalStructuredLightModuleNo>
    <socialSecurityCardModuleNo opt="1">
      <!--ro, opt, int, dep:or,{$.AcsUpdate.type,eq,socialSecurityCardModule}, attr:opt{req, string}-->1
    </socialSecurityCardModuleNo>
    <localControllerNo min="1" max="62">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </localControllerNo>
  </AcsUpdate>
```

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14.1.5.2 Upgrade devices

Request URL

POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>&childDevID=<devIndex>&isUpdateAuxDevice=<isUpdateAuxDevice>&upgradeToken=<upgradeToken>&upgradeSubOption=<upgradeSubOption>&doubleControlType=<doubleControlType>

Query Parameter

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Parameter Name	Parameter Type	Description
type	enum	device type 1.alarm device: "keypad" (keypad), "wirelessRecv" (wireless receiver module), "wiredZone" (wired zone module), "sirenIndoor" (indoor sounder), "sirenOutdoor" (outdoor sounder), "sirenAudio" (sounder two-way audio), "repeater" (repeater), "bluetoothModule" (bluetooth module), "single" (single output module), "wallSwitch" (wall mounted switch), "smartPlug" (smart plug), "detector" (detector), "transmitter" (transmitter peripheral), "remoteCtrl" (keyfob), "subModule" (sub module), "dispModule" (display module), "netReader" (network card reader), "faceModule" (face module), "touchScreenModule" (touch screen module), "temperatureModule" (temperature measurement module), "lockControlBoard" (lock control board), "networkZoneModule" (network zone module), "userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "electricLock" (electric lock), "heatingModule" (heating module), "RS485Module" (RS-485 module), "QRCodeModule" (QR code module), "electricGenie" (electric Genie), "zigbeeModule" (zigbee module), "PMRModule" (PMR module), "networkInputAndOutputModule" (network input and output module via network TAP) 2. access control device: "cardReader" (485 card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm program of card reader), "bluetoothModule" (bluetooth module), "MCU"-MCU upgrade, "temperatureModule" (temperature measurement module), "faceModule" (face module), "touchScreenModule" (touch screen module), "netReader" (network card reader), "userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "QRCodeModule" (QR code module) 3. intelligent cabinet: "lockControlBoard" (lock control board) 4. "ledReceiverCard"-receiving card; 5. "subsystem"-sub system; 6. radar-assisted device: "warningScreenFirmware"-firmware upgrade of pre-alarm screen, "warningScreenFont"-font library upgrade of pre-alarm screen, "warningScreenVoice"-audio upgrade of pre-alarm screen; 7. "electricGenie"-electric Genie
moduleAddress	string	module address is an exclusive node for Ax Hybrid example: keypad: GET /ISAPI/SecurityCP/Configuration/outputs?format=json; relay: GET /ISAPI/SecurityCP/Configuration/outputs?format=json; sounder: GET /ISAPI/SecurityCP/Configuration/wirelessSiren?format=json; output module: GET /ISAPI/SecurityCP/Configuration/outputModules?format=json; extension module: GET /ISAPI/SecurityCP/Configuration/extensionModule?format=json; zone: GET /ISAPI/SecurityCP/Configuration/zones?format=json
indexID	string	device No. 1. AX Hybrid Pro and wireless security control upgrade by ID; 2. access control devices upgrade by ID; 3. intelligent cabinet upgrade by IS
devIndex	string	--
isUpdateAuxDevice	string	--
upgradeToken	string	--
upgradeSubOption	string	--
doubleControlType	enum	--

Request Message

Binary Data

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:it describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information, range:[0,1024], desc:this node is used for debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

14.1.5.3 Get the device upgrade progress

Request URL

GET /ISAPI/System/upgradeStatus?format=json&type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
type	string	--

Request Message

None

Response Message

```
{
    "requestURL":  "/ISAPI/Streaming/channels/1",
    /*ro, opt, string, request URL*/
    "statusCode":  "test",
    /*ro, req, string, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, This field is required when the value of statusCode is not 1, and it corresponds to subStatusCode.*/
    "errorMsg":  "ok",
    /*ro, opt, string, This field is required when the value of statusCode is not 1. Detailed error description of a certain parameter can be provided*/
    "upgrading":  "TRUE",
    /*ro, opt, string, whether the device is upgrading: “TRUE" (upgrading), "FALSE" (not in upgrading)*/
    "percent":  22,
    /*ro, opt, int, upgrade progress (% complete)*/
    "idList": [
    /*ro, opt, array, ID list, subType:object*/
        {
            "id":  "test",
            /*ro, req, string, analysis unit ID*/
            "percent":  22,
            /*ro, opt, int, upgrade progress (% complete)*/
            "status":  "test"
            /*ro, opt, string, "backingUp" (backing up upgrade)*/
        }
    ]
}
```

14.1.5.4 Get the device upgrading status and progress

Request URL

GET /ISAPI/System/upgradeStatus?type=<type>&childDevID=<childDevID>&id=<id>

Query Parameter

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Parameter Name	Parameter Type	Description
type	string	"usbmic" (array microphone module)
childDevID	string	--
id	string	The id indicates the peripheral No. to be upgraded.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<upgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, upgrade status and result, attr:version{req, string, protocolVersion}-->
  <upgrading>
    <!--ro, req, bool, upgrade status-->true
  </upgrading>
  <percent>
    <!--ro, req, int, upgrade progress (% complete), range:[0,100]-->1
  </percent>
  <reboot>
    <!--ro, opt, bool, dep:and,{$.upgradeStatus.percent,eq,100}-->false
  </reboot>
  <upgradeStage>
    <!--ro, opt, enum, subType:string-->software
  </upgradeStage>
</upgradeStatus>
```

14.1.6 Time Settings

14.1.6.1 Get device time synchronization management parameters

Request URL

GET /ISAPI/System/time

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--ro, req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--ro, opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}-->2019-02-28T10:50:44+08:30
  </localTime>
  <timeZone>
    <!--ro, opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:daylight saving time configuration in time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the device time returned currently is in DST (Daylight Saving Time) system, desc:whether the device time returned currently is in DST (Daylight Saving Time) system-->true
  </isSummerTime>
  <platformType>
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform-->EZVIZ
  </platformType>
  <platformNo>
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--ro, opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--ro, opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--ro, opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

14.1.6.2 Set device time synchronization management parameters

Request URL

PUT /ISAPI/System/time

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}, desc:the time difference between local time and coordinated universal time (UTC) is configured via the timeZone field-->2019-02-28T10:50:44
  </localTime>
  <timeZone>
    <!--opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the time returned by the current device is that in the DST (daylight saving time), desc:whether the time returned by the current device is that in the DST (daylight saving time)-->true
  </isSummerTime>
  <platformType>
    <!--opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform, related URI: /ISAPI/System/Network/EZVIZ-->EZVIZ
  </platformType>
  <platformNo>
    <!--opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <FailedNodeInfoList>
    <!--ro, opt, object, information list of failed nodes, desc:for the manual time synchronization of central analysis cluster, this field is returned if time synchronization failed-->
    <FailedNodeInfo>
      <!--ro, opt, object, information of failed nodes-->
      <nodeID>
        <!--ro, req, string, node ID, range:[0,64]-->test
      </nodeID>
      <nodeIP>
        <!--ro, req, string, node IP, range:[0,20]-->test
      </nodeIP>
      <reason>
        <!--ro, opt, string, reason why the node failed to synchronize time, range:[0,128]-->test
      </reason>
    </FailedNodeInfo>
  </FailedNodeInfoList>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>
```

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14.1.6.3 Get the capability of device time synchronization management

Request URL

GET /ISAPI/System/time/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management capability set, attr:version{opt, string, protocolVersion}-->
  <timeMode opt="NTP,manual,satellite,SDK,28181,ONVIF,ALL(任何支持的校时方式都允许校时),NONE(不允校时或无校时源),platform,PTP">
    <!--ro, req, enum, time synchronization mode, subType:string, attr:opt{opt, string}, desc:“NTP” (NTP time synchronization), “manual” (manual time
synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or
no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime min="0" max="256">
    <!--ro, opt, string, local time, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTime>
  <timeZone min="0" max="256">
    <!--ro, opt, string, time zone, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </timeZone>
  <satelliteInterval min="0" max="3600">
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, attr:min{opt, string},max{opt, string}, desc:unit: minute-->60
  </satelliteInterval>
  <timeType opt="local,UTC">
    <!--ro, opt, enum, time type, subType:string, attr:opt{opt, string}, desc:“local” (local time), “UTC” (UTC time)-->local
  </timeType>
  <platformType opt="EZVIZ">
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, attr:opt{opt, string}, desc:platform type-->EZVIZ
  </platformType>
  <platformNo min="1" max="2">
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, attr:min{req, int},max{req, int}, desc:it is the only ID, which is
configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <isSupportHistoryTime>
    <!--ro, opt, bool, supported capability of the historical time synchronization list, desc:related URI: /ISAPI/System/time/historyInfo?format=json-->true
  </isSupportHistoryTime>
  <isSupportTimeFilter>
    <!--ro, opt, bool, supported capability of filtering time synchronization, desc:related URI: /ISAPI/System/time/filter/capabilities?format=json-->true
  </isSupportTimeFilter>
  <displayFormat opt="MM/dd/yyyy hh:mm,mm,dd-MM-yyyy hh,MM-dd-yyyy hh:mm,yyyy-MM-dd hh:mm,MM/dd/yyyy hh:mm:ss,dd-MM-yyyy hh:mm:ss,MM-dd-yyyy hh:mm:ss,yyyy-
MM-dd hh:mm:ss,yyyy-MM-dd,dd-MM-yyyy,MM-dd-yyyy">
    <!--ro, opt, enum, time display format, subType:string, attr:opt{req, string}, desc:if this node is returned, it indicates that the device supports
configuring time display format, related URI: /ISAPI/System/time/timeType?format=json-->MM/dd/yyyy hh:mm
  </displayFormat>
  <isSupportSyncDeviceNTPInfoToCamera>
    <!--ro, opt, bool, the capability of synchronizing device’s NTP service information with the camera, desc:related URI:
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json-->true
  </isSupportSyncDeviceNTPInfoToCamera>
  <isSupportNTPService>
    <!--ro, opt, bool-->true
  </isSupportNTPService>
  <isSupportTimeZoneListInfo>
    <!--ro, opt, bool-->true
  </isSupportTimeZoneListInfo>
  <isSupportAutoEnterSummerTime>
    <!--ro, opt, bool-->true
  </isSupportAutoEnterSummerTime>
  <ethernetPort min="0" max="255">
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}, attr:min{req, int},max{req, int}-->1
  </ethernetPort>
  <IANA
opt="Africa/Abidjan,Africa/Accra,Africa/Addis_Ababa,Africa/Algiers,Africa/Blantyre,Africa/Brazzaville,Africa/Cairo,Africa/Casablanca,Africa/Conakry,Africa/D
akar,Africa/Dar_es_Salaam,Africa/Djibouti,Africa/Freetown,Africa/Gaborone,Africa/Harare,Africa/Johannesburg,Africa/Kampala,Africa/Khartoum,Africa/Kigali,Afr
ica/Kinshasa,Africa/Lagos,Africa/Maseru,Africa/Mogadishu,Africa/Nairobi,Africa/Sao_Tome,Africa/Timbuktu,Africa/Tripoli,Africa/Tunis,America/Anchorage,Americ
a/Aruba,America/Asuncion,America/Barbados,America/Belize,America/Bogota,America/Buenos_Aires,America/Cancun,America/Caracas,America/Cayman,America/Chicago,A
merica/Curacao,America/Dawson_Creek,America/Denver,America/Detroit,America/Dominica,America/Edmonton,America/El_Salvador,America/Fortaleza,America/Grand_Tur
k,America/Grenada,America/Guatemala,America/Guyana,America/Halifax,America/Havana,America/Indiana/Indianapolis,America/Indiana/Knox,America/Indiana/Marengo,
America/Indiana/Petersburg,America/Indiana/Tell_City,America/Indiana/Vevay,America/Indiana/Vincennes,America/Indiana/Winamac,America/Jamaica,America/La_Paz,
America/Lima,America/Los_Angeles,America/Louisville,America/Managua,America/Martinique,America/Mendoza,America/Metlakatla,America/Mexico_City,America/Monter
rey,America/Montevideo,America/Montreal,America/Nassau,America/New_York,America/North_Dakota/Beulah,America/North_Dakota/Center,America/North_Dakota/New_Sal
em,America/Panama,America/Phoenix,America/Port_of_Spain,America/Port-au-
Prince,America/Puerto_Rico,America/Santo_Domingo,America/Sao_Paulo,America/St_Johns,America/St_Kitts,America/St_Lucia,America/St_Thomas,America/Tijuana,Amer
ica/Toronto,America/Vancouver,America/Winnipeg,Antarctica/South_Pole,Arctic/Longyearbyen,Asia/Almaty,Asia/Amman,Asia/Anadyr,Asia/Aqtau,Asia/Baghdad,Asia/Bah
rain,Asia/Baku,Asia/Bangkok,Asia/Beirut,Asia/Calcutta,Asia/Damascus,Asia/Dhaka,Asia/Dubai,Asia/Gaza,Asia/Hebron,Asia/Ho_Chi_Minh,Asia/Hong_Kong,Asia/Jakarta
,Asia/Jerusalem,Asia/Kabul,Asia/Karachi,Asia/Kathmandu,Asia/Kuala_Lumpur,Asia/Kuwait,Asia/Macau(China),Asia/Manila,Asia/Muscat,Asia/Phnom_Penh,Asia/Pyongyan
g,Asia/Rangoon,Asia/Riyadh,Asia/Seoul,Asia/Shanghai,Asia/Singapore,Asia/Taipei,Asia/Tehran,Asia/Tel_Aviv,Asia/Tokyo,Asia/Ulaanbaatar,Atlantic/Bermuda,Atlant
ic/Canary,Atlantic/Cape_Verde,Atlantic/Reykjavik,Atlantic/Stanley,Australia/Adelaide,Australia/Brisbane,Australia/Canberra,Australia/Darwin,Australia/Melbou
rne,Australia/NSW,Australia/Perth,Australia/Queensland,Australia/Sydney,Australia/Victoria,Canada/Newfoundland,Canada/Saskatchewan,Chile/EasterIsland,Europe
/Amsterdam,Europe/Andorra,Europe/Athens,Europe/Belfast,Europe/Belgrade,Europe/Berlin,Europe/Bratislava,Europe/Brussels,Europe/Bucharest,Europe/Budapest,Euro
pe/Chisinau,Europe/Copenhagen,Europe/Dublin,Europe/Gibraltar,Europe/Helsinki,Europe/Isle_of_Man,Europe/Istanbul,Europe/Kiev,Europe/Lisbon,Europe/London,Euro
```

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```
pe/Luxembourg,Europe/Madrid,Europe/Malta,Europe/Minsk,Europe/Monaco,Europe/Moscow,Europe/Oslo,Europe/Paris,Europe/Prague,Europe/Riga,Europe/Rome,Europe/San_
Marino,Europe/Sarajevo,Europe/Simferopol,Europe/Skopje,Europe/Sofia,Europe/Stockholm,Europe/Tallinn,Europe/Vatican,Europe/Vienna,Europe/Vilnius,Europe/Warsa
w,Europe/Zagreb,Europe/Zurich,Indian/Antananarivo,Indian/Maldives,Indian/Mauritius,Pacific/Auckland,Pacific/Fiji,Pacific/Guam,Pacific/Honolulu,Pacific/Kirit
imati,Pacific/Noumea,America/Antigua,MST,Asia/Kolkata,America/Santiago,Africa/Monrovia,Asia/Colombo,America/Chihuahua,America/Tegucigalpa,America/Paramaribo
,Europe/Ljubljana,Asia/Ashgabat,Africa/Asmara,Asia/Brunei,Africa/Bangui,Africa/Banjul,Asia/Bishkek,Africa/Bissau,Africa/Bujumbura,Asia/Dili,Asia/Dushanbe,Pa
cific/Funafuti,Pacific/Guadalcanal,Africa/Juba,America/St_Vincent,Africa/Libreville,Africa/Lome,Africa/Luanda,Africa/Lusaka,Pacific/Majuro,Africa/Malabo,Afr
ica/Maputo,Africa/Mbabane,Indian/Comoro,Africa/Ndjamena,Pacific/Palau,Africa/Niamey,Europe/Nicosia,Africa/Nouakchott,Pacific/Tongatapu,Africa/Ouagadougou,Am
erica/Lower_Princes,Europe/Podgorica,Pacific/Efate,Africa/Porto-
Novo,Asia/Tashkent,Asia/Tbilisi,Asia/Thimphu,Europe/Tirane,Europe/Vaduz,Asia/Vientiane,Africa/Windhoek,Africa/Douala,Pacific/Nauru,Asia/Yerevan,Etc/GMT+12,E
tc/GMT+11,America/Adak,Pacific/Marquesas,Etc/GMT+9,Etc/GMT+8,America/Whitehorse,Pacific/Easter,America/Regina,America/Indianapolis,America/Cuiaba,America/Ar
aguaina,America/Cayenne,America/Godthab,America/Punta_Arenas,America/Miquelon,America/Bahia,Etc/GMT+2,Atlantic/Azores,Etc/UTC,Europe/Kaliningrad,Europe/Astr
akhan,Europe/Samara,Europe/Saratov,Europe/Volgograd,Asia/Yekaterinburg,Asia/Qyzylorda,Asia/Katmandu,Asia/Omsk,Asia/Barnaul,Asia/Hovd,Asia/Krasnoyarsk,Asia/N
ovosibirsk,Asia/Tomsk,Asia/Irkutsk,Australia/Eucla,Asia/Chita,Asia/Yakutsk,Pacific/Port_Moresby,Australia/Hobart,Asia/Vladivostok,Australia/Lord_Howe,Pacifi
c/Bougainville,Asia/Srednekolymsk,Asia/Magadan,Pacific/Norfolk,Asia/Sakhalin,Asia/Kamchatka,Etc/GMT-12,Pacific/Chatham,Etc/GMT-13,Pacific/Apia">
    <!--ro, opt, enum, IANA synchronization, subType:string, attr:opt{opt, string}-->Africa/Abidjan
  </IANA>
  <isSupportSearchDeviceIANAList>
    <!--ro, opt, bool, whether the device supports searching for supported IANA list, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
  </isSupportSearchDeviceIANAList>
  <windowsZone opt="Dateline Standard Time,UTC-11,Aleutian Standard Time,Hawaiian Standard Time,Marquesas Standard Time,Alaskan Standard Time,UTC-09,Pacific
Standard Time (Mexico),UTC-08,Pacific Standard Time,US Mountain Standard Time,Mountain Standard Time (Mexico),Mountain Standard Time,Yukon Standard
Time,Central America Standard Time,Central Standard Time,Easter Island Standard Time,Central Standard Time (Mexico),Canada Central Standard Time,SA Pacific
Standard Time,Eastern Standard Time (Mexico),Eastern Standard Time,Haiti Standard Time,Cuba Standard Time,US Eastern Standard Time,Turks And Caicos Standard
Time,Paraguay Standard Time,Atlantic Standard Time,Venezuela Standard Time,Central Brazilian Standard Time,SA Western Standard Time,Pacific SA Standard
Time,Newfoundland Standard Time,Tocantins Standard Time,E. South America Standard Time,SA Eastern Standard Time,Argentina Standard Time,Greenland Standard
Time,Montevideo Standard Time,Magallanes Standard Time,Saint Pierre Standard Time,Bahia Standard Time,UTC-02,Azores Standard Time,Cape Verde Standard
Time,UTC,GMT Standard Time,Greenwich Standard Time,Sao Tome Standard Time,Morocco Standard Time,W. Europe Standard Time,Central Europe Standard Time,Romance
Standard Time,Central European Standard Time,W. Central Africa Standard Time,Jordan Standard Time,GTB Standard Time,Middle East Standard Time,Egypt Standard
Time,E. Europe Standard Time,Syria Standard Time,West Bank Standard Time,South Africa Standard Time,FLE Standard Time,Israel Standard Time,South Sudan
Standard Time,Kaliningrad Standard Time,Sudan Standard Time,Libya Standard Time,Namibia Standard Time,Arabic Standard Time,Turkey Standard Time,Arab
Standard Time,Belarus Standard Time,Russian Standard Time,E. Africa Standard Time,Iran Standard Time,Arabian Standard Time,Astrakhan Standard
Time,Azerbaijan Standard Time,Russia Time Zone 3,Mauritius Standard Time,Saratov Standard Time,Georgian Standard Time,Volgograd Standard Time,Caucasus
Standard Time,Afghanistan Standard Time,West Asia Standard Time,Ekaterinburg Standard Time,Pakistan Standard Time,Qyzylorda Standard Time,India Standard
Time,Sri Lanka Standard Time,Nepal Standard Time,Central Asia Standard Time,Bangladesh Standard Time,Omsk Standard Time,Myanmar Standard Time,SE Asia
Standard Time,Altai Standard Time,W. Mongolia Standard Time,North Asia Standard Time,N. Central Asia Standard Time,Tomsk Standard Time,China Standard
Time,North Asia East Standard Time,Singapore Standard Time,W. Australia Standard Time,Taipei Standard Time,Ulaanbaatar Standard Time,Aus Central W. Standard
Time,Transbaikal Standard Time,Tokyo Standard Time,North Korea Standard Time,Korea Standard Time,Yakutsk Standard Time,Cen. Australia Standard Time,AUS
Central Standard Time,E. Australia Standard Time,AUS Eastern Standard Time,West Pacific Standard Time,Tasmania Standard Time,Vladivostok Standard Time,Lord
Howe Standard Time,Bougainville Standard Time,Russia Time Zone 10,Magadan Standard Time,Norfolk Standard Time,Sakhalin Standard Time,Central Pacific
Standard Time,Russia Time Zone 11,New Zealand Standard Time,UTC+12,Fiji Standard Time,Chatham Islands Standard Time,UTC+13,Tonga Standard Time,Samoa
Standard Time,Line Islands Standard Time">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string}-->Dateline Standard Time
  </windowsZone>
  <isSupportTimeStateParam>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeStateParam?format=json-->true
  </isSupportTimeStateParam>
  <localTimeDST min="0" max="256">
    <!--ro, opt, string, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTimeDST>
  <isSupportStandard>
    <!--ro, opt, bool-->true
  </isSupportStandard>
  <isSupportPTPParams>
    <!--ro, opt, bool, desc:GET/PUT /ISAPI/System/time/PTPParams?format=json-->true
  </isSupportPTPParams>
  <isSupportTimeZoneLock>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeZoneLock?format=json-->true
  </isSupportTimeZoneLock>
</Time>
```


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14.1.6.4 Set parameters of all NTP servers

Request URL

PUT /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--opt, object, NTP server information-->
    <id>
      <!--req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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14.1.6.5 Get the parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{req, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--ro, req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--ro, opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--ro, opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

14.1.6.6 Set the parameters of a NTP server

Request URL

PUT /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, NTP server domain name地址去掉, range:[1,64]-->12345
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,32], desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--opt, bool-->false
  </enabled>
</NTPServer>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.6.7 Get the parameters of a NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	NTP server No.

Request Message

None

Response Message

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HTA

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--ro, req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, NTP server domain name, range:[1,64], dep:and,{$.NTPServer.addressingFormatType,eq,hostname}-->xxx12345
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,32], dep:and,{$.NTPServer.addressingFormatType,eq,ipAddress}-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--ro, opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--ro, opt, bool-->false
  </enabled>
  <devicePortNoValid>
    <!--ro, opt, enum, subType:string-->yes
  </devicePortNoValid>
  <portType>
    <!--ro, opt, enum, subType:string-->auto
  </portType>
  <customPortNo>
    <!--ro, opt, int, range:[1,65535], step:1, dep:and,{$.NTPServer.portType,eq,custom}-->1
  </customPortNo>
  <hostNameExampleList>
    <!--ro, opt, array, subType:object-->
    <hostNameExample>
      <!--ro, opt, object-->
      <hostName>
        <!--ro, opt, string-->xxx12345
      </hostName>
    </hostNameExample>
  </hostNameExampleList>
</NTPServer>
```

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14.1.6.8 Get the configuration capability of a specific NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id min="1" max="1">
    <!--ro, req, string, ID, range:[0,1], attr:min{req, int},max{req, int}-->1
  </id>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, IP address type of NTP server, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)--
>hostname
  </addressingFormatType>
  <hostName min="0" max="68">
    <!--ro, opt, string, NTP server domain name, range:[0,68], attr:min{req, int},max{req, int}-->time.windows.com
  </hostName>
  <ipAddress min="0" max="32">
    <!--ro, opt, string, IPv4 address, range:[0,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address min="0" max="128">
    <!--ro, opt, string, IPv6 address, range:[0,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo min="0" max="65535">
    <!--ro, opt, int, port No., range:[0,65535], attr:min{req, int},max{req, int}, desc:port No.-->123
  </portNo>
  <synchronizeInterval min="0" max="10800">
    <!--ro, opt, int, time synchronization interval, range:[0,10800], unit:min, attr:min{req, int},max{req, int}, desc:NTP time synchronization interval,
unit: minute-->1440
  </synchronizeInterval>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <portType opt="auto,custom" def="auto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->auto
  </portType>
  <customPortNo min="1" max="65535">
    <!--ro, opt, int, range:[0,65535], dep:and,{$.NTPServer.portType,eq,custom}, attr:min{req, int},max{req, int}-->123
  </customPortNo>
</NTPServer>
```

14.1.6.9 Get the configuration capability of parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, NTP server address type, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName min="1" max="64">
      <!--ro, opt, string, NTP server domain name, range:[1,64], attr:min{req, int},max{req, int}-->12345
    </hostName>
    <ipAddress min="1" max="32">
      <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
    </ipAddress>
    <ipv6Address min="1" max="128">
      <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo min="1" max="65535">
      <!--ro, opt, int, port No., range:[1,65535], attr:min{req, int},max{req, int}, desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval min="1" max="10800">
      <!--ro, opt, int, time synchronization interval, unit:min, attr:min{req, int},max{req, int}-->1440
    </synchronizeInterval>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable, attr:opt{req, string}, desc:true (enable), false (disable)-->true
    </enabled>
  </NTPServer>
</NTPServerList>
```

14.1.6.10 Get time zone

Request URL

GET /ISAPI/System/time/timeZone

Query Parameter

None

Request Message

None

Response Message

None

14.1.6.11 Set time zone

Request URL

PUT /ISAPI/System/time/timeZone

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2 Network Settings

14.2.1 HTTP Listening

14.2.1.1 Delete receiving server(s)

Request URL

DELETE /ISAPI/Event/notification/httpHosts

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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14.2.1.2 Delete a HTTP listening server

Request URL

DELETE /ISAPI/Event/notification/httpHosts/<hostID>

Query Parameter

Parameter Name	Parameter Type	Description
hostID	string	Add a single alarm host through POST /ISAPI/Event/notification/httpHosts?security=&iv=. It returns the alarm host ID for success.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.1.3 Set the parameters of a listening host

Request URL

PUT /ISAPI/Event/notification/httpHosts/<hostID>

Query Parameter

Parameter Name	Parameter Type	Description
hostID	string	Security control panel ID, which is returned by the URL (POST /ISAPI/Event/notification/httpHosts?security=&iv=).

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, listening host, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, listening host ID, range:[1,10]-->test
  </id>
  <url>
    <!--req, string, URL-->test
  </url>
  <protocolType>
    <!--req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
  </protocolType>
  <parameterFormatType>
    <!--req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
  </parameterFormatType>
  <addressingFormatType>
    <!--req, enum, address type, subType:string, desc:"hostname", "ipaddress"-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
  </hostName>
```

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```
    <ipAddress>
      <!--opt, string, IP address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IP address-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 Address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IPv6 Address-->test
    </ipv6Address>
    <portNo>
      <!--opt, int, port number-->1
    </portNo>
    <userName>
      <!--opt, string, user name, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,ue,none}-->test
    </userName>
    <password>
      <!--wo, opt, string, password, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,ue,none}-->test
    </password>
    <httpAuthenticationMethod>
      <!--req, enum, authentication method, subType:string, desc:"MD5digest"(MD5), "none", "base64"-->MD5digest
    </httpAuthenticationMethod>
    <ANPR>
      <!--opt, object, ANPR-->
      <detectionUpLoadPicturesType>
        <!--opt, enum, uploaded pictures type, subType:string, desc:"all", "licensePlatePicture", "detectionPicture"(detected picture)-->all
      </detectionUpLoadPicturesType>
      <videoUploadEnabled>
        <!--opt, bool, whether to upload violation pre-records, desc:when it is enabled, the device will upload the stored video via the "relationVideo"
event-->false
      </videoUploadEnabled>
    </ANPR>
    <Extensions>
      <!--opt, object, range-->
      <intervalBetweenEvents>
        <!--opt, int, event interval-->1
      </intervalBetweenEvents>
    </Extensions>
    <uploadImagesDataType>
      <!--opt, enum, picture data type, subType:string, desc:"URL", "binary" (default value). For cloud storage, only "URL" is supported-->URL
    </uploadImagesDataType>
    <httpBroken>
      <!--opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
    </httpBroken>
    <SubscribeEvent>
      <!--opt, object, picture uploading modes of all events which contain pictures, desc:picture uploading modes of all events which contain pictures-->
      <heartbeat>
        <!--opt, int, Heartbeat Interval Time-->30
      </heartbeat>
      <eventMode>
        <!--req, enum, event mode, subType:string, desc:"all" (all alarms need to be reported), "list" (only listed alarms need to be reported)-->all
      </eventMode>
      <EventList>
        <!--opt, array, event list, subType:object-->
        <Event>
          <!--opt, object, channel information linked to event-->
          <type>
            <!--req, enum, event type, subType:string, desc:see details in event types: "ADAS"(advanced driving assistance system), "ADASAlarm"(advanced
driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent", "CDsStatus", "DBD"
(driving behavior detection) "GPSUpload", "HFPD"(frequently appeared person), "IO"(I/O Alarm), "IOTD", "LES", "LFPD"(low frequency person detection),
"PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-
alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage",
"audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch",
"changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk
unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling",
"facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection", "fireEscapeDetection",
"flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"
(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature
alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person
getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection), "listSyncException"(list synchronization exception),
"loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event), "mixedTargetDetection"(multi-target-type detection),
"modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"
(staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm),
"personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection),
"personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception),
"policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection",
"radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient
recording period), "recordException", "regionEntrance", "regionExiting", "retention"(people overstay detection), "rollover", "running"(people running),
"safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"(angular acceleration alarm), "severeHDFailure"(HDD major fault
detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO",
"spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection),
"targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime
detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list
alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion
detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code
event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent",
"ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals),
"ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison"-->mixedTargetDetection
          </type>
          <minorAlarm>
            <!--opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorAlarm>
          <minorException>
            <!--opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorException>
          <minorOperation>
```

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```
        <!--opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format of the specified event, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL)-->binary
      </pictureURLType>
      <channels>
        <!--opt, string, listen to the events on the specified channel No. list, desc:if all channels are being listened to, the node shall not be applied. if some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
      </channels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, listen to the specified channel No. list, desc:if all channels are being listened to, the node shall not be applied. if some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--opt, enum, alarm picture format, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->binary
  </pictureURLType>
  <ChangedUploadSub>
    <!--opt, object, subscribe to messages-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, desc:within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->5
    </interval>
    <StatusSub>
      <!--opt, object, sub status-->
      <all>
        <!--opt, bool, whether to subscribe to all-->true
      </all>
      <channel>
        <!--opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
      </channel>
      <hd>
        <!--opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the value of <all> is true-->true
      </capability>
    </StatusSub>
  </ChangedUploadSub>
</SubscribeEvent>
<PackingSpaceRecognition>
  <!--opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event: PackingSpaceRecognition-->
  <upLoadPicturesType>
    <!--req, enum, uploaded picture type, subType:string, desc:"all", "picturesTypes" (upload specified types of pictures), "notUpload" (not upload pictures)-->all
  </upLoadPicturesType>
  <PicturesTypes>
    <!--opt, array, specified list of uploaded picture type, subType:object, dep:and, {$.HttpHostNotification.PackingSpaceRecognition.upLoadPicturesType,eq,picturesTypes}-->
    <picturesType>
      <!--opt, enum, uploaded picture type, subType:string, desc:"backgroundImage" (captured background picture), "plateImage" (license plate thumbnail)-->backgroundImage
    </picturesType>
  </PicturesTypes>
</PackingSpaceRecognition>
<fileUploadType>
  <!--opt, enum, subType:string-->cloudStrage
</fileUploadType>
<enabled>
  <!--opt, bool, whether to enable-->true
</enabled>
<netWork>
  <!--opt, enum, network, subType:int, desc:multi-NIC devices support the access via wired/wireless/mobile networks. After the network switching, the device's upload center address will be invalid. So, it is required to configure the valid network to avoid event upload failure due to invalid network 1 (wired network), 2 (mobile network: 3G/4G/5G/GPRS), 3 (wireless network)-->1
</netWork>
<method>
  <!--opt, enum, request method, subType:string, desc:"POST", "PUT", "GET"-->POST
</method>
<pictureAttachEnabled>
  <!--opt, bool, whether to enable attaching pictures, desc:true by default-->true
</pictureAttachEnabled>
<contentType>
  <!--opt, enum, HTTP content format, subType:string, desc:"multipart" (form format) by default. It is the format for all uploaded events, not limited to pictures only-->multipart
</contentType>
<eventTemplateID>
  <!--opt, int, event message template ID, desc:related URL (GET /ISAPI/Event/notification/EventTemplateParams?format=json) is to get template list-->1
</eventTemplateID>
<customTriggersEnabled>
  <!--opt, bool, whether to enable custom linkage, desc:true (listening service's linkage method is unaffected by upload center's linkage method, that is,
```

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```
independent), false or node doesn't exist (use the previous logic, that upload center sends data to listening service)-->false
  </customTriggersEnabled>
  <checkResponseEnabled>
    <!--opt, bool, whether to enable verifying listening host response, desc:when a device uploads an event to the listening host, the host should respond
with HTTP 200 OK status. Otherwise, the device assumes that the event was not received. However, some platforms' listening services fail to respond. This
parameter is used to address the actual cause. When disabled, the device skips the verification of 200 OK response from the listening host. When enabled,
failure to receive 200 OK indicates the event upload failure, and the device will log the failure or retransmit the event-->true
  </checkResponseEnabled>
</HttpHostNotification>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK"(succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot"(reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.1.4 Get the parameters of a listening host

Request URL

GET /ISAPI/Event/notification/httpHosts/<hostID>

Query Parameter

Parameter Name	Parameter Type	Description
hostID	string	Security control panel ID, which is returned by the URL (POST /ISAPI/Event/notification/httpHosts?security=&iv=).

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, listening host, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, listening host ID, range:[1,10]-->test
  </id>
  <url>
    <!--ro, req, string, URL-->test
  </url>
  <protocolType>
    <!--ro, req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"-->HTTP
  </protocolType>
  <parameterFormatType>
    <!--ro, req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
  </parameterFormatType>
  <addressingFormatType>
    <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress"-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
```

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```
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IP address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IP address-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IPv6 address-->test
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port number-->1
    </portNo>
    <userName>
      <!--ro, opt, string, user name, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
    </userName>
    <httpAuthenticationMethod>
      <!--ro, req, enum, authentication method, subType:string, desc:"MD5digest" (MD5), "none", "base64"-->MD5digest
    </httpAuthenticationMethod>
    <ANPR>
      <!--ro, opt, object, ANPR-->
      <detectionUpLoadPicturesType>
        <!--ro, opt, enum, uploaded pictures type, subType:string, desc:"all", "licensePlatePicture" (license plate picture), "detectionPicture" (detected
picture)-->all
      </detectionUpLoadPicturesType>
      <videoUploadEnabled>
        <!--ro, opt, bool, whether to upload violation pre-records, desc:when it is enabled, the device will upload the stored video via the "relationVideo"
event-->false
      </videoUploadEnabled>
    </ANPR>
    <Extensions>
      <!--ro, opt, object, range-->
      <intervalBetweenEvents>
        <!--ro, opt, int, event interval-->1
      </intervalBetweenEvents>
    </Extensions>
    <uploadImagesDataType>
      <!--ro, opt, enum, picture data type, subType:string, desc:"URL", "binary" (default value). For cloud storage, only "URL" is supported-->URL
    </uploadImagesDataType>
    <httpBroken>
      <!--ro, opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
    </httpBroken>
    <SubscribeEvent>
      <!--ro, opt, object, picture uploading modes of all events which contain pictures-->
      <heartbeat>
        <!--ro, opt, int, heartbeat interval time-->30
      </heartbeat>
      <eventMode>
        <!--ro, req, enum, event mode, subType:string, desc:"all" (all alarms need to be reported), "list" (only listed alarms need to be reported)-->all
      </eventMode>
      <EventList>
        <!--ro, opt, array, event list, subType:object-->
        <Event>
          <!--ro, opt, object, channel information linked to event-->
          <type>
            <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
"faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fieldetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "LeavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "LowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent",
"IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison
event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching
consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison" (human body comparison),
"regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
          </type>
          <minorAlarm>
            <!--ro, opt, string, alarm sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorAlarm>
          <minorException>
            <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
```

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```
        </minorException>
        <minorOperation>
          <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
        </minorOperation>
        <minorEvent>
          <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
        </minorEvent>
        <pictureURLType>
          <!--ro, opt, enum, alarm picture format of the specified event, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL)-->binary
        </pictureURLType>
        <channels>
          <!--ro, opt, string, listen to the events on the specified channel No. list, desc:if all channels are being listened to, the node shall not be applied. If some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
        </channels>
      </Event>
    </EventList>
    <channels>
      <!--ro, opt, string, listen to the specified channel No. list, desc:if all channels are being listened to, the node shall not be applied. If some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
    </channels>
    <pictureURLType>
      <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->binary
    </pictureURLType>
    <ChangedUploadSub>
      <!--ro, opt, object, subscribe to messages-->
      <interval>
        <!--ro, opt, int, the lifecycle of arming GUID, desc:within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->5
      </interval>
      <StatusSub>
        <!--ro, opt, object, sub status-->
        <all>
          <!--ro, opt, bool, whether to subscribe to all-->true
        </all>
        <channel>
          <!--ro, opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
        </channel>
        <hd>
          <!--ro, opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
        </hd>
        <capability>
          <!--ro, opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the value of <all> is true-->true
        </capability>
      </StatusSub>
    </ChangedUploadSub>
  </SubscribeEvent>
  <PackingSpaceRecognition>
    <!--ro, opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event: PackingSpaceRecognition-->
    <upLoadPicturesType>
      <!--ro, req, enum, uploaded picture type, subType:string, desc:"all", "picturesTypes"(upload specified types of pictures), "notUpload"(not upload pictures)-->all
    </upLoadPicturesType>
    <PicturesTypes>
      <!--ro, opt, array, specified list of uploaded picture types, subType:object, dep:and, {$.HttpHostNotification.PackingSpaceRecognition.upLoadPicturesType,eq,picturesTypes}-->
      <picturesType>
        <!--ro, opt, enum, uploaded picture type, subType:string, desc:"backgroundImage"(captured background picture), "plateImage"(license plate thumbnail)-->backgroundImage
      </picturesType>
    </PicturesTypes>
  </PackingSpaceRecognition>
  <enabled>
    <!--ro, opt, bool, whether to enable-->true
  </enabled>
  <netWork>
    <!--ro, opt, enum, network, subType:int, desc:1 (wired network), 2 (mobile network: 3G/4G/5G/GPRS), 3 (wireless network)-->1
  </netWork>
  <method>
    <!--ro, opt, enum, request method, subType:string, desc:"POST", "PUT", "GET"-->POST
  </method>
  <pictureAttachEnabled>
    <!--ro, opt, bool, whether to enable attaching pictures, desc:true by default-->true
  </pictureAttachEnabled>
  <contentType>
    <!--ro, opt, enum, HTTP content format, subType:string, desc:"multipart" (form format) by default. It is the format for all uploaded events, not limited to pictures only-->multipart
  </contentType>
  <eventTemplateID>
    <!--ro, opt, int, event message template ID, desc:related URL (GET /ISAPI/Event/notification/EventTemplateParams?format=json) is to get template list-->1
  </eventTemplateID>
  <customTriggersEnabled>
    <!--ro, opt, bool, whether to enable custom linkage, desc:true (listening service's linkage method is unaffected by upload center's linkage method, that is, independent), false or node doesn't exist (use the previous logic, that upload center sends data to listening service)-->false
  </customTriggersEnabled>
  <checkResponseEnabled>
```

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```
    <checkResponseEnabled>
      <!--ro, opt, bool, whether to enable verifying listening host response, desc:when a device uploads an event to the listening host, the host should
respond with HTTP 200 OK status. Otherwise, the device assumes that the event was not received. However, some platforms' listening services fail to respond.
This parameter is used to address the actual cause. When disabled, the device skips the verification of 200 OK response from the listening host. When
enabled, failure to receive 200 OK indicates the event upload failure, and the device will log the failure or retransmit the event-->false
    </checkResponseEnabled>
</HttpHostNotification>
```

14.2.1.5 Get the capabilities of listening hosts parameters

Request URL

GET /ISAPI/Event/notification/httpHosts/capabilities?type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
type	enum	custom refers to that the listening service is independent and unaffected by the upload center linkage method. default refers to that the listening service is controlled by the linkage method of the uploading center. When the type is empty or default, the listening service is the default.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<HttpHostNotificationCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, capabilities of subscribing to event types and event channels, attr:version{req, string, protocolVersion}-->
  <hostNumber>
    <!--ro, req, int, the number of listening hosts-->0
  </hostNumber>
  <urlLen min="1" max="10">
    <!--ro, req, string, URL length, attr:min{opt, int},max{req, int}-->test
  </urlLen>
  <protocolType opt="HTTP,HTTPS,EHome">
    <!--ro, req, enum, protocol type, subType:string, attr:opt{req, string}, desc:"HTTP", "HTTPS", "EHome"-->HTTP
  </protocolType>
  <parameterFormatType opt="XML,querystring,JSON">
    <!--ro, req, enum, alarm parameter format type, subType:string, attr:opt{req, string}, desc:"xml" (XML format), "querystring", "json" (JSON format)--
>XML
  </parameterFormatType>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, address format type, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName min="1" max="64">
    <!--ro, opt, string, domain name, attr:min{req, int},max{req, int}, desc:it is valid when addressingFormatType is "hostname"-->test
  </hostName>
  <ipAddress opt="ipv4,ipv6">
    <!--ro, opt, string, IP address, attr:opt{req, string}, desc:it is valid when addressingFormatType is "ipaddress"-->test
  </ipAddress>
  <portNo min="0" max="10">
    <!--ro, opt, string, port number, range:[0,65535], attr:min{req, int},max{req, int}-->1
  </portNo>
  <userNameLen min="0" max="10">
    <!--ro, opt, string, user name length, attr:min{req, int},max{req, int}-->1
  </userNameLen>
  <passwordLen min="0" max="10">
    <!--ro, opt, string, password length, attr:min{req, int},max{req, int}-->1
  </passwordLen>
  <httpAuthenticationMethod opt="MD5digest,none,base64,basic" def="MD5digest">
    <!--ro, req, enum, HTTP authentication method, subType:string, attr:opt{req, string},def{opt, string}, desc:"MD5digest"(MD5), "none", "base64"--
>MD5digest
  </httpAuthenticationMethod>
  <SubscribeEventCap>
    <!--ro, opt, object, subscribe to changing status of capability set-->
    <heartbeat min="0" max="180" def="10">
      <!--ro, opt, string, heartbeat interval time, attr:min{req, int},max{req, int},def{opt, int}-->1
    </heartbeat>
    <eventMode opt="all,list">
      <!--ro, opt, enum, event mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all channels and events on the device), "list"
(subscribe to all channels and events on the device)-->all
    </eventMode>
    <EventList>
      <!--ro, opt, array, event list, subType:object-->
      <Event>
        <!--ro, opt, object, event subscription information-->
        <type>
          <!--ro, req, enum, see details in event type list, subType:string, desc:see details in event type list-->AccessControllerEvent
        </type>
        <minorAlarm opt="0x400,0x401,0x402,0x403">
          <!--ro, opt, string, minor alarm type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401
        </minorAlarm>
        <minorException opt="0x400,0x401,0x402,0x403">
          <!--ro, opt, string, minor exception type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401
        </minorException>
        <minorOperation opt="0x400,0x401,0x402,0x403">
          <!--ro, opt, string, minor operation type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401
        </minorOperation>
        <minorEvent opt="0x01,0x02,0x03,0x04">
          <!--ro, opt, string, minor event type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401
        </minorEvent>
        <pictureURLType opt="binary,localURL,cloudStorageURL,EZVIZURL" def="cloudStorageURL">
          <!--ro, opt, enum, alarm picture format, subType:string, attr:opt{req, string},def{req, string}, desc:"binary", "localURL" (local URL),
"cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL)-->cloudStorageURL
        </pictureURLType>
      </Event>
    </EventList>
  </SubscribeEventCap>
</HttpHostNotificationCap>
```

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14.2.1.6 Set IP address of receiving server(s)

Request URL

PUT /ISAPI/Event/notification/httpHosts

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, listening host list, subType:object, attr:version{req, string, protocolVersion}-->
  <HttpHostNotification>
    <!--opt, object, listening host-->
    <id>
      <!--req, string, ID-->test
    </id>
    <url>
      <!--req, string, URL-->test
    </url>
    <protocolType>
      <!--req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"-->HTTP
    </protocolType>
    <parameterFormatType>
      <!--req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
    </parameterFormatType>
    <addressingFormatType>
      <!--req, enum, address types, subType:string, desc:"hostname" (host name), "ipaddress" (ip address)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:only one of IPv4 and IPv6 address exists when
the address type is "ipaddress"-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:only one of IPv4 and IPv6 address exists when
the address type is "ipaddress"-->test
    </ipv6Address>
    <portNo>
      <!--opt, int, port No.-->1
    </portNo>
    <userName>
      <!--opt, string, user name, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,ue,none}-->test
    </userName>
    <password>
      <!--wo, opt, string, password, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,ue,none}-->test
    </password>
    <httpAuthenticationMethod>
      <!--req, enum, authentication method, subType:string, desc:"MD5digest" (MD5), "none"-->MD5digest
    </httpAuthenticationMethod>
    <videoUploadEnabled>
      <!--opt, bool, whether to upload violation pre-records, desc:when it is enabled, the device will upload the stored video via the "relationVideo"
event-->false
    </videoUploadEnabled>
    <ANPR>
      <!--opt, object, ANPR-->
      <detectionUpLoadPicturesType>
        <!--opt, enum, uploaded pictures type, subType:string, desc:uploaded pictures type-->all
      </detectionUpLoadPicturesType>
    </ANPR>
    <Extensions>
      <!--opt, object, range-->
      <intervalBetweenEvents>
        <!--opt, int, event interval-->1
      </intervalBetweenEvents>
    </Extensions>
    <uploadImagesDataType>
      <!--opt, enum, picture data type, subType:string, desc:picture data type-->URL
    </uploadImagesDataType>
    <httpBroken>
      <!--opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
    </httpBroken>
    <SubscribeEvent>
      <!--opt, object, picture uploading modes of all events which contain pictures-->
      <heartbeat>
        <!--opt, int, heartbeat interval-->30
      </heartbeat>
      <eventMode>
        <!--req, enum, event mode, subType:string, desc:event mode-->all
      </eventMode>
      <EventList>
        <!--opt, array, event list, subType:object-->
        <Event>
          <!--opt, object, channel information linked to event-->
          <type>
            <!--req, enum, event type, subType:string, desc:event type-->mixedTargetDetection
          </type>
          <minorAlarm>
            <!--opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorAlarm>
          <minorException>
            <!--opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorException>
          <minorOperation>
```

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```
            <!--opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorOperation>
          <minorEvent>
            <!--opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
          </minorEvent>
          <pictureURLType>
            <!--opt, enum, alarm picture format of the specified event, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL)-->binary
          </pictureURLType>
          <channels>
            <!--opt, string, channel No., desc:if all channels are being listened to, the node shall not be applied. If some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
          </channels>
        </Event>
      </EventList>
      <channels>
        <!--opt, string, listen to the specified channel No. list, desc:if all channels are being listened to, the node shall not be applied. If some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
      </channels>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->binary
      </pictureURLType>
      <ChangedUploadSub>
        <!--opt, object, subscribe to messages-->
        <interval>
          <!--opt, int, the lifecycle of arming GUID, desc:within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->5
        </interval>
        <StatusSub>
          <!--opt, object, sub status-->
          <all>
            <!--opt, bool, whether to subscribe to all-->true
          </all>
          <channel>
            <!--opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
          </channel>
          <hd>
            <!--opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
          </hd>
          <capability>
            <!--opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the value of <all> is true-->true
          </capability>
        </StatusSub>
      </ChangedUploadSub>
    </SubscribeEvent>
    <PackingSpaceRecognition>
      <!--opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event: PackingSpaceRecognition-->
      <upLoadPicturesType>
        <!--req, enum, uploaded picture type, subType:string, desc:uploaded picture type-->all
      </upLoadPicturesType>
      <PicturesTypes>
        <!--opt, array, specified list of uploaded picture types, subType:object, dep:and, {$.HttpHostNotificationList[*].HttpHostNotification.PackingSpaceRecognition.upLoadPicturesType,eq,picturesTypes}-->
        <picturesType>
          <!--opt, enum, uploaded picture type, subType:string, desc:"backgroundImage" (captured background picture), "plateImage" (license plate thumbnail)-->backgroundImage
        </picturesType>
      </PicturesTypes>
    </PackingSpaceRecognition>
    <fileUploadType>
      <!--opt, enum, subType:string-->cloudStrage
    </fileUploadType>
    <enabled>
      <!--opt, bool, whether to enable-->true
    </enabled>
    <netWork>
      <!--opt, enum, network, subType:int, desc:1 (wired network), 2 (mobile network: 3G/4G/5G/GPRS), 3 (wireless network)-->1
    </netWork>
    <method>
      <!--opt, enum, request method, subType:string, desc:"POST", "PUT", "GET"-->POST
    </method>
    <pictureAttachEnabled>
      <!--opt, bool, whether to enable attaching pictures, desc:true by default-->true
    </pictureAttachEnabled>
    <contentType>
      <!--opt, enum, HTTP content format, subType:string, desc:"multipart" (form format) by default. It is the format for all uploaded events, not limited to pictures only-->multipart
    </contentType>
    <eventTemplateID>
      <!--opt, int, event message template ID, desc:related URL (GET /ISAPI/Event/notification/EventTemplateParams?format=json) is to get template list-->1
    </eventTemplateID>
    <customTriggersEnabled>
      <!--opt, bool, whether to enable custom linkage, desc:true (listening service's linkage method is unaffected by upload center's linkage method, that is, independent), false or node doesn't exist (use the previous logic, that upload center sends data to listening service)-->false
    </customTriggersEnabled>
    <checkResponseEnabled>
```

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```
      <!--opt, bool, whether to enable verifying listening host response, desc:when a device uploads an event to the listening host, the host should respond with HTTP 200 OK status. Otherwise, the device assumes that the event was not received. However, some platforms' listening services fail to respond. This parameter is used to address the actual cause. When disabled, the device skips the verification of 200 OK response from the listening host. When enabled, failure to receive 200 OK indicates the event upload failure, and the device will log the failure or retransmit the event-->true
    </checkResponseEnabled>
  </HttpHostNotification>
</HttpHostNotificationList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:detailed information of custom error returned by device applications, used for fast debugging-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:none-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.1.7 Add a listening host

Request URL

POST /ISAPI/Event/notification/httpHosts

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, listening host, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, listening host ID-->test
  </id>
  <url>
    <!--req, string, URL-->test
  </url>
  <protocolType>
    <!--req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
  </protocolType>
  <parameterFormatType>
    <!--req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
  </parameterFormatType>
  <addressingFormatType>
    <!--req, enum, address type, subType:string, desc:"hostname", "ipaddress"-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IP address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IP address-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 Address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IPv6 Address-->test
  </ipv6Address>
  <portNo>
    <!--opt, int, port number-->1
  </portNo>
  <userName>
    <!--opt, string, user name, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
  </userName>
```

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```
    <password>
      <!--opt, string, password, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
    </password>
    <httpAuthenticationMethod>
      <!--req, enum, authentication method, subType:string, desc:"MD5digest"(MD5), "none", "base64"-->MD5digest
    </httpAuthenticationMethod>
    <ANPR>
      <!--opt, object, ANPR-->
      <detectionUpLoadPicturesType>
        <!--opt, enum, uploaded pictures type, subType:string, desc:"all", "licensePlatePicture", "detectionPicture"(detected picture)-->all
      </detectionUpLoadPicturesType>
      <videoUploadEnabled>
        <!--opt, bool, whether to upload violation pre-records, desc:when it is enabled, the device will upload the stored video via the "relationVideo"
event-->false
      </videoUploadEnabled>
    </ANPR>
    <Extensions>
      <!--opt, object, range-->
      <intervalBetweenEvents>
        <!--opt, int, event interval-->1
      </intervalBetweenEvents>
    </Extensions>
    <uploadImagesDataType>
      <!--opt, enum, picture data type, subType:string, desc:"URL", "binary" (default value). For cloud storage, only "URL" is supported-->URL
    </uploadImagesDataType>
    <httpBroken>
      <!--opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
    </httpBroken>
    <SubscribeEvent>
      <!--opt, object, picture uploading modes of all events which contain pictures-->
      <heartbeat>
        <!--opt, int, Heartbeat Interval Time, unit:s-->30
      </heartbeat>
      <eventMode>
        <!--req, enum, event mode, subType:string, desc:"all" (all alarms need to be reported), "list" (only listed alarms need to be reported)-->all
      </eventMode>
      <EventList>
        <!--opt, array, event list, subType:object-->
        <Event>
          <!--opt, object, channel information linked to event-->
          <type>
            <!--req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
"faceCapture", "faceSnapModeling", "facedetection", "fallDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "LeavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent",
"IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison
event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching
consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison" (human body comparison),
"regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
          </type>
          <minorAlarm>
            <!--opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorAlarm>
          <minorException>
            <!--opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorException>
          <minorOperation>
            <!--opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorOperation>
          <minorEvent>
            <!--opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x01,0x02,0x03,0x04
          </minorEvent>
          <pictureURLType>
            <!--opt, enum, alarm picture format of the specified event, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud
storage URL), "EZVIZURL" (EZ URL)-->binary
```

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```
        </pictureURLType>
        <channels>
          <!--opt, string, listen to the events on the specified channel No. list, desc:if all channels are being listened to, the node shall not be
applied. If some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
        </channels>
      </Event>
    </EventList>
    <channels>
      <!--opt, string, listen to the specified channel No. list, desc:if all channels are being listened to, the node shall not be applied. If some channels
are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
    </channels>
    <pictureURLType>
      <!--opt, enum, alarm picture format, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ
URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not
applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices
is local URL of the device-->binary
    </pictureURLType>
    <ChangedUploadSub>
      <!--opt, object, subscribe to messages-->
      <interval>
        <!--opt, int, the lifecycle of arming GUID, desc:within the interval, if the client software does not reconnect to the device, a new GUID will be
generated by the device-->5
      </interval>
      <StatusSub>
        <!--opt, object, sub status-->
        <all>
          <!--opt, bool, whether to subscribe to all-->true
        </all>
        <channel>
          <!--opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
        </channel>
        <hd>
          <!--opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
        </hd>
        <capability>
          <!--opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the
value of <all> is true-->true
        </capability>
      </StatusSub>
    </ChangedUploadSub>
  </SubscribeEvent>
  <PackingSpaceRecognition>
    <!--opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event:
PackingSpaceRecognition-->
    <upLoadPicturesType>
      <!--req, enum, uploaded picture type, subType:string, desc:"all", "picturesTypes" (upload specified types of pictures), "notUpload" (not upload
pictures)-->all
    </upLoadPicturesType>
    <PicturesTypes>
      <!--opt, array, specified list of uploaded picture type, subType:object, dep:and,
{$.HttpHostNotification.PackingSpaceRecognition.upLoadPicturesType,eq,picturesTypes}-->
      <picturesType>
        <!--opt, enum, uploaded picture type, subType:string, desc:"backgroundImage" (captured background picture), "plateImage" (license plate thumbnail)--
>backgroundImage
      </picturesType>
    </PicturesTypes>
  </PackingSpaceRecognition>
  <fileUploadType>
    <!--opt, enum, subType:string-->cloudStrage
  </fileUploadType>
  <enabled>
    <!--opt, bool, whether to enable-->true
  </enabled>
  <netWork>
    <!--opt, enum, network, subType:int, desc:multi-NIC devices support the access via wired/wireless/mobile networks. After the network switching, the
device's upload center address will be invalid. So, it is required to configure the valid network to avoid event upload failure due to invalid network 1
(wired network), 2 (mobile network: 3G/4G/5G/GPRS), 3 (wireless network)-->1
  </netWork>
  <method>
    <!--opt, enum, request method, subType:string, desc:"POST", "PUT", "GET"-->POST
  </method>
  <pictureAttachEnabled>
    <!--opt, bool, whether to enable attaching pictures, desc:true by default-->true
  </pictureAttachEnabled>
  <contentType>
    <!--opt, enum, HTTP content format, subType:string, desc:"multipart" (form format) by default. It is the format for all uploaded events, not limited to
pictures only-->multipart
  </contentType>
  <eventTemplateID>
    <!--opt, int, event message template ID, desc:related URL (GET /ISAPI/Event/notification/EventTemplateParams?format=json) is to get template list-->1
  </eventTemplateID>
  <customTriggersEnabled>
    <!--opt, bool, whether to enable custom linkage, desc:true (listening service's linkage method is unaffected by upload center's linkage method, that is,
independent), false or node doesn't exist (use the previous logic, that upload center sends data to listening service)-->false
  </customTriggersEnabled>
  <checkResponseEnabled>
    <!--opt, bool, whether to enable verifying listening host response, desc:when a device uploads an event to the listening host, the host should respond
with HTTP 200 OK status. Otherwise, the device assumes that the event was not received. However, some platforms' listening services fail to respond. This
parameter is used to address the actual cause. When disabled, the device skips the verification of 200 OK response from the listening host. When enabled,
failure to receive 200 OK indicates the event upload failure, and the device will log the failure or retransmit the event-->true
  </checkResponseEnabled>
</HttpHostNotification>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK"(succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot"(reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.1.8 Get parameters of all listening hosts

Request URL

GET /ISAPI/Event/notification/httpHosts

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, listening host list, subType:object, attr:version{req, string, protocolVersion}-->
  <HttpHostNotification>
    <!--ro, opt, object, listening host-->
    <id>
      <!--ro, req, string, subscribe to ID, range:[1,10]-->test
    </id>
    <url>
      <!--ro, req, string, URL-->test
    </url>
    <protocolType>
      <!--ro, req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
    </protocolType>
    <parameterFormatType>
      <!--ro, req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
    </parameterFormatType>
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress"-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IP address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipaddress}, desc:only one of IPv4 and IPv6 address exists when
the address type is "ipaddress"-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 Address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipaddress}, desc:only one of IPv4 and IPv6 address exists
when the address type is "ipaddress"-->test
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port number-->1
    </portNo>
    <userName>
      <!--ro, opt, string, user name, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
    </userName>
```

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```
    <httpAuthenticationMethod>
      <!--ro, req, enum, authentication method, subType:string, desc:"MD5digest"(MD5), "none", "base64"-->MD5digest
    </httpAuthenticationMethod>
    <ANPR>
      <!--ro, opt, object, ANPR-->
      <detectionUpLoadPicturesType>
        <!--ro, opt, enum, uploaded pictures type, subType:string, desc:"all", "licensePlatePicture", "detectionPicture"(detected picture)-->all
      </detectionUpLoadPicturesType>
      <videoUploadEnabled>
        <!--ro, opt, bool, whether to upload violation pre-records, desc:when it is enabled, the device will upload the stored video via the "relationVideo"
event-->false
      </videoUploadEnabled>
    </ANPR>
    <Extensions>
      <!--ro, opt, object, range-->
      <intervalBetweenEvents>
        <!--ro, opt, int, event interval-->1
      </intervalBetweenEvents>
    </Extensions>
    <uploadImagesDataType>
      <!--ro, opt, enum, picture data type, subType:string, desc:"URL", "binary" (default value). For cloud storage, only "URL" is supported-->URL
    </uploadImagesDataType>
    <httpBroken>
      <!--ro, opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
    </httpBroken>
    <SubscribeEvent>
      <!--ro, opt, object, picture uploading modes of all events which contain pictures-->
      <heartbeat>
        <!--ro, opt, int, heartbeat interval time-->30
      </heartbeat>
      <eventMode>
        <!--ro, req, enum, event mode, subType:string, desc:"all" (all alarms need to be reported), "list" (only listed alarms need to be reported)-->all
      </eventMode>
      <EventList>
        <!--ro, opt, array, event list, subType:object-->
        <Event>
          <!--ro, opt, object, channel information linked to event-->
          <type>
            <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system),
"ADASAlarm"(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent"
(access controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently
appeared person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection),
"PALMismatch" (video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing
up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving",
"advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection),
"blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis",
"databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status
monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"
(intrusion detection), "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people
gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high
temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access),
"ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection),
"linedetection"(line crossing detection), "listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low
temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node
disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection),
"peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting",
"personQueueDetection", "personQueueRealtime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"
(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection",
"radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection),
"raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance",
"regionExiting", "retention"(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection),
"scenechangedetection", "sensorAlarm"(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm),
"shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"
(storaging full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature
difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information
report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"
(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay
detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control),
"CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent",
"TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison"
(human body comparison), "regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
          </type>
          <minorAlarm>
            <!--ro, opt, string, alarm sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorAlarm>
          <minorException>
            <!--ro, opt, string, exception sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorException>
          <minorOperation>
            <!--ro, opt, string, operation sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorOperation>
          <minorEvent>
            <!--ro, opt, string, event sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
          </minorEvent>
          <pictureURLType>
            <!--ro, opt, enum, alarm picture format of the specified event, subType:string, desc:alarm picture format of the specified event-->binary
          </pictureURLType>
          <channels>
            <!--ro, opt, string, listen to the events on the specified channel No. list, desc:if all channels are being listened to, the node shall not be
applied. if some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
```

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```
applied. if some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
          </channels>
        </Event>
      </EventList>
      <channels>
        <!--ro, opt, string, listen to the specified channel No. list, desc:if all channels are being listened to, the node shall not be applied. if some
channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
      </channels>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:the node indicates the picture types of all events which contain pictures to be
uploaded. if the node is applied, the pictureURLType of the Event will not take effect. if the node is not applied, the pictures reported from the device by
default mode: the default data type of uploaded pictures captured from front-end devices is binary. the default data type of uploaded pictures reported from
storage devices is local URL of the device.-->binary
      </pictureURLType>
      <ChangedUploadSub>
        <!--ro, opt, object, subscribe to messages-->
        <interval>
          <!--ro, opt, int, the lifecycle of arming GUID, desc:if GUID is not reconnected in the internal, a new GUID will be generated since the device
start a new arm period-->5
        </interval>
        <StatusSub>
          <!--ro, opt, object, sub status-->
          <all>
            <!--ro, opt, bool, subscribe to all?-->true
          </all>
          <channel>
            <!--ro, opt, bool, subscribe to channel status, desc:reporting is not required if all is true-->true
          </channel>
          <hd>
            <!--ro, opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
          </hd>
          <capability>
            <!--ro, opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
          </capability>
        </StatusSub>
      </ChangedUploadSub>
    </SubscribeEvent>
    <PackingSpaceRecognition>
      <!--ro, opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event:
PackingSpaceRecognition-->
      <upLoadPicturesType>
        <!--ro, opt, enum, uploaded picture type, subType:string, desc:"all", "picturesTypes" (upload specified types of pictures), "notUpload" (not upload
pictures)-->all
      </upLoadPicturesType>
      <PicturesTypes>
        <!--ro, opt, array, specified list of uploaded picture type, subType:object, dep:and,
{$.HttpHostNotificationList[*].HttpHostNotification.PackingSpaceRecognition.upLoadPicturesType,eq,picturesTypes}-->
        <picturesType>
          <!--ro, opt, enum, uploaded picture type, subType:string, desc:"backgroundImage" (captured background picture), "plateImage" (license plate
thumbnail)-->backgroundImage
        </picturesType>
      </PicturesTypes>
    </PackingSpaceRecognition>
    <fileUploadType>
      <!--ro, opt, enum, subType:string-->cloudStrage
    </fileUploadType>
    <enabled>
      <!--ro, opt, bool, whether to enable-->true
    </enabled>
    <netWork>
      <!--ro, opt, enum, network, subType:int, desc:1 (wired network), 2 (mobile network: 3G/4G/5G/GPRS), 3 (wireless network)-->1
    </netWork>
    <method>
      <!--ro, opt, enum, request method, subType:string, desc:"POST", "PUT", "GET"-->POST
    </method>
    <pictureAttachEnabled>
      <!--ro, opt, bool, whether to enable attaching pictures, desc:true by default-->true
    </pictureAttachEnabled>
    <contentType>
      <!--ro, opt, enum, HTTP content format, subType:string, desc:"multipart" (form format) by default. It is the format for all uploaded events, not
limited to pictures only-->multipart
    </contentType>
    <eventTemplateID>
      <!--ro, opt, int, event message template ID, desc:related URL (GET /ISAPI/Event/notification/EventTemplateParams?format=json) is to get template list-
->1
    </eventTemplateID>
    <customTriggersEnabled>
      <!--ro, opt, bool, whether to enable custom linkage, desc:true (listening service's linkage method is unaffected by upload center's linkage method,
that is, independent), false or node doesn't exist (use the previous logic, that upload center sends data to listening service)-->false
    </customTriggersEnabled>
    <checkResponseEnabled>
      <!--ro, opt, bool, whether to enable verifying listening host response, desc:when a device uploads an event to the listening host, the host should
respond with HTTP 200 OK status. Otherwise, the device assumes that the event was not received. However, some platforms' listening services fail to respond.
This parameter is used to address the actual cause. When disabled, the device skips the verification of 200 OK response from the listening host. When
enabled, failure to receive 200 OK indicates the event upload failure, and the device will log the failure or retransmit the event-->true
    </checkResponseEnabled>
  </HttpHostNotification>
</HttpHostNotificationList>
```

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14.3 Face Capture

14.3.1 Face Capture Settings

14.3.1.1 Clear the face picture library

Request URL

DELETE /ISAPI/Intelligent/FDLib/<FDID>/picture?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
identityKey	string	--
FDType	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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14.4 VCA

14.4.1 Face Picture Library Management

14.4.1.1 Get face picture library capability

Request URL

GET /ISAPI/Intelligent/FDLib/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, this field is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, detailed error description, this field is required when the value of statusCode is not 1*/
    "FDNameMaxLen":  64,
    /*ro, req, int, maximum length of face picture library name*/
```

```
    "customInfoMaxLen":  192,
    /*ro, req, int, maximum length of custom information*/
    "FDMaxNum":  3,
    /*ro, req, int, maximum number of face picture libraries*/
    "FDRecordDataMaxNum":  12345,
    /*ro, req, int, maximum face records supported by face picture library*/
    "supportFDFunction":  "post,delete,put,get,setUp",
    /*ro, req, string, the supported operations on face picture library, desc:"post" (create), "delete" (delete), "put" (edit), "get" (search), "setUp"
(set)*/
    "isSuportFDSearch":  true,
    /*ro, req, bool, whether supports searching in face picture library, desc:whether supports searching in face picture library*/
    "isDisplayCaptureNum": {
    /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, opt, array, subType:bool*/
    },
    "isSupportFDSearchDataPackage":  true,
    /*ro, req, bool, whether supports packaging the found data in the face picture library*/
    "isSuportFSsearchByPic":  true,
    /*ro, req, bool, whether supports searching by picture in the face picture library, desc:whether supports searching by picture in the face picture
library*/
    "isSuportFSsearchByPicGenerate":  true,
    /*ro, req, bool, whether supports exporting results of using picture to search picture from the face picture library*/
    "isSuportFDSearchDuplicate":  true,
    /*ro, req, bool, whether supports duplication checking*/
    "isSuportFDSearchDuplicateGenerate":  true,
    /*ro, req, bool, whether supports exporting the duplication checking results*/
    "isSuportFCSearch":  true,
    /*ro, req, bool, whether supports searching face picture comparison alarms*/
    "isSupportFCSearchDataPackage":  true,
    /*ro, req, bool, whether supports packaging the search results of face picture comparison alarms*/
    "isSupportFDExecuteControl":  true,
    /*ro, req, bool, whether supports creating relation between face picture libraries and cameras*/
    "generateMaxNum":  1234,
    /*ro, opt, int, maximum face records can be exported from face picture library*/
    "faceLibType":  "blackFD,staticFD,infraredFD",
    /*ro, opt, string, face picture library type*/
    "modelMaxNum":  1000,
    /*ro, opt, int, the maximum number of search results*/
    "isSupportModelData":  true,
    /*ro, opt, bool, whether it supports applying model data*/
    "isSuportFDLibArmingType":  true,
    /*ro, opt, bool, whether it supports face picture library arming type*/
    "isSuportFDLibSearch":  true,
    /*ro, opt, bool, whether it supports searching face picture library*/
    "FDArmingRecordDataMaxNum":  12345,
    /*ro, opt, int, the supported maximum number of face records in the face picture arming library*/
    "isSupportControlPersonRecordByHumanId":  true,
    /*ro, opt, bool, whether it supports modifying and deleting the face record by humanId*/
    "isSupportControlPersonRecordByRowKey":  true,
    /*ro, opt, bool, whether it supports modifying and deleting the face record by rowKey*/
    "isSupportFaceLibRebuildCfg":  true,
    /*ro, opt, bool, whether it supports recreating face picture library information and configuration*/
    "isSupportFDMove":  true,
    /*ro, opt, bool, whether it supports moving face data in the face picture library in a batch, desc:the related URI is
/ISAPI/Intelligent/FDLib/FDMove/capabilities?format=json*/
    "faceURLLen":  1024,
    /*ro, opt, int, the maximum size of the face picture URL, desc:if this node is not returned, the default size of the face picture URL supported by the
device is 256 bytes; otherwise, the device should support that the value of this node is greater than or equal to 256*/
    "isSupportArmingLibCfg":  true,
    /*ro, opt, bool, whether it supports configuring parameters of the armed face picture library, desc:related URI:
/ISAPI/Intelligent/FDLib/armingLibCfg/capabilities?format=json*/
    "isSupportModelTransformation":  true,
    /*ro, opt, bool, whether it supports converting face picture models in the face picture list library, desc:related URI:
/ISAPI/Intelligent/FDLib/model/transformation/capabilities?format=json*/
    "featurePointTypeList": ["face", "leftEye", "rightEye", "leftMouthCorner", "rightMouthCorner", "nose"],
    /*ro, opt, array, feature point types of face pictures supported by the device, subType:string*/
    "isSupportPresort":  true,
    /*ro, opt, bool, whether it supports configuring pre-categorization, desc:related URI: /ISAPI/Intelligent/FDLib/presort/capabilities?format=json*/
    "libAttribute": {
    /*ro, opt, object, library attribute type, desc:"general" (normal library), "blockList" (blocklist library), "VIP" (VIP library), "passerby" (allowlist
library, which cannot be deleted)*/
        "@opt": ["general", "blackList", "VIP", "passerby"]
        /*ro, opt, array, options, subType:string*/
    },
    "faceType": {
    /*ro, opt, object, face picture type*/
        "@opt": ["normalFace", "patrolFace", "hijackFace", "superFace"]
        /*ro, opt, array, options, subType:string*/
    },
    "saveFacePic": {
    /*ro, opt, object, whether to save the registered face picture*/
        "@opt": [true, false]
        /*ro, opt, array, options, subType:bool*/
    },
    "leaderPermission": {
    /*ro, opt, object, first authentication permission*/
        "@size":  4,
        /*ro, opt, int, the maximum number of array members, range:[1,4]*/
        "@min":  1,
        /*ro, opt, int, member value range, range:[1,4]*/
        "@max":  4
        /*ro, opt, int, member value range, range:[1,4]*/
    }
```

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```
    },
    "facePicFormat": {
    /*ro, opt, object*/
        "@opt": ["jpg", "png", "bmp"]
        /*ro, req, array, subType:string*/
    }
}
```

14.4.1.2 Search face picture library information by arming mode

Request URL

POST /ISAPI/Intelligent/FDLib/search?format=json

Query Parameter

None

Request Message

```
{
    "searchResultPosition": 0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults": 0,
    /*req, int, the maximum number of search results this time*/
    "libArmingType": "armingLib"
    /*opt, enum, arming type, subType:string, desc:"armingLib" (armed library (default)), "nonArmingLib" (not armed library)*/
}
```

Response Message

```
{
    "responseStatusStrg": "OK",
    /*ro, opt, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data).*/
    "numOfMatches": 1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches": 1,
    /*ro, opt, int, total number of matched results*/
    "FDLib": [
    /*ro, opt, array, face picture library information, subType:object*/
        {
            "FDID": "test",
            /*ro, opt, string, face picture library ID*/
            "faceLibType": "blackFD",
            /*ro, opt, enum, face picture library type, desc:"blackFD" (list library), "staticFD" (static library)*/
            "name": "test",
            /*ro, opt, string, name of the face picture library, range:[0,48]*/
            "customInfo": "test",
            /*ro, opt, string, custom information, range:[0,192]*/
            "libArmingType": "armingLib",
            /*ro, opt, enum, arming type, subType:string, desc:"armingLib" (armed library (default)), "nonArmingLib" (not armed library)*/
            "libAttribute": "blackList"
            /*ro, opt, enum, library attribute type, subType:string, desc:"blackList" (blocklist), "VIP" (VIP library), "passerby" (passer-by library, which cannot be deleted)*/
        }
    ]
}
```

14.4.1.3 Delete all face picture libraries

Request URL

DELETE /ISAPI/Intelligent/FDLib?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

14.4.1.4 Create a face picture library

Request URL

POST /ISAPI/Intelligent/FDLib?format=json

Query Parameter

None

Request Message

```
{
    "faceLibType":  "blackFD,staticFD",
    /*req, string, face picture library type, desc:face picture library type*/
    "name":  "test",
    /*req, string, name of the face picture library, range:[0,48], desc:name of the face picture library*/
    "customInfo":  "test",
    /*opt, string, custom information, range:[0,192], desc:custom information*/
    "libArmingType":  "armingLib",
    /*opt, enum, subType:string*/
    "libAttribute":  "blackList",
    /*opt, enum, subType:string*/
    "FDID":  "test"
    /*opt, string, face picture library ID, desc:face picture library ID*/
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, req, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "FDID":  "test"
    /*ro, opt, string, face picture library ID, desc:face picture library ID*/
}
```

14.4.1.5 Delete a specified face picture library

Request URL

DELETE /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD" (list library), "staticFD" (static library)
terminalNo	string	Terminal No. It is an integer starting from 1.

Request Message

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None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

14.4.1.6 Edit the information of a specific face picture library information, including name and custom information

Request URL

PUT /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	--
terminalNo	string	--

Request Message

```
{
    "name":  "test",
    /*opt, string, face picture library name, range:[0,48]*/
    "customInfo":  "test",
    /*opt, string, custom information, range:[0,192]*/
    "libArmingType":  "armingLib",
    /*opt, enum, arming type of the list library, subType:string, desc:“armingLib” (armed face picture library), “nonArmingLib” (not armed face picture library). The default value is "armingLib”*/
    "libAttribute":  "blackList"
    /*opt, enum, library type, subType:string, desc:“blackList” (blocklist library), “VIP” (VIP library), “passerby” (passerby library). The passerby library cannot be deleted*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode when statusCode is not 1*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

14.4.1.7 Get the information, including library ID, library type, name, and custom information, of all face picture libraries

Request URL

GET /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

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Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD” (list library), "staticFD” (static library), "infraredFD" (IR face picture library)
terminalNo	string	Terminal No. It is an integer starting from 1.

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:status code*/
    "statusString":  "OK",
    /*ro, req, string, status description, range:[1,64], desc:status description*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:sub status code*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:error description*/
    "faceLibType":  "blackFD",
    /*ro, opt, enum, face picture library type, subType:string, desc:"blackFD” (list library), "staticFD” (static library)*/
    "name":  "test",
    /*ro, opt, string, face picture library name, range:[1,48]*/
    "customInfo":  "test",
    /*ro, opt, string, custom information, range:[0,192]*/
    "libArmingType":  "armingLib",
    /*ro, opt, enum, library arming type, subType:string, desc:library arming type*/
    "libAttribute":  "blackList",
    /*ro, opt, enum, subType:string*/
    "personnelFileEnabled":  true
    /*ro, opt, bool*/
}
```

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14.4.2 Face Picture Library Record Management

14.4.2.1 Get the total number of face records in all face picture libraries

Request URL

GET /ISAPI/Intelligent/FDLib/Count?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "requestURL":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, see the description of this node and above nodes in the message of JSON_ResponseStatus, desc:this node is required when the value of
statusCode is not 1*/
    "FDRecordDataInfo": [
    /*ro, opt, array, information of face records in face picture library, subType:object, desc:this node is valid when errorCode is 1 and errorMsg is
"ok"*/
        {
            "FDID":  "test",
            /*ro, opt, string, face picture library ID, desc:the maximum size is 63 bytes*/
            "faceLibType":  "blackFD",
            /*ro, opt, enum, face picture library type, subType:string, desc:face picture library type "blackFD" list library,"staticFD" static library, the
maximum size is 32 bytes*/
            "name":  "test",
            /*ro, opt, string, face picture library name, desc:the maximum size is 48 bytes*/
            "recordDataNumber":  123,
            /*ro, opt, int, number of records*/
            "libArmingType":  "armingLib",
            /*ro, opt, enum, arming type, subType:string, desc:arming type*/
            "libAttribute":  "general",
            /*ro, opt, enum, subType:string*/
            "personnelFileEnabled":  true
            /*ro, opt, bool*/
        }
    ],
    "totalRecordDataNumber":  500000,
    /*ro, opt, int, total number of records, desc:total number of records*/
    "FDCapacity": {
    /*ro, opt, object*/
        "total":  0.000,
        /*ro, req, float*/
        "use":  0.000,
        /*ro, req, float*/
        "remain":  0.000,
        /*ro, req, float, remaining space, desc:remaining space*/
        "maxRecordDataNumber":  0,
        /*ro, opt, int*/
        "useRecordDataNumber":  0,
        /*ro, opt, int*/
        "remainRecordDataNumber":  0
        /*ro, opt, int*/
    }
}
```

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14.4.2.2 Get the total number of face records in a face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/Count?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD" (list library), "staticFD" (static library), "infraredFD" (IR face picture library)
terminalNo	string	--

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "FDID":  "test",
    /*ro, opt, string, face picture library ID, range:[1,64]*/
    "faceLibType":  "blackFD",
    /*ro, opt, enum, face picture library type, subType:string, desc:face picture library type*/
    "name":  "test",
    /*ro, opt, string, name of the face picture library, desc:the maximum length is 48*/
    "recordDataNumber":  123
    /*ro, opt, int, total number of records*/
}
```

14.4.2.3 Add a face record to the face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/FaceDataRecord?format=json

Query Parameter

None

Request Message

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```
{
    "faceURL":  "test",
    /*opt, string, picture storage URL inputted when uploading the face picture by URL, desc:select this field or modelData*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture library type*/
    "FDID":  "test",
    /*req, string, face picture library ID, range:[1,64]*/
    "FPID":  " test",
    /*opt, string, face record ID, range:[1,1000], desc:face record ID,it can be generated by device or inputted. If it is inputted,it should be the unique
ID with the combination of letters and digits,and the maximum length is 63 bytes; if it is generated by the device automatically,it is the same as the
employee No. (person ID)*/
    "name":  "张三",
    /*req, string, name of person in the face picture, range:[1,1000], desc:the maximum size is 96 bytes*/
    "gender":  "male",
    /*opt, enum, gender of person in the face picture, subType:string, desc:“male”, “female”, “unknown”; the maximum size is 32 bytes*/
    "bornTime":  "2004-05-03",
    /*req, string, birthday of person in the face picture, desc:the maximum size is 20 bytes*/
    "city":  "130100",
    /*opt, string, city code of birth for the person in the face picture, desc:the maximum size is 32 bytes*/
    "certificateType":  "ID",
    /*opt, enum, certificate type, subType:string, desc:"ID" (ID card), "officerID" (officer ID), "other" (other type), "LawyerCertificate" (lawyer
qualification), "birthCertificate", "householdRegister", "judgeCertificate", "studentIDCard", "workCard" (work permit),  "policeID",
"foreignPermanentResidentIDCard", "passport", "pressCard", "prosecutorCard", "temporaryIDCard", "temporaryResidencePermit"*/
    "certificateNumber":  "test",
    /*opt, string, certificate No., range:[1,64], desc:the maximum size is 32 bytes*/
    "caseInfo":  "test",
    /*opt, string, case information, desc:the maximum size is 192 bytes; it is valid when faceLibType is "blackFD”*/
    "tag":  "aa,bb,cc,dd",
    /*opt, string, custom tag, desc:up to 4 tags, which are separated by commas. The maximum size is 195 bytes. It is valid when faceLibType is "blackFD”*/
    "address":  "test",
    /*opt, string, person address, desc:the maximum size is 192 bytes. It is valid when faceLibType is "staticFD”*/
    "customInfo":  "test",
    /*opt, string, custom information, desc:the maximum size is 192 bytes. It is valid when faceLibType is "staticFD"*/
    "modelData":  "test",
    /*opt, string, target modeling data, desc:target model data,non-modeled binary data needs to be encrypted by Base64 during transmission*/
    "transfer":  true,
    /*opt, bool, whether to enable transfer, desc:whether to enable transfer*/
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal” (operate by terminal)*/
    "terminalNoList": [1],
    /*opt, array, terminal ID list, subType:int, desc:this node is required when operation type is "byTerminal"; currently, only one terminal is supported*/
    "PicFeaturePoints": [
    /*opt, array, feature points to be applied, subType:object*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of
mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, coordinates of the feature point, desc:object,coordinates of the feature point*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized X-coordinate which is between 0 and 1000*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized Y-coordinate which is between 0 and 1000*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:this node is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:this node is required when featurePointType is "face"*/
            }
        }
    ],
    "cutOutRect": {
    /*opt, object, picture cropping frame, desc:the origin is the upper-left corner of the screen, which is used for the target frame in matting*/
        "x":  0.000,
        /*req, float, x-coordinate, range:[0,1], desc:normalize coordinate*/
        "y":  0.000,
        /*req, float, y-Coordinate, range:[0,1], desc:normalize coordinate*/
        "width":  0.000,
        /*req, float, width, range:[0,1]*/
        "height":  0.000
        /*req, float, height, range:[0,1]*/
    },
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string, desc:"normalFace" (normal face, default value), "patrolFace" (patrol person face), "hijackFace" (duress
person face), "superFace" (super person face)*/
    "saveFacePic":  true,
    /*opt, bool, whether to save face pictures*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, first authentication permission, subType:int, range:[1,4], desc:if the array is empty, it indicates clearing first authentication
permission of the person; if not, it indicates that the person have first authentication permission for the door 1, 2, 3, and 4*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceURL	[Message content]	application/json	--	--	--
img	[Binary picture data]	image/jpeg		facePic.jpg	--

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Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:this field is required when the value of statusCode is not 1*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:see the description of this node and above nodes in the message of JSON_ResponseStatus*/
    "FPID":  "test",
    /*ro, opt, string, face record ID, range:[1,64], desc:face record ID returned when the face record is added,it is unique,and the maximum size is 63
bytes. This node is valid when errorCode is "1" and errorMsg is "ok"*/
    "rowKey":  "test"
    /*ro, opt, string, rowKey of face record, desc:rowKey of face record is returned after face data is added. It is unique and in string format. The
maximum length is 64 bytes.*/
}
```

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14.4.2.4 Edit face records in the face picture library in a batch

Request URL

PUT /ISAPI/Intelligent/FDLib/FDModify?format=json

Query Parameter

None

Request Message

```
{
    "faceURL":  "test",
    /*opt, string, picture storage URL inputted when uploading the face picture by URL, the maximum length is 256 bytes*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture library type: "blackFD"-list library,"staticFD"-static library,the maximum
length is 32 bytes*/
    "FDID":  "test",
    /*req, string, face picture library ID, desc:face picture library ID,the maximum length is 63 bytes,multiple face picture libraries should be separated
by commas*/
    "FPID":  " ",
    /*opt, string, face record ID*/
    "name":  "test",
    /*opt, string, name*/
    "gender":  "any",
    /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
    "bornTime":  "2004-05-03",
    /*req, time, date of birth of the person in the face picture in ISO8601 time format, the maximum length is 20 bytes, desc:date of birth of the person in
the face picture in ISO8601 time format,the maximum length is 20 bytes*/
    "city":  "130100",
    /*opt, string, code of the city of birth for the person in the face picture, the maximum length is 32 bytes, desc:code of the city of birth for the
person in the face picture,the maximum length is 32 bytes*/
    "certificateType":  "ID",
    /*req, enum, certificate type, subType:string*/
    "certificateNumber":  "test",
    /*opt, string, ID No., the maximum length is 32 bytes*/
    "caseInfo":  "test",
    /*opt, string, case information, it is valid when faceLibType is "blackFD", range:[0,192]*/
    "tag":  "aa,bb,cc,dd",
    /*opt, string, custom tag*/
    "address":  "test",
    /*opt, string, person address, it is valid when faceLibType is "staticFD"*/
    "customInfo":  "test",
    /*opt, string, custom information, it is valid when faceLibType is "staticFD"*/
    "modelData":  "test",
    /*opt, string, target model data, non-modeled binary data needs to be encrypted by base64 during transmission*/
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal"-by terminal*/
    "terminalNoList": [1],
    /*opt, array, terminal ID list, subType:int, desc:this node is required when operation type is "byTerminal"; currently, only one terminal is supported*/
    "PicFeaturePoints": [
    /*opt, array, feature points to be applied. If the device only supports three types of feature points, when the platform applies more than three types
of feature points,the device will not return error information, subType:object*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of
mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, point coordinates*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized X-coordinate which is between 0 and 1000*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized Y-coordinate which is between 0 and 1000*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:width which is between 0 and 1000. This node is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:height which is between 0 and 1000. This node is required when featurePointType is "face"*/
            }
        }
    ],
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string*/
    "saveFacePic":  true,
    /*opt, bool, whether to save face pictures*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, subType:int, range:[1,4]*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceURL	[Message content]	application/json	--	--	--
img	[Binary picture data]	image/jpeg		faceImage.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error details, desc:this field is required when statusCode is not 1*/
}
```

14.4.2.5 Move face data in the face picture library in a batch

Request URL

POST /ISAPI/Intelligent/FDLib/FDMove?format=json

Query Parameter

None

Request Message

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```
{
    "FDMove": [
    /*req, array, subType:object*/
        {
            "id":  1,
            /*req, int, serial No. of the element in the array,it starts from 1 and is used to match returned results*/
            "FDSrc": {
            /*opt, object, face records to be moved*/
                "FDID":  "test",
                /*req, string, face picture library ID*/
                "FPID":  "test",
                /*req, string, face record ID*/
                "humanId":  "humanIDXXX"
                /*opt, string, person ID*/
            },
            "FDDst": {
            /*opt, object, destination face picture library (face records will be moved to this face picture library)*/
                "FDID":  "test",
                /*req, string, face picture library ID, desc:the maximum length is 63 bytes. Multiple libraries are separated by commas*/
                "FPID":  " ",
                /*req, string, face record ID*/
                "humanId":  "humanIDXXX",
                /*opt, string, person ID*/
                "name":  "test",
                /*opt, string, name*/
                "gender":  "any",
                /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used for search), "male", "female", "unknown"*/
                "bornTime":  "2004-05-03",
                /*req, string, birth date of the person in the face picture, range:[0,20]*/
                "city":  "130100",
                /*opt, string, city code, desc:city of birth*/
                "certificateType":  "ID",
                /*req, enum, certificate type, subType:string, desc:certificate type*/
                "certificateNumber":  "test",
                /*opt, string, certificate No.*/
                "caseInfo":  "test",
                /*opt, string, remarks*/
                "tag":  "aa,bb,cc,dd",
                /*opt, string, custom tag*/
                "address":  "test",
                /*opt, string, person address*/
                "customInfo":  "test",
                /*opt, string, custom information*/
                "rowKey ":  "test",
                /*opt, string, key of the face record database,the maximum size is 64 bytes. Searching by rowKey can be efficient*/
                "transfer":  true
                /*opt, bool, whether to enable transferring, desc:whether to enable transferring*/
            }
        }
    ]
}
```

Response Message

```
{
    "FDMoveResult": [
    /*ro, req, array, result of batch moving, subType:object*/
        {
            "id":  1,
            /*ro, req, int, serial No. of the element in the array,it starts from 1 and is used to match returned results, desc:serial No. of the element in
the array,it starts from 1 and is used to match returned results*/
            "errorCode":  1,
            /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
            "errorMsg":  "ok"
            /*ro, req, string, error description, desc:this node is required when the value of statusCode is not 1*/
        }
    ]
}
```

14.4.2.6 Delete the face record(s) in the face picture library

Request URL

PUT /ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&FDID=<FDID>&faceLibType=<FDType>

Query Parameter

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Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID, and the maximum length is 64 bytes. Create a face picture library: POST /ISAPI/Intelligent/FDLib?format=json, and it is returned for success.
FDType	enum	Face picture library type: "blackFD" (list library), "staticFD" (static library)

Request Message

```
{
    "FPID": [
    /*req, array, list of face record ID, subType:object, desc:list of face record ID*/
        {
            "value":  "test",
            /*req, string, face record ID, range:[0,64]*/
            "rowKey":  "test"
            /*opt, string, main key of face picture library, range:[0,64], desc:main key of face picture library*/
        }
    ],
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal" (by terminal)*/
    "terminalNoList": [1]
    /*opt, array, terminal ID list, subType:int, desc:terminal ID list*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

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14.4.2.7 Edit a face record in a specific face picture library

Request URL

PUT /ISAPI/Intelligent/FDLib/FDSearch?format=json&FDID=<FDID>&FPID=<FPID>&faceLibType=<faceLibType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID, and the maximum length is 63 bytes. Create a face picture library: POST /ISAPI/Intelligent/FDLib?format=json, and it is returned for success.
FPID	string	Face record ID, and the maximum length is 63 bytes. Add face record: POST /ISAPI/Intelligent/FDLib?format=json, and it is returned for success.
faceLibType	string	Face picture library type: "blackFD" (list library), "staticFD" (static library).

Request Message

```
{
    "faceURL":  "test",
    /*opt, string, face picture URL, range:[0,256]*/
    "name":  "张三",
    /*req, string, name of person in the face picture, range:[0,96]*/
    "gender":  "male",
    /*opt, enum, gender of person in the face picture, subType:string, desc:"male", "female", "unknown"*/
    "bornTime":  "1970-01-01+08:00",
    /*req, date, birthday of person in the face picture, ISO8601 time format, the maximum size is 20 bytes*/
    "city":  "130100",
    /*opt, string, city code of birth for the person in the face picture, the maximum size is 32 bytes*/
    "certificateType":  "officerID",
    /*opt, enum, certificate type, subType:string, desc:certificate type*/
    "certificateNumber":  "test",
    /*opt, string, certificate No., the maximum size is 32 bytes*/
    "caseInfo":  "test",
    /*opt, string, remarks, range:[0,192], desc:case information,the maximum size is 192 bytes,it is valid when faceLibType is blackFD*/
    "tag":  "aa,bb,cc,dd",
    /*opt, string, custom tag, range:[0,195], desc:up to 4 tags, which are separated by commas. It is valid when faceLibType is blackFD*/
    "address":  "test",
    /*opt, string, person address, range:[0,192], desc:person address,the maximum size is 192 bytes,it is valid when faceLibType is staticFD.*/
    "customInfo":  "test",
    /*opt, string, custom information, range:[0,192], desc:custom information,the maximum size is 192 bytes,it is valid when faceLibType is staticFD.*/
    "modelData":  "test",
    /*opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method*/
    "rowKey":  "test",
    /*opt, string, main key of face picture library, range:[0,64], desc:searching by rowKey can be more efficient*/
    "transfer":  true,
    /*opt, bool, whether to enable transfer, desc:whether to enable transfer*/
    "PicFeaturePoints": [
    /*opt, array, feature points to be applied., subType:object, desc:If the device only supports three types of feature points, when the platform applies
more than three types of feature points, the device will not return error information*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of
mouth), "rightMouthCorner" (right corner of mouth), "nose"”*/
            "coordinatePoint": {
            /*opt, object, region, desc:object,coordinates of the feature point*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized X-coordinate which is between 0 and 1000*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized Y-coordinate which is between 0 and 1000*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:width which is between 0 and 1000. This node is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:height which is between 0 and 1000. This node is required when featurePointType is "face"*/
            }
        }
    ],
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string, desc:face picture type*/
    "saveFacePic":  true,
    /*opt, bool, whether to save face pictures*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, subType:int, range:[1,4]*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

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14.4.2.8 Search for the face records in the a face picture library or multiple face picture libraries

Request URL

POST /ISAPI/Intelligent/FDLib/FDSearch?format=json

Query Parameter

None

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults":  100,
    /*req, int, the maximum number of search results this time*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture library type: "blackFD"-list library,"staticFD"-static library,the maximum
size is 32 bytes*/
    "FDID":  "test",
    /*req, string, face picture library ID, desc:face picture library ID,the maximum size is 63*/
    "FPID":  " ",
    /*opt, string, face record ID*/
    "startTime":  "2004-05-03",
    /*opt, string, start date of birth*/
    "endTime":  "2004-05-03",
    /*opt, string, end time of birth*/
    "name":  "test",
    /*opt, string, name*/
    "gender":  "any",
    /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
    "city":  "130100",
    /*opt, string, city code, desc:city code of birth for the person in the face picture,the maximum size is 32 bytes*/
    "certificateType":  "ID",
    /*req, enum, certificate type, subType:string, desc:the maximum size is 10 bytes,certificate type: "officerID"-officer ID,"ID"-identify
card,passport,other*/
    "certificateNumber":  "test",
    /*opt, string, certificate No.*/
    "isInLibrary":  "yes",
    /*opt, enum, whether the picture is in library (whether modeling is successful), subType:string, desc:whether modeling is successful: unknown, no, yes*/
    "isDisplayCaptureNum":  true,
    /*opt, bool, whether to display number of captured pictures*/
    "rowKey":  "test",
    /*opt, string, main key of face picture library, desc:searching by rowKey can be more efficient*/
    "transfer":  true
    /*opt, bool, Transferring, desc:whether to enable transfer*/
}
```

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "activeNumMax",
    /*ro, req, enum, sub status code, subType:string*/
    "errorCode":  1,
    /*ro, opt, int, detailed error description, this field is required when the value of statusCode is not 1*/
    "errorMsg":  "ok",
    /*ro, opt, string, detailed error description, this field is required when the value of statusCode is not 1*/
    "responseStatusStrg":  "OK",
    /*ro, opt, enum, status search, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data).*/
    "numOfMatches":  1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results*/
    "MatchList": [
    /*ro, opt, array, list of matched records, subType:object*/
        {
            "FDID":  "test",
            /*ro, opt, string, face picture library ID*/
            "FDName":  "名单库A",
            /*ro, opt, string*/
            "FPID":  "test",
            /*ro, opt, string, face record ID*/
            "faceURL":  "test",
            /*ro, opt, string, face picture URL*/
            "name":  "test",
            /*ro, opt, string, name*/
            "gender":  "any",
            /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
            "bornTime":  "2004-05-03",
            /*ro, opt, string, birth date of the person in the face picture*/
            "city":  "130100",
            /*ro, opt, string*/
            "certificateType":  "ID",
            /*ro, req, enum, certificate type, subType:string*/
            "certificateNumber":  "test",
            /*ro, opt, string, certificate No.*/
            "caseInfo":  "test",
            /*ro, opt, string, remarks*/
            "tag":  "aa,bb,cc,dd",
            /*ro, opt, string, custom tag*/
            "address":  "test",
            /*ro, opt, string, person address*/
            "customInfo":  "test",
            /*ro, opt, string, custom information*/
            "isInLibrary":  "yes",
            /*ro, opt, enum, subType:string, desc:whether modeling is successful: unknown, no, yes*/
            "captureNum":  12,
            /*ro, opt, int, number of captured pictures*/
            "rowKey":  "test",
            /*ro, opt, string, main key of face picture library, desc:searching by rowKey can be more efficient*/
            "modelData":  "test",
            /*ro, opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64
method*/
            "faceType":  "normalFace",
            /*ro, opt, enum, face picture library type, subType:string*/
            "saveFacePic":  true,
            /*ro, opt, bool*/
            "leaderPermission": [1, 2, 3, 4],
            /*ro, opt, array, subType:int, range:[1,4]*/
            "numberOfArrivalsAtSiteLastDays": [
            /*ro, opt, array, subType:object*/
                {
                    "day":  1,
                    /*ro, opt, int, range:[1,7]*/
                    "numberOfArrivalsAtSite":  3
                    /*ro, opt, int, range:[1,99]*/
                }
            ],
            "dwellTimeOnPreviousDay":  4,
            /*ro, opt, int, range:[0,86400], unit:min*/
            "totalNumberOfArrivalsAtSite":  4
            /*ro, opt, int, range:[1,99]*/
        }
    ]
}
```

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14.4.2.9 Set the face picture data in the face picture library

Request URL

PUT /ISAPI/Intelligent/FDLib/FDSetUp?format=json

Query Parameter

None

Request Message

```
{
    "faceURL":  "http",
    /*opt, string, picture storage URL, which is inputted when uploading the face picture by URL*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:"blackFD" (list library), "staticFD" (static library), the maximum length is 32 bytes*/
    "FDID":  "test",
    /*req, string, face picture library ID, desc:the maximum length is 63 bytes. Multiple libraries are separated by commas*/
    "FPID":  " ",
    /*opt, string, face record ID, desc:it can be generated by the device or inputted. If it is inputted, it should be the unique ID with the combination of letters and digits, and the maximum length is 63 bytes; if it is generated by the device automatically, it is the same as the employee No. (person ID)*/
    "deleteFP":  true,
    /*opt, bool, whether to delete the face record, desc:true (yes). This field required when the face record needs to be deleted; for adding or editing the face record, it should not be configured*/
    "modelData":  "test",
    /*opt, string, target model data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method*/
    "PicFeaturePoints": [
    /*opt, array, feature picture coordinates, subType:object, desc:if the device only supports three types of feature points, when the platform applies more than three types of feature points, the device will not return error information*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, coordinates of the feature point*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized coordinates*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized coordinates*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:it is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:it is required when featurePointType is "face"*/
            }
        }
    ],
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string, desc:"normalFace" (normal face, default value), "patrolFace" (patrol person face), "hijackFace" (duress person face), "superFace" (super person face)*/
    "saveFacePic":  true,
    /*opt, bool, whether to save the registered face picture*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, first authentication permission, subType:int, range:[1,4], desc:first authentication permission*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceURL	[Message content]	application/json	--	--	--
img	[Binary picture data]	image/jpeg		faceImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.5 Two-Way Audio

14.5.1 Two-Way Audio

14.5.1.1 Get the parameters of all two-way audio channels

Request URL

GET /ISAPI/System/TwoWayAudio/channels

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, parameters configuration of the two-way audio channel, subType:object, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--ro, opt, object, configuration of the two-way audio channel-->
    <id>
      <!--ro, req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable or not-->true
    </enabled>
    <audioCompressionType>
      <!--ro, req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM",
"MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType>
      <!--ro, opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM",
"MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume>
      <!--ro, opt, int, input volume, range:[1,100]-->100
    </speakerVolume>
    <microphoneVolume>
      <!--ro, opt, int, output volume, range:[1,100]-->100
    </microphoneVolume>
    <noisereduce>
      <!--ro, opt, bool, whether to enable the environmental noise filter or not-->true
    </noisereduce>
    <audioBitRate>
      <!--ro, opt, int, audio frame rate, unit:kbs-->100
    </audioBitRate>
    <audioInputType>
      <!--ro, opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-
adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--ro, opt, object, linked video channel-->
      <enabled>
        <!--ro, req, bool, whether to enable the linked video input channel or not-->true
      </enabled>
      <videoInputChannelList>
        <!--ro, req, array, list of linked video channels, subType:object-->
        <videoInputChannelID>
          <!--ro, opt, string, linked video channel-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate>
      <!--ro, opt, float, audio sampling rate, unit:kHz-->48.00
    </audioSamplingRate>
    <lineOutForbidden>
      <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
    </lineOutForbidden>
    <muteDuringPanTilt>
      <!--ro, opt, bool, whether to mute during motion-->true
    </muteDuringPanTilt>
    <audioOutputType>
      <!--ro, opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <matrixPickUp>
      <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it
is valid only when the audio input type is "Linein")-->true
    </matrixPickUp>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

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14.5.1.2 Set audio parameters for all two-way audio channels

Request URL

PUT /ISAPI/System/TwoWayAudio/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameters configuration of the two-way audio channel, subType:object, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--opt, object, configuration of the two-way audio channel-->
    <id>
      <!--req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function or not-->true
    </enabled>
    <audioCompressionType>
      <!--req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType>
      <!--req, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume>
      <!--opt, int, input volume-->100
    </speakerVolume>
    <microphoneVolume>
      <!--opt, int, output volume-->100
    </microphoneVolume>
    <noisereduce>
      <!--opt, bool, whether to enable the environmental noise filter or not-->true
    </noisereduce>
    <audioBitRate>
      <!--opt, int, audio frame rate, unit:kbs-->100
    </audioBitRate>
    <audioInputType>
      <!--req, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--opt, object, linked video channel-->
      <enabled>
        <!--req, bool, whether to enable the linked video input channel or not-->true
      </enabled>
      <videoInputChannelList>
        <!--req, array, list of linked video channels, subType:object-->
        <videoInputChannelID>
          <!--opt, string, linked video channel-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate>
      <!--opt, float, audio sampling rate, unit:kHz-->48.00
    </audioSamplingRate>
    <lineOutForbidden>
      <!--opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
    </lineOutForbidden>
    <muteDuringPanTilt>
      <!--opt, bool, whether to mute during motion-->true
    </muteDuringPanTilt>
    <audioOutputType>
      <!--opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <matrixPickUp>
      <!--opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
    </matrixPickUp>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.1.3 Receive two-way audio data

Request URL

GET /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData?sessionId=<audioSessionID>

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--
audioSessionID	string	--

Request Message

Binary Data

Response Message

Binary Data

14.5.1.4 Send two-way audio data

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData?sessionId=<audioSessionID>&type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--
audioSessionID	string	Two-way audio session ID
type	enum	If this filed does not exist, it indicates two-way audio via channel, and the audioID is the channel ID. When the type is childDevID, it indicates two-way audio with a single sub-device, and the audioID is the gateway intercom resource index (obtained from /ISAPI/System/DynamicResource/SearchResources?format=json); when the type is childDevList, it indicates broadcasting to multiple sub-devices, with multiple childDevID values configured in /ISAPI/System/TwoWayAudio/channels//open.

Request Message

Binary Data

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, Custom Error Information Description, range:[0,1024], desc:Custom detailed error information returned by device application, used
for quick positioning and diagnosis.-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.5.1.5 Set audio parameters of a specified two-way audio channel

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, configuration of the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType>
    <!--req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume>
    <!--opt, int, input volume, range:[1,100]-->100
  </speakerVolume>
  <microphoneVolume>
    <!--opt, int, output volume, range:[1,100]-->100
  </microphoneVolume>
  <noisereduce>
    <!--opt, bool, whether to enable the environmental noise filter or not-->true
  </noisereduce>
  <audioBitRate>
    <!--opt, int, audio frame rate, unit:kbs-->100
  </audioBitRate>
  <audioInputType>
    <!--opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--opt, object, linked video channel-->
    <enabled>
      <!--req, bool, whether to enable the linked video input channel or not-->true
    </enabled>
    <videoInputChannelList>
      <!--req, array, list of linked video channels, subType:object-->
      <videoInputChannelID>
        <!--opt, string, linked video channel-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate>
    <!--opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <lineOutForbidden>
    <!--opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the value
is "true", audio output is not supported-->true
  </lineOutForbidden>
  <micInForbidden>
    <!--opt, bool, whether the audio input is not supported. If this node is not returned or its value is "false", audio input is supported;  if the value
is "true", audio input is not supported-->true
  </micInForbidden>
  <muteDuringPanTilt>
    <!--opt, bool, whether to mute during motion-->true
  </muteDuringPanTilt>
  <audioOutputType>
    <!--opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <timeOut>
    <!--opt, int, timeout during two-way audio, range:[10,600], unit:s, desc:when the Web Client initiates the two-way audio actively towards the radar
device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some noise.
When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client stops
speaking or not.-->10
  </timeOut>
  <matrixPickUp>
    <!--opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
  </matrixPickUp>
</TwoWayAudioChannel>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.1.6 Get the capability of configuring audio parameters for a specified two-way audio channel

Request URL

GET /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/capabilities?audioInputType=<audioInputType>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
audioInputType	string	Audio input type, corresponding to the "audioInputType" in the message. This parameter is mainly used to obtain the parameters corresponding to the specified audio input type.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, capability of configuring audios for the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2,Opus,G.722.1">
    <!--ro, req, enum, encoding type of the audio output, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711uLaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,G.722.1">
    <!--ro, opt, enum, encoding type of the audio input, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711uLaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume min="0" max="100">
    <!--ro, opt, int, input volume, attr:min{opt, int},max{opt, int}-->100
  </speakerVolume>
  <microphoneVolume min="0" max="100">
    <!--ro, opt, int, output volume, attr:min{opt, int},max{opt, int}-->100
  </microphoneVolume>
  <noisereduce opt="true,false">
    <!--ro, opt, bool, whether to enable the environmental noise filter or not, attr:opt{opt, string}-->true
  </noisereduce>
  <audioBitRate min="1" max="100">
    <!--ro, opt, int, audio frame rate, unit:kbs, attr:min{opt, int},max{opt, int}-->100
  </audioBitRate>
  <audioInputType opt="MicIn,LineIn,selfAdaptive,wirelessPickUp,doubleMic,networkPickUp">
    <!--ro, opt, enum, audio input type, subType:string, attr:opt{opt, string}, desc:"MicIn" (microphone-level input), "LineIn" (line-level input),
"selfAdaptive" (self-adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--ro, opt, object, linked video channel-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the linked video input channel or not, attr:opt{opt, string}-->true
    </enabled>
    <videoInputChannelList>
      <!--ro, req, array, list of linked video channels, subType:object-->
      <videoInputChannelID min="1" max="2048">
        <!--ro, opt, string, linked video channel, attr:min{opt, string},max{opt, string}-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate min="1.00" max="48.00">
    <!--ro, opt, float, audio sampling rate, unit:kHz, attr:min{opt, float},max{opt, float}-->48.00
  </audioSamplingRate>
  <lineOutForbidden opt="true,false">
    <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported, attr:opt{opt, string}-->true
  </lineOutForbidden>
  <muteDuringPanTilt opt="true,false">
    <!--ro, opt, bool, whether to mute during motion, attr:opt{opt, string}-->true
  </muteDuringPanTilt>
  <audioOutputType opt="Close,LineOut,Speaker,selfAdaptive,LineOut_Speaker,HDMI" def="LineOut">
    <!--ro, opt, enum, audio output type, subType:string, attr:opt{opt, string},def{opt, string}, desc:"Close" (output closed), "LineOut" (line-level
output), "Speaker" (speaker output), "selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <isSupportEchoCancellation>
    <!--ro, opt, bool, whether the device supports echo cancellation-->true
  </isSupportEchoCancellation>
  <timeOut min="10" max="600">
    <!--ro, opt, int, timeout during two-way audio, attr:min{req, int},max{req, int}, desc:when the Web Client initiates the two-way audio actively towards
the radar device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some
noise. When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client
stops speaking or not.-->300
  </timeOut>
  <matrixPickUp opt="true,false">
    <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it is
valid only when the audio input type is "Linein"), attr:opt{opt, string}-->false
  </matrixPickUp>
  <isSupportRestore>
    <!--ro, opt, bool, whether the device supports restoring audio parameters of the two-way audio channel, desc:corresponding API:
/ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/restore?format=json-->true
  </isSupportRestore>
  <networkPickUpStatus opt="online,offline">
    <!--ro, opt, enum, network pick-up status, subType:string, dep:and,{$.TwoWayAudioChannel.audioInputType,eq,networkPickUp}, attr:opt{opt, string},
desc:network status of the network pickup linked with the audio channel: "online", "offline"-->online
  </networkPickUpStatus>
  <noisereduceType opt="normal,smart">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->normal
  </noisereduceType>
</TwoWayAudioChannel>
```

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14.5.1.7 Stop two-way audio

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/close?sessionId=<audioSessionID>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-Way Audio Channel ID
audioSessionID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, Custom Error Information Description, range:[0,1024], desc:Custom detailed error information returned by device application, used
for quick positioning and diagnosis.-->badXmlFormat
  </description>
</ResponseStatus>
```

14.5.1.8 Start two-way audio

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/open?type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
type	enum	If this filed does not exist, it indicates two-way audio via channel, and the audioID is the channel ID. When the type is childDevID, it indicates two-way audio with a single sub device, and the audioID is the gateway intercom resource index (obtained from /ISAPI/System/DynamicResource/SearchResources?format=json); when the type is childDevList, it indicates broadcasting to multiple sub devices, with multiple childDevID values specified in the message, and the audioID in the URL is invalid.

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudio xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters for this two-way audio, attr:version{req, string, protocolVersion}-->
  <audioLevel>
    <!--opt, int, audio level, range:[0,15], desc:range: [0,15]-->0
  </audioLevel>
  <microphoneVolume>
    <!--opt, int, output volume-->100
  </microphoneVolume>
  <audioCompressionType>
    <!--opt, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioSamplingRate>
    <!--opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <isBroadcast>
    <!--opt, bool-->true
  </isBroadcast>
  <broadcastType>
    <!--opt, enum, subType:string-->alarm
  </broadcastType>
  <childList>
    <!--opt, array, subType:object, range:[0,8], dep:and,{$.TwoWayAudio.isBroadCast,eq,true}-->
    <childDevID>
      <!--opt, string, range:[9,32]-->test
    </childDevID>
    <voiceTalkIndex>
      <!--opt, int, range:[1,256]-->test
    </voiceTalkIndex>
  </childList>
</TwoWayAudio>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioSession xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, result returned when starting two-way audio succeeds, attr:version{opt, string, protocolVersion}-->
  <sessionId>
    <!--ro, req, string, two-way audio session ID-->1
  </sessionId>
</TwoWayAudioSession>
```

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14.5.1.9 Get the audio parameters of a specified two-way audio channel

Request URL

GET /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>?audioInputType=<audioInputType>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
audioInputType	string	Audio input type, corresponding to the "audioInputType" in the message. This parameter is mainly used to obtain the parameters corresponding to the specified audio input type.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, configuration of the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType>
    <!--ro, req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--ro, opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume>
    <!--ro, opt, int, input volume, range:[1,100]-->100
  </speakerVolume>
  <microphoneVolume>
    <!--ro, opt, int, output volume, range:[1,100]-->100
  </microphoneVolume>
  <noisereduce>
    <!--ro, opt, bool, whether to enable the environmental noise filter or not-->true
  </noisereduce>
  <audioBitRate>
    <!--ro, opt, int, audio frame rate, unit:kbs-->100
  </audioBitRate>
  <audioInputType>
    <!--ro, opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--ro, opt, object, linked video channel-->
    <enabled>
      <!--ro, req, bool, whether to enable the linked video input channel or not-->true
    </enabled>
    <videoInputChannelList>
      <!--ro, req, array, list of linked video channels, subType:object-->
      <videoInputChannelID>
        <!--ro, opt, string, linked video channel-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate>
    <!--ro, opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <lineOutForbidden>
    <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
  </lineOutForbidden>
  <micInForbidden>
    <!--ro, opt, bool, whether the audio input is not supported. If this node is not returned or its value is "false", audio input is supported;  if the
value is "true", audio input is not supported-->true
  </micInForbidden>
  <muteDuringPanTilt>
    <!--ro, opt, bool, whether to mute during motion-->true
  </muteDuringPanTilt>
  <audioOutputType>
    <!--ro, opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <timeOut>
    <!--ro, opt, int, timeout during two-way audio, range:[10,600], unit:s, desc:when the Web Client initiates the two-way audio actively towards the radar
device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some noise.
When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client stops
speaking or not.-->10
  </timeOut>
  <matrixPickUp>
    <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
  </matrixPickUp>
  <networkPickUpStatus>
    <!--ro, opt, enum, subType:string, dep:and,{$.TwoWayAudioChannel.audioInputType,eq,networkPickUp}-->online
  </networkPickUpStatus>
  <noisereduceType>
    <!--ro, opt, enum, subType:string-->normal
  </noisereduceType>
</TwoWayAudioChannel>
```

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14.5.1.10 Get the capability of configuring parameters for all two-way audio channels

Request URL

GET /ISAPI/System/TwoWayAudio/channels/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, capability of configuring audios for all two-way audio channels, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--ro, opt, object, capability of configuring audios for the two-way audio channel-->
    <id>
      <!--ro, req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable or not-->true
    </enabled>
    <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2,Opus">
      <!--ro, req, enum, encoding type of the audio output, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729",
"G.729a", "G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM">
      <!--ro, opt, enum, encoding type of the audio input, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume min="0" max="100">
      <!--ro, opt, int, input volume, attr:min{opt, int},max{opt, int}-->100
    </speakerVolume>
    <microphoneVolume min="0" max="100">
      <!--ro, opt, int, output volume, attr:min{opt, int},max{opt, int}-->100
    </microphoneVolume>
    <noisereduce opt="true,false">
      <!--ro, opt, bool, whether to enable the environmental noise filter or not, attr:opt{opt, string}-->true
    </noisereduce>
    <audioBitRate min="1" max="100">
      <!--ro, opt, int, audio frame rate, unit:kbs, attr:min{opt, string},max{opt, string}-->100
    </audioBitRate>
    <audioInputType opt="MicIn,LineIn,selfAdaptive,wirelessPickUp">
      <!--ro, opt, enum, audio input type, subType:string, attr:opt{opt, string}, desc:"MicIn" (microphone-level input), "LineIn" (line-level input),
"selfAdaptive" (self-adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--ro, opt, object, linked video channel-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the linked video input channel or not, attr:opt{opt, string}-->true
      </enabled>
      <videoInputChannelList>
        <!--ro, req, array, list of linked video channels, subType:object-->
        <videoInputChannelID min="1" max="2048">
          <!--ro, opt, string, linked video channel, attr:min{opt, string},max{opt, string}-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate min="1.00" max="48.00">
      <!--ro, opt, float, audio sampling rate, unit:kHz, attr:min{opt, float},max{opt, float}-->48.00
    </audioSamplingRate>
    <lineOutForbidden opt="true,false">
      <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported, attr:opt{opt, string}-->true
    </lineOutForbidden>
    <muteDuringPanTilt opt="true,false">
      <!--ro, opt, bool, whether to mute during motion, attr:opt{opt, string}-->true
    </muteDuringPanTilt>
    <audioOutputType opt="Close,LineOut,Speaker,selfAdaptive,LineOut_Speaker" def="LineOut">
      <!--ro, opt, enum, audio output type, subType:string, attr:opt{opt, string},def{opt, string}, desc:"Close" (output closed), "LineOut" (line-level
output), "Speaker" (speaker output), "selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <isSupportEchoCancellation>
      <!--ro, opt, bool, whether the device supports echo cancellation-->true
    </isSupportEchoCancellation>
    <matrixPickUp opt="true,false">
      <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it
is valid only when the audio input type is "Linein"), attr:opt{opt, string}-->false
    </matrixPickUp>
    <isSupportRestore>
      <!--ro, opt, bool, whether the device supports restoring audio parameters of the two-way audio channel, desc:corresponding API:
/ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/restore?format=json-->true
    </isSupportRestore>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

14.6 Access Control (General)

14.6.1 Access Control Event Management

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14.6.1.1 Get the capability of searching for access control events

Request URL

GET /ISAPI/AccessControl/AcsEvent/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AcsEvent": {
    /*ro, req, object, access control events*/
        "AcsEventCond": {
        /*ro, opt, object, search conditions*/
            "searchID": {
            /*ro, req, object, search ID, it is used to check whether the current search requester is the same as the previous one. If they are the same,
the search record will be stored in the device to speed up the next search*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "searchResultPosition": {
            /*ro, req, object, the start position of the search result in the result list*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "maxResults": {
            /*ro, req, object, the maximum number of search results that can be obtained by calling this URL*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "major": {
            /*ro, opt, object, major alarm type (the type value should be transformed to the decimal number)*/
                "@opt":  "0,1,2,3,5"
                /*ro, req, string, major type*/
            },
            "minorAlarm": {
            /*ro, opt, object, minor alarm type (the type value should be transformed to the decimal number)*/
                "@opt":  "1024,1025,1026,1027…"
                /*ro, req, string, minor alarm type*/
            },
            "minorException": {
            /*ro, opt, object, minor exception type (the type value should be transformed to the decimal number)*/
                "@opt":  "39,58,59,1024…"
                /*ro, req, string, minor exception type*/
            },
            "minorOperation": {
            /*ro, opt, object, minor operation type (the type value should be transformed to the decimal number)*/
                "@opt":  "80,90,112,113…"
                /*ro, req, string, minor operation type*/
            },
            "minorEvent": {
            /*ro, opt, object, minor event type (the type value should be transformed to the decimal number)*/
                "@opt":  "1,2,3,4…"
                /*ro, req, string, minor event type*/
            },
            "startTime": {
            /*ro, opt, object, start time*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,32]*/
                "@max":  32
                /*ro, req, int, the maximum value, range:[0,32]*/
            },
            "endTime": {
            /*ro, opt, object, end time*/
                "@min":  0,
                /*ro, req, int, the minimum value, range:[0,32]*/
                "@max":  32
                /*ro, req, int, the maximum value, range:[0,32]*/
            },
            "cardNo": {
            /*ro, opt, object, card No.*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "name": {
            /*ro, opt, object, name of the card holder*/
```

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```
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "picEnable":  "true,false",
        /*ro, opt, string, whether to include pictures*/
        "beginSerialNo": {
        /*ro, opt, object, start serial No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "endSerialNo": {
        /*ro, opt, object, end serial No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "employeeNoString": {
        /*ro, opt, object, employee No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
    },
    "InfoList": {
    /*ro, opt, object, information list*/
        "maxSize":  10,
        /*ro, opt, int, the maximum value*/
        "time": {
        /*ro, opt, object, time (UTC time)*/
            "@min":  0,
            /*ro, req, int, the minimum value, range:[0,32]*/
            "@max":  32
            /*ro, req, int, the maximum value, range:[0,32]*/
        },
        "netUser": {
        /*ro, opt, object, user name*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "remoteHostAddr": {
        /*ro, opt, object, remote host address*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "cardNo": {
        /*ro, opt, object, card No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "cardType": {
        /*ro, opt, object, card type*/
            "@opt":  "1,2,3,4,5,6,7,8"
            /*ro, req, string, 1 (normal card), 2 (disability card), 3 (blocklist card), 4 (patrol card), 5 (duress card), 6 (super card), 7 (visitor
card), 8 (dismiss card)*/
        },
        "cardReaderNo": {
        /*ro, opt, object, card reader No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "doorNo": {
        /*ro, opt, object, door (floor) No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "employeeNo": {
        /*ro, opt, object, employee No. (person ID)*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "serialNo": {
        /*ro, opt, object, event serial No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
```

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            /*ro, req, int, the maximum value*/
        },
        "userType": {
        /*ro, opt, object, person types*/
            "@opt":  "normal,visitor,blackList,administrators"
            /*ro, req, string, "normal" (normal person (household)), "visitor" (visitor), "blacklist" (person in blocklist), "administrators"
(administrator)*/
        },
        "currentVerifyMode": {
        /*ro, opt, object, current authentication mode of the card reader*/
            "@opt":
"cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpOrCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employe
eNoAndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,faceOrfaceAndCard,fpOrface,cardOrfaceOrPw,iris,faceOrFpOrCardOrPwOrIris,face
OrCardOrPwOrIris"
            /*ro, req, string, "cardAndPw"-card+password, "card", "cardOrPw"-card or password, "fp"-fingerprint, "fpAndPw"-fingerprint+password,
"fpOrCard"-fingerprint or card, "fpAndCard"-fingerprint+card, "fpAndCardAndPw"-fingerprint+card+password, "faceOrFpOrCardOrPw"-face or fingerprint or card
or password, "faceAndFp"-face+fingerprint, "faceAndPw"-face+password, "faceAndCard"-face+card, "face", "employeeNoAndPw"-employee No.+password, "fpOrPw"-
fingerprint or password, "employeeNoAndFp"-employee No.+fingerprint, "employeeNoAndFpAndPw"-employee No.+fingerprint+password, "faceAndFpAndCard"-
face+fingerprint+card, "faceAndPwAndFp"-face+password+fingerprint, "employeeNoAndFace"-employee No.+face, "faceOrfaceAndCard"-face or face+card, "fpOrface"-
fingerprint or face, "cardOrfaceOrPw"-card or face or password, "cardOrFpOrPw"-card or fingerprint or password*/
        },
        "QRCodeInfo": {
        /*ro, opt, object, QR code information*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "thermometryUnit": {
        /*ro, opt, object, temperature unit*/
            "@opt": ["celsius", "fahrenheit", "kelvin"]
            /*ro, req, array, "celsius" (Celsius (default)), "fahrenheit" (Fahrenheit), "kelvin" (Kelvin), subType:string*/
        },
        "currTemperature": {
        /*ro, opt, object, skin-surface temperature*/
            "@min":  1,
            /*ro, req, int, skin-surface temperature, which is accurate to one decimal place*/
            "@max":  1
            /*ro, req, int, skin-surface temperature, which is accurate to one decimal place*/
        },
        "isAbnomalTemperature": {
        /*ro, opt, object, whether the skin-surface temperature is abnormal*/
            "@opt": [true, false]
            /*ro, req, array, whether the skin-surface temperature is abnormal (true-yes), subType:bool*/
        },
        "RegionCoordinates": {
        /*ro, opt, object, coordinates of the skin-surface temperature*/
            "positionX": {
            /*ro, opt, object, X-coordinate*/
                "@min":  0,
                /*ro, req, int, the minimum value, normalized X-coordinate which is between 0 and 1000*/
                "@max":  1000
                /*ro, req, int, the maximum value, normalized X-coordinate which is between 0 and 1000*/
            },
            "positionY": {
            /*ro, opt, object, Y-coordinate*/
                "@min":  0,
                /*ro, req, int, the minimum value, normalized Y-coordinate which is between 0 and 1000*/
                "@max":  1000
                /*ro, req, int, the maximum value, normalized Y-coordinate which is between 0 and 1000*/
            }
        },
        "mask": {
        /*ro, opt, object, whether the person is wearing mask*/
            "@opt": ["unknown", "yes", "no"]
            /*ro, req, array, "unknown", "yes", "no", subType:string*/
        },
        "pictureURL": {
        /*ro, opt, object, URL of the captured picture*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "attendanceStatus": {
        /*ro, opt, object, attendance status, desc:"undefined", "checkIn" (check-in), "checkOut" (check-out), "breakOut" (start of break), "breakIn"
(end of break), "overtimeIn" (start of overtime), "overTimeOut" (end of overtime)*/
            "@opt":  "undefined,checkIn,checkOut,breakOut,breakIn,overtimeIn,overtimeOut"
            /*ro, req, string, options*/
        },
        "label": {
        /*ro, opt, object, self-defined attendance name*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "statusValue": {
        /*ro, opt, object, status value*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        }
```

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```
            },
            "helmet": {
            /*ro, opt, object, whether the person is wearing hard hat, desc:"unknown", "yes", "no"*/
                "@opt":  "unknown,yes,no"
                /*ro, req, string, options*/
            },
            "thermalPicUrl": {
            /*ro, opt, object, URL of the thermal imaging picture*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "HealthInfo": {
            /*ro, opt, object, health information*/
                "healthCode": {
                /*ro, opt, object, health code status*/
                    "@opt": [0, 1, 2, 3, 4, 5, 6, 7]
                    /*ro, req, array, options, subType:int, desc:0 (no request), 1 (no health code), 2 (green QR code), 3 (yellow QR code), 4 (red QR code),
5 (no such person), 6 (other error, e.g., searching failed due to API exception), 7 (searching for the health code timed out)*/
                },
                "NADCode": {
                /*ro, opt, object, nucleic acid test result, desc:0 (no result), 1 (negative, which means normal), 2 (positive, which means diagnosed), 3
(the result has expired)*/
                    "@opt": [0, 1, 2, 3, 4]
                    /*ro, req, array, options, subType:int*/
                },
                "travelCode": {
                /*ro, opt, object, trip code, desc:0 (no trip in the past 14 days), 1 (has left the current area left in the past 14 days), 2 (has been to
the high-risk area in the past 14 days), 3 (other)*/
                    "@opt": [0, 1, 2, 3, 4]
                    /*ro, req, array, options, subType:int*/
                },
                "travelInfo": {
                /*ro, opt, object*/
                    "@min":  0,
                    /*ro, req, int, step:1*/
                    "@max":  15
                    /*ro, req, int, step:1*/
                },
                "vaccineStatus": {
                /*ro, opt, object, whether the person is vaccinated, desc:0 (not vaccinated), 1 (vaccinated)*/
                    "@opt": [0, 1, 2, 3, 4]
                    /*ro, req, array, options, subType:int*/
                },
                "vaccineNum": {
                /*ro, opt, object*/
                    "@min":  0,
                    /*ro, req, int, step:1*/
                    "@max":  3
                    /*ro, req, int, step:1*/
                }
            },
            "FaceRect": {
            /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                "height": {
                /*ro, req, object, height*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value*/
                },
                "width": {
                /*ro, req, object, width*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value*/
                },
                "x": {
                /*ro, req, object, X-coordinate of the upper-left corner of the frame*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value*/
                },
                "y": {
                /*ro, req, object, Y-coordinate of the upper-left corner of the frame*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value*/
                }
            },
            "currentAuthenticationTimes": {
            /*ro, req, object*/
                "@min":  0,
                /*ro, req, int*/
                "@max":  255
                /*ro, req, int*/
            },
            "allowAuthenticationTimes": {
            /*ro, req, object*/
                "@min":  0,
```

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```
                /*ro, req, int*/
                "@max":  255
                /*ro, req, int*/
            }
        }
    }
}
```

14.6.1.2 Get the configuration capability of storing access control events

Request URL

GET /ISAPI/AccessControl/AcsEvent/StorageCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "EventStorageCfgCap": {
    /*ro, req, object, configuration capability of event storing*/
        "mode": {
        /*ro, req, object, event storage method, desc:"regular" (delete old events periodically), "time" (delete old events by specified time), "cycle"
(overwriting)*/
            "@opt": ["regular", "time", "cycle"]
            /*ro, opt, array, subType:string*/
        },
        "checkTime": {
        /*ro, opt, object, check time, desc:this node is valid when mode is "time". Events that occurred before the check time will be deleted*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "period": {
        /*ro, opt, object, time period for deleting old events, desc:this node is valid when mode is "regular"*/
            "@min":  10,
            /*ro, opt, int, the minimum value*/
            "@max":  10
            /*ro, opt, int, the maximum value*/
        }
    }
}
```

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14.6.1.3 Get the storage parameters of access control events

Request URL

GET /ISAPI/AccessControl/AcsEvent/StorageCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "EventStorageCfg": {
    /*ro, req, object*/
        "mode":  "regular",
        /*ro, req, enum, event storage mode, subType:string, desc:"regular" (delete old events periodically), "time" (delete old events by specified time),
"cycle" (overwriting)*/
        "checkTime":  "1970-01-01 00:00:00",
        /*ro, opt, string, check time. This node is required when the storage mode is "time"*/
        "period":  10
        /*ro, opt, int, time period for deleting old events. This node is required when the storage mode is "regular", unit:min, desc:unit: minute*/
    }
}
```

14.6.1.4 Set storage parameters of access control events

Request URL

PUT /ISAPI/AccessControl/AcsEvent/StorageCfg?format=json

Query Parameter

None

Request Message

```
{
    "EventStorageCfg": {
    /*wo, req, object*/
        "mode":  "regular",
        /*wo, req, enum, event storage method, subType:string, desc:"regular" (delete old events periodically), "time" (delete old events by specified
time), "cycle" (overwriting);*/
        "checkTime":  "1970-01-01 00:00:00",
        /*wo, opt, string, check time; this node is valid when mode is "time"*/
        "period":  10
        /*wo, opt, int, time period for deleting old events; this node is valid when mode is "regular", unit:min, desc:time period for deleting old events;
this node is valid when mode is "regular"*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

14.6.1.5 Search for access control events

Request URL

POST /ISAPI/AccessControl/AcsEvent?format=json

Query Parameter

None

Request Message

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```
{
    "AcsEventCond": {
    /*req, object, access control events*/
        "searchID":  "test",
        /*req, string, search ID, desc:it is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device to speed up the next search*/
        "searchResultPosition":  0,
        /*req, int, the start position of the search result in the result list:, desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search. If the maximum number of totalMatches supported by the device is M and the number of totalMatches stored in the device now is N (N<=M), the valid range of this node is 0 to N-1*/
        "maxResults":  30,
        /*req, int, the maximum number of search results, which is defined by the device capability, will be returned if the value of maxResults reaches the limit, desc:if maxResults exceeds the range returned by the device capability, the device will return the maximum number of search results according to the device capability and will not return error message*/
        "major":  1,
        /*req, int, major type, desc:the type value should be transformed to the decimal number; see Access Control Event Types for details*/
        "minor":  1024,
        /*req, int, minor type, desc:the type value should be transformed to the decimal number; see Access Control Event Types for details*/
        "startTime":  "1970-01-01T00:00:00+08:00",
        /*opt, datetime, start time (UTC time)*/
        "endTime":  "1970-01-01T00:00:00+08:00",
        /*opt, datetime, end time (UTC time)*/
        "cardNo":  "test",
        /*opt, string, card No.*/
        "name":  "test",
        /*opt, string, name of the card holder*/
        "videoChannel":  1,
        /*opt, int, video channel No., range:[1,86400], desc:this node is newly added to DeepinMind devices for attendance*/
        "picEnable":  true,
        /*opt, bool, whether to upload the picture along with the event information, desc:false (no), true (yes, default value); (1. all matched events will be uploaded without pictures; 2. all matched events will be uploaded with pictures if there are any; 3. if this node is not configured, the default value is true)*/
        "beginSerialNo":  1,
        /*opt, int, start serial No.*/
        "endSerialNo":  1,
        /*opt, int, end serial No.*/
        "employeeNoString":  "test",
        /*opt, string, employee No. (person ID)*/
        "timeReverseOrder":  true,
        /*opt, bool, whether to return events in descending order of time (later events will be returned first), desc:true (yes), false or this node is not returned (no)*/
        "isAbnomalTemperature":  true,
        /*opt, bool, whether the skin-surface temperature is abnormal*/
        "temperatureSearchCond":  "all",
        /*opt, enum, temperature search condition, subType:string, desc:when this node and isAbnormalTemperature both exist, isAbnormalTemperature is invalid; "all" (event with temperature), "normal" (event with normal temperature), "abnormal" (event with abnormal temperature)*/
        "isAttendanceInfo":  true,
        /*opt, bool, whether it contains attendance records, desc:this node is newly added to HEOP protocol; if this node is true, main type, minor type, employee No., name, and time will be returned*/
        "hasRecordInfo":  true
        /*opt, bool*/
    }
}
```

Response Message

```
{
    "AcsEvent": {
    /*ro, req, object, access control events*/
        "searchID":  "test",
        /*ro, req, string, search ID, it is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device to speed up the next search*/
        "responseStatusStrg":  "OK",
        /*ro, req, string, searching status description*/
        "numOfMatches":  1,
        /*ro, req, int, number of results returned this time*/
        "totalMatches":  1,
        /*ro, req, int, total number of matched results*/
        "InfoList": [
        /*ro, opt, array, information list, subType:object*/
            {
                "major":  1,
                /*ro, req, int, major alarm type*/
                "minor":  1,
                /*ro, req, int, minor alarm type*/
                "time":  "2016-12-12T17:30:08+08:00",
                /*ro, req, string, time (UTC time)*/
                "netUser":  "test",
                /*ro, opt, string, user name*/
                "remoteHostAddr":  "test",
                /*ro, opt, string, remote host address*/
                "videoChannel":  1,
                /*ro, opt, int, video channel No., range:[1,86400], desc:this node is newly added to DeepinMind devices for attendance*/
                "cardNo":  "test",
                /*ro, opt, string, card No.*/
                "cardType":  1,
                /*ro, opt, enum, card type, subType:int, desc:1 (normal card), 2 (disability card), 3 (blocklist card), 4 (patrol card), 5 (duress card), 6 (super card), 7 (visitor card), 8 (dismiss card)*/
                "whiteListNo":  1,
```

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```
                /*ro, opt, int, allowlist No.*/
                "reportChannel": 1,
                /*ro, opt, int, channel type for uploading alarm/event*/
                "cardReaderKind": 1,
                /*ro, opt, int, card reader type: 1 (IC card reader)*/
                "cardReaderNo": 1,
                /*ro, opt, int, card reader No.*/
                "doorNo": 1,
                /*ro, opt, int, door or floor No.*/
                "verifyNo": 1,
                /*ro, opt, int, multi-factor authentication No.*/
                "alarmInNo": 1,
                /*ro, opt, int, alarm input No.*/
                "alarmOutNo": 1,
                /*ro, opt, int, alarm output No.*/
                "caseSensorNo": 1,
                /*ro, opt, int, event trigger No.*/
                "RS485No": 1,
                /*ro, opt, int, RS-485 channel No.*/
                "multiCardGroupNo": 1,
                /*ro, opt, int, group No.*/
                "accessChannel": 1,
                /*ro, opt, int, RS-485 channel No.*/
                "deviceNo": 1,
                /*ro, opt, int, device No.*/
                "distractControlNo": 1,
                /*ro, opt, int, distributed controller No.*/
                "employeeNoString": "test",
                /*ro, opt, string, employee No. (person ID)*/
                "localControllerID": 1,
                /*ro, opt, int, distributed controller No.*/
                "InternetAccess": 1,
                /*ro, opt, int, network interface No.*/
                "type": 1,
                /*ro, opt, int, zone type, desc:0 (instant alarm zone), 1 (24-hour zone), 2 (delayed zone), 3 (internal zone), 4 (key zone), 5 (fire alarm zone), 6 (perimeter zone), 7 (24-hour silent zone), 8 (24-hour auxiliary zone), 9 (24-hour shock zone), 10 (emergency door open zone), 11 (emergency door closed zone), 255 (none)*/
                "MACAddr": "test",
                /*ro, opt, string, MAC address*/
                "swipeCardType": 1,
                /*ro, opt, enum, card swiping type, subType:int, desc:0 (invalid), 1 (QR code)*/
                "serialNo": 1,
                /*ro, opt, int, event serial No.*/
                "channelControllerID": 1,
                /*ro, opt, int, lane controller ID*/
                "channelControllerLampID": 1,
                /*ro, opt, int, light board ID of lane controller, range:[1,255]*/
                "channelControllerIRAdaptorID": 1,
                /*ro, opt, int, IR adapter ID of lane controller, range:[1,255]*/
                "channelControllerIREmitterID": 1,
                /*ro, opt, int, active infrared intrusion detector No. of lane controller, range:[1,255]*/
                "userType": "normal",
                /*ro, opt, string, person type*/
                "currentVerifyMode": "cardAndPw",
                /*ro, opt, enum, current authentication mode of the card reader, subType:string, desc:"cardAndPw" (card + password); "card", "cardOrPw" (card or password), "fp" (fingerprint), "fpAndPw" (fingerprint + password), "fpOrCard" "fingerprint or card", "fpAndCard" (fingerprint + card), "fpAndCardAndPw" (fingerprint + card + password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face + fingerprint), "faceAndPw" (face + password), "faceAndCard" (face + card), "face", "employeeNoAndPw" (emplyee No. +password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee No. + fingerprint), "employeeNoAndFpAndPw" (employee No. + fingerprint + password), "faceAndFpAndCard" (face + fingerprint + card), "faceAndPwAndFp" (face + password + fingerprint), "employeeNoAndFace" (employee No. + face), "faceOrfaceAndCard" (face or face + card), "fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or password), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card or password or iris), "sleep", "invalid"*/
                "QRCodeInfo": "test",
                /*ro, opt, string, QR code information*/
                "thermometryUnit": "celsius",
                /*ro, opt, enum, temperature unit, subType:string, desc:"celsius" (Celsius, default value), "fahrenheit" (Fahrenheit), "kelvin" (Kelvin)*/
                "currTemperature": 36.5,
                /*ro, opt, float, skin-surface temperature, which is accurate to one decimal place*/
                "isAbnomalTemperature": true,
                /*ro, opt, bool, whether the skin-surface temperature is abnormal (true-yes)*/
                "RegionCoordinates": {
                /*ro, opt, object, coordinates of the skin-surface temperature*/
                    "positionX": 254,
                    /*ro, opt, int, normalized X-coordinate which is between 0 and 1000*/
                    "positionY": 133
                    /*ro, opt, int, normalized Y-coordinate which is between 0 and 1000*/
                },
                "mask": "unknown",
                /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"unknown"*/
                "pictureURL": "test",
                /*ro, opt, string, picture URL*/
                "filename": "picture1",
                /*ro, opt, string, file name, desc:if multiple pictures are returned at a time, filename of each picture should be unique*/
                "attendanceStatus": "undefined",
                /*ro, opt, enum, attendance status, subType:string, desc:"undefined", "checkIn" (check-in), "checkOut" (check-out), "breakOut" (start of break), "breakIn" (end of break), "overtimeIn" (start of overtime), "overTimeOut" (end of overtime)*/
                "label": "test",
                /*ro, opt, string, custom attendance name*/
                "statusValue": 1,
                /*ro, opt, int, status value*/
                "helmet": "unknown",
                /*ro, opt, enum, whether the person wears a hard hat, subType:string, desc:"unknown", "yes", "no"*/
                "visibleLightPicUrl": "test",
                /*ro, opt, string, visible light picture URL*/
```

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```
                /*ro, opt, string, visible light picture URL*/
                "thermalPicUrl":  "test",
                /*ro, opt, string, URL of the thermal imaging picture*/
                "appType":  "attendance",
                /*ro, opt, enum, application type, subType:string, desc:"attendance" (Time & Attendance module), "signIn" (Check-In module, which is only
used for FocSign products)*/
                "HealthInfo": {
                /*ro, opt, object, health information*/
                    "healthCode":  1,
                    /*ro, opt, enum, health code status, subType:int, desc:0 (no request), 1 (no health code), 2 (green QR code), 3 (yellow QR code), 4 (red
QR code), 5 (no such person), 6 (other error, e.g., searching failed due to API exception), 7 (searching for the health code timed out)*/
                    "NADCode":  1,
                    /*ro, opt, enum, nucleic acid test result, subType:int, desc:0 (no result), 1 (negative, which means normal), 2 (positive, which means
diagnosed), 3 (the result has expired)*/
                    "travelCode":  1,
                    /*ro, opt, enum, trip code, subType:int, desc:0 (no trip in the past 14 days), 1 (has left the current area in the past 14 days), 2 (has
been to the high-risk area in the past 14 days), 3 (other)*/
                    "travelInfo":  "test",
                    /*ro, opt, string*/
                    "vaccineStatus":  1,
                    /*ro, opt, enum, whether the person is vaccinated, subType:int, desc:0 (not vaccinated), 1 (vaccinated)*/
                    "vaccineNum":  1
                    /*ro, opt, int, step:1*/
                },
                "meetingID":  "test",
                /*ro, opt, string, meeting ID*/
                "PersonInfoExtends": [
                /*ro, opt, array, additional person information, subType:object, desc:this node displays additional person information on the device*/
                    {
                        "id":  1,
                        /*ro, opt, int, extended ID of the additional person information, range:[1,32], desc:related URL:
/ISAPI/AccessControl/personInfoExtendName?format=json; this node is used for displaying the name of value; if ID does not exists, it starts from 1*/
                        "value":  "test"
                        /*ro, opt, string, extended content of the additional person information*/
                    }
                ],
                "name":  "test",
                /*ro, opt, string, name, desc:person name*/
                "FaceRect": {
                /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate of the upper-left corner of the frame, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate of the upper-left corner of the frame, range:[0.000,1.000]*/
                },
                "RecordInfo": {
                /*ro, opt, object*/
                    "startTime":  "1970-01-01T00:00:00+08:00",
                    /*ro, opt, datetime, recording start time*/
                    "endTime":  "1970-01-01T00:00:00+08:00",
                    /*ro, opt, datetime, recording end time*/
                    "playbackURL":  "rtsp://10.65.130.168:554/ISAPI/Streaming/tracks/201/?starttime=20190213T091134Z&amp;endtime=20190213T092116Z"
                    /*ro, opt, string, range:[0,256]*/
                },
                "currentAuthenticationTimes":  1,
                /*ro, opt, int, range:[0,255], step:1*/
                "allowAuthenticationTimes":  1
                /*ro, opt, int, range:[0,255], step:1*/
            }
        ]
    }
}
```

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14.6.1.6 Get the capability of getting total number of access control events by specific conditions

Request URL

GET /ISAPI/AccessControl/AcsEventTotalNum/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AcsEvent": {
    /*ro, opt, object*/
        "AcsEventTotalNumCond": {
        /*ro, opt, object, search conditions*/
            "major": {
```

```
            /*ro, req, object, major alarm type*/
                "@opt":  "0,1,2,3,5"
                /*ro, opt, string, major alarm type*/
            },
            "minorAlarm": {
            /*ro, req, object, minor alarm type*/
                "@opt":  "1024,1025,1026,1027"
                /*ro, opt, string, minor alarm type*/
            },
            "minorException": {
            /*ro, req, object, minor exception type*/
                "@opt":  "39,58,59,1024"
                /*ro, opt, string, minor exception type*/
            },
            "minorOperation": {
            /*ro, req, object, minor operation type*/
                "@opt":  "80,90,112,113"
                /*ro, opt, string, minor operation type*/
            },
            "minorEvent": {
            /*ro, opt, object, minor event type*/
                "@opt":  "1,2,3,4"
                /*ro, opt, string, minor event type*/
            },
            "startTime": {
            /*ro, opt, object, start time*/
                "@min":  1,
                /*ro, opt, int, start time (UTC time)*/
                "@max":  1
                /*ro, opt, int, end time (UTC time)*/
            },
            "endTime": {
            /*ro, opt, object, end time*/
                "@min":  1,
                /*ro, opt, int, start time (UTC time)*/
                "@max":  1
                /*ro, opt, int, end time (UTC time)*/
            },
            "cardNo": {
            /*ro, opt, object, card No.*/
                "@min":  1,
                /*ro, opt, int, card No.*/
                "@max":  32
                /*ro, opt, int, card No.*/
            },
            "name": {
            /*ro, opt, object, name of the card holder*/
                "@min":  1,
                /*ro, opt, int, name of the card holder*/
                "@max":  32
                /*ro, opt, int, name of the card holder*/
            },
            "picEnable":  "true,false",
            /*ro, opt, string*/
            "beginSerialNo": {
            /*ro, opt, object, start serial No.*/
                "@min":  1,
                /*ro, opt, int, start serial No.*/
                "@max":  1
                /*ro, opt, int, start serial No.*/
            },
            "endSerialNo": {
            /*ro, opt, object, end serial No.*/
                "@min":  1,
                /*ro, opt, int, end serial No.*/
                "@max":  1
                /*ro, opt, int, end serial No.*/
            },
            "employeeNoString": {
            /*ro, opt, object, employee No. (person ID)*/
                "@min":  1,
                /*ro, opt, int, employee No. (person ID)*/
                "@max":  32
                /*ro, opt, int, employee No. (person ID)*/
            }
        },
        "totalNum": {
        /*ro, req, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        }
    }
}
```

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14.6.1.7 Get the total number of access control events by specific conditions

Request URL

POST /ISAPI/AccessControl/AcsEventTotalNum?format=json

Query Parameter

None

Request Message

```
{
    "AcsEventTotalNumCond": {
    /*req, object*/
        "major": 1,
        /*req, int, major alarm type, desc:(the type value should be transformed to the decimal number), refer to Access Control Event Types for details*/
        "minor": 1024,
        /*req, int, sub type, step:1, desc:(the type value should be transformed to the decimal number),refer to Access Control Event Types for details*/
        "startTime": "1970-01-01+08:00",
        /*opt, date, start time (UTC time)*/
        "endTime": "1970-01-01+08:00",
        /*opt, date, end time (UTC time)*/
        "cardNo": "test",
        /*opt, string, card No.*/
        "name": "test",
        /*opt, string, name of the card holder*/
        "picEnable": true,
        /*opt, bool, whether to upload the picture along with the event information, desc:whether to contain pictures: "true"-yes,"false"-no*/
        "beginSerialNo": 1,
        /*opt, int, start serial No.*/
        "endSerialNo": 100,
        /*opt, int, end serial No.*/
        "employeeNoString": "test"
        /*opt, string, employee No. (person ID), range:[1,32]*/
    }
}
```

Response Message

```
{
    "AcsEventTotalNum": {
    /*ro, req, object*/
        "totalNum": 1,
        /*ro, req, int, total number of events that match the search conditions*/
        "existedEventNum": 1
        /*ro, opt, int*/
    }
}
```

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14.6.1.8 Get the capability of clearing event and card linkage parameters

Request URL

GET /ISAPI/AccessControl/ClearEventCardLinkageCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "ClearEventCardLinkageCfg": {
    /*ro, opt, object, clear event and card linkage parameters*/
        "ClearFlags": {
        /*ro, opt, object*/
            "eventCardLinkage": "true,false"
            /*ro, req, string, event and card linkage parameters*/
        }
    }
}
```

14.6.1.9 Clear event card linkage configurations

Request URL

PUT /ISAPI/AccessControl/ClearEventCardLinkageCfg?format=json

Query Parameter

None

Request Message

```
{
    "ClearEventCardLinkageCfg": {
    /*req, object*/
        "ClearFlags": {
        /*opt, object*/
            "eventCardLinkage":  true
            /*req, bool, whether to clear event and card linkage parameters*/
        }
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

14.6.1.10 Getting arming information

Request URL

GET /ISAPI/AccessControl/DeployInfo

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeployInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, arming information, attr:version{req, string, protocolVersion}-->
  <DeployList size="5">
    <!--ro, opt, array, arming list, subType:object, attr:size{req, int}-->
    <Content>
      <!--ro, opt, object, subscribe to messages-->
      <deployNo>
        <!--ro, req, int, arming No.-->1
      </deployNo>
      <deployType>
        <!--ro, req, enum, arming type, subType:int-->1
      </deployType>
      <protocolType>
        <!--ro, opt, enum, protocol type, subType:string, dep:or,{$.DeployInfo.DeployList[*].Content.deployType,eq,2},
{$.DeployInfo.DeployList[*].Content.deployType,eq,3}, desc:"HTTP", "HTTPS"-->HTTP
      </protocolType>
      <ipAddr>
        <!--ro, req, string, IP address-->test
      </ipAddr>
      <port>
        <!--ro, opt, int, port No., range:[1,65535]-->1
      </port>
      <eventType>
        <!--ro, opt, enum, subType:string-->AccessController
      </eventType>
    </Content>
  </DeployList>
</DeployInfo>
```

14.6.1.11 Getting arming information capability

Request URL

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GET /ISAPI/AccessControl/DeployInfo/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeployInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, arming Information, attr:version{req, string, protocolVersion}-->
  <DeployList size="5">
    <!--ro, opt, array, arming list, subType:object, attr:size{req, int}-->
    <Content>
      <!--ro, opt, object-->
      <deployNo min="1" max="10">
        <!--ro, req, int, arming No., attr:min{req, int},max{req, int}-->1
      </deployNo>
      <deployType opt="0,1,2,3">
        <!--ro, req, int, arming type, attr:opt{req, string}-->1
      </deployType>
      <protocolType opt="HTTP,HTTPS">
        <!--ro, opt, enum, protocol type, subType:string, attr:opt{req, string}, desc:"HTTP", "HTTPS"-->HTTP
      </protocolType>
      <ipAddr min="1" max="10">
        <!--ro, req, string, IP address, attr:min{req, int},max{req, int}-->test
      </ipAddr>
      <port min="0" max="10">
        <!--ro, opt, int, port No., range:[1,65535], attr:min{req, int},max{req, int}-->1
      </port>
      <eventType opt="AccessController,Consumer,AccessControllerAndConsumer">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->AccessController
      </eventType>
    </Content>
  </DeployList>
</DeployInfo>
```

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14.6.1.12 Get the event and card linkage configuration parameters

Request URL

GET /ISAPI/AccessControl/EventCardLinkageCfg/<ACEID>?format=json&filterValidFloor=<filterValidFloor>

Query Parameter

Parameter Name	Parameter Type	Description
ACEID	string	--
filterValidFloor	string	"filterValidFloor" (filter by the maximum number of valid floors), and if this parameter does not exist, it is false by default and indicates returning the maximum number of floors supported by the device.

Request Message

None

Response Message

```
{
    "EventCardLinkageCfg": {
    /*ro, req, object*/
        "proMode":  "event",
        /*ro, req, enum, linkage type, subType:string, desc:"event"-event linkage, "card"-card linkage, "mac"-MAC address linkage, "employee"-employee No.
(person ID)*/
        "EventLinkageInfo": {
        /*ro, opt, object, event linage parameters, dep:and,{$.EventCardLinkageCfg.proMode,eq,event}*/
            "mainEventType":  0,
            /*ro, req, enum, major event type, subType:int, desc:0-device event,1-alarm input event,2-access control point event,3-authentication unit (card
reader, fingerprint module) event*/
            "subEventType":  54
            /*ro, req, int, event sub type, desc:minor event type,refer to Event Linkage Types for details*/
        },
        "EmployeeInfo": {
        /*ro, opt, object, employee No. (person ID) linkage parameters, dep:and,{$.EventCardLinkageCfg.proMode,eq,employee}*/
            "employeeNo":  "test"
            /*ro, req, string, employee No. (person ID), range:[1,32]*/
        },
        "eventSourceID":  1,
        /*ro, opt, int, event source ID, dep:or,{$.EventCardLinkageCfg.proMode,eq,event},{$.EventCardLinkageCfg.proMode,eq,card},
{$.EventCardLinkageCfg.proMode,eq,employee}, desc:it is valid when proMode is "event",65535-all. For device event (mainEventType is 0),this field is
invalid; for access control point event (mainEventType is 2),this field refers to the access control point No.; for authentication unit event (mainEventType
is 3,this field refers to the authentication unit No.; for alarm input event (mainEventType is 1),this field refers to the zone alarm input ID or the event
alarm input ID*/
        "alarmout": [1, 3, 5],
        /*ro, opt, array, linked alarm output No., subType:int, desc:[1,3,5]: 1-linked alarm output No.1; 3-linked alarm output No.3; 5-linked alarm output
No.5*/
        "openDoor": [1, 3, 5],
        /*ro, opt, array, linked door No. to open, subType:int, desc:[1,3,5]: 1-linked door No.1; 3-linked door No.3; 5-linked door No.5*/
        "closeDoor": [1, 3, 5],
        /*ro, opt, array, linked door No. to close, subType:int, desc:[1,3,5]: 1-linked door No.1; 3-linked door No.3; 5-linked door No.5*/
        "alwaysOpen": [1, 3, 5],
        /*ro, opt, array, linked door No. to remain unlocked, subType:int, desc:e.g.,[1,3,5]: 1-linked door No.1; 3-linked door No.3; 5-linked door No.5*/
        "alwaysClose": [1, 3, 5],
        /*ro, opt, array, linked door No. to remain locked, subType:int, desc:[1,3,5]: 1-linked door No.1; 3-linked door No.3; 5-linked door No.5*/
        "capturePic":  true,
        /*ro, opt, bool, whether to enable capture linkage, desc:"false"-no, "true"-yes*/
        "alarmOutClose": [1, 3, 5],
        /*ro, opt, array, linked alarm output No. to disable, subType:int, desc:[1,3,5]: 1-alarm output No.1, 3-alarm output No.3,5-alarm output No.5*/
    }
}
```

14.6.1.13 Set the event card linkage parameters

Request URL

PUT /ISAPI/AccessControl/EventCardLinkageCfg/<ACEID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
ACEID	string	--

Request Message

```
{
    "EventCardLinkageCfg": {
    /*req, object, event card linkage parameters*/
        "proMode":  "event",
        /*req, enum, linkage type, subType:string, desc:"event" (event linkage), "card" (card linkage), "mac" (MAC address linkage), "employee" (employee
No., i.e., person ID)*/
        "EmployeeInfo": {
        /*opt, object, employee No. (person ID) linkage parameters, desc:it is valid when proMode is "employee"*/
            "employeeNo":  "test"
            /*opt, string, employee No. (person ID)*/
        },
        "eventSourceID":  1,
        /*opt, int, event source ID, desc:it is valid when proMode is "event". For device event (mainEventType is 0), this field is invalid; for access
control point event (mainEventType is 2), this field refers to the access control point No.; for authentication unit event (mainEventType is 3), this field
refers to the authentication unit No.; for alarm input event (mainEventType is 1), this field refers to the zone alarm input ID or the event alarm input ID
65535-all*/
        "alarmout": [1, 3, 5],
        /*opt, array, linked alarm output No., subType:int, desc:[1,3,5]: 1-linked alarm output No.1; 3-linked alarm output No.3; 5-linked alarm output
No.5*/
        "openDoor": [1, 3, 5],
        /*opt, array, linked door No. to open, subType:int, desc:[1,3,5]: 1-linked door No.1, 3-linked door No.3, 5-linked door No.5*/
        "mainDevBuzzer":  true,
        /*opt, bool, whether to enable buzzer linkage of the access controller (start buzzing):, desc:false-no, true-yes*/
        "recordVideo":  true,
        /*opt, bool, whether to enable recording linkage, desc:false-no, true-yes*/
    }
}
```

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Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code*/
    "statusString":  "test",
    /*ro, opt, string, status description*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code*/
    "errorCode":  1,
    /*ro, req, int, error code*/
    "errorMsg":  "ok"
    /*ro, req, string, error description*/
}
```

14.6.1.14 Get the configuration capability of the event and card linkage

Request URL

GET /ISAPI/AccessControl/EventCardLinkageCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "EventCardLinkageCfg": {
    /*ro, req, object, parameters of the event and card linkage*/
        "eventID": {
        /*ro, opt, object, event ID*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "proMode": {
        /*ro, req, object, linkage type*/
            "@opt":  "event,card,mac,employee"
            /*ro, opt, string, linkage method*/
        },
        "EventLinkageInfo": {
        /*ro, opt, object, event linkage information*/
            "mainEventType": {
            /*ro, opt, object, event main type*/
                "@opt":  "0,1,2,3"
                /*ro, opt, string, event main type*/
            },
            "devSubEventType": {
            /*ro, opt, object, minor event type*/
                "@opt":  "0,1,2,3,54..."
                /*ro, opt, string, minor event type*/
            },
            "alarmSubEventType": {
            /*ro, opt, object, minor type of alarm input event*/
                "@opt":  "0,1,2,3,52..."
                /*ro, opt, string, minor type of alarm input event*/
            },
            "doorSubEventType": {
            /*ro, opt, object, minor type of access control point event*/
                "@opt":  "0,1,2,3..."
                /*ro, opt, string, minor type of access control point event*/
            },
            "cardReaderSubEventType": {
            /*ro, opt, object, minor type of authentication unit event*/
                "@opt":  "0,1,2,3..."
                /*ro, opt, string, minor type of authentication unit event*/
            }
        },
        "CardNoLinkageInfo": {
        /*ro, opt, object, card linkage parameters*/
            "cardNo": {
            /*ro, opt, object, card No.*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  32
                /*ro, opt, int*/
            }
        },
        "EmployeeInfo": {
```

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```
        /*ro, opt, object, person ID*/
            "employeeNo": {
            /*ro, opt, object, person ID*/
                "@min":  1,
                /*ro, opt, int, employee No. (person ID)*/
                "@max":  32
                /*ro, opt, int*/
            }
        },
        "eventSourceID": {
        /*ro, opt, object, event source ID*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "alarmout": {
        /*ro, opt, object, linked alarm output No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "openDoor": {
        /*ro, opt, object, linked door No. to open*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "closeDoor": {
        /*ro, opt, object, linked door No. to close*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "alwaysOpen": {
        /*ro, opt, object, array,linked door No. to remain unlocked*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "alwaysClose": {
        /*ro, opt, object, linked door No. to remain locked*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "capturePic":  "true,false",
        /*ro, opt, string, whether to enable capture linkage*/
        "alarmOutClose": {
        /*ro, opt, object, array,linked alarm output No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
    }
}
```

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14.6.1.15 Get the capability of the list of event and card linkage ID

Request URL

GET /ISAPI/AccessControl/EventCardNoList/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "EventCardNoList": {
    /*ro, opt, object*/
        "id": {
        /*ro, opt, object, range of event ID that can be configured*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        }
    }
}
```

14.6.1.16 Get the list of event and card linkage ID

Request URL

GET /ISAPI/AccessControl/EventCardNoList?format=json&alarmOutID=<alarmOutID>

Query Parameter

Parameter Name	Parameter Type	Description
alarmOutID	string	--

Request Message

None

Response Message

```
{
    "EventCardNoList": {
    /*ro, opt, object*/
        "id": [1, 2, 3]
        /*ro, req, array, list of configured event and card linkage ID, subType:int, desc:[1, 2, 3] indicates that the device is configured with event
linkage 1, 2, and 3*/
    }
}
```

14.6.1.17 Get the configuration capability of event optimization

Request URL

GET /ISAPI/AccessControl/EventOptimizationCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "EventOptimizationCfg": {
    /*ro, opt, object*/
        "enable":  "true,false",
        /*ro, opt, string, whether to enable event optimization*/
        "isCombinedLinkageEvents":  "true,false"
        /*ro, opt, string, whether to enable linked event combination*/
    }
}
```

14.6.1.18 Set the event optimization parameters

Request URL

PUT /ISAPI/AccessControl/EventOptimizationCfg?format=json

Query Parameter

None

Request Message

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```
{
    "EventOptimizationCfg": {
    /*opt, object*/
        "enable":  true,
        /*opt, bool, whether to enable event optimization*/
        "isCombinedLinkageEvents":  true
        /*opt, bool, whether to enable linked event combination*/
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code*/
    "statusString":  "test",
    /*ro, opt, string, status description*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code*/
    "errorCode":  1,
    /*ro, req, int, error code*/
    "errorMsg":  "ok"
    /*ro, req, string, error details*/
}
```

14.6.1.19 Get the event optimization configuration parameters

Request URL

GET /ISAPI/AccessControl/EventOptimizationCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "EventOptimizationCfg": {
    /*ro, opt, object*/
        "enable":  true,
        /*ro, opt, bool, whether to enable event optimization*/
        "isCombinedLinkageEvents":  true
        /*ro, opt, bool, whether to enable linked event combination*/
    }
}
```

14.6.1.20 Access control event

EventType:AccessControllerEvent

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, req, int, times that the same alarm has been uploaded, desc:times that the same alarm has been uploaded*/
    "eventType":  "AccessControllerEvent",
    /*ro, req, string, event type, desc:"AccessControllerEvent" (access control event)*/
    "eventState":  "active",
```

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```
    "eventState": "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid event or event starts), "inactive" (invalid event or the event ends). For the heartbeat, the node value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription": "AccessControllerEvent",
    /*ro, req, string, event description, desc:"AccessControllerEvent" (access control event)*/
    "deviceID": "test0123",
    /*ro, opt, string, device ID (PUID), desc:this node must be returned when ISAPI event information is transmitted via ISUP*/
    "AccessControllerEvent": {
    /*ro, req, object, access control event information*/
        "deviceName": "test",
        /*ro, opt, string, device name*/
        "majorEventType": 1,
        /*ro, req, int, major alarm type, desc:the type value should be transformed to the decimal number; see Access Control Alarm Types for details*/
        "subEventType": 1,
        /*ro, req, int, minor alarm type, desc:the type value should be transformed to the decimal number; see Access Control Alarm Types for details*/
        "inductiveEventType": "authenticated",
        /*ro, opt, enum, inductive event type, subType:string, desc:this node is used by storage devices; for access control devices, this node is invalid; "authenticated", "authenticationFailed", "openingDoor", "closingDoor", "doorException", "remoteOperation", "timeSynchronization", "deviceException", "deviceRecovered", "alarmTriggered", "alarmRecovered" (arming restoring event), "callCenter"*/
        "netUser": "test",
        /*ro, opt, string, user name for network operations*/
        "remoteHostAddr": "test",
        /*ro, opt, string, remote host address*/
        "cardNo": "test",
        /*ro, opt, string, card No.*/
        "cardType": 1,
        /*ro, opt, enum, card type, subType:int, desc:1 (normal card), 2 (disability card), 3 (blocklist card), 4 (patrol card), 5 (duress card), 6 (super card), 7 (visitor card), 8 (dismiss card)*/
        "name": "test",
        /*ro, opt, string, person name*/
        "sex": "male",
        /*ro, opt, enum, subType:string, desc:"male", "female"*/
        "whiteListNo": 1,
        /*ro, opt, int, allowlist No.*/
        "reportChannel": 1,
        /*ro, opt, enum, channel type for uploading alarms/events, subType:int, desc:1 (uploading in arming mode), 2 (uploading by central group 1), 3 (uploading by central group 2)*/
        "cardReaderKind": 1,
        /*ro, opt, enum, card reader type, subType:int, desc:1 (IC card reader), 2 (ID card reader), 3 (QR code scanner), 4 (fingerprint module)*/
        "cardReaderNo": 1,
        /*ro, opt, int, card reader No., step:1, desc:card reader No.*/
        "doorNo": 1,
        /*ro, opt, int, door (floor) No.*/
        "verifyNo": 1,
        /*ro, opt, int, multiple authentication No.*/
        "alarmInNo": 1,
        /*ro, opt, int, alarm input No.*/
        "alarmOutNo": 1,
        /*ro, opt, int, alarm output No.*/
        "caseSensorNo": 1,
        /*ro, opt, int, event trigger No.*/
        "RS485No": 1,
        /*ro, opt, int, RS-485 channel No.*/
        "multiCardGroupNo": 1,
        /*ro, opt, int, group No.*/
        "accessChannel": 1,
        /*ro, opt, int, turnstile No.*/
        "deviceNo": 1,
        /*ro, opt, int, device No.*/
        "distractControlNo": 1,
        /*ro, opt, int, distributed access controller No.*/
        "employeeNo": 1,
        /*ro, opt, int, employee No. (person ID)*/
        "employeeNoString": "test",
        /*ro, opt, string, employee No. (person ID), desc:if the node employeeNo exists or the value of employeeNoString can be converted to that of employeeNo, this node is required. For the upper-layer platform or client software, the node employeeNoString will be parsed in priority; if employeeNoString is not configured, the node employeeNo will be parsed*/
        "employeeName": "test",
        /*ro, opt, string, person name, desc:this node is only used for FocSign products*/
        "localControllerID": 1,
        /*ro, opt, int, distributed access controller No., desc:0 (access controller), 1 to 64 (distributed access controller No. 1 to distributed access controller No. 64)*/
        "InternetAccess": 1,
        /*ro, opt, enum, network interface No., subType:int, desc:1 (upstream network interface No. 1), 2 (upstream network interface No. 2), 3 (downstream network interface No. 1)*/
        "type": 1,
        /*ro, opt, enum, zone type, subType:int, desc:0 (instant zone), 1 (24-hour zone), 2 (delayed zone), 3 (internal zone), 4 (key zone), 5 (fire alarm zone), 6 (perimeter zone), 7 (24-hour silent zone), 8 (24-hour auxiliary zone), 9 (24-hour shock zone), 10 (emergency door open zone), 11 (emergency door closed zone), 255 (none)*/
        "MACAddr": "test",
        /*ro, opt, string, MAC address*/
        "swipeCardType": 1,
        /*ro, opt, enum, card swiping types, subType:int, desc:0 (invalid), 1 (QR code)*/
        "serialNo": 1,
        /*ro, opt, int, event serial No., range:[1,100000], desc:it starts at 1 and each record increases by 1. It will be overwritten repeatedly when reaching the maximum value supported by the device*/
        "channelControllerID": 1,
        /*ro, opt, enum, lane controller ID, subType:int, desc:1 (main lane controller), 2 (sub-lane controller)*/
        "channelControllerLampID": 1,
        /*ro, opt, int, light board ID of lane controller, range:[1,255]*/
        "channelControllerIRAdaptorID": 1,
        /*ro, opt, int, IR adaptor ID of the lane controller, range:[1,255]*/
        "channelControllerIREmitterID": 1,
        /*ro, opt, int, active infrared intrusion detector No. of the lane controller, range:[1,255]*/
```

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```
        "userType":  "normal",
        /*ro, opt, enum, person type, subType:string, desc:"normal" (normal person (resident)), "visitor" (visitor), "blacklist" (person in the blocklist),
"administrators" (administrator)*/
        "currentVerifyMode":  "cardAndPw",
        /*ro, opt, enum, current authentication mode of the card reader, subType:string, desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card
or password), "fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face, fingerprint, card, password, or iris), "faceOrCardOrPwOrIris"
(face, card, password, or iris)*/
        "currentEvent":  true,
        /*ro, opt, bool, whether it is a real-time event*/
        "QRCodeInfo":  "test",
        /*ro, opt, string, QR code information*/
        "thermometryResult":  "success",
        /*ro, opt, enum, temperature screening result, subType:string, desc:"success", "fail"*/
        "thermometryUnit":  "celsius",
        /*ro, opt, enum, temperature unit, subType:string, desc:"celsius" (Celsius, default value), "fahrenheit" (Fahrenheit), "kelvin" (Kelvin)*/
        "currTemperature":  36.1,
        /*ro, opt, float, skin-surface temperature, which is accurate to one decimal place*/
        "isAbnomalTemperature":  true,
        /*ro, opt, bool, whether the skin-surface temperature is abnormal*/
        "RegionCoordinates": {
        /*ro, opt, object, coordinates of the skin-surface temperature*/
            "positionX":  0,
            /*ro, opt, int, normalized X-coordinate which is between 0 and 1000, range:[0,1000]*/
            "positionY":  0
            /*ro, opt, int, normalized Y-coordinate which is between 0 and 1000, range:[0,1000]*/
        },
        "remoteCheck":  true,
        /*ro, opt, bool, whether remote verification is required: true-yes, false-no (default)*/
        "mask":  "unknown",
        /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"unknown", "yes", "no"*/
        "frontSerialNo":  1,
        /*ro, opt, int, serial No. of the previous event, desc:if this node does not exist, the platform will check whether the event loss occurred
according to the node serialNo. If both the serialNo and frontSerialNo are returned, the platform will check whether the event loss occurred according to
both nodes. It is mainly used to solve the problem that the serialNo is inconsistent after subscribing events or alarms*/
        "attendanceStatus":  "checkIn",
        /*ro, opt, enum, attendance status, subType:string, desc:"checkIn" (check-in), "checkOut" (check-out), "breakOut" (start of break), "breakIn" (end
of break), "overtimeIn" (start of overtime), "overTimeOut" (end of overtime)*/
        "label":  "test",
        /*ro, opt, string, self-defined attendance name*/
        "statusValue":  1,
        /*ro, opt, int, status value*/
        "pictureURL":  "test",
        /*ro, opt, string, URL of the captured picture, range:[0,256]*/
        "visibleLightURL":  "test",
        /*ro, opt, string, visible light picture URL of the thermal imaging camera, range:[0,256]*/
        "thermalURL":  "test",
        /*ro, opt, string, URL of the thermal picture, range:[0,256]*/
        "faceBasemapURL":  "test",
        /*ro, opt, string, range:[0,256]*/
        "picturesNumber":  1,
        /*ro, opt, int, number of captured pictures*/
        "unlockType":  "password",
        /*ro, opt, enum, unlocking type, subType:string, desc:this node is returned when the minor type is MINOR_UNCLOCK_RECORD; "password" (unlock by
password), "hijcking" (unlock by duress), "card" (unlock by card), "householder" (unlock by householder), "centerplatform" (unlock by management center),
"bluetooth" (unlock by bluetooth), "qrcode" (unlocked via QR code), "face" (unlock by recognizing face), "fingerprint" (unlock by fingerprint)*/
        "classroomId":  "test",
        /*ro, opt, string, class ID*/
        "classroomName":  "test",
        /*ro, opt, string, class name*/
        "analysisModule":  "signageApp",
        /*ro, opt, enum, analysis module, subType:string, desc:this node is not returned, and the value is report via signage App; "signageApp" (signage
App), "faceSDK" (face picture SDK)*/
        "customInfo":  "test",
        /*ro, opt, string, custom information*/
        "helmet":  "unknown",
        /*ro, opt, enum, whether the person is wearing hard hat, subType:string, desc:"unknown", "yes", "no"*/
        "purePwdVerifyEnable":  true,
        /*ro, opt, bool, whether the device supports opening the door only by password,
        desc:opening the door only by password:
        the password in authentication method is person password; checking the repetition of person password is not supported by the device, it should be
performed by the upper-layer platform; adding, deleting, editing, and searching for person password locally is not supported by the device*/
        "appType":  "attendance",
        /*ro, opt, enum, application type (for FocSign products), subType:string, desc:"attendance" (Time & Attendance module), "signIn" (Check-In module)*/
        "HealthInfo": {
        /*ro, opt, object, health information*/
            "healthCode":  1,
            /*ro, opt, enum, health code status, subType:int, desc:0 (no request), 1 (no health code), 2 (green QR code), 3 (yellow QR code), 4 (red QR
code), 5 (no such person), 6 (other error, e.g., searching failed due to API exception), 7 (searching for the health code timed out)*/
            "NADCode":  1,
            /*ro, opt, enum, nucleic acid test result, subType:int, desc:0 (no result), 1 (negative, which means normal), 2 (positive, which means
diagnosed), 3 (the result has expired)*/
            "NADMsg":  "test",
            /*ro, opt, string, range:[0,64]*/
            "NADTime":  1,
            /*ro, opt, enum, subType:int*/
            "travelCode":  1,
            /*ro, opt, enum, trip code, subType:int, desc:0 (no trip in the past 14 days), 1 (has left the current area left in the past 14 days), 2 (has
been to the high-risk area in the past 14 days), 3 (other)*/
            "travelInfo":  "test"
```

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            "travelInfo": "test",
            /*ro, opt, string, trip information, desc:the empty string indicates that searching trip failed*/
            "vaccineStatus": 1,
            /*ro, opt, enum, whether the person is vaccinated, subType:int, desc:0 (not vaccinated), 1 (vaccinated)*/
            "vaccineNum": 1,
            /*ro, opt, int, step:1*/
            "vaccineMsg": "test",
            /*ro, opt, string, range:[0,64]*/
            "ANTCode": 1,
            /*ro, opt, enum, subType:int*/
            "ANTMsg": "test"
            /*ro, opt, string, range:[0,64]*/
        },
        "PhysicalInfo": {
        /*ro, opt, object, BMI information, desc:this node is obtained after authentication by BMI scales which is connected to MinMoe terminals*/
            "weight": 7000,
            /*ro, opt, int, weight, unit:kg*/
            "height": 18000
            /*ro, opt, int, height, unit:cm*/
        },
        "meetingID": "test",
        /*ro, opt, string, meeting ID, range:[1,32]*/
        "PersonInfoExtends": [
        /*ro, opt, array, additional person information, subType:object, desc:this node displays additional person information on the device*/
            {
                "id": 1,
                /*ro, opt, int, extended ID of the additional person information, range:[1,32], desc:related URL: /ISAPI/AccessControl/personInfoExtendName?format=json; this node is used for displaying the name of value; if ID does not exists, it starts from 1*/
                "value": "test"
                /*ro, opt, string, extended content of the additional person information*/
            }
        ],
        "customPrompt": "test",
        /*ro, opt, string, custom prompt message, range:[1,128], desc:this node is displayed when the authentication result is authenticated, authentication failed, or stranger*/
        "FaceRect": {
        /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
            "height": 1.000,
            /*ro, req, float, height, range:[0.000,1.000]*/
            "width": 1.000,
            /*ro, req, float, width, range:[0.000,1.000]*/
            "x": 0.000,
            /*ro, req, float, X-coordinate of the upper-left corner of the frame, range:[0.000,1.000]*/
            "y": 0.000
            /*ro, req, float, Y-coordinate of the upper-left corner of the frame, range:[0.000,1.000]*/
        },
        "faceSimilarity": 90,
        /*ro, opt, int, Similarity, range:[0,100]*/
        "faceRecognitionDistance": 0.1,
        /*ro, opt, float, unit:m*/
        "eyesDistance": 20,
        /*ro, opt, int, range:[0,100], step:1*/
        "faceRecognitionFailedReason": "attackBlacklist",
        /*ro, opt, enum, subType:string*/
        "currentAuthenticationTimes": 1,
        /*ro, opt, int, range:[0,255], step:1*/
        "allowAuthenticationTimes": 1,
        /*ro, opt, int, range:[0,255], step:1*/
        "LocalAttendanceData": {
        /*ro, opt, object*/
            "attendanceResult": [
            /*ro, opt, array, subType:object*/
                {
                    "date": "1970-01-01",
                    /*ro, opt, date*/
                    "week": 1,
                    /*ro, opt, enum, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)*/
                    "personalAttendanceStatus": "normal"
                    /*ro, opt, enum, subType:string*/
                }
            ]
        },
        "hasRecord": true
        /*ro, opt, bool*/
    },
    "URLCertificationType": "digest"
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (no authentication, it is used for the cloud protocol), "digest" (digest authentication, it is used for local picture URL returned by NVR or DVR)*/
}
```


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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
AccessControllerEvent	[Message content]	application/json	--	--	--
Picture	[Binary picture data]	image/jpeg	pictureImage	Picture.jpg	--
VisibleLight	[Binary picture data]	image/jpeg	visibleLight_image	VisibleLight.jpg	--
Thermal	[Binary picture data]	image/jpeg	thermal_image	Thermal.jpg	--
HandPicture	[Binary picture data]	image/jpeg	hand_image	HandPicture.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

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14.6.2 Access Control Schedule Management

14.6.2.1 Get the capability of clearing access control schedule configuration.

Request URL

GET /ISAPI/AccessControl/ClearPlansCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "ClearPlansCfg": {
    /*ro, req, object*/
        "ClearFlags": {
        /*ro, opt, object*/
            "doorStatusWeekPlan":  "true,false",
            /*ro, opt, string, whether to clear the week schedule of the door control*/
            "cardReaderWeekPlan":  "true,false",
            /*ro, opt, string, whether to clear the week schedule of the card reader authentication mode control*/
            "userRightWeekPlan":  "true,false",
            /*ro, opt, string, whether to clear the week schedule of the access permission control*/
            "doorStatusHolidayPlan":  "true,false",
            /*ro, opt, string, whether to clear the holiday schedule of the door control*/
            "cardReaderHolidayPlan":  "true,false",
            /*ro, opt, string, whether to clear the holiday schedule of the card reader authentication mode control*/
            "userRightHolidayPlan":  "true,false",
            /*ro, opt, string, whether to clear the holiday schedule of the access permission control*/
            "doorStatusHolidayGroup":  "true,false",
            /*ro, opt, string, whether to clear the holiday group of the door control*/
            "cardReaderHolidayGroup":  "true,false",
            /*ro, opt, string, whether to clear the holiday group of the card reader authentication mode control*/
            "userRightHolidayGroup":  "true,false",
            /*ro, opt, string, whether to clear the holiday group of the access permission control*/
            "doorStatusTemplate":  "true,false",
            /*ro, opt, string, whether to clear the schedule template of the door control*/
            "cardReaderTemplate":  "true,false",
            /*ro, opt, string, whether to clear the control schedule template of the card reader authentication mode*/
            "userRightTemplate":  "true,false"
            /*ro, opt, string, whether to clear the schedule template of the access permission control*/
        }
    }
}
```

14.6.2.2 Clear access control schedule configuration parameters

Request URL

PUT /ISAPI/AccessControl/ClearPlansCfg?format=json

Query Parameter

None

Request Message

```
{
    "ClearPlansCfg": {
    /*opt, object*/
        "ClearFlags": {
        /*opt, object*/
            "doorStatusWeekPlan":  true,
            /*opt, bool, whether to clear the week schedule of the door control*/
            "cardReaderWeekPlan":  true,
            /*opt, bool, whether to clear the week schedule of the card reader authentication mode control*/
            "userRightWeekPlan":  true,
            /*opt, bool, whether to clear the week schedule of the access permission control*/
            "doorStatusHolidayPlan":  true,
            /*opt, bool, whether to clear the holiday schedule of the door control*/
            "cardReaderHolidayPlan":  true,
            /*opt, bool, whether to clear the holiday schedule of the card reader authentication mode control*/
            "userRightHolidayPlan":  true,
            /*opt, bool, whether to clear the holiday schedule of the access permission control*/
            "doorStatusHolidayGroup":  true,
            /*opt, bool, whether to clear the holiday group of the door control*/
            "cardReaderHolidayGroup":  true,
            /*opt, bool, whether to clear the holiday group of the card reader authentication mode control*/
            "userRightHolidayGroup":  true,
            /*opt, bool, whether to clear the holiday group of the access permission control*/
            "doorStatusTemplate":  true,
            /*opt, bool, whether to clear the schedule template of the door control*/
            "cardReaderTemplate":  true,
            /*opt, bool, whether to clear the control schedule template of card reader authentication mode*/
            "userRightTemplate":  true
            /*opt, bool, whether to clear the schedule template of access permission control*/
        }
    }
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

14.6.2.3 Set holiday group parameters of door control schedule

Request URL

PUT /ISAPI/AccessControl/DoorStatusHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
holidayGroupID	string	--

Request Message

```
{
    "DoorStatusHolidayGroupCfg": {
    /*opt, object*/
        "enable":  true,
        /*req, bool, whether to enable, desc:whether to enable*/
        "groupName":  "test",
        /*req, string, holiday group name*/
        "holidayPlanNo":  "1,3,5"
        /*opt, string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

14.6.2.4 Get the holiday group configuration parameters of the door control schedule

Request URL

GET /ISAPI/AccessControl/DoorStatusHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
holidayGroupID	string	Holiday group No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

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```
{
    "DoorStatusHolidayGroupCfg": {
    /*ro, opt, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable: "true"-enable, "false"-disable*/
        "groupName":  "test",
        /*ro, req, string, holiday group name*/
        "holidayPlanNo":  "1,3,5"
        /*ro, req, string*/
    }
}
```

14.6.2.5 Get the configuration capability of door status parameters of holiday group

Request URL

GET /ISAPI/AccessControl/DoorStatusHolidayGroupCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DoorStatusHolidayGroupCfg": {
    /*ro, opt, object*/
        "groupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:true (enable)*/
        "groupName": {
        /*ro, opt, object, length of holiday group name*/
            "@min":  1,
            /*ro, opt, int, the minimum length*/
            "@max":  32
            /*ro, opt, int, the maximum length*/
        },
        "holidayPlanNo": {
        /*ro, opt, object, holiday group plan No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        }
    }
}
```

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14.6.2.6 Get the configuration parameters of the door control holiday schedule

Request URL

GET /ISAPI/AccessControl/DoorStatusHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
holidayPlanID	string	Holiday schedule No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

```
{
    "DoorStatusHolidayPlanCfg": {
    /*ro, req, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (enable), false (disable)*/
        "beginDate":  "2017-10-01",
        /*ro, req, date, start date of the holiday*/
        "endDate":  "2017-10-08",
        /*ro, req, date, end date of the holiday*/
        "HolidayPlanCfg": [
        /*ro, req, array, holiday schedule parameters, subType:object*/
            {
                "id":  1,
                /*ro, req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*ro, req, bool, whether to enable, desc:true (enable), false (disable)*/
                "doorStatus":  "remainClosed",
                /*ro, req, enum, door status, subType:string, desc:“remainOpen”-remain open (access without authentication), “remainClosed”-remain closed
(access is not allowed), “normal”-access by authentication, "sleep", "invalid”, “induction”, “barrierFree”*/
                "TimeSegment": {
                /*ro, opt, object, time*/
                    "beginTime":  "00:00:00",
                    /*ro, req, time, start time of the time period, desc:device local time*/
                    "endTime":  "10:00:00"
                    /*ro, req, time, end time of the time period, desc:device local time*/
                }
            }
        ]
    }
}
```

14.6.2.7 Set parameters of door control holiday schedule

Request URL

PUT /ISAPI/AccessControl/DoorStatusHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
holidayPlanID	string	--

Request Message

```
{
    "DoorStatusHolidayPlanCfg": {
    /*ro, req, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
        "beginDate":  "2017-10-01",
        /*ro, req, date, start date of the holiday*/
        "endDate":  "2017-10-08",
        /*ro, req, date, end data of the holiday*/
        "HolidayPlanCfg": [
        /*ro, req, array, holiday schedule parameters, subType:object*/
            {
                "id":  1,
                /*ro, req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*ro, req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
                "doorStatus":  "remainClosed",
                /*ro, req, enum, door status, subType:string, desc:"remainOpen"-remain open (access without authentication), "remainClosed"-remain closed
(access is not allowed), "normal"-access by authentication, "sleep", "invalid”*/
                "TimeSegment": {
                /*ro, opt, object, time*/
                    "beginTime":  "00:00:00",
                    /*ro, req, time, start time, desc:device local time*/
                    "endTime":  "10:00:00"
                    /*ro, req, time, end time, desc:device local time*/
                }
            }
        ]
    }
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

14.6.2.8 Get the configuration capability of the door control holiday schedule

Request URL

GET /ISAPI/AccessControl/DoorStatusHolidayPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DoorStatusHolidayPlanCfg": {
    /*ro, opt, object*/
        "planNo": {
        /*ro, opt, object, holiday schedule No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable: "true"-enable,"false"-disable, desc:"true" (enable), "false" (disable)*/
        "beginDate":  "1970-01-01",
        /*ro, opt, date, start date of the holiday*/
        "endDate":  "1971-01-01",
        /*ro, opt, date, end date of the holiday*/
        "HolidayPlanCfg": {
        /*ro, opt, object*/
            "maxSize":  8,
            /*ro, opt, int*/
            "id": {
            /*ro, opt, object, time period No.*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  8
                /*ro, opt, int*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether to enable: "true"-enable,"false"-disable, desc:"true" (enable), "false" (disable)*/
            "doorStatus": {
            /*ro, opt, object, door status*/
                "@opt":  "remainOpen,remainClosed,normal,sleep,invlid,induction,barrierFree"
                /*ro, opt, string, desc:"remainOpen" (remain open (access without authentication)), "remainClosed" (remain closed (access is not allowed)),
"normal" (access by authentication), "sleep", "invalid"*/
            },
            "TimeSegment": {
            /*ro, opt, object, time*/
                "beginTime":  "00:00:00",
                /*ro, opt, time, start time of the time period (device local time), desc:device local time*/
                "endTime":  "00:00:00",
                /*ro, opt, time, end time of the time period (device local time), desc:device local time*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:"hour", "minute", "second"; if this node is not returned, the default time accuracy is
"minute"*/
            }
        }
    }
}
```

14.6.2.9 Get the configuration parameters of the door control schedule

Request URL

GET /ISAPI/AccessControl/DoorStatusPlan/<doorID>?format=json

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Query Parameter

Parameter Name	Parameter Type	Description
doorID	string	Door No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

```
{
    "DoorStatusPlan": {
    /*ro, req, object*/
        "templateNo":  1
        /*ro, req, int, schedule template No., desc:0-cancel linking the template with the schedule and restore to the default status (normal status)*/
    }
}
```

14.6.2.10 Set parameters of door control schedule

Request URL

PUT /ISAPI/AccessControl/DoorStatusPlan/<doorID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
doorID	string	--

Request Message

```
{
    "DoorStatusPlan": {
    /*req, object*/
        "templateNo":  1
        /*req, int, schedule template No., desc:0-cancel linking the template with the schedule and restore to the default status (normal status)*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

14.6.2.11 Get the configuration capability of the door control schedule

Request URL

GET /ISAPI/AccessControl/DoorStatusPlan/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "DoorStatusPlan": {
    /*ro, opt, object*/
        "doorNo": {
        /*ro, opt, object, door No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  4
            /*ro, opt, int*/
        },
        "templateNo": {
        /*ro, opt, object, schedule template No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        }
    }
}
```

14.6.2.12 Set parameters of door control schedule template

Request URL

PUT /ISAPI/AccessControl/DoorStatusPlanTemplate/<planTemplateID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
planTemplateID	string	--

Request Message

```
{
    "DoorStatusPlanTemplate": {
    /*ro, opt, object, door control schedule template*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
        "templateName":  "test",
        /*ro, req, string, template name*/
        "weekPlanNo":  1,
        /*ro, req, int, weekly schedule No.*/
        "holidayGroupNo":  "1,3,5"
        /*ro, req, string, holiday group No., desc:holiday group No.*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

14.6.2.13 Get parameters of door control schedule template

Request URL

GET /ISAPI/AccessControl/DoorStatusPlanTemplate/<planTemplateID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
planTemplateID	string	Schedule template No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

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Request Message

None

Response Message

```
{
    "DoorStatusPlanTemplate": {
    /*ro, req, object, door control schedule template*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
        "templateName":  "test",
        /*ro, req, string, template name*/
        "weekPlanNo":  1,
        /*ro, req, int, weekly schedule No.*/
        "holidayGroupNo":  "1,3,5",
        /*ro, req, string, holiday group No., desc:holiday group No.*/
        "doorNoList": [1, 2, 3],
        /*ro, opt, array, subType:int, range:[1,128]*/
        "doorNameList": ["door1", "door2", "door3"],
        /*ro, opt, array, subType:string*/
        "regionNodeNameList": ["区域1", "区域2", "区域3"]
        /*ro, opt, array, subType:string*/
    }
}
```

14.6.2.14 Get the configuration capability of the door control schedule template

Request URL

GET /ISAPI/AccessControl/DoorStatusPlanTemplate/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DoorStatusPlanTemplate": {
    /*ro, opt, object, schedule template*/
        "templateNo": {
        /*ro, opt, object, schedule template No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value of schedule template No.*/
            "@max":  16
            /*ro, opt, int, the maximum value of schedule template No.*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:true (enable), false (disable)*/
        "templateName": {
        /*ro, opt, object, template name length*/
            "@min":  1,
            /*ro, opt, int, the minimum value of template name length*/
            "@max":  32
            /*ro, opt, int, the maximum value of template name length*/
        },
        "weekPlanNo": {
        /*ro, opt, object, weekly schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value of weekly schedule No.*/
            "@max":  16
            /*ro, opt, int, the maximum value of weekly schedule No.*/
        },
        "holidayGroupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value of holiday group No.*/
            "@max":  16
            /*ro, opt, int, the maximum value of holiday group No.*/
        }
    }
}
```

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14.6.2.15 Set parameters of door control weekly schedule

Request URL

PUT /ISAPI/AccessControl/DoorStatusWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
weekPlanID	string	--

Request Message

```
{
    "DoorStatusWeekPlanCfg": {
    /*opt, object, weekly schedule of door control*/
        "enable":  true,
        /*req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
        "WeekPlanCfg": [
        /*req, array, weekly schedule parameters, subType:object*/
            {
                "week":  "Monday",
                /*req, enum, days of the week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
                "id":  1,
                /*req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*req, bool, whether to enable*/
                "doorStatus":  "remainClosed",
                /*req, enum, door control schedule, subType:string, desc:"remainOpen"-remain open (access without authentication), "remainClosed"-remain
closed (access is not allowed), "normal"-access by authentication, "sleep", "invalid"*/
                "TimeSegment": {
                /*req, object, time*/
                    "beginTime":  "00:00:00",
                    /*req, time, start time, desc:device local time*/
                    "endTime":  "10:00:00"
                    /*req, time, end time, desc:device local time*/
                }
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

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14.6.2.16 Get the configuration parameters of the door control week schedule

Request URL

GET /ISAPI/AccessControl/DoorStatusWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
weekPlanID	string	Weekly schedule No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

```
{
    "DoorStatusWeekPlanCfg": {
    /*ro, opt, object, door control week schedule*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (enable), false (disable)*/
        "beginDate":  "2024-01-10",
        /*ro, opt, date*/
        "endDate":  "2024-09-10",
        /*ro, opt, date*/
        "WeekPlanCfg": [
        /*ro, req, array, week schedule parameters, subType:object*/
            {
                "week":  "Monday",
                /*ro, req, enum, days of the week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
                "id":  1,
                /*ro, req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*ro, req, bool, whether to enable: "true"-enable, "false"-disable*/
                "doorStatus":  "remainClosed",
                /*ro, req, enum, door status, subType:string, desc:"remainOpen"-remain open (access without authentication), "remainClosed"-remain closed
(access is not allowed), "normal"-access by authentication, "sleep","invalid", "induction", "barrierFree"*/
                "TimeSegment": {
                /*ro, req, object, time*/
                    "beginTime":  "00:00:00",
                    /*ro, req, time, start time of the time period, desc:device local time*/
                    "endTime":  "10:00:00"
                    /*ro, req, time, end time of the time period, desc:device local time*/
                }
            }
        ]
    }
}
```

14.6.2.17 Get the configuration capability of the door control week schedule

Request URL

GET /ISAPI/AccessControl/DoorStatusWeekPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "DoorStatusWeekPlanCfg": {
    /*ro, opt, object*/
        "planNo": {
        /*ro, opt, object, week schedule No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable: "true"-enable,"false"-disable, desc:"true" (enable), "false" (disable)*/
        "WeekPlanCfg": {
        /*ro, opt, object, week schedule parameters*/
            "maxSize":  56,
            /*ro, opt, int*/
            "week": {
            /*ro, opt, object*/
                "@opt":  "Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday"
                /*ro, opt, string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
            },
            "id": {
            /*ro, opt, object, weekly schedule No.*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  8
                /*ro, opt, int*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether to enable: "true"-enable,"false"-disable, desc:"true" (enable), "false" (disable)*/
            "doorStatus": {
            /*ro, opt, object, door status*/
                "@opt":  "remainOpen,remainClosed,normal,sleep,invalid,induction,barrierFree"
                /*ro, opt, string, desc:"remainOpen" (remain open (access without authentication)), "remainClosed" (remain closed (access is not allowed)),
"normal" (access by authentication), "sleep", "invalid"*/
            },
            "TimeSegment": {
            /*ro, opt, object, time*/
                "beginTime":  "00:00:00",
                /*ro, opt, time, start time of the time period (device local time), desc:device local time*/
                "endTime":  "10:00:00",
                /*ro, opt, time, end time of the time period (device local time), desc:device local time*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:"hour", "minute", "second"; if this node is not returned, the default time accuracy is
"minute"*/
            }
        }
    }
}
```


14.6.3 Access Controller Management

14.6.3.1 Get the configuration capability of the access controller

Request URL

GET /ISAPI/AccessControl/AcsCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message


```
{
    "AcsCfg": {
    /*ro, req, object*/
        "voicePrompt":  "true,false",
        /*ro, opt, string, whether to enable voice prompt, desc:"true" (yes), "false" (no)*/
        "uploadCapPic":  "true,false",
        /*ro, opt, string, whether to upload the picture from linked capture, desc:"true" (yes), "false" (no)*/
        "saveCapPic":  "true,false",
        /*ro, opt, string, whether to save the captured picture, desc:"true" (yes), "false" (no)*/
        "protocol": {
        /*ro, opt, object, communication protocol type of the card reader, desc:"Private" (private protocol), "OSDP" (OSDP protocol)*/
            "@opt":  "Private,OSDP"
            /*ro, opt, string, optional item*/
        },
        "showPicture":  "true,false",
        /*ro, opt, string, whether to display the authenticated picture, desc:"true" (yes), "false" (no)*/
        "showEmployeeNo":  "true,false",
        /*ro, opt, string, whether to display the authenticated employee ID, desc:"true" (yes), "false" (no)*/
```


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```
        "showName":  "true,false",
        /*ro, opt, string, whether to display the authenticated name, desc:"true" (yes), "false" (no)*/
        "desensitiseEmployeeNo": {
        /*ro, opt, object, whether to enable employee No. de-identification for local UI display*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "desensitiseName": {
        /*ro, opt, object, whether to enable name de-identification for local UI display*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "thermalEnabled": {
        /*ro, opt, object, whether to enable temperature measurement*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "thermalMode": {
        /*ro, opt, object, whether to enable temperature measurement only mode: true-enable (only for temperature measurement),false-disable (default)*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "thermalPictureEnabled": {
        /*ro, opt, object, whether to enable uploading visible light pictures in temperature measurement only mode: true-enable,false-disable (default).
This field is used to control uploading captured pictures and visible light pictures, desc:whether to enable uploading visible light pictures in temperature
measurement only mode: true-enable,false-disable (default). This field is used to control uploading captured pictures and visible light pictures*/
            "@opt": [true, false]
            /*ro, opt, array, whether to enable uploading visible light pictures in temperature measurement only mode, subType:bool*/
        },
        "highestThermalThreshold": {
        /*ro, opt, object, maximum value of the temperature threshold, desc:float,upper limit of the temperature threshold which is accurate to one decimal
place,unit: Celsius*/
            "@min":  1.0,
            /*ro, opt, float*/
            "@max":  2.0
            /*ro, opt, float*/
        },
        "lowestThermalThreshold": {
        /*ro, opt, object, minimum value of the temperature threshold, desc:float,lower limit of the temperature threshold which is accurate to one decimal
place,unit: Celsius*/
            "@min":  1.0,
            /*ro, opt, float*/
            "@max":  2.0
            /*ro, opt, float*/
        },
        "thermalDoorEnabled": {
        /*ro, opt, object, whether to open the door according to the temperature threshold, desc:whether to open the door when the temperature is above the
upper limit (highestThermalThreshold) or below the lower limit (lowestThermalThreshold) of the threshold: true (open the door), false (not open the door
(default))*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "QRCodeEnabled": {
        /*ro, opt, object, whether to enable QR code function, desc:whether to enable QR code function: true-enable,false-disable (default)*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "remoteCheckDoorEnabled": {
        /*ro, opt, object, whether to enable controlling the door by remote verification, desc:whether to enable controlling the door by remote
verification: true-control,false-not control (default)*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "checkChannelType": {
        /*ro, opt, object, verification channel type, desc:verification channel type*/
            "@opt": ["Ezviz", "ISUP", "ISAPI", "PrivateSDK", "ISAPIListen"]
            /*ro, opt, array, subType:string*/
        },
        "isSupportChannelIp":  true,
        /*ro, opt, bool, whether it supports configuring IP address of the verification channel, desc:whether it supports configuring IP address of the
verification channel: true-yes,this field is not returned-no*/
        "needDeviceCheck": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "remoteCheckTimeout": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int, range:[1,10]*/
            "@max":  10
            /*ro, req, int, range:[1,10]*/
        },
        "remoteCheckVerifyMode": {
        /*ro, opt, object*/
            "@opt": [1, 2, 3, 4, 5, 6, 7]
            /*ro, req, array, subType:int*/
        },
        "offlineDevCheckOpenDoorEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "remoteCheckWithISAPIListen": {
```

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```
        /*ro, opt, object*/
            "@opt": ["async", "sync"]
            /*ro, req, array, subType:string*/
        },
        "uploadVerificationPic":  "true,false",
        /*ro, opt, string, whether to upload the authenticated picture, desc:"true"-yes, "false"-no*/
        "uploadVerificationPicType": {
        /*ro, opt, object*/
            "@opt": [0, 1]
            /*ro, req, array, subType:int*/
        },
        "saveVerificationPic":  "true,false",
        /*ro, opt, string, whether to save the authenticated picture, desc:"true" (yes), "false" (no)*/
        "saveFacePic":  "true,false",
        /*ro, opt, string, whether to save the registered face picture, desc:"true" (yes), "false" (no)*/
        "thermalUnit": {
        /*ro, opt, object, temperature unit, desc:object,temperature unit: "celsius" (default),"fahrenheit"*/
            "@opt": ["celsius", "fahrenheit"]
            /*ro, opt, array, subType:string*/
        },
        "highestThermalThresholdF": {
        /*ro, opt, object, maximum value of the temperature threshold, desc:the value is accurate to one decimal place, and the unit is Fahrenheit*/
            "@min":  1.0,
            /*ro, opt, float*/
            "@max":  1.0
            /*ro, opt, float*/
        },
        "lowestThermalThresholdF": {
        /*ro, opt, object, minimum value of the temperature threshold, desc:the value is accurate to one decimal place, and the unit is Fahrenheit*/
            "@min":  1.0,
            /*ro, opt, float*/
            "@max":  1.0
            /*ro, opt, float*/
        },
        "thermalCompensation": {
        /*ro, opt, object, temperature compensation*/
            "@min":  -99.9,
            /*ro, opt, float*/
            "@max":  99.9
            /*ro, opt, float*/
        },
        "externalCardReaderEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "combinationAuthenticationTimeout": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int, range:[1,20], step:1, unit:s*/
            "@max":  20
            /*ro, req, int, range:[1,20], step:1, unit:s*/
        },
        "combinationAuthenticationLimitOrder": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "buzzerEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "saveVPAudioFile": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "saveVPAudioFileByAuth": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "faceDuplicateCheckEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
    }
}
```

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14.6.3.2 Set the parameters of the access controller

Request URL

PUT /ISAPI/AccessControl/AcsCfg?format=json

Query Parameter

None

Request Message

```
{
    "AcsCfg": {
    /*req, object, parameters of the access controller*/
        "voicePrompt": true,
        /*opt, bool, whether to enable voice prompt*/
        "uploadCapPic": true,
        /*opt, bool, whether to upload the picture from linked capture, desc:whether to upload the picture from linked capture*/
        "saveCapPic": true,
        /*opt, bool, whether to save the captured picture, desc:whether to save the captured picture*/
        "showPicture": true,
        /*opt, bool, whether to display the authenticated picture, desc:whether to display the authenticated picture*/
        "showEmployeeNo": true,
        /*opt, bool, whether to display the authenticated employee ID*/
        "showName": true,
        /*opt, bool, whether to display the authenticated name*/
        "thermalEnabled": true,
        /*opt, bool, whether to enable temperature measurement*/
        "thermalMode": true,
        /*opt, bool, whether to enable temperature measurement only mode*/
        "thermalPictureEnabled": true,
        /*opt, bool, whether to enable uploading visible light pictures in temperature measurement only mode*/
        "highestThermalThreshold": 37.3,
        /*opt, float, maximum value of the temperature threshold*/
        "lowestThermalThreshold": 38.5,
        /*opt, float, minimum value of the temperature threshold*/
        "thermalDoorEnabled": false,
        /*opt, bool, whether to open the door when the temperature is above the upper limit (highestThermalThreshold) or below the lower limit (lowestThermalThreshold) of the threshold: true-open the door,false-not open the door (default), desc:whether to open the door when the temperature is above the upper limit (highestThermalThreshold) or below the lower limit (lowestThermalThreshold) of the threshold: true (open the door), false (not open the door (default))*/
        "QRCodeEnabled": false,
        /*opt, bool, whether to enable QR code function*/
        "remoteCheckDoorEnabled": false,
        /*opt, bool, whether to enable controlling the door by remote verification, desc:whether to enable controlling the door by remote verification*/
        "checkChannelType": "Ezviz",
        /*opt, enum, verification channel type, subType:string, dep:or,{$.AcsCfg.remoteCheckDoorEnabled,eq,true}, desc:verification channel type: "Ezviz"-EZVIZ channel,"ISUP"-ISUP channel,"ISAPI"-ISAPI channel,"PrivateSDK"-private SDK channel,"ISAPIListen"-ISAPI listening channel. This field is valid when remoteCheckDoorEnabled is true*/
        "channelIp": "test",
        /*opt, string, IP address of the verification channel, dep:and,{$.AcsCfg.checkChannelType,eq,PrivateSDK}, desc:this field is valid when checkChannelType is "PrivateSDK"*/
        "uploadVerificationPic": true,
        /*opt, bool, whether to upload the authenticated picture, desc:whether to upload the authenticated picture*/
        "saveVerificationPic": true,
        /*opt, bool, whether to save the authenticated picture, desc:whether to save the authenticated picture*/
        "saveFacePic": true,
        /*opt, bool, whether to save the registered face picture, desc:whether to save the registered face picture*/
        "thermalUnit": "celsius",
        /*opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit"*/
        "highestThermalThresholdF": 1.0,
        /*opt, float, the maximum value of the temperature threshold, desc:the value is accurate to one decimal place, and the unit is Fahrenheit this node is used to check whether to open the door when the temperature is higher than the threshold*/
        "lowestThermalThresholdF": 1.0,
        /*opt, float, the minimum value of the temperature threshold, desc:the value is accurate to one decimal place, and the unit is Fahrenheit this node is used to check whether to open the door when the temperature is higher than the threshold*/
        "thermalCompensation": 1.0,
        /*opt, float, temperature compensation, desc:float,temperature compensation,the value is accurate to one decimal place. The unit depends on the node thermalUnit. If the node thermalUnit does not exist,the default unit is Celsius*/
        "externalCardReaderEnabled": true,
        /*opt, bool*/
        "combinationAuthenticationTimeout": 1,
        /*opt, int, range:[1,20], step:1, unit:s*/
        "combinationAuthenticationLimitOrder": true,
        /*opt, bool*/
        "saveVPAudioFile": false,
        /*opt, bool*/
        "saveVPAudioFileByAuth": false,
        /*opt, bool*/
    }
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:status code*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:status description*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:error code*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:error description*/
}
```

14.6.3.3 Get the configuration parameters of the access controller

Request URL

GET /ISAPI/AccessControl/AcsCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AcsCfg": {
    /*ro, req, object*/
        "RS485Backup":  true,
        /*ro, opt, bool, whether to enable downstream RS-485 communication redundancy*/
        "showCapPic":  true,
        /*ro, opt, bool, whether to display the captured picture*/
        "showUserInfo":  true,
        /*ro, opt, bool, whether to display user information*/
        "overlayUserInfo":  true,
        /*ro, opt, bool, whether to overlay user information*/
        "voicePrompt":  true,
        /*ro, opt, bool, whether to enable voice prompt*/
        "uploadCapPic":  true,
        /*ro, opt, bool, whether to upload the picture from linked capture, desc:whether to upload the picture from linked capture*/
        "saveCapPic":  true,
        /*ro, opt, bool, whether to save the captured picture, desc:whether to save the captured picture*/
        "inputCardNo":  true,
        /*ro, opt, bool, whether to allow inputting card No. on keypad*/
        "enableWifiDetect":  true,
        /*ro, opt, bool, whether to enable Wi-Fi probe*/
        "enable3G4G":  true,
        /*ro, opt, bool, whether to enable 3G/4G*/
        "protocol":  "Private",
        /*ro, opt, enum, communication protocol type of the card reader, subType:string, desc:"Private" (private protocol), "OSDP" (OSDP protocol)*/
        "enableCaptureCertificate":  true,
        /*ro, opt, bool, whether to enable capturing the ID picture, desc:true (yes), false (no). If this node does not exist, it indicates that this
function is not supported*/
        "showPicture":  true,
        /*ro, opt, bool, whether to display the authenticated picture, desc:whether to display the authenticated picture*/
        "showPresetPicture":  true,
        /*ro, opt, bool*/
        "showEmployeeNo":  true,
        /*ro, opt, bool, whether to display the authenticated employee ID*/
        "showName":  true,
        /*ro, opt, bool, whether to display the authenticated name*/
        "showPhoneNo":  true,
        /*ro, opt, bool*/
        "desensitiseEmployeeNo":  true,
        /*ro, opt, bool, whether to enable employee No. de-identification for local UI display, dep:or,{$.AcsCfg.showEmployeeNo,eq,true}*/
        "desensitiseName":  true,
        /*ro, opt, bool, whether to enable name de-identification for local UI display, dep:or,{$.AcsCfg.showName,eq,true}*/
        "desensitisePhoneNo":  true,
        /*ro, opt, bool, dep:or,{$.AcsCfg.showPhoneNo,eq,true}*/
        "thermalEnabled":  true,
        /*ro, opt, bool, whether to enable temperature measurement*/
        "thermalMode":  true,
        /*ro, opt, bool, whether to enable temperature measurement only mode*/
        "thermalPictureEnabled":  true,
        /*ro, opt, bool, whether to enable uploading visible light pictures in temperature measurement only mode: true-enable,false-disable (default). This
field is used to control uploading captured pictures and visible light pictures*/
        "thermalIp":  "192.168.1.1",
        /*ro, opt, string, IP address of the thermography device, desc:for access control devices, each device only requires one IP address; for metal
detector doors, this field does not need to be configured*/
        "highestThermalThreshold":  37.3,
        /*ro, opt, float, upper limit of the temperature threshold*/
        "lowestThermalThreshold":  38.5,
        /*ro, opt, float, lower limit of the temperature threshold*/
```

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```
        "thermalDoorEnabled":  false,
        /*ro, opt, bool, whether to open the door according to the temperature threshold, desc:whether to open the door when the temperature is above the
upper limit (highestThermalThreshold) or below the lower limit (lowestThermalThreshold) of the threshold: true (open the door), false (not open the door
(default))*/
        "QRCodeEnabled":  false,
        /*ro, opt, bool, whether to enable QR code function*/
        "remoteCheckDoorEnabled":  false,
        /*ro, opt, bool, whether to enable controlling the door by remote verification, desc:whether to enable controlling the door by remote verification*/
        "checkChannelType":  "Ezviz",
        /*ro, opt, enum, verification channel type, subType:string, dep:or,{$.AcsCfg.remoteCheckDoorEnabled,eq,true}, desc:verification channel type*/
        "channelIp":  "test",
        /*ro, opt, string, IP address of the verification channel, dep:and,{$.AcsCfg.checkChannelType,eq,PrivateSDK}, desc:this field is valid when
checkChannelType is "PrivateSDK"*/
        "needDeviceCheck":  true,
        /*ro, opt, bool, dep:or,{$.AcsCfg.remoteCheckDoorEnabled,eq,true}*/
        "remoteCheckTimeout":  5,
        /*ro, opt, int, range:[1,10], unit:s, dep:or,{$.AcsCfg.remoteCheckDoorEnabled,eq,true}*/
        "remoteCheckVerifyMode":  1,
        /*ro, opt, enum, subType:int, dep:or,{$.AcsCfg.remoteCheckDoorEnabled,eq,true}*/
        "offlineDevCheckOpenDoorEnabled":  false,
        /*ro, opt, bool, dep:or,{$.AcsCfg.remoteCheckDoorEnabled,eq,true}*/
        "remoteCheckWithISAPIListen":  "async",
        /*ro, opt, enum, subType:string, dep:or,{$.AcsCfg.checkChannelType,eq,ISAPIListen}*/
        "uploadVerificationPic":  true,
        /*ro, opt, bool, whether to upload the authenticated picture, desc:whether to upload the authenticated picture*/
        "uploadVerificationPicType":  0,
        /*ro, opt, enum, subType:int*/
        "saveVerificationPic":  true,
        /*ro, opt, bool, whether to save the authenticated picture, desc:whether to save the authenticated picture*/
        "saveFacePic":  true,
        /*ro, opt, bool, whether to save the registered face picture, desc:whether to save the registered face picture*/
        "thermalUnit":  "celsius",
        /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit"*/
        "highestThermalThresholdF":  1.0,
        /*ro, opt, float, the maximum value of the temperature threshold, desc:the value is accurate to one decimal place, and the unit is Fahrenheit this
node is used to check whether to open the door when the temperature is higher than the threshold*/
        "lowestThermalThresholdF":  1.0,
        /*ro, opt, float, the minimum value of the temperature threshold, desc:the value is accurate to one decimal place, and the unit is Fahrenheit this
node is used to check whether to open the door when the temperature is higher than the threshold*/
        "enable5G":  true,
        /*ro, opt, bool, whether to enable 5G*/
        "thermalCompensation":  1.0,
        /*ro, opt, float, temperature compensation, desc:float,temperature compensation,the value is accurate to one decimal place. The unit depends on the
node thermalUnit. If the node thermalUnit does not exist,the default unit is Celsius*/
        "externalCardReaderEnabled":  true,
        /*ro, opt, bool*/
        "combinationAuthenticationTimeout":  1,
        /*ro, opt, int, range:[1,20], step:1, unit:s*/
        "combinationAuthenticationLimitOrder":  true,
        /*ro, opt, bool*/
        "passwordEnabled":  true,
        /*ro, opt, bool*/
        "showGender":  true,
        /*ro, opt, bool, whether to display gender*/
        "showSignInTime":  true,
        /*ro, opt, bool*/
        "showsCustomInfo":  true,
        /*ro, opt, bool*/
        "showMobileWebQRCode":  true,
        /*ro, opt, bool*/
        "fireAlarmInputType":  "alwaysOpen",
        /*ro, opt, enum, subType:string*/
        "buzzerEnabled":  true,
        /*ro, opt, bool*/
        "saveVPAudioFile":  false,
        /*ro, opt, bool*/
        "saveVPAudioFileByAuth":  false,
        /*ro, opt, bool*/
        "faceDuplicateCheckEnabled":  false,
        /*ro, opt, bool*/
        "localControllerBackupMode":  1,
        /*ro, opt, enum, subType:int*/
        "maxlocalControllerNum":  64,
        /*ro, opt, int*/
        "saveFpPicByCollectionMode":  false,
        /*ro, opt, bool*/
        "faceBatchModelingMode":  "recognitionPriority",
        /*ro, opt, enum, subType:string*/
        "externalAuthResultDisplayEnabled":  false
        /*ro, opt, bool*/
    }
}
```

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14.6.3.4 Get the capability of getting the working status of

the access controller **Request URL**

GET /ISAPI/AccessControl/AcsWorkStatus/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AcsWorkStatus": {
    /*ro, req, object*/
        "doorLockStatus": {
        /*ro, opt, object, door lock status (relay status): 0-normally close, 1-normally open, 2-short-circuit alarm, 3-broken-circuit alarm, 4-exception alarm*/
            "@opt":  "0,1,2,3,4"
            /*ro, opt, string*/
        },
        "doorStatus": {
        /*ro, opt, object, door (floor) status: 1-sleep, 2-remain unlocked (free), 3-remain locked (disabled), 4-normal status (controlled)*/
            "@opt":  "1,2,3,4"
            /*ro, opt, string*/
        },
        "magneticStatus": {
        /*ro, opt, object, magnetic contact status: 0-normally close,1-normally open, 2-short-circuit alarm, 3-broken-circuit alarm, 4-exception alarm*/
            "@opt":  "0,1,2,3,4"
            /*ro, opt, string, magnetic contact status*/
        },
        "caseStatus": {
        /*ro, opt, object, event trigger status*/
            "@min":  1,
            /*ro, opt, int, event trigger status*/
            "@max":  1
            /*ro, opt, int, event trigger status*/
        },
        "antiSneakStatus": {
        /*ro, opt, object, anti-passback status: "close"-disabled,"open"-enabled*/
            "@opt":  "close,open"
            /*ro, opt, string*/
        },
        "hostAntiDismantleStatus": {
        /*ro, opt, object, tampering status of the access control device: "close"-disabled, "open"-enabled*/
            "@opt":  "close,open"
            /*ro, opt, string*/
        },
        "cardReaderOnlineStatus": {
        /*ro, opt, object, online status of the authentication unit*/
            "@min":  1,
            /*ro, opt, int, the minimum value, range:[1,8]*/
            "@max":  1
            /*ro, opt, int, the maximum value, range:[1,8]*/
        },
        "cardReaderAntiDismantleStatus": {
        /*ro, opt, object, tampering status of the authentication unit*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "cardReaderVerifyMode": {
        /*ro, opt, object, current authentication mode of the authentication unit, desc:1-sleep,2-card+password,3-card,4-card or password,5-fingerprint,6-fingerprint+password,7-fingerprint or card,8-fingerprint+card,9-fingerprint+card+password,10-face or fingerprint or card or password,11-face+fingerprint,12-face+password,13-face+card,14-face,15-employee No.+password,16-fingerprint or password,17-employee No.+fingerprint,18-employee No.+fingerprint+password,19-face+fingerprint+card,20-face+password+fingerprint,21-employee No.+face,22-face or face+card,23-fingerprint or face,24-card or face or password,25-card or face,26-card or face or fingerprint,27-card or fingerprint or password*/
            "@opt":  "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34"
            /*ro, opt, string*/
        },
        "alarmOutStatus": {
        /*ro, opt, object, No. of output port with alarms*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "cardNum": {
        /*ro, opt, object, number of added cards*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        },
        "netStatus": {
        /*ro, opt, object, network connection status*/
            "@opt": ["connect", "disconnect", "ipConflict"]
            /*ro, opt, array, subType:string*/
        },
        "InterfaceStatusList": {
        /*ro, opt, object*/
            "@size":  2,
            /*ro, opt, int*/
            "id": {
                /*ro, opt, object*/
```

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```
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  2
                /*ro, opt, int*/
            },
            "netStatus": {
            /*ro, opt, object*/
                "@opt": ["connect", "disconnect", "ipConflict"]
                /*ro, opt, array, subType:string*/
            }
        },
        "sipStatus": {
        /*ro, opt, object*/
            "@opt": ["connect", "disconnect", "unregistered"]
            /*ro, opt, array, subType:string*/
        },
        "ezvizStatus": {
        /*ro, opt, object*/
            "@opt": ["unregistered", "notAdd", "connect"]
            /*ro, opt, array, subType:string*/
        },
        "voipStatus": {
        /*ro, opt, object*/
            "@opt": ["unregistered", "connect"]
            /*ro, opt, array, subType:string*/
        },
        "wifiStatus": {
        /*ro, opt, object*/
            "@opt": ["connect", " disconnect", "connecting", "noModule"]
            /*ro, opt, array, subType:string*/
        },
    }
}
```

14.6.3.5 Get the working status of the access controller

Request URL

GET /ISAPI/AccessControl/AcsWorkStatus?format=json&serialNumber=<serialNumber>

Query Parameter

Parameter Name	Parameter Type	Description
serialNumber	string	--

Request Message

None

Response Message

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```
{
    "AcsWorkStatus": {
    /*ro, req, object*/
        "doorLockStatus":  [1, 2, 1, 2],
        /*ro, opt, enumarray, door lock status (relay status), subType:int, desc:door lock status (relay status): 0 (normally close), 1 (normally open), 2 (short-circuit alarm), 3 (broken-circuit alarm), 4 (exception alarm). For example, [1,2,1,2] indicates that door lock 1 is normally open, door lock 2 triggers short-circuit alarm, door lock 3 is normally open, and door lock 4 triggers short-circuit alarm*/
        "doorStatus":  [1, 2, 1, 2],
        /*ro, opt, enumarray, door (floor) status, subType:int, desc:door (floor) status: 1 (sleep), 2 (remain unlocked (free)), 3 (remain locked (disabled)), 4 (normal status (controlled)). For example, [1,2,1,2] indicates that door 1 is sleeping, door 2 remains unlocked, door 3 is sleeping, and door 4 remains unlocked*/
        "magneticStatus":  [1, 2, 1, 2],
        /*ro, opt, enumarray, magnetic contact status, subType:int, desc:magnetic contact status: 0 (normally close), 1 (normally open), 2 (short-circuit alarm), 3 (broken-circuit alarm), 4 (exception alarm). For example, [1,2,1,2] indicates that magnetic contact No.1 is normally open, magnetic contact No.2 triggers short-circuit alarm, magnetic contact No.3 is normally open, and magnetic contact No.4 triggers short-circuit alarm*/
        "antiSneakStatus":  "open",
        /*ro, opt, enum, anti-passback status, subType:string, desc:"close" (disabled), "open" (enabled)*/
        "hostAntiDismantleStatus":  "open",
        /*ro, opt, enum, tampering status of the access control device, subType:string, desc:"close" (disabled), "open" (enabled)*/
        "cardReaderOnlineStatus": [1, 3, 5],
        /*ro, opt, array, online status of the authentication unit, subType:int, desc:online status of the authentication unit, e.g., [1,3,5] indicates that authentication unit No.1, No.3, and No.5 are online*/
        "cardReaderAntiDismantleStatus": [1, 3, 5],
        /*ro, opt, array, tampering status of the authentication unit, subType:int, desc:tampering status of the authentication unit, e.g., [1,3,5] indicates that the tampering function of authentication unit No.1, No.3, and No.5 is enabled*/
        "cardReaderVerifyMode":  [3, 5, 3, 5],
        /*ro, opt, enumarray, current authentication mode of the authentication unit, subType:int, desc:1 (sleep), 2 (card + password), 3 (card), 4 (card or password), 5 (fingerprint), 6 (fingerprint + password), 7 (fingerprint or card), 8 (fingerprint + card), 9 (fingerprint + card + password),10 (face or fingerprint or card or password), 11 (face + fingerprint), 12 (face + password), 13 (face + card), 14 (face), 15 (employee No. + password), 16 (fingerprint or password), 17 (employee No. + fingerprint), 18 (employee No. + fingerprint + password), 19 (face + fingerprint + card), 20 (face + password + fingerprint), 21 (employee No. + face), 22 (face or face + card), 23 (fingerprint or face), 24 (card or face or password), 25 (card or face), 26 (card or face or fingerprint), 27 (card or fingerprint or password). For example, [3,5,3,5] indicates that the authentication mode of authentication unit 1 is "card", the authentication mode of authentication unit 2 is "fingerprint", the authentication mode of authentication unit 3 is "card", and the authentication mode of authentication unit 4 is "fingerprint"*/
        "cardNum":  3,
        /*ro, opt, int, number of added cards, desc:number of added cards*/
        "netStatus":  "connect",
        /*ro, opt, enum, subType:string*/
        "InterfaceStatusList": [
        /*ro, opt, array, subType:object*/
            {
                "id":  1,
                /*ro, opt, int*/
                "netStatus":  "connect"
                /*ro, opt, enum, subType:string*/
            }
        ],
        "sipStatus":  "connect",
        /*ro, opt, enum, subType:string*/
        "ezvizStatus":  "unregistered",
        /*ro, opt, enum, subType:string*/
        "voipStatus":  "unregistered",
        /*ro, opt, enum, subType:string*/
        "wifiStatus":  "connect",
        /*ro, opt, enum, subType:string*/
    }
}
```

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14.6.3.6 Get the capability of clearing all pictures in the device

Request URL

GET /ISAPI/AccessControl/ClearPictureCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "ClearPictureCfgCap": {
    /*ro, req, object, clear all pictures in the device*/
        "ClearFlags": {
        /*ro, opt, object, clear status*/
            "facePicture": {
            /*ro, opt, object, clear registered face pictures*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool*/
            },
            "capOrVerifyPicture": {
            /*ro, opt, object, clear authenticated or captured face pictures*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool*/
            },
            "irisPicture": {
            /*ro, opt, object, clear registered iris pictures*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool*/
            }
        }
    }
}
```

14.6.3.7 Clear all pictures stored in the device

Request URL

PUT /ISAPI/AccessControl/ClearPictureCfg?format=json

Query Parameter

None

Request Message

```
{
    "ClearPictureCfg": {
    /*req, object, clear picture configuration*/
        "ClearFlags": {
        /*opt, object, clear status*/
            "facePicture":  true,
            /*opt, bool, whether it supports clearing registered face pictures in the device*/
            "capOrVerifyPicture":  true,
            /*opt, bool, whether it supports clearing authenticated or captured face pictures stored in the device*/
            "irisPicture":  true
            /*opt, bool, whether to clear registered iris pictures in the device*/
        }
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

14.6.4 Access Point Unit Management

14.6.4.1 Get the capability of configuring the door lock status when the device is powered off

Request URL

GET /ISAPI/AccessControl/Configuration/lockType/capabilities?format=json

Query Parameter

None

Request Message

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None

Response Message

```
{
    "LockTypeCap": {
    /*ro, req, object, the capability of configuring the door lock status when the device is powered off*/
        "status": {
        /*ro, opt, object, door lock status when the device is powered off*/
            "@opt": ["alwaysOpen", "alwaysClose"]
            /*ro, req, array, options, subType:string*/
        }
    }
}
```

14.6.4.2 Set door lock status when the device is powered off

Request URL

PUT /ISAPI/AccessControl/Configuration/lockType?format=json&doorID=<doorID>

Query Parameter

Parameter Name	Parameter Type	Description
doorID	string	--

Request Message

```
{
    "LockType": {
    /*opt, object, status*/
        "status":  "alwaysOpen"
        /*req, enum, door lock status when the device is powered off, subType:string, desc:"alwaysOpen"-remain open, "alwaysClose"-remain closed*/
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code*/
    "statusString":  "test",
    /*ro, opt, string, status description*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code*/
    "errorCode":  1,
    /*ro, req, int, error code*/
    "errorMsg":  "ok"
    /*ro, req, string, error information*/
}
```

14.6.4.3 Get the configuration of the door lock status when the device is powered off

Request URL

GET /ISAPI/AccessControl/Configuration/lockType?format=json&doorID=<doorID>

Query Parameter

Parameter Name	Parameter Type	Description
doorID	string	--

Request Message

None

Response Message

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```
{
    "LockType": {
    /*ro, opt, object, status*/
        "status":  "alwaysOpen"
        /*ro, opt, enum, door lock status when the device is powered off, subType:string, desc:door lock status when the device is powered off:
"alwaysOpen"-remain open,"alwaysClose"-remain closed*/
    }
}
```

14.6.4.4 Get the door configuration capability

Request URL

GET /ISAPI/AccessControl/Door/param/<doorID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
doorID	string	Door No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DoorParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <doorNo min="1" max="128">
    <!--ro, opt, int, door No., attr:min{req, int},max{req, int}, desc:door No.-->1
  </doorNo>
  <doorName min="0" max="32">
    <!--ro, opt, string, door name, attr:min{req, int},max{req, int}-->test
  </doorName>
  <magneticType opt="alwaysClose,alwaysOpen">
    <!--ro, opt, enum, magnetic contact type, subType:string, attr:opt{req, string}, desc:"alwaysClose" (remain locked), "alwaysOpen" (remain unlocked)--
>alwaysClose
  </magneticType>
  <openButtonType opt="alwaysClose,alwaysOpen">
    <!--ro, opt, enum, door button type, subType:string, attr:opt{req, string}, desc:"alwaysClose" (remain locked), "alwaysOpen" (remain unlocked)--
>alwaysClose
  </openButtonType>
  <openDuration min="1" max="255">
    <!--ro, opt, int, door open duration (door relay action time), attr:min{req, int},max{req, int}-->1
  </openDuration>
  <disabledOpenDuration min="1" max="255">
    <!--ro, opt, int, door open duration by disability card (delay duration of closing the door), attr:min{req, int},max{req, int}-->1
  </disabledOpenDuration>
  <magneticAlarmTimeout min="0" max="255">
    <!--ro, opt, int, alarm time of magnetic contact detection timeout, range:[0,255], unit:s, attr:min{req, int},max{req, int}, desc:alarm time of magnetic
contact detection timeout,which is between 0 and 255,0 refers to not triggering alarm,unit: second-->1
  </magneticAlarmTimeout>
  <enableDoorLock opt="true,false">
    <!--ro, opt, bool, whether to enable locking door when the door is closed, attr:opt{req, string}-->true
  </enableDoorLock>
  <enableLeaderCard opt="true,false">
    <!--ro, opt, bool, whether to enable remaining open with first card, attr:opt{req, string}-->true
  </enableLeaderCard>
  <leaderCardMode opt="disable,alwaysOpen,authorize">
    <!--ro, opt, enum, first card mode, subType:string, attr:opt{req, string}, desc:"disable", "alwaysOpen" (remain open with first card), "authorize"
(first card authentication)-->disable
  </leaderCardMode>
  <leaderCardOpenDuration min="1" max="1440">
    <!--ro, opt, int, duration of remaining open with first card, attr:min{req, int},max{req, int}-->1
  </leaderCardOpenDuration>
  <stressPassword min="1" max="8">
    <!--ro, opt, string, duress password, attr:min{req, int},max{req, int}, desc:the maximum length is 8 bytes, and the duress password should be encoded by
Base64 for transmission-->test
  </stressPassword>
  <superPassword min="1" max="8">
    <!--ro, opt, string, super password, attr:min{req, int},max{req, int}, desc:the maximum length is 8 bytes, and the duress password should be encoded by
Base64 for transmission-->test
  </superPassword>
  <unlockPassword min="1" max="8">
    <!--ro, opt, string, dismiss password,t, attr:min{req, int},max{req, int}, desc:the maximum length is 8 bytes, and the duress password should be encoded
by Base64 for transmission-->test
  </unlockPassword>
  <useLocalController opt="true,false">
    <!--ro, opt, bool, whether it is connected to the distributed controller, attr:opt{req, string}-->true
  </useLocalController>
  <localControllerID min="0" max="64">
    <!--ro, opt, int, distributed controller No., range:[0,64], attr:min{req, int},max{req, int}, desc:ro,distributed controller No.,which is between 1 and
```

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```
64,0-unregistered-->1
  </localControllerID>
  <localControllerDoorNumber min="0" max="4">
    <!--ro, opt, int, distributed controller door No., range:[0,4], attr:min{req, int},max{req, int}, desc:ro,distributed controller door No.,which is
between 1 and 4,0-unregistered-->1
  </localControllerDoorNumber>
  <localControllerStatus opt="0,1,2,3,4,5,6,7,8,9">
    <!--ro, opt, enum, online status of the distributed controller, subType:int, attr:opt{req, string}, desc:0-offline, 1-network online, 2-RS-485 serial
port 1 on loop circuit 1, 3-RS-485 serial port 2 on loop circuit 1, 4-RS-485 serial port 1 on loop circuit 2, 5-RS-485 serial port 2 on loop circuit 2, 6-
RS-485 serial port 1 on loop circuit 3, 7-RS-485 serial port 2 on loop circuit 3,8-RS-485 serial port 1 on loop circuit 4,  9-RS-485 serial port 2 on loop
circuit 4-->1
  </localControllerStatus>
  <lockInputCheck opt="true,false">
    <!--ro, opt, bool, whether to enable door lock input detection, attr:opt{req, string}-->true
  </lockInputCheck>
  <lockInputType opt="alwaysClose,alwaysOpen">
    <!--ro, opt, enum, door lock input type, subType:string, attr:opt{req, string}, desc:"alwaysClose" (remain locked), "alwaysOpen" (remain unlocked)--
>alwaysClose
  </lockInputType>
  <doorTerminalMode opt="preventCutAndShort,preventCutAndShort,common">
    <!--ro, opt, enum, working mode of door terminal, subType:string, attr:opt{req, string}, desc:working mode of door terminal: "preventCutAndShort"-
prevent from broken-circuit and short-circuit (default),"common"-->preventCutAndShort
  </doorTerminalMode>
  <openButton opt="true,false">
    <!--ro, opt, bool, whether to enable door button, attr:opt{req, string}, desc:whether to enable door button: "true"-yes (default),"false"-no-->true
  </openButton>
  <ladderControlDelayTime min="1" max="255">
    <!--ro, opt, int, elevator control delay time (for visitor), attr:min{req, int},max{req, int}-->1
  </ladderControlDelayTime>
  <Leader>
    <!--ro, opt, object-->
    <continuousVerificationTimes min="1" max="10">
      <!--ro, opt, int, range:[1,10], attr:min{req, int},max{req, int}-->1
    </continuousVerificationTimes>
    <continuousVerificationDuration min="1" max="10">
      <!--ro, opt, int, range:[5,60], unit:s, attr:min{req, int},max{req, int}-->20
    </continuousVerificationDuration>
    <effectiveTimeEnabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </effectiveTimeEnabled>
    <dayBeginTime>
      <!--ro, opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </dayBeginTime>
    <beginEffectiveTime>
      <!--ro, opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </beginEffectiveTime>
    <endEffectiveTime>
      <!--ro, opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </endEffectiveTime>
  </Leader>
  <verificationPassOpenDoor opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </verificationPassOpenDoor>
  <relayReverseEnabled opt="true,false" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </relayReverseEnabled>
  <regionNodeID min="0" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </regionNodeID>
  <method opt="batchPut">
    <!--ro, opt, string, attr:opt{req, string}-->batchPut
  </method>
  <doorCfgType opt="all,byDoor,byRegion,byDoorSeparate">
    <!--ro, opt, string, attr:opt{req, string}-->all
  </doorCfgType>
  <hostDoorNo min="1" max="4">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </hostDoorNo>
  <doorAddedFlag opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </doorAddedFlag>
  <preCloseDoorEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </preCloseDoorEnabled>
  <preCloseDoorTime min="0" max="255">
    <!--ro, opt, int, unit:s, attr:min{req, int},max{req, int}-->1
  </preCloseDoorTime>
  <passingDetectEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </passingDetectEnabled>
  <doorNotClosedAlarmEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </doorNotClosedAlarmEnabled>
  <doorNotClosedAlarmText min="0" max="128">
    <!--ro, opt, string, range:[0,128], attr:min{req, int},max{req, int}-->test
  </doorNotClosedAlarmText>
  <nonStopFloorEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->false
  </nonStopFloorEnabled>
  <localExtensionDoorInfoList>
    <!--ro, opt, object-->
    <localExtensionDoorInfo>
      <!--ro, req, object-->
      <devType>
```

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```
        <!--ro, req, enum, subType:string-->2doors
      </devType>
      <devNum>
        <!--ro, req, int-->2
      </devNum>
    </localExtensionDoorInfo>
  </localExtensionDoorInfoList>
  <closeDoorMode opt="auto,manual">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
  </closeDoorMode>
</DoorParam>
```

14.6.4.5 Set the door (floor) parameters

Request URL

PUT /ISAPI/AccessControl/Door/param/<doorID>

Query Parameter

Parameter Name	Parameter Type	Description
doorID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DoorParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, door parameter, attr:version{req, string, protocolVersion}-->
  <doorName>
    <!--opt, string, door name-->test
  </doorName>
  <magneticType>
    <!--opt, enum, door magnetic sensor type, subType:string, desc:"alwaysClose" (remain locked), "alwaysOpen" (remain unlocked)-->alwaysClose
  </magneticType>
  <openButtonType>
    <!--opt, enum, open door button type, subType:string, desc:"alwaysClose" (remain locked), "alwaysOpen" (remain unlocked)-->alwaysClose
  </openButtonType>
  <openDuration>
    <!--opt, int, door open duration, range:[1,255], unit:s-->1
  </openDuration>
  <disabledOpenDuration>
    <!--opt, int, door open duration by disability card (delay duration of closing the door), range:[1,255], unit:s-->1
  </disabledOpenDuration>
  <magneticAlarmTimeout>
    <!--opt, int, alarm time of magnetic contact detection timeout, range:[0,255], unit:s, desc:0 refers to not triggering alarm-->1
  </magneticAlarmTimeout>
  <enableDoorLock>
    <!--opt, bool, lock door when door closed-->true
  </enableDoorLock>
  <enableLeaderCard>
    <!--opt, bool, whether to enable remaining open with first card-->true
  </enableLeaderCard>
  <leaderCardMode>
    <!--opt, enum, first card mode, subType:string, desc:first card mode: "disable","alwaysOpen"-remain open with first card,"authorize"-first card
authentication. If this node is configured,the node <enableLeaderCard> is invalid-->disable
  </leaderCardMode>
  <leaderCardOpenDuration>
    <!--opt, int, duration of remaining open with first card, range:[0,1440], unit:min, dep:and,{$.DoorParam.leaderCardMode,eq,alwaysOpen}-->1
  </leaderCardOpenDuration>
  <stressPassword>
    <!--opt, string, duress password, desc:the maximum length is 8 bytes, and the duress password should be encoded by Base64 for transmission-->test
  </stressPassword>
  <superPassword>
    <!--opt, string, super password, desc:the maximum length is 8 bytes, and the duress password should be encoded by Base64 for transmission-->test
  </superPassword>
  <unlockPassword>
    <!--opt, string, unlock password, desc:the maximum length is 8 bytes, and the duress password should be encoded by Base64 for transmission-->test
  </unlockPassword>
  <useLocalController>
    <!--opt, bool, whether it is connected to the distributed controller-->true
  </useLocalController>
  <localControllerID>
    <!--opt, int, distributed controller No., range:[0,64], desc:ro,distributed controller No.,which is between 1 and 64,0-unregistered-->1
  </localControllerID>
  <localControllerDoorNumber>
    <!--opt, int, distributed controller door No., range:[0,4], desc:ro,distributed controller door No.,which is between 1 and 4,0-unregistered-->1
  </localControllerDoorNumber>
  <localControllerStatus>
    <!--opt, enum, online status of the distributed controller, subType:int, desc:ro,online status of the distributed controller: 0-offline,1-network
online,2-RS-485 serial port 1 on loop circuit 1,3-RS-485 serial port 2 on loop circuit 1,4-RS-485 serial port 1 on loop circuit 2,5-RS-485 serial port 2 on
loop circuit 2,6-RS-485 serial port 1 on loop circuit 3,7-RS-485 serial port 2 on loop circuit 3,8-RS-485 serial port 1 on loop circuit 4,9-RS-485 serial
port 2 on loop circuit 4-->1
  </localControllerStatus>
  <lockInputCheck>
    <!--opt, bool, whether to enable door lock input detection-->true
```

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```
  </lockInputCheck>
  <lockInputType>
    <!--opt, enum, door lock input type, subType:string, desc:"alwaysClose" (remain locked), "alwaysOpen" (remain unlocked), lt remains locked by default-->alwaysClose
  </lockInputType>
  <doorTerminalMode>
    <!--opt, enum, working mode of door terminal, subType:string, desc:"preventCutAndShort" (prevent from broken-circuit and short-circuit (default)),"common"-->preventCutAndShort
  </doorTerminalMode>
  <openButton>
    <!--opt, bool, whether to enable door button-->true
  </openButton>
  <ladderControlDelayTime>
    <!--opt, int, elevator control delay time (for visitor), range:[1,255], unit:min-->1
  </ladderControlDelayTime>
  <Leader>
    <!--opt, object-->
    <continuousVerificationTimes>
      <!--opt, int, range:[1,10]-->1
    </continuousVerificationTimes>
    <continuousVerificationDuration>
      <!--opt, int, range:[5,60], unit:s-->20
    </continuousVerificationDuration>
    <effectiveTimeEnabled>
      <!--req, bool-->true
    </effectiveTimeEnabled>
    <dayBeginTime>
      <!--opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </dayBeginTime>
    <beginEffectiveTime>
      <!--opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </beginEffectiveTime>
    <endEffectiveTime>
      <!--opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </endEffectiveTime>
  </Leader>
  <verificationPassOpenDoor>
    <!--opt, bool-->true
  </verificationPassOpenDoor>
  <relayReverseEnabled>
    <!--opt, bool-->true
  </relayReverseEnabled>
</DoorParam>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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14.6.4.6 Get the door (floor) configuration parameters

Request URL

GET /ISAPI/AccessControl/Door/param/<doorID>

Query Parameter

Parameter Name	Parameter Type	Description
doorID	string	Door No. (floor No.), which starts from 1.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DoorParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <doorName>
    <!--ro, opt, string, door name-->test
  </doorName>
  <magneticType>
    <!--ro, opt, enum, magnetic contact type, subType:string, desc:"alwaysClose"-remain locked, "alwaysOpen"-remain unlocked-->alwaysClose
  </magneticType>
  <openButtonType>
    <!--ro, opt, enum, door button type, subType:string, desc:"alwaysClose"-remain locked, "alwaysOpen"-remain unlocked-->alwaysClose
  </openButtonType>
  <openDuration>
    <!--ro, opt, int, door open duration, range:[1,255], unit:s-->1
  </openDuration>
  <disabledOpenDuration>
    <!--ro, opt, int, door open duration by disability card (delay duration of closing the door), range:[1,255], unit:s-->1
  </disabledOpenDuration>
  <magneticAlarmTimeout>
    <!--ro, opt, int, alarm time of magnetic contact detection timeout, range:[0,255], unit:s, desc:0 refers to not triggering alarm-->1
  </magneticAlarmTimeout>
  <enableDoorLock>
    <!--ro, opt, bool, whether to enable locking door when the door is closed-->true
  </enableDoorLock>
  <enableLeaderCard>
    <!--ro, opt, bool, whether to enable remaining open with first card. This node is invalid when leaderCardMode is configured-->true
  </enableLeaderCard>
  <leaderCardMode>
    <!--ro, opt, enum, first card mode, subType:string, desc:"disable","alwaysOpen"-remain open with first card,"authorize"-first card authentication. If
this node is configured,the node <enableLeaderCard> is invalid-->disable
  </leaderCardMode>
  <leaderCardOpenDuration>
    <!--ro, opt, int, duration of remaining open with first card, range:[0,1440], unit:min, dep:and,{$.DoorParam.LeaderCardMode,eq,alwaysOpen}-->1
  </leaderCardOpenDuration>
  <stressPassword>
    <!--ro, opt, string, duress password, desc:the maximum length is 8 bytes, and the duress password should be encoded by Base64 for transmission-->test
  </stressPassword>
  <superPassword>
    <!--ro, opt, string, super password, desc:wo,super password,the maximum length is 8 bytes,and the super password should be encoded by Base64 for
transmission-->test
  </superPassword>
  <unlockPassword>
    <!--ro, opt, string, dismiss password, desc:the maximum length is 8 bytes, and the dismiss password should be encoded by Base64 for transmission-->test
  </unlockPassword>
  <useLocalController>
    <!--ro, opt, bool, whether it is connected to the distributed controller-->true
  </useLocalController>
  <localControllerID>
    <!--ro, opt, int, distributed controller No., which is between 1 and 64, range:[0,64], desc:0-unregistered-->1
  </localControllerID>
  <localControllerDoorNumber>
    <!--ro, opt, int, distributed controller door No., range:[0,4], desc:0-unregistered-->1
  </localControllerDoorNumber>
  <localControllerStatus>
    <!--ro, opt, enum, online status of the distributed controller, subType:int, desc:0-offline,1-network online,2-RS-485 serial port 1 on loop circuit 1,3-
RS-485 serial port 2 on loop circuit 1,4-RS-485 serial port 1 on loop circuit 2,5-RS-485 serial port 2 on loop circuit 2,6-RS-485 serial port 1 on loop
circuit 3,7-RS-485 serial port 2 on loop circuit 3,8-RS-485 serial port 1 on loop circuit 4,9-RS-485 serial port 2 on loop circuit 4-->1
  </localControllerStatus>
  <lockInputCheck>
    <!--ro, opt, bool, whether to enable door lock input detection-->true
  </lockInputCheck>
  <lockInputType>
    <!--ro, opt, enum, door lock input type, subType:string, desc:"alwaysClose"-remain locked (default), "alwaysOpen"-remain unlocked-->alwaysClose
  </lockInputType>
  <doorTerminalMode>
    <!--ro, opt, enum, working mode of door terminal, subType:string, desc:"preventCutAndShort"-prevent from broken-circuit and short-circuit
(default),"common"-->preventCutAndShort
  </doorTerminalMode>
  <openButton>
    <!--ro, opt, bool, whether to enable door button: "true"-yes (default), "false"-no-->true
  </openButton>
  <ladderControlDelayTime>
    <!--ro, opt, int, elevator control delay time (for visitor), range:[1,255], unit:min-->1
  </ladderControlDelayTime>
  <Leader>
    <!--ro, opt, object-->
    <continuousVerificationTimes>
      <!--ro, opt, int, range:[1,10]-->1
    </continuousVerificationTimes>
    <continuousVerificationDuration>
      <!--ro, opt, int, range:[5,60], unit:s-->20
    </continuousVerificationDuration>
    <effectiveTimeEnabled>
      <!--ro, req, bool-->true
    </effectiveTimeEnabled>
    <dayBeginTime>
      <!--ro, opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </dayBeginTime>
    <beginEffectiveTime>
      <!--ro, opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
    </beginEffectiveTime>
    <endEffectiveTime>
```

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```
        <!--ro, opt, time, dep:and,{$.DoorParam.Leader.effectiveTimeEnabled,eq,true}-->00:00:00
      </endEffectiveTime>
    </Leader>
    <verificationPassOpenDoor>
      <!--ro, opt, bool-->true
    </verificationPassOpenDoor>
    <relayReverseEnabled>
      <!--ro, opt, bool-->true
    </relayReverseEnabled>
    <regionNodeID>
      <!--ro, opt, string, range:[1,32]-->test
    </regionNodeID>
    <doorAddedFlag>
      <!--ro, opt, bool-->true
    </doorAddedFlag>
    <preCloseDoorEnabled>
      <!--ro, opt, bool-->true
    </preCloseDoorEnabled>
    <preCloseDoorTime>
      <!--ro, opt, int, range:[0,255], unit:s, dep:and,{$.DoorParam.preCloseDoorEnabled,eq,true}-->1
    </preCloseDoorTime>
    <passingDetectEnabled>
      <!--ro, opt, bool-->true
    </passingDetectEnabled>
    <doorNotClosedAlarmEnabled>
      <!--ro, opt, bool-->true
    </doorNotClosedAlarmEnabled>
    <doorNotClosedAlarmText>
      <!--ro, opt, string, range:[0,128], dep:or,{$.DoorParam.doorNotClosedAlarmEnabled,eq,true}-->test
    </doorNotClosedAlarmText>
    <nonStopFloorEnabled>
      <!--ro, opt, bool-->false
    </nonStopFloorEnabled>
    <closeDoorMode>
      <!--ro, opt, enum, subType:string-->auto
    </closeDoorMode>
  </DoorParam>
```

14.6.4.7 Get the status of the secure door control unit

Request URL

GET /ISAPI/AccessControl/DoorSecurityModule/moduleStatus

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ModuleStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, module status, attr:version{req, string, protocolVersion}-->
  <securityModuleNo min="1" max="256">
    <!--ro, req, string, secure door control unit No., attr:min{req, int},max{req, int}-->test
  </securityModuleNo>
  <onlineStatus opt="0,1">
    <!--ro, req, enum, online status, subType:int, attr:opt{req, string}, desc:0 (offline), 1(online)-->1
  </onlineStatus>
  <desmantelStatus opt="0,1">
    <!--ro, req, enum, tamper-proof status, subType:int, attr:opt{req, string}, desc:0 (the unit is not tampered), 1(the unit is tampered)-->1
  </desmantelStatus>
</ModuleStatus>
```

14.6.4.8 Get the capability of getting the status of the secure door control unit

Request URL

GET /ISAPI/AccessControl/DoorSecurityModule/moduleStatus/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ModuleStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, module status, attr:version{req, string, protocolVersion}-->
  <securityModuleNo min="1" max="256">
    <!--ro, req, string, secure door control unit No., attr:min{req, int},max{req, int}-->test
  </securityModuleNo>
  <onlineStatus opt="0,1">
    <!--ro, req, enum, online status, subType:int, attr:opt{req, string}, desc:0 (offline), 1 (online)-->1
  </onlineStatus>
  <desmantelStatus opt="0,1">
    <!--ro, req, enum, tampering status, subType:int, attr:opt{req, string}, desc:0 (the unit is not tampered), 1 (the unit is tampered)-->1
  </desmantelStatus>
</ModuleStatus>
```

14.6.4.9 Get lock status

Request URL

GET /ISAPI/VideoIntercom/WorkStatus

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WorkStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, status information, attr:version{req, string, protocolVersion}-->
  <lockStatusList>
    <!--ro, opt, object, lock status list, desc:for the lock of the door station, it is always online, so this node may not be returned; for the lock of the external module,this node is required-->
    <node>
      <!--ro, opt, object, node-->
      <lockID>
        <!--ro, req, int, lock ID, range:[0,1], step:1-->0
      </lockID>
      <lockStatus>
        <!--ro, req, enum, lock status, subType:string, desc:"online", "offline"-->online
      </lockStatus>
    </node>
  </lockStatusList>
</WorkStatus>
```

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14.6.5 Anti-Passback

14.6.5.1 Get the anti-passing back configuration capability

Request URL

GET /ISAPI/AccessControl/AntiSneakCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AntiSneakCfg": {
    /*ro, req, object*/
        "enable":  "true,false",
        /*ro, req, string, whether to enable anti-passing back*/
        "startCardReaderNo": {
        /*ro, opt, object, first card reader No., desc:first card reader No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  4,
            /*ro, opt, int, the maximum value*/
            "@opt": [1, 4]
            /*ro, opt, array, subType:int*/
        }
    }
}
```

14.6.5.2 Get the parameters of anti-passback configuration

Request URL

GET /ISAPI/AccessControl/AntiSneakCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AntiSneakCfg": {
    /*ro, req, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable anti-passback*/
        "startCardReaderNo":  1
        /*ro, opt, int, first card reader No., desc:first card reader No.,0-no first card reader*/
    }
}
```

14.6.5.3 Set the anti-passing back parameters

Request URL

PUT /ISAPI/AccessControl/AntiSneakCfg?format=json

Query Parameter

None

Request Message

```
{
    "AntiSneakCfg": {
    /*req, object*/
        "enable":  true,
        /*req, bool, whether to enable anti-passing back*/
        "startCardReaderNo":  1
        /*opt, int, first card reader No., desc:0-no first card reader*/
    }
}
```

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.6.5.4 Get the anti-passing back configuration parameters of a specified card reader

Request URL

GET /ISAPI/AccessControl/CardReaderAntiSneakCfg/<cardReaderID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
cardReaderID	string	Card reader ID

Request Message

None

Response Message

```
{
    "CardReaderAntiSneakCfg": {
    /*ro, req, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable the anti-passing back function of the card reader, desc:"true" (enable), "false" (disable)*/
        "followUpCardReader": [2, 3, 4]
        /*ro, opt, array, following card reader No. after the first card reader, subType:int, desc:[2,3,4] indicates that card reader No. 2, No. 3, or No. 4
can be swiped after the first card reader*/
    }
}
```

14.6.5.5 Set anti-passing back parameters of a card reader

Request URL

PUT /ISAPI/AccessControl/CardReaderAntiSneakCfg/<cardReaderID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
cardReaderID	string	--

Request Message

```
{
    "CardReaderAntiSneakCfg": {
    /*req, object, anti-passing back parameters of a card reader*/
        "enable":  true,
        /*req, bool, whether to enable the anti-passing back function of the card reader, desc:"true"-enable, "false"-disable*/
        "followUpCardReader": [2, 3, 4]
        /*opt, array, following card reader No. after the first card reader, subType:int, desc:e.g., [2,3,4] indicates that card reader No. 2, No. 3, and
No. 4 can be swiped after the first card reader*/
    }
}
```

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.6.5.6 Get the configuration capability of anti-passing back parameters of card readers

Request URL

GET /ISAPI/AccessControl/CardReaderAntiSneakCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CardReaderAntiSneakCfg": {
    /*ro, req, object*/
        "cardReaderNo": {
        /*ro, opt, object, card reader No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  512,
            /*ro, opt, int, the maximum value*/
            "@opt": [1, 4]
            /*ro, opt, array, subType:int*/
        },
        "enable":  "true,false",
        /*ro, req, string, whether to enable the anti-passing back function of the card reader*/
        "followUpCardReader": {
        /*ro, opt, object, array,following card reader No. after the first card reader*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  512,
            /*ro, opt, int*/
            "@opt": [1, 4]
            /*ro, opt, array, subType:int*/
        }
    }
}
```

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14.6.5.7 Get the capability of clearing anti-passback records

Request URL

GET /ISAPI/AccessControl/ClearAntiSneak/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "ClearAntiSneak": {
    /*ro, req, object*/
        "EmployeeNoList": {
        /*ro, opt, object*/
            "maxSize":  32,
            /*ro, opt, int*/
            "employeeNo": {
            /*ro, opt, object, employee No. (person ID)*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  32
                /*ro, opt, int*/
            }
        },
        "clearMode": {
        /*ro, opt, object*/
            "@opt": ["employeeNo", "all"]
            /*ro, opt, array, subType:string*/
        }
    }
}
```

14.6.5.8 Clear anti-passback records in the device

Request URL

PUT /ISAPI/AccessControl/ClearAntiSneak?format=json

Query Parameter

None

Request Message

```
{
    "ClearAntiSneak": {
    /*req, object, clear anti-passback records, desc:clear anti-passback records*/
        "EmployeeNoList": [
        /*opt, array, person ID list, subType:object, desc:person ID list*/
            {
                "employeeNo":  "test"
                /*req, string, employee No. (person ID)*/
            }
        ],
        "clearMode":  "employeeNo"
        /*opt, enum, clearing mode, subType:string, desc:clearing mode*/
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.6.5.9 Get the capability of clearing anti-passback parameters

Request URL

GET /ISAPI/AccessControl/ClearAntiSneakCfg/capabilities?format=json

Query Parameter

None

Request Message

None

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Response Message

```
{
    "ClearAntiSneakCfg": {
    /*ro, req, object, the capability of clearing anti-passback*/
        "ClearFlags": {
        /*ro, req, object*/
            "antiSneak":  "true,false"
            /*ro, req, string, whether to clear the anti-passback parameter*/
        }
    }
}
```

14.6.5.10 Clear anti-passing back parameters

Request URL

PUT /ISAPI/AccessControl/ClearAntiSneakCfg?format=json

Query Parameter

None

Request Message

```
{
    "ClearAntiSneakCfg": {
    /*req, object*/
        "ClearFlags": {
        /*req, object*/
            "antiSneak":  true
            /*req, bool, whether to clear the anti-passing back parameters*/
        }
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

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14.6.6 Authentication Schedule Management

14.6.6.1 Set control schedule parameters of card reader authentication mode

Request URL

PUT /ISAPI/AccessControl/CardReaderPlan/<cardReaderID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
cardReaderID	string	--

Request Message

```
{
    "CardReaderPlan": {
    /*req, object*/
        "templateNo":  1
        /*req, int, schedule template No., desc:0-cancel linking the template to the schedule and restore to the default status (normal status)*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

14.6.6.2 Get the control schedule configuration parameters of the card reader authentication mode

Request URL

GET /ISAPI/AccessControl/CardReaderPlan/<cardReaderID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
cardReaderID	string	Card reader No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

```
{
    "CardReaderPlan": {
    /*ro, req, object, schedule template structure*/
        "templateNo":  1
        /*ro, req, int, schedule template number, desc:0-cancel linking the template to the schedule and restore to the default status (normal status)*/
    }
}
```

14.6.6.3 Get the control schedule configuration capability of the card reader authentication mode

Request URL

GET /ISAPI/AccessControl/CardReaderPlan/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "CardReaderPlan": {
    /*ro, opt, object, card reader No. node*/
        "cardReaderNo": {
        /*ro, opt, object, card reader No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  4
            /*ro, opt, int*/
        },
        "templateNo": {
        /*ro, opt, object, schedule template No. node*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16,
            /*ro, opt, int*/
            "@opt": [65535, 65534, 65533]
            /*ro, opt, array, subType:int*/
        },
        "verifyMode": {
        /*ro, opt, object, authentication mode, dep:and,{$.CardReaderPlan.templateNo.@opt[*],eq,65535}*/
            "@opt":
"cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpOrCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employe
eNoAndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,faceOrfaceAndCard,fpOrface,cardOrfaceOrPw,cardOrFace,cardOrFaceOrFp,cardOrFp
OrPw,faceOrPw,employeeNoAndFaceAndPw,cardOrFaceOrFaceAndCard,iris,faceOrFpOrCardOrPwOrIris,faceOrCardOrPwOrIris,sleep,invalid"
            /*ro, opt, string, desc:"cardAndPw" (card + password), "card" (card), "cardOrPw" (card or password), "fp" (fingerprint), "fpAndPw" (fingerprint
+ password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint + card), "fpAndCardAndPw" (fingerprint + card + password), "faceOrFpOrCardOrPw"
(face or fingerprint or card or password), "faceAndFp" (face + fingerprint), "faceAndPw" (face + password), "faceAndCard" (face + card), "face" (face),
"employeeNoAndPw" (employee No. + password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee No. + fingerprint), "employeeNoAndFpAndPw"
(employee No. + fingerprint + password), "faceAndFpAndCard" (face + fingerprint + card), "faceAndPwAndFp" (face + password + fingerprint),
"employeeNoAndFace" (employee No. + face), "faceOrfaceAndCard" (face or face + card), "fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or
password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or fingerprint or password), "faceOrPw" (face
or password), "employeeNoAndFaceAndPw" (employee No. + face + card), "cardOrFaceOrFaceAndCard" (card or face or face + card), "iris",
"faceOrFpOrCardOrPwOrIris"(face or fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or  card or password or iris), "sleep",
"invalid"*/
        }
    }
}
```

14.6.6.4 Get the holiday group configuration parameters of the control schedule of the card reader authentication mode

Request URL

GET /ISAPI/AccessControl/VerifyHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
holidayGroupID	string	Holiday group No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

```
{
    "VerifyHolidayGroupCfg": {
    /*ro, opt, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (yes), false-no (default)*/
        "groupName":  "test",
        /*ro, req, string, holiday group name*/
        "holidayPlanNo":  "1,3,5"
        /*ro, opt, string, holiday group schedule No., desc:holiday group schedule No.*/
    }
}
```

14.6.6.5 Set holiday group parameters of control schedule of card reader authentication mode

Request URL

PUT /ISAPI/AccessControl/VerifyHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

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Parameter Name	Parameter Type	Description
holidayGroupID	string	--

Request Message

```
{
    "VerifyHolidayGroupCfg": {
    /*opt, object, holiday group parameters of control schedule of card reader authentication mode*/
        "enable":  true,
        /*req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
        "groupName":  "test",
        /*req, string, holiday group name*/
        "holidayPlanNo":  "1,3,5"
        /*opt, string, holiday group schedule No., desc:holiday group schedule No.*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

14.6.6.6 Get the holiday group configuration capability of the control schedule of the card reader authentication mode

Request URL

GET /ISAPI/AccessControl/VerifyHolidayGroupCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "VerifyHolidayGroupCfg": {
    /*ro, opt, object*/
        "groupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:whether to enable: "true"-enable,"false"-disable*/
        "groupName": {
        /*ro, opt, object, length of holiday group name*/
            "@min":  1,
            /*ro, opt, int, the minimum length*/
            "@max":  32
            /*ro, opt, int, the maximum length*/
        },
        "holidayPlanNo": {
        /*ro, opt, object, holiday group schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        }
    }
}
```

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14.6.6.7 Get holiday schedule parameters of the card reader authentication mode

Request URL

GET /ISAPI/AccessControl/VerifyHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
holidayPlanID	string	Holiday schedule No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

```
{
    "VerifyHolidayPlanCfg": {
    /*ro, opt, object, holiday schedule parameters of the card reader authentication mode*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (enable), false (disable)*/
        "beginDate":  "1970-01-01",
        /*ro, req, date, start date of the holiday, desc:device local time*/
        "endDate":  "1970-01-02",
        /*ro, req, date, end date of the holiday, desc:device local time*/
        "HolidayPlanCfg": [
        /*ro, req, array, holiday schedule parameters, subType:object*/
            {
                "id":  1,
                /*ro, req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*ro, req, bool, whether to enable the holiday schedule, desc:true (enable), false (disable)*/
                "verifyMode":  "cardAndPw",
                /*ro, req, enum, authentication mode, subType:string, desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password), "fp"
(fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password), "cardOrFpOrIrisOrPw" (card or
fingerprint or iris or password), "cardOnIrisOrPw" (card or iris or password), "cardAndIris" (card+ris), "fpAndIri" (fingerprint+iris), "faceAndIris"
(face+iris), "irisAndPw" (iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw" (face+iris+password), "cardAndFaceAndIris"
(card+face+iris)*/
                "TimeSegment": {
                /*ro, opt, object, time*/
                    "beginTime":  "00:00:00",
                    /*ro, req, time, start time of the time period, desc:device local time*/
                    "endTime":  "10:00:00"
                    /*ro, req, time, end time of the time period, desc:device local time*/
                }
            }
        ]
    }
}
```

14.6.6.8 Set holiday schedule parameters of card reader authentication mode

Request URL

PUT /ISAPI/AccessControl/VerifyHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
holidayPlanID	string	--

Request Message

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```
{
    "VerifyHolidayPlanCfg": {
    /*opt, object, holiday schedule parameters of the card reader authentication mode*/
        "enable":  true,
        /*req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
        "beginDate":  "1970-01-01",
        /*req, date, start date of the holiday, desc:device local time*/
        "endDate":  "1970-01-02",
        /*req, date, end date of the holiday, desc:device local time*/
        "HolidayPlanCfg": [
        /*req, array, holiday schedule parameters, subType:object*/
            {
                "id":  1,
                /*req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
                "verifyMode":  "cardAndPw",
                /*req, enum, authentication mode, subType:string, desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password), "fp"
(fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password), "cardOrFpOrIrisOrPw" (card or
fingerprint or iris or password), "cardOrIrisOrPw" (card or iris or password), "cardAndIris" (card+ris), "fpAndIri" (fingerprint+iris), "faceAndIris"
(face+iris), "irisAndPw" (iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw" (face+iris+password), "cardAndFaceAndIris"
(card+face+iris)*/
                "TimeSegment": {
                /*opt, object, time*/
                    "beginTime":  "00:00:00",
                    /*req, time, start time of the time period, desc:device local time*/
                    "endTime":  "10:00:00"
                    /*req, time, end time of the time period, desc:device local time*/
                }
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node must be returned when the value of statusCode is not 1*/
}
```

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14.6.6.9 Get the holiday schedule configuration capability of the card reader authentication mode

Request URL

GET /ISAPI/AccessControl/VerifyHolidayPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "VerifyHolidayPlanCfg": {
    /*ro, opt, object, holiday schedule of card reader authentication mode*/
        "planNo": {
        /*ro, opt, object, holiday schedule template No.*/
            "@min":  1,
            /*ro, opt, int, maximum value*/
            "@max":  16
            /*ro, opt, int, minimum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:true (enable), false (disable)*/
        "beginDate":  "1970-01-01",
        /*ro, opt, date, start date of the holiday (device local time), desc:(device local time)*/
        "endDate":  "1970-01-02",
        /*ro, opt, date, end date of the holiday (device local time), desc:(device local time)*/
        "HolidayPlanCfg": {
        /*ro, opt, object, holiday schedule parameters*/
            "maxSize":  8,
            /*ro, opt, int, maximum value*/
            "id": {
            /*ro, opt, object, time period No.*/
                "@min":  1,
                /*ro, opt, int, minimum value*/
                "@max":  8
                /*ro, opt, int, maximum value*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether to enable, desc:true (enable), false (disable)*/
            "verifyMode": {
            /*ro, opt, object, authentication mode*/
                "@opt":
"cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpOrCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employe
eNoAndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,faceOrfaceAndCard,fpOrface,cardOrfaceOrPw,cardOrFace,cardOrFaceOrFp,faceOrPw
,employeeNoAndFaceAndPw,cardOrFaceOrFaceAndCard,iris,faceOrFpOrCardOrPwOrIris,faceOrCardOrPwOrIris,sleep,invalid,cardOrFpOrFaceOrIris,fpOrFaceOrIrisOrPw,car
dOrFpOrIrisOrPw,cardOrIrisOrPw,cardAndIris,fpAndIris,faceAndIris,irisAndPw,cardAndIrisAndPw,faceAndIrisAndPw,cardAndFaceAndIris,Pw,cardOrFpOrPw,faceOrFpOrPw
,cardAndVp,faceAndVp,fpAndVp,pwAndVp"
                /*ro, opt, string, authentication mode, desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password), "fp" (fingerprint),
"fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw" (fingerprint+card+password),
"faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password), "faceAndCard" (face+card),
"face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee No.+fingerprint),
"employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp" (face+password+fingerprint),
"employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or
password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or fingerprint or password), "iris" (iris),
"faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card or password or iris), "cardOrFace" (card
or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw" (employee No.+face+password),
"cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint or password), "cardOrFpOrFaceOrIris" (card or fingerpriont or
face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password), "cardOrFpOrIrisOrPw" (card or fingerprint or iris or password),
"cardOrIrisOrPw" (card or iris or password), "cardAndIris" (card+ris), "fpAndIri" (fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw"
(iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw" (face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
            },
            "TimeSegment": {
            /*ro, opt, object, time*/
                "beginTime":  "00:00:00",
                /*ro, opt, time, start time, desc:(device local time)*/
                "endTime":  "10:00:00",
                /*ro, opt, time, end time, desc:(device local time)*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:"hour", "minute", "second". If this node is not returned, the default time accuracy is
"minute"*/
            }
        },
        "purePwdVerifyEnable":  true
        /*ro, opt, bool*/
    }
}
```

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14.6.6.10 Get the schedule template parameters of card reader authentication mode

Request URL

GET /ISAPI/AccessControl/VerifyPlanTemplate/<planTemplateID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
planTemplateID	string	Schedule template No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

```
{
    "VerifyPlanTemplate": {
    /*ro, opt, object, the schedule template parameters of card reader authentication mode*/
        "enable": true,
        /*ro, req, bool, whether to enable, desc:"true"-enable, "false"-disable*/
        "templateName": "test",
        /*ro, req, string, template name*/
        "weekPlanNo": 1,
        /*ro, req, int, week schedule No.*/
        "holidayGroupNo": "1,3,5"
        /*ro, req, string, holiday group No., desc:holiday group No.*/
    }
}
```

14.6.6.11 Set the schedule template parameters of the card reader authentication mode

Request URL

PUT /ISAPI/AccessControl/VerifyPlanTemplate/<planTemplateID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
planTemplateID	string	Schedule template No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

```
{
    "VerifyPlanTemplate": {
    /*opt, object*/
        "enable": true,
        /*req, bool, whether to enable, desc:true (yes), false (no)*/
        "templateName": "test",
        /*req, string, template name*/
        "weekPlanNo": 1,
        /*req, int, week schedule No.*/
        "holidayGroupNo": "1,3,5"
        /*req, string, holiday group No., desc:holiday group No.*/
    }
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode": "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx": "0x00000000"
    /*ro, opt, string*/
}
```

14.6.6.12 Get the schedule template configuration capability of the card reader authentication mode

Request URL

GET /ISAPI/AccessControl/VerifyPlanTemplate/capabilities?format=json

Query Parameter

None

Request Message

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None

Response Message

```
{
    "VerifyPlanTemplate": {
    /*ro, opt, object*/
        "templateNo": {
        /*ro, opt, object, schedule template No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:true (yes), false (no)*/
        "templateName": {
        /*ro, opt, object, template name length*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  32
            /*ro, opt, int*/
        },
        "weekPlanNo": {
        /*ro, opt, object, weekly schedule No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        },
        "holidayGroupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        }
    }
}
```

14.6.6.13 Get the week schedule configuration parameters of the card reader authentication mode

Request URL

GET /ISAPI/AccessControl/VerifyWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
weekPlanID	string	Weekly schedule No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

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```
{
    "VerifyWeekPlanCfg": {
    /*ro, opt, object, the week schedule configuration parameters of the card reader authentication mode*/
        "enable":  true,
        /*ro, req, bool, whether to enable the week schedule configuration parameters of the card reader authentication mode, desc:true (enable), false
(disable)*/
        "WeekPlanCfg": [
        /*ro, req, array, week schedule parameters, subType:object*/
            {
                "week":  "Monday",
                /*ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
                "id":  1,
                /*ro, req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*ro, req, bool, whether to enable week schedule*/
                "verifyMode":  "cardAndPw",
                /*ro, req, enum, authentication mode, subType:string, desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password), "fp"
(fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password), "cardOrFpOrIrisOrPw" (card or
fingerprint or iris or password), "cardOrIrisOrPw" (card or iris or password), "cardAndIris" (card+ris), "fpAndIri" (fingerprint+iris), "faceAndIris"
(face+iris), "irisAndPw" (iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw" (face+iris+password), "cardAndFaceAndIris"
(card+face+iris)*/
                "TimeSegment": {
                /*ro, req, object, time*/
                    "beginTime":  "00:00:00",
                    /*ro, req, time, start time, desc:device local time*/
                    "endTime":  "10:00:00"
                    /*ro, req, time, end time, desc:device local time*/
                }
            }
        ]
    }
}
```

14.6.6.14 Set the week schedule parameters of the card reader authentication mode

Request URL

PUT /ISAPI/AccessControl/VerifyWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
weekPlanID	string	Weekly schedule No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

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```
{
    "VerifyWeekPlanCfg": {
    /*opt, object, the week schedule configuration parameters of the card reader authentication mode*/
        "enable":  true,
        /*req, bool, whether to enable, desc:"true" (enable), "false" (disable)*/
        "WeekPlanCfg": [
        /*req, array, week schedule parameters, subType:object*/
            {
                "week":  "Monday",
                /*req, enum, days of the week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
                "id":  1,
                /*req, int, time period No., range:[1,8]*/
                "enable":  true,
                /*req, bool, whether to enable week schedule*/
                "verifyMode":  "cardAndPw",
                /*req, enum, authentication mode, subType:string, desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password), "fp"
(fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or
fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card
or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw"
(employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint or password),
"cardOrFpOrFaceOrIris" (card or fingerprint or face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password), "cardOrFpOrIrisOrPw" (card or
fingerprint or iris or password), "cardOrIrisOrPw" (card or iris or password), "cardAndIris" (card+ris), "fpAndIri" (fingerprint+iris), "faceAndIris"
(face+iris), "irisAndPw" (iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw" (face+iris+password), "cardAndFaceAndIris"
(card+face+iris)*/
                "TimeSegment": {
                /*req, object, time*/
                    "beginTime":  "00:00:00",
                    /*req, time, start time of the time period, desc:device local time*/
                    "endTime":  "10:00:00"
                    /*req, time, end time of the time period, desc:device local time*/
                }
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

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14.6.6.15 Get the weekly schedule configuration capability of the card reader authentication mode

Request URL

GET /ISAPI/AccessControl/VerifyWeekPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "VerifyWeekPlanCfg": {
    /*ro, opt, object, weekly schedule parameters of the card reader authentication mode*/
        "planNo": {
        /*ro, opt, object, weekly schedule No.*/
            "@min":  1,
            /*ro, opt, int, minimum value*/
            "@max":  16
            /*ro, opt, int, maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:"true"-enable, "false"-disable*/
        "WeekPlanCfg": {
        /*ro, opt, object, week schedule parameters*/
            "maxSize":  56,
            /*ro, opt, int, maximum value*/
            "week": {
            /*ro, opt, object, day of a week*/
                "@opt":  "Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday"
                /*ro, opt, string, day of a week, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
            },
            "id": {
            /*ro, opt, object, weekly schedule No.*/
                "@min":  1,
                /*ro, opt, int, minimum value*/
                "@max":  8
                /*ro, opt, int, maximum value*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether to enable, desc:true (enable), false (disable)*/
            "verifyMode": {
            /*ro, opt, object, authentication mode*/
                "@opt":
"cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpOrCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employe
eNoAndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,faceOrfaceAndCard,fpOrface,cardOrfaceOrPw,cardOrFace,cardOrFaceOrFp,faceOrPw
,employeeNoAndFaceAndPw,cardOrFaceOrFaceAndCard,iris,faceOrFpOrCardOrPwOrIris,faceOrCardOrPwOrIris,sleep,invalid,cardOrFpOrFaceOrIris,fpOrFaceOrIrisOrPw,car
dOrFpOrIrisOrPw,cardOrIrisOrPw,cardAndIris,fpAndIris,faceAndIris,irisAndPw,cardAndIrisAndPw,faceAndIrisAndPw,cardAndFaceAndIris,Pw,cardOrFpOrPw,faceOrFpOrPw
,cardAndVp,faceAndVp,fpAndVp,pwAndVp"
                /*ro, opt, string, authentication mode, desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password), "fp" (fingerprint),
"fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw" (fingerprint+card+password),
"faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password), "faceAndCard" (face+card),
"face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee No.+fingerprint),
"employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp" (face+password+fingerprint),
"employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or
password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or fingerprint or password), "iris" (iris),
"faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card or password or iris), "cardOrFace" (card
or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw" (employee No.+face+password),
"cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint or password), "cardOrFpOrFaceOrIris" (card or fingerprion or
face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password), "cardOrFpOrIrisOrPw" (card or fingerprint or iris or password),
"cardOrIrisOrPw" (card or iris or password), "cardAndIris" (card+ris), "fpAndIri" (fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw"
(iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw" (face+iris+password), "cardAndFaceAndIris" (card+face+iris), "sleep",
"invalid"*/
            },
            "TimeSegment": {
            /*ro, opt, object, time*/
                "beginTime":  "00:00:00",
                /*ro, opt, time, start time, desc:start time of the time period (device local time)*/
                "endTime":  "10:00:00",
                /*ro, opt, time, end time, desc:end time of the time period (device local time)*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:If this node is not returned, it indicates that the time accuracy is "minute". "hour",
"minute", "second"*/
            }
        },
        "purePwdVerifyEnable":  true
        /*ro, opt, bool*/
    }
}
```

14.6.7 Credential Recognition Unit Management

14.6.7.1 Set the card reader parameters

Request URL

PUT /ISAPI/AccessControl/CardReaderCfg/<cardReaderID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
cardReaderID	string	--

Request Message

vasilyk.gnivkinghta.kz

```
{
    "CardReaderCfg": {
    /*req, object, card reader information*/
        "enable":  true,
        /*req, bool, whether to enable, desc:true-yes, false-no*/
        "okLedPolarity":  "cathode",
        /*opt, enum, OK LED polarity, subType:string, desc:"cathode", "anode”*/
        "errorLedPolarity":  "cathode",
        /*opt, enum, Error LED polarity, subType:string, desc:"cathode", "anode"*/
        "buzzerPolarity":  "cathode",
        /*opt, enum, buzzer polarity, subType:string, desc:"cathode", "anode"*/
        "swipeInterval":  1,
        /*opt, int, time interval of repeated authentication, unit:s, desc:it is valid for authentication modes such as fingerprint, card, face, etc.*/
        "pressTimeout":  1,
        /*opt, int, timeout to reset entry on keypad, unit:s*/
        "enableFailAlarm":  true,
        /*opt, bool, whether to enable excessive failed authentication attempt alarm*/
        "maxReadCardFailNum":  1,
        /*opt, int, maximum number of failed authentication attempts*/
        "enableTamperCheck":  true,
        /*opt, bool, whether to enable tampering detection*/
        "offlineCheckTime":  1,
        /*opt, int, time to detect after the card reader is offline, unit:s*/
        "fingerPrintCheckLevel":  1,
        /*opt, enum, fingerprint recognition level, subType:int, desc:1-1/10 false acceptance rate (FAR), 2-1/100 false acceptance rate (FAR), 3-1/1000 false acceptance rate (FAR), 4-1/10000 false acceptance rate (FAR), 5-1/100000 false acceptance rate (FAR), 6-1/1000000 false acceptance rate (FAR), 7-1/10000000 false acceptance rate (FAR), 8-1/100000000 false acceptance rate (FAR), 9-3/100 false acceptance rate (FAR), 10-3/1000 false acceptance rate (FAR), 11-3/10000 false acceptance rate (FAR), 12-3/100000 false acceptance rate (FAR), 13-3/1000000 false acceptance rate (FAR), 14-3/10000000 false acceptance rate (FAR), 15-3/100000000 false acceptance rate (FAR), 16-Automatic Normal, 17-Automatic Secure, 18-Automatic More Secure (currently not support)*/
        "useLocalController":  true,
        /*opt, bool, whether it is connected to the distributed controller*/
        "localControllerID":  1,
        /*opt, int, distributed controller No., range:[0,64], dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:0-unregistered. This field is valid only when useLocalController is "true"*/
        "localControllerReaderID":  1,
        /*opt, int, card reader ID of the distributed controller, dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:0-unregistered. This field is valid only when useLocalController is "true"*/
        "cardReaderChannel":  1,
        /*opt, enum, communication channel No. of the card reader, subType:int, dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:0-Wiegand or offline, 1-RS-485A, 2-RS-485B. This field is valid only when useLocalController is "true"*/
        "fingerPrintImageQuality":  1,
        /*opt, enum, fingerprint image quality, subType:int, desc:1-low quality (V1), 2-medium quality (V1), 3-high quality (V1), 4-highest quality (V1), 5-low quality (V2), 6-medium quality (V2), 7-high quality (V2), 8-highest quality (V2)*/
        "fingerPrintContrastTimeOut":  1,
        /*opt, enum, fingerprint comparison timeout, subType:int, desc:it is between 1 and 20, unit: second, 255-infinite*/
        "fingerPrintRecogizeInterval":  1,
        /*opt, enum, fingerprint scanning interval, subType:int, desc:it is between 1 and 10, unit: second, 255-no delay*/
        "fingerPrintMatchFastMode":  1,
        /*opt, enum, fingerprint matching quick mode, subType:int, desc:1-quick mode 1, 2-quick mode 2, 3-quick mode 3, 4-quick mode 4, 5-quick mode 5, 255-automatic*/
        "fingerPrintModuleSensitive":  1,
        /*opt, enum, fingerprint module sensitivity, subType:int, desc:fingerprint module sensitivity,which is between 1 and 8*/
        "fingerPrintModuleLightCondition": "outdoor",
        /*opt, enum, fingerprint module light condition, subType:string, desc:"outdoor", "indoor”*/
        "faceMatchThresholdN":  1,
        /*opt, int, threshold of face picture 1:N comparison, range:[0,100], desc:threshold of face picture 1:N comparison,which is between 0 and 100*/
        "faceQuality":  1,
        /*opt, int, face picture quality, range:[0,100]*/
        "faceRecogizeTimeOut":  1,
        /*opt, enum, face recognition timeout, subType:int, desc:it is between 1 and 20, unit: second, 255-infinite*/
        "faceRecogizeInterval":  1,
        /*opt, enum, face recognition interval, subType:int, desc:it is between 1 and 10, unit: second, 255-no delay*/
        "cardReaderFunction":  ["fingerPrint", "face", "fingerVein", "iris", "card"],
        /*opt, enumarray, card reader type, subType:string, desc:"fingerPrint"-fingerprint, "face", "fingerVein"-finger vein, “iris”. For example, ["fingerPrint","face"] indicates that the card reader supports both fingerprint and face*/
        "cardReaderDescription":  "Wiegand\u000485Offline",
        /*opt, string, card reader description, desc:if the card reader is the Wiegand card reader or if offline, this field will be set to "Wiegand" or "485Offline”*/
        "faceImageSensitometry":  1,
        /*opt, int, face picture exposure, range:[0,655535]*/
        "livingBodyDetect":  true,
        /*opt, bool, whether to enable human detection*/
        "faceMatchThreshold1":  1,
        /*opt, int, threshold of face picture 1:1 comparison, range:[0,100], desc:threshold of face picture 1:1 comparison,which is between 0 and 100*/
        "buzzerTime":  1,
        /*opt, int, buzzing duration, range:[0,59999], unit:s, desc:0-long buzzing*/
        "faceMatch1SecurityLevel":  1,
        /*opt, enum, security level of face 1:1 recognition, subType:int, desc:1-normal, 2-high, 3-higher*/
        "faceMatchNSecurityLevel":  1,
        /*opt, enum, security level of face 1:N recognition, subType:int, desc:1-normal, 2-high, 3-higher*/
        "envirMode":  "indoor",
        /*opt, enum, environment mode of face recognition, subType:string, desc:"indoor", "other”*/
        "liveDetLevelSet":  "low",
        /*opt, enum, threshold level of liveness detection, subType:string, desc:"low", "middle"-medium, "high"*/
        "liveDetThreshold":  1,
        /*opt, int, range:[0,100]*/
        "liveDetAntiAttackCntLimit":  1,
        /*opt, int, number of anti-attacks of liveness detection,, range:[1,255], desc:this value should be configured as the same one on both client and device*/
        "enableLiveDetAntiAttack":  true,
        /*opt, bool, whether to enable anti-attack for liveness detection*/
        "liveDetAntiAttackLockedTime":  1,
```


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```
        /*opt, int, range:[0,300], unit:s*/
        "supportDelFPByID": true,
        /*opt, bool, whether the card reader supports deleting fingerprint by fingerprint ID, desc:true-yes, false-no*/
        "fingerPrintCapacity": 1,
        /*opt, int, fingerprint capacity*/
        "fingerPrintNum": 1,
        /*opt, int, number of added fingerprints*/
        "defaultVerifyMode": "cardAndPw",
        /*opt, enum, default authentication mode of the fingerprint and card reader (factory defaults):, subType:string, desc:factory defaults; "cardAndPw"
(card+password), "card" (card), "cardOrPw" (card or password), "fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card),
"fpAndCard" (fingerprint+card), "fpAndCardAndPw" (fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp"
(face+fingerprint), "faceAndPw" (face+password), "faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint
or password), "employeeNoAndFp" (employee No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard"
(face+fingerprint+card), "faceAndPwAndFp" (face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card),
"fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint),
"cardOrFpOrPw" (card or fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris),
"faceOrCardOrPwOrIris" (face or card or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or
password), "employeeNoAndFaceAndPw" (employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint
or password), "cardOrFpOrFaceOrIris" (card or fingerprint or face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw" (card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
        "faceRecogizeEnable": 1,
        /*opt, enum, whether to enable facial recognition, subType:int, desc:1-enable, 2-disable, 3-attendence checked in/out by recognition of multiple
faces*/
        "FPAlgorithmVersion": "test",
        /*opt, string, fingerprint algorithm library version, range:[1,32]*/
        "cardReaderVersion": "test",
        /*opt, string, card reader version, range:[1,32]*/
        "enableReverseCardNo": true,
        /*opt, bool, whether to enable reversing the card No.*/
        "independSwipeIntervals": 0,
        /*opt, int, time interval of person authentication, unit: second. This time interval is calculated for each person separately and is different from
swipeInterval, desc:time interval of person authentication,unit: second. This time interval is calculated for each person separately and is different from
swipeInterval*/
        "maskFaceMatchThresholdN": 1,
        /*opt, int, 1:N face picture (face with mask and normal background) comparison threshold, range:[0,100], desc:1:N face picture (face with mask and
normal background) comparison threshold,value range: [0,100]*/
        "maskFaceMatchThreshold1": 1,
        /*opt, int, 1:1 face picture (face with mask and normal background) comparison threshold, range:[0,100], desc:1:1 face picture (face with mask and
normal background) comparison threshold,value range: [0,100]*/
        "faceMotionDetLevel": "low",
        /*opt, enum, face motion detection level, subType:string, desc:"high", "medium", "low"*/
        "showMode": "normal",
        /*opt, enum, display mode, subType:string, desc:this node is not valid; simple mode indicates that the device displays authentication results
exclude employee No., name, etc.; the device applies normal mode by default; advertisement mode indicates that the device displays both advertisement and
authentication results; meeting mode indicates that the device displays check-in page of the conference; custom mode indicates that the device displays
layout of the interface customized by users; "concise"-simple mode, "normal"-normal mode (default), "advertising"-advertisement mode, "meeting"-meeting
mode, "selfDefine"-custom mode*/
        "enableScreenOff": true,
        /*opt, bool, whether to enable auto locking the screen*/
        "screenOffTimeout": 1
        /*opt, int, time, step:1, unit:s*/
    }
}
```

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Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok"
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
}
```

14.6.7.2 Get the card reader configuration parameters

Request URL

GET /ISAPI/AccessControl/CardReaderCfg/<cardReaderID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
cardReaderID	string	--

Request Message

None

Response Message

```
{
    "CardReaderCfg": {
    /*ro, req, object, card reader information*/
        "enable": true,
        /*ro, req, bool, whether to enable, desc:true (enable), false (disable)*/
        "cardReaderName": "test",
        /*ro, opt, string, range:[0,64]*/
        "okLedPolarity": "cathode",
        /*ro, opt, enum, OK LED polarity, subType:string, desc:"cathode", "anode"*/
        "errorLedPolarity": "cathode",
        /*ro, opt, enum, Error LED polarity, subType:string, desc:"cathode", "anode"*/
        "buzzerPolarity": "cathode",
        /*ro, opt, enum, buzzer polarity, subType:string, desc:"cathode", "anode"*/
        "swipeInterval": 1,
        /*ro, opt, int, time interval of repeated authentication, unit:s, desc:which is valid for authentication modes such as fingerprint, card, face,
etc.*/
        "pressTimeout": 1,
        /*ro, opt, int, timeout to reset entry on keypad, unit:s*/
        "enableFailAlarm": true,
        /*ro, opt, bool, whether to enable excessive failed authentication attempts alarm*/
        "maxReadCardFailNum": 1,
        /*ro, opt, int, maximum number of failed authentication attempts*/
        "enableTamperCheck": true,
        /*ro, opt, bool, whether to enable tampering detection*/
        "offlineCheckTime": 1,
        /*ro, opt, int, time to detect after the card reader is offline, unit:s*/
        "fingerPrintCheckLevel": 1,
        /*ro, opt, enum, fingerprint recognition level, subType:int, desc:1-1/10 false acceptance rate (FAR), 2-1/100 false acceptance rate (FAR), 3-1/1000
false acceptance rate (FAR), 4-1/10000 false acceptance rate (FAR), 5-1/100000 false acceptance rate (FAR), 6-1/1000000 false acceptance rate (FAR), 7-
1/10000000 false acceptance rate (FAR), 8-1/100000000 false acceptance rate (FAR), 9-3/100 false acceptance rate (FAR), 10-3/1000 false acceptance rate
(FAR), 11-3/10000 false acceptance rate (FAR), 12-3/100000 false acceptance rate (FAR), 13-3/1000000 false acceptance rate (FAR), 14-3/10000000 false
acceptance rate (FAR), 15-3/100000000 false acceptance rate (FAR), 16-Automatic Normal, 17-Automatic Secure, 18-Automatic More Secure (currently not
support)*/
        "useLocalController": true,
        /*ro, opt, bool, whether it is connected to the distributed controller*/
        "localControllerID": 1,
        /*ro, opt, int, distributed controller No., range:[0,64], dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:which is between 1 and 64, 0-
unregistered. This field is valid only when useLocalController is "true"*/
        "localControllerReaderID": 1,
        /*ro, opt, int, card reader ID of the distributed controller, 0-unregistered, dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:this field
is valid only when useLocalController is "true"*/
        "cardReaderChannel": 1,
        /*ro, opt, enum, communication channel No. of the card reader, subType:int, dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:0-Wiegand or
offline, 1-RS-485A, 2-RS-485B. This field is valid only when useLocalController is "true"*/
        "fingerPrintImageQuality": 1,
        /*ro, opt, enum, fingerprint image quality, subType:int, desc:1-low quality (V1), 2-medium quality (V1), 3-high quality (V1), 4-highest quality
(V1), 5-low quality (V2), 6-medium quality (V2), 7-high quality (V2), 8-highest quality (V2)*/
        "fingerPrintContrastTimeOut": 1,
        /*ro, opt, enum, fingerprint comparison timeout, subType:int, desc:fingerprint comparison timeout,which is between 1 and 20,unit: second,255-
infinite*/
        "fingerPrintRecogizeInterval": 1,
        /*ro, opt, enum, fingerprint scanning interval, subType:int, desc:fingerprint scanning interval,which is between 1 and 10,unit: second,255-no
delay*/
        "fingerPrintMatchFastMode": 1,
        /*ro, opt, enum, fingerprint matching quick mode, subType:int, desc:1-quick mode 1, 2-quick mode 2, 3-quick mode 3, 4-quick mode 4, 5-quick mode 5,
255-automatic*/
        "fingerPrintModuleSensitive": 1,
        /*ro, opt, enum, fingerprint module sensitivity, subType:int, desc:fingerprint module sensitivity,which is between 1 and 8*/
        "fingerPrintModuleLightCondition": "outdoor",
        /*ro, opt, enum, fingerprint module light condition, subType:string, desc:"outdoor", "indoor"*/
        "faceMatchThresholdN": 1,
        /*ro, opt, int, threshold of face picture 1:N comparison,which is between 0 and 100, range:[0,100]*/
        "faceQuality": 1,
        /*ro, opt, int, face picture quality, range:[0,100]*/
        "faceRecogizeTimeOut": 1,
        /*ro, opt, enum, face recognition timeout, subType:int, desc:face recognition timeout,which is between 1 and 20,unit: second,255-infinite*/
        "faceRecogizeInterval": 1,
        /*ro, opt, enum, face recognition interval, subType:int, desc:face recognition interval,which is between 1 and 10,unit: second,255-no delay*/
        "cardReaderFunction": ["fingerPrint", "face", "card", "voiceprint", "PPAndPV"],
        /*ro, opt, enumarray, card reader type, subType:string, desc:"fingerPrint", "face", "fingerVein". For example, ["fingerPrint", "face"] indicates
that the card reader supports both fingerprint and face*/
        "cardReaderDescription": "WiegandOrOffline",
        /*ro, opt, string, card reader description, desc:if the card reader is the Wiegand card reader or if offline, this field will be set to "Wiegand" or
"485Offline"*/
        "faceImageSensitometry": 1,
        /*ro, opt, int, face picture exposure, range:[0,655535]*/
        "faceMatchThreshold1": 1,
        /*ro, opt, int, threshold of face picture 1:1 comparison, range:[0,100]*/
        "buzzerTime": 1,
        /*ro, opt, int, buzzing duration, range:[0,59999], unit:s, desc:buzzing duration,which is between 0 and 5999,unit: second,0-long buzzing*/
        "faceMatch1SecurityLevel": 1,
        /*ro, opt, enum, security level of face 1:1 recognition, subType:int, desc:1 (normal), 2 (high), 3 (higher)*/
        "faceMatchNSecurityLevel": 1,
        /*ro, opt, enum, security level of face 1:N recognition: 1-normal,2-high,3-higher, subType:int, desc:1 (normal), 2 (high), 3 (higher)*/
        "envirMode": "other",
        /*ro, opt, enum, environment mode of face recognition, subType:string, desc:"indoor", "other"*/
        "livingBodyDetect": true,
        /*ro, opt, bool, whether to enable human detection*/
```

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```
        "liveDetLevelSet":  "low",
        /*ro, opt, enum, threshold level of liveness detection, subType:string, desc:"low", "middle", "high"*/
        "liveDetAntiAttackCntLimit":  1,
        /*ro, opt, int, number of anti-attacks of liveness detection, range:[1,255], desc:this value should be configured as the same one on both client and
device*/
        "enableLiveDetAntiAttack":  true,
        /*ro, opt, bool, whether to enable anti-attack for liveness detection*/
        "supportDelFPByID":  true,
        /*ro, opt, bool, whether the card reader supports deleting fingerprint by fingerprint ID, desc:"true"-yes, "false"-no*/
        "fingerPrintCapacity":  1,
        /*ro, opt, int, fingerprint capacity*/
        "fingerPrintNum":  1,
        /*ro, opt, int, number of added fingerprints*/
        "defaultVerifyMode":  "cardAndPw",
        /*ro, opt, enum, default authentication mode of the fingerprint and card reader (factory defaults), subType:string, desc:factory defaults;
"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password), "fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw" (fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password),
"faceAndFp" (face+fingerprint), "faceAndPw" (face+password), "faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw"
(fingerprint or password), "employeeNoAndFp" (employee No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard"
(face+fingerprint+card), "faceAndPwAndFp" (face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card),
"fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint),
"cardOrFpOrPw" (card or fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris),
"faceOrCardOrPwOrIris" (face or card or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or
password), "employeeNoAndFaceAndPw" (employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint
or password), "cardOrFpOrFaceOrIris" (card or fingerpriont or face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw" (card or iris or password), "cardAndIris" (card+ris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
        "faceRecogizeEnable":  1,
        /*ro, opt, enum, whether to enable facial recognition, subType:int, desc:1 (enable), 2 (disable), 3 (attendence checked in/out by recognition of
multiple faces)*/
        "FPAlgorithmVersion":  "test",
        /*ro, opt, string, fingerprint algorithm library version, range:[1,32]*/
        "cardReaderVersion":  "test",
        /*ro, opt, string, card reader version, range:[1,32]*/
        "enableReverseCardNo":  true,
        /*ro, opt, bool, whether to enable reversing the card No.*/
        "independSwipeIntervals":  0,
        /*ro, opt, int, time interval of person authentication, desc:unit: second. This time interval is calculated for each person separately and is
different from swipeInterval*/
        "maskFaceMatchThresholdN":  1,
        /*ro, opt, int, 1:N face picture (face with mask and normal background) comparison threshold, range:[0,100]*/
        "maskFaceMatchThreshold1":  1,
        /*ro, opt, int, 1:1 face picture (face with mask and normal background) comparison threshold, range:[0,100]*/
        "faceMotionDetLevel":  "low",
        /*ro, opt, enum, face motion detection level, subType:string, desc:"high", "medium", "low"*/
        "showMode":  "normal",
        /*ro, opt, enum, display mode, subType:string, desc:this node is not valid; simple mode indicates that the device displays authentication results
exclude employee No., name, etc.; the device applies normal mode by default; advertisement mode indicates that the device displays both advertisement and
authentication results; meeting mode indicates that the device displays check-in page of the conference; custom mode indicates that the device displays
layout of the interface customized by users; "concise"-simple mode, "normal"-normal mode (default), "advertising"-advertisement mode, "meeting"-meeting
mode, "selfDefine"-custom mode*/
        "enableScreenOff":  true,
        /*ro, opt, bool, whether to enable auto locking the screen*/
        "screenOffTimeout":  1,
        /*ro, opt, int, time, step:1, unit:s*/
        "QRCodeEnabled":  false,
        /*ro, opt, bool*/
        "swipeCardEnabled":  true,
        /*ro, opt, bool*/
        "pressEnabled":  true,
        /*ro, opt, bool*/
        "buzzerEnabled":  true,
        /*ro, opt, bool*/
        "elevatorCarID":  1
        /*ro, opt, int, range:[1,8]*/
    }
}
```


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14.6.7.3 Get the configuration capability of the card reader

Request URL

GET /ISAPI/AccessControl/CardReaderCfg/capabilities?format=json&cardReaderID=<cardReaderID>

Query Parameter

Parameter Name	Parameter Type	Description
cardReaderID	string	--

Request Message

None

Response Message

```
{
    "CardReaderCfg": {
    /*ro, req, object, Set Card Reader Parameters*/
        "cardReaderNo": {
        /*ro, opt, object, card reader No., desc:card reader No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  512,
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable, desc:"true"-yes,"false"-no*/
        "cardReaderName": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int*/
            "@max":  64
            /*ro, req, int*/
        },
        "okLedPolarity": {
        /*ro, opt, object, OK LED polarity*/
            "@opt":  "cathode,anode"
            /*ro, req, string, options, desc:"cathode", "anode"*/
        },
        "errorLedPolarity": {
        /*ro, opt, object, error LED polarity*/
            "@opt":  "cathode,anode"
            /*ro, req, string, options, desc:"cathode", "anode"*/
        },
        "swipeInterval": {
        /*ro, opt, object, time interval of repeated authentication, desc:it is valid for authentication modes such as fingerprint, card, face, etc., unit:
second*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  255
            /*ro, req, int, the maximum value*/
        },
        "pressTimeout": {
        /*ro, opt, object, timeout to reset entry on keypad, desc:unit: second*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  255
            /*ro, req, int, the maximum value*/
        },
        "enableFailAlarm":  "true,false",
        /*ro, opt, string, whether to enable excessive failed authentication attempts alarm*/
        "maxReadCardFailNum": {
        /*ro, opt, object, maximum number of failed authentication attempts*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  255
            /*ro, req, int, the maximum value*/
        },
        "enableTamperCheck":  "true,false",
        /*ro, opt, string, whether to enable tampering detection*/
        "offlineCheckTime": {
        /*ro, opt, object, time to detect after the card reader is offline, desc:unit: second*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  255
            /*ro, req, int, the maximum value*/
        },
        "fingerPrintCheckLevel": {
        /*ro, opt, object, fingerprint recognition level*/
            "@opt":  "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18"
            /*ro, req, string, options*/
        },
        "faceMatchThresholdN": {
        /*ro, opt, object, threshold of face picture 1:N comparison, desc:it is between 0 and 100*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "faceRecogizeTimeOut": {
        /*ro, opt, object, face recognition timeout, desc:it is between 1 and 20, unit: second, 0 (infinite)*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  20
            /*ro, req, int, the maximum value*/
        },
        "faceRecogizeInterval": {
        /*ro, opt, object, face recognition interval, desc:it is between 1 and 10, unit: second, 0 (no delay)*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  10
            /*ro, req, int, the maximum value*/
        },
        "cardReaderFunction": {
        /*ro, opt, object, card reader type*/
            "@opt":  "fingerPrint,face,fingerVein,iris,card,voiceprint,PPAndPV"
            /*ro, req, string, options, desc:"fingerPrint" (fingerprint), "face", "fingerVein" (finger vein), "iris"*/
        },
```

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```
        "cardReaderDescription": {
        /*ro, opt, object, card reader description, desc:if the card reader is the Wiegand card reader or if offline, this field will be set to "Wiegand" or
"485Offline"*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  16
            /*ro, req, int, the maximum value*/
        },
        "faceMatchThreshold1": {
        /*ro, opt, object, threshold of face picture 1:1 comparison, desc:it is between 0 and 100*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "livingBodyDetect":  "true,false",
        /*ro, opt, string, whether to enable human detection*/
        "liveDetLevelSet": {
        /*ro, opt, object, threshold level of liveness detection*/
            "@opt":  "low,middle,high,general,enhancive,professional"
            /*ro, req, string, options*/
        },
        "liveDetAntiAttackCntLimit": {
        /*ro, opt, object, number of anti-attacks of liveness detection, desc:it is between 1 and 255. This value should be configured as the same one on
both client and device*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  255
            /*ro, req, int, the maximum value*/
        },
        "enableLiveDetAntiAttack":  "true,false",
        /*ro, opt, string, whether to enable anti-attack for liveness detection, desc:whether to enable anti-attack for liveness detection*/
        "fingerPrintCapacity": {
        /*ro, opt, object, maximum number of fingerprints that can be added*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "fingerPrintNum": {
        /*ro, opt, object, number of added fingerprints*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "defaultVerifyMode": {
        /*ro, opt, object, default authentication mode of the card reader, desc:default authentication mode of the card reader (factory defaults)*/
            "@opt":
"cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpOrCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employe
eNoAndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,faceOrfaceAndCard,fpOrface,cardOrfaceOrPw,cardOrFace,cardOrFaceOrFp,faceOrPw
,employeeNoAndFaceAndPw,cardOrFaceOrFaceAndCard,iris,faceOrFpOrCardOrPwOrIris,faceOrCardOrPwOrIris,cardOrFpOrFaceOrIris,fpOrFaceOrIrisOrPw,cardOrFpOrIrisOrP
w,cardOrIrisOrPw,cardAndIris,fpAndIris,faceAndIris,irisAndPw,cardAndIrisAndPw,faceAndIrisAndPw,cardAndFaceAndIris,Pw,cardOrFpOrPw,faceOrFpOrPw,cardAndVp,fac
eAndVp,fpAndVp,pwAndVp,remoteControl,remoteControlAndFace,remoteControlOrFace,remoteControlOrCard,remoteControlOrFp,remoteControlOrPw,remoteControlOrCardOrF
p,remoteControlOrCardOrFace,remoteControlOrCardOrPw,remoteControlOrFpOrFace,remoteControlOrFpOrPw,remoteControlOrFaceOrPw,remoteControlOrCardOrFpOrFace,remo
teControlOrCardOrFpOrPw,remoteControlOrFpOrFaceOrPw,remoteControlOrCardOrFaceOrPw,remoteControlOrCardOrFpOrFaceOrPw,faceOrPPPV,cardOrPPPV,pwOrPPPV,faceOrCar
dOrPPPV,faceOrPwOrPPPV,faceOrCardOrPwOrPPPV,faceAndPPPV,cardAndPPPV,pwAndPPPV,faceAndCardAndPPPV,faceAndCardAndPwAndPPPV,PPPV,faceAndCardAndPw"
            /*ro, req, string, options*/
        },
        "faceRecogizeEnable": {
        /*ro, opt, object, whether to enable facial recognition*/
            "@opt":  "1,2,3"
            /*ro, req, string, options, desc:1 (enable), 2 (disable), 3 (attendence checked in/out by recognition of multiple faces)*/
        },
        "enableReverseCardNo":  "true,false",
        /*ro, opt, string, whether to enable reversing the card No.*/
        "independSwipeIntervals": {
        /*ro, opt, object, time interval of person authentication, desc:unit: second. This time interval is calculated for each person separately and is
different from swipeInterval*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "maskFaceMatchThresholdN": {
        /*ro, opt, object, 1:N face picture (face with mask and normal background) comparison threshold, desc:value range: [0,100]*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "maskFaceMatchThreshold1": {
        /*ro, opt, object, 1:1 face picture (face with mask and normal background) comparison threshold, desc:value range: [0,100]*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
    }
}
```

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14.6.7.4 Get the capability of configuring card No. authentication rule

Request URL

GET /ISAPI/AccessControl/CardVerificationRule/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CardVerificationRuleCap": {
    /*ro, req, object*/
        "cardNoLenMode": {
        /*ro, opt, object, length mode of card No. authentication (comparison), desc:length mode of card No. authentication (comparison)*/
            "@opt": ["full", "4Bytes", "3Bytes", "wiegand27", "wiegand35", "Corporate1000_35", "Corporate1000_48", "H10302_37", "H10304_37",
"wiegand_26CSN", "H103130_32CSN", "wiegand_56CSN", "wiegand_58"]
            /*ro, opt, array, options, subType:string*/
        },
        "CardVerificationRuleRes": {
        /*ro, opt, object*/
            "checkStatus": {
            /*ro, opt, object, status of switching card No. authentication (comparison) mode, desc:"continue" (switching result can be searched for later),
"ok" (switching completed), "duplicate" (duplicate data exist and switching failed)*/
                "@opt": ["continue", "ok", "duplicate"]
                /*ro, opt, array, options, subType:string*/
            },
            "progress": {
            /*ro, opt, object, switching progress in percentage, desc:which is between 0 and 100,and 100 indicates that the card No. authentication
(comparison) mode is switched*/
                "@min":  0,
                /*ro, opt, int, the minimum value*/
                "@max":  0
                /*ro, opt, int, the maximum value*/
            }
        },
        "reverseCardNoEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, options, subType:bool, range:[1,2]*/
        }
    }
}
```

14.6.7.5 Get the switching progress and configuration result of card No. authentication mode

Request URL

GET /ISAPI/AccessControl/CardVerificationRule/progress?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CardVerificationRuleRes": {
    /*ro, req, object*/
        "checkStatus":  "continue",
        /*ro, opt, enum, status of switching card No. authentication (comparison) mode, subType:string, desc:"continue" (switching result can be searched
for later), "ok" (switching succeeded), "duplicate" (duplicate data exist and switching failed)*/
        "progress":  100
        /*ro, opt, int, switching progress in percentage, range:[0,100], desc:100 indicates that card No. authentication (comparison) mode is switched*/
    }
}
```

14.6.7.6 Get the parameters of card No. authentication mode

Request URL

GET /ISAPI/AccessControl/CardVerificationRule?format=json

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Query Parameter

None

Request Message

None

Response Message

```
{
    "CardVerificationRule": {
    /*ro, req, object*/
        "cardNoLenMode":  "full",
        /*ro, req, enum, length mode of card No. authentication (comparison), subType:string, desc:"full", "3Bytes", "4Bytes". After the card No.
authentication (comparison) mode is switched, the device should check the card No. compatibility*/
        "reverseCardNoEnabled":  true
        /*ro, opt, bool*/
    }
}
```

14.6.7.7 Set the parameters of card No. authentication mode

Request URL

PUT /ISAPI/AccessControl/CardVerificationRule?format=json

Query Parameter

None

Request Message

```
{
    "CardVerificationRule": {
    /*req, object*/
        "cardNoLenMode":  "full",
        /*req, enum, length mode of card No. authentication (comparison), subType:string, desc:"full", "3Bytes", "4Bytes". After the card No. authentication
(comparison) mode is switched, the device should check the card No. compatibility*/
        "reverseCardNoEnabled":  true
        /*opt, bool*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.6.7.8 Get the configuration capability of enabling NFC (Near-Field Communication) function

Request URL

GET /ISAPI/AccessControl/Configuration/NFCCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "NFCCfgCap": {
    /*ro, opt, object, configuration capability of enabling NFC (Near-Field Communication) function*/
        "enable":  "true,false"
        /*ro, req, string, whether to enable NFC function, desc:true-yes, false-no (default)*/
    }
}
```

14.6.7.9 Get the parameters of enabling NFC (Near-Field Communication) function

Request URL

GET /ISAPI/AccessControl/Configuration/NFCCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "NFCCfg": {
    /*ro, req, object*/
        "enable":  true
        /*ro, req, bool, whether to enable NFC function, desc:true (yes), false (no). The value of this node is "false" by default*/
    }
}
```

14.6.7.10 Set the parameters of enabling NFC (Near-Field Communication) function

Request URL

PUT /ISAPI/AccessControl/Configuration/NFCCfg?format=json

Query Parameter

None

Request Message

```
{
    "NFCCfg": {
    /*req, object, configuration capability of enabling NFC (Near-Field Communication) function*/
        "enable":  true
        /*req, bool, whether to enable NFC function, desc:true-yes, false-no (default)*/
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "test",
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.6.7.11 Get the configuration capability of enabling RF (Radio Frequency) card recognition

Request URL

GET /ISAPI/AccessControl/Configuration/RFCardCfg/capabilities?format=json

Query Parameter

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None

Request Message

None

Response Message

```
{
    "RFCardCfgCap": {
    /*ro, req, object*/
        "cardType": {
        /*ro, opt, object, card type, desc:"EMCard"-EM card, "M1Card"-M1 card, "CPUCard"-CPU card, "IDCard"-ID card, "DesfireCard"-DESFire card,
"FelicaCard"-FeliCa card*/
            "@opt": ["EMCard", "M1Card", "CPUCard", "IDCard", "FelicaCard"]
            /*ro, req, array, options, subType:string*/
        },
        "enabled": {
        /*ro, opt, object, whether to enable RF card recognition*/
            "@opt": [true, false]
            /*ro, req, array, options, subType:bool*/
        }
    }
}
```

14.6.7.12 Get the parameters of enabling RF (Radio Frequency) card recognition

Request URL

GET /ISAPI/AccessControl/Configuration/RFCardCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "RFCardCfg": [
    /*ro, req, array, subType:object*/
        {
            "cardType":  "EMCard",
            /*ro, req, enum, card type, subType:string, desc:"EMCard” (EM card), "M1Card” (M1 card), "CPUCard” (CPU card), "IDCard” (ID card), "DesfireCard”
(DESFire card), "FelicaCard” (FeliCa card)*/
            "enabled":  true
            /*ro, req, bool, whether to enable RF card recognition, desc:true-yes, false-no*/
        }
    ]
}
```

14.6.7.13 Set the parameters of enabling RF (Radio Frequency) card recognition

Request URL

PUT /ISAPI/AccessControl/Configuration/RFCardCfg?format=json

Query Parameter

None

Request Message

```
{
    "RFCardCfg": [
    /*req, array, the parameters of enabling RF (Radio Frequency) card recognition, subType:object*/
        {
            "cardType":  "EMCard",
            /*req, enum, card type, subType:string, desc:"EMCard"(EM card), "M1Card"(M1 card), "CPUCard"(CPU card), "IDCard"(ID card), "DesfireCard"(DESFire
card), "FelicaCard"(FeliCa card)*/
            "enabled":  true
            /*req, bool, whether to enable RF card recognition, desc:"true"(yes), "false"(no)*/
        }
    ]
}
```

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "test",
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.6.7.14 Set the condition parameters of face picture comparison

Request URL

PUT /ISAPI/AccessControl/FaceCompareCond

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceCompareCond xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, facial recognition parameters, attr:version{opt, string, protocolVersion}-->
  <faceWidthLowerLimit>
    <!--opt, int, face width threshold, range:[0,100], desc:when the detected face width is larger than this threshold, the following conditions will be ignored and the face picture comparison will be executed it has the highest priority-->50
  </faceWidthLowerLimit>
  <pitch>
    <!--opt, int, face raising or bowing angle, range:[0,90], desc:the smaller the better-->0
  </pitch>
  <yaw>
    <!--opt, int, face siding left or right angle, range:[0,90], desc:the smaller the better-->0
  </yaw>
  <width>
    <!--opt, int, face width, range:[0,100]-->50
  </width>
  <height>
    <!--opt, int, face height, range:[0,100]-->50
  </height>
  <leftBorder>
    <!--opt, int, left border of face, range:[0,100]-->50
  </leftBorder>
  <rightBorder>
    <!--opt, int, right border of face, range:[0,100]-->50
  </rightBorder>
  <upBorder>
    <!--opt, int, top border of face, range:[0,100]-->50
  </upBorder>
  <bottomBorder>
    <!--opt, int, bottom border of face, range:[0,100]-->50
  </bottomBorder>
  <interorbitalDistance>
    <!--opt, int, pupillary distance, range:[0,100]-->50
  </interorbitalDistance>
  <faceScore>
    <!--opt, int, face score (face picture quality), range:[0,100], desc:the valid face score must be larger than this value-->50
  </faceScore>
  <maxDistance>
    <!--opt, enum, maximum recognition distance, subType:string, desc:"0.5" (0.5 m), "1" (1 m), "1.5" (1.5 m), "2" (2m), "auto" (automatic)-->0.5
  </maxDistance>
  <similarity>
    <!--opt, float, face picture comparison similarity-->50
  </similarity>
  <antiFake>
    <!--opt, int, face anti- spoofing parameters-->50
  </antiFake>
  <identifyType>
    <!--opt, enum, face recognition type, subType:string, desc:highest (the highest similarity, default), single (one picture whose similarity exceeds the threshold), multiple (multiple pictures whose similarity exceeds the threshold). If the value of this node is highest, it indicates all face pictures in the whole face picture library will be compared and the one with the highest similarity will be returned. If the value of this node is single, it indicates that the first picture whose similarity exceeds the threshold will be returned. If the value of this node is multiple, the first 30 pictures whose similarity exceeds the threshold will be returned-->highest
  </identifyType>
  <chooseType>
    <!--opt, enum, face recognition area, subType:string, desc:if "middle" is set, the device will recognize the middle face on the picture; if "biggest" is set, the device will recognize the biggest face on the picture; if "all" is set, the device will recognize all faces on the picture; "middle", "biggest", "all"; the default value is all-->all
  </chooseType>
  <enabled>
    <!--opt, enum, whether to enable face recognition, subType:string, desc:singleFace (recognizing a single face, default), close (disable facial
```

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```
    <!--opt, enum, whether to enable face recognition, subType:string, desc:singleFace (recognizing a single face, default), close (disable facial
recognition), multiFace (recognizing multiple faces). This node can be used to enable facial recognition for the device and it will take effect in all
readers-->singleFace
  </enabled>
  <faceScoreEnabled>
    <!--opt, bool, whether to enable face scoring, desc:whether to enable face scoring-->true
  </faceScoreEnabled>
  <ageFaceMatchEnabled>
    <!--opt, bool, whether to enable age groups matching-->true
  </ageFaceMatchEnabled>
  <ageFaceMatchList>
    <!--opt, array, list of different age groups, subType:object, range:[0,5], dep:or,{$.FaceCompareCond.ageFaceMatchEnabled,eq,true}-->
    <ageFaceMatch>
      <!--opt, object, age groups matching-->
      <ageLevel>
        <!--opt, enum, age groups, subType:int, desc:0 (child who are 0~6 years old), 1 (teenager who are 7~17years old), 2 (youth and prime who are 18~40
years old), 3 (middle age who are 41~65 years old), 4 (elderly who are more than 66 years old)-->1
      </ageLevel>
      <ageFaceMatchThreshold1>
        <!--opt, int, matching threshold when authenticating via age groups (1:1), range:[0,100]-->1
      </ageFaceMatchThreshold1>
      <ageFaceMatchThresholdN>
        <!--opt, int, matching threshold when authenticating via age groups (1:N), range:[0,100]-->1
      </ageFaceMatchThresholdN>
    </ageFaceMatch>
  </ageFaceMatchList>
  <faceScoreThreshold1>
    <!--opt, int, matching threshold when face scoring via 1:N matching mode, range:[0,100], desc:it can rate the captured face pictures and is available
for person and ID comparison and authentication without ID card-->1
  </faceScoreThreshold1>
</FaceCompareCond>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

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14.6.7.15 Get the facial recognition parameters.

Request URL

GET /ISAPI/AccessControl/FaceCompareCond

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceCompareCond xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, facial recognition parameters, attr:version{opt, string, protocolVersion}-->
  <faceWidthLowerLimit>
    <!--ro, opt, int, face width threshold, range:[0,100], desc:when the detected face width is larger than this threshold, the following conditions will be
ignored and the face picture comparison will be executed-->50
  </faceWidthLowerLimit>
  <pitch>
    <!--ro, opt, int, face pitching angle, range:[0,90], desc:the smaller the better-->0
  </pitch>
  <yaw>
    <!--ro, opt, int, face yawing angle, range:[0,90], desc:The smaller the better-->0
```

```
    <!--ro, opt, int, face yawing angle, range:[0,90], desc:the smaller, the better.-->0
  </yaw>
  <width>
    <!--ro, opt, int, face width, range:[0,100]-->50
  </width>
  <height>
    <!--ro, opt, int, face height, range:[0,100]-->50
  </height>
  <leftBorder>
    <!--ro, opt, int, left border of the face, range:[0,100]-->50
  </leftBorder>
  <rightBorder>
    <!--ro, opt, int, right border of the face, range:[0,100]-->50
  </rightBorder>
  <upBorder>
    <!--ro, opt, int, upper border of the face, range:[0,100]-->50
  </upBorder>
  <bottomBorder>
    <!--ro, opt, int, lower border of the face, range:[0,100]-->50
  </bottomBorder>
  <interorbitalDistance>
    <!--ro, opt, int, pupillary distance, range:[0,100]-->50
  </interorbitalDistance>
  <faceScore>
    <!--ro, opt, int, face score, range:[0,100], desc:the face to be detected is valid when its score is higher than the value of this node-->50
  </faceScore>
  <maxDistance>
    <!--ro, opt, enum, maximum recognition distance, subType:string, desc:0.5 (0.5m), 1 (1m), 1.5 (1.5m), 2 (2m), auto (automatic)-->0.5
  </maxDistance>
  <similarity>
    <!--ro, opt, float, face picture comparison similarity-->50
  </similarity>
  <antiFake>
    <!--ro, opt, int, face anti-spoofing parameters-->50
  </antiFake>
  <identifyType>
    <!--ro, opt, enum, facial recognition type, subType:string, desc:highest (the highest similarity, default), single (one picture whose similarity exceeds
the threshold), multiple (multiple pictures whose similarity exceeds the threshold). If the value of this node is highest, it indicates all face pictures in
the whole face picture library will be compared and the one with the highest similarity will be returned. If the value of this node is single, it indicates
that the first picture whose similarity exceeds the threshold will be returned. If the value of this node is multiple, the first 30 pictures whose
similarity exceeds the threshold will be returned-->highest
  </identifyType>
  <chooseType>
    <!--ro, opt, enum, face selection type, subType:string, desc:middle (the face in the middle of the picture), biggest (the biggest face), all (all faces,
default). If the value of this node is middle, only the face in the middle of a picture will be recognized. If the value of this node is biggest, the
biggest face in a picture will be recognized. If the value of this node is all, all faces in a picture will be recognized-->all
  </chooseType>
  <enabled>
    <!--ro, opt, enum, whether to enable facial recognition, subType:string, desc:singleFace (recognizing a single face, default), close (disable facial
recognition), multiFace (recognizing multiple faces). This node can be used to enable facial recognition for the device and it will take effect in all
readers-->singleFace
  </enabled>
  <faceScoreEnabled>
    <!--ro, opt, bool, whether to enable face grading, desc:face grading is used to select the high-quality pictures during person and ID comparison or face
picture collection. This field is used together with the faceScore-->true
  </faceScoreEnabled>
  <ageFaceMatchEnabled>
    <!--ro, opt, bool, whether to enable different age groups matching-->true
  </ageFaceMatchEnabled>
  <ageFaceMatchList>
    <!--ro, opt, array, list of different age groups, subType:object, range:[0,5], dep:or,{$.FaceCompareCond.ageFaceMatchEnabled,eq,true}-->
    <ageFaceMatch>
      <!--ro, opt, object, age groups matching-->
      <ageLevel>
        <!--ro, opt, enum, age groups, subType:int, desc:0 (child who are 0~6 years old), 1 (teenager who are 7~17years old), 2 (youth and prime who are
18~40 years old), 3 (middle age who are 41~65 years old), 4 (elderly who are more than 66 years old)-->1
      </ageLevel>
      <ageFaceMatchThreshold1>
        <!--ro, opt, int, matching threshold when authenticating via age groups (1:1), range:[0,100]-->1
      </ageFaceMatchThreshold1>
      <ageFaceMatchThresholdN>
        <!--ro, opt, int, matching threshold when authenticating via age groups (1:N), range:[0,100]-->1
      </ageFaceMatchThresholdN>
    </ageFaceMatch>
  </ageFaceMatchList>
  <faceScoreThreshold1>
    <!--ro, opt, int, matching threshold when face scoring via 1:N matching mode, range:[0,100], desc:it can rate the captured face pictures and is
available for person and ID comparison and authentication without ID card-->1
  </faceScoreThreshold1>
</FaceCompareCond>
```

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14.6.7.16 Get the capability of configuring facial recognition parameters.

Request URL

GET /ISAPI/AccessControl/FaceCompareCond/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceCompareCond xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, capability of configuring facial recognition parameters, attr:version{req, string, protocolVersion}-->
  <faceWidthLowerLimit min="1" max="10">
    <!--ro, opt, int, face width threshold, range:[0,100], attr:min{req, int},max{req, int}, desc:When the width of the face to be detected exceeds the
threshold, the following conditions will be ignored and the face picture comparison will be executed. That is, this parameter has the highest priority.-->1
  </faceWidthLowerLimit>
  <pitch min="1" max="10">
    <!--ro, opt, int, face pitching angle, range:[0,90], attr:min{req, int},max{req, int}, desc:The smaller, the better.-->1
  </pitch>
  <yaw min="1" max="10">
    <!--ro, opt, int, face yawing angle, range:[0,90], attr:min{req, int},max{req, int}, desc:The smaller, the better.-->1
  </yaw>
  <width min="1" max="10">
    <!--ro, opt, int, face width, range:[0,100], attr:min{req, int},max{req, int}-->1
  </width>
  <height min="1" max="10">
    <!--ro, opt, int, face height, range:[0,100], attr:min{req, int},max{req, int}-->1
  </height>
  <leftBorder min="1" max="10">
    <!--ro, opt, int, left border of the face, range:[0,100], attr:min{req, int},max{req, int}-->1
  </leftBorder>
  <rightBorder min="1" max="10">
    <!--ro, opt, int, right border of the face, range:[0,100], attr:min{req, int},max{req, int}-->1
  </rightBorder>
  <upBorder min="1" max="10">
    <!--ro, opt, int, upper border of the face, range:[0,100], attr:min{req, int},max{req, int}-->1
  </upBorder>
  <bottomBorder min="1" max="10">
    <!--ro, opt, int, lower border of the face, range:[0,100], attr:min{req, int},max{req, int}-->1
  </bottomBorder>
  <interorbitalDistance min="1" max="10">
    <!--ro, opt, int, pupillary distance, range:[0,100], attr:min{req, int},max{req, int}-->1
  </interorbitalDistance>
  <faceScore min="1" max="10">
    <!--ro, opt, int, face score, range:[0,100], attr:min{req, int},max{req, int}, desc:The face to be detected is valid when its score is higher than the
value of this node.-->1
  </faceScore>
  <maxDistance opt="0.5,0.75,1,1.5,2,auto">
    <!--ro, opt, string, maximum recognition distance, attr:opt{req, string}, desc:This field takes precedence over interorbitalDistance, unit: m.-->test
  </maxDistance>
  <similarity min="0.0" max="1.0">
    <!--ro, opt, float, face picture comparison similarity, attr:min{req, int},max{req, int}-->0.000
  </similarity>
  <antiFake min="1" max="10">
    <!--ro, opt, int, face anti-spoofing parameters, attr:min{req, int},max{req, int}-->1
  </antiFake>
  <identifyType opt="highest,single,multipl">
    <!--ro, opt, enum, facial recognition type, subType:string, attr:opt{req, string}, desc:highest (the highest similarity, default), single (one picture
whose similarity exceeds the threshold), multiple (multiple pictures whose similarity exceeds the threshold). If the value of this node is highest, it
indicates all face pictures in the whole face picture library will be compared and the one with the highest similarity will be returned. If the value of
this node is single, it indicates that the first picture whose similarity exceeds the threshold will be returned. If the value of this node is multiple, the
first 30 pictures whose similarity exceeds the threshold will be returned.-->highest
  </identifyType>
  <chooseType opt="middle,biggest,all">
    <!--ro, opt, enum, face selection type, subType:string, attr:opt{req, string}, desc:middle (the face in the middle of the picture), biggest (the biggest
face), all (all faces, default). If the value of this node is middle, only the face in the middle of a picture will be recognized. If the value of this node
is biggest, the biggest face in a picture will be recognized. If the value of this node is all, all faces in a picture will be recognized.-->middle
  </chooseType>
  <enabled opt="singleFace,close,multiFace">
    <!--ro, opt, enum, whether to enable facial recognition, subType:string, attr:opt{req, string}, desc:singleFace (recognizing a single face, default),
close (disable facial recognition), multiFace (recognizing multiple faces). This node can be used to enable facial recognition for the device and it will
take effect in all readers.-->singleFace
  </enabled>
  <faceScoreEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable face grading, attr:opt{req, string}, desc:Face grading is used to select the high-quality pictures during person
and ID comparison or face picture collection. This field is used together with the faceScore.-->true
  </faceScoreEnabled>
  <ageFaceMatchEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable different age groups matching, attr:opt{req, string}-->true
  </ageFaceMatchEnabled>
  <ageFaceMatchList size="5">
    <!--ro, opt, array, list of different age groups, subType:object, range:[0,5], dep:or,{$.FaceCompareCond.ageFaceMatchEnabled,eq,true}, attr:size{req,
int}-->
    <ageFaceMatch>
      <!--ro, opt, object, age groups matching-->
      <ageLevel opt="0,1,2,3,4">
        <!--ro, opt, enum, age groups, subType:int, attr:opt{req, string}, desc:0 (child who are 0~6 years old), 1 (teenager who are 7~17years old), 2
(youth and prime who are 18~40 years old), 3 (middle age who are 41~65 years old), 4 (elderly who are more than 66 years old)-->1
      </ageLevel>
      <ageFaceMatchThreshold1 min="0" max="100">
        <!--ro, opt, int, matching threshold when authenticating via age groups (1:1), range:[0,100], attr:min{req, int},max{req, int}-->1
      </ageFaceMatchThreshold1>
      <ageFaceMatchThresholdN min="0" max="100">
        <!--ro, opt, int, matching threshold when authenticating via age groups (1:N), range:[0,100], attr:min{req, int},max{req, int}-->1
```

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```
        <!--ro, opt, int, matching threshold when authenticating via age groups (1:N), range:[0,100], attr:min{req, int},max{req, int}-->1
      </ageFaceMatchThresholdN>
    </ageFaceMatch>
  </ageFaceMatchList>
  <faceScoreThreshold1 min="0" max="100">
    <!--ro, opt, int, matching threshold when face scoring via 1:N matching mode, range:[0,100], attr:min{req, int},max{req, int}-->1
  </faceScoreThreshold1>
</FaceCompareCond>
```

14.6.7.17 Get the capability of configuring parameters of the facial recognition mode.

Request URL

GET /ISAPI/AccessControl/FaceRecognizeMode/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FaceRecognizeMode": {
    /*ro, opt, object, facial recognition mode*/
        "mode": {
        /*ro, req, object, mode*/
            "@opt":  "normalMode"
            /*ro, opt, enum, mode options, subType:string, desc:"normalMode" (normal mode), "deepMode" (deep mode)*/
        }
    }
}
```

14.6.7.18 Get the parameters of the facial recognition mode.

Request URL

GET /ISAPI/AccessControl/FaceRecognizeMode?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FaceRecognizeMode": {
    /*ro, opt, object, facial recognition mode*/
        "mode":  "normalMode"
        /*ro, req, enum, facial recognition mode, subType:string, desc:"normalMode" (normal mode), "deepMode" (deep mode)*/
    }
}
```

14.6.7.19 Set the facial recognition mode parameters

Request URL

PUT /ISAPI/AccessControl/FaceRecognizeMode?format=json

Query Parameter

None

Request Message

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```
{
    "FaceRecognizeMode": {
    /*opt, object, facial recognition mode*/
        "mode":  "normalMode"
        /*req, enum, facial recognition mode, subType:string, desc:"normalMode" (normal mode), "deepMode" (deep mode)*/
    }
}
```

Response Message

```
{
    "requestURL":  "/ISAPI/Intelligent/FDLib/asyncImportDatas?format=json",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.6.7.20 Get the parameters of face recognition terminal

Request URL

GET /ISAPI/AccessControl/IdentityTerminal

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IdentityTerminal xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parameters of face recognition terminal, attr:version{req, string, protocolVersion}-->
  <terminalMode>
    <!--ro, opt, enum, terminal mode, subType:string, desc:"authMode" (authentication mode), "registerMode" (registration mode)-->authMode
  </terminalMode>
  <idCardReader>
    <!--ro, opt, enum, ID card reader model, subType:string, desc:"iDR210", "DS-K1F110-I", "DS-K1F1110-B", "DS-K1F1110-AB", "none", "DS-K1F1001-I(USB)",
"DS-K1F1002-I(USB)"-->iDR210
  </idCardReader>
  <camera>
    <!--ro, opt, enum, camera, subType:string, desc:camera model: C270,DS-2CS5432B-S-->C270
  </camera>
  <fingerPrintModule>
    <!--ro, opt, enum, fingerprint module type, subType:string, desc:"ALIWARD", ""-->ALIWARD
  </fingerPrintModule>
  <videoStorageTime>
    <!--ro, opt, int, time for saving video, range:[0,10], desc:unit: second-->1
  </videoStorageTime>
  <faceContrastThreshold>
    <!--ro, opt, int, face picture comparison threshold, range:[0,100]-->1
  </faceContrastThreshold>
  <twoDimensionCode>
    <!--ro, opt, enum, whether to enable QR code recognition, subType:string, desc:"enable", "disable"-->enable
  </twoDimensionCode>
  <blackListCheck>
    <!--ro, opt, enum, whether to enable blocklist verification, subType:string, desc:"enable", "disable"-->enable
  </blackListCheck>
  <idCardCheckCenter>
    <!--ro, opt, enum, ID card comparison mode, subType:string, desc:"local" (compare with ID card of local storage), "server" (compare with ID card of
remote server storage)-->local
  </idCardCheckCenter>
  <faceAlgorithm>
    <!--ro, opt, enum, face picture algorithm, subType:string, desc:"DeepLearn" (deep learning algorithm), "Tradition" (third-party algorithm)-->DeepLearn
  </faceAlgorithm>
  <comNo>
    <!--ro, opt, int, COM No., range:[1,9]-->1
  </comNo>
  <memoryLearning>
    <!--ro, opt, enum, whether to enable learning and memory function, subType:string, desc:"enable", "disable"-->enable
  </memoryLearning>
  <saveCertifiedImage>
    <!--ro, opt, enum, whether to enable saving authenticated picture, subType:string, desc:"enable", "disable"-->enable
```

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```
  </saveCertifiedImage>
  <MCUVersion>
    <!--ro, opt, string, MCU version information-->test
  </MCUVersion>
  <usbOutput>
    <!--ro, opt, enum, whether to enable USB output of ID card reader, subType:string, desc:“enable”, “disable”-->enable
  </usbOutput>
  <serialOutput>
    <!--ro, opt, enum, whether to enable serial port output of ID card reader, subType:string, desc:“enable”, “disable”-->enable
  </serialOutput>
  <readInfoOfCard>
    <!--ro, opt, enum, set content to be read from CPU card, subType:string, desc:“serialNo” (read serial No.), “file” (read file)-->serialNo
  </readInfoOfCard>
  <workMode>
    <!--ro, opt, enum, authentication mode, subType:string, desc:“passMode”, “accessControlMode”-->passMode
  </workMode>
  <ecoMode>
    <!--ro, opt, object, ECO mode-->
    <eco>
      <!--ro, opt, enum, whether to enable ECO mode, subType:string, desc:“enable”, “disable”-->enable
    </eco>
    <faceMatchThreshold1>
      <!--ro, opt, int, 1V1 face picture comparison threshold of ECO mode, range:[0,100]-->1
    </faceMatchThreshold1>
    <faceMatchThresholdN>
      <!--ro, opt, int, 1:N face picture comparison threshold of ECO mode, range:[0,100]-->1
    </faceMatchThresholdN>
    <changeThreshold>
      <!--ro, opt, int, switching threshold of ECO mode, range:[0,8], desc:switching threshold of ECO mode,which is between 0 and 8-->0
    </changeThreshold>
    <maskFaceMatchThresholdN>
      <!--ro, opt, int, 1:N face picture (face with mask and normal background picture) comparison threshold of ECO mode, range:[0,100]-->1
    </maskFaceMatchThresholdN>
    <maskFaceMatchThreshold1>
      <!--ro, opt, int, 1:1 face picture (face with mask and normal background picture) comparison threshold of ECO mode, range:[0,100]-->1
    </maskFaceMatchThreshold1>
    <alwaysInfrared>
      <!--ro, opt, bool, whether to enable infrared recognition, desc:if this node exists and changeThreshold is 8, it indicates that the device will not
always enable the infrared recognition-->true
    </alwaysInfrared>
    <ageFaceMatchList>
      <!--ro, opt, array, list of different age groups in ECO mode, subType:object, range:[0,5], desc:list of different age groups in ECO mode-->
      <ageFaceMatch>
        <!--ro, opt, object, age group matching in ECO mode-->
        <ageLevel>
          <!--ro, opt, enum, age groups, subType:int, desc:0 (child who are 0~6 years old), 1 (teenager who are 7~17years old), 2 (youth and prime who are
18~40 years old), 3 (middle age who are 41~65 years old), 4 (elderly who are more than 66 years old)-->1
        </ageLevel>
        <ageFaceMatchThreshold1>
          <!--ro, opt, int, matching threshold when authenticating via age groups in ECO mode (1:1), range:[0,100]-->1
        </ageFaceMatchThreshold1>
        <ageFaceMatchThresholdN>
          <!--ro, opt, int, matching threshold when authenticating via age groups in ECO mode (1:N), range:[0,100]-->1
        </ageFaceMatchThresholdN>
      </ageFaceMatch>
    </ageFaceMatchList>
  </ecoMode>
  <readCardRule>
    <!--ro, opt, enum, card No. setting rule, subType:string, desc:"wiegand26", "wiegand34"-->wiegand26
  </readCardRule>
  <enableScreenOff>
    <!--ro, opt, bool, whether the device enters the sleep mode when there is no operation after the configured sleep time-->true
  </enableScreenOff>
  <screenOffTimeout>
    <!--ro, opt, int, sleep time, range:[0,3600], unit:s, dep:or,{$.IdentityTerminal.enableScreenOff,eq,true}, desc:unit: second-->1
  </screenOffTimeout>
  <enableScreensaver>
    <!--ro, opt, bool, whether to enable the screen saver function-->true
  </enableScreensaver>
  <faceModuleVersion>
    <!--ro, opt, string, face recognition module version, range:[1,32]-->test
  </faceModuleVersion>
  <showMode>
    <!--ro, opt, enum, display mode, subType:string, desc:"concise" (simple mode,only the authentication result will be displayed), "normal" (normal mode).
The default mode is normal mode. If this node does not exist, the default mode is normal mode-->concise
  </showMode>
  <popUpPreviewWindow>
    <!--ro, opt, bool, whether to pop up live view window, dep:or,{$.IdentityTerminal.showMode,eq,advertising}-->true
  </popUpPreviewWindow>
  <needDeviceCheck>
    <!--ro, opt, bool, whether it need device check in permission free mode, dep:or,{$.IdentityTerminal.workMode,eq,passMode}-->true
  </needDeviceCheck>
  <previewShowTime>
    <!--ro, opt, int, display duration in live view, range:[1,99], unit:s, dep:or,{$.IdentityTerminal.popUpPreviewWindow,eq,true}-->1
  </previewShowTime>
  <screensaverTimeout>
    <!--ro, opt, int, range:[0,3600], unit:s, dep:or,{$.IdentityTerminal.enableScreensaver,eq,true}-->1
  </screensaverTimeout>
  <screensaverDuration>
    <!--ro, opt, int, range:[0,3600], unit:s, dep:or,{$.IdentityTerminal.enableScreensaver,eq,true}-->1
  </screensaverDuration>
  <standbyTimeout>
    <!--ro, opt, int, range:[30,1800], unit:s-->30
  </standbyTimeout>
```

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```
  <advertisingDisplayType>
    <!--ro, opt, enum, subType:string, dep:or,{$.IdentityTerminal.showMode,eq,advertising}-->full
  </advertisingDisplayType>
</IdentityTerminal>
```

14.6.7.21 Set the parameters of face recognition terminal

Request URL

PUT /ISAPI/AccessControl/IdentityTerminal

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IdentityTerminal xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, the parameters of face recognition terminal, attr:version{req, string, protocolVersion}-->
  <terminalMode>
    <!--opt, enum, terminal mode, subType:string, desc:"authMode" (authentication mode), "registerMode" (registration mode)-->authMode
  </terminalMode>
  <idCardReader>
    <!--opt, enum, ID card reader, subType:string, desc:"iDR210", "DS-K1F110-I", "DS-K1F1110-B", "DS-K1F1110-AB", "none", "DS-K1F1001-I (USB)", "DS-K1F1002-
I (USB)"-->iDR210
  </idCardReader>
  <camera>
    <!--opt, enum, camera, subType:string, desc:"C270", "DS-2CS5432B-S"-->C270
  </camera>
  <fingerPrintModule>
    <!--opt, enum, fingerprint module, subType:string, desc:fingerprint module type: ALIWARD,-->ALIWARD
  </fingerPrintModule>
  <videoStorageTime>
    <!--opt, int, time for saving video, range:[0,10], desc:unit: second-->1
  </videoStorageTime>
  <faceContrastThreshold>
    <!--opt, int, face picture comparison threshold, range:[0,100]-->1
  </faceContrastThreshold>
  <twoDimensionCode>
    <!--opt, enum, whether to enable QR code recognition, subType:string, desc:"enable", "disable"-->enable
  </twoDimensionCode>
  <blackListCheck>
    <!--opt, enum, whether to enable blocklist verification, subType:string, desc:"enable", "disable"-->enable
  </blackListCheck>
  <idCardCheckCenter>
    <!--opt, enum, ID card comparison mode, subType:string, desc:"local" (compare with ID card of local storage), "server" (compare with ID card of remote
server storage)-->local
  </idCardCheckCenter>
  <faceAlgorithm>
    <!--opt, enum, face picture algorithm, subType:string, desc:"DeepLearn" (deep learning algorithm), "Tradition" (third-party algorithm)-->DeepLearn
  </faceAlgorithm>
  <comNo>
    <!--opt, int, COM No., range:[1,9]-->1
  </comNo>
  <memoryLearning>
    <!--opt, enum, whether to enable learning and memory function, subType:string, desc:"enable", "disable"-->enable
  </memoryLearning>
  <saveCertifiedImage>
    <!--opt, enum, whether to enable saving authenticated picture, subType:string, desc:"enable", "disable"-->enable
  </saveCertifiedImage>
  <MCUVersion>
    <!--opt, string, MCU Version-->test
  </MCUVersion>
  <usbOutput>
    <!--opt, enum, whether to enable USB output of ID card reader, subType:string, desc:"enable", "disable"-->enable
  </usbOutput>
  <serialOutput>
    <!--opt, enum, whether to enable serial port output of ID card reade, subType:string, desc:"enable", "disable"-->enable
  </serialOutput>
  <readInfoOfCard>
    <!--opt, enum, set content to be read from CPU card, subType:string, desc:"serialNo" (read serial No.), "file" (read file)-->serialNo
  </readInfoOfCard>
  <workMode>
    <!--opt, enum, authentication mode, subType:string, desc:"passMode", "accessControlMode"-->passMode
  </workMode>
  <ecoMode>
    <!--opt, object, ECO mode-->
    <eco>
      <!--opt, enum, whether to enable ECO mode, subType:string, desc:"enable", "disable"-->enable
    </eco>
    <faceMatchThreshold1>
      <!--opt, int, 1V1 face picture comparison threshold of ECO mode, range:[0,100]-->1
    </faceMatchThreshold1>
    <faceMatchThresholdN>
      <!--opt, int, 1VN face picture comparison threshold of ECO mode, range:[0,100]-->1
    </faceMatchThresholdN>
    <changeThreshold>
      <!--opt, int, ECO mode threshold, range:[0,8], desc:switching threshold of ECO mode,which is between 0 and 8-->0
```

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```
    </changeThreshold>
    <maskFaceMatchThresholdN>
      <!--opt, int, 1:N face picture (face with mask and normal background picture) comparison threshold of ECO mode, range:[0,100]-->1
    </maskFaceMatchThresholdN>
    <maskFaceMatchThreshold1>
      <!--opt, int, 1:1 face picture (face with mask and normal background picture) comparison threshold of ECO mode, range:[0,100]-->1
    </maskFaceMatchThreshold1>
    <alwaysInfrared>
      <!--opt, bool, whether to enable infrared recognition, desc:if this node exists and changeThreshold is 8, it indicates that the device will not always
enable the infrared recognition-->true
    </alwaysInfrared>
    <ageFaceMatchList>
      <!--opt, array, list of different age groups in ECO mode, subType:object, range:[0,5], desc:before set this node, age group matching should be
enabled, see URL: PUT /ISAPI/AccessControl/FaceCompareCond, and ageFaceMatchEnabled should be true-->
      <ageFaceMatch>
        <!--opt, object, age group matching in ECO mode-->
        <ageLevel>
          <!--opt, enum, age groups, subType:int, desc:0 (child who are 0~6 years old), 1 (teenager who are 7~17years old), 2 (youth and prime who are 18~40
years old), 3 (middle age who are 41~65 years old), 4 (elderly who are more than 66 years old)-->1
        </ageLevel>
        <ageFaceMatchThreshold1>
          <!--opt, int, matching threshold when authenticating via age groups in ECO mode (1:1), range:[0,100]-->1
        </ageFaceMatchThreshold1>
        <ageFaceMatchThresholdN>
          <!--opt, int, matching threshold when authenticating via age groups in ECO mode (1:N), range:[0,100]-->1
        </ageFaceMatchThresholdN>
      </ageFaceMatch>
    </ageFaceMatchList>
  </ecoMode>
  <readCardRule>
    <!--opt, enum, card No. setting rule, subType:string, desc:"wiegand26", "wiegand34"-->wiegand26
  </readCardRule>
  <enableScreenOff>
    <!--opt, bool, whether the device enters the sleep mode when there is no operation after the configured sleep time-->true
  </enableScreenOff>
  <screenOffTimeout>
    <!--opt, int, sleep time, range:[0,3600], unit:s, dep:or,{$.IdentityTerminal.enableScreenOff,eq,true}, desc:unit: second-->1
  </screenOffTimeout>
  <enableScreensaver>
    <!--opt, bool, whether to enable the screen saver function-->true
  </enableScreensaver>
  <faceModuleVersion>
    <!--opt, string, face recognition module version, range:[1,32]-->test
  </faceModuleVersion>
  <showMode>
    <!--opt, enum, display mode, subType:string, desc:simple mode indicates that the device displays authentication results exclude employee No., name,
etc.; the device applies normal mode by default; advertisement mode indicates that the device displays both advertisement and authentication results;
meeting mode indicates that the device displays check-in page of the conference; custom mode indicates that the device displays layout of the interface
customized by users; "concise"-simple mode, "normal"-normal mode (default), "advertising"-advertisement mode, "meeting"-meeting mode, "selfDefine"-custom
mode-->concise
  </showMode>
  <popUpPreviewWindow>
    <!--opt, bool, whether to pop up live view window, dep:or,{$.IdentityTerminal.showMode,eq,advertising}-->true
  </popUpPreviewWindow>
  <needDeviceCheck>
    <!--opt, bool, whether it need device check in permission free mode, dep:or,{$.IdentityTerminal.workMode,eq,passMode}-->true
  </needDeviceCheck>
  <previewShowTime>
    <!--opt, int, display duration in live view, range:[1,99], unit:s, dep:or,{$.IdentityTerminal.popUpPreviewWindow,eq,true}-->1
  </previewShowTime>
  <screensaverTimeout>
    <!--opt, int, range:[0,3600], unit:s, dep:or,{$.IdentityTerminal.enableScreensaver,eq,true}-->1
  </screensaverTimeout>
  <screensaverDuration>
    <!--opt, int, range:[0,3600], unit:s, dep:or,{$.IdentityTerminal.enableScreensaver,eq,true}-->1
  </screensaverDuration>
  <standbyTimeout>
    <!--opt, int, range:[30,1800], unit:s-->30
  </standbyTimeout>
  <advertisingDisplayType>
    <!--opt, enum, subType:string, dep:or,{$.IdentityTerminal.showMode,eq,advertising}-->full
  </advertisingDisplayType>
</IdentityTerminal>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response status, attr:version{req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, desc:request URL-->test
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:read-only,status code: 0,1-OK,2-Device Busy,3-Device Error,4-Invalid Operation,5-Invalid XML Format,6-Invalid XML Content,7-Reboot Required,9-Additional Error-->1
  </statusCode>
  <statusString>
    <!--ro, req, enum, read-only,status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:read-only,describe the error reason in detail-->test
  </subStatusCode>
</ResponseStatus>
```

14.6.7.22 Get configuration capability of face recognition terminal

Request URL

GET /ISAPI/AccessControl/IdentityTerminal/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IdentityTerminal xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parameters of face recognition terminal, attr:version{req, string, protocolVersion}-->
  <terminalMode opt="authMode,registerMode">
    <!--ro, opt, enum, terminal mode, subType:string, attr:opt{req, string}, desc:"authMode” (authentication mode), "registerMode” (registration mode)-->authMode
  </terminalMode>
  <idCardReader opt="iDR210,DS-K1F110-I,DS-K1F1110-B,DS-K1F1110-AB,none ">
    <!--ro, opt, enum, ID card reader model, subType:string, attr:opt{req, string}, desc:“iDR210”, “DS-K1F110-I”, “DS-K1F1110-B”, “DS-K1F1110-AB”, “none”, “DS-K1F1001-I (USB)”, “DS-K1F1002-I (USB)”-->iDR210
  </idCardReader>
  <camera opt="C270,DS-2CS5432B-S">
    <!--ro, opt, enum, camera, subType:string, attr:opt{req, string}, desc:"C270", "DS-2CS5432B-S"-->C270
  </camera>
  <fingerPrintModule opt="ALIWARD,">
    <!--ro, opt, enum, fingerprint module, subType:string, attr:opt{req, string}, desc:fingerprint module-->ALIWARD
  </fingerPrintModule>
  <videoStorageTime min="0" max="10">
    <!--ro, opt, int, time for saving video (unit: second), range:[0,10], attr:min{req, int},max{req, int}, desc:unit: second-->1
  </videoStorageTime>
  <faceContrastThreshold min="0" max="100">
    <!--ro, opt, int, face picture comparison threshold, range:[0,100], attr:min{req, int},max{req, int}-->1
  </faceContrastThreshold>
  <twoDimensionCode opt="enable,disable">
    <!--ro, opt, enum, whether to enable QR code recognition, subType:string, attr:opt{req, string}, desc:“enable”, “disable”-->enable
  </twoDimensionCode>
  <blackListCheck opt="enable,disable">
    <!--ro, opt, enum, whether to enable blocklist verification, subType:string, attr:opt{req, string}, desc:“enable”, “disable”-->enable
  </blackListCheck>
  <idCardCheckCenter opt="local,server">
    <!--ro, opt, enum, ID card comparison mode, subType:string, attr:opt{req, string}, desc:“local” (compare with ID card of local storage), “server” (compare with ID card of remote server storage)-->local
  </idCardCheckCenter>
  <faceAlgorithm opt="DeepLearn,Tradition">
    <!--ro, opt, enum, face picture algorithm, subType:string, attr:opt{req, string}, desc:"DeepLearn" (deep learning algorithm), "Tradition" (third-party algorithm)-->DeepLearn
  </faceAlgorithm>
  <comNo min="1" max="9">
    <!--ro, opt, int, COM No., range:[1,9], attr:min{req, int},max{req, int}-->1
  </comNo>
  <memoryLearning opt="enable,disable">
    <!--ro, opt, enum, whether to enable learning and memory function, subType:object, attr:opt{req, string}, desc:“enable”, “disable”-->enable
  </memoryLearning>
  <saveCertifiedImage opt="enable,disable">
    <!--ro, opt, enum, whether to enable saving authenticated picture, subType:string, attr:opt{req, string}, desc:“enable”, “disable”-->enable
  </saveCertifiedImage>
  <MCUVersion min="1" max="10">
    <!--ro, opt, string, MCU version information, attr:min{req, int},max{req, int}-->test
  </MCUVersion>
  <usbOutput opt="enable,disable">
    <!--ro, opt, enum, whether to enable USB output of ID card reader, subType:string, attr:opt{req, string}, desc:“enable”, “disable”-->enable
  </usbOutput>
```

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```
<serialOutput opt="enable,disable">
<!--ro, opt, enum, whether to enable serial port output of ID card reader, subType:string, attr:opt{req, string}, desc:“enable”, “disable”-->enable
</serialOutput>
<readInfoOfCard opt="serialNo,file">
<!--ro, opt, enum, set content to be read from CPU card, subType:string, attr:opt{req, string}, desc:"serialNo" (read the serial No.), "file" (read the file)-->serialNo
</readInfoOfCard>
<workMode opt="passMode,accessControlMode">
<!--ro, opt, enum, authentication mode, subType:string, attr:opt{req, string}, desc:authentication mode-->passMode
</workMode>
<ecoMode>
<!--ro, opt, object, ECO mode-->
<eco opt="enable,disable">
<!--ro, opt, enum, whether to enable ECO mode, subType:string, attr:opt{req, string}, desc:“enable”, “disable”-->enable
</eco>
<faceMatchThreshold1 min="0" max="100">
<!--ro, opt, int, 1V1 face picture comparison threshold of ECO mod, range:[0,100], attr:min{req, int},max{req, int}-->1
</faceMatchThreshold1>
<faceMatchThresholdN min="0" max="100">
<!--ro, opt, int, 1:N face picture comparison threshold of ECO mode, range:[0,100], attr:min{req, int},max{req, int}-->1
</faceMatchThresholdN>
<changeThreshold min="0" max="8">
<!--ro, opt, int, switching threshold of ECO mode, range:[0,8], attr:min{req, int},max{req, int}, desc:switching threshold of ECO mode,which is between 0 and 8-->0
</changeThreshold>
<maskFaceMatchThresholdN min="0" max="100">
<!--ro, opt, int, 1:N face picture (face with mask and normal background picture) comparison threshold of ECO mode, range:[0,100], attr:min{req, int},max{req, int}-->1
</maskFaceMatchThresholdN>
<maskFaceMatchThreshold1 min="0" max="100">
<!--ro, opt, int, 1:1 face picture (face with mask and normal background picture) comparison threshold of ECO mode, range:[0,100], attr:min{req, int},max{req, int}-->1
</maskFaceMatchThreshold1>
<alwaysInfrared opt="true,false">
<!--ro, opt, bool, whether to enable infrared recognition, attr:opt{req, string}, desc:if this node exists and changeThreshold is 8, it indicates that the device will not always enable the infrared recognition-->true
</alwaysInfrared>
<ageFaceMatchList size="5">
<!--ro, opt, array, list of different age groups in ECO mode, subType:object, range:[0,5], attr:size{req, int}-->
<ageFaceMatch>
<!--ro, opt, object, age group matching in ECO mode-->
<ageLevel opt="0,1,2,3,4">
<!--ro, opt, enum, age groups, subType:int, attr:opt{req, string}, desc:0 (child who are 0~6 years old), 1 (teenager who are 7~17years old), 2 (youth and prime who are 18~40 years old), 3 (middle age who are 41~65 years old), 4 (elderly who are more than 66 years old)-->1
</ageLevel>
<ageFaceMatchThreshold1 min="0" max="100">
<!--ro, opt, int, matching threshold when authenticating via age groups in ECO mode (1:1), range:[0,100], attr:min{req, int},max{req, int}-->1
</ageFaceMatchThreshold1>
<ageFaceMatchThresholdN min="0" max="100">
<!--ro, opt, int, matching threshold when authenticating via age groups in ECO mode (1:N), range:[0,100], attr:min{req, int},max{req, int}-->1
</ageFaceMatchThresholdN>
</ageFaceMatch>
</ageFaceMatchList>
</ecoMode>
<readCardRule opt="wiegand26,wiegand34">
<!--ro, opt, enum, card No. setting rule, subType:string, attr:opt{req, string}, desc:card No. setting rule: "wiegand26","wiegand34"-->wiegand26
</readCardRule>
<enableScreenOff opt="true,false">
<!--ro, opt, bool, whether the device enters the sleep mode when there is no operation after the configured sleep time, attr:opt{req, string}-->true
</enableScreenOff>
<screenOffTimeout min="0" max="3600">
<!--ro, opt, int, sleep time, range:[0,3600], unit:s, attr:min{req, int},max{req, int}-->1
</screenOffTimeout>
<enableScreensaver opt="true,false">
<!--ro, opt, bool, whether to enable the screen saver function, attr:opt{req, string}-->true
</enableScreensaver>
<faceModuleVersion min="0" max="10">
<!--ro, opt, string, face recognition module version, attr:min{req, int},max{req, int}-->test
</faceModuleVersion>
<showMode opt="concise,normal,advertising,meeting,selfDefine,boxStatus,clock">
<!--ro, opt, enum, display mode, subType:string, attr:opt{req, string}, desc:"concise" (simple mode,only the authentication result will be displayed), "normal" (normal mode). The default mode is normal mode. If this node does not exist, the default mode is normal mode-->concise
</showMode>
<popUpPreviewWindow opt="true,false">
<!--ro, opt, bool, whether to pop up live view window, dep:or,{$.IdentityTerminal.showMode,eq,advertising}, attr:opt{req, string}-->true
</popUpPreviewWindow>
<needDeviceCheck opt="true,false">
<!--ro, opt, bool, whether it need device check in permission free mode, dep:or,{$.IdentityTerminal.workMode,eq,passMode}, attr:opt{req, string}-->true
</needDeviceCheck>
<previewShowTime min="1" max="99">
<!--ro, opt, int, display duration in live view, range:[1,99], unit:s, attr:min{req, int},max{req, int}-->1
</previewShowTime>
<screensaverTimeout min="0" max="3600">
<!--ro, opt, int, range:[0,3600], unit:s, attr:min{req, int},max{req, int}-->1
</screensaverTimeout>
<screensaverDuration min="0" max="3600">
<!--ro, opt, int, range:[0,3600], unit:s, attr:min{req, int},max{req, int}-->1
</screensaverDuration>
<standbyTimeout min="30" max="1800">
<!--ro, opt, int, range:[30,1800], unit:s, attr:min{req, int},max{req, int}-->30
</standbyTimeout>
<advertisingDisplayType opt="full,split">
<!--ro, opt, enum, subType:string, attr:opt{req, string}-->full
</advertisingDisplayType>
```

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```
</IdentityTerminal>
```

14.6.7.23 Set the parameters of M1 card encryption verification

Request URL

PUT /ISAPI/AccessControl/M1CardEncryptCfg

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<M1CardEncryptCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, configuration capability of the M1 card encryption verification, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--req, bool, whether to enable the function-->true
  </enable>
  <sectionID>
    <!--req, int, sector ID, desc:only one sector can be configured at a time-->1
  </sectionID>
</M1CardEncryptCfg>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response status, attr:version{req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, desc:request URL-->test
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:string, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->1
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK,Device Busy,Device Error,Invalid Operation,Invalid XML Format,Invalid XML
Content,Reboot"-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->test
  </subStatusCode>
</ResponseStatus>
```

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14.6.7.24 Get the configuration parameters of M1 card encryption verification

Request URL

GET /ISAPI/AccessControl/M1CardEncryptCfg

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<M1CardEncryptCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, configuration capability of the M1 card encryption verification, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--ro, req, bool, whether to enable the function-->true
  </enable>
  <sectionID>
    <!--ro, req, int, sector ID, desc:sector ID,only one sector can be configured at a time-->1
  </sectionID>
</M1CardEncryptCfg>
```

14.6.7.25 Get the configuration capability of the M1 card encryption verification

Request URL

GET /ISAPI/AccessControl/M1CardEncryptCfg/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<M1CardEncryptCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, configuration capability of the M1 card encryption verification, attr:version{req, string, protocolVersion}-->
  <enable opt="true,false">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enable>
  <sectionID min="0" max="100">
    <!--ro, req, int, sector ID, range:[0,100], attr:min{req, int},max{req, int}-->1
  </sectionID>
</M1CardEncryptCfg>
```

14.6.7.26 Get the capability of Wiegand parameters

Request URL

GET /ISAPI/AccessControl/WiegandCfg/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WiegandCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, configuration of Wiegand parameters, attr:version{req, string, protocolVersion}-->
  <wiegandNo min="1" max="4" opt="1,4">
    <!--ro, req, int, Wiegand interface No., range:[0,4], step:1, attr:min{opt, int},max{opt, int},opt{opt, string}, desc:Wiegand interface No.-->1
  </wiegandNo>
  <communicateDirection opt="receive,send">
    <!--ro, req, enum, communication direction, subType:string, attr:opt{req, string}, desc:"receive", "send"-->receive
  </communicateDirection>
  <wiegandMode
opt="wiegand26,wiegand34,wiegand27,wiegand35,Corporate1000_35,Corporate1000_48,H10302_37,H10304_37,wiegand_26CSN,H103130_32CSN,wiegand_56CSN,wiegand_58">
    <!--ro, opt, enum, Wiegand mode, subType:string, attr:opt{req, string}, desc:Wiegand mode-->wiegand26
  </wiegandMode>
  <inputWiegandMode>
    <!--ro, opt, enum, subType:string, dep:or,{$.WiegandCfg.communicateDirection,eq,receive}-->wiegand26
  </inputWiegandMode>
  <signalInterval min="1" max="20">
    <!--ro, opt, int, it is between 1 and 20,unit: ms, range:[1,20], attr:min{req, int},max{req, int}, desc:it is between 1 and 20,unit: ms-->1
  </signalInterval>
  <enable opt="true,false">
    <!--ro, opt, bool, whether to enable Wiegand parameters, attr:opt{req, string}-->true
  </enable>
  <pulseDuration min="1" max="10">
    <!--ro, opt, int, pulse duration, range:[1,10], attr:min{req, int},max{req, int}, desc:pulse duration-->1
  </pulseDuration>
  <facilityCodeEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable facilityCode, attr:opt{req, string}-->true
  </facilityCodeEnabled>
  <facilityCode min="0" max="65535">
    <!--ro, opt, int, range:[0,65535], dep:and,{$.WiegandCfg.facilityCodeEnabled,eq,true}, attr:min{req, int, range:[0,65535]},max{req, int, range:
[0,65535]}-->1
  </facilityCode>
  <dataType opt="employeeNo,cardNo">
    <!--ro, opt, enum, data type, subType:string, dep:or,{$.WiegandCfg.wiegandMode,eq,send}, attr:opt{req, string}, desc:data type-->employeeNo
  </dataType>
</WiegandCfg>
```

14.6.7.27 Set Wiegand parameters

Request URL

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PUT /ISAPI/AccessControl/WiegandCfg/wiegandNo/<wiegandID>

Query Parameter

Parameter Name	Parameter Type	Description
wiegandID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WiegandCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, req, object, Wiegand parameters, attr:version{req, string, protocolVersion}-->
  <communicateDirection>
    <!--wo, req, enum, communication direction, subType:string, desc:"receive", "send"-->receive
  </communicateDirection>
  <wiegandMode>
    <!--wo, opt, enum, Wiegand mode, subType:string, dep:or,{$.WiegandCfg.communicateDirection,eq,send}, desc:"wiegand26", "wiegand34", "wiegand27",
"wiegand35", "Corporate1000_35", "Corporate1000_48", "H10302_37", "H10304_37", "wiegand_26CSN", "H103130_32CSN", "wiegand_56CSN", "wiegand_58"-->wiegand26
  </wiegandMode>
  <inputWiegandMode>
    <!--wo, opt, enum, subType:string, dep:or,{$.WiegandCfg.communicateDirection,eq,receive}-->wiegand26
  </inputWiegandMode>
  <signalInterval>
    <!--wo, opt, int, it is between 1 and 20,unit: ms, range:[1,20], desc:unit: ms-->1
  </signalInterval>
  <enable>
    <!--wo, opt, bool, whether to enable the function or not-->true
  </enable>
  <pulseDuration>
    <!--wo, opt, int, range:[1,10]-->1
  </pulseDuration>
  <facilityCodeEnabled>
    <!--opt, bool-->true
  </facilityCodeEnabled>
  <facilityCode>
    <!--opt, int, range:[0,65535], dep:and,{$.WiegandCfg.facilityCodeEnabled,eq,true}-->1
  </facilityCode>
  <dataType>
    <!--opt, enum, data type, subType:string, dep:or,{$.WiegandCfg.wiegandMode,eq,send}, desc:data type-->employeeNo
  </dataType>
</WiegandCfg>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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14.6.7.28 Get Wiegand parameters

Request URL

GET /ISAPI/AccessControl/WiegandCfg/wiegandNo/<wiegandID>

Query Parameter

Parameter Name	Parameter Type	Description
wiegandID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WiegandCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, Wiegand parameters, attr:version{req, string, protocolVersion}-->
  <communicateDirection>
    <!--ro, req, enum, communication direction, subType:string, desc:"receive", "send"-->receive
  </communicateDirection>
  <wiegandMode>
    <!--ro, opt, enum, Wiegand mode, subType:string, dep:or,{$.WiegandCfg.communicateDirection,eq,send}, desc:"wiegand26", "wiegand34", "wiegand27",
"wiegand35", "Corporate1000_35", "Corporate1000_48", "H10302_37", "H10304_37", "wiegand_26CSN", "H103130_32CSN", "wiegand_56CSN", "wiegand_58"-->wiegand26
  </wiegandMode>
  <sequenceOrder>
    <!--ro, opt, enum, subType:string-->normal
  </sequenceOrder>
  <inputWiegandMode>
    <!--ro, opt, enum, subType:string, dep:or,{$.WiegandCfg.communicateDirection,eq,receive}-->wiegand26
  </inputWiegandMode>
  <signalInterval>
    <!--ro, opt, int, Wiegand signal sending interval, range:[1,20], desc:unit: ms-->1
  </signalInterval>
  <enable>
    <!--ro, opt, bool, whether to enable the function or not-->true
  </enable>
  <pulseDuration>
    <!--ro, opt, int, range:[1,10]-->1
  </pulseDuration>
  <facilityCodeEnabled>
    <!--ro, opt, bool-->true
  </facilityCodeEnabled>
  <facilityCode>
    <!--ro, opt, int, range:[0,65535], dep:and,{$.WiegandCfg.facilityCodeEnabled,eq,true}-->1
  </facilityCode>
  <dataType>
    <!--ro, opt, enum, data type, subType:string, dep:or,{$.WiegandCfg.wiegandMode,eq,send}, desc:data type-->employeeNo
  </dataType>
  <MSBEnabled>
    <!--ro, opt, bool, dep:or,{$.WiegandCfg.wiegandMode,eq,wiegand26Sha1},{$.WiegandCfg.wiegandMode,eq,wiegand34Sha1}-->true
  </MSBEnabled>
</WiegandCfg>
```

14.6.7.29 Get configuration parameters of Wiegand rule

Request URL

GET /ISAPI/AccessControl/WiegandRuleCfg

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<WiegandRuleCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <name>
    <!--ro, req, string, Wiegand name-->test
  </name>
  <CustomerCardIn>
    <!--ro, opt, object-->
    <totalLength>
      <!--ro, req, int, total Wiegand length, desc:when this node is set to 0, the custom Wiegand rule settings will be cleared-->1
    </totalLength>
    <checkMethod>
      <!--ro, req, enum, parity mode, subType:string, desc:parityCheck, xorCheck, noCheck;-->parityCheck
    </checkMethod>
    <ParityCheck>
      <!--ro, opt, object, configuration rule of odd-even parity, desc:this node is valid when <checkMethod> is "parityCheck"-->
      <oddBeginBit>
        <!--ro, opt, int, start bit of odd parity-->1
      </oddBeginBit>
      <oddLength>
        <!--ro, opt, int, odd parity length-->1
      </oddLength>
      <evenBeginBit>
        <!--ro, opt, int, start bit of even parity-->1
      </evenBeginBit>
      <evenLength>
        <!--ro, opt, int, even parity length-->1
      </evenLength>
    </ParityCheck>
    <XorCheck>
      <!--ro, opt, object, configuration rule of XOR parity, desc:this node is valid when <checkMethod> is "xorCheck"-->
      <xorBeginBit>
        <!--ro, opt, int, start bit of XOR parity-->1
      </xorBeginBit>
      <xorPerLength>
        <!--ro, opt, int, length of each XOR parity group-->1
      </xorPerLength>
      <xorTotalLength>
        <!--ro, opt, int, total length of XOR parity data-->1
      </xorTotalLength>
    </XorCheck>
    <cardIdBeginBit>
      <!--ro, req, int, start bit of the card No.-->1
    </cardIdBeginBit>
    <cardIdLength>
      <!--ro, req, int, card No. length-->1
    </cardIdLength>
    <siteCodeBeginBit>
      <!--ro, req, int, start bit of the site code-->1
    </siteCodeBeginBit>
    <siteCodeLength>
      <!--ro, req, int, site code length-->1
    </siteCodeLength>
    <oemBeginBit>
      <!--ro, req, int, start bit of OEM-->1
    </oemBeginBit>
    <oemLength>
      <!--ro, req, int, OEM length-->1
    </oemLength>
    <manufacturerCodeBeginBit>
      <!--ro, req, int, start bit of the manufacturer code-->1
    </manufacturerCodeBeginBit>
    <manufacturerCodeLength>
      <!--ro, req, int, manufacturer code length-->1
    </manufacturerCodeLength>
  </CustomerCardIn>
  <CustomerCardOut>
    <!--ro, opt, object-->
    <CardContentList size="4">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <Action>
        <!--ro, opt, object-->
        <No>
          <!--ro, req, int, No.-->1
        </No>
        <type>
          <!--ro, req, enum, type, subType:string, desc:type-->cardId
        </type>
        <length>
          <!--ro, req, int-->1
        </length>
      </Action>
    </CardContentList>
  </CustomerCardOut>
</WiegandRuleCfg>
```

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14.6.7.30 Set Wiegand rule

Request URL

PUT /ISAPI/AccessControl/WiegandRuleCfg

Query Parameter

None

Request Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<WiegandRuleCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <name>
    <!--req, string, Wiegand name-->test
  </name>
  <CustomerCardIn>
    <!--opt, object-->
    <totalLength>
      <!--req, int, total Wiegand length, desc:when this node is set to 0, the custom Wiegand rule settings will be cleared-->1
    </totalLength>
    <checkMethod>
      <!--req, enum, parity mode, subType:string, desc:parityCheck, xorCheck, noCheck;-->parityCheck
    </checkMethod>
    <ParityCheck>
      <!--opt, object, configuration rule of odd-even parity, desc:this node is valid when <checkMethod> is "parityCheck”-->
      <oddBeginBit>
        <!--opt, int, start bit of odd parity-->1
      </oddBeginBit>
      <oddLength>
        <!--opt, int, odd parity length-->1
      </oddLength>
      <evenBeginBit>
        <!--opt, int, start bit of even parity-->1
      </evenBeginBit>
      <evenLength>
        <!--opt, int, even parity length-->1
      </evenLength>
    </ParityCheck>
    <XorCheck>
      <!--opt, object, configuration rule of XOR parity, desc:this node is valid when <checkMethod> is "xorCheck”-->
      <xorBeginBit>
        <!--opt, int, start bit of XOR parity-->1
      </xorBeginBit>
      <xorPerLength>
        <!--opt, int, length of each XOR parity group-->1
      </xorPerLength>
      <xorTotalLength>
        <!--opt, int, total length of XOR parity data-->1
      </xorTotalLength>
    </XorCheck>
    <cardIdBeginBit>
      <!--req, int, start bit of the card No.-->1
    </cardIdBeginBit>
    <cardIdLength>
      <!--req, int, card No. length-->1
    </cardIdLength>
    <siteCodeBeginBit>
      <!--req, int, start bit of the site code-->1
    </siteCodeBeginBit>
    <siteCodeLength>
      <!--req, int, site code length-->1
    </siteCodeLength>
    <oemBeginBit>
      <!--req, int, start bit of OEM-->1
    </oemBeginBit>
    <oemLength>
      <!--req, int, OEM length-->1
    </oemLength>
    <manufacturerCodeBeginBit>
      <!--req, int, start bit of the manufacturer code-->1
    </manufacturerCodeBeginBit>
    <manufacturerCodeLength>
      <!--req, int, manufacturer code length-->1
    </manufacturerCodeLength>
  </CustomerCardIn>
  <CustomerCardOut>
    <!--opt, object-->
    <CardContentList size="4">
      <!--opt, array, subType:object, attr:size{req, int}-->
      <Action>
        <!--opt, object-->
        <No>
          <!--req, int, No.-->1
        </No>
        <type>
          <!--req, enum, module type, subType:string, desc:module type-->cardId
        </type>
        <length>
          <!--req, int-->1
        </length>
      </Action>
    </CardContentList>
  </CustomerCardOut>
</WiegandRuleCfg>
```

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Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->test
  </requestURL>
  <statusCode>
    <!--ro, req, int, status code-->1
  </statusCode>
  <statusString>
    <!--ro, req, string, status description-->test
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code-->test
  </subStatusCode>
</ResponseStatus>
```

14.6.7.31 Get the configuration capability of the custom Wiegand rule

Request URL

GET /ISAPI/AccessControl/WiegandRuleCfg/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<WiegandRuleCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <name min="1" max="10">
    <!--ro, req, string, Wiegand name, range:[1,32], attr:min{req, int},max{req, int}-->test
  </name>
  <CustomerCardIn>
    <!--ro, opt, object-->
    <totalLength min="1" max="10">
      <!--ro, req, int, total Wiegand length., attr:min{req, int},max{req, int}-->1
    </totalLength>
    <checkMethod opt="parityCheck,xorCheck,noCheck,">
      <!--ro, req, enum, parity mode, subType:string, attr:opt{req, string}, desc:parity mode-->parityCheck
    </checkMethod>
    <ParityCheck>
      <!--ro, opt, object, configuration rule of odd-even parity, desc:configuration rule of odd-even parity-->
      <oddBeginBit min="1" max="10">
        <!--ro, opt, int, start bit of odd parity, attr:min{req, int},max{req, int}-->1
      </oddBeginBit>
      <oddLength min="1" max="10">
        <!--ro, opt, int, odd parity length, attr:min{req, int},max{req, int}-->1
      </oddLength>
      <evenBeginBit min="1" max="10">
        <!--ro, opt, int, start bit of even parity, attr:min{req, int},max{req, int}-->1
      </evenBeginBit>
      <evenLength min="1" max="10">
        <!--ro, opt, int, even parity length, attr:min{req, int},max{req, int}-->1
      </evenLength>
    </ParityCheck>
    <XorCheck>
      <!--ro, opt, object, configuration rule of XOR parity, desc:configuration rule of XOR parity-->
      <xorBeginBit min="1" max="10">
        <!--ro, opt, int, start bit of XOR parity, attr:min{req, int},max{req, int}-->1
      </xorBeginBit>
      <xorPerLength min="1" max="10">
        <!--ro, opt, int, length of each XOR parity group, attr:min{req, int},max{req, int}-->1
      </xorPerLength>
      <xorTotalLength min="1" max="10">
        <!--ro, opt, int, total length of XOR parity data, attr:min{req, int},max{req, int}-->1
      </xorTotalLength>
    </XorCheck>
    <cardIdBeginBit min="1" max="10">
      <!--ro, req, int, start bit of the card No., attr:min{req, int},max{req, int}-->1
    </cardIdBeginBit>
    <cardIdLength min="1" max="10">
      <!--ro, req, int, card No. length, attr:min{req, int},max{req, int}-->1
    </cardIdLength>
    <siteCodeBeginBit min="1" max="10">
      <!--ro, req, int, start bit of the site code, attr:min{req, int},max{req, int}-->1
    </siteCodeBeginBit>
    <siteCodeLength min="1" max="10">
      <!--ro, req, int, site code length, attr:min{req, int},max{req, int}-->1
    </siteCodeLength>
    <oemBeginBit min="1" max="10">
      <!--ro, req, int, start bit of OEM, attr:min{req, int},max{req, int}-->1
    </oemBeginBit>
    <oemLength min="1" max="10">
      <!--ro, req, int, OEM length, attr:min{req, int},max{req, int}-->1
    </oemLength>
    <manufacturerCodeBeginBit min="1" max="10">
      <!--ro, req, int, start bit of the manufacturer code, attr:min{req, int},max{req, int}-->1
    </manufacturerCodeBeginBit>
    <manufacturerCodeLength min="1" max="10">
      <!--ro, req, int, manufacturer code length, attr:min{req, int},max{req, int}-->1
    </manufacturerCodeLength>
  </CustomerCardIn>
  <CustomerCardOut>
    <!--ro, opt, object-->
    <CardContentList size="4">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <Action>
        <!--ro, opt, object, card information-->
        <No min="1" max="10">
          <!--ro, req, int, No., attr:min{req, int},max{req, int}-->1
        </No>
        <type opt="cardId,siteCode,oem,manufacturerCode">
          <!--ro, req, enum, type, subType:string, attr:opt{req, string}, desc:type-->cardId
        </type>
        <length min="1" max="10">
          <!--ro, req, int, attr:min{req, int},max{req, int}-->1
        </length>
      </Action>
    </CardContentList>
  </CustomerCardOut>
</WiegandRuleCfg>
```

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14.6.8 Multi-Factor Authentication Management

14.6.8.1 Get the capability of clearing group parameters

Request URL

GET /ISAPI/AccessControl/ClearGroupCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "ClearGroupCfg": {
    /*ro, opt, object*/
        "ClearFlags": {
        /*ro, opt, object*/
            "groupCfg":  "true,false"
            /*ro, req, string, group parameters*/
        }
    }
}
```

14.6.8.2 Clear group parameters

Request URL

PUT /ISAPI/AccessControl/ClearGroupCfg?format=json

Query Parameter

None

Request Message

```
{
    "ClearGroupCfg": {
    /*opt, object, group parameters*/
        "ClearFlags": {
        /*opt, object*/
            "groupCfg":  true
            /*req, bool, whether to clear group parameters*/
        }
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code*/
    "statusString":  "test",
    /*ro, opt, string, status description*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code*/
    "errorCode":  1,
    /*ro, req, int, error code*/
    "errorMsg":  "ok"
    /*ro, req, string, error description*/
}
```

14.6.8.3 Get the group configuration parameters

Request URL

GET /ISAPI/AccessControl/GroupCfg/<groupID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
groupID	string	Group No., which starts from 1.

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Request Message

None

Response Message

```
{
    "GroupCfg": {
    /*ro, req, object, group parameters*/
        "enable":  true,
        /*ro, req, bool, whether to enable the function*/
        "ValidPeriodCfg": {
        /*ro, req, object, validity period of the group*/
            "enable":  true,
            /*ro, req, bool, whether to enable validity period, desc:whether to enable validity period*/
            "beginTime":  "1970-01-01T00:00:00+08:00",
            /*ro, req, datetime, start time of the validity period (UTC time)*/
            "endTime":  "1970-01-01T00:00:00+08:00"
            /*ro, req, datetime, end time of the validity period (UTC time)*/
        },
        "groupName":  "test"
        /*ro, opt, string, range:[0,32]*/
    }
}
```

14.6.8.4 Set the group parameters

Request URL

PUT /ISAPI/AccessControl/GroupCfg/<groupID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
groupID	string	--

Request Message

```
{
    "GroupCfg": {
    /*opt, object, group parameters*/
        "enable":  true,
        /*req, bool, whether to enable the group*/
        "ValidPeriodCfg": {
        /*opt, object, validity period of the group*/
            "enable":  true,
            /*req, bool, whether to enable validity period*/
            "beginTime":  "1970-01-01T00:00:00+08:00",
            /*req, datetime, start time of the validity period (UTC time)*/
            "endTime":  "1970-01-01T00:00:00+08:00"
            /*req, datetime, end time of the validity period (UTC time)*/
        },
        "groupName":  "test"
        /*opt, string, group name*/
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  "test",
    /*ro, opt, string, status code*/
    "statusString":  "test",
    /*ro, opt, string, status description*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code*/
    "errorCode":  1,
    /*ro, req, int, error code*/
    "errorMsg":  "ok"
    /*ro, req, string, error information*/
}
```

14.6.8.5 Get the group configuration capability

Request URL

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GET /ISAPI/AccessControl/GroupCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "GroupCfg": {
    /*ro, opt, object, group parameters*/
        "groupNo": {
        /*ro, opt, object, group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  1
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, req, string, whether to enable the group*/
        "ValidPeriodCfg": {
        /*ro, req, object, whether to enable validity period parameters of the group*/
            "enable":  "true,false",
            /*ro, req, string, whether to enable validity period*/
            "beginTime": {
            /*ro, req, object, start time of the validity period (UTC time)*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  32
                /*ro, opt, int, the maximum value*/
            },
            "endTime": {
            /*ro, req, object, end time of the validity period (UTC time)*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  32
                /*ro, opt, int, the maximum value*/
            }
        },
        "groupName": {
        /*ro, opt, object, group name*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        }
    }
}
```

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14.6.8.6 Get the configuration parameters of multi-factor authentication mode

Request URL

GET /ISAPI/AccessControl/MultiCardCfg/<doorID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
doorID	string	Door No.

Request Message

None

Response Message

```
{
    "MultiCardCfg": {
    /*ro, opt, object*/
        "doorName":  "test",
        /*ro, opt, string, range:[0,64]*/
        "doorRegionName":  "test",
        /*ro, opt, string, range:[0,64]*/
        "enable":  true,
        /*ro, req, bool, whether to enable multi-factor authentication*/
        "swipeIntervalTimeout":  10,
        /*ro, opt, int, timeout of swiping (authentication) interval*/
        "GroupCfg": [
        /*ro, opt, array, multi-factor authentication parameters, subType:object*/
            {
                "id":  1,
                /*ro, opt, int, multi-factor authentication No.*/
                "name":  "test",
                /*ro, opt, string, range:[0,64]*/
                "enable":  true,
                /*ro, opt, bool, whether to enable multi-factor authentication*/
                "enableOfflineVerifyMode":  true,
                /*ro, opt, bool, whether to enable verification mode when the access control device is offline (the super password will replace opening door
remotely)*/
                "templateNo":  1,
                /*ro, opt, int, schedule template No. to enable the multi-factor authentication*/
                "templateName":  "test",
                /*ro, opt, string, range:[0,64]*/
                "GroupCombination": [
                /*ro, opt, array, multi-factor authentication parameters, subType:object*/
                    {
                        "enable":  true,
                        /*ro, opt, bool, whether to enable multi-factor authentication*/
                        "memberNum":  3,
                        /*ro, opt, int, number of members swiping cards*/
                        "sequenceNo":  1,
                        /*ro, opt, int, serial No. of swiping cards of the multi-factor authentication group*/
                        "groupNo":  1,
                        /*ro, opt, int, group No., desc:group No.*/
                        "groupName":  "test"
                        /*ro, opt, string, range:[0,64]*/
                    }
                ]
            }
        ]
    }
}
```

14.6.8.7 Set parameters of multi-factor authentication mode

Request URL

PUT /ISAPI/AccessControl/MultiCardCfg/<doorID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
doorID	string	--

Request Message

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```
{
    "MultiCardCfg": {
    /*opt, object, parameters of multi-factor authentication mode*/
        "enable":  true,
        /*req, bool, whether to enable multi-factor authentication*/
        "swipeIntervalTimeout":  10,
        /*opt, int, timeout of swiping (authentication) interval*/
        "GroupCfg": [
        /*opt, array, multi-factor authentication parameters, subType:object*/
            {
                "id":  1,
                /*opt, int, multi-factor authentication No.*/
                "enable":  true,
                /*opt, bool, whether to enable multi-factor authentication*/
                "enableOfflineVerifyMode":  true,
                /*opt, bool, whether to enable verification mode when the access control device is offline (the super password will replace opening door
remotely)*/
                "templateNo":  1,
                /*opt, int, schedule template No. to enable the multi-factor authentication*/
                "GroupCombination": [
                /*opt, array, parameters of the multi-factor authentication group, subType:object*/
                    {
                        "enable":  true,
                        /*opt, bool, whether to enable multi-factor authentication*/
                        "memberNum":  3,
                        /*opt, int, number of members swiping cards*/
                        "sequenceNo":  1,
                        /*opt, int, serial No. of swiping cards of the multi-factor authentication group*/
                        "groupNo":  1
                        /*opt, int, group No.*/
                    }
                ]
            }
        ]
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code*/
    "statusString":  "test",
    /*ro, opt, string, status description*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code*/
    "errorCode":  1,
    /*ro, req, int, error code*/
    "errorMsg":  "ok"
    /*ro, req, string, error information*/
}
```

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14.6.8.8 Get the configuration capability of multi-factor authentication mode

Request URL

GET /ISAPI/AccessControl/MultiCardCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "MultiCardCfg": {
    /*ro, opt, object*/
        "doorNo": {
        /*ro, opt, object, door No.*/
            "@min":  1,
            /*ro, opt, int, minimum value*/
            "@max":  256
            /*ro, opt, int, maximum value*/
        },
        "enable":  "true,false",
        /*ro, req, string, whether to enable multi-factor authentication*/
        "swipeIntervalTimeout": {
        /*ro, opt, object, timeout of swiping (authentication) interval*/
            "@min":  1,
            /*ro, opt, int, minimum value*/
            "@max":  255
            /*ro, opt, int, maximum value*/
        },
        "GroupCfg": {
        /*ro, opt, object, multi-factor authentication parameters*/
            "maxSize":  20,
            /*ro, opt, int, maximum value*/
            "id": {
            /*ro, opt, object, multi-factor authentication No.*/
                "@min":  1,
                /*ro, opt, int, minimum value*/
                "@max":  20
                /*ro, opt, int, maximum value*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether to enable multi-factor authentication*/
            "enableOfflineVerifyMode":  "true,false",
            /*ro, opt, string, whether to enable verification mode when the access control device is offline (the super password will replace opening door
remotely)*/
            "templateNo": {
            /*ro, opt, object, schedule template No. to enable the multi-factor authentication*/
                "@min":  1,
                /*ro, opt, int, minimum value*/
                "@max":  20
                /*ro, opt, int, maximum value*/
            },
            "GroupCombination": {
            /*ro, opt, object, parameters of the multi-factor authentication group*/
                "maxSize":  8,
                /*ro, opt, int, maximum value*/
                "enable":  "true,false",
                /*ro, opt, string, whether to enable multi-factor authentication*/
                "memberNum": {
                /*ro, opt, object, number of members swiping cards*/
                    "@min":  1,
                    /*ro, opt, int, minimum value*/
                    "@max":  20
                    /*ro, opt, int, maximum value*/
                },
                "sequenceNo": {
                /*ro, opt, object, serial No. of swiping cards of the multi-factor authentication group*/
                    "@min":  1,
                    /*ro, opt, int, minimum value*/
                    "@max":  8
                    /*ro, opt, int, maximum value*/
                },
                "groupNo": {
                /*ro, opt, object, group No.*/
                    "@min":  1,
                    /*ro, opt, int, minimum value*/
                    "@max":  20
                    /*ro, opt, int, the maximum value*/
                }
            }
        }
    }
}
```

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14.6.9 Permission Schedules for Persons and Access Points

14.6.9.1 Set the holiday group parameters of the access permission control schedule

Request URL

PUT /ISAPI/AccessControl/UserRightHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
holidayGroupID	string	--

Request Message

```
{
    "UserRightHolidayGroupCfg": {
    /*req, object, the holiday group parameters of the access permission control schedule*/
        "enable": true,
        /*req, bool, whether to enable, desc:true (yes), false (no)*/
        "groupName": "test",
        /*req, string, holiday group name*/
        "holidayPlanNo": "1,3,5",
        /*req, string, holiday group schedule No., desc:holiday group schedule No.*/
        "operateType": "byTerminal",
        /*opt, enum, operation type, subType:string, desc:"byTerminal" (by terminal), "byOrg" (by organization), "byTerminalOrg" (by terminal
organization)*/
        "terminalNoList": [1, 2, 3, 4],
        /*opt, array, terminal ID list, subType:int, desc:this node is required when operation type is "byTerminal" or "byTerminalOrg"*/
        "orgNoList": [1, 2, 3, 4]
        /*opt, array, organization ID list, subType:int, desc:this node is required when operation type is "byOrg" or "byTerminalOrg"*/
    }
}
```

Response Message

```
{
    "requestURL": "test",
    /*ro, opt, string, URI*/
    "statusCode": "test",
    /*ro, opt, string, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "test",
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok"
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

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14.6.9.2 Get the holiday group configuration parameters of the access permission control schedule

Request URL

GET /ISAPI/AccessControl/UserRightHolidayGroupCfg/<holidayGroupID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
holidayGroupID	string	It starts from 1. The maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

```
{
    "UserRightHolidayGroupCfg": {
    /*ro, req, object*/
        "enable": true,
        /*ro, req, bool, whether to enable, desc:true (yes), false (no)*/
        "groupName": "test",
        /*ro, req, string, holiday group name*/
        "holidayPlanNo": "1,3,5",
        /*ro, req, string, holiday group schedule No., desc:holiday group schedule No.*/
        "holidayPlanName": ["test1", "test3", "test5"]
        /*ro, opt, array, subType:string*/
    }
}
```

14.6.9.3 Get the holiday group configuration capability of the access permission control

Request URL

GET /ISAPI/AccessControl/UserRightHolidayGroupCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "UserRightHolidayGroupCfg": {
    /*ro, req, object*/
        "groupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether it is enabled, desc:"true" (enabled), "false" (disabled)*/
        "groupName": {
        /*ro, opt, object, holiday group name*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        },
        "holidayPlanNo": {
        /*ro, opt, object, holiday group schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        }
    }
}
```

14.6.9.4 Set the holiday schedule parameters of the access permission control

Request URL

PUT /ISAPI/AccessControl/UserRightHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
holidayPlanID	string	--

Request Message

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```
{
    "UserRightHolidayPlanCfg": {
    /*req, object, the holiday schedule parameters of the access permission control*/
        "enable":  true,
        /*req, bool, whether to enable, desc:"true" (enable), "false" (disable)*/
        "beginDate":  "1970-01-01",
        /*req, date, start date of the holiday, desc:device local time*/
        "endDate":  "1970-01-01",
        /*req, date, end date of the holiday, desc:device local time*/
        "HolidayPlanCfg": [
        /*req, array, subType:object*/
            {
                "id":  1,
                /*req, int, time period No., range:[1,8], desc:it is between 1 and 8*/
                "enable":  true,
                /*req, bool, whether to enable, desc:"true" (enable), "false" (disable)*/
                "TimeSegment": {
                /*opt, object, time period*/
                    "beginTime":  "00:00:00",
                    /*req, time, start time of the time period, desc:device local time*/
                    "endTime":  "00:00:00"
                    /*req, time, end time of the time period, desc:device local time*/
                },
                "authenticationTimesEnabled":  true,
                /*opt, bool*/
                "authenticationTimes":  10
                /*opt, int, range:[1,255], step:1*/
            }
        ]
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "test",
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

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14.6.9.5 Get holiday schedule configuration parameters

Request URL

GET /ISAPI/AccessControl/UserRightHolidayPlanCfg/<holidayPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
holidayPlanID	string	Holiday schedule No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

```
{
    "UserRightHolidayPlanCfg": {
    /*ro, req, object, holiday schedule configuration parameters*/
        "enable": true,
        /*ro, req, bool, whether to enable, desc:true (enable), false (disable)*/
        "holidayPlanName": "test",
        /*ro, opt, string, range:[0,64]*/
        "beginDate": "1970-01-01",
        /*ro, req, date, start date of the holiday, desc:device local time*/
        "endDate": "1970-01-01",
        /*ro, req, date, end date of the holiday, desc:device local time*/
        "cycleByYear": true,
        /*ro, opt, bool*/
        "HolidayPlanCfg": [
        /*ro, req, array, holiday schedule parameters, subType:object*/
            {
                "id": 1,
                /*ro, req, int, time period No., range:[1,8], desc:it is between 1 and 8*/
                "enable": true,
                /*ro, req, bool, whether to enable, desc:true (enable), false (disable)*/
                "TimeSegment": {
                /*ro, req, object, time period*/
                    "beginTime": "00:00:00",
                    /*ro, req, time, start time of the time period, desc:device local time*/
                    "endTime": "00:00:00"
                    /*ro, req, time, end time of the time period, desc:device local time*/
                },
                "authenticationTimesEnabled": true,
                /*ro, opt, bool*/
                "authenticationTimes": 10
                /*ro, opt, int, range:[0,255], step:1*/
            }
        ]
    }
}
```

14.6.9.6 Get the holiday schedule configuration capability of the access permission control

Request URL

GET /ISAPI/AccessControl/UserRightHolidayPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "UserRightHolidayPlanCfg": {
    /*ro, req, object*/
        "planNo": {
        /*ro, opt, object, holiday schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether it is enabled, desc:"true" (enabled), "false" (disabled)*/
        "beginDate":  "1970-01-01",
        /*ro, opt, date, start date of the holiday, desc:(device local time)*/
        "endDate":  "1970-01-01",
        /*ro, opt, date, end date of the holiday, desc:(device local time)*/
        "HolidayPlanCfg": {
        /*ro, opt, object, holiday schedule parameter*/
            "maxSize":  8,
            /*ro, opt, int, the maximum value*/
            "id": {
            /*ro, opt, object, time period No.*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  8
                /*ro, opt, int, the maximum value*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether it is enabled, desc:"true" (enabled), "false" (disabled)*/
            "TimeSegment": {
            /*ro, opt, object, time period*/
                "beginTime":  "00:00:00",
                /*ro, opt, time, start time, desc:(device local time)*/
                "endTime":  "00:00:00",
                /*ro, opt, time, end time, desc:(device local time)*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:"hour", "minute", "second". If this node is not returned, the default time accuracy is
"minute"*/
            },
            "authenticationTimesEnabled": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, req, array, subType:bool*/
            },
            "authenticationTimes": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int, range:[1,255], step:1*/
                "@max":  255
                /*ro, req, int, range:[1,255], step:1*/
            }
        }
    }
}
```

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14.6.9.7 Get the schedule template configuration parameters of the access permission control

Request URL

GET /ISAPI/AccessControl/UserRightPlanTemplate/<planTemplateID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
planTemplateID	string	Schedule template No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

```
{
    "UserRightPlanTemplate": {
    /*ro, req, object*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (yes), false (no)*/
        "templateName":  "test",
        /*ro, req, string, template name*/
        "weekPlanNo":  1,
        /*ro, req, int, week schedule No., desc:week schedule No.*/
        "holidayGroupNo":  "1,3,5"
        /*ro, req, string, holiday group No., desc:holiday group No.*/
    }
}
```

14.6.9.8 Set the schedule template parameters of the access permission control

Request URL

PUT /ISAPI/AccessControl/UserRightPlanTemplate/<planTemplateID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
planTemplateID	string	--

Request Message

```
{
    "UserRightPlanTemplate": {
    /*req, object, the schedule template of the access permission control*/
        "enable":  true,
        /*req, bool, whether to enable, desc:true (yes), false (no)*/
        "templateName":  "test",
        /*req, string, template name*/
        "weekPlanNo":  1,
        /*req, int, week schedule No.*/
        "holidayGroupNo":  "1,3,5",
        /*req, string, holiday group No., desc:holiday group No.*/
        "operateType":  "byTerminal",
        /*opt, enum, operation type, subType:string, desc:"byTerminal" (by terminal), "byOrg" (by organization), "byTerminalOrg" (by terminal
organization)*/
        "terminalNoList": [1, 2, 3, 4],
        /*opt, array, terminal ID list, subType:int, desc:this node is required when operation type is "byTerminal" or "byTerminalOrg"*/
        "orgNoList": [1, 2, 3, 4]
        /*opt, array, organization ID list, subType:int, desc:this node is required when operation type is "byOrg" or "byTerminalOrg"*/
    }
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "test",
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.6.9.9 Get the schedule template configuration capability of the access permission control

Request URL

GET /ISAPI/AccessControl/UserRightPlanTemplate/capabilities?format=json

Query Parameter

None

Request Message

None

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Response Message

```
{
    "UserRightPlanTemplate": {
    /*ro, opt, object*/
        "templateNo": {
        /*ro, opt, object, schedule template No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether it is enabled, desc:"true" (enabled), "false" (disabled)*/
        "templateName": {
        /*ro, opt, object, template name*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        },
        "weekPlanNo": {
        /*ro, opt, object, weekly schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "holidayGroupNo": {
        /*ro, opt, object, holiday group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        }
    }
}
```

14.6.9.10 Get weekly schedule configuration parameters

Request URL

GET /ISAPI/AccessControl/UserRightWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
weekPlanID	string	Weekly schedule No., which starts from 1, and the maximum value supported by the device is obtained from the capability set.

Request Message

None

Response Message

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```
{
    "UserRightWeekPlanCfg": {
    /*ro, req, object, weekly schedule configuration parameters*/
        "enable":  true,
        /*ro, req, bool, whether to enable, desc:true (enable), false (disable)*/
        "WeekPlanCfg": [
        /*ro, req, array, weekly schedule parameters, subType:object*/
            {
                "week":  "Monday",
                /*ro, req, enum, day of the week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
                "id":  1,
                /*ro, req, int, time period No., range:[1,8], desc:it is between 1 and 8*/
                "enable":  true,
                /*ro, req, bool, whether to enable, desc:true (enable), false (disable)*/
                "TimeSegment": {
                /*ro, req, object, time period*/
                    "beginTime":  "10:10:00",
                    /*ro, req, string, start time of the time period, desc:device local time*/
                    "endTime":  "12:10:00"
                    /*ro, req, string, end time of the time period, desc:device local time*/
                },
                "authenticationTimesEnabled":  true,
                /*ro, opt, bool*/
                "authenticationTimes":  10
                /*ro, opt, int, range:[0,255], step:1*/
            }
        ]
    }
}
```

14.6.9.11 Set the week schedule parameters of the access permission control

Request URL

PUT /ISAPI/AccessControl/UserRightWeekPlanCfg/<weekPlanID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
weekPlanID	string	--

Request Message

```
{
    "UserRightWeekPlanCfg": {
    /*opt, object, the week schedule parameters of the access permission control*/
        "enable":  true,
        /*req, bool, whether to enable, desc:"true" (enable), "false" (disable)*/
        "WeekPlanCfg": [
        /*req, array, week schedule parameters, subType:object*/
            {
                "week":  "Monday",
                /*req, enum, days of the week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
                "id":  1,
                /*req, int, time period No., range:[1,8], desc:it is between 1 and 8*/
                "enable":  true,
                /*req, bool, whether to enable, desc:"true" (enable), "false" (disable)*/
                "TimeSegment": {
                /*req, object, time period*/
                    "beginTime":  "10:10:00",
                    /*req, string, start time of the time period, desc:(device local time)*/
                    "endTime":  "12:10:00"
                    /*req, string, end time of the time period, desc:(device local time)*/
                },
                "authenticationTimesEnabled":  true,
                /*opt, bool*/
                "authenticationTimes":  10
                /*opt, int, range:[1,255], step:1*/
            }
        ]
    }
}
```

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "test",
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.6.9.12 Get the weekly schedule configuration capability of the access permission control

Request URL

GET /ISAPI/AccessControl/UserRightWeekPlanCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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HTA

```
{
    "UserRightWeekPlanCfg": {
    /*ro, opt, object*/
        "planNo": {
        /*ro, opt, object, weekly schedule No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether it is enabled, desc:"true" (enabled), "false" (disabled)*/
        "WeekPlanCfg": {
        /*ro, opt, object, weekly schedule parameters*/
            "maxSize":  56,
            /*ro, opt, int, the maximum value*/
            "week": {
            /*ro, opt, object, week*/
                "@opt":  "Monday"
                /*ro, opt, enum, days of the week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"*/
            },
            "id": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  8
                /*ro, opt, int, the maximum value*/
            },
            "enable":  "true,false",
            /*ro, opt, string, whether it is enabled, desc:"true" (enabled), "false" (disabled)*/
            "TimeSegment": {
            /*ro, opt, object, time period*/
                "beginTime":  "test",
                /*ro, opt, string, start time, desc:(device local time)*/
                "endTime":  "test",
                /*ro, opt, string, end time, desc:(device local time)*/
                "validUnit":  "minute"
                /*ro, opt, enum, time accuracy, subType:string, desc:"hour", "minute", "second". If this node is not returned, the default time accuracy is
"minute"*/
            },
            "authenticationTimesEnabled": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, req, array, subType:bool*/
            },
            "authenticationTimes": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int, range:[1,255], step:1*/
                "@max":  255
                /*ro, req, int, range:[1,255], step:1*/
            }
        }
    }
}
```


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14.6.10 Person and Credential Management

14.6.10.1 Get the capability of deleting person information (including linked cards, fingerprints, and faces) and permissions

Request URL

GET /ISAPI/AccessControl/UserInfoDetail/Delete/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "UserInfoDetail": {
    /*ro, opt, object, user Information*/
        "mode": {
        /*ro, req, object*/
            "@opt":  "all,byEmployeeNo"
            /*ro, opt, string, deleting mode, desc:all (delete all), byEmployeeNo (delete by employee No. (person ID))*/
        },
        "EmployeeNoList": {
        /*ro, opt, object, person ID list, desc:person ID list*/
            "maxSize":  50,
            /*ro, opt, int*/
            "employeeNo": {
            /*ro, opt, object, employee No. (person ID)*/
                "@min":  1,
                /*ro, opt, int, the maximum value*/
                "@max":  32
                /*ro, opt, int, the minimum value*/
            }
        }
    }
}
```

14.6.10.2 Start deleting all person information (including linked cards, fingerprints, and faces) and permissions by employee No.

Request URL

PUT /ISAPI/AccessControl/UserInfoDetail/Delete?format=json

Query Parameter

None

Request Message

```
{
    "UserInfoDetail": {
    /*opt, object, user Information*/
        "mode":  "all",
        /*req, enum, deleting mode, subType:string, desc:deleting mode*/
        "EmployeeNoList": [
        /*opt, array, person ID list, subType:object*/
            {
                "employeeNo":  "test"
                /*opt, string, employee No.*/
            }
        ],
        "operateType":  "byTerminal",
        /*opt, enum, operation mode, subType:string, desc:"byTerminal" (by terminal), "byOrg" (by organization), "byTerminalOrg" (by terminal
organization)*/
        "terminalNoList": [1, 2, 3, 4],
        /*opt, array, terminal list, subType:int, dep:and,{$.UserInfoDetail.operateType,eq,byTerminal}*/
        "orgNoList": [1, 2, 3, 4]
        /*opt, array, organization list, subType:int, dep:or,{$.UserInfoDetail.operateType,eq,byOrg},{$.UserInfoDetail.operateType,eq,byTerminalOrg}*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

14.6.10.3 Get the status of deleting all person information (including linked cards, fingerprints, and faces) and permissions by employee No

Request URL

GET /ISAPI/AccessControl/UserInfoDetail/DeleteProcess?format=json

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Query Parameter

None

Request Message

None

Response Message

```
{
    "UserInfoDetailDeleteProcess": {
    /*ro, req, object*/
        "status":  "processing",
        /*ro, req, enum, status, subType:string, desc:status*/
        "percent":  100
        /*ro, opt, int, range:[0,100], dep:or,{$.UserInfoDetailDeleteProcess.status,eq,processing}*/
    }
}
```

14.6.11 Person and Credential Management

14.6.11.1 Get the capability of deleting person information (including linked cards, fingerprints, and faces) and permissions

Request URL

GET /ISAPI/AccessControl/UserInfoDetail/Delete/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "UserInfoDetail": {
    /*ro, opt, object, user Information*/
        "mode": {
        /*ro, req, object*/
            "@opt":  "all,byEmployeeNo"
            /*ro, opt, string, deleting mode, desc:all (delete all), byEmployeeNo (delete by employee No. (person ID))*/
        },
        "EmployeeNoList": {
        /*ro, opt, object, person ID list, desc:person ID list*/
            "maxSize":  50,
            /*ro, opt, int*/
            "employeeNo": {
            /*ro, opt, object, employee No. (person ID)*/
                "@min":  1,
                /*ro, opt, int, the maximum value*/
                "@max":  32
                /*ro, opt, int, the minimum value*/
            }
        }
    }
}
```

14.6.11.2 Start deleting all person information (including linked cards, fingerprints, and faces) and permissions by employee No.

Request URL

PUT /ISAPI/AccessControl/UserInfoDetail/Delete?format=json

Query Parameter

None

Request Message

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```
{
    "UserInfoDetail": {
    /*opt, object, user Information*/
        "mode":  "all",
        /*req, enum, deleting mode, subType:string, desc:deleting mode*/
        "EmployeeNoList": [
        /*opt, array, person ID list, subType:object*/
            {
                "employeeNo":  "test"
                /*opt, string, employee No.*/
            }
        ],
        "operateType":  "byTerminal",
        /*opt, enum, operation mode, subType:string, desc:"byTerminal" (by terminal), "byOrg" (by organization), "byTerminalOrg" (by terminal
organization)*/
        "terminalNoList": [1, 2, 3, 4],
        /*opt, array, terminal list, subType:int, dep:and,{$.UserInfoDetail.operateType,eq,byTerminal}*/
        "orgNoList": [1, 2, 3, 4]
        /*opt, array, organization list, subType:int, dep:or,{$.UserInfoDetail.operateType,eq,byOrg},{$.UserInfoDetail.operateType,eq,byTerminalOrg}*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

14.6.11.3 Get the status of deleting all person information (including linked cards, fingerprints, and faces) and permissions by employee No

Request URL

GET /ISAPI/AccessControl/UserInfoDetail/DeleteProcess?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "UserInfoDetailDeleteProcess": {
    /*ro, req, object*/
        "status":  "processing",
        /*ro, req, enum, status, subType:string, desc:status*/
        "percent":  100
        /*ro, opt, int, range:[0,100], dep:or,{$.UserInfoDetailDeleteProcess.status,eq,processing}*/
    }
}
```

14.6.12 Person and Credential Task Management

14.6.12.1 Get the status of a specified task of applying person data asynchronously

Request URL

GET /ISAPI/AccessControl/UserInfo/asyncImportDatasTasks/<taskID>/status?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

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Request Message

None

Response Message

```
{
    "AsyncImportDatasTask": {
    /*ro, req, object*/
        "taskID":  "test",
        /*ro, req, string, range:[1,64]*/
        "URL":  "test",
        /*ro, opt, string, range:[1,256]*/
        "status":  0,
        /*ro, req, enum, subType:int*/
        "totalNum":  1,
        /*ro, opt, int, dep:and,{$.AsyncImportDatasTask.status,eq,3}*/
        "successNum":  1,
        /*ro, opt, int, dep:and,{$.AsyncImportDatasTask.status,eq,3}*/
        "failedNum":  1
        /*ro, opt, int, dep:and,{$.AsyncImportDatasTask.status,eq,3}*/
    }
}
```

14.6.12.2 Delete a specified task of applying person data asynchronously

Request URL

DELETE /ISAPI/AccessControl/UserInfo/asyncImportDatasTasks/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "OK",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok",
    /*ro, opt, string*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

14.6.12.3 Get the capabilities of applying person data asynchronously

Request URL

GET /ISAPI/AccessControl/UserInfo/asyncImportDatasTasks/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AsyncImportDatasCap": {
    /*ro, req, object*/
        "taskID": {
        /*ro, req, object*/
```

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```
            "@min":  1,
            /*ro, opt, int*/
            "@max":  64
            /*ro, opt, int*/
        },
        "taskNum":  1,
        /*ro, req, int*/
        "URL": {
        /*ro, req, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  256
            /*ro, opt, int*/
        },
        "singleFileMaxSize":  1,
        /*ro, req, int*/
        "employeeNo": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  32
            /*ro, opt, int*/
        },
        "deleteUser": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "name": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  32
            /*ro, opt, int*/
        },
        "userType": {
        /*ro, opt, object*/
            "@opt": ["normal", "visitor", "blackList"]
            /*ro, opt, array, subType:string*/
        },
        "Valid": {
        /*ro, opt, object*/
            "enable": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, opt, array, subType:bool*/
            },
            "beginTime":  "1970-01-01T00:00:00+08:00",
            /*ro, opt, string*/
            "endTime":  "2037-12-31T23:59:59+08:00"
            /*ro, opt, string*/
        },
        "password": {
        /*ro, opt, object*/
            "@min":  8,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        },
        "RightPlan": {
        /*ro, opt, object*/
            "doorNo": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  1
                /*ro, opt, int*/
            },
            "planTemplateNo": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  1
                /*ro, opt, int*/
            },
            "planTemplateNum": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, opt, int*/
                "@max":  1
                /*ro, opt, int*/
            }
        },
        "localUIRight": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "userVerifyMode": {
        /*ro, opt, object*/
            "@opt": ["cardAndPw", "card", "cardOrPw", "fp", "fpAndPw", "fpOrCard", "fpAndCard", "fpAndCardAndPw", "faceOrFpOrCardOrPw", "faceAndFp",
"faceAndPw", "faceAndCard", "face", "employeeNoAndPw", "fpOrPw", "employeeNoAndFp", "employeeNoAndFpAndPw", "faceAndFpAndCard", "faceAndPwAndFp",
"employeeNoAndFace", "faceOrfaceAndCard", "fpOrface", "cardOrfaceOrPw", "cardOrFace", "cardOrFaceOrFp"]
            /*ro, opt, array, subType:string*/
```

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```
        },
        "AccessControl": {
        /*ro, opt, object*/
            "doorRightNum": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, opt, int*/
                "@max":  1
                /*ro, opt, int*/
            },
            "closeDelayEnabled": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, opt, array, subType:bool*/
            },
            "belongGroupNo": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  1
                /*ro, opt, int*/
            },
            "belongGroupNum": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, opt, int*/
                "@max":  1
                /*ro, opt, int*/
            },
            "maxOpenDoorTime": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, opt, int*/
                "@max":  1
                /*ro, opt, int*/
            },
            "openDoorTime": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  1
                /*ro, opt, int*/
            },
            "roomNumber": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  1
                /*ro, opt, int*/
            },
            "floorNumber": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  1
                /*ro, opt, int*/
            }
        },
        "FaceInfo": {
        /*ro, opt, object*/
            "deleteAllFace": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, opt, array, subType:bool*/
            },
            "List": {
            /*ro, opt, object*/
                "FDID":  "test",
                /*ro, opt, string*/
                "faceID": {
                /*ro, opt, object*/
                    "@min":  1,
                    /*ro, opt, int*/
                    "@max":  1
                    /*ro, opt, int*/
                },
                "deleteFace": {
                /*ro, opt, object*/
                    "@opt": [true, false]
                    /*ro, opt, array, subType:bool*/
                },
                "isSupportModelData":  true
                /*ro, opt, bool*/
            }
        },
        "CardInfo": {
        /*ro, opt, object*/
            "deleteAllCard": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, opt, array, subType:bool*/
            },
            "List": {
```

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```
            /*ro, opt, object*/
                "cardNo": {
                /*ro, opt, object*/
                    "@min":  1,
                    /*ro, opt, int*/
                    "@max":  32
                    /*ro, opt, int*/
                },
                "deleteCard": {
                /*ro, opt, object*/
                    "@opt": [true, false]
                    /*ro, opt, array, subType:bool*/
                },
                "cardType": {
                /*ro, opt, object*/
                    "@opt": ["normalCard", "patrolCard", "hijackCard", "superCard", "dismissingCard", "emergencyCard"]
                    /*ro, opt, array, subType:string*/
                },
                "AccessControl": {
                /*ro, opt, object*/
                    "leaderCardNum": {
                    /*ro, opt, object*/
                        "@min":  0,
                        /*ro, opt, int*/
                        "@max":  1
                        /*ro, opt, int*/
                    }
                },
                "numberPerPerson":  50
                /*ro, opt, enum, subType:int*/
            }
        },
        "FPInfo": {
        /*ro, opt, object*/
            "deleteAllFP": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, opt, array, subType:bool*/
            },
            "List": {
            /*ro, opt, object*/
                "fingerID": {
                /*ro, opt, object*/
                    "@min":  1,
                    /*ro, opt, int*/
                    "@max":  10
                    /*ro, opt, int*/
                },
                "deleteFP": {
                /*ro, opt, object*/
                    "@opt": [true, false]
                    /*ro, opt, array, subType:bool*/
                },
                "enableCardReaderNo": {
                /*ro, opt, object*/
                    "@min":  1,
                    /*ro, opt, int*/
                    "@max":  1
                    /*ro, opt, int*/
                },
                "enableCardReaderNum": {
                /*ro, opt, object*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  1
                    /*ro, opt, int*/
                },
                "fingerType": {
                /*ro, opt, object*/
                    "@opt": ["normalFP", "hijackFP", "patrolFP", "superFP", "dismissingFP"]
                    /*ro, opt, array, subType:string*/
                },
                "fingerData": {
                /*ro, opt, object*/
                    "@opt": [768]
                    /*ro, opt, array, subType:int*/
                },
                "AccessControl": {
                /*ro, opt, object*/
                    "leaderFPNum": {
                    /*ro, opt, object*/
                        "@min":  0,
                        /*ro, opt, int*/
                        "@max":  1
                        /*ro, opt, int*/
                    }
                }
            }
        }
    }
}
```

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14.6.12.4 Get the status of all tasks of applying person data asynchronously

Request URL

GET /ISAPI/AccessControl/UserInfo/asyncImportDatasTasks/status?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "TasksStatusList": [
    /*ro, req, array, subType:object*/
        {
            "AsyncImportDatasTask": {
            /*ro, opt, object*/
                "taskID":  "test",
                /*ro, req, string, range:[1,64]*/
                "URL":  "test",
                /*ro, opt, string, range:[1,256]*/
                "status":  0,
                /*ro, req, enum, subType:int*/
                "totalNum":  1,
                /*ro, opt, int*/
                "successNum":  1,
                /*ro, opt, int*/
                "failedNum":  1,
                /*ro, opt, int*/
                "taskMissingReason":  "deviceShutDown"
                /*ro, opt, enum, subType:string, dep:and,{$.TasksStatusList[*].AsyncImportDatasTask.status,eq,4}*/
            }
        }
    ]
}
```

14.6.12.5 Get the status of a specified task of applying person pictures asynchronously

Request URL

GET /ISAPI/AccessControl/UserPic/asyncImportDatasTasks/<taskID>/status?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "AsyncImportDatasTask": {
    /*ro, req, object*/
        "taskID":  "test",
        /*ro, req, string, range:[1,64]*/
        "URL":  "test",
        /*ro, opt, string, range:[1,256]*/
        "status":  1,
        /*ro, req, enum, subType:int*/
        "totalNum":  1,
        /*ro, opt, int, dep:and,{$.AsyncImportDatasTask.status,eq,2}*/
        "successNum":  1,
        /*ro, opt, int, dep:and,{$.AsyncImportDatasTask.status,eq,2}*/
        "failedNum":  1
        /*ro, opt, int, dep:and,{$.AsyncImportDatasTask.status,eq,2}*/
    }
}
```

14.6.12.6 Delete a specified task of applying person pictures asynchronously

Request URL

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DELETE /ISAPI/AccessControl/UserPic/asyncImportDatasTasks/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "OK",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok",
    /*ro, opt, string*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

14.6.12.7 Get the capabilities of applying person pictures asynchronously

Request URL

GET /ISAPI/AccessControl/UserPic/asyncImportDatasTasks/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "AsyncImportDatasCap": {
    /*ro, opt, object*/
        "taskID": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  64
            /*ro, req, int*/
        },
        "taskNum":  1,
        /*ro, req, int*/
        "URL": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  256
            /*ro, req, int*/
        },
        "singleFileMaxSize":  1,
        /*ro, req, int*/
        "employeeNo": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  32
            /*ro, opt, int*/
        },
        "FDID":  "test",
        /*ro, opt, string*/
        "deleteFacePic": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "picURL": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  1
            /*ro, opt, int*/
        }
    }
}
```

14.6.12.8 Get the status of all tasks of applying person pictures asynchronously

Request URL

GET /ISAPI/AccessControl/UserPic/asyncImportDatasTasks/status?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "TasksStatusList": [
    /*ro, req, array, subType:object*/
        {
            "AsyncImportDatasTask": {
            /*ro, opt, object*/
                "taskID":  "test",
                /*ro, req, string, range:[1,64]*/
                "URL":  "test",
                /*ro, opt, string, range:[1,256]*/
                "status":  1,
                /*ro, req, enum, subType:int*/
                "totalNum":  1,
                /*ro, opt, int*/
                "successNum":  1,
                /*ro, opt, int*/
                "failedNum":  1
                /*ro, opt, int*/
            }
        }
    ]
}
```

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14.6.13 QR Code Management

14.6.13.1 Get the capability of actively getting QR code scanning events

Request URL

GET /ISAPI/AccessControl/QRCodeEvent/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "QRCodeEventCap": {
    /*ro, req, object, capability of actively getting QR code scanning events*/
        "QRCodeEventCond": {
        /*ro, opt, object, condition of actively getting QR code scanning events*/
            "searchID": {
            /*ro, req, object, search ID, desc:search ID,which is used to check whether the current search requester is the same as the previous one. If they are the same,the search record will be stored in the device to speed up the next search*/
                "@min":  1,
                /*ro, opt, int, minimum value*/
                "@max":  1
                /*ro, opt, int, maximum value*/
            },
            "searchResultPosition": {
            /*ro, req, object, the start position of the search result in the result list, desc:the start position of search result in the result list. In a single search,if you cannot get all the records in the result list,you can mark the end position and get the following records after the marked position in the next search. If the maximum number of totalMatches supported by the device is M and the number of totalMatches stored in the device now is N (N<=M),the valid range of this node is 0 to N-1*/
                "@min":  1,
                /*ro, opt, int, minimum value*/
                "@max":  1
                /*ro, opt, int, maximum value*/
            },
            "maxResults": {
            /*ro, req, object, the maximum number of search results that can be obtained by calling the URI this time, desc:if maxResults exceeds the range returned by the device capability, the device will return the maximum number of search results according to the device capability and will not return error message*/
                "@min":  1,
                /*ro, opt, int, minimum value*/
                "@max":  1
                /*ro, opt, int, maximum value*/
            },
            "startTime":  "1970-01-01T00:00:00+08:00",
            /*ro, opt, datetime, start time (UTC time)*/
            "endTime":  "1970-01-01T00:00:00+08:00",
            /*ro, opt, datetime, end time (UTC time)*/
            "picEnable": {
            /*ro, opt, object, whether to upload the picture along with the event information, desc:false (not upload the picture along with the event information), true (upload the picture along with the event information). false (all matched events will be uploaded without pictures) true (all matched events will be uploaded with pictures if there are any) If this node is not configured,the default value is true*/
                "@opt": [true, false]
                /*ro, opt, array, optional, subType:bool*/
            },
            "beginSerialNo": {
            /*ro, opt, object, start serial No.*/
                "@min":  1,
                /*ro, opt, int, minimum value*/
                "@max":  1
                /*ro, opt, int, maximum value*/
            },
            "endSerialNo": {
            /*ro, opt, object, end serial No.*/
                "@min":  1,
                /*ro, opt, int, minimum value*/
                "@max":  1
                /*ro, opt, int, maximum value*/
            }
        },
    }
}
```

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14.6.13.2 Search QR code scanning events

Request URL

POST /ISAPI/AccessControl/QRCodeEvent?format=json

Query Parameter

None

Request Message

```
{
    "QRCodeEventCond": {
    /*req, object*/
        "searchID":  "test",
        /*req, string, search ID, desc:it is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device to speed up the next search*/
        "searchResultPosition":  0,
        /*req, int, the start position of the search result in the result list, desc:the start position of the search result in the result list*/
        "maxResults":  30,
        /*req, int, the maximum number of search results that can be obtained by calling the URI this time, desc:if maxResults exceeds the range returned by the device capability, the device will return the maximum number of search results according to the device capability and will not return error message*/
        "startTime":  "1970-01-01T00:00:00+08:00",
        /*opt, datetime, start time (UTC time)*/
        "endTime":  "1970-01-01T00:00:00+08:00",
        /*opt, datetime, end time (UTC time)*/
        "picEnable":  true,
        /*opt, bool, whether to upload the picture along with the event information*/
        "beginSerialNo":  1,
        /*opt, int, start serial No.*/
        "endSerialNo":  1
        /*opt, int, end serial No.*/
    }
}
```

Response Message

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```
{
    "QRCodeEvent": {
    /*ro, req, object*/
        "searchID":  "test",
        /*ro, req, string, search ID, desc:it is used to check whether the current search requester is the same as the previous one. If they are the same,
the search record will be stored in the device to speed up the next search*/
        "responseStatusStrg":  "OK",
        /*ro, req, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no
matched data).*/
        "numOfMatches":  1,
        /*ro, req, int, number of results returned this time*/
        "totalMatches":  1,
        /*ro, req, int, total number of matched results*/
        "InfoList": [
        /*ro, opt, array, subType:object*/
            {
                "deviceName":  "test",
                /*ro, opt, string, device name*/
                "serialNo":  1,
                /*ro, opt, int, event serial No.*/
                "QRCodeInfo":  "test",
                /*ro, req, string, QR code information*/
                "thermometryUnit":  "celsius",
                /*ro, opt, enum, temperature unit, subType:string, desc:"celsius" (Celsius), "fahrenheit" (Fahrenheit), "kelvin" (Kelvin), the default value
is "celsius"*/
                "currTemperature":  1.0,
                /*ro, opt, float, face temperature, desc:it should be accurate to one decimal place*/
                "isAbnomalTemperature":  true,
                /*ro, opt, bool, whether the face temperature is abnormal, desc:true (yes), false (no)*/
                "RegionCoordinates": {
                /*ro, opt, object, face temperature's coordinates*/
                    "positionX":  1,
                    /*ro, opt, int, X-coordinate, range:[0,1000], desc:the value is normalized to a number between 0 and 1000*/
                    "positionY":  1
                    /*ro, opt, int, Y-coordinate, range:[0,1000], desc:the value is normalized to a number between 0 and 1000*/
                },
                "mask":  "unknown",
                /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"unknown" (unknown), "yes" (wearing a mask), "no" (no mask)*/
                "visibleLightPicUrl":  "test",
                /*ro, opt, string, URL of the visible light picture*/
                "thermalPicUrl":  "test",
                /*ro, opt, string, URL of the thermal picture*/
                "helmet":  "unknown",
                /*ro, opt, enum, whether the person is wearing hard hat, subType:string, desc:"unknown", "yes", "no"*/
                "HealthInfo": {
                /*ro, opt, object*/
                    "healthCode":  1,
                    /*ro, opt, enum, subType:int*/
                    "NADCode":  1,
                    /*ro, opt, enum, subType:int*/
                    "travelCode":  1,
                    /*ro, opt, enum, subType:int*/
                    "travelInfo":  "test",
                    /*ro, opt, string*/
                    "vaccineStatus":  1
                    /*ro, opt, enum, subType:int*/
                },
                "dateTime":  "1970-01-01T00:00:00+08:00"
                /*ro, req, datetime, the time (UTC time) when the alarm is triggered, desc:the maximum size is 32*/
            }
        ]
    }
}
```

14.6.13.3 QR code event of access control

EventType:QRCodeEvent

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The
value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video
channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI
protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:number of times that the same alarm has been uploaded*/
```

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```
    "eventType":  "QRCodeEvent",
    /*ro, req, string, event type, desc:"QRCodeEvent" (QR code event of access control)*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event
ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "QR Code Event",
    /*ro, req, string, event description, desc:event description*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID (PUID), desc:it should be returned in ISUP alarm*/
    "QRCodeEvent": {
    /*ro, opt, object, QR code event of access control*/
        "deviceName":  "test",
        /*ro, opt, string, device name*/
        "serialNo":  "test",
        /*ro, opt, string, event serial No.*/
        "currentEvent":  true,
        /*ro, opt, bool, whether it is a real-time event*/
        "QRCodeInfo":  "test",
        /*ro, req, string, QR code information*/
        "thermometryUnit":  "celsius",
        /*ro, opt, enum, temperature unit, subType:string, desc:"celsius" (Celsius, default value), "fahrenheit" (Fahrenheit), "kelvin" (Kelvin)*/
        "currTemperature":  1.0,
        /*ro, opt, float, face temperature, desc:it should be accurate to one decimal place*/
        "isAbnomalTemperature":  true,
        /*ro, opt, bool, whether the face temperature is abnormal*/
        "RegionCoordinates": {
        /*ro, opt, object, face temperature's coordinates*/
            "positionX":  1,
            /*ro, opt, int, X-coordinate, range:[0,1000]*/
            "positionY":  1
            /*ro, opt, int, Y-coordinate, range:[0,1000]*/
        },
        "remoteCheck":  true,
        /*ro, opt, bool, whether remote verification is required, desc:it is not required by default*/
        "mask":  "test",
        /*ro, opt, string, whether the person wears a mask*/
        "visibleLightURL":  "test",
        /*ro, opt, string, visible light picture URL of the thermal imaging camera*/
        "thermalURL":  "test",
        /*ro, opt, string, thermal imaging picture URL*/
        "helmet":  "unknown",
        /*ro, opt, enum, whether the person is wearing hard hat, subType:string, desc:"unknown", "yes", "no"*/
        "picturesNumber":  1,
        /*ro, opt, int, number of pictures, range:[0,2], step:1*/
        "HealthInfo": {
        /*ro, opt, object, health information*/
            "healthCode":  1,
            /*ro, opt, enum, health code status, subType:int, desc:0 (no request), 1 (no health code), 2 (green QR code), 3 (yellow QR code), 4 (red QR
code), 5 (no such person), 6 (other error, e.g., searching failed due to API exception), 7 (searching for the health code timed out)*/
            "NADCode":  1,
            /*ro, opt, enum, nucleic acid test result, subType:int, desc:0 (no result), 1 (negative, which means normal), 2 (positive, which means
diagnosed), 3 (the result has expired)*/
            "NADMsg":  "test",
            /*ro, opt, string, range:[0,64]*/
            "travelCode":  1,
            /*ro, opt, enum, trip code, subType:int, desc:0 (no trip in the past 14 days), 1 (once left in the past 14 days), 2 (has been to the high-risk
area in the past 14 days), 3 (other)*/
            "travelInfo":  "test",
            /*ro, opt, string, trip information, desc:the empty string indicates that searching trip failed*/
            "vaccineStatus":  1,
            /*ro, opt, enum, whether the person is vaccinated, subType:int, desc:0 (not vaccinated), 1 (vaccinated)*/
            "vaccineNum":  1,
            /*ro, opt, int, step:1*/
            "vaccineMsg":  "test",
            /*ro, opt, string, range:[0,64]*/
            "ANTCode":  1,
            /*ro, opt, enum, subType:int*/
            "ANTMsg":  "test"
            /*ro, opt, string, range:[0,64]*/
        },
        "employeeNo":  "test",
        /*ro, opt, string, range:[1,32]*/
        "name":  "test",
        /*ro, opt, string, range:[1,64]*/
        "IDNum":  "test"
        /*ro, opt, string, range:[1,128]*/
    }
}
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
QRCodeEvent	[Message content]	application/json	--	--	--
VisibleLight	[Binary picture data]	image/jpeg	visibleLight_image	VisibleLight.jpg	--
Thermal	[Binary picture data]	image/jpeg	thermal_image	Thermal.jpg	--
Picture	[Binary picture data]	image/jpeg		Picture.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

14.6.14 Remote Door Control

14.6.14.1 Remotely control the door or elevator

Request URL

PUT /ISAPI/AccessControl/RemoteControl/door/<doorID>

Query Parameter

Parameter Name	Parameter Type	Description
doorID	string	Door No., and 65535 represents all doors.

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<RemoteControlDoor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <cmd>
    <!--wo, req, enum, command, subType:string, desc:"open" (open the door), "close" (close the door (controlled)), "alwaysOpen" (remain unlocked (free)),
"alwaysClose" (remain open (disabled)), "visitorCallLadder" (call elevator (visitor)), "householdCallLadder" (call elevator (resident))-->open
  </cmd>
  <password>
    <!--wo, opt, string, range:[8,16]-->test
  </password>
  <employeeNo>
    <!--wo, opt, string, employee No., range:[0,32], desc:employee No.-->test
  </employeeNo>
  <channelNo>
    <!--wo, opt, int, range:[0,256], step:1-->1
  </channelNo>
  <controlType>
    <!--wo, opt, enum, subType:string-->monitor
  </controlType>
  <personnelChannelGroupInfoList>
    <!--wo, opt, array, subType:object-->
    <personnelChannelGroupInfo>
      <!--wo, req, array, subType:object-->
      <personnelChannelGroupID>
        <!--wo, req, int, range:[1,6]-->1
      </personnelChannelGroupID>
      <personnelChannelInfoList>
        <!--wo, opt, array, subType:object-->
        <personnelChannelInfo>
          <!--wo, req, object-->
          <personnelChannelID>
            <!--wo, req, int, range:[1,16]-->1
          </personnelChannelID>
        </personnelChannelInfo>
      </personnelChannelInfoList>
    </personnelChannelGroupInfo>
  </personnelChannelGroupInfoList>
  <lastOpenDoorFlag>
    <!--opt, bool-->true
  </lastOpenDoorFlag>
  <callNumberList>
    <!--opt, object, dep:or,{$.RemoteControlDoor.cmd,eq,open},{$.RemoteControlDoor.cmd,eq,householdCallLadder}-->
    <callNumber>
      <!--req, int-->101
    </callNumber>
  </callNumberList>
  <callElevatorType>
    <!--opt, enum, subType:string, dep:or,{$.RemoteControlDoor.cmd,eq,householdCallLadder}-->up
  </callElevatorType>
  <throughDoorID>
    <!--opt, enum, subType:int-->1
  </throughDoorID>
</RemoteControlDoor>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.6.14.2 Get the capability set of remote door status control

Request URL

GET /ISAPI/AccessControl/RemoteControl/door/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RemoteControlDoor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, remote door status control, attr:version{req, float, protocolVersion}-->
  <doorNo min="1" max="10">
    <!--ro, opt, int, range of the door No., attr:min{req, int},max{req, int}-->1
  </doorNo>
  <cmd opt="open,close,alwaysOpen,alwaysClose,visitorCallLadder,householdCallLadder,resume">
    <!--ro, req, enum, command, subType:string, attr:opt{req, string}, desc:"open" (open the door), "close" (close the door (controlled)), "alwaysOpen"
(remain unlocked (free)), "alwaysClose" (remain locked (disabled)), "visitorCallLadder" (call elevator (visitor)), "householdCallLadder" (call elevator
(resident))-->open
  </cmd>
  <password min="8" max="16">
    <!--ro, opt, string, door opening password, range:[8,16], attr:min{req, int, range:[8,16]},max{req, int, range:[8,16]}-->test
  </password>
  <employeeNo min="0" max="32">
    <!--ro, opt, string, employee No., range:[0,32], attr:min{req, int, range:[0,32]},max{req, int, range:[0,32]}-->test
  </employeeNo>
  <channelNo min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </channelNo>
  <controlType opt="monitor,calling">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->monitor
  </controlType>
</RemoteControlDoor>
```

14.6.15 Remotely Verification

14.6.15.1 Get the remote verification capability of access control

Request URL

GET /ISAPI/AccessControl/remoteCheck/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "RemoteCheck": {
    /*ro, req, object*/
        "serialNo": {
        /*ro, opt, object, event serial No., desc:it should be the same as that in the uploaded event message*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  2000000000
            /*ro, opt, int, the maximum value*/
        },
        "checkResult": {
        /*ro, opt, object, verification result, desc:"success"-verified, "failed"-verification failed*/
            "@opt": ["success", "failed"]
            /*ro, opt, array, options, subType:string*/
        },
        "info": {
        /*ro, opt, object, additional information*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  32
            /*ro, opt, int, the maximum value*/
        }
    }
}
```

14.6.15.2 Verify the access control event remotely to control opening or closing the door

Request URL

PUT /ISAPI/AccessControl/remoteCheck?format=json

Query Parameter

None

Request Message

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```
{
    "RemoteCheck": {
    /*req, object*/
        "serialNo":  1,
        /*req, int, event serial No., desc:it should be the same as that in the uploaded event message*/
        "checkResult":  "success",
        /*req, enum, verification result, subType:string, desc:"success"-verified, "failed"-verification failed*/
        "info":  "test",
        /*opt, string, additional information, range:[1,1024], desc:additional information*/
        "customFileID":  1,
        /*opt, int, range:[1,32]*/
        "successCustomInfo": {
        /*opt, object, dep:or,{$.RemoteCheck.checkResult,eq,success}*/
            "boardingGateInfo": {
            /*opt, object*/
                "flightNum":  "test",
                /*req, string, range:[1,32]*/
                "seatNum":  "25A",
                /*req, string, range:[1,16]*/
                "departureTime":  "2020-01-01T10:00:00+08:00",
                /*req, datetime*/
                "departureCode":  "sss",
                /*req, string, range:[1,16]*/
                "arrivalTime":  "2020-01-01T12:00:00+08:00",
                /*req, datetime*/
                "arrivalCode":  "sss"
                /*req, string, range:[1,16]*/
            },
            "subwayInfo": {
            /*opt, object*/
                "consumeMode":  "amount",
                /*req, enum, subType:string*/
                "amountInfo": {
                /*opt, object, dep:and,{$.RemoteCheck.successCustomInfo.subwayInfo.consumeMode,eq,amount}*/
                    "currencyType":  "USD",
                    /*req, enum, subType:string*/
                    "endTime":  "2024-12-31",
                    /*req, date*/
                    "balance":  "11",
                    /*req, string, range:[0,6]*/
                    "actualPayment":  "1"
                    /*req, string, range:[0,6]*/
                },
                "countInfo": {
                /*opt, object, dep:and,{$.RemoteCheck.successCustomInfo.subwayInfo.consumeMode,eq,count}*/
                    "endTime":  "2024-12-31",
                    /*req, date*/
                    "remainingTimes":  9
                    /*req, int, range:[0,999999]*/
                }
            }
        },
        "userInfo": {
        /*opt, object*/
            "name":  "test",
            /*opt, string, range:[1,64]*/
            "employeeNo":  "test",
            /*opt, string, range:[1,32]*/
            "cardNo":  "test",
            /*opt, string, range:[1,32]*/
            "filePathType":  "URL",
            /*opt, enum, subType:string*/
            "filePath":  "test"
            /*opt, string, range:[1,10240]*/
        }
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

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14.7 Credentials Collection

14.7.1 Card Online Adding

14.7.1.1 Get the capability of collecting card information

Request URL

GET /ISAPI/AccessControl/CaptureCardInfo/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CardInfoCap": {
    /*ro, req, object*/
        "cardNo": {
        /*ro, opt, object, card No.*/
            "@min":  1,
            /*ro, req, int, the minimum length, range:[1,32]*/
            "@max":  32
            /*ro, req, int, the maximum length, range:[1,32]*/
        },
        "cardType": ["TypeA_M1", "TypeA_CPU", "TypeB", "ID_125K", "FelicaCard", "DesfireCard"],
        /*ro, opt, array, card type, subType:string, range:[1,6]*/
        "readerID": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  8
            /*ro, req, int*/
        }
    }
}
```

14.7.1.2 Collect card information by the card reading module of the device

Request URL

GET /ISAPI/AccessControl/CaptureCardInfo?format=json&readerID=<readerID>&timeout=<timeout>&biometricKeyMode=<biometricKeyMode>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	--
timeout	string	--
biometricKeyMode	enum	--

Request Message

None

Response Message

```
{
    "CardInfo": {
    /*ro, req, object, card information*/
        "cardNo":  "abcd1234",
        /*ro, req, string, card No.*/
        "cardType":  "TypeA_M1",
        /*ro, opt, enum, card type, subType:string, desc:"TypeA_M1", "TypeA_CPU", "TypeB", "ID_125K", "FelicaCard" (FeliCa card), "DesfireCard" (DESFire card)*/
        "readerID":  1
        /*ro, opt, int, range:[1,8]*/
    }
}
```

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14.7.2 Face Picture Online Adding

14.7.2.1 Collect face picture information

Request URL

POST /ISAPI/AccessControl/CaptureFaceData

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFaceDataCond xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <captureInfrared>
    <!--opt, bool, whether to collect infrared face pictures simultaneously, desc:"true"-yes, "false"-no-->true
  </captureInfrared>
  <dataType>
    <!--opt, enum, data type of collected face pictures, subType:string, desc:"url" (default), "binary”-->url
  </dataType>
  <readerID>
    <!--opt, int, range:[1,8]-->1
  </readerID>
</CaptureFaceDataCond>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFaceData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <CaptureFaceDataCond>
    <!--ro, opt, object-->
    <captureInfrared opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </captureInfrared>
    <dataType opt="url,binary">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->url
    </dataType>
  </CaptureFaceDataCond>
  <faceDataUrl min="1" max="768">
    <!--ro, opt, string, face data URL, if this node does not exist, it indicates that there is no face data, attr:min{req, int},max{req, int}-->test
  </faceDataUrl>
  <captureProgress min="1" max="10">
    <!--ro, opt, int, collection progress, range:[0,100], attr:min{req, int},max{req, int}-->1
  </captureProgress>
  <infraredFaceDataUrl min="1" max="100">
    <!--ro, opt, string, infrared face data URL, if this node does not exist, it indicates that there is no infrared face data, attr:min{req, int},max{req,
int}-->test
  </infraredFaceDataUrl>
  <modelData>
    <!--ro, opt, string-->test
  </modelData>
  <score>
    <!--ro, opt, int, face score (face picture quality), range:[0,100]-->0
  </score>
  <thermometryUnit>
    <!--ro, opt, string-->test
  </thermometryUnit>
  <currTemperature>
    <!--ro, opt, float-->0.0
  </currTemperature>
  <visibleLightURL>
    <!--ro, opt, string-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
</CaptureFaceData>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
CaptureFace	[Message content]	application/xml	--	--	--
FaceData	[Binary picture data]	image/jpeg		FaceDatac	--
InfraredFaceData	[Binary picture data]	image/jpeg		InfraredFaceData.jpg	--
faceMatting	[Binary picture data]	image/jpeg		faceMatting.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

14.7.2.2 Get the capability of collecting face picture information.

Request URL

GET /ISAPI/AccessControl/CaptureFaceData/capabilities

Query Parameter

None

Request Message

None

Response Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFaceData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, capability of collecting face picture information, attr:version{req, string, protocolVersion}-->
  <CaptureFaceDataCond>
    <!--ro, req, bool, whether to collect the infrared face data-->true
    <captureInfrared opt="true,false">
      <!--ro, opt, bool, whether to collect the infrared face data, attr:opt{req, string}-->true
    </captureInfrared>
    <dataType opt="url,binary">
      <!--ro, opt, enum, data type of collected face pictures, subType:string, attr:opt{req, string}, desc:url (URL, default),binary (binary)-->url
    </dataType>
  </CaptureFaceDataCond>
  <faceDataUrl min="1" max="768">
    <!--ro, opt, string, face data URL, range:[1,768], attr:min{req, int},max{req, int}-->1
  </faceDataUrl>
  <captureProgress min="1" max="10">
    <!--ro, req, int, collection progress, range:[1,10], attr:min{req, int},max{req, int}-->1
  </captureProgress>
  <infraredFaceDataUrl min="1" max="100">
    <!--ro, req, string, infrared picture URL, range:[1,100], attr:min{req, int},max{req, int}-->test
  </infraredFaceDataUrl>
  <modelData min="1" max="10">
    <!--ro, opt, string, face modeling data encoded by Base64, attr:min{req, int},max{req, int}-->test
  </modelData>
  <score min="0" max="100">
    <!--ro, opt, int, face score, range:[0,100], attr:min{req, int},max{req, int}-->80
  </score>
  <thermometryUnit opt="celsius,fahrenheit,kelvin">
    <!--ro, opt, string, temperature unit: celsius (Celsius, default), attr:opt{req, string}-->test
  </thermometryUnit>
  <currTemperature min="0" max="10">
    <!--ro, opt, float, face temperature which is accurate to one decimal place, attr:min{req, int},max{req, int}-->0.000
  </currTemperature>
  <visibleLightURL min="0" max="10">
    <!--ro, opt, string, URL of the visible light picture captured by the thermal camera, attr:min{req, int},max{req, int}-->test
  </visibleLightURL>
  <thermalURL min="0" max="10">
    <!--ro, opt, string, thermal picture URL, attr:min{req, int},max{req, int}-->test
  </thermalURL>
  <readerID min="1" max="8">
    <!--ro, opt, int, range:[1,8], attr:min{req, int},max{req, int}-->1
  </readerID>
</CaptureFaceData>
```

14.7.2.3 Get capability of getting face picture collection progress

Request URL

GET /ISAPI/AccessControl/CaptureFaceData/Progress/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFaceData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <faceDataUrl min="0" max="10">
    <!--ro, opt, string, face data URL, range:[1,32], attr:min{req, int},max{req, int}, desc:face data URL,if this node does not exist,it indicates that there is no face data-->test
  </faceDataUrl>
  <captureProgress min="0" max="100">
    <!--ro, req, int, collection progress, range:[0,100], attr:min{req, int},max{req, int}, desc:collection progress,which is between 0 and 100,0-no face data collected,100-collected,the face data URL can be parsed only when the progress is 100-->1
  </captureProgress>
  <isCurRequestOver opt="true,false">
    <!--ro, opt, bool, whether the current collection request is completed, attr:opt{req, string}-->true
  </isCurRequestOver>
  <infraredFaceDataUrl min="0" max="10">
    <!--ro, opt, string, infrared face data URL, range:[1,32], attr:min{req, int},max{req, int}, desc:if this node does not exist, it indicates that there is no infrared face data-->test
  </infraredFaceDataUrl>
  <readerID min="1" max="8">
    <!--ro, opt, int, range:[1,8], attr:min{req, int},max{req, int}-->1
  </readerID>
  <requireReaderID>
    <!--ro, opt, bool-->true
  </requireReaderID>
</CaptureFaceData>
```

14.7.2.4 Get the progress of collecting face picture information

Request URL

GET /ISAPI/AccessControl/CaptureFaceData/Progress?readerID=<readerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	The intelligent host supports capturing faces by card reader.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFaceData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <faceDataUrl>
    <!--ro, opt, string, face data URL, range:[1,32]-->test
  </faceDataUrl>
  <captureProgress>
    <!--ro, req, int, collection progress, range:[0,100], desc:collection progress,which is between 0 and 100,0-no face data collected,100-collected,the face data URL can be parsed only when the progress is 100-->100
  </captureProgress>
  <isCurRequestOver>
    <!--ro, opt, bool, whether the current collection request is completed, desc:whether the current collection request is completed: "true"-yes,"false"-no-->true
  </isCurRequestOver>
  <infraredFaceDataUrl>
    <!--ro, opt, string, infrared face data URL, range:[1,32], desc:if this node does not exist, it indicates that there is no infrared face data-->test
  </infraredFaceDataUrl>
  <faceMattingURL>
    <!--ro, opt, string, range:[1,32]-->test
  </faceMattingURL>
</CaptureFaceData>
```

14.7.3 Fingerprint Online Adding

14.7.3.1 Collect fingerprint information

Request URL

POST /ISAPI/AccessControl/CaptureFingerPrint

Query Parameter

None

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Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFingerPrintCond xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, Collect fingerprint information conditions, attr:version{req, string, protocolVersion}-->
  <fingerNo>
    <!--req, int, finger No., range:[1,10]-->1
  </fingerNo>
</CaptureFingerPrintCond>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFingerPrint xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <fingerData>
    <!--ro, opt, string, fingerprint data, range:[1,768], desc:which, should be encoded by Base64-->test
  </fingerData>
  <fingerNo>
    <!--ro, req, int, finger No., range:[1,10]-->1
  </fingerNo>
  <fingerPrintQuality>
    <!--ro, req, int, fingerprint quality, range:[1,100]-->1
  </fingerPrintQuality>
</CaptureFingerPrint>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
CaptureFingerPrint	[Message content]	application/xml	--	--	--
fingerPrintPic	[Binary picture data]	image/jpeg		fingerPrintPic.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

14.7.3.2 Get the fingerprint collection capability

Request URL

GET /ISAPI/AccessControl/CaptureFingerPrint/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFingerPrint xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, collect fingerprint information, attr:version{req, string, protocolVersion}-->
  <CaptureFingerPrintCond>
    <!--ro, req, object, finger No.-->
    <fingerNo min="1" max="10">
      <!--ro, opt, int, fingerprint No., range:[1,10], attr:min{req, int},max{req, int}-->1
    </fingerNo>
  </CaptureFingerPrintCond>
  <fingerData min="1" max="768">
    <!--ro, opt, string, fingerprint data, range:[1,768], attr:min{req, int},max{req, int}-->test
  </fingerData>
  <fingerNo min="1" max="10">
    <!--ro, opt, int, fingerprint No., range:[1,10], attr:min{req, int},max{req, int}-->1
  </fingerNo>
  <fingerPrintQuality min="1" max="100">
    <!--ro, opt, int, fingerprint quality, range:[1,100], attr:min{req, int},max{req, int}-->1
  </fingerPrintQuality>
</CaptureFingerPrint>
```

14.8 Attendance Check

14.8.1 Local Attendance Basic Configuration

14.8.1.1 Get the capability of configuring the rule parameters of local time and attendance

Request URL

GET /ISAPI/AccessControl/LocalAttendance/rule/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "RuleCap": {
    /*ro, opt, object*/
        "NormalShift": {
        /*ro, opt, object, time and attendance rule of the normal shift*/
            "earliestSignInTime": {
            /*ro, opt, object, time available for early check-in*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  1
                /*ro, req, int, unit:minutes*/
            },
            "latestSignInTime": {
            /*ro, opt, object, time available for late check-in*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  1
                /*ro, req, int, unit:minutes*/
            },
            "allowLateTime": {
            /*ro, opt, object, the maximum time of late arrival allowed*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  1
                /*ro, req, int, unit:minutes*/
            },
            "earliestSignOutTime": {
            /*ro, opt, object, time available for early check-out*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  1
                /*ro, req, int, unit:minutes*/
            },
            "latestSignOutTime": {
            /*ro, opt, object, time available for late check-out*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  1
                /*ro, req, int, unit:minutes*/
            },
            "allowLeaveEarlyTime": {
            /*ro, opt, object, the maximum time of early leave allowed*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  1
                /*ro, req, int, unit:minutes*/
            }
        },
        "supportedMethods": ["get", "put"]
        /*ro, req, array, array of string,supported operations on /ISAPI/AccessControl/LocalAttendance/rule?format=json, subType:string*/
    }
}
```

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14.8.1.2 Get the rule parameters of local time and attendance

Request URL

GET /ISAPI/AccessControl/LocalAttendance/rule?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "Rule": {
    /*ro, opt, object, rule information*/
        "NormalShift": {
        /*ro, opt, object, time and attendance rule of the normal shift, desc:cross-day attendance is not supported. The valid period of the latest check-out time is between 0 and 1440 minutes. Combined with the normal shift period, if the latest check-out time is after 24:00, the attendance record will not be saved*/
            "earliestSignInTime":  1,
            /*ro, opt, int, time available for early check-in, unit:minutes*/
            "latestSignInTime":  1,
            /*ro, opt, int, time available for late check-in, unit:minutes*/
            "allowLateTime":  1,
            /*ro, opt, int, the maximum time of late arrival allowed, unit:minutes*/
            "earliestSignOutTime":  1,
            /*ro, opt, int, time available for early check-out, unit:minutes*/
            "latestSignOutTime":  1,
            /*ro, opt, int, time available for late check-out, unit:minutes*/
            "allowLeaveEarlyTime":  1
            /*ro, opt, int, the maximum time of early leave allowed, unit:minutes*/
        }
    }
}
```

14.8.1.3 Set the rule parameters of local time and attendance

Request URL

PUT /ISAPI/AccessControl/LocalAttendance/rule?format=json

Query Parameter

None

Request Message

```
{
    "Rule": {
    /*opt, object, rule information*/
        "NormalShift": {
        /*opt, object, attendance rule of the normal shift, desc:the cross-day attendance is not supported. The valid period of latest check-out time is between 0 and 1440 minutes. Combined with the normal shift period, if the latest check-out time is after 24:00, the attendance record will not be saved*/
            "earliestSignInTime":  1,
            /*opt, int, advanced check-in time, unit:minutes*/
            "latestSignInTime":  1,
            /*opt, int, latest check-in time, unit:minutes*/
            "allowLateTime":  1,
            /*opt, int, late time (duration after the latest check-in time), unit:minutes*/
            "earliestSignOutTime":  1,
            /*opt, int, earliest check-out time, unit:minutes*/
            "latestSignOutTime":  1,
            /*opt, int, latest check-out time, unit:minutes*/
            "allowLeaveEarlyTime":  1
            /*opt, int, absence time (duration before the earliest check-out time), unit:minutes*/
        }
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

14.8.2 Local Attendance Schedule Management

14.8.2.1 Get the capability of attendance weekly schedule parameters

Request URL

GET /ISAPI/AccessControl/LocalAttendance/weekPlan/capabilities?format=json

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Query Parameter

None

Request Message

None

Response Message

```
{
    "WeekPlanCap": {
    /*ro, req, object*/
        "WeekPlanId": {
        /*ro, opt, object, weekly schedule ID of local time and attendance*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  1
            /*ro, req, int*/
        },
        "weekPlanName": {
        /*ro, opt, object, weekly schedule name of local time and attendance*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  1
            /*ro, req, int*/
        },
        "WeekPlanCfg": {
        /*ro, opt, object, weekly schedule*/
            "@size":  7,
            /*ro, req, int*/
            "week": {
            /*ro, opt, object, day of the week*/
                "@opt": ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"]
                /*ro, req, array, subType:string*/
            },
            "shiftId": {
            /*ro, opt, object, shift No.*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  1
                /*ro, req, int*/
            }
        },
        "supportedMethods": ["post", "put", "delete"]
        /*ro, req, array, operation type supported by the device, "post" (add), "put" (edit), "delete", subType:string*/
    }
}
```

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14.9 Elevator Control

14.9.1 Elevator Parameters

14.9.1.1 Get the capability of elevator control parameters

Request URL

GET /ISAPI/VideoIntercom/Elevators/<elevatorID>/ControlCfg/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
elevatorID	string	--

Request Message

None

Response Message

```
{
    "ElevatorControlCfg": {
    /*ro, req, object*/
        "elevatorNo": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  8
            /*ro, req, int*/
        },
        "enable": {
```

```
        /*ro, opt, object*/
            "@opt":  "true,false"
            /*ro, req, string*/
        },
        "numOfNegFloors": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int*/
            "@max":  10
            /*ro, req, int*/
        },
        "interfaceType": {
        /*ro, opt, object*/
            "@opt": [1, 2]
            /*ro, req, array, subType:int*/
        },
        "protocolType": {
        /*ro, opt, object*/
            "@opt": ["private", "ISAPI"]
            /*ro, req, array, subType:string*/
        },
        "transportProtocol": {
        /*ro, opt, object*/
            "@opt": ["HTTP", "HTTPS"]
            /*ro, req, array, subType:string*/
        },
        "deviceType": {
        /*ro, opt, object*/
            "@opt": ["DS-K2201", "DS-K2210", "DS-K2220", "custom"]
            /*ro, req, array, subType:string*/
        },
        "ServerAddress": {
        /*ro, opt, object*/
            "addressingFormatType": {
            /*ro, opt, object*/
                "@opt": ["ipaddress", "hostname"]
                /*ro, req, array, subType:string*/
            },
            "hostName": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  128
                /*ro, req, int*/
            },
            "ipAddress": {
            /*ro, opt, object*/
                "@min":  7,
                /*ro, req, int*/
                "@max":  15
                /*ro, req, int*/
            },
            "ipv6Address": {
            /*ro, opt, object*/
                "@min":  15,
                /*ro, req, int*/
                "@max":  39
                /*ro, req, int*/
            }
        },
        "serverPort": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int*/
            "@max":  65535
            /*ro, req, int*/
        },
        "userName": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  64
            /*ro, req, int*/
        },
        "password": {
        /*ro, opt, object*/
            "@min":  8,
            /*ro, req, int*/
            "@max":  16
            /*ro, req, int*/
        },
        "isSupportRS485": {
        /*ro, opt, object*/
            "@opt": ["DS-K2201", "DS-K2210", "DS-K2220", "custom"]
            /*ro, req, array, subType:string*/
        },
        "isSupportNetWork": {
        /*ro, opt, object*/
            "@opt": ["DS-K2201", "DS-K2210", "DS-K2220", "custom"]
            /*ro, req, array, subType:string*/
        },
        "isSupportNegativeFloor": {
        /*ro, opt, object*/
```

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```
            "@opt": ["DS-K2201", "DS-K2210", "DS-K2220", "custom"]
            /*ro, req, array, subType:string*/
        }
    }
}
```

14.9.1.2 Get the elevator control parameters

Request URL

GET /ISAPI/VideoIntercom/Elevators/<elevatorID>/ControlCfg?format=json

Query Parameter

Parameter Name	Parameter Type	Description
elevatorID	string	It starts from 1.

Request Message

None

Response Message

```
{
    "ElevatorControlCfg": {
    /*ro, req, object*/
        "enable": true,
        /*ro, opt, bool*/
        "numOfNegFloors": 1,
        /*ro, opt, int*/
        "interfaceType": 1,
        /*ro, opt, enum, subType:int*/
        "protocolType": "ISAPI",
        /*ro, opt, enum, subType:string, dep:or,{$.ElevatorControlCfg.interfaceType,eq,2}*/
        "transportProtocol": "HTTP",
        /*ro, opt, enum, subType:string, dep:or,{$.ElevatorControlCfg.protocolType,eq,ISAPI}*/
        "deviceType": "DS-K2201",
        /*ro, opt, enum, subType:string*/
        "ServerAddress": {
        /*ro, opt, object, dep:and,{$.ElevatorControlCfg.interfaceType,eq,2}*/
            "addressingFormatType": "ipaddress",
            /*ro, req, enum, subType:string*/
            "hostName": "test",
            /*ro, opt, string, dep:and,{$.ElevatorControlCfg.ServerAddress.addressingFormatType,eq,hostname}*/
            "ipAddress": "test",
            /*ro, opt, string, dep:and,{$.ElevatorControlCfg.ServerAddress.ipAddress,eq,ipaddress}*/
            "ipv6Address": "test"
            /*ro, opt, string, dep:or,{$.ElevatorControlCfg.ServerAddress.addressingFormatType,eq,ipaddress}*/
        },
        "serverPort": 0,
        /*ro, opt, int, dep:and,{$.ElevatorControlCfg.interfaceType,eq,2}*/
        "userName": "test",
        /*ro, opt, string, dep:and,{$.ElevatorControlCfg.interfaceType,eq,2}*/
        "password": "test"
        /*ro, opt, string, dep:and,{$.ElevatorControlCfg.interfaceType,eq,2}*/
    }
}
```

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14.9.1.3 Set the elevator control parameters

Request URL

PUT /ISAPI/VideoIntercom/Elevators/<elevatorID>/ControlCfg?format=json

Query Parameter

Parameter Name	Parameter Type	Description
elevatorID	string	--

Request Message

```
{
    "ElevatorControlCfg": {
    /*opt, object*/
        "enable":  "true",
        /*opt, string*/
        "numOfNegFloors":  1,
        /*opt, int*/
        "interfaceType":  1,
        /*opt, enum, subType:int*/
        "deviceType":  "DS-K2201",
        /*opt, enum, subType:string*/
        "ServerAddress": {
        /*opt, object*/
            "addressingFormatType":  "ipaddress",
            /*req, enum, subType:string*/
            "hostName":  "test",
            /*opt, string*/
            "ipAddress":  "test",
            /*opt, string*/
            "ipv6Address":  "test"
            /*opt, string*/
        },
        "serverPort":  0,
        /*opt, int*/
        "userName":  "test",
        /*opt, string*/
        "password":  "test"
        /*opt, string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

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14.9.2 Elevator Remote Control

14.9.2.1 Get the capability of elevator remote calling

Request URL

GET /ISAPI/VideoIntercom/callElevator/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CallElevatorCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <floorNumber min="0" max="999">
    <!--ro, req, int, attr:min{req, int},max{req, int}-->0
  </floorNumber>
  <authorizedFloorList size="10">
    <!--ro, req, object, attr:size{req, int}-->
    <number min="0" max="999">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->0
    </number>
  </authorizedFloorList>
</CallElevatorCfg>
```

14.10 Video Intercom (General)

14.10.1 Call Interaction Management

14.10.1.1 Get the capability of getting the caller information

Request URL

GET /ISAPI/VideoIntercom/callerInfo/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CallerInfo": {
    /*ro, req, object*/
        "buildingNo": {
        /*ro, opt, object, building No.*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "floorNo": {
        /*ro, opt, object, floor No.*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "zoneNo": {
        /*ro, opt, object, project No.*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "unitNo": {
        /*ro, opt, object, unit No.*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "devNo": {
        /*ro, opt, object, device No.*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "devType": {
        /*ro, req, object, device type, desc:1-door station, 2-main station, 3-indoor station, 4-outer door station, 5-villa door station, 6-doorphone, 7-
Infosight Client Software, 8-iVMS-4200 Client Software, 9-APP, 10-doorbell, 11-VOIP Client Software, 12-network camera, 13-access control device*/
            "@opt":  "1,2,3,4,5,6,7,8,9,10,11,12,13,14"
            /*ro, opt, string, options*/
        },
        "lockNum": {
        /*ro, opt, object, number of locks*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "voipNo": {
        /*ro, opt, object, VOIP Client Software number, desc:this node is valid when devType is 11*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  16
            /*ro, opt, int, the maximum value*/
        },
        "status": {
        /*ro, req, object, status, desc:"idle", "ring", "onCall"-the call is in progress*/
            "@opt":  "idle,ring,onCall"
            /*ro, opt, string*/
        }
    }
}
```

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14.10.1.2 Get the caller information

Request URL

GET /ISAPI/VideoIntercom/callerInfo?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CallerInfo": {
    /*ro, req, object*/
        "buildingNo":  1,
        /*ro, opt, int, building No., range:[0,16]*/
        "floorNo":  1,
        /*ro, opt, int, floor No., range:[0,16]*/
        "zoneNo":  1,
        /*ro, opt, int, project No., range:[0,16]*/
        "unitNo":  1,
        /*ro, opt, int, unit No., range:[0,16]*/
        "devNo":  1,
        /*ro, opt, int, device No., range:[0,16]*/
        "devType":  1,
        /*ro, req, int, device type, desc:1-door station, 2-main station, 3-indoor station, 4-outer door station, 5-villa door station, 6-doorphone, 7-
Infosight Client Software, 8-iVMS-4200 Client Software, 9-APP, 10-doorbell, 11-VOIP Client Software, 12-network camera, 13-access control device*/
        "lockNum":  1,
        /*ro, opt, int, number of locks, range:[0,16]*/
        "voipNo":  "test",
        /*ro, opt, string, VOIP Client Software number, desc:this node is valid when devType is 11*/
        "status":  "idle"
        /*ro, req, string, status, desc:"idle", "ring", "onCall"-the call is in progress*/
    }
}
```

14.10.1.3 Get the capability of the call signaling interaction

Request URL

GET /ISAPI/VideoIntercom/callSignal/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CallSignal": {
    /*ro, req, object*/
        "cmdType": {
        /*ro, req, object, signaling type*/
            "@opt": ["request", "cancel", "answer", "reject", "bellTimeout", "hangUp", "deviceOnCall"]
            /*ro, req, array, range, subType:string, range:[0,7], desc:"request" request for call, "cancel"-cancel the call, "answer"-answer the call,
"reject"-reject the call, "bellTimeout"-callee ringing timed out, "hangUp"-end call, "deviceOnCall"-the device is busy*/
        },
        "industryType": {
        /*ro, opt, object*/
            "@opt": ["builidings", "prison", "medicalTreatment", "broadcasting"]
            /*ro, req, array, range, subType:string, range:[0,4]*/
        },
    }
}
```

14.10.1.4 Set the call signaling interaction parameters

Request URL

PUT /ISAPI/VideoIntercom/callSignal?format=json

Query Parameter

None

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Request Message

```
{
    "CallSignal": {
    /*req, object, call signaling interaction parameters*/
        "cmdType":  "request",
        /*req, enum, operation type, subType:string, desc:operation type: "request","cancel","answer","reject","bellTimeout","hangUp","deviceOnCall"*/
        "sessionId":  "test",
        /*opt, string*/
        "src": {
        /*opt, object*/
            "periodNumber":  1,
            /*opt, int, community number*/
            "buildingNumber":  1,
            /*opt, int, building number*/
            "unitNumber":  1,
            /*opt, int, unit number*/
            "floorNumber":  1,
            /*opt, int, floor No.*/
            "roomNumber":  1,
            /*opt, int, room No.*/
            "devIndex":  1,
            /*opt, int, device No.*/
            "communityNumber":  "test",
            /*opt, string, community No.*/
            "callNumber":  "12312312311",
            /*opt, string, range:[1,11]*/
            "unitType":  "indoor",
            /*opt, enum, type, subType:string, desc:device type: "indoor"-indoor station,"villa"-door station (V series),"confirm"-doorphone,"outdoor"-door
station,"fence"-outer door station,"doorbell"-doorbell,"manage"-main station,"acs"-access control device*/
            "personUUID":  "test",
            /*opt, string*/
            "personType":  "student",
            /*opt, enum, person type, subType:string*/
            "industryType":  "builidings",
            /*opt, enum, industry type, subType:string*/
            "callType":  "voice",
            /*opt, enum, subType:string*/
            "model":  "DS-KD9403-A",
            /*opt, string, device model, range:[1,64]*/
            "serialNumber":  "test"
            /*opt, string, device serial No., range:[9,48]*/
        },
        "target": {
        /*opt, object, target information*/
            "periodNumber":  1,
            /*opt, int, community number*/
            "buildingNumber":  1,
            /*opt, int, building number*/
            "unitNumber":  1,
            /*opt, int, unit number*/
            "floorNumber":  1,
            /*opt, int, floor No.*/
            "roomNumber":  1,
            /*opt, int, room No.*/
            "devIndex":  1,
            /*opt, int, device No.*/
            "communityNumber":  "test",
            /*opt, string, community No.*/
            "callNumber":  "12312312311",
            /*opt, string, range:[1,11]*/
            "unitType":  "indoor",
            /*opt, enum, type, subType:string, desc:device type: "indoor"-indoor station,"villa"-door station (V series),"confirm"-doorphone,"outdoor"-door
station,"fence"-outer door station,"doorbell"-doorbell,"manage"-main station,"acs"-access control device*/
            "personUUID":  "test",
            /*opt, string*/
            "personType":  "student",
            /*opt, enum, person type, subType:string*/
            "industryType":  "builidings",
            /*opt, enum, industry type, subType:string*/
            "callType":  "voice",
            /*opt, enum, subType:string*/
            "callWaitingDelayTime":  10,
            /*opt, int, range:[10,60], unit:s*/
            "serialNumber":  "test"
            /*opt, string, device serial No., range:[9,48]*/
        },
        "CallInfo": {
        /*opt, object, call information*/
            "STSPort":  7000,
            /*opt, int*/
            "STSIpV4Address":  "test",
            /*opt, string*/
            "clientID":  "test",
            /*opt, string, client ID, range:[1,32]*/
            "VCPort":  7000,
            /*opt, int*/
            "VCIpV4Address":  "test",
            /*opt, string*/
            "roomID":  1,
            /*opt, int, room No.*/
            "password":  "test",
```

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```
            /*opt, string, password, range:[8,16]*/
            "conferenceMaxCount":  2
            /*opt, int*/
        },
        "messageID":  "test",
        /*opt, string, range:[1,64]*/
        "exceptionInfo":  "test"
        /*opt, string, range:[1,128]*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
}
```

14.10.1.5 Get the capability of getting the call status

Request URL

GET /ISAPI/VideoIntercom/callStatus/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CallStatus": {
    /*ro, opt, object, call status*/
        "status": {
        /*ro, req, object, status*/
            "@opt": ["idle", "ring", "onCall"]
            /*ro, req, array, options, subType:string*/
        }
    }
}
```

14.10.1.6 Get the call status

Request URL

GET /ISAPI/VideoIntercom/callStatus?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CallStatus": {
    /*ro, opt, object, call status*/
        "status":  "idle"
        /*ro, req, enum, status, subType:string, desc:"idle", "ring" (ringing), "onCall" (busy)*/
    }
}
```

14.10.2 Intercom Basic Management

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14.10.2.1 Get the operation time parameters

Request URL

GET /ISAPI/VideoIntercom/operationTime/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OperationTime xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, operation time capability, attr:version{req, string, protocolVersion}-->
  <messageTime min="0" max="120">
    <!--ro, opt, int, max. message duration, attr:min{req, int},max{req, int}-->0
  </messageTime>
  <talkTime min="0" max="120">
    <!--ro, opt, int, max. call duration, attr:min{req, int},max{req, int}-->0
  </talkTime>
  <maxRingTime min="0" max="120">
    <!--ro, opt, int, max. ring duration, attr:min{req, int},max{req, int}-->0
  </maxRingTime>
  <maxMonitoringTime min="0" max="120">
    <!--ro, opt, int, max. live view duration, attr:min{req, int},max{req, int}-->0
  </maxMonitoringTime>
  <callForwardingTime min="0" max="120">
    <!--ro, opt, int, call forwarding duration, attr:min{req, int},max{req, int}-->0
  </callForwardingTime>
  <ringDurationTime min="0" max="120">
    <!--ro, opt, int, ring timeout duration, attr:min{req, int},max{req, int}-->0
  </ringDurationTime>
  <readerID min="1" max="8">
    <!--ro, opt, int, whether it supports authentication unit ID in request parameters, attr:min{req, int, range:[1,8]},max{req, int, range:[1,8]}-->0
  </readerID>
  <requireReaderID>
    <!--ro, opt, bool-->true
  </requireReaderID>
  <callWaitingTime min="40" max="80">
    <!--ro, opt, int, call waiting duration, attr:min{req, int},max{req, int}-->10
  </callWaitingTime>
  <streamTimeout min="0" max="120">
    <!--ro, opt, int, range:[0,120], step:1, unit:s, attr:min{req, int},max{req, int}-->10
  </streamTimeout>
  <configureTimeout min="0" max="120">
    <!--ro, opt, int, range:[0,120], step:1, unit:s, attr:min{req, int},max{req, int}-->10
  </configureTimeout>
</OperationTime>
```

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14.10.2.2 Get the operation time parameters

Request URL

GET /ISAPI/VideoIntercom/operationTime?readerID=<readerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	The intelligent host has multiple card readers and needs to support obtaining by ID.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OperationTime xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, operation time, attr:version{req, string, protocolVersion}-->
  <messageTime>
    <!--ro, opt, int, max. message duration, range:[0,120], unit:s, desc:max. message duration-->0
  </messageTime>
  <talkTime>
    <!--ro, opt, int, max. call duration, range:[0,300], unit:s-->0
  </talkTime>
  <maxRingTime>
    <!--ro, opt, int, max. ring duration, range:[0,3600], unit:s, desc:when the device is being called and not picked up within the max. ring duration, the
call will be declined automatically-->0
  </maxRingTime>
  <maxMonitoringTime>
    <!--ro, opt, int, max. live view duration, range:[0,120], unit:s-->0
  </maxMonitoringTime>
  <callForwardingTime>
    <!--ro, opt, int, call forwarding duration, range:[0,120], unit:s-->0
  </callForwardingTime>
  <ringDurationTime>
    <!--ro, opt, int, ring timeout duration, range:[0,120], unit:s,
desc:1. when the device is being called and not picked up within the configured ring duration, it will stop ringing but is still in calling status
2. the configured ringing timeout should be no longer than the max. ring duration-->0
  </ringDurationTime>
  <callWaitingTime>
    <!--ro, opt, int, call waiting duration, range:[40,80], unit:s, desc:when the device starts a call and it is not picked up by the center (platform or
Web Client) within the configured call waiting duration, the device will end the call-->40
  </callWaitingTime>
  <callWaitingDelayTime>
    <!--ro, opt, int, extended call waiting duration, range:[10,60], unit:s, desc:when the device starts a call, the center (platform) can choose to extend
the call waiting duration, e.g., when the center is busy receiving calls from multiple devices-->10
  </callWaitingDelayTime>
  <streamTimeout>
    <!--ro, opt, int, range:[0,120], step:1, unit:s-->10
  </streamTimeout>
  <configureTimeout>
    <!--ro, opt, int, range:[0,120], step:1, unit:s-->10
  </configureTimeout>
  <returnCallInterfaceTimeout>
    <!--ro, opt, int, range:[0,120], step:1, unit:s-->10
  </returnCallInterfaceTimeout>
  <autoCallTimeout>
    <!--ro, opt, int, range:[0,120], step:1, unit:s-->10
  </autoCallTimeout>
</OperationTime>
```

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14.10.2.3 Set the operation time parameters

Request URL

PUT /ISAPI/VideoIntercom/operationTime?readerID=<readerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OperationTime xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, operation Time, attr:version{req, string, protocolVersion}-->
  <messageTime>
    <!--opt, int, max. message duration, range:[0,120], unit:s-->0
  </messageTime>
  <talkTime>
    <!--opt, int, max. call duration, range:[0,120], unit:s-->0
  </talkTime>
  <maxRingTime>
    <!--opt, int, max. ring duration, range:[0,120], unit:s, desc:when the device is being called and not picked up within the max. ring duration, the call
will be declined automatically-->0
  </maxRingTime>
  <maxMonitoringTime>
    <!--opt, int, max. live view duration, range:[0,120], unit:s-->0
  </maxMonitoringTime>
  <callForwardingTime>
    <!--opt, int, call forwarding time, range:[0,120], unit:s-->0
  </callForwardingTime>
  <ringDurationTime>
    <!--opt, int, ringing timeout, range:[0,120], unit:s,
desc:1. when the device is being called and not picked up within the configured ring duration, it will stop ringing but is still in calling status
2. the configured ringing timeout should be no longer than the max. ring duration-->0
  </ringDurationTime>
  <callWaitingTime>
    <!--opt, int, call waiting duration, range:[40,80], unit:s, desc:when the device starts a call and it is not picked up by the center (platform or Web
Client) within the configured call waiting duration, the device will end the call-->40
  </callWaitingTime>
  <callWaitingDelayTime>
    <!--opt, int, extended call waiting duration, range:[10,60], unit:s, desc:when the device starts a call, the center (platform) can choose to extend the
call waiting duration, e.g., when the center is busy receiving calls from multiple devices-->10
  </callWaitingDelayTime>
  <streamTimeout>
    <!--ro, opt, int, range:[0,120], step:1, unit:s-->10
  </streamTimeout>
  <configureTimeout>
    <!--ro, opt, int, range:[0,120], step:1, unit:s-->10
  </configureTimeout>
</OperationTime>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

14.10.3 Linked Device Management

14.10.3.1 Get the configuration capability of communication parameters of the device network

Request URL

GET /ISAPI/VideoIntercom/deviceCommunication/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DeviceCommunicationCap": {
```

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```
    /*ro, req, object, communication parameters of the device network*/
        "regNumber": {
        /*ro, req, object, registration No.*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  0
            /*ro, req, int, the maximum value*/
        },
        "regPassword": {
        /*ro, req, object, registration password*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  0
            /*ro, req, int, the maximum value*/
        },
        "localPort": {
        /*ro, opt, object, port No. of the local SIP server*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  0
            /*ro, req, int, the maximum value*/
        },
        "ServerAddressList": {
        /*ro, opt, object, service address list*/
            "maxSize":  0,
            /*ro, req, int, maximum number of addresses that can be configured*/
            "id": {
            /*ro, req, object, No.*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  0
                /*ro, req, int, the maximum value*/
            },
            "unitType": {
            /*ro, req, object, type, desc:"indoor" (indoor station), "outdoor" (door station), "confirm" (doorphone), "manage" (main station), "sip"
(private SIP server), "fence" (outer door station), "netAudio" (network sound box), "analogIndoor" (analog indoor station), "decoder", "advancedSip"
(advanced SIP server)*/
                "@opt": ["indoor", "outdoor", "confirm", "manage", "sip", "netAudio", "analogIndoor", "decoder", "advancedSip"]
                /*ro, req, array, options, subType:string*/
            },
            "addressingFormatType": {
            /*ro, req, object, address type, desc:"ipaddress", "hostname"*/
                "@opt": ["ipaddress", "hostname"]
                /*ro, req, array, options, subType:string*/
            },
            "hostName": {
            /*ro, opt, object, domain name*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  0
                /*ro, req, int, the maximum value*/
            },
            "ipAddress": {
            /*ro, opt, object, IPv4 address*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  0
                /*ro, req, int, the maximum value*/
            },
            "ipv6Address": {
            /*ro, opt, object, IPv6 address*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  0
                /*ro, req, int, the maximum value*/
            },
            "portNo": {
            /*ro, opt, object, port No.*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  0
                /*ro, req, int, the maximum value*/
            },
            "registerInterval": {
            /*ro, opt, object, registration period*/
                "@min":  1,
                /*ro, req, int, the minimum value, range:[1,30]*/
                "@max":  30
                /*ro, req, int, the maximum value, range:[1,30]*/
            },
            "registerStatus": {
            /*ro, opt, object, registration status*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool*/
            }
        },
        "deviceID": {
        /*ro, opt, object, device ID*/
            "@min":  1,
            /*ro, req, int, the minimum value, range:[1,6]*/
            "@max":  6
            /*ro, req, int, the maximum value, range:[1,6]*/
        },
        "deviceLocation": {
```

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```
        /*ro, opt, object, device location*/
            "@min":  1,
            /*ro, req, int, the minimum value, range:[1,32]*/
            "@max":  32
            /*ro, req, int, the maximum value, range:[1,32]*/
        },
        "networkMode": {
        /*ro, opt, object, network mode*/
            "@opt": [1, 2, 3]
            /*ro, req, array, options, subType:int*/
        }
    }
}
```

14.10.3.2 Get communication parameters of the device network

Request URL

GET /ISAPI/VideoIntercom/deviceCommunication?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DeviceCommunication": {
    /*ro, req, object, communication parameters of the device network*/
        "regNumber":  "test",
        /*ro, req, string, registration No.*/
        "localPort":  1,
        /*ro, opt, int, port No. of local SIP client*/
        "ServerAddressList": [
        /*ro, opt, array, server address list, subType:object*/
            {
                "id":  1,
                /*ro, req, int, No.*/
                "unitType":  "indoor",
                /*ro, req, enum, type, subType:string, desc:"indoor" (indoor station), "outdoor" (door station), "confirm" (doorphone), "manage" (main
station), "sip" (private SIP server), "villa" (villa door station), "fence" (outer door station), "netAudio" (network sound box), "analogIndoor" (analog
indoor station), "decoder", "advancedSip" (advanced SIP server)*/
                "addressingFormatType":  "ipaddress",
                /*ro, req, enum, address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)*/
                "hostName":  "test",
                /*ro, opt, string, domain name*/
                "ipAddress":  "test",
                /*ro, opt, string, IPv4 address*/
                "ipv6Address":  "test",
                /*ro, opt, string, IPv6 address*/
                "portNo":  0,
                /*ro, opt, int, port No.*/
                "registerInterval":  1,
                /*ro, opt, int, registration period, range:[1,30], unit:min*/
                "registerStatus":  true
                /*ro, opt, bool, registration status, desc:true (registered), false (unregistered)*/
            }
        ],
        "isPasswordConfigured":  true,
        /*ro, opt, bool, whether registration password is configured*/
        "deviceID":  "121000",
        /*ro, opt, string, device ID, range:[1,6], desc:no more than 6 digits and multi-digit ID cannot start from 0; the ID should contain numbers from 0
(by default) to 9*/
        "deviceLocation":  "test",
        /*ro, opt, string, device location, range:[1,32], desc:fill in the actual device location for registration*/
        "networkMode":  1
        /*ro, opt, enum, network mode, subType:int, desc:1-wired NIC 1 (default), 2-wired NIC 2, 3- wireless network*/
    }
}
```

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14.10.3.3 Set communication parameters of the device network

Request URL

PUT /ISAPI/VideoIntercom/deviceCommunication?format=json

Query Parameter

None

Request Message

```
{
    "DeviceCommunication": {
    /*req, object, communication parameters of the device network*/
        "regNumber":  "test",
        /*ro, req, string, registration No.*/
        "regPassword":  "test",
        /*wo, req, string, registration password*/
        "localPort":  1,
        /*opt, int, port No. of local SIP client*/
        "ServerAddressList": [
        /*opt, array, server address list, subType:object*/
            {
                "id":  1,
                /*req, int, No.*/
                "unitType":  "indoor",
                /*req, enum, type, subType:string, desc:"indoor" (indoor station), "outdoor" (door station), "confirm" (doorphone), "manage" (main station),
"sip" (private SIP server), "villa" (villa door station), "fence" (outer door station), "netAudio" (network sound box), "analogIndoor" (analog indoor
station), "decoder", "advancedSip" (advanced SIP server)*/
                "addressingFormatType":  "ipaddress",
                /*req, enum, address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)*/
                "hostName":  "test",
                /*opt, string, domain name*/
                "ipAddress":  "test",
                /*opt, string, IPv6 address*/
                "ipv6Address":  "test",
                /*opt, string, IPv6 address*/
                "portNo":  0,
                /*opt, int, port No.*/
                "registerInterval":  1
                /*opt, int, registration period, range:[1,30], unit:min*/
            }
        ],
        "deviceID":  "121000",
        /*opt, string, device ID, range:[1,6], desc:no more than 6 digits and multi-digit ID cannot start from 0; the ID should contain numbers from 0 (by
default) to 9*/
        "deviceLocation":  "test",
        /*opt, string, device location, range:[1,32], desc:fill in the actual device location for registration*/
        "networkMode":  1
        /*opt, enum, network mode, subType:int, desc:1-wired NIC 1 (default), 2-wired NIC 2, 3- wireless network*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode when statusCode is not 1*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

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14.10.3.4 Set the linked device address (linked network)

Request URL

PUT /ISAPI/VideoIntercom/relatedDeviceAddress

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RelatedDeviceAddress xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <unitType>
    <!--opt, enum, device type, subType:string, desc:device type-->outdoor
  </unitType>
  <MainOutdoorAddress>
    <!--opt, object, address of main door station-->
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </MainOutdoorAddress>
  <SIPServerAddress>
    <!--opt, object, SIP server address-->
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </SIPServerAddress>
  <centerPort>
    <!--opt, int, port No. of center platform, range:[0,65535]-->1
  </centerPort>
  <CenterAddress>
    <!--opt, object, center platform address-->
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </CenterAddress>
  <MainRoomAddress>
    <!--opt, object, IP address of indoor station-->
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </MainRoomAddress>
  <ManageAddress>
    <!--opt, object, IP address of the main station-->
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </ManageAddress>
</RelatedDeviceAddress>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.10.3.5 Get the linked device address (linked network)

Request URL

GET /ISAPI/VideoIntercom/relatedDeviceAddress

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<RelatedDeviceAddress xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <unitType>
    <!--ro, opt, enum, device type, subType:string, desc:device type-->outdoor
  </unitType>
  <MainOutdoorAddress>
    <!--ro, opt, object, address of main door station-->
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </MainOutdoorAddress>
  <SIPServerAddress>
    <!--ro, opt, object, SIP server address-->
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </SIPServerAddress>
  <centerPort>
    <!--ro, opt, int, port No. of center platform, range:[0,65535]-->1
  </centerPort>
  <CenterAddress>
    <!--ro, opt, object, center platform address-->
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </CenterAddress>
  <MainRoomAddress>
    <!--ro, opt, object, IP address of indoor station-->
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </MainRoomAddress>
  <ManageAddress>
    <!--ro, opt, object, IP address of the main station-->
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </ManageAddress>
</RelatedDeviceAddress>
```

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14.10.3.6 Get the configuration capability of the linked device address (linked network)

Request URL

GET /ISAPI/VideoIntercom/relatedDeviceAddress/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<RelatedDeviceAddress xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <unitType opt="outdoor,fence ">
    <!--ro, opt, enum, device type, subType:string, attr:opt{req, string}, desc:device type-->outdoor
  </unitType>
  <MainOutdoorAddress>
    <!--ro, opt, object, address of main door station-->
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, address type, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </MainOutdoorAddress>
  <SIPServerAddress>
    <!--ro, opt, object, SIP server address-->
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, address type, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </SIPServerAddress>
  <centerPort>
    <!--ro, opt, int, port No. of center platform, range:[0,65535]-->0
  </centerPort>
  <CenterAddress>
    <!--ro, opt, object, center platform address-->
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, address type, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </CenterAddress>
  <MainRoomAddress>
    <!--ro, opt, object, IP address of indoor station-->
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, address type, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </MainRoomAddress>
  <ManageAddress>
    <!--ro, opt, object, IP address of main station-->
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, address type, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host domain name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,128]-->255.255.255.255
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
    </ipv6Address>
  </ManageAddress>
</RelatedDeviceAddress>
```

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14.10.4 Sub Module Management

14.10.4.1 Get the configuration capability of pressing the button to call

Request URL

GET /ISAPI/VideoIntercom/keyCfg/<keyID>/capabilities?moduleId=<subModuleID>

Query Parameter

Parameter Name	Parameter Type	Description
keyID	string	--
subModuleID	string	The sub-module ID corresponds to that obtained by sub-module list. If this parameter is not included, configure the main module.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<KeyCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, button No., attr:version{req, string, protocolVersion}-->
  <id min="1" max="10">
    <!--ro, req, int, it corresponds to the <ID> in the request URI, attr:min{req, int},max{req, int}-->1
  </id>
  <module opt="main,sub" def="main">
    <!--ro, opt, string, module to be configured: "main"-main module (default),"sub"-sub module, attr:opt{req, string},def{req, string}-->main
  </module>
  <moduleId min="1" max="10">
    <!--ro, opt, int, sub module ID, attr:min{req, int},max{req, int}-->1
  </moduleId>
  <callMethod opt="callNumber,manageCenter,app,manualCallNumber,callStandardSIPNumber,callCenter" def="manageCenter">
    <!--ro, opt, string, calling method, attr:opt{req, string},def{req, string}-->manageCenter
  </callMethod>
  <callNumber min="1" max="99999">
    <!--ro, opt, string, called number, attr:min{req, int},max{req, int}, desc:called number-->101
  </callNumber>
  <enableCallCenter opt="true,false" def="true">
    <!--ro, opt, bool, whether to call the management center, attr:opt{req, string},def{req, string}-->true
  </enableCallCenter>
  <readerID min="1" max="256">
    <!--ro, opt, int, attr:min{req, int, range:[1,8]},max{req, int, range:[1,8]}-->1
  </readerID>
  <requireReaderID>
    <!--ro, opt, bool-->true
  </requireReaderID>
  <DifferentKeystrokesList size="2">
    <!--ro, opt, object, attr:size{req, int}-->
    <Item>
      <!--ro, opt, object-->
      <keypress opt="short,long">
        <!--ro, req, string, attr:opt{req, string}-->test
      </keypress>
      <callNumber min="1" max="32">
        <!--ro, opt, string, called number, range:[1,32], attr:min{req, int},max{req, int}-->test
      </callNumber>
      <enableCallCenterByLong opt="true,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </enableCallCenterByLong>
    </Item>
  </DifferentKeystrokesList>
  <templateNo min="1" max="32">
    <!--ro, opt, int, range:[1,32], attr:min{req, int},max{req, int}-->1
  </templateNo>
  <customInfo min="0" max="128">
    <!--ro, opt, string, custom information, range:[0,128], attr:min{req, int},max{req, int}-->test
  </customInfo>
  <callType opt="normal,alarm,bedheadTerminal,bedsideTerminal" def="normal">
    <!--ro, opt, string, attr:opt{req, string},def{req, string}-->normal
  </callType>
  <linkageTemplateEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </linkageTemplateEnabled>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <terminalNo>
    <!--ro, opt, object-->
    <periodNumber min="1" max="999">
      <!--ro, opt, int, range:[1,999], attr:min{req, int},max{req, int}-->1
    </periodNumber>
    <buildingNumber min="1" max="999">
      <!--ro, opt, int, range:[1,999], attr:min{req, int},max{req, int}-->1
```

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```
      </buildingNumber>
      <unitNumber min="1" max="999">
        <!--ro, opt, int, range:[1,999], attr:min{req, int},max{req, int}-->1
      </unitNumber>
      <floorNumber min="1" max="999">
        <!--ro, opt, int, range:[1,999], attr:min{req, int},max{req, int}-->1
      </floorNumber>
      <roomNumber min="1" max="999">
        <!--ro, opt, int, range:[1,999], attr:min{req, int},max{req, int}-->1
      </roomNumber>
      <deviceIndex min="1" max="999">
        <!--ro, opt, int, range:[1,999], attr:min{req, int},max{req, int}-->1
      </deviceIndex>
    </terminalNo>
    <customFunctionType opt="preview,call,callElevator,openLock,answerAndReject,callMethod,doorbell,openMainDoorStationLock">
      <!--ro, opt, string, attr:opt{req, string}-->preview
    </customFunctionType>
    <previewParams>
      <!--ro, opt, object-->
      <deviceType opt="ipc,outdoor,confirm">
        <!--ro, req, string, attr:opt{req, string}-->outdoor
      </deviceType>
      <channelID min="1" max="16">
        <!--ro, opt, int, range:[1,16], attr:min{req, int},max{req, int}-->1
      </channelID>
      <deviceIndex min="0" max="99">
        <!--ro, opt, int, range:[0,99], attr:min{req, int},max{req, int}-->1
      </deviceIndex>
    </previewParams>
    <callParams>
      <!--ro, opt, object-->
      <deviceType opt="indoor,manage">
        <!--ro, req, string, attr:opt{req, string}-->indoor
      </deviceType>
      <subDeviceType opt="indoorStation,indoorExtension">
        <!--ro, opt, string, attr:opt{req, string}-->test
      </subDeviceType>
      <periodNumber min="1" max="999">
        <!--ro, opt, int, range:[1,999], attr:min{req, int},max{req, int}-->1
      </periodNumber>
      <buildingNumber min="1" max="999">
        <!--ro, opt, int, range:[1,999], attr:min{req, int},max{req, int}-->1
      </buildingNumber>
      <unitNumber min="1" max="999">
        <!--ro, opt, int, range:[1,999], attr:min{req, int},max{req, int}-->1
      </unitNumber>
      <roomNumber min="1" max="999">
        <!--ro, opt, int, range:[1,999], attr:min{req, int},max{req, int}-->1
      </roomNumber>
      <deviceIndex min="1" max="999">
        <!--ro, opt, int, range:[1,999], attr:min{req, int},max{req, int}-->1
      </deviceIndex>
    </callParams>
    <openLockParams>
      <!--ro, opt, object-->
      <lockID min="1" max="2">
        <!--ro, req, int, range:[1,2], attr:min{req, int},max{req, int}-->1
      </lockID>
    </openLockParams>
  </KeyCfg>
```

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14.10.4.2 Get the parameters of pressing the button to call for a specific button

Request URL

GET /ISAPI/VideoIntercom/keyCfg/<keyID>?readerID=<readerID>&moduleId=<subModuleID>

Query Parameter

Parameter Name	Parameter Type	Description
keyID	string	--
readerID	string	The intelligent host has multiple card readers and needs to support obtaining by ID.
subModuleID	string	The sub-module ID corresponds to that obtained by sub-module list. If this parameter is not included, configure the main module.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<KeyCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, button No., desc:button No.-->1
  </id>
  <module>
    <!--ro, opt, enum, module to be configured, subType:string, desc:"main"(main module, by default), "sub" (sub module)-->main
  </module>
  <moduleId>
    <!--ro, opt, int, sub module ID, dep:and,{$.KeyCfg.module,eq,sub}-->1
  </moduleId>
  <callMethod>
    <!--ro, opt, enum, calling method, subType:string, desc:"callNumber" (call by number), "manageCenter" (call the management center), "app" (call APP),
"manualCallNumber" (manually call by number)-->callNumber
  </callMethod>
  <callNumber>
    <!--ro, opt, string, called number, this field is valid when <callMethod> is "callNumber", dep:or,{$.KeyCfg.callMethod,eq,callNumber}-->101
  </callNumber>
  <enableCallCenter>
    <!--ro, opt, bool, whether to call the management center, dep:or,{$.KeyCfg.callMethod,eq,manageCenter}-->true
  </enableCallCenter>
  <DifferentKeystrokesList>
    <!--ro, opt, object-->
    <Item>
      <!--ro, opt, object-->
      <keyPress>
        <!--ro, req, enum, subType:string-->long
      </keyPress>
      <callNumber>
        <!--ro, opt, string, called number, this field is valid when <callMethod> is "callNumber", range:[1,32]-->test
      </callNumber>
      <enableCallCenterByLong>
        <!--ro, opt, bool-->true
      </enableCallCenterByLong>
    </Item>
  </DifferentKeystrokesList>
  <templateNo>
    <!--ro, opt, int, range:[1,32]-->1
  </templateNo>
  <customInfo>
    <!--ro, opt, string, custom information, range:[0,128]-->test
  </customInfo>
  <callType>
    <!--ro, opt, enum, subType:string-->normal
  </callType>
  <linkageTemplateEnabled>
    <!--ro, opt, bool-->true
  </linkageTemplateEnabled>
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <terminalNo>
    <!--ro, opt, object, dep:and,{$.KeyCfg.callType,eq,bedheadTerminal},{$.KeyCfg.callType,eq,bedsideTerminal}-->
    <periodNumber>
      <!--ro, opt, int, range:[1,999]-->1
    </periodNumber>
    <buildingNumber>
      <!--ro, opt, int, range:[1,999]-->1
    </buildingNumber>
    <unitNumber>
      <!--ro, opt, int, range:[1,999]-->1
    </unitNumber>
    <floorNumber>
      <!--ro, opt, int, range:[1,999]-->1
    </floorNumber>
    <roomNumber>
      <!--ro, opt, int, range:[1,999]-->1
    </roomNumber>
    <deviceIndex>
      <!--ro, opt, int, range:[1,999]-->1
    </deviceIndex>
  </terminalNo>
  <customFunctionType>
    <!--ro, opt, enum, subType:string-->preview
  </customFunctionType>
  <previewParams>
    <!--ro, opt, object, dep:and,{$.KeyCfg.customFunctionType,eq,preview}-->
    <deviceType>
      <!--ro, req, enum, subType:string-->ipc
    </deviceType>
    <channelID>
      <!--ro, opt, int, range:[1,16], dep:and,{$.KeyCfg.previewParams.deviceType,eq,ipc}-->1
    </channelID>
    <deviceIndex>
      <!--ro, opt, int, range:[0,99], dep:and,{$.KeyCfg.previewParams.deviceType,eq,outdoor},{$.KeyCfg.previewParams.deviceType,eq,confirm}-->1
    </deviceIndex>
  </previewParams>
  <callParams>
    <!--ro, opt, object, dep:and,{$.KeyCfg.customFunctionType,eq,call}-->
    <deviceType>
```

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```
        <!--ro, req, enum, subType:string-->indoor
      </deviceType>
      <subDeviceType>
        <!--ro, opt, enum, subType:string, dep:and,{$.KeyCfg.callParams.deviceType,eq,indoor}-->indoorExtension
      </subDeviceType>
      <periodNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </periodNumber>
      <buildingNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </buildingNumber>
      <unitNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </unitNumber>
      <roomNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </roomNumber>
      <deviceIndex>
        <!--ro, opt, int, range:[1,999]-->1
      </deviceIndex>
    </callParams>
    <openLockParams>
      <!--ro, opt, object, dep:and,{$.KeyCfg.customFunctionType,eq,openLock},{$.KeyCfg.customFunctionType,eq,openMainDoorStationLock}-->
      <lockID>
        <!--ro, req, int, range:[1,2]-->1
      </lockID>
    </openLockParams>
    <keyType>
      <!--ro, opt, enum, subType:string-->call
    </keyType>
  </KeyCfg>
```

14.10.4.3 Set the parameters of pressing the button to call for a specific button

Request URL

PUT /ISAPI/VideoIntercom/keyCfg/<keyID>?readerID=<readerID>&moduleId=<subModuleID>

Query Parameter

Parameter Name	Parameter Type	Description
keyID	string	--
readerID	string	--
subModuleID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<KeyCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, button No.-->0
  </id>
  <module>
    <!--opt, enum, module to be configured, subType:string, desc:"main" (main module (default)), "sub" (sub module)-->main
  </module>
  <moduleId>
    <!--opt, int, sub module ID, desc:this node is valid when <module> is "sub" and is used to specify that the button information of which module will be
configured-->1
  </moduleId>
  <callMethod>
    <!--opt, enum, calling method, subType:string, desc:"callNumber" (call by number), "manageCenter" (call the management center), "app" (call APP),
"manualCallNumber" (manually call by number)-->callNumber
  </callMethod>
  <callNumber>
    <!--opt, string, called number, dep:and,{$.KeyCfg.callMethod,eq,callNumber}-->101
  </callNumber>
  <enableCallCenter>
    <!--opt, bool, whether to call the management center, dep:and,{$.KeyCfg.callMethod,eq,manageCenter}-->true
  </enableCallCenter>
  <DifferentKeystrokesList>
    <!--opt, object-->
    <Item>
      <!--opt, object-->
      <keyPress>
        <!--req, enum, subType:string-->long
      </keyPress>
      <callNumber>
        <!--opt, string, called number, range:[1,32], desc:this node is valid when <callMethod> is "callNumber"-->test
      </callNumber>
      <enableCallCenterByLong>
        <!--opt, bool-->true
      </enableCallCenterByLong>
    </Item>
  </DifferentKeystrokesList>
  <templateNo>
    <!--opt, int, range:[1,32]-->1
  </templateNo>
  <customInfo>
    <!--opt, string, custom information, range:[0,128]-->test
  </customInfo>
  <callType>
    <!--opt, enum, subType:string-->normal
  </callType>
</KeyCfg>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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14.10.4.4 Get configuration of all keys

Request URL

GET /ISAPI/VideoIntercom/keyCfg?readerID=<readerID>&type=virtualKey

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	The intelligent host has multiple card readers and needs to support being obtained by ID.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<KeyCfgList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <KeyCfg>
    <!--ro, req, object-->
    <id>
      <!--ro, opt, int-->1
    </id>
    <module>
      <!--ro, opt, enum, subType:string-->main
    </module>
    <moduleId>
      <!--ro, opt, int, dep:and,{$.KeyCfgList.KeyCfg.module,eq,sub}-->1
    </moduleId>
    <callMethod>
      <!--ro, opt, enum, subType:string-->callNumber
    </callMethod>
    <callNumber>
      <!--ro, opt, string, dep:or,{$.KeyCfgList.KeyCfg.callMethod,eq,callNumber}-->101
    </callNumber>
    <enableCallCenter>
      <!--ro, opt, bool, dep:or,{$.KeyCfgList.KeyCfg.callMethod,eq,manageCenter}-->true
    </enableCallCenter>
    <DifferentKeystrokesList>
      <!--ro, opt, array, subType:object-->
      <Item>
        <!--ro, opt, object-->
        <keyPress>
          <!--ro, req, enum, subType:string-->long
        </keyPress>
        <callNumber>
          <!--ro, opt, string, range:[1,32]-->test
        </callNumber>
        <enableCallCenterByLong>
          <!--ro, opt, bool-->true
        </enableCallCenterByLong>
      </Item>
    </DifferentKeystrokesList>
    <templateNo>
      <!--ro, opt, int, range:[1,32]-->1
    </templateNo>
    <customInfo>
      <!--ro, opt, string, range:[0,128]-->test
    </customInfo>
    <callType>
      <!--ro, opt, enum, subType:string-->normal
    </callType>
    <linkageTemplateEnabled>
      <!--ro, opt, bool-->true
    </linkageTemplateEnabled>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <terminalNo>
      <!--ro, opt, object, dep:and,{$.KeyCfgList.KeyCfg.callType,eq,bedheadTerminal},{$.KeyCfgList.KeyCfg.callType,eq,bedsideTerminal}-->
      <periodNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </periodNumber>
      <buildingNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </buildingNumber>
      <unitNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </unitNumber>
      <floorNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </floorNumber>
      <roomNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </roomNumber>
      <deviceIndex>
        <!--ro, opt, int, range:[1,999]-->1
      </deviceIndex>
    </terminalNo>
    <customFunctionType>
```

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```
      <!--ro, opt, enum, subType:string-->preview
    </customFunctionType>
    <previewParams>
      <!--ro, opt, object, dep:and,{$.KeyCfgList.KeyCfg.customFunctionType,eq,preview}-->
      <deviceType>
        <!--ro, req, enum, subType:string-->ipc
      </deviceType>
      <channelID>
        <!--ro, opt, int, range:[1,16], dep:and,{$.KeyCfgList.KeyCfg.previewParams.deviceType,eq,ipc}-->1
      </channelID>
      <deviceIndex>
        <!--ro, opt, int, range:[0,99], dep:and,{$.KeyCfgList.KeyCfg.previewParams.deviceType,eq,outdoor},
{$.KeyCfgList.KeyCfg.previewParams.deviceType,eq,confirm}-->1
      </deviceIndex>
    </previewParams>
    <callParams>
      <!--ro, opt, object, dep:and,{$.KeyCfgList.KeyCfg.customFunctionType,eq,call}-->
      <deviceType>
        <!--ro, req, enum, subType:string-->indoor
      </deviceType>
      <subDeviceType>
        <!--ro, opt, enum, subType:string, dep:and,{$.KeyCfgList.KeyCfg.callParams.deviceType,eq,indoor}-->indoorExtension
      </subDeviceType>
      <periodNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </periodNumber>
      <buildingNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </buildingNumber>
      <unitNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </unitNumber>
      <roomNumber>
        <!--ro, opt, int, range:[1,999]-->1
      </roomNumber>
      <deviceIndex>
        <!--ro, opt, int, range:[1,999]-->1
      </deviceIndex>
    </callParams>
    <openLockParams>
      <!--ro, opt, object, dep:and,{$.KeyCfgList.KeyCfg.customFunctionType,eq,openLock},
{$.KeyCfgList[*].KeyCfg.customFunctionType,eq,openMainDoorStationLock}-->
      <lockID>
        <!--ro, req, int, range:[1,2]-->1
      </lockID>
    </openLockParams>
    <keyType>
      <!--ro, opt, enum, subType:string-->call
    </keyType>
  </KeyCfg>
</KeyCfgList>
```

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14.10.5 Target Management Calling

14.10.5.1 Get the configuration capability of the video intercom device ID

Request URL

GET /ISAPI/VideoIntercom/deviceId/capabilities?unitType=<unitType>

Query Parameter

Parameter Name	Parameter Type	Description
unitType	enum	false

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceId xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device No.: 0-main station, other value-sub station. The device will reboot no matter <deviceIndex> is changed from 0 to other value
or from other value to 0, attr:version{req, string, protocolVersion}-->
  <unitType
opt="villa,confirm,outdoor,fence,doorbell,manage,indoor,indoorStation,indoorExtension,analogIndoor,decoder,netAudio,amplifier,pagingMicrophone,interactive,b
edheadTerminal,bedsideTerminal,doorTerminal,visitTerminal,mobileInterative">
    <!--ro, req, string, device type: "villa"-door station (V series), "confirm"-doorphone, "outdoor"-door station, "fence"-outer door station, "doorbell"-
doorbell, "manage"-main station, "acs"-intelligent access control. If this node is set to "confirm", the following nodes do not need to be configured,
attr:opt{req, string}-->test
  </unitType>
  <periodNumber min="1" max="999">
    <!--ro, opt, int, community number, attr:min{req, int},max{req, int}-->0
```

```
    </periodNumber>
    <buildingNumber min="1" max="999">
      <!--ro, opt, int, building number, attr:min{req, int},max{req, int}-->0
    </buildingNumber>
    <unitNumber min="1" max="999">
      <!--ro, opt, int, unit number, attr:min{req, int},max{req, int}-->0
    </unitNumber>
    <floorNumber min="1" max="999">
      <!--ro, opt, int, floor No., attr:min{req, int},max{req, int}-->0
    </floorNumber>
    <roomNumber min="1" max="999">
      <!--ro, opt, int, room No., attr:min{req, int},max{req, int}-->0
    </roomNumber>
    <deviceIndex min="0" max="999">
      <!--ro, opt, int, device No., attr:min{req, int},max{req, int}-->0
    </deviceIndex>
    <readerDeviceIndex min="0" max="99">
      <!--ro, opt, int, step:1, attr:min{req, int},max{req, int}-->0
    </readerDeviceIndex>
    <communityNumber min="0" max="32">
      <!--ro, opt, string, community No., attr:min{req, int},max{req, int}-->test
    </communityNumber>
    <customDeviceId min="0" max="32">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </customDeviceId>
    <periodNumberName min="0" max="32">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </periodNumberName>
    <buildingNumberName min="0" max="32">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </buildingNumberName>
    <unitNumberName min="0" max="32">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </unitNumberName>
    <floorNumberName min="0" max="32">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </floorNumberName>
    <roomNumberName min="0" max="32">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </roomNumberName>
    <deviceIndexName min="0" max="32">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </deviceIndexName>
    <communityNumberName min="0" max="32">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </communityNumberName>
    <enabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </enabled>
    <industryType opt="buildings,prison,medicalTreatment,broadcasting">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </industryType>
    <readerID min="1" max="8">
      <!--ro, opt, int, range:[1,8], attr:min{req, int, range:[1,8]},max{req, int, range:[1,8]}-->1
    </readerID>
    <requireReaderID>
      <!--ro, opt, bool-->true
    </requireReaderID>
    <isSupportZeroFloorNumber>
      <!--ro, opt, bool-->true
    </isSupportZeroFloorNumber>
    <periodNumberRealName min="0" max="64">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </periodNumberRealName>
    <buildingNumberRealName min="0" max="32">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </buildingNumberRealName>
    <unitNumberRealName min="0" max="32">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </unitNumberRealName>
    <roomNumberRealName min="0" max="32">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </roomNumberRealName>
    <deviceIndexRealName min="0" max="32">
      <!--ro, opt, string, range:[0,32], attr:min{req, int},max{req, int}-->test
    </deviceIndexRealName>
  </DeviceId>
```

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14.10.5.2 Get the video intercom device ID

Request URL

GET /ISAPI/VideoIntercom/deviceId?readerID=<readerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	The intelligent host has multiple card readers, which need to support configuration by ID. When the readerID does not exist, it indicates configuration for the host. The host supports SIP service and SIP client, and the SIP client is used for registration to the management host.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceId xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device No., attr:version{req, string, protocolVersion}-->
  <unitType>
    <!--ro, req, enum, device type, subType:string, desc:"villa"-door station (V series), "confirm"-doorphone, "outdoor"-door station, "fence"-outer door station, "doorbell"-doorbell, "manage"-main station, "acs"-intelligent access control. If this node is set to "confirm", the following nodes do not need to be configured-->villa
  </unitType>
  <periodNumber>
    <!--ro, opt, int, community number, range:[1,999]-->1
  </periodNumber>
  <buildingNumber>
    <!--ro, opt, int, building number, range:[1,999]-->1
  </buildingNumber>
  <unitNumber>
    <!--ro, opt, int, unit number, range:[1,999]-->1
  </unitNumber>
  <floorNumber>
    <!--ro, opt, int, floor No., it is between -10 and 10 and it cannot be 0, range:[1,999], desc:floor No., it is between -10 and 10 and it cannot be 0-->1
  </floorNumber>
  <roomNumber>
    <!--ro, opt, int, room No., range:[1,999]-->1
  </roomNumber>
  <deviceIndex>
    <!--ro, opt, int, device No., 0-main station, other value (from 0 to 99)-sub station. The device will reboot no matter <deviceIndex> is changed from 0 to other value or from other value to 0, range:[0,999]-->1
  </deviceIndex>
  <communityNumber>
    <!--ro, opt, string, community No.-->test
  </communityNumber>
  <customDeviceId>
    <!--ro, opt, string-->test
  </customDeviceId>
  <periodNumberName>
    <!--ro, opt, string, range:[0,32]-->test
  </periodNumberName>
  <buildingNumberName>
    <!--ro, opt, string, range:[0,32]-->test
  </buildingNumberName>
  <unitNumberName>
    <!--ro, opt, string, range:[0,32]-->test
  </unitNumberName>
  <floorNumberName>
    <!--ro, opt, string, range:[0,32]-->test
  </floorNumberName>
  <roomNumberName>
    <!--ro, opt, string, range:[0,32]-->test
  </roomNumberName>
  <deviceIndexName>
    <!--ro, opt, string, range:[0,32]-->test
  </deviceIndexName>
  <communityNumberName>
    <!--ro, opt, string, range:[0,128]-->test
  </communityNumberName>
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <industryType>
    <!--ro, opt, enum, subType:string-->builidings
  </industryType>
  <periodNumberRealName>
    <!--ro, opt, string, range:[0,64]-->test
  </periodNumberRealName>
  <buildingNumberRealName>
    <!--ro, opt, string, range:[0,32]-->test
  </buildingNumberRealName>
  <unitNumberRealName>
    <!--ro, opt, string, range:[0,32]-->test
  </unitNumberRealName>
  <roomNumberRealName>
    <!--ro, opt, string, range:[0,32]-->test
  </roomNumberRealName>
  <deviceIndexRealName>
    <!--ro, opt, string, range:[0,32]-->test
  </deviceIndexRealName>
</DeviceId>
```

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14.10.5.3 Set the video intercom device ID

Request URL

PUT /ISAPI/VideoIntercom/deviceId?readerID=<readerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceId xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, device No., attr:version{req, string, protocolVersion}-->
  <unitType>
    <!--req, enum, device type, subType:string, desc:device type-->villa
  </unitType>
  <periodNumber>
    <!--opt, int, community number-->0
  </periodNumber>
  <buildingNumber>
    <!--opt, int, building number-->0
  </buildingNumber>
  <unitNumber>
    <!--opt, int, unit number-->0
  </unitNumber>
  <floorNumber>
    <!--opt, int, floor No., desc:floor No.-->0
  </floorNumber>
  <roomNumber>
    <!--opt, int, room No.-->0
  </roomNumber>
  <deviceIndex>
    <!--opt, int, device No.-->0
  </deviceIndex>
  <communityNumber>
    <!--opt, string, community No.-->test
  </communityNumber>
  <customDeviceId>
    <!--opt, string-->test
  </customDeviceId>
  <periodNumberName>
    <!--opt, string, range:[0,32]-->test
  </periodNumberName>
  <buildingNumberName>
    <!--opt, string, range:[0,32]-->test
  </buildingNumberName>
  <unitNumberName>
    <!--opt, string, range:[0,32]-->test
  </unitNumberName>
  <floorNumberName>
    <!--opt, string, range:[0,32]-->test
  </floorNumberName>
  <roomNumberName>
    <!--opt, string, range:[0,32]-->test
  </roomNumberName>
  <deviceIndexName>
    <!--opt, string, range:[0,32]-->test
  </deviceIndexName>
  <communityNumberName>
    <!--opt, string, range:[0,128]-->test
  </communityNumberName>
  <enabled>
    <!--opt, bool-->true
  </enabled>
  <industryType>
    <!--opt, enum, subType:string-->builidings
  </industryType>
</DeviceId>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.11 Security Control Device (General)

14.11.1 Event Message Push Management

14.11.1.1 Get the parameters of the report uploading method

Request URL

GET /ISAPI/SecurityCP/ReportCenterCfg/<centerID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
centerID	string	--

Request Message

None

Response Message

```
{
    "ReportCenterCfg": {
    /*ro, req, object*/
        "enable":  true,
        /*ro, opt, bool, whether to enable the function*/
        "ChanAlarmMode": [
        /*ro, opt, array, alarm channel of the center group, subType:object*/
            {
                "id":  1,
                /*ro, opt, enum, channel ID, subType:int, desc:channel ID: 1-main channel,2-backup channel 1,3-backup channel 2,4-backup channel 3*/
                "chanAlarmMode":  "T1"
                /*ro, opt, enum, alarm channel mode, subType:string, desc:"T1" (T1 channel), "T2" (T2 channel), "N1" (N1 channel), "N2" (N2 channel), "G1"
(G1 channel), "G2" (G2 channel), "N3" ()N3 channel, ”N4" (N4 channel)*/
            }
        ]
    }
}
```

14.11.1.2 Set the parameters of the report uploading method

Request URL

PUT /ISAPI/SecurityCP/ReportCenterCfg/<centerID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
centerID	string	--

Request Message

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```
{
    "ReportCenterCfg": {
    /*req, object*/
        "enable":  true,
        /*opt, bool, whether to enable uploading report*/
        "ChanAlarmMode": [
        /*opt, array, alarm channel of the center group, subType:object*/
            {
                "id":  1,
                /*opt, enum, channel ID, subType:int, desc:1 (main channel), 2 (backup channel 1), 3 (backup channel 2), 4 (backup channel 3)*/
                "chanAlarmMode":  "T1"
                /*opt, enum, alarm channel mode, subType:string, desc:"T1" (T1 channel), "T2" (T2 channel), "N1" (N1 channel), "N2" (N2 channel), "G1" (G1
channel), "G2" (G2 channel), "N3" ()N3 channel, ”N4" (N4 channel)*/
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

14.11.1.3 Get the configuration capability of the report uploading method

Request URL

GET /ISAPI/SecurityCP/ReportCenterCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "ReportCenterCfg": {
    /*ro, req, object*/
        "CenterID": {
        /*ro, opt, object, center group No.*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  1
            /*ro, opt, int, the maximum value*/
        },
        "enable":  "true,false",
        /*ro, opt, string, whether to enable uploading report*/
        "ChanAlarmMode": {
        /*ro, opt, object, alarm channel of the center group*/
            "maxSize":  1,
            /*ro, opt, int, the maximum number of channels*/
            "id": {
            /*ro, opt, object, channel ID, desc:1 (main channel), 2 (backup channel 1), 3 (backup channel 2), 4 (backup channel 3)*/
                "@min":  1,
                /*ro, opt, int, the minimum value*/
                "@max":  2
                /*ro, opt, int, the maximum value*/
            },
            "chanAlarmMode": {
            /*ro, opt, object, alarm channel mode, desc:"T1" (T1 channel), "T2" (T2 channel), "N1" (N1 channel), "N2" (N2 channel), "G1" (G1 channel), "G2"
(G2 channel), "N3" (N3 channel), "N4" (N4 channel)*/
                "@opt":  "T1,T2,N1,N2,G1,G2,N3,N4"
                /*ro, opt, string*/
            }
        }
    }
}
```

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14.12 Zone Alarm

14.12.1 Security Control Panel Management

14.12.1.1 Get the capability of security control panel

Request URL

GET /ISAPI/SecurityCP/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "SecurityCPCap": {
    /*ro, req, object, capability node*/
        "partitionNum": 64,
        /*ro, opt, int, number of partitions that can be set, desc:1 by default*/
        "localZoneNum": 16,
        /*ro, req, int, number of local zones*/
        "extendZoneNum": 64,
        /*ro, opt, int, number of extended zones*/
        "wirelessZoneNum": 64,
        /*ro, opt, int, number of wireless zones*/
        "localRelayNum": 1,
        /*ro, req, int, number of local triggers*/
        "extendRelayNum": 1,
        /*ro, opt, int, number of extended triggers*/
        "wirelessRelayNum": 1,
        /*ro, opt, int, number of wireless triggers*/
        "repeater": 4,
        /*ro, opt, int, number of repeaters*/
        "sirenNum": 8,
        /*ro, opt, int, number of sounders*/
        "userNum": 1,
        /*ro, req, int, number of users*/
        "adminNum": 2,
        /*ro, opt, int, number of administrators*/
        "installerNum": 21,
        /*ro, opt, int, number of installers*/
        "operatorNum": 41,
        /*ro, opt, int, number of operators*/
        "ARCNum": 4,
        /*ro, opt, int, number of alarm receiving centers*/
        "phoneNum": 3,
        /*ro, opt, int, number of phone numbers*/
        "outputModNum": 8,
        /*ro, opt, int, number of output modules*/
        "cardNum": 64,
        /*ro, opt, int, number of cards*/
        "keypadNum": 8,
        /*ro, opt, int, number of keypads*/
        "cardReaderNum": 8,
        /*ro, opt, int, number of card readers*/
        "protocolOptimizationVersion": "v1.0",
        /*ro, opt, string, supported protocol optimizing version*/
        "remoteCtrlNum": 64,
        /*ro, opt, int, number of remote controls*/
        "alarmLampNum": 8,
        /*ro, opt, int, number of strobe lights*/
        "electricLockNum": 64,
        /*ro, opt, int, number of electric locks*/
        "detectorModel": {
        /*ro, opt, object, supported detector models*/
            "@opt": ["0x00001", "0x00002"]
            /*ro, opt, array, options, subType:string*/
        },
        "sirenModel": {
        /*ro, opt, object, supported sounder models*/
            "@opt": ["0x7A001", "0x7A011"]
            /*ro, opt, array, options, subType:string*/
        },
        "keypadModel": {
        /*ro, opt, object, supported keypad models*/
            "@opt": ["0x92000", "0x92010"]
            /*ro, opt, array, options, subType:string*/
        },
        "cardReaderModel": {
        /*ro, opt, object, supported card reader models*/
            "@opt": ["0x90000", "0x90010"]
            /*ro, opt, array, options, subType:string*/
```

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```
        },
        "remoteCtrlModel": {
        /*ro, opt, object, supported remote control models*/
            "@opt": ["0x81000", "0x81010"]
            /*ro, opt, array, options, subType:string*/
        },
        "repeaterModel": {
        /*ro, opt, object, supported repeater models*/
            "@opt": ["0x80000", "0x80010"]
            /*ro, opt, array, options, subType:string*/
        },
        "outputModuleModel": {
        /*ro, opt, object, supported output module models*/
            "@opt": ["0x71001", "0x71011"]
            /*ro, opt, array, options, subType:string*/
        },
        "isSptLogSearch": true,
        /*ro, opt, bool, whether it supports log search*/
        "isSptLocalUser": true,
        /*ro, opt, bool, whether it supports local users*/
        "isSptConfiguration": true,
        /*ro, opt, bool, whether it supports configuring security control panel, desc:/ISAPI/SecurityCP/Configuration*/
        "isSptControl": true,
        /*ro, opt, bool, whether it supports security control panel control, desc:/ISAPI/SecurityCP/control*/
        "isSptStatus": true,
        /*ro, opt, bool, whether it supports monitoring security control panel status, desc:/ISAPI/SecurityCP/status*/
        "isSptZoneInfo": true,
        /*ro, opt, bool, whether it supports getting the information of a specific zone, desc:/ISAPI/SecurityCP/Info/zones/<ID>?format=json*/
        "isSptSirenInfo": true,
        /*ro, opt, bool, whether it supports getting the information of a specific sounder, desc:/ISAPI/SecurityCP/Info/siren/<ID>?format=json*/
        "isSptPaceTest": true,
        /*ro, opt, bool, whether it supports pacing, desc:/ISAPI/SecurityCP/paceTest*/
        "isSptStandardCfg": true,
        /*ro, opt, bool, whether it supports standard configuration for security control panel, desc:/ISAPI/SecurityCP/standardCfg*/
        "isSptPircamCapture": true,
        /*ro, opt, bool, whether it supports pircam (detector equipped with camera) capture, desc:/ISAPI/SecurityCP/pircam/channels/<ID>/picture?
format=json*/
        "isSptManualControlCapture": true,
        /*ro, opt, bool*/
        "isSptGetPictureByURL": true,
        /*ro, opt, bool*/
        "isSptOneKeyAlarm": true,
        /*ro, opt, bool, whether it supports one-push alarm, desc:/ISAPI/SecurityCP/control/oneKeyAlarm. For compatibility, this node will also be returned
in the capability message JSON_HostControlCap after calling the URI /ISAPI/SecurityCP/control/capabilities?format=json by GET method*/
        "isSptOneKeyAlarmSoundCtrl": true,
        /*ro, opt, bool*/
        "transmitterNum": 8,
        /*ro, opt, int, the number of transmitters*/
        "transmitterModel": {
        /*ro, opt, object, supported model of transmitters*/
            "@opt": ["0x71001"]
            /*ro, opt, array, options, subType:string*/
        },
        "localAccessModuleType": {
        /*ro, opt, object, onboard access module type*/
            "@opt": ["localTransmitter", "localZone", "localRelay", "localSiren"]
            /*ro, opt, array, options, subType:string, desc:"localTransmitter" (onboard transmitter), "localZone" (onboard zone module), "localRelay"
(onboard relay module), "localSiren" (onboard sounder module)*/
        },
        "networkZoneModuleNum": 8
        /*ro, opt, int, number of network zone modules*/
    }
}
```

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14.12.1.2 Get the configuration capability of security control panel

Request URL

GET /ISAPI/SecurityCP/Configuration/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "HostConfigCap": {
    /*ro, req, object, capability node*/
        "ExDevice": {
        /*ro, opt, object, peripheral node*/
            "isSptOutput":  true,
            /*ro, opt, bool, whether it supports relay management, desc:/ISAPI/SecurityCP/Configuration/outputs*/
        },
        "isSptReportCenterCfg":  true,
        /*ro, opt, bool, whether it supports configuring the report uploading method, desc:/ISAPI/SecurityCP/ReportCenterCfg/<ID>?format=json*/
        "isSptAlarmOutCfg":  true,
        /*ro, opt, bool, whether it supports configuring alarm output parameters, desc:/ISAPI/SecurityCP/AlarmOutCfg/<ID>?format=json*/
        "isSptSetAlarmHostOut":  true,
        /*ro, opt, bool, whether it supports setting alarm output, desc:/ISAPI/SecurityCP/SetAlarmHostOut?format=json*/
    }
}
```

15 How-To Video Guidance

If you need access to corresponding video guidance for device integration, please register on https://tpp.hikvision.com and visit our Training Center: https://tpp.hikvision.com/tpp/Training. The Training Center is specifically designed to provide technical training and guidance resources for our partners. On this platform, you can find integration video tutorials for various devices, enabling better understanding and learning of the integration process. To offer more personalized service, our Training Center also supports filtering by integration protocols, devices, and applications.

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